Financial Modeling

Simon Benninga
and Tal Mofkadi

fifth edition

Uses **EXCEL** and **R**

FINANCIAL MODELING

Simon Benninga and Tal Mofkadi

FIFTH EDITION

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d_r0

This book is dedicated first and foremost to the memory of Simon Benninga, who is considered by many to be the father of financial modeling. Simon Benninga was professor of finance and director of the Sofaer International MBA program at the Faculty of Management at Tel Aviv University. For many years he was a visiting professor at the Wharton School of the University of Pennsylvania. In addition to his many academic achievements, Simon was a wonderful teacher and a kind and generous man. His knowledge, warmth, and humor inspired his students, colleagues, finance professionals, and countless others throughout the years. There is no doubt that his passion for life and finance spread like fire and caught many curious minds. May his fire never go out.

This book is also dedicated to the Benninga family, Terry, Noah, Sara, and Zvi, and to the Mofkadi family, Lizzie, Danielle, Daphne, and Emma.

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Preface and Acknowledgments

The four previous editions of *Financial Modeling* have established this book as a "must have" for finance academics and professionals. Its straightforward, hands-on approach, mixing explanation and implementation using Excel and R, has fulfilled a need in both academic and practitioner markets for readers who realize that the implementation of the finance basics typically studied in an introductory course requires another, more heavily computational and implementational approach. Excel, the most widely used computational tool in finance, is a natural vehicle for deepening our understanding of the materials, while R and Python are the programming languages most commonly used by academics and practitioners.

We are proud of the success of the previous editions of *Financial Modeling*, and we have done our best to update and improve the fifth edition, making it (or hoping to make it) even better. Users of previous editions of this book will find the fifth edition similar in style while better addressing the needs and tools required today because of advances in financial analysis.

Since Excel is considered the best way to understand implementation of financial theory, we address the limitations of Excel (mainly scalability and efficiency with big data) with two approaches: the first one is with Visual Basic for Applications (VBA) code, which can be implemented in Excel (VBA is Excel's programming language), and the second one is with R. All data-intensive chapters now show an implementation in R, and on the auxiliary website accompanying this book you will find Python implementations of the materials.

In addition, we have updated all examples in the book with current data to make them more relevant. Advancements in finance are also included, such as second-order and third-order Greeks for options and Monte Carlo methods.

Financial Modeling consists of seven sections. Each of the first five sections of the book presents a specific area of finance. These sections are independent of each other, though the reader should realize that they all assume some familiarity with the finance area. *Financial Modeling* is not an introductory text.

Section I (Chapters 1–6) deals with corporate finance topics. Section II (Chapters 7–9) covers bond and yield curve models. Section III (Chapters 10–15) is about portfolio theory. Section IV (Chapters 16–20) addresses options and derivatives topics, and Section V (Chapters 21–27) presents Monte Carlo methods and their implementation in finance.

The last two sections of *Financial Modeling* are technical in nature. Section VI (Chapters 28–33) relates to various Excel and R topics that are used throughout the book. Chapters in this section can be read and accessed as necessary. Section VII (Chapters 34–37) was written by Benjamin Czaczkes and deals with VBA. To make the book more streamlined, this section can be found on the auxiliary website accompanying this book. The book is structured such that the reader can understand the materials in all of the other chapters of *Financial Modeling* without needing the VBA or R chapters.

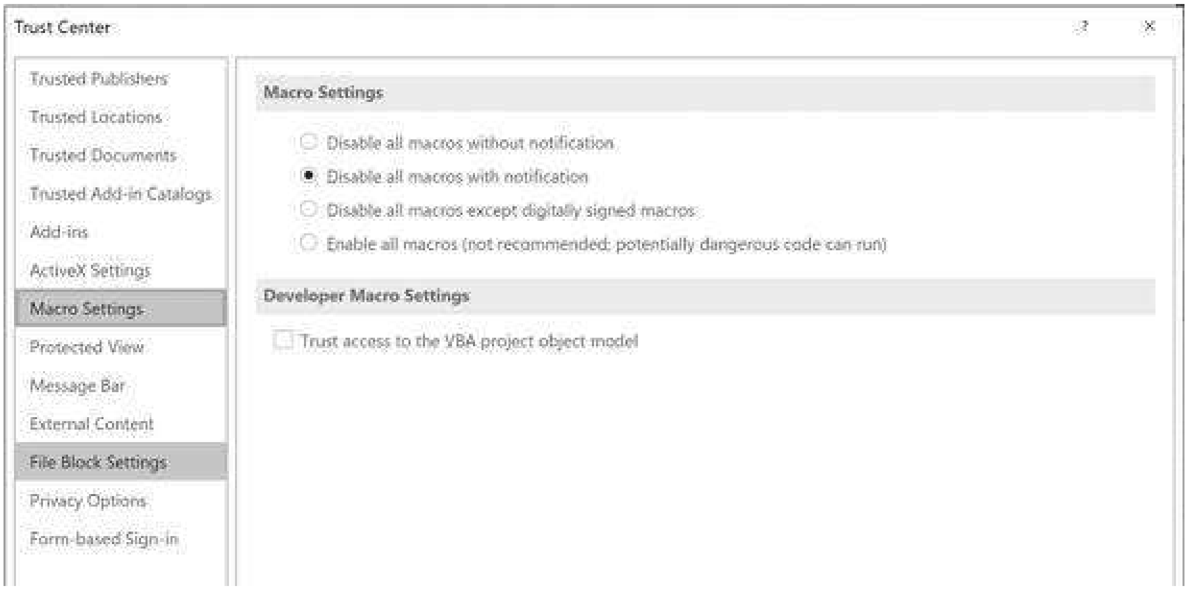
New Materials and Updates

This edition of *Financial Modeling* contains much new and updated material. We have already mentioned the introduction of R and Python. All of the chapters have been reviewed and the materials tweaked where they could be improved. Chapters 2 ("Corporate Valuation Overview"), 19 ("Option Greeks"), 25 ("Value at Risk [VaR]"), and 27 ("Using Monte Carlo Methods for Option Pricing") have been completely revised.

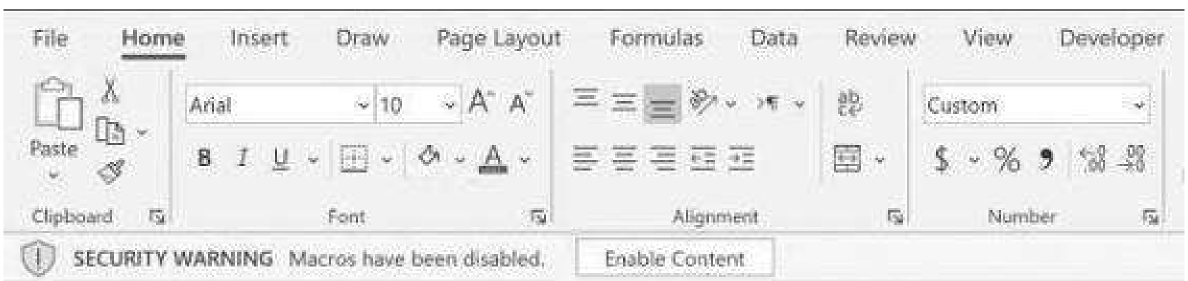
Getformula and Formulatext

The Excel files with this edition include two equivalent functions, called **Getformula** and **Formulatext**, that enable the user to track cell contents. The two functions are discussed in Chapter 0 ("Before All Else"). **Getformula** is a user-defined function (using VBA), and in order for it to work, you will need to enable it. To do so, go to **File|Options|Trust Center**:

In the **Trust Center Settings**, we recommend the following setting:



If you select the settings as shown, then when opening an Excel notebook for the first time you will be met by the following warning:



For notebooks that come with this book, you can safely click **Enable Content** so that you can use the formulas on the notebook.

Excel Versions

Throughout this book, we have used Excel 365 (2019 edition). To the best of our knowledge, all spreadsheets will work in Excel versions as varied as Excel 2007, 2010, and 2013, although some minor and obvious adaptation by the reader may be called for.

Auxiliary Website

Purchasers of *Financial Modeling* get access to all the Excel files for the chapters and exercises, Python implementation, VBA section, presentations, and other relevant materials through the auxiliary website that accompanies this book.

Using *Financial Modeling* in a University Course

Financial Modeling has become the book of choice in many advanced finance classes that emphasize a combination of modeling skills like Excel, R, and Python and a deeper understanding of the underlying financial models. *Financial Modeling* is often used in third- or fourth-year undergraduate or second-year MBA courses. These courses can vary greatly and include much instructor-specific input, but they seem to have a few general features:

- • A typical course starts with two or three classes that stress the skills needed for financial modeling (Excel / R / Python). Often these courses are held when students are using their laptops and implementing the material alongside the professor. Though almost all business school students know Excel, they often do not know the finesses of data tables (Chapter 28), some of the basic financial functions (Chapters 1 and 30), and array functions (Chapter 31).
- • Most one-semester courses cover at most one of the *Financial Modeling* sections. If we assume that in a typical university course, covering one chapter per week is an upper limit (and many chapters will require two weeks), then a typical course might concentrate on either corporate finance (Chapters 1–6); bonds, yield curve models, and risky debt (Chapters 7–9); portfolio theory (Chapters 10–15); options and derivatives (Chapters 16–20); or Monte Carlo methods (Chapters 21–27).

A major problem with a computer-based course is how to structure the final examination. Two solutions seem to work well. One alternative is to have students (whether alone or in teams) submit a final project; examples might be a corporate valuation if the course is based on Section I of the book, an event study from Section III, an option-based project from Section IV, or the computation of a complex derivative value using Monte Carlo approach (Section V). A second alternative is to have students submit, by e-mail, a spreadsheet-based examination with severe time limits. One instructor using this book sends his class the final exam (a compendium of spreadsheet problems) at 9:00 in the morning and requires an e-mail with a spreadsheet answer by noon.

Acknowledgments

We thank a number of people who have made materially significant comments to this edition:

Michael Antel, Shlomi Ben-Yehuda, Rafi Eldor, Iddo Eliazar, Aharon (Roni) Ofer, Roy Shalem, Itay Sharony. We would also like to thank our reviewers from MIT Press for excellent suggestions, comments and improvements. Special thanks to Sagi Haim, who was in charge of compiling the implementation of this book on R and Python.

Finally, we would like to thank our copy editors, Karen Brogno and John Donohue of Westchester Publishing Services, who have been unfailingly helpful and patient.

Disclaimer

The materials in this book are intended for instructional and educational purposes only, to illustrate situations similar to those encountered in the real world. They may not directly apply to real-world situations. The author and MIT Press disclaim any responsibility for the consequences of implementation.

From the Preface to the Fourth Edition

The three previous editions of *Financial Modeling* have received a gratifyingly positive response from readers. The combination of a “cookbook,” mixing explanation and implementation using Excel, has fulfilled a need in both the academic and the practitioner market from readers who realize that the implementation of the finance basics typically studied in an introductory finance course requires another, more heavily computational and implementational, approach. Excel, the most widely used computational tool in finance, is a natural vehicle for deepening our understanding of the materials.

In this fourth edition of *Financial Modeling*, I have added a section (Chapters 24–30) on Monte Carlo methods. The intention is to add a focus on the simulation of financial models. I have become convinced that a statistical understanding of modeling (“What is the mean and sigma of the portfolio return?”) understates the impact of the uncertainty. Only by simulating the models and the return processes can we get a good feel for the dimensions of the uncertainty.

With the added section on Monte Carlo, *Financial Modeling* now consists of seven sections. Each of the first five sections of the book relates to a specific area of finance. These sections are independent of each other, though the reader should realize that they all assume some familiarity with the finance area—*Financial Modeling* is not an introductory text. Section I (Chapters 1–7) deals with corporate finance topics; Section II (Chapters 8–14) with portfolio models; Section III (Chapters 15–19) with option models; and Section IV (Chapters 20–23) with bond-related topics. Section V, as discussed above, introduces the reader to Monte Carlo methods in finance.

The last two sections of *Financial Modeling* are technical in nature. Section VI (Chapters 31–35) relates to various Excel topics which are used throughout the book. Chapters in this Section VI can be read and accessed as necessary. Section VII (Chapters 36–40) deals with Excel’s programming language, Visual Basic for Applications (VBA). VBA is used throughout *Financial Modeling* to create functions and routines which make life easier, but it is never intrusive—in principle the reader can understand the materials in all of the other chapters of *Financial Modeling* without needing the VBA chapters.

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I thank a number of people have made materially significant comments to this edition:

Meni Abudy, Zvika Afik, Javierma Bedoya, Lisa Bergé, Elizabeth Caulk, Sharon Garyn-Tal, Victor Lampe, Jongdoo Lee, Erez Levy, Warren Miller, Tal Mofkadi, Roger

Myerson, Siddhartha Sarkar, Maxim Sharov, Permjit Singh, Sondre Aarseth Skjerven, Alexander Suhov, Kien-Quoc Van Pham, Chao Wang, Tim Wuu.

Finally, I would like to thank: my editor John Covell of MIT Press, Ellen Faran, the Director of MIT Press, and Nancy Benjamin and her editorial team at Books By Design. They have all been unfailingly helpful and patient.

From the Preface to the Third Edition

The two previous editions of *Financial Modeling* have received a gratifyingly positive response from readers. The combination of a "cookbook," mixing explanation and implementation using Excel, has fulfilled a need in both the academic and the practitioner market from readers who realize that the implementation of the finance basics typically studied in an introductory finance course requires another, more heavily computational and implementational, approach. Excel, the most widely used computational tool in finance, is a natural vehicle for deepening our understanding of the materials.

Acknowledgments

I want to start by thanking a group of wonderful editors: John Covell, Nancy Lombardi, Elizabeth Murry, Ellen Pope, and Peter Reinhart. My next thanks go to a dedicated group of colleagues who read the typescripts for *Financial Modeling*: Michael Chau, Jaksa Cvitanic, Arindam Bandopadhyaya, Richard Harris, Aurele Houngbedji, Iordanis Karagiannidis, Yvan Lengwiler, Nejat Seyhun, Gökçe Soydemir, David Y. Suk.

Many of the changes in this edition of *Financial Modeling* are due to the comments of readers, who have been assiduous in offering suggestions and improvements in the book. I follow a tradition started with the first two editions of *Financial Modeling* by acknowledging those readers whose comments have been incorporated into this edition:

Meni Abudy, Zvika Afik, Gordon Alexander, Apostol Bakalov, Naomi Belfer, David Biere, Vitaliy Bilyk, Oded Braverman, Roeland Brinkers, Craig Brody, Salvio Cardozo, Sharad Chaudhary, Israel Dac, Jeremy Darhansoff, Toon de Bakker, Govind vyas Dharwada, Davey Disatnik, Kevin P. Dowd, Brice Dupoyet, Cederik Engel, Orit Eshel, Yaara Geyra, Rana P. Ghosh, Bjarne Jensen, Marek Jochec, Milton Joseph, Erez Kamer, Saggi Katz, Emir Kiamilev, Brennan Lansing, Paul Ledin, Paul Legerer, Quinn Lewis, David Martin, Tom McCurdy, Tsahi Melamed, Tal Mofkadi, Geoffrey Morrisett, Sandip Mukherji, Max Nokhrin, Michael Oczkowski, David Pedersen, Mikael Petitjean, Georgio Questo, Alex Riahi, Arad Rostampour, Joseph Rubin, Andres Rubio, Ofir Shatz, Natalia Simakina, Ashutosh Singh, Permjit Singh, Gerald Strever, Shavkat Sultanbekov, Ilya Talman, Mel Tukman, Daniel Vainder, Guy Vishnia, Torben Voetmann, Chao Wang, James Ward, Roberto Wessels, Geva Yaniv, Richard Yeh, and Werner Zitzman.

Finally, I want to thank my very patient wife Terry, who has maintained her own and my equilibrium through two books and a business school deanship in the past five years.

From the Preface to the Second Edition

The purpose of this book remains to provide a “cookbook” for implementing common financial models in Excel. This edition has been expanded by six additional chapters, covering financial calculations, cost of capital, value at risk (VaR), real options, early exercise boundaries, and term-structure modeling. There is also an additional technical chapter containing a potpourri of Excel hints.

I am indebted to a number of people (in addition to those mentioned in the previous preface) for help and suggestions: Andrew A. Adamovich, Alejandro Sanchez Arevalo, Yoni Aziz, Thierry Berger-Helmchen, Roman Weissman Bermann, Michael Giacomo Bertolino, John Bollinger, Enrico Camerini, Manuel Carrera, Roy Carson, John Carson, Lydia Cassorla, Philippe Charlier, Michael J. Clarke, Alvaro Cobo, Beni Daniel, Ismail Dawood, Ian Dickson, Moacyr Dutra, Hector Tassinari Eldridge, Shlomy Elias, Peng Eng, Jon Fantell, Erik Ferning, Raz Gilad, Nir Gluzman, Michael Gofman, Doron Greenberg, Phil Hamilton, Morten Helbak, Hitoshi Hibino, Foo Siat Hong, Marek Jochec, Russell W. Judson, Tiffani Kaliko, Boris Karasik, Rick Labs, Allen Lee, Paul Legerer, Guoli Li, Moti Marcus, Gershon Mensher, Tal Mofkadi, Stephen O’Neil, Steven Ong, Oren Ossad, Jackie Rosner, Steve Rubin, Dvir Sabah, Ori Salinger, Meir Shahar, Roger Shelor, David Siu, Maja Sliwinski, Bob Taggart, Maurry Tamarkin, Mun Hon Tham, Efrat Tolkowsky, Mel Tukman, Sandra van Balen, Michael Verhofen, Lia Wang, Roberto Wessels, Ethan Weyand, Ubbo Wiersema, Weiqin Xie, Ke Yang, Ken Yook, George Yuan, Khurshid Zaynutdinov, Ehud Ziegelman, and Eric Zivot. I also want to thank my editors, who again have been a great help: Nancy Lombardi, Peter Reinhart, Victoria Richardson, and Terry Vaughn.

From the Preface to the First Edition

Like its predecessor, *Numerical Techniques in Finance,* the aim of this book is to present some important financial models and to show how they can be solved numerically and/or simulated using Excel. In this sense this is a finance "cookbook"; like any cookbook, it gives recipes with a list of ingredients and instructions for making and baking. As any cook knows, a recipe is just a starting point; having followed the recipe a number of times, you can think of your own variations and make the results suit your tastes and needs.

Financial Modeling covers standard financial models in the areas of corporate finance, financial statement simulation, portfolio problems, options, portfolio insurance, duration, and immunization. The aim in each case has been to explain clearly and concisely the implementation of the models using Excel. Very little theory is offered except where necessary to understand the numerical implementations.

While Excel is often not the tool to use for high-level, industrial-strength calculations (portfolios are an example), it is an excellent tool for understanding the computational intricacies involved in financial modeling. It is often the case that the fullest understanding of the models comes by calculating them, and Excel is one of the most accessible and powerful tools available for this purpose.

Along the way a lot of students, colleagues, and friends (these are nonexclusive categories) have helped me with advice and comments. In particular I would like to thank Olivier Blechner, Miryam Brand, Elizabeth Caulk, John Caulk, Benjamin Czaczkes, John Ferrari, John P. Flagler, Dan Fylstra, Kunihiko Higashi, Julia Hynes, Don Keim, Anthony Kim, Ken Kunimoto, Rick Labs, Adrian Lawson, Philippe Nore, Isidro Sanchez Alvarez, Nir Sharabi, Edwin Strayer, Robert Taggart, Mark Thaler, Terry Vaughn, and Xiaoge Zhou.

Finally, my thanks go to a wonderful set of editors: Nancy Lombardi, Peter Reinhart, Victoria Richardson, and Terry Vaughn.

0 Before All Else

0.1 Data Tables

Financial Modeling makes extensive use of data tables. We advise readers of the book to first make sure that they understand data tables (read Chapter 28, especially sections 28.1–28.5). Data tables are absolutely critical in the sensitivity analysis that is part of most financial models. They are a little bit complicated but an invaluable addition to the financial modeler's arsenal.

In the remainder of this short chapter, we discuss **Getformula, Formulatext,** and **R**.

0.2 What Is Getformula?

The Excel notebooks in this edition of *Financial Modeling* contain a function called **Getformula** that aids in annotating your spreadsheets. In the example below, cell C5 shows the formula contained in cell B5; the formula in question computes the annual repayment of a loan of $165,000 for seven years at 8%. Cell C5 contains the function =**Getformula(B5)**.

	A	B	C
2	Principal	165,000	
3	Interest	8%	
4	Term	7	<-- years
5	Annual payment	31,691.95	<-- =PMT(B3,B4,-B2)

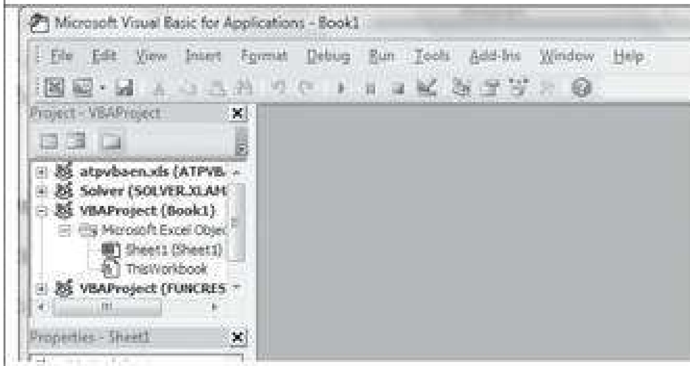
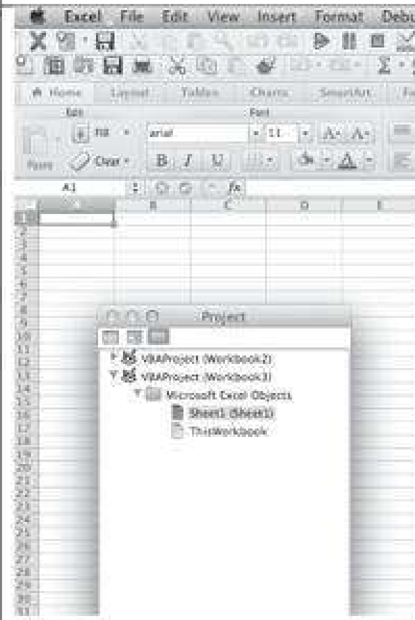
In this short document, we describe how to add this formula to your Excel notebook. Mac users: These instructions work only in Excel 2011 and above.

0.3 How to Put Getformula into Your Excel Notebook

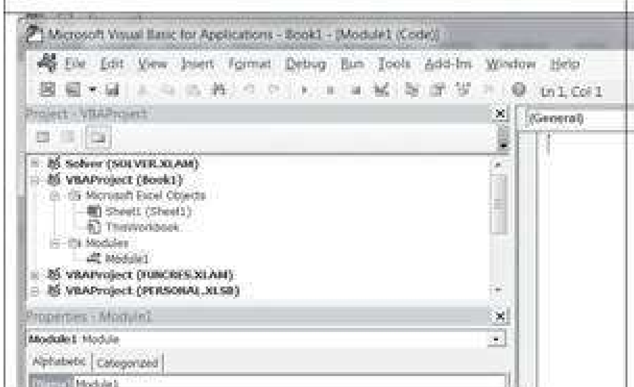
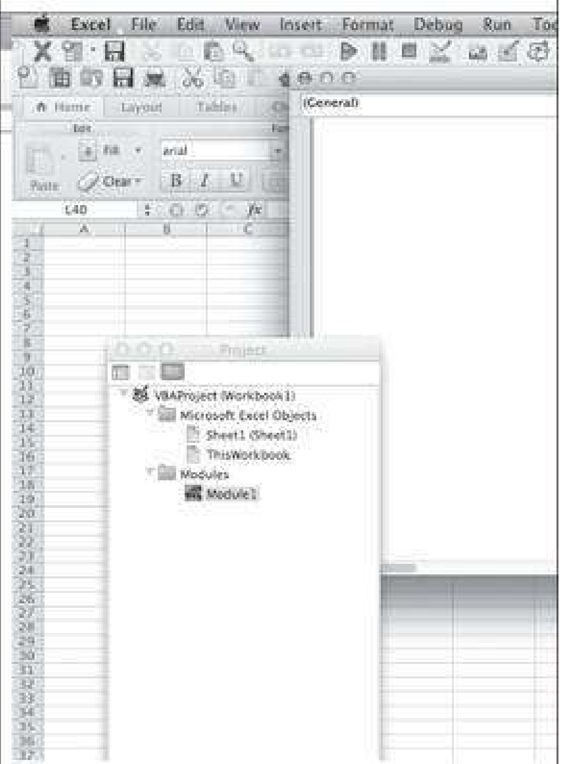
1. Open the Excel workbook in which you want the formula to work.
2. Open the VBA editor.

- On Windows computers: [Alt] + F11
- On Mac (Excel 2011): **Tools|Macro|Visual Basic Editor**

3. This will open the VBA editor.

Windows screen	Mac screen
	

4. Hit **Insert|Module** at the top of the screen.

Windows screen	Mac screen
	

5. Now insert the following text into the Module window (where it says **General**).

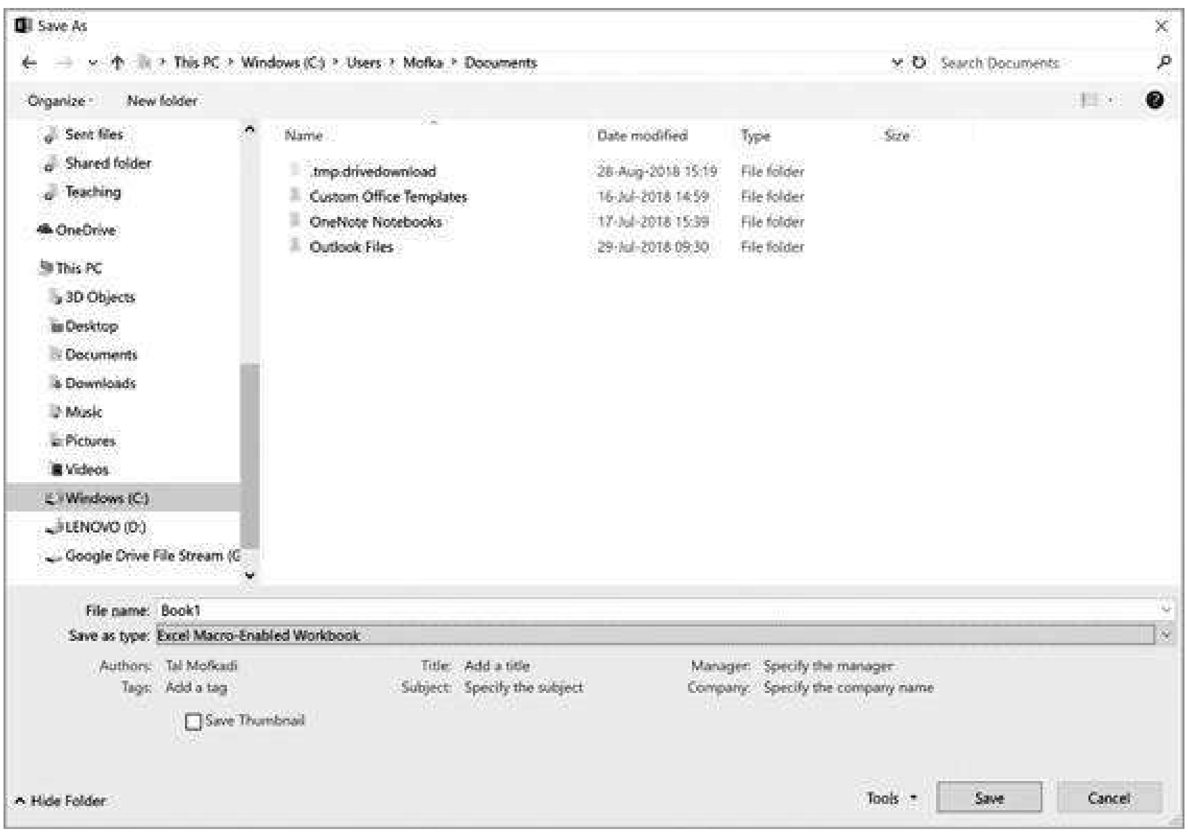
```
'8/5/2006 Thanks to Maja Sliwinski and Beni Czaczkes
Function getformula(r As Range) As String
    Application.Volatile
```

```
        If r.HasArray Then
        getformula = "<——" & " {" & r.FormulaArray & "}"
        Else
        getformula = "<——" & " " & r.FormulaArray
        End If
    End Function
```

In Windows, close the VBA window (no need to save). In the Mac, just continue to work on the spreadsheet. The formula is now part of the spreadsheet and will be saved along with it.

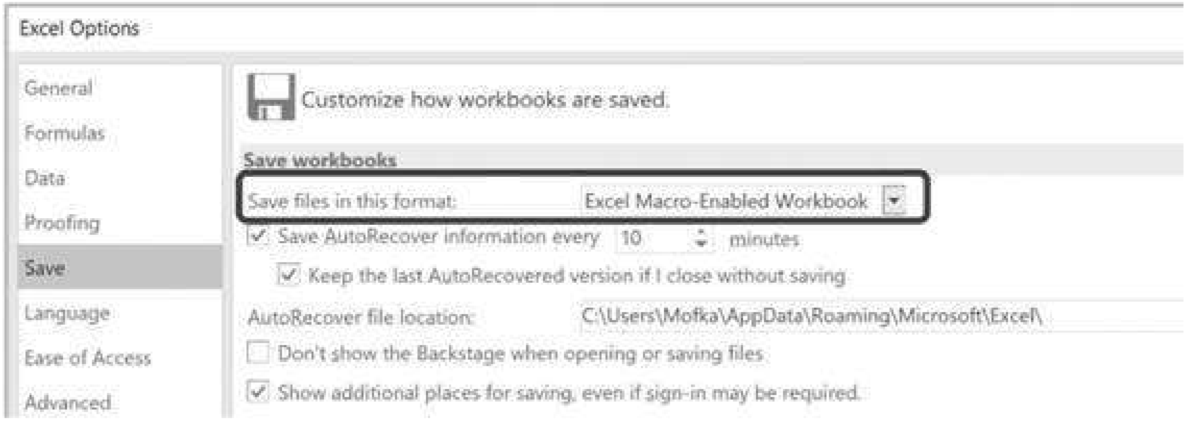
0.4 Saving the Excel Workbook: Windows

In order to save the notebook with the **Getformula** macro in VBA, you will have to save it as a **Macro-Enabled Workbook**.



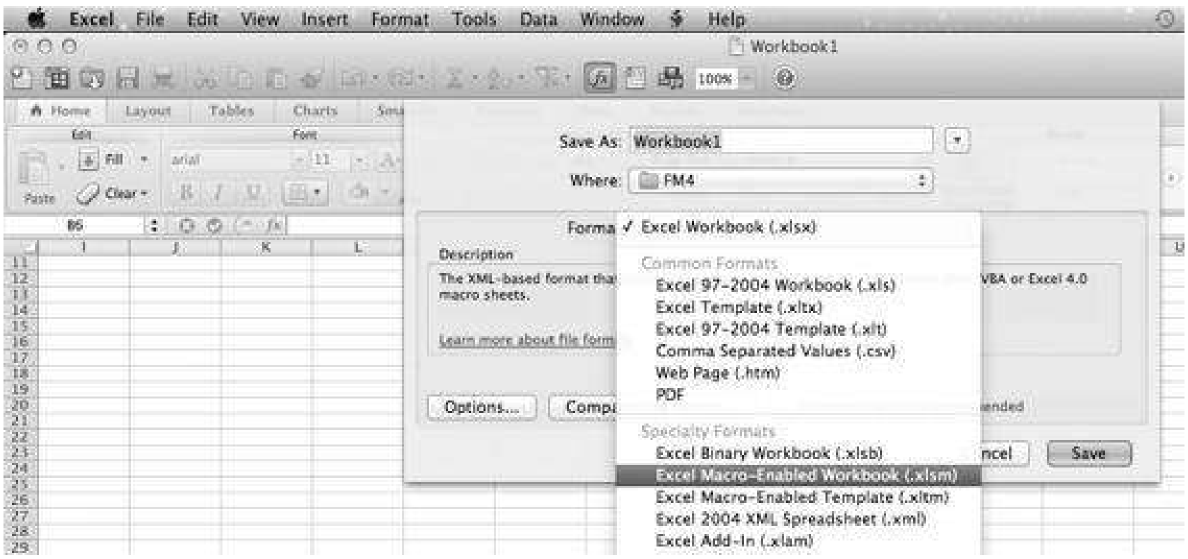
Macro-enabled workbooks have extension .xlsm, whereas regular Excel workbooks have extension .xlsx. Your users will never know the difference.

We have changed our Excel settings (**File|Options|Save**) to make the Macro-Enabled Workbook our default:



0.5 Saving the Excel Workbook: Mac

The Mac screen for saving as a Macro-Enabled Workbook looks like this:



0.6 Do You Have to Put Getformula into Each Excel Workbook?

The short answer is "yes." You could create an add-in to Excel that contains **Getformula** (see Chapter 37 on the auxiliary website that accompanies this book), but this will make it more difficult for you to share your workbooks. We prefer to put **Getformula** in each new spreadsheet we create.

0.7 Using Formulatext() Instead of Getformula

Excel 2013 and later versions contain a built-in function called **Formulatext**, which is basically doing the same thing that **Getformula** is doing. There are a few minor differences between Formulatext and Getformula that are important to note:

1. **Formulatext** does not contain the arrow pointing to the left. This can easily be solved by concatenating the string "<—" (less than and two minus signs) to **Formulatext**, as shown here.

	A	B	C
1	GETFORMULA vs. FORMULATEXT		
2	1. Formulatext and "<--"&Formulatext		
3	16	=2*8	<--=FORMULATEXT(A3)
4	16	<--=2*8	<--="<--"&FORMULATEXT(A4)

2. **Formulatext** will return an error (#N/A) when pointing to a cell without a formula while Getformula will simply return the values.

	A	B	C	D	E
6	2. Formulatext returns "N/A" when there is no formula				
7	Cell	Getformula		Formulatext	
8	20	<-- =5*4	=getformula(A8)	<--=5*4	="<--"&FORMULATEXT(A8)
9	5*4	<-- 5*4	=getformula(A9)	#N/A	="<--"&FORMULATEXT(A9)
10	20	<-- 20	=getformula(A10)	#N/A	="<--"&FORMULATEXT(A10)
11	Lizzie	<-- Lizzie	=getformula(A11)	#N/A	="<--"&FORMULATEXT(A11)

3. **Formulatext** will not work on Excel versions that are older than Excel 2013. If you are sharing your work with many users, this will frustrate them.
4. However, **Formulatext** is a native function in later versions of Excel, so you do not need to add it manually to every Excel file.
5. To use **Getformula**, you must enable macros on the user's machine. It may cause issues sometimes, specifically in large organizations. Since this book extensively uses VBA, this should not be an effective constraint.

0.8 A Shortcut to Use Getformula and Formulatext

Once you have put **Getformula** into your Excel workbook, you will have to use it! The vast majority of our uses of this function explains the content of the cell to the left of the formula itself:

	A	B	C	D	E	F
1	AUTOMATICALLY ADD FORMULATEXT AND GETFORMULA					
2	Item	Value				
3	A	5		This cell contains (=getformula(B7))		
4	B	6				
5	C	8				
6	D	9				
7	Average	7	#NAME?			

We've put a short macro into our **Personal notebook** that automates this procedure. The remainder of this section describes how to automate the **Getformula** (or **Formulatext**) procedure.

Automating the Procedure

We want to automate this procedure of putting **Getformula** (or **Formulatext**) into a cell:

- Turn it into a macro.

- Attach a key sequence (in our case **[Ctrl]** + **e**) to the macro.
- Make the macro and key sequence available in your Excel spreadsheets.

We will save the macro to our **Personal.xlsb** file. This file activates each time you start Excel. It is yours only—other readers of your spreadsheets will not see it. Below we describe the steps, for both Windows and the Mac.

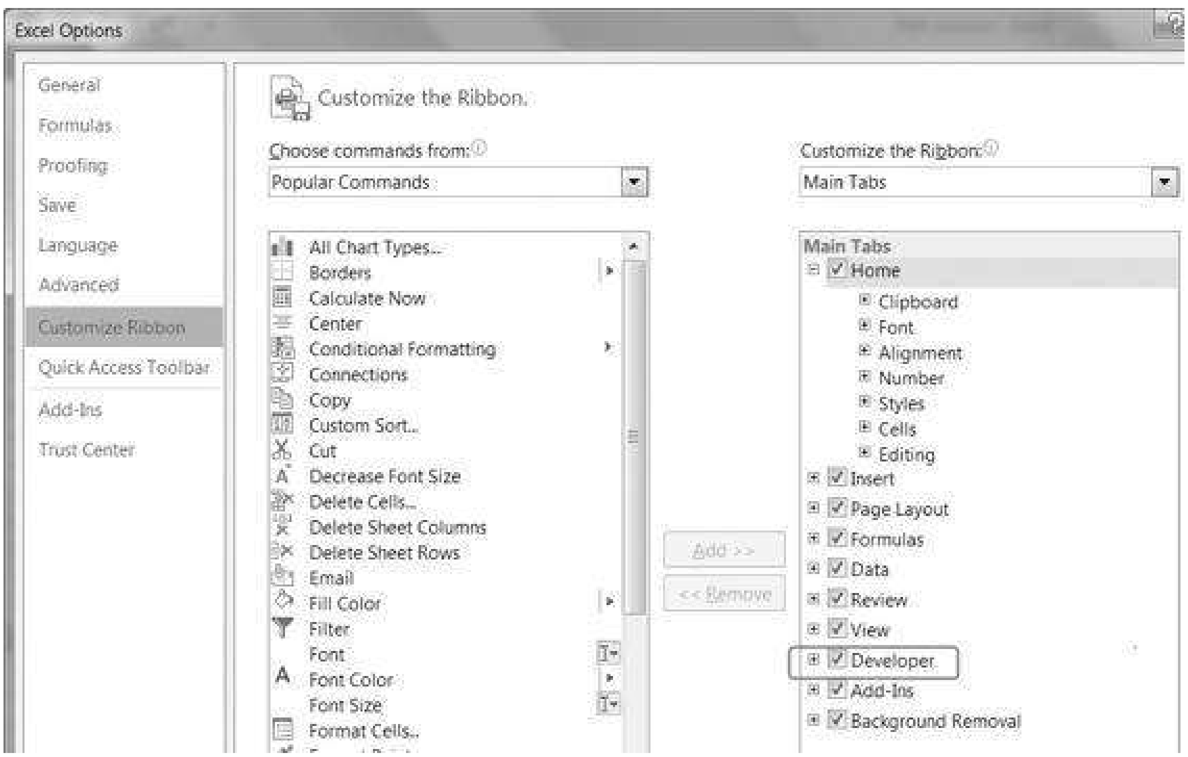
0.9 Recording Getformula: The Windows Case

Here are the steps for recording the macro in Windows:

- Activate the **Developer** tab on the menu bar.
- Use **Record Macro** to save a macro as a personal notebook.

Activate the Developer Tab

Go to **File|Options|Customize Ribbon** and activate the Developer tab as shown below:

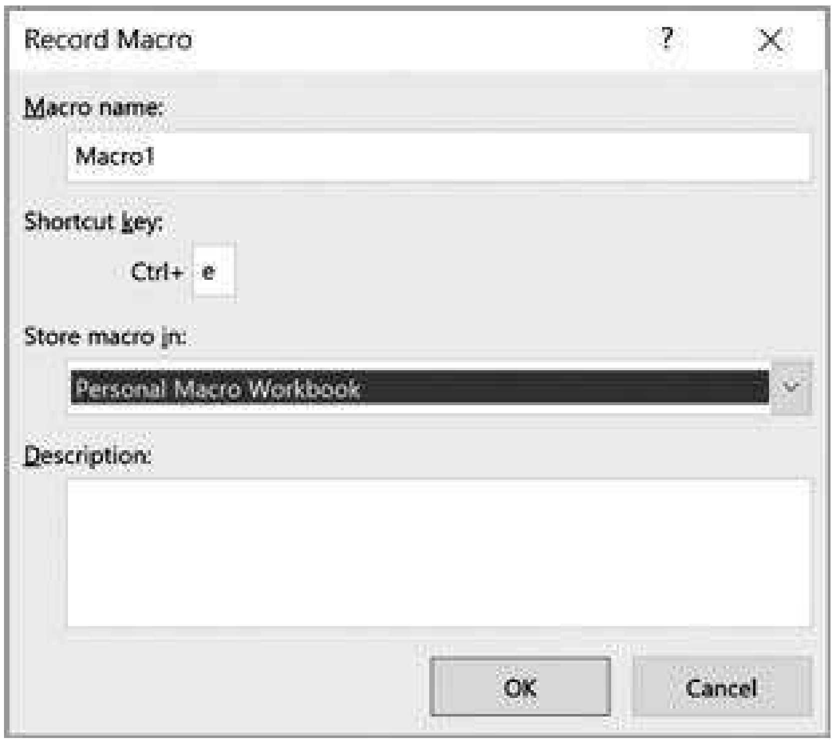


Use Record Macro

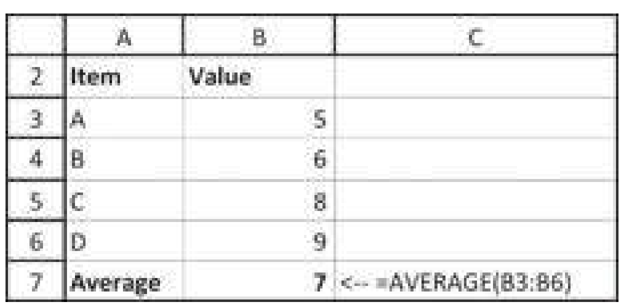
The Developer tab allows you to record a macro and save it as part of the Personal.xlsb notebook.

1. Open a blank Excel notebook and click on the **Developer** tab and then on **Record Macro**:

Excel will ask for details of the recording. Here's what we wrote. We will save this as a **Personal Macro Workbook** and then use the shortcut **[Ctrl]** + **e**:

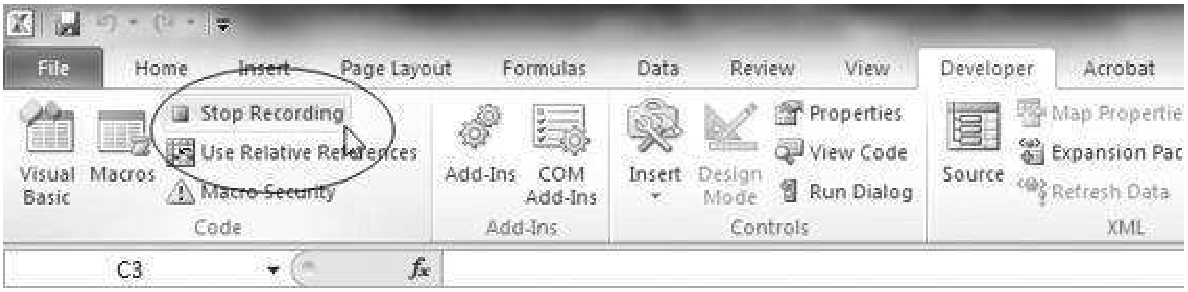


2. Now go to your spreadsheet and use **Getformula**, pointing to the cell to the left of where you want **Getformula** to appear. In the spreadsheet below, we have typed =Getformula(B7) into cell C7:



	A	B	C
2	Item	Value	
3	A	5	
4	B	6	
5	C	8	
6	D	9	
7	Average	7	<-- =AVERAGE(B3:B6)

3. Go back to the **Developer** tab and stop the recording:



4. Close down Excel. Excel will ask you if you want to save the Personal notebook. The answer is, of course, yes:

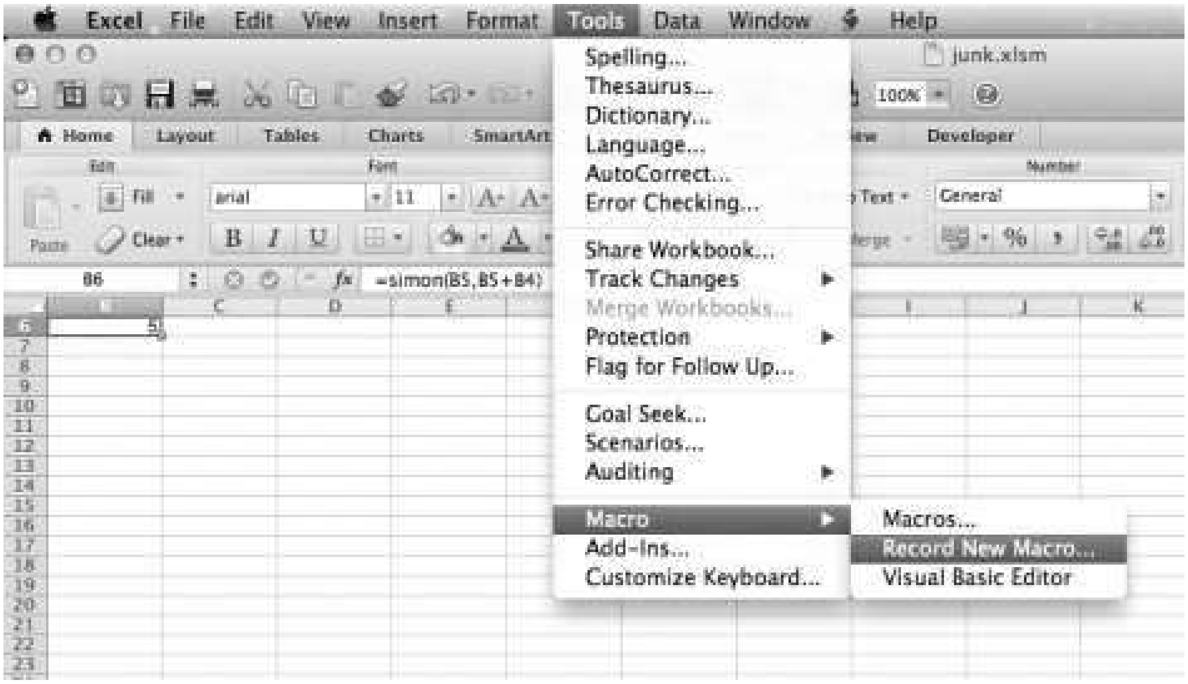
This creates the following file: C:\Users\Tal_Mofkadi\AppData\Roaming\Microsoft\Excel\XLSTART\PERSONAL.XLSB ("Tal_Mofkadi" is, of course, the user name of one of the authors on his computer—you will substitute your user name for your case).

Using the Macro

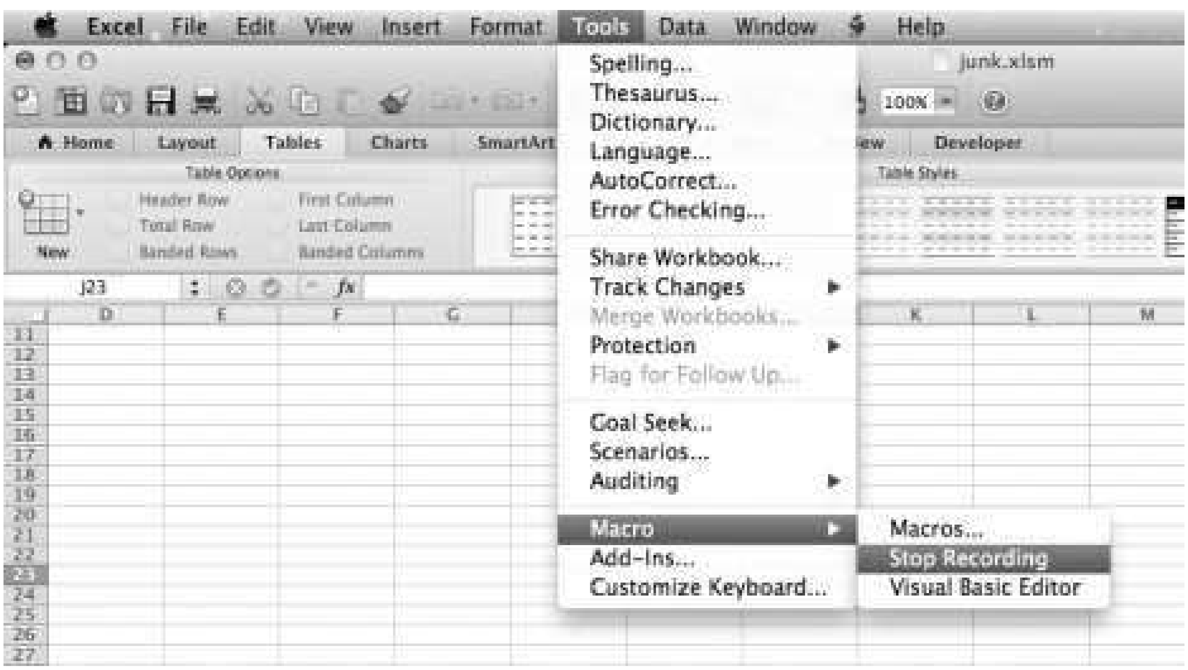
From now on, whenever you open an Excel file on *your computer*, you can use **Ctrl** + **e** to automatically add **Getformula** that points to the cell to the left of it.

0.10 Recording Getformula: The Mac Case

To record a macro in Excel, use **Tools|Macro|Record New Macro**:



To stop the recording:



As in the Windows case, you will be prompted to name the macro and assign it a control key sequence. When you save the spreadsheet you will be asked if you want to save the Personal notebook. Done!

0.11 Using R

Modeling financial situations will sometimes require tools that are more systematic than Excel. Though Excel is a great tool to use, specifically when combined with VBA, many practitioners use other statistical programs like R and Python.[1] Sections I and II of this book do not require the use of these tools, but Sections III–V, which are much more technical, do. Throughout these chapters we show an implementation in Excel and R. On the auxiliary website of this book (http://mitpress.mit.edu/FM5) you can find the Excel files, R files, and also the corresponding Python files. You can also find there PowerPoint files that may assist lecturers and students in the teaching and learning process.

The development environment chosen to implement programming (coding) in R is RStudio. We highly recommend you get used to working with it because this is also the industry standard.[2]

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1. 1. To understand R better, see https://www.r-project.org/. To understand Python better, see https://www.python.org/.
2. 2. See https://rstudio.com/.

I CORPORATE FINANCE

1 Basic Financial Analysis

1.1 Overview

This chapter aims to give you some finance basics and their Excel implementation. If you have had a good introductory course in finance, this chapter is likely to be at best a refresher.[1]

This chapter covers:

- Net present value (NPV)
- Internal rate of return (IRR)
- Payment schedules and loan tables
- Future value
- Pension and accumulation problems
- Continuously compounded interest
- Time-dated cash flows (Excel functions **XNPV** and **XIRR**)

Almost all financial problems are centered around finding the *present value* of a series of *cash receipts over time*. The cash receipts (or cash flows, as we will call them) may be certain or uncertain. The *present value* of a cash flow *Ft* anticipated to be received at time t is $\frac{CF_t}{(1+r)^t}$. The numerator of this expression is usually understood to be the *expected cash flow at time t*, and the discount rate r in the denominator is adjusted for the riskiness of this expected cash flow—the higher the risk, the higher the discount rate.

The basic concept in present value calculations is the concept of *opportunity cost*. Opportunity cost is the return that would be required of an investment to make it a viable alternative to other investments with similar risk characteristics. In the financial literature, there are many synonyms for opportunity cost; among them are discount rate, cost of capital, and interest rate. When applied to risky cash flows, we will sometimes call the opportunity cost the risk-adjusted discount rate (RADR) or the weighted average cost of capital (WACC). It goes without saying that this discount rate should be risk-adjusted (i.e., the cost of capital should reflect the same risk as the expected cash flow), and much of the standard finance literature discusses how to do this. As illustrated below, when we calculate the net present value, we use the investment's opportunity cost as a discount rate. When we calculate the internal rate of return, we compare the calculated return to the investment's opportunity cost to judge its value.

1.2 Present Value and Net Present Value

Both of these concepts are related to the value *today* of a set of future anticipated cash flows. As an example, suppose we are valuing an investment that promises \$100 per year at the end of this and the next four years. We suppose that these cash flows are risk-free: There is no doubt that this series of five payments of \$100 each will actually be paid. If a bank pays an annual interest rate of 4% on a 5-year deposit, then this 4% is the investment's opportunity cost, the alternative benchmark return to which we want to compare the investment. We can calculate the value of the investment by discounting its cash flows using this opportunity cost as a discount rate.

	A	B	C	D
1	COMPUTING THE PRESENT VALUE			
2	Discount rate	4%		
3	Year	Cash flow	Present value	
4	0			
5	1	100	96.15	<--=B5/(1+$B$2)^A5
6	2	100	92.46	<--=B6/(1+$B$2)^A6
7	3	100	88.90	<--=B7/(1+$B$2)^A7
8	4	100	85.48	<--=B8/(1+$B$2)^A8
9	5	100	82.19	<--=B9/(1+$B$2)^A9
10				
11	Present value			
12	Summing cells C5:C9	445.18	<--=SUM(C5:C9)	
13	Using Excel's NPV function	445.18	<--=NPV(B2,B5:B9)	
14	Using Excel's PV function	445.18	<--=PV(B2,5,-100)	

The **present value**, 445.18, is the *value today* of the investment. In a competitive market, the present value should correspond to the market price of the cash flows. The spreadsheet above illustrates three ways of obtaining this value:

- • Summing the individual present values in cells C5:C9. Note the use of "^" to represent the power. To simplify the copying, note the use of both the relative and absolute references; for example, =B5/(1+$B$2)^A5 in cell C5.
- • Using the Excel **NPV** function. As we show in the subsection following, Excel's **NPV** function is unfortunately misnamed—it actually computes the present value and not the net present value.
- • Using the Excel **PV** function. This function computes the present value of a series of constant payments. **PV(B2,5,-100)** is the present value of five payments of 100 each at the discount rate in cell B2. The **PV** function returns a negative value for positive cash flows; to prevent this unfortunate occurrence, we have made the cash flows negative.[2]

The Difference between Excel's PV and NPV Functions

The spreadsheet above may leave the misimpression that **PV** and **NPV** perform exactly the same computation. But this is not true—whereas **NPV** can handle any series of cash flows, **PV** can handle only constant cash flows.

	A	B	C	D
1	**COMPUTING THE PRESENT VALUE** **In this example the cash flows are not equal** **Either discount each cash flow separately or use Excel's NPV function** **Excel's PV doesn't work for this case**			
2	Discount rate	4%		
3	**Year**	**Cash**	**Present value**	
4	0			
5	1	100	96.15	<--=B5/(1+$B$2)^A5
6	2	200	184.91	<--=B6/(1+$B$2)^A6
7	3	300	266.70	<--=B7/(1+$B$2)^A7
8	4	400	341.92	<--=B8/(1+$B$2)^A8
9	5	500	410.96	<--=B9/(1+$B$2)^A9
10				
11	Present value			
12	Summing cells C5:C9	1,300.65	<--=SUM(C5:C9)	
13	Using Excel's NPV function	1,300.65	<--=NPV(B2,B5:B9)	

Excel's NPV Function Is Misnamed!

In standard finance terminology, the *present value* of a series of cash flows is the value today of the future cash flows:

$$Present\ value = \sum_{t=1}^{N} \frac{CF_t}{(1+r)^t}$$

The *net present value* is the present value minus the cost of acquiring the asset (the cash flow at time zero):

$$Net\ present\ value = \sum_{t=0}^{N} \frac{CF_t}{(1+r)^t} = \underbrace{CF_0}_{\substack{\text{In many cases} \\ CF_0 < 0\text{, meaning} \\ \text{that it represents the} \\ \text{price paid for the asset.}}} + \underbrace{\sum_{t=1}^{N} \frac{CF_t}{(1+r)^t}}_{\substack{\text{This is the present} \\ \text{value, given by Excel's} \\ \text{NPV function}}}$$

Excel's language about discounted cash flows differs somewhat from the standard finance nomenclature. To calculate the finance *net present value* of a series of cash flows using Excel, we have to calculate the *present value* of the future cash flows (using the Excel **NPV** function), taking into account the time zero cash flow (this is often the cost of the asset in question).

The Net Present Value (NPV)

Suppose that the above investment is sold for $1,500. Clearly it would not be worth its purchase price since—given the alternative return (discount rate) of 4%—the investment is worth only $1,300.65. The *net present value* (NPV) is the applicable concept here. Denoting by r the discount rate applicable to the investment, the NPV is calculated as

$$NPV = CF_0 + \sum_{t=1}^{N} \frac{CF_t}{(1+r)^t},$$

where CF_t is the investment's cash flow at time t and CF_0 is today's cash flow.

Suppose, for example, that the series of five cash flows of $100 is sold for $250. Then, as shown below, the NPV = 195.18.

	A	B	C	D
1	COMPUTING THE NET PRESENT VALUE			
2	Discount rate	4%		
3	Year	Cash flow	Present value	
4	0	-250	-250.00	<--=B4/(1+$B$2)^A4
5	1	100	96.15	<--=B5/(1+$B$2)^A5
6	2	100	92.46	<--=B6/(1+$B$2)^A6
7	3	100	88.90	<--=B7/(1+$B$2)^A7
8	4	100	85.48	<--=B8/(1+$B$2)^A8
9	5	100	82.19	<--=B9/(1+$B$2)^A9
10				
11	Net present value			
12	Summing cells C4:C9	195.18	<--=SUM(C4:C9)	
13	Using Excel's NPV function	195.18	<--=B4+NPV(B2,B5:B9)	

The NPV represents the *wealth increment* that accrues to the purchaser of the cash flows. If you buy the series of five cash flows of 100 for 250, then you have gained 195.18 in wealth today. In a competitive market, the NPV of a series of cash flows ought to be zero. Since the present value should correspond to the market price of the cash flows, the NPV should be zero. In other words, in a competitive market, assuming that 4% is the correct risk-adjusted discount rate, the market price of our five cash flows of 100 ought to be 445.18.

The Present Value of an Annuity and a Perpetuity—Some Useful Formulas[3]

An *annuity* is a security that pays a constant cash flow in each period in the future. Annuities may have a finite series of payments. When the series is infinite, it is called *perpetuity.* When the periodic cash flow is "C" and the appropriate discount rate is r, then the value today of the annuity, paying n periods its present value, is:

$$PV\ of\ annuity = \frac{C}{(1+r)^1} + \frac{C}{(1+r)^2} + \cdots + \frac{C}{(1+r)^n} = \frac{C}{r}\left(1 - \frac{1}{(1+r)^n}\right)$$

If cash stream promises an infinite series of constant future payments ($n \to \infty$), then this formula reduces to:

$$PV\ of\ perpetuity = \frac{C}{(1+r)^1} + \frac{C}{(1+r)^2} + \cdots = \frac{C}{r}$$

Both formulas can be computed with Excel. Below we compute the value of a finite annuity in three ways—using the formula (cell B6), using Excel's **PV** function (cell B7), and using Excel's **NPV** function (cell B17).

	A	B	C
1	COMPUTING THE VALUE OF AN ANNUITY		
2	Periodic payment, C	1,000	
3	Number of future periods paid, n	5	
4	Discount rate, r	6%	
5	Present value of annuity		
6	Using formula	4,212.36	<--=B2*(1-1/(1+B4)^B3)/B4
7	Using Excel's **PV** function	4,212.36	<--=PV(B4,B3,-B2)
8			
9	Period	Annuity payment	
10	0		
11	1	1,000.00	<--=B2
12	2	1,000.00	
13	3	1,000.00	
14	4	1,000.00	
15	5	1,000.00	
16			
17	Present value using Excel's **NPV** function	4,212.36	<--=NPV(B4,B11:B15)

Computing the value of a perpetuity (an infinite annuity) is even simpler, as shown in the spreadsheet below.

	A	B	C	D	E
1	THE VALUE OF A PERPETUITY (AN INFINITE ANNUITY)				
2	Periodic payment, C	1,000			
3	Discount rate, r	6%			
4	Present value of annuity	16,666.67	<--=B2/B3		

The Value of a Growing Annuity

A *growing annuity* pays out a sum *C*, which grows at a periodic growth rate *g*. If the annuity is finite, its value today is given by:

$$
\begin{aligned}
PV\ of\ growing\ annuity &= \frac{C}{(1+r)^1} + \frac{C(1+g)}{(1+r)^2} + \frac{C(1+g)^2}{(1+r)^3} + \cdots + \frac{C(1+g)^{n-1}}{(1+r)^n} \\
&= \frac{C}{r-g}\left(1-\left(\frac{1+g}{1+r}\right)^n\right)
\end{aligned}
$$

Taking this formula and letting n → ∞, we can compute the value of a growing perpetuity (*infinite growing annuity*) as follows:

$$
\begin{aligned}
PV\ of\ growing\ perpetuity &= \frac{C}{(1+r)^1} + \frac{C(1+g)}{(1+r)^2} + \frac{C(1+g)^2}{(1+r)^3} + \cdots \\
&= \frac{C}{r-g},\ provided\ g < r
\end{aligned}
$$

These formulas can easily be implemented in Excel. Below we compute the value of a finite growing annuity using the above formula (cell B7) and using Excel's **NPV** function (cell B17).

	A	B	C
1	COMPUTING THE VALUE OF A FINITE GROWING ANNUITY		
2	First payment, C	1,000	
3	Growth rate of payments, g	3%	
4	Number of future periods paid, n	5	
5	Discount rate, r	6%	
6	Present value of annuity		
7	Using formula	4,457.43	<--=B2*(1-((1+B3)/(1+B5))^B4)/(B5-B3)
8			
9	Period	Annuity payment	
10	0		
11	1	1,000.00	<--=B2
12	2	1,030.00	<--=$B$2*(1+$B$3)^(A12-$A$11)
13	3	1,060.90	<--=$B$2*(1+$B$3)^(A13-$A$11)
14	4	1,092.73	<--=$B$2*(1+$B$3)^(A14-$A$11)
15	5	1,125.51	<--=$B$2*(1+$B$3)^(A15-$A$11)
16			
17	Present value using Excel's **NPV** function	4,457.43	<--=NPV(B5,B11:B15)

When the growing annuity has an infinite life, it is called *growing perpetuity.* The next spreadsheet shows how its value is computed.

	A	B	C
1	COMPUTING THE VALUE OF A GROWING PERPETUITY (AN INFINITE GROWING ANNUITY)		
2	Periodic payment, C	1,000	<-- Starting at date 1
3	Growth rate of payments, g	3%	
4	Discount rate, r	6%	
5	Present value of annuity	33,333.33	<--=IF(B3<B4,B2/(B4-B3),"NA")

The Gordon Formula

The Gordon formula values a stock by discounting its future anticipated dividends at the cost of equity r_E. Letting P_0 be the current stock price, Div_0 the current dividend, and g the growth rate of future dividends, then

$$P_0 = \sum_{t=1}^{\infty} \frac{Div_0(1+g)^t}{(1+r_E)^t}.$$

Using the formula for an infinite growing annuity, we can write this as

$$P_0 = \frac{Div_0(1+g)}{r_E - g}, \textit{ provided } g < r_E.$$

The Gordon formula is used in Chapters 2, 4, 5, and 6 to model the firm's terminal value and in Chapter 3 to model the firm's cost of equity r_E by inverting the formula.

1.3 The Internal Rate of Return (IRR) and Loan Tables

The *internal rate of return* (IRR) is defined as the compound rate of return r, which makes the NPV equal to zero:

$$IRR = (r \mid NPV(r) = 0) \rightarrow CF_0 + \sum_{t=1}^{N} \frac{CF_t}{(1+IRR)^t} = 0$$

To illustrate, consider the example given in rows 3–8 in the spreadsheet below: a project costing 800 in year zero returns a variable series of cash flows at the end of years 1 to 5. The IRR of the project (cell B10) is 19.54%.

	A	B	C
1	**COMPUTING THE INTERNAL RATE OF RETURN**		
2	**Year**	**flow**	
3	0	-800	
4	1	100	
5	2	200	
6	3	300	
7	4	400	
8	5	500	
9			
10	Internal rate of return (IRR)	19.54%	<-- =IRR(B3:B8)

Note that the Excel **IRR** function includes as arguments *all* of the cash flows of the investment, including the first (in this case negative) cash flow of −800.

Determining the IRR by Trial and Error

There is no simple mathematical formula to compute IRR. Excel's **IRR** function uses trial and error, which can be simulated by using trial and error in a spreadsheet, as illustrated below.

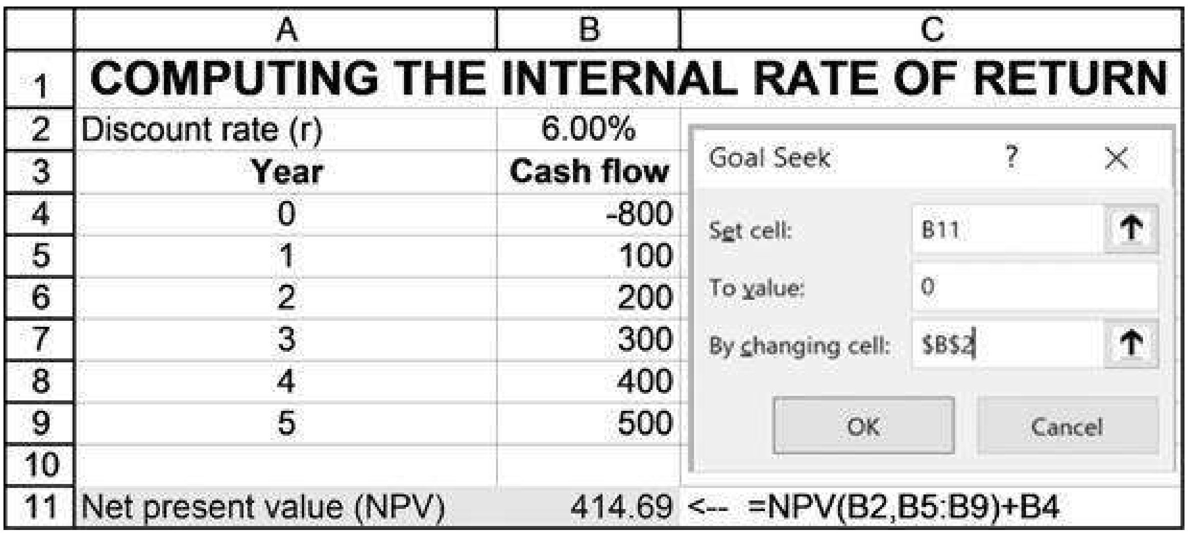
We start with a discount rate (6% in the example below) and check the project's NPV:

	A	B	C
1	**COMPUTING THE INTERNAL RATE OF RETURN**		
2	Discount rate (r)	6%	
3	**Year**	**Cash flow**	
4	0	-800	
5	1	100	
6	2	200	
7	3	300	
8	4	400	
9	5	500	
10			
11	Net present value (NPV)	414.69	<-- =NPV(B2,B5:B9)+B4

By playing with the discount rate or by using Excel's **Goal Seek** (under **Data|What-if analysis**, see Chapter 31), we can determine that at 19.54% the NPV in cell B11 is zero.

	A	B	C
1	COMPUTING THE INTERNAL RATE OF RETURN		
2	Discount rate (r)	19.54%	
3	Year	Cash flow	
4	0	-800	
5	1	100	
6	2	200	
7	3	300	
8	4	400	
9	5	500	
10			
11	Net present value (NPV)	0.00	<-- =NPV(B2,B5:B9)+B4

We allow Excel to change the interest rate to set the NPV to zero by pointing to the NPV as the Set cell by changing the interest rate cell. Here's the way the **Goal Seek** screen looked before we got the correct answer:



	A	B	C
1	COMPUTING THE INTERNAL RATE OF RETURN		
2	Discount rate (r)	6.00%	
3	Year	Cash flow	
4	0	-800	
5	1	100	
6	2	200	
7	3	300	
8	4	400	
9	5	500	
10			
11	Net present value (NPV)	414.69	<-- =NPV(B2,B5:B9)+B4

Investment Tables and the Internal Rate of Return

The IRR is the compound *rate of return paid by the investment*. To understand this fully, it helps to make an *investment table*, which shows annual investment value after accumulation of interest rate and the cash payment.

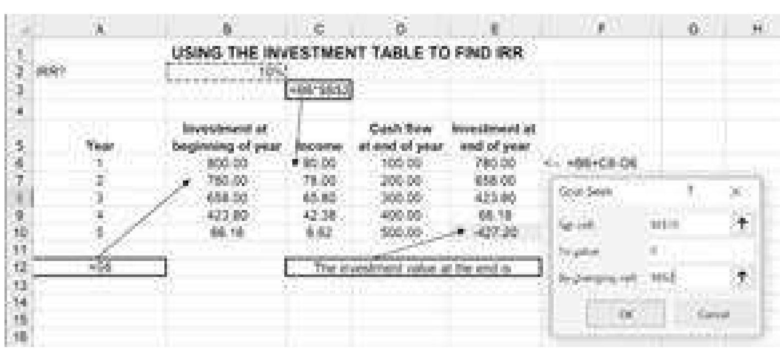
	A	B	C	D	E	F
1	INTERNAL RATE OF RETURN					
2	Year	Cash flow				
3	0	-800				
4	1	100				
5	2	200				
6	3	300				
7	4	400				
8	5	500				
9						
10	Internal rate or return (IRR)	19.54%	<-- =IRR(B3:B8)			
11						
12		USING THE IRR IN AN INVESTMENT TABLE				
13			=B17*$B$10			
14	=B3				=B4	
15						
16	Year	Investment at beginning of year	Income	Cash flow at end of year	Investment at end of year	
17	1	800.00	156.31	100.00	856.31	<-- =B17+C17-D17
18	2	856.31	167.31	200.00	823.61	
19	3	823.61	160.92	300.00	684.53	
20	4	684.53	133.75	400.00	418.28	
21	5	418.28	81.72	500.00	0.00	
22						
23	=E17		The remaining investment principal in the year after the last cash flow is zero, indicating that all the principal has been repaid			
24						
25						

The *investment table* shows how each year's saving balance is increased by the interest income and decreased by the annual cash flow. The income component at the end of each year is IRR times the principal balance at the beginning of that year.

We can use the investment table to find the internal rate of return. Consider an investment costing 800 today that pays off the cash flows indicated below at the end of years 1, 2, …, 5. At a rate of 10% (cell B2 in the spreadsheet below), the principal at the end of year 5 is negative, indicating that too little has been paid out in income. Thus, the IRR must be larger than 10%.

	A	B	C	D	E	F
1		USING THE INVESTMENT TABLE TO FIND IRR				
2	IRR?	10%				
3			=B6*$B$2			
4						
5	Year	Investment at beginning of year	Income	Cash flow at end of year	Investment at end of year	
6	1	800.00	80.00	100.00	780.00	<-- =B6+C6-D6
7	2	780.00	78.00	200.00	658.00	
8	3	658.00	65.80	300.00	423.80	
9	4	423.80	42.38	400.00	66.18	
10	5	66.18	6.62	500.00	-427.20	
11						
12	=E6		The investment value at the end is			

If the interest rate in cell B2 is indeed the IRR, then cell E10 should be 0. We can use Excel's **Goal Seek** (found under **Data|What-if analysis**) to calculate the IRR, as shown in the spreadsheet below.



As shown below, the IRR is 19.54%.

	A	B	C	D	E	F
1		USING THE INVESTMENT TABLE TO FIND IRR				
2	IRR?	19.54%				
3			=B6*$B$2			
4						
5	Year	Investment at beginning of year	Income	Cash flow at end of year	Investment at end of year	
6	1	800.00	156.31	100.00	856.31	<-- =B6+C6-D6
7	2	856.31	167.31	200.00	823.61	
8	3	823.61	160.92	300.00	684.53	
9	4	684.53	133.75	400.00	418.28	
10	5	418.28	81.72	500.00	0.00	
11						
12	=E6		The investment value at the end is zero			

The investment table is an effective illustration that the IRR is the interest rate that pays off an investment over its term. Of course, we could have simplified life by just using the **IRR** function, as shown above.

Excel's Rate Function

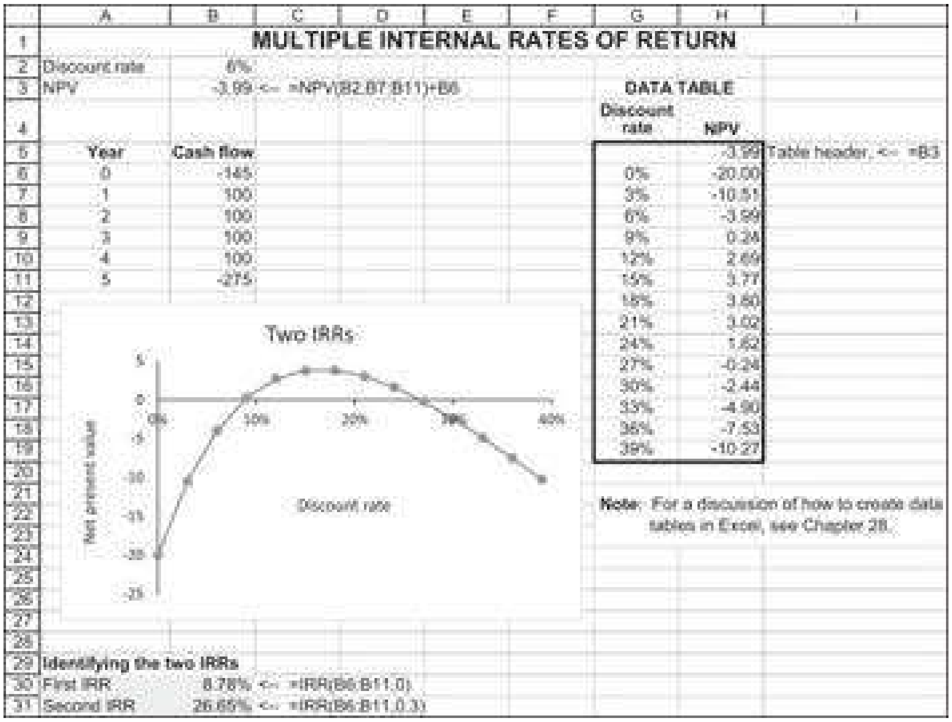
Excel's **Rate** function computes the IRR of a series of constant future payments (i.e., the IRR of an annuity). In the example below we pay $1,000 today for an annual payment of $100 for the next 30 years. **Rate** shows that the IRR is 9.307%.

	A	B	C
1	USING EXCEL'S RATE FUNCTION TO COMPUTE THE IRR		
2	Initial investment	1,000	
3	Periodic cash flow	100	
4	Number of payments	30	
5	IRR	9.307%	<-- =RATE(B4,B3,-B2)

Note: **Rate** works much like **PMT** and **PV** discussed elsewhere in this chapter; it requires a sign change between the initial investment and the periodic cash flow (we have used −B2 in cell B5). It also has switches to allow for payments that start today and payments that start one period from now (not shown in the example above).

1.4 Multiple Internal Rates of Return

Sometimes a series of cash flows has more than one IRR. In the example below we can tell that the cash flows in cells B6:B11 have two IRRs, since the NPV graph crosses the x-axis twice.



Excel's **IRR** function allows us to add an extra argument that will help us find both IRRs. Instead of writing **=IRR(B6:B11)**, we write **=IRR(B6:B11,guess)**. The argument **guess** is a starting point for the algorithm that Excel uses to find the IRR; by adjusting the **guess**, we can identify both the IRRs. Cells B30 and B31 give an illustration.

There are two things to note about this procedure:

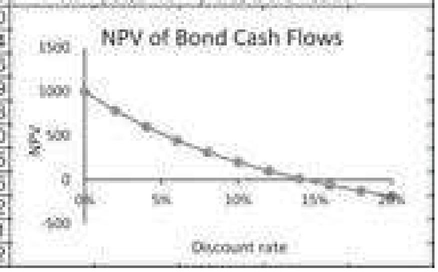
- • The argument **guess** merely has to be close to the IRR; it is not unique. For example, by setting the guesses equal to 0.1 and 0.5, we will still get the same IRRs, as shown below.

	A	B	C	D
29	**Identifying the two IRRs**			
30	First IRR	8.78%	<-- =IRR(B6:B11,0.1)	
31	Second IRR	26.65%	<-- =IRR(B6:B11,0.5)	

- • In order to identify the number and the approximate value of the IRRs, it helps greatly to graph (as we did above) the NPV of the investment as a function of various discount rates. The internal rates of return are then the points where the graph crosses the x-axis, and the approximate location of these points should be used as the guesses in the IRR function.[4]

From a purely technical point of view, a set of cash flows can have multiple IRRs only if it has at least two changes of sign. Many typical cash flows have only one change of sign. Consider, for example, the cash flows from purchasing a bond having a 10% coupon, a face value of $1,000, and eight more years to maturity. If the current market price of the bond is $800, then the stream of cash flows changes signs only once (from negative in year zero to positive in years 1 to 8). Thus, there is only one IRR:

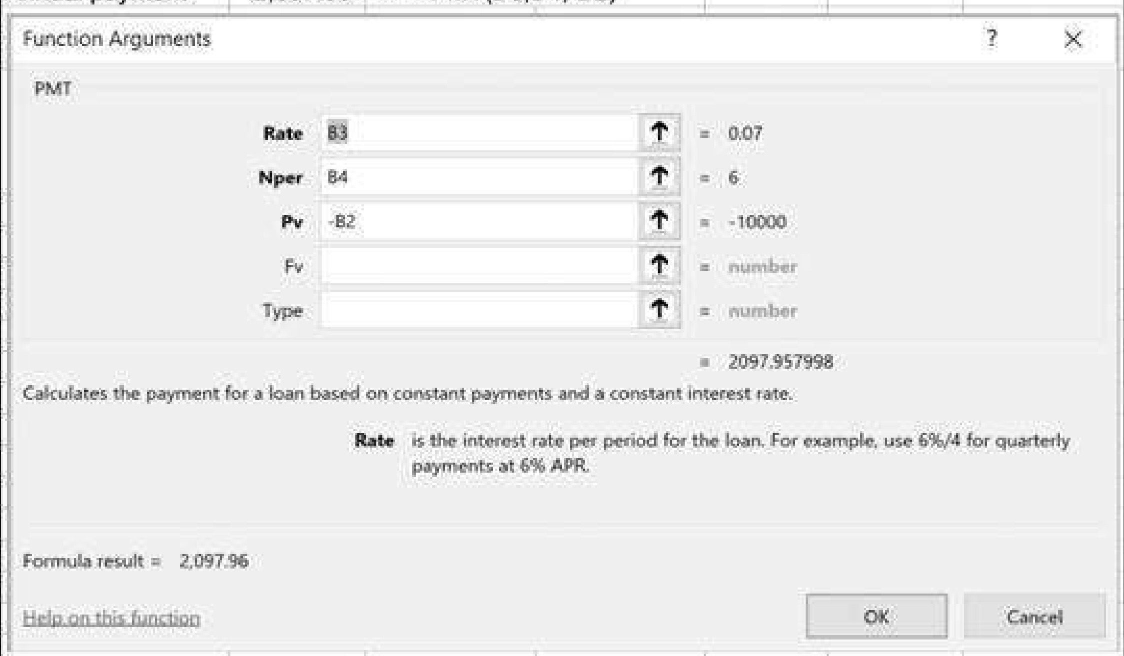
	A	B	C	D	E	F	G
1	BOND CASH FLOWS: NPV CROSSES X-AXIS ONLY ONCE, SO THERE IS ONLY ONE IRR						
2	Year	Cash flow			Data table: Effect of discount rate on NPV		
3	0	-800					
4	1	100				1,000.00	<-- =NPV(E4,B4:B11)+B3, table header
5	2	100			0%	1,000.00	
6	3	100			2%	786.04	
7	4	100			4%	603.96	
8	5	100			6%	448.39	
9	6	100			8%	314.93	
10	7	100			10%	200.00	
11	8	1,100			12%	100.65	
12					14%	14.45	
13	IRR	14.36%	<-- =IRR(B3:B11)		16%	-60.62	
14					18%	-126.21	
15					20%	-183.72	



1.5 Flat Payment Schedules

Another common problem is to compute a “flat” repayment for a loan. For example: You take a loan for $10,000 at an interest rate of 7% per year. The bank wants you to make a series of payments that will pay off the loan and the interest over 10 years. We can use Excel’s **PMT** function to determine how much each annual payment should be.

	A	B	C
1	FLAT PAYMENT SCHEDULES		
2	Loan principal	10,000	
3	Interest rate	7%	
4	Loan term	6	<-- Number of years over which loan is repaid
5	Annual payment	2,097.96	<-- =PMT(B3,B4,-B2)



Notice that we have put “PV”—Excel’s nomenclature for the initial loan principal—with a minus sign. As discussed previously, if we do not do this, Excel returns a negative payment (a minor irritant). You can confirm that the answer of 2,097.96 is correct by creating a loan table.

	A	B	C	D	E	F
1		FLAT PAYMENT SCHEDULES				
2	Loan principal	10,000				
3	Interest rate	7%				
4	Loan term	6	<-- Number of years over which loan is repaid			
5	Annual payment	2,097.96	<-- =PMT(B3,B4,-B2)			
6						
7	=B2		=$B$3*B10			
8				Split payment into:		
9	Year	Principal at beginning of year	Payment at end of year	Interest	Return of principal	
10	1	10,000.00	2,097.96	700.00	1,397.96	<-- =C10-D10
11	2	8,602.04	2,097.96	602.14	1,495.82	
12	3	7,106.23	2,097.96	497.44	1,600.52	
13	4	5,505.70	2,097.96	385.40	1,712.56	
14	5	3,793.15	2,097.96	265.52	1,832.44	
15	6	1,960.71	2,097.96	137.25	1,960.71	
16	7	0.00				
17						
18	=B10-E10					

The zero in cell B16 indicates that the loan is fully repaid over its term of six years. You can easily confirm that the present value of the payments over the six years is the initial principal of 10,000.

1.6 Future Values and Applications

We start with a triviality. Suppose you deposit $1,000 in an account today, leaving it there for 10 years. Suppose the account accumulates an annual interest of 10%. How much will you have at the end of 10 years? The answer, as shown in the spreadsheet below, is $2,593.74.

	A	B	C	D	E
1		SIMPLE FUTURE VALUE			
2	Interest	10%			
3					
4	Year	Account balance, beginning of year	Interest earned during year	Total in account, end of year	
5	1	1,000.00	100.00	1,100.00	<-- =C5+B5
6	2	1,100.00	110.00	1,210.00	<-- =C6+B6
7	3	1,210.00	121.00	1,331.00	
8	4	1,331.00	133.10	1,464.10	
9	5	1,464.10	146.41	1,610.51	=$B$2*B5
10	6	1,610.51	161.05	1,771.56	
11	7	1,771.56	177.16	1,948.72	
12	8	1,948.72	194.87	2,143.59	
13	9	2,143.59	214.36	2,357.95	=D5
14	10	2,357.95	235.79	2,593.74	
15					
16	A simpler way		2,593.74	<-- =B5*(1+B2)^10	

As cell D16 shows, you don't need all these complicated calculations. The *future value* of 1,000 in 10 years at 10% per year is given by

$FV = 1{,}000 * (1 + 10\%)^{10} = 2{,}593.74.$

Now consider the following, slightly more complicated problem: Again, you intend to open a savings account. Your initial deposit of $1,000 today (the beginning of year 1) will be followed by a similar deposit at the beginning of years 2, …, 10 (total of 10 deposits at the beginning of each year). If the account earns 10% per year, how much will you have in the account at the end of year 10?

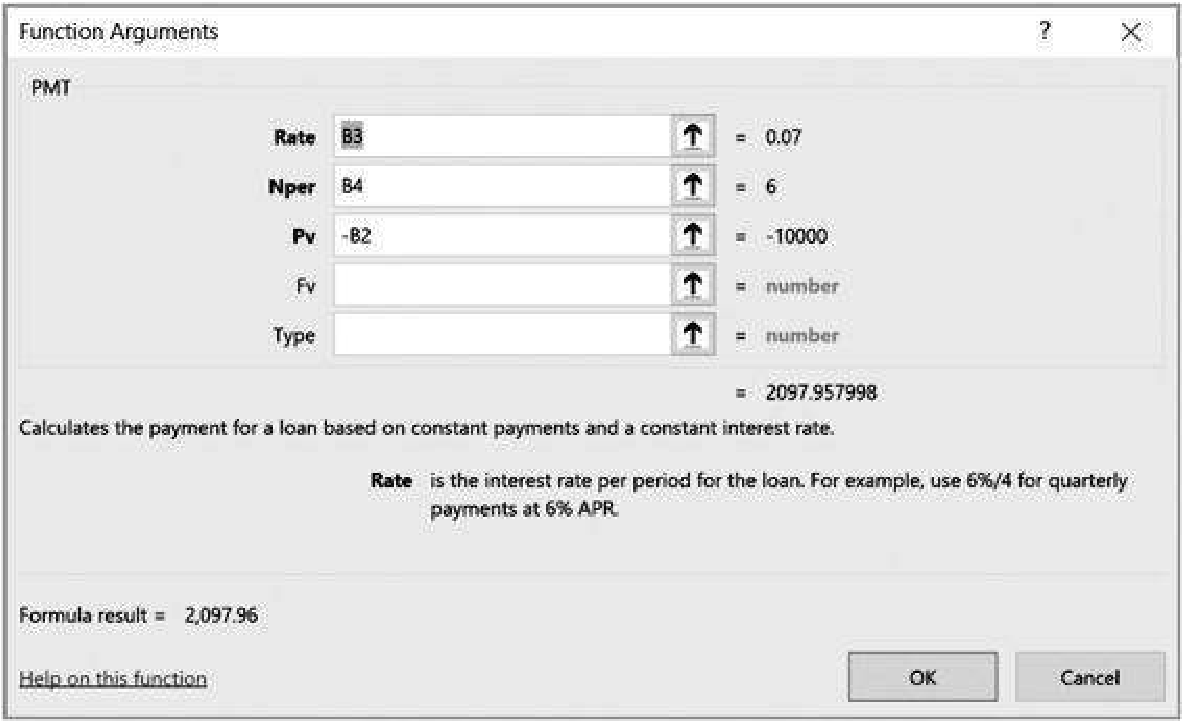
This problem is easily modeled in Excel, as follows:

	A	B	C	D	E	F
1		**FUTURE VALUE WITH ANNUAL DEPOSITS**				
2	Interest	10%				
3	Annual deposit	1,000	<-- Made today **and** at beginning of each of next 9 years			
4	Number of deposits	10				
5						
6	**Year**	**Account balance, beginning of year**	**Deposit at beginning of year**	**Interest earned during year**	**Total in account, end year**	
7	1	0.00	1,000	100.00	1,100.00	<-- =D7+C7+B7
8	2	1,100.00	1,000	210.00	2,310.00	<-- =D8+C8+B8
9	3	2,310.00	1,000	331.00	3,641.00	
10	4	3,641.00	1,000	464.10	5,105.10	
11	5	5,105.10	1,000	610.51	6,715.61	=$B$2*(C7+B7)
12	6	6,715.61	1,000	771.56	8,487.17	
13	7	8,487.17	1,000	948.72	10,435.89	
14	8	10,435.89	1,000	1,143.59	12,579.48	
15	9	12,579.48	1,000	1,357.95	14,937.42	
16	10	14,937.42	1,000	1,593.74	17,531.17	
17						
18	Future value	17,531.17	<-- =FV(B2,B4,-B3,,1)		=E7	

Thus, the answer is that we will have $17,531.17 in the account at the end of year 10. This same answer can be represented as a formula that sums the future values of each deposit (cell E16 above):

$$\begin{aligned} \textit{Total at beginning of year } 10 &= 1{,}000*(1+10\%)^{10} + 1{,}000*(1+10\%)^{9} \\ &\quad + \cdots + 1{,}000*(1+10\%)^{1} \\ &= \sum_{t=1}^{10} 1{,}000*(1+10\%)^{t} \end{aligned}$$

An Excel function: Note from cell B18 above that Excel has a function **FV** that gives this sum. The dialog box brought up by **FV** is the following:



We note three things about this function:

- • For positive deposits, **FV** returns a negative number. This is an irritating property of this function which it shares with **PV** and **PMT**. To avoid negative numbers, we have put the **PMT** in as −1,000.
- • The line **PV** in the dialog box refers to a situation where the account has some initial value other than 0 when the series of deposits is made. In the example above this has been left blank, which indicates that the initial account value is zero.
- • As noted in the picture, "Type" (either 1 or 0) refers to whether the deposit is made at the beginning or the end of each period (in our example, the former is the case).

1.7 A Pension Problem—Complicating the Future Value Problem

Consider a typical exercise: You are currently 55 years old and intend to retire at age 60. To make your retirement easier, you intend to start a retirement account:

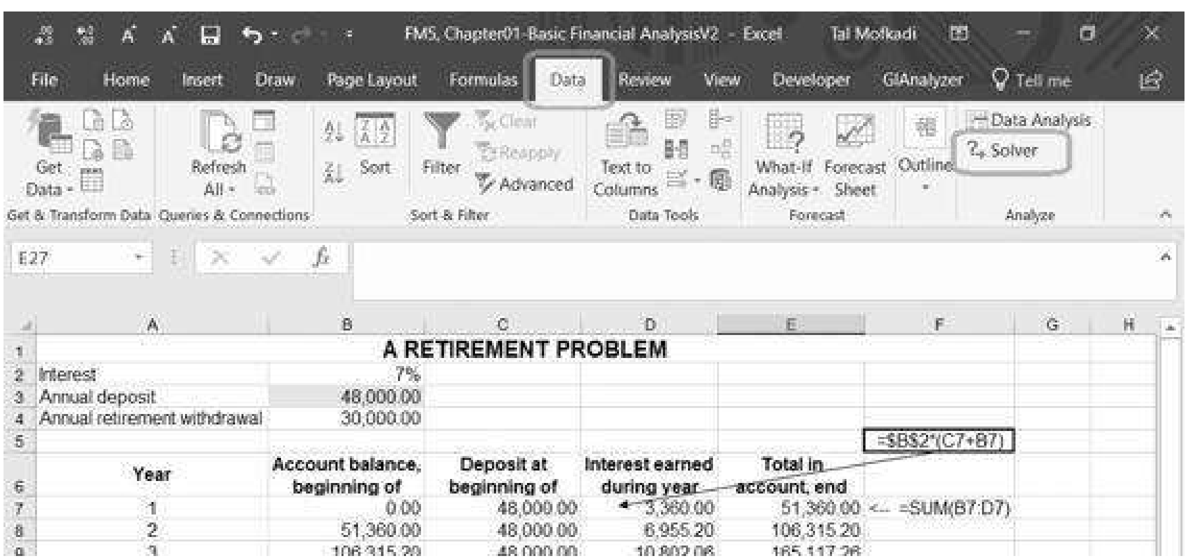
- • At the beginning of each of years 1, 2, 3, 4, 5 (that is, starting today and at the beginning of each of the next four years), you intend to make a deposit into the retirement account. You think that the account will earn 7% per year.
- • After retirement at age 60, you anticipate living eight more years.[5] At the beginning of each of these years you want to withdraw $30,000 from your retirement account. Your account balances will continue to earn 7%.

How much should you deposit annually in the account? A naive analysis is telling you that in order to provide $30,000 per year for eight years (total of $240,000), you need to contribute $240,000/5 = $48,000 in each of the first five years. As the spreadsheet shows, this is wrong. If you make annual deposits of $48,000 you'll end up with a lot of money at the end of eight years!

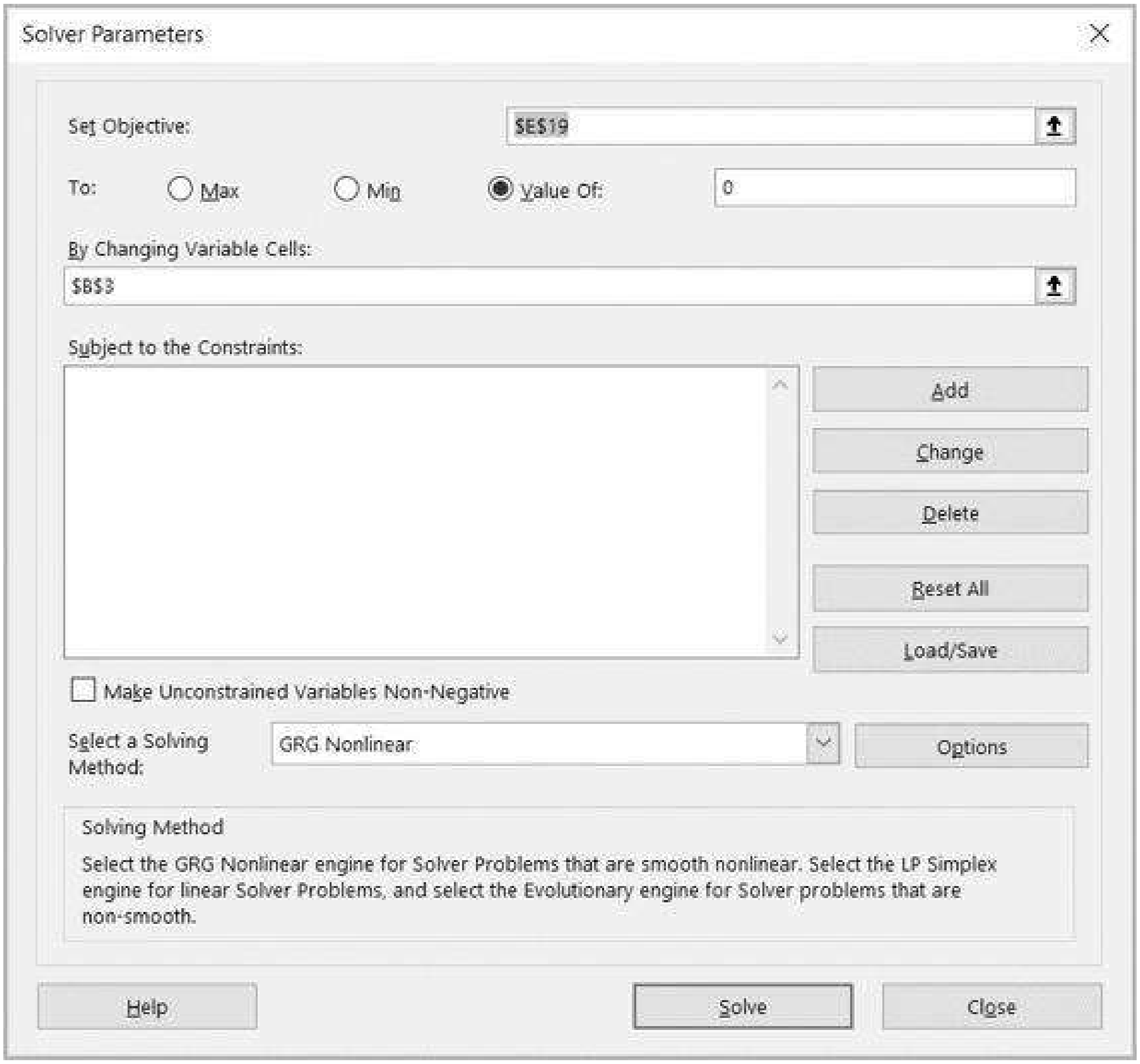
	A	B	C	D	E	F
1		A RETIREMENT PROBLEM - BASICS				
2	Interest	7%				
3	Annual deposit	48,000.00	<-- =240000/5			
4	Annual retirement withdrawal	30,000.00				
5						=$B$2*(C7+B7)
6	Year	Account balance, beginning of year	Deposit at beginning of year	Interest earned during year	Total in account, end year	
7	1	0.00	48,000.00	3,360.00	51,360.00	<-- =SUM(B7:D7)
8	2	51,360.00	48,000.00	6,955.20	106,315.20	
9	3	106,315.20	48,000.00	10,802.06	165,117.26	
10	4	165,117.26	48,000.00	14,918.21	228,035.47	
11	5	228,035.47	48,000.00	19,322.48	295,357.96	
12	6	295,357.96	-30,000.00	18,575.06	283,933.01	
13	7	283,933.01	-30,000.00	17,775.31	271,708.32	
14	8	271,708.32	-30,000.00	16,919.58	258,627.91	
15	9	258,627.91	-30,000.00	16,003.95	244,631.86	
16	10	244,631.86	-30,000.00	15,024.23	229,656.09	
17	11	229,656.09	-30,000.00	13,975.93	213,632.02	
18	12	213,632.02	-30,000.00	12,854.24	196,486.26	
19	13	196,486.26	-30,000.00	11,654.04	178,140.29	
20						
21	**Note**: This problem has 5 deposits and 8 annual withdrawals, all made at the beginning of the year. The beginning of year 13 is the last year of the retirement plan. If the annual deposit is correctly computed, the balance at the beginning (and end) of year 13 after the withdrawal should be zero.					

The reason is clear—this analysis ignores the return on the accumulated savings and the powerful effects of compound interest. (If you set the interest rate in the spreadsheet equal to 0%, you'll see that you're right.)

There are several ways to solve this problem. The first involves Excel's Solver. This can be found on the **Data** menu.[6]



Clicking on **Solver** makes a dialog box appear. The screenshot below shows how we have filled in it:



If we now click on the **Solve** box, we get the following answer:

	A	B	C	D	E	F
1	A RETIREMENT PROBLEM					
2	Interest	7%				
3	Annual deposit	31,150.60				
4	Annual retirement withdrawal	30,000.00				
5						=$B$2*(C7+B7)
6	Year	Account balance, beginning of year	Deposit at beginning of year	Interest earned during year	Total in account, end year	
7	1	0.00	31,150.60	2,180.54	33,331.14	<-- =SUM(B7:D7)
8	2	33,331.14	31,150.60	4,513.72	68,995.46	
9	3	68,995.46	31,150.60	7,010.22	107,156.28	
10	4	107,156.28	31,150.60	9,681.48	147,988.36	
11	5	147,988.36	31,150.60	12,539.73	191,678.68	
12	6	191,678.68	-30,000.00	11,317.51	172,996.19	
13	7	172,996.19	-30,000.00	10,009.73	153,005.92	
14	8	153,005.92	-30,000.00	8,610.41	131,616.34	
15	9	131,616.34	-30,000.00	7,113.14	108,729.48	
16	10	108,729.48	-30,000.00	5,511.06	84,240.55	
17	11	84,240.55	-30,000.00	3,796.84	58,037.38	
18	12	58,037.38	-30,000.00	1,962.62	30,000.00	
19	13	30,000.00	-30,000.00	0.00	0.00	

Solving the Retirement Problem Using Financial Formulas

We can solve this problem in a more intelligent fashion if we understand the discounting process. The present value of the whole series of payments, discounted at 7%, must be zero:

$$\sum_{t=0}^{4} \frac{initial\ deposit}{(1.07)^t} - \sum_{t=5}^{12} \frac{30{,}000}{(1.07)^t} = 0 \;\Rightarrow\; initial\ deposit = \frac{\sum_{t=5}^{12} \frac{30{,}000}{(1.07)^t}}{\sum_{t=0}^{4} \frac{1}{(1.07)^t}}$$

Rewrite $\sum_{t=5}^{12} \frac{30{,}000}{(1.07)^t} = \frac{1}{(1.07)^5} \sum_{t=0}^{7} \frac{30{,}000}{(1.07)^t}$. We can now use Excel's **PV** and **PMT** functions to solve the problem, as shown below:

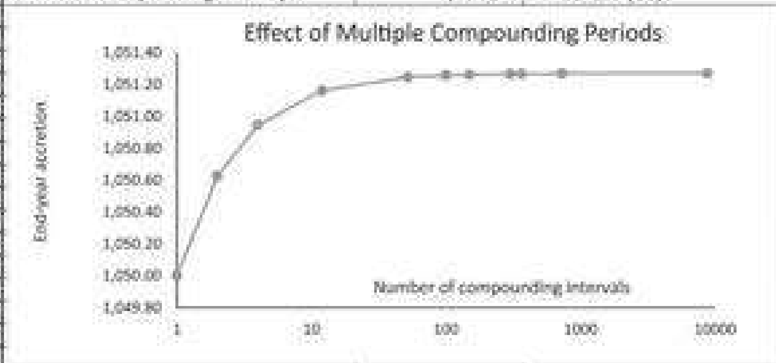
	A	B	C
1	A RETIREMENT PROBLEM - USING FINANCIAL FORMULAS		
2	Interest	7%	
3	Annual retirement withdrawal	30,000.00	
4	Years of withdrawal	8	
5	Years of deposit	5	
6	Present value of withdrawals	136,664.25	<--=-PV(B2,B4,B3,,1)/(1+B2)^B5
7	Annual deposit	31,150.60	<--=PMT(B2,B5,-B6,,1)
8			
9	Check:	31,150.60	<--=PMT(B2,B5,PV(B2,B4,B3)/(1+B2)^B5)

1.8 Continuous Compounding

Suppose you deposit $1,000 in a bank account that pays 5% per year. This means that at the end of the year you will have $1,000 $*$ (1.05) = $1,050. Now suppose that the bank interprets "5% per year" to mean that it pays you 2.5% interest twice a year. Thus after six months you'll have $1,025, and after one year you will have $\$1{,}000 * \left(1+\frac{0.05}{2}\right)^2 = \$1{,}050.625$. By this logic, if you get paid interest n times per year, your accretion at the end of the year will be $\$1{,}000 * \left(1+\frac{0.05}{n}\right)^n$. As n increases this amount gets larger, converging (rather quickly, as you will soon see) to $e^{0.05}$, which in Excel is written as the function **Exp**. When n is infinite, we refer to this as *continuous compounding*. (By typing **Exp(1)** in a spreadsheet cell, you can see that e = 2.7182818285. …)

As you can see in the next display, $1,000 continuously compounded for one year at 5% grows to $1,000 $* e^{0.05}$ = $1,051.271 at the end of the year. Continuously compounded for t years, it will grow to $1,000 $* e^{0.05 * t}$, where t need not be a whole number (for example, when $t = 4.25$, then the accumulation factor $e^{0.05 * 4.25}$ measures the growth of the initial investment at 5% annually, continuously compounded for four years and three months).

	A	B	C
1	MULTIPLE COMPOUNDING PERIODS		
2	Initial deposit	1,000	
3	Interest rate	5%	
4	Number of compounding periods per year	2	
5	Interest per compounding period	2.500%	<--=B3/B4
6	Accretion in one year	1,050.625	<--=B2*(1+B5)^B4
7	Continuous compounding with Exp	1,051.271	<--=B2*EXP(B3)



	A	B	C
24	Compounding periods per year	End-year accretion	
25	1	1,050.000	<--=$B$2*(1+$B$3/A25)^A25
26	2	1,050.625	<--=$B$2*(1+$B$3/A26)^A26
27	4	1,050.945	
28	12	1,051.162	
29	52	1,051.246	
30	100	1,051.258	
31	150	1,051.262	
32	300	1,051.267	
33	365	1,051.267	
34	730	1,051.269	
35	8760	1,051.271	

The conclusion: more compounding periods increase the future value, though there is a clear asymptotic value. As we will see below, for accretion over t years, this value is e^{rt}.

Back to Finance—Continuous Discounting

If the accretion factor for continuous compounding at interest r over t years is e^{rt}, then the discount factor for the same period is e^{-rt}. Thus, a cash flow C_t occurring in year t and discounted at continuously compounded rate r will be worth $C_t \times e^{-rt}$ today. This is illustrated below:

	A	B	C	D
1	CONTINUOUS DISCOUNTING			
2	Interest	8%		
3				
4	Year	Cash flow	Continuously discounted PV	
5	1	100	92.31	<-- =B5*EXP(-$B$2*A5)
6	2	200	170.43	<-- =B6*EXP(-$B$2*A6)
7	3	300	235.99	
8	4	400	290.46	
9	5	500	335.16	
10				
11	Present value		1,124.35	<-- =SUM(C5:C9)

Calculating the Continuously Compounded Return from Price Data

Suppose at time 0 you had $1,000 in the bank and suppose that one year later you had $1,200. What was your percentage return? Although the answer may appear obvious, it actually depends on the compounding method. If the bank paid interest only once a year, then the return would be 20%:

$$\frac{1,200}{1,000} - 1 = 20\%$$

However, if the bank paid interest twice a year, you would need to solve the following equation to calculate the return:

$$1,000 * \left(1+\frac{r}{2}\right)^2 = 1,200 \quad \Rightarrow \quad r = 2 * \left[\left(\frac{1,200}{1,000}\right)^{1/2} - 1\right] = 19.089\%$$

The annual percentage return when interest is paid twice a year is, therefore, 2 * 9.5445% = 19.089%.

In general, if there are n compounding periods per year, you have to solve

$$r = n * \left[\left(\frac{1,200}{1,000}\right)^{1/n} - 1\right].$$

If n is very large, this converges to

$$r = \ln\left(\frac{1,200}{1,000}\right) = 18.2322\%.$$

	A	B	C
1	**CALCULATING RETURNS FROM PRICES**		
2	Initial deposit	1,000	
3	End-of-year value	1,200	
4	Number of compounding periods	2	
5	Implied annual interest rate	19.09%	<-- =((B3/B2)^(1/B4)-1)*B4
6			
7	Continuous return	18.23%	<-- =LN(B3/B2)
8			
9	**Implied annual interest rate with n compounding periods**		
10	**Number of compounding periods**	**Rate**	
11		19.09%	<-- =B5, data table header
12	1	20.00%	
13	2	19.09%	
14	4	18.65%	
15	12	18.37%	
16	24	18.30%	
17	365	18.24%	
18	1000	18.23%	

Why Use Continuous Compounding?

All of this may seem somewhat esoteric. However, continuous compounding/discounting is often used in financial calculations. In this book, it is used to calculate portfolio returns (Section III) and in practically all of the options calculations (Section IV).

There's another reason to use continuous compounding—its ease of calculation. Suppose, for example, that your \$1,000 grew to \$1,500 in one year and nine months. What's the annualized rate of return? The easiest—and most consistent—way to answer this question is to calculate the continuously compounded annual return. Since one year and nine months equals 1.75 years, this return is:

$$1{,}000 * e^{r*1.75} = 1{,}500 \quad \Rightarrow \quad r = \frac{1}{1.75}\ln\left[\frac{1{,}500}{1{,}000}\right] = 23.1694\%$$

1.9 Discounting Using Dated Cash Flows

Most of the computations in this chapter consider cash flows that occur at fixed periodic intervals. Typically we look at cash flows that occur on dates 0, 1, …, n, where the period indicates an annual, semiannual, or other fixed interval. Two Excel functions, **XIRR** and **XNPV**, allow us to do computations on cash flows that occur on specific dates that need not be at even intervals.[7]

In the example below we compute the IRR of an investment of \$1,000 made on January 1, 2019, with payments on specific dates.

	A	B	C
1	USING XIRR TO COMPUTE THE ANNUALIZED INTERNAL RATE OF RETURN		
2	Date	Cash flow	
3	01-Jan-19	-1,000	
4	03-Mar-19	150	
5	04-Jul-20	100	
6	12-Oct-20	50	
7	25-Dec-20	1,000	
8			
9	IRR	16.67%	<-- =XIRR(B3:B7,A3:A7)

The function **XIRR** outputs an annualized return. It works by computing the daily IRR and annualizing it: $XIRR = (1 + dailyIRR)^{365} - 1$.

XNPV computes the net present value of a series of cash flows occurring on specific dates, as shown below:

	A	B	C
1	**USING XNPV TO COMPUTE THE NET PRESENT VALUE**		
2	Annual discount rate	8%	
3			
4	**Date**	**Cash flow**	
5	01-Jan-19	-1,000	
6	03-Mar-20	100	
7	04-Jul-20	195	
8	12-Oct-21	350	
9	25-Dec-22	800	
10			
11	Net present value	136.37	<-- =XNPV(B2,B5:B9,A5:A9)
12			
13	Note that **XNPV** has a different syntax from **NPV**! **XNPV** requires all the cash flows, including the initial cash flow, whereas **NPV** assumes that the first cash flow occurs one period hence.		

Fixing Bugs in XNPV and XIRR

Both **XNPV** and **XIRR** have bugs, which Microsoft has not fixed in several versions of Excel. The file with this chapter includes functions that fix these bugs, called **NXNPV** and **NXIRR**.[8]

- **XNPV** doesn't work with zero or negative interest rates.
- **XIRR** does not identify multiple internal rates of return.

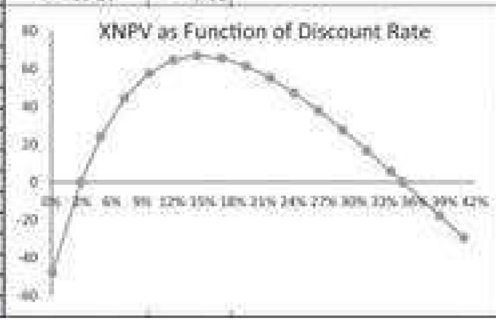
The **XNPV** relates to the failure of this function to correctly deal with zero or negative discount rates.

	A	B	C
1	**PROBLEM WITH XNPV** **XNPV does not work with zero or negative discount rates**		
2	Discount rate	-3.00%	
3	Net present value		
4	Using XNPV	#NUM!	<-- =XNPV(B2,B8:B12,A8:A12)
5	Using NXNPV	591.93	<-- =nXNPV(B2,B8:B12,A8:A12)
6			
7	**Date**	**Cash flow**	
8	01-Jan-19	-1,000	
9	03-Mar-20	100	
10	04-Jul-20	195	
11	12-Oct-21	350	
12	25-Dec-22	800	

The **NXNPV** function fixes this problem.

The bug in **XIRR** is that the **Guess** switch on **XIRR** doesn't work. Consider the following problem:

	A	B	C	D	E	F	G
1	PROBLEMS WITH XIRR - XIRR cannot be calculated when there are multiple IRRs						
2	Discount rate	22.00%					
3	Net present value	54.22927	<--=XNPV(B2,B11:B17,A11:A17)				
4					Data table: XNPV as function of discount rate		
5	IRR						
6	No Guess	#NUM!	<--=XIRR(B11:B17,A11:A17)		Rate	54.229	<--=B3, data table header
7	Guess = 35%	#NUM!	<--=XIRR(B11:B17,A11:A17,35%)		0.1%	-47.942	
8	Guess = 5%	#NUM!	<--=XIRR(B11:B17,A11:A17,5%)		2.9%	0.000	
9					4.9%	24.134	
10	Date	Cash flow			7.3%	44.620	
11	30-Jun-19	-500			9.7%	57.578	
12	14-Feb-20	100			12.1%	64.597	
13	14-Jan-21	300			14.5%	66.947	
14	14-Apr-22	400			16.9%	65.641	
15	14-Jun-23	600			19.3%	61.492	
16	14-Mar-24	800			21.7%	55.149	
17	14-Feb-25	-1,750			24.1%	47.135	
18					26.5%	37.867	
19					28.9%	27.682	
20					31.3%	16.847	
21					33.7%	5.579	
22					34.9%	0.000	
23					38.5%	-17.609	
24					40.9%	-29.283	



From the data table, it is evident that there are two internal rates of return (around 2.9% and around 34.9%). But the **XIRR** function does not identify either (see cells B6:B8).

The function **NXIRR** fixes this bug.

	A	B	C
1	NXIRR FIXES THE XIRR BUG		
2	Discount rate	-3.00%	
3			
4	IRR		
5	No Guess	2.91%	<-- =nXIRR(B10:B16,A10:A16) , no Guess
6	Guess = 35%	34.87%	<-- =nXIRR(B10:B16,A10:A16,35%), Guess = 35%
7	Guess = 5%	2.91%	<-- =nXIRR(B10:B16,A10:A16,5%) , Guess = 5%
8			
9	Date	Cash flow	
10	30-Jun-19	-500	
11	14-Feb-20	100	
12	14-Jan-21	300	
13	14-Apr-22	400	
14	14-Jun-23	600	
15	14-Mar-24	800	
16	14-Feb-25	-1,750	

Exercises

1. You are offered an asset costing $600 that has cash flows of $100 at the end of each of the next 10 years.
   1. If the appropriate discount rate for the asset is 8%, should you purchase it?
   2. What is the IRR of the asset?

2. You just took a $10,000, 5-year mortgage loan. Payments at the end of each year are flat (equal in every year) at an interest rate of 15%. Calculate the appropriate loan amortization table, showing the breakdown in each year between principal and interest.

3. You are offered an investment with the following conditions:

- The cost of the investment is \$1,000.
- The investment pays out a sum X at the end of the first year; this payout grows at the rate of 10% per year for 11 years.

If your discount rate is 15%, calculate the smallest X that would entice you to purchase the asset. For example, as you can see in the following display, X = \$100 is too small—the NPV is negative:

	A	B	C
1	Discount rate	15%	
2	Initial payment	129.2852	
3	NPV	-226.52	<-- =B6+NPV(B1,B7:B17)
4			
5	**Year**	**Cash flow**	
6	0	-1000.00	
7	1	100.00	<-- 100
8	2	110.00	<-- =B7*1.1
9	3	121.00	<-- =B8*1.1
10	4	133.10	
11	5	146.41	
12	6	161.05	
13	7	177.16	
14	8	194.87	
15	9	214.36	
16	10	235.79	
17	11	259.37	

4. The following cash flow pattern has two IRRs. Use Excel to draw a graph of the NPV of these cash flows as a function of the discount rate. Then use the IRR function to identify the two IRRs. Would you invest in this project if the opportunity cost were 20%?

	A	B
4	**Year**	**Cash flow**
5	0	-500
6	1	600
7	2	300
8	3	300
9	4	200
10	5	-1,000

5. In this exercise we solve iteratively for the internal rate of return. Consider an investment that costs 800 and has cash flows of 300, 200, 150, 122, and 133 in years 1 to 5. Setting up the loan amortization table below shows that 10% is greater than the IRR (since the return of principal at the end of year 5 is less than the principal at the beginning of the year).

	A	B	C	D	E	F	G	H
1	IRR?	10.00%						
2				LOAN TABLE			Division of payment between:	
3	Year	Cash flow		Year	Principal at beginning of year	Payment at end of year	Interest	Principal
4	0	-800		1	800.00	300.00	80.00	220.00
5	1	300		2	580.00	200.00	58.00	142.00
6	2	200		3	438.00	150.00	43.80	106.20
7	3	150		4	331.80	122.00	33.18	88.82
8	4	122		5	242.98	133.00	24.30	108.70
9	5	133		6	134.28	<-- Should be zero for IRR		

Setting cell B1 to 3% shows that 3% is less than the IRR, since the return of principal at the end of year 5 is greater than the principal at the beginning of year 5.

By changing the IRR in cell B1, find the internal rate of return of the investment.

	A	B	C	D	E	F	G	H
1	IRR?	3.00%						
2				LOAN TABLE			Division of payment between:	
3	Year	Cash flow		Year	Principal at beginning of year	Payment at end of year	Interest	Principal
4	0	-800		1	800.00	300.00	24.00	276.00
5	1	300		2	524.00	200.00	15.72	184.28
6	2	200		3	339.72	150.00	10.19	139.81
7	3	150		4	199.91	122.00	6.00	116.00
8	4	122		5	83.91	133.00	2.52	130.48
9	5	133		6	-46.57	<-- Should be zero for IRR		

6. An alternative definition of the IRR is the rate that makes the principal at the beginning of year 6 equal to zero.[9] This is shown in the printout above, in which cell E9 gives the principal at the beginning of year 6. Using the **Goal Seek** function of Excel, find this rate (below we illustrate how the screen should look).

(Of course, you should check your calculations by using the Excel **IRR** function.)

7. Calculate the flat annual payment required to pay off a 13%, 5-year mortgage loan of $100,000.
8. You have just taken a car loan of $15,000. The loan is for 48 months at an annual interest rate of 15% (which the bank translates to a monthly rate of 15%/12 = 1.25%). The 48 payments (to be made at the end of each of the next 48 months) are all equal.
    1. Calculate the monthly payment on the loan.
    2. In a loan table calculate, for each month, the principal remaining on the loan at the beginning of the month and the split of that month's payment between interest and repayment of principal.
    3. Show that the principal at the beginning of each month is the present value of the remaining loan payments at the loan interest rate (use either **NPV** or the **PV** functions).

9. You are considering buying a car from a local auto dealer. The dealer offers you one of two payment options:

- You can pay $30,000 cash.
- You can choose a "deferred payment plan," paying the dealer $5,000 cash today and a payment of $1,050 at the end of each of the next 30 months.

As an alternative to the dealer financing, you have approached a local bank, which is willing to give you a car loan of $25,000 at the rate of 1.25% per month.

1. Assuming that 1.25% is the opportunity cost, calculate the present value of all the payments on the dealer's deferred payment plan.
2. What is the effective interest rate being charged by the dealer? Do this calculation by preparing a spreadsheet like the one below (only part of the spreadsheet is shown—you have to do this calculation for all 30 months).

	D	E	F	G	H
2	Month	Cash payment	Payment under deferred payment plan	Difference	
3	0	30,000	5,000	25,000	<-- =E3-F3
4	1	0	1,050	-1,050	<-- =E4-F4
5	2	0	1,050	-1,050	
6	3	0	1,050	-1,050	
7	4	0	1,050	-1,050	
8	5	0	1,050	-1,050	
9	6	0	1,050	-1,050	
10	7	0	1,050	-1,050	
11	8	0	1,050	-1,050	

Now calculate the IRR of the difference column; this is the monthly *effective interest rate* on the deferred payment plan.

10. You are considering a savings plan that calls for a deposit of $15,000 at the end of each of the next five years. If the plan offers an interest rate of 10%, how much will you accumulate at the end of year 5? Do this calculation by completing the spreadsheet below. This spreadsheet does the calculation twice—once using the **FV** function and once using a simple table that shows the accumulation at the beginning of each year.

	A	B	C	D
1	Annual payment	15,000		
2	Interest rate	10%		
3	Number of years	5		
4	Total value	$91,576.50	<-- =FV(B2,B3,-B1,,0)	
5				
6	Year	Accumulation at beginning of year	Payment at end of year	Annual interest
7	1	0	15,000	0.00
8	2	15,000	15,000	1,500.00
9	3	31,500		
10	4			
11	5			
12	6			

11. Redo the previous calculation, this time assuming that you make five deposits at the *beginning* of this year and the following four years. How much will you accumulate by the end of year 5?

12. A mutual fund has been advertising that, had you deposited $250 per month in the fund for the last 10 years, you would now have accumulated $85,000. Assuming that these deposits were made at the beginning of each month for a period of 120 months, calculate the effective annual return fund investors got.

Hint: Set up the spreadsheet below and then use **Goal Seek**.

	A	B	C
1	Monthly payment	250	
2	Number of months	120	
3			
4	Effective monthly return?		
5	Accumulation		<-- =FV(B4,B2,-B1,,1)

The effective annual return can then be calculated in one of two ways:

- $(1 + \textit{monthly return})^{12} - 1$: This is the compound annual return, which is preferable, since it makes allowance for the reinvestment of each month's earnings.
- 12 * *monthly return*: This method is often used by banks.

13. You have just turned 35, and you intend to start saving for your retirement. Once you retire in 30 years (when you turn 65), you would like to have an income of $100,000 per year for the next 20 years. Calculate how much you would have to save between now and age 65 in order to finance your retirement income. Make the following assumptions:

- All savings draw compound interest of 10% per year.
- You make the first payment today and the last payment on the day you turn 64 (30 payments).
- You make the first withdrawal when you turn 65 and the last withdrawal when you turn 84 (20 payments).

14. You currently have $25,000 in the bank, in a savings account that draws 5% interest. Your business needs $25,000, and you are considering two options: (a) Use the money in your savings account or (b) borrow the money from the bank at 6%, leaving the money in the savings account.

Your financial analyst suggests that solution (b) is better. The logic: The sum of the interest paid on the 6% loan is lower than the interest earned at the same time on the $25,000 deposit. The calculations are illustrated below. Show that this logic is wrong. (If you think about it, it couldn't be preferable to take a 6% loan when you are getting 5% interest from the bank. However, the explanation for this may not be trivial.)

	A	B	C	D	E	F
1	EXERCISE 14, financial analyst's calculations					
2	Interest earned	5%				
3	Interest paid	6%				
4	Initial deposit	25,000				
5					=PMT($B$3,2,-$B$4)	
6		THE 6% LOAN				
7	Year	Principal at beginning of year	Payment at end of year	Interest paid	Repayment of principal	
8	1	25,000.00	13,635.92	1,500.00	12,135.92	<-- =C8-D8
9	2	12,864.08	13,635.92	771.84	12,864.08	
10		Total interest paid		2,271.84		
11						
12		Savings Account				
13	Year	In savings account at beginning of year	End-year interest earned	In account at end of year		
14	1	25,000.00	1,250.00	26,250.00		
15	2	26,250.00	1,312.50	27,562.50		
16		Interest earned	2,562.50			

15. Use **XIRR** to compute the internal rate of return of the following investment:

	A	B
1	Date	Cash flow
2	30-Jun-20	-899
3	14-Feb-21	70
4	14-Feb-22	70
5	14-Feb-23	70
6	14-Feb-24	70
7	14-Feb-25	70
8	14-Feb-26	1,070

16. Use **XNPV** to value the following investment. Assume that the annual discount rate is 15%.

	A	B
4	Date	Cash flow
5	30-Jun-20	-500
6	14-Feb-21	100
7	14-Feb-22	300
8	14-Feb-23	400
9	14-Feb-24	600
10	14-Feb-25	800
11	14-Feb-26	-1,800

17. Identify the two internal rates of return of the investment in exercise 16.

---

1. 1. In our book *Principles of Finance with Excel*, 3rd ed. (Oxford University Press, 2017), we have discussed many basic Excel/finance topics at greater length.
2. 2. This strange property—returning negative values for positive cash flows—is shared by a number of otherwise impeccable Excel functions such as PMT and PV. The somewhat convoluted logic that led Microsoft to write these functions this way is not worth explaining.
3. 3. All the formulas in this subsection depend on some well-known but oft-forgotten high school algebra.
4. 4. If you don't put in a guess (as we did in the previous section), Excel defaults to a guess of 0.1. Thus, in the current example, IRR(B34:B39) will return 8.78%.
5. 5. Of course, you're going to live much longer! And we wish you good health! The dimensions of this problem have been chosen to make it fit nicely on a page.
6. 6. If the Solver does not appear on the Tools menu, then you must load it. Go to **File|Options|Add-ins** and click **Solver Add-In** on the list of programs. Note that you could also use the **Goal Seek** tool to solve this problem. For simple problems such as this one, there is not much difference between the **Solver** and **Goal Seek**; the one (not inconsiderable) advantage of the **Solver** is that it remembers its previous arguments, so that if you bring it up again on the same spreadsheet, you can see what you did in the previous iteration. In later chapters we will illustrate problems that cannot be solved by **Goal Seek** and where the use of the **Solver** is a necessity.
7. 7. If you do not see these functions, add them in by going to **Tools|Add-ins** on the toolbar and checking **Analysis ToolPak**.
8. 8. These bug fixes were developed by our colleague Benjamin Czaczkes.
9. 9. In general, of course, the IRR is the rate of return that makes the principal in the year *following* the last payment equal to zero.

2 Corporate Valuation Overview

2.1 Overview

Chapters 3–6 discuss various aspects of corporate valuation, one of the trickiest topics in finance. This chapter aims to provide a short overview of this complicated topic. In succeeding sections and chapters we turn to the components of valuation: the comparable approach, also called multiples valuation (see section 2.4 of this chapter); the computation of the cost of capital (Chapter 3); and the direct valuation of corporate free cash flows from the firm's consolidated statement of cash flows and the projection of firm pro forma financial statements (Chapter 4).

What Is Corporate Valuation About?

When we discuss the valuation of a company, we may be referring to any of the following:

- • Enterprise value: Valuing the company's productive activities.
- • Equity: Valuing the shares of a company, whether for the purpose of buying or selling a single share or valuing all of the equity for purposes of a corporate acquisition.
- • Debt: Valuing the company's debt. When debt is risky, its value depends on the value of the company that has issued the debt.
- • Other: We may want to value other securities related to the company—for example, the firm's warrants or options, employee stock options, etc.

In Chapters 2–6 we discuss the first two of these topics, leaving the valuation of debt and other securities for later chapters.[1]

2.2 Three Methods to Compute Enterprise Value (EV)

The key concept in corporate valuation is *enterprise value*. The enterprise value (EV) of the firm is the value of the firm's core business activities. EV forms the basis of most corporate valuation models. We distinguish between three approaches to computing the enterprise value:

1. The accounting approach to EV moves items on the balance sheet so that all operating items are on the left-hand side of the balance sheet and all financial items are on the right-hand side. Although most academics sneer at this approach, it is often a useful starting point for thinking about enterprise value.

2. The efficient markets approach to EV revalues—to the extent possible—present items on the accounting EV balance sheet at market values. An obvious revaluation is to replace the firm's book value of equity with the market value of the equity. To the extent that we know the market value of other firm liabilities—debt, pension obligations, etc.—this market value will also replace the book value. The efficient markets approach is the basis of the "comparables approach," also called "multiples approach of valuation."
3. The discounted cash flow (DCF) approach values the EV as the present value of the firm's future anticipated free cash flows (FCFs) discounted at the weighted average cost of capital (WACC). The FCFs can best be thought of as the cash flows produced by the firm's operating assets—its working capital, fixed assets, goodwill, etc.

    In this book we use two implementations of the DCF approach. These approaches differ in their derivation of the firm's free cash flows.

    - In Chapter 4 we base our projections of future anticipated FCFs on an analysis of the firm's consolidated statement of cash flows.
    - In Chapters 5 and 6 we base our projections of future anticipated FCFs on a pro forma model for the firm's financial statements.

2.3 Using Accounting Book Values to Value a Company: The Firm's Accounting Enterprise Value

While we would rarely use accounting numbers to value a company, the balance sheet of a company is a useful starting framework for the valuation process. In this section we show how accounting statements can help us define the concept of *enterprise value* (EV). As a starting point, consider the balance sheet for XYZ Corp.

	A	B	C	D	E
1	**XYZ CORP BALANCE SHEET**				
2	**Assets**			**Liabilities and equity**	
3	Short-term assets			Short-term liabilities	
4	Cash	1,000		Accounts payable	1,500
5	Marketable securities	1,500		Taxes payable	200
6	Inventories	1,500		Current portion of long-term debt	1,000
7	Accounts receivable	3,000		Short-term debt	500
8					
9	Fixed assets			Long-term debt	1,500
10	Land	150		Pension liabilities	800
11	Plant, property and equipment at cost	2,000			
12	Minus accumulated depreciation	-700		Preferred stock	200
13	**Net fixed assets**	**1,450**		Minority interest	100
14					
15	Goodwill	1,000		Equity	
16				Stock at par	1,000
17	Discontinued operation	500		Accumulated retained earnings	3,500
18				Stock repurchases	-350
19	**Total assets**	**9,950**		**Total liabilities and equity**	**9,950**

We identify the operational, excess, and financial items on the balance sheet.
We separate the operational versus excess assets (also called *nonoperating assets*).

- • Operational assets generate the expected cash flows.
- • Excess assets are not part of the cash-generating assets and hence not part of the enterprise value. The excess assets may be financial assets (e.g., cash and cash equivalents) or tangible assets (e.g., real estate that is not used for the operations, discontinued operations, net operating losses, joint ventures, or net pension assets).

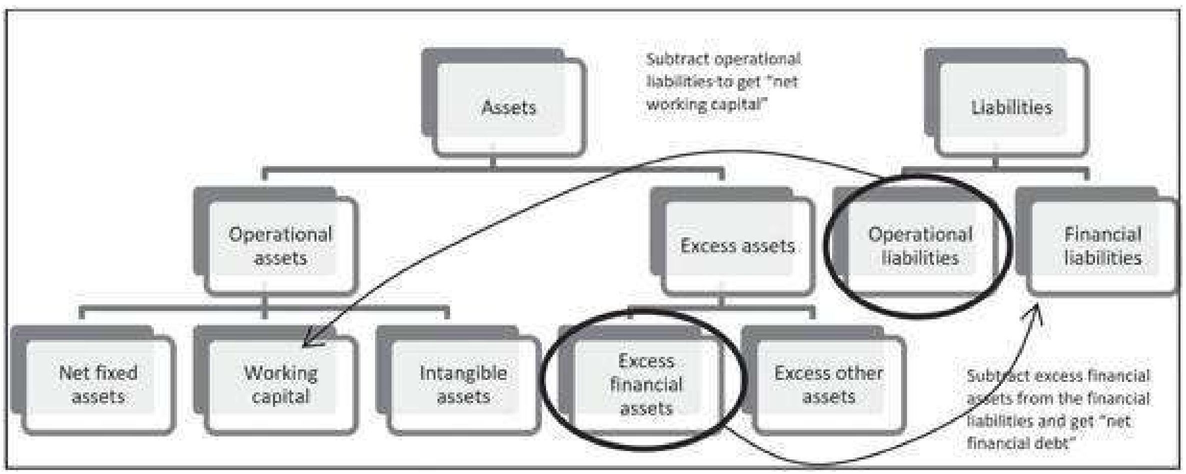
We also split the liabilities into operational versus financial liabilities:

- • Operational liabilities mainly include account payables and other payables.
- • Financial liabilities include all the interest-bearing liabilities.

	A	B	C	D	E
1	**XYZ CORP BALANCE SHEET** The balance sheet items are categorized to main groups by their economic nature				
2	**Assets**			**Liabilities and equity**	
3	Excess financial assets			Operational liabilities (negative working capital)	
4	Cash	1,000		Accounts payable	1,500
5	Marketable securities	1,500		Taxes payable	200
6	**Total excess financial assets**	**2,500**		**Total operating liabilities**	**1,700**
7					
8	Net fixed assets			Financial liabilities	
9	Land	150		Current portion of long-term debt	1,000
10	Plant, property and equipment at cost	2,000		Short-term debt	500
11	Minus accumulated depreciation	-700		Long-term debt	1,500
12	**Net fixed assets**	**1,450**		Pension liabilities	800
13				Preferred stock	200
14	Working capital			Minority interest	100
15	Inventories	1,500		**Total financial liabilities**	**4,100**
16	Accounts receivable	3,000			
17				Equity	
18	Goodwill	1,000		Stock at par	1,000
19				Accumulated retained earnings	3,500
20	Excess assets			Stock repurchases	-350
21	Discontinued operation	500		**Total equity**	**4,150**
22					
23	**Total assets**	**9,950**		**Total liabilities and equity**	**9,950**
24					
25	=SUM(B6,B12,B15:B21)			=SUM(E21,E16,E6)	

Once we categorize each item in the balance sheet, we create the operational balance sheet in two steps:

1. We move the operational current assets to the left side of the balance sheet.
2. We move all the debt (short-term debt, current portion of long-term debt, and long-term) into one debt item.



When we finish these steps, we have all of the firm's productive assets on the left side of the balance sheet and all of its financing on the right side. The left-hand side of the resulting balance sheet includes the firm's *enterprise value*, defined as the value of the firm's operational net assets. These are the assets that provide the cash flows for the firm's actual business activities.

	A	B	C	D	E
1	**XYZ CORP OPERATIONAL BALANCE SHEET** **Operational current liabilities moved to left side, all financial liabilities on right side**				
2	**Assets**			**Liabilities and equity**	
3	Net fixed assets	1,450		Net financial debt	
4				Financial liabilities	4,100
5	Working capital			Minus excess financial assets	-2,500
6	Inventories	1,500		**Net financial debt**	**1,600**
7	Accounts receivable	3,000			
8	Minus accounts payable	-1,500			
9	Minus taxes payable	-200			
10	**Net working capital**	**2,800**			
11				Equity	4,150
12	Goodwill	1,000			
13					
14	**Total net operational assets**	**5,250**			
15					
16	Excess assets	500			
17					
18	**Total assets**	**5,750**		**Total liabilities and equity**	**5,750**
19					
20	='Balance sheet categorized'!B21			=SUM(E11,E6)	
21					
22	=SUM(B14,B16)				

Caterpillar Inc.[2]

To show the rewriting of the balance sheet in practice, here is the December 31, 2019, balance sheet for Caterpillar Inc. (CAT):

	A	B	C	D	E
1	**CATERPILLAR INC., BALANCE SHEET** **31 December 2019 (Dollars in Millions)**				
2	**Assets:**			**Liabilities:**	
3				**Current liabilities**	
4	**Current assets**			Accounts payable	5,957
5	Cash and short-term investments	8,284		Accrued expenses	3,750
6	Receivables - trade and other	8,568		Accrued wages, salaries and employee benefits	1,629
7	Receivables - finance	9,336		Customer advances	1,187
8	Prepaid expense and other assets	1,739		Dividends payable	567
9	Inventories	11,266		Other current liabilities	2,155
10	**Total current assets**	**39,193**		Machinery, energy & transportation	21
11				Financial products	11,356
12	Property, plant and equipment, net	12,904		**Total current liabilities**	**26,621**
13	Long-term receivables - trade and other	1,193			
14	Long-term receivables - finance	12,651		**Long-term debt due after one year:**	
15	Noncurrent deferred and refundable income taxes	1,411		Machinery, energy & transportation	9,141
16	Intangible assets	1,565		Financial products	17,140
17	Goodwill	6,196		Liability for postemployment benefits	6,599
18	Other assets	3,340		Other liabilities	4,323
19				**Total liabilities**	**63,824**
20					
21				**Total stockholders' equity**	**14,629**
22					
23	**Total assets**	**78,453**		**Total liabilities and stockholders' equity**	**78,453**

To get to the enterprise value balance sheet for Caterpillar, we move financial items from the left side of the balance sheet to the right, and we move operating current liabilities from the right side of the balance sheet to the left. Notice that we netted out liquid assets (cash and marketable securities) from the financial debts of the company. The assumption is that these assets are not needed for the core business activities of Caterpillar.

The book value of Caterpillar's enterprise value is $54,285 million ($54.285 billion). When calculating the enterprise value using the accounting data, it is usually called *invested capital* or *capital employed*.

	A	B	C	D	E
1	**CATERPILLAR INC., OPERATIONAL BALANCE SHEET 31 December 2019, (Dollars in Millions)**				
2	**Enterprise value (accounting values)**			**Net financial debt**	
3	Property, plant and equipment, net	12,904		Short-term financial products	11,355
4	Net working capital	33,620		Dividends payable	567
5	Goodwill and intangible	7,761		Other current liabilities	2,155
6	**Accounting enterprise value (invested capital)**	**54,285**		Machinery, energy and transportation	9,141
7				Long-term financial products	17,140
8	Other assets	3,340		Liability for postemployment benefits	6,599
9				Other liabilities	4,323
10	**Total**	**57,625**		Less cash and short-term investments	-8,284
11				**Net financial debt**	**42,996**
12					
13				**Equity**	14,629
14					
15				**Total**	**57,625**
16					
17	**Where net working capital is:**				
18	Receivables - trade and other	8,568			
19	Receivables - finance	9,336			
20	Prepaid expense and other assets	1,739			
21	Inventories	11,266			
22	Long-term receivables - trade and other	1,193			
23	Long-term receivables - finance	12,651			
24	Noncurrent deferred and refundable income taxes	1,411			
25	Accounts payable	-5957			
26	Accrued expenses	-3750			
27	Accrued wages, salaries and employee benefits	-1629			
28	Customer advances	-1187			
29	Machinery, energy and transportation	-21			
30	**Net working capital**	**33,620**			

2.4 The Efficient Markets Approach to Corporate Valuation

The Caterpillar example assumes that the book value is a correct valuation of the company. But a simple calculation shows how problematic this is: At the end of 2019, Caterpillar had 552.66 million shares outstanding, and the market price per share was $147.68. This suggests that Caterpillar's enterprise value is priced at $121.3 billion—a far cry from the book value of the enterprise value of $54.3 billion.

	A	B	C	D	E
1	**CATERPILLAR VALUATION OF EQUITY: EFFICIENT MARKETS APPROACH**				
2				Shares outstanding (Dec 2019, M)	552.66
3	Formula in cell $B$6 is: =E7+E6-B7			Share price (Dec 2019, $)	147.68
4				Market capitalization ($M) - equity value	81,617
5					
6	**Enterprise value efficient markets approach**	**121,273**		Net financial debt	42,996
7	Other assets	3,340		Equity	81,617
8					
9	**Total**	**124,613**		**Total**	**124,613**

The efficient markets approach to the valuation of Caterpillar's equity and financial liabilities assumes that the market value of a company's shares or debt is simply the market value at the time of valuation. This approach is better than the accounting approach from the previous section and much simpler than the DCF valuations illustrated in succeeding sections and in Chapters 5 and 6. Moreover, it has the power of logic and much academic research behind it. If markets work—in the sense that there are many participants trading the corporate securities, that there is a lot of information about the company in question, and that the valuator has no special information—why not accept the market price as the true value of the company?[3]

Using the Efficient Markets Approach to Value Specific Balance Sheet Items

Applying the efficient markets approach to the Caterpillar enterprise value balance sheet gives 124,613 for the right-hand side of the balance sheet. This means, of course, that we must revalue the left-hand side of the balance sheet. One approach for bringing this EV balance sheet into balance is to assume that the book value of net working capital and the intangible assets is a reasonable approximation of market value. We can then recompute the market value of the firm's long-term assets to bring the balance sheet into balance.

	A	B	C	D	E
1	**CATERPILLAR INC., ENTERPRISE VALUE BALANCE SHEET** **Right-hand side revalued at market values** **Left-hand side brought into balance with right-hand side by adjusting long-term assets**				
2				Shares outstanding (Dec 2019, M)	553
3				Share price (Dec 2019, $)	147.68
4				Market capitalization ($M) - equity value	81,617
5	**Enterprise value**				
6	Net fixed assets (market value)	79,892	<-- =B9-B8-B7	Net financial debt	42,996
7	Net working capital	33,620			
8	Goodwill and intangible	7,761			
9	**Enterprise value efficient markets approach**	**121,273**	<-- =B12-B10		
10	Other assets	3,340		Equity	81,617
11					
12	**Total**	**124,613**	<-- =E12	**Total**	**124,613**

Note that we could also apply the market valuation to other components of the right-hand side of the balance sheet—we could try to revalue the firm's financial obligations, its other liabilities, and minority interest. This is usually not done, unless there is a convincing case that the book value for these liabilities differs materially from their market value.

Using the Efficient Markets Approach to Value Non-Listed Companies

In many cases, we are required to value a non-listed company. The efficient markets approach can easily help us get a rough estimate relatively quickly. Remember that the efficient markets approach states that listed companies with enough trading volume are fairly priced. We will observe market prices for listed companies and infer from them the value of our non-listed company. This is very similar to my guessing what my apartment value is by using what my next-door neighbor sold her apartment for a week ago.

This method is called "valuation by multiples." There is a vast discussion in the literature about the cons and pros of this "quick and dirty" valuation method (spoiler: many cons, one major pro—this method is super easy). We leave this discussion to other textbooks.[4]

Caterpillar is operating in the large-cap, international Construction, Agriculture, and Machinery industry. Somewhat similar companies include Deere & Company (DE) and others. For the simplicity of this discussion, we will use DE as the only comparable firm. Practitioners usually take the average or the median of several players in the industry. Below you can find key financial information for DE:

	A	B	C	D	E	F
1		**DEERE & COMPANY KEY FINANCIALS, 2019** (MILLIONS OF DOLLARS)				
2	**Assets**			**Liabilities**		
3	Cash and cash equivalents	3,857		Short-term borrowings	10,784	
4	Marketable securities	581		Short-term securitization borrowings	4,321	
5	Receivables from unconsolidated affiliates	46		Payables to unconsolidated affiliates	142	
6	Trade accounts and notes receivable - net	5,230		Accounts payable and accrued expenses	9,656	
7	Financing receivables - net	29,195		Deferred income taxes	495	
8	Financing receivables securitized – net	4,383		Long-term borrowings	30,229	
9	Other receivables	1,487		Retirement benefits and other liabilities	5,953	
10	Equipment on operating leases - net	7,567		**Total liabilities**	**61,580**	<-- =SUM(E3:E9)
11	Inventories	5,975		Redeemable noncontrolling interest	14	
12	Property and equipment - net	5,973				
13	Investments in unconsolidated affiliates	215		**Total stockholders' equity**	**11,417**	
14	Goodwill	2,917				
15	Other intangible assets - net	1,380				
16	Retirement benefits	840				
17	Deferred income taxes	1,466				
18	Other assets	1,899				
19	**Total assets**	**73,011**	<-- =SUM(B3:B18)	**Total liabilities and stockholders' equity**	**72,997**	
20						
21	**Key income statement numbers**					
22	Net sales and revenues	39,258				
23	Cost of sales	28,370				
24	**Gross profit**	**10,888**	<-- =B22-B23			
25	Earnings before interest, taxes, depreciation and amortization (EBITDA)	6,107				
26	Depreciation and amortization	2,019				
27	Earnings before interest and taxes (EBIT)	4,088				
28	**Net income**	**3,257**				

Rearranging DE's accounting balance sheet to an operational balance sheet, and using the efficient markets approach to value DE's enterprise value, produces the following spreadsheet:

	A	B	C	D	E	F
1		**DEERE & COMPANY KEY FINANCIALS, 2019** Balance sheet was rearranged to operational balance sheet (MILLIONS OF DOLLARS)				
2	**Enterprise value (accounting values)**			**Net financial debt**		
3	Property and equipment - net	13,755		Short-term borrowings	10,784	
4	Net working capital	34,008		Long-term borrowings	30,229	
5	Goodwill and intangible	4,297		Redeemable noncontrolling interest	14	
6	**Accounting enterprise value (invested capital)**	**52,060**	<-- =SUM(B3:B5)	Retirement benefits and other liabilities	5,953	
7				Less cash and cash equivalents	-3,857	
8				Less marketable securities	-581	
9	Other assets	1,899		**Net financial debt**	**42,542**	<-- =SUM(E3:E8)
10				**Equity**	**11,417**	<-- =B12-E9
11						
12	**Total assets**	**53,959**	<-- =SUM(B6:B11)	**Total**	**53,959**	<-- =SUM(E9:E10)
13						
14		**DEERE & COMPANY, ENTERPRISE VALUE BALANCE SHEET**				
15				Shares outstanding (Oct 2019, M)	313.14	
16				Share price (Oct 2019, $)	176.11	
17				Market capitalization ($M) - equity value	55,147	<-- =E15*E16
18						
19	**Enterprise value efficient markets approach**	**95,790**	<-- =B21-B20	Net financial debt	42,542	
20	Other assets	1,899		Equity	55,147	
21	**Total**	**97,689**	<-- =E21	**Total**	**97,689**	<-- =SUM(E19:E20)

We can now use the insights from DE to assess CAT's enterprise value. Note that a multiple is defined as the enterprise value divided by the source of value. For example, sales multiple is defined as $\frac{\textit{Enterprise value}}{\textit{sales}}$, earnings before interest and taxes (EBIT) multiple is defined as $\frac{\textit{Enterprise value}}{\textit{EBIT}}$, and so on. In our example we compute CAT's enterprise value by using the EBIT and the earnings before interest, taxes, depreciation, and amortization (EBITDA) multiples deduced from DE.

	A	B	C	D
31	**CATERPILLAR INC., VALUATION USING MULTIPLES BASED ON DEERE & COMPANY**			
32	**Step 1 - calculate the multiples from DE**			
33	Net sales and revenues multiple	2.44	<--=$B$19/B25	
34	Gross profit multiple	8.80	<--=$B$19/B26	
35	Earnings before interest, taxes, depreciation and amortization (EBITDA) multiple	15.69	<--=$B$19/B27	
36	Earnings before interest and taxes (EBIT) multiple	23.43	<--=$B$19/B28	
37	Net income multiple	29.41	<--=$B$19/B29	
38				
39	**Step 2 - Enterprise value estimation of CAT**			
40		**CAT's enterprise value**		
41	Based on net sales and revenues multiple	131,273	<--=B33*C25	
42	Based on gross profit multiple	151,058	<--=B34*C26	
43	Based on EBITDA multiple	170,452	<--=B35*C27	
44	Based on EBIT multiple	194,252	<--=B36*C28	
45	Based on net income multiple	178,404	<--=B37*C29	
46	**Average**	**165,088**	**<--=AVERAGE(B41:B45)**	
47				
48	Plus: excess assets	3,340		
49	Less net financial debt	-42,996		
50	**Equity value**	**125,432**	**<--=SUM(B46:B49)**	

Note that in this case, the multiples approach yields a higher valuation to CAT than the market directly assigns to it ($125.4 billion using the multiples valuation versus $81.6 billion as reflected in the market).

2.5 Enterprise Value as the Present Value of the Free Cash Flows: DCF "Top Down" Valuation

In the previous section we valued the EV by using the market value of the right-hand side of the firm's enterprise value balance sheet—using the market value of its equity and perhaps the market valuation of other elements of the firm's financing. In this section we concentrate on the left-hand side of the enterprise value balance sheet.

The discounted cash flow (DCF) method focuses on two central concepts:

1. The firm's *free cash flows* (FCFs) are defined as the cash created by the firm's operating activities.
2. The firm's *weighted average cost of capital* (WACC) is the risk-adjusted discount rate appropriate to the risk of the FCFs (Chapter 3).

The firm's *enterprise value* (EV) is the present value of the future FCFs discounted at the WACC:

$$EV = \sum_{t=1}^{\infty} \frac{FCF_t}{(1+WACC)^t}$$

The idea is to value a company by considering the present value of the FCFs, where FCF is defined as the cash flow to the firm from its assets (the word *assets* is used broadly and can be fixed assets, intellectual and trademark assets, and net working capital). The next subsection discusses this concept in more detail.

In these chapters we often make two additional assumptions: We assume a limited number of predictive periods for the FCFs, and we assume that cash flows occur evenly throughout the year. This leads to the following formula:

$$EV = \sum_{t=1}^{N} \frac{FCF_t}{(1+WACC)^{t-0.5}} + \frac{Terminal\ value}{(1+WACC)^{N-0.5}}$$

$$= \underbrace{\left[\sum_{t=1}^{N} \frac{FCF_t}{(1+WACC)^{t}} + \frac{Terminal\ value}{(1+WACC)^{N}}\right]}_{\text{Computed with Excel's NPV function}} (1+WACC)^{0.5}$$

The assumption that, for valuation purposes, cash flows occur approximately in mid-year is meant to capture the fact that most corporate cash flows occur throughout the year, and that it is therefore a mistake to value them as if they occur at year-end. As can be seen from the above formula, the mid-year assumption is easy to compute in Excel: We simply take the Excel **NPV** and multiply it by $(1 + \text{WACC})^{0.5}$.

Defining the Free Cash Flow (FCF)

The free cash flow (FCF) is a measure of how much cash is produced by the firm's operations. There are two accepted definitions of the FCF (both of which, of course, ultimately boil down to the same thing).

Free Cash Flow Based on the Income Statement

Net income (profit after taxes)	This is the basic measure of the profitability of the business, but it is an accounting measure that includes financing flows (such as interest), as well as noncash expenses such as depreciation. Profit after taxes does not account for either changes in the firm's working capital or purchases of new fixed assets, both of which can be important cash drains on the firm.
+ Depreciation and other non-cash expenses	Depreciation is a non-cash expense that in the FCF calculation is added back to the after-tax profit. To the extent that there are other noncash expenses in the firm's income statement, these are also added back.
− Increase in operating current assets	When the firm's sales increase, more investment is needed in inventories, accounts receivable, etc. This increase in current assets is not an expense for tax purposes (and is therefore ignored in the profit after taxes), but it is a cash drain on the company. When adjusting the FCF for current assets, we take care not to include financial current assets that are not directly related to sales, such as cash and marketable securities.
+ Increase in operating current liabilities	An increase in sales often causes an increase in financing related to sales (e.g., accounts payable or taxes payable). This increase in current liabilities—when related to sales—provides cash to the firm. Since it is directly related to sales, we include this cash in the free cash flow

	calculations. When adjusting the FCF for current liabilities, we take care only to include financial current liabilities that are directly related to sales, such as account payables, and exclude financial liabilities that are not operating, such as changes in short-term debt and the current portion of long-term debt.
− Increase in fixed assets at cost (CAPEX)	An increase in fixed assets (the long-term productive assets of the company) is a use of cash, which reduces the firm's free cash flow.
+ After-tax interest payments (net)	FCF is an attempt to measure the cash produced by the business activity of the firm. To neutralize the effect of interest payments on the firm's profits, we: • Add back the after-tax cost of interest on debt (*after-tax* since interest payments are tax deductible). • Subtract the after-tax interest payments on cash and marketable securities.

An alternative, equivalent definition is based on the firm's earnings before interest and taxes:

Free Cash Flow Based on the EBIT

EBIT	The sum of $-\Delta CA + \Delta CL$ is the change in the firm's net working capital ΔNWC.
− Taxes on EBIT	
+ Depreciation and other noncash expenses	
− Increase in operating current assets	
+ Increase in operating current liabilities	
− Increase in fixed assets at cost (CAPEX)	

How Can We Predict Future FCFs?

In valuing a company, it is critical to project its anticipated future free cash flows. In this book we explore two ways of deriving these cash flows. Both methods are primarily based on accounting data. Since accounting data is historical data, we need to add some judgment about how this data will develop in the future.

- • One method is to base our estimates of future free cash flows on the consolidated statement of cash flows (CSCF) of the firm (see section 2.6 below). This is a simplified approach for valuation.
- • The second method is to estimate a set of pro forma financial statements for the firm and to derive the free cash flows from these statements (Chapter 4).

Both methods are briefly reviewed in succeeding sections.

2.6 Free Cash Flows Based on Consolidated Statement of Cash Flows

Using the consolidated statement of cash flows (CSCF) is considered as a simplified approach for valuation. As will be presented below, by assuming four numbers, we'll be able to estimate the economic value of the enterprise value. The main input for this approach is the operating free cash flow of the firm, which can be extracted from the firm's consolidated statement of cash flows.

The CSCF is part of every financial statement. It is the accountant's explanation of how much cash was generated by the business and how this cash was generated. It is composed of three sections: operating cash flows, investment cash flows, and financing cash flows. When we use the CSCF to determine FCFs, we use the following general procedure (explanations are given in subsequent subsections and again in Chapter 4):

- • We accept the operating cash flows as reported by the firm.
- • We carefully examine the investment cash flows, leaving for the FCF the investment cash flows related to productive activities and eliminating those investment cash flows relating to investment by the firm in financial assets.
- • We do not count toward the FCF any of the financial cash flows.
- • In all cases we are careful about whether specific items are one time or recurring, eliminating from consideration the one-time items.
- • We adjust the totals of the adjusted CSCF numbers by adding back net interest paid.

Let us assume that in 2020, ABC Corp. has generated an operating free cash flow of \$481,091. In Chapter 4, we discuss in detail how the historical free cash flows are derived and how they can be used as the basis for cash flow projections. One valuation of a company might look like this.

	A	B	C	D	E	F	G	H
1	**ABC CORP. VALUATION** **Based on consolidated statement of cash flows (CSCF)**							
2	Free cash flow (FCF) year ending 31 Dec. 2020	481,091						
3	Growth rate of FCF, years 1-5	8.00%	<-- Optimistic about short-term growth					
4	Long-term FCF growth rate	5.00%	<-- More pessimistic about long-term growth					
5	Weighted average cost of capital, WACC	10.70%						
6								
7	**Year**	**2020**	**2021**	**2022**	**2023**	**2024**	**2025**	
8	FCF		519,578	561,144	606,036	654,518	706,880	<-- =F8*(1+$B$3)
9	Terminal value						13,021,473	<-- =G8*(1+B4)/(B5-B4)
10	**Total**		**519,578**	**561,144**	**606,036**	**654,518**	**13,728,353**	**<-- =SUM(G8:G9)**
11								
12	Enterprise value	10,592,904	<-- =NPV(B5,C10:G10)*(1+B5)^0.5					
13	Add back initial cash and marketable securities	73,697	<-- From current balance sheet					
14	Subtract 2020 financial liabilities	-1,379,106	<-- From current balance sheet					
15	**Equity value**	**9,287,495**	**<-- =SUM(B12:B14)**					
16	Per share (1 million shares outstanding)	9.29	<-- =B15/1000000					

2.7 Free Cash Flows Based on Pro Forma Financial Statements

Another way to project free cash flows is to build a set of *predictive* financial statements based on our understanding of the company and its financial statements. We discuss the construction of such a model in Chapter 4 and give a fully worked-out example for Merck in Chapter 5. A typical model might look like the following:

	A	B	C	D	E	F	G	H
1	**PRO FORMA FINANCIAL MODEL**							
2	Sales growth	10%						
3	Current assets/Sales	15%						
4	Current liabilities/Sales	8%						
5	Net fixed assets/Sales	77%						
6	Costs of goods sold/Sales	50%						
7	Depreciation rate	10%						
8	Interest rate on debt	10%						
9	Interest paid on cash and marketable securities	8%						
10	Tax rate	27%						
11	Dividend payout ratio	40%						
12								
13	**Income statement**							
14	**Year**	**0**	**1**	**2**	**3**	**4**	**5**	
15	Sales	1,000	1,100	1,210	1,331	1,464	1,611	<-- =F15*(1+$B$2)
16	Costs of goods sold	(500)	(550)	(605)	(666)	(732)	(805)	<-- =-G15*$B$6
17	**Gross profit**	**500**	**550**	**605**	**666**	**732**	**805**	**<-- =SUM(G15:G16)**
18	Interest payments on debt	(32)	(32)	(32)	(32)	(32)	(32)	<-- =-$B$8*(F38+G38)/2
19	Interest earned on cash and marketable securities	6	20	49	82	118	160	<-- =$B$9*(F29+G29)/2
20	Depreciation	(100)	(117)	(137)	(161)	(189)	(220)	<-- =-$B$7*(G32+F32)/2
21	**Profit before tax**	**874**	**971**	**1,090**	**1,219**	**1,362**	**1,518**	**<-- =SUM(G15:G20)**
22	Taxes	(236)	(262)	(294)	(329)	(368)	(410)	<-- =-G21*$B$10
23	**Profit after tax**	**638**	**709**	**795**	**890**	**994**	**1,108**	**<-- =G22+G21**
24	Dividends	(255)	(284)	(318)	(356)	(398)	(443)	<-- =-$B$11*G23
25	Retained earnings	383	425	477	534	597	665	<-- =G24+G23
26								
27	**Balance sheet**							
28	**Year**	**0**	**1**	**2**	**3**	**4**	**5**	
29	Cash and marketable securities	80	421	806	1,239	1,724	2,268	<-- =G41-G30-G34
30	Current assets	150	165	182	200	220	242	<-- =G15*$B$3
31	Fixed assets							
32	At cost	1,070	1,264	1,486	1,740	2,031	2,364	<-- =G34-G33
33	Depreciation	(300)	(417)	(554)	(715)	(904)	(1,124)	<-- =F33+G20
34	Net fixed assets	770	847	932	1,025	1,127	1,240	<-- =G15*$B$5
35	**Total assets**	**1,000**	**1,433**	**1,920**	**2,463**	**3,071**	**3,747**	**<-- =G34+G30+G29**
36								
37	Current liabilities	80	88	97	106	117	129	<-- =G15*$B$4
38	Debt	320	320	320	320	320	320	<-- =F38
39	Stock	450	450	450	450	450	450	<-- =F39
40	Accumulated retained earnings	150	575	1,053	1,587	2,183	2,848	<-- =F40+G25
41	**Total liabilities and equity**	**1,000**	**1,433**	**1,920**	**2,463**	**3,071**	**3,747**	**<-- =SUM(G37:G40)**

Using the definition of free cash flows from section 2.5:

	A	B	C	D	E	F	G	H
43	**Free cash flow calculation**							
44	**Year**	**0**	**1**	**2**	**3**	**4**	**5**	
45	Profit after tax		709	795	890	994	1108	<-- =G23
46	Add back depreciation		117	137	161	189	220	<-- =-G20
47	Subtract increase in current assets		(15)	(17)	(18)	(20)	(22)	<-- =-(G30-F30)
48	Add back increase in current liabilities		8	9	10	11	12	<-- =G37-F37
49	Subtract increase in fixed assets at cost		(194)	(222)	(254)	(291)	(333)	<-- =-(G32-F32)
50	Add back after-tax interest on debt		23	23	23	23	23	<-- =-(1-$B$10)*G18
51	Subtract after-tax interest on cash and marketable securities		(15)	(36)	(60)	(87)	(116)	<-- =-(1-$B$10)*G19
52	**Free cash flow**		**634**	**691**	**752**	**819**	**892**	**<-- =SUM(G45:G51)**

We can now use these free cash flows to compute the enterprise value of the firm (row 64 below) and the value of its shares (cell B69):

	A	B	C	D	E	F	G	H
55	**Valuing the firm**							
56	Weighted average cost of capital	10.7%						
57	Long-term free cash flow growth rate	5%						
58								
59	Year	0	1	2	3	4	5	
60	FCF		634	691	752	819	892	<-- =G32
61	Terminal value						16,436	<-- =G60*(1+B57)/(B56-B57)
62	Total		634	691	752	819	17,329	<-- =G61+G60
63								
64	Enterprise value, present value of row 62	13,320	<-- =NPV(B56,C62:G62)*(1+B56)^0.5					
65	Add in initial (year 0) cash and marketable securities	80	<-- =B29					
66	Asset value in year 0	13,400	<-- =B65+B64					
67	Subtract out value of firm's debt today	-320	<-- =-B38					
68	Equity value	13,080	<-- =B66+B67					
69	Share value (100 shares)	130.80	<-- =B68/100					

2.8 Summary

In this chapter we have introduced four EV valuation methods:

- • The book value approach values the firm's enterprise value using its balance sheet numbers, appropriately rearranged.
- • The efficient markets approach substitutes, where possible, market values for financial assets and liabilities instead of their book values and then makes appropriate adjustments to the valuation of the firm's real assets. We can use the efficient markets approach to estimate the enterprise value using a comparable valuation approach (multiples valuation).
- • One approach to discounting the firm's free cash flows (FCFs) bases the estimates of future FCFs on the firm's consolidated statement of cash flows. These FCFs are then discounted at the appropriate weighted average cost of capital (also called WACC, see Chapter 3).
- • A second approach to discounting the firm's free cash flows constructs the FCFs from a model of the firm's projected future accounting statements (pro forma accounting statements). As in the previous bullet, these FCFs are discounted at the WACC.

Exercises

1. Three years of balance sheets for Apple are given on the companion site of this book. Required:
   1. Restate these balance sheets so that the accounting enterprise value is on the left side.
   2. Use the efficient markets approach to evaluate Apple's enterprise value.

2. On the companion site of this book you'll find Cisco's consolidated statement of cash flows. Examine and transform it into a free cash flow.
3. On the companion site of this book are some year-end numbers for Cisco's equity. Restate the enterprise value in market terms.
4. Use the template for the ABC Corp. valuation in section 2.6 to value Amazon's stock (AMZN). Assume that the weighted average cost of capital for AMZN is 10%, the growth rate for years 1–5 is 4%, and that the long-term growth rate is 0%. (Details and a template are on the website that accompanies this book.)

1. 1. Debt valuation is discussed in Section II, “Bonds.” The valuation of derivative securities is discussed in Section IV, “Options.”
2. 2. We revisit CAT in Chapter 3, where we compute its weighted average cost of capital, and in Chapters 4 and 6.
3. 3. We must be careful: Market prices change over time, often precipitously. The efficient markets hypothesis only says that it is impossible to predict market prices beyond the information compounded into current market information.
4. 4. Good references are Benninga and Sarig (1997); Damodaran (2012); and Goedhart, Koller, and Wessels (2015).

3 Calculating the Weighted Average Cost of Capital (WACC)

3.1 Overview

In this chapter we discuss the calculation of the firm's weighted average cost of capital (WACC). The WACC has two important uses in finance:

- • When used as the discount rate for a firm's anticipated free cash flows (FCFs), the WACC gives the enterprise value of the firm. FCF is discussed in Chapter 2 and again at length in Chapters 4 and 5. At this point, it suffices to say that the FCF is the cash flow generated by the firm's core business activities. These chapters also show how to apply the WACC to the valuation of firms.
- • The WACC is also the appropriate risk-adjusted discount rate for firm projects whose riskiness is similar to the average riskiness of the firm's cash flows. When used in this context, the WACC is often referred to as the firm's "hurdle rate."

The WACC is a weighted average of the firm's cost of equity r_E and its cost of debt r_D, with the weights created by the market values of the firm's equity (E) and debt (D):

$$WACC = \frac{E}{E+D} r_E + \frac{D}{E+D} r_D (1 - T_C)$$

where

E = market value of the firm's equity
D = market value of the firm's debt
r_E = firm's cost of equity
r_D = firm's cost of debt
T_C = firm's corporate tax rate

This chapter discusses the computation of the five components of the WACC—the market value of the firm's equity and debt E and D, the firm's tax rate T_C, the firm's expected cost of debt r_D, and the expected cost of equity r_E. We finish by showing detailed examples of how to compute the firm's WACC. The reader should be warned that the application of the models discussed requires a good deal of judgment—computing the WACC is part science and part art!

The main technical problem is the computation of the firm's cost of equity r_E. We consider two models for calculating the cost of equity r_E, the discount rate applied to equity cash flows.

- • The Gordon model calculates the cost of equity based on the anticipated cash flows paid to the shareholders of the firm. In the implementation of the Gordon model, dividends growing at a constant future rate are most commonly used as the anticipated shareholder cash flows. We explore two variations of the model: multiple future growth rates and total equity cash flows.
- • The capital asset pricing model (CAPM) calculates the cost of equity based on the correlation between the firm's equity returns and the returns of a large, diversified market portfolio. Variations on this model include the tax framework in which the model is defined.

The other problematic component of the cost of capital is the cost of debt r_D, the expected future cost of the firm's borrowing. This book contains three models to calculate the cost of debt; two of these models are discussed in this chapter and a third method is discussed separately in Chapter 9.

- • The cost of debt r_D is often computed by using the firm's current net interest payments divided by its average net debt (where net debt is debt minus cash and marketable securities).
- • An alternative method is to compute r_D by imputing the firm's cost of debt from a rating-adjusted yield curve.
- • Finally, we can compute the expected return on the firm's bonds as a proxy for its cost of debt; we discuss this method separately in Chapter 9.

A terminological note: As noted in Chapter 1, "cost of capital" is a synonym for the "appropriate discount rate" to be applied to a series of cash flows. In finance "appropriate" is most often a synonym for "risk adjusted." Hence another name for the cost of capital is the risk-adjusted discount rate (RADR).

The Remainder of This Chapter

In the following sections we discuss the various components of the WACC with examples on how they might be computed:

- • Section 3.2: Computing the equity value, E.
- • Section 3.3: Computing the value of the firm's debt, D.
- • Section 3.4: Computing the firm's corporate tax rate T_C.
- • Section 3.5: Computing the firm's cost of debt, r_D.
- • Section 3.6: Computing the firm's cost of equity, r_E. We show how to use both the Gordon dividend model and the CAPM to compute r_E. Each model has a number of twists and variations, discussed in these sections.

- • Sections 3.7 and 3.8: Computing the expected return on the market $E(r_M)$ and the risk-free rate r_f in the CAPM.
- • Section 3.9: We present four worked-out cases of the computation of the WACC and a unified template that will help you make sense of the WACC computation.
- • Section 3.10: We discuss problems with using the dividend model and the CAPM, including the case of the determination of the WACC for nonmarketed firms.

3.2 Computing the Value of the Firm's Equity, *E*

Of all the computations related to the WACC, computing the value of the firm's equity is the easiest: As long as the company is publicly listed, take *E* to be the product of the number of shares outstanding times the current value per share.

As an example, consider Caterpillar Inc. (CAT), a New York Stock Exchange company that manufactures and sells construction and mining equipment, diesel and natural gas engines, and industrial gas turbines. On December 31, 2019, CAT has 205.7 million shares outstanding, each trading at $33.80. The equity value of the company is $6.953 billion.[1]

	A	B	C
1	**COMPUTING THE VALUE OF EQUITY - E, CATERPILLAR INC. (CAT)**		
2	Shares outstanding (millions)	575.54	
3	Share price, 31 December 2019	147.68	
4	Equity value - "market cap" (millions, $)	84,996	<--=B3*B2

3.3 Computing the Value of the Firm's Debt, *D*

We compute the value of the firm's debt by the market value of its financial debt minus the market value of its excess financial assets. A common approximation for this number is to take the balance sheet value of the firm's debt minus the value of the firm's cash balances and minus the value of its marketable securities. Here is an example for CAT:

	A	B	C	D
1	**CATERPILLAR INC. (CAT), COMPUTING NET DEBT (millions $)**			
2		Dec. 31, 2019	Dec. 31, 2018	
3	Short-term financial products	11,355	2,332	
4	Dividends payable	567	495	
5	Other current liabilities	2,155	1,919	
6	Machinery, energy and transportation	9,141	8,005	
7	Financial products	17,140	16,995	
8	Liability for postemployment benefits	6,599	7,455	
9	Other liabilities	4,323	3,756	
10	Cash and short-term investments	-8,284	-7,857	
11	**Net debt**	**42,996**	**33,100**	<--=SUM(C3:C10)

For purposes of computing the weighted average cost of capital, our definition of debt excludes other debt-like items such as pension liabilities and deferred taxes.

Though we consider these items as debts, it is hard to attach a cost to them; we prefer to approximate the WACC by using only financial obligations net of liquid assets.

It is not uncommon for a company to have "negative net debt"—this occurs when the company has more cash and marketable securities than debt. When this occurs, we set D in the WACC computation to be zero and put the negative net debt as an excess financial asset. Apple is an example:

	A	B	C	D
1	**APPLE HAS NEGATIVE NET DEBT OR POSITIVE EXCESS FINANCIAL ASSETS (million $)**			
2		**2018**	**2017**	
3	**Liabilities**			
4	Other current liabilities	32,687	30,551	
5	Commercial paper	11,964	11,977	
6	Term debt (current)	8,784	6,496	
7	Term debt (non-current)	93,735	97,207	
8	Other non-current liabilities	45,180	40,415	
9	**Excess financial assets**			
10	Cash and cash equivalents	-25,913	-20,289	
11	Marketable securities	-40,388	-53,892	
12	Other current assets	-12,087	-13,936	
13	Marketable securities	-170,799	-194,714	
14	**Net debt**	**-56,837**	**-96,185**	<-- =SUM(C4:C13)

In Apple's case, we say that the firm is unlevered (operating without effective debt) and that it holds net excess financial assets of $225,187 million as of the end of fiscal year 2018.

3.4 Computing the Firm's Tax Rate, T_C

In the WACC formula, T_C should measure the firm's *marginal* tax rate, but it is common to measure it by computing the firm's *reported* tax rate. Usually this should cause no problems, as the following example shows for Apple (AAPL):

	A	B	C	D
13	**APPLE TAX RATE**			
14		**Sep. 28, 2019**	**Sep. 29, 2018**	
15	Profit before tax	65,737	72,903	
16	Taxes	10,481	13,372	
17	**Tax rate, T_C**	**15.9%**	**18.3%**	<--=C16/C15

The tax rate for Apple is somewhat stable. In our WACC computation we would most likely use the current tax rate or the average over the past several years or the expected tax rate, if we have specific knowledge about it. Companies like Apple are very good at placing their income in comfortable tax venues, and it appears that a reasonable estimate for the tax rate is somewhere between 16% and 18%.

Sometimes, however, this doesn't work, as the following example shows for Deere & Company (DE):

	A	B	C	D
7	DEERE & COMPANY TAX RATE			
8		Nov. 03, 2019	Oct. 28, 2018	
9	Profit before tax	4,088	4,071	
10	Taxes	852	1,727	
11	**Tax rate, T_C**	**20.8%**	**42.4%**	<--=C10/C9

Applying the same technique to CAT shows a somewhat normative rate for 2019 at about 22%.

	A	B	C	D
1	CATERPILLAR TAX RATE			
2		Dec. 31, 2019	Dec. 31, 2018	
3	Profit before tax	7,812	7,822	
4	Taxes	1,746	1,698	
5	**Tax rate, T_C**	**22.35%**	**21.71%**	<--=C4/C3

Corporate tax in the United States has changed dramatically over the years. Since January 1, 2018, the nominal federal corporate tax rate is a flat 21%, which went into effect with the passage of the Tax Cuts and Jobs Act of 2017. State and local taxes and rules vary by jurisdiction but average around 7%. This brings the total average statutory tax rate to about 28%.

3.5 Computing the Firm's Cost of Debt, r_D

We now turn to calculating the cost of debt r_D. In principle, r_D is the marginal cost to the firm (before corporate taxes) of borrowing an additional dollar. There are at least three ways of calculating the firm's cost of debt. We will state them briefly below and then go on to illustrate the application of two of the methods. Although the methods may not be theoretically perfect, they are often used in practice.

- • As a practical matter, the cost of debt can be approximated by taking the *average cost* of the firm's existing debt. The problem with this method is that it runs the danger of confusing the *past costs* with the *future anticipated* cost of debt that we actually want to measure.
- • We can use the yield of similar-risk, newly issued corporate securities. If a company is rated A and has mostly medium-term debt, then we can use the average yield on medium-term, A-rated debt as the firm's cost of debt. Note that this method is somewhat problematic because the yield on a bond is its *promised return*, whereas the cost of debt is the *expected return* on a firm's debt. Since there is usually a risk of default, the promised return is generally higher than the expected return. Nevertheless, despite the problematics, this method is often a good compromise, specifically for relatively safe debt.

- • We can use a model that estimates the cost of debt from data about the firm's bond prices, the estimated probabilities of default, and the estimated payoffs to bondholders in case of default. This method requires a lot of work and is mathematically nontrivial; we postpone its discussion until Chapter 9. For cost of capital calculations, it would be used in practice only if the firm we are analyzing has significant amounts of risky debt.

The first two methods above are relatively easy to apply, and in many cases the problems or errors that are encountered in these methods are not critical.[2] As a matter of theory, however, both methods fail to make proper risk adjustments for the cost of the firm's debt. The third method, which involves computing the expected return on a firm's debt, is more in line with standard financial theory, but it is also more difficult to apply. It may not, therefore, be worth the effort.

In the remainder of this section, we apply the first two of these methods to calculate the cost of debt for Caterpillar.

Method 1: Caterpillar's Average Cost of Debt

For Caterpillar, we compute the average cost of debt as follows:

$$r_D = \frac{\text{Current year's net interest paid}}{\text{Average net debt over this and previous year}}$$

	A	B	C	D
1	CATERPILLAR INC. (CAT), COST OF DEBT - r_D			
2	Year ending December	2019	2018	
3	Short-term financial products	11,355	2,332	
4	Dividends payable	567	495	
5	Other current liabilities	2,155	1,919	
6	Machinery, energy and transportation (debt)	9,141	8,005	
7	Financial products	17,140	16,995	
8	Liability for postemployment benefits	6,599	7,455	
9	Other liabilities	4,323	3,756	
10	Cash and short-term investments	-8,284	-7,857	
11	Net debt	42,996	33,100	<--=SUM(C3:C10)
12	Interest expense of financial products	754	722	
13	Implied cost of debt, r_D	1.98%	<--=B12/(AVERAGE(B11:C11))	

There are several aspects of our calculations worth noting:

- • When calculating the average cost of debt r_D from the financial statements, it is important to include all financial debt, without distinguishing between short-term and long-term items.
- • We treat liquid assets such as cash and cash equivalents as *negative debt* and subtract them from the firm's debt. The idea here is that the firm could use its cash to pay off part of its debt, so that the effective debt financing of the firm is its financial debt minus cash. However, the implementation of this particular piece of

theory is largely a judgment call—we may not want to attribute all cash to the possibility of paying off debt, and we may want to compute the firm's cost of borrowing as opposed to the interest it earns on cash.

Were we to use the average cost of debt for Caterpillar as a prediction of its future cost of debt r_D, we would most likely use the current cost $r_D \approx 2\%$ in the WACC computation. This is because we believe that historical costs of debt have little predictive power for future costs.

Cash Raises the Cost of Debt

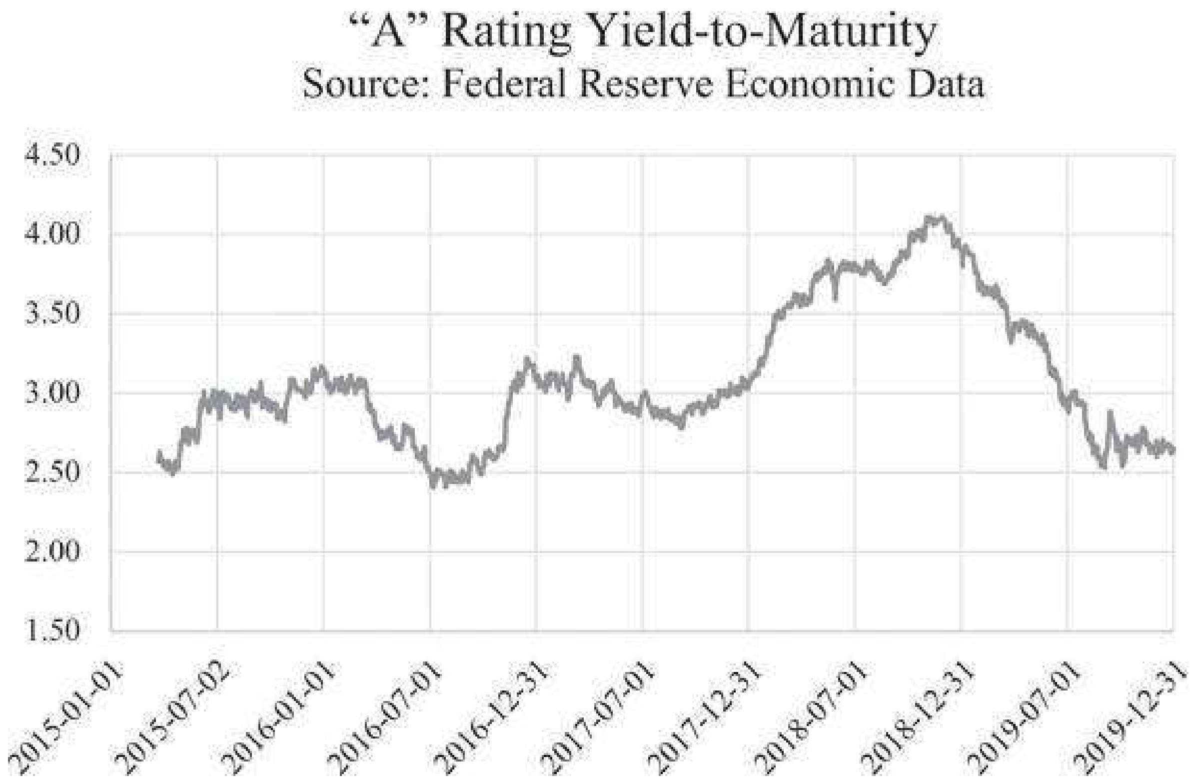
When a firm has cash balances that earn less interest than the cost of borrowing, the average cost of debt, based on net interest and net debt, is higher than the cost of borrowing. To see this, assume that the interest rate on cash is less by ε than the interest rate paid on debt:

$$\begin{aligned}
\begin{matrix}\text{Average} \\ \text{cost of debt}\end{matrix} &= \frac{\text{Interest paid} - \text{Interest earned}}{Debt - Cash} = \frac{Debt * i_{Debt} - Cash * i_{Cash}}{Debt - Cash} \\
&= \frac{Debt * i_{Debt} - Cash * (i_{Debt} - \varepsilon)}{Debt - Cash} = \frac{(Debt - Cash) * i_{Debt} + \varepsilon * Cash}{Debt - Cash} \\
&= i_{Debt} + \frac{Cash}{Debt - Cash} * \varepsilon > i_{Debt}
\end{aligned}$$

The estimate of r_D reflects the costs of holding reserves of liquid assets that provide low financial returns. From a purely financial point of view, the company would have benefited its shareholders by using the liquid assets to repurchase its debt or by paying them out as either dividends or share repurchases.[3]

Method 2: r_D as the Rating-Adjusted Yield for Caterpillar

There is another way to measure the cost of debt for Caterpillar's borrowing: we can impute the marginal cost of Caterpillar's debt from a yield curve for the appropriate debt. As of January 2019, CAT bonds are rated A by Standard & Poor's, A by Fitch, and A3 by Moody's (equivalent to A by Standard & Poor's). From the Federal Reserve Economic Data (https://fred.stlouisfed.org), we find that as of December 31, 2019, the Yield-to-Maturity (YTM) on a bond with a rating of "A" was 2.65%:



3.6 Two Approaches to Computing the Firm's Cost of Equity, r_E

As you may remember, the equation for the weighted average cost of capital is $WACC = \frac{E}{(E+D)} * r_E + \frac{D}{(E+D)} * r_D * (1 - T_C)$. Thus far in this chapter we have discussed the estimation of four of the five parameters of the WACC: E, D, T_C, r_D. We now come to the most problematic of the computations related to the WACC parameters—the computation of the cost of equity r_E. There are two approaches to r_E that can readily be computed:

- • The Gordon dividend model computes r_E based on current dividend Div_0, current stock price P_0, and the anticipated growth of future dividends g:

$$r_E = \frac{Div_0(1+g)}{P_0} + g$$

- • The CAPM computes r_E based on the risk-free rate r_f , the expected return on the market $E(r_M)$, and a firm-specific risk measure β:

$$r_E = r_f + \beta[E(r_M) - r_f]$$

where

r_f = *the market risk − free rate of interest*

$E(r_M)$ = *the expected return on the market portfolio*

$$\beta = a\ firm - specific\ risk\ measure = \frac{Cov(r_{stock}, r_M)}{Var(r_M)}$$

Each model has its variations and problems, which are discussed in the next two sections.

Approach 1: Implementing the Gordon Model for r_E

The Gordon dividend model derives the cost of equity from the following deceptively simple statement:

The value of a share is the present value of the future anticipated dividend stream from the share, where the future anticipated dividends are discounted at the appropriate risk-adjusted cost of equity r_E.[4]

The simplest application of the Gordon model is the case where the anticipated future growth rate of dividends is constant. Suppose that the current stock price is P_0, the current dividend is Div_0, and the anticipated growth rate of future dividends is g. The Gordon model states that the stock price equals the discounted (at the appropriate cost of equity r_E) future dividends:

$$P_0 = \frac{Div_0 * (1+g)}{(1+r_E)^1} + \frac{Div_0 * (1+g)^2}{(1+r_E)^2} + \frac{Div_0 * (1+g)^3}{(1+r_E)^3} + \frac{Div_0 * (1+g)^4}{(1+r_E)^4} \cdots$$
$$= \sum_{t=1}^{\infty} \frac{Div_0 * (1+g)^t}{(1+r_E)^t}$$

Thus, given a constant anticipated dividend growth rate, we derive the Gordon model cost of equity:[5]

$$P_0 = \frac{Div_0 * (1+g)}{r_E - g}, \ provided\ g < r_E$$

Note the proviso at the end of this formula: In order for the infinite sum on the first line of the formula to have a finite solution, the growth rates of the dividends must be less than the discount rate. In our discussion of the Gordon model with abnormal growth rates (see below), we return to the case where this is not true.

To apply this formula, consider a firm that paid over the last year a dividend of \$3 (i.e., Div_0 = \$3 per share), whose share price is traded at \$130 per share (i.e., P_0 = \$50). Suppose the dividend is anticipated to grow constantly by 7% per year. Then the firm's cost of equity r_E is 9.47%:

	A	B	C
1	THE GORDON MODEL COST OF EQUITY		
2	Current share price, P_0	130	
3	Dividend paid over last year, Div_0	3	
4	Anticipated dividend growth rate, g	7%	
5	Gordon model cost of equity, r_E	9.47%	<-- =B3*(1+B4)/B2+B4

Using the Gordon Model to Compute the Cost of Equity for Caterpillar

We apply the Gordon model to Caterpillar, whose 10-year dividend history is given in the spreadsheet below (note that some of the data has been hidden).

	A	B	C	D	E	F	G
1	CATERPILLAR'S DIVIDEND HISTORY						
2	Date	Dividend per share ($)		Year	Total dividend	Annual dividend growth rate	
3	18-Oct-19	1.03		2019	3.78	14.19%	<--=LN(E3/E4)
4	19-Jul-19	1.03		2018	3.28	5.64%	<--=LN(E4/E5)
5	18-Apr-19	0.86		2017	3.10	-21.67%	
6	18-Jan-19	0.86		2016	3.85	26.97%	
7	19-Oct-18	0.86		2015	2.94	-8.47%	
8	19-Jul-18	0.86		2014	3.20	62.08%	
9	20-Apr-18	0.78		2013	1.72	-36.59%	
10	19-Jan-18	0.78		2012	2.48	32.05%	
11	20-Oct-17	0.78		2011	1.80	4.55%	
12	18-Jul-17	0.78		2010	1.72	2.35%	<--=LN(E12/E13)
13	20-Apr-17	0.77		2009	1.68		
14	18-Jan-17	0.77					
15	20-Oct-16	0.77		Average (method I)		8.11%	<--=AVERAGE(F3:F12)
16	18-Jul-16	0.77		Average (method II)		8.45%	<--=(E3/E13)^(1/10)-1
17	21-Apr-16	0.77					
18	18-Jan-16	0.77		Formula in cell E4 is:			
19	15-Jan-16	0.77		=SUMIFS($B$3:$B$52,$A$3:$A$52,">="&DATE(D4,1,1),$A$3:$A$52,"<="&DATE(D4,12,31))			
20	22-Oct-15	0.77					
21	16-Jul-15	0.77					
45	22-Oct-09	0.42					
46	16-Jul-09	0.42					
47	16-Apr-09	0.42					
48	15-Jan-09	0.42					

The annualized growth rate of Caterpillar's historical dividends may be either 8.11% or 8.45%, depending on the calculation method. For purposes of computing the cost of equity r_E, the question is which of these rates better predicts future anticipated dividend growth rates.[6] In the spreadsheet below we allow for both possibilities. The calculations use Caterpillar's stock price at the end of December 2019, P_0 = $147.68.

	A	B	C
1	USING GORDON MODEL COMPUTING CATERPILLAR'S R_E		
2	Caterpillar's stock price P_0, 31-Dec-2019	147.68	
3	Current dividend, Div_0	3.78	
4	Dividend growth rate, g		
5	Average (method I)	8.11%	
6	Average (method II)	8.45%	
7			
8	Gordon model cost of equity, r_E		
9	Using method I	10.88%	<--=$B$3*(1+B5)/$B$2+B5
10	Using method II	11.22%	<--=$B$3*(1+B6)/$B$2+B6

For all practical purposes, given the margin of error in our estimates of the future, these numbers are identical—remember that we are trying to predict future dividend growth based on past dividend payouts.

Adjusting the Gordon Model to Account for All Cash Flows to Equity

As illustrated above, the Gordon model is computed on a per-share basis and for dividends only. However, for purposes of valuing the firm's equity, the Gordon model should be extended to include all cash flows to equity. In addition to dividends, cash flows to equity include at least two additional components:

- • Share repurchases now account for around 50% of the total cash disbursed by American corporations to their shareholders.[7]
- • The issuance of stock by the firm is an important *negative cash flow to equity*. In many firms the most important instance of stock issuance occurs when employees exercise their stock options.

To account for these additional cash flows to equity, we must rewrite the Gordon model in terms of total equity value. The basic valuation model of Gordon now becomes:

$$\textit{Market value of equity} = \sum_{t=1}^{\infty} \frac{\textit{Cash flow to equity}_0 * (1+g)^t}{(1+r_E)^t}$$

where

g = *Anticipated growth rate of cash flow to equity*

This gives the formula for the cost of equity r_E as:

$$r_E = \frac{\textit{Cash flow to equity}_0 (1+g)}{\textit{Market value of equity}} + g, \textit{ if } g < r_E$$

As an example, we consider the data below for Caterpillar:

	A	B	C	D	E
1	**GORDON MODEL FOR CATERPILLAR'S EQUITY PAYOUTS**				
2	Share repurchase rate of total payout		50%		
3		Dividends	Total equity payout	Growth rate	
4	2019	3.78	7.56	14.19%	<--=LN(C4/C5)
5	2018	3.28	6.56	5.64%	<--=LN(C5/C6)
6	2017	3.10	6.20	-21.67%	<--=LN(C6/C7)
7	2016	3.85	7.70	26.97%	
8	2015	2.94	5.88	<--=B8/(1-$C$2)	
9					
10	Average growth rate	6.28%	<--=AVERAGE(D4:D7)		
11					
12	Computing the Gordon model cost of equity r_E based on total equity payouts				
13	Price per share (Dec 2019)	147.68			
14	Gordon model cost of equity, r_E	11.72%	<--=C4*(1+B10)/B13+B10		

If we assume that Caterpillar's historic growth rate of cash flow to equity, 6.28%, will persist in the indefinite future, then its cost of equity is r_E = 11.72%.[8] In the next subsection we present another variation of the Gordon model that flexes this assumption.

"Abnormal Growth" and the Gordon Model

A basic condition of the Gordon formula $r_E = \frac{Div_0(1+g)}{P_0} + g$ is the condition $g < r_E$.[9] In finance examples, violations of $g < r_E$ usually occur for fast-growing firms for which—at least for short periods of time—we anticipate very high growth rates, so that $g > r_E$. If such "supernormal" growth or "abnormal" growth were the case in the long run, the original dividend discount formula shows that P_0 would have an infinite value because, when $g > r_E$, the expression is $\sum_{t=1}^{\infty} \frac{Div_0 * (1+g)^t}{(1+r_E)^t} \to \infty$. Thus, a period of very high dividend growth rates (where $g > r_E$) must be followed by a period in which the long-term growth rate of dividends is less than the cost of equity, $g < r_E$.

Suppose that the firm is anticipated to pay high-growth dividends during periods *1*, …, *m*, and that for subsequent periods the growth rate of dividends will be lower. We can write the discounted value of these anticipated future dividends as:

$$\textit{Share value today} = \textit{Present value of dividends}$$
$$= \underbrace{\sum_{t=1}^{m} \frac{Div_0 * (1+g_1)^t}{(1+r_E)^t}}_{\substack{\textit{PV of m years} \\ \textit{of high-growth } g_1 \\ \textit{dividends}}} + \underbrace{\sum_{t=m+1}^{\infty} \frac{Div_m * (1+g_2)^{t-m}}{(1+r_E)^t}}_{\substack{\textit{PV of remaining} \\ \textit{normal-growth } g_2 \\ \textit{dividends}}}$$

The problem is usually determining the cost of equity r_E from predicted growth rates. In the example below we use a VBA function **TwoStageGordon** to compute r_E that equalizes the values of both sides of the above equation. The construction of **TwoStageGordon** is discussed at the end of this section. The assumption is that a firm's share price is currently $P_0 = 60$, that its current dividend is $Div_0 = 3$, and that after five years of 20% growth in dividends, the dividend growth rate will slow to 6%. As you can see below, $r_E = 15.11\%$.

	A	B	C
1	**THE GORDON MODEL WITH TWO GROWTH RATES Using the TwoStageGordon function**		
2	Current dividend, Div_0	3.00	
3	Growth rate g_1, years 1-->m ("abnormal")	20%	
4	Growth rate g_2, years 6-->∞	6%	
5	Number of abnormal growth years, *m*	5	
6	Share price	60.00	
7	Cost of equity	15.11%	<-- =twostagegordon(B6,B2,B3,B5,B4)

Implementing the Two-Stage Gordon Model for Caterpillar

Below we use the two-stage model to compute the cost of equity for Caterpillar. We assume that the 5-year abnormal growth rate of total equity payouts of 0.62% will hold only for the next three years and that afterward the growth rate of equity payouts will be 4%. The cost of equity is $r_E = 9.66\%$.

	A	B	C	D	E
1	**CATERPILLAR'S COST OF EQUITY USING THE TWO-STAGE GORDON MODEL**				
2	Share repurchase rate of total payout		50%		
3		**Dividends**	**Total equity payout**	**Growth rate**	
4	2019	3.78	7.56	14.19%	<--=LN(C4/C5)
5	2018	3.28	6.56	5.64%	<--=LN(C5/C6)
6	2017	3.10	6.20	-21.67%	
7	2016	3.85	7.70	26.97%	
8	2015	2.94	5.88	<--=B8/(1-$C$2)	
9					
10	Average growth rate	6.28%	<--=AVERAGE(D4:D7)		
11					
12	**Cost of equity r_E based on total equity payouts and the two-stage Gordon model**				
13	Share price, 31 December 2019	147.68			
14					
15	Abnormal growth rate, g_{high}	6.28%	<--=B10		
16	Number of abnormal-growth years, m	3	<-- Author guess		
17	Normal growth rate, g_{normal}	4.00%	<-- Author guess		
18					
19	Cost of equity, r_E	9.66%	<--=TwoStageGordon(B13,C4,B15,B16,B17)		

Technical Note

The function **TwoStageGordon** (which you will find on the book's auxiliary website in the Excel files for this chapter, "FM5, Chapter 03–WACC.xlsm") computes the cost of equity r_E for a two-stage Gordon model. The function computes the discount rate r_E, which equates the current share price to the present value of future equity cash flows:

```
Function TwoStageGordon(P0, Div0, Highgrowth, Highgrowthyrs,
Normalgrowth)
    high = 4
    low = 0

    Do While (high - low) > 0.00001
    Estimate = (high + low) / 2
    Term1 = Div0 * (1 + Highgrowth) / (Estimate - Highgrowth) * _
    (1 - ((1 + Highgrowth) / (1 + Estimate)) ^ Highgrowthyrs)

    Term2 = Div0 * (1 + Highgrowth) ^ Highgrowthyrs * (1 +
    Normalgrowth) _
    / (Estimate - Normalgrowth) / (1 + Estimate) ^ Highgrowthyrs

    If (Term1 + Term2) > P0 Or Term2 < 0 Then
        low = (high + low) / 2
        Else: high = (high + low) / 2
    End If
```

```
        Loop
        TwoStageGordon = (high + low) / 2
    End Function
```

Approach 2: The CAPM: Computing the Beta, β

The capital asset pricing model (CAPM) is the only viable alternative to the Gordon model for calculating the cost of capital. It is also the most widely used cost of equity model, the reasons being both its theoretical elegance and its implementational simplicity. The CAPM derives the firm's cost of capital from its covariance with the market return.[10] The classic CAPM formula for the firm's cost of equity is:

$$r_E = r_f + \beta_E(E(r_M) - r_f)$$

where

$$r_f = \textit{the market risk} - \textit{free rate of interest}$$
$$E(r_M) = \textit{the expected return on the market portfolio}$$
$$\beta_E = \textit{a firm} - \textit{specific risk measure} = \frac{Cov(r_{stock}, r_M)}{Var(r_M)}$$

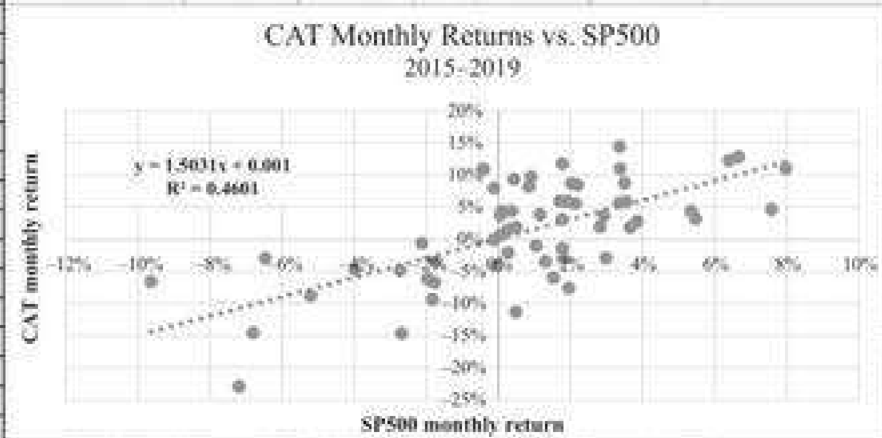
In the remainder of this section we focus on measuring the firm's β_E; the next section shows how to apply the CAPM to find the firm's cost of equity r_E.

Beta Is the Regression Coefficient of the Firm's Stock Returns on the Market Returns

In the spreadsheet below we show the five years of monthly prices and returns for Caterpillar and for the Standard & Poor's 500 (SP500), which we take to proxy for the stock market as a whole. In cells B2:B4 we regress the Caterpillar returns on those of the SP500:

$$r_{CAT,t} = \alpha_{CAT} + \beta_{CAT} r_{SP500,t} = -0.0010 + 1.5031 * r_{SP500,t}, \; R^2 = 0.4601$$

	A	B	C	D	E	F	G
1	**COMPUTING THE BETA FOR CATERPILLAR** **Monthly returns for Caterpillar and SP500, 2015-2019**						
2	Alpha	0.0010	<--=INTERCEPT(E10:E68,F10:F68)				
3	Beta	1.5031	<--=SLOPE(E10:E68,F10:F68)				
4	R-squared	0.4601	<--=RSQ(E10:E68,F10:F68)				
5	t-statistic for alpha	0.1307	<--=K13/K14				
6	t-statistic for beta	6.9691	<--=J13/J14				
7							
8		Prices			Returns		
9	**Date**	**CAT**	**SP500**		**CAT**	**SP500**	
10	01-Dec-19	146.65	3,230.78		2.02%	2.82%	<--=LN(C10/C11)
11	01-Nov-19	143.72	3,140.98		5.69%	3.35%	<--=LN(C11/C12)
12	01-Oct-19	135.77	3,037.56		8.71%	2.02%	<--=LN(C12/C13)
13	01-Sep-19	124.45	2,976.74		5.96%	1.70%	<--=LN(C13/C14)



	A	B	C	D	E	F	G
29	01-May-18	144.88	2,705.27		5.60%	2.14%	
68	01-Feb-15	70.03	2,104.50		4.41%	5.34%	
69	01-Jan-15	67.00	1,994.99				

Here's what we can learn from this regression:

- • CAT's beta, β_{CAT}, shows the sensitivity of its stock return to the market return. It is calculated by the following formula:

$$\beta_{CAT} = \frac{Covariance(SP500\ returns, CAT\ returns)}{Variance(SP500\ returns)}$$

We can compute β either by using the formula in the spreadsheet below (cell J5) or by using the Excel **Slope** function in the spreadsheet above (cell B3). Over the period covered, a 1% increase (decrease) in the monthly returns of the SP500 was accompanied by a 1.5031% increase (decrease) in CAT's returns. The t-statistic for the slope (cell B6) shows that the β_{CAT} is highly significant (see below for how this function was constructed).[11]

In our example above, implementing the direct formula looks like this:

	I	J	K	L
2	Computing beta using **Covar** and **Var**			
3	Covar(CAT,SP500)	0.0018	<--=COVARIANCE.S(E10:E68,F10:F68)	
4	Variance(SP500)	0.0012	<--=VAR.S(F10:F68)	
5	**Beta**	**1.5031**	<--=J3/J4	

- • CAT's alpha, α_{CAT}, shows that irrespective of changes in the SP500, the monthly return on Caterpillar over the period was $\alpha_{CAT} = 0.1\%$. On an annual basis, this is $12 * (0.1\%) = 1.2\%$, which seems to indicate that, in the jargon of financial markets, Caterpillar had positive performance over the period.[12] Note, however, the low *t*-statistic for alpha (cell B5). This function (its construction in Excel is discussed below) shows that the positive intercept is not significantly different from zero.
- • The R^2 of the regression shows that 46.01% of the variation in CAT's returns is accounted for by variability in the SP500. An R^2 of 46% may seem low, but in the CAPM literature this is not uncommon and actually quite high. It says that roughly 46% of the variation in CAT's returns is explicable by the variation in the SP500 return. The rest of the variability in the Caterpillar returns can be diversified away by including Caterpillar shares in a diversified portfolio of shares. The average R^2 for stocks is approximately 30% to 40%, meaning that market factors account for approximately this percentage of a stock's variability, with factors idiosyncratic to the stock accounting for the rest.

The spreadsheet that accompanies this chapter shows three ways of doing the regression: One way is to use the functions **Intercept**, **Slope**, **Rsq**. A second method involves using the Excel functions **Covariance.s** and **Var.s**. A third way involves Excel's **Trendline** ability (in charts). Having graphed the returns of CAT and the SP500 on an **XY Scatterplot**, we do the following:

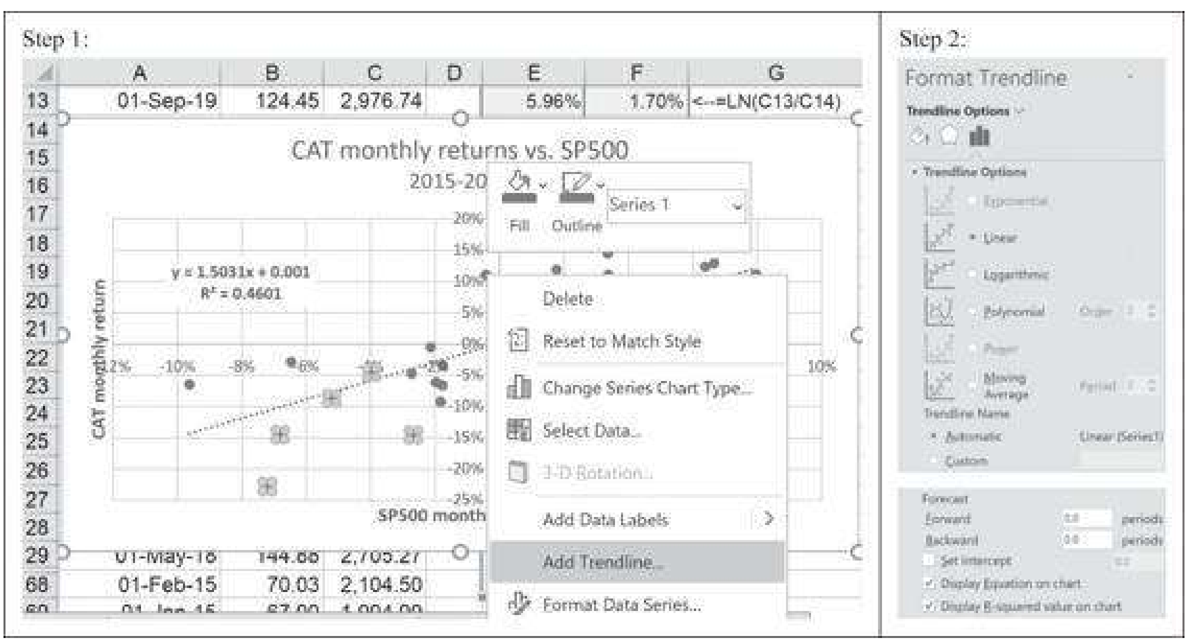


Figure 3.1
The sequence of commands for producing regression results from the **XY Scatterplot** in Excel. Having marked the

points, we right-click to select **Add Trendline** (left panel). We choose the linear regression (right panel) and indicate that the regression equation and the R^2 should be printed on the graph.

The Homemade Functions TIntercept and TSlope

The spreadsheet above computes the *t*-statistics for the intercept and slope. This is built on the **Linest** function discussed in Chapters 30 and 31. Applying **Linest** to the return data, we get:

	I	J	K	L
9		Cells J13:K17created with the		
10		array formula		
11		{=LINEST(E10:E68,F10:F68,,1)}		
12		Slope	Intercept	
13	Slope -->	1.5031	0.0010	<-- Intercept
14	Standard error of slope -->	0.2157	0.0076	<-- Standard error of intercept
15	R-squared -->	0.4601	0.0569	<-- Standard error of y values
16	F statistic -->	48.5677	57.0000	<-- Degrees of freedom
17	SS_{xy} -->	0.1570	0.1842	<-- SSE = Residual sum of squares
18				
19	t-statistic for alpha	0.1307	<--=K13/K14	
20	t-statistic for alpha	0.1307	<--=tintercept(E10:E68,F10:F68)	
21	t-statistic for beta	6.9691	<--=J13/J14	
22	t-statistic for beta	6.9691	<--=tslope(E10:E68,F10:F68)	

By using the Excel function **Index** we can define a VBA function **TIntercept** that divides the value of the intercept term produced by **Linest** (first row, second column of **Linest** output) by the standard error of the intercept (second row, second column). Here's the code for the *t*-statistic for the intercept function:

```
Function tintercept(yarray, xarray)
    tintercept = Application.Index(Application. _
    LinEst(yarray, xarray,, 1), 1, 2) / _
    Application.Index(Application.LinEst (yarray, _
    xarray,, 1), 2, 2)
End Function
```

Similarly we can define a function **TSlope** that gives the *t*-statistic for the slope:

```
Function tslope(yarray, xarray)
    tslope = Application.Index(Application. _
    LinEst(yarray, xarray,, 1), 1, 1) / _
    Application.Index(Application.LinEst (yarray, _
    xarray,, 1), 2, 1)
End Function
```

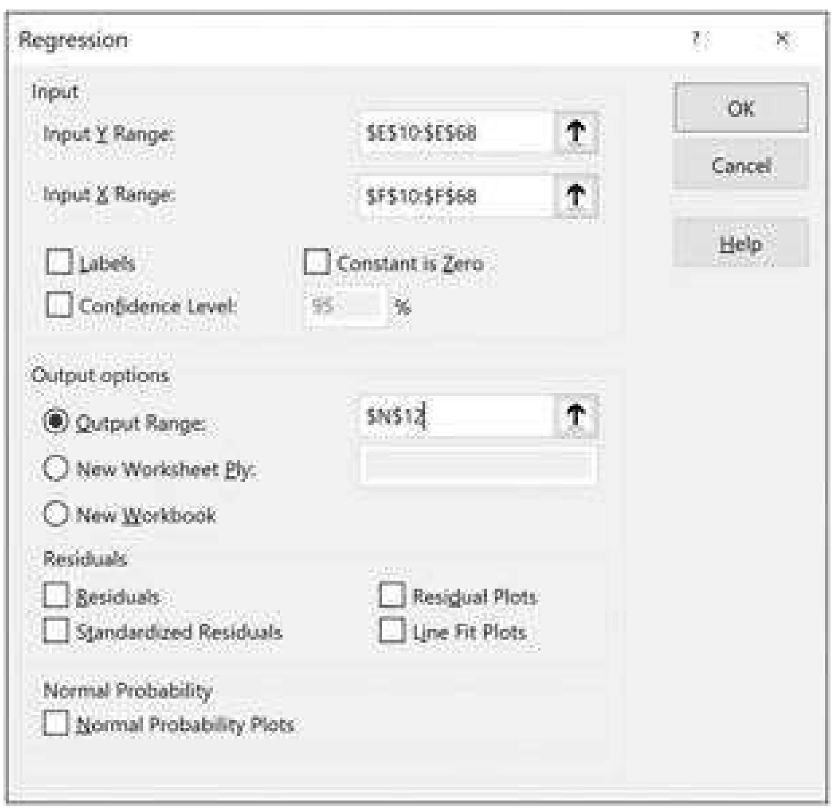
Both of these functions are embedded in the spreadsheet that accompanies this chapter.

Using Excel's Data Analysis Add-In to Estimate the Beta

There's a fourth way to produce the regression output: By clicking on **Tools|Data Analysis|Regression**, we can use a sophisticated Excel routine that computes more statistics, including the *t*-statistics. The output produced by this routine is illustrated below:

	N	O	P	Q	R	S	T	U	V
12	SUMMARY OUTPUT								
13									
14	*Regression Statistics*								
15	Multiple R	0.65							
16	R Square	0.43							
17	Adjusted R Square	0.42							
18	Standard Error	0.06							
19	Observations	59							
20									
21	ANOVA								
22		*df*	*SS*	*MS*	*F*	*Significance F*			
23	Regression	1.00	0.15	0.15	42.72	0.00			
24	Residual	57.00	0.20	0.00					
25	Total	58.00	0.34						
26									
27		*Coefficients*	*Standard Error*	*t Stat*	*P-value*	*Lower 95%*	*Upper 95%*	*Lower 95.0%*	*Upper 95.0%*
28	Intercept	0.00	0.01	-0.15	0.88	-0.02	0.01	-0.02	0.01
29	X Variable 1	1.59	0.24	6.54	0.00	1.11	2.08	1.11	2.08

We used **Data|Data Analysis|Regression** to produce this output. The settings are shown below:



While **Data|Data Analysis|Regression** produces a lot of data, it has one major drawback: The output is not dynamically updated when the underlying data changes. For this reason we prefer to use the other methods illustrated.

Using the Security Market Line (SML) to Calculate Caterpillar's Cost of Equity, r_E

In the capital asset pricing model (CAPM), the security market line (SML) is used to calculate the risk-adjusted cost of capital. The classic CAPM formula uses an SML equation:

Cost of equity, $r_E = r_f + \beta_E[E(r_M) - r_f]$

Here r_f is the risk-free rate of return in the economy and $E(R_M)$ is the expected rate of return on the market. The choice of values for the SML parameters is often problematic. A common approach is to choose:

- • r_f equal to the risk-free interest rate in the economy (for example, the yield on Treasury bonds). For the moment, for illustrative purposes, we use $r_f = 2\%$.
- • $E(r_M)$ equal to the historic average of the market return, defined as the average return of a broad-based market portfolio. There is an alternative approach based on market multiples; both approaches are discussed below. For the current section, we use $E(r_M) = 8\%$.

The spreadsheet below illustrates the classic CAPM cost of equity computation for Caterpillar's cost of equity:

	A	B	C
1	**COMPUTING THE COST OF EQUITY FOR CATERPILLAR** **Classic CAPM: $r_E = r_f + \beta*[E(r_M) - r_f]$**		
2	Caterpillar's equity beta, β_E	1.5031	
3	Risk-free rate, r_f	2.00%	
4	Expected market return, $E(r_M)$	8.00%	
5	Caterpillar cost of equity, r_E	11.02%	<-- =B3+B2*(B4-B3)

Computing the Cost of Equity for a Nonlisted Company

When the company is not listed and we have no available stock data, we often use comparable firms to estimate the cost of equity. The main issue to remember is that the cost of equity is combining two sources of risk:

1. Business/industry risk. This is the risk derived from the industry and operating activity of the firm. The higher the risk in the industry, the higher the cost of capital in the industry.
2. Leverage risk. This is the risk to the stockholders derived from the leverage decision of the specific firm. The higher the leverage, the higher the cost of capital to the equity holders of the firm.

Since we observe the cost of equity of our comparable firms, and since the leverage is firm-specific, we need to first cancel the effect of the leverage on the cost of equity for our sample firms to get the industry unlevered cost of capital and then to re-lever the cost of capital and calculate the cost of equity for our firm. The equation that links the cost of equity to the unlevered cost of capital is as follows:[13]

- $r_E = r_A + \frac{D}{E} * (1 - t_C) * (r_A - r_D)$; if we assume a constant debt over time
- $r_E = r_A + \frac{D}{E} * (r_A - r_D)$; if we assume a constant debt to equity ratio over time

Here is the example for Caterpillar:

CATERPILLAR'S COST OF EQUITY USING COMPARABLE FIRMS
December 2019

Cell 'G6' contains the formula =$B$2+F6*$B$3

	Equity ($M)	Net debt ($M)	Corporate tax rate (t_C)	Cost of debt (r_D)	Equity beta (β_E)	Cost of equity (r_E)	Unlevered cost of capital (r_A)	
Risk-free rate (r_f)	2%							
Market risk premium (MRP)	8%							
Assuming constant debt (D)								
Astec Industries	946	0	12%	0.00%	1.66	13.28%	13.28%	<-- =(G6+D6*C6/B6*(1-D6))/(1+C6/B6*(1-D6))
John Deere	54,840	47,300	21%	3.09%	1.02	8.12%	6.07%	
Manitowoc Crane Group	621	830	21%	2.94%	2.32	15.92%	9.73%	
Terex	2,120	1,100	19%	7.80%	1.85	13.10%	11.47%	
			Industry unlevered cost of capital (r_A)				10.13%	<-- =AVERAGE(H6:H9)
CAT's data								
Normative debt to equity (D/E)	30%							
Cost of debt (r_D)	3.90%							
Corporate tax rate (t_C)	23.30%							
Cost of equity (r_E)	12.84%	<-- =H10+B13*(1-B15)*(H10-B14)						

3.7 Three Approaches to Computing the Expected Return on the Market, $E(r_M)$

Two critical questions remain in the computation of the cost of equity r_E using the CAPM:

- • What is the expected return on the market, $E(r_M)$? Should it be computed from historical data? (And if so, how long should the data series be?) Or perhaps it can be computed from current market data, based on a forward-looking approach, without resorting to historical data?
- • What is the risk-free rate, r_f?

In this section we deal with the first question, leaving the computation of r_f for section 3.8. There are three major approaches to computing $E(r_M)$ using:

1. The historical return on a major market index
2. The historical market risk premium on the market index
3. The Gordon model

All three approaches are illustrated in this section, and their effect on computing Caterpillar's cost of equity is illustrated at the end of this section.

$E(r_M)$ as the Historical Average Return on a Market Portfolio

A simple approach to computing $E(r_M)$ is to take it as the average of the historical returns of a major market index. In the computation below we illustrate this approach by using the SP500 as a proxy for the market. The annualized return on this fund since 1950 is 6.64%. We can take this as a reliable proxy for the historical annual average return from holding the SP500:

	A	B	C	D
1	**MEASURING E(r_M) USING HISTORICAL DATA** **Derived from prices for the SP500 Index Fund (symbol: ^GSPC)** **1960-2019 (some lines are hidden)**			
2	Annualized return	6.64%	<--=AVERAGE(C:C)	
3	Annualized standard deviation	16.08%	<--=STDEV.S(C:C)	
4				
5	**Date**	**Price**	**Return**	
6	31/Dec/19	3,221.29	25.08%	<--=LN(B6/B7)
7	31/Dec/18	2,506.85	-6.44%	<--=LN(B7/B8)
8	31/Dec/17	2,673.61	17.75%	
9	31/Dec/16	2,238.83	9.11%	
10	31/Dec/15	2,043.94	-0.73%	
11	31/Dec/14	2,058.90	10.79%	
12	31/Dec/13	1,848.36	25.93%	
13	31/Dec/12	1,426.19	12.58%	
63	31/Dec/62	63.10	-12.57%	
64	31/Dec/61	71.55	20.81%	
65	31/Dec/60	58.11	-3.02%	
66	1/Jan/60	59.89		

Computing the Market Risk Premium $E(r_M) - r_f$ Directly

We can also compute the market risk premium directly. This requires a bit more work. In the spreadsheet below we show the annual returns on the SP500 and annual interest paid on U.S. Treasury bills. The average annual risk premium on the SP500 is 3.64%.

	A	B	C	D	E	F
1	THE MARKET RISK PREMIUM E (r_M) – r_f USING HISTORICAL DATA SP500 Index (symbol: ^GSPC) minus Treasury Bills 1990–2019					
2	Annual risk premium	3.64%	<-- =AVERAGE(E7:E65)			
3	Annual standard deviation	17.41%	<-- =STDEV.S(E7:E65)			
4						
5	Date	Price	Return	1-year T-bill rate	Market risk premium	
6	31/Dec/19	3,221.29	25.08%	2.63%	22.45%	
7	31/Dec/18	2,506.85	-6.44%	1.76%	-8.20%	<--=C7-D7
8	31/Dec/17	2,673.61	17.75%	0.85%	16.90%	<--=C8-D8
9	31/Dec/16	2,238.83	9.11%	0.65%	8.46%	<--=C9-D9
10	31/Dec/15	2,043.94	-0.73%	0.25%	-0.98%	
11	31/Dec/14	2,058.90	10.79%	0.13%	10.66%	
12	31/Dec/13	1,848.36	25.93%	0.16%	25.77%	
13	31/Dec/12	1,426.19	12.58%	0.12%	12.46%	
14	31/Dec/11	1,257.60	0.00%	0.29%	-0.29%	
15	31/Dec/10	1,257.64	12.03%	0.47%	11.56%	
16	31/Dec/09	1,115.10	21.07%	0.37%	20.70%	
17	31/Dec/08	903.25	-48.59%	3.34%	-51.93%	
18	31/Dec/07	1,468.36	3.47%	5.00%	-1.53%	
19	31/Dec/06	1,418.30	12.77%	4.38%	8.39%	
20	31/Dec/05	1,248.29	2.96%	2.75%	0.21%	
21	31/Dec/04	1,211.92	8.61%	1.26%	7.35%	
22	31/Dec/03	1,111.92	23.41%	1.32%	22.09%	
23	31/Dec/02	879.82	-26.61%	2.17%	-28.78%	
24	31/Dec/01	1,148.08	-13.98%	5.32%	-19.30%	
25	31/Dec/00	1,320.28	-10.69%	5.98%	-16.67%	
26	31/Dec/99	1,469.25	17.84%	4.53%	13.31%	
27	31/Dec/98	1,229.23	23.64%	5.51%	18.13%	
28	31/Dec/97	970.43	27.01%	5.51%	21.50%	
29	31/Dec/96	740.74	18.45%	5.18%	13.27%	
30	31/Dec/95	615.93	29.35%	7.20%	22.15%	
31	31/Dec/94	459.27	-1.55%	3.63%	-5.18%	
32	31/Dec/93	466.45	6.82%	3.61%	3.21%	
33	31/Dec/92	435.71	4.37%	4.12%	0.25%	
34	31/Dec/91	417.09	23.35%	6.82%	16.53%	
35	31/Dec/90	330.22	-6.78%	7.76%	-14.54%	
36	31/Dec/89	353.40				

Applying the risk premium directly to the computation of Caterpillar's cost of equity gives a cost of equity r_E close to 8.2% (note that we still haven't settled the question of r_f):

	A	B	C
1	COMPUTING THE COST OF EQUITY FOR CATERPILLAR USING THE MARKET RISK PREMIUM E(r_M) – r_f		
2	Caterpillar's equity beta, β_i	1.5031	
3	E(r_M) - r_f estimated from historical data	3.64%	
4	Risk-free rate, r_f	2.70%	<-- Still to be discussed
5	Caterpillar's cost of equity, $r_{E,CAT}$	8.17%	<--=B4+B2*B3

Calculating the Expected Return on the Market Using the Gordon Model

Setting $E(r_M)$ = 6.64% approximates the historic market return in the United States from 1960 to 2019. Historic averages are appropriate if we think that the future anticipated rates of return will correspond to the historic average. On the other hand, we may want to take current market data to calculate directly the future anticipated market yield.

We can do this computation by using the Gordon model. Recall from section 3.6 that the model says that the cost of equity r_E is given by:

$$E(r_E) = \frac{Div_0(1+g)}{P_0} + g$$

This formula also applies to the market portfolio, so we can write:

$$E(r_M) = \frac{Div_0(1+g)}{P_0} + g,$$

interpreting Div_0, P_0, and g to be the current dividend, price, and growth rate of the market portfolio. Assume that the firm pays out a constant proportion a of its earnings as dividends; then, indicate by EPS_0 the current earnings per share, $Div_0 = a * EPS_0$. Interpreting g to be the earnings growth of the firm, we can write:

$$E(r_M) = \frac{a * EPS_0(1+g)}{P_0} + g = \frac{a*(1+g)}{P_0/EPS_0} + g$$

The term on the right-hand side of this equation, P_0 / EPS_0, is the price earnings ratio of the market. We can use this formula to compute $E(r_M)$ and thus tie the cost of equity to currently observable market parameters. Here is an implementation:

	A	B	C
1	COMPUTING E(r_M) USING MARKET MULTIPLE		
2	Market price/earnings multiple, December 2019	24.67	
3	Equity cash flow payout ratio	50%	<-- Approx. U.S.: Dividends + repurchases
4	Anticipated growth of market equity cash flow	5.0%	<-- Analyst's estimate
5	Expected market return, E(r_M)	7.13%	<-- =B3*(1+B4)/B2+B4

3.8 What's the Risk-Free Rate r_f in the CAPM?

Opinions here seem to differ widely. Some authors suggest using a short-term rate, while others use a middle- or long-term rate. Practitioners often use a 10- to 15-year rate. For the examples in this chapter, here is some data from the U.S. Department of the Treasury:

	A	B	C	D	E	F	G	H	I	J	K	L	M
1	THE RISK-FREE YIELD CURVE												
2	Source:												
3	Date	1 Mo	2 Mo	3 Mo	6 Mo	1 Yr	2 Yr	3 Yr	5 Yr	7 Yr	10 Yr	20 Yr	30 Yr
4	31/Dec/19	1.48%	1.51%	1.55%	1.60%	1.59%	1.58%	1.62%	1.69%	1.83%	1.92%	2.25%	2.39%

Financially speaking, we should discount the expected future cash flows with a discount rate that matches the duration of the cash flows. In Chapter 7, we explain that the duration of a perpetual cash flow is $Duration = \frac{1+WACC}{WACC}$. In the table below, we can see that a WACC ranging between 7% to 12% would correspond to duration between 15 and 9 years.

	A	B	C	D	E
7	WACC	Duration			
8	7%	15.29	<--=(1+A8)/A8		
9	8%	13.50	<--=(1+A9)/A9		
10	9%	12.11			
11	10%	11.00			
12	11%	10.09			
13	12%	9.33			

For the sake of our example, we use the 10-year risk-free rate.

3.9 Computing the WACC

In the succeeding sections we compute the WACC for Caterpillar. The example we use is $E(r_M) = 7.13\%$ (computed in section 3.7 from the SP500 price/earnings multiple) and the 10-year Treasury bond rate is $r_f = 2.69\%$ (as discussed in section 3.8).

We have discussed Caterpillar's cost of equity r_E and cost of debt r_D in previous sections. Our template summarizes these computations:

	A	B	C
1	COMPUTING THE WACC FOR CATERPILLAR (DECEMBER 2019)		
2	Shares outstanding (millions)	575.54	
3	Share price, 31 December 2019	147.68	
4	Equity value, E (millions)	84,996	<-- =B2*B3
5	Net debt, D (millions)	33,100	
6	Cost of debt, r_D	2.65%	YTM on "A rating" bonds
7	Tax rate, T_C	22.35%	Effective tax rate
8			
9	**WACC based on CAPM**		
10	Risk-free rate, r_f	1.92%	
11	Expected market return, $E(r_M)$	7.13%	Using Gordon model
12	Equity beta, β	1.5031	
13	Cost of equity, r_E	9.75%	<-- =B10+B12*(B11-B10)
14	WACC	7.59%	<-- =$B$5/SUM($B$4:$B$5)*$B$6*(1-$B$7)+$B$4/SUM($B$4:$B$5)*B13
15			
16	**WACC based on two-stage Gordon model**		
17	Cost of equity, r_E	9.68%	Using two-stage Gordon model
18	WACC	7.53%	<-- =$B$5/SUM($B$4:$B$5)*$B$6*(1-$B$7)+$B$4/SUM($B$4:$B$5)*B17
19			
20	**Average WACC**	**7.56%**	<-- =AVERAGE(B18,B14)

The estimated WACC for Caterpillar (cell B20) includes a judgment call: We have averaged two of the WACC estimates.

3.10 When Don't the Models Work?

All models have problems, and nothing is perfect. In this section we discuss some of the potential problems with the Gordon model and with the CAPM.

Problems with the Gordon Model

Obviously, the Gordon model does not work if a firm doesn't pay dividends and appears to have no intention—in the immediate future—of paying dividends.[14] But even for dividend-paying firms, it may be difficult to apply the model. Particularly problematic, in many cases, is the extraction of the future dividend payout rate from past dividends.

Consider, for example, the dividend history of Ford Motor Company in the years 2007–2018:

	A	B	C	D
1	FORD MOTOR CO. DIVIDEND HISTORY 2007–2018			
2	Year	Dividend	Growth rate	
3	2007-2011	0.00		
4	2012	0.20	NA	
5	2013	0.40	69.31%	<--=LN(B5/B4)
6	2014	0.50	22.31%	<--=LN(B6/B5)
7	2015	0.60	18.23%	
8	2016	0.60	0.00%	
9	2017	0.60	0.00%	
10	2018	0.60	0.00%	
11	Growth rate, 2012-2018		18.31%	<--=AVERAGE(C5:C10)
12	Growth rate, 2013-2018		8.11%	<--=AVERAGE(C6:C10)
13	Growth rate, 2015-2018		0.00%	<--=AVERAGE(C8:C10)

The problem here is easily identifiable: Ford, which did not pay any dividend from 2007 to 2011, started paying 20-cent dividends in 2012 and steadily increased them to 0.6 cents by 2015. Since then Ford kept the annual dividend at 60 cents per share. If we use history to predict the future, what should our expectations be for the long run?

It appears that the history of Ford's dividends is not, perhaps, the best guide to its future dividend payout. There are several solutions to those wishing to use the Gordon model:

- • We can use Ford's total payouts to equity, as illustrated in this chapter. This method does not mean, however, that we can get away from judgment calls (witness our extensive use of the two-stage Gordon model).
- • Another alternative to finding Ford's cost of capital is to predict its future dividends by doing a full-blown financial model for the company. Such models—illustrated in the succeeding two chapters—are often used by analysts. Though they are complicated and time-consuming to build, they take into account all of the firm's productive and financial activities. Potentially they are, therefore, a more accurate predictor of the dividend.

Problems with the CAPM

We discuss the CAPM properties and test of the CAPM in details in Chapters 10–13. It is sufficient to point out in this section that there are potentially some issues in implementing the CAPM model, as detailed below:

- • Some stocks have negative beta, but they are clearly risky. This indicates that the stocks, in a portfolio context, have *negative risk*. Were this true, it would mean that adding those stocks to a portfolio would lower the portfolio variance enough to justify a below-risk-free return for those stocks. While this might be true for some stocks, it is hard to believe that, in the long run, the β of many stocks is indeed negative.[15]

- • In some cases, we'll get that the R^2 of the regression between a stock's returns and the SP500 is essentially zero, meaning that the SP500 simply doesn't explain any of the variation in the stock's returns. For statistics enthusiasts: The *t*-statistics of the intercept and the slope indicate that neither differs significantly from zero. In short, the regression of the stock's historic returns on the SP500 indicates no connection between the two whatsoever.

What are we to make of these situations? How should we calculate the cost of capital for those stocks? There are several alternatives:

- • We could assume that the β of the stock is in fact 0; given the standard deviation of the β estimate for the stock, the β is not statistically different from zero, so this assumption makes sense. This means that all of a stock's risk is diversifiable and that the correct cost of equity for the stock is the riskless rate of interest.
- • We could ignore the observed β and calculate the cost of equity with comparable companies, as if the company is not listed, as explained above.

For what it's worth, we would follow the latter case.

3.11 Summary

In this chapter we have illustrated in detail the application of two models for calculating the cost of equity: the Gordon dividend model and the CAPM. We have also considered two of the three practicable models for calculating the cost of debt. Because the application of these models allows for many judgment calls, our advice is as follows:

- • If possible, try to use several models to calculate the cost of capital to triangulate your results.
- • If you have time, try to calculate the cost of capital not only for the firm you are analyzing, but also for other firms in the same industry.
- • From your analysis, try to pick out a *consensus* estimate of the cost of capital. Don't hesitate to exclude numbers that strike you as unreasonable.

In sum, the calculation of the cost of capital is not just a mechanistic exercise!

Exercises

1. ABC Corp. has a stock price P_0 = 50. The firm has just paid a dividend of \$3 per share, and intelligent shareholders think that this dividend will grow by a rate of 5% per year. Use the Gordon dividend model to calculate the cost of equity of ABC.

2. Unheardof, Inc., has just paid a dividend of $5 per share. This dividend is anticipated to increase at a rate of 15% per year. If the cost of equity for Unheardof is 25%, what is the market value of a share of the company?

3. Dismal.Com is a producer of depressing Internet products. The company is currently not paying dividends, but its chief financial officer thinks that starting in three years it can pay a dividend of $15 per share, and that this dividend will grow by 20% per year. Assuming that the cost of equity of Dismal.Com is 35%, value a share based on the discounted dividends.

4. The current stock price of TransContinental Airways is $65 per share. TCA currently pays an annual per-share dividend of $3. Over the past five years this dividend has grown annually at a rate of 23%. A respected analyst assumes that the current growth rate of dividends will hold up for the next five years, after which dividend growth will slow to 5% annually. Use the **TwoStageGordon** function to compute the cost of equity.[16]

5. ABC Corp. has just paid a dividend of $3 per share. You—an experienced analyst—feel quite sure that the growth rate of the company's dividends over the next 10 years will be 15% per year. After 10 years you think that the company's dividend growth rate will slow to the industry average, which is about 5% per year. If the cost of equity for ABC is 12%, what is the value today of one share of the company?

6. Consider a company that has $\beta_{equity} = 1.5$ and $\beta_{debt} = 0.4$. Suppose that the risk-free rate of interest is 6%, the expected return on the market $E(r_M) = 15\%$, and that the corporate tax rate is 40%. If the company has 40% equity and 60% debt in its capital structure, calculate its weighted average cost of capital using the CAPM model.

7. On the companion website for this book, you will find the following monthly data for Cisco's stock price and the SP500 index. Compute the equation $r_{csco,\ t} = \alpha_{csco} + \beta_{csco} * r_{SP500,\ t}$ and include the R^2 and t-statistics for the equation and its coefficients.[17]

	A	B	C
1	**CISCO (CSCO) AND SP500 PRICES 2015–2020**		
2	Date	CSCO	SP500
3	01-Dec-19	47.61	3,230.78
4	01-Nov-19	44.98	3,140.98
5	01-Oct-19	46.81	3,037.56
6	01-Sep-19	48.68	2,976.74
7	01-Aug-19	46.12	2,926.46
8	01-Jul-19	54.24	2,980.38
9	01-Jun-19	53.58	2,941.76
10	01-May-19	50.94	2,752.06
11	01-Apr-19	54.43	2,945.83
12	01-Mar-19	52.53	2,834.40
13	01-Feb-19	50.37	2,784.49
14	01-Jan-19	45.66	2,704.10
51	01-Dec-15	23.74	2,043.94
52	01-Nov-15	23.82	2,080.41
53	01-Oct-15	25.02	2,079.36
54	01-Sep-15	22.76	1,920.03
55	01-Aug-15	22.44	1,972.18
56	01-Jul-15	24.46	2,103.84
57	01-Jun-15	23.63	2,063.11
58	01-May-15	25.22	2,107.39
59	01-Apr-15	24.81	2,085.51
60	01-Mar-15	23.51	2,067.89
61	01-Feb-15	25.20	2,104.50
62	01-Jan-15	22.37	1,994.99

8. It is January 1, 2020. Normal America, Inc. (NA), has paid a year-end dividend in each of the last 10 years, as shown by the table below:

	A	B	C	D
1	NORMAL AMERICA, INC.			
2	Year	Dec. 31 stock price	Dec. 15 dividend per share	SP500 return
3	2010	33.00		
4	2011	30.69	2.50	4.7%
5	2012	35.38	2.50	16.2%
6	2013	42.25	3.00	31.4%
7	2014	34.38	3.00	-3.3%
8	2015	36.25	1.60	30.2%
9	2016	32.25	1.40	7.4%
10	2017	43.00	0.80	9.9%
11	2018	42.13	0.80	1.2%
12	2019	52.88	1.10	37.4%
13	2020	55.75	1.60	22.9%

1. Calculate NA's β with respect to the SP500.
2. Suppose that the Treasury bill rate is 5.5% and that the expected return on the market is $E(r_M) = 13\%$. If the corporate tax rate $T_C = 3\%$, calculate NA's cost of equity using the CAPM model.
3. Assume that NA's cost of debt is 8%. If the company is financed by one-third equity and two-thirds debt, what is its weighted average cost of capital using the CAPM model?

9. Suppose that the SP500 price/earnings ratio is 17.5 and its dividend payout ratio is 55%, and that you estimate a future growth of dividends of 8%. What is $E(r_M)$?

	A	B	C
1	COMPUTING $E(r_M)$ FROM MARKET P/E		
2	Current SP500 P/E	17.5	
3	Dividend payout ratio	55%	
4	Growth rate of dividends	7%	
5	$E(r_M)$		

10. At the end of December 2020, the price/earnings ratio of the SP500 was 19.24. Assume that the index is a proxy for the market and that it has a 60% dividend payout ratio. Also assume that dividends are expected to grow at 7%. Compute $E(r_M)$.
11. The template for this exercise (see this book's auxiliary website) gives the prices of the Vanguard Index 500 Fund (VFINX). This fund's prices replicate the SP500 with dividends reinvested. Use this data to estimate the expected return on the SP500 in two variations: (1) all the data and (2) the last two years. (This exercise shows the problematics of using historical market data to estimate the expected returns.)
12. The template for this exercise (see this book's auxiliary website) gives the 10-year history of Intel's quarterly dividends. Compute Intel's cost of equity r_E using the Gordon dividend model. Compare the cost of equity computed on the basis of 10 years of growth with that computed on the last five years of growth.
13. You are considering buying the bonds of a very risky company. A bond with a \$100 face value, a 1-year maturity, and a coupon rate of 22% is selling for \$95. You consider the probability that the company will survive to pay off the bond by 80%. With 20% probability, you think that the company will default, in which case you think that you will be able to recover \$40.

1. What is the expected return on the bond?

2. If the company has cost of equity $r_E = 25\%$, tax rate $T_C = 35\%$, and 40% of its capital structure is equity, what is its weighted average cost of capital?

---

1. 1. Most market traders refer to this number as the "market capitalization" or "market cap."
2. 2. It bears repeating that calculating the cost of capital requires a large number of assumptions and does not necessarily give a precise answer. Users of cost of capital estimates should always do a sensitivity analysis around the numbers calculated. Given the data on the company you are analyzing, some sloppiness in the cost of capital calculations (with its accompanying savings in time) may be expedient.
3. 3. This statement ignores the *option value* of the liquid assets and their ability to provide the company with financial flexibility.
4. 4. This model is named after M. J. Gordon, who first published this formula in a paper entitled "Dividends, Earnings, and Stock Prices." See Gordon (1959), 99–105.
5. 5. Provided that the growth rate is constant and that $g < r_E$, the expression $\sum_{t=1}^{\infty} \frac{Div_0 * (1+g)^t}{(1+r_E)^t}$ is reduced to $\frac{Div_0 * (1+g)}{r_E - g}$. We will spare you this derivation, which is based on a formula for geometric series usually studied in high school.
6. 6. Or perhaps neither does! Perhaps we are better off using another story altogether to predict future anticipated dividend growth. We could use a pro forma model (discussed in Chapter 4) to predict the firm's anticipated dividend payout policy.
7. 7. See DeAngelo, DeAngelo, and Skinner (2008).
8. 8. Most firms only report stock repurchases and employee stock exercises annually. Thus, the only data available for these number is annual data, whereas dividends are reported quarterly and stock price data—used to compute the CAPM beta, discussed later in section 3.6—are available daily.
9. 9. In this section we interpret Div_0 in the Gordon formula to denote either dividends per share or total equity payouts.
10. 10. The CAPM is discussed in detail in Chapters 10–13. At this point we outline the application of the model to finding the cost of capital without entering into the theory.
11. 11. For the precise meaning of a *t*-statistic, you should refer to a good statistics text. For our purposes, a *t*-statistic over 1.96 indicates that with 95% probability the variable under discussion (either the intercept or the slope) is significantly different from zero. Thus, the *t*-statistic for the intercept of 0.001 indicates that the intercept is not significantly different from zero, whereas the *t*-statistic for the slope of 6.9691 indicates that the slope is significantly different from zero.
12. 12. To be precise, the CAPM model states: $E(r_i) = r_f + \beta_i * (E(r_M) - r_f)$, or: $E(r_i) = r_f(1 - \beta_i) + \beta_i E(r_M)$. So alpha should be compared to $r_f(1 - \beta_i)$ and not to 0.
13. 13. There is an implicit assumption about the risk of the tax shields that is implied in these formulas. When we assume constant debt in dollar value, the risk of tax shields is similar to the risk of debt; and when we assume a constant debt to equity ratio, the risk of tax shields is similar to the risk of assets.
14. 14. Firms cannot intend *never* to pay dividends, because such an intention would rationally mean that the value of the shares is zero.

15. 15. A more plausible explanation is that—for the period covered—the stock's return has *nothing whatsoever* to do with the market return.

16. 16. To do this problem you will have to copy the formula from the chapter spreadsheet to your answer spreadsheet. See Chapter 0 for details.

17. 17. To do this problem you will have to copy the functions **tintercept** and **tslope** from the spreadsheet **fm3_chapter02.xls** to your answer spreadsheet. See Chapter 0 for details.

4 Pro Forma Analysis and Valuation Based on the Discounted Cash Flow Approach

4.1 Overview

Chapter 2 of this book defines three approaches to corporate valuation. All are based on computing the firm's enterprise value (EV), defined as the economic value (market value) of the firm's operating assets.

- • The accounting approach to EV moves items on the balance sheet so that all operating items are on the left-hand side of the balance sheet and all financial items are on the right-hand side. Often, the EV using the accounting approach is called "capital employed," "invested capital," or simply "capital."
- • The efficient markets approach to EV revalues—to the extent possible—items on the accounting EV balance sheet at market values. An obvious revaluation is to replace the firm's book value of equity with the market value of the equity.
- • The discounted cash flow (DCF) approach values the EV as the present value of the firm's future anticipated free cash flows (FCFs) discounted at the weighted average cost of capital (WACC). The FCFs are the cash flows produced by the firm's productive assets—its working capital, fixed assets, goodwill, and so on. In this chapter we use two implementations of the DCF approach. These approaches differ in their derivation of the firm's free cash flows.

In the first section of this chapter we base our projections of future anticipated FCFs on an analysis of the firm's consolidated statement of cash flows. This method is easy to implement and relatively simple (for a valuation method, all of which take a lot of time). In the second section we base our projections of future anticipated FCFs on a pro forma model for the firm's financial statements. Pro forma statements are powerful tools that can be used for business plans as well as valuations, but they are difficult and time-consuming to implement.

Although the two sections in this chapter (simplified approach and pro forma approach) differ by their method for deriving the free cash flows, both sections boil down to the information presented in the following template:

	A	B	C	D	E	F	G	H
1	**BASIC CASH FLOW VALUATION TEMPLATE**							
2	Long-term FCF growth rate	4%						
3	Weighted average cost of capital (WACC)	11%						
4								
5	**Year**	**0**	**1**	**2**	**3**	**4**	**5**	
6	Forecasted future FCFs		1,000	1,080	1,166	1,260	1,360	
7	Discounting period		0.50	1.50	2.50	3.50	4.50	
8	Discounting factor		0.95	0.86	0.77	0.69	0.63	<--=1/(1+$B$3)^G7
9	Discounted FCF		949	924	899	874	851	<--=G8*G6
10								
11	PV short-term forecast	4,496	<--=SUM(C9:G9)					
12	PV long-term forecast (Terminal value)	12,638	<--=G6*(1+B2)/(B3-B2)*G8					
13	**Enterprise value**	**17,134**	<--=SUM(B11:B12)					
14	Add excess assets	2,000						
15	Subtract net debt	-10,000						
16	**Equity value**	**9,134**	<--=SUM(B13:B15)					
17	**Per share value (1,000 shares)**	**9.13**	<--=B16/1000					

The difference between the two DCF approaches is in the derivation of the future FCFs. In the first section of this chapter, we examine the firm's consolidated statement of cash flows (CSCF) and use it as a basis for estimating the future FCFs. We then discuss issues related to estimating the short-term growth rate (8% above) and the long-term growth rate (4% above), assuming that you have learned from Chapter 3 how to compute the WACC (11% above).

We focus on several important technical issues:

- • Adjustments that need to be made in the passage from the CSCF to the FCF. These adjustments involve:

- Financing adjustments
- Corrections for the vagaries of accounting rules
- Eliminating non-forward-looking items

- • Dates that don't match. Quite often the dates are not evenly spaced. We may be, for example, projecting from annual statements that end on December 31, but the current valuation date may be September. How do we make our valuation appropriate in this case? One of the answers is to use **XNPV**, as we shall see.
- • Estimating the return on assets versus the return on equity. **XIRR** can provide us with the answer. Finally, we discuss the methodology for making reality fit our template. (Or is it vice versa? Sometimes it's hard to tell!)

4.2 Setting the Stage—Discounting the Free Cash Flow (FCF)

The FCF from Operating Activities

The *free cash flow* is defined as the cash produced by a business without taking into account the way the business is financed. It is the best measure of the cash produced by a business. We discussed the definition of the FCF in Chapter 2 and here only recall the definition:

Defining the FCF

Profit after taxes	Accounting measure of firm profitability. It is not a cash flow.	
Add after-tax net interest payments	FCF measures the cash produced by the business activity of the firm. Add back after-tax net interest payments to neutralize the interest component of the profit after taxes.	
Add back depreciation	This noncash expense is added back to the profit after tax.	
Subtract increase in operating current assets	Increase in sales-related current assets is not an expense for tax purposes (and is therefore ignored in the profit after taxes), but it is a cash drain on the company.	For purposes of FCF, our definitions of current assets and current liabilities exclude financing items such as cash and debt.
Add increase in operating current liabilities	Increase in sale-related current liabilities provides cash to the firm.	
Subtract increase in fixed assets at cost	An increase in fixed assets (the long-term productive assets of the company) is a use of cash that reduces the firm's free cash flow.	

In this chapter we base our computations of the free cash flow on the firm's consolidated statement of cash flows.

Mid-Year Discounting

Rewriting the enterprise value (EV) to include the terminal value gives:

$$EV = \underbrace{\sum_{t=1}^{5} \frac{FCF_t}{(1+WACC)^t}}_{\text{PV short-term forecast}} + \underbrace{\frac{1}{(1+WACC)^5} \frac{FCF_5(1+g_{LT})}{(WACC - g_{LT})}}_{\text{Terminal value}}$$

This formulation assumes that all cash flows occur at year-end. In fact, most corporate cash flows occur throughout the year. If, for example, we generate \$120 over the year, with 10% discount rate, an end-year discounting gives:

$$PV = \frac{120}{(1+10\%)^1} = 109.09$$

Whereas if we assume that the $120 is generated evenly over the year ($10 at the end of each month), we get:

$$PV = \frac{10}{(1+10\%)^{1/12}} + \frac{10}{(1+10\%)^{2/12}} + \cdots + \frac{10}{(1+10\%)^{12/12}} = 114.00$$

And if we assume that the cash flows are received every day (n = 365) we get 114.44.

	A	B	C
1	**END-YEAR vs. MID-YEAR DISCOUNTING**		
2	WACC=	10%	
3	CF	120	
4			
5	End-year discounting		
6	T	1	
7	PV	109.09	<--=B3/(1+$B$2)^B6
8			
9	Mid-year discounting		
10	n (times a year)	12	
11	PV (exact)	114.00	<--=-PV((1+B2)^(1/B10)-1,B10,$B$3/B10)
12	Mid-year discounting	114.42	<--=$B$3/(1+$B$2)^0.5
13			
14	Sensitivity analysis – PV of CFs paid *n* periods a year evenly		
15	n	PV	
17	1	109.09	
18	2	111.75	
19	4	113.10	
20	12	114.00	
21	365	114.44	

If we approximate this fact by assuming that, on average, the year-t cash flow occurs in the middle of the year (cell B11 above), we can rewrite the EV equation as:

$$EV = \underbrace{\sum_{t=1}^{5} \frac{FCF_t}{(1+WACC)^{t-0.5}}}_{\text{PV short-term forecast}} + \underbrace{\frac{1}{(1+WACC)^{4.5}} \frac{FCF_5(1+g_{LT})}{(WACC-g_{LT})}}_{\text{Terminal value}}$$

It is this version of EV that is used throughout this section.

Computing the Terminal Value

The EV is defined as the present value of all future FCFs:

$$EV = \sum_{t=1}^{\infty} \frac{FCF_t}{(1+WACC)^{t-0.5}}.$$

Our valuation model assumes a growth rate for the short term (years 1–5) and another growth rate for the long term (year 6 and subsequent). Denoting the short-term growth rate by g_{ST} and the long-term growth rate by g_{LT}:

$$EV = \underbrace{\sum_{t=1}^{5} \frac{FCF_0(1+g_{ST})^t}{(1+WACC)^{t-0.5}}}_{\substack{\text{The present value}\\ \text{of cash flows, t = 1, ..., 5,}\\ \text{growing at the}\\ \text{short-term growth rate}}} + \underbrace{\frac{1}{(1+WACC)^{4.5}}}_{\substack{\text{Discounting the}\\ \text{terminal value to}\\ \text{time 0}}} \underbrace{\sum_{t=1}^{\infty} \frac{FCF_5(1+g_{LT})^t}{(1+WACC)^t}}_{\substack{\text{Terminalvalue:}\\ \text{The present value in year 4.5}\\ \text{of cash flows, t > 5,}\\ \text{growing at long-term}\\ \text{growth rate}}}$$

Using a standard technique, we can show that:

$$\underbrace{Terminal\ value}_{\text{At time 4.5}} = \sum_{t=1}^{\infty} \frac{FCF_5 * (1+g_{LT})^t}{(1+WACC)^t} = \begin{cases} \dfrac{FCF_5 * (1+g_{LT})}{WACC - g_{LT}} & if\ WACC > g_{LT} \\ undefined & if\ WACC > g_{LT} \end{cases}$$

4.3 Simplified Approach Based on Consolidated Statement of Cash Flows

The consolidated statement of cash flows for ABC Corp. for the years 2014–2018 is given below:

	A	B	C	D	E	F	G
1	**ABC CORPORATION Consolidated Statement of Cash Flows, 2014-2018**						
2		2014	2015	2016	2017	2018	
3	**Operating Activities:**						
4	Net accounting earnings	479,355	495,597	534,268	505,856	520,273	
5	Adjustments to reconcile net earnings to net cash provided by operating activities						
6	Add back depreciation and amortization	41,583	47,647	48,438	45,839	46,622	
7	Changes in operating assets and liabilities						
8	Subtract increase in accounts receivable	9,387	25,951	-12,724	1,665	-2,153	
9	Subtract increase in inventories	-37,630	-22,780	-16,247	-15,780	-5,517	
10	Subtract increase in prepaid expenses and other assets	-52,191	13,573	18,255	14,763	-2,975	
11	Add increase in accounts payable, accrued expenses, pensions, and other liabilities	29,612	51,172	6,757	40,541	60,255	
12	**Net cash provided by operating activities**	**470,116**	**611,160**	**574,747**	**592,844**	**616,505**	<--=SUM(F4:F11)
13							
14	**Investing Activities:**						
15	Short-term investments, net	-8,000	-55,000	50,000	-10,000	20,000	
16	Purchases of property, plant and equipment	-48,944	-70,326	-89,947	-37,044	-88,426	
17	Proceeds from dispositions of property, plant and equipment	197	6,956	22,542	6,179	28,693	
18	**Net cash used in investing activities**	**-53,747**	**-118,370**	**-17,005**	**-40,865**	**-39,733**	<--=SUM(F15:F17)
19							
20	**Financing Activities:**						
21	Repayment of debt	0	0	-300,000	0	-7,095	
22	Proceeds from revolving credit facility borrowings	1,242,431	0	0	0	250,000	
23	Proceeds from the issuance of stock	48,286	114,276	69,375	68,214	37,655	
24	Dividends paid	-332,986	-344,128	-361,208	-367,499	-378,325	
25	Stock repurchased	-150,095	-200,031	-200,038	-200,003	-597,738	
26	**Net cash used in financing activities**	**807,636**	**-429,883**	**-791,871**	**-499,288**	**-695,303**	<--=SUM(F21:F25)
27							
28	**Changes in cash balances**	**1,224,005**	**62,907**	**-234,129**	**52,691**	**-118,531**	<--=F12+F18+F26
29							
30	**Supplemental disclosure of cash flow information**						
31	Cash paid during the period for						
32	Income taxes	255,043	175,972	314,735	283,618	305,094	
33	Interest	83,553	83,551	70,351	57,151	57,910	
34							
35	Income tax rate	34.73%	26.20%	37.07%	35.93%	36.96%	<--=F32/(F4+F32)

In order to turn this CSCF into FCFs, we eliminate all the financial items. We also adjust the figures by adding back after-tax net interest. Since we want to use the CSCF to produce forward-looking FCFs, we may also want to eliminate operating or investment items that are not expected to recur.[1] In most cases these adjustments amount to the following:

- • Accept all operating activities in the CSCF
- • Dispense with all financing activities
- • Carefully examine the investing activities, eliminating those that are financial but keeping the operational items
- • Add back after-tax interest

In the current example:

	A	B	C	D	E	F	G
1	**ABC CORPORATION - Consolidated Statement of Cash Flows, 2018-2022**						
2		**2018**	**2019**	**2020**	**2021**	**2022**	
3	**Operating Activities:**						
4	Net earnings	479,355	495,597	534,268	505,856	520,273	
5	Adjustments to reconcile net earnings to net cash provided by operating activities						
6	Add back depreciation and amortization	41,583	47,647	46,438	45,839	46,622	
7	Changes in operating assets and liabilities						
8	Subtract increase in accounts receivable	9,387	25,951	-12,724	1,685	-2,153	
9	Subtract increase in inventories	-37,630	-22,780	-16,247	-15,780	-5,517	
10	Subtract increase in prepaid expenses and other assets	-52,191	13,573	16,255	14,703	-2,975	
11	Add increase in accounts payable, accrued expenses, pensions, and other liabilities	29,612	51,172	6,757	40,541	60,255	
12	**Net cash provided by operating activities**	**470,116**	**611,160**	**574,747**	**592,844**	**616,505**	<--=SUM(F4:F11)
13							
14	**Investing Activities:**						
15	Short-term investments, net						
16	Purchases of property, plant and equipment	-48,944	-70,326	-89,947	-37,044	-88,426	
17	Proceeds from dispositions of property, plant and equipment	197	6,956	22,942	6,179	28,693	
18	**Net cash used in investing activities**	**-53,747**	**-118,370**	**-67,005**	**-30,865**	**-59,733**	<--=SUM(F15:F17)
19							
20	**Financing Activities:**						
21	Repayment of debt						
22	Proceeds from revolving credit facility borrowings						
23	Proceeds from the issuance of stock						
24	Dividends paid						
25	Stock repurchased						
26	**Net cash used in financing activities**						
27							
28	**Free cash flow before interest adjustment**	**416,369**	**492,790**	**507,742**	**561,979**	**556,772**	<--=F12+F18+F26
29	**Add back after-tax net interest**	**54,537**	**61,658**	**44,271**	**36,620**	**36,504**	<--=(1-F37)*F35
30	**Free cash flow (FCF)**	**470,906**	**554,448**	**552,013**	**598,599**	**593,276**	<--=F28+F29
31							
32	**Supplemental disclosure of cash flow information**						
33	Cash paid during the period for						
34	Income taxes	255,043	175,972	314,735	283,618	305,094	
35	Interest	83,553	83,551	70,351	57,151	57,910	
36							
37	Income tax rate	34.73%	26.20%	37.07%	35.92%	36.96%	<--=F34/(F4+F34)

The Enterprise Value and the Value per Share

At this point we have estimates for the firm's historical FCFs. In order to apply the methodology, we need to estimate three parameters:

- • The short-term growth rate of ABC cash flows (g_{ST})

- • The long-term growth rate of ABC cash flows (g_{LT})
- • The weighted average cost of capital (WACC)

Some analysis, combined with seat-of-the-pants intuition, leads to the following parameter choices and valuation. The **Data Table** does sensitivity analysis on the long-term growth rate and the WACC.

	A	B	C	D	E	F	G	H
1	**ABC CORP. VALUATION BASED ON CONSOLIDATED CF STATEMENT**							
2	Free cash flow (FCF) 2022	593,276						
3	Short-term FCF growth rate (g_{ST})	8%						
4	Long-term FCF growth rate (g_{LT})	5%						
5	Weighted average cost of capital (WACC)	11%						
6								
7	**Year**	**0**	**1**	**2**	**3**	**4**	**5**	
8	Forecasted Future FCFs	593,276	640,738	691,997	747,357	807,145	871,717	<--=F8*(1+$B$3)
9	Discounting period		0.50	1.50	2.50	3.50	4.50	
10	Discounting factor		0.95	0.86	0.77	0.69	0.63	<--=1/(1+$B$5)^G9
11	Discounted FCF		608,161	591,725	575,732	560,172	545,032	<--=G10*G8
12								
13	PV short term forecast	2,880,822	<--=SUM(C11:G11)					
14	PV long term forecast (Terminal value)	9,538,060	<--=G8*(1+B4)/(B5-B4)*G10					
15	**Enterprise value**	**12,418,882**	<--=SUM(B13:B14)					
16	Add excess assets	2,000,000	<-- From 2018 balance sheet					
17	Subtract net debt	-10,000,000	<-- From 2018 balance sheet					
18	**Equity value**	**4,418,882**	<--=SUM(B15:B17)					
19	**Per share value (100,000 shares)**	**44.19**	<--=B18/100000					
20								
21	**Data table: Share value vs g_{LT} and WACC**				data table header,			
22	↓WACC →g_{LT}				=IF(B5>B4,B19,"NA")			
23	44.19	0%	2.5%	**5%**	7.5%	10.0%		
24	6%	64.09	148.72	**656.50**	NA	NA		
25	8%	27.90	65.73	**186.62**	1,276.41	NA		
26	9%	15.85	43.40	**105.40**	374.04	NA		
27	**11%**	**-1.64**	**14.53**	**44.19**	**116.21**	**548.34**		
28	13%	-13.73	-3.32	**13.60**	45.89	132.00		
29	16%	-26.15	-20.14	**-11.42**	2.45	27.87		
30	18%	-32.11	-27.73	**-21.67**	-12.72	1.81		

The data table header (our nomenclature for what the **Data Table** actually computes) includes an **If** statement. The reason is that, as shown above, the terminal value formula is valid only if the long-term growth rate is less than the WACC.

Implementing the Model for Caterpillar: Reverse-Engineering the Market Value

The method discussed in this chapter can often be used to extract market-implied expectations of growth. As an example, consider Caterpillar. Manipulation of the CSCF produces a 2019 FCF of $5.59 billion. Using the WACC of 7.65% computed in Chapter 3 and some arbitrary values for the short- and long-term FCF growth rates gives us the valuation template for Caterpillar below.[2]

	A	B	C	D	E	F	G	H
1	**CATERPILLAR VALUATION BASED ON CONSOLIDATED CF STATEMENT**							
2	Free cash flow (FCF) 2019	5,592,479						
3	Short-term FCF growth rate (g_{ST})	4%						
4	Long-term FCF growth rate (g_{LT})	2%						
5	Weighted average cost of capital (WACC)	7.56%						
6								
7	**Year**	**0**	**1**	**2**	**3**	**4**	**5**	
8	Forecasted future FCFs	5,592,479	5,816,178	6,048,826	6,290,779	6,542,410	6,804,106	<-- =F8*(1+$B$3)
9	Discounting period		0.50	1.50	2.50	3.50	4.50	
10	Discounting factor		0.96	0.90	0.83	0.77	0.72	<-- =1/(1+$B$5)^G9
11	Discounted FCF		5,607,992	5,422,256	5,242,672	5,069,036	4,901,151	<-- =G10*G8
12								
13	PV short-term forecast	26,243,107	<-- =SUM(C11:G11)					
14	PV long-term forecast (Terminal value)	89,873,643	<-- =G8*(1+B4)/(B5-B4)*G10					
15	**Enterprise value**	**116,116,750**	<-- =SUM(B13:B14)					
16	Add excess assets	1,251,000	<-- From 2018 balance sheet					
17	Subtract net debt	-25,836,000	<-- From 2018 balance sheet					
18	**Equity value**	**91,531,750**	<-- =SUM(B15:B17)					
19	Shares outstanding (thousands)	575,540						
20	**Per share value**	**159.04**	<-- =B18/B19					

The WACC for Caterpillar was discussed in Chapter 3, where we saw that plausible estimates are in the range of 7% to 12%. The WACC = 7.65% was determined in Chapter 3 as an average of plausible estimates. If the short-term growth rate is 4% and the long-term growth is 2%, then our model produces a share value of $159.04, which is somewhat close to Caterpillar's stock price of $147.68. We use a **Data Table** to find what current market price represents in terms of long-term growth and WACC:

	A	B	C	D	E	F	G	H	I	J	K	L
18	**Equity value**	**91,531,750**	<-- =SUM(B15:B17)									
19	Shares outstanding (thousands)	575,540										
20	**Per share value**	**159.04**	<-- =B18/B19									
21	Share price, 31 December 2018	147.68										
22	**Data table: Share value vs g_{LT} and WACC**				data table header! =IF(B5>B4,B20,"NA")							
23	WACC ↓ g_{LT} →											
24		159.04	0%	1.0%	2%	3%	4.0%	5%	6%	7.0%	8%	9%
25		6.0%	79.77	92.16	110.51	135.05	**172.49**	234.96	359.32	734.79	NA	NA
26		6.5%	71.59	83.16	98.29	119.93	**148.73**	195.57	279.69	476.60	1,460.22	NA
27		7.0%	65.21	75.29	88.25	105.54	**129.73**	166.03	226.52	347.51	710.48	NA
28		**7.6%**	**59.50**	**68.35**	**79.55**	**94.21**	**114.19**	**143.85**	**186.41**	**276.86**	**490.57**	**1,413.15**
29		8.0%	54.57	62.16	71.94	84.55	**101.24**	124.64	159.83	214.42	335.61	687.34
30		9.0%	45.51	51.79	59.27	68.73	**80.99**	97.11	119.82	153.88	210.64	324.17
31		10.0%	38.14	43.14	49.14	56.47	**65.64**	77.43	93.16	115.15	148.16	203.17

We have used Excel's **Conditional Formatting** to highlight Caterpillar's share valuations that are +/–20% of the current market price. It is clear that the implied long-term growth rate is ranging between 3% to 6%.

Interim Summary

So far we have illustrated a relatively simple valuation technique. Starting with free cash flows derived from the consolidated statement of cash flows, we build a simple valuation template that is a function of only four parameters: the current FCF, the short-term FCF growth rate, the long-term FCF growth rate, and the WACC. Our technique allows us to focus on the main valuation parameters and also allows us to reverse-engineer the growth and WACC expectations built into the current market price.

4.4 Pro Forma Financial Statement Modeling

The usefulness of financial statement projections for corporate financial management is undisputed. Such projections, termed *pro forma financial statements*, are the bread and butter for much corporate financial analysis. In this section we focus on the use of pro formas for valuing the firm and its component securities. However, pro formas also form the basis for many credit analyses; by examining pro forma financial statements we can predict how much financing a firm will need in future years. We can play the usual "what if" games of simulation models, and we can use pro formas to ask what strains on the firm may be caused by changes in financial and sales parameters.

In this section we present a variety of financial models. All the models are sales driven, in that they assume that many of the balance sheet and income statement items are directly or indirectly related to sales. The mathematical structure of solving the models involves finding the solution to a set of simultaneous linear equations predicting both the balance sheets and the income statements for the coming years. However, the user of a spreadsheet need never worry about the solution of the model; the fact that spreadsheets can solve—by iteration—the financial relations of the model means that we only must worry about correctly stating the relevant accounting relations in our Excel model.

How Financial Models Work: Theory and an Initial Example

Almost all financial statement models are *sales driven*; this term means that as many as possible of the most important financial statement variables are assumed to be functions of the sales level of the firm. For example, accounts receivable is often taken as a direct percentage of the sales of the firm. A slightly more complicated example might postulate that the fixed assets (or some other account) are a step function of the level of sales:

$$Fixed\ assets = \begin{cases} a & if\ sales < A \\ b & if\ A \le sales < B \\ etc. & \end{cases}$$

In order to solve a financial planning model, we must distinguish between those financial statement items that are functional relationships of sales and perhaps of other financial statement items and those items that involve policy decisions. The asset side of the balance sheet is usually assumed to be dependent only on functional relationships. The current liabilities may also be taken to involve functional relationships only, leaving the mix between long-term debt and equity as a policy decision.

A simple example is the following. We wish to predict the financial statements for a firm whose current balance sheet and income statement are as follows:

	A	B
2		**0** actual
3	Sales growth	
4	Costs of goods sold/Sales	50%
5	Interest rate on debt	10%
6	Annual growth rate in SG&A	
7	Current assets/Sales	15%
8	Current liabilities/Sales	8%
9	Net fixed assets/Sales	77%
10	Depreciation rate	13%
11	Interest paid on cash and marketable securities	8.0%
12	Tax rate	35%
13	Dividend payout ratio	35%
14		
15	**Year**	**0**
16	**Income statement**	
17	Sales	1,000
18	Costs of goods sold	(500)
19	**Gross profit**	**400**
20	*Gross profit %*	*40%*
21	Selling general and administrative	(100)
22	Interest payments on debt	(32)
23	Interest earned on cash and marketable securities	6.4
24	Depreciation	(100)
25	**Profit before tax**	**174**
26	Taxes	(61)
27	**Profit after tax (net profit)**	**113**
28	*Net profit %*	11%
29	Dividends	(40)
30	**Retained earnings**	**73**
31		
32	**Balance sheet**	
33	Cash and marketable securities	80
34	Current assets	150
35	Fixed assets	
36	At cost	1,070
37	Accumulated depreciation	(300)
38	Net fixed assets	770
39	**Total assets**	**1,000**
40		
41	Current liabilities	80
42	Debt	320
43	Stock	450
44	Accumulated retained earnings	150
45	**Total liabilities and equity**	**1,000**

The current (year 0) level of sales is 1,000. The firm expects its sales to grow at a rate of 10% per year, and it anticipates the following financial statement relations:

Item	**Assumption**
Current assets	Assumed to be 15% of end-of-year sales.

Current liabilities	Assumed to be 8% of end-of-year sales.
Depreciation	10% of the average value of assets during the year.
Fixed assets at cost	Sum of net fixed assets plus accumulated depreciation.
Debt	The firm neither repays any existing debt nor borrows any more money over the five-year horizon of the pro formas.
Cash and marketable securities	Average balances of cash and marketable securities are assumed to earn 8% interest.

The "Absorbing Item"

Perhaps the most important financial policy variable in the financial statement modeling is the "absorbing item." This relates to the decision about which balance sheet item will "close" the model:

- • It guarantees that assets and liabilities are equal (this is "closure" in the accounting sense).
- • It determines how the firm finances its investments (this is "financial closure").

In general, the absorbing item in a pro forma model will be one of three financial balance sheet items: (1) cash and marketable securities, (2) debt, or (3) stock.[3] As an example, consider the balance sheet of our first pro forma model:

Assets	**Liabilities and Equity**
Cash and marketable securities	Current liabilities
Current assets	Debt
Fixed assets	Equity
Fixed assets at cost − accumulated depreciation	Stock (net funds directly provided by shareholders)
Net fixed assets	Accumulated retained earnings (profits not paid out)
Total assets	**Total liabilities and equity**

In the current example, we assume that net fixed assets will be the absorbing item. This assumption has two meanings:

1. The *mechanical* meaning of the absorbing item: Formally, we define:

$$\begin{matrix}\textit{Net fixed}\\ \textit{assets}\end{matrix} = \begin{matrix}\textit{Current}\\ \textit{liabilities}\end{matrix} + \textit{Debt} + \textit{Equity} - \begin{matrix}\textit{Cash and}\\ \textit{marketable securities}\end{matrix} - \begin{matrix}\textit{Current}\\ \textit{assets}\end{matrix}$$

By using this definition, we guarantee that assets and liabilities will always be equal.

2. The *financial* meaning of the absorbing item: By defining the absorbing item to be net fixed assets, we are also making a statement about how the firm finances itself. In our model below, for example, the firm sells no additional stock, does not pay back any of its existing debt, and does not raise any more debt. This definition means that all incremental financing (if needed) for the firm will come from the net fixed assets account; it also means that if the firm has additional cash, it will use it to buy more fixed assets.

Projecting Next Year's Balance Sheet and Income Statement

Above we have given the financial statement for year 0. We now project the financial statement for year 1:

	A	B	C	D
1	**SETTING UP THE FINANCIAL STATEMENT MODEL - next year forecast**			
2		0 actual	1 assumption	
3	Sales growth		10%	
4	Costs of goods sold/Sales	50%	50%	<-- =B4
5	Interest rate on debt	10%	10%	<-- =B5
6	Annual growth rate in SG&A		3%	<-- =3%
7	Current assets/Sales	15%	15%	<-- =B7
8	Current liabilities/Sales	8%	8%	<-- =B8
9	Net fixed assets/Sales	77%	Absorbing item	
10	Depreciation rate	13%	13%	<-- =B10
11	Interest paid on cash and marketable securities	8.0%	8.0%	<-- =B11
12	Tax rate	35%	35%	<-- =B12
13	Dividend payout ratio	35%	35%	<-- =B13
14				
15	**Year**	**0**	**1**	
16	**Income statement**			
17	Sales	1,000	1,100	<-- =B17*(1+C3)
18	Costs of goods sold	(500)	(550)	<-- =-C17*C4
19	**Gross profit**	**400**	**550**	**<-- =SUM(C17:C18)**
20	*Gross profit %*	40%	50%	<-- =C19/C17
21	Selling, general and administrative	(100)	(103)	<-- =B21*(1+C6)
22	Interest payments on debt	(32)	(32)	<-- =-C5*AVERAGE(B42:C42)
23	Interest earned on cash and marketable securities	6.4	6.4	<-- =C11*AVERAGE(B33:C33)
24	Depreciation	(100)	(116)	<-- =-C38*C10
25	**Profit before tax**	**174**	**306**	**<-- =C19+SUM(C21:C24)**
26	Taxes	(61)	(107)	<-- =-C25*C12
27	**Profit after tax (net profit)**	**113**	**199**	**<-- =C26+C25**
28	*Net profit %*	11%	18%	<-- =C27/C17
29	Dividends	(40)	(70)	<-- =-C13*C27
30	**Retained earnings**	**73**	**129**	**<-- =C29+C27**
31				
32	**Balance sheet**			
33	Cash and marketable securities	80	80	<-- =B33
34	Current assets	150	165	<-- =C17*C7
35	Fixed assets			
36	At cost	1,070	1,307	<-- =C38-C37
37	Accumulated depreciation	(300)	(416)	<-- =B37+C24
38	Net fixed assets	770	892	<-- =C45-SUM(C33:C34)
39	**Total assets**	**1,000**	**1,137**	**<-- =C38+C34+C33**
40				
41	Current liabilities	80	88	<-- =C17*C8
42	Debt	320	320	<-- =B42
43	Stock	450	450	<-- =B43
44	Accumulated retained earnings	150	279	<-- =B44+C30
45	**Total liabilities and equity**	**1,000**	**1,137**	**<-- =SUM(C41:C44)**

The formulas are mostly obvious. Use the formula in each year (model parameters are in bold):

- Income Statement Equations:
  - $Sales_t = Sales_{t-1} * (1 + \textbf{\textit{Sales growth}})$
  - Costs of goods sold = sales * **costs of goods sold/sales**

The assumption is that the only expenses related to sales are costs of goods sold:

- $SG\&A_T = SG\&A_{T-1} * (1 + SG\&\textit{A growth})$, where SG&A is Selling, general and administrative expense
- Interest payments on debt = **Interest rate on debt** * Average debt over the year

This formula allows us to accommodate changes in the model for repayment of debt and rollover of debt at different interest rates. Note that in the current version of the model, debt stays constant; but in other versions of the model discussed below, debt will vary over time:

- Interest earned on cash and marketable securities = **Interest rate on cash** * Average cash and marketable securities over the year
- Depreciation = **Depreciation rate** * Net fixed assets at year end

This is a simplified way to model depreciation. We discuss another alternative in Chapter 5.

- Profit before taxes = Sales − Costs of goods sold − Interest payments on debt + Interest earned on cash and marketable securities − Depreciation
- Taxes = **Tax rate** * Profit before taxes
- Profit after taxes = Profit before taxes − Taxes
- Dividends = **Dividend payout ratio** * Profit after taxes

The firm is assumed to pay out a fixed percentage of its profits as dividends. An alternative would be to assume that the firm has a target for its dividends per share.

- Retained earnings = Profit after taxes − Dividends

- Balance Sheet Equations:
  - Cash and marketable securities = Same as last year
  - Current assets = **Current assets/sales** * Sales
  - $\begin{pmatrix}\text{Net fixed}\\ \text{assets}\end{pmatrix} = \begin{pmatrix}\text{Total liabilities}\\ \text{and equity}\end{pmatrix} - \begin{pmatrix}\text{Cash and marketable}\\ \text{securities}\end{pmatrix} - \begin{pmatrix}\text{Current}\\ \text{assets}\end{pmatrix}$

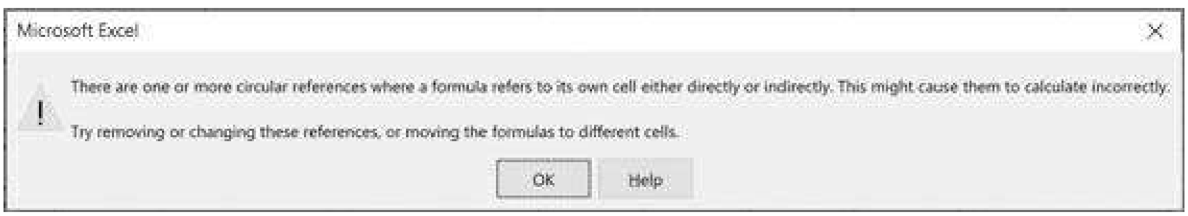
- Accumulated depreciation = Previous year's accumulated depreciation + **Depreciation rate** * Average fixed assets at cost over the year
- Fixed assets at cost = Net fixed assets + Accumulated depreciation

Note that this model does not distinguish between property, plant, and equipment (PP&E) and other fixed assets such as land.

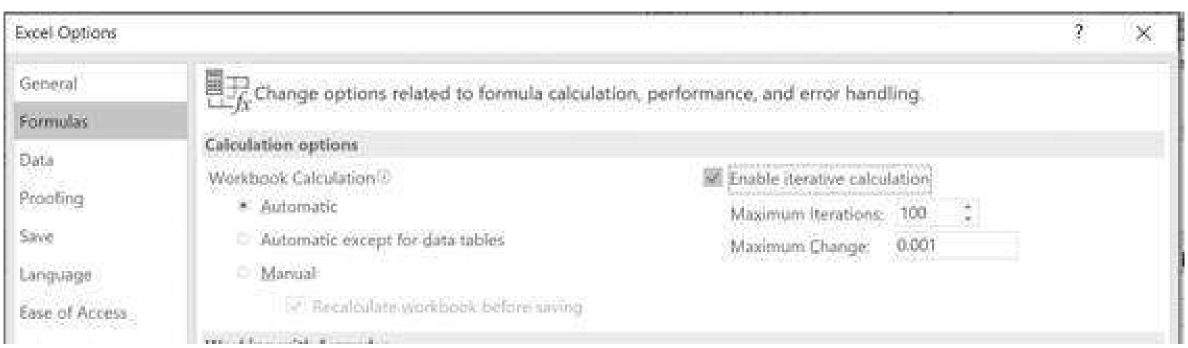
- Current liabilities = **Current liabilities/Sales** * Sales
- Debt is assumed to be unchanged. An alternative model, which we will explore later, assumes that debt is the balance sheet absorbing item.
- Stock doesn't change (shareholders provide no additional direct financing; the company is assumed to issue no new stock or repurchase any stock).
- Accumulated retained earnings = Previous year's accumulated retained earnings + Current year's additions to retained earnings

Circular References in Excel

Financial statement models in Excel almost always involve cells that are mutually dependent. As a result, the solution of the model depends on Excel's ability to solve circular references. If you open a spreadsheet that involves iteration and if your spreadsheet is not set up for circular references, you will see the following Excel error message:



To make sure your spreadsheet recalculates, you have to go to the **File|Options|Formulas** box and click **Enable iterative calculation**:



Extending the Model to Year 2 and Beyond

Now that you have the model set up, you can extend it by copying the columns:

	A	B	C	D	E	F	G	H
1	SETTING UP THE FINANCIAL STATEMENT MODEL - 5 year forecast							
2		0 actual	1 assumption	2 assumption	3 assumption	4 assumption	5 assumption	
3	Sales growth		10%	10%	10%	10%	10%	
4	Costs of goods sold/Sales	50%	50%	50%	50%	50%	50%	
5	Interest rate on debt	10%	10%	10%	10%	10%	10%	
6	Annual growth rate in SG&A		3%	3%	3%	3%	3%	
7	Current assets/Sales	15%	15%	15%	15%	15%	15%	
8	Current liabilities/Sales	8%	8%	8%	8%	8%	8%	
9	Net fixed assets/Sales	77%			Accounting item			
10	Depreciation rate	13%	13%	13%	13%	13%	13%	
11	Interest earned on cash and marketable securities	8%	8%	8%	8%	8%	8%	
12	Tax rate	35%	35%	35%	35%	35%	35%	
13	Dividend payout ratio	35%	35%	35%	35%	35%	35%	
14								
15	**Year**	**0**	**1**	**2**	**3**	**4**	**5**	
16	**Income statement**							
17	Sales	1,000	1,100	1,210	1,331	1,464	1,611	<-- =F17*(1+G3)
18	Costs of goods sold	(500)	(550)	(605)	(666)	(732)	(805)	<-- =-G17*G4
19	**Gross profit**	**500**	**550**	**605**	**666**	**732**	**805**	<-- =SUM(G17:G18)
20	Gross profit %	50%	50%	50%	50%	50%	50%	<-- =G19/G17
21	Selling, general and administrative	(100)	(103)	(106)	(109)	(113)	(116)	<-- =F21*(1+G6)
22	Interest payments on debt	(32)	(32)	(32)	(32)	(32)	(32)	<-- =-G5*AVERAGE(F42:G42)
23	Interest earned on cash and marketable securities	6.4	6.4	6.4	6.4	6.4	6.4	<-- =G11*AVERAGE(F33:G33)
24	Depreciation	(100)	(116)	(133)	(153)	(175)	(199)	<-- =-G10*G38
25	**Profit before tax**	**174**	**306**	**340**	**378**	**419**	**465**	**<-- =G19+SUM(G21:G24)**
26	Taxes	(61)	(107)	(119)	(132)	(147)	(163)	<-- =-G25*G12
27	**Profit after tax (net profit)**	**113**	**199**	**221**	**246**	**273**	**302**	**<-- =G26+G25**
28	Net profit %	11%	18%	18%	18%	19%	19%	<-- =G27/G17
29	Dividends	(40)	(70)	(78)	(87)	(96)	(107)	<-- =-G13*G27
30	**Retained earnings**	**73**	**129**	**143**	**159**	**176**	**196**	**<-- =G29+G27**
31								
32	**Balance sheet**							
33	Cash and marketable securities	80	80	80	80	80	80	<-- =F33
34	Current assets	150	165	182	200	220	242	<-- =G17*G7
35	Fixed assets							
36	At cost	1,070	1,307	1,576	1,879	2,221	2,605	<-- =G38-G37
37	Accumulated depreciation	(300)	(416)	(549)	(702)	(877)	(1,075)	<-- =F37+G24
38	Net fixed assets	770	892	1,027	1,177	1,344	1,530	<-- =G45-SUM(G33:G34)
39	**Total assets**	**1,000**	**1,137**	**1,288**	**1,457**	**1,644**	**1,851**	**<-- =G38+G34+G33**
40								
41	Current liabilities	80	88	97	106	117	129	<-- =G17*G8
42	Debt	320	320	320	320	320	320	<-- =F42
43	Stock	450	450	450	450	450	450	<-- =F43
44	Accumulated retained earnings	150	279	422	580	757	952	<-- =F44+G30
45	**Total liabilities and equity**	**1,000**	**1,137**	**1,288**	**1,457**	**1,644**	**1,851**	**<-- =SUM(G41:G44)**

FCF: Measuring the Cash Produced by the Business

Now that we have the model, we can use it to make financial predictions using the FCF calculation. For reference, we briefly repeat the definition of FCF as first given in Chapter 2:

Defining the Free Cash Flow

Profit after taxes	Accounting measure of firm profitability. It is not a cash flow.	
+ Depreciation	This noncash expense is added back to the profit after tax.	
− Increase in operating current assets	Increase in sales-related current assets is not an expense for tax purposes (and is therefore ignored in the profit after taxes), but it is a cash drain on the company.	For purposes of FCF, our definitions of current assets and current liabilities exclude financing items such as cash and debt.
+ Increase in operating current liabilities	Increase in sale-related current liabilities provides cash to the firm.	
− Increase in	An increase in fixed assets (the long-term productive assets of the	

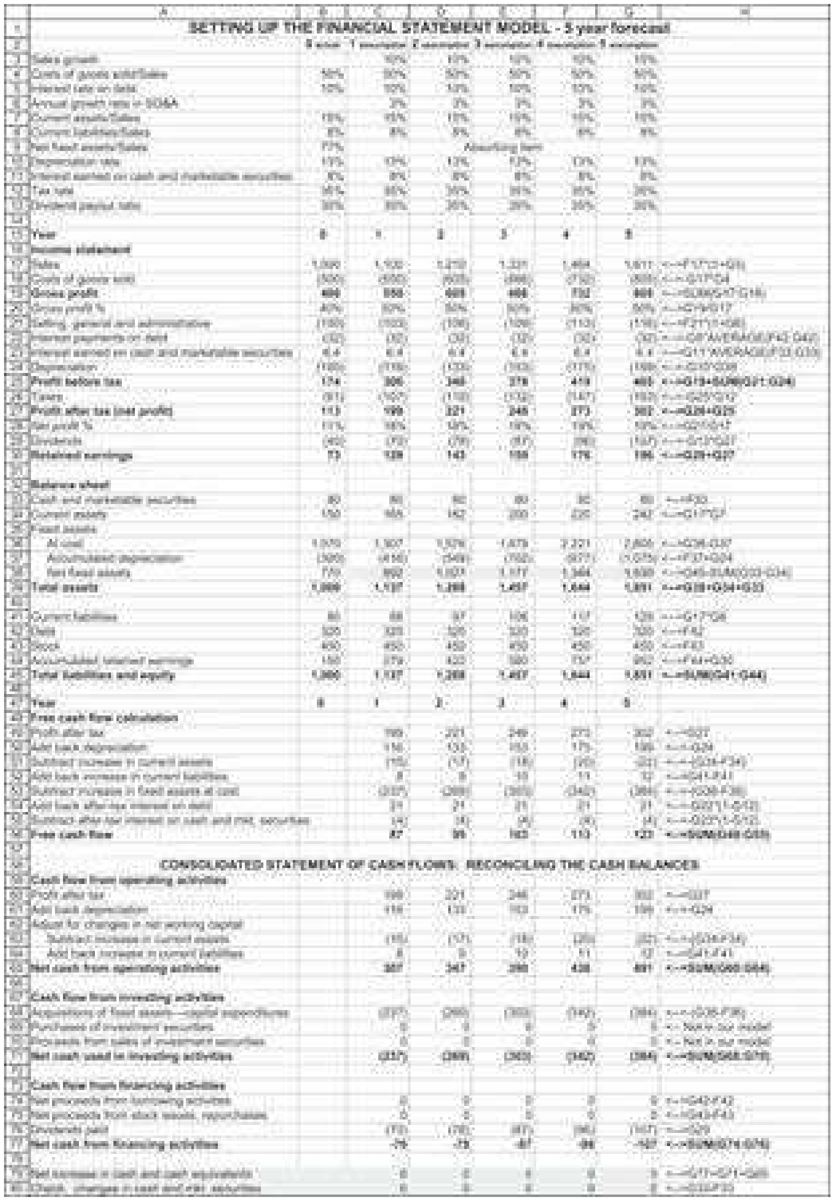
fixed assets at cost	company) is a use of cash, which reduces the firm's free cash flow.
+ After-tax net interest payments	FCF measures the cash produced by the business activity of the firm. Add back after-tax net interest payments to neutralize the interest component of the profit after taxes.

Here is the calculation for our firm:

	A	B	C	D	E	F	G	H
47	**Year**	0	1	2	3	4	5	
48	**Free cash flow calculation**							
49	Profit after tax		196	221	246	273	302	<-- =G27
50	Add back depreciation		118	133	153	175	199	<-- =G24
51	Subtract increase in current assets		(15)	(17)	(18)	(20)	(22)	<-- =-(G34-F34)
52	Add back increase in current liabilities		8	9	10	11	12	<-- =G41-F41
53	Subtract increase in fixed assets at cost		(237)	(269)	(303)	(342)	(384)	<-- =-(G36-F36)
54	Add back after-tax interest on debt		21	21	21	21	21	<-- =G22*(1-G12)
55	Subtract after-tax interest on cash and mkt. securities		(4)	(4)	(4)	(4)	(4)	<-- =-G23*(1-G12)
56	**Free cash flow**		**87**	**95**	**103**	**113**	**123**	<-- =SUM(G49:G55)

Reconciling the Cash Balances

The free cash flow calculation is different from the "consolidated statement of cash flows" that is a part of every accounting statement. The purpose of the consolidated statement of cash flows is to explain the increase in the cash accounts in the balance sheet as a function of the cash flows from the firm's operating, investing, and financing activities. In the pro forma example of this section, it can be derived from a standard accounting statement of cash flows:



SETTING UP THE FINANCIAL STATEMENT MODEL - 5 year forecast

Line 80 checks that the changes in the cash accounts derived through the consolidated statement of cash flows match those derived in the financial model (which in our case assumes no change in cash and marketable securities). As you can see, the model works, in the sense that changes in cash balances from the consolidated statement of cash flows in fact match those in the projected balance sheets of the pro forma model.

4.5 Using the FCF to Value the Firm and Its Equity

Most financial analysts consider it presumptuous to project an infinite number of free cash flows; therefore, the projected cash flow stream is often cut off at some arbitrary date and a *terminal value* is substituted for the cash flows beyond this date:

$$Enterprise\ Value = \frac{FCF_1}{(1+WACC)^{0.5}} + \frac{FCF_2}{(1+WACC)^{1.5}} + \cdots + \frac{FCF_5}{(1+WACC)^{4.5}} + \frac{Terminal\ Value_{year\ 5}}{(1+WACC)^{4.5}}$$

In this formula, the *year-5 terminal value* is a proxy for the present value of all FCFs from year 6 onward. Instead of projecting the FCFs from year 6 onward, we use the most common terminal value model, assuming that year-5 FCFs grow at a long-term growth rate g_{LT}:

$$Terminal\ Value\ at\ period\ 4.5 = \sum_{t=1}^{\infty} \frac{FCF_{5+t}}{(1+WACC)^t} = \sum_{t=1}^{\infty} \frac{FCF_5 * (1+g_{LT})^t}{(1+WACC)^t}$$
$$= \frac{FCF_5 * (1+g_{LT})}{WACC - g_{LT}}$$
$$provided\ g_{LT} < WACC$$

This model (based on the formula for the present value of a growing annuity, see section 1.1) assumes that the year-5 cash flow will continue to grow at a constant long-term growth rate. Note that the formula is pointless if the long-term growth rate g_{LT} is greater than or equal to the WACC; if this were the case, then the terminal value would be infinite (clearly impossible).

Here is an example that uses our projections:

	A	B	C	D	E	F	G	H
58			VALUING THE FIRM					
59	Long-term FCF growth rate (g_{LT})	3%	Assumption					
60	Weighted average cost of capital (WACC)	12%	Assumption					
61								
62	Year	0	1	2	3	4	5	
63	Forecasted Future FCFs		87	95	103	112	123	<-- [illegible]
64	Discounting period		0.50	1.50	2.50	3.50	4.50	
65	Discounting factor		0.94	0.84	0.75	0.67	0.60	<-- =1/(1+$B$60)^G64
66	Discounted FCF		82	80	78	75	74	<-- =G65*G63
67								
68	PV short term forecast	389	<-- =SUM(C66:G66)					
69	PV long term forecast (Terminal value)	847	<-- =G63*(1+B59)/(B60-B59)*G65					
70	**Enterprise value**	**1,237**	**<-- =SUM(B68:B69)**					
71	Add excess assets	0	Not relevant in this example					
72	Subtract net debt	-240	<-- [illegible]					
73	**Equity value**	**997**	**<-- =SUM(B70:B72)**					

Note that the long-term FCF growth rate in cell B59 is different from the sales growth in row 3 in our setup of the financial statement model presented above. The sales growth is the anticipated growth over the years 1–5; the long-term growth rate is probably better estimated by making a more realistic estimate of the growth of the firm's market sector. For firms operating in a mature market, we often estimate the long-term FCF growth as the sum of real growth plus anticipated inflation.

Some Notes on the Valuation Procedure

In this section we cover some issues related to the valuation procedure outlined in this section.

Terminal Value

In determining the terminal value, we used a version of the growing annuity model described in Chapter 1. We have assumed that—after the year-5 projection horizon—the cash flows will grow at a long-term rate of growth of 3%. This gives the terminal value as:

$$\textit{Terminal Value at period } 4.5 = \frac{FCF_5 * (1 + \textit{long-term FCF growth})}{WACC - \textit{long-term FCF growth}}$$

As noted in the previous section, this formula is only valid if the long-term FCF growth is less than the WACC.

There are other ways of calculating the terminal value. All of the following are common variations that can be implemented in the framework of our model (see the end-of-chapter exercises):

- • Terminal value = Year-5 book value of debt + Equity

This calculation assumes that the book value correctly predicts the market value.

- • Terminal value = (Enterprise market/book multiple) * (Year-5 book value of debt + Equity)
- • Terminal value = P/E multiple * Year-5 profits + Year-5 book value of debt
- • Terminal value = EBITDA multiple * Year-5 anticipated EBITDA
  (EBITDA = Earnings before interest, taxes, depreciation, and amortization)

The Treatment of Cash and Marketable Securities in the Valuation

We have added the initial cash balances back to the present value of the projected FCFs to get the enterprise value. This procedure assumes the following:

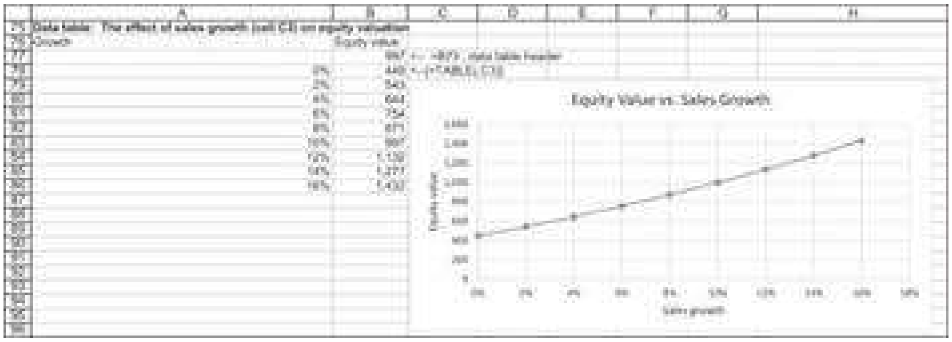
- • Year-0 balances of cash and marketable securities are not needed to produce the FCFs in subsequent years. In other words, the cash and marketable securities are excess financial assets.
- • Year-0 balances of cash and marketable securities are "surpluses" that could be drawn down or paid out by shareholders without affecting the future economic performance of the firm.

Mid-Year Discounting

As explained above, for discounting purposes, we should therefore discount cash flows as if, on average, they occur in the middle of the year.

Sensitivity Analysis

As in any Excel model, we can perform extensive sensitivity analysis on our valuation. Taking the example in this section above as our base case, we can ask, for example, what is the effect of the sales growth rate on the equity value of the firm:



Cells B77:B86 contain a data table (see Chapter 28 if you are unsure of how to construct these tables).

Another variation is to calculate the effect on equity valuation of both the long-term FCF growth and the WACC. Here, however, you have to be careful: Examining the terminal value equation $Terminal\ Value_{t=4.5} = \dfrac{FCF_5 * (1 + g_{LT})}{WACC - g_{LT}}$ will show you that this calculation only makes sense if the WACC is greater than the long-term growth rate.[4] To overcome this problem, we define the data table cell A99 (this is the calculation on which the data table does its sensitivity analysis) in the following way:

	A	B	C	D	E	F	G
97	The effect of long-term growth rate and WACC on equity valuation						
98			Data table header, =IF(B60<=B59,"NA",B73)				
99	997	8%	10%	12%	16%	18%	20%
100	0%	1,276	970	767	513	429	362
101	2%	1,669	1,191	905	579	477	398
102	3%	1,983	1,349	997	619	506	419
103	5%	3,239	1,854	1,260	721	577	470
104	7%	9,518	3,031	1,734	870	673	537
105	9%	NA	8,920	2,848	1,102	813	628
106	10%	NA	NA	4,222	1,277	909	688
107	12%	NA	NA	NA	1,888	1,197	850
108	14%	NA	NA	NA	3,722	1,772	1,122
109		Note: Data tables are discussed in Chapter 28.					

4.6 Setting the Debt to Be the Absorbing Item and Incorporating Target Debt/Equity Ratio into the Pro Forma

Another change we might want to make in our model relates to the absorbing item. Suppose that we set the debt as the absorbing item and that the firm has a target ratio of debt to equity. In each of the years 1–5, it wants the ratio of debt/equity on the balance sheet to conform to a certain ratio. This situation is illustrated in the following example:

	A	B	C	D	E	F	G	H
1			TARGET DEBT/EQUITY RATIO Cash is fixed, ratio of debt/equity changes in each year					
2		0 actual	1 assumption	2 assumption	3 assumption	4 assumption	5 assumption	
3	Sales growth		10%	10%	10%	10%	10%	
4	Costs of goods sold/Sales	60%	50%	50%	50%	50%	50%	
5	Interest rate on debt	10%	10%	10%	10%	10%	10%	
6	Annual growth rate in SG&A		3%	3%	3%	3%	3%	
7	Current assets/Sales	15%	15%	15%	15%	15%	15%	
8	Current liabilities/Sales	8%	8%	8%	8%	8%	8%	
9	Net fixed assets/Sales	77%	77%	77%	77%	77%	77%	
10	Depreciation rate	13%	13%	13%	13%	13%	13%	
11	Interest earned on cash and marketable securities	8%	8%	8%	8%	8%	8%	
12	Tax rate	35%	35%	35%	35%	35%	35%	
13	Dividend payout ratio	35%	35%	35%	35%	35%	35%	
14	D/E ratio	53%	45%	40%	30%	30%	30%	
15		Initial (year 0) debt/equity ratio =B43/SUM(B44:B45)						
16	**Year**	0	1	2	3	4	5	
17	**Income statement**							
18	Sales	1,000	1,100	1,210	1,331	1,464	1,611	<-- =F18*(1+G3)
19	Costs of goods sold	(600)	(550)	(605)	(666)	(732)	(805)	<-- =-G18*G4
20	**Gross profit**	**400**	**550**	**605**	**666**	**732**	**805**	<-- =SUM(G18:G19)
21	Gross profit %	40%	50%	50%	50%	50%	50%	<-- =G20/G18
22	Selling, general and administrative	(100)	(103)	(106)	(109)	(113)	(116)	<-- =F22*(1+G6)
23	Interest payments on debt	(32)	(32)	(31)	(29)	(29)	(32)	<-- =-G5*AVERAGE(F43:G43)
24	Interest earned on cash and marketable securities	6.4	6.4	6.4	6.4	6.4	6.4	<-- =G11*AVERAGE(F34:G34)
25	Depreciation	(100)	(110)	(121)	(133)	(146)	(161)	<-- =-G10*G39
26	**Profit before tax**	**174**	**312**	**353**	**400**	**451**	**503**	**<-- =G20+SUM(G22:G25)**
27	Taxes	(61)	(109)	(124)	(140)	(158)	(176)	<-- =-G26*G12
28	**Profit after tax (net profit)**	**113**	**203**	**229**	**260**	**293**	**327**	**<-- =G27+G26**
29	Net profit %	11%	18%	19%	20%	20%	20%	<-- =G28/G18
30	Dividends	(40)	(72)	(81)	(92)	(103)	(115)	<-- =-G13*G28
31	**Retained earnings**	**73**	**131**	**149**	**169**	**190**	**212**	**<-- =G30+G28**
32								
33	**Balance sheet**							
34	Cash and marketable securities	80	80	80	80	80	80	<-- =F34
35	Current assets	150	165	182	200	220	242	<-- =G18*G7
36	Fixed assets							
37	At cost	1,070	1,257	1,463	1,689	1,938	2,212	<-- =G39-G38
38	Accumulated depreciation	(300)	(410)	(531)	(664)	(811)	(972)	<-- =F38+G25
39	Net fixed assets	770	847	932	1,025	1,127	1,240	<-- =G9*G18
40	**Total assets**	**1,000**	**1,092**	**1,193**	**1,305**	**1,427**	**1,562**	**<-- =G39+G35+G34**
41								
42	Current liabilities	80	88	97	106	117	129	<-- =G18*G8
43	Debt	325	312	313	276	302	331	<-- =G14*SUM(G44:G45)
44	Stock	450	411	353	324	229	103	<-- =G40-G42-G43-G45
45	Accumulated retained earnings	150	281	430	598	787	999	<-- =F45+G31
46	**Total liabilities and equity**	**1,000**	**1,092**	**1,193**	**1,305**	**1,427**	**1,562**	**<-- =SUM(G42:G45)**

Row 14 of the spreadsheet shows the target debt/equity (D/E) ratio in each of years 1 to 5. The firm wants to lower its current D/E ratio of 53% to 30% over the next three years. The relevant changes to the equations of our initial model are the following:

- • Debt = **Target debt/equity ratio** $*$ (Stock + Retained earnings)
- • Stock = Total assets − Current liabilities − Debt − Accumulated retained earnings

Note that the firm will issue new debt in years 4 and 5; in years 1 to 5 the stock account decreases, which is indicating a repurchase of equity.

4.7 Calculating the Return on Invested Capital

A good consistency check used by many practitioners is to calculate the return on invested capital (ROIC). The ROIC is defined as

$$ROIC_t = \frac{NOPAT_t}{Invested\ capital_{t-1}},$$

where

- • NOPAT is the net operating profit less adjusted tax. As explained before, this can be calculated directly by presenting the income statement without finance expenses (or income) or indirectly by:

$$NOPAT = Net\ profit + Interest\ expenses * (1 - Tax)$$

- • The invested capital is defined as:

$$Invested\ capital = Net\ fixed\ Assets + Net\ working\ capital + Intangible\ assets$$

The ROIC is telling you how the firm is using its operational assets. In most cases it should not fluctuate significantly over time. This is the calculation of ROIC in our example above:

	A	B	C	D	E	F	G	H
[illegible]	Calculating the return on invested capital (ROIC)							
[illegible]		0 actual	1 assumption	2 assumption	3 assumption	4 assumption	5 assumption	
[illegible]	Net operating profit after tax (NOPAT)	130	218	248	275	308	343	[illegible]
[illegible]	Invested capital	840	924	1,016	1,118	1,230	1,353	[illegible]
[illegible]	Return on invested capital (ROIC)		[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]

In our model, the ROIC is about 26%–28%.

4.8 Project Finance: Debt Repayment Schedules

Here is another use for pro forma modeling: In a typical case of "project finance," the firm borrows money in order to finance a project. The borrowing often comes with strings attached:

- • The firm is not allowed to pay any dividends until the debt is paid off.
- • The firm is not allowed to issue any new equity.
- • The firm must pay back the debt over a specified period.

The following simplified example assumes that the absorbing item is the cash balances. A new firm or project is set up. In year 0:

- • The firm has assets of 2,200, which are financed with 100 of current liabilities, 1,100 of equity, and 1,000 of debt.
- • The debt must be paid off in equal installments of principal over the next five years. Until the debt is paid off, the firm is not allowed to pay dividends (if there is extra cash, it will go into a cash and marketable securities account).

	A	B	C	D	E	F	G	H
1	PROJECT FINANCE No dividends, debt repayment schedule fixed, net fixed assets constant							
2		0 actual	1 assumption	2 assumption	3 assumption	4 assumption	5 assumption	
3	Sales growth		15%	15%	15%	15%	15%	
4	Costs of goods sold/Sales		45%	45%	45%	45%	45%	
5	Interest rate on debt		10%	10%	10%	10%	10%	
6	Annual growth rate in SG&A			3%	3%	3%	3%	
7	Current assets/Sales		15%	15%	15%	15%	15%	
8	Current liabilities/Sales		8%	8%	8%	8%	8%	
9	Net fixed assets	2,000	2,000	2,000	2,000	2,000	2,000	
10	Depreciation rate		10%	10%	10%	10%	10%	
11	Interest earned on cash and equiv.		8%	8%	8%	8%	8%	
12	Tax rate		35%	35%	35%	35%	35%	
13	Dividend payout ratio		0%	0%	0%	0%	0%	<-- No dividends until all the debt is paid off
14	Debt repayment		200	200	200	200	200	<-- debt is paid over 5 years
15								
16	**Year**	0	1	2	3	4	5	
17	**Income statement**							
18	Sales		1,150	1,323	1,521	1,749	2,011	<--=F18*(1+G3)
19	Costs of goods sold		(518)	(595)	(684)	(787)	(905)	<--=-G18*G4
20	**Gross profit**		**633**	**727**	**836**	**962**	**1,106**	<--=SUM(G18:G19)
21	Gross profit %		55%	55%	55%	55%	55%	<--=G20/G18
22	Selling, general and administrative		(50)	(52)	(53)	(55)	(56)	<--=F22*(1+G6)
23	Interest payments on debt		(90)	(70)	(50)	(30)	(10)	<--=-G5*AVERAGE(F43:G43)
24	Interest earned on cash & marketable securities		0.4	2.9	10.7	25.9	50.0	<--=G11*AVERAGE(F34:G34)
25	Depreciation		(200)	(200)	(200)	(200)	(200)	<--=-G10*G39
26	**Profit before tax**		**293**	**409**	**544**	**703**	**890**	**<--=G20+SUM(G22:G25)**
27	Taxes		(103)	(143)	(190)	(246)	(311)	<--=-G26*G12
28	**Profit after tax (net profit)**		**190**	**266**	**354**	**457**	**578**	**<--=G27+G26**
29	Net profit %		17%	20%	23%	26%	29%	<--=G28/G18
30	Dividends		-	-	-	-	-	<--=-G13*G28
31	**Retained earnings**		**190**	**266**	**354**	**457**	**578**	**<--=G30+G28**
32								
33	**Balance sheet**							
34	Cash and marketable securities	-	10	64	203	444	805	<--=G46-G39-G35
35	Current assets	200	173	198	228	262	302	<--=G18*G7
36	Fixed assets							
37	At cost	2,000	2,200	2,400	2,600	2,800	3,000	<--=G39-G38
38	Accumulated depreciation	-	(200)	(400)	(600)	(800)	(1,000)	<--=F38+G25
39	Net fixed assets	2,000	2,000	2,000	2,000	2,000	2,000	<--=G9
40	**Total assets**	**2,200**	**2,182**	**2,262**	**2,431**	**2,707**	**3,106**	**<--=G39+G35+G34**
41								
42	Current liabilities	100	92	106	122	140	161	<--=G18*G8
43	Debt	1,000	800	600	400	200	-	<--=F43-G14
44	Stock	1,100	1,100	1,100	1,100	1,100	1,100	<--=F44
45	Accumulated retained earnings	-	190	456	810	1,267	1,845	<--=F45+G31
46	**Total liabilities and equity**	**2,200**	**2,182**	**2,262**	**2,431**	**2,707**	**3,106**	**<--=SUM(G42:G45)**

The debt repayment terms are incorporated into the model by simply specifying the debt balances at the end of each year. Since the firm is assumed to issue no new equity (in accordance with the covenants on the lending), it follows that the model's absorbing item cannot be on the liabilities side of the balance sheet. In our model the absorbing item is the cash and marketable securities account.

The model incorporates one other assumption often made about fixed assets: It assumes that the *net* fixed assets stay constant over the life of the project. Essentially, this means that the depreciation accurately reflects the capital maintenance of the fixed assets. As you can see from looking at rows 37–39 above, this means that the fixed assets at cost grow each year by the increase in asset depreciation. It also means that there is no net cash flow from depreciation:

	A	B	C	D	E	F	G	H
48	**Year**	0	1	2	3	4	5	
49	**Free cash flow calculation**							
50	Profit after tax		190	266	354	457	578	<-- =G28
51	Add back depreciation		200	200	200	200	200	<-- =-G25
52	Subtract increase in current assets		28	(26)	(30)	(34)	(39)	<-- Cash flow generated by depreciation equals capital expenditures
53	Add back increase in current liabilities		(8)	14	16	18	21	<--
54	Subtract increase in fixed assets at cost		(200)	(200)	(200)	(200)	(200)	<--
55	Add back after-tax interest on debt		59	46	33	20	7	<-- =-G23*(1-G12)
56	Subtract after-tax interest on cash + equiv.		(0)	(2)	(7)	(17)	(32)	<-- =-G24*(1-G12)
57	**Free cash flow**		**268**	**297**	**365**	**444**	**534**	<-- =SUM(G50:G56)

In this example the firm has no problem in making its debt principal repayments. As credit analysts, we might be interested in how the firm's ability to meet its payments is affected by the various parameter values.

4.9 Calculating the Return on Equity

Practitioners define return on equity (ROE) as:

$$ROE_t = \frac{Net\ Income_t}{Average(Equity_{t-1}, Equity_t)}$$

In our example above, we get an average annual ROE of 19.6%:

	A	B	C	D	E	F	G	H
60	**RETURN ON EQUITY (ROE)**							
61	**Year**	**0**	**1**	**2**	**3**	**4**	**5**	
62	Profit after tax (net profit)		190	266	354	457	578	<--=G28
63	Equity	1,100	1,290	1,556	1,910	2,367	2,945	<--=SUM(G44:G45)
64	**Return on equity**		**16%**	**19%**	**20%**	**21%**	**22%**	<--=G62/AVERAGE(F63:G63)
65	**Average ROE over the period**	**19.6%**	<--=AVERAGE(C64:G64)					

However, this is not the actual effective annual return for equity holders. We can use the pro forma models illustrated in this chapter to compute the anticipated return on equity. Look at the previous example: Equity owners in the project must pay 1,100 in year 0. During years 1–4 they get no payoffs, but in year 5 they own the company. Suppose that the book value of the assets accurately reflects the market value. Then at the end of year 5 the equity in the firm is worth stock + accumulated retained earnings = 2,945. The effective annual return on the equity investment is calculated as follows:

	A	B	C	D	E	F	G	H
67	**EFFECTIVE ANNUAL RETURN ON EQUITY**							
68	**Year**	**0**	**1**	**2**	**3**	**4**	**5**	
69	Equity cash flow	-1,100	-	-	-	-	2,945	<-- =G23+G37+G38
70	**Effective Return on Equity**	**21.8%**	<-- =IRR(B69:G69)					

Note that this equity return increases as the equity investment decreases.[5] Consider the case where the firm initially borrows 1,500 as debt and the equity owners invest 600 (change cell B43 in our original project finance spreadsheet above to 1,500):

	A	B	C	D	E	F	G	H
67	**EFFECTIVE ANNUAL RETURN ON EQUITY**							
68	**Year**	**0**	**1**	**2**	**3**	**4**	**5**	
69	Equity cash flow	-600	-	-	-	-	2,277	<-- =G23+G37+G38
70	**Effective Return on Equity**	**30.8%**	<-- =IRR(B69:G69)					

As the following data table and graph show, the less the initial equity investment, the greater the equity return:

	A	B
72	**Data table: ROE as a function of initial**	
73	**Equity investment**	**Effective return on equity**
74		21.8% <--Data table header, =B70
75	2,000	15.3%
76	1,800	16.3%
77	1,600	17.5%
78	1,400	18.9%
79	1,200	20.7%
80	1,000	23.0%
81	800	26.3%
82	600	31.0%
83	400	38.8%
84	200	55.3%

4.10 Tax Loss Carryforwards

Firms can apply accumulated losses to reduce their current tax liabilities. In this section we show how to model such tax loss carryforwards in our pro forma model. We make several changes in our basic model, highlighted below:

- • We assume that the cost of goods sold are 55% of sales.
- • We assume that the selling, general and administrative (SG&A) cost last year was 350.

	A	B	C	D	E	F	G	H
1	MODELING TAX LOSS CARRYFORWARDS							
2		0 actual	1 assumption	2 assumption	3 assumption	4 assumption	5 assumption	
3	Sales growth		10%	10%	10%	10%	10%	
4	Costs of goods sold/Sales	55%	55%	55%	55%	55%	55%	
5	Interest rate on debt	10%	10%	10%	10%	10%	10%	
6	Annual growth rate in SG&A		3%	3%	3%	3%	3%	
7	Current assets/Sales	15%	15%	15%	15%	15%	15%	
8	Current liabilities/Sales	8%	8%	8%	8%	8%	8%	
9	Net fixed assets/Sales	77%			Absorbing item			
10	Depreciation rate	13%	13%	13%	13%	13%	13%	
11	Interest earned on cash and marketable securities	8%	8%	8%	8%	8%	8%	
12	Tax rate	35%	35%	35%	35%	35%	35%	
13	Dividend payout ratio	-53%	-53%	-53%	-53%	-53%	-53%	
14								
15	Year	0	1	2	3	4	5	
16	Income statement							
17	Sales	1,000	1,100	1,210	1,331	1,464	1,611	<--=F17*(1+G3)
18	Costs of goods sold	(550)	(605)	(666)	(732)	(805)	(886)	<--=-G17*G4
19	Gross profit	450	495	545	599	659	725	<--=SUM(G17:G18)
20	Gross profit %	45%	45%	45%	45%	45%	45%	<--=G19/G17
21	Selling, general and administrative	(350)	(361)	(371)	(382)	(394)	(406)	<--=F21*(1+G6)
22	Interest payments on debt	(32)	(32)	(32)	(32)	(32)	(32)	<--=-G5*AVERAGE(F46:G46)
23	Interest earned on cash and marketable securities	6.4	6.4	6.4	6.4	6.4	6.4	<--=G11*AVERAGE(F37:G37)
24	Depreciation	(100)	(101)	(108)	(118)	(131)	(148)	<--=-G10*G42
25	Profit before tax	(76)	8	40	73	109	145	<--=G19+SUM(G21:G24)
26	Taxes	-	-	-	(16)	(38)	(51)	<--=-MAX(G12*G33,0)
27	Profit after tax (net profit)	(76)	8	40	57	71	94	<--=G26+G25
28	Net profit %	-8%	1%	3%	4%	5%	6%	<--=G27/G17
29	Dividends	(40)	4	21	30	37	50	<--=-G13*G27
30	Retained earnings	(116)	12	61	87	108	144	<--=G29+G27
31								
32	Tax loss carryforward from previous years	-	(76)	(67)	(27)	-	-	<-- =MIN(F32+F25,0)
33	Taxable profit	0	0	0	46	109	145	<-- =MAX(G25+G32,0)
34	Effective tax rate	0%	0%	0%	22%	35%	35%	<-- =-G26/G25
35								
36	Balance sheet							
37	Cash and marketable securities	80	80	80	80	80	80	<--=F37
38	Current assets	150	165	182	200	220	242	<--=G17*G7
39	Fixed assets							
40	At cost	1,070	1,176	1,337	1,534	1,763	2,046	<--=G42-G41
41	Accumulated depreciation	(300)	(401)	(508)	(626)	(757)	(905)	<--=F41+G24
42	Net fixed assets	770	776	829	908	1,006	1,140	<--=G49-SUM(G37:G38)
43	Total assets	1,000	1,021	1,090	1,187	1,306	1,462	<--=G42+G38+G37
44								
45	Current liabilities	80	88	97	106	117	129	<--=G17*G8
46	Debt	320	320	320	320	320	320	<--=F46
47	Stock	450	450	450	450	450	450	<--=F47
48	Accumulated retained earnings	150	163	224	311	419	563	<--=F48+G30
49	Total liabilities and equity	1,000	1,021	1,090	1,187	1,306	1,462	<--=SUM(G45:G48)

In each year we model the accumulated tax loss (row 32). If the previous year exhibited a loss, then the accumulated loss increased by that amount. If the previous year exhibits a gain (year 2, with a gain of 40), then the accumulated tax loss carryforward gets closer to zero (cell E32). At some point (year 4 in our example), tax losses may be completely used up.

In any year, the taxable profits are given in row 33. If the accumulated tax loss carryforward is greater than that year's profit, then there is no tax. Otherwise only the difference in income is taxed (consider year 4, for example). These differences are reflected in the effective tax rate in row 34.

The effective tax rate is used in computing the free cash flows, as shown below:

	A	B	C	D	E	F	G	H
51	Year	0	1	2	3	4	5	
52	Free cash flow calculation							
53	Profit after tax		8	40	57	71	94	<--=G27
54	Add back depreciation		101	108	118	131	148	<--=-G24
55	Subtract increase in current assets		(15)	(17)	(18)	(20)	(22)	<--=-(G38-F38)
56	Add back increase in current liabilities		8	9	10	11	12	<--=G45-F45
57	Subtract increase in fixed assets at cost		(106)	(161)	(197)	(229)	(282)	<--=-(G40-F40)
58	Add back after-tax interest on debt		32	32	26	21	21	<--=-G22*(1-G34)
59	Subtract after-tax interest on cash and mkt. securities		(6)	(6)	(5)	(4)	(4)	<--=-G23*(1-G34)
60	Free cash flow		21	4	(18)	(21)	(33)	<--=SUM(G53:G59)

4.11 Conclusion

Pro forma modeling is one of the basic skills of corporate financial analysis, a devious combination of finance, the implementation of accounting rules, and spreadsheet skills. In order to be useful, financial models must match the situation at hand, but they must also be simple enough so that the user can easily understand *why* the results happen (be they valuations, creditworthiness, or simply commonsense predictions of how a firm or project might look several years down the road).

Exercises

1. On the website that accompanies this book, you'll find information for Kellogg. Use this information and the template in section 4.1 to value Kellogg.
2. Choose a company with the following characteristics:

   - It is listed in the United States.
   - It has positive operating cash flows.
   - It has positive net debts.

   Download the balance sheet, income statement, and the cash flow statement for the chosen company. We recommend using the U.S. Securities and Exchange Commission (SEC) website called EDGAR to pull the latest 10-K filling: https://www.sec.gov/edgar/searchedgar/companysearch.html.

   For the company you chose, please calculate the following values (list your assumptions):

   1. The economic balance sheet (or operational balance sheet).
   2. The enterprise value using the efficient markets assumption.
   3. The weighted average cost of capital (WACC). Explicitly write down your assumptions.
   4. Use the WACC to calculate the value per share using the simplified valuation approach.
   5. Use the WACC to calculate the value per share using the "full pro forma analysis" approach.

   Note: On the companion website of this chapter you can find the author's suggestion for Amazon as of December 2019. Also, Chapter 5 illustrates the analysis for Merck.

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1. [1]. The current example has no such items.
2. [2]. We have skipped many of the details, which are available in the Excel file for this chapter.
3. [3]. As noted in Chapter 3, cash can often be considered negative debt and vice versa. We return to this point in section 4.5.
4. [4]. If the $growth\ rate \geq WACC$, then $Terminal\ Value \rightarrow \infty$. Thus, the WACC puts an effective bound on the long-term growth rate.
5. [5]. This is an interesting but unsurprising observation: As the equity investment goes down, the project becomes more leveraged and hence riskier for the equity investors. The increased return should compensate the equity holders for this extra risk. The really interesting question (not answered here) is whether the increased return is in fact a compensation for the riskiness.

5 Building a Pro Forma Model: The Case of Merck

5.1 Overview

In this chapter we implement the pro forma modeling techniques discussed in Chapter 4 to build a financial model for Merck & Co., Inc. (MRK). Merck, as most readers of this book will surely know, is a global healthcare company (known outside North America as “MSD”). The company manufactures and sells prescription medicines, vaccines, biologic therapies, and animal health products. The company’s operations can be categorized into four operating segments: Pharmaceutical, Animal Health, Healthcare Services, and Alliances.

Our efforts in this chapter will be to understand Merck’s financial statements for the years 2015–2018 in order to fit them into the format illustrated in Chapter 4. This is not a trivial task, and it requires an understanding of Merck’s business in combination with modeling skills and a small dose of financial artfulness (otherwise we would never get to the end of the exercise).

Warning: This Chapter May Be Deleterious to Your Mental Health[1]

The material in this chapter is *irritating* and *complicated* (and sometimes *tedious*) but *not difficult*. Why irritating? Because pro forma financial models require a lot of assumptions and analysis. Everything is related to almost everything else. And because pro forma models require you to recall some basic accounting concepts.

However, the case in this chapter illustrates one of the most important corporate finance applications: the valuation of a company in the framework of its accounting and financial parameters. Such valuations are the core of most business plans, corporate financial planning models, and (intelligent) analyst valuations.

5.2 Merck’s Financial Statements, 2015–2018

Merck’s financial statements for the four years from 2015 to 2018 are given below:

	A	B	C	D	E
1	**MERCK & CO. STATEMENT OF INCOME, YEARS ENDED DECEMBER 31** **($ IN MILLIONS EXCEPT PER SHARE AMOUNTS)**				
2		2015	2016	2017	2018
3	Sales	39,498	39,807	40,122	42,294
4	Cost of sales	14,934	14,030	12,912	13,509
5	**Gross profit**	**24,564**	**25,777**	**27,210**	**28,785**
6	Other costs, expenses and other				
7	Selling, general and administrative	10,313	10,017	10,074	10,102
8	Research and development	6,704	10,261	10,339	9,752
9	Restructuring costs	619	651	776	632
10	Interest income	-289	-328	-385	-343
11	Interest expense	672	693	754	772
12	Other (income) expense, net	1,144	-176	-869	-831
13	**Total other costs, expenses and other**	**19,163**	**21,118**	**20,689**	**20,084**
14	**Income before taxes**	**5,401**	**4,659**	**6,521**	**8,701**
15	Taxes on income	942	718	4,103	2,508
16	**Net income**	**4,459**	**3,941**	**2,418**	**6,193**
17	Less net (Loss) income attributable to noncontrolling interests	17	21	24	-27
18	**Net income attributable to Merck & Co., Inc. shareholders**	**4,442**	**3,920**	**2,394**	**6,220**
19	Earnings per common share attributable to common shareholders	1.56	1.41	0.87	2.32

	A	B	C	D	E
1	**MERCK CONSOLIDATED BALANCE SHEET, DECEMBER 31** **($ IN MILLIONS EXCEPT PER SHARE AMOUNTS)**				
2		2015	2016	2017	2018
3	**Assets**				
4	Current assets				
5	Cash and cash equivalents	8,524	6,515	6,092	7,965
6	Short-term investments	4,903	7,826	2,406	899
7	Accounts receivable	6,484	7,018	6,873	7,071
8	Inventories	4,700	4,866	5,096	5,440
9	Other current assets	5,140	4,389	4,299	4,500
10	**Total current assets**	**29,751**	**30,614**	**24,766**	**25,875**
11	Investments	13,039	11,416	12,125	6,233
12	Property, plant and equipment (at cost)	28,430	27,775	29,041	29,615
13	Less accumulated depreciation	15,923	15,749	16,602	16,324
14	**Net PP&E**	**12,507**	**12,026**	**12,439**	**13,291**
15	Goodwill	17,723	18,162	18,284	18,253
16	Other intangibles, net	22,602	17,305	14,183	11,431
17	Other assets	6,055	5,854	6,075	7,554
18	**Total assets**	**101,677**	**95,377**	**87,872**	**82,637**
19					
20	**Liabilities and equity**				
21	Current liabilities				
22	Loans payable and current portion of long-term debt	2,583	568	3,057	5,308
23	Trade accounts payable	2,533	2,807	3,102	3,318
24	Accrued and other current liabilities	11,216	10,274	10,427	10,151
25	Income taxes payable	1,560	2,239	708	1,971
26	Dividends payable	1,309	1,316	1,320	1,458
27	**Total current liabilities**	**19,201**	**17,204**	**18,614**	**22,206**
28	Long-term debt	23,829	24,274	21,353	19,806
29	Deferred income taxes	6,535	5,077	2,219	1,702
30	Other noncurrent liabilities	7,345	8,514	11,117	12,041
31	**Total liabilities**	**56,910**	**55,069**	**53,303**	**55,755**
32					
33	**Merck & Co., Inc. stockholders' equity**				
34	Common stock	1,788	1,788	1,788	1,788
35	Other paid-in capital	40,222	39,939	39,902	38,808
36	Retained earnings	45,348	44,133	41,350	42,579
37	Accumulated other comprehensive loss	-4,148	-5,226	-4,910	-5,545
38	Less treasury stock, at cost	-38,534	-40,546	-43,794	-50,929
39	**Total Merck & Co., Inc. stockholders' equity**	**44,676**	**40,088**	**34,336**	**26,701**
40	Noncontrolling interests	91	220	233	181
41	**Total equity**	**44,767**	**40,308**	**34,569**	**26,882**
42	**Total liabilities and equity**	**101,677**	**95,377**	**87,872**	**82,637**
43	**Check (assets = liabilities + equity)?**	TRUE	TRUE	TRUE	TRUE

	A	B	C	D	E
1	**MERCK CONSOLIDATED STATEMENT OF CASH FLOWS, YEARS ENDED DECEMBER 31 ($ IN MILLIONS)**				
2		2015	2016	2017	2018
3	**Cash flows from operating activities**				
4	Net income	4,459	3,941	2,418	6,193
5	Adjustments to reconcile net income to net cash provided by operating activities:				
6	Depreciation and amortization	6,375	5,471	4,676	4,519
7	Other	1,972	3,565	4,374	1,763
8	Net changes in assets and liabilities:				
9	Accounts receivable	-480	-619	297	-418
10	Inventories	805	206	-145	-911
11	Trade accounts payable	-37	278	254	230
12	Accrued and other current liabilities	-8	-2,018	-922	-341
13	Income taxes payable	-266	124	-3,291	827
14	Noncurrent liabilities	-277	-809	-123	-266
15	Other	-5	237	-1,087	-674
16	**Net cash provided by operating activities**	**12,538**	**10,376**	**6,451**	**10,922**
17					
18	**Cash flows from investing activities**				
19	Capital expenditures	-1,283	-1,614	-1,888	-2,615
20	Purchases of securities and other investments	-16,681	-15,651	-10,739	-7,994
21	Proceeds from sales of securities and other investments	20,413	14,353	15,664	15,252
22	Acquisitions, net of cash acquired	-7,428	-780	-396	-431
23	Other	221	482	38	102
24	**Net cash provided by (used in) investing activities**	**-4,758**	**-3,210**	**2,679**	**4,314**
25					
26	**Cash flows from financing activities**				
27	Net change in short-term borrowings	-1,540	0	-26	5,124
28	Payments on debt	-2,906	-2,386	-1,103	-4,287
29	Proceeds from issuance of debt	7,938	1,079	0	0
30	Purchases of Treasury stock	-4,186	-3,434	-4,014	-9,091
31	Dividends paid to stockholders	-5,117	-5,124	-5,167	-5,172
32	Proceeds from exercise of stock options	485	939	499	591
33	Other	-61	-118	-195	-325
34	**Net cash used in financing activities**	**-5,387**	**-9,044**	**-10,006**	**-13,160**
35	Effect of exchange rate	-1310	-131	457	-205
36	**Net increase (decrease) in cash, cash equivalents**	**1,083**	**-2,009**	**-419**	**1,871**

Rewriting the Balance Sheet

The first step in analyzing any firm is to convert its accounting balance sheet to an economic balance sheet. This is done by rewriting the balance sheets to combine the short-term and long-term financial debt items. We also write all the operating (i.e., nonfinancial) current assets and operating current liabilities as one item.

	H	I	J	K	L	M
1	**MERCK OPERATIONAL BALANCE SHEET, DECEMBER 31 ($ IN MILLIONS EXCEPT PER SHARE AMOUNTS)**					
2		2015	2016	2017	2018	
3	**Operational assets**					
4	Fixed assets	25,546	23,442	24,564	19,524	<--=E14+E11
5	Net working capital	12,231	11,227	12,458	11,722	<--=E7+E8+E9-E23-E25
6	Intangible assets	40,325	35,467	32,467	29,684	<--=E15+E16
7	**Total operational assets**	**78,102**	**70,136**	**69,489**	**60,930**	**<--=SUM(L4:L6)**
8	Excess assets	6,055	5,854	6,075	7,554	<--=E17
9	**Total assets**	**84,157**	**75,990**	**75,564**	**68,484**	**<--=SUM(L7:L8)**
10						
11	**Net financial debt**					
12	Interest bearing liabilities	51,599	48,927	48,406	49,189	<--=SUM(E22,E24,E28,E29,E30,E40)
13	Less cash and equivalents	-13,427	-14,341	-8,498	-8,864	<--=-(E5+E6)
14	**Net financial debt**	**38,172**	**34,586**	**39,908**	**40,325**	**<--=SUM(L12:L13)**
15						
16	**Equity**	**45,985**	**41,404**	**35,656**	**28,159**	**<--=L9-L14**
17	Check:	TRUE	TRUE	TRUE	TRUE	<--=L16=(E39+E26)

5.3 Analyzing the Financial Statements

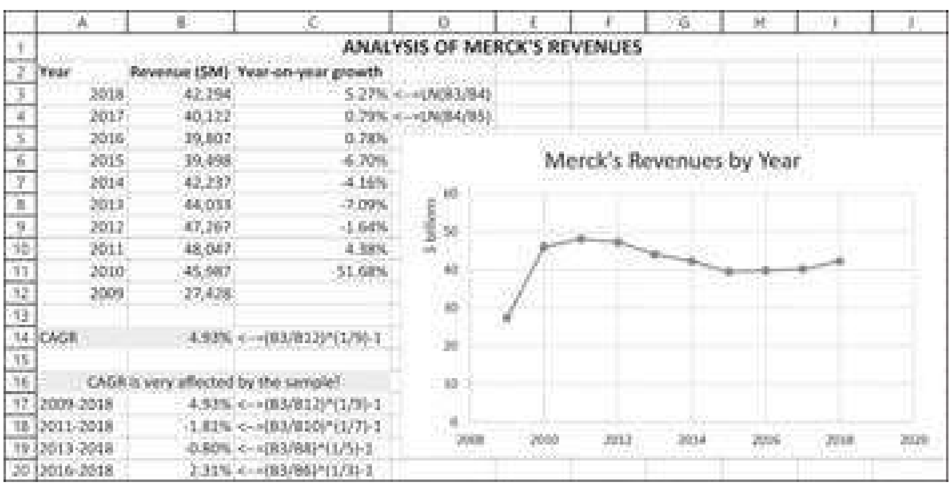
We now build a set of financial statements with the technology of Chapter 4 but in roughly the same format as Merck's historical statements. Our model is forward looking but based on Merck's historical financial statements. Our choice of model parameters is based on our analysis of the historical performance and on a number of judgment calls.

Sales Projections

The sales projection is one of the most critical elements in our model. Should it be based on a historical analysis of Merck sales? Should it be based on expert opinion about the potential growth of the pharmaceutical industry? Should we differentiate between short-term (next five years) growth and long-term growth?

None of these questions has a satisfactory answer. We illustrate several methods and suggest that you use the pro forma model to experiment with the various answers to see whether they have a value impact. In the Excel screenshot below we show:

- Year-on-year sales growth
- Cumulative average growth rates (CAGR)



	A	B	C
1	ANALYSIS OF MERCK'S REVENUES		
2	Year	Revenue ($M)	Year-on-year growth
3	2018	42,294	5.27% <--=LN(B3/B4)
4	2017	40,122	0.79% <--=LN(B4/B5)
5	2016	39,807	0.78%
6	2015	39,498	-6.70%
7	2014	42,237	-4.16%
8	2013	44,033	-7.09%
9	2012	47,267	-1.64%
10	2011	48,047	4.38%
11	2010	45,987	51.68%
12	2009	27,428	
13			
14	CAGR	4.93%	<--=(B3/B12)^(1/9)-1
15			
16	CAGR is very affected by the sample!		
17	2009-2018	4.93%	<--=(B3/B12)^(1/9)-1
18	2011-2018	-1.81%	<--=(B3/B10)^(1/7)-1
19	2013-2018	-0.80%	<--=(B3/B8)^(1/5)-1
20	2016-2018	2.31%	<--=(B3/B6)^(1/3)-1

The CAGR in cell B14 is 4.93%, which is calculated as follows:

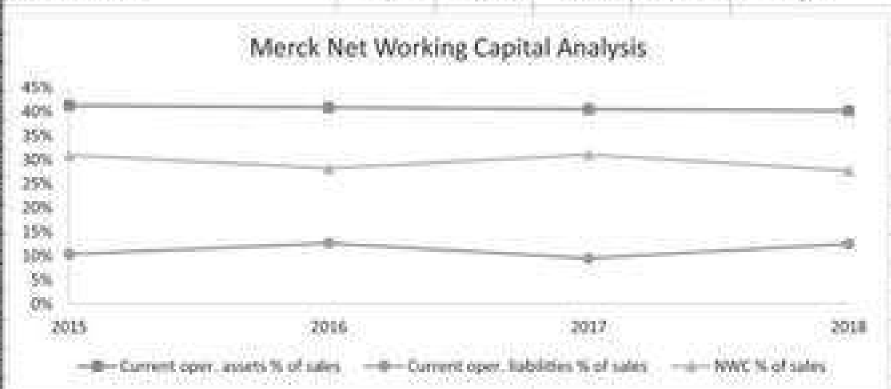
$$CAGR = \left(\frac{Sales_{2018}}{Sales_{2009}}\right)^{\left(\frac{1}{9}\right)} - 1 = 4.93\%$$

CAGR is often used, but because of its dependence on the sample, it is a problematic number. We find it a suspect measure of long-term growth unless verified by other time spans. As the data show, the year-on-year growth of Merck exhibits a lot of variability. In the Excel above we have played with the samples and find that the CAGR can vary from −1.81% to 4.93%.

Current Assets and Current Liabilities

The ratio of current assets to sales and current liabilities to sales shows a lot of stability over the four years. As model parameter values, we have chosen the average of the four years.

	A	B	C	D	E	F
1	**MERCK WORKING CAPITAL ANALYSIS**					
2		2015	2016	2017	2018	
3	Sales	39,498	39,807	40,122	42,294	
4						
5	**Current operational assets**					
6	Accounts receivable	6,484	7,018	6,873	7,071	
7	Inventories	4,700	4,866	5,096	5,440	
8	Other current assets	5,140	4,389	4,299	4,500	
9	**Total current operational assets**	**16,324**	**16,273**	**16,268**	**17,011**	<--=SUM(E6:E8)
10	Current oper. assets % of sales	41.33%	40.88%	40.55%	40.22%	<--=E9/E3
11						
12	**Current operational liabilities**					
13	Trade accounts payable	2,533	2,807	3,102	3,318	
14	Income taxes payable	1,560	2,239	708	1,971	
15	**Total current operational liabilities**	**4,093**	**5,046**	**3,810**	**5,289**	<--=SUM(E13:E14)
16	Current oper. liabilities % of sales	10.36%	12.68%	9.50%	12.51%	<--=E15/E3
17						
18	**Net working capital (NWC)**	**12,231**	**11,227**	**12,458**	**11,722**	<--=E9-E15
19	NWC % of sales	30.97%	28.20%	31.05%	27.72%	<--=E18/E3



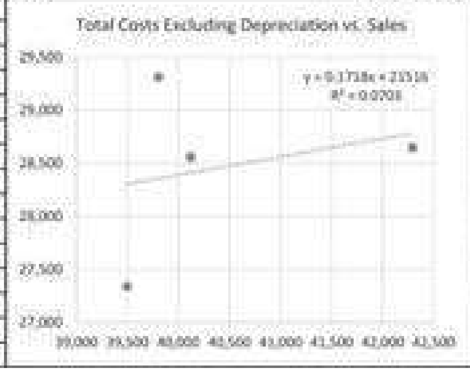
Operating Costs

Like most industrial companies, Merck's operating costs include depreciation. Following the model in Chapter 4, we want to separate depreciation from other operating costs, so we refer to the consolidated statement of cash flows to compute the operating costs without depreciation. Operating costs have varied considerably over the four years from 2015 to 2018.

We also note that the operating cost ratio is an increasing function of sales (shown below). This indicates a large fixed-cost component. Modeling this and running a regression, we chose to model the operating costs as a linear function of sales:

$$Costs_t = 21{,}516 + 0.1718 * Sales_t$$

	A	B	C	D	E	F
1	MERCK COST ANALYSIS					
2		2015	2016	2017	2018	
3	Sales	39,498	39,807	40,122	42,294	<--='Merck Income Statement'!E3
4	Cost of sales	14,934	14,030	12,912	13,509	<--='Merck Income Statement'!E4
5	Other costs, expenses and other	19,163	21,118	20,689	20,084	<--='Merck Income Statement'!E13
6	Interest expense	-672	-693	-754	-772	<--=-'Merck Income Statement'!E11
7	Interest income	289	328	385	343	<--=-'Merck Income Statement'!E10
8	Depreciation and amortization	-6,375	-5,471	-4,676	-4,519	<--=-'Merck Cash-flows'!E6
9	**Total costs excluding depreciation**	**27,339**	**29,312**	**28,556**	**28,645**	**<--=SUM(E4:E8)**
10	Costs excluding depreciation % of sales	69.2%	73.6%	71.2%	67.7%	<--=E9/E3
11						
12	Intercept	21,516	<--=INTERCEPT(B9:E9,B3:E3)			
13	Slope	0.17	<--=SLOPE(B9:E9,B3:E3)			



Though the regression explanation power is quite weak (R-squared of ~7%), we use this regression in our pro forma model for Merck.

Fixed Assets and Sales

We need to determine a model for the fixed assets of Merck. This involves two decisions:

- • Are fixed assets (FAs) best modeled by assuming that the critical model ratio is Net FA/Sales or Gross FA/Sales? We have a slight predilection toward the former, believing it to make more economic sense.[2]
- • What is the applicable average depreciation rate?

In this subsection we deal with the first question. The depreciation rate is discussed in the next section.

	A	B	C	D	E	F
1	MERCK FIXED ASSET ANALYSIS					
2		2015	2016	2017	2018	
3	Property, plant and equipment (at cost) (PP&E)	28,430	27,775	29,041	29,615	
4	Less accumulated depreciation	15,923	15,749	16,602	16,324	
5	**Net PP&E**	**12,507**	**12,026**	**12,439**	**13,291**	
6						
7	Sales	39,498	39,807	40,122	42,294	
8						
9	Gross fixed asset/sales	72.0%	69.8%	72.4%	70.0%	<--=E3/E7
10	Net fixed asset/sales	31.7%	30.2%	31.0%	31.4%	<--=E5/E7
11						
12	4-year average					
13	Gross fixed asset/sales	71.0%	<--=AVERAGE(B9:E9)			
14	Net fixed asset/sales	31.1%	<--=AVERAGE(B10:E10)			

The Net property, plant, and equipment (PP&E) to Sales ratio is relatively stable at Merck; for our model, we will use the average value over the period, which is 31.1%.

Depreciation

Analyzing the depreciation schedule over the period, we find that the normative depreciation schedule is about six years.

	A	B	C	D	E	F
1	MERCK DEPRECIATION ANALYSIS					
2		2015	2016	2017	2018	
3	Property, plant and equipment (at cost) (PP&E)	28,430	27,775	29,041	29,615	
4	Less accumulated depreciation	15,923	15,749	16,602	16,324	
5	Net PP&E	12,507	12,026	12,439	13,291	
6						
7	Depreciation	6,375	5,471	4,676	4,519	
8	Implied depreciation schedule	4.46	5.08	6.21	6.55	<-- =E3/E7
9						
10	Average depreciation schedule	6	<-- =ROUND(AVERAGE(B8:E8),0)			

Dividends

Merck's total dividends are quite stable in dollar terms over the years—much more stable than the net income. In the financial world, this phenomenon is called "dividend smoothing." Over this four-year period, the average annual growth rate was 0.36%. We will use this number in our model.

	A	B	C	D	E
1	MERCK DIVIDEND ANALYSIS				
2		2015	2016	2017	2018
3	Net income attributable to Merck & Co., Inc. shareholders	4,442	3,920	2,394	6,220
4	Dividends paid to stockholders	5,117	5,124	5,167	5,172
5	YoY growth rate		0.14%	0.84%	0.10%
6					
7	Average growth rate	0.36%	<-- =(E4/B4)^(1/3)-1		

Financial Liabilities

It seems that the financial liabilities are quite stable at about $50 billion over the years. We will use this financial debt in our model:

	A	B	C	D	E	F
1	MERCK FINANCIAL LIABILITIES ANALYSIS					
2		2015	2016	2017	2018	
3	Financial liabilities					
4	Loans payable and current portion of long-term debt	2,583	568	3,057	5,308	
5	Accrued and other current liabilities	11,216	10,274	10,427	10,151	
6	Long-term debt	23,829	24,274	21,353	19,806	
7	Deferred income taxes	6,535	5,077	2,219	1,702	
8	Other noncurrent liabilities	7,345	8,514	11,117	12,041	
9	Noncontrolling interests	91	220	233	181	
10	Total financial liabilities	51,599	48,927	48,406	49,189	<-- =SUM(E4:E9)

Merck's Tax Rate

Over the years, Merck's average tax rate was 31.1% or 32.7% (depending on whether you use a simple average or a weighted average). We will use 32% in our model:

	A	B	C	D	E	F
1	**MERCK TAX RATE ANALYSIS**					
2		**2015**	**2016**	**2017**	**2018**	
3	Income before taxes	5,401	4,659	6,521	8,701	
4	Taxes on income	942	718	4,103	2,508	
5	Effective tax rate	17.4%	15.4%	62.9%	28.8%	<--=E4/E3
6						
7	Average tax rate					
8	Simple average	31.1%	<--=AVERAGE(B5:E5)			
9	Weighted average	32.7%	<--=SUM(B4:E4)/SUM(B3:E3)			

Note that since January 1, 2018, and the passage of the Tax Cuts and Jobs Act of 2017, the nominal federal corporate tax rate in the United States of America is a flat 21%. State and local taxes and rules vary by jurisdiction but average at around 7%. This brings the total average statutory tax rate to about 28%.

Merck's Cost of Debt

Merck is a highly rated borrower in financial markets. As of December 2018, it was rated A1 by Moody's, A+ by Fitch, and A by Standard & Poor's. On average, bonds rated "A" carried a risk premium of 1.4%.[3]

The 10-year risk-free rate is 2.69% (using data from the U.S. Department of the Treasury):

	A	B	C	D	E	F	G	H	I	J	K	L	M
1	**THE RISK-FREE YIELD CURVE**												
2	Source:	https://www.treasury.gov/resource-center/data-chart-center/interest-rates/Pages/default.aspx											
3	**Date**	**1 Mo**	**2 Mo**	**3 Mo**	**6 Mo**	**1 Yr**	**2 Yr**	**3 Yr**	**5 Yr**	**7 Yr**	**10 Yr**	**20 Yr**	**30 Yr**
4	31-Dec-18	2.44%	2.45%	2.45%	2.56%	2.63%	2.48%	2.46%	2.51%	2.59%	2.69%	2.87%	3.02%

Thus, the cost of debt is 2.69% + 1.4% = 4.09%.

Interest Earned on Cash and Short-Term Securities

At the time this chapter is being written, short-term interest rates are relatively low. We will assume that Merck earns 1% on its short-term cash investments.

What's the Plug Number?

It is natural to set the plug number as one of the following financial items. We use the cash and cash equivalents (plus short-term investments) in our model.

	A	B	C	D	E	F
1	**MERCK CASH AND CASH EQUIVALENTS**					
2		**2015**	**2016**	**2017**	**2018**	
3	Cash and cash equivalents	8,524	6,515	6,092	7,965	
4	Short-term investments	4,903	7,826	2,406	899	
5	**Total cash and equivalents**	**13,427**	**14,341**	**8,498**	**8,864**	<--=SUM(E3:E4)

5.4 A Model for Merck

Using the values computed for Merck and the assumptions detailed above, we arrive at the following pro forma model for the company:

MERCK PRO FORMA MODEL - 5 year forecast							
	0 actual	1 assumption	2 assumption	3 assumption	4 assumption	5 assumption	
Sales growth		4.93%	4.93%	4.93%	4.93%	4.93%	
Total operating costs intercept		21,516	21,516	21,516	21,516	21,516	<--=F4
Total operating costs slope		0.1718	0.1718	0.1718	0.1718	0.1718	<--=F5
Interest rate on debt		4.1%	4.1%	4.1%	4.1%	4.1%	<--=2.69%+1.4%
Current assets/Sales		40.7%	40.7%	40.7%	40.7%	40.7%	<--=F7
Current liabilities/Sales		11.26%	11.3%	11.3%	11.3%	11.3%	<--=F8
Net fixed assets/Sales		31.08%	31.1%	31.1%	31.1%	31.1%	<--=F9
Depreciation schedule		8	8	8	8	8	<--=F10
Interest earned on cash and marketable securities		1%	1%	1%	1%	1%	
Tax rate		32%	32%	32%	32%	32%	<--=F12
Dividend annual growth rate		0.36%	0.36%	0.36%	0.36%	0.36%	<--=F13
Year	**0**	**1**	**2**	**3**	**4**	**5**	
Income statement							
Sales	42,294	44,379	46,567	48,862	51,271	53,799	<--=F17*(1+G3)
Total operating costs excluding depreciation	(28,645)	(29,141)	(29,517)	(29,912)	(30,326)	(30,760)	<--=-(G4+G5*G17)
Earnings before interest taxes depreciation (EBITDA)	**13,649**	**15,237**	**17,049**	**18,950**	**20,945**	**23,039**	<--=SUM(G17:G18)
EBITDA %	32%	34%	37%	39%	41%	43%	<--=G19/G17
Interest payments on debt	(772)	(2,026)	(2,045)	(2,045)	(2,045)	(2,045)	<--=-G6*AVERAGE(F44:G44)
Interest earned on cash and marketable securities	343	85.8	81.8	81.7	85.9	93.8	<--=G11*AVERAGE(F32:G32)
Depreciation	(4,519)	(4,936)	(5,842)	(6,929)	(8,202)	(9,694)	<--=-F35/G10
Profit before tax	**8,701**	**8,359**	**9,244**	**10,058**	**10,784**	**11,393**	<--=G19+SUM(G21:G23)
Taxes	(2,508)	(2,675)	(2,958)	(3,219)	(3,451)	(3,646)	<--=-G24*G12
Profit after tax (net profit)	**6,193**	**5,684**	**6,286**	**6,840**	**7,333**	**7,747**	<--=G25+G24
Net profit %	15%	13%	13%	14%	14%	14%	<--=G26/G17
Dividends	(5,172)	(5,190)	(5,209)	(5,228)	(5,246)	(5,265)	<--=F28*(1+G13)
Retained earnings	**1,021**	**494**	**1,077**	**1,612**	**2,087**	**2,482**	<--=G28+G26
Balance sheet							
Cash and marketable securities	8,864	8,306	9,089	8,290	8,906	9,860	<--=G47-SUM(G37:G40,G33)
Current assets	17,011	18,062	18,973	19,908	20,890	21,920	<--=G17*G7
Fixed assets							
At cost	29,615	35,051	41,573	49,215	58,166	68,646	<--=G37-G36
Depreciation	(16,324)	(21,260)	(27,102)	(34,030)	(42,233)	(51,927)	<--=F36+G23
Net fixed assets	13,291	13,791	14,471	15,184	15,933	16,718	<--=G9*G17
Investments	6,233	6,233	6,233	6,233	6,233	6,233	<--=F38
Goodwill and other intangibles	29,684	29,684	29,684	29,684	29,684	29,684	<--=F39
Other assets	7,554	7,554	7,554	7,554	7,554	7,554	<--=F40
Total assets	**82,637**	**83,650**	**84,973**	**86,844**	**89,202**	**91,969**	<--=SUM(G32:G33,G37:G40)
Current liabilities	5,289	4,997	5,243	5,502	5,773	6,058	<--=G17*G8
Debt	49,189	50,000	50,000	50,000	50,000	50,000	<--=F44
Equity excluding retained earnings	(8,875)	(8,875)	(8,875)	(8,875)	(8,875)	(8,875)	<--=F45
Accumulated retained earnings	37,034	37,528	38,605	40,217	42,304	44,786	<--=F46+G29
Total liabilities and equity	**82,637**	**83,650**	**84,973**	**86,844**	**89,202**	**91,969**	<--=SUM(G43:G46)

This model gives the following free cash flow (FCF) projections and return on invested capital (ROIC) calculation:

Year	**0**	**1**	**2**	**3**	**4**	**5**	
Free cash flow calculation							
Profit after tax	6,193	5,684	6,286	6,840	7,333	7,747	<--=G26
Add back depreciation	4,519	4,936	5,842	6,929	8,202	9,694	<--=-G23
Subtract increase in current assets	-743	-1,071	-891	-935	-981	-1,030	<--=-(G33-F33)
Add back increase in current liabilities	1,479	-292	246	259	271	285	<--=G43-F43
Subtract increase in fixed assets at cost	-574	-5,436	-6,522	-7,642	-8,951	-10,480	<--=-(G35-F35)
Add back after-tax interest on debt	772	1,378	1,391	1,391	1,391	1,391	<--=-G21*(1-G12)
Less after-tax interest on cash and marketable securities	-233	-58	-56	-56	-58	-64	<--=-G22*(1-G12)
Free cash flow	**11,413**	**5,142**	**6,296**	**6,785**	**7,206**	**7,543**	<--=SUM(G51:G57)
Return on invested capital calculation (ROIC)							
	0	1	2	3	4	5	
Net operating profit less adjusted tax (NOPAT)		9,680	10,579	11,393	12,116	12,720	<--=(G24-SUM(G21:G22))*(1-G12)
Invested capital	60,630	62,750	64,118	65,508	66,967	68,497	<--=G37+G38+G39+(G33-G43)
Return on invested capital (ROIC)		15.69%	16.85%	17.77%	18.50%	18.99%	<--=G62/F63

5.5 Using the Model to Value Merck

In Chapter 3 we discussed the cost of capital for Caterpillar. Using the same methods for Merck brings us to the conclusion that weighted average cost of capital (WACC) = 8.81%.

	A	B	C
1	**COMPUTING THE WACC FOR Merck (DECEMBER 2018)**		
2	Shares outstanding (millions)	2,580	
3	Share price, 31 December 2018	76.41	
4	Equity value, E (millions)	197,138	<-- =B2*B3
5	Net debt, D (millions)	40,325	
6	Cost of debt, r_D	4.0900%	
7	Tax rate, T_C	32.00%	
8			
9	**WACC based on CAPM**		
10	Risk-free rate, r_f	2.69%	10 year US gov. bonds
11	Expected market return, $E(r_M)$	7.69%	Using Gordon model
12	Equity beta, β	1.47	Pharmaceutical industry beta. Source: Professor Damodaran's website
13	Cost of equity, r_E	10.04%	<-- =B10+B12*(B11-B10)
14	WACC	8.81%	<-- =$B$5/SUM($B$4:$B$5)*$B$6*(1-$B$7)+$B$4/SUM($B$4:$B$5)*B13

Using this number in our model and assuming that the long-term FCF growth is 5.5% produces the following FCF valuation:

	A	B	C	D	E	F	G	H
66			**VALUING THE FIRM**					
67	Long-term FCF growth rate (g_L)	5.5%	Assumption					
68	Weighted average cost of capital (WACC)	8.81%	Assumption					
69								
70	Year	0	1	2	3	4	5	
71	Forecasted future FCFs		5,142	6,296	6,785	7,206	7,543	<-- =G58
72	Discounting period		0.50	1.50	2.50	3.50	4.50	
73	Discounting factor		0.92	0.84	0.78	0.71	0.66	<-- =1/(1+$B$68)^G72
74	Discounted FCF		4,726	5,318	5,266	5,141	4,946	<-- =G71*G73
75								
76	PV short-term forecast	25,397	<-- =SUM(C74:G74)					
77	PV long-term forecast (terminal value)	157,625	<-- =G71*(1+B67)/(B68-B67)*G73					
78	**Enterprise value**	**183,021**	**<-- =SUM(B76:B77)**					
79	Add excess assets	13,787	<-- =SUM(B38:B40)					
80	Subtract net debt	-40,325	<-- =-(B44-B32)					
81	**Equity value**	**156,483**	**<-- =SUM(B78:B80)**					
82	**Value per share**	**60.65**	**<-- =B81/WACC!B2**					

According to these model values, Merck is currently overvalued (traded at $76.41 per share).

A **Data Table** on the short-term and long-term growth rates produces the following table.

	A	B	C	D	E	F	G	H
84	Data table: The effect of short-term growth rate (cell C3), and long-term growth rate (cell B67) on equity valuation							
85			Data table header					
86	Short-term sales growth rate ↓ Long-term growth rate →		=B81					
87	156,483		0%	2%	4%	5.5%	6%	7%
88		0%	13,363	20,667	33,535	53,668	65,155	107,170 <-- {=TABLE(B67,C3)}
89		2%	25,598	40,325	61,674	85,696	111,856	178,787
90		4%	46,721	62,503	91,408	138,069	161,455	254,525
91		**5.83%**	**54,394**	**73,532**	**105,787**	**156,483**	**185,409**	**291,109**
92		6%	64,770	85,265	122,793	180,795	213,748	334,598
93		7%	74,192	97,096	138,125	203,972	240,973	376,296
94		8%	83,773	109,241	155,869	227,855	268,334	419,133
95		9%	93,642	121,702	173,696	252,367	297,643	463,124
96		10%	103,763	134,485	190,753	277,874	327,114	508,282

5.6 Valuation Model for Merck Using Multiples

Implementing the multiple valuation approach for Merck, using a peer group of 11 companies in the drug industry, yields a valuation range of $124 billion to $186.6 billion:

	A	B	C	D	E	F	G
1		**MERCK MULTIPLE VALUATION MODEL (DEC 2018)**					
2	**Symbol**	**Name**	**EV/Sales**	**EV/EBITDA**	**EV/EBIT**	**EV/FCFF**	
3	ABBV	Abbvie Inc	5.32	21.39	27.31	12.98	
4	AGN	Allergan Plc	4.24	127.83	NA	11.87	
5	AMGN	Amgen Inc	5.61	10.92	12.99	11.80	
6	BIIB	Biogen Inc	4.76	9.28	10.88	10.35	
7	BMY	Bristol-Myers Squibb Company	3.70	12.64	13.99	14.05	
8	LLY	Eli Lilly and Company	5.62	23.47	34.16	21.85	
9	GILD	Gilead Sciences Inc	3.59	8.25	9.69	9.46	
10	JNJ	Johnson & Johnson	4.34	13.34	18.05	15.95	
11	PFE	Pfizer Inc	5.04	13.26	19.31	17.08	
12	REGN	Regeneron Pharmaceuticals	5.69	14.24	15.07	17.40	
13	ZTS	Zoetis Inc Cl A	7.91	20.90	24.30	25.74	
14		**Average**	**5.07**	**25.05**	**18.58**	**15.32**	<--=AVERAGE(F3:F13)
15		**Median**	**5.04**	**13.34**	**16.56**	**14.05**	<--=MEDIAN(F3:F13)
16							
17	**MRK**	**Merck & Company data for 2018 ($, millions)**					
18			**Sales**	**EBITDA**	**EBIT**	**FCFF**	
19		Indicator	42,294	13,649	9,130	11,413	
20		**Estimated enterprise value**	**213,162**	**182,078**	**151,193**	**160,349**	**<--=F19*F15**
21		Add excess assets	13,787	13,787	13,787	13,787	
22		Subtract net debt	-40,325	-40,325	-40,325	-40,325	
23		**Equity value**	**186,624**	**155,540**	**124,655**	**133,811**	**<--=SUM(F20:F22)**
24		**Value per share**	72.33	60.29	48.32	51.86	<--=F23/WACC!$B$2

Where EV/Sales is enterprise value to sales, EV/EBITDA is enterprise value to EBITDA, and so on.

5.7 Summary

On one hand, the pro forma financial model is an extraordinarily labor-intensive method of valuing the firm. On the other hand, in modeling Merck, we have discovered much about the way the firm operates and about how much we understand (and don't) about its financial statements.

The pro forma modeling technique cannot be undertaken lightly, but if an intensive investigation into the firm's workings is needed, this is the way to go.

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1. 1. It certainly affected the authors' mental stability!
2. 2. If depreciation has any economic meaning, then producing more sales should require more net fixed assets.
3. 3. Federal Reserve Economic Data (https://fred.stlouisfed.org/).

6 Financial Analysis of Leasing

6.1 Overview

A lease is a contractual arrangement by which the owner of an asset (the *lessor*) rents the assets to a *lessee*. In this chapter we analyze long-term leases, in which the asset spends most of its useful life with the lessee. In economic terms, the leases we examine are considered by the lessees as alternatives to purchasing an asset. This analysis fits many long-term equipment leases but not short-term leasing (car rentals, for example). Financial theory regards such leases as being essentially debt contracts: For the lessee, the lease is an alternative to purchasing the asset with debt, and the lessor is essentially providing financing for the lessee.

In the example that follows we consider a company that is faced with the choice of either purchasing or leasing a piece of equipment. We assume that the operating inflows and outflows from the equipment are not affected by its ownership—irrespective of how the asset is held (whether owned or leased), the owner/lessee will have the same sales and must bear the responsibility for maintaining the equipment. According to Statement 13 of the Financial Accounting Standards Board (FASB 13), the lease we are considering is one that "transfers substantially all of the benefits and risks incident to the ownership of property" to the lessee.[1]

The analysis in this chapter concentrates exclusively on the *cash flows* from the lease. It is assumed that the lessor pays taxes on the income from the lease rentals and gets a tax shield on the depreciation of the asset, and that the lessee can claim the rent as an expense. The analysis assumes that the tax authorities treat the lessor as the owner of the asset and the lessee as the user.[2]

6.2 A Simple but Misleading Example

The essence of our analysis can be understood from the following simple example: A company has decided to acquire the use of a machine costing $600,000. If purchased, the machine will be depreciated on a straight-line basis to a residual value of zero. The machine's estimated life is six years, and the company's tax rate T_C is 30%.

The company's alternative to purchasing the machine is to lease it for six years. A lessor has offered to lease the machine to the company for $115,000 annually, with the first payment to be made today and with five additional payments to be made at the start of each of the next five years.

One way of analyzing this problem (**a misleading way, as it turns out**) is to compare the present values of the cash flows to the company of leasing and of buying the asset.

The company believes that the lease payment and the tax shield from depreciation are riskless. Suppose, furthermore, that the risk-free rate is 5%. Based on the following calculation, the company should lease the asset because the leasing alternative has higher (less negative) net present value (NPV).[3]

$$NPV(leasing) = -\sum_{t=0}^{5} \frac{lease\ rental(1-T_c)}{(1+r)^t} = -\sum_{t=0}^{5} \frac{115{,}000 * (1-30\%)}{(1+5\%)^t} = -429{,}023$$

$$NPV(buying) = -asset\,cost + PV(tax\ shields\ on\ depreciations)$$

$$= -600{,}000 + \sum_{t=1}^{6} \frac{100{,}000 * 30\%}{(1+5\%)^t} = -447{,}729$$

Presented in spreadsheet format:

	A	B	C
1	HOW NOT TO ANALYZE A LEASE		
2	Asset cost	600,000	
3	Interest rate	5%	
4	Lease rental payment	115,000	
5	Annual depreciation	100,000	<--=(B2-0)/6
6	Tax rate	30%	
7			
8	NPV (leasing)	-429,023	<--=PV(B3,6,B4*(1-B6),,1)
9	NPV (buying)	-447,729	<--=-B2-PV(B3,6,B6*B5)
10	Decision?	Lease	<--=IF(B8>B9,"Lease","Buy")

This analysis suggests that leasing the asset is preferable to buying it. However, it is misleading because it ignores the fact that leasing is very much like buying the asset with a loan. The financial risks are thus different when we compare a lease (implicitly a purchase with loan financing) against a straightforward purchase without loan financing. If the company is willing to lease the asset, then perhaps it should also be willing to borrow money to buy the assets. This borrowing will change the cash flow patterns and could also produce tax benefits. Hence, our decision about the leasing decision could change if we were to take the loan potential into account.

In the following section we present a method of analyzing leases that deals with this problem by imagining what kind of loan would produce cash flows (and hence financial risks) equivalent to those produced by the lease. This method of lease analysis is called the *equivalent-loan method.*

6.3 Leasing and Firm Financing—the Equivalent-Loan Method

The idea behind the equivalent-loan method is to devise a hypothetical loan that is somehow equivalent to the lease.[4] It then becomes easy to see whether the lease or the purchase of an asset is preferable.

The easiest way to understand the equivalent-loan method is with an example. We return to the previous example:

	A	B	C	D	E	F	G	H	I
1	EQUIVALENT-LOAN METHOD—THE LESSEE POINT OF VIEW								
2	Asset cost	600,000							
3	Interest rate	5%							
4	Lease rental payment	115,000							
5	Annual depreciation	100,000	<--=B2/6						
6	Tax rate	30%							
7									
8	Year	0	1	2	3	4	5	6	
9	After-tax cash flows from leasing								
10	After-tax lease rental	-80,500	-80,500	-80,500	-80,500	-80,500	-80,500		=-$B$4*(1-$B$6)
11									
12	After-tax cash flows from buying the asset								
13	Asset cost	-600,000							
14	Depreciation tax shield		30,000	30,000	30,000	30,000	30,000	30,000	<--=$B$5*$B$6
15	Net cash from buying	-600,000	30,000	30,000	30,000	30,000	30,000	30,000	<--=SUM(H13:H14)
16									
17	Differential cash flow: Lease saves lessee								
18	Lease minus buy	519,500	-110,500	-110,500	-110,500	-110,500	-110,500	-30,000	<--=H10-H15
19									
20	IRR of differential cash flow	3.75%	<--=IRR(B18:H18,0)						
21	After-tax interest rate	3.50%	<--=B3*(1-B6)						
22	Decision?	Buy	<--=IF(B20<B21,"Lease","Buy")						

Rows 2–6 give the various parameters of the problem. The spreadsheet then compares two *after-tax* cash flows: that of the lease and that of the buy. We write outflows with a minus sign and inflows (such as the tax shield from the depreciation) with a plus sign.

- • The cash flow from leasing the asset in each of years 0–5 is: $(1 - \textit{tax rate}) * \textit{lease payment}$.
- • The cash flow from buying the asset is the asset cost in year 0 (an outflow, hence negative) and the tax shield on the asset's depreciation, $\textit{tax rate} * \textit{depreciation}$, in years 1–6 (an inflow, hence written here with a positive sign).

The *differential cash flow* between the lease and the buy decision line (line 18) shows that leasing the asset, instead of buying it, results in the following cash flows to the lessor:

- • A cash inflow of $519,500 in year 0. This inflow is the *cash saved at time 0* by the lease: Purchasing the asset costs $600,000, whereas leasing the asset costs at time 0 only $80,500 on an after-tax basis. Thus, the lease initially saves the lessee $519,500.
- • A cash outflow of $110,500 in years 1–5 and an outflow of $30,000 in year 6. This outflow corresponds to the *after-tax cost of the lease versus the buy* in these years. This cost has two components: the after-tax lease payment ($80,500) and the fact that when leasing, the lessee does not get the tax shield on the asset's depreciation ($30,000).

Thus, leasing instead of purchasing the asset is like getting a loan of $519,500 with after-tax repayments of $110,500 in years 1–5 and an after-tax repayment of $30,000 in year 6. The lease, in other words, can be viewed as an alternative method of financing the asset. *In order to compare the lease to the buy, we should compare the cost of this financing with the cost of alternative financing*. The internal rate of return (IRR) of the differential cash flows (3.75% in cell B20) gives the after-tax cost of the financing implicit in the lease; this is larger than the after-tax cost of firm borrowing, since in this case (where the firm's tax rate is 30% and its borrowing cost is 5%), this cost is 3.5%. Our conclusion: Buying is preferable to leasing.

Why Do We Decide against the Lease?

Not everyone is fully convinced by the preceding argument. We therefore present an alternative argument in this subsection. We show that if the firm can borrow at 5%, it can borrow *more money* with the *same schedule of after-tax repayments* as that which resulted from the lease versus the buy. This hypothetical loan is shown in the following table:

	A	B	C	D	E	F	G	H	I
24	Year	Principal at beginning of year	Interest payment	Principal payment	After-tax loan repayment	Lease minus buy cash flows, years 1-6			
25	1	523,318	26,166	92,184	110,500	110,500	<-- {=TRANSPOSE(C18:H18)}		
26	2	431,134	21,557	95,410	110,500	110,500			
27	3	335,724	16,786	98,750	110,500	110,500			
28	4	236,974	11,849	102,206	110,500	110,500			
29	5	134,768	6,738	105,783	110,500	110,500			
30	6	28,986	1,449	28,986	30,000	30,000			
31									
32	=NPV($B$3*(1-$B$6),F25:$F$30)			=B25-B26					
33									
34		=$B$3*B25			=(1-$B$6)*C25+D25				

The table (a version of the loan tables discussed in Chapter 1) shows the principal of a hypothetical bank loan bearing a 5% interest rate. At the beginning of year 0 (that is, at the time when the firm either purchases or leases the asset), for example, the firm borrows \$523,318 from the bank. At the end of the year, the firm repays \$118,350 to the bank, of which \$26,166 is interest (since \$26,166 = 5% $*$ \$523,318) and the remainder, \$92,184, is repayment of principal. The net after-tax repayment in year 1—assuming full tax deductibility of the interest payment—is $(1 - 30\%) * \$26{,}166 + \$92{,}184 = \$110{,}500$, which is, of course, the same after-tax differential cash flow calculated in our original spreadsheet.

Payments in subsequent years are calculated similarly to the illustration in the preceding paragraph. At the beginning of year 6, there is still \$28,986 of principal outstanding; this is fully paid off at the end of the year with an after-tax payment of \$30,000.

What is the point of this example? If the firm is considering leasing the asset in order to get the financing of \$519,500, which the lease gives, it should instead borrow \$532,318 from the bank at 5%; it can repay this larger loan with the same after-tax cash flows as are implicit in the lease. The bottom line: Purchasing is still preferable to leasing the asset.

The alternative loan table shown above was constructed in the following way.

The principal at the beginning of each of years 1–6 is the present value of the lease versus buy outflows, discounted at (1 − 30%) * 5%. Thus, for example:

$$523{,}318=\sum_{t=1}^{5}\frac{110{,}500}{[1+5\%*(1-30\%)]^t}+\frac{30{,}000}{[1+5\%*(1-30\%)]^6}$$

$$431{,}134=\sum_{t=1}^{4}\frac{110{,}500}{[1+5\%*(1-30\%)]^t}+\frac{30{,}000}{[1+5\%*(1-30\%)]^5}$$

$$\vdots$$

$$28{,}986=\frac{30{,}000}{[1+5\%*(1-30\%)]^1}$$

Once the principal at the start of each year is known, it is an easy matter to construct the rest of the columns.

Interest = 5% * *Principal at the beginning of the year*

Repayment of principal = *Principal at the beginning of this year*
− *Principal at the beginning of next year*

After-tax payment = (1 − *Tax rate*) * *Interest* + *Repayment of principal*

6.4 The Lessor's Problem: Calculating the Highest Acceptable Lease Rental

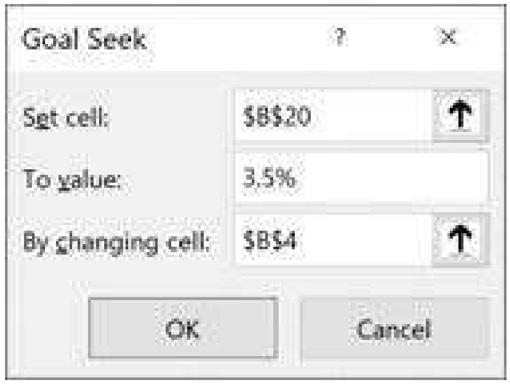
The lessor's problem is the opposite of that of the lessee:

- • The lessee must decide whether—given a rental rate on the leased asset—it is preferable to buy the asset or lease it.
- • The lessor has to decide what *minimum rental rate* justifies the purchase of the asset in order to lease it out.

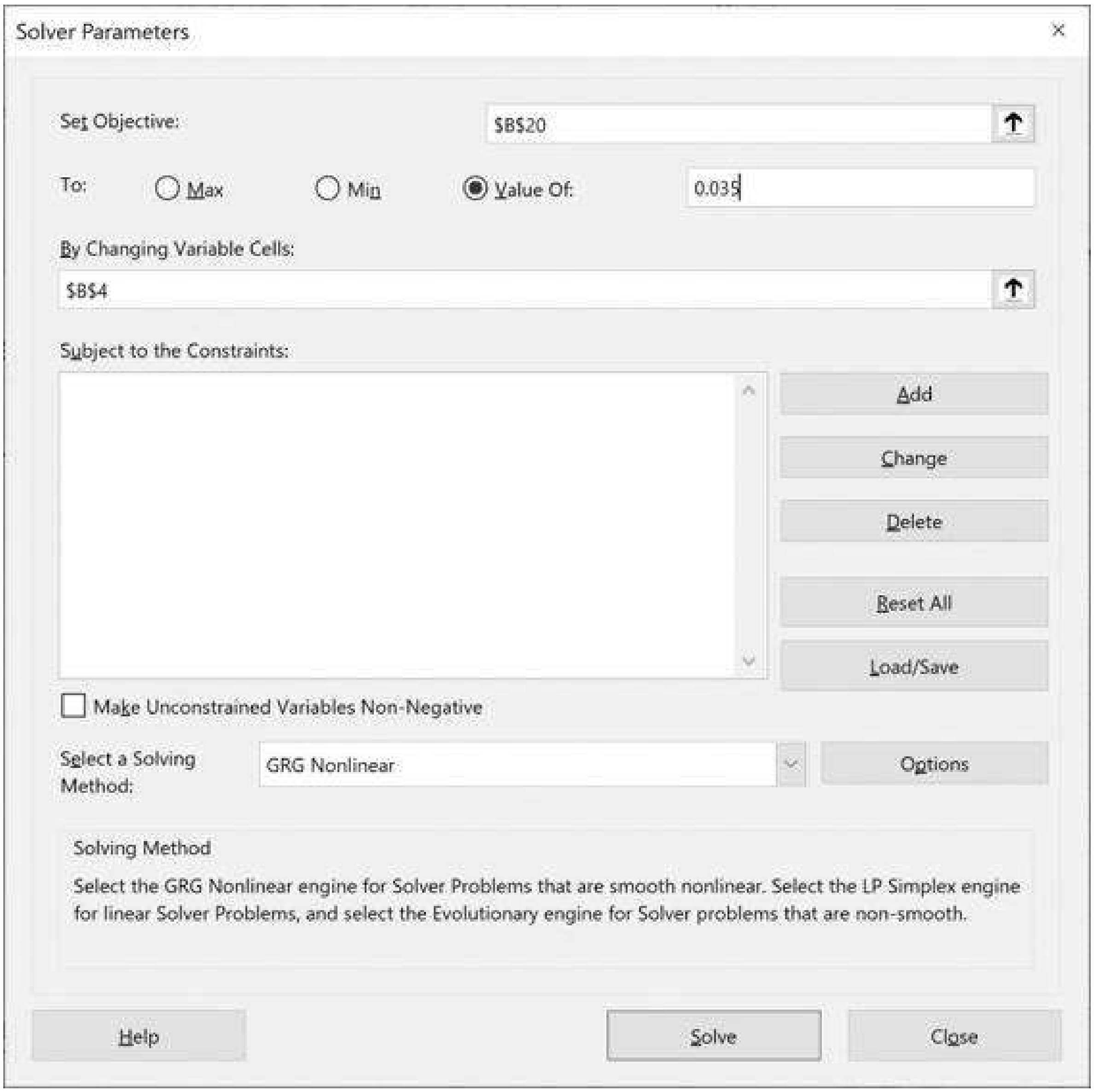
One way of solving the lessor's problem is to turn the above analysis around. We use the Excel **Goal Seek** (**Data|What-If Analysis|Goal Seek**) to get $114,011 as the lessor's minimum acceptable rental:

	A	B	C	D	E	F	G	H
1	THE LESSOR'S PROBLEM Calculating the lowest acceptable lease rate							
2	Asset cost	600,000						
3	Interest rate	5%						
4	Lowest acceptable lease payment	114,011	<-- Computed either with **Goal Seek** or **Solver**					
5	Annual depreciation	100,000	<--=(B2-0)/6					
6	Tax rate	30%						
7								
8	**Year**	**0**	**1**	**2**	**3**	**4**	**5**	**6**
9	**Lessor after-tax cash flows from leasing**							
10	After-tax lease rental	79,808	79,808	79,808	79,808	79,808	79,808	
11								
12	**Lessor after-tax cash flows from buying the asset**							
13	Asset cost	-600,000						
14	Depreciation tax shield		30,000	30,000	30,000	30,000	30,000	30,000
15	Net cash from buying	-600,000	30,000	30,000	30,000	30,000	30,000	30,000
16								
17	**Lessor cash flows**							
18	Lease − buy	-520,192	109,808	109,808	109,808	109,808	109,808	30,000
19								
20	IRR of differential cash flow	3.50%	<-- =IRR(B18:H18)					

Here's what the **Goal Seek** settings look like:



If you're using **Data|Solver** to do this problem, the settings would look like this:



6.5 Asset Residual Value and Other Considerations

In the above example we have ignored the residual value of the asset—its anticipated market value at the end of the lease term. In a mechanical sense, it is easy to include the residual value. Suppose, for example, you think that the asset will have a market value of $50,000 in year 7; assuming that this value is fully taxed (after all, we've depreciated the asset to zero value over the first six years), the after-tax residual value will be (1 − *tax rate*) ∗ $100,000 = $60,000.

	A	B	C	D	E	F	G	H	I
1		RESIDUAL VALUES IN LEASE ANALYSIS							
2	Asset cost	600,000							
3	Interest rate	5%							
4	Lease rental payment	115,000							
5	Annual depreciation	100,000	<-- =(B2-0)/6						
6	Tax rate	30%							
7	Residual value	50,000	<-- Anticipated to be realized in year 6; fully taxed						
8									
9	Year	0	1	2	3	4	5	6	
10	**After-tax cash flows from leasing**								
11	After-tax lease rental	-80,500	-80,500	-80,500	-80,500	-80,500	-80,500		=-$B$4*(1-$B$6)
12									
13	**After-tax cash flows from buying the asset**								
14	Asset cost	-600,000							
15	Depreciation tax shield		30,000	30,000	30,000	30,000	30,000	30,000	<-- =$B$5*$B$6
16	After-tax residual							35,000	<-- =(1-B6)*B7
17	**Net cash from buying**	**-600,000**	**30,000**	**30,000**	**30,000**	**30,000**	**30,000**	**65,000**	<-- =SUM(H14:H16)
18									
19	**Differential cash flow**								
20	Lease minus buy	519,500	-110,500	-110,500	-110,500	-110,500	-110,500	-65,000	<-- =H11-H17
21									
22	IRR of differential cash flow	5.47%	<-- =IRR(B20:H20,0)						
23	After-tax interest rate	3.50%	<-- =B3*(1-B6)						
24	Decision?	Buy	<-- =IF(B22<B23,"Lease","Buy")						

Not surprisingly, the possibility of realizing an extra cash flow from asset ownership makes the lease even less attractive than before (you can see this difference by noting that the return rate in cell B22, the IRR of the differential cash flows, has increased from 3.75% in our original example to 5.47%).

Be a bit careful here, however; the spreadsheet treats the residual value as if it has the same certainty of realization as the depreciation tax shields and the lease rentals. This can be far from the truth! There is no good practical solution to this problem; an ad hoc way of dealing with it might be to reduce the $50,000 by a factor that expresses the uncertainty about its realization. The finance technical jargon for this is "certainty equivalent factor," and you can find it referenced in any finance text.[5] The spreadsheet snapshot below assumes that you've decided that the certainty-equivalence factor for the residual value is 0.7:

RESIDUAL VALUES IN LEASE ANALYSIS
Estimated residual value multiplied by certainty-equivalence factor which represents uncertainty about realizing residual

	A	B	C	D	E	F	G	H	I
2	Asset cost	600,000							
3	Interest rate	5%							
4	Lease rental payment	115,000							
5	Annual depreciation	100,000	<-- =(B2-0)/6						
6	Tax rate	30%							
7	Residual value	50,000	<-- Anticipated to be realized in year 6, fully taxed						
8	Certainty-equivalence factor for residual	0.70							
9									
10	**Year**	0	1	2	3	4	5	6	
11	**After-tax cash flows from leasing**								
12	After-tax lease rental	-80,500	-80,500	-80,500	-80,500	-80,500	-80,500		=$B$4*(1-$B$6)
13									
14	**After-tax cash flows from buying the asset**								
15	Asset cost	-600,000							
16	Depreciation tax shield		30,000	30,000	30,000	30,000	30,000	30,000	<-- =$B$5*$B$6
17	After-tax residual							24,500	<-- =(1-B6)*B7*B8
18	**Net cash from buying**	**-600,000**	**30,000**	**30,000**	**30,000**	**30,000**	**30,000**	**54,500**	<-- =SUM(H15:H17)
19									
20	**Differential cash flow**								
21	Lease minus buy	519,500	-110,500	-110,500	-110,500	-110,500	-110,500	-54,500	<-- =H12-H18
22									
23	IRR of differential cash flow	4.97%	<-- =IRR(B21:H21,0)						
24	After-tax interest rate	3.50%	<-- =B3*(1-B6)						
25	Decision?	Buy	<-- =IF(B23>B24,"Lease","Buy")						

6.6 Mini-Case: When Is Leasing Profitable for Both the Lessor and the Lessee?

The symmetry between the lessee's problem and the lessor's problem suggests that if the lessee wants to lease, it will not be profitable for the lessor to purchase the asset in order to lease it out.

In some cases, however, it may be that the differences in tax rates between the lessee and the lessor make it profitable for both to enter into a leasing arrangement. Here is an example: Greenville Electric Corp. is a public utility that pays no taxes. Its credit rating is of the highest order since all of Greenville's debts are guaranteed by the city of Greenville. Greenville Electric has decided that it requires a new turbine. The turbine costs \$12 million and will be depreciated to zero salvage value over three years. Greenville Electric borrows at 6%.

Greenville can either lease or buy the plant. The lease offer it has is \$2.3 million per year for six years (starting today); the lessee is a captive leasing subsidiary of United Turbine Corp., the manufacturer of the turbine. United Turbine Leasing also borrows at 6% and has a tax rate of 30%.

As can be seen below, the lease is profitable to both lessor and lessee. Greenville Electric gets financing at a cost of 5.96%, compared to its borrowing cost of 6%; United Turbine gets an after-tax return of 4.41%, compared to its after-tax borrowing cost of 4.2%. Both Greenville Electric and United Turbine profit.[6]

	A	B	C	D	E	F	G	H
1	GREENVILLE ELECTRIC CORP.							
2	Turbine cost	12,000,000						
3	Greenville's borrowing rate	6.00%						
4	Lease payment	2,300,000						
5	Greenville Electric's tax rate	0%						
6								
7	**Year**	**0**	**1**	**2**	**3**	**4**	**5**	
8	Lessee after-tax lease costs	-2,300,000	-2,300,000	-2,300,000	-2,300,000	-2,300,000	-2,300,000	
9								
10	**Lessee after-tax purchase costs**							
11	Asset cost	-12,000,000						
12	Depreciation tax shield		0	0	0	0	0	<--=$B$2/5*$B$5
13	Net cash from buying	-12,000,000	0	0	0	0	0	<--=SUM(G11:G12)
14								
15	**Cash saved by leasing**							
16	Lease - purchase cash flows	9,700,000	-2,300,000	-2,300,000	-2,300,000	-2,300,000	-2,300,000	<--=G8+G13
17								
18	IRR of differential cash flow	5.96%	<-- =IRR(B16:G16,0)					
19	Greenville's after-tax borrowing cost	6.00%	<-- =B3					
20								
21	UNITED TURBINE LEASING CORPORATION							
22	Turbine cost	12,000,000						
23	Lease payment	2,300,000						
24	Depreciation (straight line, 3 years)	4,000,000	<--=B22/3					
25	United Turbine's borrowing rate	6.00%						
26	United Turbine's corporate tax rate	30%						
27								
28	**Year**	**0**	**1**	**2**	**3**	**4**	**5**	
29	**Lessor cash flows**							
30	Equipment cost	-12,000,000						
31	Lease payment, after tax	1,610,000	1,610,000	1,610,000	1,610,000	1,610,000	1,610,000	<--=(1-$B$26)*$B$23
32	Depreciation tax shield		1,200,000	1,200,000	1,200,000	<--=$B$24*$B$26		
33	**Total lessor cash flow**	**-10,390,000**	**2,810,000**	**2,810,000**	**2,810,000**	**1,610,000**	**1,610,000**	**<--=SUM(G30:G32)**
34								
35	IRR of lessor cash flows	4.41%	<-- =IRR(B33:G33)					
36	United Turbine's after-tax borrowing cost	4.20%	<-- =B25*(1-B26)					

6.7 Leveraged Leasing

Up to this point we analyzed the lease versus purchase decision from both the points of view of the lessee (the long-term user of the asset) and the lessor (the asset's owner, who rents it out to the lessee). Now we'll analyze leveraged leasing. In a leveraged lease, the lessor finances the purchase of the asset to be leased with debt. From the point of view of the lessee, there is no difference in the analysis of a leveraged or a non-leveraged lease. From the lessor's point of view, however, the cash flows of a leveraged lease present some interesting problems.

At least six parties are typically involved in a leveraged lease: the lessee, the equity partners in the lease, the lenders to the equity partners, an owner trustee, an indenture trustee, and the manufacturer of the asset. In most cases, a seventh party is also involved: a lease packager (a broker or leasing company). Figure 6.1 illustrates the arrangements among the six parties of a typical leveraged lease.

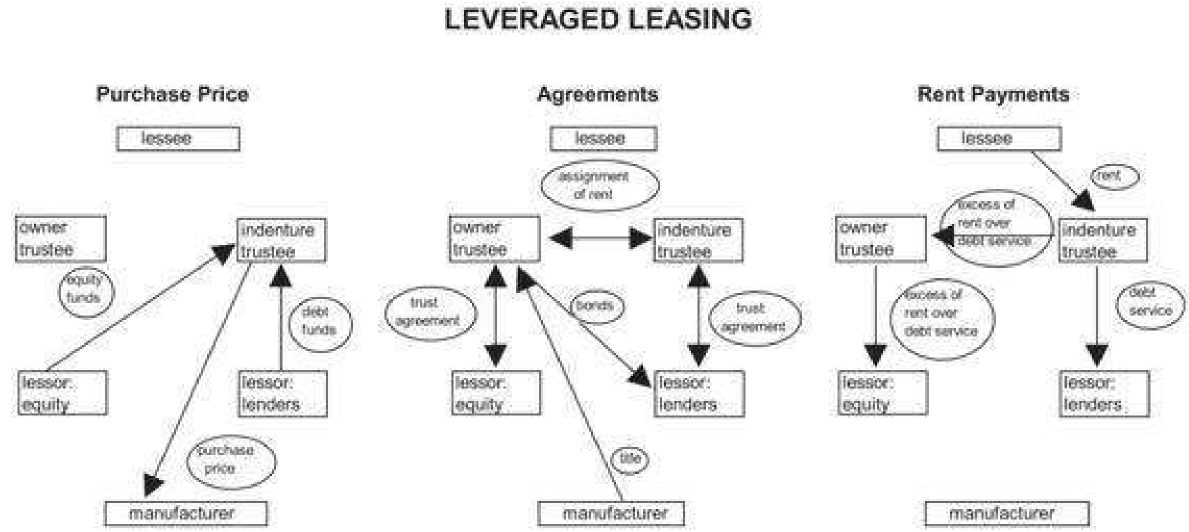


Figure 6.1
Leveraged leasing.

The two major problems related to the analysis of leveraged leases are these:

- • *The straightforward financial analysis of the lease from the point of view of the lessor*. This concerns the calculation of the cash flows obtained by the lessor and a computation of these cash flows' net present value (NPV) or internal rate of return (IRR).
- • *The accounting analysis of the lease*. Accountants use a method called the *multiple phases method* (MPM) to calculate a rate of return on leveraged leases. The MPM rate of return is different from the IRR. In an ordinary financial context this should be of no concern, since the efficient markets hypothesis tells us that only cash flows matter. However, in a less than efficient world, people tend to get very concerned about how things look on their financial statements. Since the accounting rate of return on the lease is difficult to compute, we will use Excel to calculate it; then we will analyze the results.

6.8 A Leveraged Lease Example

We can explore these issues by considering an example, roughly based on an example given in Appendix E of FASB 13, the accounting profession's magnum opus on accounting for leases.

A leasing company is considering the purchase of an asset whose cost is $1,000,000. The asset will be purchased with $200,000 of the company's equity and with $800,000 of debt. The interest on the debt is 10%, so that the annual payment of interest and principal over the 15-year term of the debt is $105,179.[7]

The company will lease the asset out for $110,000 per year, payable at the end of each year. The lease term is 15 years. The asset will be depreciated over a period of eight years, using a standard Internal Revenue Service (IRS) depreciation schedule for assets with a 7-year life.[8] The depreciation schedule for such assets is:

Year	Depreciation
1	14.28%
2	24.49%
3	17.49%
4	12.50%
5	8.92%
6	8.92%
7	8.92%
8	4.48%

Because the asset will be fully depreciated at the time it is sold (year 16), the whole anticipated residual value ($300,000) will be taxable. Since the company's tax rate is 30%, this means that the after-tax cash flow from the residual is $(1 - 30\%) * 300{,}000 =$ $210,000.

These facts are summarized in the spreadsheet below, which also derives the lessor's cash flows:

	A	B	C	D	E	F	G	H	I
1				BASIC LEVERAGED LEASE EXAMPLE					
2	Cost of asset	1,000,000							
3	Lease term	15							
4	Residual value	300,000	<-- Realized year 16						
5	Equity	200,000							
6	Debt	800,000	<-- 15-year term loan, equal payments of interest and principal						
7	Interest	10%							
8	Annual debt payment	105,179	<-- =PMT(B7,B3,-B6)				=(1-tax)*C14+tax*D14-(1-tax)*G14-H14		
9	Annual rent received	110,000							
10	Tax rate	30%							
11									
12	Year	Equity invested	Rental or salvage	Depreciation	Principal at start of year	Loan payment	Interest	Repayment of principal	Cash flow to equity
13	0	-200,000							-200,000
14	1		110,000	142,800	800,000	105,179	80,000	25,179	38,661
15	2		110,000	244,900	774,821	105,179	77,482	27,697	68,536
16	3		110,000	174,900	747,124	105,179	74,712	30,467	46,705
17	4		110,000	125,000	716,657	105,179	71,666	33,513	30,821
18	5		110,000	89,200	683,144	105,179	68,314	36,865	19,275
19	6		110,000	89,200	646,280	105,179	64,628	40,551	17,969
20	7		110,000	89,200	605,728	105,179	60,573	44,606	16,753
21	8		110,000	44,800	561,122	105,179	56,112	49,067	2,095
22	9		110,000		512,056	105,179	51,206	53,973	-12,617
23	10		110,000		458,082	105,179	45,808	59,371	-14,437
24	11		110,000		398,711	105,179	39,871	65,308	-16,218
25	12		110,000		333,403	105,179	33,340	71,839	-18,177
26	13		110,000		261,565	105,179	26,156	79,023	-20,332
27	14		110,000		182,542	105,179	18,254	86,925	-22,703
28	15		110,000		95,617	105,179	9,562	95,617	-25,311
29	16		300,000						210,000
30						IRR of cash flows			8.04%
31							=IRR(I13:I29)		

The last column gives the cash flow to the equity owners of the asset. A typical year's cash flow for the equity owner is calculated as follows:

$$Cash\ flow = (1 - tax) * lease\ payment + tax * depreciation$$
$$- (1 - tax) * interest - principal\ repayment$$

The cash flows of the typical long-lived leveraged lease are usually positive at the beginning of the lease term and then decline over time, turning positive again at the end, when the residual value is received. There are three reasons for this phenomenon:

- • The cash flow that stems from depreciation typically ends or falls off rapidly before the end of the lease term. The more accelerated the depreciation method, the larger will be the depreciation allowances (and hence the larger the depreciation tax shields) at the beginning of the asset's life.
- • In the later years of the lease, the portion of the annual debt payments devoted to interest (tax-deductible) falls, while the portion of the annual debt payments that constitutes a repayment of principal (not tax-deductible) rises.
- • Finally, of course, we anticipated a large cash flow from the realization of the asset's residual value at the end of the lease term.

6.9 Summary

This chapter has looked at the lease-purchase decision. We have examined the decision to lease as a purely financing decision, assuming that (1) all operational factors between leasing and buying are equivalent and (2) the essential firm decision to acquire the use of the asset has already been made. On the basis of these assumptions, the lease-purchase decision can be made using the equivalent-loan method.

A leveraged lease is an arrangement whereby lessors—the owners of the asset—finance their investment with a combination of debt and equity. In this chapter we analyzed the equity income of the lessor in a leveraged lease. The economic analysis of the lease cash flows shows that at some point in the lease life, the equity owner has negative equity value.

Exercises

1. Your company is considering either purchasing or leasing an asset that costs \$1,000,000. The asset, if purchased, will be depreciated on a straight-line basis over six years to a zero-residual value. A leasing company is willing to lease the asset for \$300,000 per year; the first payment on the lease is due at the time the lease is undertaken (i.e., year 0), and the remaining five payments are due at the beginning of years 1–5. Your company has a tax rate T_C = 40% and can borrow at 10% from its bank.
   1. Should your company lease or purchase the asset?
   2. What is the maximum lease payment it will agree to pay?

2. ABC Corp. is considering leasing an asset from XYZ Corp. Here are the relevant facts:

Asset cost	**\$1,000,000**

Depreciation schedule	year 1:	20%
	year 2:	32%
	year 3:	19.20%
	year 4:	11.52%
	year 5:	11.52%
	year 6:	5.76%
Lease term	6 years	
Lease payment	$200,000 per year, at the beginning of years 0, 1, …, 5.	
Asset residual value	Zero	
Tax rates	ABC: T_C = 0% (ABC has tax-loss carryforwards that prevent it from using any additional tax shields) XYZ: T_C = 40%	

ABC's interest costs are 10% and XYZ's interest costs are 7%. Show that it will be advantageous both for ABC to lease the asset and for XYZ to purchase the asset in order to lease it out to ABC.

3. Continuing with the above example: Find the *maximum rental* that ABC will pay and the *minimum rental* that XYZ will accept.
4. Perform a sensitivity analysis (using **Data|Table**) on the certainty-equivalence (CE) factor in section 6.5, showing how the IRR of the differential cash flows varies with the CE factor.
5. Hemp Airlines (HA, "we fly high") is about to buy five CFA3000 commuter jets. Each airplane costs $50 million. A friendly bank has put together a consortium to finance the deal. The consortium includes a 20% equity investment and an 80% debt component. The debt has an interest rate of 8% annually and is a term loan over 10 years. At the end of each of the next 10 years, HA will pay a lease payment of $35 million. At the end of the 10-year lease term, Hemp has the option to buy the aircraft for $10 million each; it is anticipated that it will exercise this option in which the planes are priced at their anticipated fair market value. The airplanes will be depreciated on a straight-line basis over five years to zero salvage value.

   If the equity partner in the lease has a tax rate of 35%, what is its expected compound rate of return?

---

1. 1. International Financial Reporting Standards (IFRS) 16 and U.S. Generally Accepted Accounting Principles (U.S. GAAP) address accounting for leases.
2. 2. From an economic point of view, this assumption is not innocuous: If the lessee has true economic ownership of the asset, why doesn't the lessee get to take the depreciation? But the facts are otherwise.
3. 3. At this point we assume that the residual value of the asset at the end of its life is zero. In section 6.5 we drop this assumption.
4. 4. This method is attributed to Myers, Dill, and Bautista (1976). A somewhat more accessible explanation can be found in Levy and Sarnat (1979).
5. 5. For further references on certainty equivalents, see, for example, Brealey, Myers, and Allen (2013, chapter 9). However, note that neither this work nor the present text (nor anyone else) can tell you precisely how to calculate the certainty-equivalence factor. It depends on your attitude toward risk.

6. 6. Who loses? The government, of course! What makes the lease profitable is the utilization of otherwise unused depreciation tax shields.
7. 7. Using Excel: =PMT(10%,15,-800000) gives 105,179.
8. 8. The depreciation schedule we use is referred to as the modified cost recovery system (MACRS) depreciation. More information can be obtained from an introductory finance text or from many websites (one example, www.real-estate-owner.com/depreciation-chart.html).

II BONDS

7 Bond's Duration

7.1 Overview

Duration is used as a measure of the sensitivity of the price of a bond to changes in the interest rate at which the bond is discounted. It is widely used as a risk measure for bonds—the higher a bond's duration, the riskier it is. In this chapter we consider a basic duration measure—Macaulay duration—which is defined for the case when the term structure is flat. In section 7.7 we examine the uses of duration in immunization strategies.

Consider a bond with payments C_t, where $t = 1, \ldots, N$. Typically, the first $N - 1$ payments will be interest payments, and C_N will be the sum of the repayment of principal and the last interest payment. If the term structure is flat and the discount rate for all of the payments is the yield to maturity (denoted by YTM), then the bond's market price today will be:

$$P = \sum_{t=1}^{N} \frac{C_t}{(1+YTM)^t}$$

In the spreadsheet below we calculate the price and YTM in two approaches. The direct approach (lines 9 and 16) uses Excel's NPV and IRR equations. In this approach we need to lay out the individual payments. Excel has also functions designed for bonds. In lines 10 and 12, we use the indirect approach using Excel's functions PRICE() and YIELD().[1]

	A	B	C	D	E	F	G	H
1		THE PRICE AND YTM USING PRICE() AND YIELD()						
2	Face value ($)	100						
3	Annual coupon	5%						
4	Coupon frequency	2	twice a year					
5	YTM	6%						
6		31-Dec-19	30-Jun-20	31-Dec-20	30-Jun-21	31-Dec-21	30-Jun-22	
7	Cash flows		2.50	2.50	2.50	2.50	102.50	<--=$B$3*$B$2/$B$4+B2
8								
9	Price	97.71	<--=NPV(B5/2,C7:G7)					
10	Price	97.71	<--=PRICE(B6,G6,B3,B5,B2,B4,3)					
11								
12	YTM	6.00%	<--=YIELD(B6,G6,B3,B9,100,B4,3)					
13								
14		31-12-19	30-06-20	31-12-20	30-06-21	31-12-21	30-06-22	
15		-97.71	2.50	2.50	2.50	2.50	102.50	
16	YTM	6.00%	<--=IRR(B15:G15)*2					

Note that the YTM is considered as stated rates by Excel's PRICE() and YIELD() and not as the effective interest rate.

The Macaulay duration measure (throughout this chapter and the next, whenever we use the word *duration* we are always referring to this measure) is meant to represent the time-weighted average maturity of the payments received from the bond. It is defined as:

$$D = \frac{1}{P}\sum_{t=1}^{N} \frac{t \cdot C_t}{(1+YTM)^t}$$

In section 7.3 we go deeper into the meaning of this formula. First, we want to show how to calculate the duration in Excel.

7.2 Two Examples

Consider two bonds. Bond A has just been issued. Its face value is $1,000, it bears the current market interest rate of 7%, and it will mature in 10 years. Bond B was issued five years ago, when interest rates were higher. This bond has $1,000 face value and bears a 13% coupon rate. When issued, this bond had a 15-year maturity, so its remaining maturity is 10 years. Since the current market rate of interest is 7%, bond B's market price is given as follows:

$$\$1,421.41 = \sum_{t=1}^{10} \frac{\$130}{(1.07)^t} + \frac{\$1,000}{(1.07)^{10}}$$

It is worthwhile calculating the duration of each of the two bonds (just once!) the long way. We set up a table in Excel. In the spreadsheet below, rows 19 through 24 show alternatives to the long computation of the duration.

	A	B	C	D	E	F	G
1		BASIC DURATION CALCULATION					
2	YTM	7%					
3	Year	$C_{t,A}$	$t^*C_{t,A}$		$C_{t,B}$	$t^*C_{t,B}$	
4	1	70	70	<--=B4*A4	130	130	<--=E4*A4
5	2	70	140		130	260	
6	3	70	210		130	390	
7	4	70	280		130	520	
8	5	70	350		130	650	
9	6	70	420		130	780	
10	7	70	490		130	910	
11	8	70	560		130	1,040	
12	9	70	630		130	1,170	
13	10	1,070	10,700	<--=B13*A	1,130	11,300	<--=E13*A13
14							
15	Bond price	1,000.00	<--=NPV(B2,B4:B13)		1,421.41	<--=NPV(B2,E4:E13)	
16		1,000.00	<--=-PV(B2,A13,B4,1000)		1,421.41	<--=-PV(B2,A13,E4,1000)	
17	Duration	7.52	<--=NPV(B2,C4:C13)/B15		6.75	<--=NPV(B2,F4:F13)/E15	
18							
19	Using the Excel function **Duration** and the "home-made" function **Dduration**						
20	Bond A	7.52	<--=DURATION(DATE(2020,12,31),DATE(2030,12,31),7%,B2,1)				
21		7.52	<--=dduration(A13,7%,B2,1)				
22							
23	Bond B	6.75	<--=DURATION(DATE(2020,12,31),DATE(2030,12,31),13%,B2,1)				
24		6.75	<--=dduration(A13,13%,7%,1)				

As might be expected, the duration of bond A is longer than that of bond B, since the average payoff of bond A takes longer than that of bond B. To look at this another way, the net present value of bond A's first-year payoff ($70) represents 6.54% of the bond's price $\left(\frac{70}{(1+7\%)^1}\Big/1{,}000=6.54\%\right)$, whereas the net present value of bond B's first-year payoff ($130) is 8.55% of its price $\left(\frac{130}{(1+7\%)^1}\Big/1{,}421=8.55\%\right)$. The figures for the second-year payoffs are 6.11% $\left(\frac{70}{(1+7\%)^2}\Big/1{,}000=6.11\%\right)$ and 7.99% $\left(\frac{130}{(1+7\%)^2}\Big/1{,}421=7.99\%\right)$, respectively.

Using the Excel Duration Formula

Excel has two duration formulas, **Duration()** and **MDuration()**. **MDuration** is defined as follows:

$$MDuration=\frac{Duration}{\left(1+\dfrac{YTM}{number\ of\ coupon\ payments\ per\ year}\right)}$$

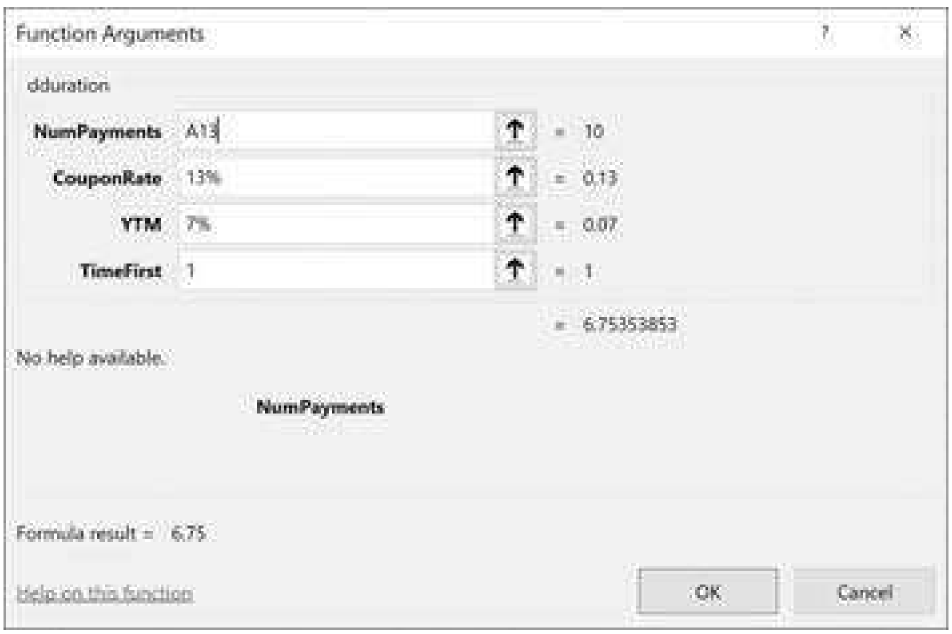
Both formulas have the same syntax; for example, for **Duration()** the syntax is:
Duration(settlement, maturity, coupon, yield, frequency, basis)
where

- • **settlement** is the settlement date (i.e., the purchase date) of the bond
- • **maturity** is the bond's maturity date
- • **coupon** is the bond's coupon
- • **yield** is the bond's yield to maturity
- • **frequency** is the number of coupon payments per year
- • **basis** is the "day count basis" (i.e., the number of days in a year), which is a code between 0 and 4:

- 0 or omitted US (NASD) 30/360
- 1 Actual/actual
- 2 Actual/360
- 3 Actual/365
- 4 European 30/360

The **Duration** formula gives the standard Macaulay duration. The **MDuration** formula can be used in calculating the price elasticity of the bond (see section 7.3). These two duration formulas may require a bit of trickery to implement because they demand a date serial number for both the settlement and the maturity. In the spreadsheet above, the Excel formula is implemented in cells B20 and B23 by assuming that bond A's settlement date is December 31, 2020, and that the bond's maturity date is December 31, 2030. The choice of dates is arbitrary. The last parameter of the Excel duration formula, which gives the basis, is optional and could be omitted.

The insertion of serial date formats in the Excel **Duration** formula is often unhandy. Later in this chapter we use VBA to define a simpler duration formula that overcomes this problem and also computes the duration of a bond when bond payments are unevenly spaced. This "homemade" duration formula is called **Dduration**. The programming aspects of this function are discussed in section 7.6. The spreadsheet above illustrates this function in cells B20 and B23. The dialog box for **Dduration**'s computation of the duration of bond B is given below:



The parameter **TimeFirst** is the time from the bond purchase date until the first payment. For the examples of bond A and bond B, this parameter is 1.

7.3 What Does Duration Mean?

In this section we present three different meanings of duration. Each is interesting and important in its own right.

Duration as the Time-Weighted Average of the Bond's Payments

This was the original definition of Macaulay (1938). Rewrite the duration formula as follows:

$$D = \frac{1}{P}\sum_{t=1}^{N}\frac{t \cdot C_t}{(1+YTM)^t} = \sum_{t=1}^{N}\frac{C_t/P}{(1+YTM)^t} \cdot t$$

The terms $\frac{C_t/P}{(1+YTM)^t}$ sum to 1. This follows from the definition of the bond price; each of these terms is the proportion of the bond's price represented by the payment at time *t*. In the duration formula, each of the terms $\frac{C_t/P}{(1+YTM)^t}$ is multiplied by its time of occurrence: Thus, *the duration is the weighted average of the payments' time, weighted by the relative contribution of each payment to the bond's price.*

Duration as the Bond's Price Elasticity with Respect to Its Discount Rate

It turns out that the duration is highly correlated to the sensitivity of the bond's price to changes in the YTM. This way of viewing duration explains why the duration measure can be used to measure the bond's price volatility; it also shows why duration is often used as a risk measure for bonds. To derive this interpretation, we take the derivative of the bond's price with respect to the current interest rate:

$$P = \sum_{t=1}^{N} \frac{C_t}{(1+YTM)^t}$$

So:

$$\frac{dP}{dYTM} = \sum_{t=1}^{N} \frac{-t \cdot C_t}{(1+YTM)^{t+1}}$$

A little algebra shows that:

$$\frac{dP}{dYTM} = \sum_{t=1}^{N} \frac{-t \cdot C_t}{(1+YTM)^{t+1}} = -\frac{D \cdot P}{1+YTM}$$

Or:

$$\underbrace{\frac{\Delta P}{P}}_{\text{% change in price}} = -\underbrace{\frac{D}{1+YTM}}_{\text{Modified duration}} \cdot \underbrace{\Delta YTM}_{\text{Change in YTM}}$$

This transforms into two useful interpretations of duration:

• First, duration can be regarded as the *elasticity of the bond price with respect to the discount factor*. By "discount factor" we mean $1 + r$:

$$\frac{dP/P}{dYTM/(1+YTM)} = \frac{\text{% change in bond price}}{\text{% change in discount factor}} = -D$$

• Second, we can use duration to measure the *price volatility* of a bond, by rewriting the previous equation as:

$$\frac{dP}{P} = -D \cdot \frac{dYTM}{1+YTM}$$

To show this interpretation of duration in a spreadsheet, we go back to the examples of the previous section. Suppose that the market interest rate rises by 1%, from 7% to 8% (this is a 100 basis points change). What will happen to the bond prices? The price of bond A will be:

$$\sum_{t=1}^{10} \frac{\$70}{(1.08)^t} + \frac{\$1,000}{(1.08)^{10}} = \$932.9$$

A similar calculation shows the price of bond B to be:

$$\sum_{t=1}^{10} \frac{\$130}{(1.08)^t} + \frac{\$1,000}{(1.08)^{10}} = \$1,335.50$$

As predicted by the price volatility formula, the changes in the bond prices are approximated by $\Delta P \cong -D \cdot P \cdot \Delta YTM/(1+YTM)$. To see this, work out the numbers for each bond:

	A	B	C	D
1	**DURATION AS PRICE ELASTICITY** **The change in the bond price is approximately ΔP = - Duration*Price*ΔYTM/(1+YTM)**			
2	Discount rate (YTM)	7%		
3		Bond A	Bond B	
4	Coupon rate	7%	13%	
5	Face value	1,000	1,000	
6	Maturity	10	10	
7	Bond price	1,000.00	1,421.41	<-- =-PV($B$2,C6,C4*C5,C5)
8	Duration	7.515	6.754	<-- =DURATION(DATE(2020,12,31),DATE(2030,12,31),C4,$B$2,1)
9	Modified duration (direct)	7.024	6.312	<-- =C8/(1+$B$2)
10	Modified duration (excel)	7.024	6.312	<-- =MDURATION(DATE(2020,12,31),DATE(2030,12,31),C4,$B$2,1)
11				
12	New discount rate	8.00%		
13	New price (direct)	932.90	1,335.50	<-- =-PV($B$12,C6,C4*C5,C5)
14	% change in 1+YTM: ΔYTM/(1+YTM)	0.93%		<-- =(B12-B2)/(1+B2)
15	**Change in price**			
16	Actual	-67.10	-85.91	<-- =C13-C7
	Approximated by duration			
17	ΔP = - Duration *Price*ΔYTM/(1+YTM)	-70.24	-89.72	<-- =-C8*$B$14*C7
18	Using **MDuration**	-70.24	-63.12	<-- =-($B$12-$B$2)*C5*C10

Note row 18 in the spreadsheet above: Instead of using the Excel **Duration** function and multiplying by $\Delta YTM/(1+YTM)$, we could have used the **MDuration** function and multiplied by ΔYTM.

Babcock's Formula: Duration as the Convex Combination of Bond Yields

A third interpretation of duration is Babcock's (1985) formula, which shows that duration is a weighted average of two factors:

$$D = \left(1 - \frac{y}{YTM}\right) \cdot N + \frac{y}{YTM} \cdot PAF(YTM, N) \cdot (1+YTM),$$

where

$$y = \frac{bond\ coupon}{bond\ price}, \text{the current yield of the bond}$$

$$PAF(YTM, N) = \sum_{i=1}^{N} \frac{1}{(1+YTM)^i}, \text{the present value of an } N\text{-period annuity}$$

This formula gives two useful insights into the duration measure:

- • Duration is a weighted average of the maturity of the bond (N) and of (1 + *YTM*) times the present value annuity factor (PAF) associated with the bond. (Note that the PAF is given by the Excel formula −**PV(r,N,1)**.)
- • In many cases the current yield of the bond, y, is not greatly different from its yield to maturity r. In these cases, duration is not very different from $(1 + r) * PAF$.

Unlike the two previous interpretations, Babcock's formula holds only for the case of a bond with constant coupon payments and single repayment of principal at time N; that is, the formula does not extend to the case where the payments C_t differ over time.

Here is an implementation of Babcock's formula for bond B:

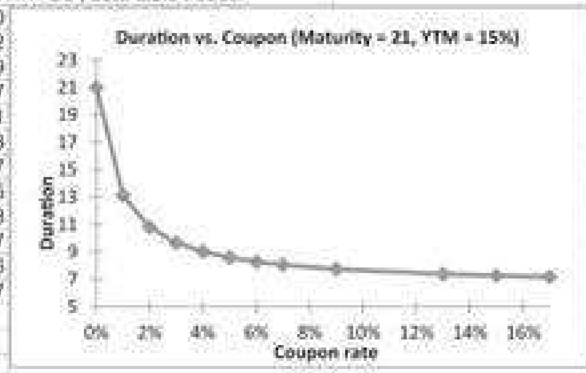
	A	B	C	D
1	**BABCOCK'S FORMULA FOR DURATION** **Duration is convex combination of current yield and present value factor:** **D = N*y/YTM + (1 - y/YTM) * PV(YTM,N,-1)*(1+YTM)**			
2		**Bond A**	**Bond B**	
3	N, bond maturity	10	10	
4	Yield to maturity (YTM)	7%	7%	
5	C, bond coupon	7%	13%	
6	Face value	1,000	1,000	
7	Price	1,000.00	1,421.41	<-- =-PV(C4,C3,C6*C5,C6)
8	Current coupon yield	7.00%	9.15%	<-- =C5*C6/C7
9	Present value annuity factor PAF(r,N)	7.0236	7.0236	<-- =-PV(C4,C3,1)
10				
11	**Two duration formulas**			
12	Babcock's formula	7.5152	6.7535	<-- =C3*(1-C8/C4)+C8/C4*C9*(1+C4)
13	Standard formula	7.5152	6.7535	<-- =DURATION(DATE(2020,12,31),DATE(2030,12,31),C5,C4,1)

7.4 Duration Patterns

Intuitively, we would expect that duration is a decreasing function of a bond's coupon and an increasing function of a bond's maturity. The first of these intuitions is correct, but the second is not.

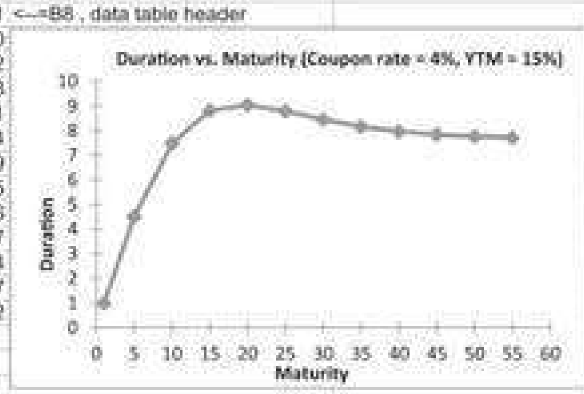
The spreadsheet below shows the effect of increasing the coupon on a bond's duration, which—as our intuition indicated—indeed declines as the coupon increases.

	A	B	C
1	**EFFECT OF COUPON ON DURATION**		
2	Current date	31-12-20	<--=DATE(2020,12,31)
3	Maturity, in years	21	
4	Maturity date	31-12-41	<--=DATE(2020+B3,12,31)
5	YTM	15%	Yield to maturity (i.e., discount rate)
6	Coupon	4%	
7	Face value	1,000	
8	Duration	9.0110	<--=DURATION(B2,B4,B6,B5,1)
9			
10	**Data table: Effect of coupon on duration**		
11	**Bond coupon**	**Duration**	
12		9.01	<--=B8 , data table header
13	0%	21.00	
14	1%	13.12	
15	2%	10.79	
16	3%	9.67	
17	4%	9.01	
18	5%	8.58	
19	6%	8.27	
20	7%	8.05	
21	9%	7.73	
22	13%	7.37	
23	15%	7.26	
24	17%	7.17	
25			
26			
27			



It is not true, however, that the duration is always an increasing function of the bond maturity:

	A	B	C
1	**EFFECT OF MATURITY ON DURATION**		
2	Current date	31-12-20	<--=DATE(2020,12,31)
3	Maturity, in years	21	
4	Maturity date	31-12-41	<--=DATE(2020+B3,12,31)
5	YTM	15%	Yield to maturity (i.e., discount rate)
6	Coupon	4%	
7	Face value	1,000	
8	Duration	9.0110	<--=DURATION(B2,B4,B6,B5,1)
9			
10	**Data table: Effect of maturity on duration**		
11	**Bond maturity**	**Duration**	
12		9.01	<--=B8 , data table header
13	1	1.00	
14	5	4.52	
15	10	7.48	
16	15	8.81	
17	20	9.04	
18	25	8.79	
19	30	8.45	
20	35	8.16	
21	40	7.97	
22	45	7.84	
23	50	7.77	
24	55	7.72	
25			
26			
27			



7.5 The Duration of a Bond with Uneven Payments

The duration formulas discussed above assume that bond payments are evenly spaced. This is almost invariably the case for bonds, *except for the first payment*. For example, consider a bond that pays interest on May 1 of each of the years 2021, 2022, …, 2035, with repayment of its face value on the last date. All the payments are spaced one year apart; however, if this bond was purchased on September 1, 2020, then the time to the first payment is eight months (September to May), not one year. We shall refer to such a bond as a *bond with uneven payments*. In this section we discuss two aspects of this (extremely common) problem:

- • The calculation of the duration of such a bond, when the YTM is known. We show that the duration has a very simple formula, related to the duration of a bond with even payments (i.e., the standard duration formula). In the process of the discussion we develop a simpler duration formula in Excel.
- • The calculation of the YTM of a bond with uneven payments. This requires a bit of trickery and ultimately leads us to another VBA function.

Duration of a Bond with Uneven Payments

Consider a bond with N payments, the first of which occurs at time $\alpha < 1$ and the rest of which are evenly spaced. In the derivation that follows, we show that the duration of such a bond is given by the sum of two terms:

- • First term: The duration of a bond with N payments spaced at even intervals (i.e., the standard duration discussed above).
- • Second term: $\alpha - 1$, which is a negative term since $\alpha < 1$.

It is relatively simple to show why this is so. Denote the payments on the bond by $C_\alpha, C_{\alpha+1}, C_{\alpha+2}, \ldots, C_{\alpha+N-1}$, where $0 < \alpha < 1$. The price of the bond is given by

$$P = \sum_{t=1}^{N} \frac{C_{t-1+\alpha}}{(1+YTM)^{t-1+\alpha}} = (1+YTM)^{1-\alpha} \sum_{t=1}^{N} \frac{C_{t-1+\alpha}}{(1+YTM)^{t}}$$

The duration of this bond is given by

$$D = \frac{1}{P} \sum_{t=1}^{N} \frac{(t-1+\alpha)C_{t-1+\alpha}}{(1+YTM)^{t-1+\alpha}}$$

Rewrite this last expression as follows:

$$D=\frac{1}{P}(1+YTM)^{1-\alpha}\left\{\sum_{t=1}^{N}\frac{t\cdot C_{t+\alpha-1}}{(1+YTM)^{t}}-\sum_{t=1}^{N}\frac{(1-\alpha)C_{t+\alpha-1}}{(1+YTM)^{t}}\right\}$$

$$=\frac{1}{(1+YTM)^{1-\alpha}\sum_{t=1}^{N}\frac{C_{t+\alpha-1}}{(1+YTM)^{t}}}(1+YTM)^{1-\alpha}\left\{\sum_{t=1}^{N}\frac{t\cdot C_{t+\alpha-1}}{(1+YTM)^{t}}-\sum_{t=1}^{N}\frac{(1-\alpha)C_{t+\alpha-1}}{(1+YTM)^{t}}\right\}$$

$$=\underbrace{\left(\frac{\sum_{t=1}^{N}\frac{tC_{t+\alpha-1}}{(1+r)^{t}}}{\sum_{t=1}^{N}\frac{C_{t+\alpha-1}}{(1+r)^{t}}}\right)}_{\substack{\text{Duration assuming}\\ \text{full period before coupon}}}+\underbrace{(\alpha-1)}_{\substack{\text{Adjustment}\\ \text{to period }\alpha}}$$

Here is an example of the calculation of the duration of a bond with uneven periods. Recall that when there is α until the first payment, the duration formula is given by:

$$D=\frac{1}{P}\sum_{t=1}^{N}\frac{(t+\alpha-1)C_{t+\alpha-1}}{(1+r)^{t+\alpha-1}}$$

Each of the cells C10:C14 calculates the value of a term of this formula:

	A	B	C	D	E
1	**DURATION OF BOND WITH UNEVEN PERIODS** **Direct Calculation and DDuration Function**				
2	Alpha	0.25	Time until first coupon payment (in years)		
3	N	5	Number of payments		
4	YTM	6%			
5	Coupon	10%			
6	Face	1,000			
7	Bond price	1,221	<--=NPV(B4,B10:B14)*(1+B4)^(1-B2)		
8					
9	**Period**	**Payment**	**$t^*C_t/(1+YTM)^t$**		
10	0.25	100	24.64	<--=(B10*A10)/(1+$B$4)^A10	
11	1.25	100	116.22		
12	2.25	100	197.35		
13	3.25	100	268.93		
14	4.25	1,100	3,649.49		
15					
16	**Duration calculation**				
17	Direct	3.4871	<--=SUM(C10:C14)/B7		
18	Formula	3.4871	<--=DURATION(DATE(2020,12,31),DATE(2020+B3,12,31),B5,B4,1)-(1-B2)		
19	Newly defined VBA function	3.4871	<--=dduration(B3,B5,B4,B2)		

As noted in section 7.2, the built-in Excel formula **Duration()** is somewhat difficult to use because of the insertion of the dates. We have already introduced a simpler duration formula using VBA; the syntax of this formula is **Dduration(numPayments, couponRate, YTM, timeFirst)**:

```
Function DDuration(NumPayments, CouponRate, YTM, TimeFirst)
    price = 0
    DDuration = 0
```

```
    For Index = 1 To NumPayments
        price = CouponRate / (1 + YTM) ^ Index + price
        DDuration = CouponRate * Index / (1 + YTM) ^ Index +
        DDuration
        If Index = NumPayments Then
        price = price + 1 / (1 + YTM) ^ NumPayments
        DDuration = DDuration + NumPayments / (1 + YTM) ^
        NumPayments
        End If
    Next Index
    DDuration = DDuration / price + TimeFirst - 1
End Function
```

Our homemade formula **DDuration** requires only the number of payments on the bond, the coupon rate, and the time to the first payment α. The use of the formula is illustrated in the spreadsheet above, in cell B19.

Calculating the YTM for Uneven Periods

As the discussion above shows, the calculation of duration requires us to know the bond's yield to maturity (YTM); this YTM is just the internal rate of return (IRR) of the bond's payments and its initial price. Often the YTM is given, but when it is not, we cannot use Excel's **IRR** function but must instead use the **XIRR** function.

Consider a bond that currently costs \$1,123 and pays a coupon of \$89 on January 1 for each of the next five years. On January 1, 2025, the bond will pay \$1,089, the sum of its annual coupon and its face value. The current date is March 31, 2021. The problem in finding the YTM of this bond is that while most of the bond payments are spaced one year apart, there is only 0.7562 of a year until the first coupon payment. Thus, we wish to use Excel to solve the following equation:

$$-1{,}123+\sum_{t=0}^{3}\frac{89}{(1+YTM)^{t+0.7562}}+\frac{1{,}089}{(1+YTM)^{4+0.7562}}=0$$

To solve this problem, we can use the Excel function **XIRR**:

	A	B	C
1	USING XIRR TO CALCULATE THE IRR WITH UNEVEN PAYMENTS		
2	Current date	31-Mar-2021	
3	Annual coupon	89	Paid January 1 for each of next 5 years
4	Maturity date	1-Jan-2025	
5	Face value	1,000	
6	Price of bond today	1,123	
7	Time to first payment	0.7562	<--=YEARFRAC(B2,A11,1)
8			
9	Date	Payment	
10	31-Mar-2021	-1,123	
11	1-Jan-2022	89	
12	1-Jan-2023	89	
13	1-Jan-2024	89	
14	1-Jan-2025	89	
15	1-Jan-2026	1,089	
16	YTM	6.350%	<--=XIRR(B10:B15,A10:A15)

As in the case of the Excel function **IRR**, you can also provide a guess for the IRR, although this may be left out.[2]

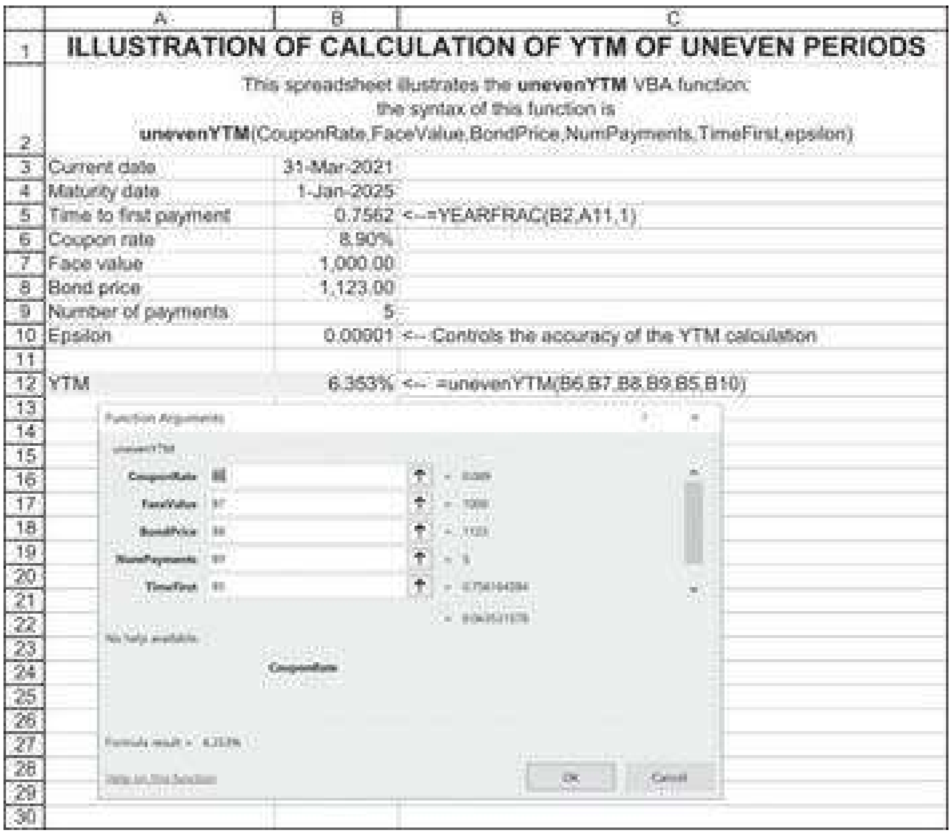
Calculating the YTM for Uneven Payments Using a VBA Program

If you do not know the payment dates, you can use VBA to calculate the YTM for a series of uneven payments. The program below is composed of two functions. The first function, **annuityvalue**, calculates the value $\sum_{t=1}^{N} \frac{1}{(1+r)^t}$. The second function, **unevenYTM**, uses the simple bisection technique to calculate the YTM of a series of uneven payments, leaving you to choose the accuracy **epsilon** of the desired result:

```
Function annuityvalue(interest, numberPeriods)
    annuityvalue = 0
    For Index = 1 To numberPeriods
        annuityvalue = annuityvalue + 1 / (1 + interest) ^ Index
    Next Index
End Function

Function unevenYTM(couponRate, faceValue, bondPrice, _
numberPayments, timeToFirstPayment, epsilon)
    Dim YTM As Double
    high = 1
    low = 0
    While Abs(annuityvalue(YTM, numberPayments) * couponRate * _
            faceValue + faceValue / (1 + YTM) ^ numberPayments - _
            bondPrice / (1 + YTM) ^ (1 - timeToFirstPayment)) >=
            epsilon
        YTM = (high + low) / 2
        If annuityvalue(YTM, numberPayments) * couponRate * _
            faceValue + faceValue / (1 + YTM) ^ numberPayments - _
            bondPrice / (1 + YTM) ^ (1 - timeToFirstPayment) > 0
            Then
                low = YTM
```

```
        Else
                high = YTM
        End If
        Wend
    unevenYTM = (high + low) / 2
End Function
```

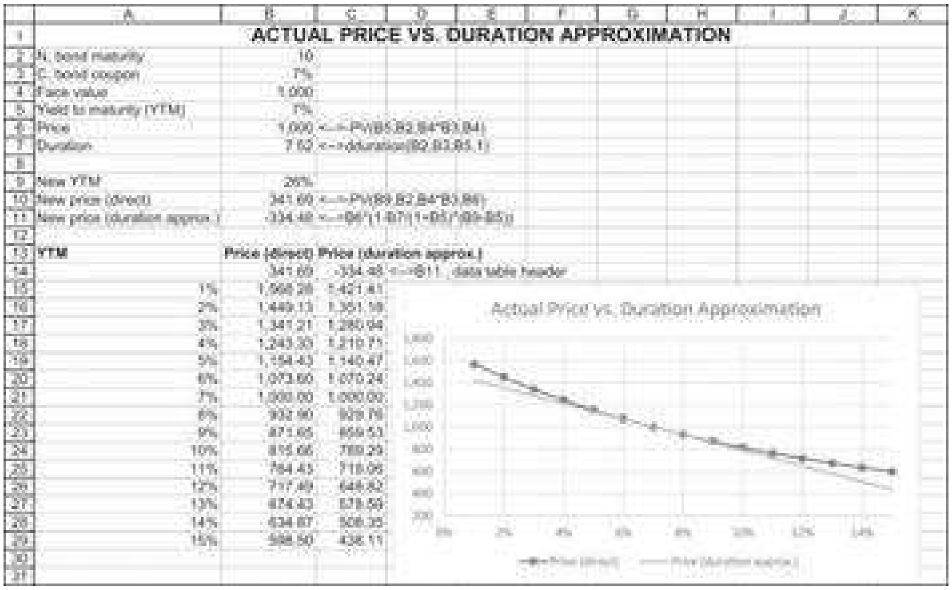
Here's an illustration of the use of this function:



	A	B	C
1	ILLUSTRATION OF CALCULATION OF YTM OF UNEVEN PERIODS		
2	This spreadsheet illustrates the **unevenYTM** VBA function: the syntax of this function is **unevenYTM**(CouponRate,FaceValue,BondPrice,NumPayments,TimeFirst,epsilon)		
3	Current date	31-Mar-2021	
4	Maturity date	1-Jan-2025	
5	Time to first payment	0.7562	<--=YEARFRAC(B2,A11,1)
6	Coupon rate	8.90%	
7	Face value	1,000.00	
8	Bond price	1,123.00	
9	Number of payments	5	
10	Epsilon	0.00001	<-- Controls the accuracy of the YTM calculation
11			
12	YTM	6.353%	<-- =unevenYTM(B6,B7,B8,B9,B5,B10)

7.6 Convexity of a Bond

As already explained, duration is a *first-order approximation* to the bond's price change (in percentage) to the change in YTM while the actual bond price is convex with respect to YTM. As can be seen below, the duration is relatively good in approximating the bond price for small YTM changes, but this approximation may not be good for large YTM changes.



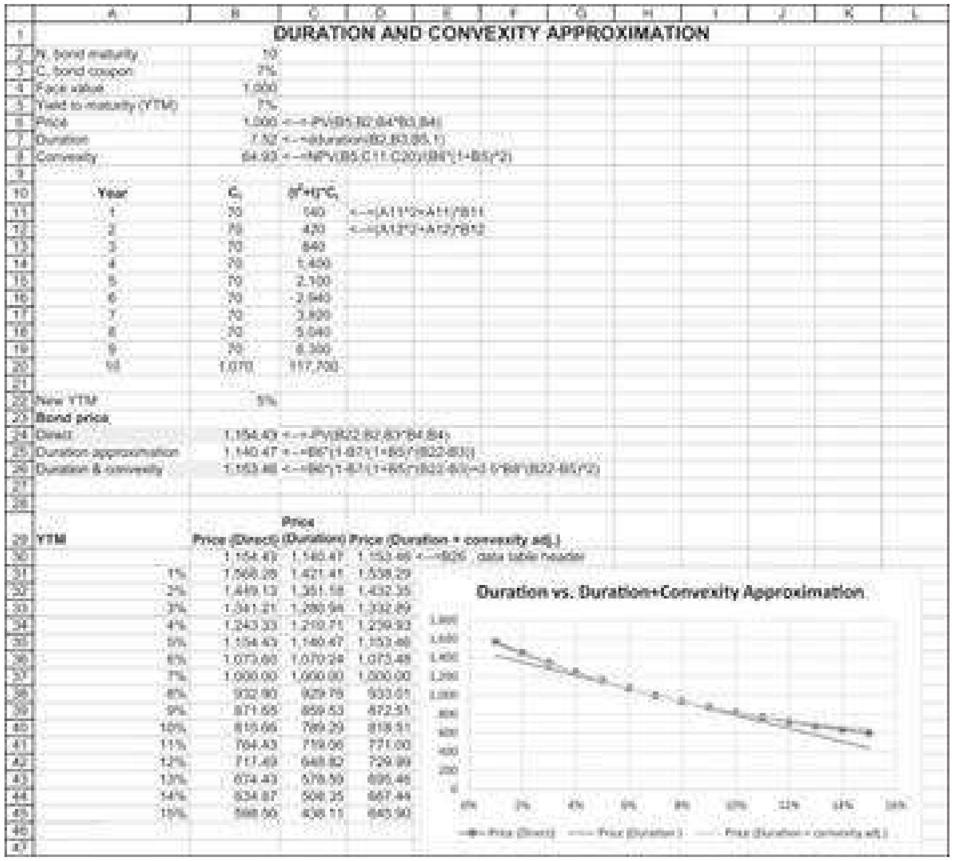
	A	B	C	D
1		ACTUAL PRICE VS. DURATION APPROXIMATION		
2	N, bond maturity	10		
3	C, bond coupon	7%		
4	Face value	1,000		
5	Yield to maturity (YTM)	7%		
6	Price	1,000	<-- =PV(B5,B2,B4*B3,B4)	
7	Duration	7.52	<-- =duration(B2,B3,B5,1)	
8				
9	New YTM	26%		
10	New price (direct)	341.69	<-- =PV(B9,B2,B4*B3,B4)	
11	New price (duration approx.)	-334.48	<-- =B6*(1-B7/(1+B5)*(B9-B5))	
12				
13	YTM	Price (direct)	Price (duration approx.)	
14		341.69	-334.48	<-- =B11 , data table header
15	1%	1,568.28	1,421.41	
16	2%	1,449.13	1,351.18	
17	3%	1,341.21	1,280.94	
18	4%	1,243.33	1,210.71	
19	5%	1,154.43	1,140.47	
20	6%	1,073.60	1,070.24	
21	7%	1,000.00	1,000.00	
22	8%	932.90	929.76	
23	9%	871.65	859.53	
24	10%	815.66	789.29	
25	11%	764.43	719.06	
26	12%	717.49	648.82	
27	13%	674.43	578.59	
28	14%	634.87	508.35	
29	15%	598.50	438.11	

In many cases a second-order approximation, sometimes called *convexity correction*, is often used. The convexity formula is:

$$Convexity = \frac{1}{P \cdot (1+YTM)^2} \sum_{t=1}^{N} \left[\frac{C_t}{(1+YTM)^t} (t^2 + t) \right]$$

And the convexity adjustment to the price should be:

$0.5 \cdot Convexity \cdot (\Delta YTM)^2$



DURATION AND CONVEXITY APPROXIMATION

N, bond maturity	10
C, bond coupon	7%
Face value	1,000
Yield to maturity (YTM)	7%
Price	1,000
Duration	7.52
Convexity	64.93

Year	C_t	$(t^2+t)*C_t$
1	70	140
2	70	420
3	70	840
4	70	1,400
5	70	2,100
6	70	2,940
7	70	3,920
8	70	5,040
9	70	6,300
10	1,070	117,700

New YTM	5%
Bond price	
Direct	1,154.43
Duration approximation	1,140.47
Duration & convexity	1,153.46

YTM	Price (Direct)	Price (Duration)	Price (Duration + convexity adj.)
	1,154.43	1,140.47	1,153.46
1%	1,568.28	1,421.41	1,538.29
2%	1,449.13	1,351.18	1,432.35
3%	1,341.21	1,280.94	1,332.89
4%	1,243.33	1,210.71	1,239.93
5%	1,154.43	1,140.47	1,153.46
6%	1,073.60	1,070.24	1,073.48
7%	1,000.00	1,000.00	1,000.00
8%	932.90	929.76	933.01
9%	871.65	859.53	872.51
10%	815.66	789.29	818.51
11%	764.43	719.06	771.00
12%	717.49	648.82	729.99
13%	674.43	578.59	695.46
14%	634.87	508.35	667.44
15%	598.50	438.11	645.90

As can be seen, the convexity adjustment is approximating the direct (and actual) price relatively well.

7.7 Immunization Strategies

A bond portfolio's value in the future depends on the interest-rate structure prevailing up to and including the date at which the portfolio is liquidated. If a portfolio has the same payoff at some specific future date, no matter what interest-rate structure prevails, then it is said to be *immunized*. This section discusses immunization strategies, which are closely related to the conception of duration.

A Basic Model of Immunization

Consider the following situation: A firm has a known future obligation, Q (a good example would be a financial firm, which knows that it has to make a payment in the future). The discounted value of this obligation is

$$V_0 = \frac{Q}{(1+r)^N},$$

where r is the appropriate discount rate. It is easy to show that the duration of this obligation is N.

Suppose that this future obligation is hedged by a bond held by the firm. By this we mean that the firm currently holds a bond whose value V_B is equal to the discounted value of the future obligation, V_0. If $C_1, C_2, \ldots, C_M$ is the stream of anticipated payments made by the bond, then the bond's present value is given by

$$V_B = \sum_{t=1}^{M} \frac{C_t}{(1+r)^t}.$$

Now suppose that the underlying interest rate, r, changes to $r + \Delta r$. Using a first-order linear approximation, we find that the new value of the future obligation is given by

$$V_0 + \Delta V_0 \approx V_0 + \frac{dV_0}{dr}\Delta r = V_0 + \Delta r\left[\frac{-N \cdot Q}{(1+r)^{N+1}}\right].$$

On the other hand, the new value of the bond is given by

$$V_B + \Delta V_B \approx V_B + \frac{dV_B}{dr}\Delta r = V_B + \Delta r \sum_{t=1}^{M} \frac{-t \cdot P_t}{(1+r)^{t+1}}.$$

If these two expressions are equal, a change in r will not affect the hedging properties of the company's portfolio. Setting the expressions equal gives us the condition:

$$V_B + \Delta r \cdot \sum_{t=1}^{M} \frac{-t \cdot P_t}{(1+r)^{t+1}} = V_0 + \Delta r\left[\frac{-N \cdot Q}{(1+r)^{N+1}}\right]$$

Recalling that

$$V_B = V_0 = \frac{Q}{(1+r)^N},$$

we can simplify this expression to get

$$\frac{1}{V_B}\sum_{t=1}^{M} \frac{t \cdot P_t}{(1+r)^t} = N.$$

The last equation is worth restating as a formal proposition: Suppose that the term structure of interest rates is always flat (that is, the discount rate for cash flows occurring at all future times is the same) or that the term structure moves up or down in parallel movements. Then, for the market value of an asset to be equal (under all changes of the discount rate r) to the market value of a future obligation Q, a necessary and sufficient condition is that the duration of the asset equal the duration of the obligation. Here we understand the word *equal* to mean equal in the sense of a first-order approximation.

An obligation against which an asset of this type is held is said to be *immunized*. The above statement has two critical limitations:

- • The immunization discussed applies only to first-order approximations. When we get to a numerical example in the succeeding sections, we shall see that there is a big difference between first-order equality and "true" equality.[3]
- • We have assumed that the term structure is either flat or that the term structure moves up or down in parallel movements. At best, this might be considered to be a poor approximation of reality. Alternative theories of the term structure lead to alternative definitions of duration and immunization (for alternatives, see Bierwag et al. 1981; Bierwag, Kaufman, and Toevs 1983a, 1983b; Cox, Ingersoll, and Ross 1979; Vasicek 1977). In an empirical investigation of these alternatives, Gultekin and Rogalski (1984) found that the simple Macauley duration we use in this chapter works at least as well as any of the alternatives.

A Numerical Example

In this section we consider a basic numerical immunization example. Suppose you are trying to immunize an obligation at year 10 whose present value is \$1,000 (this means that, at the current interest rate of 6%, its future value is $\$1{,}000 \cdot (1.06)^{10} = \$1{,}790.85$. You intend to immunize the obligation by purchasing \$1,000 worth of a bond or a combination of bonds.

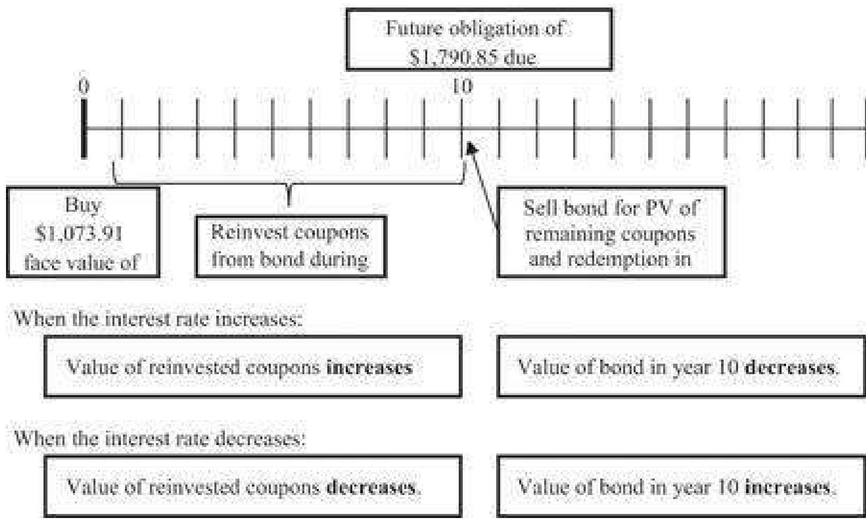
You consider three bonds:

- • Bond 1 has 10 years remaining until maturity, a coupon rate of 6.5%, and a face value of \$1,000.
- • Bond 2 has 15 years until maturity, a coupon rate of 7%, and a face value of \$1,000.
- • Bond 3 has 30 years until maturity, a coupon rate of 5.5% and a face value of \$1,000.

At the existing yield to maturity of 6 percent, the prices of the bonds differ. Bond 1, for example, is worth $\$1{,}036.80 = \sum_t^{10} \frac{65}{(1.06)^t} + \frac{1{,}000}{(1.06)^{10}}$; thus, in order to purchase \$1,000 worth of this bond, you have to purchase \$964.51 = \$1,000/\$1,036.80 of *face value* of the bond.

On the other hand, bond 3 is currently worth \$931.18, so that in order to buy \$1,000 of market value of this bond, you will have to buy \$1,073.91 of face value. If you intend to use this bond to finance a \$1790.85 obligation 10 years from now, here's a schematic of the problem you face:

THE IMMUNIZATION PROBLEM
Illustrated for the 30-year bond.



As we will see below, the 30-year bond will exactly finance the future obligation of $1790.85 only for the case in which the current market interest rate of 6% remains unchanged.

Here is a summary of price and duration information for the three bonds:

	A	B	C	D	E
1	**BASIC IMMUNIZATION EXAMPLE WITH 3 BONDS**				
2	Yield to maturity	6%			
3		**Bond 1**	**Bond 2**	**Bond 3**	
4	Coupon rate	6.50%	7.000%	5.50%	
5	Maturity	10	15	30	
6	Face value	1,000	1,000	1,000	
7	Bond price	$1,036.80	$1,097.12	$931.18	<--=PV($B$2,D5,D4*D6, D6)
8	Face value equal to $1,000 of market value	$ 964.51	$ 911.48	$1,073.91	<--=D6/D7*D6
9	Duration	7.7030	9.9967	14.8308	<--=dduration(D5,D4,$B$2,1)

Note that to calculate the duration, we have used the "homemade" **Dduration** function defined above.

If the yield to maturity doesn't change, then you will be able to reinvest each coupon at 6%. Thus, bond 3, for example, will give a terminal wealth at the end of 10 years that is calculated as follows:

$$\sum_{t=1}^{10} 55 \cdot (1.06)^{t-1} + \left[\sum_{t=1}^{30-10} \frac{55}{(1.06)^t} + \frac{1{,}000}{(1.06)^{20}} \right] = 1{,}667.59$$

The first term in this expression, $\sum_{t=1}^{10} 55 \cdot (1.06)^{t-1} = 724.94$, is the sum of the reinvested coupons. The second and third terms, $\sum_{t=1}^{30-10} \frac{55}{(1.06)^t} + \frac{1{,}000}{(1.06)^{20}} = 942.65$, represent the market value of the bond in year 10, when the bond has 20 more years until maturity. Since we will be buying $1,073.91 of face value of this bond (107.39%), we have, at the end of 10 years, 1.07.36 $*$ $1,667.59 = $1,790.85. This is exactly the amount we wanted to have at this date. The results of this calculation for all three bonds, provided there is no change in the yield to maturity, are given in the following table:

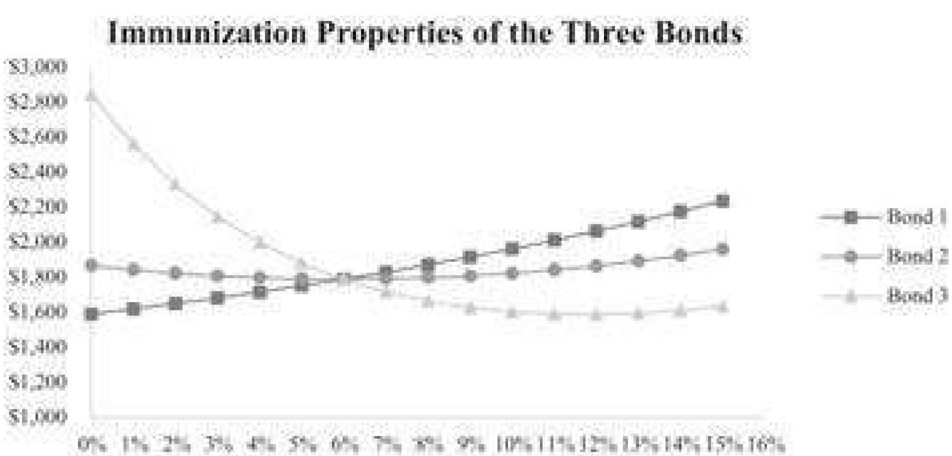
	A	B	C	D	E
11	In year 10:				
12	New yield to maturity	6%			
13		**Bond 1**	**Bond 2**	**Bond 3**	
14	Bond price	$1,000.00	$1,042.12	$942.65	<--=PV($B$12,D5-10,D4*D6,D6)
15	Reinvested coupons	$856.75	$922.66	$724.94	<--=FV($B$12,10,D4*D6)
16	**Total**	$1,856.75	$1,964.78	$1,667.59	<--=SUM(D14:D15)
17	Multiply by percent of face value bought	96.45%	91.15%	107.39%	<--=D8/D6
18	Product	$1,790.85	$1,790.85	$1,790.85	<--=D17*D16

The upshot of this table is that purchasing $1,000 of any of the three bonds will provide —10 years from now—funding for your future obligation of $1,790.85, *provided the market interest rate of 6% doesn't change.*

Now suppose that, immediately after you purchase the bonds, the yield to maturity changes to some new value and stays there. This will obviously affect the calculation we did above. For example, if the yield falls to 5%, the table will now look like this:

	A	B	C	D	E
11	In year 10:				
12	New yield to maturity	5%			
13		**Bond 1**	**Bond 2**	**Bond 3**	
14	Bond price	$1,000.00	$1,086.59	$1,062.31	<--=PV($B$12,D5-10,D4*D6,D6)
15	Reinvested coupons	$817.56	$880.45	$691.78	<--=FV($B$12,10,D4*D6)
16	**Total**	$1,817.56	$1,967.04	$1,754.10	<--=SUM(D14:D15)
17	Multiply by percent of face value bought	96.45%	91.15%	107.39%	<--=D8/D6
18	Product	$1,753.05	$1,792.91	$1,883.74	<--=D17*D16

Thus, if the yield falls, bond 1 will no longer fund our obligation, whereas bond 3 will overfund it. Bond 2's ability to fund the obligation—not surprisingly, in view of the fact that its duration is approximately 10 years—hardly changes. We can repeat this calculation for any new yield to maturity. The results are shown in the figure below; the figure was produced by running a **Data|Table** (see Chapter 28).



Clearly, if you want an immunized strategy, you should buy bond 2!

Linearity of Duration

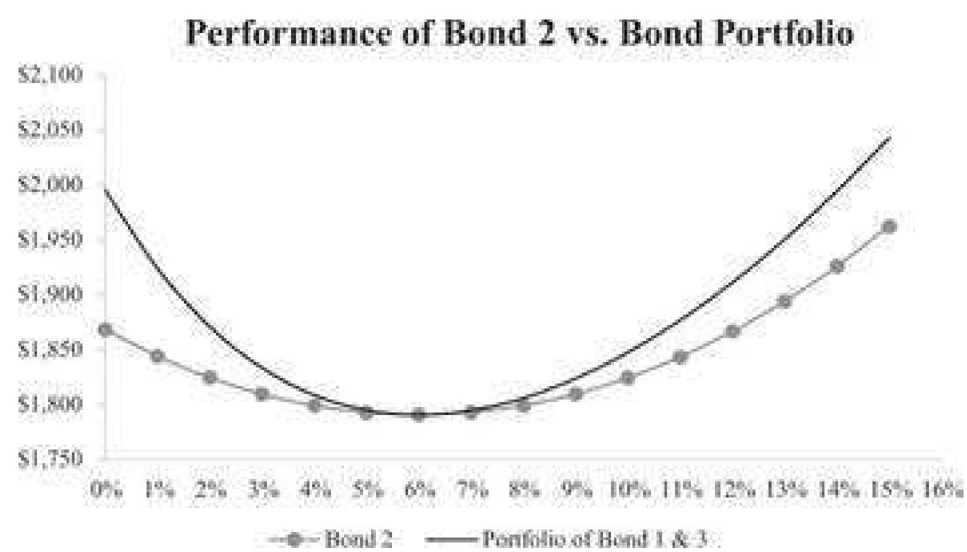
The duration of a portfolio is the weighted average duration of the assets in the portfolio. This means that there is another way to get a bond investment with a duration of 10: If we invest \$678 in bond 1 and \$322 in bond 3, the resulting portfolio also has a duration of 10. These weights are calculated as follows:

$$\lambda * Duration_{Bond1} + (1-\lambda) * Duration_{Bond3} = 7.703 \cdot \lambda + 14.83\,(1-\lambda) = 10$$

Suppose we repeat our experiment with this portfolio of bonds. Starting in row 15 of the spreadsheet below, we repeat the experiment of the previous section (varying the YTM) but add in the portfolio of bond 1 and bond 3. The results below show that the value in year 10 (row 22) does not vary for the portfolio.

	A	B	C	D	E	F
1	**IMMUNIZATION EXAMPLE - DURATION LINEARITY**					
2	Yield to maturity	6%				
3		**Bond 1**	**Bond 2**	**Bond 3**		
4	Coupon rate	6.50%	7.000%	5.50%		
5	Maturity	10	15	30		
6	Face value	1,000	1,000	1,000		
7	Bond price	\$1,036.80	\$1,097.12	\$931.18	<--=PV(\$B\$2,D5,D4*D6,D6)	
8	Face value equal to \$1,000 of market value	\$ 964.51	\$ 911.48	\$1,073.91	<--=D6/D7*D6	
9	Duration	7.7030	9.9967	14.8308	<--=duration(D5,D4,\$B\$2,1)	
10						
11	Investment in Bond 1 (λ)	0.678	<--=(10-D9)/(B9-D9)			
12	Investment in Bond 3 (1-λ)	0.322	<--=1-B11			
13	**Portfolio duration**	**10.000**	<--=B11*B9+B12*D9			
14						
15	In year 10:					
16	New yield to maturity	5%				
17		**Bond 1**	**Bond 2**	**Bond 3**	**Portfolio of Bond 1 & 3**	
18	Bond price	\$1,000.00	\$1,086.59	\$1,062.31		
19	Reinvested coupons	\$817.56	\$880.45	\$691.78		
20	**Total**	\$1,817.56	\$1,967.04	\$1,754.10		
21	Multiply by percent of face value bought	96.45%	91.15%	107.39%		
22	Product	\$1,753.05	\$1,792.91	\$1,883.74	\$ 1,796.17	<--=B11*B22+B12*D22

Building a data table based on this experiment and graphing the results shows that the portfolio's performance is better than that of bond 2 by itself:



Look again at the graph: Notice that, while for both bond 2 and the bond portfolio the terminal value is somewhat convex in the yield to maturity, the terminal value of the portfolio is *more convex* than that of the single bond. Redington (1952), one of the influential propagators of the concept of duration and immunization, thought this convexity very desirable, and we can see why in the above example: No matter what the change in the yield to maturity, the portfolio of bonds provides *more overfunding* of the future obligation than the single bond. Since this is a desirable property for an immunized portfolio, it leads us to formulate the following rule:

In a comparison between two immunized portfolios, both of which are to fund a known future obligation, the portfolio whose terminal value is more convex with respect to changes in the yield to maturity is preferable.[4]

7.8 Summary

In this chapter we have summarized the basics of duration, a commonly used risk measure for bonds. The Macaulay duration measure was originally developed to measure the time-weighted average of a bond's payments. It can also represent the bond's price elasticity to a change in its discount rate.

This chapter also discussed immunization of bonds portfolio. The value of an immunized portfolio of bonds is insensitive to small changes in the underlying yield to maturity of the bonds. Immunization involves setting the bond portfolio's duration equal to the duration of the underlying liability against which the portfolio is held. This chapter shows how to implement immunization strategies to bond portfolios. Needless to say, Excel is an excellent tool for immunization calculations.

Exercises

1. In the spreadsheet below, create a **Data Table** in which the duration is computed as a function of the coupon rate (coupon = 0%, 1%, …, 11%). Comment on the relation between the coupon rate and the duration.

	A	B	C
1	**CHANGING THE COUPON RATE** **Effect on duration**		
2	Current date	31-Dec-20	
3	Maturity, in years	21	
4	Maturity date	31-Dec-41	
5	YTM	15%	
6	Coupon	4%	
7	Face value	1,000	
8			
9	Duration	9.01103	<--=DURATION(B2,B4,B6,B5,1)

2. What is the effect on a bond's duration of increasing the bond's maturity? As in the previous example, use a numerical example and plot the answer. Note that as $N \to \infty$, the bond becomes a consol (a bond that has no repayment of principal but an infinite stream of coupon payments). The duration of a consol is given by $(1 + YTM)/YTM$. Show that your numerical answers converge to this formula.
3. "Duration can be viewed as a proxy for the riskiness of a bond. All other things being equal, the riskier of two bonds should have lower duration." Check this claim with an example. What is its economic logic?
4. A pure discount bond with maturity N is a bond with *no payments* at times $t = 1, \ldots, N - 1$; at time $t = N$, a pure discount bond has a single terminal payment of both principal and interest. What is the duration of such a bond?
5. Replicate the two graphs in section 7.4.
6. On January 23, 2020, the market price of a West Jefferson Development Bond was \$1,122.32. The bond pays \$59 in interest on March 1 and September 1 of each of the years 2020–2026. On September 1, 2026, the bond is

redeemed at its face value of $1,000. Calculate the yield to maturity of the bond and then calculate its duration.

7. Rewrite the formula **Dduration** in section 7.5, so that if the **timeToFirstPayment** α is not inserted, then α automatically defaults to 1.
8. Prove that the duration of a portfolio is the weighted average duration of the portfolio assets.
9. Using the example of section 7.7:
    1. Find a combination of bonds 1 and 3 with a duration of 8.
    2. Find a combination of bonds 1 and 2 with a duration of 8.
    3. Which portfolio (a or b) would you prefer to immunize an obligation with a duration of 8?
10. Using the example of section 7.7:
    1. Find a combination of bonds 1 and 3 with a duration of 12.
    2. Find a combination of bonds 1 and 2 with a duration of 12.
    3. Which portfolio (a or b) would you prefer to immunize an obligation with a duration of 12?

---

1. 1. The syntax for PRICE() is PRICE(settlement date, maturity date, coupon rate, YTM, redemption per $100 face value, coupon frequency per year, day count basis). The syntax for YIELD(): YIELD(settlement date, maturity date, coupon rate, price, redemption per $100 face value, coupon frequency per year, day count basis).
2. 2. There is also a function **XNPV** for finding the present value of a series of payments paid out at uneven dates. This function is discussed in Chapter 30.
3. 3. In *Animal Farm*, George Orwell made the same observation about the barnyard: “All animals are equal, but some animals are more equal than others.”
4. 4. There is another interpretation of the convexity shown in this example: It shows the impossibility of parallel changes in the term structure! If such changes describe the uncertainty relating to the term structure, a bond position can be chosen that always benefits from changes in the term structure. This is an arbitrage, and therefore impossible. We thank Zvi Wiener for pointing this out to us.

8 Modeling the Term Structure*

8.1 Overview

This chapter discusses the term structure of interest rates and the problem of fitting an equation to the term structure of interest rate. The main problem looks like this: We are given a set of bond prices. Can we say something intelligent about the yields on these bonds, for example, in terms of time to maturity or in terms of yields versus duration?

The problem is surprisingly complicated. It is obvious that we need to take into account the riskiness of the bonds (the set of bonds should be of equal risk) and their time pattern of interest payments. There is an additional complicating factor: The most common interest rate measure for bonds is the yield to maturity (YTM), essentially the internal rate of return of the bond price and the future promised payments on the bond. For analytical purposes, however, it is more meaningful to attach a discount factor d_t to payments made in each time period. These discount factors define the so-called *pure discount yields* on the bonds.[1]

This chapter starts with the simplest term structure problem, where there is only one bond for each time period. We then go on to discuss more complicated situations. The functional form we fit to the term structure is the Nelson-Siegel model (see Nelson and Siegel 1987). We also show a variation of this model from Svensson (1994, 1995).

8.2 The Term Structure of Interest Rates

The term structure of interest rates, also called *the yield curve*, is a table or a chart showing yields or interest rates across different maturity periods.

A Simple Example

Assume that four bonds are traded in the market. Each has face value of 1,000, different maturity (T = 1, 2, 3, 4), and an annual coupon of 5%. In the table below we present the market price of each bond and the YTM (calculated using the **Rate()** Excel function):

	A	B	C	D	E	F
1	THE TERM STRUCTURE OF INTEREST RATES					
2	Coupon rate	5%				
3	Face value	1,000				
4						
5	T	1	2	3	4	
6	Market price	1,000	995.49	989.08	982.63	
7	Yield to maturity (YTM)	5.00%	5.24%	5.40%	5.50%	<--=RATE(E5,$B$3*$B$2,-E6,$B$3)

Spot Interest Rate

Spot interest rate for maturity of *t* years refers to the yield to maturity on a zero-coupon bond with *t* years until maturity. It is also the average annual interest rate from time 0 to time *t*. We denote the spot interest rate as $r_{0\to t}^{spot}$. For each time to maturity we calculate:

T Spot rate

1 $$1{,}000 = \frac{1{,}050}{(1+r_{0\to 1}^{spot})^1} \Rightarrow r_{0\to 1}^{spot} = 5.00\%$$

2 $$995.49 = \frac{50}{(1+5\%)^1} + \frac{1{,}050}{(1+r_{0\to 2}^{spot})^2} \Rightarrow r_{0\to 2}^{spot} = 5.25\%$$

3 $$989.08 = \frac{50}{(1+5\%)^1} + \frac{50}{(1+5.25\%)^2} + \frac{1{,}050}{(1+r_{0\to 3}^{spot})^3} \Rightarrow r_{0\to 3}^{spot} = 5.42\%$$

4 $$982.63 = \frac{50}{(1+5\%)^1} + \frac{50}{(1+5.25\%)^2} + \frac{50}{(1+5.42\%)^2} + \frac{1{,}050}{(1+r_{0\to 4}^{spot})^3} \Rightarrow r_{0\to 4}^{spot} = 5.51\%$$

The technical approach in solving the spot rate involves *coupon stripping*. Coupon stripping is a technique of converting a coupon-paying bond into a zero-coupon bond. This is done by subtracting the present value of coupons (from time 0 to $t-1$) from the price of the bond.

	A	B	C	D	E	F	G
1	THE TERM STRUCTURE OF INTEREST RATES						
2	Coupon rate	5%					
3	Face value	1,000				<--=C6-$B$2*$B$3/(1+B7)^B5	
4							
5	T	1	2	3	4		
6	Market price	1,000	995.49	989.08	982.63		
7	Yield to maturity (YTM)	5.00%	5.24%	[illegible]	5.50%	<--=RATE(E5,$B$3*$B$2,-E6,$B$3)	
8	Price-PV(coupons,..)	1,000	947.87	898.32	847.17	<--=E6-$B$2*$B$3/(1+B7)^B5-$B$2*$B$3/(1+C7)^C5-$B$2*$B$3/(1+D7)^D5	
9	r(spot, 0→t)	5.00%	5.25%	5.42%	5.51%	<--=(($B$3+$B$3*$B$2)/E8)^(1/E5)-1	

In order to infer the pure yield curve from coupon bonds is to treat each coupon payment as a separate zero-coupon bond (with FV = coupon). A coupon bond then becomes just a “portfolio” of many zero-coupon bonds. By determining the price of each of these “zeros,” we can calculate the spot rate as the YTM of a single-payment security and thereby construct the *pure yield curve*. This procedure is usually referred to as *coupon stripping* or *bootstrapping*. From the table above, we learn that the average annual rate in the market for three years, for example, is 5.42% (cell D9).

Forward Interest Rate

The *forward interest rate* is the year *t* rate that you can secure today. It is the specific yearly rate. From the above definition we learn that the spot rate is basically an average of the forward rates.

The method to solve the forward rate is to construct an investment for *t* periods; the investment can be performed in two alternatives: the first one—a direct investment in the spot rate for t periods, and the second one—an indirect investment for t periods by investing in the spot rate for $t-1$ periods and then one more period in the forward rate:

$$(1+r_{0\to t-1}^{spot})^{t-1}(1+r_t^{forward})^1 = (1+r_{0\to t}^{spot})^t$$

Hence the forward rate in year t is therefore:

$$r_t^{forward} = \frac{(1+r_{0\to t}^{spot})^t}{(1+r_{0\to t-1}^{spot})^{t-1}} - 1$$

In our example:

	A	B	C	D	E	F
1	THE TERM STRUCTURE OF INTEREST RATES					
2	Coupon rate	5%				
3	Face value	1,000				<--=C8-$B$2*$B$3/(1+B7)^B5
4						
5	T	1	2	3	4	
6	Market price	1,000	995.49	989.86	982.63	
7	Yield to maturity (YTM)	5.00%	5.24%	5.40%	5.50%	<--=RATE(E5,$B$3*$B$2,-E6,$B$3)
8	Price-PV(coupons,...)	1,000	947.87	896.32	847.17	<--=E6-$B$2*$B$3/(1+B7)^B5-$B$2*$B$3/(1+C7)^C5-$B$2*$B$3/(1+D7)^D5
9	r(spot, 0-->t)	5.00%	5.25%	5.42%	5.51%	<--=(($B$3+$B$3*$B$2)/E8)^(1/E5)-1
10	r(forward, t)	5.00%	5.50%	5.75%	5.80%	<--=(1+E9)^E5/(1+D9)^D5-1
11						
12	=(1+C9)^C5/(1+B9)^B5-1					
13						

Determining the Bond's Forward Price

The forward interest rate is the interest rate that can be locked in today for a loan in the future. In the example below, the current 7-year rate is 6% and the 4-year rate is 5%. By simultaneously creating a deposit in one maturity and a loan in the other maturity, we create a security that has zero cash flows everywhere except at years 4 and 7:

	A	B	C	D	E	F	G	H	I	J
1	DETERMINING THE FORWARD RATE FROM T1=4 TO T2=7									
2	Bond maturity, T2	7								
3	Interim period, T1	4								
4	Year T2 pure discount rate	6%								
5	Year T1 pure discount rate	5%								
6										
7	Discretely compounded interest rates									
8		0	1	2	3	4	5	6	7	
9	7-year deposit at 6.00%	100.00							-150.36	<--=-B9*(1+$B$4)^I8
10	4-year loan at 5.00%	-100.00				121.55 <--=-B10*(1+$B$5)^F8				
11	Sum of above: A 3-year deposit at year 4	0.00	0.00	0.00	0.00	121.55	0.00	0.00	-150.36	<--=SUM(I9:I10)
12	Forward rate from year 4 to year 7 (discrete)	7.35% <--=(-I11/F11)^(1/(B2-B3))-1								
13										
14	Continuously-compounded interest rates									
15		0	1	2	3	4	5	6	7	
16	7-year deposit at 6.00%	100.00							-152.20	<--=-$B$16*EXP($B$4*I15)
17	4-year loan at 5.00%	-100.00				122.14 <--=-$B$17*EXP($B$5*F15)				
18	Sum of above: A 3-year deposit at year 4	0.00	0.00	0.00	0.00	122.14	0.00	0.00	-152.20	<--=SUM(I16:I17)
19	Forward rate from year 4 to year 7 (continuous)	7.33% <--=LN(-I18/F18)/(B2-B3)								

The spreadsheet above shows two forward rate computations. If the interest rates are discretely compounded, then the forward rate from year 4 to year 7 is given by:

$$\begin{aligned} \textit{Discretely compounded forward rate, year 4 to 7} &= \left[\frac{(1+r_7)^7}{(1+r_4)^4}\right]^{(1/3)} - 1 \\ &= \left[\frac{(1+6\%)^7}{(1+5\%)^4}\right]^{(1/3)} - 1 = \left(\frac{1.5036}{1.2155}\right)^{(1/3)} - 1 = 7.35\% \end{aligned}$$

If the rates are continuously compounded (as in the Black model and as many applications in finance assume):

$$\textit{Continously compounded forward rate, year 4 to 7} = \left(\frac{1}{3}\right)\mathrm{Ln}\left[\frac{e^{r_7*7}}{e^{r_4*4}}\right]$$

$$= \left(\frac{1}{3}\right)\mathrm{Ln}\left[\frac{e^{r_7*7}}{e^{r_4*4}}\right] = \left(\frac{1}{3}\right)\left(\frac{1.5220}{1.2214}\right) = 7.33\%$$

We apply the forward interest rate to the example in the previous subsection to determine the forward price at year 4 of a $1,000 face-value zero-coupon bond, maturing in seven years.

	A	B	C
1	**DETERMINING THE FORWARD PRICE OF THE BOND**		
2	Bond's maturity, T2	7	
3	Option maturity, T1	4	
4	Bond maturity value	1,000	
5	Interest rate to T2	6%	
6	Interest rate to T1	5%	
7			
8	**Forward rate from year T1=4 to year T2=7**		
9	Discrete approach	7.35%	<--=((1+B5)^B2/(1+B6)^B3)^(1/(B2-B3))-1
10	Continuous approach	7.33%	<--=1/(B2-B3)*LN(EXP(B2*B5)/EXP(B3*B6))
11	**Bond forward price to T1**		
12	Discrete approach	808.38	<--=B4/(1+B9)^(B2-B3)
13	Continuous approach	802.52	<--=B4*EXP(-B10*(B2-B3))

8.3 Bond Pricing Using the Equivalent Single Bond Approach

In this section we introduce two methods of pricing bonds in the case where each bond maturity is associated with a single bond.[2] The two methods are:

- • Computing the yield to maturity of each bond. Each bond's YTM is the internal rate of return of the bond price and the future payments.
- • Computing a set of unique, time-dependent, discount factors for the bonds. Labeling the discount factor for time t by d_t, a bond's price is computed by $Price = \sum_{t=1}^{N} C_t d_t$, where C_t is the promised bond payment at time t.

To illustrate these two methods, consider 15 bonds, each of which pays an annual coupon until maturity and each of which has a face value of 100. The bonds have a maturity of 1, 2, …, 15. In the spreadsheet clip below we show the table of bonds, their coupon rates, and their YTMs:

	A	B	C	D	E	F
1	**INITIAL EXAMPLE**					
2	Bond	Price	Maturity	Coupon rate		YTM
3	A	96.60	1	2.0%		5.59%
4	B	93.71	2	2.5%		5.93%
5	C	91.56	3	3.0%		6.17%
6	D	90.24	4	3.5%		6.34%
7	E	89.74	5	4.0%		6.47%
8	F	90.04	6	4.5%		6.56%
9	G	91.09	7	5.0%		6.63%
10	H	92.82	8	5.5%		6.69%
11	I	95.19	9	6.0%		6.73%
12	J	98.14	10	6.5%		6.76%
13	K	101.60	11	7.0%		6.79%
14	L	105.54	12	7.5%		6.81%
15	M	109.90	13	8.0%		6.83%
16	N	114.64	14	8.5%		6.84%
17	O	119.73	15	9.0%		6.85%

The YTMs are computed by putting the bond payments into a triangular matrix, part of which is shown below.

	F	G	H	I	J	K	L	M	N	O	P	Q	R
2	YTM	Interest rate y_t	0	1	2	3	4	5	6	7	8	9	10
3	5.59%	1	-96.60	102.0	<--=IF(I$2<$C3,$D3*100,IF(I$2=$C3,(1+$D3)*100,""))								
4	5.93%	2	-93.71	2.5	102.5								
5	6.17%	3	-91.56	3.0	3.0	103.0							
6	6.34%	4	-90.24	3.5	3.5	3.5	103.5						
7	6.47%	5	-89.74	4.0	4.0	4.0	4.0	104.0					
8	6.56%	6	-90.04	4.5	4.5	4.5	4.5	4.5	104.5				
9	6.63%	7	-91.09	5.0	5.0	5.0	5.0	5.0	5.0	105.0			
10	6.69%	8	-92.82	5.5	5.5	5.5	5.5	5.5	5.5	5.5	105.5		
11	6.73%	9	-95.19	6.0	6.0	6.0	6.0	6.0	6.0	6.0	6.0	106.0	
12	6.76%	10	-98.14	6.5	6.5	6.5	6.5	6.5	6.5	6.5	6.5	6.5	106.5
13	6.79%	11	-101.60	7.0	7.0	7.0	7.0	7.0	7.0	7.0	7.0	7.0	7.0
14	6.81%	12	-105.54	7.5	7.5	7.5	7.5	7.5	7.5	7.5	7.5	7.5	7.5
15	6.83%	13	-109.90	8.0	8.0	8.0	8.0	8.0	8.0	8.0	8.0	8.0	8.0
16	6.84%	14	-114.64	8.5	<--=IRR(H17:W17)				8.5	8.5	8.5	8.5	8.5
17	6.85%	15	-119.73	9.0	9.0	9.0	9.0	9.0	9.0	9.0	9.0	9.0	9.0

There is another way to interpret the data in our basic example. Suppose that each time period t has its own discount factor d_t. Then the prices of the bonds could be written as the discounted prices of bond payments:

Bond1: $96.6 = 102d_1$

Bond2: $93.71 = 2.5d_1 + 102.5d_2$

Bond3: $91.56 = 3d_1 + 3d_2 + 103d_3$

…

In general: $price = \sum_{t=1}^{N} C_t d_t$

Using matrices, we can solve for the discount factors d_t:

$$\begin{bmatrix} C_{11} & 0 & 0 & 0 & 0 & \cdots & 0 \\ C_{21} & C_{22} & 0 & 0 & 0 & \cdots & 0 \\ \cdots & & & & & \cdots & \cdots \\ C_{N1} & C_{N2} & & & & & C_{NN} \end{bmatrix} \cdot \begin{bmatrix} d_1 \\ d_2 \\ \vdots \\ d_N \end{bmatrix} = \begin{bmatrix} Price_1 \\ Price_2 \\ \vdots \\ Price_N \end{bmatrix}$$

Solving this system of equations gives:

$$\begin{bmatrix} d_1 \\ d_2 \\ \vdots \\ d_N \end{bmatrix} = \begin{bmatrix} C_{11} & 0 & 0 & 0 & 0 & \cdots & 0 \\ C_{21} & C_{22} & 0 & 0 & 0 & \cdots & 0 \\ \cdots & & & & & \cdots & \cdots \\ C_{N1} & C_{N2} & & & & & C_{NN} \end{bmatrix}^{-1} \cdot \begin{bmatrix} Price_1 \\ Price_2 \\ \vdots \\ Price_N \end{bmatrix}$$

This computation is easily done in Excel:

	F	G	H	I	J	K	L	M	N	O	P	Q	R
2	Discount factor	Interest rate y_t	**0**	**1**	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**
3	0.9471	1	-96.60	102.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
4	0.8911	2	-93.71	2.5	102.5	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
5	0.8354	3	-91.56	3.0	3.0	103.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
6	0.7815	4	-90.24	3.5	3.5	3.5	103.5	0.0	0.0	0.0	0.0	0.0	0.0
7	0.7300	5	-89.74	4.0	4.0	4.0	4.0	104.0	0.0	0.0	0.0	0.0	0.0
8	0.6814	6	-90.04	4.5	4.5	4.5	4.5	4.5	104.5	0.0	0.0	0.0	0.0
9	0.6358	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	5.0	105.0	0.0	0.0	0.0
10	0.5930	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	5.5	5.5	105.5	0.0	0.0
11	0.5530	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	6.0	6.0	6.0	106.0	0.0
12	0.5157	10	-98.14	6.5	6.5	6.5	6.5	6.5	6.5	6.5	6.5	6.5	106.5
13	0.4809	11	-101.60	7.0	7.0	7.0	7.0	7.0	7.0	7.0	7.0	7.0	7.0
14	0.4484	12	-105.54	7.5	7.5	7.5	7.5	7.5	7.5	7.5	7.5	7.5	7.5
15	0.4181	13	-109.90	8.0	8.0	8.0	8.0	8.0	8.0	8.0	8.0	8.0	8.0
16	0.3898	14	-114.64	8.5	8.5	8.5	8.5	8.5	8.5	8.5	8.5	8.5	8.5
17	0.3635	15	-119.73	9.0	9.0	9.0	9.0	9.0	9.0	9.0	9.0	9.0	9.0

<--{=MMULT(MINVERSE(I3:W17),-H3:H17)}

Pricing Advantages of the Discount Factors

The advantage of the discount factors is that they allow the accurate pricing of any other bond with the same time pattern of payments. Consider, for example, a five-period bond with a coupon of 3%. The price of this bond, using the current term structure, is 85.55.

	G	H	I	J	K	L	M
19	Time -->	**0**	**1**	**2**	**3**	**4**	**5**
20	New bond	85.55	3	3	3	3	103
21			=MMULT(I20:M20,F3:F7)				
22							

From Discount Factors to a Term Structure

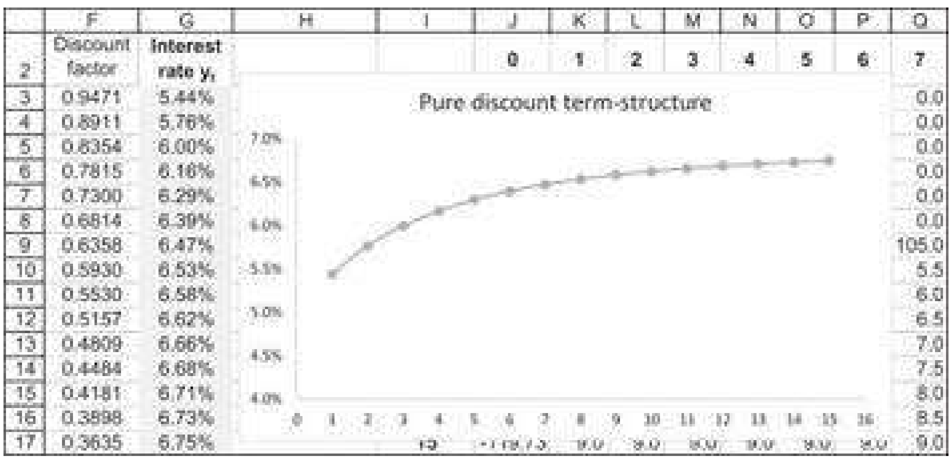
The discount factors determine a *zero-coupon term structure*: $d_t = e^{-y_t * t}$, where y_t is the continuously compounded pure discount rate for time t. Solving the discount factors for the zero-coupon interest rate gives:

$$y_t = In(d_t)/t$$

This means that we can write our bond pricing equation in terms of discount rates instead of discount factors:

$$Price = \sum_{t=1}^{N} C_t d_t = \sum_{t=1}^{N} C_t e^{-y_t t}.$$

Applying this to our example:



	F	G	J	K	L	M	N	O	P	Q
2	Discount factor	Interest rate y_t	0	1	2	3	4	5	6	7
3	0.9471	5.44%								0.0
4	0.8911	5.76%								0.0
5	0.8354	6.00%								0.0
6	0.7815	6.16%								0.0
7	0.7300	6.29%								0.0
8	0.6814	6.39%								0.0
9	0.6358	6.47%								105.0
10	0.5930	6.53%								5.5
11	0.5530	6.58%								6.0
12	0.5157	6.62%								6.5
13	0.4809	6.66%								7.0
14	0.4484	6.68%								7.5
15	0.4181	6.71%								8.0
16	0.3898	6.73%								8.5
17	0.3635	6.75%								9.0

8.4 Pricing with Several Bonds at the Same Maturity

In the previous section there was only one bond for each maturity date. In real data situations, there are often a number of bonds with similar maturities and possibly inconsistent pricing. In bond markets, which are typified by thin trading and often misreported prices, this is a common occurrence. Suppose, as in the next example, we have several bonds with similar maturities. In the example below there are two bonds for 3-, 6-, and 9-year maturities, each with a slightly different YTM.

	A	B	C	D	E	F
1	**MULTIPLE BONDS, SAME MATURITY**					
2	Bond #	Price	Maturity (years)	Coupon rate		YTM
3	A	91.897	1	2.0%		10.99%
4	B	83.256	2	2.5%		12.47%
5	C	76.000	3	3.0%		13.20%
6	D	76.235	3	3.2%		13.32%
7	E	71.211	4	3.5%		13.22%
8	F	67.967	5	4.0%		13.14%
9	G	66.000	6	4.5%		13.01%
10	H	66.163	6	4.2%		12.56%
11	I	65.488	7	5.0%		12.74%
12	J	65.700	8	5.5%		12.53%
13	K	64.000	9	5.8%		12.75%
14	L	66.616	9	6.0%		12.35%
15	M	68.099	10	6.5%		12.19%
16	N	70.048	11	7.0%		12.06%
17	O	72.386	12	7.5%		11.95%
18						
19					=IRR(H17:T17)	

The matrix of cash flows is no longer square, since there are now 15 bonds with 12 maturities. In section 8.3 we found the discount factors d_t by inverting the matrix of payments, but in this case this matrix is not invertible because it is not square:

	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T
2	YTM	Time → Bond ↓	0	1	2	3	4	5	6	7	8	9	10	11	12
3	10.99%	1	-91.90	102	0	0	0	0	0	0	0	0	0	0	0
4	12.47%	2	-83.26	2.5	102.5	0	0	0	0	0	0	0	0	0	0
5	13.20%	3	-76.00	3	3	103	0	0	0	0	0	0	0	0	0
6	13.32%	4	-76.23	3.2	3.2	103.2	0	0	0	0	0	0	0	0	0
7	13.22%	5	-71.21	3.5	3.5	3.5	103.5	0	0	0	0	0	0	0	0
8	13.14%	6	-67.97	4	4	4	4	104	0	0	0	0	0	0	0
9	13.01%	7	-66.00	4.5	4.5	4.5	4.5	4.5	104.5	0	0	0	0	0	0
10	12.56%	8	-66.16	4.2	4.2	4.2	4.2	4.2	104.2	0	0	0	0	0	0
11	12.74%	9	-65.49	5	5	5	5	5	5	105	0	0	0	0	0
12	12.53%	10	-65.70	5.5	5.5	5.5	5.5	5.5	5.5	5.5	105.5	0	0	0	0
13	12.75%	11	-64.00	5.8	5.8	5.8	5.8	5.8	5.8	5.8	5.8	105.8	0	0	0
14	12.35%	12	-66.62	6	6	6	6	6	6	6	6	106	0	0	0
15	12.19%	13	-68.10	6.5	6.5	6.5	6.5	6.5	6.5	6.5	6.5	6.5	106.5	0	0
16	12.06%	14	-70.05	7	7	7	7	7	7	7	7	7	7	107	0
17	11.95%	15	-72.39	7.5	7.5	7.5	7.5	7.5	7.5	7.5	7.5	7.5	7.5	7.5	107.5

We can find the discount factors by using a least-squares approximation (LS): For each maturity t we find a factor d_t, so that the square of the pricing errors for the set of bonds is minimized. The least-squares approximation is usually employed to determine the approximate solution of overdetermined systems (i.e., sets of equations in which the equations outnumber the unknowns). "Least squares" means that the overall solution minimizes the sum of the squares of the errors made in the results of every equation.

We want to solve for a vector of discount factors d that best approximates the pricing of the bonds:

$$\begin{bmatrix} \textit{Cashflows} \\ \textit{matrix} \\ \ \end{bmatrix} \cdot \begin{bmatrix} \textit{Discount} \\ \textit{factors} \\ \textit{column} \end{bmatrix} = \begin{bmatrix} \textit{Bond} \\ \textit{prices} \\ \textit{column} \end{bmatrix}$$

$$\begin{bmatrix} 102 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 2.5 & 102.5 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 3 & 3 & 103 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ 5.8 & 5.8 & & & \cdots & & & 5.8 & 105.8 & 0 & 0 & 0 \\ 6 & 6 & & & \cdots & & & 6 & 106 & 0 & 0 & 0 \\ 6.5 & 6.5 & 6.5 & & & \cdots & & & 6.6 & 106.5 & 0 & 0 \\ 7 & 7 & 7 & 7 & & & \cdots & & & 7 & 107 & 0 \\ 7.5 & 7.5 & 7.5 & 7.5 & & & \cdots & & & 7.5 & 7.5 & 107.5 \end{bmatrix} \cdot \begin{bmatrix} d_1 \\ d_2 \\ d_3 \\ \vdots \\ d_8 \\ d_9 \\ d_{10} \\ d_{11} \\ d_{12} \end{bmatrix} = \begin{bmatrix} 91.897 \\ 83.256 \\ 76.000 \\ \vdots \\ 64.000 \\ 66.616 \\ 68.099 \\ 70.048 \\ 72.386 \end{bmatrix}$$

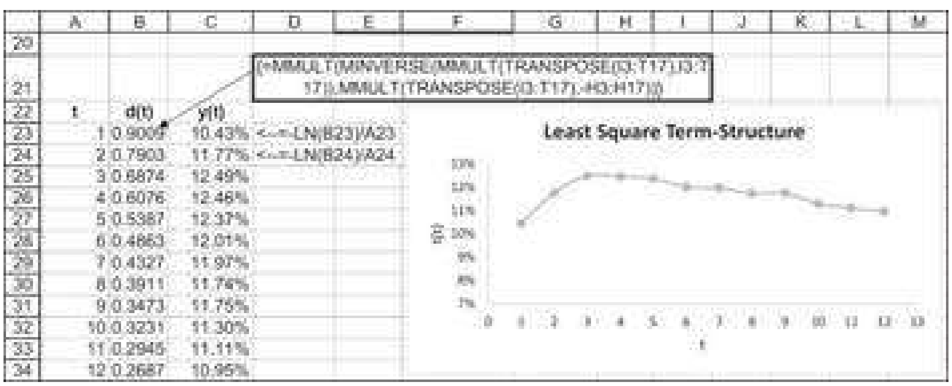
The formula for the least-squares approximation is

$$d = (Cashflows^T * Cashflows)^{-1} * (Cashflows^T * Prices)$$

In Excel this formula becomes

$$\begin{pmatrix} d_1 \\ \vdots \\ d_N \end{pmatrix} = \text{MMult}\begin{pmatrix} \text{Minverse(MMult(Transpose(}\textit{Cashflows}\text{), }\textit{Cashflows}\text{)),} \\ \text{MMult(Transpose(}\textit{Cashflows}\text{), }\textit{Prices}\text{)} \end{pmatrix}$$

Implementing this on our example:



The fit of discount factors to bond prices will no longer be precise (as in the previous section). Below we compare the actual bond prices to the fitted prices:

	O	P	Q	R
21	Bond	Maturity (years)	Actual price	Fitted price
22	A	1	91.897	91.8969
23	B	2	83.256	83.2556
24	C	3	76.000	75.8791
25	D	3	76.235	76.3548
26	E	4	71.211	71.2104
27	F	5	67.967	67.9665
28	G	6	66.000	66.6826
29	H	6	66.163	65.4793
30	I	7	65.488	65.4895
31	J	8	65.700	65.7017
32	K	9	64.000	64.7904
33	L	9	66.616	65.8269
34	M	10	68.099	68.0989
35	N	11	70.048	70.0480
36	O	12	72.386	72.3857

8.5 The Nelson-Siegel Approach of Fitting a Functional Form to the Term Structure

In many cases we may want to fit a functional form to the term structure. The advantage of a functional form is that it allows us to interpolate interest rates for time periods and coupons that are not in our data set. It also allows us to determine the sensitivity of the term structure to structural factors.

One popular fitted form for the term structure is the *Nelson-Siegel (NS) term structure*, which postulates that

$$y(t) = \alpha_1 + \alpha_2\left(\beta\left(\frac{1-e^{-t/\beta}}{t}\right)\right) + \alpha_3\left(\beta\left(\frac{1-e^{-t/\beta}}{t}\right) - e^{-t/\beta}\right)$$

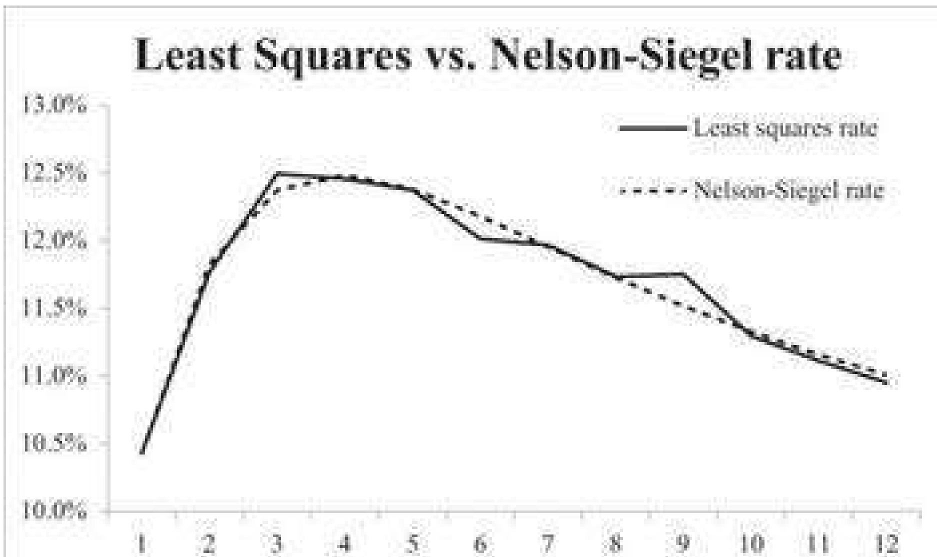
This model can also be written as

$$y(t) = \alpha_1 + (\alpha_2 + \alpha_3)\beta\left(\frac{1-e^{-t/\beta}}{t}\right) - \alpha_3 e^{-t/\beta}$$

In section 8.6 we analyze the NS term structure model and discuss the meaning and the plausible values of its parameters (α_1, α_2, α_3, β). In this section we skip these analytics of the model and show how to fit NS term structure to the discount factors of the example of the previous section.[3] Our end result is given below (explanations to follow).

	A	B	C	D	E	F	G	H	I	J	K	L	M	N
1							FITTING NELSON-SIEGEL							
2	Bond	Price	Maturity (years)	Coupon rate		Maturity	$d^{LS}(t)$	$y^{LS}(t)$	NS rate	Error				
3	A	91.8967	1	2.0%		1	0.9009	10.43%	10.43%	2.5E-09		α_1	0.09215	
4	B	83.2564	2	2.5%		2	0.7903	11.77%	11.82%	3.1E-07		α_2	-0.01773	
5	C	78.0000	3	3.0%		3	0.6874	12.49%	12.37%	1.4E-06		α_3	0.13573	
6	D	78.2947	3	3.2%		4	0.6076	12.46%	12.48%	6.7E-08		β	1.64857	
7	E	71.2110	4	3.5%		5	0.5387	12.37%	12.38%	2.4E-09		Error	0.00001	<--=SUM(J3:J14)
8	F	67.9672	5	4.0%		6	0.4863	12.01%	12.18%	2.7E-06				
9	G	66.0000	6	4.5%		7	0.4327	11.97%	11.96%	2.1E-08		=$M$3+($M$4+$M$5)*$M$6*(1-EXP(-F14/$M$6))/F14-$M$5*EXP(-F14/$M$6)		
10	H	66.1625	6	4.2%		8	0.3911	11.74%	11.73%	7.6E-09				
11	I	65.4881	7	5.0%		9	0.3473	11.75%	11.52%	5.5E-06				
12	J	65.7003	8	5.5%		10	0.3231	11.30%	11.33%	6.4E-08				
13	K	64.0000	9	5.8%		11	0.2945	11.11%	11.16%	1.9E-07	<--=(H13-I13)^2			
14	L	66.6158	9	6.0%		12	0.2687	10.95%	11.01%	3.4E-07	<--=(H14-I14)^2			
15	M	68.0969	10	6.5%										
16	N	70.0490	11	7.0%				=-LN(G14)/F14						
17	O	72.3657	12	7.5%										
18							{=MMULT(MINVERSE(MMULT(TRANSPOSE(C23:N37),C23:N37)),MMULT(TRANSPOSE(C23:N37),B3:B17))}							
19														

The graph below shows the discount factors and the fitted NS term structure:



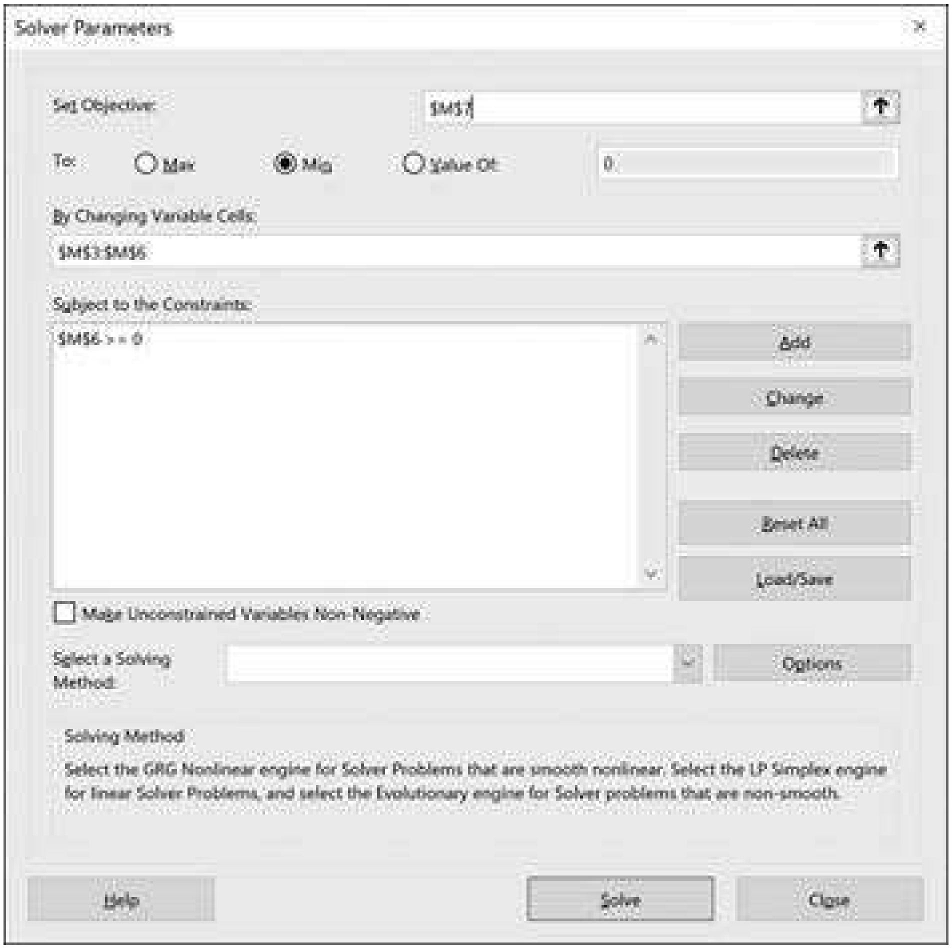
Computational Procedure

Here is how we computed the NS term structure above:

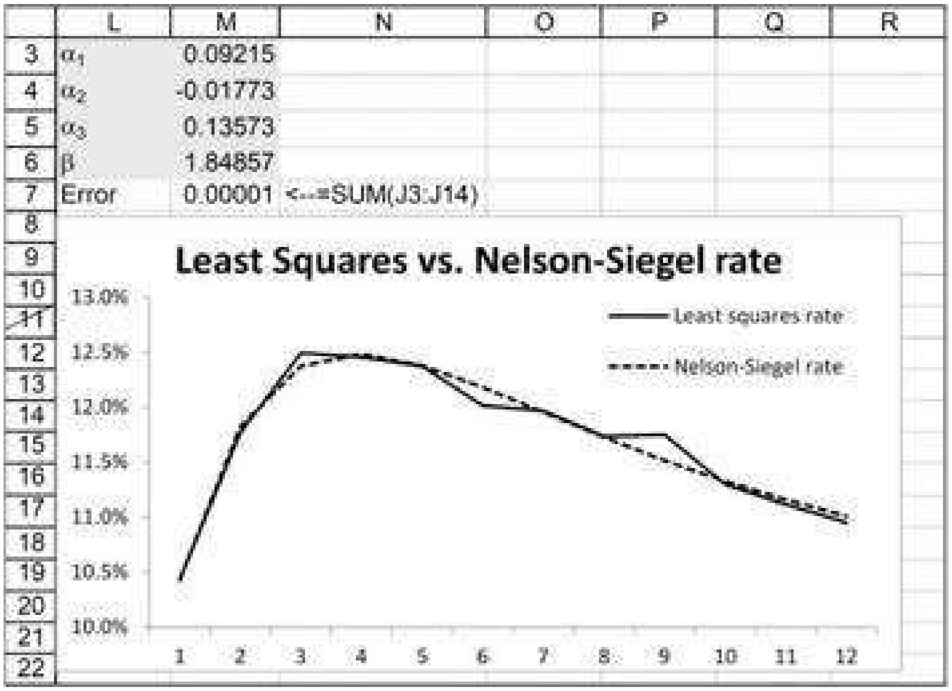
- **Step 1**: We start by computing the discount factors d_t and the corresponding yields $y(t)$ for the data in a method similar to that discussed in sections 8.3–8.4. In the example above, we label these $d^{LS}(t)$ and $y^{LS}(t)$.
- **Step 2**: We now assume arbitrary but reasonable values for α_1, α_2, α_3, and β. In the next section we discuss what these values might be.
- **Step 3**: Given the arbitrary starting values for α_1, α_2, α_3, and β, the NS discount zero coupon yields

$$y^{NS}(t) = \alpha_1 + (\alpha_2 + \alpha_3)\beta\left(\frac{1-e^{-t/\beta}}{t}\right) - \alpha_3 e^{-t/\beta}.$$

- **Step 4**: We now optimize α_1, α_2, α_3, and β to minimize the sum of the squares between the NS and LS: $\sum_{i=1}^{N}[y^{LS}(t) - y^{NS}(t)]^2$, where N is the number of bonds in our sample. This is performed by using Excel's **Solver** to find (α_1, α_2, α_3, β) that minimizes this summed squared difference. Here's the dialog box:



As we will see in the next section, beta controls the "hump" in the term structure and must be positive. Here's the solution produced by **Solver**:



8.6 The Properties of the Nelson-Siegel Term Structure

Fitting the NS term structure to data requires that we determine reasonable initial values for the parameters (α_1, α_2, α_3, β). In this section we examine the NS to explain what these values might be.

The NS zero-coupon yield is:

$$y(t) = \alpha_1 + \alpha_2\left(\beta\left(\frac{1-e^{-t/\beta}}{t}\right)\right) + \alpha_3\left(\beta\left(\frac{1-e^{-t/\beta}}{t}\right) - e^{-t/\beta}\right)$$
$$= \alpha_1 + (\alpha_2+\alpha_3)\left(\beta\left(\frac{1-e^{-t/\beta}}{t}\right)\right) - \alpha_3 e^{-t/\beta}$$

NS Property 1: The Shortest-Term Rate $y(0)$

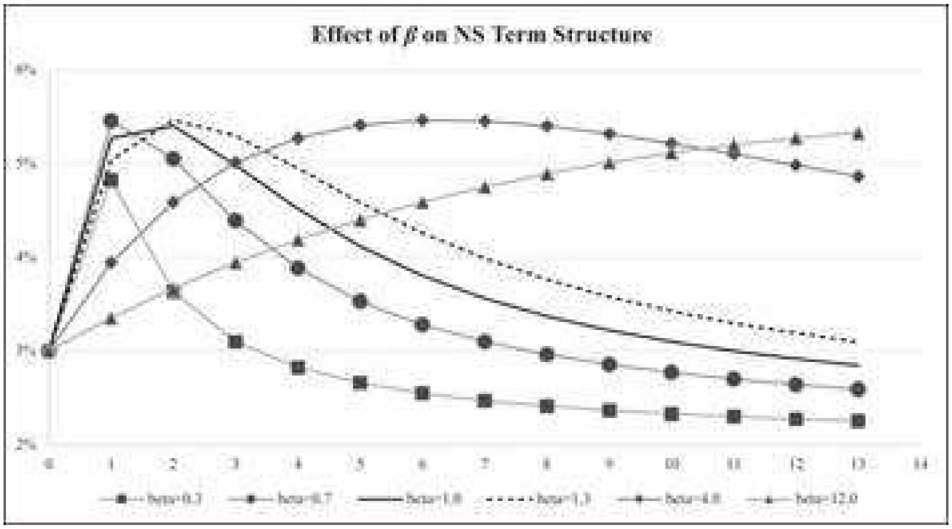
Setting $t = 0$ gives $y(0) = \alpha_1 + \alpha_2$. This is the NS shortest-term rate. It follows that for most term structures $\alpha_1 + \alpha_2 > 0$.

NS Property 2: The Longest-Term Rate $y(\infty)$

Setting $t \to \infty$ gives $y(\infty) = \alpha_1$. This is the asymptotic long-term interest rate in the NS model.

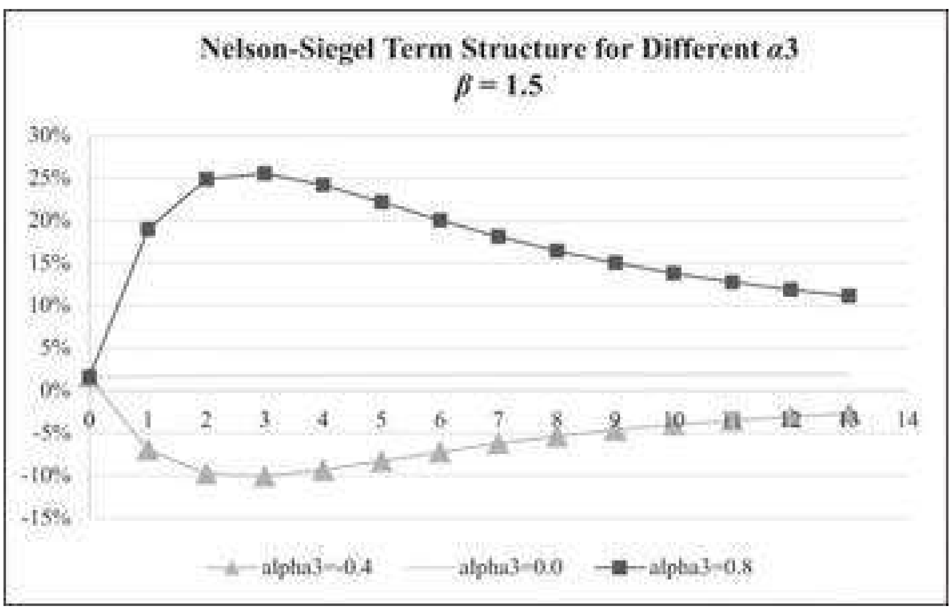
NS Property 3: β Controls the Location of the Term Structure Hump

The NS term structure is defined only if $\beta > 0$. If the term structure has a “hump,” this can be controlled by the β. In the graph below we illustrate that large betas (e.g., β = 12) produce upward-sloping term structures, whereas lower βs produce humpy term structures. Increasing the beta pushes the hump out. Roughly speaking, β is the location of the hump.



NS Property 4: α_3 Affects the Hump

The factor α_3 has no influence at either very short or very long yields and contributes to the term structure at medium-term yields only. Very roughly speaking, $\alpha_3 > 0$ produces a concave term structure, $\alpha_3 < 0$ produces a convex term structure, and $\alpha_3 = 0$ produces a flat term structure.



Nelson-Siegel: Summing Up

Fitting a term structure to the NS model involves setting initial values for $(\alpha_1, \alpha_2, \alpha_3, \beta)$. Plausible initial values can be set by "eyeballing" the empirical term structure and setting:

- • $\alpha_1 + \alpha_2$ = approximate zero-term interest rate
- • α_1 = approximate long-term rate
- • β = approximate location of the hump
- • α_3 affects the concavity/convexity of the term structure

8.7 Term Structure for Treasury Notes

In this section we fit the NS model to the prices of Treasury notes. This is a much larger data set than the one discussed in the previous section, and it allows us to illustrate a different optimizing technique.

We start with the data set of prices for Treasury notes (i.e., bonds with maturities of less than 10 years). The date of the data is March 31, 1989.

	A	B	C	D	E	F	G
1	309 Treasury Notes on 30-September-2019						
2	Current date	30-Sep-19					
3	Maturity	Coupon	Clean price	Yield	Accrued interest	Invoice price	
4	15-Oct-19	1.000	99.31	1.817	0.459	99.769	<--=E4+C4
5	31-Oct-19	1.250	99.31	1.631	0.520	99.830	<--=E5+C5
6	31-Oct-19	1.500	99.31	1.880	0.624	99.934	
7	15-Nov-19	1.000	99.29	1.768	0.375	99.665	
8	15-Nov-19	3.375	100.06	1.758	1.266	101.328	
9	30-Nov-19	1.000	99.27	1.860	0.333	99.607	
310	15-Feb-49	3.000	119.07	2.118	0.375	119.447	
311	15-May-49	2.875	116.21	2.116	1.078	117.284	
312	15-Aug-49	2.250	102.27	2.121	0.281	102.547	

=COUPDAYBS($B$2,A4,2,1)/COUPDAYS($B$2,A4,2,1)*(B4/2)

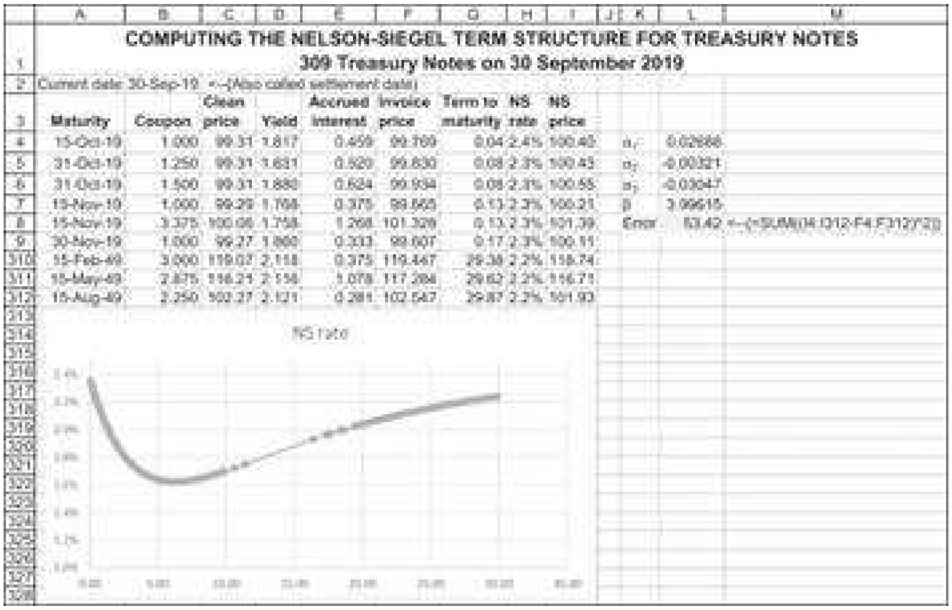
The "clean price" of column C above is the quoted price of the bond. In U.S. markets, the actual price paid by the purchaser of the bond, the so-called invoice price, is the sum of the clean price plus the relative portion of the current coupon payment.[4] This latter term, the "accrued interest," is computed as follows:

$$\textit{Accrued interest} = \frac{\textit{Current date} - \textit{Last interest date}}{\textit{Next interest date} - \textit{Last interest date}} * \textit{Periodic interest}$$

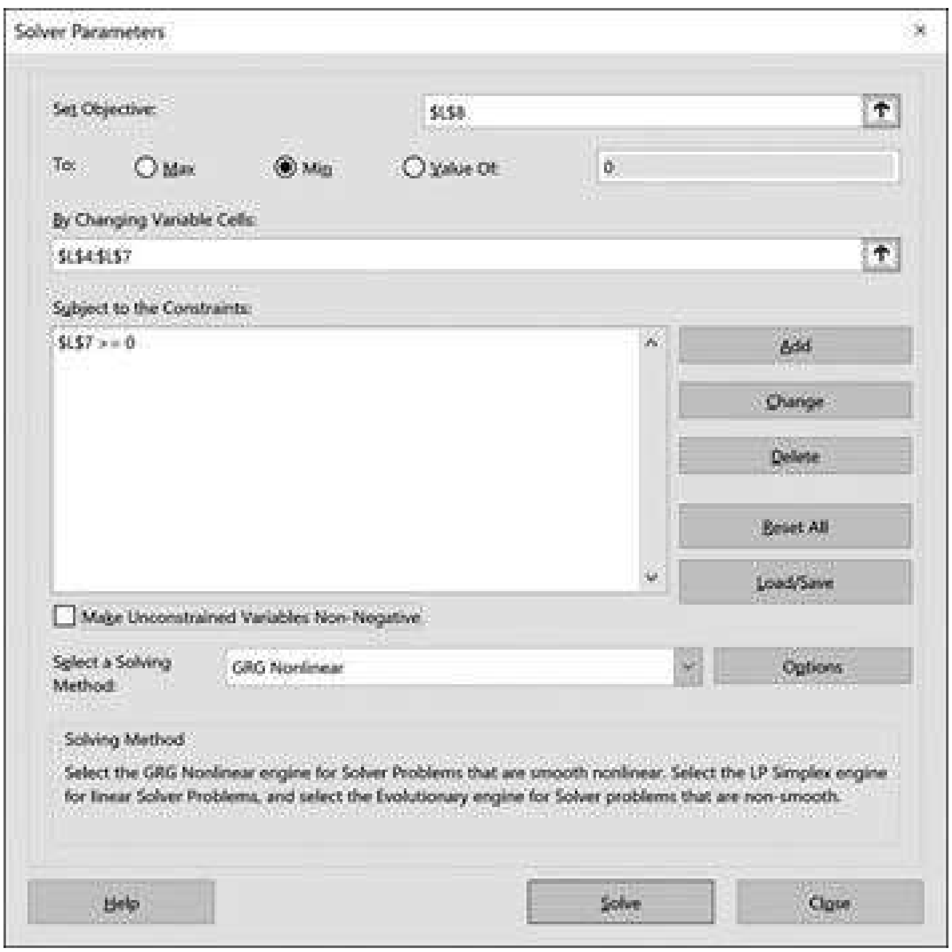
The Excel file computes the accrued interest using the formulas **Coupdaybs()** and **Coupdays()**.

Computing the NS Term Structure

To compute the NS term structure for this data, we define a VBA function **NSprice** that accepts $(\alpha_1, \alpha_2, \alpha_3, \beta)$ as input and computes the Nelson-Siegel price of a bond. We now optimize using **Solver** to minimize the sum of the absolute differences between the **NSprice** and the bond's invoice price. Here is the outcome of our optimization after asking **Solver** to minimize the error term in cell L8. The result is:



And this is the Solver dialog box:



8.8 Summary

In this chapter we present several simple and powerful mathematical techniques for term structure calculation. The Nelson-Siegel approximation was introduced in detail and its applications were discussed. We began with the simplest case, in which there is only one bond for each maturity so that the term structure can be found uniquely. Next, we extended the method to a set in which several bonds can have the same maturity. Finally, we discuss a real-life case of Treasury notes that can have any maturity date.

Appendix: VBA Functions Used in This Chapter

The function **NSrate** computes the Nelson-Siegel rate for time *t* based on specific values of the four NS parameters α_1, α_2, α_3, β:

```
Function NSrate(alpha1, alpha2, alpha3, beta, t)
    If t = 0 Then
        NSrate = alpha1 + alpha2
    Else
        NSrate = alpha1 + (alpha2 + alpha3) * (beta / t) * _ (1 -
        Exp(-t / beta)) - alpha3 * Exp(-t / beta)
    End If
End Function
```

The function **NSdiscount** computes the NS discount factors.

```
Function NSdiscount(alpha1, alpha2, alpha3, beta, t)
    If t = 0 Then
        NSdiscount = 1
    Else
        NSdiscount = Exp(-t * (alpha1 + (alpha2 + alpha3) * (beta /
        t) * _
        (1 - Exp(-t / beta)) - alpha3 * Exp(-t / beta)))
    End If
End Function
```

The function **NSprice** computes the price of a standard coupon bond using the Nelson-Siegel term structure:

```
Function NSprice(alpha1, alpha2, alpha3, beta, j, no_payments,
rate, _ frequency)
    temp = 0
    rrate = rate / frequency
    For i = 0 To no_payments - 1
    temp = temp + rrate * 100 * _
```

```
        NSdiscount(alpha1, alpha2, alpha3, beta, j + i /
        frequency)
    Next i
    NSprice = temp + 100 * _
        NSdiscount(alpha1, alpha2, alpha3, beta, j + _ no_payments
        - 1) / frequency)
End Function
```

The function **NSImpliedRate** uses Excel functions to compute the implied rates.

```
Function NSImpliedRate (Maturity, Settlement, couponRate,
faceValue, bondPrice, frequency, alpha1, alpha2, alpha3, beta,
Basis)
Dim termStruct, NSalpha(), numberPayments, timeToMaturity,
discountSum, n As Variant
numberPayments = WorksheetFunction.CoupNum(Settlement, Maturity,_
frequency, Basis)
timeToFirstPayment = WorksheetFunction.CoupDaysNc(Settlement,_
Maturity, frequency, Basis) /
WorksheetFunction.CoupDays(Settlement,_ Maturity, frequency,
Basis) / frequency
timeToMaturity = timeToFirstPayment + (numberPayments - 1) /
frequency
discountSum = 0
    For i = 1 To numberPayments
        discountSum = discountSum + NSdiscount (alpha1, alpha2,_
        alpha3, beta, timeToFirstPayment + (i - 1) / frequency)
    Next i
    termStruct = -WorksheetFunction.Ln((bondPrice - couponRate *_
    discountSum / frequency) / faceValue) / timeToMaturity
NSImpliedRate = termStruct
End Function
```

---

1. * This chapter was coauthored by Dr. Alexander Suhov of Tel Aviv University, a.y.suhov@gmail.com.
2. 1. Those familiar with the Treasury bond strip market will recognize the factors d_t as the discount factors associated with Treasury strips. See http://www.treasurydirect.gov/instit/marketables/strips/strips.htm.

3. 2. In the next section we discuss the case where there are multiple bonds for the same maturity with possibly inconsistent pricing.
4. 3. Fitting the NS term structure to a set of bond data is an art! Subsequent sections of this chapter discuss alternative methods to the one described in this section.
5. 4. In most European bond markets, the quoted bond price is the actual price paid for the bond, and there is no separate accrued interest calculation.

9 Calculating Default-Adjusted Expected Bond Returns

9.1 Overview

In this chapter we discuss the effects of default risk on the returns from holding bonds to maturity. The *expected return* on a bond that may possibly default is different from the bond's *promised return*. The latter is defined as the bond's *yield to maturity*, the internal rate of return calculated from the bond's current market price and its *promised* coupon payments and *promised eventual return* of principal in the future. The bond's expected return is less easily calculated: We need to take into account both the bond's probability of future default and the *recovery rate*, the percentage of its principal that holders can expect to recover in the case of default. To complicate matters still further, default can happen in stages, through the gradual degradation of the issuing company's creditworthiness.[1]

Here we use a Markov model to solve for the expected return on a risky bond. Our adjustment procedure takes into account all three of the factors mentioned: (1) the probability of default, (2) the transition of the issuer from one state of creditworthiness to another, and (3) the percentage recovery of face value when the bond defaults. In sections 9.2–9.4 we first use Excel to solve a relatively small-scale problem. We then use some publicly available statistics to program a fuller spreadsheet model. Finally, we show that this model can be used to derive bond betas, the capital asset pricing model's risk measure for securities (discussed in Chapters 10–13).

Some Preliminaries

Before proceeding, we need to define a number of terms:

- • A bond is issued with a given amount of *principal* or *face value*. When the bond matures, the bondholder is promised the return of this principal. If the bond is issued *at par*, then it is sold for the principal amount.
- • A bond bears an interest rate called the *coupon rate*. The periodic payment promised to the bondholders is the product of the coupon rate times the bond's face value.
- • At any given moment, a bond will be sold in the market for a *market price*. This price may differ from the bond's coupon rate.[2]
- • The bond's *yield to maturity* (YTM) is the internal rate of return (IRR) of the bond, assuming that it is held to maturity and that it does not default.

American corporate bonds are rated by various agencies on the basis of the bond issuer's ability to make repayment on the bonds. The classification scheme for two of the major rating agencies, Standard & Poor's (S&P) and Moody's, is given below:

Long-Term Senior Debt Ratings

Investment-Grade Ratings			Speculative-Grade Ratings		
S&P	Moody's	Interpretation	S&P	Moody's	Interpretation
AAA	Aaa	Highest quality	BB+ BB BB	Ba1 Ba2 Ba3	Likely to fulfill obligations; ongoing uncertainty
AA+ AA AA–	Aa1 Aa2 AA3	High quality	B+ B B–	B1 B2 B3	High-risk obligations
A+ A A–	A1 A2 A3	Strong payment capacity	CCC+ CCC CCC–	Caa	Current vulnerability to default
BBB+ BBB BBB–	Baa1 Baa2 Baa3	Adequate payment capacity	C D	Ca D	In bankruptcy or default, or other marked shortcomings

When a bond defaults, its holders will typically receive some payoff, though less than the promised bond coupon rate and return of principal. We refer to the percentage of face value paid off in default as the *recovery percentage*.[3]

9.2 Calculating the Expected Return in a One-Period Framework

The bond's yield to maturity is *not* its expected return. It is clear that both a bond's rating and the anticipated payoff to bondholders in the case of bond default should affect its expected return. All other things being equal, we would expect that if two newly issued bonds have the same term to maturity, then the lower-rated bond (having the higher default probability) should have a higher coupon rate or lower price (or both). Similarly, we would expect that an issued and traded bond whose rating has been lowered would experience a decrease in price. We might also expect that the lower the anticipated payoff in the case of default, the lower will be the bond's expected return.

As a simple illustration, we calculate the expected return of a one-year bond that can default at maturity. We use the following symbols:

- F = face value of the bond
- P = price of bond

- Q = annual coupon rate of the bond
- π = probability that the bond will *not* default at end of year
- λ = fraction of bond's value that bondholders collect upon default (also called *recovery rate*)

The bond's expected end-of-year cash flow is $\pi \cdot (1 + Q) \cdot F + (1 - \pi) \cdot \lambda \cdot F$, and its *expected return* is given as follows:

$$\text{One-year bond expected return} = \frac{\text{Expected year-end cash flow}}{\text{Initial bond price}} - 1$$

$$E(r) = \frac{\pi \cdot (1+Q) \cdot F + (1-\pi) \cdot \lambda \cdot F}{P} - 1$$

This calculation is illustrated in the spreadsheet below:

	A	B	C
1	**EXPECTED RETURN ON A ONE-YEAR BOND WITH AN ADJUSTMENT FOR DEFAULT PROBABILITY**		
2	Face value, F	100	
3	Price, P	90	
4	Annual coupon rate, Q	8%	
5	Default probability, (1-π)	20%	
6	Recovery percentage, λ	40%	
7			
8	Expected period 1 cash flow	94.4	<--=B2*(1+B4)*(1-B5)+B2*B6*B5
9	Expected return	4.89%	<--=B8/B3-1

9.3 Calculating the Bond Expected Return in a Multi-period Framework

We now introduce multiple periods into the above problem. In this section we define a basic Markov model that uses a ratings transition matrix to compute a bond's expected return. The model is illustrated using a very simple set of ratings, much simpler than the complex rating system illustrated in section 9.1. Later, in section 9.6 we present more realistic data.

We suppose that at any date there are four possible bond "ratings":

A, B, C	**Bond ratings of solvent bonds in decreasing order of credit worthiness.**
D	The bond is in default for the first time and pays off recovery rate λ of the face value.
E	The bond was in default in the previous period; it therefore pays off 0 in the current period and in any future periods.

The *transition probability* matrix Π is given as follows:

$$\Pi = \begin{bmatrix} \pi_{AA} & \pi_{AB} & \pi_{AC} & \pi_{AD} & 0 \\ \pi_{BA} & \pi_{BB} & \pi_{BC} & \pi_{BD} & 0 \\ \pi_{CA} & \pi_{CB} & \pi_{CC} & \pi_{CD} & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}$$

The probabilities in each row ($\pi_{i,j}$) of the matrix Π indicate the probability that in *one period* the bond will go from a rating of i to a rating of j. In the numerical examples in this and the following two sections, we use the following Π:

	A	B	C	D	E	F
2	One-period transition matrix					
3		A	B	C	D	E
4	A	0.97	0.02	0.01	0.00	0.00
5	B	0.05	0.80	0.15	0.00	0.00
6	C	0.01	0.02	0.75	0.22	0.00
7	D	0.00	0.00	0.00	0.00	1.00
8	E	0.00	0.00	0.00	0.00	1.00

What does this matrix Π mean?

- • If a bond is rated A in the current period, there is a probability of 0.97 that it will still be rated A in the next period. There is a probability of 0.02 that it will be rated B in the next period and a probability of 0.01 that it will be rated C. It is impossible for the bond to be rated A today and D or E in the subsequent period.
- • A bond that starts off with a rating of B can—in a subsequent period—be rated A (with a probability of 0.05), B (with a probability of 0.8), or C (probability 0.15). Bonds rated B in the current period do not default (rating D) in the next period. The transition probabilities from state C to states A, B, C, and D are 0.01, 0.02, 0.75, and 0.22, respectively.
- • While it is possible to go from ratings A, B, or C to any of ratings A, B, C, or D, it is *not* possible to go from A, B, or C to E. This is so because E denotes that default took place in the *previous period*.
- • A bond that is currently in state D (i.e., first-time default) will necessarily be in E in the next period. Thus, the fourth row of our matrix Π will always be [0 0 0 0 1].
- • Once the rating is in E, it remains there permanently. This means that the fifth row of the matrix Π will also always be [0 0 0 0 1].

The Multi-period Transition Matrix

The matrix Π defines the transition probabilities over one period. The two-period transition probabilities are given by the matrix product $\Pi * \Pi$. The spreadsheet below uses the array function **MMult**.[4] It shows that the product $\Pi * \Pi$ is:

Two-period Transition probability $= \Pi * \Pi$

And the result is:

	A	B	C	D	E
A	0.9420	0.0356	0.0202	0.0022	0.0000
B	0.0900	0.6440	0.2330	0.0330	0.0000
C	0.0182	0.0312	0.5656	0.1650	0.2200
D	0.0000	0.0000	0.0000	0.0000	1.0000
E	0.0000	0.0000	0.0000	0.0000	1.0000

Thus, if a bond is rated “B” today, there is a probability of 9% that in two periods it will be rated “A,” a probability of 64.4% that in two periods it will be rated B, a probability of 23.3% that in two periods it will rated C, and a probability of 3.3% that in two periods it will default (and hence be rated D).

Here is the spreadsheet:

	A	B	C	D	E	F
1	**USING THE MMULT FUNCTION To compute two-period transition matrices**					
2	**One-period transition matrix**					
3		**A**	**B**	**C**	**D**	**E**
4	**A**	0.97	0.02	0.01	0.00	0.00
5	**B**	0.05	0.80	0.15	0.00	0.00
6	**C**	0.01	0.02	0.75	0.22	0.00
7	**D**	0.00	0.00	0.00	0.00	1.00
8	**E**	0.00	0.00	0.00	0.00	1.00
9						
10	**Two-period transition matrix**					
11		**A**	**B**	**C**	**D**	**E**
12	**A**	0.9420	0.0356	0.0202	0.0022	0.0000
13	**B**	0.0900	0.6440	0.2330	0.0330	0.0000
14	**C**	0.0182	0.0312	0.5656	0.1650	0.2200
15	**D**	0.0000	0.0000	0.0000	0.0000	1.0000
16	**E**	0.0000	0.0000	0.0000	0.0000	1.0000
17	**Cells B12:F16 contain the array formula {=MMULT(B4:F8,B4:F8)}**					

In general, the year t transition matrix is given by the matrix power Π^t. Calculating these matrix powers by the procedure illustrated above is cumbersome, so we define a VBA function **Matrixpower** to compute powers of matrices:

```
Function Matrixpower(matrix, n)
    If n = 1 Then
        Matrixpower = matrix
```

```
        Else: Matrixpower = Application.MMult(Matrixpower(matrix,
        n - 1), matrix)
    End If
End Function
```

The use of this function is illustrated below. The function **Matrixpower** allows a one-step computation of the power of any transition matrix:

	A	B	C	D	E	F
1	**USING THE FUNCTION MATRIXPOWER** **To compute multi-period transition matrices**					
2	One-period transition matrix					
3		A	B	C	D	E
4	A	0.97	0.02	0.01	0.00	0.00
5	B	0.05	0.80	0.15	0.00	0.00
6	C	0.01	0.02	0.75	0.22	0.00
7	D	0.00	0.00	0.00	0.00	1.00
8	E	0.00	0.00	0.00	0.00	1.00
9						
10	t	10				
11						
12	t-period transition matrix					
13		A	B	C	D	E
14	A	0.7648	0.0799	0.0699	0.0148	0.0706
15	B	0.2123	0.1429	0.1747	0.0432	0.4269
16	C	0.0450	0.0250	0.0755	0.0208	0.8338
17	D	0.0000	0.0000	0.0000	0.0000	1.0000
18	E	0.0000	0.0000	0.0000	0.0000	1.0000
19	Cells B14:F18 contain the array formula {=matrixpower(B4:F8,B10)}					

From the above example it follows that if a bond started out with an A rating, there is a probability of 1.48% that the bond will be in default at the end of 10 periods and a 7.06% probability that it will have defaulted in a previous period (rating E).

The Bond Payoff Vector

Recall that Q denotes the bond's coupon rate and λ denotes the recovery percentage payoff of face value if the bond defaults. The payoff vector of the bond depends on whether the bond is currently in its last period N or whether $t < N$:

$$Payoff\,(t, t<N) = \begin{cases} F*Q \\ F*Q \\ F*Q \\ F*(Q+\lambda) \\ 0 \end{cases} \qquad Payoff\,(t, t=N) = \begin{cases} F*(1+Q) \\ F*(1+Q) \\ F*(1+Q) \\ F*(Q+\lambda) \\ 0 \end{cases}$$

The first three elements of each vector denote the payoff in non-defaulted states; the fourth element, $F * (Q + \lambda)$, is the payoff if the rating is D (recovery of λ from the principal plus a coupon of $F * Q$); and the fifth element, 0, is the payoff if the bond rating is E. Recall that E is the rating for the period *after* the bond defaults—in our model the payoff in rating E is always zero. The distinction between the two vectors depends, of course, on the repayment of principal in the terminal period.

Before we can define the expected payoffs, we need to define one further vector, which will denote the *initial state of the bond*. This current state vector is a vector with a 1 for the current rating of the bond and zeros elsewhere. Thus, for example, if the bond has rating A at date 0, then *initial* = [1 0 0 0 0]; if it has a date 0 rating of B, then *initial* = [0 1 0 0 0].

We can now define the expected bond payoff in period t:

$$E[payoff(t)] = initial \cdot \Pi^t \cdot payoff(t)$$

9.4 A Numerical Example

We continue using the numerical Π from the previous section to price a bond with the following characteristics:

- • The bond is currently rated B.
- • The face value is 1,000.
- • Its coupon rate Q = 7%.
- • The bond has five more years to maturity.
- • The bond's current market price is traded at par (price = face value).
- • The bond's recovery percentage λ = 40%.

The spreadsheet below shows the facts listed above, as well as the payoff vectors of the bond at dates before maturity (in cells E3:E7) and on the maturity date (cells F3:F7). The transition matrix is given in cells B11:F15, and the initial vector is given in B17:F17.

The expected bond payoffs are given in row 20. Before we explain how they were calculated, we note the important economic fact that—if the expected payoffs are used —then the *bond's expected return* is calculated by the Excel **IRR** function. As cell B21 shows, this expected return is 4.18%. The actual formula in cell B21 is **IRR(20:20)**.

	A	B	C	D	E	F	G	H	I
1	**CALCULATING THE EXPECTED BOND RETURN**								
2	Face value, F	1,000		Rating	Payoff (t<N)	Payoff (N)			
3	Bond price, P	1,000		A	70	1,070	<--=$B$2*(1+$B$4)		
4	Coupon rate, Q	7%		B	70	1,070			
5	Recovery rate, λ	40%		C	70	1,070			
6	Bond term, N	5		D	470	470	<--=$B$2*($B$5+$B$4)		
7	Initial rating	B		E	0	0			
8			=$B$2*($B$5+$B$4)				=$B$4*$B$2		
9	**Transition matrix**								
10		**A**	**B**	**C**	**D**	**E**			
11	**A**	0.9700	0.0200	0.0100	0.0000	0.0000			
12	**B**	0.0500	0.8000	0.1500	0.0000	0.0000			
13	**C**	0.0100	0.0200	0.7500	0.2200	0.0000			
14	**D**	0.0000	0.0000	0.0000	0.0000	1.0000			
15	**E**	0.0000	0.0000	0.0000	0.0000	1.0000			
16									
17	**Initial vector**	0	1	0	0	0	<--=IF(UPPER($B$7)=F10,1,0)		
18									
19	**Year**	**0**	**1**	**2**	**3**	**4**	**5**	**6**	**7**
20	Expected payoffs	-1,000	70.00	83.20	88.19	88.06	878.31	0.00	0.00
21	Expected yield (IRR of expected payoff)	4.48%	<--=IRR(20:20)						
22									
23			=IF(E19>$B$6,0,MMULT($B$17:$F$17,MMULT(matrixpower($B$11:$F$15,E19),IF(E19=$B$6,$F$3:$F$7,$E$3:$E$7))))						
24									

Note the use of the **IF** statement in translating the bond's initial rating (cell B6) to the initial vector given in row 17. To avoid confusion, we write this as **IF(Upper(B7)="A",1,0)**. This guarantees that even if the bond's rating is entered as a lowercase letter, the initial vector will come out correctly.

How to Calculate the Expected Bond Payoffs

As indicated in section 9.3 above, the period t expected bond payoff is given by the following formula: $E[payoff\,(t)] = initial \cdot \Pi^t \cdot payoff\,(t)$. The formula in row 20 uses two **IF** statements to implement this formula:

= ***IF*** [***Current Year*** **>** *Bond term, 0,*
MMULT (***Initial, MMULT*** (***Matrixpower*** (***transition, year***), ***IF***(***Current year= Bond term, Payoff*** (*N*), ***Payoff*** (***t*** < *N*))))] =

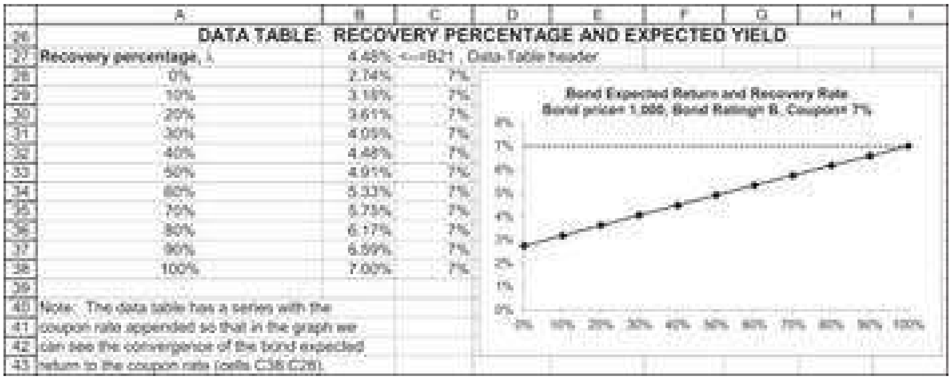
Here's what this means:

- • **First IF**: If the current year is greater than the bond term N (in our example N = 5), then the payoff on the bond is 0.
- • **On every period:** we use the equation: $E[payoff\,(t)] = initial \cdot \Pi^t \cdot payoff\,(t)$.
- • **Second IF**: If the current year is equal to the bond term N, then the expected payoff on the bond is using the payoff vector where $t = N$, otherwise it is using the payoff vector where $t < N$.

Copying this formula gives the whole vector of expected bond payoffs.

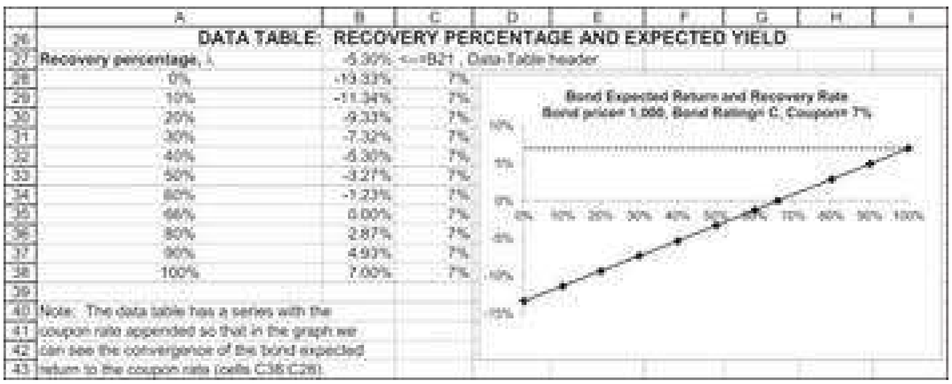
9.5 Experimenting with the Example

We can gain some insight into the relation between a bond's expected return, its coupon rate, and its YTM by constructing some data tables. In the data table below we compute the bond's expected return as a function of its recovery rate (λ):



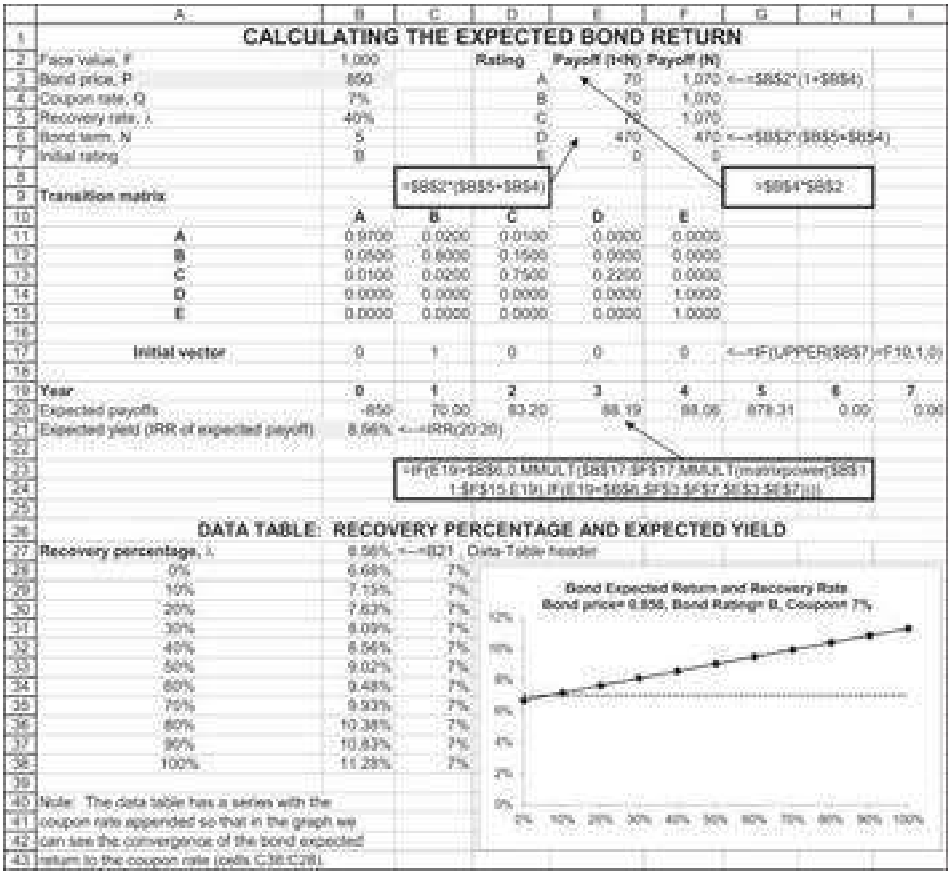
	A	B	C
26	DATA TABLE:	RECOVERY PERCENTAGE AND EXPECTED YIELD	
27	Recovery percentage, λ	4.48%	<--=B21 , Data-Table header
28	0%	2.74%	7%
29	10%	3.18%	7%
30	20%	3.61%	7%
31	30%	4.05%	7%
32	40%	4.48%	7%
33	50%	4.91%	7%
34	60%	5.33%	7%
35	70%	5.75%	7%
36	80%	6.17%	7%
37	90%	6.59%	7%
38	100%	7.00%	7%
39			
40	Note: The data table has a series with the		
41	coupon rate appended so that in the graph we		
42	can see the convergence of the bond expected		
43	return to the coupon rate (cells C38:C28).		

We conclude that a bond selling at par will, every recovery percentage $\lambda < 100\%$, have an expected return less than the coupon rate. If the bond's initial rating is lower, then the expected return is even negative for $\lambda < 66\%$:



	A	B	C
26	DATA TABLE:	RECOVERY PERCENTAGE AND EXPECTED YIELD	
27	Recovery percentage, λ	-5.30%	<--=B21 , Data-Table header
28	0%	-13.33%	7%
29	10%	-11.34%	7%
30	20%	-9.33%	7%
31	30%	-7.32%	7%
32	40%	-5.30%	7%
33	50%	-3.27%	7%
34	60%	-1.23%	7%
35	66%	0.00%	7%
36	80%	2.87%	7%
37	90%	4.93%	7%
38	100%	7.00%	7%
39			
40	Note: The data table has a series with the		
41	coupon rate appended so that in the graph we		
42	can see the convergence of the bond expected		
43	return to the coupon rate (cells C38:C28).		

When the bond price is below par (meaning that the bond is sold at less than 100% of its face value), and as shown in the example below, where p = 850, the bond's expected return can be both below and above its coupon rate.



	A	B	C	D	E	F	G	H	I
1	CALCULATING THE EXPECTED BOND RETURN								
2	Face value, F	1,000		Rating	Payoff (t<N)	Payoff (N)			
3	Bond price, P	850		A	70	1,070	<-- =$B$2*(1+$B$4)		
4	Coupon rate, Q	7%		B	70	1,070			
5	Recovery rate, λ	40%		C	70	1,070			
6	Bond term, N	5		D	470	470	<-- =$B$2*($B$5+$B$4)		
7	Initial rating	B		E	0	0			
8			=$B$2*($B$5+$B$4)				=$B$4*$B$2		
9	Transition matrix								
10		A	B	C	D	E			
11	A	0.9700	0.0200	0.0100	0.0000	0.0000			
12	B	0.0500	0.8000	0.1500	0.0000	0.0000			
13	C	0.0100	0.0200	0.7500	0.2200	0.0000			
14	D	0.0000	0.0000	0.0000	0.0000	1.0000			
15	E	0.0000	0.0000	0.0000	0.0000	1.0000			
16									
17	Initial vector	0	1	0	0	0	<-- =IF(UPPER($B$7)=F10,1,0)		
18									
19	Year	0	1	2	3	4	5	6	7
20	Expected payoffs	-850	70.00	83.20	88.19	88.06	878.31	0.00	0.00
21	Expected yield (IRR of expected payoff)	8.56%	<-- =IRR(20:20)						
22									
23			=IF(E19>$B$6,0,MMULT($B$17:$F$17,MMULT(matrixpower($B$11:$F$15,E19),IF(E19=$B$6,$F$3:$F$7,$E$3:$E$7))))						
24									
25									
26	DATA TABLE: RECOVERY PERCENTAGE AND EXPECTED YIELD								
27	Recovery percentage, λ	8.56%	<-- =B21 Data-Table header						
28	0%	6.68%	7%						
29	10%	7.15%	7%						
30	20%	7.63%	7%						
31	30%	8.09%	7%						
32	40%	8.56%	7%						
33	50%	9.02%	7%						
34	60%	9.48%	7%						
35	70%	9.93%	7%						
36	80%	10.38%	7%						
37	90%	10.83%	7%						
38	100%	11.28%	7%						
39									
40	Note: The data table has a series with the								
41	coupon rate appended so that in the graph we								
42	can see the convergence of the bond expected								
43	return to the coupon rate (cells C38:C28).								

9.6 Computing the Bond Expected Return for an Actual Bond

In this section we illustrate the computation of the expected return for an actual bond. Although the principles used are the same as those discussed above, we introduce three innovations:

- • We compute the bond's actual price using its quoted price and the accrued interest. The *accrued interest* is jargon for the unpaid part of the bond coupon since the last interest payment. In U.S. bond markets, the accrued interest is added to the quoted bond price to compute the amount actually paid for the bond. In most European bond markets, the quoted bond price is the actual price paid for the bond and there is no separate accrued interest calculation. The accrued interest is defined as:

$$\text{Accrued interest} = \frac{\text{Current date} - \text{Last interest date}}{\text{Next interest date} - \text{Last interest date}} * \text{Periodic interest}$$

- • We use actual payment dates for the bond and use the **XIRR** function to compute the bond's expected yield.
- • We use an actual transition matrix for bond ratings.

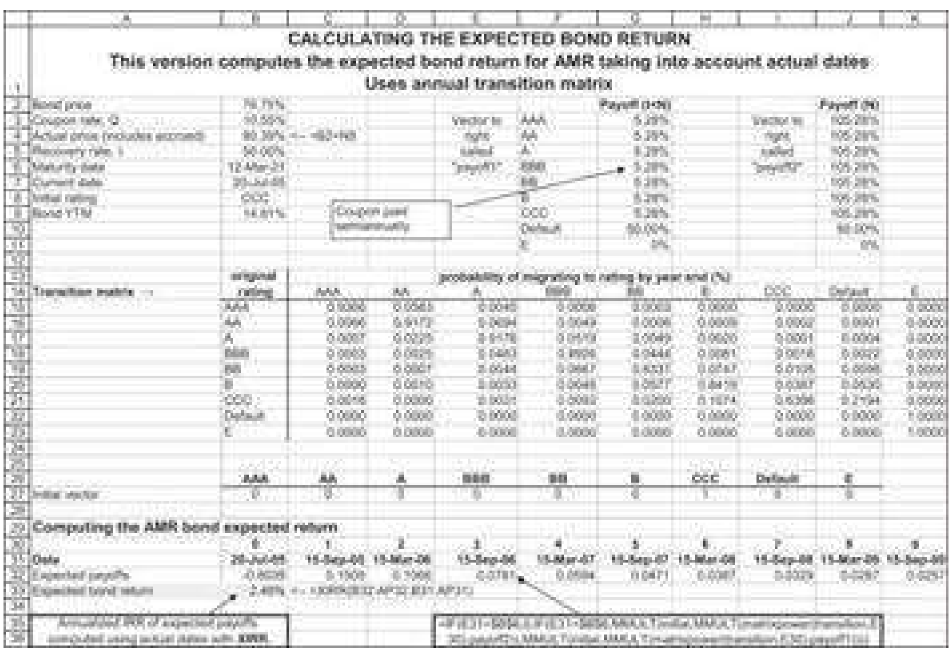
The bond we analyze is a CCC-rated bond issued by Hertz Corp. It was originally issued on August 1, 2019, and matures on August 1, 2026. The bond has a coupon of 7.125% payable semiannually on May 15 and November 15.

When we looked up the bond on October 15, 2019, its price was \$103.66. To the quoted price of \$103.66, we must add the bond's accrued interest:

	M	N	O
2	**Accrued interest and actual price calculation**		
3	Face value, F	100	
4	Coupon rate, Q	7.125%	
5	Last payment date	01-Aug-19	
6	Next payment date	01-Feb-20	
7	Current date	15-Oct-19	
8	Percentage of period	0.41	<--=(N7-N5)/(N6-N5)
9	Accrued interest	1.452	<--=N3*N4*N8/2
10	Quoted price	103.66	
11	Actual bond price, P	105.112	<--=N10+N9

It follows that the actual price paid for the bond is 103.66 + 1.452 = 105.112.

In the spreadsheet below we compute the AMR expected return. Assuming a recovery rate of 50%, the bond has an expected return of 2.49%.It follows that the actual price paid for the bond is 76.75% + 3.64% = 80.39%.



What Are the Recovery Rates?

The recovery rate is obviously a critical factor in computing the bond's expected return. Using historical data on the recovery rates in bankruptcy from various industries, the table below shows that the average recovery rate from a variety of industries was 41%.

Recovery Rates by Industry: Defaulted Bonds by Three-Digit SIC Code, 1971–1995

			Recovery Rate			
Industry	SIC Code	Number of Observations	Average	Weighted Observations	Median Average	Standard Deviation Weighted
Public utilities	490	56	70.47	65.48	79.07	19.46
Chemicals, petroleum, rubber and plastic products	280, 290, 300	35	62.73	80.39	71.88	27.10
Machinery, instruments, and related products	350, 360, 380	36	48.74	44.75	47.50	20.13
Services—business and personal	470, 632, 720, 730	14	46.23	50.01	41.50	25.03
Food and kindred products	200	18	45.28	37.40	41.50	21.67
Wholesale and retail trade	500, 510, 520	12	44.00	48.90	37.32	22.14
Diversified manufacturing	390, 998	20	42.29	29.49	33.88	24.98
Casino, hotel, and recreation	770, 790	21	40.15	39.74	28.00	25.66
Building materials, metals, and fabricated products	320, 330, 340	68	38.76	29.64	37.75	22.86
Transportation and transportation equipment	370, 410, 420, 450	52	38.42	41.12	37.13	27.98
Communication, broadcasting,	270, 480, 780	65	37.08	39.34	34.50	20.79

movies, printing, publishing						
Financial institutions	600, 610, 620, 630,	66	35.69	35.44	32.15	25.72
Construction and real estate	670 150, 650	35	35.27	28.58	24.00	28.69
General merchandise stores	530, 540, 560, 570, 580, 590	89	33.16	29.35	30.00	20.47
Mining and petroleum drilling	100, 103	45	33.02	31.83	32.00	18.01
Textile and apparel products	220, 230	31	31.66	33.72	31.13	15.24
Wood, paper, and leather products	240, 250, 260, 310	11	29.77	24.30	18.25	24.38
Lodging, hospitals, and nursing facilities	700 through 890	22	26.49	19.61	16.00	22.65
Total		**696**	**41.00**	**39.11**	**36.25**	**25.56**

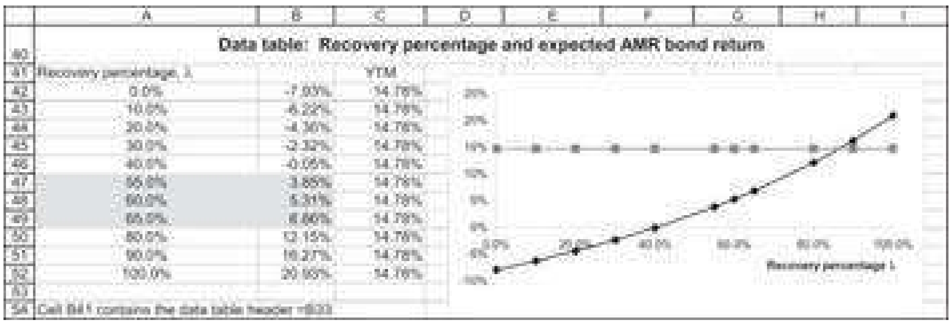
Source: Altman and Kishore (1996).

Using the Altman and Kishore numbers, the average recovery percentage for transportation companies is 38.42%, with a standard deviation of 27.98%. Taking one standard deviation on either side of the average, we can conclude that the recovery percentage for a transportation company is somewhere between (38.42% − 27.98%, 38.42% + 27.98%) = (~10%, ~66%).

In the spreadsheet below we have “backward engineered” a plausible set of recovery ratios, from 55% to 65%, for the AMR bond. These “guesstimates” for the AMR recovery is based on two assumptions:

- • The AMR bond should not have an expected return significantly more than the riskless rate of return, which at the time of our calculations was around ~4%.
- • The AMR bond expected return should be significantly less than its YTM of ~15%. The YTM is based on promised payments, and we find it implausible that these should correspond to the expected returns.

This gives (highlighted area below) an expected bond yield of between 3.85% and 6.86% for the AMR bond:



Data table: Recovery percentage and expected AMR bond return

Recovery percentage, λ		YTM
0.0%	-7.93%	14.78%
10.0%	-6.22%	14.78%
20.0%	-4.36%	14.78%
30.0%	-2.32%	14.78%
40.0%	-0.05%	14.78%
55.0%	3.85%	14.78%
60.0%	5.31%	14.78%
65.0%	6.86%	14.78%
80.0%	12.15%	14.78%
90.0%	16.27%	14.78%
100.0%	20.93%	14.78%

Cell B41 contains the data table header =B33

9.7 Semiannual Transition Matrices

The analysis of the AMR bond in the previous section assumes that the annual transition probabilities are also valid for bonds paying semiannual coupons. We could refine this assumption by computing a semiannual transition matrix from S&P data. Such a matrix would be the square root of the Π matrix. This is not a calculation that can be easily done in Excel. In the spreadsheet below we have used a computation from R to find the semiannual transition matrix.[5]

	A	B	C	D	E	F	G	H	I	J
1	**COMPUTING THE SQUARE ROOT OF THE TRANSITION MATRIX**									
2										
3					**The one-year transition matrix**					
4	**Original**			**Probability of migrating to rating by year end (%)**						
5	**rating**	AAA	AA	A	BBB	BB	B	CCC	Default	E
6	AAA	0.9366	0.0583	0.0040	0.0008	0.0003	0.0000	0.0000	0.0000	0.0000
7	AA	0.0066	0.9172	0.0694	0.0049	0.0006	0.0009	0.0002	0.0001	0.0000
8	A	0.0007	0.0225	0.9176	0.0519	0.0049	0.0020	0.0001	0.0004	0.0000
9	BBB	0.0003	0.0025	0.0483	0.8926	0.0444	0.0081	0.0016	0.0022	0.0000
10	BB	0.0003	0.0007	0.0044	0.0667	0.8331	0.0747	0.0105	0.0098	0.0000
11	B	0.0000	0.0010	0.0033	0.0046	0.0577	0.8419	0.0387	0.0530	0.0000
12	CCC	0.0016	0.0000	0.0031	0.0093	0.0200	0.1074	0.6396	0.2194	0.0000
13	Default	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	1.0000
14	E	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	1.0000
15										
16	**The square root of the one-year transition matrix as computed by R** **Note the negative entries**									
17	**Original**			**Probability of migrating to rating by year end (%)**						
18	**rating**	AAA	AA	A	BBB	BB	B	CCC	Default	E
19	AAA	0.9677	0.0303	0.0015	0.0004	0.0001	0.0000	0.0000	0.0000	0.0000
20	AA	0.0034	0.9574	0.0362	0.0021	0.0002	0.0004	0.0001	0.0000	0.0000
21	A	0.0003	0.0117	0.9573	0.0272	0.0023	0.0010	0.0000	0.0003	-0.0001
22	BBB	0.0001	0.0012	0.0254	0.9439	0.0238	0.0038	0.0008	0.0017	-0.0007
23	BB	0.0002	0.0003	0.0018	0.0359	0.9115	0.0406	0.0056	0.0068	-0.0026
24	B	0.0000	0.0005	0.0017	0.0018	0.0314	0.9161	0.0225	0.0510	-0.0248
25	CCC	0.0009	-0.0001	0.0016	0.0050	0.0105	0.0624	0.7988	0.2706	-0.1495
26	Default	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	1.0000
27	E	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	1.0000
28										
29	**The semiannual transition matrix. To eliminate the negative entries in the R matrix above:** **a. We assume that a transition from AAA, ... , CCC --> E is impossible (i.e.: last column is zero except for last two entries).** **b. We set all other negative entries to zero.** **c. We set default probability so that each row sums to 1.**									
30	**Original**			**Probability of migrating to rating by year end (%)**						
31	**rating**	AAA	AA	A	BBB	BB	B	CCC	Default	E
32	AAA	0.9677	0.0303	0.0015	0.0004	0.0001	0.0000	0.0000	0.0000	0.0000
33	AA	0.0034	0.9574	0.0362	0.0021	0.0002	0.0004	0.0001	0.0001	0.0000
34	A	0.0003	0.0117	0.9573	0.0272	0.0023	0.0010	0.0000	0.0001	0.0000
35	BBB	0.0001	0.0012	0.0254	0.9439	0.0238	0.0038	0.0008	0.0010	0.0000
36	BB	0.0002	0.0003	0.0018	0.0359	0.9115	0.0406	0.0056	0.0041	0.0000
37	B	0.0000	0.0005	0.0017	0.0018	0.0314	0.9161	0.0225	0.0261	0.0000
38	CCC	0.0009	0.0000	0.0016	0.0050	0.0105	0.0624	0.7988	0.1208	0.0000
39	Default	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	1.0000
40	E	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	1.0000

If we use the semiannual transition matrix to compute the expected bond returns we get:

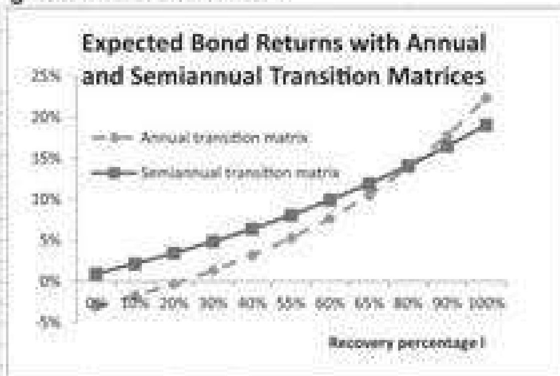
CALCULATING THE EXPECTED BOND RETURN
This version computes the expected bond return for AMR taking into account actual dates
Uses semiannual transition matrix

	A	B	C	D	E	F	G	H	I	J
2	Bond price	[illegible]					Payoff (t<N)			Payoff (N)
3	Coupon rate, C	10.55%			Vector to	AAA	5.28%		Vector to	105.28%
4	Actual price (includes accrued)	80.39% <-- [illegible]			right	AA	5.28%		right	105.28%
5	Recovery rate, λ	50.00%			called	A	5.28%		called	105.28%
6	Maturity date	12-Mar-21			"payoff1"	BBB	5.28%		"payoff2"	105.28%
7	Current date	20-Jul-05				BB	5.28%			105.28%
8	Maturity (years)	15.65				B	5.28%			105.28%
9	Number of semiannual payments	30.00	Coupon paid semiannually			CCC	5.28%			105.28%
10	Initial rating	CCC				Default	50.00%			50.00%
11	Bond YTM	14.81%				E	0%			0%

	A	B	C	D	E	F	G	H	I	J	K
13		Original			Probability of migrating to rating by year end (%)						
14	Transition matrix -->	rating	AAA	AA	A	BBB	BB	B	CCC	Default	E
15		AAA	0.9677	0.0303	0.0015	0.0004	0.0001	0.0000	0.0000	0.0000	0.0000
16		AA	0.0034	0.9574	0.0362	0.0021	0.0002	0.0004	0.0001	0.0001	0.0000
17		A	0.0003	0.0117	0.9573	0.0272	0.0023	0.0010	0.0000	0.0001	0.0000
18		BBB	0.0001	0.0012	0.0254	0.9439	0.0238	0.0038	0.0008	0.0010	0.0000
19		BB	0.0002	0.0003	0.0018	0.0359	0.9115	0.0406	0.0056	0.0041	0.0000
20		B	0.0000	0.0005	0.0017	0.0018	0.0314	0.9161	0.0225	0.0261	0.0000
21		CCC	0.0009	-0.0001	0.0016	0.0050	0.0105	0.0624	0.7988	0.1208	0.0000
22		Default	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	1.0000
23		E	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	1.0000
24											
25											
26		**AAA**	**AA**	**A**	**BBB**	**BB**	**B**	**CCC**	**Default**	**E**	
27	Initial vector	0	0	0	0	0	0	1	0	0	
28											
29	**Period**	**0**	**1**	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**
30	**Date**	**20-Jul-05**	**15-Sep-05**	**15-Mar-06**	**15-Sep-06**	**15-Mar-07**	**15-Sep-07**	**15-Mar-08**	**15-Sep-08**	**15-Mar-09**	**15-Sep-09**
31	Expected payoffs	-0.8039	0.1068	0.0963	0.0771	0.0664	0.0576	0.0509	0.0451	0.0405	0.0367
32	Expected yield	8.07% <-- =XIRR(B31:AP31,B30:AP30)									

The semiannual transition matrix gives expected bond returns that are, in general, higher than those given by the annual transition matrix:

	A	B	C
1	COMPARING THE EXPECTED RETURNS WITH SEMIANNUAL VS. ANNUAL TRANSITION MATRICES Using actual dates and XIRR		
2		Expected bond return	
3	Recovery percentage, λ	Annual transition matrix	Semiannual transition matrix
4	0%	-3.05%	0.93%
5	10%	-1.75%	2.14%
6	20%	-0.30%	3.45%
7	30%	1.32%	4.86%
8	40%	3.15%	6.40%
9	55%	5.25%	8.07%
10	60%	7.68%	9.90%
11	65%	10.51%	11.89%
12	80%	13.83%	14.07%
13	90%	17.73%	16.45%
14	100%	22.33%	19.04%



9.8 Computing Bond Beta

A vexatious problem in corporate finance is the computation of bond betas. The model presented in this chapter can be easily used to compute the beta of a bond. The capital asset pricing model's *security market line* (SML) is given as follows:

$$E(r_d) = r_f + \beta_d[E(r_m) - r_f]$$

where

$E(r_d)$ = *expected return on debt*
r_f = *return on riskless debt*
$E(r_m)$ = *return on equity market portfolio*

If we know the expected return on debt, we can calculate the debt β, provided we know the risk-free rate r_f and the expected rate of return on the market $E(r_m)$. Suppose, for example, that the market risk premium $E(r_m) - rf = 5\%$, and that $r_f = 1\%$. Then a bond having an expected return of 2% will have a β of 0.2:

	A	B	C
1	CALCULATING A BOND'S BETA		
2	Market risk premium, $E(r_m) - r_f$	5.00%	
3	r_f	1%	
4	Expected bond return	2.00%	
5	Implied bond beta	0.200	<--=(B4-B3)/B2

Using our data for AMR, we get, for the SML model:

	A	B	C
7	CALCULATING AMR's BOND BETA		
8	Market risk premium, $E(r_m) - r_f$	5.00%	
9	r_f	1%	
10	Expected bond return	2.49%	Using annual transition matrix
11	Implied bond beta	0.298	<--=(B10-B9)/B8

9.9 Summary

In this chapter we have shown how to compute the expected return on a risky bond using a simple technique involving rating transitions. Computing a bond's expected returns puts the bond analysis on the same footing as the analysis of stocks. Expected returns—common in the analysis of stocks—are rarely computed for bonds, where the common analysis is in terms of yields to maturity. But the yield to maturity of a bond, essentially the bond's **IRR** based on its promised future payments, includes an ill-defined premium for the bond's default.

Having computed a bond's expected return, we can then compute its beta using the SML. Compared to the vast efforts to compute and calibrate stock betas, relatively little research energy has been expended on bond betas. The technique illustrated in this chapter, based on the transition matrix of the bond ratings, is relatively new. This technique still has to be refined and thoroughly tested by academic research. Several refinements to the rating-based technique for computing expected bond returns still need to be explored, including:

- • *Better transition matrices*. Transition matrices need to be refined and perhaps made industry-specific. (The problem with industry-specific data is that the number of observations drops dramatically. Nevertheless, there are examples of such data—see, for example, Standard & Poor's 2020.)
- • *Time-dependent transition matrices*. Our technique assumes that transition matrices are stationary—constant through time. Perhaps better techniques can be developed that allow for matrices to change with time. For example, we would expect that in difficult economic conditions, the ratings transition matrix would "shift to the right"—that the probabilities of a given rating getting worse over any period would increase.
- • *More data on recovery ratios*. Our technique assumes a constant recovery rate ratio. Real-life situations are probably more dynamic and have recovery rates that vary over time and industry, and are also firm specific.

Improving the model may result in better understanding of risk and better estimation of the expected cost of debt.

Exercises

1. A newly issued bond with one year to maturity has a price of 100, which equals its face value. The coupon rate on the bond is 15%; the probability of default in one year is 35%; and the bond's payoff in default will be 65% of its face value.
    1. Calculate the bond's expected return.
    2. Create a data table showing the expected return as a function of the recovery percentage and the price of the bond.

2. Consider the case of five possible rating states: A, B, C, D, and E. A, B, and C are initial bond ratings, D symbolizes first-time default, and E indicates default in the previous period. Assume that the transition matrix Π is:

$$\Pi = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0.06 & 0.90 & 0.03 & 0.01 & 0 \\ 0.02 & 0.05 & 0.88 & 0.05 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}$$

A 10-year bond issued today at par with an A rating is assumed to bear a coupon rate of 7%.

- If a bond is issued today at par with a B rating and with a recovery percentage of 50%, what should be its coupon rate so that its expected return will also be 7%?
- If a bond is issued today at par with a C rating and with a recovery percentage of 50%, what should be its coupon rate so that its expected return will be 7%?

3. Using the transition matrix of the previous problem, assume a C-rated bond is selling at par on July 18, 2007. The bond's maturity is July 17, 2017; it has a coupon (paid annually on July 17) of 11%; and it has a recovery percentage of $\lambda = 67\%$. What is the bond's expected return?
4. An underwriter issues a new 7-year B-rated bond with a coupon rate 9%. If the expected rate of return on the bond is 8%, what is the bond's implied recovery percentage λ? Assume the transition matrix given in section 9.3.
5. An underwriter issues a new 7-year C-rated bond at par. The anticipated recovery rate in default of the bond is expected to be 55%. What should be the coupon rate on the bond so that its expected return is 9%? Assume the transition matrix of exercise 2.

---

1. 1. Besides default risk, bonds are also subject to term structure risk: The prices of bonds may show significant variations over time as a result of changing term structure. This statement will be especially true for long-term bonds. In this chapter we abstract from term structure risk, confining ourselves only to a discussion of the effects of default risk on bond expected returns.
2. 2. Just to complicate matters, in the United States bond prices are quoted as "*clean prices*"—that is, without the accumulated coupon between the time of the last coupon payment and the price quotation dated. Thus, the convention is to add to a bond's listed price the *prorated coupon* (the accrued interest) between the time of the last coupon payment and the purchase date. The sum of these two is termed the *invoice price* (or "*dirty price*") of the bond; the invoice price is the actual cost at any moment to a purchaser of buying the bond. In our discussion in this chapter, we use the term *market price* to denote the invoice price. The computation of the accrued interest is illustrated in section 9.5.
3. 3. The bond's recovery percentage is not, as you might think, the payoff to the bondholders in the final settlement of a bankruptcy. Instead, it is usually computed as the price of the bond in the period immediately following a financial distress event.
4. 4. See discussion of matrix products and array functions in Chapter 29.
5. 5. R is an open source high-powered computational program. See the R Project for Statistical Computing (https://www.r-project.org/).

III PORTFOLIO THEORY

10 Portfolio Models—Introduction

10.1 Overview

In this chapter we review the basic mechanics of portfolio calculations. We start with a simple example of two assets, showing how to derive the return distributions from historical price data. We then discuss the general case of N assets; for this case, it becomes convenient to use matrix notation and to exploit Excel's matrix-handling capabilities.

It is useful before going on to review some basic notation: Each asset i (assets may be stocks, bonds, real estate, or whatever, although our numerical examples will be largely confined to stocks) is characterized by several statistics: $E(r_i)$, the expected return on asset i; $Var(r_i)$, the variance of asset i's return; and $Cov(r_i, r_j)$, the covariance of asset i's and asset j's returns. Occasionally we will use μ_i to denote the expected return on asset i. In addition, it will often be convenient to write $Cov(r_i, r_j)$ as σ_{ij} and $Var(r_i)$ as σ_{ii} (instead of σ_i^2, as usual). Since the covariance of an asset's returns with itself, $\sigma_{ii} = Cov(r_i, r_i)$, is in fact the variance of the asset's returns, this notation is not only economical but also logical.

Throughout this chapter, the example shows an implementation in Excel and in R. We start by setting the working directory in R:

```
# We thank Sagi Haim for developing this script

########################################
# Chapter 10 - Portfolio Introduction
########################################

# Set working directory
workdir <-readline(prompt="working directory?")
setwd(workdir)
```

10.2 Computing Descriptive Statistics for Stocks

In this section we compute the return statistics for two stocks: Apple (stock symbol AAPL) and Kellogg (K). Here is the price and return data. The returns include the dividends; see Appendix 10.2 for more details. Note that some lines are hidden.

	A	B	C	D	E	F	G
1	PRICES AND RETURNS FOR APPLE AND KELLOGG Jan 2009 - Jan 2019						
2	Date	AAPL	Kellogg		AAPL	Kellogg	
3	01-Dec-18	157.07	56.45		-12.06%	-10.13%	<--=LN(C3/C4)
4	01-Nov-18	177.20	62.46		-20.34%	-2.83%	<--=LN(C4/C5)
5	01-Oct-18	217.17	64.26		-3.10%	-6.70%	
6	01-Sep-18	223.99	68.72		-0.48%	-1.71%	
119	01-Apr-09	11.99	30.97		17.98%	13.94%	
120	01-Mar-09	10.01	26.94		16.30%	-5.18%	
121	01-Feb-09	8.51	28.37		-0.91%	-11.56%	
122	01-Jan-09	8.59	31.85				

These data give the closing price at the end of each month for each stock. We define the return as $r_t = ln\left(\frac{P_t}{P_{t-1}}\right)$.

At the top of the Excel sheet we calculate the return statistics for each stock. The *monthly return* is the percentage return that would be earned by an investor who bought the stock at the end of a particular month $t-1$ and sold it at the end of the following month.

Note that we use the continuously compounded return on the stock, $r_t^{continuous} = ln\left(\frac{P_t}{P_{t-1}}\right)$. An alternative would have been to use the discrete return, $r_t^{Discrete} = \frac{P_t}{P_{t-1}-1}$. Appendix 10.1 discusses the reasons for our choice of the continuously compounded return.

Below please find the R code:

```
# Read the csv with AAPL and K stocks
monthly_prices <- read.csv("Chapt_10_data.csv", row.names = 1,
                           stringsAsFactors = FALSE,
                           colClasses=c("character", "double", "double"))
# Sort the data by date
monthly_prices <- monthly_prices[order(
  as.Date(row.names(monthly_prices), format="%d-%b-%y")),]

# Compute Continuous Returns
monthly_returns <- data.frame(apply(monthly_prices
                                    ,2,function(x) diff(log(x)) ))
# Note: the "apply" function utilizes a function over the rows (1) or columns (2)

# Sneak peak to the first five lines in our returns table
head(monthly_returns)
```

Return and Volatility for Apple (AAPL) and Kellogg (K)

We now make an heroic assumption: We assume that the return data for the 120 months represent the distribution of the returns for the coming month. We thus assume that the past gives us some information about the way returns will behave in the future. This assumption allows us to assume that the average of the historic data represents the *expected monthly return* from each stock. It also allows us to assume that we may learn from the historic data what is the variance of the future returns. Using the **Average, Var.s**, and **Stdev.s** functions in Excel, we calculate the statistics for the return distribution:

	A	B	C	D	E	F
124		AAPL	Kellogg			
125	Monthly mean	2.44%	0.48%	<-- =AVERAGE(F3:F121)		
126	Monthly variance	0.0054	0.0021	<-- =VAR.S(F3:F121)		
127	Monthly standard deviation	7.33%	4.54%	<-- =STDEV.S(F3:F121)		
128						
129	Annual mean	29.31%	5.77%	<-- =12*C125		
130	Annual variance	0.0645	0.0247	<-- =12*C126		
131	Annual standard deviation	25.39%	15.72%	<-- =SQRT(12)*C127		

Note: Sample versus Population Statistics and Excel

We interrupt this discussion of portfolio computations with a short note on the statistical computations. In statistics it is common to distinguish between sample statistics and population statistics. If we examine the whole range of possibilities for a given random variable, then we are dealing with the *population*. If we are dealing with a set of outcomes for the random variable, then we are dealing with a *sample*. In the case of portfolio return statistics, we will almost always be dealing with a *sample* rather than the whole population.

The table below gives the definitions of the population and sample statistics and the Excel functions that compute them.

	Notation	Population	Sample
Mean, average	μ, $\bar{r}$	$\text{Average} = \frac{1}{N}\sum_{i=1}^{N} r_i$	$\text{Average} = \frac{1}{N}\sum_{i=1}^{N} r$
Variance	var(r_i), σ_{ii}, σ_i^2	$\text{Var.p} = \frac{1}{N}\sum_{i=1}^{N}(r_i - \bar{r})^2$	$\text{Var.s} = \frac{1}{N-1}\sum_{i=1}^{N}(r_i - \bar{r})^2$
Standard deviation	σ_i	$\text{Stdev.p} = \sqrt{\frac{1}{N}\sum_{i=1}^{N}(r_i - \bar{r})^2}$	$\text{Stdev.s} = \sqrt{\frac{1}{N-1}\sum_{i=1}^{N}(r_i - \bar{r})^2}$
Covariance	*cov*(r_i, r_j), σ_{ij}	$\text{Covariance.p} = \frac{1}{N}\sum_{t=1}^{N}(r_{it} - \bar{r}_i)(r_{jt} - \bar{r}_j)$	$\text{Covariance.s} = \frac{1}{N-1}\sum_{t=1}^{N}(r_{it} - \bar{r}_i)(r_{jt} - \bar{r}_j)$
Correlation	ρ_{ij}	$\text{Correl}(i, j) = \frac{\text{Covar.p}(i, j)}{\text{Stdev.p}(i) * \text{Stdev.p}(j)} =$	$\frac{\text{Covar.s}(i, j)}{\text{Stdev.s}(i) * \text{Stdev.s}(j)}$
Regression slope	β_i	$\text{Slope}(i\_\text{data}, M\_\text{data}) = \frac{\text{Covar.p}(i, M)}{\text{Var.p}(i, M)} =$	$\frac{\text{Covar.S}(i, M)}{\text{Var.S}(i, M)}$

Sample or Population? Does It Matter?

All of this discussion about population versus sample statistics may not matter much. We like the point of view stated by William Press and colleagues in their splendid book *Numerical Recipes*: "There is a long story about why the denominator is $N - 1$ instead of *N*. If you have never heard that story, you may consult any good statistics text. We might also comment that if the difference between *N* and *N* − 1 ever matters to you, then you are up to no good anyway—e.g., trying to substantiate a questionable hypothesis with marginal data."[1]

This is the R code to do the same:

```
# Mean Returns
mean_returns <- apply(monthly_returns, 2,mean) # here we can also use the colMeans function
mean_returns

# Variance (Sample)
monthly_var <- apply(monthly_returns, 2,var)
monthly_var

# Standard Deviation (Sample)
monthly_sigma <- apply(monthly_returns, 2,sd)
monthly_sigma

# Annual Statistics
mean_returns * 12
monthly_var * 12
monthly_sigma * sqrt(12)
```

Computing Covariance and Correlation Coefficient

Next, we want to calculate the co-movement of the two stocks. This co-movement is measured by *covariance* and the *correlation coefficient* of the returns.

	A	B	C	D	E	F	G	H
1		COMPUTING COVARIANCE AND CORRELATION						
2		Stock returns			Return minus average			
3	Date	AAPL	Kellogg		AAPL	Kellogg	Product	
4	01-Dec-18	-12.06%	-10.13%	=B4-$B$124-->	-14.50%	-10.61%	1.54%	<--=E4*F4
5	01-Nov-18	-20.34%	-2.83%	=B5-$B$124-->	-22.78%	-3.32%	0.76%	<--=E5*F5
6	01-Oct-18	-3.10%	-6.70%		-5.54%	-7.18%	0.40%	
7	01-Sep-18	-0.48%	-1.71%		-2.93%	-2.19%	0.06%	
120	01-Apr-09	17.98%	13.94%		15.54%	13.46%	2.09%	
121	01-Mar-09	16.30%	-5.18%		13.86%	-5.66%	-0.78%	
122	01-Feb-09	-0.91%	-11.56%		-3.36%	-12.04%	0.40%	
123								
124	Average	2.44%	0.48%	<--=AVERAGE(C4:C122)				
125	Variance	0.0054	0.0021	<-- =VAR.S(C4:C122)				
126	Standard deviation	7.33%	4.54%	<-- =STDEV.S(C4:C122)				
127	Standard deviation	7.33%	4.54%	<-- =SQRT(C125)				
128								
129	Covariance computation							
130	Method 1	0.000852	<--=COVARIANCE.S(B4:B122,C4:C122)					
131	Method 2	0.000852	<--=AVERAGE(G4:G122)*COUNT(G4:G122)/(COUNT(G4:G122)-1)					
132								
133	Correlation computation							
134	Method 1	0.2562	<-- =CORREL(B4:B122,C4:C122)					
135	Method 2	0.2562	<-- =B130/(B126*C126)					

The column **Product** contains the multiple of the deviation from the mean in each month—that is, the terms $[r_{AAPL,\,t} - E(r_{AAPL})][r_{K,\,t} - E(r_K)]$, for $t = 1, \ldots, 120$. The population covariance is *Average* (*product*) and the sample covariance is $Average(product) * \frac{n}{n-1}$. While it is worthwhile calculating the covariance this way at least once, there is a shorter way that is also illustrated above: The Excel functions **Covariance.s(AAPL,K)** calculate the sample covariance directly.[2] There is no necessity to find the difference between the returns and the means. Simply use **Covariance.p** or **Covariance.s** directly on the columns, as illustrated above.

This is the R code to calculate covariance and correlation in our example:

```
# Calculating Covariance and Correlation
cov_mat<-cov(monthly_returns) # Covariance Matrix
cov_ab <- cov_mat["AAPL","K"]
cov_ab # Covariance of Apple and K

# Correlation Coefficient Method 1 (Correlation Matrix)
cor(monthly_returns) # Correlation Matrix
corr_ab <- cor(monthly_returns)["AAPL","K"]
corr_ab # Correlation between Apple and K
```

The covariance is a hard number to interpret, since its size depends on the units in which we measure the returns. (If we were to write the returns in percentages—for example, 4 instead of 0.04—then the covariance would be 8.52, which is 10,000 times the number we just calculated.) We can also calculate the *correlation coefficient* ρ_{AB}, which is defined as:

$$\rho_{AAPL,K} = \frac{Cov(r_{AAPL}, r_K)}{\sigma_{AAPL}\sigma_K}$$

The correlation coefficient is unit-free; calculating it for our example gives $\rho_{AAPL,\ K} = 0.2562$. As illustrated above, the correlation coefficient can be calculated directly in Excel using the function **Correl(AAPL,K).**

The correlation coefficient measures the degree of linear relation between the returns of stock A and stock B. The following facts can be proven about the correlation coefficient:

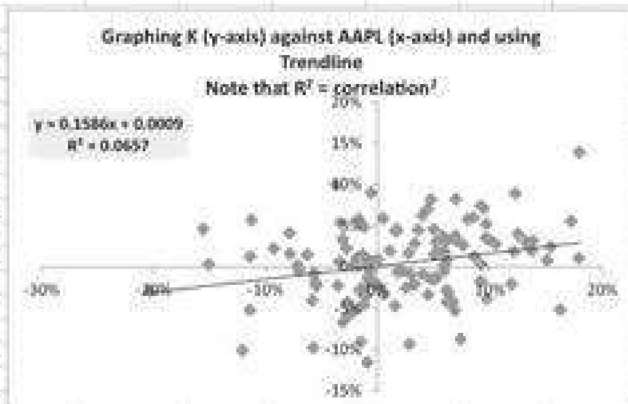
- • The correlation coefficient is always between +1 and −1: $-1 \le \rho_{AB} \le 1$.
- • If the correlation coefficient is +1, then the returns on the two assets are linearly related with a positive slope—that is, if $\rho_{AB} = 1$, then $r_{At} = c + dr_{Bt}$, *where* $d > 0$. In other words, for every movement of 1% in stock B, we will see d% movement in stock A (where d is positive).
- • If the correlation coefficient is −1, then the returns on the two assets are linearly related with a negative slope—that is, if $\rho_{AB} = -1$, then $r_{At} = c + dr_{Bt}$, *where* $d < 0$. In other words, for every movement of 1% in stock B, we will see d% movement in stock A (where d is negative).
- • If the return distributions are independent, then the correlation coefficient will be zero. (The opposite is not true: If the correlation coefficient is zero, this does not necessarily mean that the returns are independent. See the end-of-chapter exercises for an example.)

A Different View of the Correlation Coefficient

Another way to look at the correlation coefficient is to graph the Apple and Kellogg returns and then use the Excel **Trendline** facility to regress the returns of Kellogg on those of AAPL (the use of Excel's **Trendline** function—used to calculate the regression equation—is explained in Chapter 30). You can confirm from the previous calculations that the regression R^2 is the correlation coefficient squared $\left(\underbrace{0.2562^2}_{\rho} = \underbrace{0.0657}_{R^2}\right)$:

USING TRENDLINE ON AAPL AND KELLOGG RETURNS

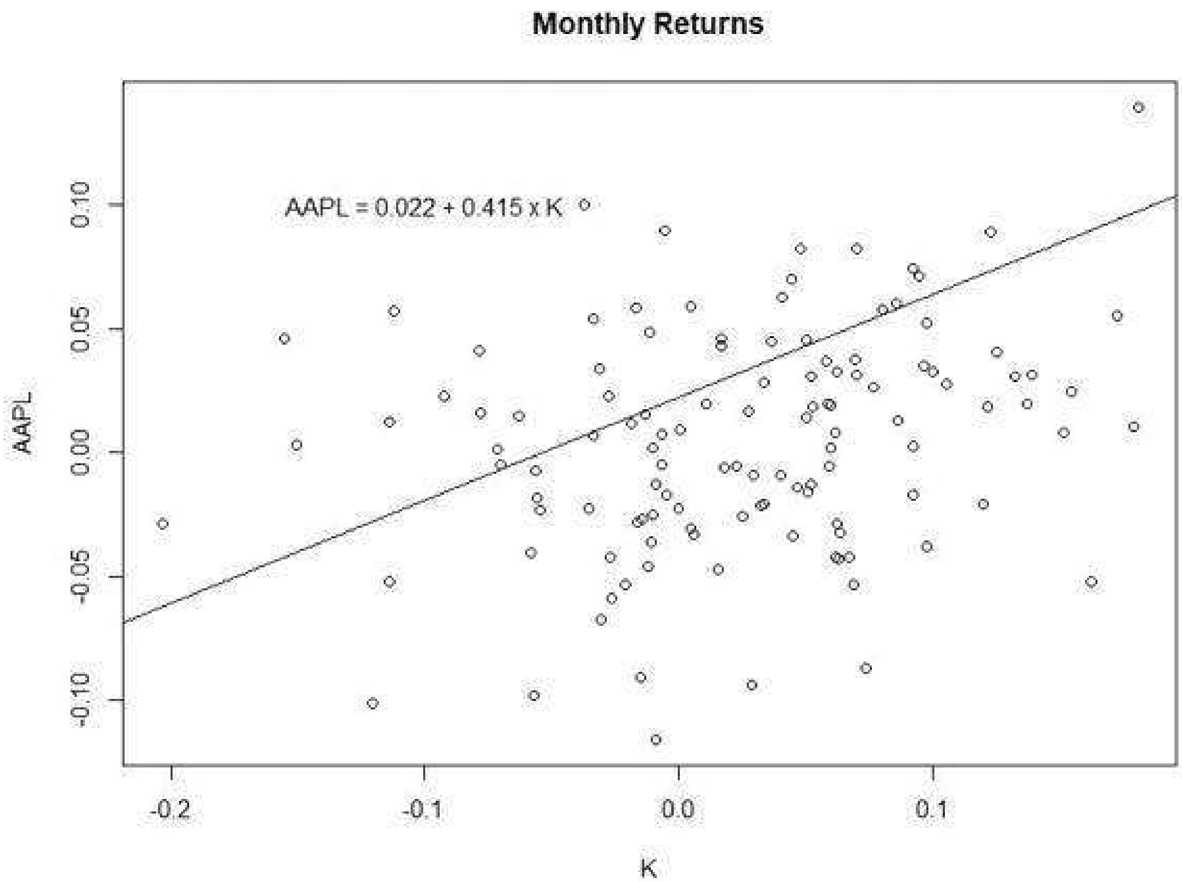
Date	AAPL	K
1-Dec-18	-12.06%	-10.13%
1-Nov-18	-20.34%	-2.83%
1-Oct-18	-3.10%	-6.70%
1-Sep-18	-0.48%	-1.71%
1-Aug-18	17.92%	1.06%
1-Jul-18	2.76%	1.65%
1-Jun-18	-0.56%	8.99%
1-May-18	12.29%	8.91%
1-Apr-18	-1.51%	-9.06%
1-Mar-18	-5.57%	-1.81%
1-Feb-18	6.19%	-2.84%
1-Jan-18	-1.07%	0.19%
1-Dec-17	-1.18%	4.86%
1-Nov-17	1.65%	4.30%
1-Oct-17	9.24%	0.28%
1-Sep-17	-5.82%	-4.02%



Here is the R code to do the same:

```
## A Different View of the Correlation Coefficient
linear_model <- lm(monthly_returns[,"AAPL"] ~ monthly_returns[,"K"],
                   data = monthly_returns)

# Scatter Plot
plot.ts(monthly_returns[,"AAPL"],monthly_returns[,"K"],  xy.labels = FALSE,
        main = "Monthly Returns",
        ylab = "AAPL",
        xlab = "K")

# Add Trendline
abline(linear_model)

# Add Equation
equation <- paste(as.character(colnames(monthly_returns)[1]),
                  "=", round(linear_model$coefficients[1], digits = 3),
                  "+", round(linear_model$coefficients[2], digits = 3),
                  "x", as.character(colnames(monthly_returns)[2]),
                  sep = " ", collapse = NULL)
text(-0.1,0.1, equation)
```

And the plot appears as follows:



10.3 Calculating Portfolio Means and Variances

In this section we show how to do the basic calculations for a portfolio's mean and variance. Suppose we form a portfolio invested equally in AAPL and K. What will be the mean and the variance of this portfolio? It is worth doing the brute-force calculations at least once in Excel:

	A	B	C	D	E
1	CALCULATING THE MEAN AND STANDARD DEVIATION OF A PORTFOLIO				
2	Proportion of AAPL	50%			
3	Proportion of K	50%	<--=1-B2		
4					
5		AAPL return	K return	Portfolio return	
6	1-Dec-18	-12.06%	-10.13%	-11.09%	<--=$B$2*B6+$B$3*C6
7	1-Nov-18	-20.34%	-2.83%	-11.59%	<--=$B$2*B7+$B$3*C7
8	1-Oct-18	-3.10%	-6.70%	-4.90%	
122	1-Apr-09	17.98%	13.94%	15.96%	
123	1-Mar-09	16.30%	-5.18%	5.56%	
124	1-Feb-09	-0.91%	-11.56%	-6.24%	

Computing the statistics for this example looks like this:

	A	B	C	D	E
1	**CALCULATING THE MEAN AND STANDARD DEVIATION OF A PORTFOLIO**				
2	Proportion of AAPL	50%			
3	Proportion of K	50%	<--=1-B2		
4					
5		**AAPL return**	**K return**	**Portfolio return**	
6	1-Dec-18	-12.06%	-10.13%	-11.09%	<--=$B$2*B6+$B$3*C6
7	1-Nov-18	-20.34%	-2.83%	-11.59%	<--=$B$2*B7+$B$3*C7
8	1-Oct-18	-3.10%	-6.70%	-4.90%	
122	1-Apr-09	17.98%	13.94%	15.96%	
123	1-Mar-09	16.30%	-5.18%	5.56%	
124	1-Feb-09	-0.91%	-11.56%	-6.24%	
125					
126	**Asset returns**	**AAPL**	**Kellogg**		
127	Mean return	2.44%	0.48%	<--=AVERAGE(C6:C124)	
128	Variance	0.0054	0.0021	<--=VAR.S(C6:C124)	
129	Standard deviation	7.33%	4.54%	<--=STDEV.S(C6:C124)	
130	Covariance	0.000852	<--=COVARIANCE.S(B6:B124,C6:C124)		
131					
132	**Portfolio mean return**				
133	Method 1:	1.46%	<--=AVERAGE(D6:D124)		
134	Method 2:	1.46%	<--=B2*B127+B3*C127		
135					
136	**Portfolio return variance**				
137	Method 1:	0.0023	<--=VAR.S(D6:D124)		
138	Method 2:	0.0023	<--=B2^2*B128+B3^2*C128+2*B2*B3*B130		
139					
140	**Portfolio return standard deviation**				
141	Method 1:	0.0478	<--=STDEV.S(D6:D124)		
142	Method 2:	0.0478	<--=SQRT(B138)		

The mean portfolio return is exactly the average of the mean returns of the two assets:

Expected portfolio return = $0.5 \cdot E(r_{AAPL}) + 0.5 \cdot E(r_K)$

In general, the mean return of the portfolio is the *weighted average return* of the component stocks. If we denote by x the proportion invested in AAPL and by $(1 - x)$ the proportion invested in stock K, then the expected portfolio return is given by:

$$E(r_p) = x \cdot E(r_{AAPL}) + (1 - x) \cdot E(r_K)$$

However, the portfolio's variance is *not* the average of the two variances of the stocks! The formula for the variance is:

$$Var(r_p) = x^2 Var(r_{AAPL}) + (1 - x)^2 Var(r_K) + 2x(1 - x) Cov(r_{AAPL}, r_K)$$

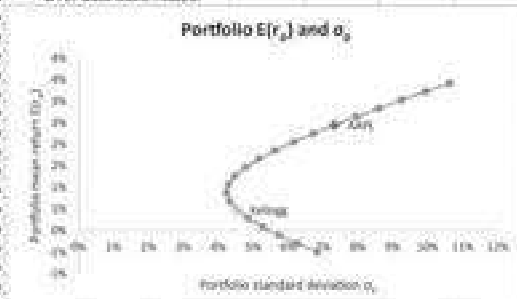
Another way of writing this is:

$$\sigma_p^2 = x^2 \sigma_{AAPL}^2 + (1 - x)^2 \sigma_K^2 + 2x(1 - x) \rho_{AAPL,K} \sigma_{AAPL} \sigma_K$$

A frequently performed exercise is to plot the means and standard deviations for various portfolio proportions x. To do this, we build a table using Excel's **Data|What**

If|DataTable command (see Chapter 28); cells B16 and C16 contain the data table's header, which refers to cells B12 and B10, respectively.

	A	B	C	D
1	CALCULATING THE MEAN AND STANDARD DEVIATION OF A PORTFOLIO			
2	Asset returns	AAPL	Kellogg	
3	Mean return	2.44%	0.48%	
4	Variance	0.00537	0.00206	
5	Standard deviation	7.33%	4.54%	
6	Covariance	0.000852		
7				
8	Proportion of AAPL	50%		
9				
10	Portfolio mean return	1.46%	<--=B8*B3+(1-B8)*C3	
11	Portfolio return variance	0.0023	<--=B8^2*B4+(1-B8)^2*C4+2*B8*(1-B8)*B6	
12	Portfolio return standard deviation	4.78%	<--=SQRT(B11)	
13				
14	Data table, varying the proportion of AAPL			
15		Portfolio standard deviation	Portfolio mean return	
16	Proportion of AAPL	4.78%	1.46%	<-- =B10, data table header
17	-50%	6.85%	-0.50%	
18	-40%	6.28%	-0.30%	
19	-30%	5.74%	-0.11%	
20	-20%	5.26%	0.09%	
21	-10%	4.85%	0.28%	
22	0%	4.54%	0.48%	
23	10%	4.33%	0.68%	
24	20%	4.25%	0.87%	
25	30%	4.30%	1.07%	
26	40%	4.48%	1.27%	
27	50%	4.78%	1.46%	
28	60%	5.17%	1.66%	
29	70%	5.63%	1.85%	
30	80%	6.16%	2.05%	
31	90%	6.73%	2.25%	
32	100%	7.33%	2.44%	
33	110%	7.96%	2.64%	
34	120%	8.61%	2.83%	
35	130%	9.27%	3.03%	
36	140%	9.95%	3.22%	
37	150%	10.64%	3.42%	



Here is the R code to do the same:

```
Xa <- 0.5 # change if you want a different portfolio
proportions <- c(Xa, 1 - Xa)

# Portfolio mean returns
port_returns <- sum(mean_returns * proportions)
port_returns

# Portfolio Return Variance
port_var <- sum(proportions * proportions * monthly_var,2 * prod(proportions) * cov_ab)

# alternative and more elegant approach to calculate portfolio variance
port_var <- proportions %*% cov_mat %*% proportions

# Portfolio standard deviation
port_sigma <- sqrt(port_var)
```

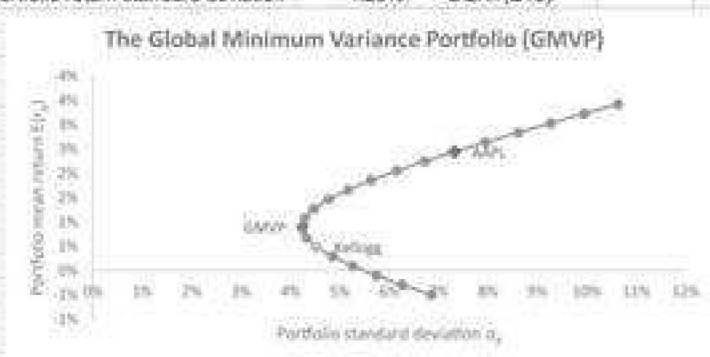
Calculating the Minimum Variance Portfolio

The *global minimum variance portfolio* (GMVP) is the portfolio with the lowest possible risk. This can be easily calculated either by brute-force method (**Solver**) or by solving the mathematical equation:

$$\min_{x} \sigma_p^2 = x^2 \cdot \sigma_{AAPL}^2 + (1-x)^2 \cdot \sigma_K^2 + 2 \cdot x \cdot (1-x) \cdot \rho_{AAPL,K} \cdot \sigma_{AAPL} \cdot \sigma_K$$

$$\Rightarrow x_{AAPL}^{GMVP} = \frac{\sigma_K^2 - \rho_{AAPL,K}\sigma_{AAPL}\sigma_K}{\sigma_K^2 + \sigma_{AAPL}^2 - 2\rho_{AAPL,K}\sigma_{AAPL}\sigma_K} = \frac{Var(r_K) - Cov(r_{AAPL}, r_K)}{Var(r_{AAPL}) + Var(r_K) - 2Cov(r_{AAPL}, r_K)}$$

	A	B	C	D	E	F	G
1	CALCULATING THE GLOBAL MINIMUM VARIANCE PORTFOLIO						
2	Asset returns	AAPL	Kellogg				
3	Mean return	2.44%	0.48%				
4	Variance	0.00537	0.00206				
5	Standard deviation	7.33%	4.54%				
6	Covariance	0.000652					
7							
8	**The global minimum variance portfolio (GMVP), two approaches**						
9	Brute-force approach (using Solver)						
10	Proportion of AAPL	21.07%	<-- solved using solver				
11	Portfolio mean return	0.89%	<--=B10*B3+(1-B10)*C3				
12	Portfolio return variance	0.0018	<--=B10^2*B4+(1-B10)^2*C4+2*B10*(1-B10)*B6				
13	Portfolio return standard deviation	4.25%	<--=SQRT(B12)				
14							
15	Math approach						
16	Proportion of AAPL	21.07%	<--=(C4-B6)/(B4+C4-2*B6)				
17	Portfolio mean return	0.89%	<--=B16*B3+(1-B16)*C3				
18	Portfolio return variance	0.0018	<--=B16^2*B4+(1-B16)^2*C4+2*B16*(1-B16)*B6				
19	Portfolio return standard deviation	4.25%	<--=SQRT(B18)				



Implementing the above in R looks like this:

```
## Calculating the  Minimum-Variance Portfolio
# Analytical Approach
# This returns the weight of the 1st asset in the minimum variance portfolio:
GMVP_AAPL <- as.numeric((monthly_var["K"] - cov_ab)/
  (monthly_var["AAPL"] + monthly_var["K"] - 2 * cov_ab))
GMVP_AAPL # AAPL's weight in the GMVP


# Calculating GMVP variance:
GMVP_prop <- c(XGMVP_AAPL = GMVP_AAPL, XGMVP_K = 1 - GMVP_AAPL)
GMVP_var <- GMVP_prop %*% cov_mat %*% GMVP_prop
GMVP_sigma<-sqrt(GMVP_var)
GMVP_var
GMVP_sigma
```

And using the Solver equivalent in R looks like this:

```
## Solver Approach
# This formula calculates the variance of a 2 asset portfolio
port_var_2stocks <- function(x1, cov_mat) {
  prop<-c(x1,1-x1)
  prop %*% cov_mat %*% prop
}

# minimizing the portfolio's variance:
GMVP.Result <- optimize(port_var_2stocks, # the function to be optimized
                        interval = c(0,10), # the end-points of the interval to be searched
                        tol = 0.00001, # the desired accuracy
                        maximum = FALSE, # FALSE FOR maximum, TRUE for minimum
                        cov_mat = cov_mat) # Constant Variables

GMVP.Result # we get the same variance result as the analytic approach
x1 <- GMVP.Result$minimum
x1
```

10.4 Portfolio Mean and Variance—Case of *N* Assets

In the previous sections we discussed computation of the portfolio's mean, variance, and standard deviation for the case wherein the portfolio is composed of only two assets. In this section we extend this discussion to portfolios of more than two assets. For this case, matrix notation greatly simplifies the writing of the portfolio problem.[3] In the general case of N assets, suppose that the proportion of asset i in the portfolio is denoted by x_i. We require that $\sum_i x_i = 1$, but we place no restrictions on the signs of the x_i; if $x_i > 0$, this indicates a purchase of asset i, and if $x_i < 0$, this indicates a short sale.[4] We usually write the portfolio composition x and the vector of means $E(r)$ as column vectors (we make no pretense at consistency—when convenient we will write these as row vectors):

$$x = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_N \end{bmatrix}, \quad E(r) = \begin{bmatrix} E(r_1) \\ E(r_2) \\ \vdots \\ E(r_N) \end{bmatrix}$$

We may then write x^T and $E(r)^T$ as the transpose of these two vectors:

$$x^T = [x_1, x_2, \ldots, x_N],\ E(r)^T = [E(r_1), E(r_2), \ldots, E(r_N)]$$

The expected return of the portfolio whose proportions are given by X is the weighted average of the expected returns of the individual assets $E(r_p) = \sum_{i=1}^{N} x_i E(r_i)$, which, in matrix notation, can be written as:

$$E(r_p) = \sum_{i=1}^{N} x_i E(r_i) = x^T E(r) = E(r)^T x$$

The portfolio's variance is given by:

$$Var(r_p) = \sum_{i=1}^{N} \sum_{j=1}^{N} x_i x_j Cov(r_i, r_j) = \sum_{i=1}^{N} (x_i)^2 Var(r_i) + 2 \sum_{i=1}^{N} \sum_{j=i+1}^{N} x_i x_j Cov(r_i, r_j)$$

This looks bad, but it is really a straightforward extension of the expression for the variance of a portfolio of two assets, which we had before: Each asset's variance appears once, multiplied by the square of the asset's proportion in the portfolio; the covariance of each pair of assets appears once, multiplied by twice the product of the individual assets' proportions. Another way of writing the variance is to use the following notation:

$Var(r_i) = \sigma_{ii}, \quad Cov(r_i, r_j) = \sigma_{ij}$

In this case, the variance of portfolio x is:

$$Var(r_X) = \sum_i \sum_j x_i x_j \sigma_{ij}$$

The most economical representation of the portfolio variance uses matrix notation. It is also the easiest representation to implement for large portfolios in Excel. In this representation we call the matrix that has σ_{ij} in the i^{th} row and the j^{th} column the *variance-covariance matrix*:

$$S = \begin{bmatrix} \sigma_{11} & \sigma_{12} & \sigma_{13} & \cdots & \sigma_{1N} \\ \sigma_{21} & \sigma_{22} & \sigma_{23} & & \sigma_{2N} \\ \sigma_{31} & \sigma_{32} & \sigma_{33} & \cdots & \sigma_{3N} \\ \vdots & & \vdots & \ddots & \vdots \\ \sigma_{N1} & \sigma_{N2} & \sigma_{N3} & \cdots & \sigma_{NN} \end{bmatrix}$$

Then the portfolio variance is given by $Var(r_p) = x^T Sx$. In Excel formulas, this is written as the array function **MMult(MMult(Transpose(x),S),x)**.[5]

Computing the Covariance between Two Portfolios

If we have two portfolios represented as row vectors, $x = [x_1, x_2, \ldots, x_N]$ and $y = [y_1, y_2, \ldots, y_N]$, then the covariance of the two portfolios is given by $Cov(x, y) = xSy^T = ySx^T$. In Excel formulas, this is the array function **MMult(MMult(x,S),Transpose(y))**.

Portfolio Calculations Using Matrices: An Example

We implement the above formulas in a numerical example. Suppose that there are four risky assets that have the following expected returns and variance covariance matrix:

	A	B	C	D	E	F	G
1		A FOUR-ASSET PORTFOLIO PROBLEM					
2		Variance-covariance, S					Mean
3		Asset 1	Asset 2	Asset 3	Asset 4		returns E(r)
4	**Asset 1**	0.10	0.01	0.03	0.05		6%
5	**Asset 2**	0.01	0.30	0.06	-0.04		8%
6	**Asset 3**	0.03	0.06	0.40	0.02		10%
7	**Asset 4**	0.05	-0.04	0.02	0.50		15%

We consider two portfolios of risky assets:

	A	B	C	D	E	F	G
9		Asset 1	Asset 2	Asset 3	Asset 4		
10	Portfolio x	0.20	0.30	0.40	0.1	<--=1-SUM(B10:D10)	
11	Portfolio y	0.20	0.10	0.10	0.6	<--=1-SUM(B11:D11)	

We calculate the means, variances, and covariance of the two portfolios. We use the Excel array function **MMult** for the matrix multiplications and the array function **Transpose** to make a row vector into a column vector.[6]

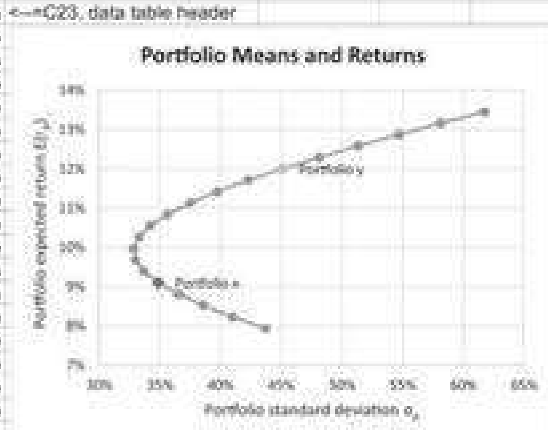
	A	B	C	D	E	F	G	H
1		**A FOUR-ASSET PORTFOLIO PROBLEM**						
2			**Variance-covariance, S**				**Mean**	
3		**Asset 1**	**Asset 2**	**Asset 3**	**Asset 4**		**returns E(r)**	
4	**Asset 1**	0.10	0.01	0.03	0.05		6%	
5	**Asset 2**	0.01	0.30	0.06	-0.04		8%	
6	**Asset 3**	0.03	0.06	0.40	0.02		10%	
7	**Asset 4**	0.05	-0.04	0.02	0.50		15%	
8								
9		**Asset 1**	**Asset 2**	**Asset 3**	**Asset 4**			
10	Portfolio x	0.20	0.30	0.40	0.1	<--=1-SUM(B10:D10)		
11	Portfolio y	0.20	0.10	0.10	0.6	<--=1-SUM(B11:D11)		
12								
13	**Portfolio x and y statistics: Mean, variance, covariance, correlation**							
14		**Portfolio x**	**Portfolio y**					
15	Mean, $E(r_p)$	9.10%	12.00%	<--{=MMULT(B11:E11,G4:G7)}				
16	Variance σ_p^2	0.1216	0.2034	<--{=MMULT(MMULT(B11:E11,B4:E7),TRANSPOSE(B11:E11))}				
17	Std-dev σ_p	34.87%	45.10%	<--=SQRT(C16)				
18	Covariance(x,y)	0.0714	<--{=MMULT(MMULT(B10:E10,B4:E7),TRANSPOSE(B11:E11))}					
19	Correlation ρ_{xy}	0.4540	<--=B18/(B17*C17)					

Implementing the above example in R looks like this:

```
# Input: Variance Covariance Matrix
var_cov_mat <- matrix(c(0.10, 0.01, 0.03, 0.05,
                        0.01, 0.30, 0.06, -0.04,
                        0.03, 0.06, 0.40, 0.02,
                        0.05, -0.04,  0.02, 0.50),
                      ncol = 4)

# Adding column names to a matrix:
row.names(var_cov_mat) <- c("Asset 1",  "Asset 2", "Asset 3", "Asset 4")
colnames(var_cov_mat) <-  row.names(var_cov_mat)

# Insert mean returns
mean_returns <- c(0.06, 0.08, 0.1, 0.15)

#Insert Portfolio weights
port_x_wei <- c(0.2, 0.3, 0.4, 0.1)
port_y_wei <- c(0.2, 0.1, 0.1, 0.6)

# Portfolios Mean Returns
port_x_ret <- mean_returns %*% port_x_wei
port_y_ret <- mean_returns %*% port_y_wei

# Portfolios variance calculation of n-assets
port_x_var <- port_x_wei %*% var_cov_mat %*% port_x_wei
port_y_var <- port_y_wei %*% var_cov_mat %*% port_y_wei

# Portfolios Standard deviation
port_x_sigma <- sqrt(port_x_var)
port_y_sigma <- sqrt(port_y_var)

# Covariance(X,Y)
cov_xy <- port_x_wei %*% var_cov_mat %*% port_y_wei

# Correlation Coefficient (X,Y)
correl_xy <- cov_xy / (port_x_sigma * port_y_sigma)
```

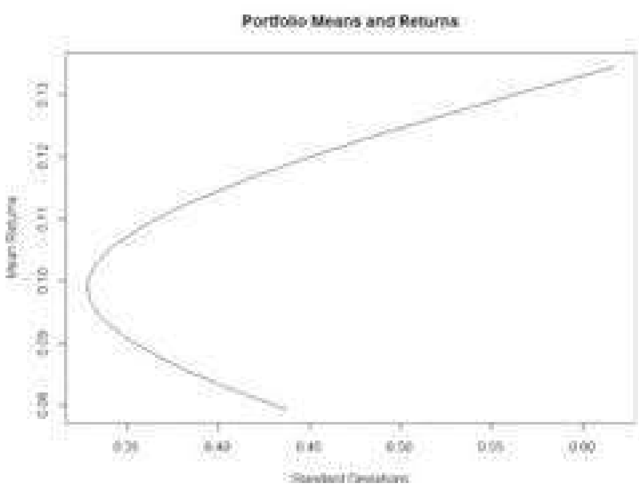
We can now calculate the standard deviation and return of combinations of portfolios x and y. Note that once we have calculated the means, variances, and the covariance of the returns of the two portfolios, the calculation of the mean and the variance of any portfolio is the same as for the two-asset case.[7]

	A	B	C	D	E	F	G	H	I
1			A FOUR-ASSET PORTFOLIO PROBLEM						
2			Variance-covariance, S				Mean		
3		Asset 1	Asset 2	Asset 3	Asset 4		returns E(r)		
4	Asset 1	0.10	0.01	0.03	0.05		6%		
5	Asset 2	0.01	0.30	0.06	-0.04		8%		
6	Asset 3	0.03	0.06	0.40	0.02		10%		
7	Asset 4	0.05	-0.04	0.02	0.50		15%		
8									
9		Asset 1	Asset 2	Asset 3	Asset 4				
10	Portfolio x	0.20	0.30	0.40	0.1	<--=1-SUM(B10:D10)			
11	Portfolio y	0.20	0.10	0.10	0.6	<--=1-SUM(B11:D11)			
12									
13	Portfolio x and y statistics: Mean, variance, covariance, correlation								
14		Portfolio x	Portfolio y						
15	Mean, $E(r_p)$	9.10%	12.00%	<--{=MMULT(B11:E11,G4:G7)}					
16	Variance σ_p^2	0.1216	0.2034	<--{=MMULT(MMULT(B11:E11,B4:E7),TRANSPOSE(B11:E11))}					
17	Std-dev σ_p	34.87%	45.10%	<--=SQRT(C16)					
18	Covariance(x,y)	0.0714	<--{=MMULT(MMULT(B10:E10,B4:E7),TRANSPOSE(B11:E11))}						
19	Correlation ρ_{xy}	0.4540	<--=B18/(B17*C17)						
20									
21	Calculating returns of combinations of Portfolio x and Portfolio y								
22	Proportion of portfolio x		0.3						
23	Mean portfolio return, $E(r_p)$		11.13%	<-- =C22*B15+(1-C22)*C15					
24	Portfolio variance σ_p^2		0.1406	<-- =C22^2*B16+(1-C22)^2*C16+2*C22*(1-C22)*B18					
25	Portfolio std-dev σ_p		37.50%	<-- =SQRT(C24)					
26									
27	Table of returns (using Data Table)								
28	Proportion of x	Standard deviation	Mean						
29		37.50%	11.13%	<--=C23, data table header					
30	-0.5	61.72%	13.45%						
31	-0.4	58.15%	13.16%						
32	-0.3	54.68%	12.87%						
33	-0.2	51.33%	12.58%						
34	-0.1	48.13%	12.29%						
35	0.0	45.10%	12.00%						
36	0.1	42.29%	11.71%						
37	0.2	39.74%	11.42%						
38	0.3	37.50%	11.13%						
39	0.4	35.63%	10.84%						
40	0.5	34.20%	10.55%						
41	0.6	33.26%	10.26%						
42	0.7	32.84%	9.97%						
43	0.8	32.99%	9.68%						
44	0.9	33.67%	9.39%						
45	1.0	34.87%	9.10%						
46	1.1	36.53%	8.81%						
47	1.2	38.60%	8.52%						
48	1.3	41.00%	8.23%						
49	1.4	43.69%	7.94%						



Implementing the above in R looks like this:

```
# Calculating returns of combinations of Portfolio X and Portfolio Y
# Z is a portfolio constructed of portfolio X and portfolio Y.
X <- 0.3 #Proportion of X
port_z_weights <- c(X, 1-X) #Proportions of X and Y

# Portfolio Returns
port_z_ret <- port_z_weights %*% c(port_x_ret, port_y_ret)

# Portfolio Variance and sigma
port_z_var <- port_z_weights[1] ^ 2 * port_x_var +
              port_z_weights[2] ^ 2 * port_y_var + 2 * prod(port_z_weights) * cov_xy

port_z_sigma <- sqrt(port_z_var)

## Plot
# Set an Array of Portfolio weights
port_x_weight <- c( 1:20 / 10 - 0.6)
port_y_weight <- 1 - port_x_weight

# Portfolios Mean Returns
port_z_ret <- port_x_weight %*% port_x_ret + port_y_weight %*% port_y_ret

# Portfolios Sigma
port_z_sigma <- sqrt(port_x_weight ^ 2 %*% port_x_var + port_y_weight ^ 2 %*%
                     port_y_var + (port_x_weight * port_y_weight) %*% cov_xy %*% 2)

# Plot
plot(data.frame(port_z_sigma, port_z_ret), type="l", xlab = "Standard Deviations",
     ylab = "Mean Returns", main = "Portfolio Means and Returns")
```



10.5 Envelope Portfolios

An *envelope portfolio* is the portfolio of risky assets that gives the lowest variance of return of all portfolios having the same expected return. An *efficient portfolio* is the portfolio that the highest expected return of all portfolios having the same variance. We may define an envelope portfolio mathematically. For a given return $\mu = E(r_p)$, an efficient portfolio $p = [x_1, x_2, \ldots, x_N]$ is one that solves:

$$\min_{x_i} Var(r_p) = \min_{x_i} \sum_i \sum_j x_i x_j \sigma_{ij} \text{ subject to } \sum_i x_i r_i = \mu = E(r_p)$$
$$\sum_i x_i = 1$$

The *envelope* is the set of all envelope portfolios, and the *efficient frontier* is the set of all efficient portfolios.[8] As shown by Black (1972), the envelope is the set of all convex combinations of any two envelope portfolios.[9] This means that if $x = [x_1, x_2, \ldots, x_N]$ and

$y = [y_1, y_2, \ldots, y_N]$ are envelope portfolios and if a is a constant, then the portfolio z is defined by:

$$z = ax + (1-a)y = \begin{bmatrix} ax_1 + (1-a)y_1 \\ ax_2 + (1-a)y_2 \\ \vdots \\ ax_N + (1-a)y_N \end{bmatrix}$$

This is also an envelope. Thus, we can compute the whole envelope frontier if we can find any two envelope portfolios.

By this theorem, once we have found two efficient portfolios x and y, we know that any other efficient portfolio is a convex combination of x and y. If we denote the mean and variance of x and y by $[E(r_x), \sigma_x^2]$ and $[E(r_y), \sigma_y^2]$, and if $z = ax + (1 - a)y$, then:

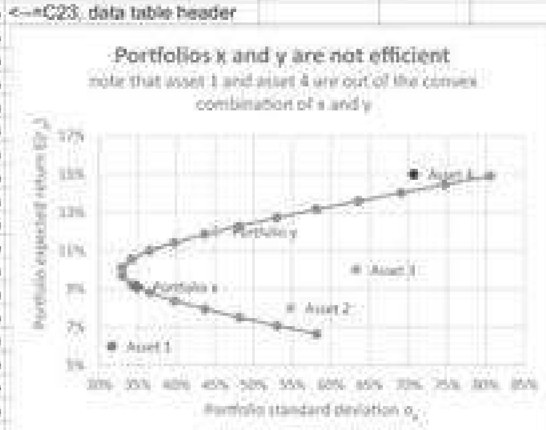
$$\begin{aligned} E(r_z) &= aE(r_x) + (1-a)E(r_y) \\ \sigma_z^2 &= a^2\sigma_x^2 + (1-a)^2\sigma_y^2 + 2a(1-a)Cov(x, y) \\ &= a^2\sigma_x^2 + (1-a)^2\sigma_y^2 + 2a(1-a)(xS\,y^T) \end{aligned}$$

Further details of the calculation of efficient portfolios are discussed in Chapter 11.

Portfolios x and y Are Not on the Envelope

To show that the concepts of envelope and efficient portfolio is nontrivial, we show that the two portfolios whose combinations are graphed in the previous example are not either envelope or efficient. This is easy to see if we extend the data table to include numbers for the individual assets:

	A	B	C	D	E	F	G
1		A FOUR-ASSET PORTFOLIO PROBLEM					
2			Variance-covariance, S				Mean
3		Asset 1	Asset 2	Asset 3	Asset 4		returns E(r)
4	Asset 1	0.10	0.01	0.03	0.05		6%
5	Asset 2	0.01	0.30	0.06	-0.04		8%
6	Asset 3	0.03	0.06	0.40	0.02		10%
7	Asset 4	0.05	-0.04	0.02	0.50		15%
8							
9		Asset 1	Asset 2	Asset 3	Asset 4		
10	Portfolio x	0.20	0.30	0.40	0.1	<--=1-SUM(B10:D10)	
11	Portfolio y	0.20	0.10	0.10	0.6	<--=1-SUM(B11:D11)	
12							
13	Portfolio x and y statistics: Mean, variance, covariance, correlation						
14		Portfolio x	Portfolio y				
15	Mean, $E(r_p)$	9.10%	12.00%	<--{=MMULT(B11:E11,G4:G7)}			
16	Variance σ_p^2	0.1216	0.2034	<--{=MMULT(MMULT(B11:E11,B4:E7),TRANSPOSE(B11:E11))}			
17	Std-dev σ_p	34.87%	45.10%	<--=SQRT(C16)			
18	Covariance(x,y)	0.0714	<--{=MMULT(MMULT(B10:E10,B4:E7),TRANSPOSE(B11:E11))}				
19	Correlation ρ_{xy}	0.4540	<--=B18/(B17*C17)				
20							
21	Calculating returns of combinations of Portfolio x and Portfolio y						
22	Proportion of portfolio x		0.3				
23	Mean portfolio return, $E(r_p)$		11.13%	<-- =C22*B15+(1-C22)*C15			
24	Portfolio variance σ_p^2		0.1406	<-- =C22^2*B16+(1-C22)^2*C16+2*C22*(1-C22)*B18			
25	Portfolio std-dev σ_p		37.50%	<-- =SQRT(C24)			
26							
27	Table of returns (using Data Table)						
28	Proportion of x	Standard deviation	Mean				
29		37.50%	11.13%	<--=C23, data table header			
30	-1.00	80.60%	14.90%				
31	-0.85	74.80%	14.47%				
32	-0.70	69.10%	14.03%				
33	-0.55	63.54%	13.60%				
34	-0.40	58.15%	13.16%				
35	-0.25	52.99%	12.73%				
36	-0.10	48.13%	12.29%				
37	0.05	43.68%	11.86%				
38	0.20	39.74%	11.42%				
39	0.35	36.51%	10.99%				
40	0.50	34.20%	10.55%				
41	0.65	32.98%	10.12%				
42	0.80	32.99%	9.68%				
43	0.95	34.21%	9.25%				
44	1.10	36.53%	8.81%				
45	1.25	39.76%	8.38%				
46	1.40	43.69%	7.94%				
47	1.55	48.16%	7.51%				
48	1.70	53.02%	7.07%				
49	1.85	58.19%	6.64%				



Were the two portfolios on the envelope, then all individual assets would fall on or inside the curved line in the graph. In our case, two of the stock returns (stock 1 and stock 4) fall outside the frontier created by combinations of portfolios *x* and *y*. Thus, *x* and *y* cannot be efficient portfolios. In Chapter 11 you will learn to compute efficient and envelope portfolios, and as you will see there, this requires considerably more computation.

10.6 Summary

In this chapter we have reviewed the basic concepts and mathematics of portfolios. In succeeding chapters, we shall describe how to compute the variance-covariance matrix from asset returns and how to calculate efficient portfolios.

Exercises

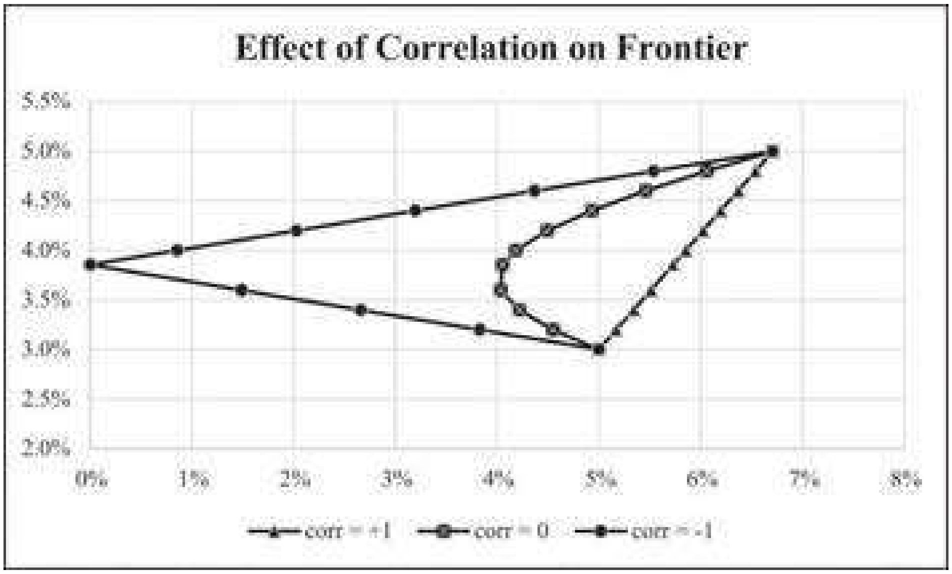
1. The exercise file for this chapter (see auxiliary website) contains monthly data for stock prices of Kellogg and IBM. Compute the return statistics and graph a frontier of combinations of the two stocks.
2. Consider the two stocks below. Graph the frontier of combinations of the two stocks. Show the effect on the frontier of varying the correlation from −1 to +1.

	A	B	C	D	E	F
1	TWO STOCKS Varying the correlation coefficient					
2		Stock A	Stock B			
3	Mean	3.00%	8.00%			
4	Sigma	15.00%	22.00%			
5	Correlation	0.3000				

3. The Excel file for these exercises (which can be found on the website accompanying this book) gives five years of monthly prices for two Vanguard funds—the Vanguard Index 500 Fund (VFINX) and the Vanguard High-Yield Corporate Bond Fund (VWEHX). The first of these funds tracks the Standard & Poor's 500, and VWEHX is a junk-bond fund. Compute the monthly returns and the frontier of combinations of these two funds.
4. Consider the two random variables X and Y whose values are given below. Note that X and Y are perfectly correlated, though perhaps not linearly correlated. Compute their correlation coefficient.

	A	B
1	X	Y
2	-5	25
3	-4	16
4	-3	9
5	-2	4
6	-1	1
7	0	0
8	1	1
9	2	4
10	3	9
11	4	16
12	5	25

5. Assets A and B have the means and variances indicated in the exercise file. Graph three cases, $\rho_{AB} = -1, 0, +1$ on one set of axes, producing the following chart:



6. Three assets have the following means and variance-covariance matrix.

	A	B	C	D	E	F
1		Variance-covariance matrix				Means
2		0.30	0.02	-0.05		10%
3		0.02	0.40	0.06		12%
4		-0.05	0.06	0.60		14%
5						
6		Portfolio1	Portfolio2			
7	Asset1	30%	50%			
8	Asset2	20%	40%			
9	Asset3	50%	10%			

1. Calculate the statistics—mean, variance, standard deviation, covariance, correlation—for the portfolios.
2. Create a chart of the mean and standard deviation of combinations of the portfolios.
3. Add the individual asset returns to the chart. Are the two portfolios on the efficient frontier?

7. Consider the data below and find the portfolio weights so that the expected return of the portfolio is 14%. What is the corresponding portfolio standard deviation?

	A	B	C
1		Mean return	Standard deviation of return
2	Stock 1	12%	35%
3	Stock 2	18%	50%
4	Covariance(r_1,r_2)	0.08350	

8. In the previous problem, find two portfolios whose standard deviation is 45%. (There is an analytical solution to this problem, but it can also be solved by **Solver**.)

Appendix 10.1: Continuously Compounded versus Geometric Returns

Using the continuously compounded return assumes that $P_t = P_{t-1}e^{r_t}$, where r_t is the rate of return during the period ($t - 1$, t). Suppose that r_1, r_2, …, r_{12} are the returns for 12 periods (a period could be a month or it could be a year), then the price of the stock at the end of the 12 periods will be $P_{12} = P_0 e^{r_1+r_2+\cdots+r_{12}}$. This representation of prices and returns allows us to assume that the *average periodic return* is $r = (r_1 + r_2 + \ldots + r_{12})/12$. Since we wish to assume that the return data for the 12 periods represent the distribution of the returns for the coming period, it follows that the continuously compounded return is the appropriate return measure and not the discretely compounded return $r_t = \dfrac{(P_t - P_{t-1})}{P_{t-1}}$.

How Different Are Continuously Compounded and Discretely Compounded Returns?

The continuously compounded return will always be smaller than the discretely compounded return, but the difference is usually not large. The following table shows the differences for the example in section 10.2:

	A	B	C	D	E	F	G	H	I	J
1	APPLE AND KELLOGG CONTINUOUS AND DISCRETE RETURNS Jan 2009 - Jan 2019									
2		Adjusted close price			Continuous return			Discrete return		
3	Date	AAPL	Kellogg		AAPL	Kellogg		AAPL	Kellogg	
4	01-Dec-18	157.07	56.45		-12.06%	-10.13%	<--=LN(C4/C5)	-11.36%	-9.63%	<--=C4/C5-1
5	01-Nov-18	177.20	62.46		-20.34%	-2.83%	<--=LN(C5/C6)	-18.40%	-2.79%	<--=C5/C6-1
6	01-Oct-18	217.17	64.26		-3.10%	-6.70%		-3.05%	-6.48%	
7	01-Sep-18	223.99	68.72		-0.48%	-1.71%		-0.48%	-1.70%	
120	01-Apr-09	11.99	30.97		17.98%	13.94%		19.70%	14.96%	
121	01-Mar-09	10.01	26.94		16.30%	-5.18%		17.70%	-5.05%	
122	01-Feb-09	8.51	28.37		-0.91%	-11.56%		-0.91%	-10.92%	
123	01-Jan-09	8.59	31.85							
124										
125	Monthly mean	=AVERAGE(E4:E122)-->			2.44%	0.48%		2.74%	0.58%	<-- =AVERAGE(I4:I122)
126	Monthly variance	=VAR.S(E4:E122)-->			0.0054	0.0021		0.0056	0.0021	<-- =VAR.S(I4:I122)
127	Monthly standard deviation	=STDEV.S(E4:E122)-->			7.33%	4.54%		7.46%	4.58%	<-- =STDEV.S(I4:I122)
128										
129	Annual mean	=12*E125-->			29.31%	5.77%		38.38%	7.25%	<-- =(1+I125)^12-1
130	Annual variance	=12*E126-->			0.0645	0.0247		0.0667	0.0249	<-- =12*I126
131	Annual standard deviation	=SQRT(12)*E127-->			25.39%	15.72%		25.83%	15.79%	<-- =SQRT(12)*I127

Calculating Annual Returns and Variances from Periodic Returns

Suppose we calculate a series of continuously compounded *monthly* rates of return r_1, r_2, …, r_n and we wish to then calculate the mean and the variance of the *annual* rate of return. Clearly, the mean annual return is given by:[10]

$$\textit{Mean annual return} = 12\left[\frac{1}{n}\sum_{t=1}^{n} r_t\right]$$

To calculate the variance of the annual rate of return, we assume that the monthly rates of return are independent identically distributed random variables. If we use the continuously compounded returns, it then follows that $Var(r) = 12\left[\frac{1}{n}\sum_{t=1}^{n} Var(r_t)\right] = 12\sigma^2_{monthly}$, and that the standard deviation of the annual rate of return is given by $\sigma = \sqrt{12}\sigma_{monthly}$.

Appendix 10.2: Adjusting for Dividends

When downloading data from Yahoo or other sources, the “adjusted price” includes an adjustment for dividends. In this appendix we discuss two ways of making this adjustment.[11] The first, and simplest, method of adjusting for dividends is to add them to the annual change in price. In the following example, if you purchased Kellogg’s stock (K) at the 2007 year-end price of \$52.43 per share and held it for one year, you would, at the end of the year, have made 13.9%.

$$\textit{Discretely compounded return, 2008} = \frac{43.85 + 1.30}{52.43} - 1 = -13.9\%$$

The continuously compounded return is calculated by:

$$\textit{Continuously compounded return, 2008} = ln\left[\frac{43.85 + 1.30}{52.43}\right] = -14.9\%$$

(The choice between discrete and continuous compounding is discussed further in subsequent sections of this appendix.)

	A	B	C	D	E	F
1	KELLOGG (K) STOCK ADJUSTING FOR DIVIDENDS					
2	Year	Share price at year end	Dividend per share	Discretely compounded return	Continuously compounded return	
3	2007	52.43				=(B4+C4)/B3-1
4	2008	43.85	1.30	-13.9%	-14.9%	<--=LN(SUM(B4:C4)/B3)
5	2009	53.20	1.43	24.6%	22.0%	
6	2010	51.08	1.56	-1.1%	-1.1%	
7	2011	50.57	1.67	2.3%	2.2%	
8	2012	55.85	1.74	13.9%	13.0%	
9	2013	61.07	1.80	12.6%	11.8%	
10	2014	65.44	1.90	10.3%	9.8%	
11	2015	72.27	1.98	13.5%	12.6%	
12	2016	73.71	2.04	4.8%	4.7%	
13	2017	67.98	2.12	-4.9%	-5.0%	
14	2018	57.01	2.20	-12.9%	-13.8%	
15			Average return	4.5%	3.8%	<--=AVERAGE(E4:E14)
16	Standard deviation of returns			11.98%	11.66%	<--=STDEV.S(E4:E14)

Dividend Reinvestment

Another way of calculating returns is to assume that the dividends are reinvested in the stock:

	A	B	C	D	E	F	G	H
19	KELLOGG (K) REINVESTING THE DIVIDENDS IN SHARES							
20	Year	Share price at year end	Dividend per share	Effective shares held at year end	Value of shares at end of year	Discretely compounded return	Continuously compounded return	
21	2007	52.43		1.00	52.43			=E22/E21-1
22	2008	43.85	1.30	1.03	45.15	-13.9%	-14.9%	<--=LN(E22/E21)
23	2009	53.20	1.43	1.06	56.21	24.5%	21.9%	
24	2010	51.08	1.56	1.09	55.53	-1.2%	-1.2%	
25	2011	50.57	1.67	1.12	56.64	2.0%	2.0%	
26	2012	55.85	0.43	1.13	62.99	11.2%	10.6%	
27	2013	61.07	0.00	1.13	68.87	9.3%	8.9%	
28	2014	65.44	0.00	1.13	73.80	7.2%	6.9%	
29	2015	72.27	0.00	1.13	81.51	10.4%	9.9%	
30	2016	73.71	2.04	1.16	85.17	4.5%	4.4%	
31	2017	67.98	2.12	1.19	80.67	-5.3%	-5.4%	
32	2018	57.01	2.20	1.23	69.85	-13.4%	-14.4%	
33					Average return	3.2%	2.6%	<--=AVERAGE(G22:G32)
34					Standard deviation of returns	11.34%	11.09%	<--=STDEV.S(G22:G32)
35								
36		=D21+C22/B22			=D22*B22			
37								

Consider first 2008: Since we purchased the share at the end of 2007, we own one share at the end of 2007. If the 2008 dividend is turned into shares at the 2008 year-end price, we can use it to buy 0.03 additional shares:

$$\textit{New shares purchased at year-end } 2008 = \frac{1.30}{43.85} = 0.03$$

Thus, we start 2009 with 1.03 shares. Since the 2009 dividend per share was \$1.43, the total dividend received on the shares is 1.03 $*$ \$1.43 = \$1.47. Reinvesting these dividends in shares gives:

$$\textit{New shares purchased at year-end } 2009 = \frac{1.47}{53.20} = 0.03$$

Thus, at the end of 2009, the holder of Kellogg shares will have accumulated 1 + 0.03 + 0.03 = 1.06 shares (cell D23 in the spreadsheet above).

As the spreadsheet fragment above shows, this reinvestment of dividends will produce a holding of 1.09 shares at the end of 2010, worth \$55.66.

We can calculate the return on this investment in one of two ways:

$$Avg\begin{pmatrix} \textit{Continuously} \\ \textit{compounded return} \end{pmatrix} = \frac{1}{11}\sum_{t=1}^{11} ln\left(\frac{\textit{Value of shares at end year}_t}{\textit{Value of shares at end year}_{t-1}}\right)$$

Note that this continuously compounded return (the method preferred in this book) is the same as that calculated in the first spreadsheet fragment in this appendix from the annual returns (E15 = G33 and E16 = G34).

An alternative is to calculate the geometric return:

$$Avg\begin{pmatrix} \textit{Discretely} \\ \textit{compounded return} \end{pmatrix} = \frac{1}{11}\sum_{t=1}^{11}\left(\frac{\textit{Value of shares at end year}_t}{\textit{Value of shares at end year}_{t-1}} - 1\right)$$

1. 1. Press et al. (2007), 722.
2. 2. The function **Covar** was used in pre-Excel 2013 and calculates the population covariance; it still works in current versions. Previous Excel versions had functions **VarP**, **StdevP**, **VarS**, and **StdevS** (though still working, these are now replaced by **Var.P**, **Stdev.P**, **Var.S**, and **Stdev.S**).
3. 3. Chapter 29 gives an introduction to matrices sufficient to deal with all the problems encountered in this book. The Excel matrix functions **MMult** and **MInverse** used in portfolio problems are discussed in this chapter.
4. 4. In Chapter 11 we discuss portfolio optimization when there are short-sale restrictions. In the meantime, we assume that short-selling is unrestricted.
5. 5. Array functions are discussed in Chapter 31. The many examples below illustrate their use for portfolio optimization.
6. 6. We explain in Chapter 34 that **MMult** and **Transpose** are array functions and must be entered by pushing [Ctrl] + [Shift] + [Enter] simultaneously.
7. 7. This sentence is critical. As we shall see in the next chapter, all portfolio problems of *N* assets ultimately boil down to the two-asset case.
8. 8. Chapter 13 discusses the difference between envelope and efficient portfolios. In a sentence, the efficient frontier is a subset of the envelope containing only the optimal portfolios.
9. 9. We discuss Fischer Black's theorem more extensively in Chapter 11.
10. 10. Note that this is not true for discretely computed returns. Thus, the computation of the variance and the standard deviation are simpler when using continuous compounding.
11. 11. It might be argued that since the free sources available on the web make these adjustments automatically, the details in this appendix are superfluous. Nevertheless, we think they offer some interesting insights. (If you disagree, turn the page!)

11 Efficient Portfolios and the Efficient Frontier

11.1 Overview

This chapter covers the theory and calculations necessary for both versions of the classical capital asset pricing model (CAPM)—the one that is based on a risk-free asset (also known as the Sharpe-Lintner-Mossin model) and Black's (1972) zero-beta CAPM (which does not require the assumption of a risk-free asset). You will find that using a spreadsheet enables you to do the necessary calculations easily.

The structure of the chapter is as follows: We begin with some preliminary definitions and notation. We then state the major results (proofs are given in the appendix to the chapter). In succeeding sections, we implement these results, showing you:

- How to calculate efficient portfolios
- How to calculate the efficient frontier

This chapter includes more theoretical material than most chapters in this book: Section 11.2 contains the propositions on portfolios that underlie the calculations of both efficient portfolios and the security market line in Chapter 13. If you find the theoretical material in section 11.2 difficult, skip it at first and try to follow the illustrative calculations in section 11.3. This chapter assumes that the variance-covariance matrix is given; we delay a discussion of various methods of computing the variance-covariance matrix until Chapter 12.

11.2 Some Preliminary Definitions and Notation

Throughout this chapter we use the following notation: There are N risky assets, each of which has expected return $E(r_i)$. The matrix $E(r)$ is the column vector of expected returns of these assets:

$$E(r) = \begin{bmatrix} E(r_1) \\ E(r_2) \\ \vdots \\ E(r_N) \end{bmatrix}$$

S is the $N \times N$ variance-covariance matrix:

$$S = \begin{bmatrix} \sigma_{11} & \sigma_{21} & \cdots & \sigma_{N1} \\ \sigma_{12} & \sigma_{22} & & \sigma_{N2} \\ \vdots & & \ddots & \vdots \\ \sigma_{1N} & \sigma_{2N} & \cdots & \sigma_{NN} \end{bmatrix}$$

A *portfolio of risky assets* (when our intention is clear, we shall just use the word *portfolio*) is a column vector x whose coordinates sum to 1:

$$x = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_N \end{bmatrix}, \sum_{i=1}^{N} x_i = 1$$

Each coordinate x_i represents the proportion of the portfolio invested in risky asset i.

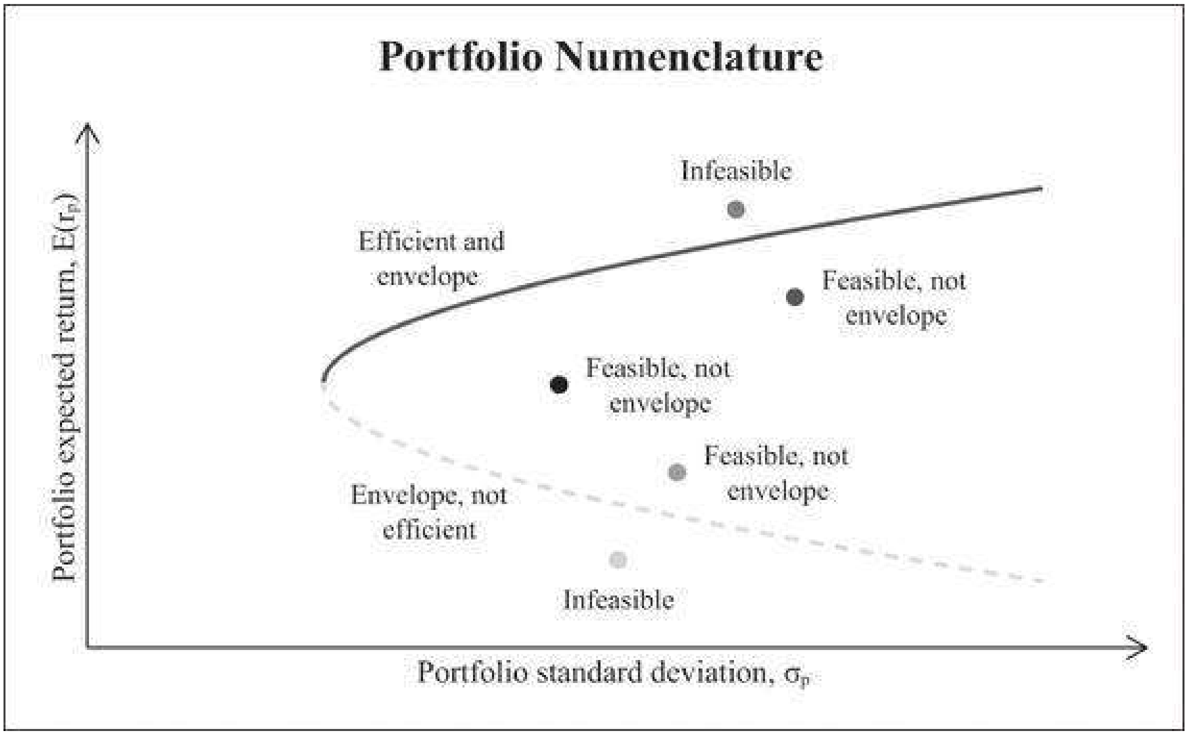
The *expected portfolio return* $E(r_p)$ of a portfolio x is given by the product of x and R:

$$E(r_x) = x^T \cdot R \equiv \sum_{i=1}^{N} x_i E(r_i)$$

The *variance of portfolio* x's *return*, $\sigma_x^2 \equiv \sigma_{xx}$, is given by the product $x^T S x = \sum_{i=1}^{N} \sum_{j=1}^{N} x_i x_j \sigma_{ij}$.

The *covariance between the return of two portfolios x and y*, $Cov(r_x, r_y)$, is defined by the product $\sigma_{xy} = x^T S y = \sum_{i=1}^{N} \sum_{j=1}^{N} x_i y_j \sigma_{ij}$. Note that $\sigma_{xy} = \sigma_{yx}$.

The following graph illustrates four concepts. A *feasible* portfolio is any portfolio whose proportions sum to 1. The *feasible set* is the set of portfolio means and standard deviations generated by the feasible portfolios; this feasible set is the area inside and to the right of the curved line. A feasible portfolio is on the *envelope* of the feasible set if, for a given mean return, it has minimum variance. Finally, a portfolio x is *an efficient portfolio* if it maximizes the return given the portfolio variance (or standard deviation). That is, x is efficient if there is no other portfolio y such that $E(r_y) > E(r_x)$ and $\sigma_y < \sigma_x$. The set of all efficient portfolios is called the *efficient frontier*; this frontier is the heavier line in the graph.



11.3 Five Propositions on Efficient Portfolios and the CAPM

In the appendix to this chapter we prove the following results, which are basic to the calculations of the CAPM. All of these propositions are used in deriving the efficient frontier and the security market line; numerical illustrations are given in the next section and in succeeding chapters.

Proposition 1—How to Find Portfolio on the Envelope

Let c be a constant. We use the notation $E(r) - c$ to denote the following column vector:

$$E(r) - c = \begin{bmatrix} E(r_1) - c \\ E(r_2) - c \\ \vdots \\ E(r_N) - c \end{bmatrix}$$

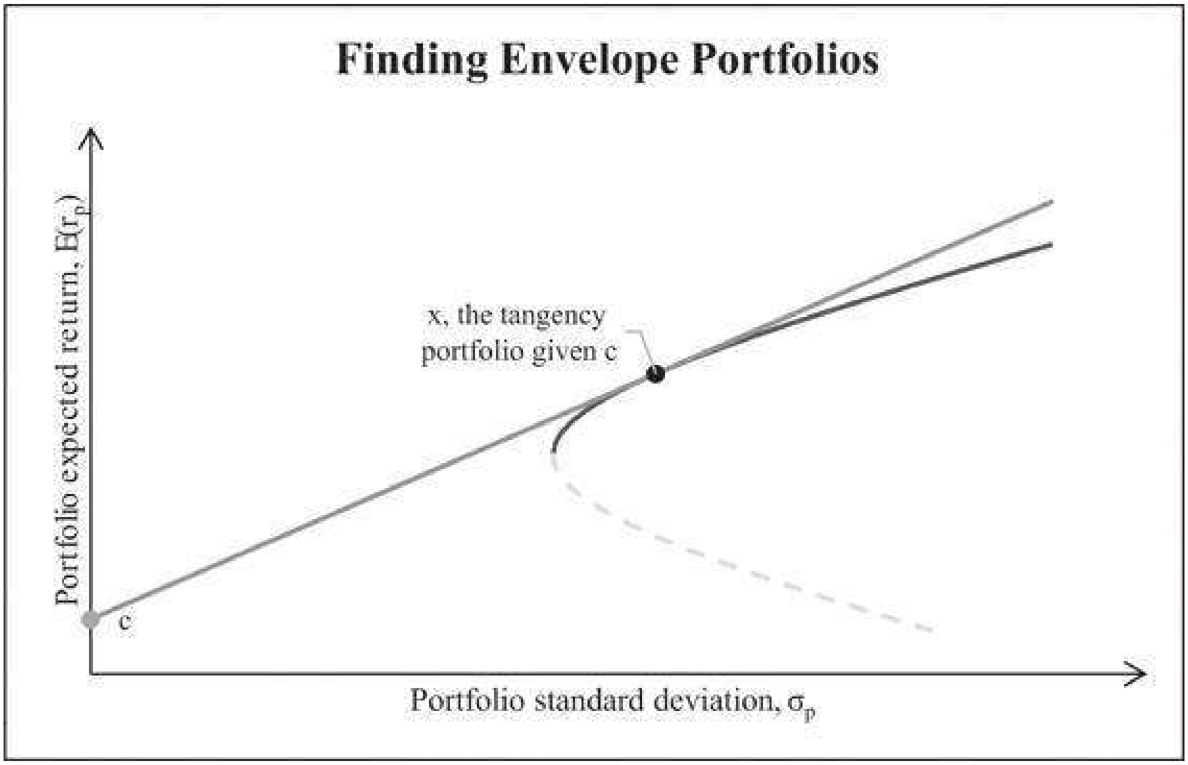
Let the vector z solve the system of simultaneous linear equations $E(r) - c = Sz$. Then this solution produces a portfolio x on the envelope of the feasible set in the following manner:

$$z = \{z_1, z_2, \ldots, z_N\} = S^{-1}\{E(r) - c\}$$

$$x = \{x_1, x_2, \ldots, x_N\}, \quad \text{where} \quad x_i = \frac{z_i}{\sum_{j=1}^{N} z_j}$$

Furthermore, all envelope portfolios are of this form.

Intuition: A formal proof of the proposition is given in this chapter's mathematical appendix, but the intuition is simple and geometric. Suppose we pick a constant c and we try to find an efficient portfolio x for which there is a tangency between c and the feasible set:



Proposition 1 gives a procedure for finding x; furthermore, the proposition states that all envelope portfolios (in particular, all efficient portfolios) are the result of the procedure outlined in the proposition. That is, if x is any envelope portfolio, then there exists a constant c and a vector z such that $S \cdot z = E(r) - c$ and $x_i = \frac{z_i}{\sum_i z_i}$.

Proposition 2—Use Two Envelope Portfolios to Establish the Whole Envelope

By a theorem first proved by Black (1972), any two envelope portfolios are enough to establish the whole envelope. Given any two envelope portfolios $x = \{x_1, \ldots, x_N\}$ and $y = \{y_1, \ldots, y_N\}$, all envelope portfolios are convex combinations of x and y. This means that given any constant a, the portfolio

$$ax + (1-a)y = \begin{bmatrix} ax_1 + (1-a)y_1 \\ ax_2 + (1-a)y_2 \\ \vdots \\ ax_N + (1-a)y_N \end{bmatrix}$$

is an envelope portfolio.

Proposition 3—Black's Zero-Beta Model

If y is any envelope portfolio, then for any other portfolio (envelope or not) x, we have the relationship:

$E(r_x) = c + \beta_x [E(r_y) - c]$

where

$$\beta_x = \frac{Cov(x, y)}{\sigma_y^2}$$

Furthermore, c is the expected return of all portfolios z whose covariance with y is zero:

$c = E(r_z)$, where $cov(y, z) = 0$

Notes: If y is on the envelope, the regression of any and all portfolios x on y gives a linear relationship. In this version of the CAPM (usually known as "Black's zero-beta CAPM," in honor of Fischer Black, whose 1972 paper proved this result) the Sharpe-Lintner-Mossin security market line (SML) is replaced with an SML in which the role of the risk-free asset is played by a portfolio with a zero-beta with respect to the particular envelope portfolio y. Note that this result is true for any envelope portfolio y.

The converse of Proposition 3 is also true.

Proposition 4—First Step to the Security Market Line (SML)

Suppose that there exists a portfolio y such that for any portfolio x the following relation holds:

$E(r_x) = c + \beta_x [E(r_y) - c]$

where

$$\beta_x = \frac{Cov(x, y)}{\sigma_y^2}$$

Then the portfolio y is an envelope portfolio.

Propositions 3 and 4 show that *an SML relation holds if and only if we regress all portfolio returns on an envelope portfolio with an* R^2 = 100%. As Roll (1977, 1978) has forcefully pointed out, these propositions show that it is not enough to run a test of the CAPM by showing that the SML holds.[1] The only real test of the CAPM is *whether the true market portfolio is mean-variance efficient*. We shall return to this topic in Chapter 13.

The Market Portfolio

The *market portfolio M* is a portfolio composed of *all the risky assets in the economy*, with each asset taken in proportion to its value. To make this concept more specific, suppose that there are N risky assets and that the market value of asset i is V_i. Then the market portfolio has the following weights:

$$\text{proportion of asset } i \text{ in } M = \frac{V_i}{\sum_{h=1}^{N} V_h}$$

If the market portfolio M is efficient (this is a big "if" as we shall see in Chapter 13), Proposition 3 is also true for the market portfolio. That is, the SML holds with $E(r_z)$ substituted for c:

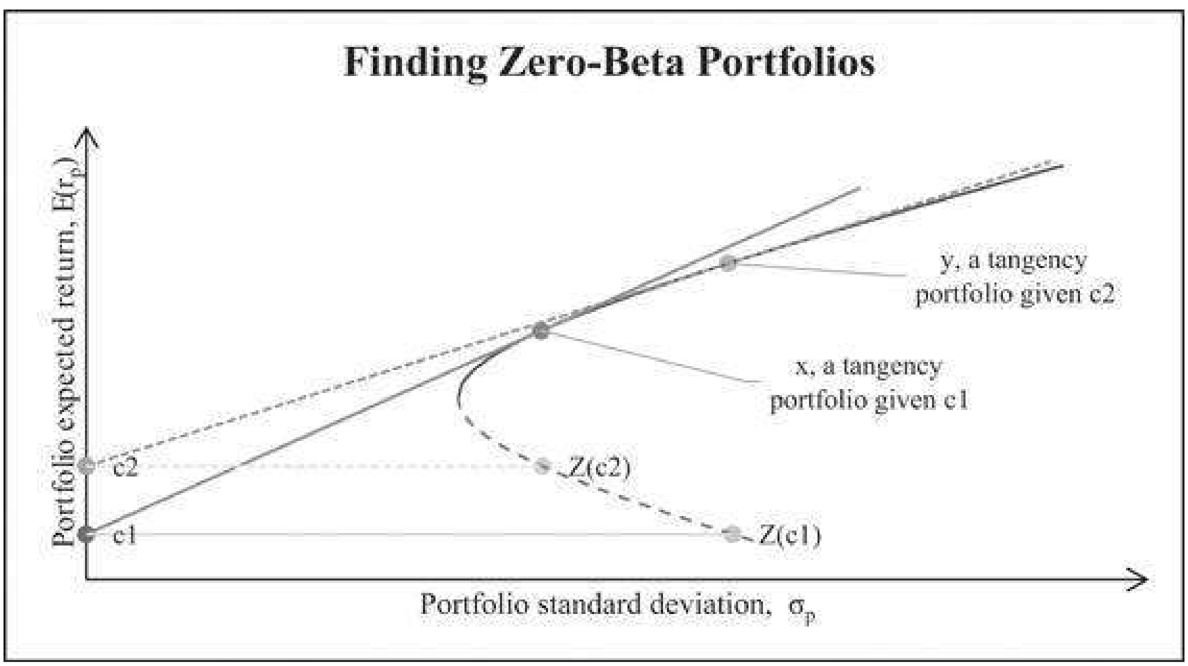
$E(r_x) = E(r_z) + \beta_x\, [E(r_M) - E(r_z)]$

where

$$\beta_x = \frac{Cov(x, M)}{\sigma_M^2}$$

$$Cov(z, M) = 0$$

This version of the SML has received the most empirical attention of all of the CAPM results. In Chapter 13 we show how to calculate β_x and how to calculate the SML; we go on to examine Roll's criticism of these empirical tests. From the following graph, it is easy to see how to locate a zero-beta portfolio on the envelope of the feasible set:



When there is a risk-free asset, Proposition 3 specializes to the security market line of the classic capital asset pricing model.

Proposition 5—The SML

If there exists a risk-free asset with return r_f, then there exists an envelope portfolio M such that for any asset i (efficient and non-efficient) we get:

$$E(r_i) = r_f + \beta_i[E(r_M) - r_f]$$

where

$$\beta_i = \frac{cov(r_i, r_M)}{\sigma_M^2}$$

As shown in the classic papers by Sharpe (1964), Lintner (1965), and Mossin (1966), if all investors choose their portfolios only on the basis of portfolio mean and standard deviation, then the portfolio x of Proposition 5 is the market portfolio M.

In the remainder of this chapter, we explore the meaning of these propositions using numerical examples worked out in Excel.

11.4 Calculating the Efficient Frontier: An Example

In this section we calculate the efficient frontier using Excel. We consider a world with four risky assets having the following expected returns and variance-covariance matrix:

	A	B	C	D	E	F	G
1	CALCULATING THE FRONTIER						
2	Variance-covariance matrix, S						Mean
3		Asset 1	Asset 2	Asset 3	Asset 4		returns E(r)
4	Asset 1	0.10	0.01	0.03	0.05		6%
5	Asset 2	0.01	0.30	0.06	-0.04		8%
6	Asset 3	0.03	0.06	0.40	0.02		10%
7	Asset 4	0.05	-0.04	0.02	0.50		15%

We separate our calculations into two parts. In the next subsection we calculate two portfolios on the envelope of the feasible set. In the subsequent subsection we calculate the efficient frontier.

The equivalent code in R will look like this:

```
## Inputs
# Variance-covariance matrix
var_cov_mat <- matrix(c(0.10, 0.01, 0.03, 0.05,
                        0.01, 0.30, 0.06, -0.04,
                        0.03, 0.06, 0.40, 0.02,
                        0.05, -0.04,0.02, 0.50),
                        nrow=4)

# Mean Returns
assets_mean_ret <- c(0.06,0.08,0.1,0.15)
```

Calculating Two Envelope Portfolios

By Proposition 2, we must find two efficient portfolios in order to identify the whole efficient frontier. By Proposition 1, each envelope portfolio solves the system $R - c = S$

· z for z. To identify two efficient portfolios, we use two different values for c. For each value of c, we solve for z and then set $x_i = \frac{z_i}{\sum_i z_i}$ to find an efficient portfolio.

The c values we solve for are arbitrary (see section 11.6), but to make life easy, we first solve this system for $c = 0$. This gives the following results:

	A	B	C	D	E	F	G	H
9	**Computing an envelope portfolio with constant = 0**							
10		**z**					**Envelope portfolio, x**	
11	**Asset 1**	0.3861	<--{=MMULT(MINVERSE(B4:E7),G4:G7-0)}				0.3553	<--=B11/$B$15
12	**Asset 2**	0.2567					0.2362	
13	**Asset 3**	0.1688					0.1553	
14	**Asset 4**	0.2752					0.2532	
15	**Sum**	**1.0868**	<--=SUM(B11:B14)			Sum	**1.0000**	<--=SUM(G11:G14)

The formulas in the cells are as follows:

- • For z we use the array function **MMult(MInverse(B4:E7),G4:G7-0)** (see the spreadsheet above). The range B4:E7 contains the variance-covariance matrix and the cells G4:G7 contain the expected returns of the assets.
- • For x, each cell contains the associated value of z divided by the sum of all the z's. Thus, for example, cell G12 contains the formula **=B12/$B$15** which is the same as writing **=B12/SUM($B$11:$B$14)**.

To find the second envelope portfolio, we now solve this system for $c = 0.04$ (cell B18).

	A	B	C	D	E	F	G	H
17	**Computing an envelope portfolio with constant = 4.00%**							
18	Constant, c	4.00%						
19		**z**					**Envelope portfolio, y**	
20	**Asset 1**	0.0404	<--	{=MMULT(MINVERSE(B4:E7),G4:G7-B18)}			0.0782	<--=B20/$B$24
21	**Asset 2**	0.1386					0.2684	
22	**Asset 3**	0.1151					0.2227	
23	**Asset 4**	0.2224					0.4307	
24	**Sum**	**0.5165**	<--=SUM(B20:B23)			Sum	**1.0000**	<-- =SUM(G20:G23)

The portfolio y in cells G20:G23 is, by the results of Proposition 1, an envelope portfolio. This vector z associated with y is calculated in a manner similar to that of the first vector, except that the array function in the cells is **MMult(MInverse(B4:E7),G4:G7-B18)**, where G4:G7-B18 contains the vector of expected returns minus the constant 0.04.

Implementing this in R looks like this:

```
# Function: Calculate Envelope Portfolio proportions in one step
envelope_portfolio <- function(var_cov_mat, assets_mean_ret, constant){
  Z <- solve(var_cov_mat) %*% (assets_mean_ret - constant) # Z
  Z_Sum <- sum(Z) # Sum of Z's
  return( as.vector(Z/Z_Sum) ) # Efficient Portfolio proportions
}

# Computing Portfolios X and Y Asset Allocation
port_x_prop <- envelope_portfolio(var_cov_mat, assets_mean_ret, 0)
port_y_prop <- envelope_portfolio(var_cov_mat, assets_mean_ret, 0.04)
```

Note that we first define a function in R and then substitute the different constants to get the proportions of each efficient portfolio.

To complete the basic calculations, we compute the means, standard deviations, and covariance of returns for portfolios *x* and *y*:

	A	B	C	D
26		Portfolio x	Portfolio y	
27	Constant, c	0%	4.00%	<--=B18
28	$E(r_p)$	9.37%	11.30%	<--{=MMULT(TRANSPOSE(G20:G23),G4:G7)}
29	$Var(r_p)$	0.0862	0.1414	<--{=MMULT(MMULT(TRANSPOSE(G20:G23),B4:E7),G20:G23)}
30	Sigma	29.37%	37.60%	<--=SQRT(C29)
31				
32	Cov(x,y)	0.1040	<--{=MMULT(MMULT(TRANSPOSE(G11:G14),B4:E7),G20:G23)}	
33	Corr(x,y)	0.9419	<--=B32/(B30*C30)	

The transpose vectors of x and of y are inserted using the array function **Transpose** (see Chapter 31 for a discussion of array functions). This now enables us to calculate the mean, variance, and covariance as follows:

- **E(r_p)** uses the array formula **MMult(Transpose(x),means)**. Note that we could have also used the function **Sumproduct(x,means)**.
- **Var(r_p)** uses the array formula **MMult(MMult(Transpose(x),var_cov),x)**.
- **Sigma** uses the formula **SQRT(Var(r_p))**.
- **Cov(x,y)** uses the array formula **MMult(MMult(Transpose(x),var_cov),y)**.
- **Corr(x,y)** uses the formula **Cov(x,y)/(Sigma(x) * Sigma(y))**.

Implementing the above in R looks like this:

```
# Mean Returns
port_x_ret <- assets_mean_ret %*% port_x_prop
port_y_ret <- assets_mean_ret %*% port_y_prop

# Portfolios Variance and Covariance
port_x_var <- port_x_prop %*% var_cov_mat %*% port_x_prop
port_y_var <- port_y_prop %*% var_cov_mat %*% port_y_prop
cov_xy <- port_x_prop %*% var_cov_mat %*% port_y_prop

# Portfolios Standard Deviations
port_x_sigma <- sqrt(port_x_var)
port_y_sigma <- sqrt(port_y_var)

# Correlation Coefficient (Rho)
port_correl<-cov_xy / (port_x_sigma * port_y_sigma)
```

The following spreadsheet illustrates everything that has been done in this subsection:

	A	B	C	D	E	F	G	H
1	**CALCULATING EFFICIENT PORTFOLIOS AND THE FRONTIER**							
2	Variance-covariance matrix, S						**Mean**	
3		**Asset 1**	**Asset 2**	**Asset 3**	**Asset 4**		**returns E(r)**	
4	**Asset 1**	0.10	0.01	0.03	0.05		6%	
5	**Asset 2**	0.01	0.30	0.06	-0.04		8%	
6	**Asset 3**	0.03	0.06	0.40	0.02		10%	
7	**Asset 4**	0.05	-0.04	0.02	0.50		15%	
8								
9	**Computing an envelope portfolio with constant = 0**							
10		**z**					**Envelope portfolio, x**	
11	**Asset 1**	0.3861	<--{=MMULT(MINVERSE(B4:E7),G4:G7-0)}				0.3553	<--=B11/$B$15
12	**Asset 2**	0.2567					0.2362	
13	**Asset 3**	0.1688					0.1553	
14	**Asset 4**	0.2752					0.2532	
15	**Sum**	**1.0868**	<--=SUM(B11:B14)			**Sum**	**1.0000**	<--=SUM(G11:G14)
16								
17	**Computing an envelope portfolio with constant = 4.00%**							
18	Constant, c	4.00%						
19		**z**					**Envelope portfolio, y**	
20	**Asset 1**	0.0404		{=MMULT(MINVERSE(B4:E7),G4:G7-B18)}			0.0782	<--=B20/$B$24
21	**Asset 2**	0.1386					0.2684	
22	**Asset 3**	0.1151					0.2227	
23	**Asset 4**	0.2224					0.4307	
24	**Sum**	**0.5165**	<--=SUM(B20:B23)			**Sum**	**1.0000**	<--=SUM(G20:G23)
25								
26		**Portfolio x**	**Portfolio y**					
27	Constant, c	0%	4.00%	<--=B18				
28	E(r_p)	9.37%	11.30%	<--{=MMULT(TRANSPOSE(G20:G23),G4:G7)}				
29	Var(r_p)	0.0862	0.1414	<--{=MMULT(MMULT(TRANSPOSE(G20:G23),B4:E7),G20:G23)}				
30	Sigma	29.37%	37.60%	<--=SQRT(C29)				
31								
32	Cov(x,y)	0.1040	<--{=MMULT(MMULT(TRANSPOSE(G11:G14),B4:E7),G20:G23)}					
33	Corr(x,y)	0.9419	<--=B32/(B30*C30)					

Calculating the Envelope

By Proposition 2 of section 11.3, convex combinations of the two portfolios calculated in the previous subsection allow us to calculate the whole envelope of the feasible set (which, of course, includes the efficient frontier). Suppose we let p be a portfolio that has proportion a invested in portfolio x and proportion $(1 - a)$ invested in y. Then, as discussed in Chapter 10, the mean and standard deviation of p's returns are

$$E(r_p) = aE(r_x) + (1 - a)\, E(r_y).$$

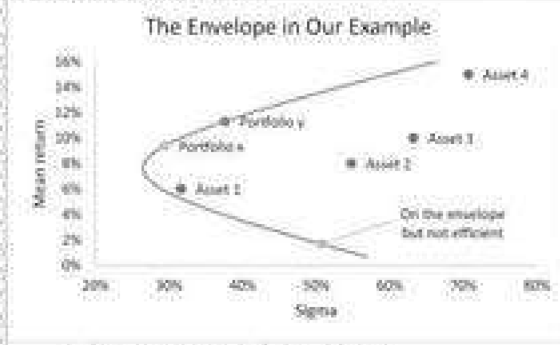
$$\sigma_p = \sqrt{a^2\sigma_x^2 + (1 - a)^2\sigma_y^2 + 2a(1 - a)Cov(x, y)}$$

Here's a sample calculation for our two portfolios:

	A	B	C	D	E	F	G
35	**A single portfolio calculation**						
36	Proportion of x	30%					
37	E(r_p)	10.72%	<--=B36*B28+(1-B36)*C28				
38	Var(r_p) = σ_p^2	0.1207	<--=B36^2*B29+(1-B36)^2*C29+2*B36*(1-B36)*B32				
39	σ_p	34.75%	<--=SQRT(B38)				

We can turn this calculation into a Data Table (see Chapter 28) to get the following table:

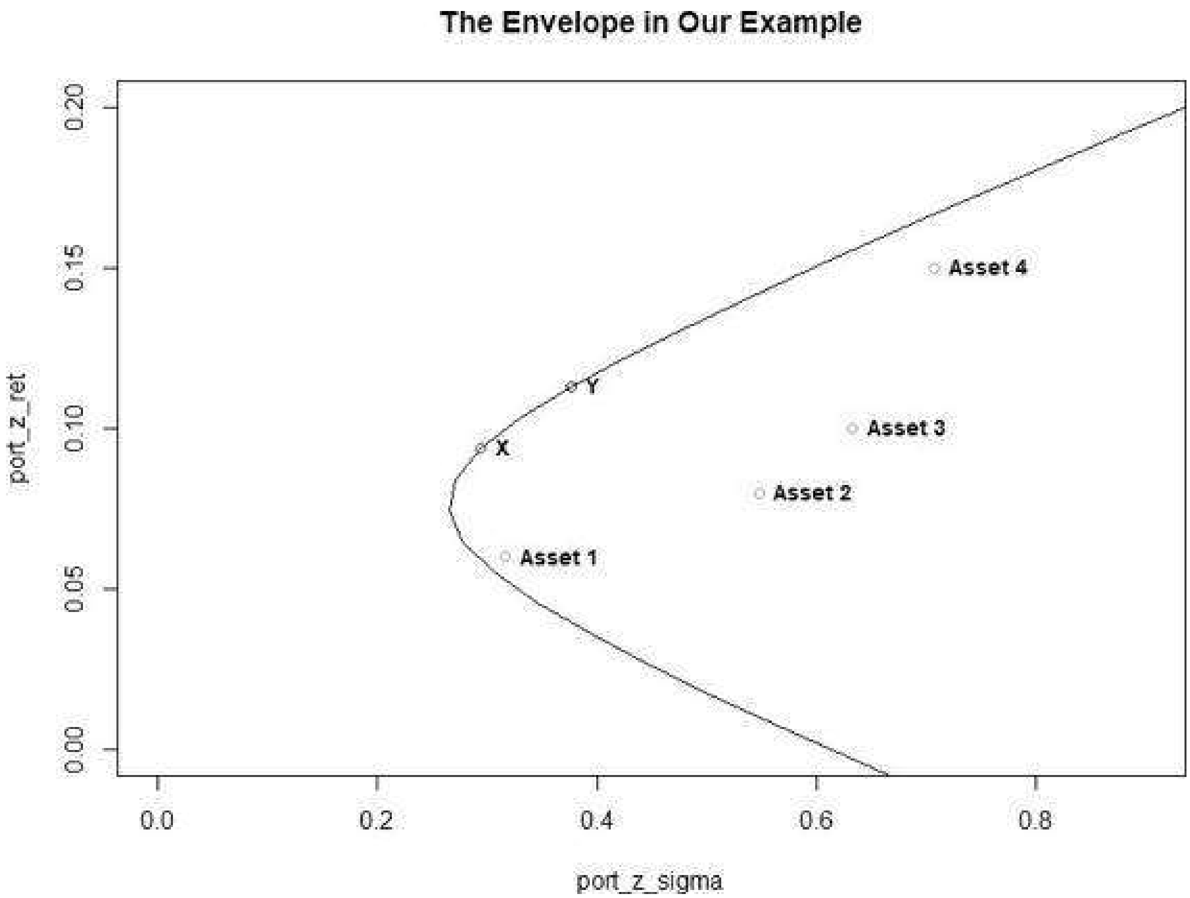
	A	B	C	D
35	**A single portfolio calculation**			
36	Proportion of x	30%		
37	$E(r_p)$	10.72%	<--=B36*B28+(1-B36)*C28	
38	$Var(r_p) = \sigma_p^2$	0.1207	<--=B36^2*B29+(1-B36)^2*C29+2*B36*(1-B36)*B32	
39	σ_p	34.75%	<--=SQRT(B38)	
40				
41	**Data table: We vary the proportion of x to produce a graph of the frontier**			
42	**Prop. of x**	**Sigma**	**Return**	
43		34.75%	10.72%	<--=B37, data table header
44	-250%	67.15%	16.13%	
45	-200%	60.78%	15.17%	
46	-150%	54.56%	14.20%	
47	-100%	48.56%	13.23%	
48	-50%	42.86%	12.27%	
49	0%	37.60%	11.30%	
50	50%	33.00%	10.34%	
51	100%	29.37%	9.37%	
52	150%	27.09%	8.41%	
53	200%	26.52%	7.44%	
54	250%	27.76%	6.48%	
55	300%	30.60%	5.51%	
56	350%	34.64%	4.54%	
57	400%	39.52%	3.58%	
58	450%	44.96%	2.61%	
59	**500%**	**50.79%**	**1.65%**	<-- On the envelope but not efficient
60	550%	56.88%	0.68%	



The two portfolios, *x* and *y*, whose convex combinations compose the envelope are marked. Note that the convex combinations all lie on the envelope but may not necessarily be efficient. An example is the almost last point in the data table (line 59), which invests 500% in *x* and 400% in portfolio *y*. Thus, while every efficient portfolio is a convex combination of any two efficient portfolios, it is *not true* that every convex combination of any two efficient portfolios is efficient.

Implementing the above in R should look like this:

```r
# Setting Different Proportions of X
port_x_props <- c(c(1:25)/2 - 5)
port_y_props <- 1 - port_x_props

# Calculating Mean Returns
port_z_ret <- port_x_props * as.numeric(port_x_ret) +
              port_y_props * as.numeric(port_y_ret)
port_z_sigma <- sqrt(port_x_props^2 * as.numeric(port_x_var) +
                     port_y_props^2 * as.numeric(port_y_var) +
                     (port_x_props * port_y_props) * as.numeric(cov_xy) * 2)

# Plot Envelope
plot(port_z_sigma, port_z_ret, main="The Envelope in Our Example",
     type="l", xlim=c(0, 0.9), ylim=c(0, 0.2))

# Plot X and Y
lines(port_x_sigma, port_x_ret, type = "b", col = "red")
text(port_x_ret ~ port_x_sigma, labels = c("X"),cex = 0.9, font = 2, pos = 4)
lines(port_y_sigma, port_y_ret, type = "b", col = "blue")
text(port_y_ret ~ port_y_sigma, labels = c("Y"),cex = 0.9, font = 2, pos = 4)

#Assets Sigma
assets_sd <- sqrt(diag(var_cov_mat))

# Plot Assets
points(assets_sd, assets_mean_ret, col = "Green")
text(assets_mean_ret ~ assets_sd,
     labels = c("Asset 1", "Asset 2", "Asset 3", "Asset 4"),
     cex=0.9, font=2, pos=4)
```



Finding Efficient Portfolios in One Step

The examples in this section find efficient portfolios by writing out most of the components of the portfolio separately on the spreadsheet. However, for some uses we will want to calculate the efficient portfolio in one step. For example:

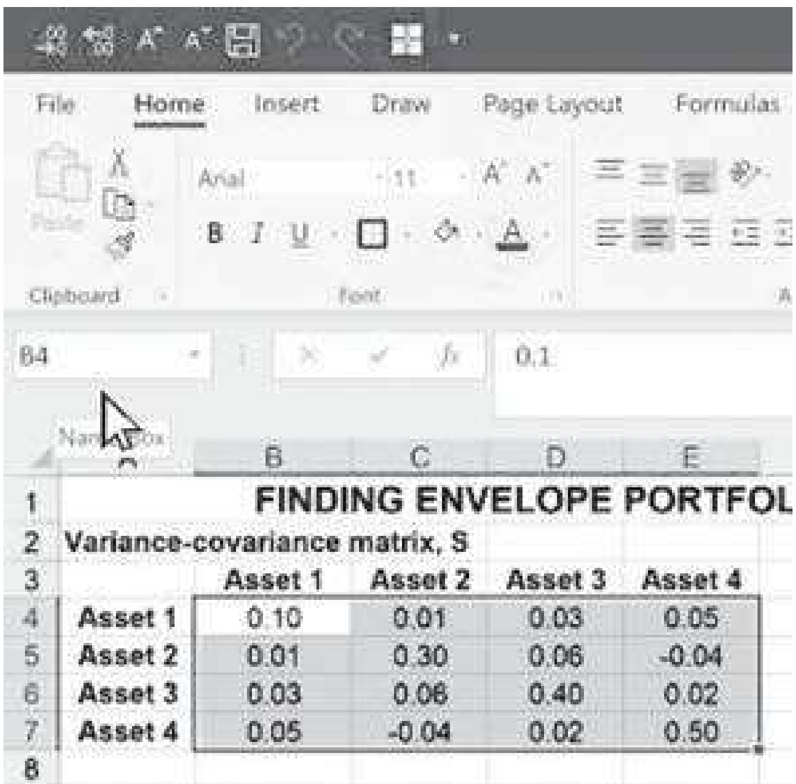
	A	B	C	D	E	F	G	H
1	FINDING ENVELOPE PORTFOLIOS IN ONE STEP							
2	Variance-covariance matrix, S						Mean	
3		Asset 1	Asset 2	Asset 3	Asset 4		returns	
4	Asset 1	0.10	0.01	0.03	0.05		6%	
5	Asset 2	0.01	0.30	0.06	-0.04		8%	
6	Asset 3	0.03	0.06	0.40	0.02		10%	
7	Asset 4	0.05	-0.04	0.02	0.50		15%	
8								
9	Constant	4%						
10		Envelope portfolio						
11	Asset 1	7.82% ←		{=MMULT(MINVERSE(B4:E7),G4:G7-B9)/SUM(MMULT(MINVERSE(B4:E7),G4:G7-B9))}				
12	Asset 2	26.84%						
13	Asset 3	22.27%						
14	Asset 4	43.07%						
15	Sum	100.00%	<--=SUM(B11:B14)					
16								
17	Portfolio mean	11.30%	<-- =SUMPRODUCT(B11:B14,G4:G7)					
18	Portfolio sigma	37.60%	<-- {=SQRT(MMULT(MMULT(TRANSPOSE(B11:B14),B4:E7),B11:B14))}					

This approach requires a number of Excel tricks, most of which relate to the correct use of array functions. The end result is that we can write the Proposition 1 expression for an envelope portfolio, $x = \dfrac{S^{-1}\{E(r)-c\}}{Sum[S^{-1}\{E(r)-c\}]}$, in one cell:

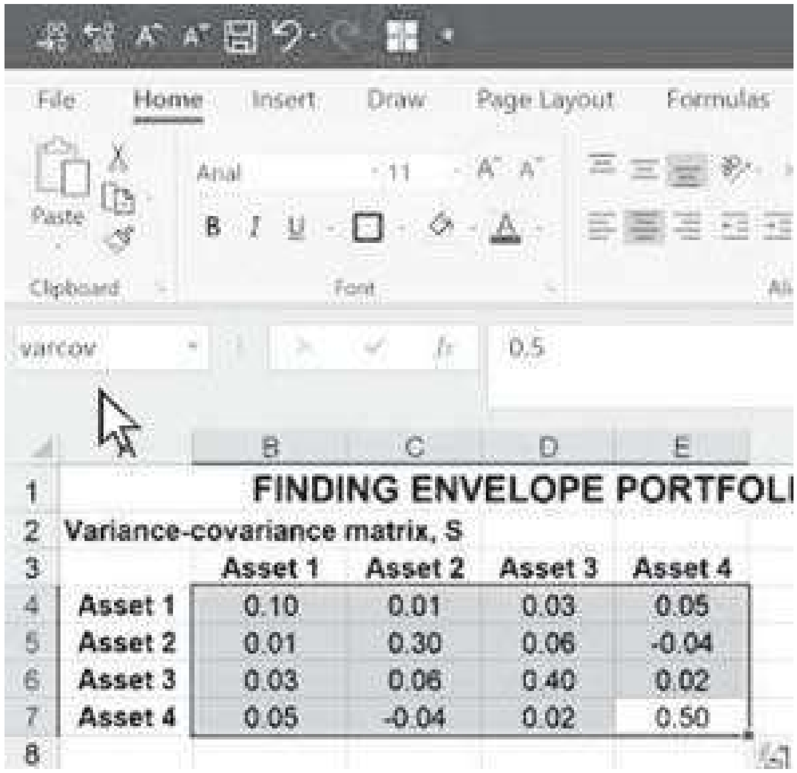
- • In cells B11:B14 we have used the array formula G4:G7-B9 to indicate the expected returns minus the constant in cell B9.
- • In these same cells we have used **Sum(MMult(Minverse(B4:E7),G4:G7-B9))** to give the denominator of the expression $x = \dfrac{S^{-1}\{E(r)-c\}}{Sum[S^{-1}\{E(r)-c\}]}$.

Using Cell Names for Clarity

We can make the whole process even clearer by using cell names. To define a cell name, simply mark the cell or a range of cells and go to the **Name Box**, as shown below:



You can now go into the box and type a name:



The name can now be used, as illustrated below:

	A	B	C	D	E	F	G	H
1	FINDING ENVELOPE PORTFOLIOS IN ONE STEP AND USING NAMES							
2	Variance-covariance matrix, S						Mean	
3		Asset 1	Asset 2	Asset 3	Asset 4		returns	
4	Asset 1	0.10	0.01	0.03	0.05		6%	
5	Asset 2	0.01	0.30	0.06	-0.04		8%	
6	Asset 3	0.03	0.06	0.40	0.02		10%	
7	Asset 4	0.05	-0.04	0.02	0.50		15%	
8								
9	Constant	4%						
10		Envelope portfolio						
11	Asset 1	7.82%		{=MMULT(MINVERSE(varcov),G4:G7-B9)/SUM(MMULT(MINVERSE(varcov),G4:G7-B9))}				
12	Asset 2	26.84%						
13	Asset 3	22.27%						
14	Asset 4	43.07%						
15	Sum	100.00%	<--=SUM(B11:B14)					
16								
17	Portfolio mean	11.30%	<-- =SUMPRODUCT(B11:B14,G4:G7)					
18	Portfolio sigma	37.60%	<-- {=SQRT(MMULT(MMULT(TRANSPOSE(B11:B14),varcov),B11:B14))}					

11.5 Three Notes on the Optimization Procedure

In this section we note three additional facts about the optimization procedure of Proposition 1, which leads to the computation of envelope portfolios.

Note 1—All Roads Lead to Rome: Envelope Is Determined by Any Two *c*'s

By Proposition 2, the envelope is determined by any two of its portfolios. This means that for the determination of the envelope it is irrelevant which two portfolios we use. To drive home this point, the spreadsheet below computes three envelope portfolios:

- • The envelope portfolio x is computed with a constant c = 0%.

- • The envelope portfolio y is computed with a constant $c = 4\%$.
- • A third envelope portfolio z is computed with a constant $c = 6\%$ (cells D11:D14). As shown in rows 21–27, portfolio z is composed of a convex combination of x and y. This is true for any x, y, and z.

This little exercise shows that the constants c, which determine the envelope, are completely arbitrary. Any two constants will determine the same envelope.

	A	B	C	D	E	F	G	H
1				CALCULATING THE ENVELOPE All constants c lead to the same envelope				
2	Variance-covariance matrix, S						Mean	
3		Asset 1	Asset 2	Asset 3	Asset 4		returns E(r)	
4	Asset 1	0.10	0.01	0.03	0.05		6%	
5	Asset 2	0.01	0.30	0.06	-0.04		8%	
6	Asset 3	0.03	0.06	0.40	0.02		10%	
7	Asset 4	0.05	-0.04	0.02	0.50		15%	
8								
9	Constant	0%	4%	6%				
10		Portfolio x	Portfolio y	Portfolio z				
11	Asset 1	36%	8%	-57%	<--{=MMULT(MINVERSE(B4:E7),G4:G7-			
12	Asset 2	24%	27%	34%	D9)/SUM(MMULT(MINVERSE(B4:E7),G4:G7-D9))}			
13	Asset 3	16%	22%	38%				
14	Asset 4	25%	43%	85%				
15	Sum	100%	100%	100%	<--=SUM(D11:D14)			
16								
17	Portfolio mean	9.37%	11.30%	15.84%	<-- =SUMPRODUCT(D11:D14,$G$4:$G$7)			
18	Portfolio sigma	29.37%	37.60%	65.20%	<--			
19					{=SQRT(MMULT(MMULT(TRANSPOSE(D11:D14),$			
20					B$4:$E$7),D11:D14))}			
21	Show: portfolio z is a linear proportion of portfolio x and portfolio y							
22	Proportion	-2.348225	<--=(D11-C11)/(B11-C11)					
23	Check							
24	Asset 1	-57%	<--=$B$22*B11+(1-$B$22)*C11					
25	Asset 2	34%	<--=$B$22*B12+(1-$B$22)*C12					
26	Asset 3	38%	<--=$B$22*B13+(1-$B$22)*C13					
27	Asset 4	85%	<--=$B$22*B14+(1-$B$22)*C14					

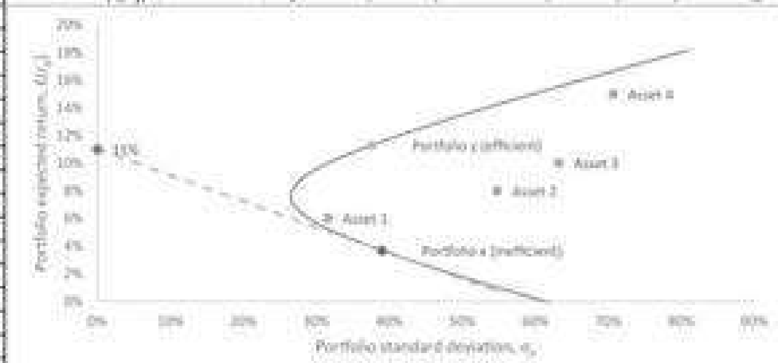
Note 2—Some Values of *c* Locate Non-efficient Envelope Portfolios

The optimization procedure of Proposition 1 locates a portfolio x, which has the following proportions:

$$x = \frac{S^{-1}\{E(r) - c\}}{Sum[S^{-1}\{E(r) - c\}]}$$

Although always on the envelope, this portfolio is not necessarily efficient, as is shown in the example below, where a constant $c = 11\%$ leads to an inefficient portfolio.

	A	B	C	D	E	F	G
1		**CALCULATING THE ENVELOPE** **SOME c's CAN LEAD TO INEFFICIENT PORTFOLIOS**					
2	**Variance-covariance matrix, S**						**Mean**
3		**Asset 1**	**Asset 2**	**Asset 3**	**Asset 4**		**returns E(r)**
4	**Asset 1**	0.10	0.01	0.03	0.05		6%
5	**Asset 2**	0.01	0.30	0.06	-0.04		8%
6	**Asset 3**	0.03	0.06	0.40	0.02		10%
7	**Asset 4**	0.05	-0.04	0.02	0.50		15%
8							
9	Constant	11%	4%				
10		**Portfolio x**	**Portfolio y (efficient)**				
11	**Asset 1**	117%	8%	<--{=MMULT(MINVERSE(B4:E7),G4:G7-			
12	**Asset 2**	14%	27%	C9)/SUM(MMULT(MINVERSE(B4:E7),G4:G7-			
13	**Asset 3**	-4%	22%	C9))}			
14	**Asset 4**	-27%	43%				
15	**Sum**	**100%**	**100%**	<-- =SUM(C11:C14)			
16							
17	Portfolio mean	3.67%	11.30%	<-- =SUMPRODUCT(C11:C14,$G$4:$G$7)			
18	Portfolio sigma	39.01%	37.60%	{=SQRT(MMULT(MMULT(TRANSPOSE(C11:C14),$B$4:$E$7),C11:C14))}			
19	Covariance(r_x,r_y)	-0.00631	<--{=MMULT(MMULT(TRANSPOSE(B11:B14),B4:E7),C11:C14)}				



Note 3—The Portfolio Associated with $c = r_f$ Is Optimal

We've said all this before in our discussion of Proposition 1, but it's worth repeating.[2] If we set c to be equal to the risk-free rate of interest, and if the resulting optimizing portfolio $x = \frac{S^{-1}\{E(r)-c\}}{Sum[S^{-1}\{E(r)-c\}]}$ is efficient, then this portfolio is the optimal investment portfolio for an investor whose preferences are defined solely in terms of the mean and standard deviation of portfolio returns. In the example below, we assume that $r_f = 4\%$. Locating the optimizing portfolio $x = \frac{S^{-1}\{E(r)-c\}}{Sum[S^{-1}\{E(r)-c\}]}$ on the envelope shows that it is efficient. Therefore, the *optimal investment portfolio* for this case is given by y in our example above.

11.6 Finding the Market Portfolio: The Capital Market Line (CML)

Suppose a risk-free asset exists and suppose that this asset has expected return r_f. Let M be the efficient portfolio that is the solution to the system of equations:

$$E(r) - r_f = S \cdot z$$

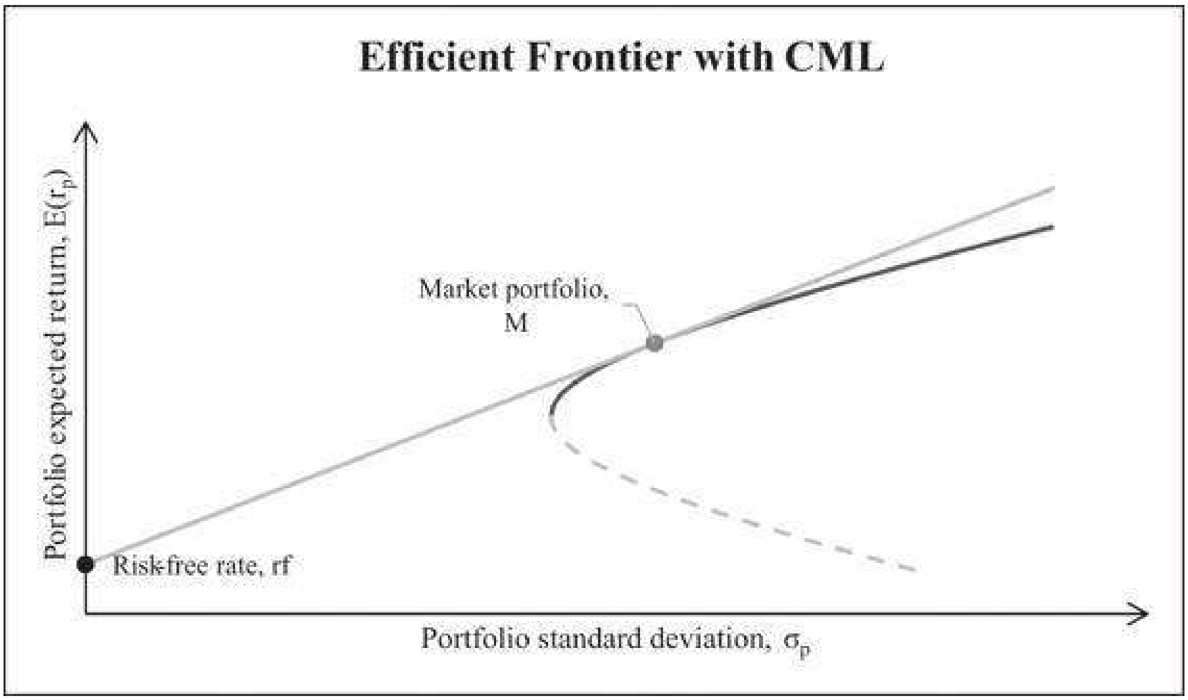
$$M_i = \frac{z_i}{\sum_{i=1}^{N} z_i}$$

Now consider a convex combination of the portfolio M and the risk-free asset r_f; for example, suppose that the weight of the risk-free asset in such a portfolio is a. It follows from the standard equations for portfolio return and standard deviation that:

$$E(r_p) = ar_f + (1-a)E(r_M)$$

$$\sigma_p = \sqrt{a^2 \underbrace{\sigma_{r_f}^2}_{=0} + (1-a)^2\sigma_M^2 + 2a(1-a)\underbrace{Cov(r_f, r_y)}_{=0}} = (1-a)\sigma_M$$

The locus of all such combinations for $a \geq 0$ is known as the *capital market line* (CML). It is graphed below along with the efficient frontier:



The portfolio M is called the *market portfolio* for several reasons:

- • Suppose investors agree about the statistical portfolio information (i.e., the vector of expected returns $E(r)$ and the variance-covariance matrix S). Suppose furthermore that investors are interested only in maximizing the expected portfolio return given portfolio standard deviation σ. Then it follows that *all optimal portfolios will lie on the CML*.
- • In the case above, it further follows that *the portfolio M is the only portfolio of risky assets included in any optimal portfolio*. M must therefore include *all the risky assets, with each asset weighted in proportion to its market value*. That is:

$$\textit{weight of risky asset i in portfolio M} = \frac{V_i}{\sum_{i=1}^{N} V_i}$$

where V_i is the market value of asset i.

It is not difficult to find M when we know r_f : We merely have to solve for the efficient portfolio given that the constant $c = r_f$. When r_f changes, we get a different "market" portfolio—this is just the efficient portfolio given a constant of r_f.

11.7 Computing the Global Minimum Variance Portfolio (GMVP)

Now we want to discuss the *global minimum variance portfolio*—the GMPV.

Suppose there are N assets having a variance-covariance matrix S. The GMVP is the portfolio $x = \{x_1, x_2, \ldots, x_N\}$ that has the lowest variance from among all feasible portfolios. The minimum variance portfolio is defined as follows:

$$x_{GMVP} = \{x_{GMVP,1}, x_{GMVP,2}, \ldots, x_{GMVP,N}\} = \frac{1_{row} \cdot S^{-1}}{1_{row} \cdot S^{-1} \cdot 1_{row}^T}$$

where

$$1_{row} = \underbrace{\{1, 1, \ldots, 1\}}_{\substack{\uparrow \\ \text{N-dimensional} \\ \text{row vector of 1s}}}$$

or

$$x_{GMVP} = \begin{Bmatrix} x_{GMVP,1} \\ x_{GMVP,2} \\ \vdots \\ x_{GMVP,N} \end{Bmatrix} = \frac{S^{-1} \cdot 1_{column}}{1_{column}^T \cdot S^{-1} \cdot 1_{column}},$$

where

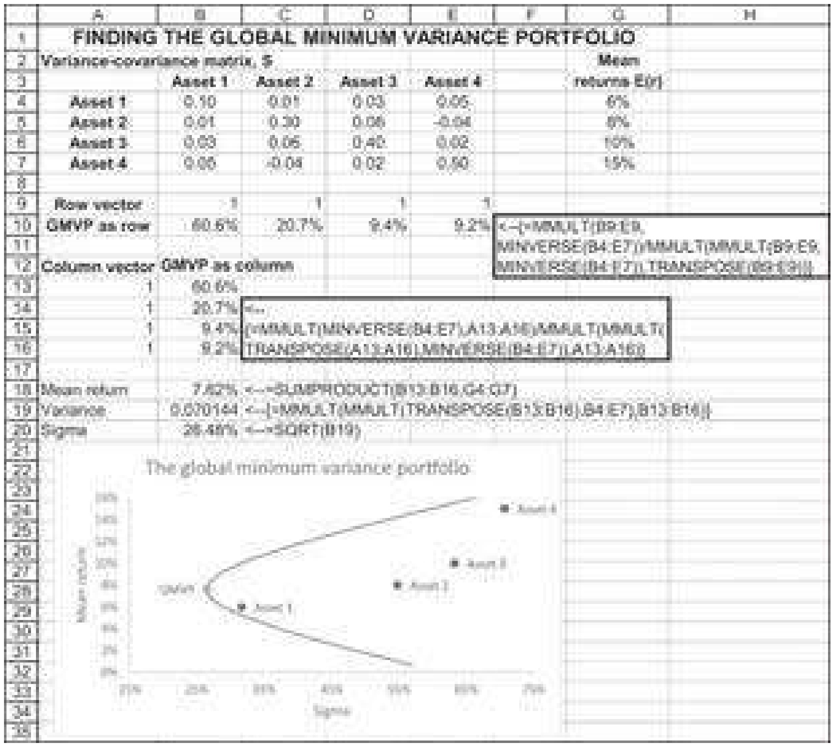
$$1_{column} = \underbrace{\begin{Bmatrix} 1 \\ 1 \\ \vdots \\ 1 \end{Bmatrix}}_{\substack{\uparrow \\ \text{N-dimensional} \\ \text{column vector of 1s}}}$$

This formula is from Merton (1972).

The particular fascination of the minimum variance portfolio is that it is the only portfolio on the efficient frontier whose computation does not require the asset expected returns. The mean μ_{GMVP} and the variance σ^2_{GMVP} of the minimum variance portfolio are given by:

$$\mu_{GMVP} = x_{GMVP} \cdot E(r), \quad \sigma^2_{GMVP} = x_{GMVP} \cdot S \cdot x^T_{GMVP}$$

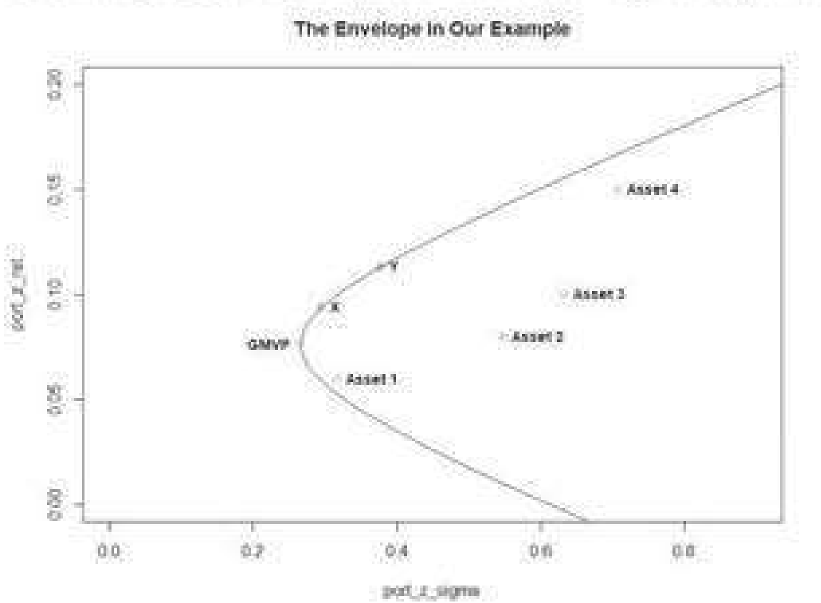
Here’s an implementation of these formulas when applied to our four-asset example:



	A	B	C	D	E	F	G	H
1	FINDING THE GLOBAL MINIMUM VARIANCE PORTFOLIO							
2	Variance-covariance matrix, S						Mean	
3		Asset 1	Asset 2	Asset 3	Asset 4		returns E(r)	
4	Asset 1	0.10	0.01	0.03	0.05		6%	
5	Asset 2	0.01	0.30	0.06	-0.04		8%	
6	Asset 3	0.03	0.06	0.40	0.02		10%	
7	Asset 4	0.05	-0.04	0.02	0.50		15%	
8								
9	Row vector	1	1	1	1			
10	GMVP as row	60.6%	20.7%	9.4%	9.2%	<--{=MMULT(B9:E9,		
11						MINVERSE(B4:E7))/MMULT(MMULT(B9:E9,		
12	Column vector	GMVP as column				MINVERSE(B4:E7)),TRANSPOSE(B9:E9))}		
13	1	60.6%						
14	1	20.7%	<--					
15	1	9.4%	{=MMULT(MINVERSE(B4:E7),A13:A16)/MMULT(MMULT(					
16	1	9.2%	TRANSPOSE(A13:A16),MINVERSE(B4:E7)),A13:A16)}					
17								
18	Mean return	7.62%	<-- =SUMPRODUCT(B13:B16,G4:G7)					
19	Variance	0.070144	<-- {=MMULT(MMULT(TRANSPOSE(B13:B16),B4:E7),B13:B16)}					
20	Sigma	26.48%	<-- =SQRT(B19)					

Using R to do the same should look like this:

```r
GMVP_as_row <- function(var_cov_mat){
  colSums( solve(var_cov_mat) ) /
    sum( solve(var_cov_mat) )
}

GMVP_prop <- GMVP_as_row(var_cov_mat)

# GMVP Mean Return
GMVP_mean_ret <- GMVP_prop %*% assets_mean_ret

# GMVP Sigma
GMVP_Sigma <- sqrt(GMVP_prop %*% var_cov_mat %*% GMVP_prop)

# Add GMVP to the envelope
lines(GMVP_Sigma, GMVP_mean_ret, type = "b", col = "orange")
text(GMVP_mean_ret ~ GMVP_Sigma, labels = c("GMVP"),cex = 0.9, font = 2, pos = 2)
```



11.8 Testing the SML—Implementing Propositions 3–5

To illustrate Propositions 3–5, consider the following data for four risky assets:

	A	B	C	D	E	F
1		ILLUSTRATING PROPOSITIONS 3-5				
2	Dates	Asset 1	Asset 2	Asset 3	Asset 4	
3	1	-6.63%	-2.49%	-4.27%	11.72%	
4	2	8.53%	2.44%	-3.15%	-8.33%	
5	3	1.79%	4.46%	1.92%	19.18%	
6	4	7.25%	17.90%	-6.53%	-7.41%	
7	5	0.75%	-8.22%	-1.76%	-1.44%	
8	6	-1.57%	0.83%	12.58%	-5.92%	
9	7	-2.10%	5.14%	13.41%	-0.46%	
10						
11	Mean	1.15%	2.67%	1.79%	1.05%	<-- =AVERAGE(E3:E9)

The asset returns on seven dates are given in rows 3–7, and the average return is given in row 11.

We use some sophisticated array functions to compute the variance-covariance matrix:

	A	B	C	D	E	F	G
13	Variance-covariance matrix						
14		Asset 1	Asset 2	Asset 3	Asset 4		
15	Asset 1	0.0024	0.0019	-0.0015	-0.0024	<--cells B15:E18 contain the formula	
16	Asset 2	0.0019	0.0056	-0.0007	-0.0016	{=MMULT(TRANSPOSE(B3:E9-	
17	Asset 3	-0.0015	-0.0007	0.0057	-0.0005	B11:E11),B3:E9-B11:E11)/7}	
18	Asset 4	-0.0024	-0.0016	-0.0005	0.0094		
19							
20	Finding an efficient portfolio w						
21	Constant	0.50%					
22							
23	Asset 1	0.3129 <—		Cells B15:E18 contain the formula			
24	Asset 2	0.2464		{=MMULT(MINVERSE(B15:E18),TRANSPOSE(B11:E11)-			
25	Asset 3	0.2690		B21)/SUM(MMULT(MINVERSE(B15:E18),TRANSPOSE(B11:E11)-B21))}			
26	Asset 4	0.1717					

The efficient portfolio given the constant $c = 0.5\%$ is given in cells B23:B26; we compute this portfolio using the method of Proposition 1.[3] We call this portfolio w. The returns of portfolio w on dates 1–7 are given in column G below:

	A	B	C	D	E	F	G
1			ILLUSTRATING PROPOSITIONS 3-5				
2	Dates	Asset 1	Asset 2	Asset 3	Asset 4	Efficient portfolio w	
3	1	-6.63%	-2.49%	-4.27%	11.72%	-1.92%	<--{=MMULT(B3:E9,B22:B25)}
4	2	8.53%	2.44%	-3.15%	-8.33%	0.99%	
5	3	1.79%	4.46%	1.92%	19.18%	5.47%	
6	4	7.25%	17.90%	-6.53%	-7.41%	3.65%	
7	5	0.75%	-8.22%	-1.76%	-1.44%	-2.51%	
8	6	-1.57%	0.83%	12.88%	-5.92%	2.16%	
9	7	-2.10%	5.14%	13.41%	-0.46%	4.14%	
10	Mean return	1.15%	2.87%	1.79%	1.05%	1.73%	<--=AVERAGE(F3:F9)
11							
12	Variance-covariance matrix						
13		Asset 1	Asset 2	Asset 3	Asset 4		
14	Asset 1	0.0028	0.0022	-0.0018	-0.0028	<--cells B14:E17 contain the formula	
15	Asset 2	0.0022	0.0065	-0.0008	-0.0019	{=MMULT(TRANSPOSE(B3:E9-B10:E10),B3:E9-	
16	Asset 3	-0.0018	-0.0008	0.0067	-0.0005	B10:E10)/6}	
17	Asset 4	-0.0028	-0.0019	-0.0005	0.0110		
18							
19	Finding an efficient portfolio w						
20	Constant	0.50%					
21							
22	Asset 1	0.3129	<-- Cells B14:E17 contain the formula				
23	Asset 2	0.2464	{=MMULT(MINVERSE(B14:E17),TRANSPOSE(B10:E10)-				
24	Asset 3	0.2690	B20)/SUM(MMULT(MINVERSE(B14:E17),TRANSPOSE(B10:E10)-B20))}				
25	Asset 4	0.1717					

We illustrate Propositions 3–5 in two steps:

- • Step 1. We regress the returns of each asset on the returns of the efficient portfolio. For assets $i = 1, \ldots, 4$ we run the regression $r_{it} = \alpha_i + \beta_i r_{wt} + \varepsilon_{it}$. This regression is often called the *first-pass regression*. The results are given below.

	A	B	C	D	E	F	G
28	**Implementing propositions 3-5 to find the SML**						
29	Step 1: Regress each asset's returns on those of the efficient portfolio w						
30		**Asset 1**	**Asset 2**	**Asset 3**	**Asset 4**		
31	Alpha	0.0024	-0.0047	-0.0002	0.0029	<-- =INTERCEPT(E3:E9,$F$3:$F$9)	
32	Beta	0.5284	1.9301	1.0490	0.4478	<-- =SLOPE(E3:E9,$F$3:$F$9)	
33	R-squared	0.0697	0.5241	0.1505	0.0167	<-- =RSQ(E3:E9,$F$3:$F$9)	

- • Step 2. We now regress the betas of the assets on their mean returns. This regression is often called the *second-pass regression.* Running this regression, $\bar{r}_i = \gamma_0 + \gamma_1 \beta_i + \varepsilon_i$, gives:

	A	B	C	D	E	F
35	Step 2: Regress the asset mean returns on their betas					
36	Intercept	0.50%	<-- =INTERCEPT(B10:E10,B32:E32)			
37	Slope	0.0123	<-- =SLOPE(B10:E10,B32:E32)			
38	R-squared	1.0000	<-- =RSQ(B10:E10,B32:E32)			

To check the results of Propositions 3–5, we run a test:

	A	B	C	D	E	F
40	**Check Propositions 3 & 4**					
41	**Step 2 coefficients should be: Intercept = c, Slope = E(rw) - c**					
42	Intercept = c ?	yes	<--=IF(ROUND(B36-B20,10)=0,"yes","no")			
43	Slope = $E(r_w)$ - c ?	yes	<--=IF(B37=F10-B20,"yes","no")			

The "perfect" regression results (note the $R^2 = 1$ in cell B38) are the results promised us by Propositions 3–5:

- • The second-pass regression intercept is equal to c and the slope is equal to $E(r_w) - c$.
- • If there is a riskless asset with return $c = r_f$, then Proposition 5 promises that in the second-pass regression $\bar{r}_i = \gamma_0 + \gamma_1 \beta_i + \varepsilon_i$, $\gamma_0 = r_f$ and $\gamma_1 = E(r_w) - r_f$.
- • If there is no riskless asset, then Proposition 3 states that in the second-pass regression $\gamma_0 = E(r_z)$ and $\gamma_1 = E(r_w) - E(r_z)$, where z is a portfolio whose covariance with w is zero.
- • Finally, if we run the two-stage regression of the type described on *any portfolio* w and get a "perfect regression," then Proposition 4 guarantees that w is in fact efficient.

To drive home the point that this technique always works, we show you all the calculations using a different value for c (cell B20, highlighted below). As proved in Propositions 3–5, the result is still a perfect regression of the means on the betas:

	A	B	C	D	E	F	G
1				ILLUSTRATING PROPOSITIONS 3-5			
2	Dates	Asset 1	Asset 2	Asset 3	Asset 4	Efficient portfolio w	
3	1	-6.63%	-2.49%	-4.27%	11.72%	-0.71%	<--{=MMULT(B3:E9,B22:B25)}
4	2	8.53%	2.44%	-3.15%	-8.33%	-1.62%	
5	3	1.79%	4.46%	1.92%	19.18%	5.78%	
6	4	7.25%	17.90%	-6.53%	-7.41%	9.60%	
7	5	0.75%	-8.22%	-1.76%	-1.44%	-7.18%	
8	6	-1.57%	0.83%	12.88%	-5.92%	4.28%	
9	7	-2.10%	5.14%	13.41%	-0.46%	8.46%	
10	Mean return	1.15%	2.87%	1.79%	1.65%	2.66%	<--=AVERAGE(F3:F9)
11							
12	Variance-covariance matrix						
13		Asset 1	Asset 2	Asset 3	Asset 4		
14	Asset 1	0.0028	0.0022	-0.0018	-0.0028	<--cells B14:E17 contain the formula	
15	Asset 2	0.0022	0.0085	-0.0008	-0.0019	{=MMULT(TRANSPOSE(B3:E9-B10:E10),B3:E9-	
16	Asset 3	-0.0018	-0.0008	0.0067	-0.0005	B10:E10)/6)}	
17	Asset 4	-0.0028	-0.0019	-0.0005	0.0110		
18							
19	Finding an efficient portfolio w						
20	Constant	1.00%					
21							
22	Asset 1	-0.1894	<-- Cells B14:E17 contain the formula				
23	Asset 2	0.7711	{=MMULT(MINVERSE(B14:E17),TRANSPOSE(B10:E10)-				
24	Asset 3	0.3095	B20)/SUM(MMULT(MINVERSE(B14:E17),TRANSPOSE(B10:E10)-B20))}				
25	Asset 4	0.1088					
26							
27							
28	Implementing propositions 3-5 to find the SML						
29	Step 1: Regress each asset's returns on those of the efficient portfolio w						
30		Asset 1	Asset 2	Asset 3	Asset 4		
31	Alpha	0.0091	-0.0012	0.0053	0.0097	<-- =INTERCEPT(E3:E9,$F$3:$F$9)	
32	Beta	0.0889	1.1244	0.4735	0.0294	<-- =SLOPE(E3:E9,$F$3:$F$9)	
33	R-squared	0.0102	0.7166	0.1235	0.0003	<-- =RSQ(E3:E9,$F$3:$F$9)	
34							
35	Step 2: Regress the asset mean returns on their betas						
36	Intercept	1.00%	<-- =INTERCEPT(B10:E10,B32:E32)				
37	Slope	0.0166	<-- =SLOPE(B10:E10,B32:E32)				
38	R-squared	1.0000	<-- =RSQ(B10:E10,B32:E32)				
39							
40	Check Propositions 3 & 4						
41	Step 2 coefficients should be: Intercept = c, Slope = E(rw) - c						
42	Intercept = c ?	yes	<--=IF(ROUND(B36-B20,10)=0,"yes","no")				
43	Slope = $E(r_w)$ - c ?	yes	<--=IF(B37=F10-B20,"yes","no")				

11.9 Efficient Portfolios without Short Sales

Solutions to the maximization problem of finding efficient portfolios allow *negative* portfolio proportions; when $x_i < 0$, this assumes that the *i*th security is sold short by the investor and that the proceeds from this short sale become immediately available to the investor.

Reality is considerably more complicated than this academic model of short sales. In particular, it is rare for all of the short-sale proceeds to become available to the investor at the time of investment, since brokerage houses typically escrow some or even all of the proceeds. It may also be that the investor is completely prohibited from making any short sales (indeed, most small investors seem to proceed on the assumption that short sales are impossible).[4]

In this section we investigate these problems. We show how to use Excel's **Solver** to find efficient portfolios of assets when we restrict short sales.[5]

A Numerical Example

We start with the problem of finding an optimal portfolio when no short sales are allowed. The problem we solve is similar to the maximization problem we used above, with the addition of the short-sale constraint $x_i > 0$ for the asset proportions

$$\max \Theta = \frac{E(r_x) - c}{\sigma_p},$$

such that

$$\sum_{i=1}^{N} x_i = 1 \text{ and } x_i \geq 0,\ i = 1, \ldots, N,$$

where

$$E(r_x) = x^T \cdot R = \sum_{i=1}^{N} x_i E(r_i)$$

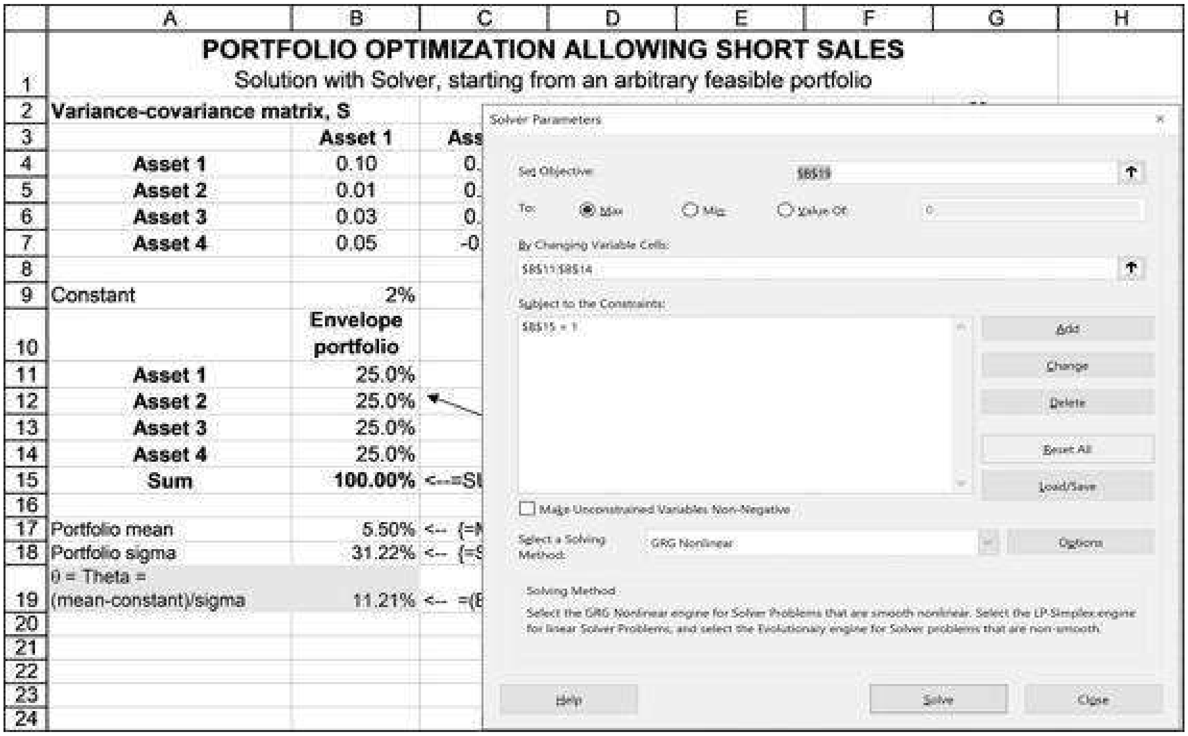
$$\sigma_p = \sqrt{x^T S x} = \sqrt{\sum_{i=1}^{N} \sum_{j=1}^{N} x_i x_j \sigma_{ij}}.$$

Solving an Unconstrained Portfolio Problem

To set the scene, we consider the optimization problem below, which we solve without any short-sale constraints. The spreadsheet shows a four-asset variance-covariance matrix and associated expected returns. Given a constant c = 2%, the optimal portfolio is given in cells B11:B14. Notice θ in cell B19: This is the *Sharpe ratio* of the portfolio, the ratio of its excess return over the constant c to its standard deviation: $\theta = \frac{E(r_x) - c}{\sigma_x}$. The optimal portfolio maximizes the Sharpe ratio θ.

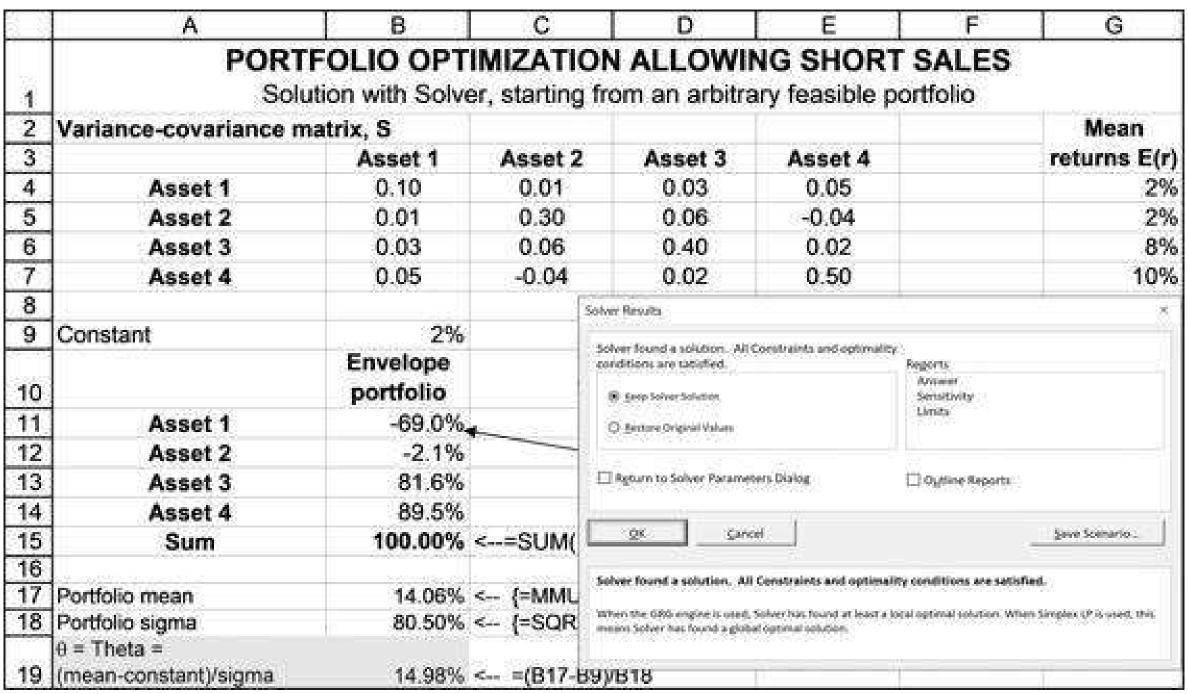
	A	B	C	D	E	F	G
1	PORTFOLIO OPTIMIZATION ALLOWING SHORT SALES						
2	Variance-covariance matrix, S						Mean
3		Asset 1	Asset 2	Asset 3	Asset 4		returns E(r)
4	Asset 1	0.10	0.01	0.03	0.05		2%
5	Asset 2	0.01	0.30	0.06	-0.04		2%
6	Asset 3	0.03	0.06	0.40	0.02		6%
7	Asset 4	0.05	-0.04	0.02	0.50		10%
8							
9	Constant	2%					
10		Envelope portfolio					
11	Asset 1	-69.02%		{=MMULT(MINVERSE(B4:E7),G4:G7-B9)/SUM(MMULT(MINVERSE(B4:E7),G4:G7-B9))}			
12	Asset 2	-2.10%					
13	Asset 3	81.65%					
14	Asset 4	89.47%					
15	Sum	100.00%	<-- =SUM(B11:B14)				
16							
17	Portfolio mean	14.06%	<-- {=MMULT(TRANSPOSE(B11:B14),G4:G7)}				
18	Portfolio sigma	80.50%	<-- {=SQRT(MMULT(TRANSPOSE(B11:B14),MMULT(B4:E7,B11:B14)))}				
19	θ = Theta = (mean-constant)/sigma	14.98%	<-- =(B17-B9)/B18				

There is another way to solve this unconstrained problem. Starting from an arbitrary portfolio (the spreadsheet below uses $x_1 = x_2 = x_3 = x_4 = 0.25$), we use **Solver** to find a solution:



	A	B	C
1	PORTFOLIO OPTIMIZATION ALLOWING SHORT SALES Solution with Solver, starting from an arbitrary feasible portfolio		
2	Variance-covariance matrix, S		
3		Asset 1	Ass
4	Asset 1	0.10	0.
5	Asset 2	0.01	0.
6	Asset 3	0.03	0.
7	Asset 4	0.05	-0
8			
9	Constant	2%	
10		Envelope portfolio	
11	Asset 1	25.0%	
12	Asset 2	25.0%	
13	Asset 3	25.0%	
14	Asset 4	25.0%	
15	Sum	100.00%	<--=SU
16			
17	Portfolio mean	5.50%	<-- {=M
18	Portfolio sigma	31.22%	<-- {=S
19	θ = Theta = (mean-constant)/sigma	11.21%	<-- =(B

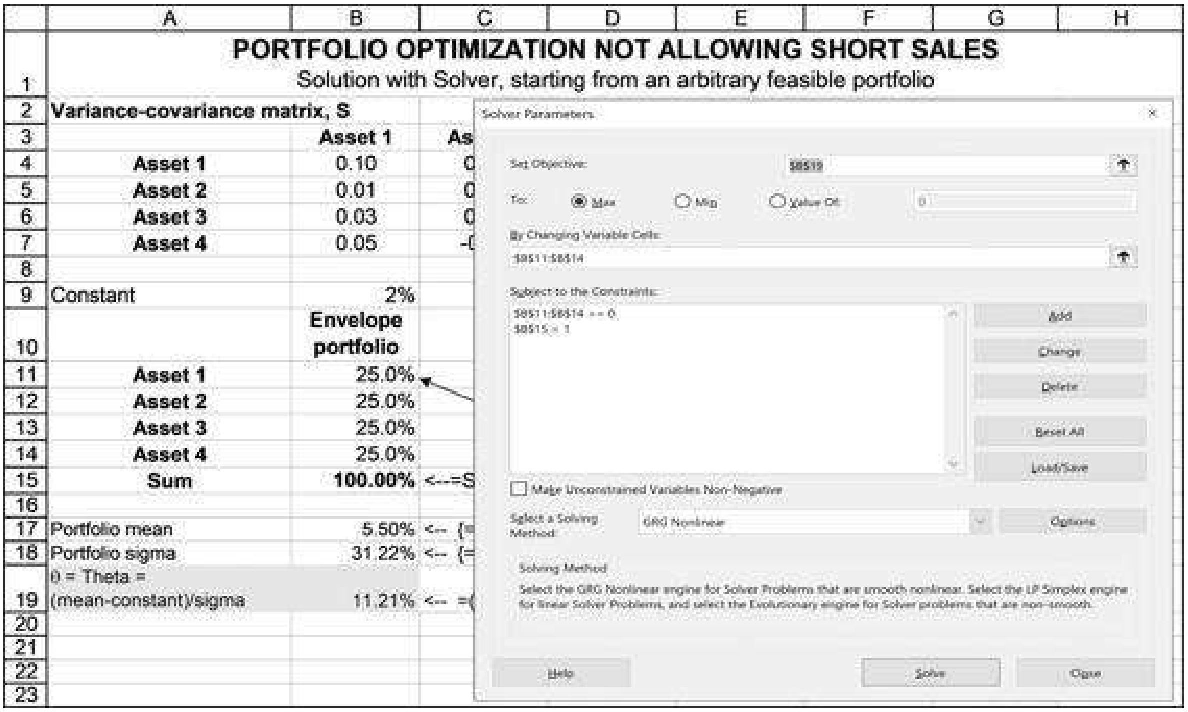
The **Solver** solution maximizes θ (cell B19) subject to the constraint that cell B15, which contains the sum of the portfolio positions, equals 1.[6] When we press **Solve**, we get the solution we achieved before:



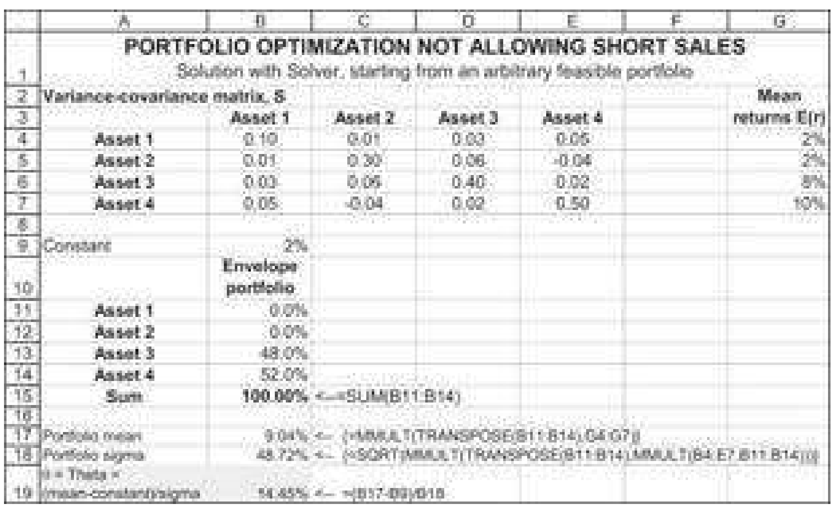
	A	B	C	D	E	F	G
1	PORTFOLIO OPTIMIZATION ALLOWING SHORT SALES Solution with Solver, starting from an arbitrary feasible portfolio						
2	Variance-covariance matrix, S						Mean
3		Asset 1	Asset 2	Asset 3	Asset 4		returns E(r)
4	Asset 1	0.10	0.01	0.03	0.05		2%
5	Asset 2	0.01	0.30	0.06	-0.04		2%
6	Asset 3	0.03	0.06	0.40	0.02		8%
7	Asset 4	0.05	-0.04	0.02	0.50		10%
8							
9	Constant	2%					
10		Envelope portfolio					
11	Asset 1	-69.0%					
12	Asset 2	-2.1%					
13	Asset 3	81.6%					
14	Asset 4	89.5%					
15	Sum	100.00%	<--=SUM(				
16							
17	Portfolio mean	14.06%	<-- {=MMU				
18	Portfolio sigma	80.50%	<-- {=SQR				
19	θ = Theta = (mean-constant)/sigma	14.98%	<-- =(B17-B9)/B18				

Solving a Constrained Portfolio Problem

The optimal solution above contains a short position in assets 1 and 2. To restrict the short selling, we add a no-short-sale constraint to **Solver**. Starting from an arbitrary solution, we bring up **Solver** as shown below:



Pressing **Solve** yields the following solution:



	A	B	C	D	E	F	G
1	PORTFOLIO OPTIMIZATION NOT ALLOWING SHORT SALES Solution with Solver, starting from an arbitrary feasible portfolio						
2	Variance-covariance matrix, S						Mean
3		Asset 1	Asset 2	Asset 3	Asset 4		returns E(r)
4	Asset 1	0.10	0.01	0.03	0.05		2%
5	Asset 2	0.01	0.30	0.06	-0.04		2%
6	Asset 3	0.03	0.06	0.40	0.02		8%
7	Asset 4	0.05	-0.04	0.02	0.50		10%
8							
9	Constant	2%					
10		Envelope portfolio					
11	Asset 1	0.0%					
12	Asset 2	0.0%					
13	Asset 3	48.0%					
14	Asset 4	52.0%					
15	Sum	100.00%	<--=SUM(B11:B14)				
16							
17	Portfolio mean	9.04%	<-- {=MMULT(TRANSPOSE(B11:B14),G4:G7)}				
18	Portfolio sigma	48.72%	<-- {=SQRT(MMULT(TRANSPOSE(B11:B14),MMULT(B4:E7,B11:B14)))}				
19	θ = Theta = (mean-constant)/sigma	14.45%	<-- =(B17-B9)/B18				

The non-negativity constraints is added by clicking on the **Add** button in the **Solver** dialog box. This brings up the following window (shown here filled in):[7]

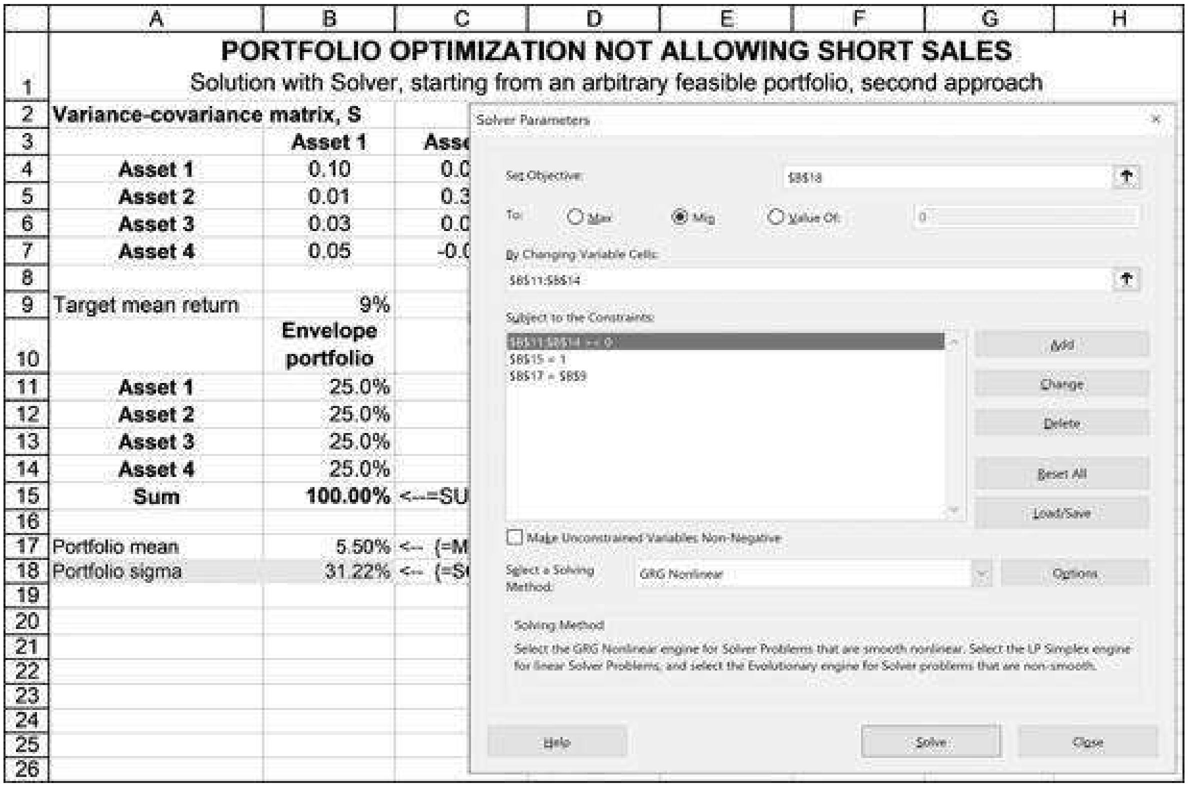
The second constraint (which constrains the portfolio proportions to sum to 1) is added in a similar fashion.

Implementing the same approach in R should look like this:

```
# Input: Assets Mean Returns
assets_mean_ret <- c(0.02, 0.02, 0.08, 0.1)

# In this part, we set the constraints for the matrix as AA >= B (linear constraints)
# the constraints are: X1 >= 0; X2 >= 0; X3 >= 0; X4 >= 0

# Setting Coefficients Matrix
A_matrix <- diag(1,nrow = 4) # identity matrix to enforce all asset proportions are > 0
A_matrix <- rbind(A_matrix, rep(1, 4)) # sum of proportions >= 1
A_matrix <- rbind(A_matrix, rep(-1, 4)) # sum of proportions <= 1

# Setting Result vector
B_vector <- rep(0, 4)
B_vector <- c(B_vector, 1, -1)

# Setting Target Function: Sharpe
sharpe <- function(proportions, var_cov_mat, assets_mean_ret, constant){
  ex_return = proportions %*% (assets_mean_ret - constant)
  sigma = sqrt(proportions %*% var_cov_mat %*% proportions)
  return(ex_return / sigma)}

# Run (Note: Initial values must be in interior of the feasible region)
result <- constrOptim(theta = c(0.25, 0.25, 0.25, 0.25), #  initial value
                      f = sharpe, # function to maximize
                      grad = NULL, # gradient of f
                      ui = A_matrix, # constraint matrix
                      ci = B_vector- 1e-05, # constraint vector minus infinitesimal
                      mu = 1e-05, # result accuracy
                      control = list(fnscale = -1), # maximization
                      outer.iterations = 100000,
                      method = "Nelder-Mead", # use this whithout gradient function
                      # target function variables:
                      var_cov_mat = var_cov_mat,
                      assets_mean_ret = assets_mean_ret,
                      constant = 0.02)
# Result
round(result$par, 4)
sharpe(result$par, var_cov_mat, assets_mean_ret, 0.02)
```

An Alternative Approach to Calculate the Efficient Portfolio without Short Sales

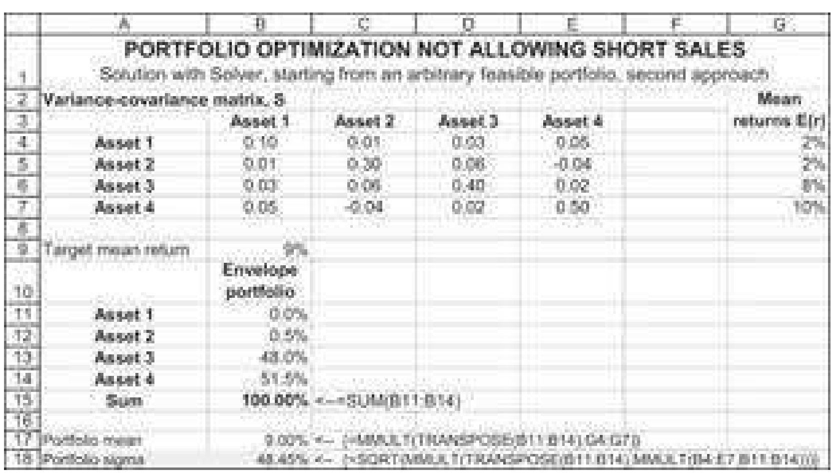
A slightly different approach to solve for an efficient portfolio as a constrained problem is to solve for a portfolio with a certain mean return that has the lowest possible risk. This ensures that we are on the efficient frontier. Below is the Solver dialog box for our example:



In this example we find the least risky portfolio that has mean return 9% while meeting the following restrictions:

- • All portfolio weights are positive.
- • The sum of all weights is 1.
- • The mean return of the portfolio is 9%.

The results are:



Implementing this in R looks like this:

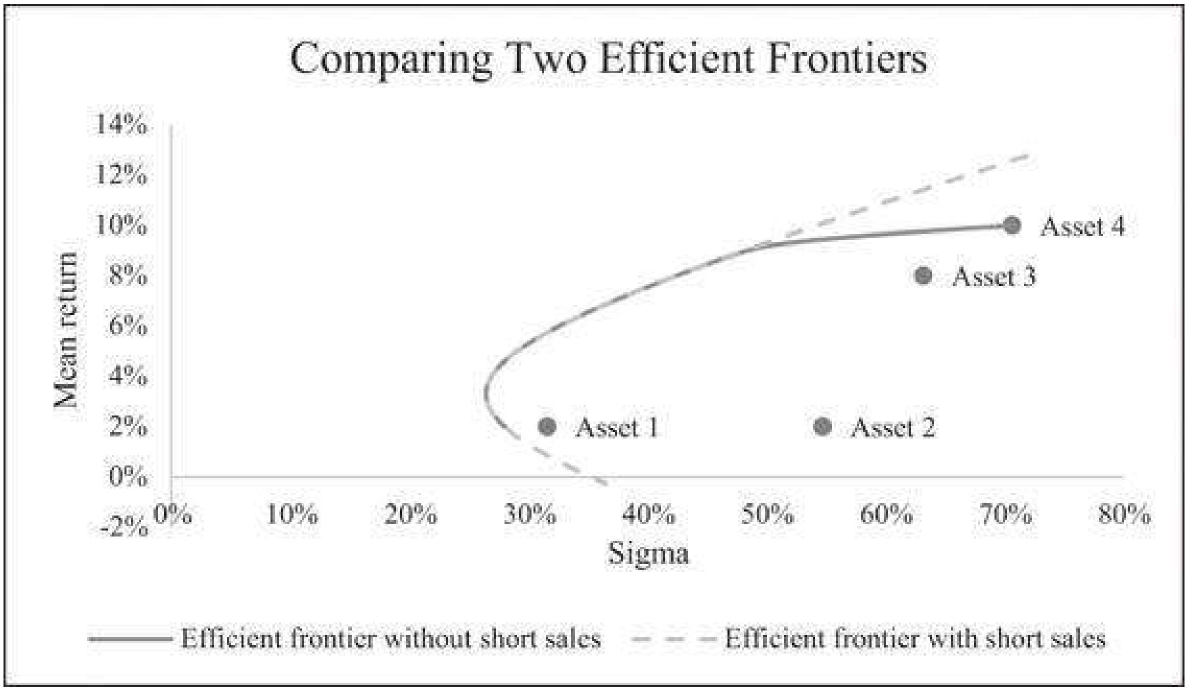
```
## No short (Solver 2nd approach)
# Adding a rP = 5% constraint
A_matrix <- rbind(A_matrix, assets_mean_ret) # Add a return = 5% constraint
B_vector <- c(B_vector, 0.05)

# Setting Target Function: Variance
port_var <- function(proportions, var_cov_mat){
  return(proportions %*% var_cov_mat %*% proportions)
}

# Run
result <- constrOptim(theta = c(0, 0, 0.25, 0.75), #  starting value
                      f = port_var, # function to maximize
                      grad = NULL, # gradient of f
                      ui = A_matrix, # constraint matrix
                      ci = B_vector - 1e-05, # constraint vector
                      mu = 1e-03, # result accuracy
                      outer.iterations = 1000000,
                      method = "Nelder-Mead", # use this without gradient function
                      # target function variables:
                      var_cov_mat = var_cov_mat)

# Result
round(result$par, 4) # Proportions
assets_mean_ret %*% result$par # Mean return
sqrt(port_var(result$par, var_cov_mat)) # Sigma
# To get more accurate results, see solve.QP function on the website
```

The R file for this chapter, which is on the companion website for this chapter, shows several other (and probably better) optimization procedures to do the same.

The Efficient Frontier with Short-Sale Restrictions

We want to graph the efficient frontier with short-sale restrictions. Recall that in the case of no short-sale restrictions, it was enough to find two efficient portfolios in order to determine the whole efficient frontier (this was proved in Proposition 2 of this chapter). When we impose short-sale restrictions, this statement is no longer true. In this case the determination of the efficient frontier requires the plotting of a large number of points. The only efficient (pardon the pun!) way of doing this is with a VBA program that automatically finds the solution to any *c* we throw at it. We can then use a data table to find the set of optimal portfolios for every constant.

In the next section we describe such a program. Once we have the program and the graph of the efficient frontier without short sales, we can compare this efficient frontier to the efficient frontier *with* short sales allowed:



Putting these two graphs on one set of axes shows that the effect of the short-sale restrictions is mainly for portfolios with higher returns and sigmas.
The relation between these two graphs is not all that surprising:

- • In general, the efficient frontier with short sales dominates the efficient frontier without short sales. This must clearly be so, since the short-sale restriction imposes an extra constraint on the maximization problem.
- • For some cases, the two efficient frontiers coincide. This is mainly in the low-sigma area of the chart.

A VBA Program for the Efficient Frontier without Short Sales

The output for the restricted short-sale case shown in section 10.9 above was produced with the following VBA program:

```
Sub plot_eff_frontier()
Range("a24:f100").ClearContents
Lowest =
WorksheetFunction.RoundUp(WorksheetFunction.Min(Range("g4:g7")) *
100, 0)
Highest = WorksheetFunction.
RoundDown(WorksheetFunction.Max(Range("g4:g7")) * 100, 0)
m = 0
For i = Lowest To Highest Step (Highest - Lowest) / 40
    Range("b9") = (i) / 100
    solverreset
```

```
        SolverOk SetCell:="$B$18", MaxMinVal:=2, ValueOf:=0,
        ByChange:="$B$11:$B$14", _
            Engine:=1, EngineDesc:="GRG Nonlinear"
        SolverAdd CellRef:="$B$11:$B$14", Relation:=3,
        FormulaText:="0"
        SolverAdd CellRef:="$B$17", Relation:=2, FormulaText:="$B$9"
        SolverAdd CellRef:="B15", Relation:=2, FormulaText:="100%"
        SolverOk SetCell:="$B$18", MaxMinVal:=2, ValueOf:=0,
        ByChange:="$B$11:$B$14", _
            Engine:=1, EngineDesc:="GRG Nonlinear"
        SolverSolve True
        m = m + 1
        Cells(23 + m, 1) = Range("b17")
        Cells(23 + m, 2) = Range("b18")
        Cells(23 + m, 3) = Range("b11")
        Cells(23 + m, 4) = Range("b12")
        Cells(23 + m, 5) = Range("b13")
        Cells(23 + m, 6) = Range("b14")
    Next i
    End Sub
```

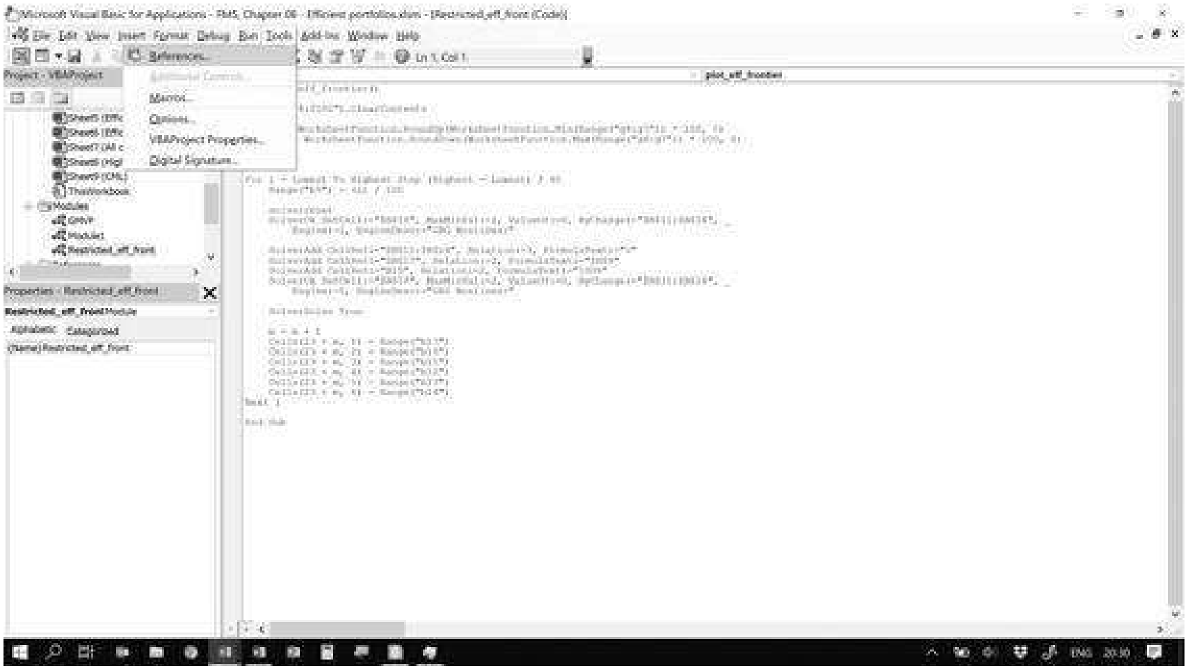
The program repeatedly calls Solver for different values of the target return (cell B9) and putting the output in cells B24:F64.

The final output looks like this:

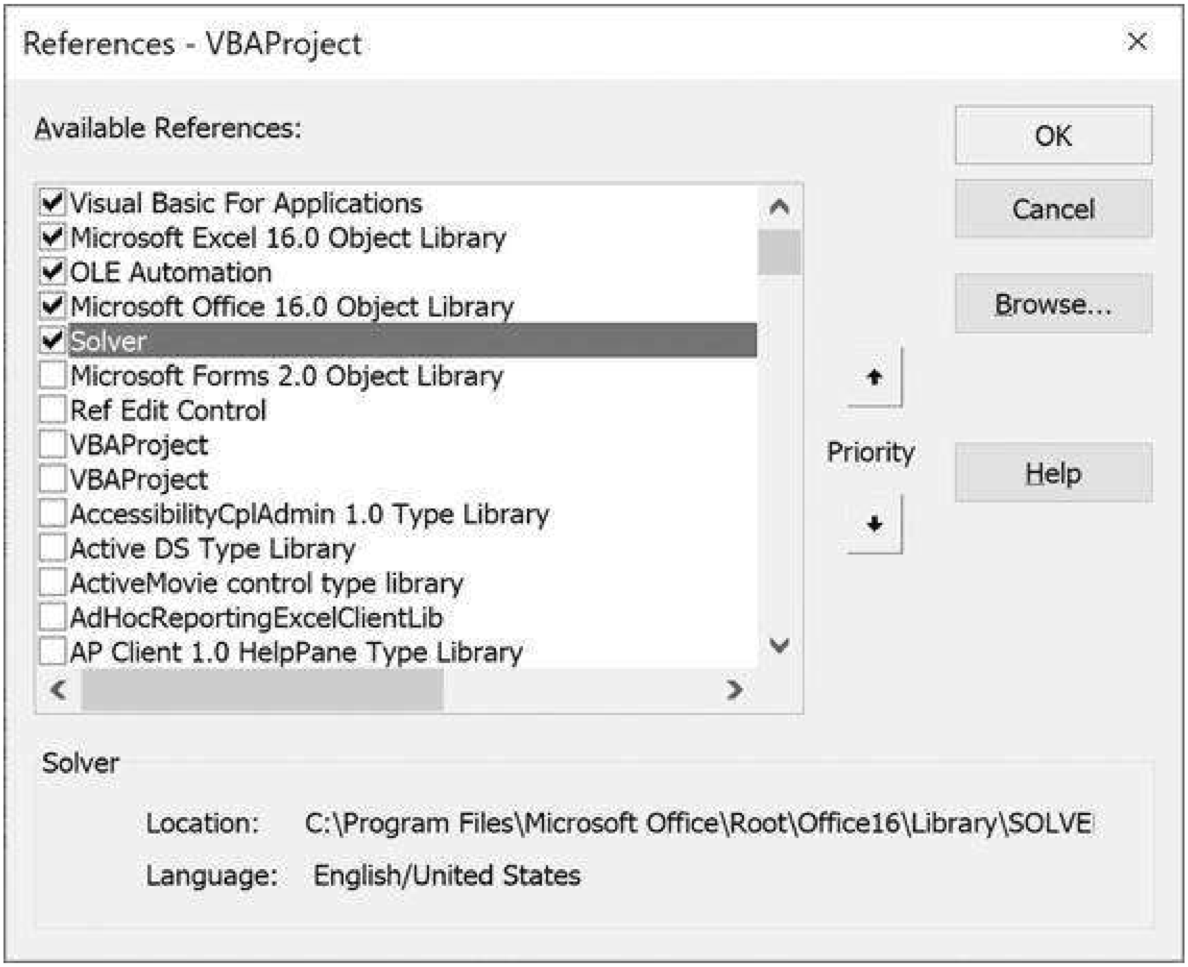
	A	B	C	D	E	F	G
1	PORTFOLIO OPTIMIZATION NOT ALLOWING SHORT SALES Solution with Solver, starting from an arbitrary feasible portfolio, second approach						
2	Variance-covariance matrix, S						Mean
3		Asset 1	Asset 2	Asset 3	Asset 4		returns E(r)
4	Asset 1	0.10	0.01	0.03	0.05		2%
5	Asset 2	0.01	0.30	0.06	-0.04		2%
6	Asset 3	0.03	0.06	0.40	0.02		8%
7	Asset 4	0.05	-0.04	0.02	0.50		10%
8							
9	Target mean return	10%		0%	2%		
10	Constrained portfolio	Envelope portfolio		Envelope portfolio A	Envelope portfolio B		
11	Asset 1	0.0%		9.43%	-69.02%	<--	
12	Asset 2	0.0%		11.70%	-2.10%	{=MMULT(MINVERSE($B$4:$E$7),$G$4:$G$7-	
13	Asset 3	0.0%		37.94%	81.65%	E9)/SUM(MMULT(MINVERSE($	
14	Asset 4	100.0%		40.92%	89.47%	B$4:$E$7),$G$4:$G$7-E9))}	
15	Sum	100.00%		100.00%	100.00%	<-- =SUM(E11:E14)	
16							
17	Portfolio mean	10.00%		7.56%	14.06%		
18	Portfolio sigma	70.71%		40.03%	80.50%		
19			Covariance	0.2983			
20			Rho	0.93			
21							Calculate
22	Efficient frontier without short-sales						
23	Mean return	Sigma	X(Asset 1)	X(Asset 2)	X(Asset 3)	X(Asset 4)	
24	2.0%	26.1%	76.3%	23.7%	0.0%	0.0%	
25	2.2%	27.6%	73.9%	23.1%	2.0%	1.0%	
26	2.4%	27.2%	71.5%	22.6%	3.3%	2.5%	
62	9.6%	58.5%	0.0%	0.0%	20.0%	80.0%	
63	9.8%	64.2%	0.0%	0.0%	10.0%	90.0%	
64	10.0%	70.7%	0.0%	0.0%	0.0%	100.0%	

Adding a Reference to Solver in VBA

If the above routine does not work, you may need to add a reference to **Solver** in the VBA editor. Press [Alt] + F11 to get to the editor and then go to **Tools|References**:

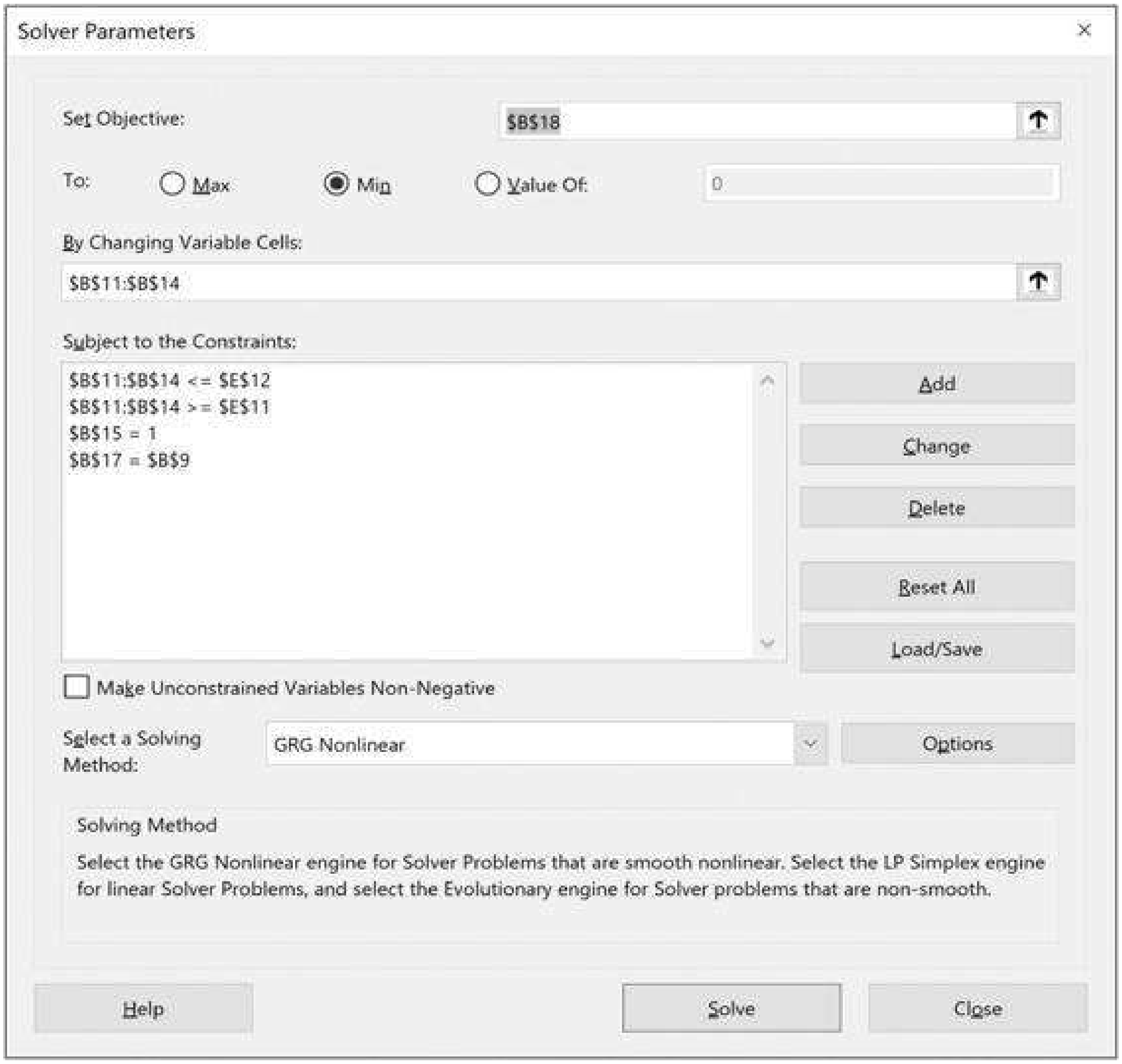


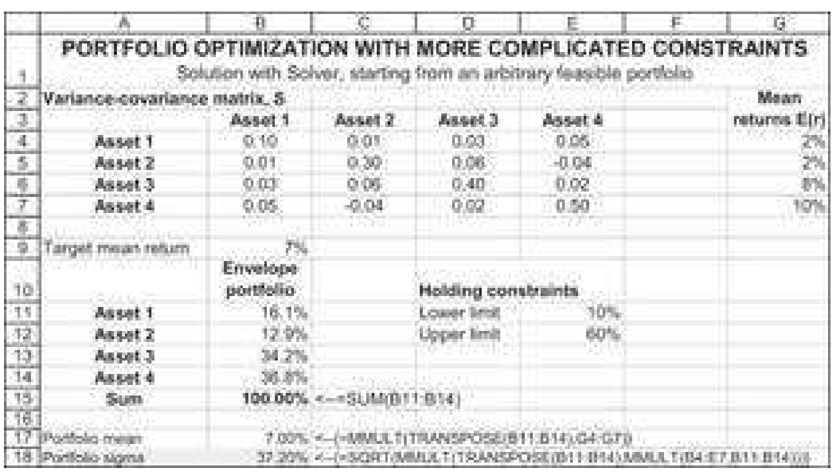
If this reference is missing, go to **Tools|References** on the VBA menu and make sure that **Solver** is checked:



Other Position Restrictions

It goes without saying that Excel and **Solver** can accommodate other position limits. Suppose, for example, that the investor wants 9% mean return, as before, and also at least 10% of the portfolio invested in any asset and no more than 60% of the portfolio invested in any single asset. This is easily set up in **Solver**:





	A	B	C	D	E	F	G
1	PORTFOLIO OPTIMIZATION WITH MORE COMPLICATED CONSTRAINTS Solution with Solver, starting from an arbitrary feasible portfolio						
2	Variance-covariance matrix, S						Mean
3		Asset 1	Asset 2	Asset 3	Asset 4		returns E(r)
4	Asset 1	0.10	0.01	0.03	0.05		2%
5	Asset 2	0.01	0.30	0.06	-0.04		2%
6	Asset 3	0.03	0.06	0.40	0.02		8%
7	Asset 4	0.05	-0.04	0.02	0.50		10%
8							
9	Target mean return	7%					
10		Envelope portfolio		Holding constraints			
11	Asset 1	16.1%		Lower limit	10%		
12	Asset 2	12.9%		Upper limit	60%		
13	Asset 3	34.2%					
14	Asset 4	36.8%					
15	Sum	100.00%	<--=SUM(B11:B14)				
16							
17	Portfolio mean	7.00%	<--{=MMULT(TRANSPOSE(B11:B14),G4:G7)}				
18	Portfolio sigma	37.20%	<--{=SQRT(MMULT(TRANSPOSE(B11:B14),MMULT(B4:E7,B11:B14)))}				

11.10 Summary

In this chapter we have presented theorems relating to efficient portfolios and then showed how to implement these theorems to find the efficient frontier. Two basic propositions allow us to derive portfolios on the envelope of the feasible set of portfolios and the envelope itself. Three further propositions relate the expected returns of any asset or portfolio to the expected returns on any efficient portfolio. Under certain circumstances, this allows us to derive the security market line (SML) and the capital market line (CML) of the classic capital asset pricing model (CAPM).

The idea of how to find the efficient portfolio, with some constraints on the portfolio holdings, is easy to understand. However, no one would claim that Excel offers a quick way to solve for portfolio maximization, with or without short-sale constraints. However, it can be used to illustrate the principles involved, and the Excel **Solver** provides an easy-to-use and intuitive interface for setting up these problems.

In subsequent chapters we discuss the implementation of the CAPM. We show how to compute the variance-covariance matrix (Chapter 12), how to test the SML (Chapter 13), and how to derive useful portfolio optimization routines from our knowledge of efficient set mathematics (Chapter 15, which discusses the Black-Litterman model).

Exercises

1. Consider the data below for six furniture companies.

	A	B	C	D	E	F	G	H	I
2	Variance-covariance matrix	La-Z-Boy	Kimball	Flexsteel	Leggett	Miller	Shaw		Means
3	La-Z-Boy	0.1152	0.0398	0.1792	0.0492	0.0568	0.0989		29.24%
4	Kimball	0.0398	0.0649	0.0447	0.0062	0.0349	0.0269		20.68%
5	Flexsteel	0.1792	0.0447	0.3334	0.0775	0.0886	0.1487		25.02%
6	Leggett	0.0492	0.0062	0.0775	0.1033	0.0191	0.0597		31.64%
7	Miller	0.0568	0.0349	0.0886	0.0191	0.0594	0.0243		15.34%
8	Shaw	0.0989	0.0269	0.1487	0.0597	0.0243	0.1653		43.87%

   1. Given this matrix, and assuming that the risk-free rate is 0%, calculate the efficient portfolio of these six firms.
   2. Repeat, assuming that the risk-free rate is 5%.
   3. Use these two portfolios to generate an efficient frontier for the six furniture companies. Plot this frontier.
   4. Is there an efficient portfolio with only positive proportions of all the assets?

2. A sufficient condition to produce positively weighted efficient portfolios is that the variance-covariance matrix be diagonal. That is, that $\sigma_{ij} = 0$, for $i \neq j$. By continuity, positively weighted portfolios will result if the off-diagonal elements of the variance-covariance matrix are sufficiently small compared to the diagonal. Consider a transformation of the above matrix in which:

$$\sigma_{ij} = \begin{cases} \varepsilon \sigma_{ij}^{original} & if\ i \neq j \\ \sigma_{ii}^{original} & \end{cases}$$

   When $\varepsilon = 1$, this transformation will give the original variance-covariance matrix, and when $\varepsilon = 0$, the transformation will give a fully diagonal matrix.

   For $r = 10\%$ find the maximum ε for which all portfolio weights are positive.

3. In the example below, use Excel to find an envelope portfolio whose β with respect to the efficient portfolio y is zero. *Hint*: Notice that because the covariance is linear, so is β: Suppose that $z = \lambda x + (1 - \lambda)y$ is a convex

combination of x and y and that we are trying to find the β_z. Then:

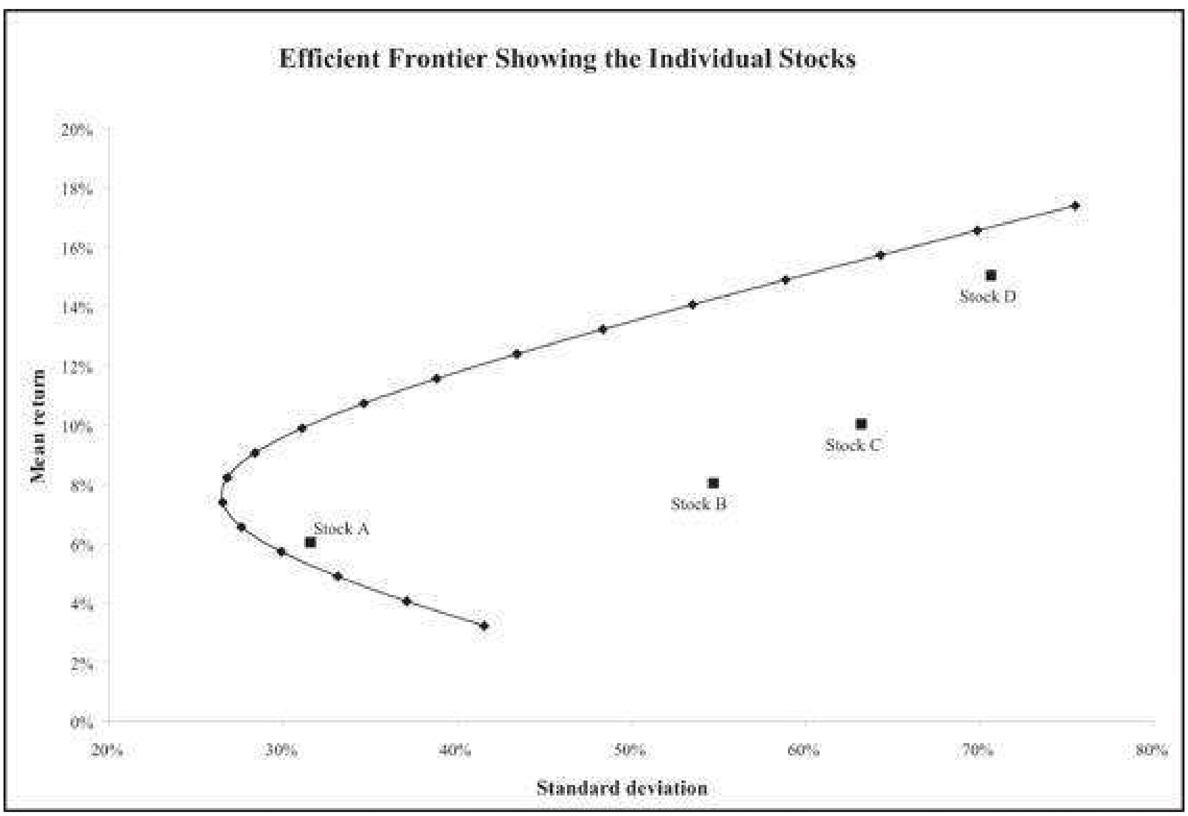
$$\beta_z = \frac{Cov(z,y)}{\sigma_y^2} = \frac{Cov(\lambda x + (1-\lambda)y, y)}{\sigma_y^2}$$
$$= \frac{\lambda\, Cov(x,y)}{\sigma_y^2} + \frac{(1-\lambda)\, Cov(y,y)}{\sigma_y^2} = \lambda \beta_x + (1-\lambda)$$

	A	B	C	D	E	F
1	Variance-covariance matrix					Mean returns
2	0.400	0.030	0.020	0.000		0.06
3	0.030	0.200	0.001	-0.060		0.05
4	0.020	0.001	0.300	0.030		0.07
5	0.000	-0.060	0.030	0.100		0.08

4. Calculate the envelope set for the four assets below and show that the individual assets all lie within this envelope set.

	A	B	C	D	E	F
1	A FOUR-ASSET PORTFOLIO PROBLEM					
2	Variance-covariance					Mean returns
3	0.10	0.01	0.03	0.05		6%
4	0.01	0.30	0.06	-0.04		8%
5	0.03	0.06	0.40	0.02		10%
6	0.05	-0.04	0.02	0.50		15%

You should get a graph that looks something like the following:



5. Given the data below:
   1. Calculate the efficient frontier assuming no short sales are allowed.
   2. Calculate the efficient frontier assuming that short sales are allowed.
   3. Graph both frontiers on the same set of axes.

	A	B	C	D	E	F	G	H	I
3		A	B	C	D	E	F		Mean returns
4	A	0.0100	0.0000	0.0000	0.0000	0.0000	0.0000		0.0100
5	B	0.0000	0.0400	0.0000	0.0000	0.0000	0.0000		0.0200
6	C	0.0000	0.0000	0.0900	0.0000	0.0000	0.0000		0.0300
7	D	0.0000	0.0000	0.0000	0.1500	0.0000	0.0000		0.0400
8	E	0.0000	0.0000	0.0000	0.0000	0.2000	0.0000		0.0500
9	F	0.0000	0.0000	0.0000	0.0000	0.0000	0.3000		0.0550

Mathematical Appendix

In this appendix we collect the various proofs of statements made in the chapter. As in the chapter, we assume that we are examining data for *N* risky assets. It is important to note that all the definitions of "feasibility" and "optimality" are made relative to this set. Thus the phrase "efficient" really means "efficient relative to the set of the *N* assets being examined."

PROPOSITION 0 The set of all feasible portfolios of risky assets is convex.

Proof of Proposition 0 A portfolio x is feasible if and only if the proportions of the portfolio add up to 1. That is, $\sum_{i=1}^{N} x_i = 1$, where N is the number of risky assets. Suppose that x and y are feasible portfolios and suppose that λ is some number between 0 and 1. Then it is clear that $z = \lambda x + (1 - \lambda)y$ is also feasible.

PROPOSITION 1 Let c be a constant and denote by R the vector of mean returns. A portfolio x is on the envelope relative to the sample set of N assets if and only if it is the normalized solution of the system:

$$R - c = Sz, \quad \text{where: } x_i = \frac{z_i}{\sum_h z_h}$$

Proof of Proposition 1 A portfolio x is on the envelope of the feasible set of portfolios if and only if it lies on the tangency of a line connecting some point c on the y-axis to the feasible set. Such a portfolio must either maximize or minimize the ratio $\frac{x(R-c)}{\sigma^2(x)}$, where $x(R - c)$ is the vector product that gives the portfolio's expected excess return over c, and $\sigma^2(x)$ is the portfolio's variance. Let this ratio's value, when maximized (or minimized), be λ. Then our portfolio must satisfy the following:

$$\frac{x(R-c)}{\sigma^2(x)} = \lambda \;\Rightarrow\; x(R-c) = \sigma^2(x)\,\lambda = xSx^T\,\lambda$$

Let h be a particular asset and differentiate this last expression with respect to x_h. This gives $R_h - c = Sx^T\lambda$. Writing $z_h = \lambda x_h$, we see that a portfolio is efficient if and only if it solves the system $R - c = Sz$. Normalizing z so that its coordinates add to 1 gives the desired result.

PROPOSITION 2 The convex combination of any two envelope portfolios is on the envelope of the feasible set.

Proof of Proposition 2 Let x and y be portfolios on the envelope. By the above theorem, it follows that there exist two vectors, z_x and z_y, and two constants c_x and c_y, such that:

- x is the normalized-to-unity vector of z_x. That is, $x_i = \frac{z_{xi}}{\sum_h z_{xh}}$, and y is the normalized-to-unity vector of z_y, i.e., $y_i = \frac{z_{yi}}{\sum_h z_{yh}}$.
- $R - c_x = Sz_x$ and $R - c_y = Sz_y$.

Furthermore, since z maximizes the ratio $\frac{z(R-c)}{\sigma^2(z)}$, it follows that any normalization of z also maximizes this ratio. With no loss in generality, therefore, we can assume that z sums to 1.

It follows that for any real number a, the portfolio $a * z_x + (1-a) * z_y$ solves the system $R - (ac_x + (1-a)c_y) = Sz$. This proves our claim.

PROPOSITION 3 Let y be any envelope portfolio of the set of N assets. Then, for any other portfolio x (including, possibly, a portfolio composed of a single asset), there exists a constant c such that the following relation holds between the expected return on x and the expected return on portfolio y:

$E(r_x) = c + \beta_x [E(r_y) - c]$

where

$$\beta_x = \frac{Cov(x, y)}{\sigma_y^2}$$

Furthermore, $c = E(r_z)$, where z is any portfolio for which $Cov(z, y) = 0$.

Proof of Proposition 3 Let y be a particular envelope portfolio and let x be any another portfolio. We assume that both portfolios x and y are column vectors. Note the following:

$$\beta_x = \frac{\text{cov}(x, y)}{\sigma_y^2} = \frac{x^T Sy}{y^T Sy}$$

Now, since y is on the envelope, we know that there exist a vector w and a constant c which solve the system $Sw = R - c$ and that $y = \frac{w}{\sum_i w_i}$. Substituting this in the expression for β_x, we get:

$$\beta_x = \frac{\text{cov}(x, y)}{\sigma_y^2} = \frac{x^T Sy}{y^T Sy} = \frac{x^T S \frac{w}{\sum_i w_i}}{\frac{y^T Sw}{\sum_i w_i}} = \frac{x^T Sw}{y^T Sw} = \frac{x^T (R-c)}{y^T (R-c)}$$

Next, note that since $\sum_i x_i = 1$, it follows that $x^T (R-c) = E(r_x) - c$ and that $y^T (R-c) = E(r_y) - c$. This shows that $\beta_x = \frac{E(r_x) - c}{E(r_y) - c}$,

which can be rewritten as:

$E(r_x) = c + \beta_x [E(r_y) - c]$

To finish the proof, let z be a portfolio that has zero covariance with y. Then the above logic shows that $c = E(rz)$. This proves the claim.

PROPOSITION 4 If, in addition to the N risky assets, there exists a risk-free asset with return r_f, then the standard SML holds:

$E(r_i) = r_f + \beta_i[E(r_M) - r_f]$

where

$$\beta_i = \frac{Cov(r_i, r_M)}{\sigma_M^2}$$

Proof of Proposition 4 If there exists a risk-free security, then the tangent line from this security to the efficient frontier dominates all other feasible portfolios. Call the point of tangency on the efficient frontier M, when the constant is r_f. Then the result follows.

Note: It is important to repeat again that the terminology “market portfolio” refers in this case to the “market portfolio relative to the sample set of N assets.”

PROPOSITION 5 Suppose that there exists a portfolio y such that for any portfolio x, the following relation holds:

$E(r_x) = c + \beta_x [E(r_y) - c]$

where

$$\beta_x = \frac{Cov(x, y)}{\sigma_y^2}$$

Then the portfolio y is on the envelope.

Proof of Proposition 5 Substituting in for the definition of β_x, it follows that for any portfolio x the following relation holds:

$$\frac{Cov(x, y)}{\sigma_y^2} = \frac{x^T S y}{y^T S y} = \frac{x^T (R-c)}{y^T (R-c)}$$

Let x be the vector composed solely of the first risky asset: $x = \{1, 0, \ldots, 0\}$. Then the above equation becomes:

$$S_1 y \frac{y^T (R-c)}{\sigma_y^2} = E(r_1) - c$$

Here we write $S_1 a y = E(r_1) - c$, where S_1 is the first row of the variance-covariance matrix S. Note that $a = \frac{y^T (R-c)}{\sigma_y^2}$ is a constant whose value is independent of the vector x. If we let x be a vector composed solely of the i^{th} risky asset, we get $S_i\, a y = E(r_i) - c$. This proves that the vector $z = ay$ solves the system $Sz = R - c$; by Proposition 1, this means that the normalization of z is on the envelope. But this normalization is simply the vector y.

---

1. 1. Roll’s 1977 paper is more often cited and is more comprehensive, but his 1978 paper is much easier to read and more intuitive. If you are interested in this literature, start there.
2. 2. And it forms the basis of our discussion of the Black-Litterman model in Chapter 15.
3. 3. Following the discussion in section 11.6, a careful reader will recall that Proposition 1 only guarantees that this portfolio is on the envelope. But it is, in fact, efficient.
4. 4. The actual procedures for implementing a short sale are not simple. For a well-written academic survey, see D’Avolio (2002). See also Surowiecki (2003).
5. 5. We do not go into the efficient set mathematics when short sales of assets are restricted. This involves the Kuhn-Tucker conditions, a discussion of which can be found in Elton et al. (2009).
6. 6. If you cannot find the Solver under **Data|Solver**, you may not have loaded the Solver add-in. To do so, go to **File|Options|Add-ins|Manage Add-ins** and check the **Solver Add-in**.
7. 7. There’s another way of doing this. **Solver** has an option to “Make Unconstrained Variables Non-Negative.”

12 Calculating the Variance-Covariance Matrix

12.1 Overview

In order to calculate efficient portfolios, we must be able to compute the variance-covariance matrix from return data for stocks. In this chapter we discuss this computation, showing how to do the calculations in Excel. The most obvious calculation is the *sample variance-covariance matrix*, which is computed directly from the historic returns. We illustrate several methods for calculating the sample variance-covariance matrix, including a direct calculation in the spreadsheet using the excess return matrix and an implementation of this method with VBA.

While the sample variance-covariance matrix may appear to be an obvious choice, a large literature recognizes that it may not be the best estimate of variances and covariances. Disappointment with the sample variance-covariance matrix stems both from its often unrealistic parameters and from its inability to predict. These issues are discussed briefly in sections 12.5 and 12.6. As an alternative to the sample matrix, sections 12.7–12.10 discuss so-called "shrinkage" methods for improving the estimate of the variance-covariance matrix.[1]

Before starting this chapter, you may want to peruse Chapter 31, which discusses *array functions*. These are Excel functions whose arguments are vectors and matrices; their implementation is slightly different from standard Excel functions. This chapter makes heavy use of the array functions **Transpose()** and **MMult()** as well as some other "homegrown" array functions.

Throughout this chapter, the examples show an implementation in Excel. However, on the companion website of this book you can find an equivalent "R" file that shows full implementation of the concepts in this chapter.

12.2 Computing the Sample Variance-Covariance Matrix

Suppose we have return data for N assets over T periods. Writing the return of asset i in period t as r_{it}, we write the *mean return* of asset i as follows:

$$\bar{r}_i = \frac{1}{T}\sum_{t=1}^{T} r_{it}$$

$$i = 1, \ldots, N$$

Then the covariance of the return of asset i and asset j is calculated as:

$$\sigma_{ij} = cov(i, j) = \frac{1}{T-1} \sum_{t=1}^{T} (r_{it} - \overline{r}_i) \cdot (r_{jt} - \overline{r}_j)$$

$$i, j = 1, \ldots, N$$

The matrix of these covariances (which includes, of course, the variances when $i = j$) is the *sample variance-covariance matrix*. Our problem is to calculate these covariances efficiently. Define the *excess return matrix* to be:

$$A = matrix\ of\ excess\ returns = \begin{bmatrix} r_{11} - \overline{r}_1 & r_{21} - \overline{r}_2 & \cdots & r_{N1} - \overline{r}_N \\ r_{12} - \overline{r}_1 & r_{22} - \overline{r}_2 & \cdots & r_{N2} - \overline{r}_N \\ \vdots & \vdots & \ddots & \vdots \\ r_{1T} - \overline{r}_1 & r_{2T} - \overline{r}_2 & \cdots & r_{NT} - \overline{r}_N \end{bmatrix}$$

Columns of matrix A subtract the mean asset return from the individual asset returns.

Multiplying A^T (the transpose of matrix A) by A and dividing through by $T - 1$ gives the sample variance-covariance matrix:

$$S = [\sigma_{ij}] = \frac{A^T \cdot A}{T-1}$$

To consider the computational aspects, we use $T = 60$ months of return data for $N = 10$ stocks. The spreadsheet below shows the price data (adjusted for dividends) and the computed returns:

	A	B	C	D	E	F	G	H	I	J	K	L
1	FIVE YEARS OF PRICES FOR 10 STOCKS AND THE SP500											
2		Microsoft	Apple	Amazon	Disney	Cisco	J.P. Morgan	Google	ExxonMobil	Procter & Gamble	Bank of America	SP500
3	Date	MSFT	AAPL	AMZN	DIS	CSCO	JPM	GOOG	XOM	PG	BAC	^GSPC
4	01-Jan-19	103.60	165.09	1,718.73	111.52	46.63	103.89	1,116.37	71.65	95.04	28.33	2,704.10
5	01-Dec-18	100.77	156.46	1,501.97	108.81	42.73	96.10	1,035.61	66.68	90.56	24.38	2,506.85
6	01-Nov-18	109.54	176.52	1,690.17	114.60	47.20	109.46	1,094.43	76.96	93.11	28.10	2,760.17
7	01-Oct-18	105.51	216.33	1,598.01	113.95	44.81	106.57	1,076.77	77.13	86.60	27.21	2,711.74
61	01-Apr-14	35.83	74.26	304.13	73.93	19.55	48.55	523.78	84.71	69.76	14.07	1,883.95
62	01-Mar-14	36.33	67.54	336.37	74.99	18.97	52.65	555.45	80.80	68.11	15.97	1,872.34
63	01-Feb-14	33.70	63.46	362.10	75.28	18.44	49.28	603.90	79.07	66.47	15.35	1,859.45
64	01-Jan-14	33.29	60.37	358.69	67.64	18.40	47.70	586.67	75.70	64.26	15.56	1,782.59

Using the Excel function **Ln(P$_t$/P$_{t-1}$)**, we compute the monthly returns:

	A	B	C	D	E	F	G	H	I	J	K	L
1	FIVE YEARS OF MONTHLY RETURNS FOR 10 STOCKS AND THE SP500											
2	Date	MSFT	AAPL	AMZN	DIS	CSCO	JPM	GOOG	XOM	PG	BAC	
3	1-Jan-19	2.78%	5.37%	13.48%	2.46%	8.75%	5.85%	7.51%	7.20%	4.83%	15.01%	
4	1-Dec-18	-8.35%	-12.06%	-11.81%	-5.19%	-9.96%	-13.02%	-5.52%	-14.34%	-2.78%	-14.20%	
5	1-Nov-18	3.75%	-20.34%	5.61%	0.57%	5.20%	2.67%	1.63%	-0.23%	7.25%	3.22%	
6	1-Oct-18	-6.64%	-3.10%	-22.59%	-1.82%	-6.15%	-3.44%	-10.29%	-6.49%	6.34%	-6.40%	
7	1-Sep-18	2.18%	-0.48%	-0.48%	4.30%	1.68%	-1.53%	-2.05%	6.90%	0.34%	-4.87%	
58	1-Jun-14	2.54%	6.69%	3.84%	2.04%	0.93%	3.62%	2.71%	0.83%	-2.76%	3.51%	
59	1-May-14	1.33%	7.02%	2.73%	5.72%	7.18%	-0.11%	6.12%	-1.85%	-1.36%	0.00%	
60	1-Apr-14	-1.45%	9.48%	-10.08%	-0.92%	3.05%	-8.11%	-5.87%	4.73%	2.39%	-12.70%	
61	1-Mar-14	7.51%	6.23%	-7.37%	-0.92%	2.80%	6.62%	-8.36%	2.16%	2.44%	3.97%	
62	1-Feb-14	1.23%	5.00%	0.95%	10.70%	0.26%	3.26%	2.89%	4.36%	3.38%	-1.32%	
63												
64	Mean	1.89%	1.68%	2.61%	0.83%	1.55%	1.26%	1.07%	-0.09%	0.65%	1.00%	<-- =AVERAGE(K3:K62)

Below we compute the excess returns and the variance-covariance matrix:

	A	B	C	D	E	F	G	H	I	J	K	L
66					Excess returns: $r_i - r_f$							
67	Date	MSFT	AAPL	AMZN	DIS	CSCO	JPM	GOOG	XOM	PG	BAC	
68	1-Jan-19	0.88%	3.69%	10.87%	1.63%	7.20%	4.58%	6.44%	7.29%	4.18%	14.01%	<--=K3-K$64
69	1-Dec-18	-10.24%	-13.74%	-14.43%	-6.02%	-11.51%	-14.28%	-6.60%	-14.25%	-3.43%	-15.20%	<--=K4-K$64
70	1-Nov-18	1.86%	-22.02%	3.00%	0.26%	3.65%	1.40%	0.55%	-0.13%	6.59%	2.22%	<--=K5-K$64
71	1-Oct-18	-8.73%	-4.77%	-25.20%	-2.65%	-7.70%	-4.71%	-11.36%	-6.40%	5.69%	-7.40%	<--=K6-K$64
124	1-May-14	-0.56%	5.34%	0.12%	4.89%	5.63%	-1.37%	5.05%	-1.76%	-2.01%	-1.00%	
125	1-Apr-14	-3.34%	7.82%	-12.69%	-1.75%	1.48%	-9.38%	-6.94%	4.82%	1.74%	-13.70%	
126	1-Mar-14	5.62%	4.55%	-9.98%	-1.75%	1.25%	5.36%	-9.44%	2.25%	1.78%	2.97%	
127	1-Feb-14	-0.66%	3.32%	-1.67%	9.87%	-1.29%	1.95%	1.82%	4.45%	2.73%	-2.32%	
128												
129					Variance-Covariance Matrix							
130		MSFT	AAPL	AMZN	DIS	CSCO	JPM	GOOG	XOM	PG	BAC	
131	MSFT	0.0034	0.0017	0.0024	0.0010	0.0016	0.0016	0.0019	0.0009	0.0005	0.0021	
132	AAPL	0.0017	0.0052	0.0018	0.0008	0.0020	0.0006	0.0012	0.0002	0.0004	0.0007	
133	AMZN	0.0024	0.0018	0.0069	0.0014	0.0023	0.0014	0.0032	0.0008	-0.0003	0.0022	
134	DIS	0.0010	0.0008	0.0014	0.0024	0.0011	0.0014	0.0007	0.0010	0.0005	0.0016	
135	CSCO	0.0016	0.0020	0.0023	0.0011	0.0033	0.0014	0.0009	0.0005	0.0004	0.0017	
136	JPM	0.0016	0.0006	0.0014	0.0014	0.0014	0.0032	0.0008	0.0009	0.0002	0.0039	
137	GOOG	0.0019	0.0012	0.0032	0.0007	0.0009	0.0008	0.0032	0.0003	0.0001	0.0015	
138	XOM	0.0009	0.0002	0.0008	0.0010	0.0005	0.0009	0.0003	0.0023	0.0004	0.0010	
139	PG	0.0005	0.0004	-0.0003	0.0005	0.0004	0.0002	0.0001	0.0004	0.0016	0.0001	
140	BAC	0.0021	0.0007	0.0022	0.0016	0.0017	0.0039	0.0015	0.0010	0.0001	0.0058	
141												
142				{=MMULT(TRANSPOSE(B68:K127),B68:K127)/59}								

A VBA Function to Compute the Variance-Covariance Matrix

To automate this procedure, we write a VBA function that computes the variance-covariance matrix using the Excel function **Covariance.S**. When Excel functions with periods, such as **Covariance.S**, are used in VBA, the period becomes an underscore: **Covariance_S**.[2]

```
Function VarCovar(rng As Range) As Variant
    Dim i As Integer
    Dim j As Integer
    Dim numcols As Integer
    numcols=rng.Columns.Count
    numrows=rng.Rows.Count
    Dim matrix() As Double
    ReDim matrix(numcols - 1, numcols - 1)
    For i = 1 To numcols
        For j = i To numcols
            matrix(i - 1, j - 1) =
            Application.WorksheetFunction.Covariance_S(rng.Column
            s(i), rng.Columns(j))
            matrix(j - 1, i - 1) = matrix(i - 1, j - 1)
        Next j
    Next i
    VarCovar = matrix
End Function
```

The VBA function computes **COVARIANCE.S** for every entry of the variance-covariance matrix. Here's the result:

	A	B	C	D	E	F	G	H	I	J	K	L
1	ESTIMATING THE VARIANCE-COVARIANCE MATRIX WITH VBA											
2	Variance-Covariance Matrix (user defined 'VarCovar' function)											
3		MSFT	AAPL	AMZN	DIS	CSCO	JPM	GOOG	XOM	PG	BAC	
4	MSFT	0.0034	0.0017	0.0024	0.0010	0.0016	0.0016	0.0019	0.0009	0.0005	0.0021	<--{=varcovar('5y 10
5	AAPL	0.0017	0.0052	0.0018	0.0008	0.0020	0.0006	0.0012	0.0002	0.0004	0.0007	stocks Returns'!B3:K62)}
6	AMZN	0.0024	0.0018	0.0069	0.0014	0.0023	0.0014	0.0032	0.0008	-0.0003	0.0022	
7	DIS	0.0010	0.0008	0.0014	0.0024	0.0011	0.0014	0.0007	0.0010	0.0005	0.0016	
8	CSCO	0.0016	0.0020	0.0023	0.0011	0.0033	0.0014	0.0009	0.0005	0.0004	0.0017	
9	JPM	0.0016	0.0006	0.0014	0.0014	0.0014	0.0032	0.0008	0.0009	0.0002	0.0039	
10	GOOG	0.0019	0.0012	0.0032	0.0007	0.0009	0.0008	0.0032	0.0003	0.0001	0.0015	
11	XOM	0.0009	0.0002	0.0008	0.0010	0.0005	0.0009	0.0003	0.0023	0.0004	0.0010	
12	PG	0.0005	0.0004	-0.0003	0.0005	0.0004	0.0002	0.0001	0.0004	0.0016	0.0001	
13	BAC	0.0021	0.0007	0.0022	0.0016	0.0017	0.0039	0.0015	0.0010	0.0001	0.0058	

Using R to Compute the Variance-Covariance Matrix

R is a great tool to use for calculating the variance-covariance matrix. It is much more efficient than Excel, as can be seen by the code below:

```
# we thank Sagi Haim for developing this script

###########################################
# CHAPTER 12 - VRIANCE-COVARIANCE MATRIX
###########################################

### 12.2 Computing the sample variance-covariance matrix
# save a csv file with historical prices of stocks in a working directory
workdir <- readline( prompt = "working directory?") # insert working directory path
setwd(workdir)

# read the csv with adjusted prices of stocks
day_ret <- na.omit(read.csv("Chapt_12_data.csv", row.names = 1))
index_ret <- day_ret[,"X.GSPC"]
assets_ret <- day_ret[,colnames(day_ret) != "X.GSPC"]
# COMMENT : the input csv should present return data as text.

# sample Variance Covariance Matrix
var_cov_mat <- cov(assets_ret)
```

Note that the heart of the code is one line consisting of "cov(assets_ret)."

Should We Divide by $M - 1$ or by M

In the above calculations, we use the sample covariance (in Excel **Covariance.S** and in VBA **Covariance_S**) to divide by $M - 1$ instead of M in order to get the unbiased estimate of the variances and covariances. We don't think this matters very much, but for reference to a higher authority, we suggest our discussion of M versus $M - 1$ in section 10.2.

12.3 The Correlation Matrix

Using the Excel function **Correl**, we can compute the correlation matrix of the returns:

```
Function CorrMatrix(rng As Range) As Variant
    Dim i As Integer
    Dim j As Integer
    Dim numcols As Integer
    numcols = rng.Columns.Count
    numrows = rng.Rows.Count
```

```
    Dim matrix() As Double
    ReDim matrix(numcols - 1, numcols - 1)

    For i = 1 To numcols
        For j = i To numcols
            matrix(i - 1, j - 1) =
            Application.WorksheetFunction.Correl(rng.Columns(i),
            rng.Columns(j))
        matrix(j - 1, i - 1) = matrix(i - 1, j - 1)
        Next j
    Next i
    CorrMatrix = matrix
End Function
```

	A	B	C	D	E	F	G	H	I	J	K	L
1	ESTIMATING THE CORRELATION MATRIX WITH VBA											
2		Correlation matrix (user defined 'CorrMatrix' function)										
3		MSFT	AAPL	AMZN	DIS	CSCO	JPM	GOOG	XOM	PG	BAC	
4	MSFT	1.0000	0.3974	0.4888	0.3565	0.4689	0.4770	0.5682	0.3061	0.2084	0.4642	<--{=CorrMatrix('5y.10
5	AAPL	0.3974	1.0000	0.3084	0.2189	0.4747	0.1367	0.2891	0.0545	0.1316	0.1240	stocks Returns'!B3:K62)}
6	AMZN	0.4888	0.3084	1.0000	0.3425	0.4822	0.2913	0.6773	0.1956	-0.0959	0.3486	
7	DIS	0.3565	0.2189	0.3425	1.0000	0.3786	0.5170	0.2682	0.4046	0.2423	0.4387	
8	CSCO	0.4689	0.4747	0.4822	0.3786	1.0000	0.4297	0.2767	0.1961	0.1875	0.3834	
9	JPM	0.4770	0.1367	0.2913	0.5170	0.4297	1.0000	0.2494	0.3445	0.0774	0.9181	
10	GOOG	0.5682	0.2891	0.6773	0.2682	0.2767	0.2494	1.0000	0.1284	0.0315	0.3415	
11	XOM	0.3061	0.0545	0.1956	0.4046	0.1961	0.3445	0.1284	1.0000	0.1856	0.2762	
12	PG	0.2084	0.1316	-0.0959	0.2423	0.1875	0.0774	0.0315	0.1856	1.0000	0.0312	
13	BAC	0.4642	0.1240	0.3486	0.4387	0.3834	0.9181	0.3415	0.2762	0.0312	1.0000	

Here's another version of the correlation matrix, this time only the upper half:

```
'Triangular correlation matrix
Function CorrMatrixTriangular(rng As Range) As Variant
Function CorrMatrixTriangular(rng As Range) As Variant
    Dim i As Integer
    Dim j As Integer
    Dim numcols As Integer
    numcols = rng.Columns.Count
    numrows = rng.Rows.Count
    Dim matrix() As Variant
    ReDim matrix(numcols - 1, numcols - 1)

    For i = 1 To numcols
        For j = 1 To numcols
        If i <= j Then
        matrix(i - 1, j - 1) = _
        Application.WorksheetFunction.Correl (rng.Columns(i),
        rng.Columns(j))
        Else
        matrix(i - 1, j - 1) = " "
        End If
```

```
        Next j
    Next i
    CorrMatrixTriangular = matrix
End Function
```

	A	B	C	D	E	F	G	H	I	J	K	L
15		Correlation matrix, upper half (user defined 'CorrMatrixTriangular' function)										
16		MSFT	AAPL	AMZN	DIS	CSCO	JPM	GOOG	XOM	PG	BAC	
17	MSFT	1.0000	0.3974	0.4888	0.3565	0.4689	0.4770	0.5682	0.3061	0.2084	0.4642	<--
18	AAPL		1.0000	0.3084	0.2189	0.4747	0.1367	0.2891	0.0545	0.1316	0.1240	{=CorrMatrixTriangular(
19	AMZN			1.0000	0.3425	0.4822	0.2913	0.6773	0.1956	-0.0959	0.3486	5y 10 stocks
20	DIS				1.0000	0.3786	0.5170	0.2682	0.4046	0.2421	0.4387	
21	CSCO					1.0000	0.4297	0.2767	0.1961	0.1875	0.3834	
22	JPM						1.0000	0.2494	0.3445	0.0774	0.9181	
23	GOOG							1.0000	0.1284	0.0315	0.3415	
24	XOM								1.0000	0.1856	0.2762	
25	PG									1.0000	0.0312	
26	BAC										1.0000	

As in the variance-covariance matrix, calculating the correlation matrix in R is very easy; it involves one line of code:

```
### 12.3 The Correlation Matrix
cor_mat <- cor(assets_ret, method = "pearson")
```

Here are some statistics for the correlations. The average correlation for our sample (0.3116) is somewhat normative (usually a sample of stocks will give average correlation of 0.2 to 0.3). The largest correlations ($\rho_{BAC,JPM}$ = 0.9181, $\rho_{GOOG,AMZN}$ = 0.6773) look quite high, though perhaps there are economic explanations.[3]

	A	B	C	D	E	F	G	H	I	J	K
28							Some correlation statistics				
29	Average	0.3116 <--=AVERAGEIF(B17:K26,"<1")									
30	Median	0.3061 <--{=MEDIAN(IF(B17:K26=1,"",B17:K26))}									
31											
32	Largest	0.918 <--=LARGE(B17:K26,11)					Smallest	-0.096 <--=SMALL(B17:K26,1)			
33	Next largest	0.677 <--=LARGE(B17:K26,12)					Next smallest	0.031 <--=SMALL(B17:K26,2)			
34	etc.	0.568 <--=LARGE(B17:K26,13)					etc.	0.032 <--=SMALL(B17:K26,3)			
35	etc.	0.517 <--=LARGE(B17:K26,14)					etc.	0.055 <--=SMALL(B17:K26,4)			
36	etc.	0.489 <--=LARGE(B17:K26,15)					etc.	0.077 <--=SMALL(B17:K26,5)			

12.4 Four Alternatives to the Sample Variance-Covariance Matrix

In succeeding sections we illustrate four alternatives to the sample variance-covariance matrix:

- • The **single-index model** assumes that the only sources of variance risk are the market variance and the betas of the assets, so that $\sigma_{ij} = \beta_i \beta_j \sigma_M$.
- • The **constant correlation model** assumes the correlation between all asset returns is constant, so that $\sigma_{ij} = \rho \sigma_i \sigma_j$.
- • **Shrinkage methods** assume that the variance-covariance matrix is a convex combination of the sample variance-covariance and a matrix with variances on the

diagonal and zeros elsewhere.

- • **Option methods** use options to derive the standard deviations of returns for the assets. In section 12.8, we combine this method with the constant correlation method to compute a variance-covariance matrix.

The first three models arise out of a distrust of return data for generating the covariances of the data in the future. The fourth method—using option data—goes further and assumes that even the sample variances are an inaccurate prediction of future variance.

12.5 Alternatives to the Sample Variance-Covariance: The Single-Index Model

The single-index model (SIM) began as an attempt to simplify some of the computational complexities of calculating the variance-covariance matrix.[4] The basic assumption of the SIM is that the returns of each asset can be linearly regressed on a market index *x*:

$$\tilde{r}_i = \alpha_i + \beta_i \tilde{r}_x + \tilde{\varepsilon}_i$$

where the correlation between ε_i and ε_j is zero. Given this assumption, it is easy to establish the following two facts:

- • $E(\tilde{r}_i) = \alpha_i + \beta_i E(\tilde{r}_x)$
- • $\sigma_{ij} = \begin{cases} \sigma_i^2 & \text{when } i = j \\ \beta_i \beta_j \sigma_x^2 & \text{when } i \neq j \end{cases}$

Essentially, the SIM involves changes in the estimates of the covariances but not the sample variance. We can automate the procedure for computing the SIM by writing some VBA code:

```
Function SIM_VarCov(assetdata As Range, marketdata As Range) _As
Variant
    Dim i As Integer
    Dim j As Integer
    Dim numcols As Integer
    numcols=assetdata.Columns.Count
    Dim matrix() As Double
    ReDim matrix(numcols - 1, numcols - 1)
    For i = 1 To numcols
    For j = i To numcols
        If i = j Then
        matrix(i - 1, j - 1) = _
```

```
            Application.WorksheetFunction.
            Var_S(assetdata.Columns(i))
            Else
            matrix(i - 1, j - 1) = _
            Application.WorksheetFunction.Slope _
            (assetdata.Columns(i), marketdata) * _
            Application.WorksheetFunction.Slope _
            (assetdata.Columns(j), marketdata) * _
            Application.WorksheetFunction.Var_S(marketdata)
            matrix(j - 1, i - 1) = matrix(i - 1, j - 1)
        End If
        Next j
        Next i
        SIM_VarCov = matrix
    End Function
```

The two arguments of this function are the asset returns and the market returns. Applying this code in our example:

COMPUTING THE SINGLE-INDEX VARIANCE-COVARIANCE MATRIX

Using the user defined 'SIM_VarCov' function

	MSFT	AAPL	AMZN	DIS	CSCO	JPM	GOOG	XOM	PG	BAC	
MSFT	0.0034	0.0013	0.0019	0.0011	0.0013	0.0013	0.0012	0.0010	0.0004	0.0017	<--{=SIM_VarCov(B16:K75,L16:L75)}
AAPL	0.0013	0.0052	0.0018	0.0011	0.0013	0.0013	0.0011	0.0010	0.0004	0.0017	
AMZN	0.0019	0.0018	0.0069	0.0016	0.0019	0.0018	0.0017	0.0015	0.0006	0.0024	
DIS	0.0011	0.0011	0.0016	0.0024	0.0011	0.0011	0.0010	0.0009	0.0004	0.0015	
CSCO	0.0013	0.0013	0.0019	0.0011	0.0035	0.0013	0.0012	0.0010	0.0004	0.0018	
JPM	0.0013	0.0013	0.0018	0.0011	0.0013	0.0032	0.0012	0.0010	0.0004	0.0017	
GOOG	0.0012	0.0011	0.0017	0.0010	0.0012	0.0012	0.0032	0.0009	0.0004	0.0016	
XOM	0.0010	0.0010	0.0015	0.0009	0.0010	0.0010	0.0009	0.0023	0.0003	0.0014	
PG	0.0004	0.0004	0.0006	0.0004	0.0004	0.0004	0.0004	0.0003	0.0016	0.0006	
BAC	0.0017	0.0017	0.0024	0.0015	0.0018	0.0017	0.0016	0.0014	0.0006	0.0058	

Date	MSFT	AAPL	AMZN	DIS	CSCO	JPM	GOOG	XOM	PG	BAC	^GSPC
1-Jan-19	2.8%	5.4%	13.5%	2.5%	8.7%	5.8%	7.5%	7.2%	4.8%	15.0%	7.6%
1-Dec-18	-8.3%	-12.1%	-11.8%	-5.2%	-10.0%	-13.0%	-5.5%	-14.3%	-2.8%	-14.2%	-9.6%
1-Nov-18	3.7%	-20.3%	5.6%	0.6%	5.2%	2.7%	1.6%	-0.2%	7.2%	3.2%	1.8%
1-Apr-14	-1.4%	9.5%	-10.1%	-0.9%	3.0%	-8.1%	-5.9%	4.7%	2.4%	-12.7%	0.6%
1-Mar-14	7.5%	6.2%	-7.4%	-0.9%	2.8%	6.6%	-8.4%	2.2%	2.4%	4.0%	0.7%
1-Feb-14	1.2%	5.0%	0.9%	10.7%	0.3%	3.3%	2.9%	4.4%	3.4%	-1.3%	4.2%

Implementing the method in R in our example looks like this:

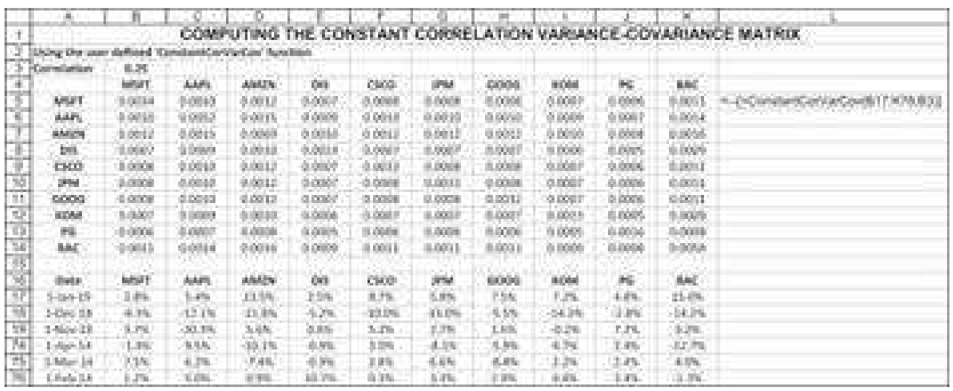
```
# Calculating beta
var_index <- var(index_ret)
assets_beta <- sapply(as.list(assets_ret) , function(x){
                lm(formula = x ~ index_ret)$coefficients[2]})
names(assets_beta) <- names(assets_ret)

# Calculating covariance(ri,rj) where i<>j
single_indx_cov_mat <- assets_beta %*% t(assets_beta) * var_index

# Calculating variance(ri) in the diagonal
diag(single_indx_cov_mat) <- diag(var_cov_mat)

# Adding names to the SIM var-cov matrix
rownames(single_indx_cov_mat) <- colnames(single_indx_cov_mat)
```

12.6 Alternatives to the Sample Variance-Covariance: Constant Correlation

The constant correlation model of Elton and Gruber (1973) computes the variance-covariance matrix by assuming that the variances of the asset returns are the sample returns, but that the covariances are all related by the same correlation coefficient, which is generally taken to be the average correlation coefficient of the assets in question. Since $Cov(r_i, r_j) = \sigma_{ij} = \rho_{ij}\sigma_i\sigma_j$, this means that in the constant correlation model:

$$\sigma_{ij} = \begin{cases} \sigma_{ii} = \sigma_i^2 & when\ i = j \\ \sigma_{ij} = \rho\sigma_i\sigma_j & when\ i \neq j \end{cases}$$

Using our data for the 10 stocks, we can implement the constant correlation model. We first compute the correlations of all the stocks:



We've written a VBA function to compute this matrix from the return data:

```
Function ConstantCorVarCov(data As Range, corr As Double) As
Variant
    Dim i As Integer
    Dim j As Integer
    Dim numcols As Integer
    numcols = data.Columns.Count
    numrows = data.Rows.Count
    Dim matrix() As Double
    ReDim matrix(numcols - 1, numcols - 1)
    If Abs(corr) >= 1 Then GoTo Out
    For i = 1 To numcols
    For j = i To numcols
        If i = j Then
        matrix(i - 1, j - 1) =_
        Application.WorksheetFunction.Var_S (data.Columns(i))
        Else
        Sigma_i = _
        Application.WorksheetFunction.StDev_S (data.Columns(i))
```

```
        Sigma_j = _
        Application.WorksheetFunction.StDev_S (data.Columns(j))
        matrix(i - 1, j - 1) = corr * Sigma_i * Sigma_j
        matrix(j - 1, i - 1) = matrix(i - 1, j - 1)
        End If
    Next j
    Next i
Out:
    If Abs(corr) >= 1 Then ConstantCorVarCov = "Error" _
    Else: ConstantCorVarCov = matrix
End Function
```

Implementing this in R for our example looks like this:

```
# Input: fixed correlation ceffficient of 0.25
const_corr <- 0.25

# Creating a vector of assets sigma
assets_sigma <- matrix(sqrt(diag(var_cov_mat)))

# Calculating covariance(ri,rj) where i<>j
const_corr_var_cov_mat <- assets_sigma %*% t(assets_sigma) * const_corr

# Calculating variance(ri) in the diagonal
diag(const_corr_var_cov_mat) <- diag(var_cov_mat)

# Name the matrix
colnames(const_corr_var_cov_mat) <- names(assets_ret)
rownames(const_corr_var_cov_mat) <- names(assets_ret)
```

12.7 Alternatives to the Sample Variance-Covariance: Shrinkage Methods

A third class of methods of estimating the variance-covariance matrix has recently achieved popularity. So-called *shrinkage methods* assume that the variance-covariance matrix is a convex combination of the sample covariance matrix and some other matrix:

$$Shrinkagervariance - covariancematrix = \lambda * Samplevar - Cov + (1 - \lambda) * Othermatrix$$

In the example below, the "other" matrix is a diagonal matrix of only variances, with zeros elsewhere. The shrinkage estimator $\lambda = 0.3$ (cell B28).

ESTIMATING THE VARIANCE-COVARIANCE MATRIX USING THE SHRINKAGE APPROACH
Gives weight 0.30 (the shrinkage factor) to sample var-cov and weight 0.70 to a diagonal matrix of only variances

Sample variance-covariance matrix (user defined "VarCovar" function)

	MSFT	AAPL	AMZN	DIS	CSCO	JPM	GOOG	XOM	PG	BAC	
MSFT	0.0034	0.0017	0.0024	0.0010	0.0016	0.0016	0.0019	0.0009	0.0005	0.0021	<-- [illegible]
AAPL	0.0017	0.0052	0.0018	0.0008	0.0020	0.0006	0.0012	0.0002	0.0004	0.0007	
AMZN	0.0024	0.0018	0.0069	0.0014	0.0023	0.0014	0.0032	0.0008	-0.0003	0.0022	
DIS	0.0010	0.0008	0.0014	0.0026	0.0011	0.0014	0.0007	0.0010	0.0005	0.0016	
CSCO	0.0016	0.0020	0.0023	0.0011	0.0083	0.0014	0.0009	0.0005	0.0004	0.0017	
JPM	0.0016	0.0006	0.0014	0.0014	0.0014	0.0032	0.0008	0.0009	0.0002	0.0039	
GOOG	0.0019	0.0012	0.0032	0.0007	0.0009	0.0008	0.0032	0.0003	0.0001	0.0015	
XOM	0.0009	0.0002	0.0008	0.0010	0.0005	0.0009	0.0003	0.0021	0.0004	0.0010	
PG	0.0005	0.0004	-0.0003	0.0005	0.0004	0.0002	0.0001	0.0004	0.0014	0.0001	
BAC	0.0021	0.0007	0.0012	0.0018	0.0017	0.0039	0.0015	0.0010	0.0001	0.0058	

Other matrix = diagonal matrix of only variances and zero elsewhere

	MSFT	AAPL	AMZN	DIS	CSCO	JPM	GOOG	XOM	PG	BAC	
MSFT	0.0034	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	<-- =IF($A17=B$16,B4,0)
AAPL	0.0000	0.0052	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	
AMZN	0.0000	0.0000	0.0069	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	
DIS	0.0000	0.0000	0.0000	0.0026	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	
CSCO	0.0000	0.0000	0.0000	0.0000	0.0083	0.0000	0.0000	0.0000	0.0000	0.0000	
JPM	0.0000	0.0000	0.0000	0.0000	0.0000	0.0032	0.0000	0.0000	0.0000	0.0000	
GOOG	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0032	0.0000	0.0000	0.0000	
XOM	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0021	0.0000	0.0000	
PG	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0014	0.0000	
BAC	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0058	

Shrinkage factor 0.30 <-- this is the weight put on the sample var-cov

	MSFT	AAPL	AMZN	DIS	CSCO	JPM	GOOG	XOM	PG	BAC	
MSFT	0.0034	0.0005	0.0007	0.0003	0.0005	0.0005	0.0006	0.0003	0.0002	0.0006	<-- =$C$28*K4+(1-$C$28)*K17
AAPL	0.0005	0.0052	0.0006	0.0002	0.0006	0.0002	0.0004	0.0001	0.0001	0.0002	
AMZN	0.0007	0.0006	0.0069	0.0004	0.0007	0.0004	0.0010	0.0002	-0.0001	0.0007	
DIS	0.0003	0.0002	0.0004	0.0026	0.0003	0.0004	0.0002	0.0003	0.0002	0.0005	
CSCO	0.0005	0.0006	0.0007	0.0003	0.0083	0.0004	0.0003	0.0002	0.0001	0.0005	
JPM	0.0005	0.0002	0.0004	0.0004	0.0004	0.0032	0.0002	0.0003	0.0001	0.0012	
GOOG	0.0006	0.0004	0.0010	0.0002	0.0003	0.0002	0.0032	0.0001	0.0000	0.0004	
XOM	0.0003	0.0001	0.0002	0.0003	0.0002	0.0003	0.0001	0.0021	0.0001	0.0003	
PG	0.0002	0.0001	-0.0001	0.0001	0.0001	0.0001	0.0000	0.0001	0.0014	0.0000	
BAC	0.0006	0.0002	0.0007	0.0005	0.0005	0.0012	0.0004	0.0003	0.0000	0.0058	

There is little theory about choosing the proper shrinkage estimator.[5] Our suggestion is to choose a shrinkage operator λ so that the global minimum variance portfolio (GMVP) is wholly positive when computed (see next section for details).

The implementation in R looks like this:

```
# Input: lambda
lambda <- 0.3

## Creating a matrix of only variances on the diagonal
other_mat <- diag(nrow = ncol(var_cov_mat)) * diag(var_cov_mat)

# Shrinkage Method Variance Covariance Matrix
shrnk_var_cov_mat <- lambda * var_cov_mat + (1 - lambda) * other_mat
```

12.8 Using Option Information to Compute the Variance Matrix

Another way to compute the variance matrix is to use the information from the options market. We use the implied volatility for each of the stocks from their corresponding at-the-money call options and then compute the variance matrix using constant correlation:

$$\sigma_{ij} = \begin{cases} \sigma^2_{i,implied} & if\ i = j \\ \rho\sigma_{i,implied}\sigma_{j,implied} & if\ i \neq j \end{cases}$$

Here's an example for our 10-stock case. We use implied volatility data from the options markets for each of the 10 stocks and a constant correlation variance-covariance matrix (where $\rho = 0.25$):

	A	B	C	D	E	F	G	H	I	J	K	L
1		CONSTANT CORRELATION MATRIX WITH IMPLIED VOLATILITIES										
2		MSFT	AAPL	AMZN	DIS	CSCO	JPM	GOOG	XOM	PG	BAC	
3	Implied volatility	33.99%	31.79%	23.54%	23.98%	21.19%	54.98%	22.76%	21.68%	17.12%	21.78%	Extracted from options prices
4												
5	Correlation	0.25										
6		MSFT	AAPL	AMZN	DIS	CSCO	JPM	GOOG	XOM	PG	BAC	
7	MSFT	0.1155	0.0270	0.0200	0.0204	0.0180	0.0467	0.0193	0.0184	0.0145	0.0185	<-- {=ImpliedVolvarcov(B3:K3,B5)}
8	AAPL	0.0270	0.1011	0.0187	0.0191	0.0168	0.0437	0.0181	0.0172	0.0136	0.0173	
9	AMZN	0.0200	0.0187	0.0554	0.0141	0.0125	0.0324	0.0134	0.0128	0.0101	0.0128	
10	DIS	0.0204	0.0191	0.0141	0.0575	0.0127	0.0330	0.0136	0.0130	0.0103	0.0131	
11	CSCO	0.0180	0.0168	0.0125	0.0127	0.0449	0.0291	0.0121	0.0115	0.0091	0.0115	
12	JPM	0.0467	0.0437	0.0324	0.0330	0.0291	0.3023	0.0313	0.0298	0.0235	0.0299	
13	GOOG	0.0193	0.0181	0.0134	0.0136	0.0121	0.0313	0.0518	0.0123	0.0097	0.0124	
14	XOM	0.0184	0.0172	0.0128	0.0130	0.0115	0.0298	0.0123	0.0470	0.0093	0.0118	
15	PG	0.0145	0.0136	0.0101	0.0103	0.0091	0.0235	0.0097	0.0093	0.0293	0.0093	
16	BAC	0.0185	0.0173	0.0128	0.0131	0.0115	0.0299	0.0124	0.0118	0.0093	0.0474	

The programming of the **ImpliedVolVarCov** is similar to previous VBAs in this chapter:

```
Function ImpliedVolVarCov(volatilities As Range, corr As Double)
As Variant
    Dim i As Integer
    Dim j As Integer
    Dim numcols As Integer
    numcols = volatilities.Columns.Count
    numrows = numcols
    Dim matrix() As Double
    ReDim matrix(numcols - 1, numcols - 1)
    If Abs(corr) >= 1 Then GoTo Out
    For i = 1 To numcols
    For j = i To numcols
        If i = j Then
        matrix(i - 1, j -1) = volatilities(i) ^ 2
        Else
        matrix(i - 1, j - 1) = corr * volatilities(i) *
        volatilities(j)
        matrix(j - 1, i - 1) = matrix(i - 1, j - 1)
        End If
    Next j
    Next i
Out:
    If Abs(corr) >= 1 Then ImpliedVolVarCov = “ERROR” _
Else ImpliedVolVarCov = matrix
End Function
```

And the implementation in R looks like this:

```
# Input: implied volatility
impl_vol <- matrix(c(0.3399, 0.3179, 0.2354, 0.2398, 0.2119, 0.3498, 0.2276,
                     0.2168, 0.1712, 0.2178))

# Using constant correlation
const_corr <- 0.25

# Calculating covariance(ri,rj) where i<>j
impl_var_cov_mat <- impl_vol %*% t(impl_vol) * const_corr

# Calculating variance(ri) in the diagonal
diag(impl_var_cov_mat) <- impl_vol^2

# Name the matrix column and row headers
colnames(impl_var_cov_mat) <- names(assets_ret)
rownames(impl_var_cov_mat) <- names(assets_ret)
```

12.9 Which Method to Compute the Variance-Covariance Matrix?

This chapter gives five alternatives to computing the variance-covariance matrix:

- • The sample variance-covariance
- • The single-index model
- • The constant correlation approach
- • Shrinkage methods
- • Implied volatility-based variance-covariance matrices

How to we compare these alternatives? Which one should we choose? We could compare the technical outcomes of using each of these methods—for example, show the alternative values of the GMVP using different methods—but this largely misses the point.

The choice of how to compute the variance-covariance matrix is largely a question of how you view capital markets. If you strongly believe that the past predicts the future, then perhaps your choice should be to use the sample VarCov matrix. We prefer to get away from history, so our preference is to use an option-based volatility model with a changing correlation: In “normal” times we would use a “normal” correlation of between 0.2 and 0.3; in times of crisis, we would use a much higher correlation—say, a correlation between 0.5 and 0.6.

12.10 Summing Up

In this chapter we considered how to compute the variance-covariance matrix, which is central to all portfolio optimization. Starting with the standard sample variance-covariance matrix, we also showed how to compute several alternatives that have appeared in the literature as perhaps improving portfolio computations.

Exercises

1. Below you will find annual return data for six furniture companies for the years 2010–2020. Use these data to calculate the variance-covariance matrix of the returns.

	A	B	C	D	E	F	G	H
1	DATA FOR 6 FURNITURE COMPANIES							
2		A	B	C	D	E	F	
3	2010	36.67%	0.20%	41.94%	21.92%	26.13%	22.50%	
4	2011	122.82%	61.43%	195.09%	62.27%	73.38%	117.89%	
5	2012	14.44%	63.51%	-38.38%	-1.27%	45.15%	7.80%	
6	2013	21.39%	28.42%	1.30%	81.17%	24.27%	38.14%	
7	2014	45.36%	-7.44%	21.89%	19.83%	10.73%	54.48%	
8	2015	20.19%	48.27%	9.11%	-10.21%	-11.92%	26.82%	
9	2016	-8.94%	-11.28%	12.65%	13.77%	7.06%	-6.24%	
10	2017	27.02%	12.86%	12.08%	32.55%	-7.55%	123.03%	
11	2018	-11.64%	2.42%	-17.13%	-6.48%	1.31%	15.48%	
12	2019	20.29%	6.90%	3.62%	50.12%	-6.54%	19.92%	
13	2020	34.06%	22.21%	33.46%	64.40%	5.71%	62.76%	
14								
15	Beta	0.80	0.96	0.65	0.85	0.86	1.40	
16	Mean returns	29.24%	20.66%	25.02%	31.64%	15.34%	43.87%	<--=AVERAGE(G3:G13)

The remaining exercises refer to the data in the tab **Price data** on the exercise spreadsheet that you'll find on the auxiliary website accompanying this book. This tab gives three years of price data for six stocks and the Standard & Poor's 500 as a surrogate for the market.

2. Compute the returns of the data and the statistics for the each of the assets (mean return, variance and standard deviation of return, beta).
3. Compute the sample variance-covariance matrix and the correlation matrix for the six stocks.
4. Use the function **SIM** defined in the chapter to compute the single-index variance-covariance matrix.
5. Compute the GMVP using the sample variance-covariance matrix.
6. Compute the GMVP using the constant correlation covariance matrix.

---

1. 1. We return to the issue of prediction in Chapter 15, which discusses the Black-Litterman model of portfolio optimization.
2. 2. Our thanks to Amir Kirsh. This function was revised in 2012 by Benjamin Czaczkes and Simon Benninga, and in 2019 by Tal Mofkadi.
3. 3. J. P. Morgan (JPM) and Bank of America (BAC) are two financial institutions in the United States. Google (GOOG) and Amazon (AMZN) are both technology companies, though in somewhat different markets and with a business focus.
4. 4. Sharpe (1963).
5. 5. Three papers by Olivier Ledoit and Michael Wolf (2003, 2004a, 2004b) may offer some guidance.

13 Estimating Betas and the Security Market Line

13.1 Overview

The capital asset pricing model (CAPM) is one of the two most influential innovations in financial theory in the latter half of the twentieth century.[1] By integrating the portfolio decision with utility theory and the statistical behavior of asset prices, the formulators of the CAPM defined the paradigm that is now generally used for the analysis of stock prices and corporate valuations.

What does the CAPM actually say? What are its empirical implications? Roughly speaking, we can differentiate between two kinds of implications of the CAPM. First, the capital market line (CML) defines the *individual optimal portfolios* for investors interested in the mean and variance of their optimal portfolio. Second, given agreement between investors on the statistical properties of asset returns and on the importance of mean-variance optimization, the security market line (SML) defines the *risk-return* relation for *each individual asset*.

It is useful to differentiate between the case wherein a risk-free asset exists and the case wherein there is no risk-free asset.[2]

Case 1: A Risk-Free Asset Exists

Suppose a risk-free asset exists and has return r_f. We can differentiate between the individual optimization of investors and the general equilibrium implications of the CAPM.

- • *Individual optimization:* Assuming that investors optimize based on the expected return and standard deviation of their portfolio returns (in the jargon of finance—they have "mean-variance" ranking), the CAPM states that each individual investor's optimal portfolio falls on the CML:

$$E(r_p) = r_f + \sigma_p \frac{[E(r_x) - r_f]}{\sigma_x},$$

where portfolio x is a portfolio that maximizes $\frac{E(r_y) - r_f}{\sigma_y}$ for all feasible portfolios y. Proposition 1 of Chapter 11 shows that the weights x can be computed by

$x = \{x_1, x_2, \ldots, x_N\} = \frac{S^{-1}[E(r) - r_f]}{\Sigma S^{-1}[E(r) - r_f]}$, where S is the variance-covariance matrix of risky asset returns and $E(r) = [E(r_1), E(r_2), \ldots, E(r_N)]$ is the vector of expected asset returns.

- • *General equilibrium*: If all investors agree about the statistical assumptions of the model—the variance-covariance matrix S and the vector of expected asset returns $E(r)$—and if a risk-free asset exists, then individual asset returns are defined by the SML:

$$E(r_i) = r_f + \frac{Cov(r_i, r_M)}{Var(r_m)}[E(r_M) - r_f],$$

where M denotes the market portfolio, the value-weighted portfolio of all risky assets. The expression $\frac{Cov(r_i, r_M)}{Var(r_m)}$ is generally termed the asset's *beta*: $\beta_i = \frac{Cov(r_i, r_M)}{Var(r_m)} = \frac{\sigma_i}{\sigma_m} \cdot \rho_{i,m}$.

Case 2: No Risk-Free Asset Exists

If there is no risk-free asset, then the implications of the CAPM for both individual optimization and for general equilibrium are defined by Black's (1972) zero-beta model (Proposition 3 of Chapter 11):

- • *Individual optimization*: In the absence of a risk-free asset, individual optimal portfolios will fall along the *efficient frontier*. As shown in Proposition 2 of Chapter 9, this frontier is the upward-sloping portion of the mean-sigma combinations created by the convex combination of any two optimizing portfolios $x = \frac{S^{-1}[E(r) - c_1]}{\Sigma S^{-1}[E(r) - c_1]}$ and $y = \frac{S^{-1}[E(r) - c_2]}{\Sigma S^{-1}[E(r) - c_2]}$, where c_1 and c_2 are two arbitrary constants.
- • *General equilibrium*: In the absence of a risk-free asset, if all investors agree about the statistical assumptions of the model—the variance-covariance matrix S and the vector of expected asset returns $E(r)$—then individual asset returns are defined by the SML:

$$E(r_i) = E(r_z) + \frac{Cov(r_i, r_x)}{\sigma_x^2}[E(r_x) - E(r_z)],$$

where x is any efficient portfolio and z is a portfolio that has zero covariance with x (the so-called zero-beta portfolio).

The case of no risk-free asset is obviously weaker than the case of a risk-free asset. If there is a risk-free asset, the general equilibrium version of the CAPM says that all portfolios are situated on a single, agreed-upon line. If there is no risk-free asset, then all optimal portfolios are on the same frontier; but in this case, asset betas can differ, since there are many efficient portfolios x that fulfill the equation $E(r_i) = E(r_z) + \frac{Cov(r_i, r_x)}{\sigma_x^2}[E(r_x) - E(r_z)]$.

The CAPM as a Prescriptive and a Descriptive Tool

As you can see from the discussion above, the CAPM is both *prescriptive* and *descriptive*.

As a prescriptive tool, the CAPM tells a mean-variance investor how to choose the optimal portfolio. By finding a portfolio of the form $\frac{S^{-1}[E(r) - c_1]}{\Sigma S^{-1}[E(r) - c_1]}$, the investor can identify an optimal portfolio from the data set.

As a descriptive tool, the CAPM gives conditions under which we can generalize about the structure of expected returns in the market. Whether or not a risk-free asset exists, these conditions assume that investors agree on the statistical structure of asset returns—the variance-covariance matrix and the expected returns. In this case, all the returns are expected to lie on a security market line of the form $E(r_i) = r_f + \frac{Cov(r_i, r_M)}{\sigma_M^2}[E(r_M) - r_f]$ (if there is a risk-free asset) or of the form $E(r_i) = E(r_z) + \frac{Cov(r_i, r_x)}{\sigma_x^2}[E(r_x) - E(r_z)]$ if there is no risk-free asset.

About This Chapter

In this chapter we look at some typical capital market data and replicate a simple test of the descriptive part of the CAPM. This means that we have to calculate the betas for a set of assets, and we then have to determine the equation of the SML. The test in this chapter is the simplest possible test of the CAPM. There is an enormous literature in which the possible statistical and methodological pitfalls of CAPM tests are discussed.[3]

13.2 Testing the SML

Typical tests of the SML start with return data on a set of risky assets. The steps in the test are as follows:

- • Determine a candidate for the market portfolio M. In our example, we will use the Standard & Poor's 500 Index (SP500) as a candidate for M. This is a critical step: In principle, the "true" market portfolio should—as pointed out in Chapter 11—contain all the market's risky assets in proportion to their value. It is clearly impossible to calculate this theoretical market portfolio, and we must therefore make do with a surrogate. As you will see in the next two sections, the propositions of Chapter 11 can shed much light on how the choice of the market surrogate affects the r-squared of our regression test of the CAPM.

- • For each of the assets in question, determine the asset beta, $\beta_i = \dfrac{Cov(r_i, r_M)}{Var(r_m)}$. This is often called the *first-pass regression*.
- • Regress the mean returns of the assets on their respective betas. This is often called the *second-pass regression*:

$$\bar{r}_i = \gamma_0 + \gamma_1 \beta_i$$

If the CAPM in its descriptive format holds, then the second-pass regression should be the SML.[4]

We illustrate the tests of the CAPM with a simple numerical example that uses data for the 30 stocks in the Dow-Jones Industrials (DJ30). We start with the prices of the SP500 (symbol ^GSPC) and the DJ30 (some of the rows and columns are not shown):

	A	B	C	D	E	F	AA	AB	AC	AD	AE
1	PRICE DATA FOR THE DOW-JONES INDUSTRIAL STOCKS AND THE SP500 June 2014 - May 2019										
2	Company	SP500	Microsoft	Apple	Visa	Johnson & Johnson	3M	Caterpillar	Goldman Sachs	Walgreens Boots Alliance	Travelers
3	Date	GSPC	MSFT	AAPL	V	JNJ	MMM	CAT	GS	WBA	TRV
4	31-May-19	2,752.06	123.68	175.07	161.33	131.15	159.75	119.81	182.49	49.34	145.57
5	01-May-19	2,788.86	125.27	177.62	162.51	131.21	159.22	121.84	186.53	50.13	145.73
6	01-Apr-19	2,945.83	130.12	199.90	164.18	140.24	187.90	138.58	205.00	53.12	143.75
7	01-Mar-19	2,834.40	117.51	189.22	155.95	138.84	206.02	134.67	191.13	62.74	136.36
8	01-Feb-19	2,784.49	111.14	171.75	147.63	134.82	204.22	136.51	195.03	70.16	132.13
60	01-Oct-14	2,018.05	42.17	99.48	57.60	94.70	136.03	86.24	177.78	58.24	90.95
61	01-Sep-14	1,972.29	41.64	92.80	50.90	93.65	125.33	84.22	171.77	53.75	84.26
62	01-Aug-14	2,003.37	40.55	93.94	50.32	90.53	126.63	92.75	167.08	54.59	84.95
63	01-Jul-14	1,930.67	38.52	87.62	49.96	87.35	123.90	85.14	161.26	62.03	80.33
64	01-Jun-14	1,960.23	37.22	85.17	49.89	91.30	125.96	91.83	156.19	66.86	83.89
65											
66	Market cap (B)		947.74	805.51	352.57	348.21	92.08	68.52	66.76	45.11	38.13

We first transform these price data to returns:

	A	B	C	D	E	F	AB	AC	AD	AE	AF
1	RETURN DATA FOR THE DOW-JONES INDUSTRIAL STOCKS AND THE SP500 June 2014 - May 2019										
2	Company	SP500	Microsoft	Apple	Visa	Johnson & Johnson	Caterpillar	Goldman Sachs	Walgreens Boots Alliance	Travelers	
3	Symbol	GSPC	MSFT	AAPL	V	JNJ	CAT	GS	WBA	TRV	
4	Mean return	0.57%	2.00%	1.20%	1.98%	0.60%	0.44%	0.26%	-0.51%	0.92%	<--=AVERAGE(AE10:AE70)
5	Alpha	0.00%	1.33%	0.56%	1.42%	0.22%	-0.43%	-0.43%	-0.92%	0.35%	<--=INTERCEPT(AE10:AE69,$B$10:$B$69)
6	Beta	1.00	1.19	1.14	0.95	0.68	1.54	1.22	0.72	1.00	<--=SLOPE(AE10:AE69,$B$10:$B$69)
7	R-squared	1.00	0.44	0.27	0.51	0.34	0.45	0.34	0.11	0.49	<--=RSQ(AE10:AE69,$B$10:$B$69)
8											
9	Date										
10	31-May-19	-1.30%	-1.27%	-1.44%	-0.73%	-0.05%	-1.68%	-2.19%	-1.58%	-0.11%	
11	01-May-19	-5.48%	-3.80%	-11.82%	-1.02%	-6.65%	-12.87%	-9.44%	-5.60%	1.37%	
12	01-Apr-19	3.86%	10.20%	5.49%	5.14%	1.00%	2.86%	7.00%	-16.64%	5.29%	
13	01-Mar-19	1.79%	5.57%	8.69%	5.48%	2.94%	-1.36%	-2.02%	-11.18%	3.15%	
14	01-Feb-19	2.90%	7.02%	3.95%	9.27%	2.64%	3.73%	-0.66%	-1.49%	5.70%	
66	01-Oct-14	2.29%	1.26%	6.95%	12.36%	1.11%	2.37%	3.44%	8.02%	7.63%	
67	01-Sep-14	-1.56%	2.65%	-1.23%	1.15%	3.39%	-9.66%	2.77%	-1.54%	-0.82%	
68	01-Aug-14	3.70%	5.13%	6.97%	0.71%	3.57%	8.57%	3.55%	-12.76%	5.59%	
69	01-Jul-14	-1.50%	3.44%	2.83%	0.14%	-4.43%	-7.57%	3.19%	-7.51%	-4.33%	
70	01-Jun-14										

The First-Pass Regression

Row 4 gives each asset's average monthly return over the 60-month period. (To annualize these returns, we would multiply by 12.) Rows 5–7 report the results of the *first-pass regression*. For each asset i, we report the regression $r_{it} = \alpha_i + \beta_i r_{SP500,\ t}$. We use the Excel function **Slope** to compute the β of each asset and the functions **Intercept** and **Rsq** to compute the α and R^2 for each regression.

As a check, we also compute the α, β, and R^2 for the SP500 index (column B). Not surprisingly, $\alpha_{SP} = 0$, $\beta_{SP} = 1$, and $R^2 = 1$.

Using R to Perform the First-Pass Regression

As you probably know by now, R is a very powerful tool to perform statistical analysis. We mainly use the linear model (R command: "lm") to calculate the alpha, beta, and *r*-squared. This is how is should look:

```r
workdir <- readline( prompt = "working directory?") # insert working directory path
setwd(workdir)

# Compute Daily Returns
day_ret <- read.csv("Chapt11_ret_data.csv", row.names = 1)
index_ret <- day_ret[,"GSPC"]
assets_ret <- day_ret[,colnames(day_ret) != "GSPC"]

### 11.2 Testing the SML
## The First-Pass Regression
mean_ret <- colMeans(assets_ret)

# First-pass regression: calculating Alpha, Beta and R-sqr vs. GSPC
alpha <- sapply(as.list(assets_ret) , function(x){
                lm(formula = x ~ index_ret)$coefficients[1]})

beta <- sapply(as.list(assets_ret) , function(x){
               lm(formula = x ~ index_ret)$coefficients[2]})

r_squared <- sapply(as.list(assets_ret) , function(x){
                    summary(lm(formula = x ~ index_ret))$r.squared})
```

The Second-Pass Regression

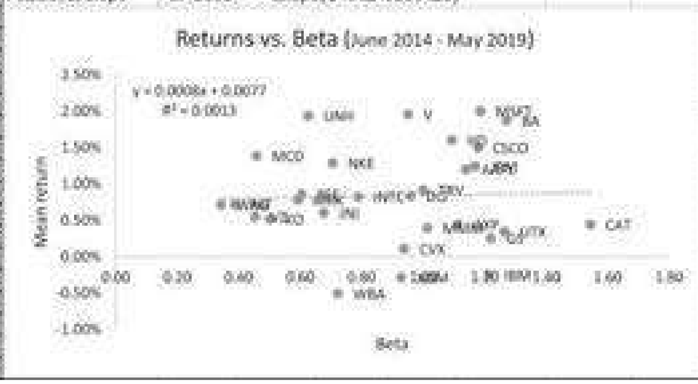
The SML postulates that the mean return of each security should be linearly related to its beta. Assuming that the historic data provide an accurate description of the distribution of future returns, we postulate that $E(R_i) = \alpha + \beta_i \Pi + \varepsilon_i$, where the definitions of α and Π depend on whether we are in Case 1 or Case 2 of section 11.1:

$$\alpha = \begin{cases} r_f & \text{Case 1: there exists } a \text{ risk-free asset} \\ E(r_z) & \text{Case 2: no risk-free asset; } z \text{ has zero} \\ & \text{correlation with efficient portfolio } y \end{cases}$$

$$\Pi = \begin{cases} E(r_M) - r_f & \text{Case 1} \\ E(r_y) - E(r_z) & \text{Case 2} \end{cases}$$

In the second step of our test of the CAPM, we examine this hypothesis by regressing the mean returns on the β's.

	A	B	C	D	E	F	AB	AC	AD	AE
1	**THE SECOND-PASS REGRESSION** June 2014 - May 2019									
2	Company	SP500	Microsoft	Apple	Visa	Johnson & Johnson	Caterpillar	Goldman Sachs	Walgreens Boots Alliance	Travelers
3	**Symbol**	**GSPC**	**MSFT**	**AAPL**	**V**	**JNJ**	**CAT**	**GS**	**WBA**	**TRV**
4	Mean return	0.57%	2.00%	1.20%	1.96%	0.60%	0.44%	0.26%	-0.51%	0.92%
5	Alpha	0.00%	1.33%	0.56%	1.42%	0.22%	-0.43%	-0.43%	-0.92%	0.35%
6	Beta	1.00	1.19	1.14	0.95	0.68	1.54	1.22	0.72	1.00
7	R-squared	1.00	0.44	0.27	0.51	0.34	0.45	0.34	0.11	0.49
8										
9	**Second-pass regression, regressing monthly returns on Beta**							**Null hypothesis**		
10	Intercept	0.768%	<--=INTERCEPT(C4:AE4,C6:AE6)					0.064%	Avg(rf) in the period	
11	Slope	0.08%	<--=SLOPE(C4:AE4,C6:AE6)					0.501%	<--=B4-AC10	
12	R-squared	0.0013	<--=RSQ(C4:AE4,C6:AE6)					1		
13										
14	t-statistic, intercept	1.96537	<--=tintercept(C4:AE4,C6:AE6)							
15	t-statistic, slope	0.18568	<--=tslope(C4:AE4,C6:AE6)							

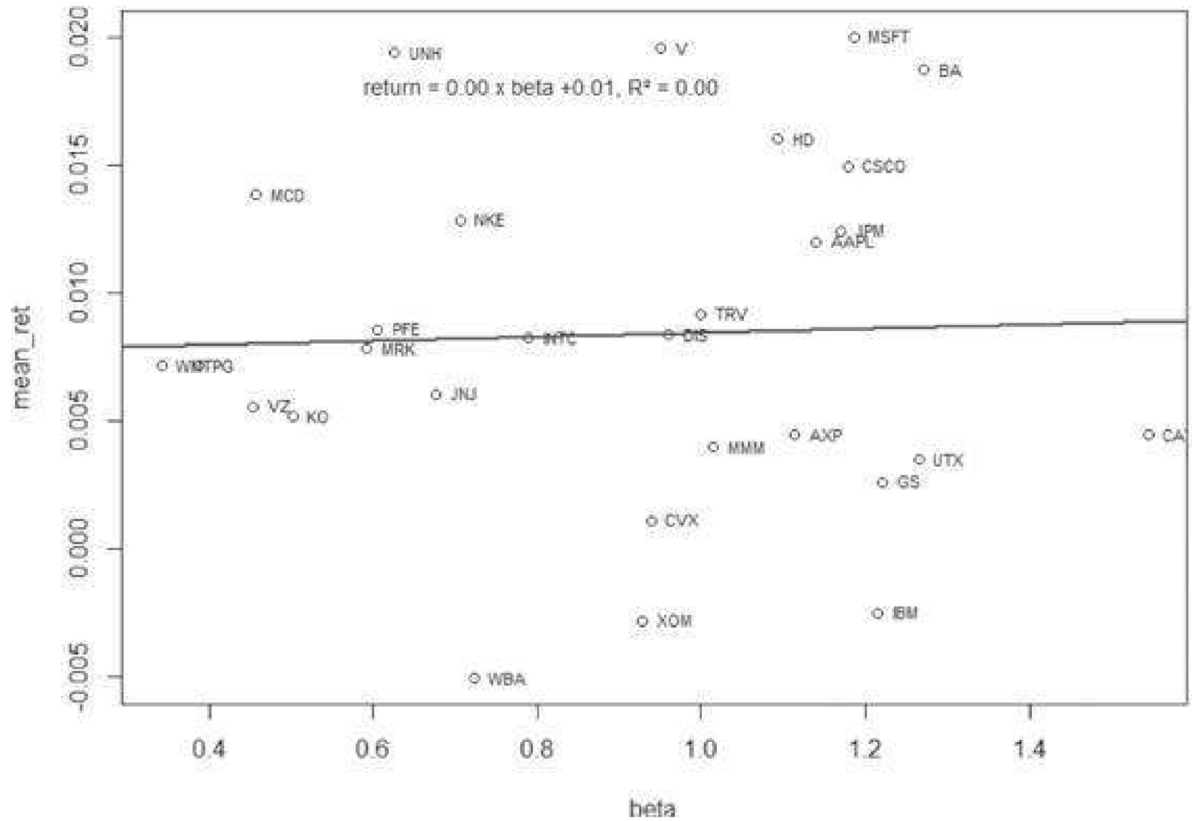


The results (cells B10:B12) are very disappointing. Our test yields the following SML:

$$Average(r_i) = \underbrace{0.0077}_{\substack{\uparrow \\ \gamma_0}} + \underbrace{0.0008}_{\substack{\uparrow \\ \gamma_1}}\beta_i,\ R^2 = 0.0013$$

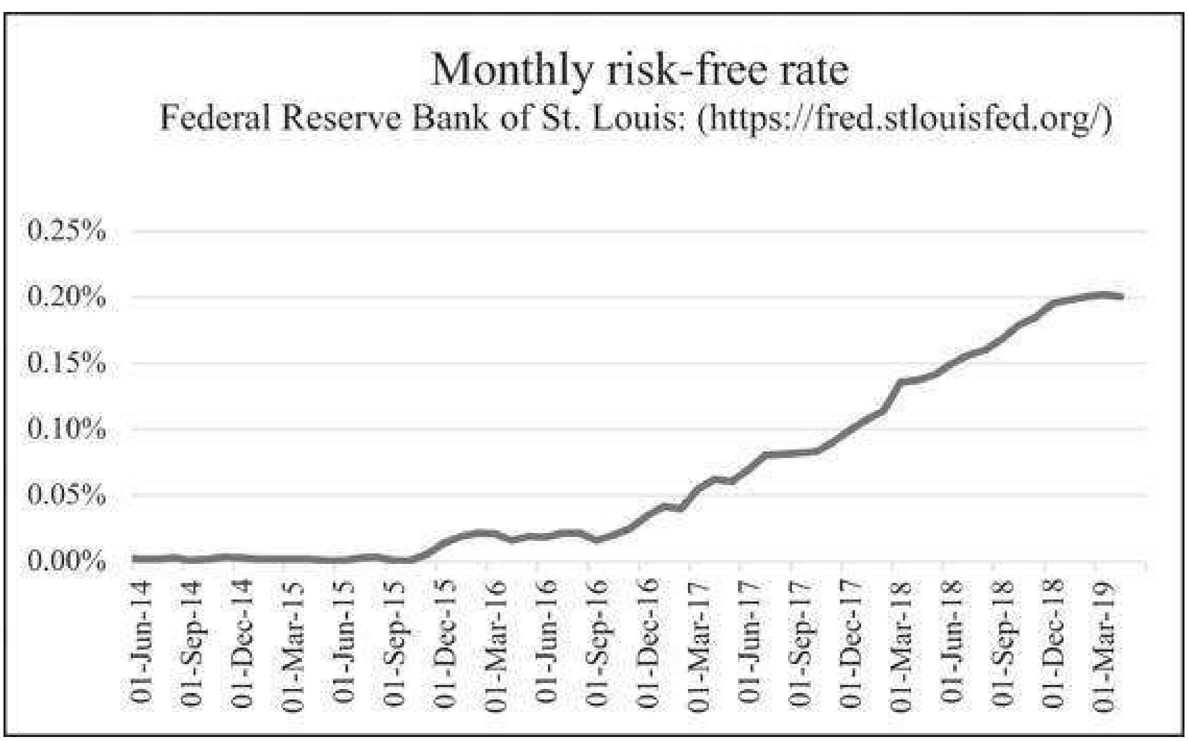
Implementing this in R should look like this:

```
## Second-Pass Regression
scnd_ps_reg <- lm(formula = mean_ret ~ beta)
summary(scnd_ps_reg)

# Plot - Returns vs. Beta
plot(beta, mean_ret)
text(beta, mean_ret, labels = names(mean_ret), cex= 0.7, pos=4)
abline(scnd_ps_reg, col="blue", lwd=2)
scnd_ps_reg_coef <- round(coef(scnd_ps_reg), 4)
rsquared <- round(summary(scnd_ps_reg)$r.squared, 4)
legend("topright", legend = sprintf("return = %3.2f x beta %+ 3.2f, R\u00b2 = %3.2f",
                                    scnd_ps_reg_coef[2], scnd_ps_reg_coef[1], rsquared),
                                    bty = "n", cex = 0.8)
```



There is nothing about these numbers that inspires confidence:

- • γ_0 should correspond to the risk-free rate over the period. The risk-free rate changed wildly over the 60 months surveyed while the average monthly risk-free interest rate was 0.064% (or almost 12 times lower than the value γ_0).



- • γ_1 should correspond to $E(r_M) - r_f$. The average monthly return of the SP500 over the period was 0.57% and the average monthly risk-free interest rate was 0.064%, so that γ_1 should be approximated by 0.501%.
- • The *t*-statistics for the intercept (cell B14) is borderline, but the *t*-statistics of the slope (cell B15) indicate that it is not statistically different from zero.[5]

Our test of the SML has failed. The CAPM may have prescriptive validity, but it does not describe our data.

Why Are the Results So Bad?

The experiment we did—checking the CAPM by plotting the SML—does not appear to have worked out very well. There does not appear to be much evidence in favor of the SML: Neither the R^2 of the regression nor the *t*-statistics give much evidence that there is a relation between expected return and portfolio β.

There are a number of reasons why these disappointing results may hold:

- • One reason is that perhaps the CAPM itself does not hold. This could be true for a variety of reasons:

    • Perhaps in the market short sales of assets are restricted. Our derivation of the CAPM (see Chapter 11 on efficient portfolios) assumes that there are no short-sale restrictions. Clearly this is an unrealistic assumption. The computation of efficient portfolios when short sales are restricted is considered in Chapter 11. In this case, however, there are no simple relations (such as those proved in Chapter 11) between the returns of assets and their betas. In particular, if short sales are restricted, there is no reason to expect the SML to hold.

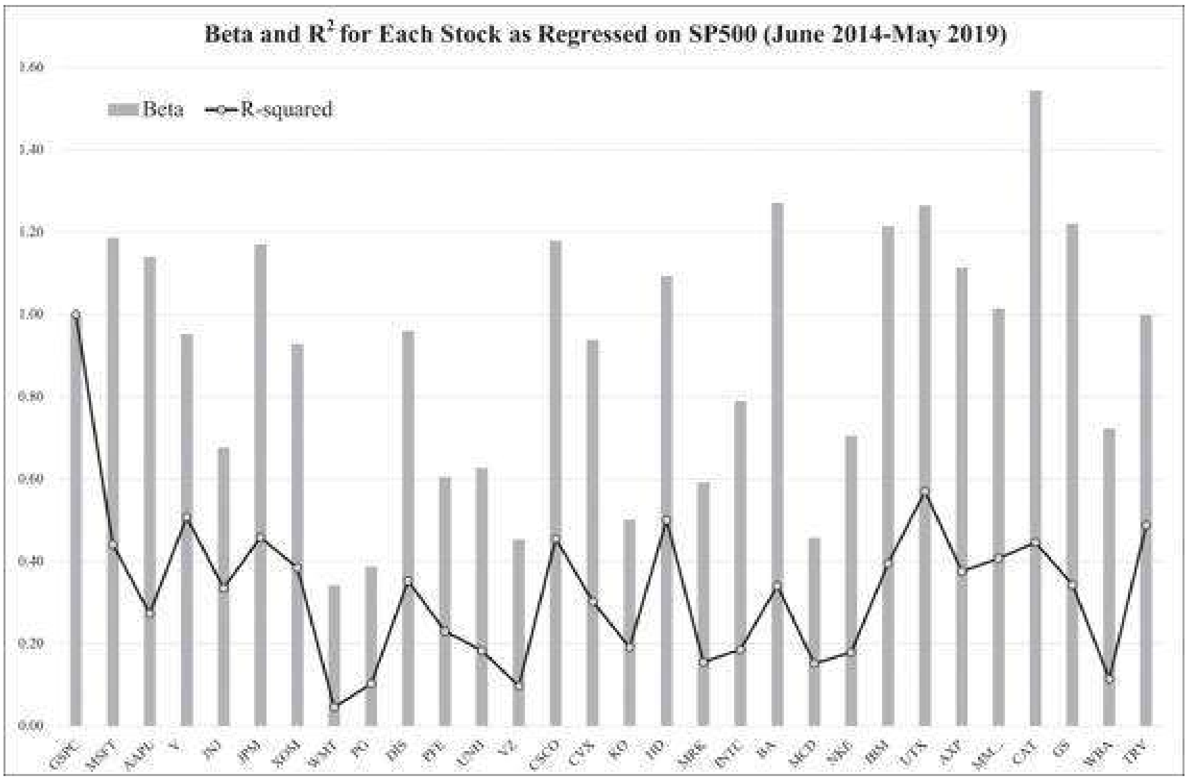
    • Perhaps individuals do not have homogeneous probability assessments, or perhaps they do not have the same expectations of asset returns, variances, and covariances.

- • Perhaps the CAPM holds only for portfolios and not for single assets.
- • Perhaps our set of assets isn't large enough. After all, the CAPM talks about *all risky assets*, whereas we have chosen—for illustrative purposes—to do our test on a very small subset of these assets. The literature on CAPM testing records tests in which the set of risky assets has been expanded to include other risky assets like bonds, real estate, and even nondiversifiable assets such as human capital.
- • Perhaps the "market portfolio" isn't efficient. This possibility is suggested by the mathematics of Chapter 11 on efficient portfolios, and it is this suggestion that we further explore in the next section.

- • Perhaps the CAPM holds only if the market returns are positive (in the period surveyed they were, on average, negative).

13.3 Did We Learn Something?

The results of our exercise in section 13.2 are quite disappointing. Did we learn anything positive from this exercise? Absolutely. For example, the regression model does a decent job of describing individual asset returns in relation to the SP500:



On average, the SP500 describes about 31% of the variability of the DJ30 stocks, which have an average beta of 0.9:

	A	B	C	D	E	F	G	AC	AD
1	**OUR SML EXERCISE: WHAT DID WE LEARN?**								
2	Average alpha	0.33%	<--=AVERAGE('Second Pass regression'!C5:AE5)						
3	Average beta	0.90	<--=AVERAGE('Second Pass regression'!C6:AE6)						
4	Average r-squared	0.3112	<--=AVERAGE('Second Pass regression'!C7:AE7)						
5									
6	**t-statistics for intercept and slope**								
7		MSFT	AAPL	V	JNJ	JPM	XOM	WBA	TRV
8	t-stat for intercept	2.2512	0.6750	3.4182	0.5226	1.0307	-1.5514	-1.0294	0.7799
9	t-stat for slope	6.7770	4.6709	7.7367	5.4106	7.0114	6.0193	2.7437	7.4487
10									
11	**Average t-statistics**								
12	For intercept (absolute)	1.1288	<--{=AVERAGE(ABS(B8:AD8))}						
13	For slope	5.1721	<--=AVERAGE(B9:AD9)						

Cell B4 above computes the average R^2. On average, the first-pass regressions are significant. The average R^2 of 31% that we got for our first-pass regressions of the basic SML is actually a respectable number in finance. Students—influenced by overenthusiastic statistics instructors and an overly linear view of the world—often feel that the R^2 of any convincing regression should be at least 90%. Finance does not appear to be a highly linear profession: A good rule of thumb is that any financial regression that gives an R^2 greater than 80% is possibly mis-specified and misleading.[6]

Another way to look at the significance of our results is to compute the t-statistics for the intercept and slope of the first-pass regressions (rows 8 and 9 above). While the intercepts are not significantly different from zero (since their t-statistic is less than |2|), the slopes are very significant.

An Excel Note: Computing the Absolute Value of an Array of Numbers

In the computations above we use a neat Excel trick related to array functions (see Chapter 31). By using **Abs** as an array function (that is, by entering the function using [Ctrl] + [Shift] + [Enter]), we can compute the average of the absolute values of a vector of numbers. A simple example is shown below:

	A	B	C	D	E	F	G
1	USING ABS FUNCTION IN ARRAY The Excel "Abs" function computes the absolute value If we use it as an array function, it can be applied to a range of numbers						
2							
3	Numbers	1	-1	2	-2	0	
4							
5	Average number	0.0000	<--=AVERAGE(B3:F3)				
6	Average absolute number	1.2000	<--{=AVERAGE(ABS(B3:F3))}				
7	The above, but not as array function	1.0000	<--=AVERAGE(ABS(B3:F3))				

Notice cell B7: Using the same function as a regular function does not produce the correct answer (it is considering only the first cell in the array—cell B3).

13.4 The Non-efficiency of the "Market Portfolio"

When we calculated the SML in section 13.2, we regressed the mean return of each asset on the returns of the market portfolio, represented by the SP500. The propositions of Chapter 11 on efficient portfolios suggest that our failure to find adequate results may stem from the fact that the SP500 portfolio is not efficient relative to the set of the assets that we have chosen. Proposition 3 of Chapter 11 states that if we had chosen to regress our asset returns on a portfolio that is efficient with respect to the asset set itself, we would get an r-squared of 100%. Proposition 4 of Chapter 11 shows that if we get an r-squared of 100% then the portfolio on which we regress the asset returns is necessarily efficient with respect to the set of assets. In this section we give a numerical illustration of these propositions.

In the spreadsheet below we create an "efficient portfolio" in column B. This portfolio (its construction is described in the next subsection) is efficient with respect to the Dow-Jones 30. As you can see in cells A10:B12, when we perform the second-pass regression—regressing the individual average returns of the assets on their betas computed with respect to the efficient portfolio—the results are perfect. The resulting regression has an intercept of 0.064% and a slope of 9.023%. Most important—it has an R^2 of 100%.

	A	B	C	D	E	AC	AD	AE
1	**SECOND-PASS REGRESSION ON THE EFFICIENT PORTFOLIO** June 2014 - May 2019							
2	**Date**	**Efficient portfolio**	**MSFT**	**AAPL**	**V**	**GS**	**WBA**	**TRV**
3								
4	Average return	9.09%	2.00%	1.20%	1.96%	0.26%	-0.51%	0.92%
5	Beta	1.00	0.21	0.13	0.21	0.02	-0.06	0.09
6	Alpha	0.00%	0.05%	0.06%	0.05%	0.06%	0.07%	0.06%
7	R-squared	1.0000	0.0902	0.0208	0.1539	0.0007	0.0055	0.0273
8								
9	**SML--regressing the average returns on the betas**							
10	Intercept	0.064%	<-- =INTERCEPT(C4:AE4,C5:AE5)					
11	Slope	9.023%	<-- =SLOPE(C4:AE4,C5:AE5)					
12	R-squared	1.0000	<-- =RSQ(C4:AE4,C5:AE5)					
13								
14	31-May-19	-2.19%	-1.27%	-1.44%	-0.73%	-2.19%	-1.58%	-0.11%
15	01-May-19	8.86%	-3.80%	-11.82%	-1.02%	-9.44%	-5.80%	1.37%
16	01-Apr-19	19.45%	10.20%	5.49%	5.14%	7.00%	-16.64%	5.28%
17	01-Mar-19	8.37%	5.57%	9.69%	5.48%	-2.02%	-11.18%	3.15%
69	01-Nov-14	8.55%	1.82%	9.64%	6.71%	-0.84%	6.61%	3.56%
70	01-Oct-14	10.03%	1.26%	6.95%	12.36%	3.44%	8.02%	7.63%
71	01-Sep-14	9.88%	2.65%	-1.23%	1.16%	2.77%	-1.54%	-0.82%
72	01-Aug-14	26.51%	5.13%	6.97%	0.71%	3.55%	-12.78%	5.59%
73	01-Jul-14	15.62%	3.44%	2.83%	0.14%	3.19%	-7.51%	-4.33%

The Mysterious Portfolio Is Efficient

The propositions of Chapter 11 leave us with only one conclusion: The “efficient portfolio” must be efficient with respect to the DJ30. And so it is. In the spreadsheet below we show the construction of this efficient portfolio, which follows the propositions of Chapter 11.

- • We first construct the variance-covariance matrix S using the function **Varcovar** defined in Chapter 12.
- • We then compute the efficient portfolio by solving $\frac{S^{-1}[E(r)-c]}{\Sigma S^{-1}[E(r)-c]}$. In the spreadsheet below we use c = 0.064%, which then turns out to be the intercept of the second-pass regression.

	A	B	C	D	E	F	G	AC	AD	AE
1	**THE EX-POST EFFICIENT PORTFOLIO FOR THE DOW-JONES STOCKS** July 2001 - July 2006									
2	**Date**	**GSPC**	**MSFT**	**AAPL**	**V**	**JNJ**	**JPM**	**GS**	**WBA**	**TRV**
3										
4	Average return	0.57%	2.00%	1.20%	1.96%	0.60%	1.24%	0.26%	-0.51%	0.92%
5	Beta	1.00	1.19	1.14	0.86	0.68	1.17	1.22	0.72	1.00
6	Alpha	0	1.33%	0.56%	1.42%	0.22%	0.58%	-0.43%	-0.92%	0.36%
7	R-squared	1	0.4419	0.2733	0.5079	0.3354	0.4587	0.3439	0.1149	0.4889
8										
9	31-May-19	-1.3%	-1.3%	-1.4%	-0.7%	0.0%	-1.0%	-2.2%	-1.6%	-0.1%
10	01-May-19	-6.5%	-3.8%	-11.8%	-1.0%	-6.7%	-7.3%	-9.4%	-6.8%	1.4%
11	01-Apr-19	3.9%	10.2%	5.5%	5.1%	1.0%	13.7%	7.0%	-16.6%	5.5%
12	01-Mar-19	1.8%	5.6%	9.7%	5.5%	2.9%	-3.0%	-2.0%	-11.2%	3.1%
13	01-Feb-19	2.9%	7.0%	4.0%	9.3%	2.8%	1.6%	-0.7%	-1.5%	5.7%
14	01-Jan-19	7.6%	2.8%	5.4%	2.3%	3.1%	5.8%	17.0%	5.6%	5.3%
15	01-Dec-18	-9.6%	-8.3%	-12.1%	-7.0%	-12.3%	-13.0%	-12.8%	-20.9%	-8.5%
67	01-Aug-14	3.7%	5.1%	7.0%	0.7%	3.8%	3.7%	3.5%	-12.8%	5.6%
68	01-Jul-14	-1.5%	3.4%	2.8%	0.1%	-4.4%	0.1%	3.2%	-7.5%	-4.3%
69										
70										
71	**The cells below compute the variance-covariance matrix for the DJ30 by using the formula {=varcovar(C9:AE68)}**									
72		MSFT	AAPL	V	JNJ	JPM	XOM	WBA	TRV	
73	MSFT	0.0036	0.0019	0.0017	0.0009	0.0017	0.0011	0.0003	0.0013	
74	AAPL	0.0019	0.0054	0.0017	0.0007	0.0008	0.0004	0.0002	0.0010	
75	V	0.0017	0.0017	0.0020	0.0006	0.0010	0.0006	0.0009	0.0012	
76	JNJ	0.0009	0.0007	0.0006	0.0015	0.0008	0.0010	0.0010	0.0010	
99	GS	0.0015	0.0017	0.0008	0.0006	0.0033	0.0011	0.0013	0.0013	
100	WBA	0.0003	0.0002	0.0009	0.0010	0.0012	0.0006	0.0051	0.0012	
101	TRV	0.0013	0.0010	0.0012	0.0010	0.0013	0.0011	0.0012	0.0023	
102										
103	**Finding an efficient portfolio**									
104	Constant	0.064%								
105										
106		**MSFT**	**AAPL**	**V**	**JNJ**	**JPM**	**XOM**	**WBA**	**TRV**	
107	Weight	14.1%	-30.3%	70.2%	73.0%	188.2%	-16.9%	-73.2%	-32.7%	
108		{=TRANSPOSE(MMULT(MINVERSE(B73:AD101),TRANSPOSE(C4:AE4)-B104)/SUM(MMULT(MINVERSE(B73:AD101),TRANSPOSE(C4:AE4)-B104)))}								
109										
110										
111	**Sum**	**100%**	<--=SUM(B107:AD107)							
112	Largest short	-112.4%	<--=MIN(B107:AD107)							
113	Largest long	188.2%	<--=MAX(B107:AD107)							

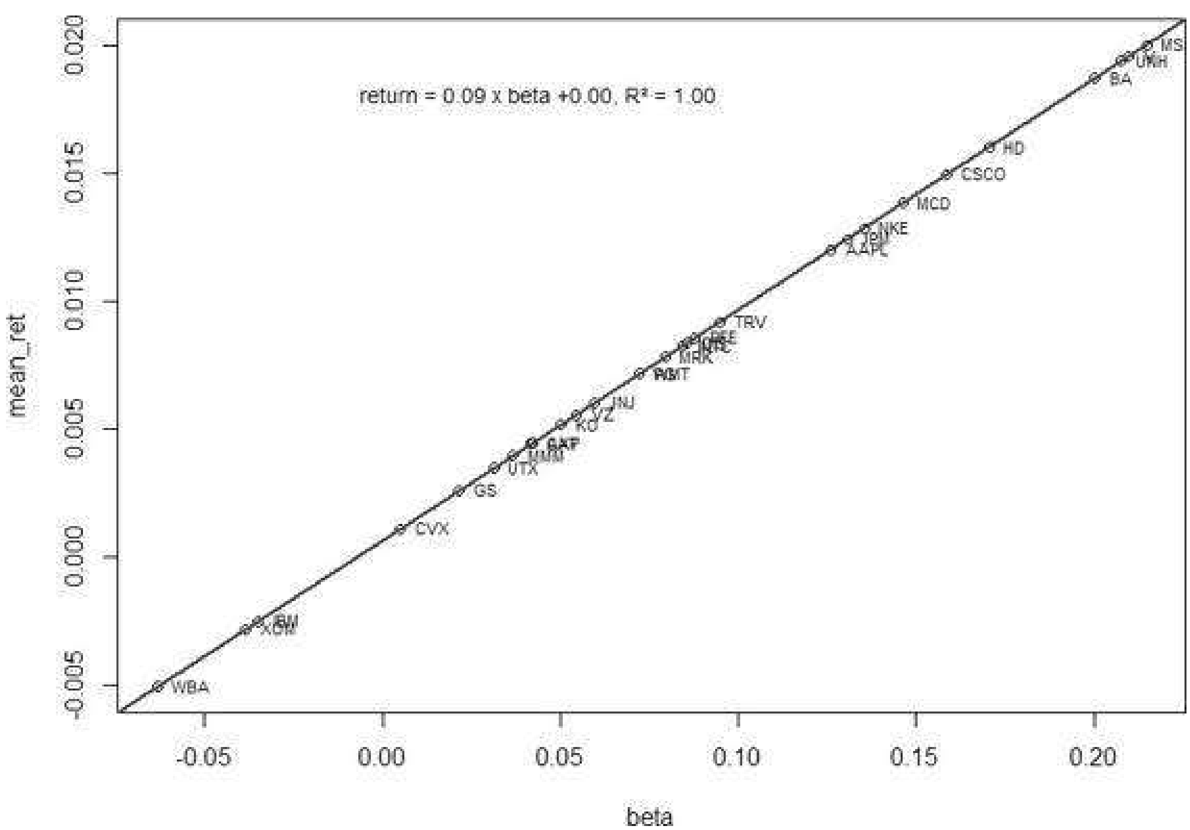
The "efficient portfolio" is unique for the constant c. A different c will give us a different efficient portfolio and a perfect new second-pass regression.

Note also that even though the R^2 of the second-pass regression is 100% (since the portfolio is efficient), the R^2's of the individual first-pass regressions are far from notable.

Performing the First-Pass Regression and the Second-Pass Regression on the Efficient Portfolio with R

The R code of finding the efficient portfolio and then performing the first-pass and second-pass regressions should look like this:

```r
## 13.4  The Non-Efficiency of the "Market Portfolio"
# Function: Calculate Envelope Portfolio proportions in one step (FROM CHAPTER 8)
envelope_portfolio <- function(var_cov_mat, assets_mean_ret, constant){
  Z <- solve(var_cov_mat) %*% (assets_mean_ret - constant) # Z
  Z_sum <- sum(Z) # Sum of Z's
  return( as.vector(Z/Z_sum) ) # Efficient Portfolio proportions
  }

# Inputs:
var_cov_mat <- cov(assets_ret)
mean_ret <- colMeans(assets_ret)
constant <- 0.000643

# Solve
eff_port_prop <- envelope_portfolio(var_cov_mat, mean_ret, constant)
eff_port_ret <- as.matrix(assets_ret) %*% eff_port_prop

# The First-Pass Regression based on the Efficient Portfolio
alpha <- sapply(as.list(assets_ret) , function(x){
  lm(formula = x ~ eff_port_ret)$coefficients[1]})

beta <- sapply(as.list(assets_ret) , function(x){
  lm(formula = x ~ eff_port_ret)$coefficients[2]})

r_squared <- sapply(as.list(assets_ret) , function(x){
  summary(lm(formula = x ~ eff_port_ret))$r.squared})

# T-statistic for intercept and slope
t_stat <- sapply(as.list(assets_ret) , function(x){
  summary(lm(formula = x ~ eff_port_ret))$coefficients[5:6]})
row.names(t_stat) <- c("t_intercept", "t_slope")

# The Second-Pass Regression on the Efficient Portfolio
scnd_ps_reg <- lm(formula = mean_ret ~ beta)
summary(scnd_ps_reg)

# Plot - Returns vs. Beta
plot(beta, mean_ret)
text(beta, mean_ret, labels = names(mean_ret), cex= 0.7, pos=4)
abline(scnd_ps_reg, col="blue", lwd=2)
scnd_ps_reg_coef = round(coef(scnd_ps_reg), 4)
rsquared <- round(summary(scnd_ps_reg)$r.squared, 4)
legend('topright', legend = sprintf("return = %3.2f x beta %+3.2f, R\u00b2 = %3.2f",
                                    scnd_ps_reg_coef[2], scnd_ps_reg_coef[1], rsquared),
                                    bty = 'n', cex = 0.8)
```



13.5 So What's the Real Market Portfolio? How Can We Test the CAPM?

A little reflection will reveal that although the "efficient" portfolio of the previous section may be efficient with respect to the 30 stocks of the Dow-Jones, it could not be the *true market portfolio*, even if the DJ30 stocks represented the whole universe of risky securities. This is because many of the stocks appear in the "efficient portfolio" with negative weights. Surely a minimal characteristic of the equilibrium market portfolio must be that all shares appear in it with *positive proportions*.

Roll (1977, 1978) suggests that the only test of the CAPM is to answer the question: *Is the true market portfolio mean-variance efficient?* If the answer to this question is "yes," then it follows from Proposition 3 of Chapter 11 that a linear relation holds between the mean of each portfolio and its β. In our example, we can shed some light on this question by building a table of the asset proportions of portfolios on the efficient frontier.

In the table below we give some evidence that all efficient portfolios for the DJ30 contain significant short positions. Using the wonders of Excel's **Data-Table**, we compute the largest short and long positions for a series of efficient portfolios, each defined by its own constant c. All of these portfolios contain large short positions (and, as you can see, also large long positions):

	A	B	C	D	E	F	G
115	Data-table: computing the largest short and long position for a given constant c						
116	Constant c	Largest short	Largest long				
117		-112%	188%	<--Data table header: =B113			
118	0.00%	-101%	169%				
119	0.03%	-106%	178%				
120	0.06%	-112%	187%				
121	0.09%	-118%	197%				
122	0.12%	-124%	208%				
123	0.15%	-132%	221%				
124	0.18%	-140%	236%				
125	0.21%	-149%	252%				
126	0.24%	-161%	271%				
127	0.27%	-173%	293%				
128	0.30%	-189%	320%				
129	0.33%	-207%	351%				
130	0.36%	-232%	389%				
131	0.39%	-265%	437%				
132	0.42%	-307%	498%				
133	0.45%	-363%	580%				
134	0.48%	-441%	693%				

Our depressing conclusion: If the data for the DJ30 and the SP500 are representative, the CAPM as a descriptive theory of capital markets appears not to work.[7]

13.6 Conclusion: Does the CAPM Have Any Uses?

Is the game lost? Do we have to give up on the CAPM? Not totally:

- • First of all, it could be that the mean returns are approximately described by their regression on a market portfolio. In this alternative description of the CAPM, we claim (with some justification) that the β of an asset (which measures the dependence of the asset's returns on the market returns) is an important measure of the asset's risk.

- • Second, the CAPM might be a good normative description of how to choose portfolios. As we showed in the appendix of Chapter 3, larger diversified portfolios are quite well described by their betas so that the average beta of a well-diversified portfolio may be a reasonable description of the portfolio's risk.

Exercises

1. In a well-known paper, Roll (1978) discusses tests of the SML in a four-asset context:

Variance-covariance matrix					Returns
0.10	0.02	0.04	0.05		0.06
0.02	0.20	0.04	0.01		0.07
0.04	0.04	0.40	0.10		0.08
0.05	0.01	0.10	0.60		0.09

1. Derive two efficient portfolios in this four-asset model and draw a graph of the efficient frontier.
2. Show that the following four portfolios are efficient by proving that each is a convex combination of the two portfolios you derived in part a above:

Security 1	0.59600	0.40700	-0.04400	-0.49600
Security 2	0.27621	0.31909	0.42140	0.52395
Security 3	0.07695	0.13992	0.29017	0.44076
Security 4	0.05083	0.13399	0.33242	0.53129

3. Suppose that the market portfolio is composed of equal proportions of each asset—that is, the market portfolio has proportions $(0.25, 0.25, 0.25, 0.25)$. Calculate the resulting SML. Is the portfolio $(0.25, 0.25, 0.25, 0.25)$ efficient?
4. Repeat this exercise but substitute one of the four portfolios of part b above as the candidate for the market portfolio.

The remaining questions relate to a data set for 10 stocks. The data are given on the exercise file with this chapter.

	A	B	C	D	E	F	G	H	I	J	K	L
1		**PRICE DATA: 10 STOCKS AND SP500, 2015-2020** **SP500 represented by Vanguard's Index 500 fund (includes dividends)**										
2		1	2	3	4	5	6	7	8	9	10	11
3		Apple	Google	Amazon	Seagate	Comcast	Merck	Johnson-Johnson	General Electric	Hewlett Packard	Goldman Sachs	SP500
4												
5	**Date**	**AAPL**	**GOOG**	**AMZN**	**STX**	**CMCSA**	**MRK**	**JNJ**	**GE**	**HPQ**	**GS**	**GSPC**
6	01-Jan-20	308.78	1,434.23	2,008.72	56.08	42.99	84.74	147.93	12.44	21.12	236.31	3,225.52
7	01-Dec-19	292.95	1,337.02	1,847.84	57.92	44.76	89.59	144.95	11.14	20.18	228.53	3,230.78
8	01-Nov-19	265.82	1,304.96	1,800.80	58.10	43.94	85.88	135.66	11.25	19.72	218.77	3,140.98
9	01-Oct-19	247.43	1,260.11	1,776.66	56.49	44.40	85.36	130.30	9.96	17.06	210.89	3,037.56
10	01-Sep-19	222.77	1,219.00	1,735.91	51.74	44.66	82.37	127.68	8.91	18.43	204.82	2,976.74
11	01-Aug-19	206.84	1,188.10	1,776.29	48.29	43.85	84.62	125.73	8.23	17.81	200.28	2,926.46
12	01-Jul-19	211.10	1,216.68	1,866.78	44.54	42.56	81.21	127.55	10.42	20.49	216.21	2,980.38
13	01-Jun-19	196.12	1,080.91	1,893.63	44.68	41.68	81.51	136.42	10.46	20.08	193.50	2,941.76
14	01-May-19	172.81	1,103.63	1,775.07	39.68	40.42	77.00	127.59	9.40	18.05	178.43	2,752.06

2. Fill in the template file below.

	A	B	C	D	E	F	G	H	I	J	K	L
1	RETURN DATA: 10 STOCKS AND SP500											
2		1	2	3	4	5	6	7	8	9	10	11
3		Apple	Google	Amazon	Seagate	Comcast	Merck	Johnson-Johnson	General Electric	Hewlett Packard	Goldman Sachs	SP500
4	Monthly statistics											
5	Mean											
6	Variance											
7	Sigma											
8												
9	Annual statistics											
10	Mean											
11	Variance											
12	Sigma											
13												
14	Regressing individual assets on the SP500											
15	Alpha											
16	Beta											
17	Rsq											
18	T-test, intercept											
19	T-test, slope											
20												
21												
22	RETURN DATA											
23	Date	AAPL	GOOG	AMZN	STX	CMCSA	MRK	JNJ	GE	HPQ	GS	SP500
24	1/Jan/20											
25	1/Dec/19											
26	1/Nov/19											
27	1/Oct/19											
28	1/Sep/19											
29	1/Aug/19											
30	1/Jul/19											
31	1/Jun/19											
81	1/Apr/15											
82	1/Mar/15											
83	1/Feb/15											

3. Perform the second-pass regression: Regress the monthly average returns on the betas of the assets. Does this confirm that the SP500 is efficient?
4. Compute the variance-covariance matrix for the 10 stocks. Using the monthly average returns and a monthly risk-free interest rate of 0.20%, compute an efficient portfolio. Here is the template:

	A	B	C	D	E	F	G	H	I	J	K	L	M
1	COMPUTING AN EFFICIENT PORTFOLIO OF THE 10 STOCKS												
2													
3						Variance-covariance matrix							
4		AAPL	GOOG	WFM	STX	CMCSA	MRK	JNJ	GE	HPQ	GS		Average returns
5	AAPL												
6	GOOG												
7	WFM												
8	STX												
9	CMCSA												
10	MRK												
11	JNJ												
12	GE												
13	HPQ												
14	GS												
15													
16													
17	Risk-free	0.20%											
18													
19		Efficient portfolio											
20	AAPL												
21	GOOG												
22	WFM												
23	STX												
24	CMCSA												
25	MRK												
26	JNJ												
27	GE												
28	HPQ												
29	GS												

5. Using the efficient portfolio instead of the SP500:
    1. Compute the monthly returns on the efficient portfolio.
    2. Regress the average monthly returns of the stocks on their betas with respect to the efficient portfolio.
    3. Explain your results considering Propositions 3 and 4 from Chapter 11.

1. 1. The other remarkable innovation is option pricing theory, which is discussed in Chapters 16–20. These two innovations together have accounted for a number of Nobel Prizes in Economics: Harry Markowitz (1990), William Sharpe (1990), Myron Scholes (1997), and Robert Merton (1997). But for their untimely demise, others associated with these theories—Jan Mossin (1936–1987), Fischer Black (1938–1995)—would doubtless also have received the Nobel.
2. 2. The existence (or nonexistence) of a risk-free asset is closely related to the investment horizon. Assets that are risk-free over a short term may not be riskless over a longer term.
3. 3. Good places to begin are Elton et al. (2009) and Bodie, Kane, and Marcus (2011). For further references, see the Selected References section at the end of this book. Our personal expositional favorite is Roll (1978).
4. 4. This is a direct consequence of Propositions 3 and 4 of Chapter 11.
5. 5. The functions **tintercept** and **tslope** were created by the author. They are attached to the spreadsheet for this chapter and are discussed in Chapter 3.
6. 6. An exception to this useful rule relates to diversified portfolios—here the R^2 increases dramatically.
7. 7. All is not lost! In Chapter 15 we examine the Black-Litterman model, which is a more positivist approach to portfolio choice.

14 Event Studies[*]

14.1 Overview

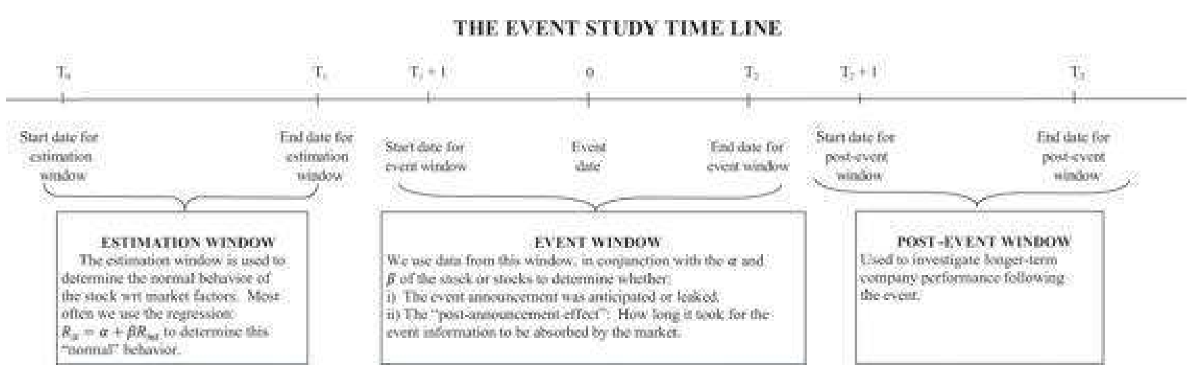
Event studies are some of the most powerful and widely used applications of the capital asset pricing model (CAPM) discussed in Chapters 10–15 of *Financial Modeling*. An event study is an attempt to determine whether a particular event in the capital market or in the life of a company affected a company's stock market performance. The event study methodology aims to separate company-specific events from market-specific and industry-specific events. The event study methodology has often been used in litigation processes and as evidence for or against market efficiency.

An event study aims to determine if an event or announcement caused an abnormal movement in a company's stock price. The *abnormal returns* (AR) are calculated as the difference between a stock's actual return and its expected return, where the stock's expected return is typically measured using *the market model*, which relies only on a stock's market index to estimate its expected return.[1] Using the market model, we can measure the correlation between an individual stock's returns and its corresponding market returns. In some cases, we sum the abnormal returns to arrive at the *cumulative abnormal return* (CAR), which measures the total impact of an event through a particular time period, also called the event window.

14.2 Outline of an Event Study

In this section we outline the methodology of an event study. In succeeding sections we apply the methodology to a number of different cases.

An event study is composed of three time frames: the *estimation window* (sometimes referred to as the control period), the e*vent window*, and the *post-event window*. The following chart illustrates these time frames.



The time line illustrates the timing sequence of an event. The length of the estimation window is represented as T_0 to T_1. The event occurs at time 0 and the event window is represented as $T_1 + 1$ to T_2. The length of the post-event window is represented as $T_2 + 1$ to T_3.

An event is defined as a point in time when a company makes an announcement or when a significant market event occurs. For example, if we are studying the impact of mergers and acquisitions on the stock market, the announcement date is normally the point of interest. If we are examining how the market reacts to earnings restatements, the event window begins on the event date when a company announces its restatement. In practice, it is common to expand the event window to two trading days—the event date and the following trading day. This is done to capture the market movement if the event was announced immediately before the market closed or after market closing.

The *event window* often starts a few trading days before the actual event day. This allows us to investigate pre-event leakage of information. The length of the event window is centered on the announcement and is normally from three to 10 days, though exceptions are not rare.

The *estimation window* is also used to determine the normal behavior of a stock's return with respect to a market or industry index. The estimation of the stock's return in the estimation window requires us to define a model of "normal" behavior: Most often we use a regression model for this purpose.[2]

The usual length of the estimation window is three years, with weekly observations of stock returns, while some use 252 trading days (or one calendar year). If you don't have this many observations in the sample, you need to determine whether the number of observations you do have is sufficient to produce robust results. As a guideline, you should have a minimum of 126 observations; if you have less than 126 observations in the estimation window it is possible that the parameters of the market model will not indicate the true stock price movements and thus the relationship between the stock returns and the market returns. The estimation window that you select is supposedly a period that was free of any problems (i.e., a period that reflects the stock's normal price movements).[3]

The *post-event window* is most often used to investigate the performance of a company following announcements such as a major acquisition or an initial public offering (IPO). The *post-event window* allows us to measure the longer-term impact of the event. The post-event window can be as short as few days and as long as several years, depending on the event.

Measuring the Stock's Behavior in the Estimation Window and the Event Window

As its name implies, the *estimation window* is used to estimate a model of the stock's returns under "normal" circumstances. The most common model used for this purpose is the market model, which is essentially a regression of the company's stock returns and the returns of the market index.[4] The market model for a stock *i* can be expressed as:

$r_{it} = \alpha_i + \beta_i r_{Mt}$

Here, r_{it} and r_{Mt} represent the stock and the market return on day t. The coefficients α_i and β_i are estimated by running an ordinary least-squares regression over the estimation window.

Note that the market model is similar to the CAPM model, $r_{it} = r_f + \beta_i\ (E(r_M) - r_f)$, with minor changes: $r_{it} = \underbrace{r_f(1-\beta_i)}_{\alpha_i} + \beta_i r_{Mt}$.[5]

The most common criteria for selecting market or industry indices are whether the company is listed on NYSE/AMEX or Nasdaq and whether any restrictions are imposed by data availability. In general, the market index should be a broad-based, value-weighted index or a float-weighted index. The industry index should be specific to the company being analyzed. For litigation purposes, it is common to construct the industry index instead of using alternative Standard & Poor's 500 (SP500) or the Morgan Stanley Capital International (MSCI) indices. Most industry indices are available from Yahoo Finance.[6]

Given the equation $r_{it} = \alpha_i + \beta_i r_{Mt}$ in the estimation window, we can now measure the impact of an event on the stock's return in the event window. For a particular day t in the event window, we define the stock's *abnormal return* as the difference between its actual return and its predicted return:

$$AR_{it} = \underbrace{r_{it}}_{\substack{\uparrow \\ \text{Actual stock} \\ \text{return in event} \\ \text{window day } t}} - \underbrace{(\alpha_i + \beta_i r_{Mt})}_{\substack{\uparrow \\ \text{Return predicted} \\ \text{by the stock's } \alpha,\ \beta, \\ \text{and market return}}}$$

We interpret the abnormal return during the event window as a measure of the impact the event had on the market value of the security. This assumes that the event is exogenous with respect to the change in the security's market value.

The cumulative abnormal return is a measure of the total abnormal returns during the event window. CAR_t is the sum of all the abnormal returns from the beginning of the event window $T_1 + 1$ until a particular day t in the window:

$$CAR_t = \sum_{j=1}^{t} AR_{T_1+j}$$

Market-Adjusted and Two-Factor Models

As mentioned above, you can use several alternative models to calculate a security's expected return. The market-adjusted model is simplest in design and is often used to get a first impression of stock price movements. When using the market-adjusted model, you calculate the abnormal return by taking the difference between the actual return of the security and the actual return of the market index. Thus, there is no need to run ordinary least squares (OLS) regressions to estimate parameters. In fact, all you need is the returns at the time of the event. However, when testing the abnormal returns for statistical significance, you still need to gather returns for the estimation period.

The two-factor model uses the returns from the market and the industry. You calculate a stock's expected return using parameters from a regression of the actual returns against the market and industry returns during the estimation period. The industry returns are included primarily to account for industry-specific information in addition to the market-specific information. To calculate the abnormal return, you subtract from the actual return the portion that can be explained by the intercept, the market, and the industry. The two-factor model is illustrated in detail in section 14.5.

The results in a large sample of events are not especially sensitive to your choice of estimation model.[7] However, if you are dealing with a small sample, you should explore alternative models.

14.3 An Initial Event Study: Procter & Gamble Buys Gillette

On January 28, 2005, Procter & Gamble (P&G) announced a bid for Gillette Company. In its press release (see below), P&G indicated that its bid valued Gillette at a premium of 18% over its market price.

P&G Signs Deal to Acquire The Gillette Company

Raises Long-Term Sales Growth Outlook

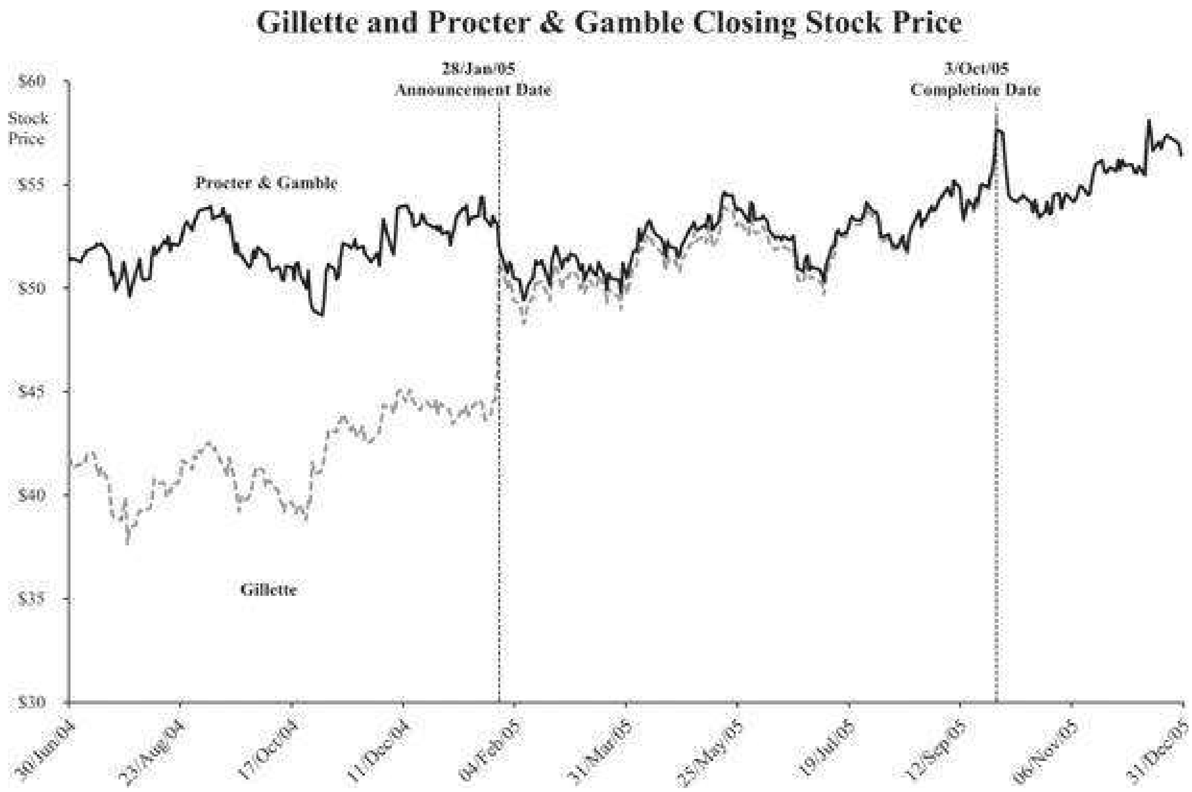
CINCINNATI; AND BOSTON, Jan. 28, 2005 /PRNewswire-FirstCall/ -- The Procter & Gamble Company (NYSE: PG) announced it has signed a deal to acquire 100% of The Gillette Company (NYSE: G). Gillette, founded in 1901 and headquartered in Boston, Mass., markets a number of category-leading consumer products such as Gillette(R) razors and blades including the Mach3(R) and Venus(R) brands, Duracell(R) CopperTop(R) batteries, Oral-B(R) manual and power toothbrushes, and Braun shavers and small appliances. The transaction is valued at approximately $57 billion (USD) making it the largest acquisition in P&G history.

Terms of the Deal

Under terms of the agreement, unanimously approved by the board of directors of both companies on January 27, P&G has agreed to issue 0.975 shares of its common stock for each share of Gillette common stock. Based on the closing share price of P&G and Gillette stock on January 27, 2005, this represents an 18% premium to Gillette shareholders.

P&G will acquire all of Gillette's business, including manufacturing, technical and other facilities. The transaction, which is subject to certain conditions including approval by Gillette's and P&G's shareholders and regulatory clearance, is expected to close in fall 2005.

As might be expected, the bid had a dramatic effect on Gillette's stock price.



From the graph it appears that there might have also been a decrease in the price of P&G stock.

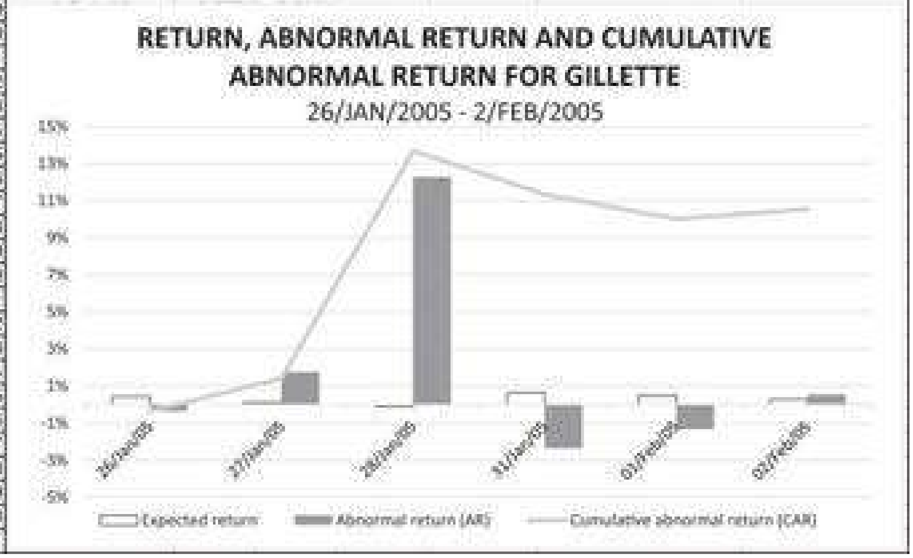
The Estimation Window

We will attempt an event study to judge the impact of the takeover announcement on the returns of Gillette and P&G. To do this, we first determine the estimation window as the 252 trading days preceding the two days before the announcement on January 28, 2005:

	A	B	C	D	E	F	G
1	**GILLETTE RETURNS: ESTIMATION WINDOW AND EVENT WINDOW**						
2	Intercept	0.0007	<-- =INTERCEPT($C$11:$C$262,$B$11:$B$262)				
3	Slope	0.5965	<-- =SLOPE($C$11:$C$262,$B$11:$B$262)				
4	R-squared	0.1267	<-- =RSQ($C$11:$C$262,$B$11:$B$262)				
5	Steyx	0.0113	<-- =STEYX($C$11:$C$262,$B$11:$B$262)				
6							
7	Days in estimation window		252	<-- =COUNT(A11:A262)			
8							
9				EVENT WINDOW			
10	Date	SP 500	Gillette	Expected return	Abnormal return (AR)	Cumulative abnormal return (CAR)	
11	27-Jan-04	-0.97%	-0.42%				
12	28-Jan-04	-1.39%	-1.28%				
13	29-Jan-04	0.39%	-0.94%				
14	30-Jan-04	-0.22%	-1.40%				
259	20-Jan-05	-0.76%	-0.56%				
260	21-Jan-05	-0.62%	-1.51%				
261	24-Jan-05	-0.39%	0.57%				
262	25-Jan-05	0.38%	1.43%				
263	26-Jan-05	0.58%	0.07%	0.42%	-0.35%	-0.35%	<-- =E263
264	27-Jan-05	0.07%	1.87%	0.11%	1.76%	1.41%	<-- =F263+E264
265	28-Jan-05	-0.30%	12.17%	-0.11%	12.28%	13.69%	<-- =F264+E265
266	31-Jan-05	0.94%	-1.72%	0.63%	-2.35%	11.34%	
267	1-Feb-05	0.70%	-0.83%	0.49%	-1.32%	10.02%	
268	2-Feb-05	0.35%	0.79%	0.28%	0.51%	10.54%	

Cell D263 contains formula =$B$2+$B$3*B263

Cell E263 contains formula =C263-D263



The regression results indicate the normal behavior of Gillette in the estimation window: $r_{Gillette,t} = 0.007 + 0.5965 \cdot r_{SP500,t}$. The **Steyx** function measures the standard error of the regression-predicted *y* values. Below we show how to use this value to measure the significance of the event's abnormal returns.

The Event Window

We define the event window as two days before and three days after the announcement. To measure the impact of the announcement effect in the event window, we use the market model $r_{Gillette,t} = 0.0007 + 0.5965 \cdot r_{SP500,t}$. The formulas in the event window are in the spreadsheet above. As you can see, the announcement of the acquisition of Gillette by P&G led to several large Gillette abnormal returns in the event window.

We can use **Steyx**, the standard error of the regression prediction to measure the significance of the abnormal returns. Only two of the abnormal returns—on the event date of January 28 and the day following—are actually significant at the 5% level:

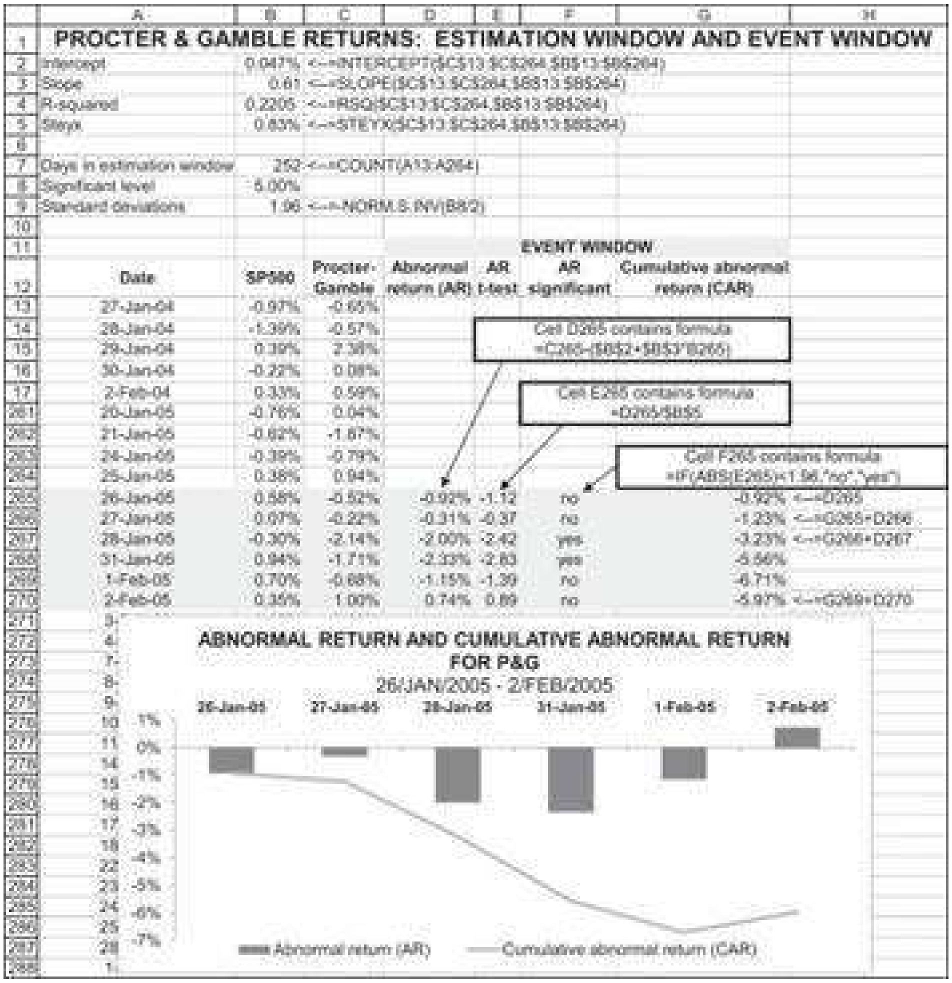
	A	B	C	D	E	F	G
1	**GILLETTE RETURNS: THE SIGNIFICANCE OF THE ABNORMAL**						
2	Intercept	0.0007	<-- =INTERCEPT($C$13:$C$264,$B$13:$B$264)				
3	Slope	0.5965	<-- =SLOPE($C$13:$C$264,$B$13:$B$264)				
4	R-squared	0.1267	<-- =RSQ($C$13:$C$264,$B$13:$B$264)				
5	Steyx	1.13%	<-- =STEYX($C$13:$C$264,$B$13:$B$264)				
6							
7	Days in estimation window		252	<-- =COUNT(A13:A264)			
8	Significant level	5.00%					
9	Standard deviations	1.96	<--=-NORM.S.INV(B8/2)				
10							
11				**EVENT WINDOW**			
12	**Date**	**SP 500**	**Gillette**	**Abnormal return (AR)**	**AR t-test**	**AR significant**	
13	27-Jan-04	-0.97%	-0.42%				
14	28-Jan-04	-1.39%	-1.28%				Cell D265 contains formula
15	29-Jan-04	0.39%	-0.94%				=C265-($B$2+$B$3*B265)
16	30-Jan-04	-0.22%	-1.40%				
261	20-Jan-05	-0.76%	-0.56%				Cell E265 contains formula
262	21-Jan-05	-0.62%	-1.51%				=D265/$B$5
263	24-Jan-05	-0.39%	0.57%				
264	25-Jan-05	0.38%	1.43%				
265	26-Jan-05	0.58%	0.07%	-0.35%	-0.31	No	<--=IF(ABS(E265)>$B$9,"Yes","No")
266	27-Jan-05	0.07%	1.87%	1.76%	1.56	No	
267	28-Jan-05	-0.30%	12.17%	12.28%	10.86	Yes	
268	31-Jan-05	0.94%	-1.72%	-2.35%	-2.08	Yes	
269	1-Feb-05	0.70%	-0.83%	-1.32%	-1.17	No	
270	2-Feb-05	0.35%	0.79%	0.51%	0.46	No	

We calculate the test statistic by dividing the abnormal returns by the **Steyx** in cell B5. Assuming that the regression residuals are normally distributed, if the absolute value of the test statistic is larger than 1.96 (cell B9), then the abnormal return is significant at the 95% level (meaning that the chances that the abnormal return is random and insignificant are less than 5%, or 2.5% in either the positive side or negative side—cell B8). If the test statistic is larger than 2.58, its significance level is 1%. As can be seen from rows 265–270 above, at the 1% level, only the announcement itself has a significant abnormal return.[8]

	A	B	C	D	E	F
1	**HOW MANY STANDARD DEVIATIONS MAKE IT SIGNIFICANT?**					
2	**Significant level**	**Standard deviations**				
3	20%	1.282	<--=-NORM.S.INV(A3/2)			
4	15%	1.440	<--=-NORM.S.INV(A4/2)			
5	10%	1.645				
6	5%	1.960				
7	2%	2.326				
8	1%	2.576				

What about P&G?

Thus far we have concentrated on the event influence on the takeover target, Gillette. Applying the same methodology to P&G's stock returns shows that the announcement had a negative impact on its stock returns. There may also have been some information leaks prior to the January 28, 2005, announcement.



	A	B	C	D	E	F	G	H
1	PROCTER & GAMBLE RETURNS: ESTIMATION WINDOW AND EVENT WINDOW							
2	Intercept	0.047%	<--=INTERCEPT($C$13:$C$264,$B$13:$B$264)					
3	Slope	0.61	<--=SLOPE($C$13:$C$264,$B$13:$B$264)					
4	R-squared	0.2205	<--=RSQ($C$13:$C$264,$B$13:$B$264)					
5	Steyx	0.83%	<--=STEYX($C$13:$C$264,$B$13:$B$264)					
6								
7	Days in estimation window	252	<--=COUNT(A13:A264)					
8	Significant level	5.00%						
9	Standard deviations	1.96	<--=-NORM.S.INV(B8/2)					
10								
11				EVENT WINDOW				
12	Date	SP500	Procter-Gamble	Abnormal return (AR)	AR t-test	AR significant	Cumulative abnormal return (CAR)	
13	27-Jan-04	-0.97%	-0.65%					
14	28-Jan-04	-1.39%	-0.57%					
15	29-Jan-04	0.39%	2.38%					
16	30-Jan-04	-0.22%	0.08%					
17	2-Feb-04	0.33%	0.59%					
261	20-Jan-05	-0.76%	0.04%					
262	21-Jan-05	-0.62%	-1.87%					
263	24-Jan-05	-0.39%	-0.79%					
264	25-Jan-05	0.38%	0.94%					
265	26-Jan-05	0.58%	-0.52%	-0.92%	-1.12	no	-0.92%	<--=D265
266	27-Jan-05	0.07%	-0.22%	-0.31%	-0.37	no	-1.23%	<--=G265+D266
267	28-Jan-05	-0.30%	-2.14%	-2.00%	-2.42	yes	-3.23%	<--=G266+D267
268	31-Jan-05	0.94%	-1.71%	-2.33%	-2.83	yes	-5.56%	
269	1-Feb-05	0.70%	-0.68%	-1.15%	-1.39	no	-6.71%	
270	2-Feb-05	0.35%	1.00%	0.74%	0.89	no	-5.97%	<--=G269+D270

Cell D265 contains formula =C265-($B$2+$B$3*B265)

Cell E265 contains formula =D265/$B$5

Cell F265 contains formula =IF(ABS(E265)<1.96,"no","yes")

Summarizing: What Happened on the Announcement Date?

On January 28, 2005, P&G announced the purchase of Gillette. Each share of Gillette was purchased for 0.975 shares of P&G. At the 5% significance level, the acquisition announcement had significant effects on the stock prices of Gillette and P&G only on the announcement date and the day after. After an initial positive impact on Gillette (a 12.28% increase over the normally anticipated stock return on January 28 and a further negative 2.35% on January 31) and an initial negative impact on P&G (negative 2.02% on January 28 and another negative 2.23% on January 31), there were no additional significant effects on the stock prices around the announcement. The cumulative effects are summarized below:

	A	B	C	D	E	F	G	H	I
1	GILLETTE PURCHASED BY PROCTER & GAMBLE Measuring the synergies in the event window								
2		Shares outstanding (thousands)	Share price, 25/Jan/05, ($)	Market value, 25/Jan/05, (billion $)					
3	Gillette	1,000,000	44.53	44.53	<--=B3*C3/1000				
4	P&G	2,741,000	53.49	146.62	<--=B4*C4/1000				
5									
6			Cell F11 contains formula				Cell G11 contains formula		
7			=$D$3*SUM($B$11:B11)/1000				=$D$4*SUM($C$11:C11)/1000		
8									
9	Date		Abnormal returns (AR)				Cumulative abnormal valuations		
10		Gillette	P&G	Sum		Gillette	P&G	Sum	
11	26-Jan-05	-0.35%	-0.92%	-1.27%		-0.16	-1.35	-1.51	<--=F11+G11
12	27-Jan-05	1.76%	-0.31%	1.45%		0.63	-1.80	-1.17	
13	28-Jan-05	12.28%	-2.00%	10.28%		6.10	-4.73	1.36	
14	31-Jan-05	-2.35%	-2.33%	-4.68%		5.05	-8.16	-3.11	
15	01-Feb-05	-1.32%	-1.15%	-2.47%		4.46	-9.84	-5.38	
16	02-Feb-05	0.51%	0.74%	1.25%		4.69	-8.76	-4.07	

The table above attempts to measure the short-term synergies of the announcement by multiplying the CAR for Gillette and P&G times their market value on the day before the event window. In the short period of time measured by this event window, the cumulative synergy appears to be negative, with the positive value creation for Gillette shareholders outweighed by the negative impact on P&G.[9]

14.4 A Fuller Event Study: Impact of Earnings Announcements on Stock Prices

In the previous section we used the event study methodology to explore the impact of a merger announcement on the returns of both the takeover target (Gillette) and the acquirer (P&G). In this section we show how to aggregate the returns of an event in order to evaluate the market response to a particular type of event. We consider the effect of earning announcements on a set of stores in the grocery industry.

An Initial Example: Safeway's Positive Earnings Surprise on July 20, 2006

To set the stage, consider the earnings announcement made by Safeway on July 20, 2006. On this date, Safeway announced earnings per share (EPS) of $0.42, a number that exceeded by 6 cents the market consensus estimates of $0.36.[10] On the same day, the SP500 declined by 0.85%, and Safeway stock rose by 8.39%. The spreadsheet below shows the example of Safeway's earnings announcement on July 20, 2006. We use the event study methodology to gauge the market reaction to this earnings surprise:

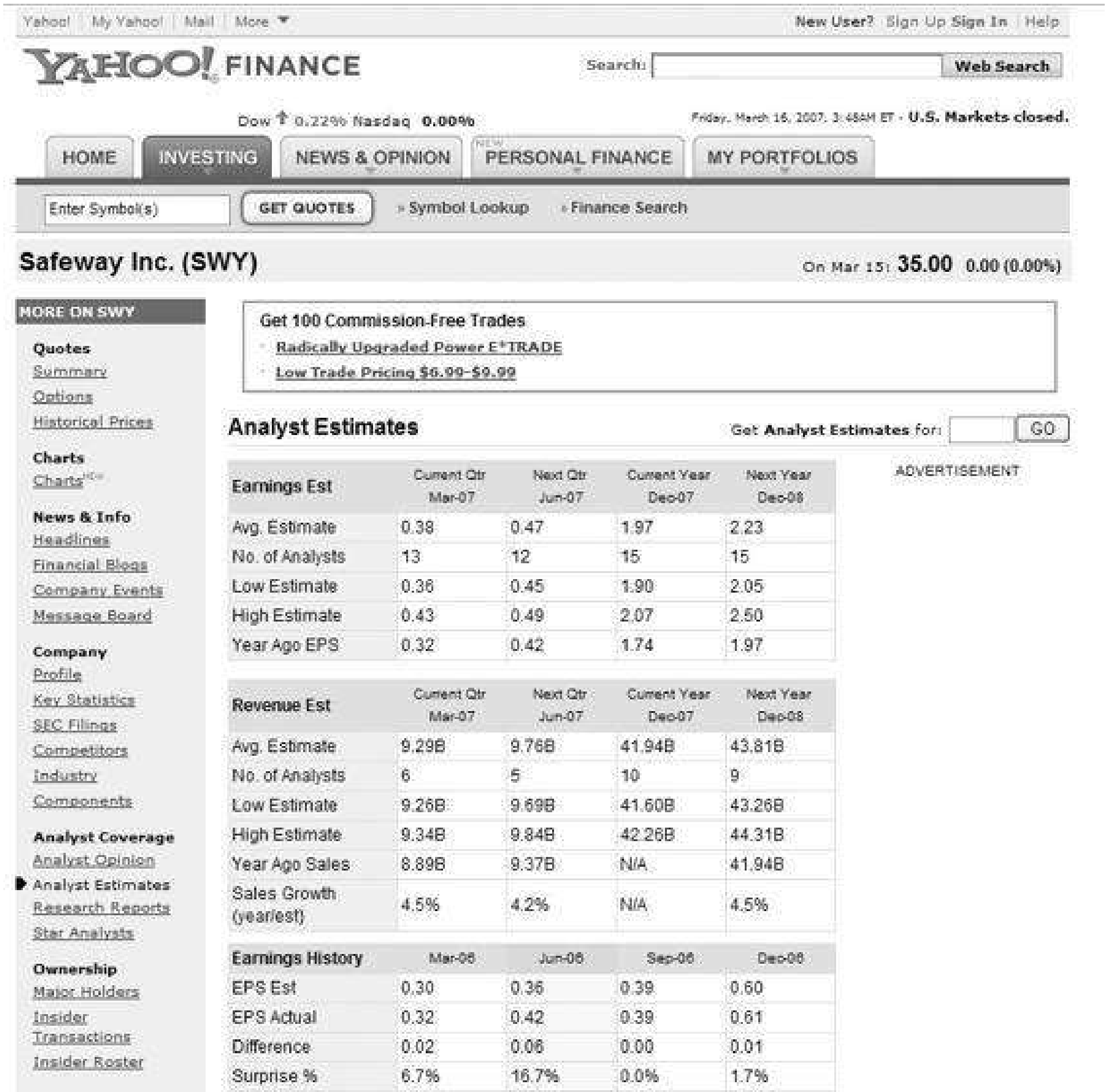
	A	B	C
1	**THE MARKET REACTION TO A POSITIVE EARNINGS SURPRISE BY SAFEWAY, 20 July 2006**		
2	Announcement date	20-Jul-06	
3	Earnings per share	$0.42	
4	Consensus earnings estimate	$0.36	
5	Earnings surprise (forecast error)	$0.06	<-- =B3-B4
6			
7	**How did the market interpret the earnings**		
8	Safeway	8.39%	
9	S&P 500	-0.85%	
10			
11	**The market model: Regressing Safeway returns on the SP500 returns (2/Jan/05 - 19/Jul/06)** $r_{Safeway}$=0.061%+0.94*r_{SP500}		
12	Intercept	0.061%	
13	Slope	0.94	
14	Steyx	1.29%	
15			
16	**The residual return: Expected stock returns versus actual returns (20/Jul/06)**		
17	Expected return	-0.74%	<-- =B13*B9+B12
18	Residual return	9.13%	<-- =B8-B17
19	t-statistic	7.07	<-- =B18/B14

In cells B12:B14 we have regressed Safeway daily returns on those of the SP500 for the 388 trading days preceding the announcement (January 2, 2005 to July 19, 2006). The regression shows that $r_{safeway} = 0.061\% + 0.94 * r_{SP500}$ and that the standard error of the estimate is 1.29%.

This data show that on the day of the announcement, the anticipated return of Safeway, given the negative 0.85% return of the SP500 and absent the earnings surprise, should have been −0.74%. This means that the abnormal return, measuring the impact of the earnings announcement, was 9.13% (cell B18, above). The *t*-statistic for the abnormal return was 7.07, showing that it is highly significant.

The Earnings Surprise Numbers

Our earnings surprise numbers are drawn from Yahoo, as shown in the following screenshot. While Yahoo is an admirable source of data, it does not provide a database of historical analyst estimates and actual earnings numbers; such data are available from Bloomberg’s Best Consensus Earnings Estimates and other commercial sources.



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Safeway Inc. (SWY) On Mar 15: **35.00** 0.00 (0.00%)

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Earnings Est	Current Qtr Mar-07	Next Qtr Jun-07	Current Year Dec-07	Next Year Dec-08
Avg. Estimate	0.38	0.47	1.97	2.23
No. of Analysts	13	12	15	15
Low Estimate	0.36	0.45	1.90	2.05
High Estimate	0.43	0.49	2.07	2.50
Year Ago EPS	0.32	0.42	1.74	1.97

Revenue Est	Current Qtr Mar-07	Next Qtr Jun-07	Current Year Dec-07	Next Year Dec-08
Avg. Estimate	9.29B	9.76B	41.94B	43.81B
No. of Analysts	6	5	10	9
Low Estimate	9.26B	9.69B	41.60B	43.26B
High Estimate	9.34B	9.84B	42.26B	44.31B
Year Ago Sales	8.89B	9.37B	N/A	41.94B
Sales Growth (year/est)	4.5%	4.2%	N/A	4.5%

Earnings History	Mar-06	Jun-06	Sep-06	Dec-06
EPS Est	0.30	0.36	0.39	0.60
EPS Actual	0.32	0.42	0.39	0.61
Difference	0.02	0.06	0.00	0.01
Surprise %	6.7%	16.7%	0.0%	1.7%

An Event Study: The Grocery Industry

We extend our Safeway study by considering 16 quarterly earnings announcements by four grocery companies for fiscal year 2006.[11]

EARNINGS ANNOUNCEMENT IN 2006-07 FOR KROGER (KR), SUPERVALU (SVU), SAFEWAY (SWY), AND WHOLE FOODS (WFMI)

	A	B	C	D	E	G	H	I	J	L	M	N	O	P	Q
2	Date	Ticker	Consensus EPS ($)	Actual EPS ($)	Surprise: =D3-C3	Starting Point	Intercept	Slope	STEXY	Actual Return	SP500 Return	Expected Return: =H3+I3*M3	AR: =L3-N3	t-stat	
3	07-Mar-06	KR	0.36	0.39	0.03	44	0.031%	0.6662	1.23%	1.36%	-0.19%	-0.09%	1.46%	1.18	<-- =O3/J3
4	20-Jun-06	KR	0.42	0.42	0.00	117	0.041%	0.6063	1.13%	5.05%	0.00%	0.04%	5.01%	**4.44**	
5	12-Sep-06	KR	0.29	0.29	0.00	175	0.065%	0.5629	1.08%	-5.67%	1.00%	0.64%	-6.31%	**-5.82**	
6	05-Dec-06	KR	0.28	0.30	0.02	234	0.029%	0.4483	1.10%	5.08%	0.40%	0.21%	4.87%	**4.43**	
7	18-Apr-06	SVU	0.56	0.55	-0.01	77	-0.080%	0.7049	1.23%	-0.28%	1.69%	1.11%	-1.39%	-1.13	<-- =O7/J7
8	26-Jul-06	SVU	0.57	0.53	-0.04	142	-0.033%	0.5416	1.22%	-7.09%	-0.04%	-0.05%	-7.03%	**-5.76**	
9	10-Oct-06	SVU	0.53	0.61	0.08	195	-0.020%	0.6016	1.28%	4.36%	0.20%	0.10%	4.26%	**3.32**	
10	09-Jan-07	SVU	0.56	0.54	-0.02	256	0.029%	0.5206	1.30%	-1.70%	-0.05%	0.00%	-1.70%	-1.31	
11	27-Apr-06	SWY	0.30	0.32	0.02	90	0.010%	1.0139	1.30%	2.88%	0.33%	0.34%	2.54%	1.95	<-- =O11/J11
12	20-Jul-06	SWY	0.36	0.42	0.06	138	0.012%	0.9089	1.18%	8.38%	-0.85%	-0.78%	9.17%	**7.75**	
13	12-Oct-06	SWY	0.39	0.39	0.00	197	0.032%	0.7633	1.34%	-1.43%	0.95%	0.75%	-2.18%	-1.63	
14	27-Feb-07	SWY	0.60	0.61	0.01	286	0.116%	0.7505	1.31%	-3.95%	-3.32%	-2.54%	-1.41%	-1.08	
15	04-May-06	WFMI	0.35	0.36	0.01	85	0.063%	0.8045	1.75%	12.50%	0.32%	0.33%	12.17%	**6.94**	<-- =O15/J15
16	01-Aug-06	WFMI	0.34	0.35	0.01	146	-0.071%	1.2329	1.64%	-12.51%	-0.45%	-0.63%	-11.88%	**-7.23**	
17	03-Nov-06	WFMI	0.29	0.29	0.00	213	-0.135%	1.3199	1.95%	-26.21%	-0.22%	-0.43%	-25.78%	**-13.22**	
18	22-Feb-07	WFMI	0.40	0.38	-0.02	286	-0.202%	1.5321	2.43%	13.13%	-0.09%	-0.33%	13.46%	**5.55**	
20	Positive surprise			2.49%	<-- =SUMIF($E$3:$E$18,">0",$O$3:$O$18)/COUNTIF($E$3:$E$18,">0")										
21	Non-positive surprise			-3.24%	<-- =SUMIF($E$3:$E$18,"<=0",$O$3:$O$18)/COUNTIF($E$3:$E$18,"<=0")										

Cell G18 contains formula:
=COUNTIF('Safeway, Market Data'!$A$3:$A$551,"<="&TEXT(A18,"0"))-252

Cell H18 contains formula: =INTERCEPT(OFFSET('Safeway, Market Data'!$A$2,$G18,10,252,1),OFFSET('Safeway, Market Data'!$A$2,$G18,2,252,1))

Cell I18 contains formula:
=SLOPE(OFFSET('Safeway, Market Data'!$A$2,$G18,10,252,1),OFFSET('Safeway, Market Data'!$A$2,$G18,2,252,1))

Cell J18 contains formula:
=STEYX(OFFSET('Safeway, Market Data'!$A$2,$G18,10,252,1),OFFSET('Safeway, Market Data'!$A$2,$G18,2,252,1))

For each announcement we have determined the intercept and slope of the market-model regression for the 252 days preceding the announcement.[12] Here's a specific example from the above spreadsheet:

	A	B	C	D	E	G	H	I	J	L	M	N	O	P
2	Date	Ticker	Consensus EPS ($)	Actual EPS ($)	Surprise: =D3-C3	Starting Point	Intercept	Slope	STEXY	Actual Return	SP500 Return	Expected Return: =H3+I3*M3	AR: =L3-N3	t-stat
3	07-Mar-06	KR	0.36	0.39	0.03	44	0.031%	0.6662	1.23%	1.36%	-0.19%	-0.09%	1.46%	1.18
18	22-Feb-07	WFMI	0.40	0.38	-0.02	286	-0.202%	1.5321	2.43%	13.13%	-0.09%	-0.33%	13.46%	**5.55**

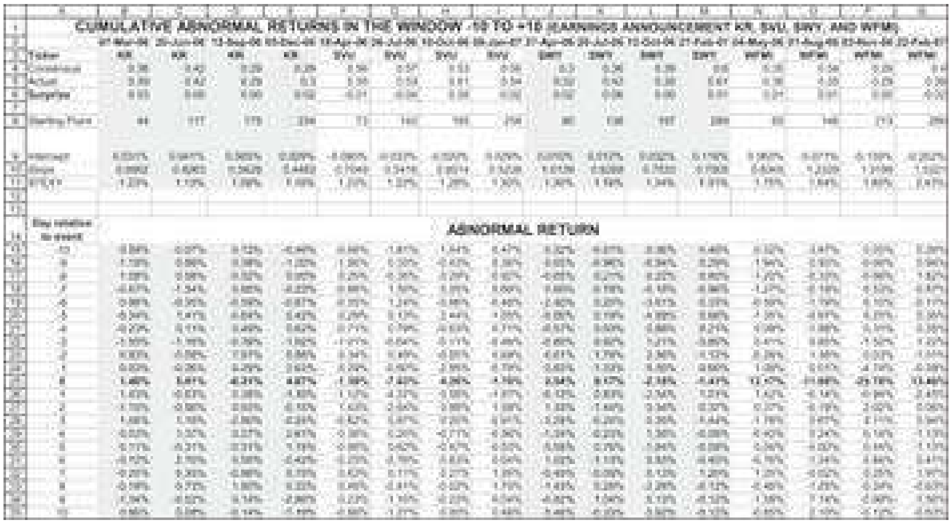
Row 3 tracks the market model of Kroger stock to the SP500 in the 252 trading days before the earnings announcement on March 7, 2006. The market model is $r_{KR} = 0.031\% + 0.662 * r_{SP500}$. Kroger's actual return on the announcement date, 1.36%, is 1.46% higher than the return that would have been predicted by its market model. However, dividing this 1.46% by the standard deviation of the abnormal returns (the regression residuals) (**Steyx** = 1.23%) gives a t-statistic of 1.18, which is not significant at the 5% level.

Row 18 tracks the market model of Whole Foods stock to the SP500 in the year before the earnings announcement after market close on February 21, 2007. The market model is $r_{WFMI} = -0.202\% + 1.5321 * r_{SP500}$, with **Steyx** = 2.43%. The abnormal return on the day of the announcement, 13.46%, is significant at the 5% level (hence the boldface in cell P18). To temper this interpretation, note that on the day of the earnings announcement, Whole Foods announced the merger with Wild Oats Markets; this makes it difficult to interpret the true market reaction to the earnings release. In general, we have to be careful interpreting abnormal returns when an event is announced with confounding information.

The Cumulative Abnormal Returns

We restate the data in slightly different form in the spreadsheet below—using the **Offset** function—and then compute the abnormal returns in an event window which goes from

−10 to +10 days around the earnings announcements:


CUMULATIVE ABNORMAL RETURNS IN THE WINDOW -10 TO +10 (EARNINGS ANNOUNCEMENT KR, SVU, SWY, AND WFMI)

ABNORMAL RETURN

We can compute the average abnormal return (AAR) and CAR for each of the days in the event window. In the following table we perform this computation separately for positive versus non-positive earnings announcements:

Cell T7 contains formula:
{=SQRT(SUMPRODUCT(IF($B$6:$Q$6<=0,$B$11:$Q$11),IF($B$6:$Q$6<=0,$B$11:$Q$11))*(1/COUNTIF($B$6:$Q$6,"<=0")^2))}

Unadjusted cross-sectional errors

Positive	0.54%
Non-Positive	0.49%

Cell T8 contains formula:
{=SQRT(SUMPRODUCT(IF($B$6:$Q$6>0,$B$11:$Q$11),IF($B$6:$Q$6>0,$B$11:$Q$11))*(1/COUNTIF($B$6:$Q$6,">0")^2))}

	Positive Earnings				**Non-Positive Earnings**		
Day relative to event	**AAR**	**T-stat**	**CAR**		**AAR**	**T-stat**	**CAR**
-10	-0.57%	-1.162	-0.57%		-0.09%	-0.163	-0.09%
-9	-0.48%	-0.993	-1.05%		0.10%	0.194	0.02%
-8	0.00%	0.003	-1.05%		0.04%	0.067	0.05%
-7	-0.39%	-0.805	-1.44%		0.01%	0.025	0.07%
-6	-0.57%	-1.178	-2.01%		-0.54%	-1.006	-0.48%
-5	0.05%	0.104	-1.96%		-0.55%	-1.028	-1.03%
-4	-0.18%	-0.375	-2.14%		0.37%	0.677	-0.67%
-3	-0.75%	-1.541	-2.89%		-0.34%	-0.638	-1.01%
-2	0.53%	1.090	-2.36%		0.54%	0.992	-0.47%
-1	0.85%	1.744	-1.51%		-0.74%	-1.369	-1.21%
0	2.65%	**5.441**	1.13%		-3.24%	**-6.006**	-4.46%
1	0.46%	0.941	1.59%		-1.63%	**-3.019**	-6.08%
2	0.02%	0.033	1.61%		0.40%	0.736	-5.69%
3	-0.63%	-1.294	0.98%		0.22%	0.412	-5.47%
4	0.02%	0.035	0.99%		0.24%	0.443	-5.23%
5	-0.30%	-0.609	0.70%		-0.52%	-0.962	-5.75%
6	-0.28%	-0.569	0.42%		0.55%	1.027	-5.19%
7	0.34%	0.700	0.76%		0.47%	0.864	-4.72%
8	-0.36%	-0.733	0.41%		0.14%	0.268	-4.58%
9	0.15%	0.315	0.56%		-0.45%	-0.838	-5.03%
10	0.12%	0.243	0.68%		-0.20%	-0.361	-5.23%

Cell T35 contains formula
=SUMIF($B$6:$Q$6,">0",B35:Q35)/COUNTIF($B$6:$Q$6,">0")

Cell U35 contains formula
=T35/$T$8

Cell X35 contains formula
=SUMIF($B$6:$Q$6,"<=0",B35:Q35)/COUNTIF($B$6:$Q$6,"<=0")

Cell V35 contains formula
=T35+V34

Cell Y35 contains formula
=X35/$T$7

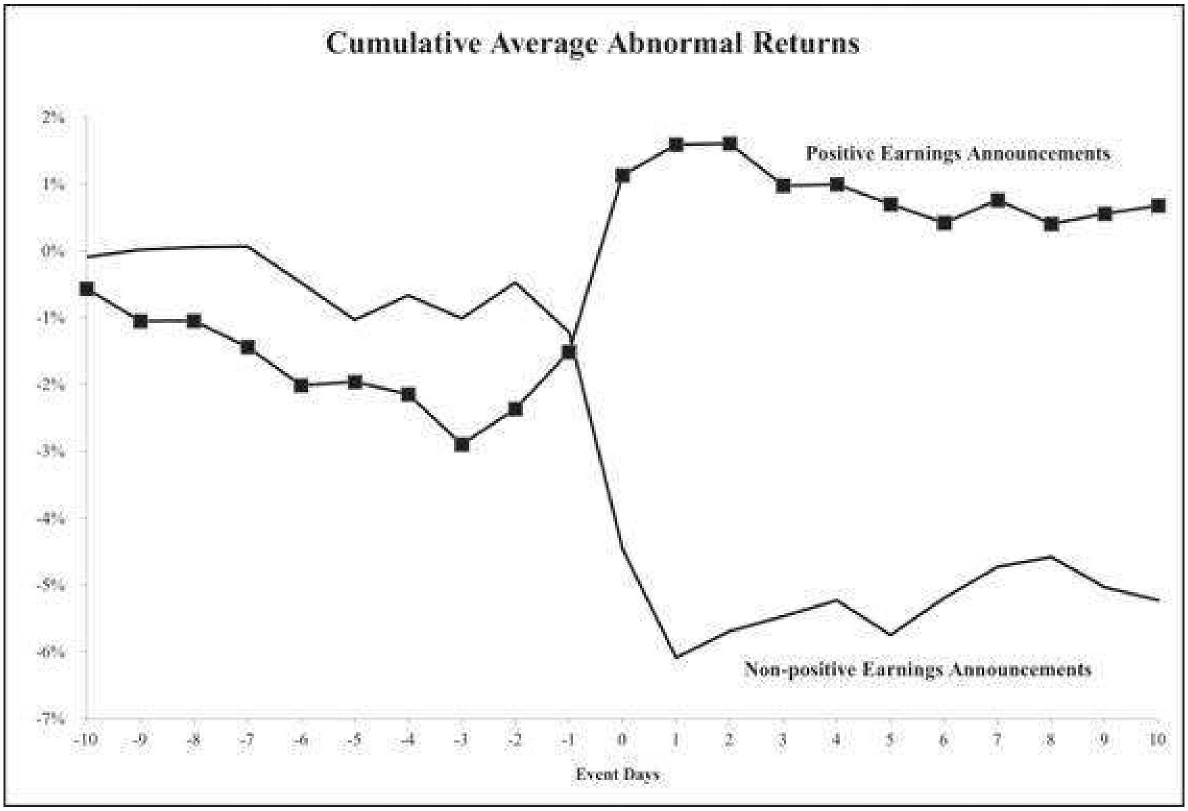
Cell Z35 contains formula
=X35+Z34

The test statistics for the positive and non-positive announcements have been computed by dividing the AAR for each day by the appropriate cross-sectional error for the specific type of return (cells T7 and T8):

$$\text{cell T7: } \sqrt{\frac{\textit{Sumproduct of Steyx for positive announcements}}{(\textit{Number of positive announcements})^2}}$$

$$\text{cell T8: } \sqrt{\frac{\textit{Sumproduct of Steyx for negative announcements}}{(\textit{Number of negative announcements})^2}}$$

Graphing the CARs gives:



On average, there appears to have been little leakage prior to the announcement date of either the good news announcements or the bad news. The market appears to have absorbed the information in the announcements rapidly—following the announcement date ("event day 0"), there appears to have been little additional response.

14.5 Using a Two-Factor Model of Returns for an Event Study

The model used in section 14.2 assumes an equilibrium model $r_{it} = \alpha_i + \beta_i r_{Mt}$. This "one factor" model assumes that the returns on the stocks in question are driven only by one market index. In this section we illustrate a two-factor model. We assume that returns are a function of both a market and an industry factor: $r_{i,\,t} = \alpha_i + \beta_{i,\,Market} r_{M,\,t} + \beta_{i,\,Industry} r_{Industry,\,t}$. We then use this model to determine if a specific event influenced the returns and in which direction.

This time our event date is November 16, 2006. On this date Wendy's announced the purchase by tender of 22,418,000 shares at a price of $35.75 per share. This share purchase represented approximately 19% of the firm's equity. Wendy's stock closed on November 16 at $35.66.

WENDY'S

Wendy's Announces Final Results of its Modified "Dutch Auction" Tender Offer

DUBLIN, Ohio (November 22, 2006) – Wendy's International, Inc. (NYSE:WEN) today announced the final results of its modified "Dutch Auction" tender offer, which expired at 5:00 p.m., Eastern Time, on November 16, 2006.

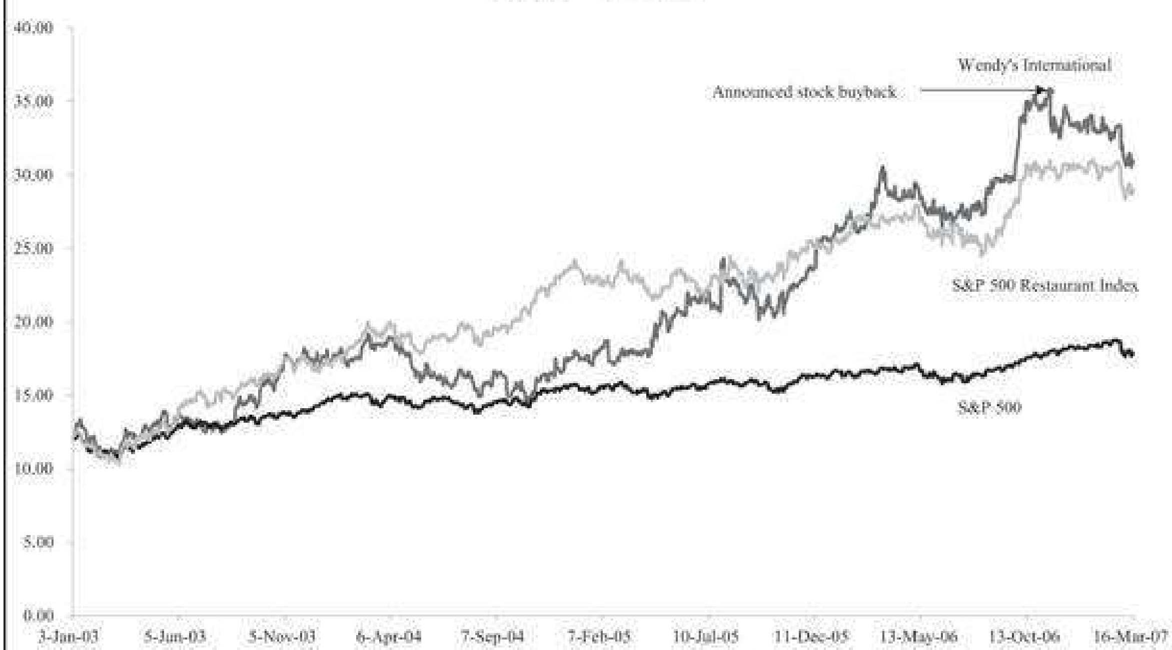
The Company has accepted for purchase 22,413,278 of its common shares at a purchase price of $35.75 per share, for a total cost of $801.3 million.

Shareholders who deposited common shares in the tender offer at or below the purchase price will have all of their tendered common shares purchased, subject to certain limited exceptions.

American Stock Transfer & Trust Company, the depositary for the tender offer, will promptly issue payment for the shares validly tendered and accepted for purchase under the tender offer.

The number of shares the Company accepted for purchase in the tender offer represents approximately 19% of its currently outstanding common shares.

Wendy's Intl vs. S&P 500 and S&P 500 Restaurant Index
1/3/03 – 3/16/07



Note: The S&P 500 and the S&P 500 Restaurant Index are indexed to $12.04 the closing price of Wendy's International on January 3, 2003.

Did the Repurchase Affect Wendy's Returns?

We start by regressing the daily returns on Wendy's on the SP500 and the SP500 Restaurant Index for the 252 days preceding the tender date of November 16, 2006. We use the array function **Linest** to do this computation.[13] The **Linest** box looks like this:

	A	B	C	D
2		Industry	Market	Intercept
3	Slope -->	0.4157	0.5095	0.0012
4	Standard Error -->	0.0851	0.1410	0.0007
5	R^2 -->	0.3140	0.0103	#N/A
6	F statistic -->	56.9738	249	#N/A
7	SS_{xy} -->	0.0122	0.0266	#N/A

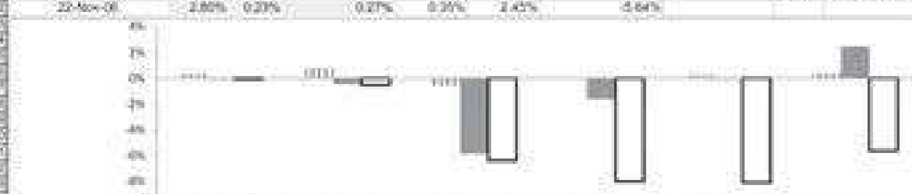
(Note that the #N/A above is produced by Excel and simply means that there is no entry for this column.)

From this box we can conclude that Wendy's return is sensitive to both the market and the industry:

$$r_{Wendys,t} = 0.0012 + \underbrace{0.5095}_{\substack{\text{Market reaction}\\ \text{coefficient.}\\ \text{Standard error:}\\ 0.1410}} * r_{M,t} + \underbrace{0.4157}_{\substack{\text{Industry reaction}\\ \text{coefficient.}\\ \text{Standard error:}\\ 0.0851}} * r_{Industry,t}$$

This **Linest** box is shown again below. Dividing the coefficients by their respective standard errors (row 9) shows that they are both significant at the 1% level. Note that cell C4 gives the standard error of the *y* estimate; we use this in the analysis to determine the significance of the abnormal returns. A further analysis follows the spreadsheet.

	A	B	C	D	E	F	G	H	I	J
1	WENDY'S RETURNS: ESTIMATION WINDOW AND EVENT WINDOW									
2		Industry	Market	Intercept						
3	Slope -->	0.4157	0.5095	0.0012						
4	Standard Error -->	0.0851	0.1410	0.0007						
5	R^2 -->	0.3140	0.0103	#N/A		Cells B3:D7 contain the array formula {=LINEST(B15:B266,C15:D266,,TRUE)}				
6	F statistic -->	56.9738	249	#N/A						
7	SS_{xy} -->	0.0122	0.0266	#N/A						
8										
9	t-stat	4.8818	3.6142	1.6967	<--=D3/D4	Cell C5 is the standard error of the y estimate, used in the t-test of the abnormal returns				
10										
11	Days in estimation window	252	<--=COUNT(A15:A266)							
12										
13						EVENT WINDOW				
14	Date	Wendy's ret	SP500	SP500 Restaurant Index	Expected return	Abnormal return (AR)	Cumulative abnormal return (CAR)		T-test of AR	
15	15-Nov-05	0.08%	-0.39%	-1.26%						
16	16-Nov-05	-0.37%	0.19%	-0.60%		Cell E267 contains formula =$D$3+$C$3*C267+$B$3*D267				
17	17-Nov-05	0.84%	0.94%	1.60%						
18	18-Nov-05	0.27%	0.44%	0.19%						
19	21-Nov-05	1.58%	0.53%	0.92%						
263	9-Nov-06	-0.69%	-0.53%	-0.35%						
264	10-Nov-06	1.46%	0.19%	0.95%			Cell F267 contains formula =B267-E267			
265	13-Nov-06	0.08%	0.25%	-0.05%						
266	14-Nov-06	0.73%	0.63%	0.03%						
267	15-Nov-06	0.25%	0.24%	0.21%	0.33%	-0.08%	-0.08%	<-- =F267	-0.0741	<--=F267/$C$5
268	16-Nov-06	0.31%	0.23%	1.26%	0.76%	-0.45%	-0.53%	<-- =G267+F268	-0.4263	
269	17-Nov-06	-6.44%	0.10%	-1.78%	-0.57%	-5.87%	-6.40%	<-- =G268+F269	-5.6805	
270	20-Nov-06	-1.59%	-0.09%	-0.19%	0.02%	-1.61%	-8.01%		-1.5582	
271	21-Nov-06	0.15%	0.16%	0.07%	0.23%	-0.08%	-8.09%		-0.0784	<--=F271/$C$5
272	22-Nov-06	2.80%	0.23%	0.27%	0.35%	2.45%	-5.64%			



	15-Nov-06	16-Nov-06	17-Nov-06	20-Nov-06	21-Nov-06	22-Nov-06
Expected return	0.33%	0.76%	-0.57%	0.02%	0.23%	0.35%
Abnormal return (AR)	-0.08%	-0.45%	-5.87%	-1.61%	-0.08%	2.45%
Cumulative abnormal return (CAR)	-0.08%	-0.53%	-6.40%	-8.21%	-8.09%	-5.64%

In rows 267–272 of the spreadsheet, we use the two-factor model to analyze the abnormal returns and the cumulative abnormal returns of the Wendy's announcement. While there is little AR or CAR on the days before the announcement, it is clear that the announcement on November 16 had a considerable impact on Wendy's returns on the day following (−5.87% abnormal return on November 17) and on the next day (−1.61% AR on November 20). Dividing the abnormal return by the standard error in C5 shows that only the AR on the event day is significant at the 5% level.

Furthermore, an analysis of the announcement broken down into the market and the industry factors shows that on both of the two days after the November 16 announcement date, the effects of the market index on Wendy's returns were slight. On November 17, however, there was a significant impact on the SP500 Restaurant Index and on Wendy's, which was lacking on November 20, as shown below:

	A	B	C	D	E	F	G	H
1	WENDY'S RETURNS: ESTIMATION WINDOW AND EVENT WINDOW							
13						EVENT WINDOW		
14	Date	Wendy's (wen)	SP500	SP500 Restaurant Index	Expected return	Abnormal return (AR)	Cumulative abnormal return (CAR)	
266	14-Nov-06	0.73%	0.60%	0.03%				
267	15-Nov-06	0.25%	0.24%	0.21%	0.33%	-0.08%	-0.08%	<--=F267
268	16-Nov-06	0.31%	0.23%	1.26%	0.76%	-0.45%	-0.53%	<--=G267+F268
269	17-Nov-06	-6.44%	0.10%	-1.78%	-0.57%	-5.87%	-6.40%	<--=G268+F269
270	20-Nov-06	-1.59%	-0.05%	-0.19%	0.02%	-1.61%	-8.01%	
271	21-Nov-06	0.15%	0.16%	0.07%	0.23%	-0.08%	-8.09%	
272	22-Nov-06	2.80%	0.23%	0.27%	0.35%	2.45%	-5.64%	

We want to first discuss the day after the event: On November 17, 2006, the SP500 rose by 0.10% and the SP500 Restaurant Index fell by 1.78%. Given the regression

$$r_{Wendys,\,t} = 0.0012 + 0.5095 * r_{M,\,t} + 0.4157 * r_{Industry,\,t},$$

the change in the SP500 would have affected Wendy's returns by approximately +0.05% (=0.5095 $*$ 0.10%) and the change in the industry index would have affected Wendy's returns by approximately −0.74% (=0.4157 $*$ (−1.78%)). But Wendy's decreased by −6.44% on the same day, well in excess of the impact of either of the two factors.

Next, we discuss the day of November 20, when the SP500 fell by 0.05% and the SP500 Restaurant Index fell by 0.19%. Given the regression

$$r_{Wendys,\,t} = 0.0012 + 0.5095 * r_{M,\,t} + 0.4157 * r_{Industry,\,t},$$

the change in the SP500 would have affected Wendy's returns by approximately −0.08% and the change in the index for the restaurant industry would have affected Wendy's returns by approximately −0.03%. But Wendy's decreased by −1.59% on the same day, which is again well in excess of the impact of either of the two factors.

The impact of the announcement was felt even on the third day after the event, but we leave this analysis to the reader.

14.6 Using Excel's Offset Function to Locate a Regression in a Data Set

The analysis in section 14.2 requires us to do a regression of a specific stock's returns on the returns of the SP500, where the starting point of the regression is the 252 trading days before a specific date. The technique in section 14.2 uses a number of Excel functions:

- • The functions **Intercept**, **Slope**, **Rsq** give the regression intercept, slope, and *r*-squared, correspondently. These functions are illustrated in Chapter 30 and in the previous portfolio chapters. The function **Steyx** gives the standard deviation of the regression residuals.
- • The function **Countif** counts the number of cells in a range which meets a specific condition. **Countif** has the syntax **Countif(data,condition)**. *However*, the **condition** must be a text condition (which means that in this example we will use the Excel function **Text** to translate a date to a text number—more on this later).
- • The function **Offset** (see also Chapter 30) allows us to specify a cell or a block of cells in an array.

To illustrate the problem, consider the following data of returns for Apple (AAPL) and the SP500. We want to run a regression of the AAPL returns on the SP500 returns for 10 dates before July 18, 2019:

	A	B	C	D	E	F	G
1	USING OFFSET, COUNTIF, AND TEXT TO LOCATE A REGRESSION IN A DATA SET						
2	Date	Apple (AAPL)	Return		SP500	Return	
3	21-Jun-19	198.78			2,950.46		
4	24-Jun-19	198.58	-0.101%	<-- =LN(B4/B3)	2,945.35	-0.17%	<-- =LN(E4/E3)
5	25-Jun-19	195.57	-1.527%		2,917.38	-0.95%	
6	26-Jun-19	199.80	2.140%		2,913.78	-0.12%	
7	27-Jun-19	199.74	-0.030%		2,924.92	0.38%	
8	28-Jun-19	197.92	-0.915%		2,941.76	0.57%	
9	01-Jul-19	201.55	1.817%		2,964.33	0.76%	
10	02-Jul-19	202.73	0.584%		2,973.01	0.29%	
11	03-Jul-19	204.41	0.825%		2,995.82	0.76%	
12	05-Jul-19	204.23	-0.088%		2,990.41	-0.18%	
13	08-Jul-19	200.02	-2.083%		2,975.95	-0.48%	
14	09-Jul-19	201.24	0.608%		2,979.63	0.12%	
15	10-Jul-19	203.23	0.984%		2,993.07	0.45%	
16	11-Jul-19	201.75	-0.731%		2,999.91	0.23%	
17	12-Jul-19	203.30	0.765%		3,013.77	0.46%	
18	15-Jul-19	205.21	0.936%		3,014.30	0.02%	
19	16-Jul-19	204.50	-0.347%		3,004.04	-0.34%	
20	17-Jul-19	203.35	-0.564%		2,984.42	-0.65%	
21	18-Jul-19	205.66	1.130%		2,995.11	0.36%	
22	19-Jul-19	202.59	-1.504%		2,976.61	-0.62%	
23	22-Jul-19	207.22	2.260%		2,985.03	0.28%	
24	23-Jul-19	208.64	0.779%		3,005.47	0.68%	
25							
26	Starting date	18-Jul-19					
27	Rows from top of data to starting date	19	<-- =COUNTIF(A3:A24,"<="&TEXT(B26,"0"))				
28	**Regression**						
29	Intercept	-0.0003	<-- =INTERCEPT(OFFSET(A3:F24,B27-11,2,10,1),OFFSET(A3:F24,B27-11,5,10,1))				
30	Slope	1.5292	<-- =SLOPE(OFFSET(A3:F24,B27-11,2,10,1),OFFSET(A3:F24,B27-11,5,10,1))				
31	R-squared	0.4964	<-- =RSQ(OFFSET(A3:F24,B27-11,2,10,1),OFFSET(A3:F24,B27-11,5,10,1))				
32							
33	**Check**						
34	Intercept	-0.0003	<-- =INTERCEPT(C11:C20,F11:F20)				
35	Slope	1.5292	<-- =SLOPE(C11:C20,F11:F20)				
36	R-squared	0.4964	<-- =RSQ(C11:C20,F11:F20)				

To run this regression, we first use **Countif(data,condition)** to count the row number of the data on which the starting date falls. Since **condition** must be a text entry, we translate the date in cell B26 to a text by using **Text(b26,″0″)**.[14] The Excel function **=Countif(A3:A24,”<=“&Text(B26,”0”))** now counts the number of cells in the column A3:A24 that are less than or equal to the date in cell B26. The answer, as you can see in cell B27, is 19.

Next, we use **Offset(A3:F24,B27-11,2,10,1)** to locate the 10 rows of AAPL returns before the nineteenth row indicated by the starting date. This is a tricky function!

$$\text{Offset}(\underbrace{\text{A3:F24}}_{\substack{\text{Reference}\\ \text{array}}},\ \underbrace{\underbrace{B27-11}_{\substack{\text{Starting}\\ \text{row in}\\ \text{reference}\\ \text{array}}},\quad \underbrace{2}_{\substack{\text{2 columns}\\ \text{over from}\\ \text{top corner}\\ \text{of reference}\\ \text{array}}}}_{\substack{\text{Fix starting}\\ \text{corner in reference}\\ \text{array from which}\\ \text{data is considered}}},\quad \underbrace{\underbrace{10}_{\substack{\text{10 rows}\\ \textit{below}\\ \text{starting}\\ \text{corner}}},\quad \underbrace{1}_{\substack{\text{Only 1}\\ \text{column}\\ \text{of data}}}}_{\text{Data considered}}\ )$$

The functions **Intercept**, **Slope**, **Rsq** can now be used with **Offset (A3:F24,B27-11,2,10,1)** and **Offset(A3:F24,B27-11,5,10,1)**:

$$= \text{Intercept}(\underbrace{\text{Offset(A3:F24, B27-11, 2, 10, 1)}}_{y\text{-data}}, \underbrace{\text{Offset(A3:F24, B27-11, 5, 10, 1)}}_{x\text{-data}})$$

14.7 Conclusion

Event studies, used to determine the impact of a particular market effect on a specific stock or a generic market effect on a set of stocks, are some of the most widely used technologies in practical finance. While Excel may not be the optimal tool for performing an event study, we have used it in this chapter to illustrate both uses of the event study. We have shown that Excel can readily be used to perform either a one-factor or a two-factor event study. The Excel techniques employed are easily acquired by a sophisticated user.

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1. * This chapter was coauthored with Dr. Torben Voetmann, principal at the Brattle Group (Torben.Voetmann@brattle.com) and adjunct professor at the University of San Francisco.
2. 1. Abnormal returns are also referred to as residual returns, and both terms are used interchangeably throughout this chapter.
3. 2. The regression model is similar to the first-pass regression discussed in Chapters 10 and 13. See further discussion below.
4. 3. Of course, something will always be going on in such a long window—quarterly earnings announcements, dividend announcements, news about the companies being considered, and so on. Our assumption is that these other events constitute at most “noise” and are not material for the event being studied.
5. 4. Financial economists most often use the market model to estimate the expected return of a security, although they sometimes use the market-adjusted model or the two-factor market model. See an example of the two-factor model in section 14.5.

6. 5. When the beta is close to one or when the risk-free rate is not volatile, the two models will yield the same results.
7. 6. Yahoo is probably not the best source for index data (and it is free). A widely used source for industry data is Bloomberg. A wonderful free source of industry portfolio data is available from the Fama/French Forum (http://mba.tuck.dartmouth.edu/pages/faculty/ken.french/data_library.html).
8. 7. Brown and Warner (1985).
9. 8. One limitation of **Steyx** is that the variance is slightly understated. The true variance of the market model is the estimation variance from **Steyx** and the additional variance due to the sampling error in α_i and β_i. However, the sampling error approaches zero as the length of the estimation window increases. Since we used 252 trading days in the estimation window, the effect of the sampling error is minimal and it is, therefore, often disregarded when calculating the variance of the abnormal returns.
10. 9. A study commissioned by the Massachusetts Secretary of the Commonwealth William F. Galvin after the merger offer suggests that the synergies of the merger, between $22 billion and $28 billion, were largely captured by P&G. See the report and an article from *BusinessWeek* (today *Bloomberg*) at the following link: https://www.bloomberg.com/news/articles/2005-06-15/in-a-lather-over-the-gillette-deal.
11. 10. Yahoo Finance is the source of our earnings surprise data (see below).
12. 11. We included only Kroger, Supervalu, Safeway, and Whole Foods in the sample. This is obviously an incomplete sample, both in terms of the firms covered and the number of announcements. However, this extended example is meant to impart the flavor of a full-bodied event study.
13. 12. The event window is defined in column G by the "Starting point," which uses **Countif** to locate a date 252 business days before the event date in the data base of stock returns. Notice that the "Starting point" is used in the **Intercept**, **Slope**, **Rsq** formulas in columns H, I, and J.
14. 13. The use of **Linest** to perform multiple regressions is discussed in Chapter 30. It is not the most user friendly of Excel functions.
15. 14. An alternative approach can be: "**=MATCH(B26,A3:A24).**"

15 The Black-Litterman Approach to Portfolio Optimization

15.1 Overview

Chapters 10–14 have set out the classic approach to portfolio optimization that was first explicated by Harry Markowitz in the 1950s and subsequently expanded by Sharpe (1964), Lintner (1965), and Mossin (1966). An enormous academic and practitioner literature, as well as several Nobel Prizes in Economics, testify to the impact of this new point of view on asset valuation and portfolio choice. It is hardly an exaggeration to say that today, no conversation about a stock's risk is complete without mentioning its beta, and discussions of portfolio performance regularly invoke the alpha (both of these topics are discussed in Chapters 10 and 13).

Markowitz, Sharpe, Lintner, and Mossin changed the paradigm of investment management. Well before that, individual investors knew that they should "diversify" and not "put all their eggs in one basket." But Markowitz and those who followed him gave statistical and implementational meaning to these clichés. Modern portfolio theory (MPT) changed the way intelligent investors discuss investment.

Nevertheless, MPT has disappointed. It is possible to come away from a standard textbook discussion of portfolio optimization with the impression that a fixed set of mechanical optimization rules, combined with a bit of knowledge about personal preferences, suffices to define an investor's optimal portfolio. Anyone who has tried to implement portfolio optimization using market data knows that the dream is often a nightmare. Implementations of portfolio theory using historical returns produce wildly unrealistic portfolios, with huge short positions and correspondingly imaginary long positions. It might be thought that limiting short sales, as we have illustrated in Chapter 11, could solve some of these problems. However, short-sale limitations severely restrict the investable asset universe. In many cases, as we show in our example in section 11.4, the computational cause of short sales involves the use of historical data as a proxy for future returns.

The main problem with the mechanical implementation of portfolio optimization is that historical asset return data produce bad predictions for future asset returns. The estimation of the covariances between asset returns and the estimation of expected returns—the underpinnings of portfolio theory—from historical data often produces unbelievable numbers.

In Chapter 9 we alluded to some of these problems in the context of estimating the variance-covariance matrix. There we showed that historical data may not be the best way to estimate this matrix—other methods, in particular the so-called shrinkage methods—may produce more reliable estimates of the covariances. In this chapter we take things a step further. We illustrate the problems of standard portfolio optimization by using a 10-asset portfolio problem. The MPT optimization for our data produces an insane “optimal” portfolio, with many huge long and short positions. The problems of portfolio optimization illustrated in our example are, unfortunately, not unusual. Using the data in a mechanical way to derive “optimal” portfolios simply doesn’t work.[1]

In 1991, Fischer Black and Robert Litterman of Goldman Sachs published an approach that deals with many of the problematics of portfolio optimization.[2] Black and Litterman start with the assumption that investors choose their optimal portfolio from among a given group of assets. This group of assets—it might be the Standard & Poor’s 500 Index (SP500), the Russell 2000, or a mix of international indices—defines the framework within which investors choose a portfolio. An investor’s universe of assets defines a *benchmark* portfolio.

The Black-Litterman (BL) model takes as its starting point the assumption that, in the absence of additional information, the benchmark cannot be outperformed. This assumption is based on much research, which shows that it is very difficult to outperform a typical well-diversified benchmark.[3]

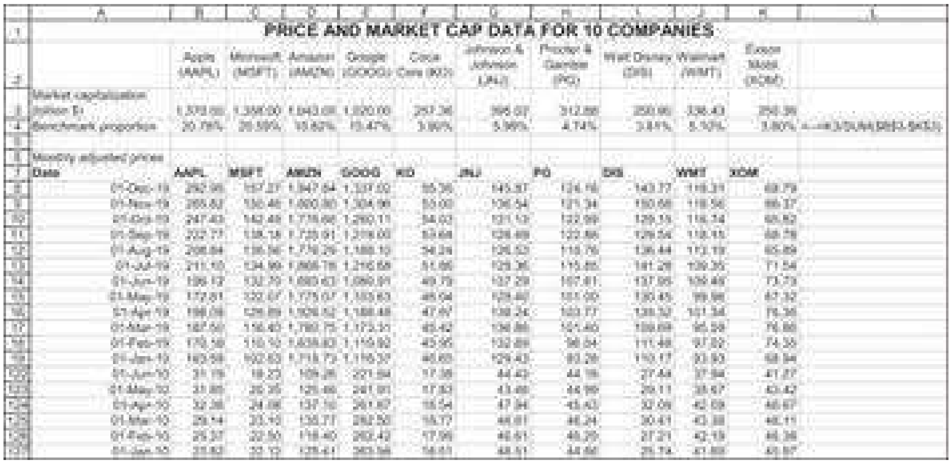
In effect, the BL model turns modern portfolio theory on its head—instead of inputting data and deriving an optimal portfolio, the BL approach assumes that a given portfolio is optimal and from this assumption derives the expected returns of the benchmark components. The implied vector of expected benchmark returns is the starting point of the BL model.

The BL-implied asset returns can be interpreted as the market’s information about the future returns of each asset in the benchmark portfolio. If our investors agree with this market assessment, they’re finished: They can then buy the benchmark, knowing that it is optimal. But what if an investor disagrees with one or more of the implied returns? BL shows how the investor’s opinions can be incorporated into the optimization problem to produce a portfolio that is better for the investor.

In this chapter we start with an illustration of the problematics of MPT. We then go on to illustrate the BL approach. Throughout this chapter, the examples show a full implementation in Excel and in R. On the companion website of this book you can also find the “R” file that is used throughout this chapter.

15.2 A Naive Problem

We start with a naive though representative problem: The Super-Duper Fund has set its benchmark portfolio to be a portfolio composed of 10 leading stocks. Joanna Roe, a new portfolio analyst for the Super-Duper Fund, has decided to use portfolio theory to recommend optimal portfolio holdings based on this benchmark. The screen below gives 10 years of monthly price data on these stocks as of December 2019 (note that some of the rows have been hidden).

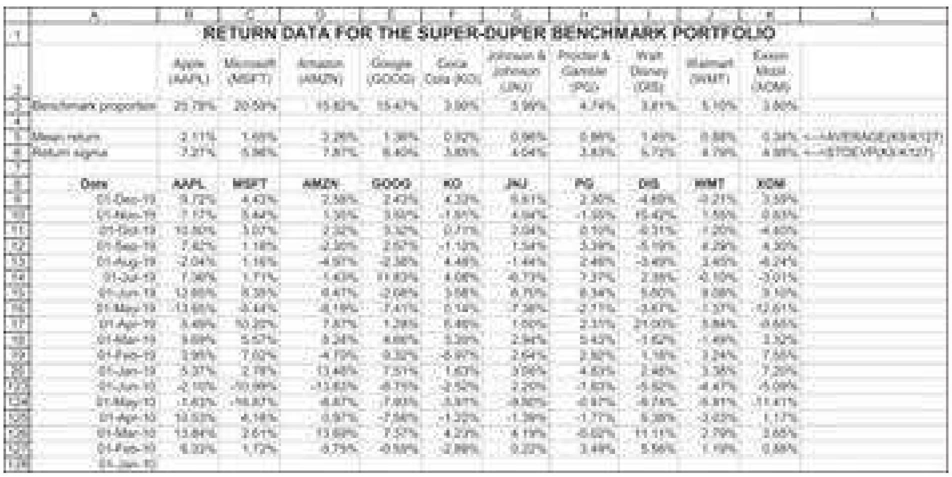


PRICE AND MARKET CAP DATA FOR 10 COMPANIES

Date	AAPL	MSFT	AMZN	GOOG	KO	JNJ	PG	DIS	WMT	XOM

Row 3 gives the current total equity value of each of the benchmark stocks, and row 4 computes the benchmark proportions—the individual equity values divided by the total market capitalization of the benchmark.

Following the procedures described in Chapters 10 and 11, Joanna first transforms the price data to returns and then computes a variance-covariance matrix for these returns:



RETURN DATA FOR THE SUPER-DUPER BENCHMARK PORTFOLIO

Date	AAPL	MSFT	AMZN	GOOG	KO	JNJ	PG	DIS	WMT	XOM

Naive Optimization

Using the return data, Joanna computes the sample variance-covariance matrix of excess returns as illustrated in Chapter 12. To implement the portfolio optimization, she needs data on the T-bill rate: The December 1, 2019, 1-month T-bill rate is 1.6% annually and 1.6%/12 = 0.133% monthly. Using the variance-covariance matrix, the T-bill rate, and the historical mean returns, Joanna computes an "optimal" portfolio by solving the equation:

$$\text{Optimal portfolio}\begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_{10} \end{bmatrix} = \frac{S^{-1}\begin{bmatrix} \overline{r}_{AAPL} - r_f \\ \overline{r}_{MSFT} - r_f \\ \vdots \\ \overline{r}_{XOM} - r_f \end{bmatrix}}{Sum\left[S^{-1}\cdot\begin{bmatrix} \overline{r}_{AAPL} - r_f \\ \overline{r}_{MSFT} - r_f \\ \vdots \\ \overline{r}_{XOM} - r_f \end{bmatrix}\right]} = \frac{S^{-1}\begin{bmatrix} \overline{r}_{AAPL} - r_f \\ \overline{r}_{MSFT} - r_f \\ \vdots \\ \overline{r}_{XOM} - r_f \end{bmatrix}}{[1,1,\ldots,1]\cdot S^{-1}\cdot\begin{bmatrix} \overline{r}_{AAPL} - r_f \\ \overline{r}_{MSFT} - r_f \\ \vdots \\ \overline{r}_{XOM} - r_f \end{bmatrix}}$$

This portfolio is shown below (highlighted):

	A	B	C	D	E	F	G	H	I	J	K
1	SUPER-DUPER BENCHMARK PORTFOLIO—NAIVE OPTIMIZATION										
2		Apple (AAPL)	Microsoft (MSFT)	Amazon (AMZN)	Google (GOOG)	Coca Cola (KO)	Johnson & Johnson (JNJ)	Procter & Gamble (PG)	Walt Disney (DIS)	Walmart (WMT)	Exxon Mobil (XOM)
3	Benchmark proportion	20.78%	20.59%	15.62%	15.47%	3.90%	5.99%	4.74%	3.81%	5.10%	3.80%
4	Mean return	2.11%	1.65%	2.26%	1.36%	0.92%	0.96%	0.86%	1.45%	0.88%	0.34%
5											
6	Variance-covariance matrix of excess returns										
7		AAPL	MSFT	AMZN	GOOG	KO	JNJ	PG	DIS	WMT	XOM
8	AAPL	0.00533	0.00174	0.00197	0.00160	0.00054	0.00056	0.00051	0.00113	0.00057	0.00084
9	MSFT	0.00174	0.00360	0.00194	0.00186	0.00062	0.00063	0.00047	0.00144	0.00052	0.00108
10	AMZN	0.00197	0.00194	0.00825	0.00239	0.00097	0.00101	-0.00009	0.00131	0.00047	0.00109
11	GOOG	0.00160	0.00186	0.00239	0.00413	0.00062	0.00055	0.00055	0.00143	0.00037	0.00094
12	KO	0.00054	0.00062	0.00097	0.00062	0.00149	0.00078	0.00058	0.00075	0.00073	0.00065
13	JNJ	0.00056	0.00063	0.00101	0.00055	0.00078	0.00165	0.00068	0.00071	0.00082	0.00102
14	PG	0.00051	0.00047	-0.00009	0.00055	0.00058	0.00068	0.00148	0.00047	0.00060	0.00050
15	DIS	0.00113	0.00144	0.00131	0.00143	0.00075	0.00071	0.00047	0.00330	0.00060	0.00133
16	WMT	0.00057	0.00052	0.00047	0.00037	0.00073	0.00082	0.00060	0.00060	0.00231	0.00056
17	XOM	0.00084	0.00108	0.00109	0.00094	0.00065	0.00102	0.00050	0.00133	0.00056	0.00251
18											
19	Current t-bill rate	0.133%	<-- =1.6%/12								
20											
21	"Optimal" portfolio										
22	AAPL	20.3%	<-- {=MMULT(MINVERSE(B8:K17),TRANSPOSE(B4:K4)-B19)/SUM(MMULT(MINVERSE(B8:K17),TRANSPOSE(B4:K4)-B19))}								
23	MSFT	14.3%									
24	AMZN	23.9%									
25	GOOG	-6.1%									
26	KO	4.0%									
27	JNJ	20.9%									
28	PG	34.6%									
29	DIS	26.4%									
30	WMT	6.5%									
31	XOM	-44.8%									
32	Sum of proportions	100%	<-- =SUM(B22:B31)								

Implementing the naive optimization in R looks like this:

```
# we thank Sagi Haim for developing this script

##################################
# Chapter 15 - Black Litterman
##################################

# Set working directory
workdir <-readline(prompt="working directory?")
setwd(workdir)

# Read the csv with adjusted prices
monthly_prices <- read.csv("Chapt_15_data.csv", row.names = 1,
                           stringsAsFactors = FALSE)

# Sort the data by date
monthly_prices <- monthly_prices[order(as.Date(row.names(monthly_prices),
                                         format="%d-%m-%y")),]

# Compute Continuous Returns
monthly_returns <- data.frame(apply(monthly_prices, 2, function(x) diff(log(x)) ))

# Returns
head(monthly_returns)

# Market Capitalization ($B)
market_cap <- c(1370, 1156, 1043, 1020, 257.156,
                395.024,  312.879,  250.902,  336.435,  250.356)
# Input: Current T-Bill rate
t_bill_r <- 0.016/12

# Compute Benchmark Proportion
bench_prop <- market_cap / sum(market_cap)

# CoputE Returns Statistics
mean_ret <- colMeans(monthly_returns)
sd_ret <- apply(monthly_returns, 2, sd)
cov_mat <- cov(monthly_returns)

# Naive Optimization
optimal_port <- (solve(cov_mat) %*% (mean_ret - t_bill_r))
                / sum( solve(cov_mat) %*% (mean_ret - t_bill_r))
```

Why Does Naive Optimization Fail?

In some sense, the strange portfolio “optimization” positions were predictable. The spreadsheet below highlights some disturbing features of the data that can partially explain the odd “optimized” portfolio.

- • Some of the historical mean returns are negative. If we ignore the effects of correlations, a negative expected return should imply a short position in the stock.[4] There is, however, a deeper philosophical question about using past returns as proxies for future expected returns: Even though the past returns are negative, there is no reason to assume that this means that the future, expected returns from a stock should be negative. This is one of the problems when we use historical data to extract anticipations about the future.
- • The correlations between asset returns are in some cases very large. Large correlations for a particular stock can lead us to prefer other stocks with smaller returns but more moderate correlations.

The next spreadsheet highlights stocks with very low historical returns and stocks whose correlations are greater than 0.5 or negative.

	Apple (AAPL)	Microsoft (MSFT)	Amazon (AMZN)	Google (GOOG)	Coca Cola (KO)	Johnson & Johnson (JNJ)	Procter & Gamble (PG)	Walt Disney (DIS)	Walmart (WMT)	Exxon Mobil (XOM)	
VERY LOW RETURNS AND HIGH/LOW CORRELATIONS											
Benchmark proportion	20.78%	20.59%	15.82%	15.47%	3.80%	3.99%	4.74%	3.81%	5.10%	3.80%	
Mean return	2.11%	1.65%	3.26%	1.36%	0.92%	0.96%	0.96%	1.45%	0.88%	0.34%	
Variance-covariance matrix of excess returns											
	AAPL	MSFT	AMZN	GOOG	KO	JNJ	PG	DIS	WMT	XOM	
AAPL	1.00000	0.39861	0.34121	0.34164	0.19139	0.19041	0.18242	0.27536	0.16150	0.23093	
MSFT	0.39861	1.00000	0.40836	0.48212	0.35267	0.34117	0.20243	0.41777	0.11862	0.30834	
AMZN	0.34121	0.40836	1.00000	0.47127	0.31464	0.31647	-0.03063	0.38796	0.12403	0.27597	
GOOG	0.34164	0.48212	0.47127	1.00000	0.30222	0.21266	0.22152	0.36793	0.11877	0.29192	[illegible]
KO	0.19139	0.35267	0.31464	0.30222	1.00000	0.49471	0.78073	0.35794	0.39223	0.33342	[illegible]
JNJ	0.19041	0.34117	0.31647	0.21266	0.49471	1.00000	0.42243	0.30569	0.42021	0.50308	[illegible]
PG	0.18242	0.20243	-0.03063	0.22152	0.78073	0.42243	1.00000	0.21138	0.32409	0.35644	
DIS	0.27536	0.41777	0.38796	0.36793	0.35794	0.30569	0.21138	1.00000	0.21599	0.46344	
WMT	0.16150	0.11862	0.12403	0.11877	0.39223	0.42021	0.32409	0.21599	1.00000	0.25274	
XOM	0.23093	0.30834	0.27597	0.29192	0.33342	0.50308	0.35644	0.46344	0.25274	1.00000	

What about Changing the Variance-Covariance Matrix?

In Chapter 12 we discussed various methods for calculating the variance-covariance matrix. If you do not feel comfortable with the historical variance-covariance matrix, you may use the constant correlation approach or the single-index approach. Another alternative is to use the shrinkage method. "Shrinkage," you will recall, is a bit of jargon for taking a convex combination of the sample variance-covariance matrix with a diagonal matrix of only the variances. Shrinkage methods have been shown to be effective at improving the performance of the global minimum variance portfolio (GMVP).

15.3 Black and Litterman's Solution to the Optimization Problem

The BL approach provides an initial solution to the optimization problem above. The BL approach is composed of two parts:

- **Step 1**: *What does the market think?* A vast amount of financial research shows that it is difficult to beat the returns of benchmark portfolios. The first step of the BL approach takes this research as a starting point. It assumes that the benchmark is ex ante optimal and derives the expected returns of each asset under this assumption. Another way of saying this is that in step 1, we compute the expected returns of the assets that would make the investor choose the benchmark using the optimization techniques in Chapter 11.
- **Step 2**: *Incorporating investor opinions.* In step 1, BL shows how to compute the benchmark asset returns based on the assumption of optimality. Suppose the investor has divergent opinions from these market-based expected returns. Step 2 shows how to incorporate these opinions into the optimization procedure. Note that —because of the correlations between asset returns—an investor's opinion about any particular asset's returns will affect all the other expected returns. A critical part in step 2 is to adjust *all* asset returns for an investor's opinion about any return.

Investors who follow the BL procedure start off by seeing what the market weights imply for the expected returns. They can then adjust these weights by adding their own opinions about any asset's expected returns. In the next two sections we discuss these two steps in detail.

15.4 BL Step 1: What Does the Market Think?

As shown in Chapter 11, an optimal portfolio must solve the equation:

$$\begin{bmatrix} \textit{Efficient} \\ \textit{portfolio} \\ \textit{proportions} \end{bmatrix} = \underbrace{\begin{bmatrix} \textit{Variance}- \\ \textit{covariance} \\ \textit{matrix} \end{bmatrix}^{-1} * \left\{ \begin{bmatrix} \textit{Expected} \\ \textit{portfolio} \\ \textit{returns} \end{bmatrix} - \begin{matrix} \textit{risk-free} \\ \textit{rate} \end{matrix} \right\}}_{\uparrow \atop \text{Normalize to sum to 1}}$$

Solving the equation for the vector of expected portfolio returns, this means that an efficient portfolio must solve the following equation:

$$\begin{bmatrix} \textit{Expected} \\ \textit{portfolio} \\ \textit{returns} \end{bmatrix} = \begin{bmatrix} \textit{Variance}- \\ \textit{covariance} \\ \textit{matrix} \end{bmatrix} \begin{bmatrix} \textit{Efficient} \\ \textit{portfolio} \\ \textit{proportions} \end{bmatrix} * \begin{matrix} \textit{Normalizing} \\ \textit{factor} \end{matrix} + \begin{matrix} \textit{risk-free} \\ \textit{rate} \end{matrix}$$

Joanna of Super-Duper Fund assumes that, in the absence of any additional knowledge or opinions about the market, the current market weights of the portfolio indicate the efficiency weights. She estimates that the expected benchmark return over the next month will be 1% and uses this estimation to set the normalizing factor.[5] We illustrate below.

Solving the first part of the last equation (without the normalizing factor) gives:

	A	B	C	D	E	F	G	H	I	J	K	L	M
1	SUPER-DUPER BENCHMARK PORTFOLIO—WHAT DOES THE MARKET THINK? No normalizing factor												
2	Anticipated benchmark return	1.00%	<--=12%/12										
3	Current T-bill rate	0.13%											
4													
5		Apple (AAPL)	Microsoft (MSFT)	Amazon (AMZN)	Google (GOOG)	Coca Cola (KO)	Johnson & Johnson (JNJ)	Procter & Gamble (PG)	Walt Disney (DIS)	Walmart (WMT)	Exxon Mobil (XOM)		
6	Market capitalization (billion $)	1,370.00	1,356.00	1,043.00	1,020.00	207.36	356.02	312.58	250.90	308.43	250.36		
7	Benchmark proportions	20.79%	20.58%	15.82%	15.47%	3.60%	5.39%	4.74%	3.81%	5.10%	3.80%	<--=K6/SUM($B$6:$K$6)	
8													
9	Variance-covariance matrix												
10		AAPL	MSFT	AMZN	GOOG	KO	JNJ	PG	DIS	WMT	XOM		Without normalizing factor
11	AAPL	0.0053	0.0017	0.0020	0.0016	0.0005	0.0006	0.0005	0.0011	0.0006	0.0008		0.22%
12	MSFT	0.0017	0.0038	0.0019	0.0019	0.0008	0.0008	0.0005	0.0014	0.0005	0.0011		0.19%
13	AMZN	0.0020	0.0019	0.0063	0.0024	0.0010	0.0010	-0.0001	0.0013	0.0005	0.0011		0.24%
14	GOOG	0.0016	0.0019	0.0024	0.0041	0.0008	0.0006	0.0005	0.0014	0.0004	0.0009		0.20%
15	KO	0.0005	0.0008	0.0010	0.0008	0.0015	0.0008	0.0006	0.0008	0.0007	0.0006		0.08%
16	JNJ	0.0006	0.0008	0.0010	0.0006	0.0008	0.0019	0.0007	0.0007	0.0006	0.0010		0.08%
17	PG	0.0005	0.0005	-0.0001	0.0005	0.0006	0.0007	0.0015	0.0005	0.0006	0.0005		0.05%
18	DIS	0.0011	0.0014	0.0013	0.0014	0.0008	0.0007	0.0005	0.0035	0.0006	0.0013		0.13%
19	WMT	0.0006	0.0005	0.0005	0.0004	0.0007	0.0006	0.0006	0.0006	0.0023	0.0006		0.08%
20	XOM	0.0008	0.0011	0.0011	0.0009	0.0006	0.0010	0.0005	0.0013	0.0006	0.0025		0.10%
21													
22	Check: The expected return of the benchmark?	0.18%	<-- {=MMULT(B7:K7,M11:M20)}										
23													
24													
25													
26	Additional check Optimal portfolio												
27	GM	-47.22%	<-- {=MMULT(MINVERSE(B11:K20),M11:M20-B3)/SUM(MMULT(MINVERSE(B11:K20),M11:M20-B3))}										
28	HD	-32.68%											
29	IP	18.80%											
30	HPQ	22.29%											
31	MO	52.57%											
32	AXP	17.64%											
33	AA	60.58%											
34	DD	4.71%											
35	MRK	25.52%											
36	MMM	10.88%											
37	Sum of proportions	100.0%	<--=SUM(B27:B36)										

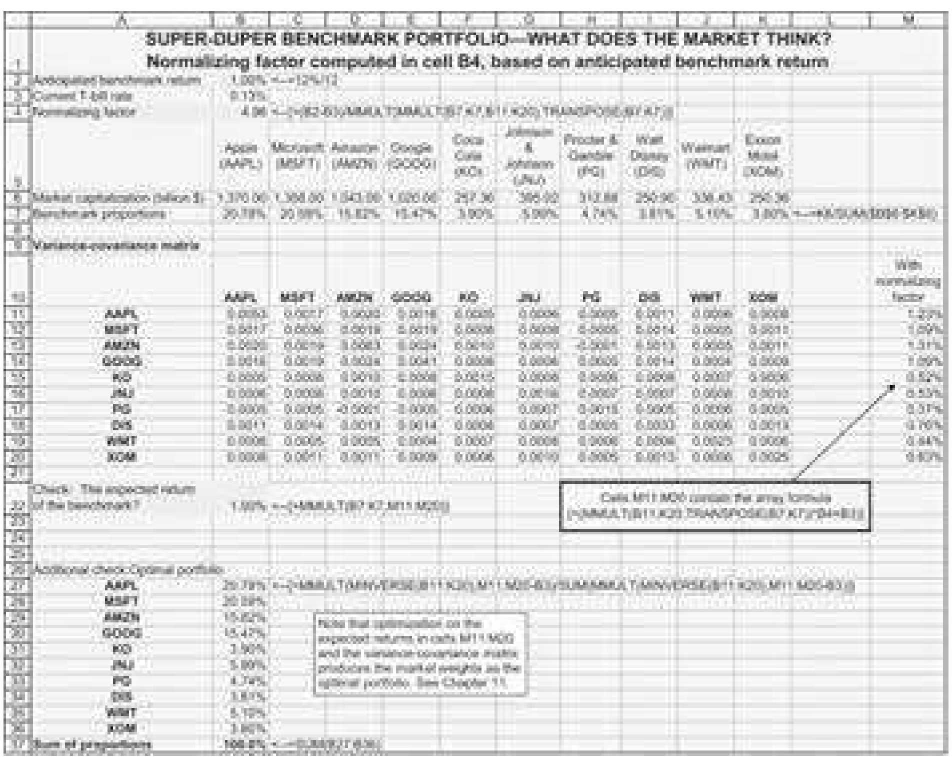
Cells M11:M20 contain the array formula {=MMULT(B11:K20,TRANSPOSE(B7:K7)-B3)}

Note that given the weights in row 7, the expected benchmark return is 0.18% per month as opposed to the 1% per month posited by Joanne (cell B2). To achieve this expected benchmark return, we multiply row 7 by a normalizing factor:

$$\begin{bmatrix} \textit{Benchmark} \\ \textit{portfolio} \\ \textit{returns} \end{bmatrix} = \begin{bmatrix} \textit{Variance}- \\ \textit{covariance} \\ \textit{matrix} \end{bmatrix} \begin{bmatrix} \textit{Benchmark} \\ \textit{portfolio} \\ \textit{proportions} \end{bmatrix} * \begin{matrix} \textit{Normalizing} \\ \textit{factor} \end{matrix} + \begin{matrix} \textit{risk-free} \\ \textit{rate} \end{matrix}$$

$$= \begin{bmatrix} \textit{Variance} \\ \textit{covariance} \\ \textit{matrix} \end{bmatrix} \begin{bmatrix} \textit{Benchmark} \\ \textit{portfolio} \\ \textit{proportions} \end{bmatrix} * \underbrace{\left(\frac{\begin{matrix} \textit{Expected} \\ \textit{benchmark} \\ \textit{return} \end{matrix} - \begin{matrix} \textit{risk-free} \\ \textit{rate} \end{matrix}}{\left(\begin{bmatrix} \textit{Benchmark} \\ \textit{portfolio} \\ \textit{proportions} \end{bmatrix}^T \begin{bmatrix} \textit{Variance}- \\ \textit{covariance} \\ \textit{matrix} \end{bmatrix} \begin{bmatrix} \textit{Benchmark} \\ \textit{portfolio} \\ \textit{proportions} \end{bmatrix} \right)} \right)}_{\text{The normalizing factor}} + \begin{matrix} \textit{risk-free} \\ \textit{rate} \end{matrix}$$

This procedure is illustrated below:



We have performed an additional check by deriving the optimal portfolio given the current T-bill rate of 0.13% and the expected returns in M11:M20. This should give us back the benchmark proportions in row 7—and it does!

Implementing the above analysis in R would look like this:

```
## Optimization with Normalizing
# Input
bench_ret <- 0.12/12 # anticipated benchmark return
t_bill_r <- 0.016/12 # Current T-Bill rate

# Normalizing Factor
norm_fact <- (bench_ret - t_bill_r) / (bench_prop %*% cov_mat %*% bench_prop)

# Market Expected Returns
exp_ret <- cov_mat %*% bench_prop * as.double(norm_fact) + t_bill_r
bench_prop %*% exp_ret # check
```

15.5 BL Step 2: Introducing Opinions—What Does Joanna Think?

Having made two assumptions—first, that the benchmark is efficient, and second, that the expected benchmark return is 1% per month—Joanna has derived the expected returns for each of the benchmark components (cells M11:M20 above). We are now ready to introduce Joanna's opinions about asset returns. The rough idea is that if she disagrees with a market return, she can use the optimization procedure from Chapter 11 to derive a portfolio whose proportions differ from those of the benchmark.

We have to be careful. Because asset returns are correlated, any opinion Joanna has about one asset's returns will translate to an opinion about all other asset returns. To illustrate this point, suppose that Joanna thinks that the return on Google (GOOG) will be 0.7% over the next month instead of the market opinion of 1.09%. Then this translates to:

	A	B	C	D	E	F	G	H	I	J	K
1	ADJUSTING THE BENCHMARK FOR AN ANALYST'S OPINION In this example the only opinion is about GOOG										
2	Anticipated benchmark return	1.00%	<-- =12%/12								
3	Current T-bill rate	0.13%									
4	Normalizing factor	4.96	<-- {=(B2-B3)/MMULT(MMULT(B7:K7,B11:K20),TRANSPOSE(B7:K7))}								
5		Apple (AAPL)	Microsoft (MSFT)	Amazon (AMZN)	Google (GOOG)	Coca Cola (KO)	Johnson & Johnson (JNJ)	Procter & Gamble (PG)	Walt Disney (DIS)	Walmart (WMT)	Exxon Mobil (XOM)
6	Market capitalization (billion $)	1,370.00	1,359.00	1,043.00	1,020.00	257.36	395.02	312.68	250.50	336.43	250.30
7	Benchmark proportions	20.78%	20.59%	15.82%	15.47%	3.90%	5.99%	4.74%	3.81%	5.10%	3.80%
8											
9	Variance-covariance matrix										
10		AAPL	MSFT	AMZN	GOOG	KO	JNJ	PG	DIS	WMT	XOM
11	AAPL	0.0053	0.0017	0.0020	0.0016	0.0005	0.0006	0.0005	0.0011	0.0006	0.0008
12	MSFT	0.0017	0.0036	0.0019	0.0019	0.0008	0.0008	0.0005	0.0014	0.0005	0.0011
13	AMZN	0.0020	0.0019	0.0063	0.0024	0.0010	0.0010	-0.0001	0.0013	0.0005	0.0011
14	GOOG	0.0016	0.0019	0.0024	0.0041	0.0008	0.0006	0.0005	0.0014	0.0004	0.0009
15	KO	0.0005	0.0008	0.0010	0.0008	0.0015	0.0009	0.0006	0.0008	0.0007	0.0006
16	JNJ	0.0006	0.0008	0.0010	0.0006	0.0009	0.0016	0.0007	0.0007	0.0008	0.0010
17	PG	0.0005	0.0005	-0.0001	0.0005	0.0006	0.0007	0.0015	0.0005	0.0006	0.0005
18	DIS	0.0011	0.0014	0.0013	0.0014	0.0008	0.0007	0.0005	0.0033	0.0006	0.0013
19	WMT	0.0006	0.0005	0.0005	0.0004	0.0007	0.0008	0.0006	0.0006	0.0023	0.0006
20	XOM	0.0008	0.0011	0.0011	0.0009	0.0006	0.0010	0.0005	0.0013	0.0006	0.0025
21											
22		Expected benchmark returns, no opinions	Analyst opinion, delta, δ	Returns adjusted for opinions						Opinion-adjusted optimized portfolio	Portfolio: benchmark, no opinions
23	AAPL	1.23%	0.00%	1.50%	<-- =B23+$C$26*E11/$E$14				AAPL	15.48%	20.78%
24	MSFT	1.09%	0.00%	1.40%	<-- =B24+$C$26*E12/$E$14				MSFT	15.34%	20.59%
25	AMZN	1.31%	0.00%	1.72%	<-- =B25+$C$26*E13/$E$14				AMZN	11.78%	15.82%
26	GOOG	1.09%	0.70%	1.79%					GOOG	37.02%	15.47%
27	KO	0.82%	0.00%	0.86%		Cells J23:J32 contain the array formula {=MMULT(MINVERSE(B11:K20),D23:D32-B3)/SUM(MMULT(MINVERSE(B11:K20),D23:D32-B3))}			KO	2.91%	3.90%
28	JNJ	0.53%	0.00%	0.62%					JNJ	4.46%	5.99%
29	PG	0.37%	0.00%	0.46%					PG	3.54%	4.74%
30	DIS	0.78%	0.00%	1.00%					DIS	2.64%	3.81%
31	WMT	0.44%	0.00%	0.50%					WMT	3.80%	5.10%
32	XOM	0.63%	0.00%	0.79%					XOM	2.83%	3.80%

The δ introduced in cell C26 above indicates Joanna's deviation from the BL base case. In the example above, Joanna thinks that GOOG will differ from the market's return of 1.09% (cell B26) by $\delta_{GOOG} = 0.7\%$. What this example illustrates is that because asset returns are correlated through the variance-covariance matrix, an opinion about one asset's returns (in this case GOOG) also affects Joanna's anticipated returns for all other assets. Because of the covariance between asset returns, for example, this means that return she expects for Apple (AAPL) is 1.5%, and so on:

$$r_{AAPL,opinion\ adjusted} = r_{AAPL,market} + \frac{Cov(r_{AAPL}, r_{GOOG})}{Var(r_{GOOG})}\delta_{GOOG} = 1.50\%$$

$$r_{MSFT,opinion\ adjusted} = r_{MSFT,market} + \frac{Cov(r_{MSFT}, r_{GOOG})}{Var(r_{GOOG})}\delta_{GOOG} = 1.40\%$$

The newly optimized portfolio is given in cells J23:J32. Joanna's positive opinion about GOOG returns has, predictably, increased the proportion of GOOG in her portfolio. But her opinion about GOOG has also affected all the other portfolio weights:

	I	J	K
22		Opinion-adjusted optimized portfolio	Portfolio benchmark, no opinions
23	AAPL	15.48%	20.78%
24	MSFT	15.34%	20.59%
25	AMZN	11.79%	15.82%
26	GOOG	37.02%	15.47%
27	KO	2.91%	3.90%
28	JNJ	4.46%	5.99%
29	PG	3.54%	4.74%
30	DIS	2.84%	3.81%
31	WMT	3.80%	5.10%
32	XOM	2.83%	3.80%

Implementing the above in R looks like this:

```
# Input: Analyst opinion - Delta
delta_GOOG <- 0.007

# Returns adjusted for opinions
opn_adj_ret <- exp_ret + delta_GOOG * cov_mat[,"GOOG"] / cov_mat["GOOG", "GOOG"]

# Opinion-adjusted optimized portfolio
opn_adj_prop <- solve(cov_mat) %*% (opn_adj_ret - t_bill_r) /
                sum(solve(cov_mat) %*% (opn_adj_ret - t_bill_r))
```

The BL Tracking Matrix

When Joanna has opinions about multiple stocks, the effect of these opinions on other stocks looks like a multivariate regression:

$$r_{AAPL,adjusted} = r_{AAPL,market} + \frac{Cov(r_{AAPL}, r_{AAPL})}{Var(r_{AAPL})}\delta_{AAPL} + \frac{Cov(r_{AAPL}, r_{MSFT})}{Var(r_{MSFT})}\delta_{MSFT} + \cdots + \frac{Cov(r_{AAPL}, r_{XOM})}{Var(r_{XOM})}\delta_{XOM}$$

$$r_{MSFT,adjusted} = r_{MSFT,market} + \frac{Cov(r_{MSFT}, r_{AAPL})}{Var(r_{AAPL})}\delta_{AAPL} + \frac{Cov(r_{MSFT}, r_{MSFT})}{Var(r_{MSFT})}\delta_{MSFT} + \cdots + \frac{Cov(r_{MSFT}, r_{XOM})}{Var(r_{XOM})}\delta_{XOM}$$

...

$$r_{XOM,adjusted} = r_{XOM,market} + \frac{Cov(r_{XOM}, r_{AAPL})}{Var(r_{AAPL})}\delta_{AAPL} + \frac{Cov(r_{XOM}, r_{MSFT})}{Var(r_{MSFT})}\delta_{MSFT} + \cdots + \frac{Cov(r_{XOM}, r_{XOM})}{Var(r_{XOM})}\delta_{XOM}$$

We define the BL tracking matrix as:

$$BL - Tracking = \begin{bmatrix} \frac{Cov(r_{AAPL}, r_{AAPL})}{Var(r_{AAPL})} & \frac{Cov(r_{AAPL}, r_{MSFT})}{Var(r_{MSFT})} & \cdots & \frac{Cov(r_{AAPL}, r_{XOM})}{Var(r_{XOM})} \\ \frac{Cov(r_{MSFT}, r_{AAPL})}{Var(r_{AAPL})} & \frac{Cov(r_{MSFT}, r_{MSFT})}{Var(r_{MSFT})} & \cdots & \frac{Cov(r_{MSFT}, r_{XOM})}{Var(r_{XOM})} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{Cov(r_{XOM}, r_{AAPL})}{Var(r_{AAPL})} & \frac{Cov(r_{XOM}, r_{MSFT})}{Var(r_{MSFT})} & \cdots & \frac{Cov(r_{XOM}, r_{XOM})}{Var(r_{XOM})} \end{bmatrix}$$

$$= \begin{bmatrix} 1 & \frac{\sigma_{AAPL,MSFT}}{\sigma^2_{MSFT}} & \cdots & \frac{\sigma_{AAPL,XOM}}{\sigma^2_{XOM}} \\ \frac{\sigma_{MSFT,AAPL}}{\sigma^2_{AAPL}} & 1 & \cdots & \frac{\sigma_{MSFT,XOM}}{\sigma^2_{XOM}} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\sigma_{XOM,AAPL}}{\sigma^2_{AAPL}} & \frac{\sigma_{XOM,MSFT}}{\sigma^2_{MSFT}} & \cdots & 1 \end{bmatrix}$$

The opinion-adjusted returns are then:

$$\begin{bmatrix} r_{AAPL,market} \\ r_{MSFT,market} \\ \vdots \\ r_{XOM,market} \end{bmatrix} + \begin{bmatrix} 1 & \frac{\sigma_{APPL,MSFT}}{\sigma^2_{MSFT}} & \cdots & \frac{\sigma_{AAPL,XOM}}{\sigma^2_{XOM}} \\ \frac{\sigma_{MSFT,APPL}}{\sigma^2_{AAPL}} & 1 & \cdots & \frac{\sigma_{MSFT,XOM}}{\sigma^2_{XOM}} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\sigma_{XOM,AAPL}}{\sigma^2_{AAPL}} & \frac{\sigma_{XOM,MSFT}}{\sigma^2_{MSFT}} & \cdots & 1 \end{bmatrix} * \begin{bmatrix} \delta_{AAPL} \\ \delta_{MSFT} \\ \vdots \\ \delta_{XOM} \end{bmatrix}$$

$$= \begin{bmatrix} r_{AAPL,opinion-adj.} \\ r_{MSFT,market,opinion-adj.} \\ \vdots \\ r_{XOM,opinton-adj.} \end{bmatrix}$$

The VBA function **BLtracking** takes the variance-covariance matrix as its argument and is defined as:

```
'BLtracking's argument is the variance-covariance matrix
Function BLtracking(rng As Range) As Variant
    Dim i As Integer
    Dim j As Integer
    Dim numcols As Integer
    numcols = rng.Columns.Count
    Dim matrix() As Double
    ReDim matrix(numcols - 1, numcols - 1)
    For i = 1 To numcols
        For j = 1 To numcols
            matrix(i - 1, j - 1) = rng(i, j) / rng(j, j)
        Next j
    Next i
    BLtracking = matrix
End Function
```

Here are the computations for our example:

BLACK-LITTERMAN TRACKING MATRIX

Tracking factors = cov(i,j)/var(i)

Variance-covariance matrix

	AAPL	MSFT	AMZN	GOOG	KO	JNJ	PG	DIS	WMT	XOM
AAPL	0.0053	0.0017	0.0020	0.0016	0.0005	0.0006	0.0005	0.0011	0.0006	0.0008
MSFT	0.0017	0.0036	0.0019	0.0019	0.0008	0.0008	0.0005	0.0014	0.0005	0.0011
AMZN	0.0020	0.0019	0.0063	0.0024	0.0010	0.0010	-0.0001	0.0013	0.0005	0.0011
GOOG	0.0016	0.0019	0.0024	0.0041	0.0008	0.0006	0.0005	0.0014	0.0004	0.0009
KO	0.0005	0.0008	0.0010	0.0008	0.0015	0.0008	0.0006	0.0008	0.0007	0.0006
JNJ	0.0006	0.0008	0.0010	0.0006	0.0008	0.0016	0.0007	0.0007	0.0008	0.0010
PG	0.0005	0.0005	-0.0001	0.0005	0.0006	0.0007	0.0015	0.0005	0.0006	0.0005
DIS	0.0011	0.0014	0.0013	0.0014	0.0008	0.0007	0.0005	0.0033	0.0006	0.0013
WMT	0.0006	0.0005	0.0005	0.0004	0.0007	0.0008	0.0006	0.0006	0.0023	0.0006
XOM	0.0008	0.0011	0.0011	0.0009	0.0006	0.0010	0.0005	0.0013	0.0006	0.0025
Variance	0.0053	0.0036	0.0063	0.0041	0.0015	0.0016	0.0015	0.0033	0.0023	0.0025

BL tracking matrix (direct)

	AAPL	MSFT	AMZN	GOOG	KO	JNJ	PG	DIS	WMT	XOM	
AAPL	1.0000	0.4826	0.3151	0.3883	0.3616	0.3427	0.3464	0.3435	0.2453	0.3368	<--=K4/K$15
MSFT	0.3260	1.0000	0.3099	0.4503	0.5474	0.5047	0.3159	0.4363	0.2230	0.4296	<--=K5/K$15
AMZN	0.3695	0.5380	1.0000	0.5799	0.6478	0.6168	-0.0630	0.3954	0.2040	0.4358	<--=K6/K$15
GOOG	0.3006	0.5162	0.3830	1.0000	0.5523	0.3368	0.3702	0.4337	0.1587	0.3746	
KO	0.1013	0.2271	0.1548	0.1998	1.0000	0.4713	0.3947	0.2273	0.3153	0.2574	
JNJ	0.1058	0.2306	0.1624	0.1343	0.5193	1.0000	0.4456	0.2158	0.3546	0.4076	
PG	0.0961	0.1297	-0.0149	0.1326	0.3907	0.4004	1.0000	0.1414	0.2594	0.1993	
DIS	0.2126	0.4001	0.2089	0.3470	0.5025	0.4330	0.3159	1.0000	0.2582	0.5320	
WMT	0.1063	0.1431	0.0754	0.0889	0.4879	0.4980	0.4054	0.1807	1.0000	0.2235	
XOM	0.1583	0.2989	0.1747	0.2275	0.4319	0.6209	0.3378	0.4037	0.2424	1.0000	<--=K13/K$15

Cells B20:K29 contain the formula =B4/B$15

BL tracking matrix (vba)

	AAPL	MSFT	AMZN	GOOG	KO	JNJ	PG	DIS	WMT	XOM
AAPL	1.0000	0.4826	0.3151	0.3883	0.3616	0.3427	0.3464	0.3435	0.2453	0.3368
MSFT	0.3260	1.0000	0.3099	0.4503	0.5474	0.5047	0.3159	0.4363	0.2230	0.4296
AMZN	0.3695	0.5380	1.0000	0.5799	0.6478	0.6168	-0.0630	0.3954	0.2040	0.4358
GOOG	0.3006	0.5162	0.3830	1.0000	0.5523	0.3368	0.3702	0.4337	0.1587	0.3746
KO	0.1013	0.2271	0.1548	0.1998	1.0000	0.4713	0.3947	0.2273	0.3153	0.2574
JNJ	0.1058	0.2306	0.1624	0.1343	0.5193	1.0000	0.4456	0.2158	0.3546	0.4076
PG	0.0961	0.1297	-0.0149	0.1326	0.3907	0.4004	1.0000	0.1414	0.2594	0.1993
DIS	0.2126	0.4001	0.2089	0.3470	0.5025	0.4330	0.3159	1.0000	0.2582	0.5320
WMT	0.1063	0.1431	0.0754	0.0889	0.4879	0.4980	0.4054	0.1807	1.0000	0.2235
XOM	0.1583	0.2989	0.1747	0.2275	0.4319	0.6209	0.3378	0.4037	0.2424	1.0000

Cells B34:K43 contain the array formula {=bltracking(B4:K13)}

We can now use the tracking matrix to discuss the situation where there is more than one opinion.

Two or More Opinions

Suppose Joanna believes the GOOG monthly return will be 0.7% higher than its market return of 1.09% and the monthly return for Walt Disney (DIS) will be 0.4% lower than its market expected return of 0.76%. Joanna also believes that the expected returns on all other assets other than GOOG and DIS do not need corrections.

How can we reflect Joanna's opinions? We use the **BLtracking** matrix to compute the unique set of deltas that solves for the expected returns:

$$
\begin{bmatrix} r_{APPL,opinion-adj.} \\ r_{MSFT,market,option-adj.} \\ \vdots \\ r_{XOM,opinion-adj.} \end{bmatrix} = \begin{bmatrix} r_{AAPL,market} \\ r_{MSFT,market} \\ \vdots \\ r_{XOM,market} \end{bmatrix} + \begin{bmatrix} Black-Litterman \\ tracking \\ Matrix \end{bmatrix} * \begin{bmatrix} \delta_{AAPL} = 0 \\ \vdots \\ \delta_{GOOG} = 0.7\% \\ \vdots \\ \delta_{DIS} = -0.4\% \\ \vdots \\ \delta_{AAPL} = 0 \end{bmatrix}
$$

Here's the implementation in Excel:

	A	B	C	D	E	F	G	H	I	J	K
1	**ADJUSTING THE BENCHMARK FOR AN ANALYST'S OPINION** Multiple opinions: Accepting Joanna's deltas, but letting them influence all the returns										
2	Anticipated benchmark return	1.00%	<-- =12%/12								
3	Current T-bill rate	0.13%									
4	Normalizing factor	0.02	<-- {=(B2-B3)/MMULT(MMULT(B7:K7,B24:K33),TRANSPOSE(B7:K7))}								
5		Apple (AAPL)	Microsoft (MSFT)	Amazon (AMZN)	Google (GOOG)	Coca Cola (KO)	Johnson & Johnson (JNJ)	Procter & Gamble (PG)	Walt Disney (DIS)	Walmart (WMT)	Exxon Mobil (XOM)
6	Market capitalization (billion $)	1,370.00	1,358.00	1,043.00	1,020.00	257.36	395.02	312.88	250.90	336.43	256.36
7	Benchmark proportions	20.78%	20.59%	15.82%	15.47%	3.90%	5.99%	4.74%	3.81%	5.10%	3.89%
8											
9	**Variance-covariance matrix**										
10		**AAPL**	**MSFT**	**AMZN**	**GOOG**	**KO**	**JNJ**	**PG**	**DIS**	**WMT**	**XOM**
11	**AAPL**	0.0053	0.0017	0.0020	0.0016	0.0005	0.0006	0.0005	0.0011	0.0006	0.0008
12	**MSFT**	0.0017	0.0036	0.0019	0.0019	0.0008	0.0008	0.0005	0.0014	0.0005	0.0011
13	**AMZN**	0.0020	0.0019	0.0063	0.0024	0.0010	0.0010	-0.0001	0.0013	0.0005	0.0011
14	**GOOG**	0.0016	0.0019	0.0024	0.0041	0.0008	0.0006	0.0005	0.0014	0.0004	0.0009
15	**KO**	0.0005	0.0008	0.0010	0.0008	0.0015	0.0008	0.0008	0.0008	0.0007	0.0006
16	**JNJ**	0.0006	0.0008	0.0010	0.0006	0.0008	0.0016	0.0007	0.0007	0.0008	0.0010
17	**PG**	0.0005	0.0005	-0.0001	0.0005	0.0008	0.0007	0.0015	0.0005	0.0008	0.0005
18	**DIS**	0.0011	0.0014	0.0013	0.0014	0.0008	0.0007	0.0005	0.0033	0.0006	0.0013
19	**WMT**	0.0006	0.0005	0.0005	0.0004	0.0007	0.0006	0.0008	0.0006	0.0023	0.0006
20	**XOM**	0.0008	0.0011	0.0011	0.0009	0.0006	0.0010	0.0005	0.0013	0.0006	0.0025
21											
22	**Black-Litterman tracking matrix**										
23		**AAPL**	**MSFT**	**AMZN**	**GOOG**	**KO**	**JNJ**	**PG**	**DIS**	**WMT**	**XOM**
24	**AAPL**	1.0000	0.4826	0.3151	0.3883	0.3610	0.3427	0.3464	0.3435	0.2453	0.3368
25	**MSFT**	0.3260	1.0000	0.3099	0.4803	0.5474	0.5047	0.3159	0.4363	0.2230	0.4296
26	**AMZN**	0.3695	0.5380	1.0000	0.5799	0.6479	0.6168	-0.0836	0.3954	0.2040	0.4358
27	**GOOG**	0.3006	0.5162	0.3830	1.0000	0.5523	0.3368	0.3702	0.4337	0.1587	0.3746
28	**KO**	0.1013	0.2271	0.1548	0.1998	1.0000	0.4713	0.3947	0.2273	0.3153	0.2574
29	**JNJ**	0.1058	0.2306	0.1624	0.1343	0.5193	1.0000	0.4456	0.2158	0.3549	0.4076
30	**PG**	0.0961	0.1297	-0.0149	0.1326	0.3807	0.4094	1.0000	0.1414	0.2594	0.1993
31	**DIS**	0.2128	0.4001	0.2089	0.3470	0.5025	0.4330	0.3159	1.0000	0.2582	0.5320
32	**WMT**	0.1063	0.1431	0.0754	0.0889	0.4879	0.4980	0.4054	0.1807	1.0000	0.2236
33	**XOM**	0.1563	0.2989	0.1747	0.2275	0.4319	0.6209	0.3378	0.4037	0.2424	1.0000
34											
35		Expected benchmark returns, no opinions	Analyst opinion, delta, δ	Returns adjusted for opinions						Opinion-adjusted optimized portfolio	Portfolio benchmark, no opinions
36	**AAPL**	1.23%	0.00%	1.36%	<-- {=B36:B45+MMULT(B24:K33,C36:C45)}				**AAPL**	18.93%	20.78%
37	**MSFT**	1.09%	0.00%	1.23%	<-- {=B36:B45+MMULT(B24:K33,C36:C45)}				**MSFT**	18.78%	20.59%
38	**AMZN**	1.31%	0.00%	1.56%	<-- {=B36:B45+MMULT(B24:K33,C36:C45)}				**AMZN**	14.41%	15.82%
39	**GOOG**	1.09%	0.70%	1.82%					**GOOG**	45.28%	15.47%
40	**KO**	0.52%	0.00%	0.57%		Cells J36:J45 contain the array formula {=MMULT(MINVERSE(B11:K20),D36:D45-B3)/SUM(MMULT(MINVERSE(B11:K20),D36:D45-B3))}			**KO**	3.58%	3.90%
41	**JNJ**	0.53%	0.00%	0.54%					**JNJ**	5.48%	5.99%
42	**PG**	0.37%	0.00%	0.40%					**PG**	4.32%	4.74%
43	**DIS**	0.76%	-0.40%	0.60%					**DIS**	-18.80%	3.81%
44	**WMT**	0.44%	0.00%	0.43%					**WMT**	4.65%	5.10%
45	**XOM**	0.83%	0.00%	0.83%					**XOM**	3.46%	3.89%

Here are three comments:

- • The δs in C36:C45 are computed so that the expected returns of GOOG and DIS reflect Joanna's opinions and so that the expected returns of all other assets are changed based of the BL's tracking matrix.
- • The revised optimal portfolio—given the two opinions about GOOG and DID—is computed in J36:J45.

And here is the implementation in R:

```
# The Black-Litterman tracking matrix
track_mat <- t( cov_mat / diag(cov_mat) )

## Two or more opinions
# Input: Analyst opinion - Delta ARRAY
delta_port <- c(0.000, 0.000, 0.000, 0.007, 0.000, 0.000, 0.000, -0.004, 0.000, 0.000)

# Returns adjusted for opinions
opn_adj_ret <- exp_ret + track_mat %*% delta_port

# Opinion-adjusted optimized portfolio
opn_adj_prop <- solve(cov_mat) %*% (opn_adj_ret - t_bill_r) /
                sum(solve(cov_mat) %*% (opn_adj_ret - t_bill_r))
```

Do You Believe in Your Opinions?

Do we really believe our own opinions? Do we actually have confidence in what we believe? There is a whole theory of Bayesian adjustments to our beliefs, which is explained in Theil (1971). An application to portfolio modeling can be found in Black and Litterman (1991) and other associated papers (e.g., Idzorek 2007). This author finds these papers dauntingly complicated and difficult to implement. A simpler approach to the confidence question is to form a portfolio based on a convex combination of the market weights and the opinion-adjusted weights:

$$\textit{Portfolio proportions} = (1-\gamma) * \begin{matrix}\textit{Market}\\ \textit{weights}\end{matrix} + \gamma * \begin{matrix}\textit{Opinion}-\textit{adjusted}\\ \textit{weights}\end{matrix}$$

Here, γ is our degree of confidence in our opinions. An application to our last example is given below:

	A	B	C	D	E	F
35			γ, opinion confidence		0.8	<-- Weight attached to analyst opinion
36		Expected benchmark returns, no opinions	Analyst opinion, delta, δ	Returns adjusted for opinions	Returns adjusted for opinions and confidence	
37	**AAPL**	1.23%	0.00%	1.36%	1.31%	<--=$E$35*D37+(1-$E$35)*B37
38	**MSFT**	1.09%	0.00%	1.23%	1.17%	<--=$E$35*D38+(1-$E$35)*B38
39	**AMZN**	1.31%	0.00%	1.56%	1.46%	
40	**GOOG**	1.09%	0.70%	1.62%	1.41%	
41	**KO**	0.52%	0.00%	0.57%	0.55%	
42	**JNJ**	0.53%	0.00%	0.54%	0.54%	
43	**PG**	0.37%	0.00%	0.40%	0.39%	
44	**DIS**	0.76%	-0.40%	0.60%	0.66%	
45	**WMT**	0.44%	0.00%	0.43%	0.44%	
46	**XOM**	0.63%	0.00%	0.63%	0.63%	
47						
48			**Optimal portfolio**			
49		Benchmark, no opinions		Opinion-adjusted	Opinion and confidence adjusted	
50	**AAPL**	20.78%		18.93%	19.63%	Cells E50:E59 contain the array formula: {=MMULT(MINVERSE(B11:K20),E37:E46-B3)/SUM(MMULT(MINVERSE(B11:K20),E37:E46-B3))}
51	**MSFT**	20.59%		18.76%	19.45%	
52	**AMZN**	15.82%		14.41%	14.94%	
53	**GOOG**	15.47%		45.26%	34.00%	
54	**KO**	3.90%		3.56%	3.69%	
55	**JNJ**	5.99%		5.46%	5.66%	
56	**PG**	4.74%		4.32%	4.48%	
57	**DIS**	3.81%		-18.80%	-10.26%	
58	**WMT**	5.10%		4.65%	4.82%	
59	**XOM**	3.80%		3.46%	3.59%	

And here is the R implementation of the above example:

```
# Input: Opinion Confidence
conf <- 0.6

# Returns adjusted for opinions and confidence
conf_adj_ret <- conf * opn_adj_ret + ( 1 - conf ) * exp_ret

# Opinion and confidence adjusted
conf_adj_prop <- solve(cov_mat) %*% (conf_adj_ret - t_bill_r) /
                 sum(solve(cov_mat) %*% (conf_adj_ret - t_bill_r))
```

15.6 Using BL for International Asset Allocation[6]

We end this chapter by implementing the BL model on data for five international indices. The spreadsheet below gives data on five major world stock market indices:

- • The SP500, a value-weighted index of the 500 largest U.S. stocks.
- • The Morgan Stanley Capital International (MSCI) World Ex-US Index, which is composed of 21 developed countries based on gross domestic product (GDP) per capita.
- • The Russell 2000 Index, which consists of the 2,000 smallest companies in the Russell 3000 Index (which is market cap weighted and captures about 98% of the investable U.S. marketplace).
- • The MSCI Emerging Markets Index, consisting of indices for 26 emerging economies.
- • The total bond market index. This Vanguard exchange traded fund (ETF) consists of the entire investment-grade bond market, with holdings in T-bills, corporates, mortgage-backed securities (MBS), and agency bonds. It holds securities of all maturity lengths and is heavily weighted toward the short end of the curve.

INDEX DATA, 2015-2020

5 years ending December 2020

Correlation	SP500	MSCI World ex-US	Russell 2000	MSCI Emerging	Total Bond Market		Weight	Standard deviation
SP500	1.0000	0.9461	0.7834	0.7103	-0.1118		30%	11.90%
MSCI World Ex-US	0.9461	1.0000	0.6977	0.8326	-0.0461		20%	12.18%
Russell 2000	0.7834	0.6977	1.0000	0.4878	-0.2807		4%	14.42%
MSCI Emerging	0.7103	0.8326	0.4878	1.0000	0.0949		4%	16.48%
Total Bond Market	-0.1118	-0.0461	-0.2807	0.0949	1.0000		42%	3.35%
							100%	

Column H gives the value weights on each index in a composite portfolio as of end of December 2020, and column I gives the standard deviation of each index.

The Variance-Covariance Matrix

We use Excel array functions (see Chapter 31) to compute the variance-covariance matrix for the five indices from the correlation matrix above:

	A	B	C	D	E	F	G	H	I
1	INDEX DATA, 2015-2020								
2	5 years ending December 2020								
3	Correlation	SP500	MSCI World ex-US	Russell 2000	MSCI Emerging	Total Bond Market		Weight	Standard deviation
4	SP500	1.0000	0.9461	0.7834	0.7103	-0.1118		30%	11.90%
5	MSCI World Ex-US	0.9461	1.0000	0.6977	0.8328	-0.0461		20%	12.18%
6	Russell 2000	0.7834	0.6977	1.0000	0.4878	-0.2807		4%	14.42%
7	MSCI Emerging	0.7103	0.8328	0.4878	1.0000	0.0949		4%	16.48%
8	Total Bond Market	-0.1118	-0.0461	-0.2807	0.0949	1.0000		42%	3.35%
9								100%	
10									
11	Variance-covariance matrix								
12		SP500	MSCI World ex-US	Russell 2000	MSCI Emerging	Total Bond Market			
13	SP500	0.0141	0.0137	0.0134	0.0139	-0.0004	<-- {=I4:I8*TRANSPOSE(I4:I8)*B4:F8}		
14	MSCI World Ex-US	0.0137	0.0148	0.0122	0.0167	-0.0002			
15	Russell 2000	0.0134	0.0122	0.0208	0.0116	-0.0014			
16	MSCI Emerging	0.0139	0.0167	0.0116	0.0271	0.0005			
17	Total Bond Market	-0.0004	-0.0002	-0.0014	0.0005	0.0011			
18									
19	Standard deviation of composite		6.91%	<-- {=SQRT(MMULT(MMULT(TRANSPOSE(H4:H8),B13:F17),H4:H8))}					
20									
21	Check: 1st row (var-cov)	0.0141	0.0137	0.0134	0.0139	-0.0004	<-- =I4*I8*F4		

The strange formula, **I4:I8 * Transpose(I4:I8) * B4:F8**, in cells B13:F17 is composed of two parts:

- • **I4:I8 * Transpose(I4:I8)** multiplies the column vector I4:I8 times its transpose. This is equivalent to multiplying

$$\begin{bmatrix} \sigma_{SP500} \\ \sigma_{MSCI\ World} \\ \sigma_{Russell\ 2000} \\ \sigma_{MSCI\ Emerging} \\ \sigma_{Total\ bond} \end{bmatrix} * [\sigma_{SP500} \quad \sigma_{MSCI\ World} \quad \sigma_{Russell\ 2000} \quad \sigma_{MSCI\ Emerging} \quad \sigma_{Total\ bond}],$$

which, in the wonderful world of array functions, gives a matrix of the covariances:

$$\begin{bmatrix} \sigma^2_{SP500} & \sigma_{SP500}\sigma_{MSCI\ World} & \sigma_{SP500}\sigma_{Russell\ 2000} & \cdots & \sigma_{SP500}\sigma_{Total\ bond} \\ \sigma_{MSCI\ World}\sigma_{SP500} & \sigma^2_{MSCI\ World} & \cdots & & \\ \vdots & & & & \vdots \\ \sigma_{Total\ bond}\sigma_{SP500} & \sigma_{Total\ bond}\sigma_{MSCI\ World} & \cdots & \cdots & \sigma^2_{Total\ bond} \end{bmatrix}$$

- • Multiplying the above by the matrix of correlations B4:F8 gives the variance-covariance matrix.
- • Of course, the whole array formula **I4:I8 * Transpose(I4:I8) * B4:F8** is entered with [Ctrl] + [Shift] + [Enter].

In row 21, perform a check on our computation. Row 21 contains brute-force computations of the first row of the variance-covariance matrix—just to make sure our array formula works as advertised.

In cell B19 we compute the standard deviation of the five-index portfolio using their relative weights.

	A	B	C	D	E	F
23	Risk-free rate	1.00%				
24	Expected return on SP500	10.00%				
25						
26	**Black-Litterman implied returns**					
27	SP500	10.0%	<--{=MMULT(B13:F17,H4:H8)*(B24-B23)/INDEX((MMULT(B13:F17,H4:H8)),1,1)+B23}			
28	MSCI World Ex-US	10.30%				
29	Russell 2000	9.22%				
30	MSCI Emerging	11.59%				
31	Total Bond Market	1.30%				

The BL implied expected returns are based on three assumptions:

- • The weighted portfolio of the five indices is mean-variance optimal.
- • The anticipated risk-free rate is 1%.
- • The expected return of the SP500 Index is 10%.

Given these assumptions, the expected returns on the five-index portfolio are given in cells B27:B31. Note the array formula given in these cells:

=MMULT(B13:F17,H4:H8) * (B24-B23)/
INDEX((MMULT(B13:F17,H4:H8)),1,1)+B23

This formula uses the expected return on the SP500 to normalize the returns. It is equivalent to the general formula:

$$\begin{bmatrix} \text{Benchmark} \\ \text{portfolio} \\ \text{returns} \end{bmatrix} = \begin{bmatrix} \text{Variance} - \\ \text{covariance} \\ \text{matrix} \end{bmatrix} \begin{bmatrix} \text{Benchmark} \\ \text{portfolio} \\ \text{proportions} \end{bmatrix} * \begin{matrix} \text{Normalizing} \\ \text{factor} \end{matrix} + \begin{matrix} \text{Risk-free} \\ \text{rate} \end{matrix}$$

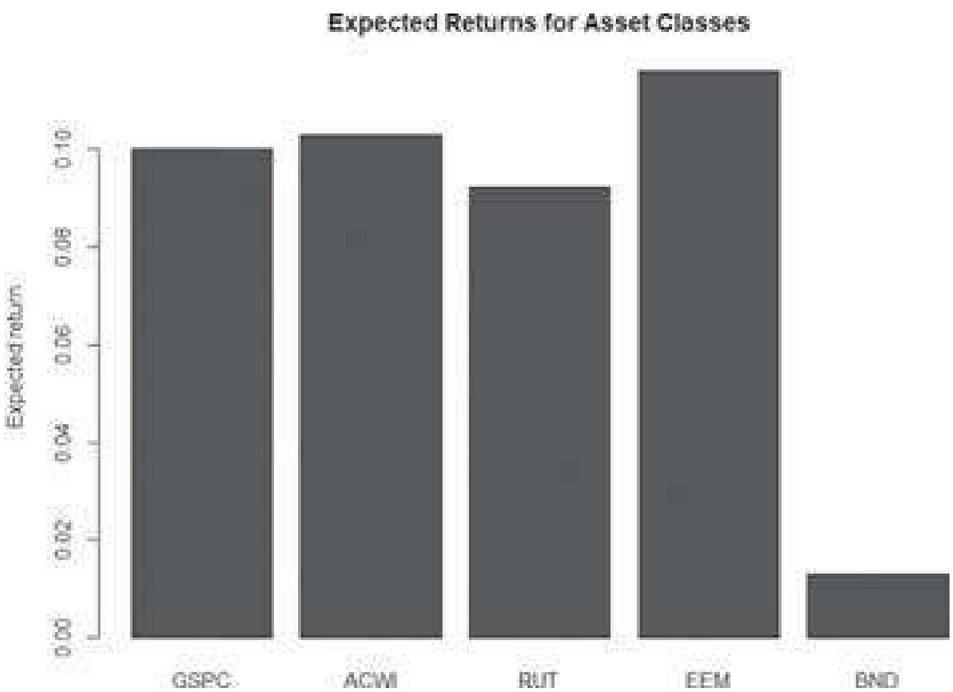
Here, the constraint is that the SP500 return is equal our assumption of 10%. So, the normalizing factor must be:

$$\frac{E(r_{SP500}) - r_f}{[SP500\ \text{row from var-cov matrix}] \begin{bmatrix} \text{Benchmark} \\ \text{portfolio} \\ \text{proportions} \end{bmatrix}}$$

Implementing the Analysis in R

Implementing the above example in R would be relatively easy, as follows:

```
# Read the csv with world indices
monthly_prices <- read.csv("Chap_15_data_2.csv", row.names = 1,stringsAsFactors = FALSE)

# Sort the data by date
monthly_prices <- monthly_prices[order(
    as.Date(row.names(monthly_prices), format="%d-%m-%y")),]

# Compute Continuous Returns
monthly_returns <- data.frame(apply(monthly_prices
                                    , 2, function(x) diff(log(x)) ))

# Input: Weights
weights <- c("GSPC" = 0.30, "ACWI" = 0.20, "RUT" = 0.04, "EEM" = 0.04, "BND" = 0.42)

# Compute statistics
cor_mat <- cor(monthly_returns) # Correlation matrix
monthly_returns <- data.frame(apply(monthly_prices
                                    , 2, function(x) diff(log(x))))
sd_ret <- apply(monthly_returns, 2,sd)*sqrt(12)
cov_mat <-  (sd_ret %*% t(sd_ret))*cor_mat # Covariance matrix (population)
row.names(cov_mat) <- colnames(cov_mat)
# Of course we can use alternatively: cov_mat <- cov(monthly_returns)*12

# Standard deviation of composite
sd_composite <- sqrt(weights %*% cov_mat %*% weights)

# Input:
rf <- 0.01 # risk free rate
r_SP <- 0.10 # Expected return on S&P 500

# Black-Litterman implied returns
implied_ret <- cov_mat %*% weights * ( r_SP - rf ) / (cov_mat %*% weights)["GSPC",] + rf
barplot (c(implied_ret),names.arg=row.names(implied_ret),legend.text= , col="blue",
         main="Expected Returns for Asset Classes",ylab="Expected return")
```



What's the Upshot?

If we believe that the world portfolio is efficient (and in the absence of further information, there is little reason to believe otherwise), then the anticipated returns from each of its components are given by the BL model, by the risk-free rate, and by an additional assumption on expected returns (in our case, the expected return of the SP500). The exercises for this chapter explore some other variations to the latter assumption.

15.7 Summary

Applying portfolio theory is not merely a matter of using historical market data to derive covariances and expected returns. Blindly applying sample data to derive optimal portfolios (as in section 11.4) usually leads to absurd results. The BL approach gets around these absurdities by first assuming that—in the absence of analyst opinions and other information—the benchmark market weights and current risk-free interest rate correctly predict the future asset returns. The resulting asset returns can then be adjusted for opinions and confidence in opinions to derive an optimal portfolio.

Exercises

1. You have decided to create your own index of higher-beta components of the Dow-Jones 30 Industrials. Using Yahoo's stock screener, you come up with the following data:
    1. Compute the variance-covariance matrix of returns.
    2. Assuming that the risk-free rate is 1.2% annually (=1.2%/12 = 0.1% monthly), and that the expected high-beta index annual return is 12% (= 1% monthly), compute the BL monthly expected returns for each stock.

	A	B	C	D	E	F	G	H	I	J	K	L	M
1						HIGH-BETA INDEX FROM 12 COMPONENTS							
2		Alibaba Group	Bank of America	NVIDIA	Citigroup Inc.	Boeing	Advanced Micro Devices	Micron Technology	Petroleo Brasileiro	Las Vegas Sands	Suncor Energy	Schlumberger	Marriott International
3	Ticker	BABA	BAC	NVDA	C	BA	AMD	MU	PBR	LVS	SU	SLB	MAR
4	Market Cap ($M)	466,698	330,733	143,996	179,688	184,185	50,033	50,119	103,964	53,250	51,136	56,677	50,341
5	Beta	1.88	1.66	1.56	1.86	1.61	2.43	1.75	1.89	1.60	1.68	2.00	1.54
6													
7	Stock prices												
8	Date	BABA	BAC	NVDA	C	BA	AMD	MU	PBR	LVS	SU	SLB	MAR
9	01-Jan-20	206.89	32.62	236.29	73.92	316.39	47.00	53.08	14.11	64.03	30.29	33.82	138.50
10	01-Dec-19	212.10	34.81	235.16	79.36	323.83	45.86	53.78	15.94	66.93	31.96	39.07	150.83
11	01-Nov-19	200.00	32.93	216.45	74.10	361.90	39.15	47.51	14.64	60.84	30.59	35.18	129.30
65	01-May-15	89.32	15.12	21.40	49.78	122.71	2.26	27.60	8.13	39.30	23.28	75.60	72.76
66	01-Apr-15	81.29	14.60	21.47	49.08	125.18	2.26	28.13	9.25	40.89	25.97	78.80	74.70
67	01-Mar-15	83.24	14.08	20.24	47.42	131.06	2.68	27.13	5.85	42.05	23.08	69.50	74.95
68	01-Feb-15	85.12	14.44	21.25	48.25	130.93	3.11	30.67	6.46	43.47	23.79	69.69	77.36
69	01-Jan-15	89.08	13.84	18.50	43.21	126.17	2.57	29.27	5.85	41.54	23.52	68.22	69.36

2. You are an analyst investing in the high-beta portfolio from the previous problem. You believe that the monthly return of Schlumberger (SLB) will be 0.5% higher than the market expected return. What are your recommended optimal portfolio proportions?
3. Another analyst believes that SLB return will be 0.5% higher than the market expected return per month over the next year and that the NVIDIA Corporation (NVDA) return will be lower by 0.2% than the market expected return. What are this analyst's recommended portfolio proportions?

---

1. [1](). DeMiguel, Garlappi, and Uppal (2009) illustrate how badly mechanical optimization works. The authors examine optimal allocations among 10 sector portfolios. They find that a naive portfolio allocation rule of investing equal proportions in each portfolio—irrespective of market values—outperforms more sophisticated data-based optimizations.
2. [2](). Black and Litterman (1991). Black and Litterman were apparently unaware of an earlier paper, which includes much of their research, by Sharpe (1974). Sharpe's paper discusses the reverse-engineering of the portfolio returns from portfolio composition.
3. [3](). "According to Lipper Analytical Services, over the 10 years ended in June 2000, more than 80% of 'general equity' mutual funds … underperformed the SP500 Index—the major benchmark for stock mutual funds." Or, from a well-known academic: "Professional investment managers, both in the U.S. and abroad, do not outperform their index benchmarks" (Malkiel 2005).

4. 4. In principle you might want to include a negative-return stock in your portfolio if it has a negative correlation with enough other stocks to lower the total portfolio variance. However, this rarely happens.
5. 5. One percent per month is equivalent to estimating an annual expected benchmark return of 12%.
6. 6. We thank Steven Schoenfeld of Northern Trust for providing us with the initial data and some suggestions.

IV OPTIONS

16 Introduction to Options

16.1 Overview

In this chapter we give a brief introduction to options. The chapter can, at best, serve as an introduction for the already informed. If you know nothing whatsoever about options, read an introduction to the topic in a basic finance text.[1] We start with the basic definitions and options terminology, go on to discuss graphs of option payoffs and "profit diagrams," and finally discuss some of the more important option arbitrage propositions (sometimes referred to as linear pricing restrictions). In subsequent chapters we discuss two methods of pricing options: The binomial option pricing model (Chapter 17) and the Black-Scholes option pricing model (Chapter 18).

16.2 Basic Option Definitions and Terminology

An *option on a stock* is a security that gives the holder the right to buy or to sell one share of the stock on or before a particular date for a predetermined price. Here is a brief glossary of terms and notation used in the field of options:

- • *Call, C*: An option that gives the holder the **right to buy** a share of stock on or before a given date at a predetermined price.
- • *Put, P*: An option that gives the holder the **right to sell** a share of stock on or before a given date at a predetermined price.
- • *Exercise price, X*: The price at which the holder can buy or sell the underlying stock; sometimes also referred to as the *strike price*.
- • *Expiration date, T*: The date on or before which the holder can buy or sell the underlying stock.
- • *Stock price, S_t*: The price at which the underlying stock is selling at date t. The current stock price is denoted S_0.
- • *Option price*: The price at which the option is sold or bought; sometimes also referred to as the *premium*.

American versus European options: In the jargon of options markets, an American option is one that can be exercised on or before the expiration date T, whereas a European option is one which can be exercised only on the expiration date T. This terminology is confusing for two reasons:

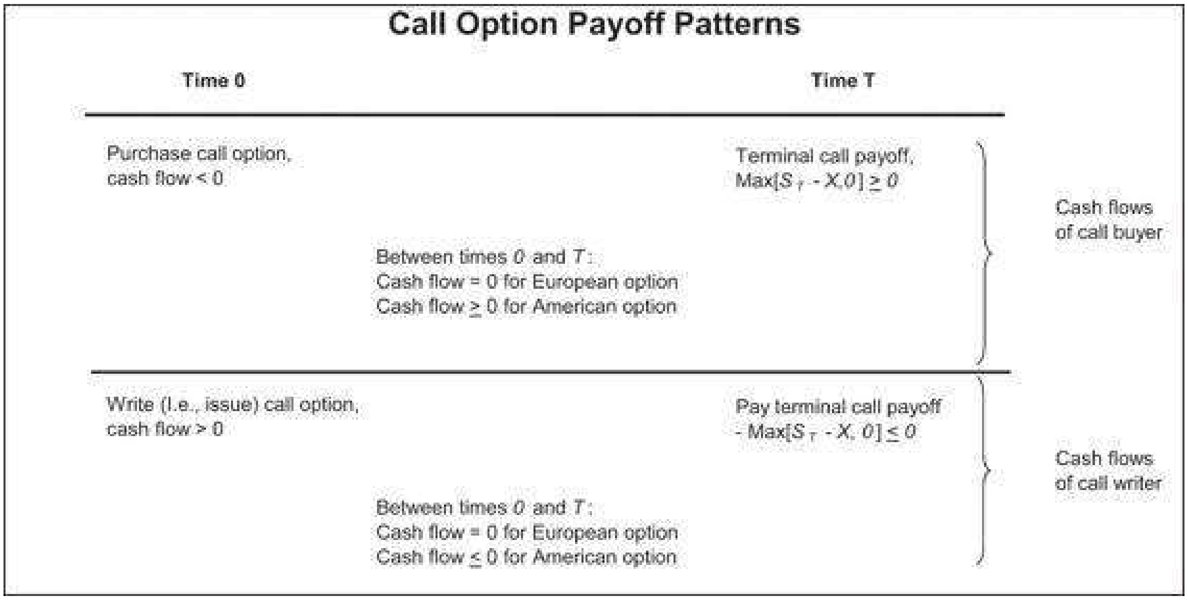
- • The options sold on both European and American options exchanges are almost invariably American options.
- • The simplest option pricing formulas (these include the famous Black-Scholes option pricing formula discussed in Chapter 18) are for *European options*. As we show in section 16.6, in many cases we can price American options as if they were European options.

We use C_t to denote the price of a European call on date t and P_t to denote the European put price. If it is clear that the option price refers to today's price, we often drop the subscript, writing C or P instead of C_0 or P_0. When we need fuller notation, we write $C_t(S_t, X, T)$ for the price of a call on date t when the price of the underlying stock is S_t, the exercise price is X, and the expiration date is T. When we wish to specify that our option pricing formula relates to an American option, we use the superscript A: $C_t^A, C_t^A(S_t, X, T)$ or $P_t^A(S_t, X, T)$. When written without superscripts, the options pricing refers to European options.

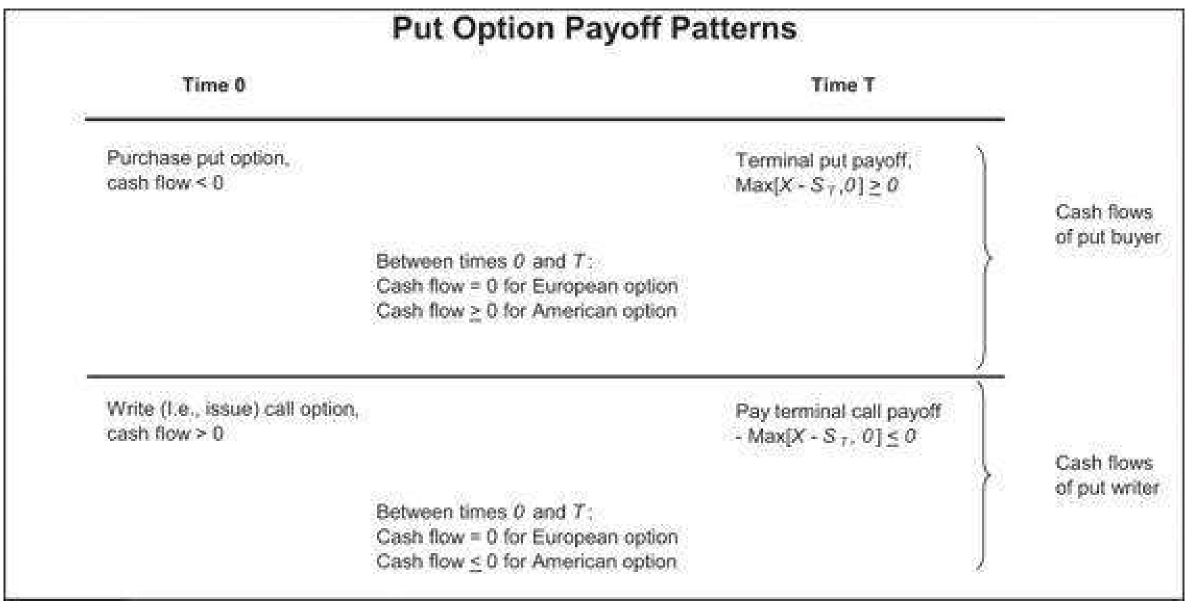
At-the-money, *in-the-money*, *out-of-the-money*: If the exercise price X of a call or a put is equal to the current price of the stock S_0, then the option is *at-the-money*. If a positive cash flow could be made by immediately exercising an American option (that is, $S_0 - X > 0$ for a call and $X - S_0 > 0$ for a put), then the option is *in-the-money*.[2]

Writing Options versus Purchasing Options: Cash Flows

The purchaser of a call option acquires the right to buy a share of stock for a given price on or before date T and pays for this right at the time of purchase. The *writer* or seller of this call option is the seller of this right: The writer collects the option price today in return for agreeing to deliver one share of stock in the future for the exercise price, if the purchaser of the call demands. In terms of cash flows, the purchaser of an option always has an initial negative cash flow (the price of the option) and a future cash flow that is at worst zero (if it is not worthwhile exercising the option) and otherwise positive (if the option is exercised). The cash flow position of the writer of the option is reversed: An initial positive cash flow is followed by a terminal cash flow, which is at best zero.



A similar payoff pattern holds for the cash flows of the purchaser and writer of a put option on a stock:



16.3 Some Examples

Below we show the most traded options (by open interest) on July 31, 2019, using data from Yahoo Finance (https://finance.yahoo.com/options/highest-open-interest). Each option represents a trade on 100 units of the underlying, so that 102.191 million (M) options on General Electric (GE) in cell K4 represent call options on 10.219 billion shares of GE with an exercise price of $20. The "open interest" is the number of outstanding contracts at the end of the day. For GE, 102M calls were traded on July 31, 2019, and at the end of the day 280,691 calls were outstanding.

MOST ACTIVE OPTIONS, 31 July 2019

Rank	Underlying Symbol	Name	Strike	Expiration Date	Price	Change	% Change	Bid	Ask	Volume	Open Interest
1	^VIX	VIX Aug 2019 20.000 call	20	20-08-19	13.7	-0.24	-1.72%	0.4	0.45	5,246	360,577
2	GE	GE Jan 2020 20.000 call/CON	20	16-01-20	10.43	-0.09	-0.85%	0	0.01	102.191M	280,691
3	CHK	CHK Jan 2020 7.000 call	7	16-01-20	1.81	0.01	0.56%	0.02	0.03	29.615M	273,541
4	^VIX	VIX Aug 2019 19.000 call	19	20-08-19	13.7	-0.24	-1.72%	0.45	0.5	2,441	262,279
5	^VIX	VIX Sep 2019 23.000 call	23	17-09-19	13.7	-0.24	-1.72%	0.6	0.75	16	249,145
6	CLDR	CLDR Jun 2021 5.000 put	5	17-06-21	5.99	-0.01	-0.08%	1.05	1.15	1.638M	249,092
7	EEM	EEM Sep 2019 36.000 put	36	19-09-19	42.13	-0.16	-0.38%	0.06	0.07	19.406M	230,398
8	HYG	HYG Sep 2019 84.000 put	84	19-09-19	87.24	0.15	0.18%	0.19	0.24	4.798M	223,963
9	^VIX	VIX Sep 2019 25.000 call	25	17-09-19	13.7	-0.24	-1.72%	0.5	0.55	6,589	218,543
10	IEF	IEF Aug 2019 98.000 put	98	15-08-19	109.81	0.22	0.21%	0	0.15	1.967M	215,006
11	^VIX	VIX Aug 2019 25.000 call	25	20-08-19	13.7	-0.24	-1.72%	0.2	0.25	6,723	213,737
12	XOP	XOP Sep 2019 25.000 put	25	19-09-19	25.38	0.41	1.62%	0.86	0.87	14.391M	212,299
13	EEM	EEM Dec 2019 39.000 put	39	19-12-19	42.13	-0.16	-0.38%	0.84	0.87	19.406M	211,524
14	GE	GE Jan 2020 15.000 call/CON	15	16-01-20	10.43	-0.09	-0.85%	0.06	0.06	102.191M	210,620
15	EEM	EEM Dec 2019 37.000 put	37	19-12-19	42.13	-0.16	-0.38%	0.5	0.53	19.406M	208,954
16	CVS	CVS Jan 2021 90.000 call	90	14-01-21	56.32	-0.21	-0.37%	0.65	0.78	2.407M	206,712
17	^VIX	VIX Sep 2019 24.000 call	24	17-09-19	13.7	-0.24	-1.72%	0.55	0.6	56	206,258
18	EEM	EEM Aug 2019 44.000 call	44	15-08-19	42.13	-0.16	-0.38%	0.06	0.07	19.406M	191,842
19	HYG	HYG Sep 2019 82.000 put	82	19-09-19	87.24	0.15	0.18%	0.09	0.14	4.798M	178,863
20	EEM	EEM Sep 2019 40.000 put	40	19-09-19	42.13	-0.16	-0.38%	0.35	0.36	19.406M	176,795

A few facts stand out about this list:

- • Many of the options are on indices. The options allow investors/speculators to make bets on broad-based market movements.
- • The list is approximately evenly divided between puts (9 out of 20) and calls. This is not always true—when investors are optimistic about future market movements, calls tend to dominate the most active, and vice versa.
- • The options in the most active list tend to be short-term options. Although longer-term options exist, these are traded less than the near term.

16.4 Option Payoff and Profit Patterns

One of the attractions of options is that they allow their owners to change the payoff patterns of the underlying assets. In this section we consider:

- • The basic payoff and profit patterns of a call and a put option and a share
- • The payoff patterns of various combinations of options and shares

Stock Profit Patterns

We start with the **payoff pattern from a purchased stock**. Suppose you buy a share of General Pills (GP) stock in July at its then-current market price of $40. If in September the price of the stock is $70, you will have made a $30 profit; if its price is $30, you will have a loss (or a negative profit) of $10.[3] Generalize this by writing the price of the stock in September as S_T and its price in July by S_0. Then we write the profit function from the stock as:

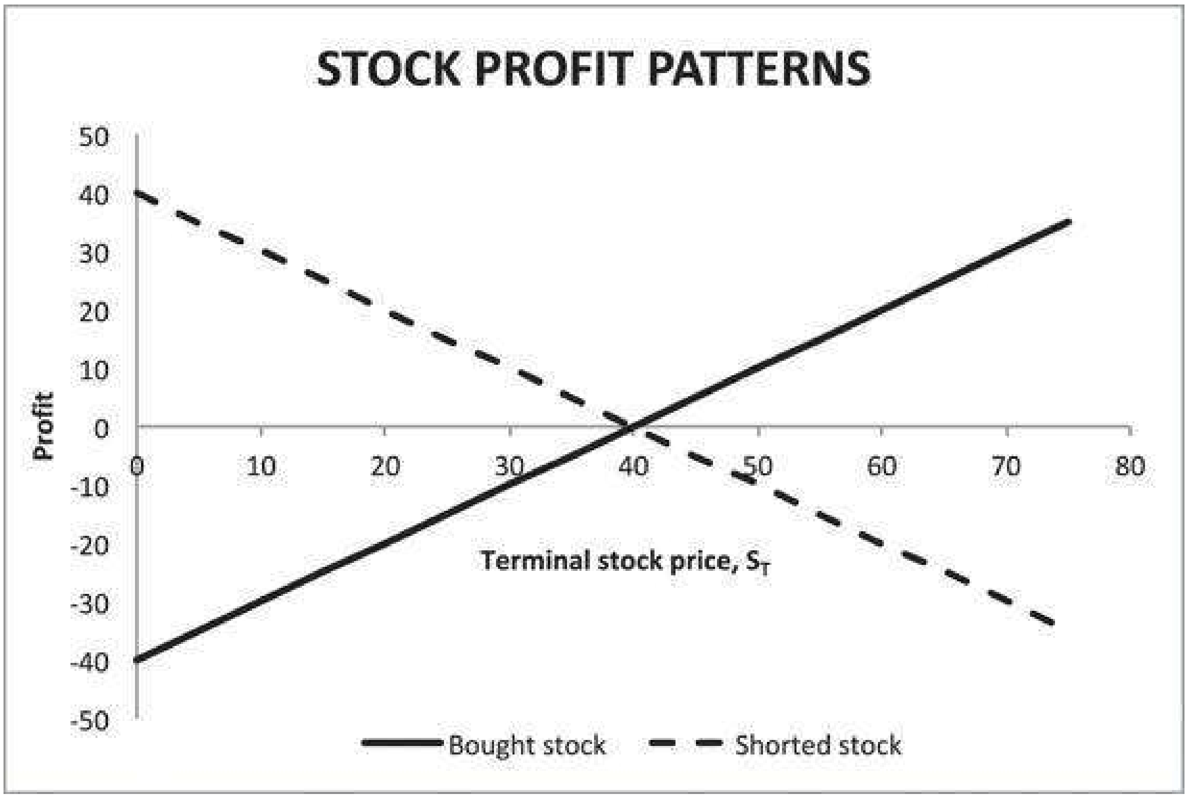
$$\textit{Profit from stock} = S_T - S_0$$

Payoff from the Short Sale of a Stock: Suppose we had sold one share of GP stock short in July, when its market price was $40. If in September the market price of GP was $70, and if at that point we undid the short sale (i.e., we purchase a share at the market price in order to return the share to the lender of the original short), then our loss will be \$30 (profit of −\$30):

$$\textit{Profit from short sale of stock} = S_0 - S_T = -1 * (\textit{Profit from purchase of stock})$$

Notice that the profit from the short sale is the *negative* of the profit from the purchase; this is always the case (also for options, considered below).

Graphing Stock Profit Patterns: The Excel graph below graphs the profit patterns from both a purchase and a short sale of the GP stock described above.



Call Option Profit Patterns

As in the case of a stock, we start with the **payoff pattern from a purchased call**. We go back to the General Pills options of the previous section. Suppose that in July you bought one GP September call with X = 40 for \$4.[4] In September you will exercise the call only if the market price of GP is higher than \$40. If we write the initial (July) call price as C_0, we can write the profit function from the call in September as follows:

$$\begin{aligned} \textit{Call profit in September} &= -C_0 + max(0, S_T - X) \\ &= Max(0, S_T - 40) - 4 \\ &= \begin{cases} S_T - 40 - 4 & \textit{if } S_T \geq 40 \\ -4 & \textit{if } S_T < 40 \end{cases} \end{aligned}$$

Payoff Pattern from a Written Call: In options markets, the purchaser of a call buys the call from a counterparty who issues the call. In the jargon of options, the issuer of the call is called the *call writer*. It is worthwhile to spend a few minutes considering the difference between the security bought by the call purchaser and the call writer:

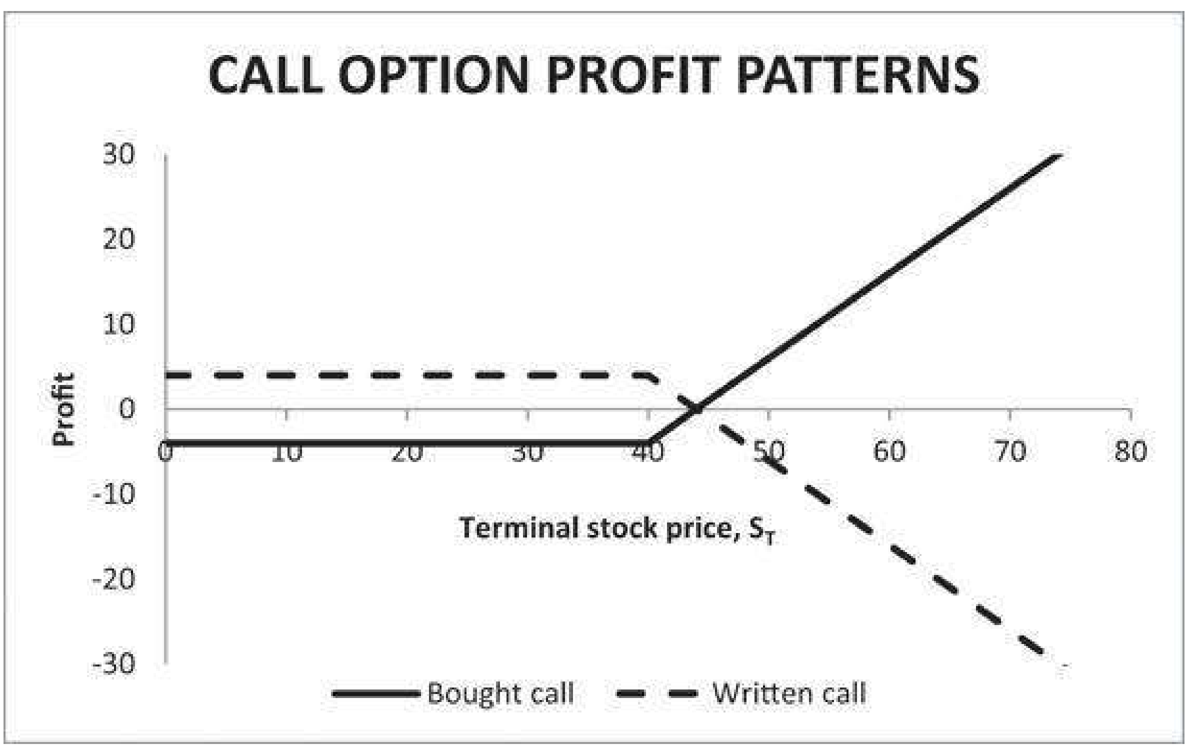
The call purchaser buys a security that *gives the right to buy a share of stock on or before date T for price X.* The cost of this privilege is the *call price* C_0, which is paid at the time of the call purchase. Thus, the call purchaser has an initial negative cash flow (the purchase price C_0); on the other hand, the purchaser's cash flow at date T is always non-negative: $max(S_T - X, 0)$.

The call writer gets C_0 at the date of the call purchase. In return for this price, the writer of the call *agrees to sell a share of the stock for price X on or before date T.* Notice that whereas the call purchaser has an option, the call writer has undertaken an obligation. Note that the cash flow pattern of the call writer is opposite to that of the call purchaser: The writer's initial cash flow is positive ($+ C_0$), and the cash flow at date T is always non-positive: $-max(S_T - X, 0)$.

The profit of a call writer is the opposite of that of the call purchaser. For the case of the GP options:

$$\begin{aligned} \text{Call writer's profit in September} &= +C_0 - max(0, S_T - X) \\ &= 4 - Max(0, S_T - 40) \\ &= \begin{cases} 4 - (S_T - 40) & \text{if } S_T \geq 40 \\ +4 & \text{if } S_T < 40 \end{cases} \end{aligned}$$

Graphing the profit patterns of the bought and the written call gives:



Put Option Profit Patterns

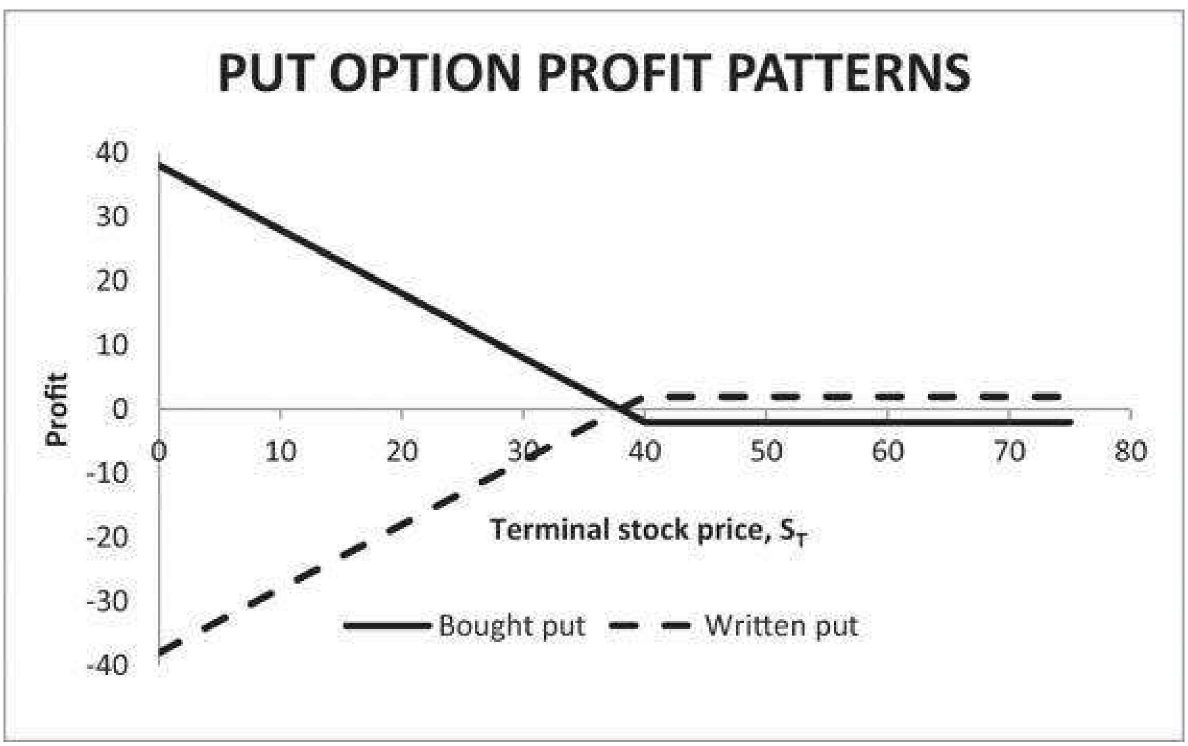
Payoff Pattern from a Purchased Put: If in July you bought one GP September put with X = 40 for $2,[5] then in September you will exercise the put only if the market price of GP is lower than $40. If we write the initial (July) put price as P_0, we can write the profit function from the put in September as follows:

$$
\begin{aligned}
\textit{Put profit in September} &= max(0, X - S_T) - P_0 \\
&= max(0, 40 - S_T) - 2 \\
&= \begin{cases} 38 - S_T & \text{if } S_T \le 40 \\ -2 & \text{if } S_T > 40 \end{cases}
\end{aligned}
$$

Payoff Pattern from a Written Put: The *put writer* is obligated to purchase one share of GP stock on or before date *T* for the put exercise price of *X*. For putting oneself in this invidious position, the writer of the put receives, at the time the put is written, the put price P_0. The payoff pattern from writing the GP September 40 put is therefore:

$$
\begin{aligned}
\textit{Put writer's profit in September} &= P_0 - max(0, X - S_T) \\
&= 2 - max(0, 40 - S_T) \\
&= \begin{cases} -38 + S_T & \text{if } S_T \le 40 \\ 2 & \text{if } S_T > 40 \end{cases}
\end{aligned}
$$

Graphing the profit patterns of the bought and the written put gives the following:



16.5 Option Strategies: Payoffs from Portfolios of Options and Stocks

There is some interest in graphing the combined profit pattern from a portfolio of options and stocks. These patterns give an indication of how options can be used to *change the payoff patterns* of “standard” securities such as stocks and bonds. Here are a few common examples.

Protective Put

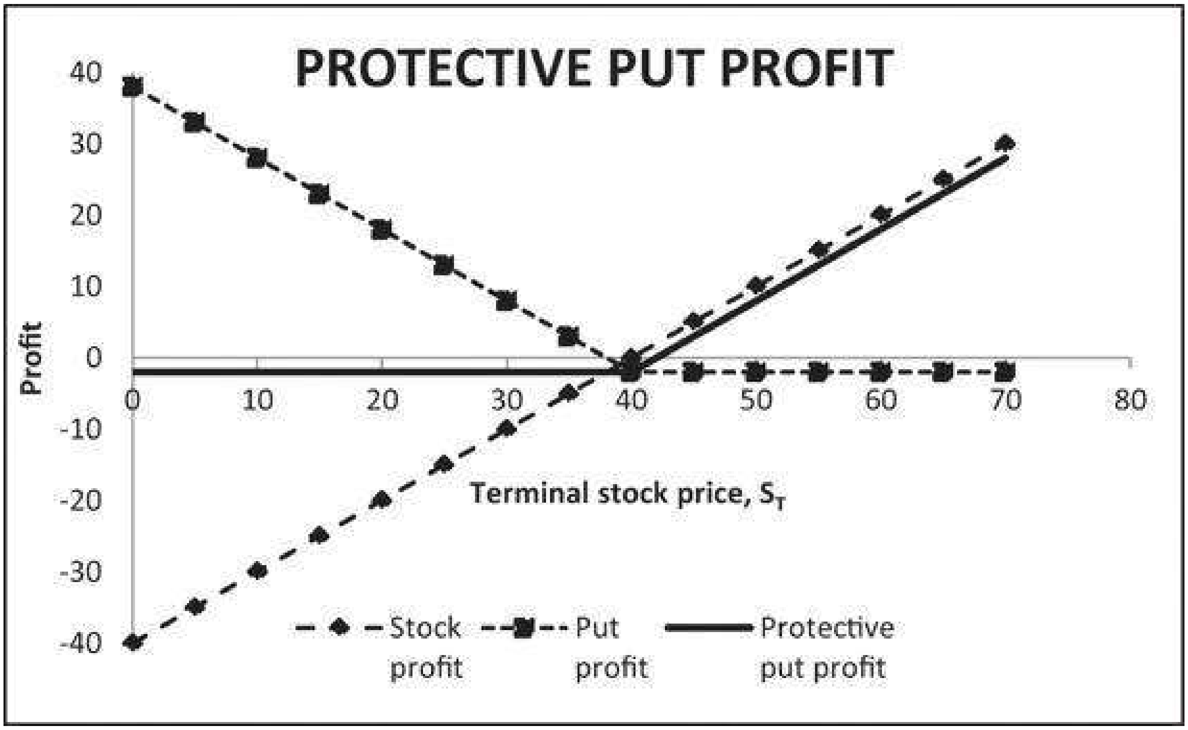
Consider the following combination:

- • One share of stock, purchased for S_0
- • One put, purchased for P with exercise price X

This option strategy is often called a “protective put” strategy or “portfolio insurance.” In Chapter 26, we return to this topic, exploring it in much further detail. The payoff pattern of the protective put is given by:

$$
\begin{aligned}
\textit{Stock profit} + \textit{put profit} &= (S_T - S_0) + [max(X - S_T, 0) - P_0] \\
&= \begin{cases} S_T - S_0 + X - S_T - P_0 & \text{if } S_T \le X \\ S_T - S_0 - P_0 & \text{if } S_T > X \end{cases} \\
&= \begin{cases} X - S_0 - P_0 & \text{if } S_T \le X \\ S_T - S_0 - P_0 & \text{if } S_T > X \end{cases}
\end{aligned}
$$

When this pattern is applied to the GP example (that is, buying a share at \$40 and a put with X = \$40 for \$2), we get the following graph:



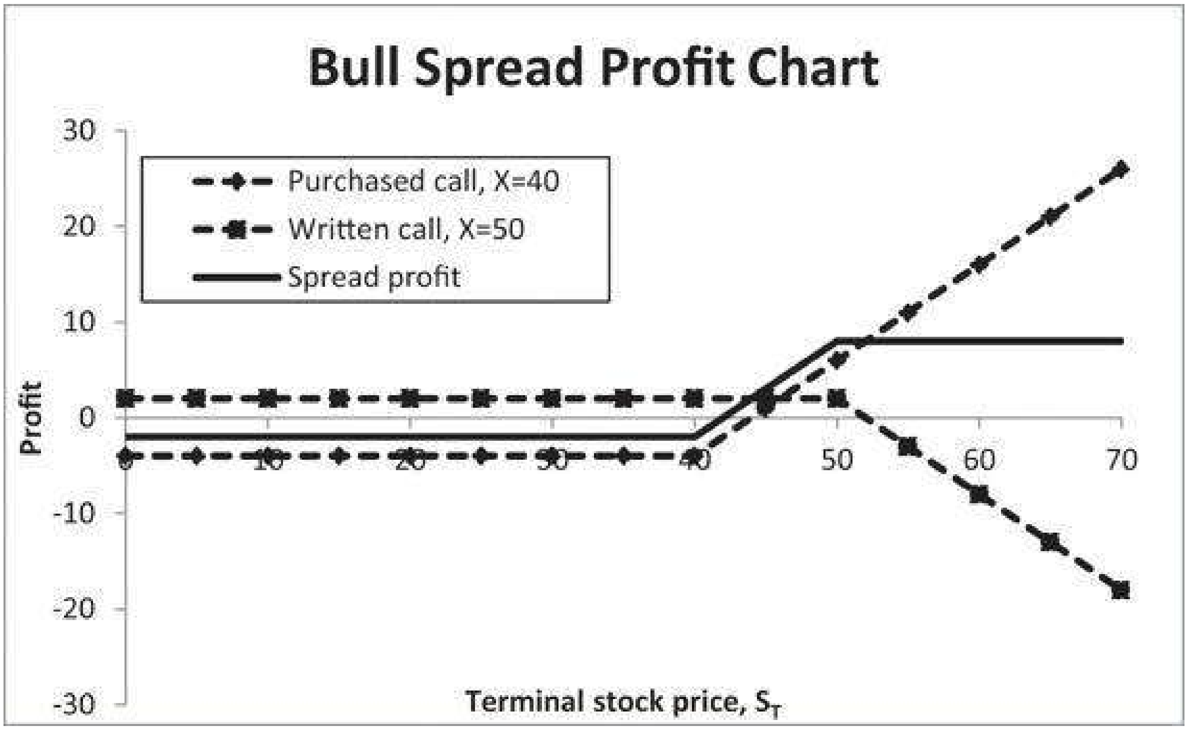
This pattern looks very much like the payoff pattern from a call.[6]

Spreads

Another combination involves buying and writing calls with different exercise prices. When the bought call has a low exercise price and the written call has a higher exercise price, the combination is called a *bull spread*. As an example, suppose you bought a call (for $4) with an exercise price of $40 and wrote a call (for $2) with an exercise price of $50. This bull spread gives a profit of:

$$\text{Long call}(40) + \text{short call}(50) = [max(S_T - 40, 0) - 4] - [max(S_T - 50, 0) - 2]$$
$$= \begin{cases} -4 + 2 = -2 & \text{if } S_T \leq 40 \\ S_T - 40 - 4 + 2 = S_T - 42 & \text{if } 40 < S_T \leq 50. \\ S_T - 40 - 4 - (S_T - 50 - 2) = 8 & \text{if } S_T > 50 \end{cases}$$

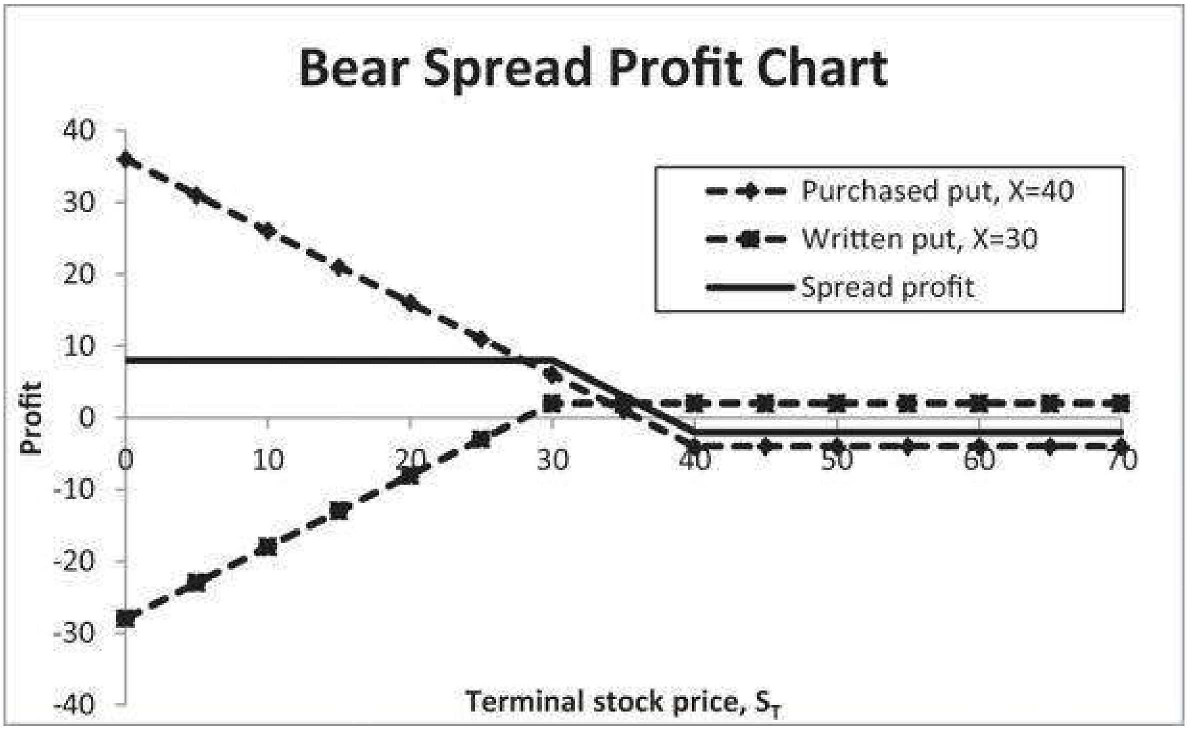
The Excel graph given below shows each of the two calls and the resulting spread profit:



Buying and writing puts with different exercise prices is yet another combination. When the bought put has a high exercise price and the written put has a lower exercise price, the combination is called a *bear spread*. As an example, suppose you bought a put for \$4 with an exercise price of \$40 and wrote a put (for \$2) with an exercise price of \$30. This bear spread gives a profit of:

$$\text{Long put}(40) + \text{short put}(30) = [max(40 - S_T, 0) - 4] - [max(30 - S_T, 0) - 2]$$
$$= \begin{cases} -4 + 2 = -2 & if\ S_T \geq 40 \\ 40 - S_T - 4 + 2 = S_T - 38 & if\ 30 \leq S_T < 40. \\ 40 - S_T - 4 - (30 - S_T - 2) = 8 & if\ S_T < 30 \end{cases}$$

The Excel graph given below shows each of the two calls and the resulting spread profit:



16.6 Option Arbitrage Propositions

In succeeding chapters we price options given specific assumptions about the probability distribution of the underlying asset (usually the stock) on which the option is written. However, there is much that can be learned about the pricing of options without making these specific probability assumptions. In this section we consider a number of these *arbitrage restrictions* on option pricing. Our list is by no means exhaustive, and we have concentrated on those propositions that provide insight into the pricing of options or that will be used in later sections.

Throughout we assume that there is a single risk-free interest rate that prices bonds; we also assume that this risk-free rate is continuously compounded so that the present value of a riskless security which pays off X at time T is given by $X * e^{-rT}$.

PROPOSITION 1 Consider a call option written on a stock that pays no dividends before the option's expiration date T. Then the lower bound on a call option price is given by:

$$C_0 \geq Max(S_0 - X * e^{-rT}, 0)$$

Comment Before proving this proposition, it will be helpful to consider its meaning. Suppose that the riskless interest rate is 5%, and suppose we have an American call option with maturity $T = 1/2$ (i.e., the expiration date of the option is one-half year from today) with $X = 80$ written on a stock whose current stock price $S_0 = 83$. A naive approach to determining a lower bound on this option's price would be to state that it is worth at least \$3, since it could be exercised immediately with a profit of \$3. Proposition 1 shows that the option's value is *at least* $83 - 80 * e^{-0.05 * 0.5} = 4.975$. Furthermore, a careful examination of the proof below will show that this fact *does not* depend on the option being an American option—it is also true for a European option.

	A	B	C
1	**PROPOSITION 1 — HIGHER LOWER BOUNDS FOR CALL PRICES**		
2	Current stock price, S_0	83	
3	Option time to maturity, T	0.5	
4	Option exercise price, X	80	
5	Interest rate, r	5%	
6			
7	Naive minimum option price, Max(S_0-X,0)	3	<--=MAX(B2-B4,0)
8	Proposition 1 lower bound on option price, Max(S_0 - X*exp(-rT),0	4.975	<--=MAX(B2-B4*EXP(-B5*B3),0)

Proof of Proposition 1 Standard arbitrage proofs are built on the consideration of the cash flows from a particular strategy. In this case, the strategy is the following:

At Time 0 (Today)

- Buy one share of the stock.
- Borrow the present value (PV) of the option exercise price X.
- Write a call on the option.

At Time T

- Exercise the option if this is profitable.
- Repay the borrowed funds.

This strategy produces the following cash flow table:

	Today	**At Time T**	
Action	Cash Flow	$S_T < X$	$S_T \geq X$
Buy stock	$-S_0$	$+S_T$	$+S_T$
Borrow PV of X	$+Xe^{-rt}$	$-X$	$-X$
Write call	$+C_0$	0	$-(S_T - X)$
Total	$-S_0 + Xe^{-rT} + C$	$S_T - X \leq 0$	0

Note that at time T, the cash flow resulting from this option is either negative (if the call is not exercised) or zero (when $S_T \geq X$). Now a financial asset (in this case, the combination of purchasing a stock, borrowing X, and writing a call) that has only non-positive payoffs in the future must have a positive initial cash flow, which shows that:

$C_0 - S_0 + Xe^{-rT} > 0$ or $C_0 > S_0 - Xe^{-rT}$

To finish the proof, we note that in no case can the value of a call be less than zero. Thus, we have that $C_0 \geq Max(S_0 - Xe^{-rT}, 0)$, which proves the proposition.

Proposition 1 has an immediate and very interesting consequence: In many cases the early-exercise feature of an American call option is worthless; this means that an

American call option can be valued as if it were a European call. The precise conditions are explained in the next theorem:

PROPOSITION 2 Consider an American call option written on a stock that will not pay any dividends before the option's expiration date T. Then it is never optimal to exercise the option before its maturity.

Proof of Proposition 2 Suppose the holder of the option is considering exercising it early, at some date $t < T$. The only reason to consider such early exercise is that $S_t - X > 0$, where S_t is the price of the underlying stock at time t. However, by Proposition 1, the market value of the option at time t is at least $S_t - X * e^{-r(T - t)}$, where r is the risk-free rate of interest. Since $S_t - X * e^{-r(T - t)} > S_t - X$, it follows that the option's holder is better off selling the option in the market than exercising it.

Proposition 2 means that many American call options can be priced as if they were European calls. Note that this is not true for American puts, even if the underlying stock pays no dividends. In Chapter 17 we give some examples in the context of a binomial model.

PROPOSITION 3 (PUT BOUNDS) The lower bound on the value of a put option is:

$P_0 \geq max(0, Xe^{-rT} - S_0)$

Proof of Proposition 3: The proof of this proposition has the same form as the proof of the previous theorem. We set up a table of strategies:

Today		**At Time *T***	
Action	Cash Flow	$S_T < X$	$S_T \geq X$
Short stock	$+S_0$	$-S_T$	$-S_T$
Lend PV of X	$-Xe^{-rT}$	$+X$	$+X$
Write put	$+P_0$	$-(X - S_T)$	0
Total	$P_0 + S_0 - Xe^{rT}$	0	$X - S_T \leq 0$

Since the strategy has only negative or zero payoffs in the future, it must have a positive cash flow today, so that we can conclude that:

$P_0 - Xe^{-rT} + S_0 \geq 0$

Combined with the fact that in no case can a put value be negative, this proves the proposition.

PROPOSITION 4 (PUT-CALL PARITY) Let C_0 be the price of a European call with exercise price X written on a stock whose current price is S_0. Let P_0 be the price of a European put on the same stock with the same exercise price X. Suppose both put and call have exercise date T, and suppose that the continuously compounded interest rate is r. Then:

$$C_0 + Xe^{-rT} = P_0 + S_0$$

Proof of Proposition 4 The proof is similar in style to that of the two previous propositions. We consider a combination of the four assets (the put, the call, the stock, and a bond) and show that the pricing relation must hold:

Today		**At Time T**	
Action	Cash Flow	$S_T < X$	$S_T \geq X$
Buy call	$-C_0$	0	$+S_T - X$
Buy a bond with payoff X at time T	$-Xe^{-rT}$	X	X
Write a put	$+P_0$	$-(X - S_T)$	0
Short one share of the stock	$+S_0$	$-S_T$	$-S_T$
Total	$-C_0 - Xe^{-rT} + P_0 + S_0$	0	0

Since the strategy has future payoffs that are zero no matter what happens to the price of the stock, it follows that the initial cash flow of the strategy must also be zero.[7] This means that

$$C_0 + X * e^{-rT} - P_0 - S_0 = 0$$

which proves the proposition.

Put-call parity states that the stock price S_0, the price of a call C_0 with exercise price X, and the price of a put P_0 with exercise price X are simultaneously determined with the interest rate r. Following is an illustration that uses the call price C_0, the option exercise price X, the current stock price S_0, and the interest rate r to compute the price of a put with exercise price X and time to maturity T:

	A	B	C
1	**PUT-CALL PARITY**		
2	Current stock price, S_0	55	
3	Option time to maturity, T	0.5	
4	Option exercise price, X	60	
5	Interest rate, r	5%	
6	Call price, C_0	3	
7	Put price, P_0	6.519	<-- =B6+B4*EXP(-B5*B3)-B2
8			
9	This spreadsheet uses put-call parity to derive the put price P_0 from the call price C_0, the interest rate r, the time to maturityT, and the exercise price X.		

PROPOSITION 5 (CALL OPTION PRICE CONVEXITY) Consider three European calls, all written on the same non-dividend-paying stock and with the same expiration date T. We suppose that the exercise prices on the calls are X_1, X_2, and X_3 and denote the associated call prices by C_1, C_2, and C_3. We further assume that $X_2 = \frac{X_1 + X_3}{2}$. Then $C_2 < \frac{C_1 + C_3}{2}$. It follows that the call option price is a convex function of the exercise price.

Proof of Proposition 5 To prove the proposition, we consider the following strategy:

At Time 0		**At Time T**			
Action	Cash Flow	$S_T < X_1$	$X_1 \le S_T < X_2$	$X_2 \le S_T < X_3$	$X_3 \le S_T$
Buy call with exercise price X_1	$-C_1$	0	$S_T - X_1$	$S_T - X_1$	$S_T - X_1$
Buy call with exercise price X_3	$-C_3$	0	0	0	$S_T - X_3$
Write two calls with exercise price X_2	$+2C_2$	0	0	$-2(S_T - X_2)$	$-2(S_T - X_2)$
Total	$2C_2 - C_1 - C_3$	0	$S_T - X_1 \ge 0$	$2X_2 - X_1 - S_T = X_3 - S_T > 0$	0

Since the payoffs in the future are all non-negative (with a positive probability of being positive), it follows that the initial cash flow from the position must be negative:

$$2C_2 - C_1 - C_3 < 0 \Rightarrow C_2 < \frac{C_1 + C_3}{2}$$

This proves the proposition. (Note that the assumption that $X_2 = \frac{X_1 + X_3}{2}$ is made for convenience and does not affect the generality of the argument.)

Without proof, we state a similar proposition for puts.

PROPOSITION 6 (PUT PRICE CONVEXITY) Consider three European puts, all written on the same non-dividend-paying stock and with the same expiration date T. We suppose that the exercise prices on the calls are X_1, X_2, and X_3 and denote the associated put prices by P_1, P_2, and P_3. We further assume that $X_2 = \frac{X_1 + X_3}{2}$. Then the put price is a convex function of the exercise price:

$$P_2 < \frac{P_1 + P_3}{2}$$

Finally, we state the following proposition, whose proof involves only a minor modification of the proof of Proposition 1:

PROPOSITION 7 (CALL OPTION BOUNDS WITH A KNOWN FUTURE DIVIDEND) Consider a call with exercise price X and maturity date T. Suppose that at some time $t < T$, the stock will, with certainty, pay a dividend D. Then the lower bound on the call option price is given by:

$$C_0 \geq Max(S_0 - De^{-rt} - Xe^{-rT}, 0)$$

Proof of Proposition 7 The proof involves only a minor modification of the proof of Proposition 1.

Today		Time t	At Time T	
Action	Cash Flow		$S_T < X$	$S_T \geq X$
Buy stock	$-S_0$	$+D$	$+S_T$	$+S_T$
Borrow the PV of the dividend D	$+De^{-rT}$	$-D$		
Borrow PV of X	$+Xe^{-rT}$		$-X$	$-X$
Write call	$+ C_0$		0	$-(S_T - X)$
Total	$-S_0 + De^{-rt} + Xe^{-rT} + C_0$	0	$S_T - X \leq 0$	0

This proves the proposition.

16.7 Summary

This chapter summarizes the basic definitions and features of options. It is, however, by no means an adequate introduction to these complex securities for those with no pre-knowledge. To decipher the mysteries of options, we recommend the introductory chapters of a good option text.

Exercises

1. When you looked at the newspaper quotes for options on ABC stock, you saw that a February call option with X = \$37.5 is priced at \$6.375, whereas the April call option with the same exercise price is priced at 6. Can you devise an arbitrage out of these prices? Do you have an explanation for the newspaper quotes?
2. An American call option is written on a stock whose price today is S = \$50. The exercise price of the call is X = \$45.

1. If the call price is 2, explain how you would use arbitrage to make an immediate profit.
2. If the option is exercisable at time $T = 1$ year and if the interest rate is 10%, what is the minimum price of the option? Use Proposition 1.

3. A European call option is written on a stock whose current price $S = \$80$. The exercise price $X = \$80$, the interest rate $r = 8\%$, and the time to option exercise $T = 1$. The stock is assumed to pay a dividend of 3 at time $t = ½$. Use Proposition 7 to determine the minimum price of the call option.
4. A put with an exercise price of €50 has a price of €6 and a call on the same stock with an exercise price of €60 has a price of €10. Both put and call have the same expiration date. On the same set of axes, draw the profit diagram for:
   1. One put bought and one call bought.
   2. Two puts bought and one call bought.
   3. Three puts bought and one call bought.
   4. All three lines cross each other for the same value of S_T. Derive this value.

5. Consider the following two calls:
   - Both calls are written on shares of ABC Corp. whose current share price is \$100. ABC does not pay any dividends.
   - Both calls have one year to maturity.
   - One call has $X_1 = 90$ and has a price of \$30; the second call has $X_2 = 100$ and has a price of \$20.
   - The riskless, continuously compounded interest rate is 10%.

   By designing a spread (i.e., buying one call and writing another) position, show that the difference between the two call prices is *too large* and that a riskless arbitrage exists.
6. A share of ABC Corp. sells for \$95. A call on the share with an exercise price \$90 sells for \$8.
   1. Graph the profit pattern from buying one share and one call on the share.
   2. Graph the profit pattern from buying one share and two calls.
   3. Consider the profit pattern from buying one share and calls. At which share price do all of the profit lines cross?

7. A European call with a maturity of six months and exercise price $X = 80$ is selling for \$12. The option is written on a stock whose current price is \$85; a European put, written on the same stock with the same maturity and with the same exercise price, is selling for \$5. If the annual interest rate (continuously compounded) is 10%, construct an arbitrage from this situation.
8. Prove Proposition 6. Then solve the following problem: Three puts on shares of XYZ with the same expiration date are selling at the following prices:
   - Exercise price 40: 6
   - Exercise price 50: 4
   - Exercise price 60: 1

   Show an arbitrage strategy that allows you to profit from these prices and prove that it works.
9. The current stock price of ABC Corp. is 50. Prices for six-month calls on ABC are given in the table below. Draw a profit diagram of the following strategy: Buy one 40 call, write two 50 calls, buy one 60 call, and write two 70 calls.

Call	Price
40	16.5
50	9.5
60	4.5
70	2

10. Consider the following option strategy, which consists only of calls:

Exercise Price	Bought/Written? Number?	Price per Call Option
20	1 written	45
30	2 bought	33
40	1 written	22
50	1 bought	18
60	2 written	17
70	1 bought	16

1. Draw the profit diagram for this strategy.
2. The prices given include one violation of an arbitrage condition. Identify this violation and explain.

11. A share of Formila Corp. is currently trading at $38.50, and a 1-year call option on Formila with X = $40 is trading at $3. The risk-free interest rate is 4.5%.
    1. What should be the price of a 1-year put option on the stock with X = $40? Why?
    2. If the price of a put is $2, construct an arbitrage strategy.
    3. If the price of a put is $4, construct an arbitrage strategy.

---

1. 1. Good chapters can be found in Hull (2015); Bodie, Kane, and Marcus (2019).
2. 2. It is of course not logical that you can ever make an immediate profit by buying an American option and immediately exercising it. Thus, for American calls, $C_0 > S_0 - X$, and for American puts, $P_0 > X - S_0$, *in-the-*

money and *out-of-the-money* refer only to the relation between S_0 and X without taking into account the option price.

3. 3. Our use of the word *profit* in this section constitutes a slight abuse of language and the standard finance concept of the word, since we are ignoring the interest costs associated with buying the asset. In the case at hand, this abuse of language is both traditional and harmless.
4. 4. Because the exercise price of this call is equal to the current market price of the stock, it is called an *at-the-money* call. When the exercise price of the call is higher than the current market price, it is called an *out-of-the-money* call, and when the exercise price is lower than the current market price, the call is an *in-the-money* call.
5. 5. Because the exercise price of this put is equal to the current market price of the stock, it is called an *at-the-money* put. When the exercise price of the put is higher than the current market price, it is called an *in-the-money* put, and when the exercise price is lower than the current market price, the call is an *out-of-the-money* put.
6. 6. In section 16.6 we prove and illustrate the *put-call parity theorem*. It follows from this theorem that a call must be priced at a price *C* such that $C = P + S_0 - X * e^{-rT}$. Thus, when calls are correctly priced according to this theorem, the payoff from a put + stock combination is the same as that from a call + bond combination. We prove this theorem and give an illustration in the next section.
7. 7. This is a fundamental fact of finance: If a financial strategy has future payoffs that are identically zero, then its current cost must also be zero. Likewise, if a financial strategy has future payoffs that are non-negative, then its time-zero payoff must be negative (that is, it must cost something).

17 The Binomial Option Pricing Model

17.1 Overview

Next to the Black-Scholes model (discussed in Chapter 18), the binomial option pricing model is the most widely used option pricing model. It has many advantages: It is a simple model, which—in addition to giving many insights into option pricing—is easily programmed and adapted to numerous, and often quite complicated, option pricing problems. When extended to many periods, the binomial model becomes one of the most powerful ways of valuing securities like options whose payoffs are contingent on the market prices of other assets.

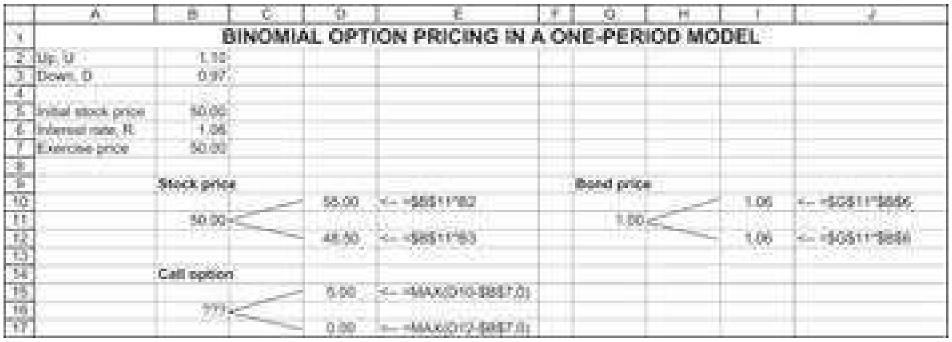
The binomial model depends on using state prices to compute the values of risky assets. When the state pricing principles underlying the model are understood, we gain deeper insight into the economics of contingent asset pricing. In this chapter, which illustrates the simple uses of the binomial model, we devote a considerable amount of space to deriving and using state prices. In Chapters 26 and 27 we return to the binomial model and use it in the Monte Carlo pricing approach of contingent securities.

17.2 Two-Date Binomial Pricing

To illustrate the use of the binomial model, we start with the following very simple example:

- • There is one period and two dates, date 0 = today and date 1 = one year from now.
- • There are two “fundamental” assets: a stock and a bond. There is also a derivative asset: a call option written on the stock.
- • The stock price today is $50 and at date 1 will either go up by 10% or go down by 3%.
- • The one-period interest rate is 6%.
- • The call option matures at date 1 and has exercise price X = $50.

Below is a clip from a spreadsheet that incorporates this model. Notice that in cells B2, B3, and B6 we have used values for 1 plus the 10% up move, 1 plus the −3% down move, and 1 plus the 6% interest. We use capital letters U, D, and R to denote these values.[1]



	A	B	C	D	E	F	G	H	I	J
1	BINOMIAL OPTION PRICING IN A ONE-PERIOD MODEL									
2	Up, U	1.10								
3	Down, D	0.97								
4										
5	Initial stock price	50.00								
6	Interest rate, R	1.06								
7	Exercise price	50.00								
8										
9		Stock price					Bond price			
10				55.00	<-- =$B$11*B2				1.06	<-- =$G$11*$B$6
11		50.00					1.00			
12				48.50	<-- =$B$11*B3				1.06	<-- =$G$11*$B$6
13										
14		Call option								
15				5.00	<-- =MAX(D10-$B$7,0)					
16		???								
17				0.00	<-- =MAX(D12-$B$7,0)					

We wish to price the call option. We do this by showing that there is a *combination of the bonds and stocks that exactly replicates the call option's payoffs*. To show this, we use some basic algebra; suppose we find *A* shares of the stock and *B* bonds such that:

$55A + 1.06B = 5$

$48.5A + 1.06B = 0$

This system of equations solves to give:

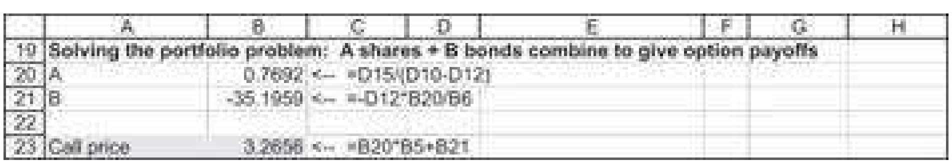
$$A = \frac{5-0}{55-48.5} = 0.7692$$

$$B = \frac{0-48.5A}{1.06} = -35.1959$$

Thus purchasing 0.77 of a share of the stock and borrowing \$35.20 at 6% for one period will give payoffs of \$5 if the stock price goes up and \$0 if the stock price goes down—the payoffs of the call option. It follows that the price of the option must be equal to the cost of replicating its payoffs. That is:

Call option price = 0.7692 * \$50 − \$35.1959 * \$1 = \$3.2656

This logic is called "pricing by arbitrage" or "the law of one price": If two assets or sets of assets (in our case, the call option and the portfolio of 0.77 of the stock and −\$35.20 of the bonds) have the same payoffs, they must have the same market price.



	A	B	C	D	E	F	G	H
19	Solving the portfolio problem: A shares + B bonds combine to give option payoffs							
20	A	0.7692	<-- =D15/(D10-D12)					
21	B	-35.1959	<-- =-D12*B20/B6					
22								
23	Call price	3.2656	<-- =B20*B5+B21					

Applying the same logic to a put gives the put price as 0.4354:

	A	B	C	D	E	F
1	BINOMIAL PUT OPTION PRICING IN A ONE-PERIOD MODEL					
2	Up, U	1.10				
3	Down, D	0.97				Solving the put price
4						55*A+1.06*B=0
5	Initial stock price	50.00				48.5*A+1.06*B=1.5
6	Interest rate, R	1.06				
7	Exercise price	50.00				A=-1.5/(55-48.5)
8		Put option				B=-55*A/1.06
9				0.00	<-- =MAX($B$7-B5*B2,0)	
10		???				
11				1.50	<-- =MAX($B$7-B5*B3,0)	
12						
13	Solving the portfolio problem: A shares + B bonds combine to give option payoffs					
14	A	-0.2308	<-- =-D11/(B5*(B2-B3))			
15	B	11.9739	<-- =-B5*B2*B14/B6			
16						
17	Put price	0.4354	<-- =B14*B5+B15			

In succeeding sections, we show that this simple arbitrage argument can be extended to multiple periods. But in the meantime, we confine ourselves in the next section to generalizing the logic.

17.3 The State Prices

There is actually a simpler (and more general) way to solve this problem. Viewed from today, there are only two possibilities for next period: Either the stock price goes up or it goes down. Think about the market determining a price q_U for \$1 in the "up" state of the world and a price q_D for \$1 in the "down" state of the world. Then both the bond and the stock (and any other asset) have to be priced using these *state prices*:

Stock: $q_U * S * U + q_D * S * D = S \Rightarrow q_U U + q_D D = 1$ *Bond*: $q_U * R + q_D * R = 1$

The state prices are thus an illustration of the linear pricing principle. If the stock price can move up in one period by a factor U and down by a factor D, and if 1 plus the one-period interest rate is R, then any other asset will be priced by discounting its payoff in the "up" state by q_U and by discounting its payoff in the "down" state by q_D.

The two equations above solve to give:

$$q_U = \frac{R - D}{R(U - D)}, \quad q_D = \frac{U - R}{R(U - D)}$$

In our case these state prices are given by:

	A	B	C
1	DERIVING THE STATE PRICES		
2	Up, U	1.10	
3	Down, D	0.97	
4	Interest rate, R	1.06	
5			
6	State prices		
7	q_U	0.6531	<--=(B4-B3)/(B4*(B2-B3))
8	q_D	0.2903	<--=(B2-B4)/(B4*(B2-B3))
9			
10	Check: Confirm that state prices actually price the stock and the bond		
11	Pricing the stock: 1 = q_U*U+q_D*D?	1	<--=B7*B2+B8*B3
12	Pricing the bond: (q_U+q_D)*R=1?	1	<--=SUM(B7:B8)*B4

In rows 11 and 12 we check that the state prices indeed give back the interest rate and the stock price.

We can now use the state prices to price the call and the put on the stock and also to establish that put-call parity holds. The call and the put options should be priced by:

$$C = q_U \; max(S * U - X, 0) + q_D \; max(S * D - X, 0)$$
$$P = q_U \; max(X - S * U, 0) + q_D \; max(X - S * D, 0)$$

or, if priced by put-call-parity

$$P = C + PV(X) - S$$

In a spreadsheet:

	A	B	C
1	**BINOMIAL PRICING WITH STATE PRICES IN A ONE-PERIOD MODEL**		
2	Up, U	1.10	
3	Down, D	0.97	
4	Interest rate, R	1.06	
5	Initial stock price, S	50.00	
6	Option exercise price, X	50.00	
7			
8	**State prices**		
9	q_U	0.6531	<--=(B4-B3)/(B4*(B2-B3))
10	q_D	0.2903	<--=(B2-B4)/(B4*(B2-B3))
11			
12	**Pricing the call and the put**		
13	Call price	3.2656	<--=B9*MAX(B5*B2-B6,0)+B10*MAX(B5*B3-B6,0)
14	Put price	0.4354	<--=B9*MAX(B6-B5*B2,0)+B10*MAX(B6-B5*B3,0)
15			
16	**Put-call parity**		
17	Stock + put	50.4354	<--=B5+B14
18	Call + PV(X)	50.4354	<--=B13+B6/B4
19			
20	**Note about PV(X) in put-call parity:** In the continuous-time framework (the standard Black-Scholes framework), PV(X) = X*Exp(-r*T). Because the framework here is discrete time, PV(X) is also discrete-time: PV(X)=X/(1+r)=X/R.		

The formulas we use (with $S = 50$, $X = 50$, $U = 1.10$, $D = 0.97$, $R = 1.06$) are

$C = q_U \; max(S * U - X, 0) + q_D \; max(S * D - X, 0)$
$= 0.6531 * max(55 - 50, 0) + 0.2903 * max(45.5 - 50, 0) = 3.2656$

for the call, and

$P = q_U \; max(X - S * U, 0) + q_D \; max(X - S * D, 0)$
$= 0.6531 * max(50 - 55,0) + 0.2903 * max(50 - 48.5, 0) = 0.4354$

for the put.

As expected, the put-call parity theorem holds for this particular put and call (cells B17:B18):

$$P + S = C + \frac{X}{R}$$
$$0.4354 + 50 = 3.27 + \frac{50}{1.06}$$

State Prices and Risk-Neutral Probabilities

Multiplying the state prices by 1 plus the interest rate, denoted by R, gives the *risk-neutral probabilities:* $\pi_U = q_U R$; $\pi_D = q_D R$. The risk-neutral probabilities look like a probability distribution of the states, since they sum to 1:

$$\pi_U + \pi_D = q_U R + q_D R = \frac{R-D}{R(U-D)} R + \frac{U-R}{R(U-D)} R = 1$$

Furthermore, there is a fundamental equivalence of pricing by the risk-neutral probabilities and pricing by the actual state probabilities (which are often unobservable). Suppose an asset has state-dependent payoffs X_U in the "up" state and X_D in the "down" state of a one-period model. Then the actual expected value of the asset, at date 1, is $Pr_U\, X_U + Pr_D\, X_D$, where Pr_i is the probability to state "i" and the date 0 price of the asset is the present value of the expected value discounted by its expected return ($E(r)$). Usually it is impractical to extract the actual probabilities and the expected return. Luckily, the date 0 price of the asset using the risk-neutral probabilities is the discounted expected asset payoff, where the expectation is computed using the risk-neutral probabilities as if they are the actual state probabilities and where the discounting is done with the risk-free rate:

$$\frac{\overbrace{\pi_U X_U + \pi_D X_D}^{=q_U X_U + q_D X_D}}{R} = \frac{\textit{Expected asset payoff using state probabilities}}{1+E(r)}$$

We prefer to use state prices, but many researchers are more comfortable using the pseudo-probabilities of the risk-neutral prices and then discounting the "expected" payoffs.

17.4 The Multi-period Binomial Model

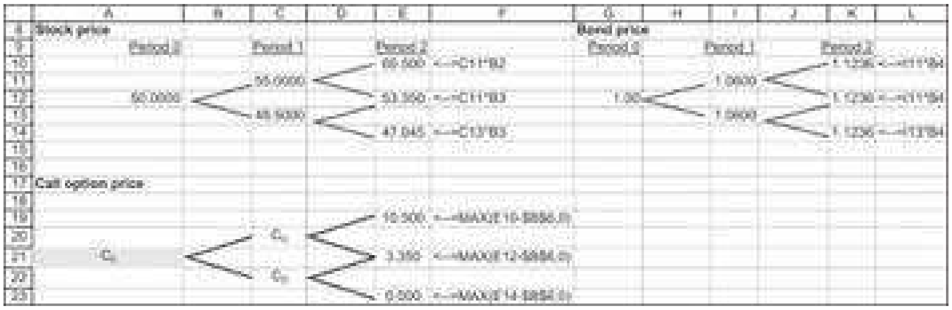
The binomial model can easily be extended to more than one period. Consider, for example, a two-period (three-date) binomial model that has the following characteristics:

- • In each period the stock price goes up by 10% or down by −3% from what it was in the previous period. This means that $U = 1.10$, $D = 0.97$.
- • In each period the interest rate is 6%, so that $R = 1.06$.

Because U, D, and R are the same in each period,

$$q_U = \frac{R-D}{R(U-D)} = 0.6531,\ q_D = \frac{U-R}{R(U-D)} = 0.2903$$

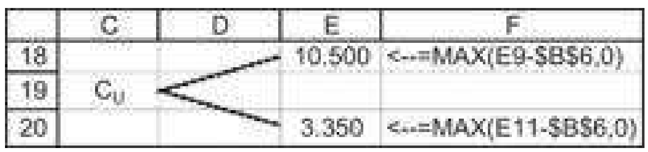
we can now use these state prices to price a call option written on the stock after two periods. As before, we assume that the stock price is $50 initially and that the call exercise price is $X = 50$ after two periods. This gives the following picture:



How would we determine the call option price today (C_0)? To do this, we go backward, starting at period two (cells E19:E23 above):

At date 2: At the end of two periods, the stock price is either $60.50 (corresponding to two "up" movements in the price), $53.35 (one "up" and one "down" movement), or $47.05 (two "down" movements in the price). Given the exercise price of $X = 50$, this means that the terminal option payoff in period 2 is either $10.50, $3.35, or $0.

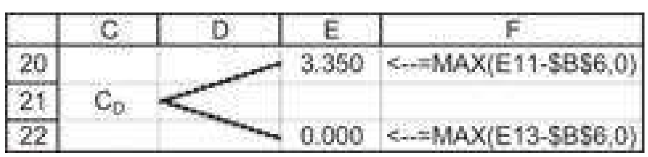
At date 1: There are two possibilities. One is that we have reached an "up" state, in which case the current stock price is $55, and the option will pay off $10.50 or $3.35 in the next period:



We use the state prices of $q_u = 0.6531$, $q_d = 0.2903$ to price the option at this state:

Option price at "up" state, date 1, $(C_U) = 0.6531 * 10.50 + 0.2903 * 3.35 = 7.8302$

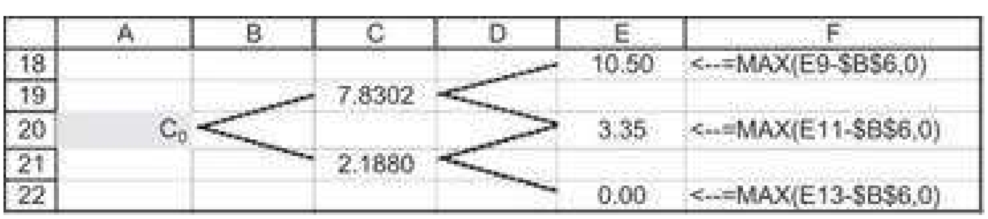
The alternative possibility is that we are in the "down" state of period 1:



Using the same state prices (which, after all, depend only on the "up" and "down" movements of the stock price and the interest rate), we get:

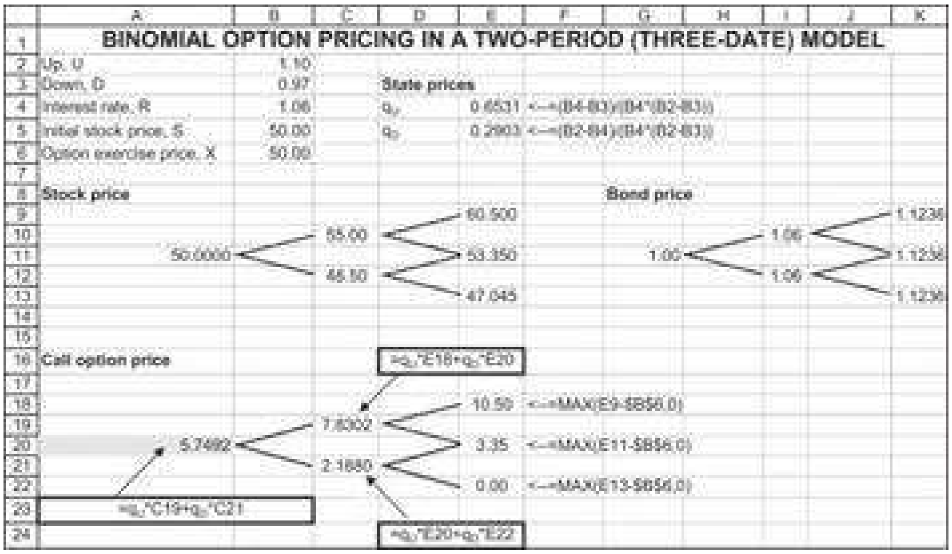
Option price at "down" state, date 1, $(C_D) = 0.6531 * 3.35 + 0.2903 * 0 = 2.188$

At date 0: Going backward in this way, we have now filled in the following picture:



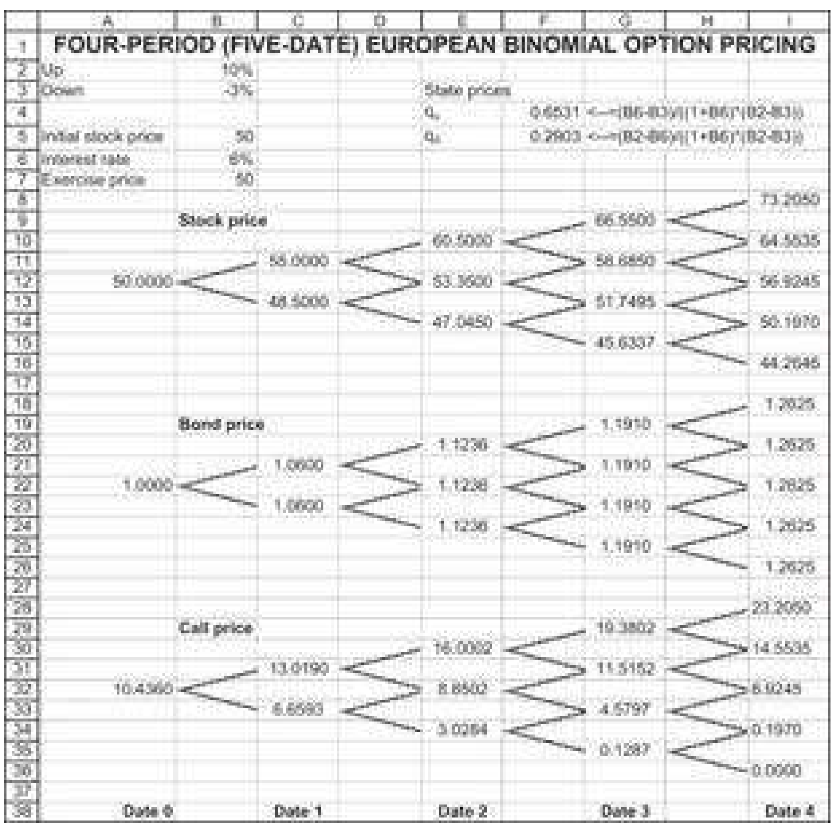
Thus, at period 0, the buyer of an option owns a security that will be worth \$7.83 if the underlying stock has an "up" movement and that will be worth \$2.19 if the stock has a "down" movement. We can again use the state prices to value this option:

Option price at date 0, (C_0) = 0.6531 * 7.830 + 0.2903 * 2.188 = 5.749



Extending the Binomial Pricing Model to Many Periods

It is clear that the logic of the above example can be extended to many periods. Here's another Excel graphic showing a four-period (five-date) model using the same "up" and "down" parameters as before:



Using R to Implement the Binomial Model of Many Periods

R is very effective in running the binomial model for many periods. It is especially much more effective than Excel when n is relatively large. Here is the implementation of the above example in R:

```
## Pricing European Call options with binomial tree
# Input:
S0 <- 50
X <- 50
U <- 1.1
D <- 0.97
R <- 1.06
m <- 4 # number of periods

# State prices
qU <- (R - D) / (R * (U - D))
qD <- (U - R) / (R * (U - D))

# Risk neutral probabilities
pi_U <- qU * R
pi_D <- qD * R

# A matrix of all binomial tree paths
paths <- sapply(1:m, function(x){rep(c(rep(U, (2^x) / 2), rep(D, (2^x) / 2)), 2^(m-x))})

# Stock price paths
stock_paths <- cbind(S0, t(S0 * apply(paths, 1, cumprod)))

# Stock price at time T
ST <- stock_paths[,m+1]

# Call option payoff
payoff <- sapply(ST - X, max, 0)

# Path risk-neutral probability
prob_mat <- ifelse(paths == U, pi_U, pi_D)
prob_vec <- apply(prob_mat, 1, prod)

# Price today = weighted average discounted payoff
Call_price <- payoff %*% prob_vec / R^m
```

Do You Really Have to Price Everything Backward?

The answer is "no." There's no necessity to price the call price payoffs "backward" at each node back from the terminal date, *as long as the call is European*.[2] It is enough to price each of the terminal payoffs by the state prices, provided you count properly the number of paths to each terminal node. Here's an illustration, using the same example:

	A	B	C	D	E	F	G	H
1	BINOMIAL OPTION PRICING WITH STATE PRICES IN A FOUR-PERIOD (FIVE-DATE) MODEL							
2	Up, U	1.10						
3	Down, D	0.97		State prices				
4	Interest rate, R	1.06		q_U		0.6531 <--=(B4-B3)/(B4*(B2-B3))		
5	Initial stock price, S	50.00		q_D		0.2903 <--=(B2-B4)/(B4*(B2-B3))		
6	Option exercise price, X	50.00						
7					=$E$4^A12*$E$5^B12			
8	=$B$5*$B$2^A12*$B$3^B12							
9				=MAX(C12-$B$6,0)			=COMBIN(4,B12)	
10	Number of "up" steps at terminal date	Number of "down" steps at terminal date	Terminal stock price	Option payoff at terminal state	State price for terminal date	Number of paths to terminal state	Value	
11	4	0	73.2050	23.2050	0.1820	1	4.2224	
12	3	1	64.5535	14.5535	0.0809	4	4.7076	<--=D12*E12*F12
13	2	2	56.9245	6.9245	0.0359	6	1.4933	
14	1	3	50.1970	0.1970	0.0160	4	0.0126	
15	0	4	44.2648	0.0000	0.0071	1	0.0000	
16						Call price	10.4360	<--=SUM(G11:G15)
17						Put price	0.0407	<--=G16+B6/B4^4-B5
18	Notes							
19	There are 5 dates in this model (0, 1, ..., 5) but only 4 periods and thus only 4 possible "up" or "down" steps.							
20	The put price in cell G17 is computed using put-call parity: put = call + PV(X) - stock							

European puts can be priced either by using the logic above or—as in cell G17 above—by using put-call parity.

To recapitulate: The price of a European call option in a binomial model with *n* periods is given by:

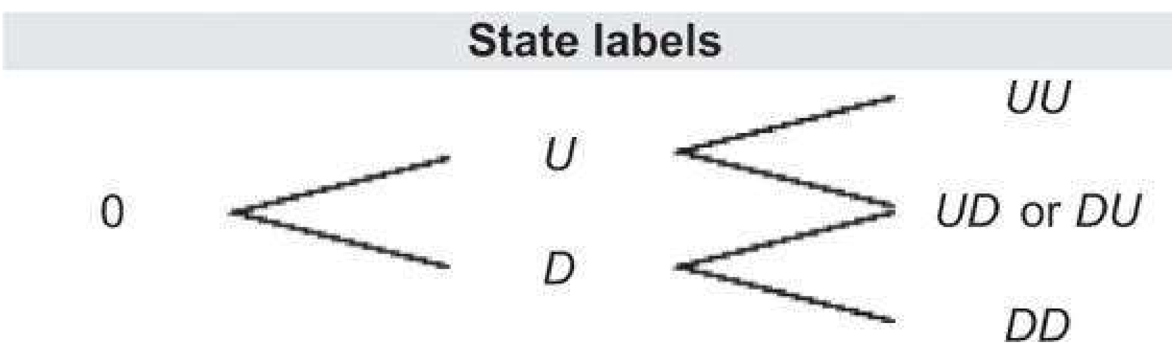
$$Call\ price = \sum_{i=0}^{n} \binom{n}{i} q_U^i q_D^{n-i}\ max(S * U^i D^{n-i} - X, 0)$$

$$Put\ price = \begin{cases} \sum_{i=0}^{n} \binom{n}{i} q_U^i q_D^{n-i}\ max(X - S * U^i D^{n-i}, 0) & direct\ pricing \\ Call\ price + \dfrac{X}{R^n} - S & using\ put\text{-}call\ parity \end{cases}$$

In section 17.6 we implement these formulas in VBA and in R.

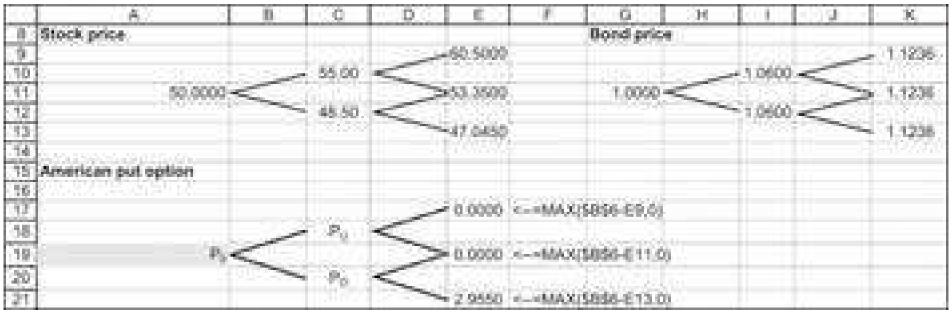
17.5 Pricing American Options Using the Binomial Pricing Model

We can use the binomial pricing model to calculate the prices of American options as well as European options. We reconsider the basic model above, in which *Up* = 1.10, *Down* = 0.97, *R* = 1.06, *S* = 50, and *X* = 50. We examine the three-date version of the model. Recall from Chapter 16 that an American call option on a non-dividend-paying stock has the same value as a European call option. It is thus more interesting to start with the pricing of an American put. The payoff patterns for the stock and the bond have been given above, and it remains only to consider the payoff patterns for a put option with *X* = 50. We reference the states of the world by using the following labels:



Put Payoffs at Date 3

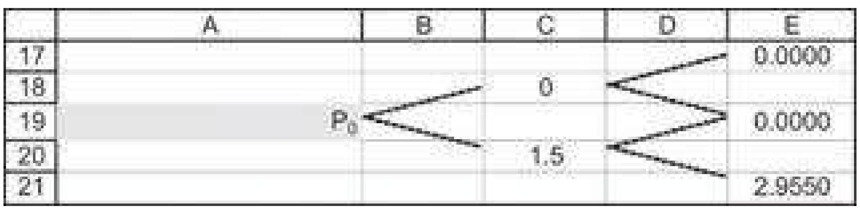
Here are the values of the stock, the bond, and the date 3 put payoffs:



At date 2, the holder of an American put can choose whether to hold the put or to exercise it. We now have the following value function for date 1:

$$\text{Put value, date 1, state } U, (P_U) = \max \begin{cases} \text{Put value if exercised} = Max(X - S_U, 0) \\ q_U * \text{Put payoff in state } UU + q_D * \text{Put payoff in state } UD \end{cases}$$

A similar function holds for the put value in state *d* at date 1. The resulting tree now looks like this:



Here's the explanation:

- • In state U, the put is valueless. When the stock price is \$55, it is not worthwhile to exercise the put early, since $Max(X - S_U, 0) = Max(50 - 55, 0) = 0$. On the other hand, since the future put payoffs from state U are zero, state-dependent present value of these future payoffs (the second line in the previous formula) is also zero.
- • In state D, on the other hand, the holder of the put gets $Max(50 - 48.5, 0) = 1.5$ if the holder exercises the put; however, if one holds the put without exercise, its market value is the state-dependent value of the future payoffs:

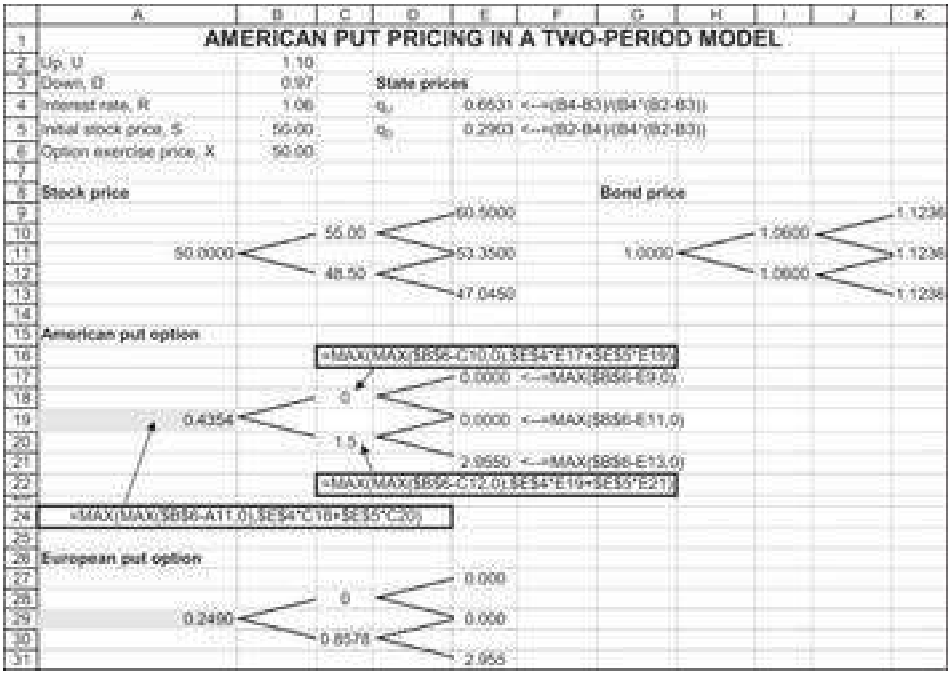
$$q_U * 0 + q_D * 2.9550 = 0.6531 * 0 + 0.2903 * 2.9550 = 0.8578$$

It is clearly preferable to exercise the put in this state rather than to hold onto it.

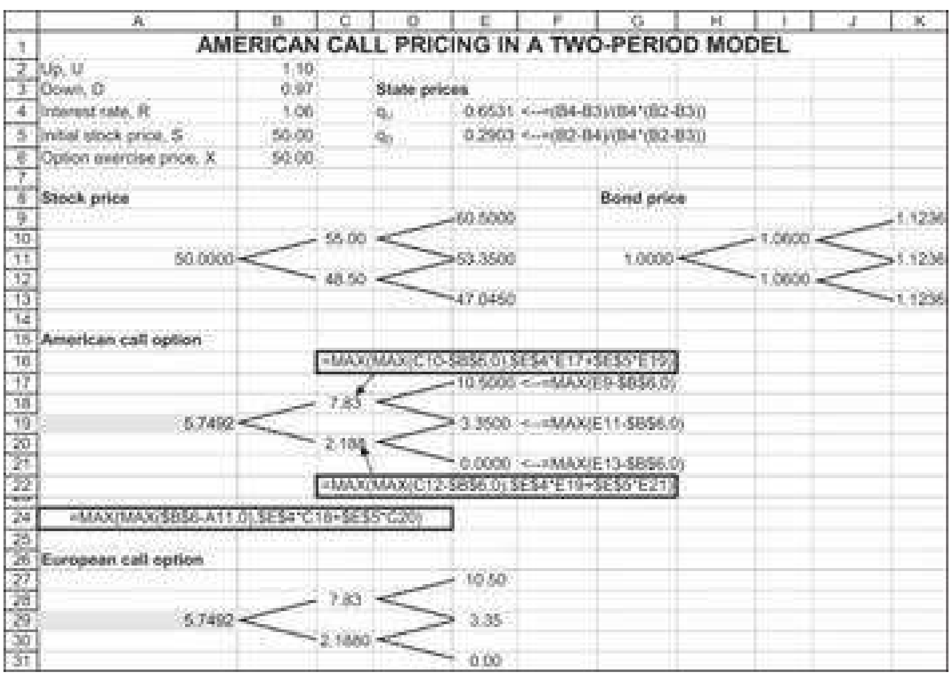
At date 0, a similar value function recurs:

$$\begin{matrix}\text{Put value} \\ \text{at date 0}\end{matrix} = \max \begin{cases} \text{Put value if exercised} = Max(X - S_0, 0) \\ q_U * \text{Put payoff in state } U + q_D * \text{Put payoff in state } D \end{cases}$$

The spreadsheet is given below:



We can use the same logic to price an American call option, though—following Proposition 2 of Chapter 16—we know that the value of an American and a European call should coincide. And so, they do:



17.6 Programming the Binomial Option Pricing Model

The pricing procedure used in the above examples can easily be programmed using Excel's VBA programming language and in R. In the binomial model, the price can move *up* or *down* in any time period. If q_U is the state price associated with an up move and if q_D is the state price associated with a down move, then the binomial European option prices are given by:

$$Binomial\ European\ call = \sum_{i=0}^{n} \binom{n}{i} q_U^i q_D^{n-i}\ max(S * U^i D^{n-i} - X, 0)$$

$$Binomial\ European\ put = \begin{cases} \sum_{i=0}^{n} \binom{n}{i} q_U^i q_D^{n-i}\ max(X - S * U^i D^{n-i}, 0) \\ \quad or\ by\ put - call\ parity \end{cases}$$

Here, U is an up move in the stock price, D is a down move in the stock price, and $\binom{n}{i}$ is the binomial coefficient (the number of up moves in n total moves):

$$\binom{n}{i} = \frac{n!}{i!(n-i)!}$$

We use Excel's **Combin(n,i)** to give values for the binomial coefficients.

Programming the Binomial Option Pricing Model: The Case of European Options

Here are two VBA functions that compute the value of binomial European calls and puts. The function **Binomial_eur_put** uses put-call parity to price the put:

```
Function Binomial_eur_call(Up, Down, Interest, Stock, Exercise,
Periods)
    q_up = (Interest - Down) / (Interest * (Up - Down))
    q_down = 1 / Interest - q_up
    Binomial_eur_call = 0
    For Index = 0 To Periods
        Binomial_eur_call = Binomial_eur_call _
            + Application.Combin(Periods, Index) * _ q_up ^ Index
            * _
            q_down ^ (Periods - Index) * _
            Application.Max(Stock * Up ^ Index * Down ^ _(Periods
            - Index) - Exercise, 0)
    Next Index
End Function
Function Binomial_eur_put(Up, Down, Interest, Stock, _
    Exercise, Periods)
    Binomial_eur_put = Binomial_eur_call(Up, Down, _
    Interest, Stock, Exercise, Periods) _
    + Exercise / Interest ^ Periods - Stock
End Function
```

When we implement this in a spreadsheet for the four-period example of section 17.4, we get the following:

	A	B	C
1	**VBA FUNCTIONS FOR CALLS AND PUTS**		
2	Up, U	1.10	
3	Down, D	0.97	
4	Interest rate, R	1.06	
5	Initial stock price, S	50.00	
6	Option exercise price, X	50.00	
7	Number of periods, n	4	
8			
9	European call	10.4360	<--=binomial_eur_call(B2,B3,B4,B5,B6,B7)
10	European put	0.0407	<--=binomial_eur_put(B2,B3,B4,B5,B6,B7)
11			
12	**Checking put-call parity**		
13	Stock + put	50.0407	<--=B5+B10
14	Call + PV(X)	50.0407	<--=B9+B6/B4^B7

The equivalent program in R should look like this:

```
## European options
# Number of Up and Down steps
ups <- c(m:0)
downs <- m - ups

# Terminal stock price
St <- S0 * U^ups * D^downs

# Option payoff at terminal state
payoffs <- pmax(St - X, 0)

# State price for terminal date
state_prices <- qU^ups * qD^downs

# Number of paths to terminal state
paths <- choose(m, 0:m)

# Value
value <- payoffs * state_prices * paths

# European Call Price
Eur_Call_price <- sum(value)

# European Put Price (using Put-Call-Parity)
Eur_Put_price <- Call_price + X / R^m - S0
```

Programming the Binomial Option Pricing Model: The Case of American Options

Proposition 2 in section 16.6 states that the price of an American call on a non-dividend-paying stock is the same as that of a European option. The pricing of an American put, however, can be different. The following VBA function below uses a binomial option pricing model like the one from section 17.5 to price American puts:

```
Function Binomial_amer_put(Up, Down, Interest, Stock, Exercise,
Periods)
    q_up = (Interest - Down) / (Interest * (Up - Down))
    q_down = 1 / Interest - q_up
    Dim OptionReturnEnd() As Double
    Dim OptionReturnMiddle() As Double
    ReDim OptionReturnEnd(Periods + 1)
    For State = 0 To Periods
        OptionReturnEnd(State) = _
        Application.Max(Exercise - Stock * Up ^ State * Down ^ _
        (Periods - State), 0)
    Next State
    For Index = Periods - 1 To 0 Step -1
        ReDim OptionReturnMiddle(Index)
        For State = 0 To Index
            OptionReturnMiddle(State) = _
            Application.Max(Exercise - Stock * Up ^ State * Down
            ^ _ (Index - State), q_down * OptionReturnEnd(State)
            + _
            q_up * OptionReturnEnd(State + 1))
        Next State
        ReDim OptionReturnEnd(Index)
        For State = 0 To Index
            OptionReturnEnd(State) = OptionReturnMiddle(State)
        Next State
    Next Index
    Binomial_amer_put = OptionReturnMiddle(0)
End Function
```

In this function we use two arrays, called **OptionReturnEnd** and **OptionReturnMiddle**. At each date t, these arrays store the option values for the date itself and the next date ($t + 1$).

Here's an implementation in a spreadsheet, using the two-period, three-date example from section 17.5:

	A	B	C
1	VBA FUNCTIONS FOR CALLS AND PUTS		
2	Up, U	1.10	
3	Down, D	0.97	
4	Interest rate, R	1.06	
5	Initial stock price, S	50.00	
6	Option exercise price, X	50.00	
7	Number of periods, n	2	
8			
9	American put	0.4354	<--=binomial_amer_put(B2,B3,B4,B5,B6,B7)
10	European put	0.2490	<--=binomial_eur_put(B2,B3,B4,B5,B6,B7)
11	American call	5.7492	<--=binomial_amer_call(B2,B3,B4,B5,B6,B7)
12	European call	5.7492	<--=binomial_eur_call(B2,B3,B4,B5,B6,B7)

The values in cells B9 and B10 are for an American and a European put; these values correspond to those given in section 17.5. In cell B11 we use a function similar to the American put function to price American calls. Unsurprisingly—given Proposition 2 of Chapter 16—this function gives the same value as the binomial European call pricing function.

The VBA function works well for many more periods.[3] In the example below we calculate the value of an American put and call for options that expire at $T = 0.75$ of a year. The stochastic process that defines the stock returns has mean $\mu = 10\%$ and standard deviation $\sigma = 35\%$. The annual continuously compounded interest rate is $r = 5\%$ and each year is divided into 40 subperiods, so that a single year has length $\Delta t = 1/40 = 0.025$. Given these numbers, Up, Down, and R are defined by $Up = e^{(\mu - 0.5*\sigma^2)\Delta t + \sigma\sqrt{\Delta t}}$, $Down = e^{(\mu - 0.5*\sigma^2)\Delta t - \sigma\sqrt{\Delta t}}$, $R = e^{r\Delta t}$.

Here is the pricing of an American and European call and put:

	A	B	C
1	VBA FUNCTIONS FOR CALLS AND PUTS n divisions per year, Δt = 1/n Up=exp((μ–0.5*σ^2)*Δt + σ*sqrt(Δt)), Down = exp((μ–0.5*σ^2)*Δt - σ*sqrt(Δt))		
2	Mean return per year, μ	10%	
3	Standard deviation of annual return, σ	35%	
4	Annual interest rate, r	5%	
5			
6	Initial stock price, S	50.00	
7	Option exercise price, X	50.00	
8	Option exercise date (years)	0.75	
9	Number of periods until maturity	30	
10	Number of divisions of 1 year, n	40	<--=B9/B8
11	Δt, the length of one division	0.025	<--=1/B10
12	Up move per Δt	1.057924	<--=EXP((B2-0.5*B3^2)*B11+B3*SQRT(B11))
13	Down move per Δt	0.947081	<--=EXP((B2-0.5*B3^2)*B11-B3*SQRT(B11))
14	Interest rate per Δt	1.001251	<--=EXP(B4*B11)
15			
16			
17	American put	5.2420	<--=binomial_amer_put(B12,B13,B14,B6,B7,B9)
18	European put	5.0626	<--=binomial_eur_put(B12,B13,B14,B6,B7,B9)
19	American call	6.9029	<--=binomial_amer_call(B12,B13,B14,B6,B7,B9)
20	European call	6.9029	<--=binomial_eur_call(B12,B13,B14,B6,B7,B9)

Notice that there are 30 periods in our model since there are 40 divisions of one year and the option's maturity is $T = 0.75$.

This procedure works well even for a very large number of periods. In the example below, the option has a maturity $T = 0.5$, and the basic one-year period is divided into

400 subperiods. Excel easily computes the value of the American put and call, even though a considerable amount of computation is involved:

	A	B	C
1	**VBA FUNCTIONS FOR CALLS AND PUTS** **n divisions per year, Δt = 1/n** **Up=exp((μ–0.5*σ²)*Δt + σ*sqrt(Δt)), Down = exp((μ–0.5*σ²)*Δt - σ*sqrt(Δt))**		
2	Mean return per year, μ	10%	
3	Standard deviation of annual return, σ	35%	
4	Annual interest rate, r	5%	
5			
6	Initial stock price, S	50.00	
7	Option exercise price, X	50.00	
8	Option exercise date (years)	0.50	
9	Number of periods until maturity	200	
10	Number of divisions of 1 year, n	400	<--=B9/B8
11	Δt, the length of one division	0.0025	<--=1/B10
12	Up move per Δt	1.017753	<--=EXP((B2-0.5*B3^2)*B11+B3*SQRT(B11))
13	Down move per Δt	0.982747	<--=EXP((B2-0.5*B3^2)*B11-B3*SQRT(B11))
14	Interest rate per Δt	1.000125	<--=EXP(B4*B11)
15			
16			
17	American put	4.3899	<--=binomial_amer_put(B12,B13,B14,B6,B7,B9)
18	European put	4.2766	<--=binomial_eur_put(B12,B13,B14,B6,B7,B9)
19	American call	5.5111	<--=binomial_amer_call(B12,B13,B14,B6,B7,B9)
20	European call	5.5111	<--=binomial_eur_call(B12,B13,B14,B6,B7,B9)

Using R to price the American put options should look like this:

```r
# Input:
S0 <- 50
X <- 50
U <- 1.1
D <- 0.97
r <- 1.06
m <- 2 # number of periods

# State prices
qU <- (r - D) / (r * (U - D))
qD <- (U - r) / (r * (U - D))

# A matrix of all binomial tree paths
paths <- sapply(1:m, function(x){rep(c(rep(U, (2^x) / 2), rep(D, (2^x) / 2)), 2^(m-x))})

# Stock price paths
stock_paths <- cbind(S0, t(S0 * apply(paths, 1, cumprod)))

# Exercise values
Put_ex_value <- matrix(pmax(0, X - stock_paths), ncol = m + 1)
Call_ex_value <- matrix(pmax(0, stock_paths - X), ncol = m + 1)
# Exercise value at maturity
Put_maturity_values <- Put_ex_value[, m + 1]
Call_maturity_values <- Call_ex_value[, m + 1]

# calculating backwards the binomial tree
put_value <- Put_maturity_values
for (i in m:1) {
    head_res <- head(put_value, length(put_value)/2) # UPs
    tail_res <-tail(put_value, length(put_value)/2) # Downs
    eur_value <- head_res * qU + tail_res * qD # Calculating one period backwards
    Put_ex_values <- Put_ex_value[1:(length(put_value)/2), i] # Exercise value
    put_value <- ifelse(Put_ex_values > eur_value, Put_ex_values, eur_value) # American Put
}
names(put_value) <- c("P0")
put_value
```

And the equivalent equation for the American call option looks like this:

```
call_value <- call_maturity_values
for (i in m:1) {
    head_res <- head(call_value, length(call_value)/2) # UPs
    tail_res <-tail(call_value, length(call_value)/2) # Downs
    eur_value <- head_res * qU + tail_res * qD # Calculating one period backwards
    call_ex_values <- call_ex_value[1:(length(call_value)/2), i] # Exercise value
    call_value <- ifelse(call_ex_values > eur_value, call_ex_values, eur_value) # American call
}
names(call_value) <- c("C0")
call_value
```

17.7 Convergence of Binomial Pricing to the Black-Scholes Price

In this section we discuss the convergence of the binomial model to the Black-Scholes pricing formula. The discussion assumes some understanding of lognormality (discussed in Chapter 23) and of the Black-Scholes option pricing formula (discussed in Chapter 18). So, you may want to skip this section and come back to it later.

Whenever we consider a finite approximation to the option pricing formulas, we must use an approximation to the up and the down movements. One widespread translation of the interest rate r and the stock's volatility σ to an "up" or "down" move necessary for the binomial model is shown here:[4]

$$\Delta t = T/n \qquad R = e^{r\Delta t}$$

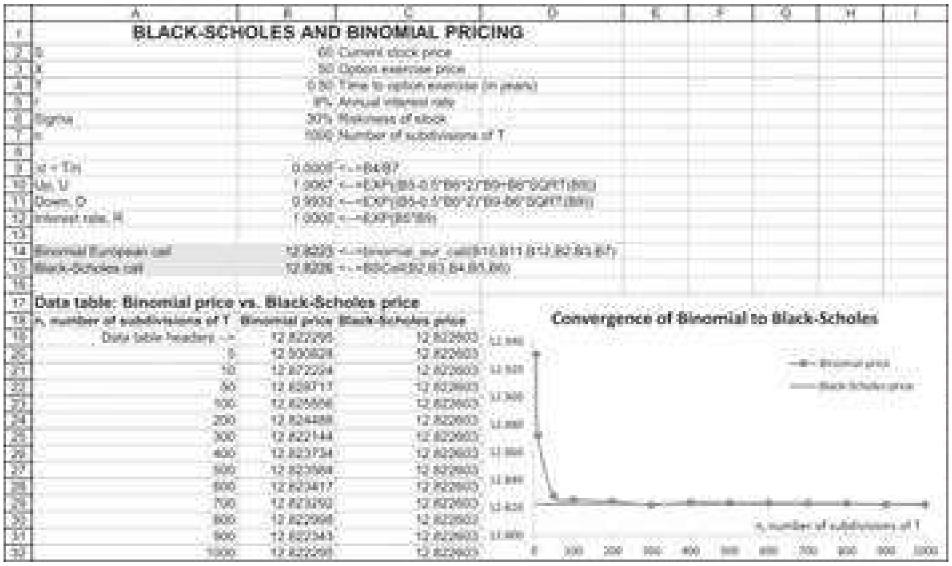
$$U = 1 + up = e^{(\mu - 0.5*\sigma^2)*\Delta t + \sigma\sqrt{\Delta t}} \qquad D = 1 + down = e^{(\mu - 0.5*\sigma^2)*\Delta t - \sigma\sqrt{\Delta t}}$$

This approximation guarantees that as $\Delta t \to 0$ (i.e., as $n \to \infty$), the resulting distribution of the stock returns approaches the lognormal distribution.[5]

Here is an implementation of this methodology in a spreadsheet. The function **Binomial_Eur_call** is the same as that defined above; the function **BSCall** is the Black-Scholes formula and is defined and discussed in Chapter 18:

	A	B	C	D
1	BLACK-SCHOLES AND BINOMIAL PRICING			
2	S	60	Current stock price	
3	X	50	Option exercise price	
4	T	0.50	Time to option exercise (in years)	
5	r	8%	Annual interest rate	
6	Sigma	30%	Riskiness of stock	
7	n	1000	Number of subdivisions of T	
8				
9	Δt = T/n	0.0005	<--=B4/B7	
10	Up, U	1.0067	<--=EXP((B5-0.5*B6^2)*B9+B6*SQRT(B9))	
11	Down, D	0.9933	<--=EXP((B5-0.5*B6^2)*B9-B6*SQRT(B9))	
12	Interest rate, R	1.0000	<--=EXP(B5*B9)	
13				
14	Binomial European call	12.8223	<--=binomial_eur_call(B10,B11,B12,B2,B3,B7)	
15	Black-Scholes call	12.8226	<--=BSCall(B2,B3,B4,B5,B6)	

The binomial model gives a good approximation to the Black-Scholes (cells B14:B15). As the n gets larger, this approximation gets better, though the convergence to the Black-Scholes price is not smooth:



Note that the converge happens quite quickly—within several dozen steps, the binomial price is within 0.01 of the Black-Scholes price.

17.8 Using the Binomial Model to Price Employee Stock Options[6]

An employee stock option (ESO) is a call option given by a company to its employees as part of their remuneration package. Like all call options, the value of an ESO depends on the current price of the stock, the option's exercise price, and the time until exercise. However, ESOs typically have several special conditions:

- • The option has a vesting period. During this period, the employee is not allowed to exercise the option. An employee leaving the company before the vesting period forfeits his option. In the model of this section we assume that a typical employee of the company leaves at exit rate *e* per year.
- • An employee leaving the company after the vesting period is forced to immediately exercise his option.
- • For tax reasons, many ESOs have exercise prices equal to the stock price on the date of issue.

In the model below, adapted from a paper by Hull and White (2004), we assume that an employee will choose to exercise her option when the price of the stock is greater than some multiple *m* of the ESO's exercise price X. We first present the model's implementation and results and then discuss the VBA program, which gives us these results:

	A	B	C
1	**A BINOMIAL EMPLOYEE STOCK OPTION PRICING MODEL** Based on Hull-White (2004)		
2	S	50	Current stock price
3	X	50	Option exercise price
4	T	10.00	Time to option exercise (in years)
5	Vesting period (years)	3.00	
6	Interest	5.00%	Annual interest rate
7	Sigma	35%	Riskiness of stock
8	Stock dividend rate	2.5%	Annual dividend rate on stock
9	Exit rate, e	10%	
10	Option exercise multiple, m	3.0	
11	n	50	Number of subdivisions of one year
12			
13	Employee stock option value	13.2921	<--=ESO(B2,B3,B4,B5,B6,B7,B8,B9,B10,B11)
14	Black-Scholes call	19.1842	<--=BSCall(B2*EXP(-B8*B4),B3,B4,B6,B7)

The **ESO** function in cell B13 depends on the 10 variables listed in cells B2:B11.

In the example above, the employee stock option is given when the stock price is \$50. The ESO has exercise price X = \$50. The option has a 10-year maturity and a 3-year vesting period. The interest rate is 5% annually, and the stock pays an annual dividend of 2.5% of its stock value. The rate at which employees leave the company is 10% per year. The model assumes that after the vesting period, employees will choose to exercise this option if the stock price is three times or more the option's exercise price.[7] The binomial model with which the computations in cell B13 were done divides each year into 50 subdivisions.

Given these assumptions, the employee stock option is valued at \$13.2921 (cell B13). A comparable Black-Scholes option on a dividend-paying stock would be valued at \$19.18.[8]

The ESO Valuation and FASB 123

The American Financial Accounting Standards Board (FASB) and the International Accounting Standards Board (IASB) agree that employee stock options should be priced using a model of the type explored above, and that the value of options awarded should be accounted for in a firm's net income. If, for example, a firm had issued 1 million options of the type that appear in the previous spreadsheet, we would value these options at \$13,292,100.

The VBA Code for the ESO Model

Below we give the VBA code for this model. A short discussion follows the code.

```
Function ESO(Stock As Double, X As Double, T As Double, Vest As
Double, _
    Interest As Double, Sigma As Double, Divrate As Double, _
    Exitrate As Double, Multiple As Double, n As Single)
    Dim Up As Double, Down As Double, R As Double, Div As Double,
    _
    piUp As Double, piDown As Double, Delta As Double, _
    i As Integer, j As Integer
    ReDim Opt(T * n, T * n)
```

```
    ReDim S(T * n, T * n)
    Up = Exp((Interest - 0.5 * (Sigma) ^ 2) * (1 / n) + Sigma *
    Sqr(1 / n))
    Down = Exp((Interest - 0.5 * (Sigma) ^ 2) * (1 / n) - Sigma *
    Sqr(1 / n))
    R = Exp(Interest / n)
    Div = Exp(-Divrate / n)
    piUp = (R * Div - Down) / (Up - Down) 'Risk-neutral up
    probability
    piDown = (Up - R * Div) / (Up - Down) 'Risk-neutral down
    probability
    'Defining the stock price
    For i = 0 To T * n
        For j = 0 To i ' j is the number of Up steps
        S(i, j) = Stock * Up ^ j * Down ^ (i - j)
        Next j
    Next i
    'Defining the option value on the last nodes of tree
    For i = 0 To T * n
        Opt(T * n, i)=Application.Max(S(T * n, i) - X, 0)
    Next i
    'Early exercise when stock price > multiple * exercise after
    vesting
    For i = T * n - 1 To 0 Step -1
        For j = 0 To i
        If i > Vest * n And S(i, j) >= Multiple * X _
        Then Opt(i, j)=Application. Max(S(i, j) - X, 0)
        If i > Vest * n And S(i, j) < Multiple * X _
        Then Opt(i, j) = ((1 - Exitrate) ^ (1 / n) * _
        (piUp * Opt(i + 1, j + 1) + piDown * _
        Opt(i + 1, j)) / R + (1 - (1 - Exitrate) ^ (1 / n)) * _
        Application.Max(S(i, j) - X, 0))
        If i <= Vest * n Then Opt(i, j) = _
        (1 - Exitrate) ^ (1 / n) * (piUp * _
        Opt(i + 1, j + 1) + piDown *
        Opt(i + 1, j)) / R
        Next j
    Next i
    ESO = Opt(0, 0)
End Function
```

Explaining the VBA Code

The VBA code has several parts. The first part defines the variables, adjusting the Up, Down, and 1 plus interest R for the n divisions of each year. Having made this adjustment, the code defines the risk-neutral probabilities π_{Up} and π_{Down}:

```
Up = Exp((Interest - 0.5 * (Sigma) ^ 2) * (1 / n) + Sigma * Sqr(1 /
n))
Down = Exp((Interest - 0.5 * (Sigma) ^ 2) * (1 / n) - Sigma *
Sqr(1 / n))
R = Exp(Interest / n)
Div = Exp(-Divrate / n)
piUp = (R * Div - Down) / (Up - Down) 'Risk-neutral up probability
piDown = (Up - R * Div) / (Up - Down) 'Risk-neutral down
probability
```

The stock price is defined as an array $S(i, j)$, where i defines the periods, $i = 0, 1, \ldots, T * n$, and j defines the number of Up steps at each period, $j = 0, 1, \ldots, i$. The next part of the code defines the stock price.

```
'Defining the stock price
For i = 0 To T * n
    For j = 0 To i ' j is the number of Up steps
    S(i, j) = Stock * Up ^ j * Down ^ (i - j)
    Next j
Next i
```

The option values are defined in the next piece of code, which is the heart of our employee stock option function. Option value is defined as an array $Opt(i, j)$:

```
'Defining the option value on the last nodes of tree
For i = 0 To T * n
    Opt(T * n, i) = Application.Max(S(T * n, i) - X, 0)
Next i
'Early exercise when stock price > multiple * exercise after
vesting
For i = T * n - 1 To 0 Step -1
    For j = 0 To i
    If i > Vest * n And S(i, j) >= Multiple * X _
    Then Opt(i, j)= Application.Max(S(i, j) - X, 0)
    If i > Vest * n And S(i, j) < Multiple * X _
```

```
        Then Opt(i, j) = ((1 - Exitrate) ^ (1 / n) * _
        (piUp * Opt(i + 1, j + 1) + piDown * _
        Opt(i + 1, j)) / R + (1 - (1 - Exitrate) ^ (1 / n)) * _
        Application.Max(S(i, j) - X, 0))
        If i <= Vest * n Then Opt(i, j) = _
        (1 - Exitrate) ^ (1 / n) * (piUp * _
        Opt(i + 1, j + 1) + piDown * Opt(i + 1, j)) / R
        Next j
    Next i
```

Here's what this piece of code says:

$$Opt(i,j) = \begin{cases} Max(S(T*n,j) - X, 0) & \text{Terminal nodes} \\ Max(S(i,j) - X, 0) & \text{After vesting, } S(i,j) \geq m*X \\ (1-Exitrate)^{(1/n)} * \dfrac{\pi_{Up}Opt(i+1,j+1) + \pi_{Down}Opt(i+1,j)}{R} + (1-(1-Exitrate)^{(1/n)} * Max(S(i,j)-X,0) & \text{After vesting, } S(i,j) < m*X \\ (1-Exitrate)^{(1/n)} * \dfrac{\pi_{Up}Opt(i+1,j+1) + \pi_{Down}Opt(i+1,j)}{R} & \text{Before vesting} \end{cases}$$

At the terminal nodes, we simply exercise the option. Before the terminal nodes and after vesting, we check to see if the stock price is greater than the desired multiple m of the exercise price. If it is, we exercise the option. If $S(i, j) < m * X$, then the ESO's payoff depends on whether the employee exits the firm or not. With a probability $(1 - Exitrate)^{(1/n)}$, the employee does not exit the firm,[9] in which case the option payoff is the discounted expected next period payoff:

$$(1-Exitrate)^{(1/n)} * \frac{\pi_{Up}Opt(i+1,j+1) + \pi_{Down}Opt(i+1,j)}{R}$$

On the other hand, if the employee exits the firm and the vesting period has passed, the employee will try to exercise the option, giving the expected payoff:

$(1 - (1 - Exitrate)^{(1/n)}) * Max\ (S(i, j) - X, 0)$

Finally, before vesting, the ESO is simply worth the expected discounted (by the risk-neutral probabilities) payoff of the next period values:

$$(1 - Exitrate)^{(1/n)} * \frac{\pi_{Up}Opt(i+1, j+1) + \pi_{Down}Opt(i+1, j)}{R}$$

The final step in the code is to define the value of the function ESO: $ESO = Opt(0, 0)$.

The R Code for the ESO Model

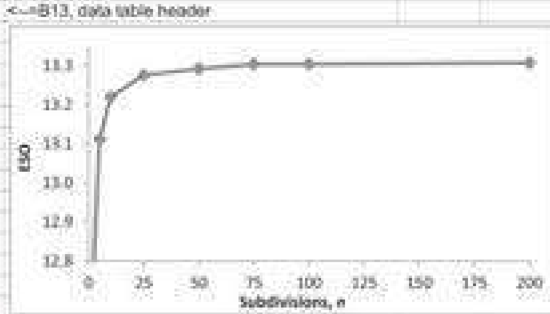
Below we present the R code for the ESO model presented above:

```
# Input
S0 <- 50 # Current stock price
X <- 50 # Option exercise price
t <- 10 #   Time to option exercise (in years)
vesting <- 3 # Vesting period (years)
interest <- 0.05 # Annual interest rate
sigma <- 0.35 # Riskiness of stock
div_rate <- 0.025 # Annual dividend rate on stock
exit_rate <- 0.1 # Annual Exit rate
ex_multiple <- 3 # Option exercise multiple
n <- 50 # Number of subdivisions of one year

fm5_ESO <- function(S0, X, t, vesting, interest, sigma, div_rate, exit_rate, ex_multiple, n){

    delta_t <- 1/n # Delta t

    # Risk neutral probabilities
    U <- exp((interest - 0.5 * sigma^2) * delta_t + sqrt(delta_t) * sigma)
    D <- exp((interest - 0.5 * sigma^2) * delta_t - sqrt(delta_t) * sigma)
    R <- exp(interest * delta_t)
    div <- exp(-div_rate / n)
    pi_U <- (R * div - D) / (U - D)
    pi_D <- (U - R * div) / (U - D)

    # Defining the stock price
    St <- matrix(nrow = n * t + 1, ncol = n * t + 1)
    for(c in 1:(n * t + 1)){
        for(r in 1:c){
            St[r,c] <- S0 * U^(r - 1) * D^(c - r)
        }
    }

    # Defining the option value on the last nodes of tree
    opt_value <- matrix(nrow = (n * t + 1), ncol = (n * t + 1))
    opt_value[, n * t + 1] <- pmax(St[, n * t + 1] - X, 0)

    # Early exercise when stock price > multiple * exercise after vesting
    for(c in (n*t):1){ # columns
        for(r in 1:c){ # rows
            if(c > (vesting * n + 1)){
                if(St[r,c] >= ex_multiple * X){opt_value[r, c] <- max(St[r, c] - X, 0)}
                if(St[r,c] < ex_multiple * X){opt_value[r, c] <-
                    (((1 - exit_rate) ^ (1 / n))* (pi_U * opt_value[r + 1, c + 1] + pi_D *
                       opt_value[r, c + 1]) / R + (1 - (1 - exit_rate) ^ (1 / n)) *
                         max(St[r,c] - X, 0))}
                                }
        if(c <= vesting * n + 1) { opt_value[r, c] <- (1 - exit_rate)^(1 / n) *
            (pi_U * opt_value[r + 1, c + 1] + pi_D * opt_value[r, c + 1]) / R }
                }}

# Result
return(opt_value[1,1])
}

fm5_ESO(S0, X, t, vesting, interest, sigma, div_rate, exit_rate, ex_multiple, n)
```

Some Sensitivity Analysis

We can use data tables to perform sensitivity analysis on our **ESO** function.

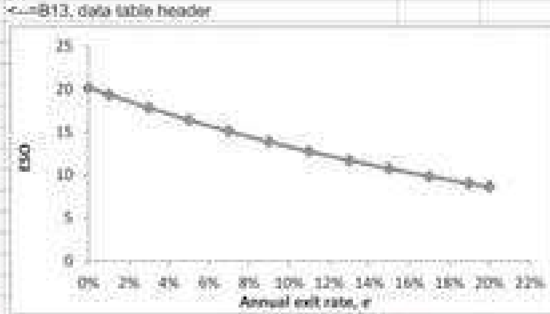
	A	B	C
1	**ESO SENSITIVITY TO NUMBER OF SUBDIVISIONS OF ONE YEAR, *n***		
2	S	50	Current stock price
3	X	50	Option exercise price
4	T	10.00	Time to option exercise (in years)
5	Vesting period (years)	3.00	
6	Interest	5.00%	Annual interest rate
7	Sigma	35%	Riskiness of stock
8	Stock dividend rate	2.50%	Annual dividend rate on stock
9	Exit rate, e	10.00%	
10	Option exercise multiple, m	3.00	
11	n	50	Number of subdivisions of one year
12			
13	Employee stock option value	13.2921	<--=ESO(B2,B3,B4,B5,B6,B7,B8,B9,B10,B11)
14	Black-Scholes call	19.1842	<--=BSCall(B2*EXP(-B8*B4),B3,B4,B6,B7)
15			
16	**Sensitivity of ESO value to number of subdivisions, *n***		
17	n	13.2921	<--=B13, data table header
18	2	12.7605	
19	5	13.1114	
20	10	13.2198	
21	25	13.2758	
22	50	13.2921	
23	75	13.3038	
24	100	13.3037	
25	200	13.3077	



The graph gives ample evidence that $n = 50$ or 75 and does well enough for valuing ESOs. Since larger values of n become time-consuming, we recommend lower numbers.

In the graph below, we show the sensitivity of the ESO value to the employee exit rate e, the rate at which employees leave the firm each year:

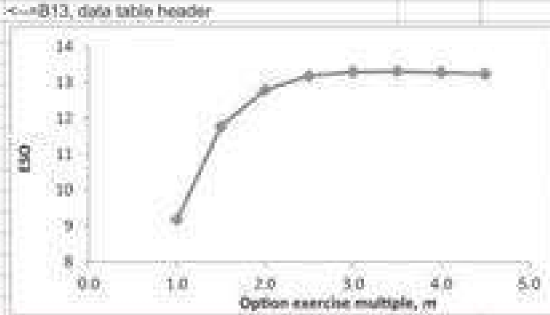
	A	B	C
1	**ESO SENSITIVITY TO ANNUAL EXIT RATE, *e***		
2	S	50	Current stock price
3	X	50	Option exercise price
4	T	10.00	Time to option exercise (in years)
5	Vesting period (years)	3.00	
6	Interest	5.00%	Annual interest rate
7	Sigma	35%	Riskiness of stock
8	Stock dividend rate	2.50%	Annual dividend rate on stock
9	Exit rate, e	10.00%	
10	Option exercise multiple, m	3.00	
11	n	50	Number of subdivisions of one year
12			
13	Employee stock option value	13.2921	<--=ESO(B2,B3,B4,B5,B6,B7,B8,B9,B10,B11)
14	Black-Scholes call	19.1842	<--=BSCall(B2*EXP(-B8*B4),B3,B4,B6,B7)
15			
16	**Sensitivity of ESO value to annual exit rate, *e***		
17	e	13.2921	<--=B13, data table header
18	0%	20.1646	
19	1%	19.3504	
20	3%	17.8151	
21	5%	16.3952	
22	7%	15.0817	
23	9%	13.8658	
24	11%	12.7400	
25	13%	11.6972	
26	15%	10.7310	
27	17%	9.8355	
28	19%	9.0054	
29	20%	8.6133	



The exit rate has a major effect on the value of the ESO: The higher the turnover of employees, the lower the value of the employee stock options. In terms of FASB 123 valuation, the exit rate e is an important valuation factor.

Finally, we do a sensitivity of the ESO value on the exit multiple *m*. Recall that the Hull-White model assumes that employees holding an ESO exercise the option when the stock price is a multiple *m* of the option exercise price *X*. Basically this locks the employee into a suboptimal strategy, since in general call options should be held to maturity (though we note that, in this case, the option is written on a stock that pays a dividend, which may make an early exercise optimal). In the example below, we clearly see the suboptimality of an early ESO exercise: the higher the multiple *m*, the higher the value of the ESO.

	A	B	C
1	**ESO FUNCTION SENSITIVITY TO OPTION EXERCISE MULTIPLE *m***		
2	S	50	Current stock price
3	X	50	Option exercise price
4	T	10.00	Time to option exercise (in years)
5	Vesting period (years)	3.00	
6	Interest	5.00%	Annual interest rate
7	Sigma	35%	Riskiness of stock
8	Stock dividend rate	2.50%	Annual dividend rate on stock
9	Exit rate, e	10.00%	
10	Option exercise multiple, m	3.00	
11	n	50	Number of subdivisions of one year
12			
13	Employee stock option value	13.2921	<--=ESO(B2,B3,B4,B5,B6,B7,B8,B9,B10,B11)
14	Black-Scholes call	19.1842	<--=BSCall(B2*EXP(-B8*B4),B3,B4,B6,B7)
15			
16	**Sensitivity of ESO value to multiple, *m***		
17	m	13.2921	<--=B13, data table header
18	1.0	9.1698	
19	1.5	11.7668	
20	2.0	12.7879	
21	2.5	13.1817	
22	3.0	13.2921	
23	3.5	13.3057	
24	4.0	13.2785	
25	4.5	13.2359	
26	5.0	13.1934	
27	5.5	13.1517	
28	6.0	13.1145	
29	6.5	13.0825	
30			
31			



Last but Not Least

The Hull-White model is a numerical approximation of the ESO option valuation, but it is not a closed-form formula. A paper by Cvitanić, Wiener, and Zapatero (2006) gives an analytical derivation of the value of employee stock options. The formula stretches over 16 pages of typescript and will not be given here. An Excel implementation of the formula exists and can be downloaded at http://pluto.mscc.huji.ac.il/~mswiener/research/ESO.htm.

17.9 Using the Binomial Model to Price Nonstandard Options: An Example

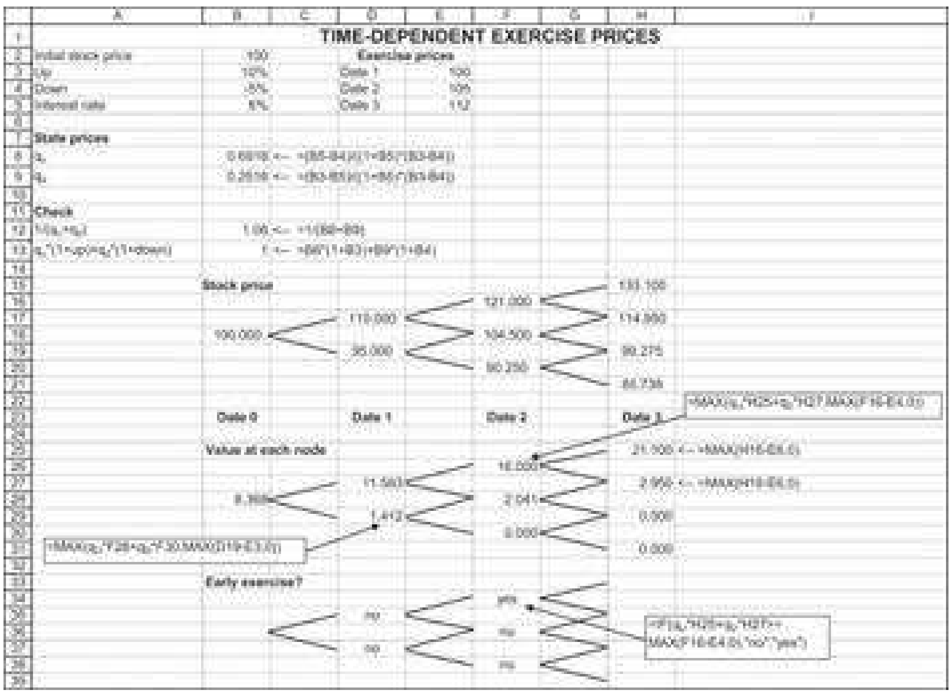
The binomial model can also be used to price nonstandard options. Consider the following example: You hold an option to buy a share of a company. The option allows for early exercise, but the exercise price varies with the time at which you choose to exercise. For the case we consider, the option has the following conditions:

- • There are *n* possible exercise dates *only* (i.e., the option is only exercisable on these dates).

- • Exercise at date t precludes exercise at all dates $s > t$. However, if you don't exercise at date s, you may still exercise at date $t > s$.
- • The exercise price at date t is X_t, meaning that the exercise price can vary with time.

We want to value this option using a binomial framework. To do so, we recognize that basically this is just an American option with three separate exercise prices.

Here's how we set up this problem in a spreadsheet, using the logic of the American option valuation described in section 17.5:



Most of this spreadsheet follows section 17.5. Cells B15:H21 describe the stock price over time, which follows a binomial process with the Up = 1.10 and Down = 0.95 (cells B3 and B4). Where things get interesting is in the option valuation (rows 23:31). As is usual for an American option, at each node of the tree, we consider whether the option is worth more whether exercised or whether held. But note that in the above picture, the exercise price varies with the date, so that the exercise price at date 3 is E5, that of date 2 is E4, and that of date 1 is E3.

As you can see in cell B28, the value of the American call option is 8.368.

17.10 Summary

The binomial model is intuitive and easy to implement. As a widespread alternative to Black-Scholes pricing, the model can easily be put into a spreadsheet and programmed in VBA. This chapter has explored both the basic uses of the binomial model and its implementation to price American and other options. A section on employee stock options has shown how to implement the Hull and White (2004) model for valuing these options. Throughout, we have stressed the role of state prices in implementing the model.

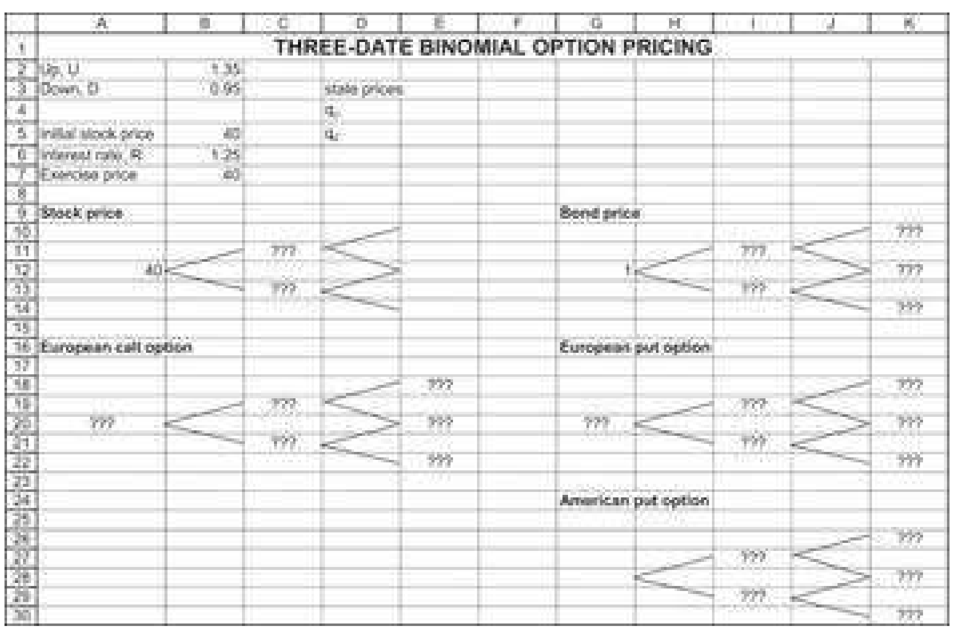
Exercises

1. A stock selling for $25 today will, in a year, be worth either $35 or $20. If the interest rate is 8%, what is the value today of a 1-year call option on the stock with exercise price $30? Use the simultaneous equation approach of section 17.2 to price the option.
2. In exercise 1, compute the state prices q_U and q_D, and use these prices to calculate the value today of a 1-year put option on the stock with exercise price $30. Show that put-call parity holds. That is, using your answer from this problem and the previous problem, show that:

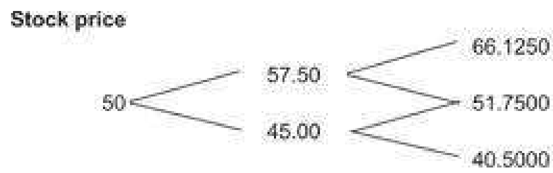
$$Call\ price + \frac{X}{1+r} = Stock\ price\ today + Put\ price$$

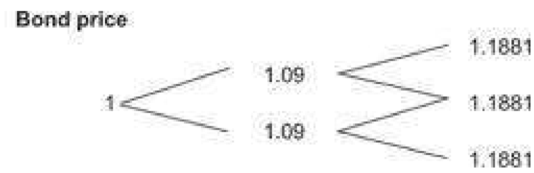
3. In a binomial model, a call option and a put option are both written on the same stock. The exercise price of the call option is 30 and the exercise price of the put option is 40. The call option's payoffs are 0 and 5 and the put option's payoffs are 20 and 5. The price of the call is 2.25 and the price of the put is 12.25.
   1. What is the riskless interest rate? Assume that the basic period is one year.
   2. What is the price of the stock today?

4. All reliable analysts agree that a share of ABC Corp., selling today for $50, will be priced at either $65 or $45 one year from now. They further agree that the probabilities of these events are 0.6 and 0.4, respectively. The market risk-free rate is 6%. What is the value of a call option on ABC whose exercise price is $50 and which matures in one year?
5. A stock is currently selling for 60. The price of the stock at the end of the year is expected either to increase by 25% or to decrease by 20%. The riskless interest rate is 5%. Calculate the price of a European put on the stock with exercise price 55. Use the binomial option pricing model.
6. Fill in all the cells labeled "???" in the following spreadsheet. Why is there no additional pricing tree for an American call option?



7. Consider the following two-period binomial model, in which the annual interest rate is 9% and in which the stock price goes up by 15% per period or down by 10%:

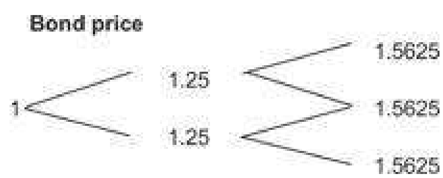




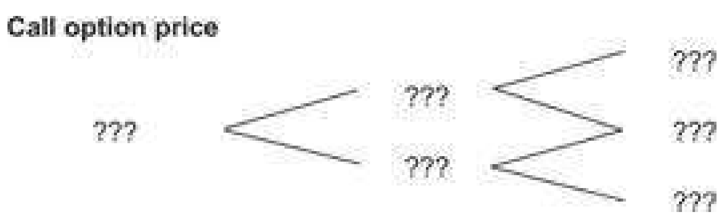
1. Price a European call on the stock with exercise price 60.
2. Price a European put on the stock with exercise price 60.
3. Price an American call on the stock with exercise price 60.
4. Price an American put on the stock with exercise price 60.

8. Consider the following three-date binomial model:

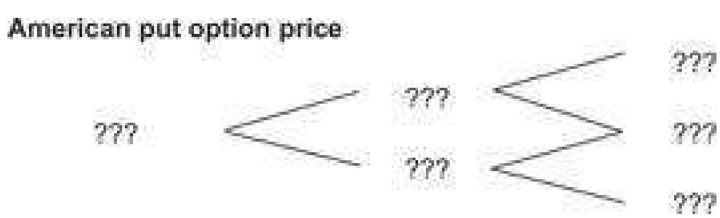
- In each period the stock price either goes up by 30% or decreases by 10%.
- The one-period interest rate is 25%.



a. Consider a European call with $X = 30$ and $T = 2$. Fill in the blanks in the tree:



b. Price a European put with $X = 30$ and $T = 2$.
c. Now consider an American put with $X = 30$ and $T = 2$. Fill in the blanks in the tree:



9. A prominent securities firm introduced a new financial product. This product, called "The Best of Both Worlds" (BOBOW for short), costs \$10. It matures in five years, at which point it repays the investor the \$10 cost *plus* 120% of any positive return in the Standard & Poor's 500 Index (SP500). There are no payments before maturity.

For example: If the SP500 is currently at 1,500, and if it is at 1,800 in five years, a BOBOW owner will receive back \$12.40 = \$10 * [1 + 1.2 * (1800/1500 − 1)]. If the SP500 is at or below 1,500 in five years, the BOBOW owner will receive back \$10.

Suppose that the annual interest rate on a 5-year, continuously compounded, pure-discount bond is 6%. Suppose further that the SP500 is currently at 1,500 and that you believe that in five years it will be at either 2,500 or 1,200. Use the binomial option pricing model to show that BOBOWs are underpriced.

10. This problem is a continuation of the discussion of section 17.6. Show that as $n \to \infty$, the binomial European put price converges to the Black-Scholes put price. (Note that, as part of the spreadsheet for this chapter, we have included a function called **BSPut,** which computes the Black-Scholes put price.)

11. Here's a simplified version of exercise 10. Consider an alternative parameterization of the binomial:

$$\Delta t = T/n \qquad R = e^{r\Delta t}$$

$$\text{Up} = e^{(r)\Delta t + \sigma\sqrt{\Delta t}} \qquad q_U = \frac{R - \text{Down}}{R * (\text{Up-Down})}$$

$$\text{Down} = e^{(r)\Delta t - \sigma\sqrt{\Delta t}} \qquad q_D = \frac{1}{R} - q_U$$

Construct binomial European call and put option pricing functions in VBA for this parameterization and show that they also converge to the Black-Scholes formula. (The message here is that the parameterization of the binomial Up and Down is not unique.)

12. A call option is written on a stock whose current price is \$50. The option has maturity of three years, and during this time the annual stock price is expected to increase by 25% or to decrease by −10%. The annual interest rate is constant at 6%. The option is exercisable at date 1 at a price of \$55, at date 2 for a price of \$60, and at date 3 for a price of \$65. What is its value today? Will you ever exercise the option early?

13. Reconsider the above problem. Show that if the date 1 exercise price is X, the date 2 exercise price is $X * (1 + r)$, and the date 3 exercise price is $X * (1 + r)^2$, you will not exercise the option early.[10]

14. An investment bank is offering a security linked to the price, two years from today, for Bisco stock, which is currently at \$3 per share. Denote Bisco's stock price in two periods by S_2. The security being offered pays off $Max(S_2^2 - 40, 0)$. You estimate that in each of the next two periods, Bisco stock will increase by either 50% or decrease by 20%. The annual interest rate is 8%. Price the security.

---

1. 1. Should there be a need to distinguish between 1.10 (1 plus the 10% up move of the stock) and 10% (the up move itself), we will use U for the former and lowercase u for the latter.
2. 2. When we discuss American options in section 17.5, we will see that backward pricing is critical.
3. 3. The discussion that follows is perhaps best read after Chapters 23 and 27.
4. 4. Many practitioners use $U = 1 + up = e^{\sigma\sqrt{\Delta t}}$ and $D = 1 + down = e^{-\sigma\sqrt{\Delta t}}$ as an approximation.
5. 5. An alternative approximation that converges to a lognormal price process is given in the next subsection. See also Benninga, Steinmetz, and Stroughair (1993); Hull (2006); and Omberg (1987).
6. 6. This section has benefited from discussions with Torben Voetmann of Cornerstone Research and Zvi Wiener of Hebrew University, Jerusalem.
7. 7. Research cited by Hull and White (2004) shows that the average stock-price-to-exercise-price ratio at which ESO owners exercise their options is between 2.2 and 2.8.
8. 8. We're getting way ahead of ourselves here! The adaptation of Black-Scholes for dividend-paying stock is given in section 18.6.
9. 9. Note that $(1 - Exit\ rate^{\text{period of } 1/n\ y})^{(1/n)} = (1 - Exit\ rate^{\text{period of 1 year}})$.
10. 10. It can also be shown that this property holds if the exercise prices grow more slowly than the interest rate. Thus, for the problem considered in section 17.5, there will be early exercise of the American call only when the exercise prices grow at a rate faster than the interest rate.

18 The Black-Scholes Model

18.1 Overview

In a path-breaking paper published in 1973, Fischer Black and Myron Scholes proved a formula for pricing European call and put options on non-dividend-paying stocks. Their model is probably the most famous model of modern finance. The Black-Scholes formula is relatively easy to use, and it is often an adequate approximation to the price of more complicated options. In this chapter we make no pretense at a full-blown development of the model; this requires a knowledge of stochastic processes and is a not-inconsiderable mathematical investment. Instead, we shall describe the mechanics of the model and show how to implement it in Excel. We also illustrate several uses of the Black-Scholes formula in the valuation of structured assets.

18.2 The Black-Scholes Model

Consider a stock whose price is lognormally distributed.[1] The Black-Scholes model uses the following formula to price a European call on a non-dividend-paying stock:

$$C = S \cdot N(d_1) - X \cdot e^{-rT} \cdot N(d_2)$$

where

$$d_1 = \frac{ln(S/X) + \left(r + \frac{\sigma^2}{2}\right)T}{\sigma\sqrt{T}}$$

$$d_2 = d_1 - \sigma\sqrt{T}$$

Here, C denotes the price of a call, S is the price of the underlying stock, X is the exercise price of the call, T is the call's time to exercise, r is the interest rate, and σ is the standard deviation of the logarithm of the stock's return. $N(d)$ denotes a value of the cumulative standard normal distribution function to the left of d. It is assumed that the stock will pay no dividends before date T.

By the put-call parity theorem (see Chapter 16), a put with the same exercise date T and exercise price X written on the same stock will have price $P = C - S + Xe^{-rT}$. Substituting for C in this equation and doing some algebra gives the Black-Scholes European put pricing formula:

$$P = Xe^{-rT} N(-d_2) - SN(-d_1)$$

In Chapter 17, we hinted at one form of the proof of the Black-Scholes formula. There it was shown numerically that the Black-Scholes formula coincides with the binomial option pricing model formula when (i) the length of a typical period → 0, (ii) the “up” and the “down” moves in the binomial model converge to a lognormal price process, and (iii) the term structure of interest rates is flat.

Implementing the Black-Scholes Formulas in a Spreadsheet

The Black-Scholes formulas for call and put pricing are easily implemented in a spreadsheet. The following example shows how to calculate the price of a call option written on a stock whose current price $S = 50$, when the exercise price $X = 45$, the annualized interest rate $r = 4\%$, and $\sigma = 30\%$. The option has $T = 0.75$ years to exercise. All three of the parameters T, r, and σ are assumed to be in annual terms.[2]

	A	B	C
1	BLACK-SCHOLES OPTION PRICING FORMULA		
2	S	50	Current stock price
3	X	45	Exercise price
4	r	4.00%	Risk-free rate of interest
5	T	0.75	Time to maturity of option (in years)
6	Sigma	30%	Stock volatility, σ
7			
8	d_1	0.6509	<--=(LN(B2/B3)+(B4+0.5*B6^2)*B5)/(B6*SQRT(B5))
9	d_2	0.3911	<--=B8-SQRT(B5)*B6
10			
11	$N(d_1)$	0.7424	<--=NORM.S.DIST(B8,1)
12	$N(d_2)$	0.6521	<--=NORM.S.DIST(B9,1)
13			
14	Call price	8.64	<--=B2*B11-B3*EXP(-B4*B5)*B12
15	Put price	2.31	<--=B3*EXP(-B4*B5)*NORMSDIST(-B9)-B2*NORMSDIST(-B8)
16	Put price (PCP)	2.31	<--=B14-B2+B3*EXP(-B4*B5)

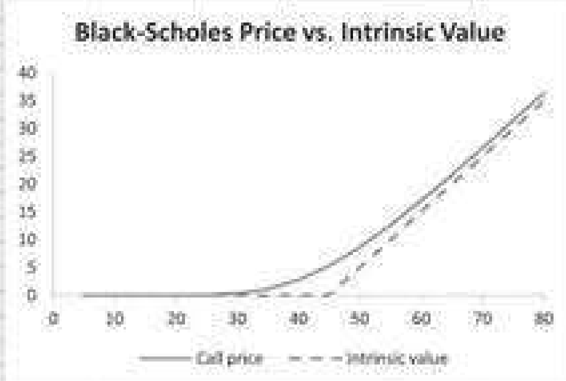
The spreadsheet calculates the put price twice. In cell B15, the put price is computed by using the Black-Scholes formula, and in cell B16 the put price is calculated by using the put-call parity.

We can use this spreadsheet to do the usual sensitivity analysis. For example, the following **Data|Table** (see Chapter 28) gives—as the stock price S varies—the Black-Scholes value of the call compared to its intrinsic value [i.e., $max(S-X, 0)$].

	A	B	C
18	Data table: Comparing the Black-Scholes to the intrinsic value		
19	Stock price	Call price	Intrinsic value
20		8.6434	5
21	5	0.0000	0
22	10	0.0000	0
23	15	0.0000	0
24	20	0.0029	0
25	25	0.0484	0
26	30	0.3101	0
27	35	1.1077	0
28	40	2.7319	0
29	45	5.2777	0
30	50	8.6434	5
31	55	12.6307	10
32	60	17.0378	15
33	65	21.7056	20
34	70	26.5256	25
35	75	31.4304	30
36	80	36.3811	35

data table header, =B14

data table header, =MAX(B2-B3,0)



18.3 Programming the Black-Scholes Option Pricing Model

Using VBA to Define a Black-Scholes Model as a Function

Although the spreadsheet implementation of the Black-Scholes formulas illustrated in the previous section is sufficient for some purposes, we are sometimes interested in having a closed form function that we can use directly in Excel. We can do so with Visual Basic for Applications (VBA). Below we define functions **dOne**, **dTwo**, and **BSCall**:

```
Function dOne(Stock, Exercise, Time, Interest, sigma)
    dOne=(Log(Stock / Exercise) + (Interest + 0.5 * (sigma)^2) *
    Time) _ / (sigma * Sqr(Time))
End Function
Function dTwo(Stock, Exercise, Time, Interest, sigma)
    dTwo = dOne(Stock, Exercise, Time, Interest, sigma) - _ sigma
    * Sqr(Time)
End Function
Function BSCall(Stock, Exercise, Time, Interest, sigma)
    BSCall = Stock * Application.NormSDist(dOne(Stock, Exercise, _
        Time, Interest, sigma)) - Exercise * Exp(-Time *
        Interest) * _
    Application.NormSDist(dTwo(Stock, Exercise, Time, _ Interest,
    sigma))
End Function
```

Note the use of the Excel function **NormSDist**, which gives the standard normal distribution.[3]

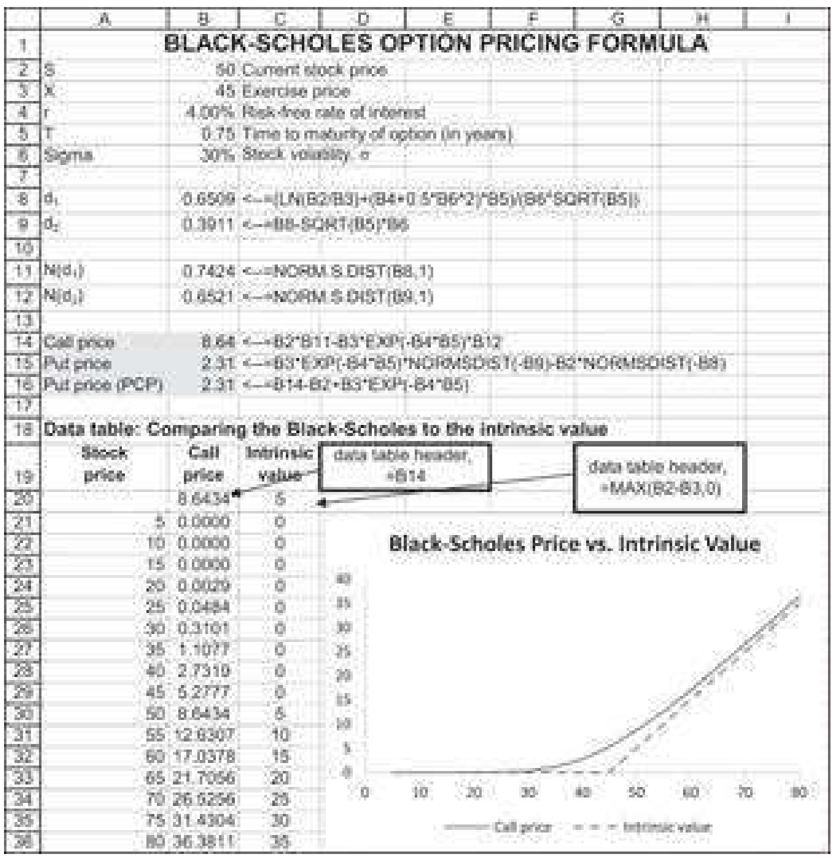
Pricing Puts

We apply the same approach in another VBA function, **BSPut,** to price the put options:

```
Function BSPut(Stock, Exercise, Time, _
Interest, sigma)
    BSPut = Exercise * Exp(-Time * Interest) * _
    Application.Norm_S_Dist(-dTwo(Stock, Exercise, _ Time,
    Interest, sigma), 1) - Stock * Application.Norm_S_Dist _
    (-dOne(Stock, Exercise, Time, Interest, _
    sigma), 1)
End Function
```

Using These Functions in an Excel Spreadsheet

Here is an example of these functions used in Excel. The graph below was created by a data table (in presentations, we usually hide the first row of such a table; here we have shown it).



	A	B	C
1	BLACK-SCHOLES OPTION PRICING FORMULA		
2	S	50	Current stock price
3	X	45	Exercise price
4	r	4.00%	Risk-free rate of interest
5	T	0.75	Time to maturity of option (in years)
6	Sigma	30%	Stock volatility, σ
7			
8	d_1	0.6509	<--=(LN(B2/B3)+(B4+0.5*B6^2)*B5)/(B6*SQRT(B5))
9	d_2	0.3911	<--=B8-SQRT(B5)*B6
10			
11	$N(d_1)$	0.7424	<--=NORM.S.DIST(B8,1)
12	$N(d_2)$	0.6521	<--=NORM.S.DIST(B9,1)
13			
14	Call price	8.64	<--=B2*B11-B3*EXP(-B4*B5)*B12
15	Put price	2.31	<--=B3*EXP(-B4*B5)*NORMSDIST(-B9)-B2*NORMSDIST(-B8)
16	Put price (PCP)	2.31	<--=B14-B2+B3*EXP(-B4*B5)
17			
18	Data table: Comparing the Black-Scholes to the intrinsic value		
19	Stock price	Call price	Intrinsic value
20		8.6434	5
21	5	0.0000	0
22	10	0.0000	0
23	15	0.0000	0
24	20	0.0029	0
25	25	0.0484	0
26	30	0.3101	0
27	35	1.1077	0
28	40	2.7319	0
29	45	5.2777	0
30	50	8.6434	5
31	55	12.6307	10
32	60	17.0378	15
33	65	21.7056	20
34	70	26.5256	25
35	75	31.4304	30
36	80	36.3811	35

Using R to Implement the Black-Scholes Model

Below we present the code in R that implements the Black-Scholes using data from the example above:

```r
# Input:
S <- 50
X <- 45
r <- 0.04
t <- 0.75
sigma_S <- 0.3

# Function: d1
fm5_bs_d1 <- function(S, X, t, r, sigma_S){
    return( ( log(S/X) + (r + 0.5 * sigma_S ^ 2) * t ) / ( sigma_S * sqrt(t)) )
    }
fm5_bs_d1(S, X, t, r, sigma_S)

# Function: d2
fm5_bs_d2 <- function(S, X, t, r, sigma_S){
    return(fm5_bs_d1(S, X, t, r, sigma_S) - sigma_S * sqrt(t))
    }
fm5_bs_d2(S, X, t, r, sigma_S)

# Function: N(d1)
fm5_bs_Nd1 <- function(S, X, t, r, sigma_S){
    return(pnorm(fm5_bs_d1(S, X, t, r, sigma_S), mean = 0, sd = 1))
    }
fm5_bs_Nd1(S, X, t, r, sigma_S)

# Function: N(d2)
fm5_bs_Nd2 <- function(S, X, t, r, sigma_S){
    return(pnorm(fm5_bs_d2(S, X, t, r, sigma_S), mean = 0, sd = 1))
    }
fm5_bs_Nd2(S, X, t, r, sigma_S)

# Function: B&S Call option
fm5_bs_call <- function(S, X, t, r, sigma_S){
    return(S * fm5_bs_Nd1(S, X, t, r, sigma_S)
           - X * exp(-r*t) * fm5_bs_Nd2(S, X, t, r, sigma_S))
           }
fm5_bs_call(S, X, t, r, sigma_S)

# Function: B&S Put option
fm5_bs_put <- function(S, X, t, r, sigma_S){
    return(X * exp(-r*t) * (1 - fm5_bs_Nd2(S, X, t, r, sigma_S)) -
               S *(1 - fm5_bs_Nd1(S, X, t, r, sigma_S)))
           }
fm5_bs_put(S, X, t, r, sigma_S)
```

18.4 Calculating the Volatility

The Black-Scholes formula depends on five parameters: the stock price S, the option exercise price X, the option's time to maturity T, the interest rate r, and the standard deviation of the returns of the stock underlying the option σ. Four of these five parameters are straightforward, but the fifth parameter, σ, is problematic. There are two common ways of computing sigma, σ.

- σ can be computed based on the *historical returns* of the stock.
- σ can be computed based on the *implied volatility* of the stock.

In the two subsections below, we illustrate both methods of computing σ and apply them to the pricing of options on the Standard & Poor's 500 (SP500)—symbol SPY, called "spiders."

The Volatility of Historical Returns

We can use historical stock returns to compute the volatility. The method is as follows:

- • For a given time frame and frequency of returns, we compute the periodic volatility. The time frame commonly used varies wildly. Some practitioners use a short term of, say, 30 days, while others use a much longer time frame (up to one year). Similarly, the frequency of returns can be daily, weekly, or sometimes monthly. Since options are mostly short term, a shorter time frame is more common.
- • We annualize the periodic volatility by multiplying by the square root of the number of periods per year. Thus:

$$\sigma_{annual} = \begin{cases} \sqrt{12} * \sigma_{monthly} \\ \sqrt{52} * \sigma_{weekly} \\ \sqrt{250} * \sigma_{daily} \end{cases}$$

The number of days per year is an open question. Most practitioners use 250 or 252 for the number of transaction dates per year. However, instances of using 365 can also be found.

In the spreadsheet below we show one year's daily prices for the SPY, which is an exchange-traded fund (ETF) that tracks the SP500. The historical prices of the SPY and the resulting historical volatility are computed below:

	A	B	C	D	E	F	G	H
1	SPY HISTORICAL PRICES, DAILY DATA (01-AUG-2018 TO 31-JUL-2019)							
2	Date	Adj Close	Return			Return statistics, one year		
3	31-Jul-19	297.43				Count	250	<--=COUNT(C:C)
4	30-Jul-19	300.72	1.10%	<--=LN(B4/B3)		Average daily return	0.03%	<--=AVERAGE(C:C)
5	29-Jul-19	301.46	0.25%	<--=LN(B5/B4)		Standard deviation of daily return	0.97%	<--=STDEV.S(C:C)
6	26-Jul-19	302.01	0.18%					
7	25-Jul-19	300.00	-0.67%			Annualized mean return	-0.37%	<--=12*G4
8	24-Jul-19	301.44	0.48%			Annualized sigma	15.39%	<--=SQRT(252)*G5
9	23-Jul-19	300.03	-0.47%					
10	22-Jul-19	297.90	-0.71%			Return statistics, last half year		
11	19-Jul-19	297.17	-0.25%			Count	124	<--=COUNT(C130:C253)
12	18-Jul-19	298.83	0.56%			Average daily return	0.03%	<--=AVERAGE(C130:C253)
13	17-Jul-19	297.74	-0.37%			Standard deviation of daily return	1.20%	<--=STDEV.S(C130:C253)
14	16-Jul-19	299.78	0.68%					
15	15-Jul-19	300.75	0.32%			Annualized mean return	0.37%	<--=12*G12
16	12-Jul-19	300.65	-0.03%			Annualized sigma	19.00%	<--=SQRT(252)*G13
252	02-Aug-18	276.91	-0.43%					
253	01-Aug-18	275.41	-0.54%					

The historical volatility based on a full year of data is 15.39%, whereas the volatility for the last six months is 19%.

The Implied Volatility

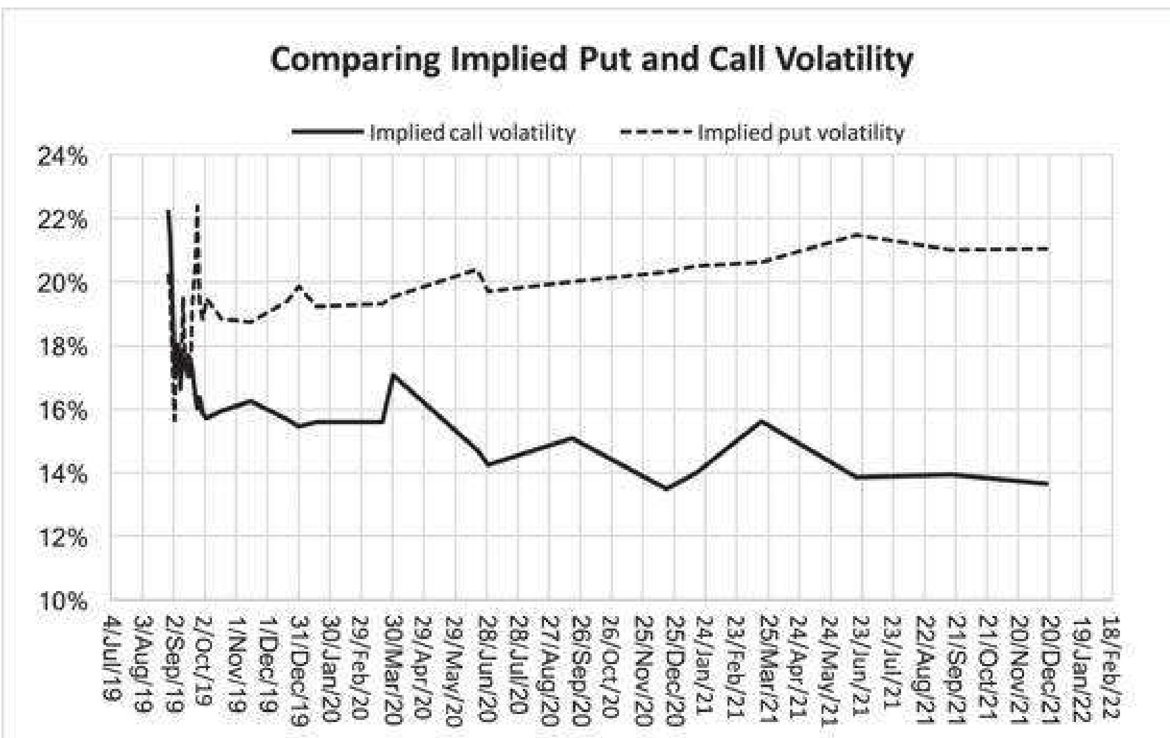
The implied volatility ignores history; instead, it determines the option σ based on actual option prices. Whereas the historical volatility is a backward-looking volatility, the implied volatility is a forward-looking estimate.[4]

To estimate the implied volatility for the SPY calls expiring January 19, 2013, we solve the Black-Scholes formula for the sigma, which gives the current market price:

	A	B	C
1	IMPLIED VOLATILITY FOR THE AUGUST 2018 SPY OPTIONS		
2	Current date	26-Aug-19	
3	Option expiration date	18-Sep-20	
4			
5	Current SPY price, S	287.5	
6	Option strike price, X	290	
7	Time to maturity, T	1.0611	<--=YEARFRAC(B2,B3,)
8	Interest rate	1.73%	1 year treasury rate (source: https://www.treasury.gov)
9			
10	Actual call price	18.32	
11	Actual put price	22.15	
12			
13	Implied call volatility	14.38%	<--=CallVolatility(B5,B6,B7,B8,B10)
14	Proof: Black-Scholes call price	18.32	<--=BSCall(B5,B6,B7,B8,B13)
15			
16	Implied put volatility	20.03%	<--=PutVolatility(B5,B6,B7,B8,B11)
17	Proof: Black-Scholes put price	22.15	<--=BSPut(B5,B6,B7,B8,B16)

The implied volatility is 14.38% for the call and 20.03% for the put. As shown in cells B14 and B17, these volatilities, when inserted in the Black-Scholes formula, give back the current market price. We have used functions **CallVolatility** and **PutVolatility** described below.

We also compute the implied volatilities of the at-the-money puts and calls:



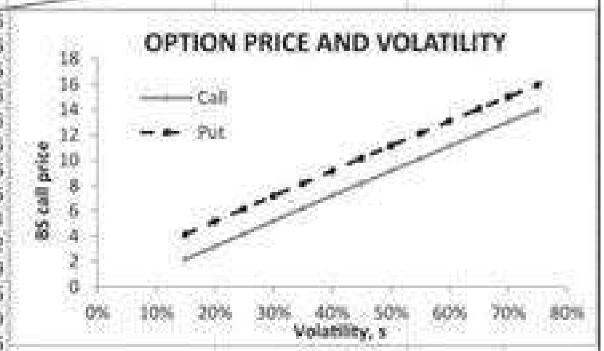
Comparing Historical with Implied Volatility

It is impossible to say which of these two methods is better for pricing options. On the one hand, it is common to attribute to the historical returns some kind of validity as a predictor of the future anticipated returns. On the other hand, the implied volatility gives a good indication of what the market is currently thinking. Our advice: Use them both and compare.

18.5 Programming a Function to Find the Implied Volatility

We first design a VBA function that computes the implied volatility. To do this, we first note that the option price is a monotonic and increasing function of the sigma. Here's a **Data|Table** from our basic Black-Scholes spreadsheet:

	A	B	C	D
1	BLACK-SCHOLES OPTION PRICE IS MONOTONIC IN SIGMA			
2	S	45	Current stock price	
3	X	50	Exercise price	
4	T	1.25	Time to maturity of option (in years)	
5	r	5.0%	Risk-free rate of interest	
6	Sigma	40%	Stock volatility	
7				
8	Call price	7.19	<--=BSCall(B2,B3,B4,B5,B6)	
9	Put price	9.16	<--=BSPut(B2,B3,B4,B5,B6)	
10				data table header, =B9
11	Data table: Option price as function of volatility σ			
12		Call	Put	
13		7.19	9.16	
14	15%	2.19	4.16	
15	20%	3.18	5.15	
16	25%	4.19	6.16	
17	30%	5.19	7.16	
18	35%	6.19	8.16	
19	40%	7.19	9.16	
20	45%	8.18	10.15	
21	50%	9.17	11.14	
22	55%	10.15	12.12	
23	60%	11.12	13.09	
24	65%	12.08	14.05	
25	70%	13.03	15.00	
26	75%	13.98	15.95	



We use VBA to define a function **ImpliedVolatility,** which finds the σ for a call option.

	A	B	C
1	BLACK-SCHOLES IMPLIED VOLATILITY		
2	The VBA module attached to this spreadsheet defines a function called **ImpliedVolatility(S,X,T,interest,target_price, OptionType)** To use this function fill in the relevant rows (in boldface). The cell labeled "Implied volatility" contains the function.		
3	**S**	**51.00**	
4	**X**	**50.00**	
5	**T**	**1.25**	
6	**Interest**	**5.00%**	
7	**Target option price**	**6.00**	
8	**Option type**	**C**	"C" for call and "P" for Put
9			
10	Implied call volatility	16.67%	<--=ImpliedVolatility(B3,B4,B5,B6,B7,B8)
11	Implied put volatility	36.20%	<--=ImpliedVolatility(B3,B4,B5,B6,B7,"P")

The function is defined as **ImpliedVolatility(Stock, Exercise, Time, Interest, Target, OptionType)**, where the definitions are:

- **Stock** is the stock price S
- **Exercise** is the option's exercise price X
- **Time** is the time to the option's maturity T
- **Interest** is the interest rate r
- **Target** is the option price
- **OptionType** is the type of option ("C" for Call or "P" for Put).

The function finds σ for which the option price using the Black-Scholes formula equals the option input price.

```
Function ImpliedVolatility(Stock, Exercise, Time, Interest, _
Target, OptionType As String)
    High = 2
    Low = 0
    If UCase(OptionType) = "C" Then
    Do While (High - Low) > 0.0001
        If BSCall(Stock, Exercise, Time, Interest, _ (High + Low)
        / 2) > Target Then
            High = (High + Low) / 2
            Else: Low = (High + Low) / 2
        End If
    Loop
    ImpliedVolatility = (High + Low) / 2
    ElseIf UCase(OptionType) = "P" Then
        Do While (High - Low) > 0.0001
        If BSPut(Stock, Exercise, Time, Interest, _
        (High + Low) / 2) > Target Then
            High = (High + Low) / 2
            Else: Low = (High + Low) / 2
        End If
    Loop
    ImpliedVolatility = (High + Low) / 2
    Else: ImpliedVolatility = "Error"
    End If
End Function
```

The technique used by the function is very similar to the technique used in trial and error. We start with two estimates for the possible σ : There is a **High** estimate of 200% and a **Low** estimate of 0%. We now do the following:

- • Plug the average of the **High** and the **Low** into the Black-Scholes formula of the corresponding function. This gives us **BSCall(Stock, Exercise, Time, Interest, (High+Low)/2)** or **BSPut(Stock, Exercise, Time, Interest, (High+Low)/2)**. (Note the function **ImpliedVolatility** assumes that the functions **BSCall()** and **BSPut()** are available to the spreadsheet.)
- • If **CallOption(Stock, Exercise, Time, Interest, (High+Low)/2** > **Target**, then the current σ estimate of (High + Low)/2) is too high and we replace High by (High + Low) / 2.
- • If **CallOption(Stock, Exercise, Time, Interest, (High+Low)/2** < **Target**, then the current σ estimate of (High + Low) / 2) is too low and we replace Low by (High + Low) / 2.

We repeat this procedure until the difference **High-Low** is less than 0.0001 (or some other arbitrary constant).

Finding the Implied Volatility Using R

Implementing the same approach in R should look like this:

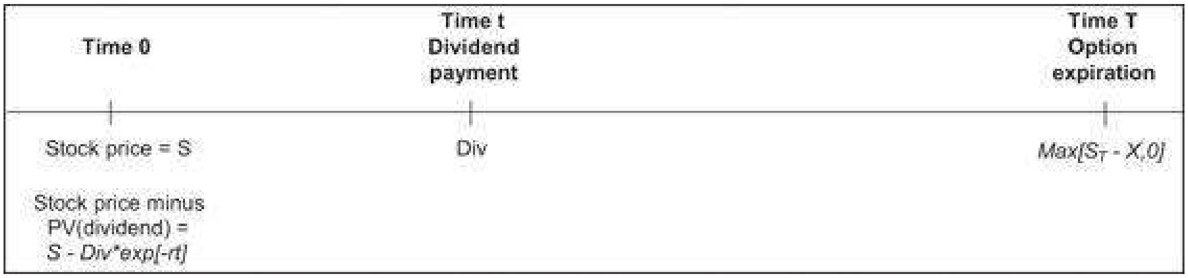
```r
## The Implied Volatility
# Input:
S <- 51
X <- 50
t <- 1.25
r <- 0.05
target <- 6.0

# Implied Volatility
fm5_bs_iv <- function(S, X, t, r, target, option_type){

    high = 2
    low = 0

    if(option_type == "C"){
        while((high - low) > 0.0001){
            ifelse(fm5_bs_call(S, X, t, r, (high + low)/2) - target > 0,
                   high <- (high + low) / 2,
                   low <- (high + low) / 2)
        }
        return((high + low) / 2)
    }

    if(option_type == "P"){
        while((high - low) > 0.0001){
            ifelse(fm5_bs_put(S, X, t, r, (high + low)/2) - target > 0,
                   high <- (high + low) / 2,
                   low <- (high + low) / 2)
        }
        return((high + low) / 2)
    }
}

# Implied Volatility
fm5_bs_iv(S, X, t, r, target, option_type = "C")
fm5_bs_iv(S, X, t, r, target, option_type = "P")
```

18.6 Dividend Adjustments to the Black-Scholes

The Black-Scholes formula assumes that the option's underlying security pays no dividends prior to the exercise date T. In certain cases, it is easy to make an adjustment to the model for dividends. This section looks at two such adjustments. We first look at option pricing when future dividends are known with certainty, and we then examine option pricing when the underlying security pays out a continuous dividend. The principle underlying both cases is the same: The options are priced on an adjusted underlying value that nets out the present value of dividends paid between the option purchase date and the option exercise date.

A Known Dividend to Be Paid before the Option Expiration

It often happens that a stock's future dividend is known at the time the option is traded. This is most commonly the case when a dividend has already been announced, but it can also happen because many stocks pay quite regular and relatively inflexible dividends. In this case, the option should be priced not on the current stock price S but on the stock price minus the present value of the dividend or dividends anticipated before the option expiration date T:



Here's an example: Coca-Cola (stock symbol KO) pays quarterly dividends in the middle of March, June, September, and November of each year. The dividends seem quite stable; in 2018 and 2019, the dividends were $0.39 and $0.4 per share.

	A	B	C
1	**COCA-COLA DIVIDENDS 2015-2019**		
2	**Date**	**Dividend**	
3	13-Jun-19	0.4	
4	14-Mar-19	0.4	
5	29-Nov-18	0.39	
6	13-Sep-18	0.39	
7	14-Jun-18	0.39	
8	14-Mar-18	0.39	
9	30-Nov-17	0.37	
10	14-Sep-17	0.37	
11	13-Jun-17	0.37	
12	13-Mar-17	0.37	
13	29-Nov-16	0.35	
14	13-Sep-16	0.35	
15	13-Jun-16	0.35	
16	11-Mar-16	0.35	
17	27-Nov-15	0.33	
18	11-Sep-15	0.33	
19	11-Jun-15	0.33	
20	12-Mar-15	0.33	

Computing the implied volatility for the February 2020 calls and puts on Coca-Cola shows that taking account of anticipated dividends makes a significant difference in the pricing. We can also deduce from the proximity of the prices, which take dividends into account (cells B19:B20), versus the distance between the prices, which do not take the dividends into account (cells B22:B23) that the former are correct.

	A	B	C	D	E
1	**PRICING THE COCA-COLA FEB 2020 CALLS AND PUTS**				
2	Current date	28-Aug-19			
3	Option expiration date	21-Feb-20			
4	Current stock price	54.99			
5	Interest rate	1.73%	<--Short term T-bill rate		
6					
7		**Date**	**Anticipated dividend**	**Present value**	
8	Mid September	13-Sep-19	0.4	0.3997	<--=C8*EXP(-$B$5*((B8-$B$2)/365))
9	End November	29-Nov-19	0.4	0.3982	<--=C9*EXP(-$B$5*((B9-$B$2)/365))
10					
11	Stock price net of PV(dividends)	54.19	<-- =B4-SUM(D8:D9)		
12	Exercise price, X	55.00	<-- at the money		
13	Time to maturity, T	0.4849	<-- =YEARFRAC(B3,B2,3)		
14	Interest rate, r	1.73%	Risk-free rate of interest		
15	Call price	2.50	<-- Call price on 28-Aug-2019		
16	Put price	3.25	<-- Put price on 28-Aug-2019		
17					
18	**Implied volatility**				
19	Call, S net of dividends	17.69%	<--=ImpliedVolatility(B11,B12,B13,B14,B15,"C")		
20	Put, S net of dividends	20.36%	<--=ImpliedVolatility(B11,B12,B13,B14,B16,"P")		
21					
22	Call, S with dividends	14.91%	<--=ImpliedVolatility(B4,B12,B13,B14,B15,"C")		
23	Put, S with dividends	22.63%	<--=ImpliedVolatility(B4,B12,B13,B14,B16,"P")		

The R code to find the implied volatility when there are dividends in known date would look like this:

```r
## A Known Dividend to Be Paid Before the Option Expiration
today <- as.Date("2019-08-28")
expiration <- as.Date("2020-02-21")
S0 <- 54.99
r <- 0.0173
div_1 <- c(date = "2019-09-12", payment = 0.4)
div_2 <- c(date = "2019-11-29", payment = 0.4)

# Calculating dividends present value
delta_t_1 <- as.double(difftime(div_1["date"], today))/365
div_1["PV"] <- as.double(div_1["payment"]) * exp(-r * delta_t_1)

delta_t_2 <- as.double(difftime(div_2["date"], today))/365
div_2["PV"] <- as.double(div_2["payment"]) * exp(-r * delta_t_2)

net_S0 <- S0 - as.double(div_1["PV"]) - as.double(div_2["PV"])

# Calculating Black and Scholes Implies Volatility
X <- 55
ex_date <- as.Date("2020-02-21")
t <- as.double(difftime(ex_date, today))/365
call_price <- 2.5
put_price <- 3.25

# Implied Volatility, net of dividends
fm5_bs_iv(S = net_S0, X = X, t = t, r = r, target = call_price, option_type = "C")
fm5_bs_iv(S = net_S0, X = X, t = t, r = r, target = put_price, option_type = "P")

# Implied Volatility, with dividends
fm5_bs_iv(S = S0, X = X, t = t, r = r, target = call_price, option_type = "C")
fm5_bs_iv(S = S0, X = X, t = t, r = r, target = put_price, option_type = "P")
```

Dividend Adjustments for Continuous Dividend Payouts: The Merton Model

In the previous subsection we looked at the case of known future dividends. This subsection discusses a model by Merton (1973) for the pricing options on a stock that pays continuous dividends. Continuous dividends may seem an odd assumption. But a basket of stocks such as the SP500 or the Dow-Jones 30 (DJ30) can best be approximated by the assumption of a continuous dividend payout because there are many stocks and because the index components more or less pay out their dividends throughout the year.

Assuming the continuous dividend yield of k, Merton proved the following option pricing formula:

$$C = S \cdot e^{-kT} \cdot N(d_1) - X \cdot e^{-rT} \cdot N(d_2),$$

$$P = X \cdot e^{-rT} \cdot N(-d_2) - S \cdot e^{-kT} \cdot N(-d_1),$$

where:

$$d_1 = \frac{ln(S/X) + \left(r - k + \frac{\sigma^2}{2}\right) \cdot T}{\sigma\sqrt{T}}$$

$$d_2 = d_1 - \sigma\sqrt{T}$$

This model is used below to price the ETF that tracks the SP500 index:

	A	B	C
1			**MERTON'S DIVIDEND-ADJUSTED OPTION PRICING MODEL used here to price SP500 Spiders (symbol: SPY)**
2	S	292.45	Current underlying asset price (01-Sep-19)
3	X	300.00	Exercise price (also called Strike price)
4	T	0.296	<-- option expires 18-Dec-19, today's date 01-Sep-19
5	r	1.95%	Risk-free rate of interest (T-bills rate)
6	k	1.70%	Dividend yield
7	Sigma	14%	Underlying asset volatility
8			
9	d_1	-0.2869	<--=(LN(B2/B3)+(B5-B6+0.5*B7^2)*B4)/(B7*SQRT(B4))
10	d_2	-0.3631	<--=B9-B7*SQRT(B4)
11			
12	$N(d_1)$	0.3871	<--=NORM.S.DIST(B9,1)
13	$N(d_2)$	0.3583	<--=NORM.S.DIST(B10,1)
14			
15	Call price	5.77	<--=B2*EXP(-B6*B4)*B12-B3*EXP(-B5*B4)*B13
16	Put price	13.06	<--=B3*EXP(-B5*B4)*NORM.S.DIST(-B10,1)-B2*EXP(-B6*B4)*NORM.S.DIST(-B9,1)
17	Check using PCP	13.06	<--=B15-B2*EXP(-B6*B4)+B3*EXP(-B5*B4)

And below, we show the implementation in R:

```r
## Pricing SPY
# Input
S <- 292.45
X <- 300
t <- 0.296
r <- 0.0195
k <- 0.017
sigma_S <- 0.14

# Function: d1
fm5_bsm_d1 <- function(S, X, t, r, sigma_S, k){
    return( ( log(S/X) + (r - k + 0.5 * sigma_S ^ 2) * t ) / ( sigma_S * sqrt(t)) )
            }
fm5_bsm_d1(S, X, t, r, sigma_S, k)

# Function: d2
fm5_bsm_d2 <- function(S, X, t, r, sigma_S, k){
    return(fm5_bsm_d1(S, X, t, r, sigma_S,k) - sigma_S * sqrt(t))
            }
fm5_bsm_d2(S, X, t, r, sigma_S, k)

# Function: N(d1)
fm5_bsm_Nd1 <- function(S, X, t, r, sigma_S, k){
    return(pnorm(fm5_bsm_d1(S, X, t, r, sigma_S, k), mean = 0, sd = 1))
            }
fm5_bsm_Nd1(S, X, t, r, sigma_S, k)

# Function: N(d2)
fm5_bsm_Nd2 <- function(S, X, t, r, sigma_S, k){
    return(pnorm(fm5_bsm_d2(S, X, t, r, sigma_S, k), mean = 0, sd = 1))
            }
fm5_bsm_Nd2(S, X, t, r, sigma_S, k)

# Function: Black-Scholes-Merton Call option
fm5_bsm_call <- function(S, X, t, r, sigma_S, k){
            return(S * fm5_bsm_Nd1(S, X, t, r, sigma_S, k) * exp(-k * t) -
           X * exp(-r*t) * fm5_bsm_Nd2(S, X, t, r, sigma_S, k))
            }
fm5_bsm_call(S, X, t, r, sigma_S, k)

# Function: Black-Scholes-Merton Put option
fm5_bsm_put <- function(S, X, t, r, sigma_S, k){
    return(X * exp(-r*t) * (1-fm5_bsm_Nd2(S, X, t, r, sigma_S, k))-
           S * (1-fm5_bsm_Nd1(S, X, t, r, sigma_S, k)) * exp(-k * t))
            }
fm5_bsm_put(S, X, t, r, sigma_S, k)
```

The Merton model is often used to price currency options. Suppose we take an option on the Australian dollar (AUD). The option specifies a dollar exchange rate for AUDs (in the example below, the call option lets us buy 10,000 AUDs in 0.2959 years for $0.7 per AUD). The asset underlying the option is a euro interest-bearing security with interest rate r_{AUD}.

	A	B	C
1	**PRICING AN OPTION TO BUY AUSTRALIAN DOLLARS IN US DOLLARS**		
2	S	0.6718	Current exchange rate: U.S. dollar price of one Australian dollar
3	X	0.700	Exercise price
4	r_{US}	1.95%	Risk-free rate of interest in U.S. (T-bills rate)
5	r_{AUD}	0.93%	Australian dollar interest rate
6	T	0.2959	<-- option expires 18-Dec-19, today's date 01-Sep-19
7	Sigma	9.70%	AUD volatility in dollars
8	d_1	-0.6957	<--=(LN(B2/B3)+(B4+0.5*B7^2-B5)*B6)/(B7*SQRT(B6))
9	d_2	-0.7485	<--=B8-SQRT(B6)*B7
10			
11	n	10,000	Number of AUD per call contract
12			
13	$N(d_1)$	0.2433	<--=NORM.S.DIST(B8,1)
14	$N(d_2)$	0.2271	<--=NORM.S.DIST(B9,1)
15			
16	Call price	49.57	<--=(B2*EXP(-B5*B6)*B13-B3*EXP(-B4*B6)*B14)*B11
17	Put price	309.76	<--=(B3*EXP(-B4*B6)*NORMSDIST(-B9)-B2*EXP(-B5*B6)*NORMSDIST(-B8))*B11

In this example, the underlying asset of the currency option is an AUD. The AUD pays a dividend, which is the AUD interest rate. Therefore, the Merton model applies, with the underlying asset price being S * $exp(-r_{AUD} * T)$. Note also the change in d_1, where $\boldsymbol{r_{US} - r_{AUD}}$ appears instead of r_{US} as in the regular Black-Scholes formula.

Implementing this in R is straightforward once our functions have been defined:

```
## Pricing an option to buy AUD in USD
# Input
S <- 0.6718
X <- 0.7
rUS <- 0.0195
rAUD <- 0.0093
t <- 0.2959
sigma_S <- 0.0970
n <- 10000

Call_price <- n * fm5_bsm_call(S, X, t, r=rUS, sigma_S, k=rAUD)
Put_price <- n* fm5_bsm_put(S, X, t, r=rUS, sigma_S, k=rAUD)
```

18.7 "Bang for the Buck" with Options

There is another application of the Black-Scholes formula. Suppose that you are convinced that a given stock will go up in a very short period of time. You want to buy calls on the stock that have a maximum "bang for the buck"—that is, you want the percentage profit on your option investment to be maximal. Using the Black-Scholes formula, it is easy to show that you should:

- • Buy calls with the shortest possible maturity.
- • Buy calls that are most highly out-of-the-money (i.e., with the highest exercise price possible).

Here's a spreadsheet illustration:

	A	B	C
1	"BANG FOR THE BUCK" WITH OPTIONS		
2	S	25	Current stock price
3	X	25	Exercise price
4	r	6.00%	Risk-free rate of interest
5	T	0.5	Time to maturity of option (in years)
6	Sigma	30%	Stock volatility
7			
8	d_1	0.2475	<--=(LN(B2/B3)+(B4+0.5*B6^2)*B5)/(B6*SQRT(B5))
9	d_2	0.0354	<--=B8-SQRT(B5)*B6
10			
11	$N(d_1)$	0.5977	<--=NORM.S.DIST(B8,1)
12	$N(d_2)$	0.5141	<--=NORM.S.DIST(B9,1)
13			
14	Call price	2.47	<--=B2*B11-B3*EXP(-B4*B5)*B12
15	Put price	1.73	<--=B14-B2+B3*EXP(-B4*B5)
16			
17	Call bang	6.0483	<--=B11*B2/B14
18	Put bang	5.8070	<--=NORMSDIST(-B8)*B2/B15

The "call bang" defined in cell B17 is simply the percentage change in the call price dividend by the percentage change in the stock price (in economics, this is known as the "price elasticity"):

$$\text{Call bang} = \frac{\partial C / C}{\partial S / S} = \frac{\partial C}{\partial S}\frac{S}{C} = N(d_1)\frac{S}{C}$$

Similarly, for a put, the "bang for the buck" is defined by the formula below (of course, the story behind the put "bang for the buck" is that you are convinced that the stock price will go down):

$$\text{Put bang} = \frac{\partial P / P}{\partial S / S} = \frac{\partial P}{\partial S}\frac{S}{P} = -N(-d_1)\frac{S}{P}$$

This is defined in cell B18 in the preceding spreadsheet. To make the numbers easier to understand, we have dropped the minus sign, making the "put bang" = $N(-d_1)\frac{S}{P}$.

If you play with the spreadsheet, you will see that the longer the time to maturity, and the more in-the-money the option is—the less the bang for the buck. Another way of saying all of this is that the riskiest options are the most out-of-the-money and the shortest-term options.

	A	B	C	D	E	F	G	H
20	Data table: Effect of S and T on "call bang"							
21	↓ Underlying asset (S); → time to exercise (T)							
22	6.0463	0.25	0.5	0.75	1			
23	15	25.86	14.18	10.17	8.11			
24	16	23.32	12.99	9.41	7.56		data table header, =B17	
25	17	20.99	11.90	8.72	7.06			
26	18	18.86	10.91	8.09	6.61			
27	19	16.91	10.01	7.52	6.19			
28	20	15.12	9.18	6.99	5.81			
29	21	13.50	8.43	6.51	5.46			
30	22	12.03	7.74	6.07	5.14			
31	23	10.71	7.12	5.67	4.84			
32	24	9.63	6.56	5.30	4.57			
33	25	8.49	6.05	4.97	4.33			
34	26	7.57	5.59	4.67	4.10			
35	27	6.77	5.18	4.39	3.89			
36	28	6.07	4.81	4.14	3.70			
37	29	5.47	4.48	3.91	3.53			
38	30	4.96	4.18	3.70	3.37			

18.8 The Black Model for Bond Option Valuation[5]

Black (see Black and Cox 1976) suggested an adaptation of the Black-Scholes model which is often used for simple valuation of options on bonds or forwards. Letting F stand for the forward price of an asset, the Black-Scholes equation given in section 18.2 is replaced by:

$$C = e^{-rT}[F \cdot N(d_1) - X \cdot N(d_2)],$$

where

$$d_1 = \frac{ln(F/X) + \dfrac{\sigma^2 \cdot T}{2}}{\sigma\sqrt{T}}$$

$$d_2 = d_1 - \sigma\sqrt{T}$$

The corresponding put price is given by:

$$P = e^{-rT}[X \cdot N(-d_2) - F \cdot N(-d_1)]$$

To use the Black model, consider the case of an option on a zero-coupon bond, where the option maturity is $T = 0.5$. The option gives the holder the opportunity to buy the bond at time T for exercise price $X = 130$. Suppose that the risk-free interest rate is $r = 2\%$. If the forward price of the bond to the exercise date is $F = 133$ and the volatility of the forward price is $\sigma = 6\%$, then the pricing of the bond option using the Black model is given below.

	A	B	C
1	THE BLACK (1976) MODEL TO PRICE BOND OPTIONS		
2	F	133.0	<-- Bond forward price
3	X	130.0	<-- Exercise price
4	r	2.00%	<--Risk-free rate of interest
5	T	0.5	
6	Sigma	6%	<-- Bond forward price volatility, σ
7			
8	d_1	0.5590	<--=(LN(B2/B3)+B6^2*B5/2)/(B6*SQRT(B5))
9	d_2	0.5165	<--=B8-SQRT(B5)*B6
10			
11	Call price	4.00	<--=EXP(-B4*B5)*(B2*NORM.S.DIST(B8,1)-B3*NORM.S.DIST(B9,1))
12	Put price	1.03	<--=EXP(-B4*B5)*(B3*NORM.S.DIST(-B9,1)-B2*NORM.S.DIST(-B8,1))

And the implementation in R is:

```
# Inputs
Fwd <- 133 # Bond Forward price
X <- 130 # Exercise price
r <- 0.02
t <- 0.5
sigma_S <- 0.06

d1 <- fm5_bs_d1(Fwd, X, t, r = 0, sigma_S)
d2 <- fm5_bs_d2(Fwd, X, t, r = 0, sigma_S)
Nd1 <- fm5_bs_Nd1(Fwd, X, t, r = 0, sigma_S)
Nd2 <- fm5_bs_Nd2(Fwd, X, t, r = 0, sigma_S)

#Options prices
Call_on_bond <- exp(-r * t)*( Fwd * Nd1 - X * Nd2)
Put_on_bond <- exp(-r * t) * (X * (1-Nd2) - Fwd * (1-Nd1))
```

Thus, a call on the bond is worth 4.00 and a put is worth 1.03. A discussion on how to calculate the bond forward price can be found in Chapter 8 (term structure).

18.9 Using the Black-Scholes Model to Price Risky Debt

The Merton (1974) model is a traditional financial theory in calculating **the value of risky** corporate debts. It also serves as the benchmark model for other models to predict default risk. A KMV model (named after Kealhofer, McQuown, and Vasicek) is one of the extensions of the Merton model that succeeded in providing assessments of a firm's likelihood to default. The basic idea behind the Merton model is that the limited liability of the firm is generating an option-like pattern to the firm's equity. Simply said, in case the firm's asset value (denoted by V_T) will fall below the debt value at maturity (denoted by FV_T), the debt holders will seize the assets and the equity holders will get nothing. On the other hand, should the firm's asset value be above the debt value at maturity, the debt holders will get FV_T and the equity holders will get the difference between the assets value and FV_T. Mathematically, the equity payoff at the debt maturity (T) is:

$$E_T = max(V_T - FV_T, 0)$$

This is exactly the payoff of a call option at maturity. Thus, we can price the equity of the firm as a call option where the underlying asset is the firm's assets, the exercise price is the debt value at maturity (principal + interest), the volatility is the asset's volatility, and the time to maturity is the time to debt maturity. The debt value is simply the difference between asset value and equity value. We can also assess the default probability at the maturity date by using the risk-neutral probability that the option will not be exercised. In the Black-Scholes (Merton) model this is $1 - N(d_2)$. Calculating the default probability is called the KMV model.

In the example below we calculate the debt value of a firm with 1,400 asset value today. The assets are following lognormal distribution and have standard deviation of 15%.[6] The firm also has an outstanding 1,150 debt to be paid two and a half years from today and the risk-free rate is 2% per year.

	A	B	C
1	THE MERTON MODEL (1974) TO PRICE RISKY CORPORATE DEBT		
2	V	1,400	<--Firm's asset value today
3	FV_T	1,150	<--Debt payment at debt maturity
4	r	2.00%	<--Risk-free rate of interest
5	T	2.5	<--Time to debt maturity
6	Sigma	20%	<-- Asset's volatility
7			
8	d_1	0.7802	<--=(LN(B2/B3)+B6^2*B5/2)/(B6*SQRT(B5))
9	d_2	0.4639	<--=B8-SQRT(B5)*B6
10			
11	Equity value	352.91	<--=B2*NORM.S.DIST(B8,1)-B3*EXP(-B4*B5)*NORM.S.DIST(B9,1)
12	Debt value	1,047.09	<--=B2-B11
13	Yield to Maturity	3.82%	<--=(B3/B12)^(1/B5)-1
14			
15	The KMV model		
16	Default probability	32.1%	<--=1-NORM.S.DIST(B9,1)

Using the Merton model, we get that a debt value today of 1,047 (cell B12) is equivalent to a yield to maturity (YTM) of 3.82%. We can also use the KMV model to assess that the default probability two and a half years from today is 32% (cell B16).

Implementing the model in R should look like this:

```r
# Inputs
V <- 1400 # firm Asset value today
FVt <- 1150 # debt payment at maturity
r <- 0.02 # risk free rate
t <- 2.5 # time to debt maturity
sigma_S <- 0.2 # assets volatility

# Result
d1 <- fm5_bs_d1(V, FVt, t, r = 0, sigma_S)
d2 <- fm5_bs_d2(V, FVt, t, r = 0, sigma_S)
Nd1 <- fm5_bs_Nd1(V, FVt, t, r = 0, sigma_S)
Nd2 <- fm5_bs_Nd2(V, FVt, t, r = 0, sigma_S)

equity_value <- V * Nd1 - FVt * exp(-r * t) * Nd2
debt_value <- V - equity_value
YTM <- (FVt / debt_value)^(1 / t) - 1 # Yield to maturity

default_prob <- 1 - Nd2 # Default probability by the KMV model
```

18.10 Using the Black-Scholes Formula to Price Structured Securities

A "structured security" is Wall Street parlance for securities that incorporate combinations of stocks, options, and bonds. In this section we give three examples of such securities and show how to price them using the Black-Scholes model.[7] In the process we also return to the discussion of Chapter 16, showing how the profit diagrams of option strategies can help us understand such securities.

A Simple Structured Security: Principal Protection Plus Participation in Market Upside Moves

A simple and popular structured security offers a guaranteed return of the investor's principal plus some participation in the upside moves of the market. Here is an example: Homeside Bank offers its customers the following "Principal-Protected, Upside Potential" security (PPUP):

- • Initial investment in the security: $1,000.
- • No interest paid on the security.
- • In five years, the PPUP pays back the $1,000 plus 50% of the increase in the SP500. Writing the index price today as S_0 and the index price in five years as S_T, the payoff on the PPUP can be written as:

$$\$1{,}000\left[1+50\% * Max\left(\frac{S_T}{S_0}-1,0\right)\right]$$

To analyze the PPUP, we first rewrite the maturity payment as:

$$\$1{,}000\left[1+50\% * Max\left(\frac{S_T}{S_0}-1,0\right)\right]$$

$$=\underbrace{\$1{,}000}_{\substack{\uparrow\\ \text{Payoff on}\\ \text{zero-coupon}\\ \text{bond}}}+\$1{,}000 * \frac{50\%}{S_0} * \underbrace{Max(S_T-S_0,0)}_{\substack{\uparrow\\ \text{Payoff on at-the-money}\\ \text{call option}}}.$$

This shows that the PPUP payoff is composed of two parts:

- • A \$1,000 return of principal. Since no interest is paid on this principal, its value today is the present value of the payment at the risk-free interest rate, \$1,000 * e^{-rT}, where r is the interest rate and $T = 5$ is the maturity of the PPUP.
- • $\$1{,}000 * \frac{50\%}{S_0}$ times the value today of an at-the-money call on the SP500.

We can use the following spreadsheet to price this security:

	A	B	C
1	ANALYZING A SIMPLE STRUCTURED PRODUCT \$1,000 deposit with 50% participation in SP500 increase over 5 years		
2	Initial SP500 price, S0	3,000	<-- The price of the SP500 at PPUP issuance
3	Structured exercise price, X	3,000	
4	Risk-free interest rate for 5 years, r	5.00%	
5	Time to maturity, T	5	
6	Volatility of SP500, σ_{SP}	20%	
7	Participation rate	50%	<-- Percentage of increase in the S&P going to PPUP owner
8			
9	Structured components, value today		
10	Bond paying \$1000 at maturity	778.80	<--=EXP(-B4*B5)*1000
11	Participation rate /S0*at-the-money call on SP500	145.69	<--=1000*B7/B2*BSCall(B2,B3,B5,B4,B6)
12	Value of structured security today	924.49	<--=SUM(B10:B11)

The value of the structured security (cell B12) is \$924.49. This valuation has two parts:

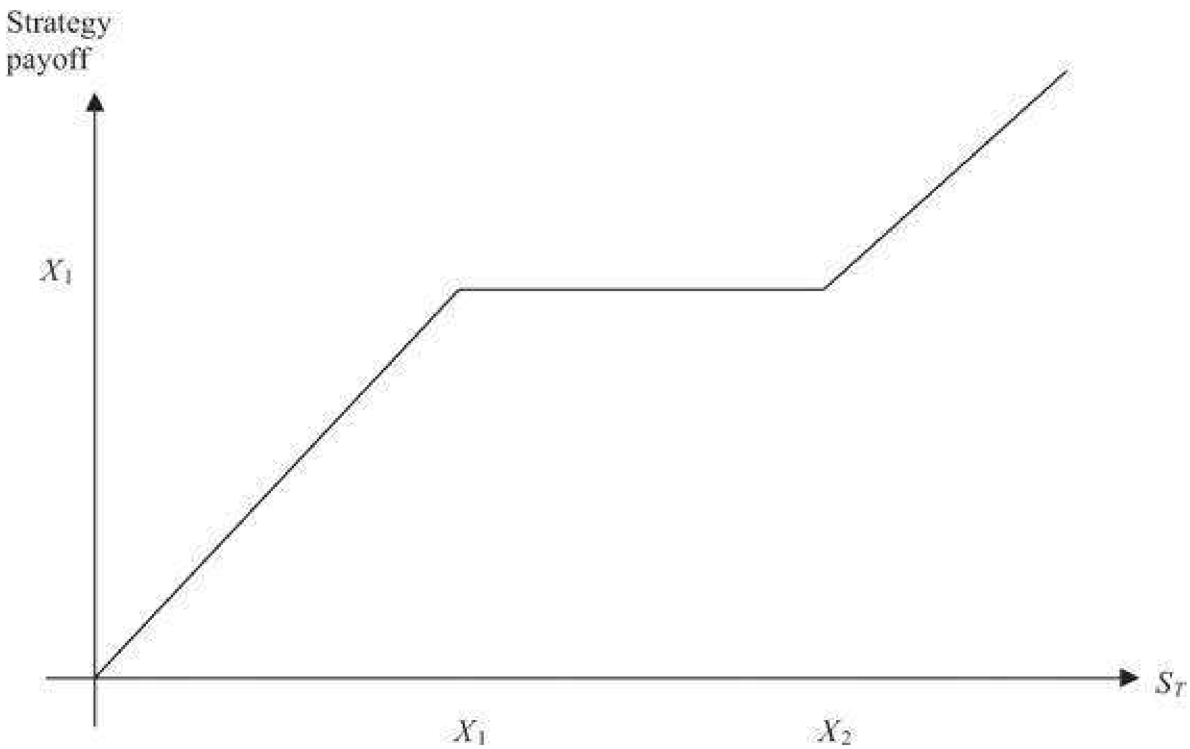
- • The present value of the bond part of the PPUP is \$778.80 (cell B10).
- • The value of $\$1{,}000 * \frac{50\%}{950}$ at-the-money calls on the SP500 is \$145.69 (cell B11).

Given the parameters in cells B2:B7, the PPUP is *overpriced*—it sells for \$1,000, whereas its market value ought to be \$924.49. Another way to think about the structured product is to compute its implied volatility: What σ_{SP} (cell B6) will give the market valuation (cell B12) of the PPUP to equal the $1,000 price being asked by the Homeside Bank? Either **GoalSeek** or **Solver** will solve this problem:

	A	B	C
1	**ANALYZING A SIMPLE STRUCTURED PRODUCT** **$1,000 deposit with 50% participation in SP500 increase over 5 years—finding sigma**		
2	Initial SP500 price, S0	3,000	<-- The price of the SP500 at PPUP issuance
3	Structured exercise price, X	3,000	
4	Risk-free interest rate for 5 years, r	5.00%	
5	Time to maturity, T	5	
6	Volatility of SP500, σ_{SP}	42.00%	<-- used goal seek / solver
7	Participation rate	50%	<-- Percentage of increase in the S&P going to PPUP owner
8			
9	Structured components, value today		
10	Bond paying $1000 at maturity	778.80	<--=EXP(-B4*B5)*1000
11	Participation rate /S0*at-the-money call on SP500	221.20	<--=1000*B7/B2*BSCall(B2,B3,B5,B4,B6)
12	Value of structured security today	1000.00	<--=SUM(B10:B11)

More Complicated Structured Products

Suppose you want to create a security with the following payoff pattern:



The payoff pattern increases (dollar for dollar) as the terminal price of the underlying asset increases from $0 \le S_T \le X_1$. Between X_1 and X_2, the payoff pattern is flat, and for $X_2 \le S_T$, the payoff again increases, dollar for dollar, with the price of the underlying. The algebraic formula for this payoff pattern is:

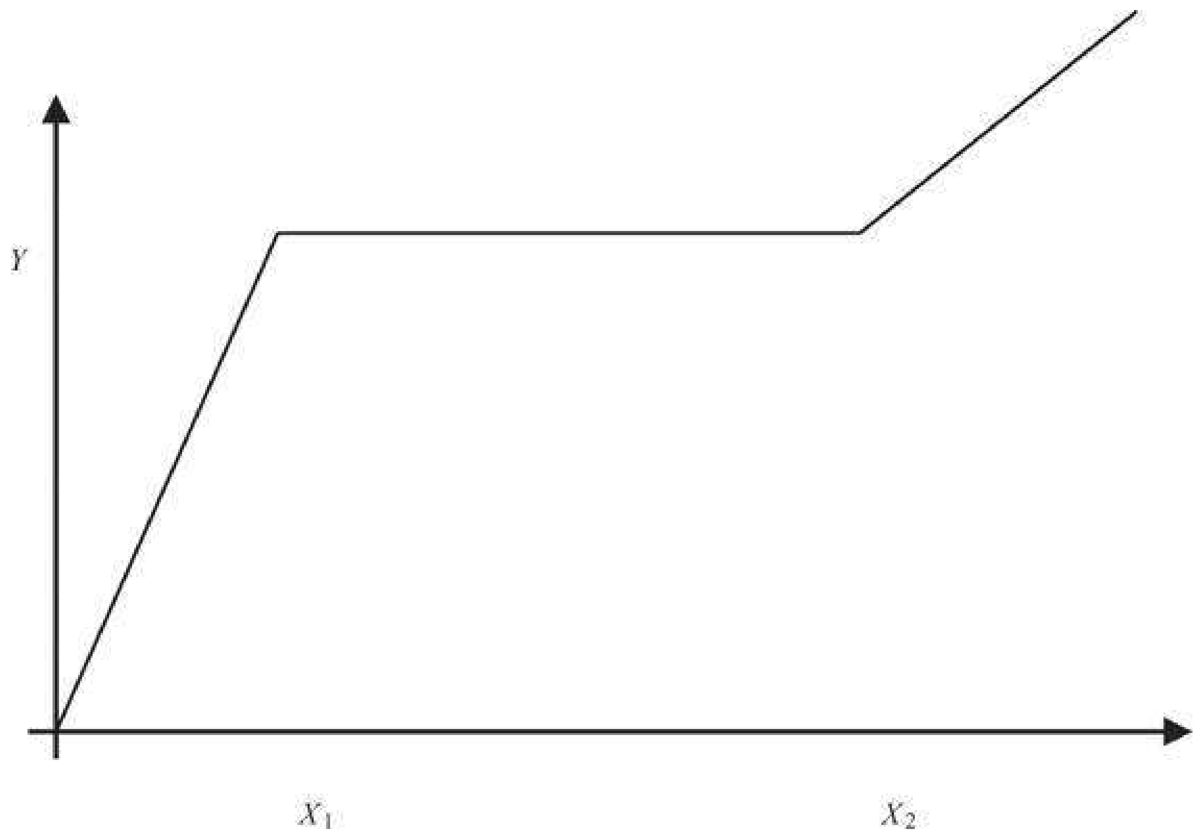
$X_1 - Max(X_1 - S_T, 0) + Max(S_T - X_2, 0)$

To prove that this formula creates the graph:

$$X_1 - \underbrace{Max(X_1 - S_T, 0)}_{\substack{\uparrow \\ \text{Payoff of written} \\ \text{put}}} + \underbrace{Max(S_T - X_2, 0)}_{\substack{\uparrow \\ \text{Payoff of bought} \\ \text{call}}}$$

$$= \begin{cases} X_1 - X_1 + S_T = S_T & S_T < X_1 \\ X_1 & X_1 \le S_T < X_2 \\ X_1 + S_T - X_2 & X_2 \le S_T \end{cases}$$

A slightly more complicated payoff pattern is the following:



The initial part of the payoff has slope $\frac{Y}{X_1}$, and the second increasing part of the payoff pattern has slope $\frac{Y}{X_2}$. This payoff pattern is created by the following formula:

$$Y - \underbrace{\frac{Y}{X_1} Max(X_1 - S_T, 0)}_{\substack{\uparrow \\ \text{Payoff of } \frac{Y}{X_1} \\ \text{written puts}}} + \underbrace{\frac{Y}{X_2} Max(S_T - X_2, 0)}_{\substack{\uparrow \\ \text{Payoff of } \frac{Y}{X_2} \\ \text{bought calls}}}$$

To prove that this is indeed the payoff:

$$Y - \underbrace{\frac{Y}{X_1} Max(X_1 - S_T, 0)}_{\substack{\uparrow \\ \text{Payoff of } \frac{Y}{X_1} \text{ written puts}}} + \underbrace{\frac{Y}{X_2} Max(S_T - X_2, 0)}_{\substack{\uparrow \\ \text{Payoff of } \frac{Y}{X_2} \text{ bought calls}}}$$

$$= \begin{cases} Y - \frac{Y}{X_1}(X_1 - S_T) = \frac{Y}{X_1} S_T & S_T < X_1 \\ Y & X_1 \le S_T < X_2 \\ Y + \frac{Y}{X_2}(S_T - X_2) = \frac{Y}{X_2} S_T & X_2 \le S_T \end{cases}$$

As an example of this kind of structured payoff security, Figure 18.1 shows the term sheet for a structured product issued by ABN-AMRO bank. The payment on this “Airbag” security depends on the value of the Stoxx50—an index of European stocks. Here are the details:

- • Issuance date: March 24, 2003
- • Terminal date: March 24, 2008
- • Cost: 1,020 euros
- • Payment at terminal date:

ABN·AMRO

AirBag on the Euro STOXX 50

17 March 2003

FINAL TERMS AND CONDITIONS

[illegible]

SUMMARY	Issuer & Lead Manager:	ABN AMRO Bank N.V. (Senior Long Term Debt Rating: Moody's Aa3, S&P AA-)
	Issue:	AirBag on the Euro STOXX 50
	Underlying:	Euro STOXX 50 (Bloomberg: SX5E)
	Spot Reference (SX5E(t)):	2158.00
	Issue Price:	EUR 1,020
	Entitlement:	1
	Issue Size:	5,000 Certificates
	AirBag Start:	100% of the Spot Reference (2158.00)
	AirBag Stop:	75% of the Spot Reference (1618.50)
	Percentage drop without any loss:	25%
	SX5E(t1):	Official closing level of the Underlying on the Valuation Date
	Redemption:	1. If SX5E(t1) is less than or equal to the AirBag Stop: EUR 1,000 x 1.33 x $\left(\frac{SX5E(t1)}{SX5E(t)}\right)$ 2. If SX5E(t1) is greater than the AirBag Stop but less than or equal to the AirBag Start: EUR 1,000 x 100% 3. If SX5E(t1) is greater than the AirBag Start: EUR 1,000 x $\left(\frac{SX5E(t1)}{SX5E(t)}\right)$
	Form:	Global bearer (permanent)
	Clearing:	Euroclear Bank SA, Clearstream Banking SA
	ISIN Code:	XS0165647966
	Valoren Code:	1578781
	Common Code:	16564796
	Minimum Trading Size:	1 AirBag Certificate
	Quoted on:	Reuters page: ABNPB15, Bloomberg page: AAPB, Internet: [illegible]
	Listing:	None
	Applicable Law:	English
	Selling Restrictions:	No sales to US persons or into the US, standard Dutch and UK selling restrictions apply.
TIMETABLE	Launch Date:	17/03/03
	Pricing Date:	17/03/03
	Issue & Payment Date:	24/03/03
	Valuation & Expiration Date:	14/03/08
	Final Settlement Date:	21/03/08

This term sheet is for information purposes only and does not constitute an offer to sell or a solicitation to buy any security or other financial instrument. All prices are indicative and dependent upon market conditions and the terms are liable to change and completion in the final documentation.

Figure 18.1
The ABN-AMRO term sheet for its Euro Stoxx50 Airbag security.

Payment at maturity

$$
= \begin{cases} 1{,}000 * 1.33 * \left(\dfrac{Stoxx50_{Maturity}}{Stoxx50_{Initial}}\right) & \text{If } Stoxx50_{Maturity} < 1{,}618.50 \\ 1{,}000 & 1{,}618.50 < Stoxx50_{Maturity} < 2{,}158 \\ 1{,}000 * \left(\dfrac{Stoxx50_{Maturity}}{Stoxx50_{Initial}}\right) & \text{If } Stoxx50_{Maturity} > 2{,}158 \end{cases}
$$

We recognize this security as one whose payoff has the form discussed earlier:

$$\underbrace{Y}_{\substack{\uparrow \\ \text{Bond payoff}}} \; - \underbrace{\frac{Y}{X_1} Max(X_1 - S_T, 0)}_{\substack{\uparrow \\ \frac{Y}{X_1} \text{ written puts with} \\ \text{exercise price } X_1}} + \underbrace{\frac{Y}{X_2} Max(S_T - X_2, 0)}_{\substack{\uparrow \\ \frac{Y}{X_2} \text{ purchased call} \\ \text{with exercise price } X_2}}$$

where

$X_1 = 1{,}618.50$

$X_2 = 2{,}158$

$Y = 1{,}000$

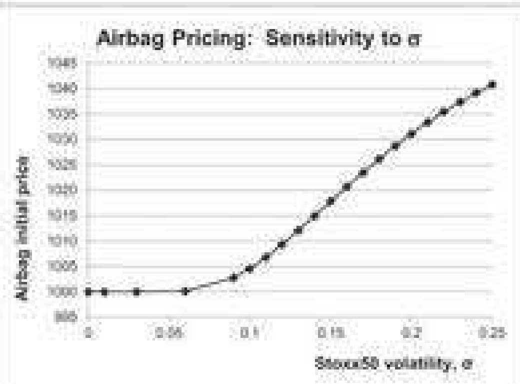
Here's the spreadsheet for the payoff. Cell B7 shows the payoff definition given by the Airbag issuer, and cell B8 shows the payoff in the option terms defined above. The data table in cells A13:B29 shows that these two definitions are equivalent:


ABN-AMRO AIRBAG

	A	B	C	D
2	Y	1,000.00		
3	X_1	1,618.50		
4	X_2	2,158.00		
5	S_T	2,373.80		
6	Airbag payoff			
7	By Airbag definition	1,100.00	<--=IF(B5<B3,B2*(B4/B3)*B5/B4,IF(B5>B4,B2*B5/B4,1000))	
8	Option formula	1,100.00	<--=B2-B2/B3*MAX(B3-B5,0)+B2/B4*MAX(B5-B4,0)	
11	Data table of payoffs			
12	S_T	Airbag definition	Option formula	
13				<-- Data table headers hidden
14	0	0.00	0.00	
15	100	61.79	61.79	
16	500	308.93	308.93	
17	750	463.39	463.39	
18	1,000	617.86	617.86	
19	1,250	772.32	772.32	
20	1,618.5	1,000.00	1,000.00	
21	1,750	1,000.00	1,000.00	
22	2,000	1,000.00	1,000.00	
23	2,158	1,000.00	1,000.00	
24	2,500	1,158.48	1,158.48	
25	2,750	1,274.33	1,274.33	
26	3,000	1,390.18	1,390.18	
27	3,250	1,506.02	1,506.02	
28	3,500	1,621.87	1,621.87	
29	3,750	1,737.72	1,737.72	

To see how the Airbag is priced, we use the Black-Scholes model and find the Stoxx50 volatility implied by the Airbag price:

	A	B	C
1		**PRICING THE ABN-AIRBAG** **Find the implied volatility**	
2	Stoxx50 price today, S_0	2,158.0	
3	X_1	1,618.50	
4	X_2	2,158.0	
5	Y	1,000.0	
6	Risk-free interest rate for 5 years, r	7.00%	
7	Time to maturity, T	5	
8	Volatility of the Stoxx50, sigma	15.75%	
9			
10	Airbag components, value today		
11	Bond paying X_1 at maturity	704.69	<--=EXP(-B6*B7)*B5
12	Y/X_1 * written puts with exercise X_1	-4.69	<--=-B5/B3*BSPut(B2,B3,B7,B6,B8)
13	Purchased call with exercise X_2	320.01	<--=B5/B4*BSCall(B2,B4,B7,B6,B8)
14	Value of structured security today	1,020.00	<--=SUM(B11:B13)
15			
16			
17	**Table: Sensitivity of Airbag to sigma**	1,020.00	<--=B14 Data table header
18	0%	1,000.00	
19	1%	1,000.00	
20	3%	1,000.00	
21	6%	1,000.16	
22	9%	1,002.76	
23	10%	1,004.57	
24	11%	1,006.80	
25	12%	1,009.34	
26	13%	1,012.09	
27	14%	1,014.95	
28	15%	1,017.84	
29	16%	1,020.70	
30	17%	1,023.49	
31	18%	1,026.16	
32	19%	1,028.70	
33	20%	1,031.11	
34	21%	1,033.35	
35	22%	1,035.45	
36	23%	1,037.39	
37	24%	1,039.19	
38	25%	1,040.84	



When the Stoxx50 σ is 15.75% (cell B8), the Airbag's price is 1,020 euros (cell B14). The table shows the sensitivity of this price to the σ. Notice that Airbag values are not very sensitive to σ : Doubling the sigma from 10% to 20% increases the Airbag value by about 17 euros. This is because of the offsetting values of the short put and the long call in the Airbag.

We can do one more exercise on the Airbag. Using a two-dimensional **Data Table**, we examine the Airbag's price sensitivity to both time to maturity T and the Stoxx50 volatility σ :

	A	B	C	D	E	F	G	H
1	ABN-AMRO AIRBAG SENSITIVITY TO TIME TO MATURITY AND SIGMA							
2	Stoxx50 price today, S_0	2,158.0						
3	X_1	1,618.50						
4	X_2	2,158.0						
5	Y	1,000.0						
6	Risk-free interest rate for 5 years, r	7.00%						
7	Time to maturity, T	5						
8	Volatility of the Stoxx50, sigma	15.75%						
9								
10	Airbag components, value today							
11	Bond paying X_1 at maturity	704.69	<--=EXP(-B6*B7)*B5					
12	Y/X_1 * written puts with exercise X_1	-4.69	<--=-B5/B3*BSPut(B2,B3,B7,B6,B8)					
13	Purchased call with exercise X_2	320.01	<--=B5/B4*BSCall(B2,B4,B7,B6,B8)					
14	Value of structured security today	1,020.00	<--=SUM(B11:B13)					
15								
16			Time to maturity, T					
17	Data table header: =B14	1020.00	5	4	3	2	1	0.0001
18		5%	1000.02	1000.07	1000.20	1000.59	1001.77	1000.20
19		10%	1004.57	1006.22	1008.40	1011.13	1013.78	1000.40
20		15%	1017.84	1021.09	1024.72	1028.28	1029.65	1000.59
21		20%	1031.11	1035.21	1039.61	1043.69	1044.54	1000.79
22	Volatility of the Stoxx50, sigma -->	25%	1040.84	1045.48	1050.44	1055.14	1056.54	1000.99
23		30%	1047.16	1052.22	1057.69	1063.09	1065.58	1001.19
24		35%	1050.86	1056.29	1062.26	1068.39	1072.19	1001.39
25		40%	1052.66	1058.44	1064.88	1071.75	1076.95	1001.59
26		45%	1053.10	1059.19	1066.10	1073.70	1080.28	1001.79
27		50%	1052.58	1058.94	1066.29	1074.59	1082.53	1001.99

The Airbag is a fairly stable security—its price varies no more than 10% for a wide variety of σ's and for nearly all times to maturity.

18.11 Summary

The Black-Scholes formula for pricing options is one of the most powerful innovations in finance. The formula is widely used both to price options and as a conceptual framework for analyzing complex securities. In this chapter we have explored the implementation of the Black-Scholes formula. Using "plain vanilla" Excel allows us to price Black-Scholes options; using VBA we are able define both the Black-Scholes price and the implied volatility for options. Finally, we showed how to use Black-Scholes to price options on bonds and how to price risky debt with the Merton model.

Exercises

1. Use the Black-Scholes model to price the following:
   - A call option on a stock whose current price is 50, with exercise price $X = 50$, $T = 0.5$, $r = 2\%$, $\sigma = 25\%$.
   - A put option with the same parameters.

2. Use the data from exercise 1 and **Data|Table** to produce graphs that show:
   - The sensitivity of the Black-Scholes call price to changes in the initial stock price S.
   - The sensitivity of the Black-Scholes put price to changes in σ.
   - The sensitivity of the Black-Scholes call price to changes in the time to maturity T.
   - The sensitivity of the Black-Scholes call price to changes in the interest rate r.
   - The sensitivity of the put price to changes in the exercise price X.

3. Use the data from exercise 1 and **Data|Table** to produce a graph comparing a call's *intrinsic value* [defined as $Max(S - X, 0)$] and its Black-Scholes price. From this graph you should be able to deduce that it is never optimal to exercise early a call priced by the Black-Scholes.

4. Use the data from exercise 1 and **Data|Table** to produce a graph comparing a put's intrinsic value [defined as $Max(X - S, 0)$] and its Black-Scholes price. From this graph you should be able to deduce that it may be optimal to exercise early a put priced by the Black-Scholes formula.
5. The table below gives prices for Apple (AAPL) options on April 10, 2020. The option with exercise price $X =$ \$270 is assumed to be the at-the-money option.

	A	B	C	D	E	F	G
1	APPLE (AAPL) OPTIONS						
2	Stock price	268					
3	Current date	10-Apr-20					
4	Expiration date	18-Sep-20					
5	Time to maturity, T	0.440	<--=YEARFRAC(B3,B4,1)				
6	Interest	0.20%					
7							
8	Strike	Call option price	Implied volatility		Strike	Put option price	Implied volatility
9	230	49.27			230	12.30	
10	235	45.19			235	13.60	
11	240	42.00			240	15.00	
12	250	35.40			250	18.30	
13	260	28.70			260	22.00	
14	270	23.48			270	26.23	
15	280	18.80			280	33.20	
16	290	13.70			290	38.15	
17	300	10.50			300	44.20	
18	305	8.95			305	47.90	
19	310	7.65			310	51.55	
20	315	6.60			315	54.96	
21	320	5.35			320	59.45	

1. Compute the implied volatility of each option (use the functions **CallVolatility** and **PutVolatility** defined in the chapter or the equivalent R code).
2. Graph these volatilities. Is there a volatility "smile"?

6. Reexamine the $X = 240$ call for AAPL in the previous exercise. What price would be necessary for this call in order for the implied volatility to be 60%?
7. Use the Excel **Solver** to find the stock price for which there is the maximum difference between the Black-Scholes call option price and the option's intrinsic value. Use the following values: $S = 45$, $X = 45$, $T = 1$, $\sigma = 40\%$, $r = 1\%$.
8. As shown in this chapter, Merton (1973) shows that for the case of an asset with price S paying a continuously compounded dividend yield k, this leads to the following call option pricing formula:

$$C = Se^{-kT}N(d_1) - Xe^{-rT}N(d_2),$$

where

$$d_1 = \frac{ln(S/X) + (r - k + \sigma^2/2)T}{\sigma\sqrt{T}}$$
$$d_2 = d_1 - \sigma\sqrt{T}$$

1. Modify the **BSCall** and **BSPut** functions defined in this chapter to fit the Merton model.
2. Use the function to price an at-the-money option on an index whose current price is $S = 1500$, when the option's maturity $T = 1$, the dividend yield is $k = 2.2\%$, its standard deviation $\sigma = 20\%$, and the interest rate $r = 1\%$.

9. On April 10, 2020, call and put options to purchase and sell 10,000 euros at \$1.0975 per euro are traded on the Chicago Mercantile Exchange. The options' expiration date is April 30, 2020. If the dollar interest rate is 0.2%, the euro interest rate is −0.6% (negative 0.6%), and the volatility of the euro is 10%, what should be the price of a call and a put?

10. Note that you can use the Black-Scholes formula to calculate the call option premium as a percentage of the exercise price in terms of *S/X*:

$$C = SN(d_1) - Xe^{-rT}N(d_2) \Rightarrow \frac{C}{X} = \frac{S}{X}N(d_1) - e^{-rT}N(d_2$$

where

$$d_1 = \frac{\ln(S/X) + (r + \sigma^2/2)T}{\sigma\sqrt{T}}$$
$$d_2 = d_1 - \sigma\sqrt{T}$$

Implement this in a spreadsheet or R.

11. Note that you can also calculate the Black-Scholes put option premium as a percentage of the exercise price in terms of S/X:

$$P = -SN(-d_1) + Xe^{-rT}N(-d_2) \Rightarrow \frac{P}{X} = e^{-rT}N(-d_2) - \frac{S}{X}N(-d_1)$$

where

$$d_1 = \frac{ln(S/X) + (r + \sigma^2/2)T}{\sigma\sqrt{T}}$$
$$d_2 = d_1 - \sigma\sqrt{T}$$

1. Implement this in a spreadsheet or R.
2. Find the ratio of *S/X* for which *C/X* and *P/X* cross when $T = 0.5$, $\sigma = 25\%$, $r = 1\%$. (You can use a graph or you can use Excel's **Solver**.) Note that this crossing point is affected by the interest rate and the option maturity, but not by σ.

12. Consider a structured security of the following type: The purchaser invests \$1,000 and in three years gets back the initial investment plus 95% of the increase in a market index whose current price is 100. The interest rate is 6% per year, continuously compounded. Assuming the security is fairly priced, what is the implied volatility of the market index?

---

1. 1. The lognormal distribution is discussed in Chapter 23, though for purposes of applying the Black-Scholes model, section 18.3 is sufficient.
2. 2. Section 23.7 discusses how to calculate the annualized σ of the lognormal process given nonannual data.

3. 3. In the spreadsheet we use a newer version of this function, **Norm.S.Dist(x,TRUE** or **FALSE). TRUE** gives the cumulative distribution, whereas **FALSE** gives the probability density.
4. 4. The nomenclature "forward-looking" versus "backward-looking" makes it sound as if the implied volatility is always better than the historical. This is, of course, not the intention.
5. 5. This section is advanced and can be skipped on first reading. A full discussion of the pricing of options on bonds is beyond the scope of the current edition of this book. However, this very useful and often-used adaptation of the Black model is simple enough to attach to this chapter.
6. 6. There are many methods of estimating the assets' volatility, and a very common approach is to use the equation $\sigma_V = \frac{\sigma_E}{(V/E) \cdot N(d1)}$.
7. 7. Not all structures can be priced using Black-Scholes. More complicated, path-dependent securities often need to be priced using the Monte Carlo methods discussed in Chapters 26 and 27.

19 Option Greeks

19.1 Overview

In this chapter we discuss the sensitivities of the Black-Scholes formula to its various parameters. The "Greeks," as they are called (because of the Greek letters used to denote most of them), are the partial derivatives of the Black-Scholes formula with respect to its arguments. They can be thought of as giving a measure of the riskiness of an option. The five most commonly used Greeks are as follows:

- • Delta, denoted by Δ, is the partial derivative of the option price with respect to the price of the underlying stock price: $\Delta_{Call} = \frac{\partial \text{Call}}{\partial S}, \Delta_{Put} = \frac{\partial \text{Put}}{\partial S}$. Delta can be thought of as a measure of the variability of the option's price when the price of the underlying changes. The call delta and the put delta in absolute value equal 1, $\Delta_{Call} + |\Delta_{Put}| = 1$.
- • Gamma, Γ, is the second derivative of the option's price with respect to the underlying stock. Gamma gives the convexity of the option price with respect to the stock price: $\Gamma_{Call} = \frac{\partial^2 \text{Call}}{\partial S^2}, \Gamma_{Put} = \frac{\partial^2 \text{Put}}{\partial S^2}$. Gamma can be thought of as a measure of the variability of the option's delta when the price of the underlying changes. For options priced by the Black-Scholes formula, the call and put have the same gamma, $\Gamma_{Call} = \Gamma_{Put}$.
- • Vega is the sensitivity of the option price to the standard deviation of the underlying stock's return σ : Since there is no Greek letter called "vega," the Greek letter Nu, v, is commonly used to denote vega. Vega can be thought of as a measure of the variability of the option's price when the sigma of the underlying asset changes: $v_{Call} = \frac{\partial \text{Call}}{\partial \sigma}, v_{Put} = \frac{\partial \text{Put}}{\partial \sigma}$.

Given the Black-Scholes formula, calls and puts have the same vega: $v_{Call} = v_{Put}$.

- • Theta, θ, is a change in the option's value as the time to maturity decreases. We generally expect that options become less valuable with the passage of time (though this is not always true). Theta can be thought of as a measure of the option's price erosion when a day goes by. Writing T as the option's remaining time to maturity, we set theta equal to the negative of the derivative of the option price with respect to T: $\theta_{\text{Call}} = -\frac{\partial \text{Call}}{\partial T}, \theta_{\text{Put}} = -\frac{\partial \text{Put}}{\partial T}$.

Given the Black-Scholes formula, the put's theta is slightly higher than the call's theta in absolute values: $\theta_{put} = \theta_{call} - rXe^{-rT}$.

- • Rho, ρ, measures the interest rate sensitivity of an option: $\rho_{Call} = -\frac{\partial \text{Call}}{\partial r}, \rho_{Put} = -\frac{\partial \text{Put}}{\partial r}$.

Given the Black-Scholes formula, the put's rho is lower than the call's rho: $\rho_{Put} = \rho_{Call} - xTe^{-rT}$.

In this chapter we show you how to measure the option's Greeks (not just the five common Greeks) and how to use them in hedging. For generality, we illustrate using the Merton model (Chapter 18, section 6), an extended version of the Black-Scholes formula that applies to both stocks paying a continuous dividend or to currencies.

19.2 Defining and Computing the Greeks

The Greeks are the sensitivities of an option price with respect to certain of its variables. In the table below we set out the Greeks for options defined on an underlying that pays a continuous dividend. As discussed in section 18.6, such options are priced using the Merton model. Of course, the standard Black-Scholes model is obtained from the Merton model by setting the dividend yield $k = 0$. Currency options can be priced by the Merton formula by setting S = current exchange rate, X = option exercise exchange rate, r = domestic interest rate, and k = foreign interest rate.

The Merton version of the Black-Scholes formula is given by:

$$C = Se^{-kT}N(d_1) - Xe^{-rT}N(d_2)$$
$$P = Xe^{-rT}N(-d_2) - Se^{-kT}N(-d_1)$$

where

$$d_1 = \frac{ln\left(\frac{S}{X}\right) + (r - k + 0.5 * \sigma^2)T}{\sigma\sqrt{T}}$$
$$d_2 = d_1 - \sigma\sqrt{T}$$

Table 19.1 gives the Greeks for this formula. The appendixes to this chapter give the VBA and R implementations of these functions.

Table 19.1: The Greeks

	Measures	Call	Put
First-order Greeks			
Delta, written as either Δ or δ	Price sensitivity of option $\frac{\partial V}{\partial S}$	$\Delta_{Call} = e^{-kT}N(d_1)$	$\Delta_{Put} = \Delta_{Call} - 1 = -e^{-kT}N(-d_1)$
Vega, written as v	Sensitivity to volatility $\frac{\partial V}{\partial \sigma}$	$Se^{-kT}N'(d_1)\sqrt{T} = \frac{S\sqrt{T}e^{-0.5(d_1)^2-kT}}{\sqrt{2\pi}}$	
Theta, written as θ	Time sensitivity $-\frac{\partial V}{\partial T}$	$-\frac{Se^{-kT}N'(d_1)\sigma}{2\sqrt{T}} + kSe^{-kT}N(d_1)$ $-rXe^{-rT}N(d_2)$	$-\frac{Se^{-kT}N'(d_1)\sigma}{2\sqrt{T}} - kSe^{-kT}N(-d_1)$ $+rXe^{-rT}N(-d_2)$
Rho, ρ	Interest rate sensitivity $\frac{\partial V}{\partial r}$	$XTe^{-rT}N(d_2)$	$-XTe^{-rT}N(-d_2)$
Second-order Greeks			
Gamma, written as Γ	The option's convexity with respect to underlying price $\frac{\partial^2 V}{\partial S^2}$	$\frac{e^{-kT}N'(d_1)}{S\sigma\sqrt{T}} = \frac{e^{0.5(d_1)^2-kT}}{S\sigma\sqrt{2T\pi}}$	
Vanna	The sensitivity of the delta to volatility $\frac{\partial^2 V}{\partial S\partial\sigma} = \frac{\partial\Delta}{\partial\sigma}$	$-e^{-kT}N'(d_1)\frac{d_2}{\sigma} = \frac{v}{S}\left(1 - \frac{d_1}{\sigma\sqrt{T}}\right)$	
Charm	The sensitivity of theta with respect to the underlying's price $\frac{\partial^2 V}{\partial T\partial S} = \frac{\partial\theta}{\partial S}$	$ke^{-kT}N(d_1)$ $-e^{-kT}N'(d_1)\frac{2(r-k)T - d_2\sigma\sqrt{T}}{2T\sigma\sqrt{T}}$	$-ke^{-kT}N(-d_1)$ $-e^{-kT}N'(d_1)\frac{2(r-k)T - d_2\sigma\sqrt{T}}{2T\sigma\sqrt{T}}$
Vomma	The sensitivity of vega to volatility changes $\frac{\partial^2 V}{\partial\sigma^2} = \frac{\partial v}{\partial\sigma}$	$Se^{-kT}N'(d_1)\frac{d_1 d_2}{\sigma}\sqrt{T} = v\frac{d_1 d_2}{\sigma}$	
Veta	The sensitivity of vega with respect to the passage of time $\frac{\partial^2 V}{\partial\sigma\partial T} = \frac{\partial v}{\partial T}$	$-Se^{-kT}N'(d_1)\sqrt{T}\left[k + \frac{(r-k)d_1}{\sigma\sqrt{T}} - \frac{1+d_1 d_2}{2T}\right]$	
Third-order Greeks			
Speed	The sensitivity of gamma to the underlying price $\frac{\partial^3 V}{\partial S^3}$	$-e^{-kT}\frac{N'(d_1)}{S^2\sigma\sqrt{T}}\left(\frac{d_1}{\sigma\sqrt{T}} + 1\right) = \frac{-\Gamma}{S}\left(\frac{d_1}{\sigma\sqrt{T}} + 1\right)$	
Zomma	The sensitivity of gamma to volatility $\frac{\partial^3 V}{\partial S^2\partial\sigma} = \frac{\partial\Gamma}{\partial\sigma}$	$e^{-kT}\frac{N'(d_1)(d_1 d_2 - 1)}{S\sigma^2\sqrt{T}} = \Gamma\left(\frac{d_1 d_2 - 1}{\sigma}\right)$	
Color	The sensitivity of gamma to time $\frac{\partial^3 V}{\partial S^2\partial T} = \frac{\partial\Gamma}{\partial T}$	$-e^{-kT}\frac{N'(d_1)}{2TS\sigma\sqrt{T}}\left[1 + 2kT + \frac{2d_1(r-k)T}{\sigma\sqrt{T}} - d_1 d_2\right]$ $= \frac{-\Gamma}{2T}\left[1 + 2kT + \frac{2d_1(r-k)T}{\sigma\sqrt{T}} - d_1 d_2\right]$	

Where:

$$d_1 = \frac{ln(S/X) + (r - k + 0.5\cdot\sigma^2)\cdot T}{\sigma\cdot\sqrt{T}};\ d_2 = d_1 - \sigma\cdot\sqrt{T} \quad N(x) = \underbrace{\int_{-\infty}^{x}\frac{1}{\sqrt{2\pi}}e^{-(0.5\cdot y^2)}dy}_{\text{Norm.S.Dist}(x,1)\text{ in Excel}};\ N'(x) = \underbrace{\frac{1}{\sqrt{2\pi}}e^{-(0.5\cdot x^2)}}_{\text{Norm.S.Dist}(x,0)\text{ in Excel}}$$

The Greeks are implemented in the spreadsheet below, which shows the direct calculation of each Greek as well as a VBA function implementation and a brute-force approach.[1]

	A	B	C
1	**BLACK-SCHOLES GREEKS Uses Merton model for a continuously dividend-paying stock**		
2	S	100	Current stock price
3	X	90	Exercise price
4	T	0.5	Time to maturity of option (in years)
5	r	5.0%	Risk-free rate of interest
6	k	1.5%	Dividend yield
7	Sigma	35%	Stock volatility
8			
9	d_1	0.6202	<--=(LN(B2/B3)+(B5-B6+0.5*B7^2)*B4)/(B7*SQRT(B4))
10	d_2	0.3727	<--=B9-SQRT(B4)*B7
11	$N(d_1)$	0.7324	<--=NORM.S.DIST(B9,1)
12	$N(d_2)$	0.6453	<--=NORM.S.DIST(B10,1)
13	$N'(d_1)$	0.3291	<--=NORM.S.DIST(B9,0)
14	$N'(d_2)$	0.3722	<--=NORM.S.DIST(B10,0)
15			
16	<u>**Call**</u>		
17	Price (direct)	16.0517	<--=B2*EXP(-B6*B4)*B11-B3*EXP(-B5*B4)*B12
18	Price (VBA)	16.0517	<--=@bsmertoncall(B2,B3,B4,B5,B6,B7)
19			
20			**Call Greeks, direct**
21	**First-order Greeks**		
22	Delta	0.7270	<--=EXP(-B6*B4)*B11
23	Vega	23.1004	<--=B2*EXP(-B6*B4)*B13*SQRT(B4)
24	Theta	-9.8269	<--=B2*EXP(-B6*B4)*B13*B7/(2*SQRT(B4))+B6*B2*EXP(-B6*B4)*B11-B5*B3*EXP(-B5*B4)*B12
25	Rho	28.3220	<--=B3*B4*EXP(-B5*B4)*B12
26			
27	**Second-order Greeks**		
28	Gamma	0.0132	<--=EXP(-B6*B4)*B13/(B2*B7*SQRT(B4))
29	Vanna	-0.3479	<--=B23/B2*(1-B9/(B7*SQRT(B4)))
30	Charm	0.0865	<--=B6*EXP(-B6*B4)*B11-EXP(-B6*B4)*B13*(2*(B5-B6)*B4-B10*B7*SQRT(B4))/(2*B4*B7*SQRT(B4))
31	Vomma	15.2549	<--=B23*B9*B10/B7
32	Veta	26.0671	<--=-B2*EXP(-B6*B4)*B13*(B6+((B5-B6)*B9)/(B7*SQRT(B4))-(1+B9*B10)/(2*B4))*SQRT(B4)
33			
34	**Third-order Greeks**		
35	Speed	-0.0005	<--=-B28/B2*(B9/(B7*SQRT(B4))+1)
36	Zomma	-0.0290	<--=B28*(B9*B10-1)/B7
37	Color	-0.0115	<--=-B28/(2*B4)*(1+2*B6*B4+(2*B9*(B5-B6)*B4/(B7*SQRT(B4))-B9*B10))

	A	B	C
39			**Call Greeks, VBA formulas**
40	**First-order Greeks**		
41	Delta	0.7270	<--=deltacall(B2,B3,B4,B5,B6,B7)
42	Vega	23.1004	<--=vega(B2,B3,B4,B5,B6,B7)
43	Theta	-9.8269	<--=Thetacall(B2,B3,B4,B5,B6,B7)
44	Rho	28.3220	<--=rhocall(B2,B3,B4,B5,B6,B7)
45			
46	**Second-order Greeks**		
47	Gamma	0.0132	<--=optiongamma(B2,B3,B4,B5,B6,B7)
48	Vanna	-0.3479	<--=vanna(B2,B3,B4,B5,B6,B7)
49	Charm	0.0865	<--=CharmCall(B2,B3,B4,B5,B6,B7)
50	Vomma	15.2549	<--=vomma(B2,B3,B4,B5,B6,B7)
51	Veta	26.0671	<--=veta(B2,B3,B4,B5,B6,B7)
52			
53	**Third-order Greeks**		
54	Speed	-0.0005	<--=Speed(B2,B3,B4,B5,B6,B7)
55	Zomma	-0.0290	<--=Zomma(B2,B3,B4,B5,B6,B7)
56	Color	-0.0115	<--=Color(B2,B3,B4,B5,B6,B7)
57			
58			**Call Greeks, brute force**
59	lim h-->0	0.00001	
60	**First-order Greeks**		
61	Delta	0.7270	<--=(bsmertoncall(B2+B59,B3,B4,B5,B6,B7)-B17)/B59
62	Vega	23.1005	<--=(bsmertoncall(B2,B3,B4,B5,B6,B7+B59)-B17)/B59
63	Theta	-9.8269	<--=-(bsmertoncall(B2,B3,B4+B59,B5,B6,B7)-B17)/B59
64	Rho	-28.3221	<--=-(bsmertoncall(B2,B3,B4,B5+B59,B6,B7)-B17)/B59
65			
66	**Second-order Greeks**		
67	Gamma	0.0132	<--=(deltacall(B2+B59,B3,B4,B5,B6,B7)-B41)/B59
68	Vanna	-0.3479	<--=(deltacall(B2,B3,B4,B5,B6,B7+B59)-B41)/B59
69	Charm	0.0865	<--=(Thetacall(B2+B59,B3,B4,B5,B6,B7)-B43)/B59
70	Vomma	15.2543	<--=(vega(B2,B3,B4,B5,B6,B7+B59)-B42)/B59
71	Veta	26.0669	<--=(vega(B2,B3,B4+B59,B5,B6,B7)-B42)/B59
72			
73	**Third-order Greeks**		
74	Speed	-0.0005	<--=(optiongamma(B2+B59,B3,B4,B5,B6,B7)-B47)/B59
75	Zomma	-0.0290	<--=(optiongamma(B2,B3,B4,B5,B6,B7+B59)-B47)/B59
76	Color	-0.0115	<--=(optiongamma(B2,B3,B4+B59,B5,B6,B7)-B47)/B59

Greek calculations for puts are given below:

	E	F	G
16	**Put**		
17	Price (direct)	4.5768	<--=B3*EXP(-B5*B4)*(1-B12)-B2*EXP(-B6*B4)*(1-B11)
18	Price (VBA)	4.5768	<--=bsmertonput(B2,B3,B4,B5,B6,B7)
19			
20			**Put Greeks, direct**
21	**First-order Greeks**		
22	Delta	-0.2656	<--=-EXP(-B6*B4)*(1-B11)
23	Vega	23.1004	<--=B2*EXP(-B6*B4)*B13*SQRT(B4)
24	Theta	-6.9268	[illegible]
25	Rho	-15.5670	<--=-B3*B4*EXP(-B5*B4)*(1-B12)
26			
27	**Second-order Greeks**		
28	Gamma	0.0132	<--=EXP(-B6*B4)*B13/(B2*B7*SQRT(B4))
29	Vanna	-0.3479	<--=B23/B2*(1-B9/(B7*SQRT(B4)))
30	Charm	0.0716	[illegible]
31	Vomma	15.2549	<--=F23*B9*B10/B7
32	Veta	26.0671	<--=-B2*EXP(-B6*B4)*B13*(B6+(B5-B6)*B9/(B7*SQRT(B4))-(1+B9*B10)/(2*B4))*SQRT(B4)
33			
34	**Third-order Greeks**		
35	Speed	-0.0005	<--=-F28/B2*(B9/(B7*SQRT(B4))+1)
36	Zomma	-0.0290	<--=F28*(B9*B10-1)/B7
37	Color	-0.0115	<--=-F28/(2*B4)*(1+2*B6*B4+(2*B9*(B5-B6)*B4/(B7*SQRT(B4))-B9*B10))
38			
39			**Put Greeks, VBA formulas**
40	**First-order Greeks**		
41	Delta	-0.2656	<--=deltaput(B2,B3,B4,B5,B6,B7)
42	Vega	23.1004	<--=vega(B2,B3,B4,B5,B6,B7)
43	Theta	-6.9268	<--=Thetaput(B2,B3,B4,B5,B6,B7)
44	Rho	-15.5670	<--=rhoput(B2,B3,B4,B5,B6,B7)
45			
46	**Second-order Greeks**		
47	Gamma	0.0132	<--=optiongamma(B2,B3,B4,B5,B6,B7)
48	Vanna	-0.3479	<--=vanna(B2,B3,B4,B5,B6,B7)
49	Charm	0.0716	<--=CharmPut(B2,B3,B4,B5,B6,B7)
50	Vomma	15.2549	<--=vomma(B2,B3,B4,B5,B6,B7)
51	Veta	26.0671	<--=veta(B2,B3,B4,B5,B6,B7)
52			
53	**Third-order Greeks**		
54	Speed	-0.0005	<--=Speed(B2,B3,B4,B5,B6,B7)
55	Zomma	-0.0290	<--=Zomma(B2,B3,B4,B5,B6,B7)
56	Color	-0.0115	<--=Color(B2,B3,B4,B5,B6,B7)

	E	F	G
58			**Put Greeks, brute force**
59			
60	**First-order Greeks**		
61	Delta	-0.266	<--=(bsmertonput(B2+B59,B3,B4,B5,B6,B7)-F17)/B59
62	Vega	23.1005	<--=(bsmertonput(B2,B3,B4,B5,B6,B7+B59)-F17)/B59
63	Theta	-6.927	<--=-(bsmertonput(B2,B3,B4+B59,B5,B6,B7)-F17)/B59
64	Rho	15.57	<--=-(bsmertonput(B2,B3,B4,B5+B59,B6,B7)-F17)/B59
65			
66	**Second-order Greeks**		
67	Gamma	0.0132	<--=(deltaput(B2+B59,B3,B4,B5,B6,B7)-F41)/B59
68	Vanna	-0.3479	<--=(deltaput(B2,B3,B4,B5,B6,B7+B59)-F41)/B59
69	Charm	0.0716	<--=(Thetaput(B2+B59,B3,B4,B5,B6,B7)-F43)/B59
70	Vomma	15.2543	<--=(vega(B2,B3,B4,B5,B6,B7+B59)-F42)/B59
71	Veta	26.0669	<--=(vega(B2,B3,B4+B59,B5,B6,B7)-F42)/B59
72			
73	**Third-order Greeks**		
74	Speed	-0.0005	<--=(optiongamma(B2+B59,B3,B4,B5,B6,B7)-F47)/B59
75	Zomma	-0.0290	<--=(optiongamma(B2,B3,B4,B5,B6,B7+B59)-F47)/B59
76	Color	-0.0115	<--=(optiongamma(B2,B3,B4+B59,B5,B6,B7)-F47)/B59

Excel can be used to examine the sensitivities of the Greeks to various parameters. We give some examples below. Figure 19.1 shows the deltas as functions of the stock price of the call option and the put option.

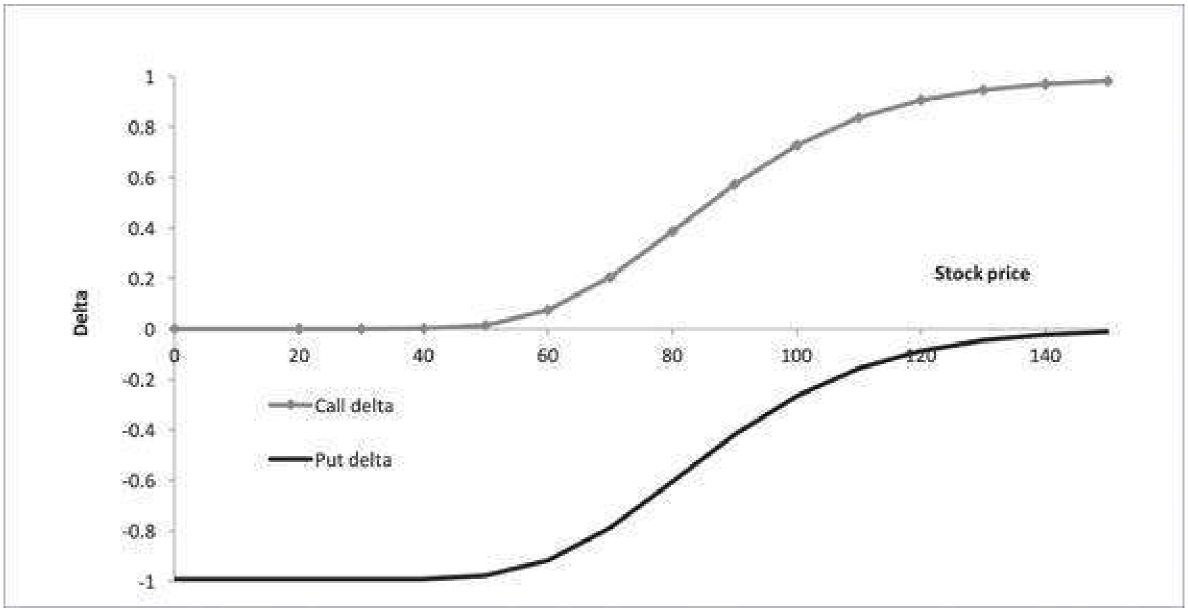


Figure 19.1
Call and put deltas as functions of stock price. As a call or a put becomes more in-the-money, the delta tends toward +1 for a call and -1 for a put. Essentially, the call or put price moves in tandem with the underlying stock price. An extremely out-of-the-money put or call has a delta = 0.

Figures 19.2 and 19.3 show the theta of a call as a function of the stock price and the time to option expiration.

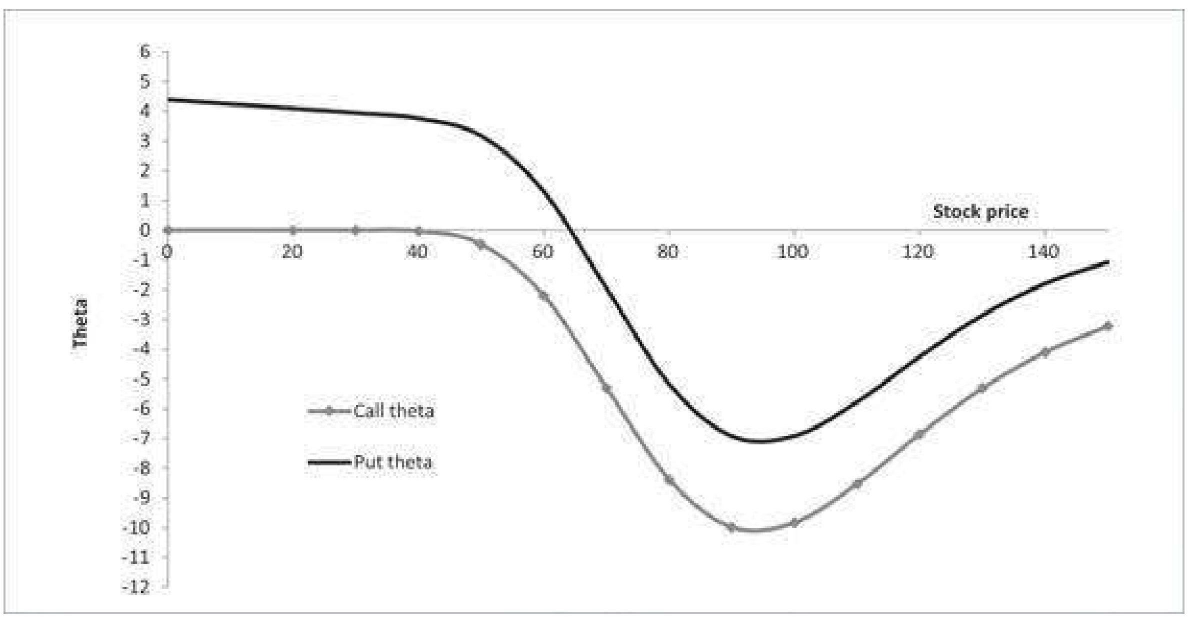


Figure 19.2
Call and put thetas as functions of stock price (option exercise price = 90). Very in-the- Money puts can have a positive theta, meaning that as the time to maturity gets shorter, the put gains in value. Other than this case, options generally have a negative theta, meaning that they lose value as the time to maturity decreases.

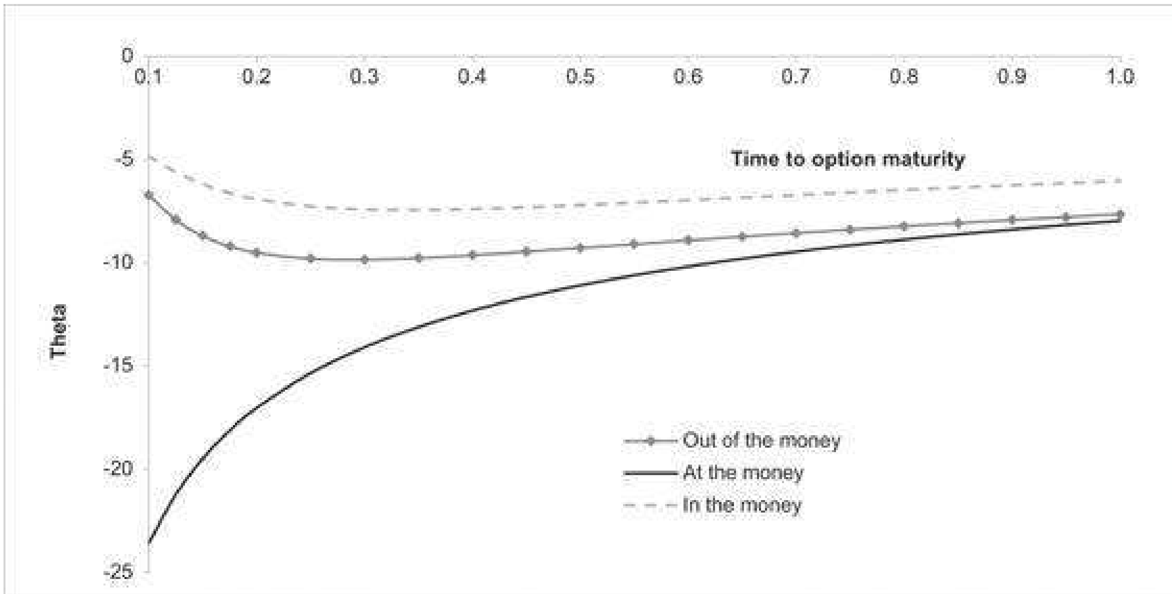


Figure 19.3
Call theta, time to maturity, and moneyness. Calls always have negative theta (meaning that they lose value as the time to maturity decreases). However, the rate at which they lose value varies with the moneyness of the call.

19.3 Delta Hedging a Call[2]

Delta hedging is a fundamental technique in option pricing. The idea is to replicate an option by a portfolio of stocks and bonds, with the portfolio proportions determined by the Black-Scholes formula.

Suppose we decide to replicate an at-the-money European call option that has eight weeks to run until expiration. The stock on which the option is written has S_0 = \$40 and exercise price X = \$35, the interest rate is r = 2%, and the stock's volatility is σ = 35%. The Black-Scholes price of this option is 5.54.

	A	B	C	D	E	F	G	H
1	**DELTA HEDGING A CALL**							
2	S, current stock price	40.00						
3	X, exercise	35.00						
4	r, interest rate	2.00%						
5	k, dividend yield	0.00%						
6	T, expiration	0.1538	<--=8/52		=(D12-D11)*C12			
7	Sigma	35%						
8			=deltacall(C12,$B$3,B12,$B$4,$B$5,$B$7)				=F11+G11+E12	
9								
10	**Weeks until expiration**	**Time until expiration**	**Stock price**	**Delta**	**Stock purchase**	**Outstanding loan (purchase stock)**	**Weekly interest**	
11	8	8/52	40.000	0.856	34.25	34.25	0.01	
12	7	7/52	38.152	0.775	-3.09	31.18	0.01	<--=F12*$B$4/52
13	6	6/52	37.932	0.775	-0.01	31.18	0.01	
14	5	5/52	38.663	0.839	2.46	33.65	0.01	
15	4	4/52	39.658	0.912	2.90	36.56	0.01	
16	3	3/52	40.499	0.963	2.09	38.67	0.01	
17	2	2/52	39.155	0.954	-0.39	38.29	0.01	
18	1	1/52	41.776	1.000	1.94	40.25	0.02	
19	0	0	39.053	1.000	0.00	40.27	0.02	
20								
21	**Value at maturity**							
22	Stock holdings	39.05	<--=C19*D19					
23	Loan repaid	-40.28	<--=-SUM(F19:G19)					
24	**Total portfolio value**	**-1.23**	**<--=SUM(B22:B23)**					
25	Short call payoff	-4.053	<--=-MAX(0,C19-B3)					
26	**Portfolio minus call**	**-5.28**	**<--=SUM(B24:B25)**					
27								
28	PV(portfolio minus call)	-5.27	<--=B26/(1+B4)^(B6)					
29	B&S call value	5.54	<--=bsmertoncall(B2,B3,B6,B4,B5,B7)					

Note that we use the formula **BSMertoncall** but with the dividend yield k = 0%, so that this is, in effect, a regular Black-Scholes call option.

In the spreadsheet above, we create this option by replicating, on a week-to-week basis, the Black-Scholes option pricing formula using delta hedging.

- • At the beginning, eight weeks before the option's expiration, we short one option and we purchase $N(d_1)$ stocks so that we have a dollar amount $S \cdot N(d_1)$ of shares in the portfolio. We finance this purchase with a loan carrying an annual interest rate of 2%. We now determine our portfolio holdings for each of the successive weeks.
- • In each successive week we set the stock holdings in the portfolio according to the formula $S \cdot N(d_1)$, but we set the portfolio borrowing so that the *net cash flow of the portfolio* is zero. Note that $S \cdot N(d_1) = S\Delta_{Call}$, hence the name "delta hedging."
- • At the end of the 8-week period, we liquidate the portfolio.

The delta hedge would be perfect if we rebalanced our portfolio continuously. However, here we have rebalanced only weekly. Below is the same analysis for the case where $S_T < X$:

	A	B	C	D	E	F	G	H
1			DELTA HEDGING A CALL					
2	S, current stock price	40.00						
3	X, exercise	35.00						
4	r, interest rate	2.00%						
5	k, dividend yield	0.00%						
6	T, expiration	0.1538	<--=8/52		=(D12-D11)*C12			
7	Sigma	35%						
8			=deltacall(C12,$B$3,B12,$B$4,$B$5,$B$7)				=F11+G11+E12	
9								
10	Weeks until expiration	Time until expiration	Stock price	Delta	Stock purchase	Outstanding loan (purchase stock)	Weekly interest	
11	8	8/52	40.000	0.856	34.25	34.25	0.01	
12	7	7/52	38.233	0.780	-2.91	31.36	0.01	<--=F12*$B$4/52
13	6	6/52	38.407	0.805	0.96	32.33	0.01	
14	5	5/52	37.189	0.736	-2.57	29.76	0.01	
15	4	4/52	34.846	0.508	-7.96	21.82	0.01	
16	3	3/52	31.339	0.104	-12.64	9.18	0.00	
17	2	2/52	31.431	0.064	-1.26	7.93	0.00	
18	1	1/52	28.851	0.000	-1.85	6.08	0.00	
19	0	0	28.458	0.000	0.00	6.08	0.00	
20								
21	**Value at maturity**							
22	Stock holdings	0.00	<--=C19*D19					
23	Loan repaid	-6.09	<--=-SUM(F19:G19)					
24	**Total portfolio value**	**-6.09**	**<--=SUM(B22:B23)**					
25	Short call payoff	0.000	<--=-MAX(0,C19-B3)					
26	**Portfolio minus call**	**-6.09**	**<--=SUM(B24:B25)**					
27								
28	PV(portfolio minus call)	-6.07	<--=B26/(1+B4)^(B6)					
29	B&S call value	5.54	<--=bsmertoncall(B2,B3,B6,B4,B5,B7)					

19.4 The Greeks of a Portfolio

Since $f'(ax + by) = af'(x) + bf'(y)$, it is easy to show that the Greek of a portfolio is simply a summation of the Greeks of the individual assets in the portfolio. That is:

$$Greek(w_1 Option_1 + w_2 Option_2 + \cdots + w_n Option_n)$$
$$= w_1 Greek_{Option_1} + w_2 Greek_{Option_2} + \cdots + w_n Greek_{Option_n}$$

In the spreadsheet below we show a portfolio with three options—two calls and a put. We use delta, vega, and gamma to show that the Greek of a portfolio is the sum of the Greek for each asset in the portfolio. For each Greek, we use two methods: the direct approach and a brute-force approach.

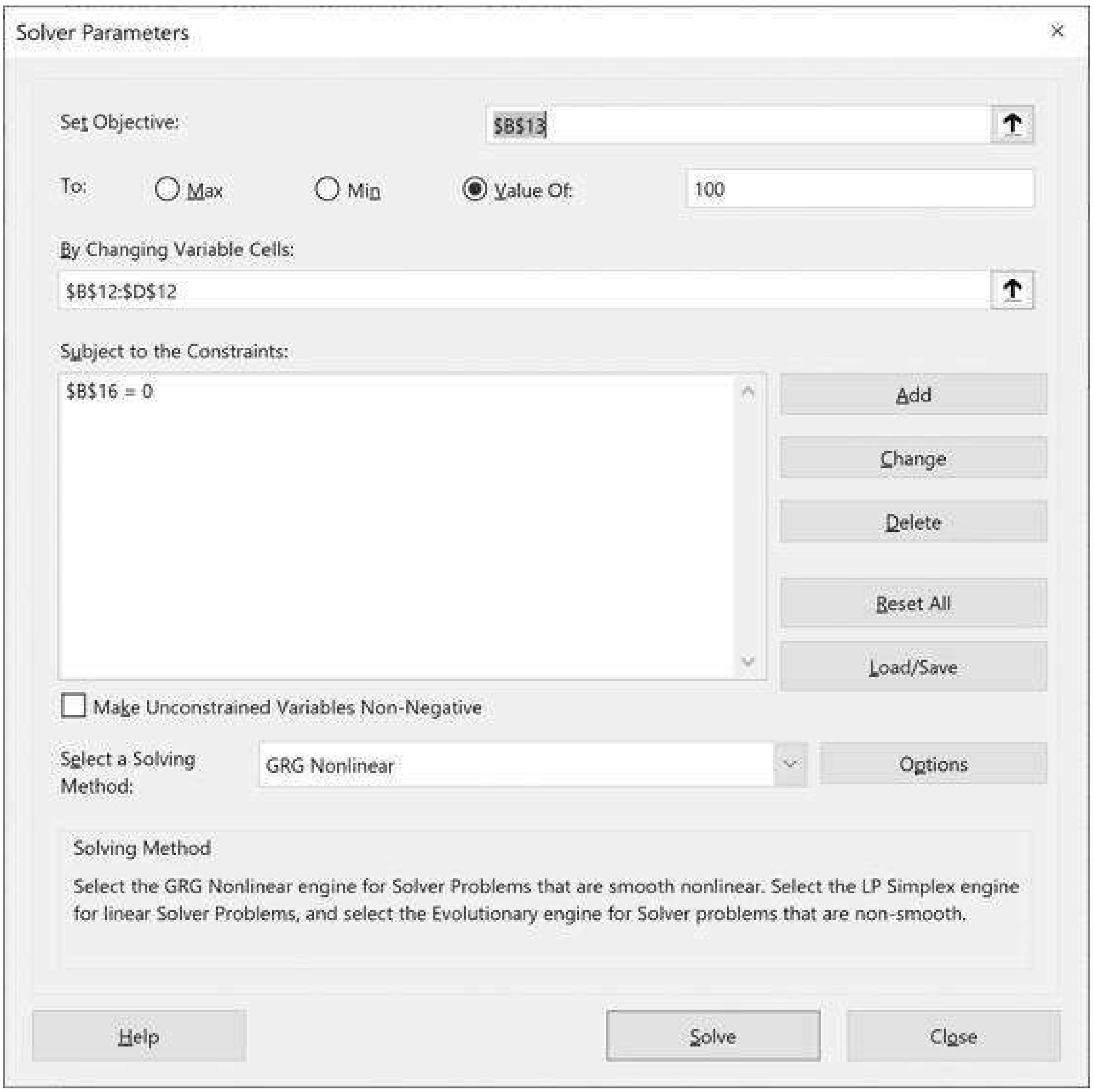
	A	B	C	D	E
1	**PORTFOLIO GREEKS** **PORTFOLIO GREEK = SUM OF INDIVIDUAL ASSET GREEK**				
2	**Option**	1	2	3	
3	Type	Call	Put	Call	
4	Stock price (S)	40.00	40.00	40.00	
5	Exercise (X)	36.00	37.00	42.00	
6	Interest rate (r)	2.00%	2.00%	2.00%	
7	Dividend yield (k)	0.00%	0.00%	0.00%	
8	Expiration (T)	0.25	0.50	0.75	
9	Sigma	35%	35%	35%	=bsmertonput(C4,C5,C8,C6,C7,C9)
10	Quantity	3	5	-1	
11	Value	5.97	2.32	4.24	<--=bsmertoncall(D4,D5,D8,D6,D7,D9)
12	**Portfolio value**	**25.28**	**<--=SUMPRODUCT(B10:D10,B11:D11)**		
13					
14	Delta	0.81	-0.316	0.516	<--=deltacall(D4,D5,D8,D6,D7,D9)
15	Portfolio Delta	0.34	<--=SUMPRODUCT(B10:D10,B14:D14)		
16	Value (S+0.01)	5.98	2.32	4.25	<--=bsmertoncall(D4+0.01,D5,D8,D6,D7,D9)
17	Portfolio value	25.29	<--=SUMPRODUCT(B16:D16,B10:D10)		
18	Portfolio Delta	0.34	<--=(B17-B12)/0.01		
19					
20	Vega	5.42	10.06	13.81	<--=vega(D4,D5,D8,D6,D7,D9)
21	Portfolio Vega	52.76	<--=SUMPRODUCT(B20:D20,B10:D10)		
22	Value (σ+0.01)	6.02	2.42	4.38	<--=bsmertoncall(D4,D5,D8,D6,D7,D9+0.01)
23	Portfolio value	25.81	<--=SUMPRODUCT(B22:D22,B10:D10)		
24	Portfolio Vega	52.98	<--=(B23-B12)/0.01		
25					
26	Gamma	0.039	0.036	0.033	<--=optiongamma(D4,D5,D8,D6,D7,D9)
27	Portfolio Gamma	0.263	<--=SUMPRODUCT(B26:D26,B10:D10)		
28	Delta (S+0.01)	0.811	-0.316	0.516	<--=deltacall(D4+0.01,D5,D8,D6,D7,D9)
29	Portfolio Delta	0.338	<--=SUMPRODUCT(B28:D28,B10:D10)		
30	Portfolio Gamma	0.263	<--=(B29-B15)/0.01		

19.5 Greek-Neutral Portfolio

In many cases we are interested in a *Greek-neutral portfolio*. A delta-neutral portfolio is a portfolio that is immuned to small movements in the underlying asset price, a vega-neutral portfolio is a portfolio that is immuned to small movements in the underlying asset sigma, and so on. Note that the Greeks themselves change when the underlying asset price is changing. That is why we have the second-order and third-order Greeks. Let us use the three-options example presented above and create a delta-neutral portfolio. Also assume that we are interested in a $100 portfolio. One of the possible portfolios will be as follows:

	A	B	C	D	E
1	**GREEK-NEUTRAL PORTFOLIO**				
2	**Option**	1	2	3	
3	Type	Call	Put	Call	
4	Stock price (S)	40.00	40.00	40.00	
5	Exercise (X)	35.00	37.00	42.00	
10	Value	5.97	2.32	4.24	<--=bsmertoncall(D4,D5,D8,D6,D7,D9)
11					
12	Quantity	4.71	20.98	5.45	
13	**Portfolio value**	**100**	**<--=SUMPRODUCT(B12:D12,B10:D10)**		
14					
15	Delta	0.81	-0.316	0.516	<--=deltacall(D4,D5,D8,D6,D7,D9)
16	Portfolio Delta	0.00	<--=SUMPRODUCT(B12:D12,B15:D15)		
17					
18	Gamma	0.039	0.036	0.033	<--=optiongamma(D4,D5,D8,D6,D7,D9)
19	Portfolio Gamma	1.115	<--=SUMPRODUCT(B18:D18,B12:D12)		

This portfolio can easily be found by using **Solver**. This is the Solver dialog box used:



Note that in this example we used three options and two constraints (portfolio value and portfolio delta), this means that there is one degree of freedom and there are infinitely many other portfolios that will satisfy our conditions. One of the possible other conditions can be X holdings in one of the options.

In figure 19.4 we show the portfolio value as a function of the underlying asset price:

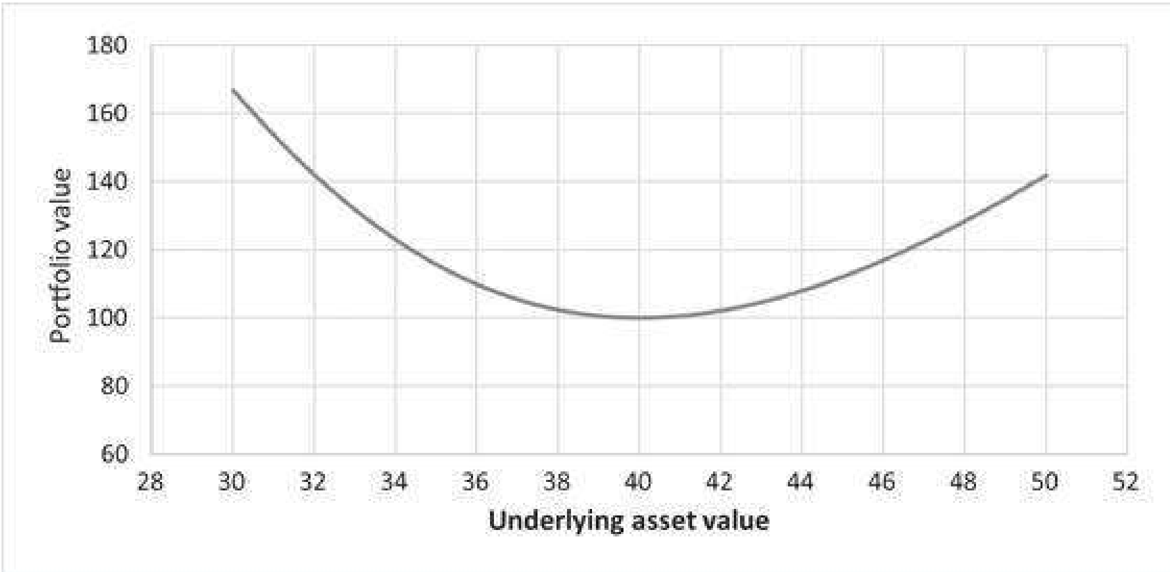


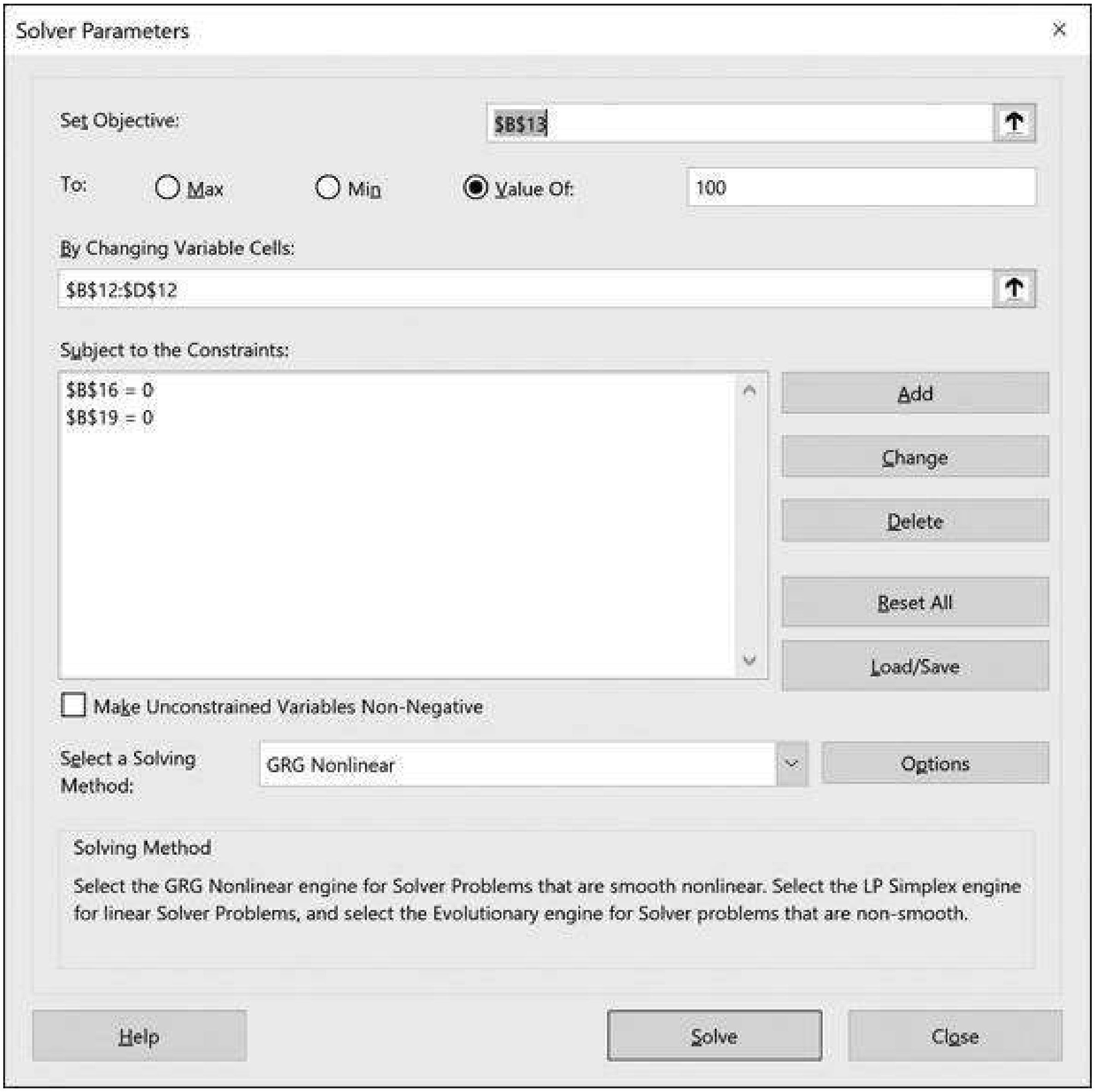
Figure 19.4

A delta-neutral portfolio is not sensitive to small movements of the underlying asset, but since the delta is changing when underlying asset is changing, it is not protected from large underlying movements.

A portfolio with both delta and gamma equal zero (a delta- and gamma-neutral portfolio) provides better protection to underlying asset movement. We can use the extra degree of freedom to achieve this goal. This is the portfolio that meets our three criteria (portfolio value = 100, delta = 0, and gamma = 0):

	A	B	C	D	E
1	**DELTA- AND GAMMA-NEUTRAL PORTFOLIO**				
2	**Option**	**1**	**2**	**3**	
3	Type	Call	Put	Call	
4	Stock price (S)	40.00	40.00	40.00	
5	Exercise (X)	35.00	37.00	42.00	
10	Value	5.97	2.32	4.24	<--=bsmertoncall(D4,D5,D8,D6,D7,D9)
11					
12	Quantity	228.41	52.59	-326.50	
13	**Portfolio value**	**100**	**<--=SUMPRODUCT(B12:D12,B10:D10)**		
14					
15	Delta	0.81	-0.316	0.516	<--=deltacall(D4,D5,D8,D6,D7,D9)
16	Portfolio Delta	0.00	<--=SUMPRODUCT(B12:D12,B15:D15)		
17					
18	Gamma	0.039	0.036	0.0329	<--=optiongamma(D4,D5,D8,D6,D7,D9)
19	Portfolio Gamma	0.00	<--=SUMPRODUCT(B18:D18,B12:D12)		

This portfolio is achieved by using the following **Solver** parameters:



As can be seen in figure 19.5, this portfolio is better immuned to underlying asset movements:

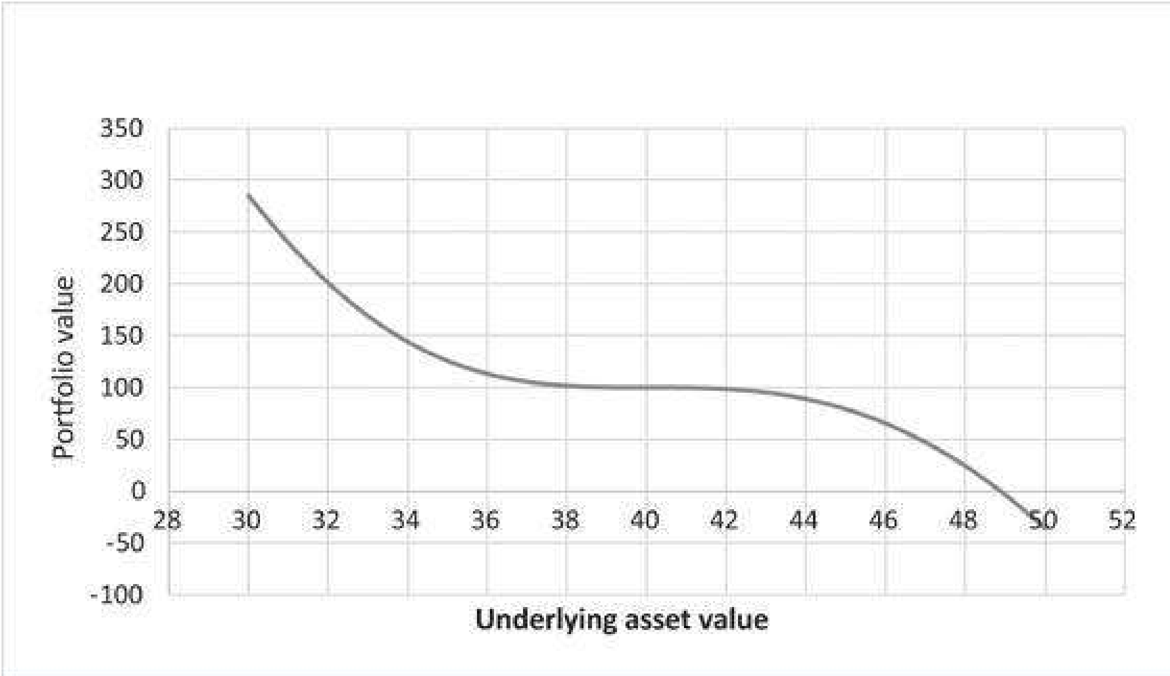


Figure 19.5

A delta-and gamma-neutral portfolio better protected from movements of the underlying asset (compared to only delta-neutral portfolio). As before, since the gamma is changing when the underlying asset is changing, it is not protected from very large underlying movements.

19.6 The Relationship between Delta, Theta, and Gamma

As presented above, one straightforward approach to create a delta-neutral portfolio is to short one option and hold delta stocks. For small underlying asset shocks, this portfolio is immuned and so, over a short period of time (Δt), we expect this portfolio to earn the risk-free rate of return. This intuition, with slightly advanced math, yields the Black-Scholes Merton differential equation:

$$\Theta + (r - d) \cdot S \cdot \Delta + 0.5 \cdot \sigma^2 S^2 \Gamma = r \cdot \text{Premium}$$

$$\Theta + (r - d) \cdot S \cdot \Delta + 0.5 \cdot \sigma^2 S^2 \Gamma = r \cdot \text{Premium}$$

The following Excel sheet shows this relationship for an arbitrary call and put:

	A	B	C	D
1	DELTA, THETA, AND GAMMA RELATIONSHIP			
2	Option	1	2	
3	Type	Call	Put	
4	Stock price (S)	40.00	40.00	
5	Exercise (X)	35.00	37.00	
6	Interest rate (r)	2.00%	2.00%	
7	Dividend yield (k)	0.00%	0.00%	
8	Expiration (T)	0.25	0.50	
9	Sigma	35%	35%	
10	Value	5.97	2.32	<--=bsmertonput(C4,C5,C8,C6,C7,C9)
11				
12	Delta	0.81	-0.316	<--=deltaput(C4,C5,C8,C6,C7,C9)
13	Theta	-4.324	-3.222	<--=Thetaput(C4,C5,C8,C6,C7,C9)
14	Gamma	0.039	0.036	<--=optiongamma(C4,C5,C8,C6,C7,C9)
15				
16	$\Theta+(r\text{-}d)\cdot S\cdot\Delta+0.5\cdot\sigma^2\cdot S^2\cdot\Gamma$	0.119	0.046	<--=C13+(C6-C7)*C4*C12+0.5*C9^2*C4^2*C14
17	r-premium	0.119	0.046	<--=C6*C10

19.7 Summary

In this chapter we have explored the sensitivities of the option pricing formula to its various parameters. Using these Greeks, we have delved into the intricacies of delta hedging, a useful technique for replicating an option position with a combination of stocks and bonds. The interested reader should know that there is much more that can be said about this topic. Good starting places for further reading are Hull (2017) and Taleb (1997). An extensive collection of option pricing formulas including Greeks can be found in Haug (2006).

Exercises

1. Produce a graph similar the second panel of figure 19.1 for puts.
2. Figure 19.2 shows the call theta as a function of time to maturity. Produce a similar graph for puts.
3. Although θ is generally negative, there are cases (typically of high interest rates) where it can be positive, including:

- An in-the-money put with a high interest rate
- An in-the-money call on a currency that has a high interest rate (or—equivalently—an in-the-money call on a stock with a very high dividend payout rate)

Find two examples.

Appendix 19.1: VBA for Greeks

Black-Scholes Functions

Throughout the chapter we use the Merton version of the Black-Scholes formula for pricing options with a continuous dividend yield (see chapter 18, section 18.6 for details). The VBA connected with this model is given below.

```
Function dOne(stock, exercise, time, _
interest, divyield, sigma)
    dOne = (Log(stock / exercise) + _
    (interest - divyield) * time) / _
    (sigma * Sqr(time)) + 0.5 * sigma * _
    Sqr(time)
End Function

Function dTwo(stock, exercise, time, _
interest, divyield, sigma)
    dTwo = dOne(stock, exercise, time, _
    interest, divyield, sigma) - sigma * _
    Sqr(time)
End Function

Function BSMertonCall(stock, exercise, time, _
interest, divyield, sigma)
```

```
        BSMertonCall = stock * Exp(-divyield * _
        time) * Application.Norm_S_Dist _
        (dOne(stock, exercise, time, _
        interest, divyield, sigma), 1) - exercise * _
        Exp(-time * interest) * Application.Norm_S_Dist _
        (dTwo(stock, exercise, time, interest, _
        divyield, sigma), 1)
    End Function
    Function BSMertonPut(stock, exercise, time, _
    'Put pricing function uses put-call
    'parity theorem
    interest, divyield, sigma)
        BSMertonPut = BSMertonCall(stock, exercise, _
        time, interest, divyield, sigma) + _
        exercise * Exp(-interest * time) - _
        stock * Exp(-divyield * time)
    End Function
```

Defining the Normal Distribution

The option pricing functions above use the old version of the Excel function **Norm.S.Dist(x,False/True)**. If the second parameter in this function is set to **False**, it computes the normal density; with the parameter set to **True**, it computes the normal distribution function.[3]

The VBA writing of these functions is **Application.Norm_S_Dist(x,0 or 1)**. Without regard for consistency, we have used both versions of this function in our VBA for Greeks.

Sometimes it is convenient to use a homemade function for the normal probability density defined below:

```
    Function normaldf(x)
    'The standard normal probability density,
    'this is N'(x) and it is equivalent to:
    'nromaldf = Application.Norm_S_Dist(_
    dOne(stock, exercise, time, interest, divyield, sigma), 0)
        normaldf = Exp(-x ^ 2 / 2) / _
        (Sqr(2 * Application.Pi()))
```

Defining the Greeks

Below we give the Greeks in VBA:

```
Function DeltaCall(stock, exercise, time, _
interest, divyield, sigma)
    DeltaCall = Exp(-divyield * time) * _
    Application.Norm_S_Dist(dOne(stock, exercise, _
    time, interest, divyield, sigma), 1)
End Function

Function DeltaPut(stock, exercise, time, _
interest, divyield, sigma)
    DeltaPut = -Exp(-divyield * time) * _
    Application.Norm_S_Dist(-dOne(stock, _
    exercise, time, interest, divyield, _
    sigma), 1)
End Function

Function OptionGamma(stock, exercise, time, _
interest, divyield, sigma)
    OptionGamma = Exp(-divyield * time) * _
    Application.Norm_S_Dist(dOne(stock, exercise, _
    time, interest, divyield, sigma), 0) / _
    (stock * sigma * Sqr(time))
End Function

Function Vega(stock, exercise, time, _
interest, divyield, sigma)
    Vega = stock * Sqr(time) * _
    Application.Norm_S_Dist(dOne(stock, exercise, _
    time, interest, divyield, sigma), 0) _
    * Exp(-divyield * time)
End Function

Function ThetaCall(stock, exercise, time, _
interest, divyield, sigma)
    ThetaCall = -stock * normaldf _
    (dOne(stock, exercise, time, _
    interest, divyield, sigma)) * _
    sigma * Exp(-divyield * time) / _
    (2 * Sqr(time)) + divyield * stock * _
    Application.Norm_S_Dist(dOne(stock, _
    exercise, time, interest, _
    divyield, sigma), 1) * Exp(-divyield * time) _
    - interest * exercise * Exp(-interest * _
    time) * Application.Norm_S_Dist _
    (dTwo(stock, exercise, time, _
    interest, divyield, sigma), 1)
End Function

Function ThetaPut(stock, exercise, time, _
interest, divyield, sigma)
    ThetaPut = -stock * normaldf _
    (dOne(stock, exercise, _
    time, interest, divyield, sigma)) * _
    sigma * Exp(-divyield * time) / _
```

```
    (2 * Sqr(time)) - divyield * stock _
    * Application.Norm_S_Dist(-dOne(stock, _
    exercise, time, interest, divyield, _
    sigma), 1) * Exp(-divyield * time) _
    + interest * exercise * Exp _
    (-interest * time) * Application.Norm_S_Dist _
    (-dTwo(stock, exercise, time, _
    interest, divyield, sigma), 1)
End Function

Function RhoCall(stock, exercise, time, _
interest, divyield, sigma)
    RhoCall = exercise * time * _
    Exp(-interest * time) * _
    Application.Norm_S_Dist(dTwo _
    (stock, exercise, time, interest, _
    divyield, sigma), 1)
End Function

Function RhoPut(stock, exercise, time, _
interest, divyield, sigma)
    RhoPut = -exercise * time * _
    Exp(-interest * time) * _
    Application.Norm_S_Dist(-dTwo _
    (stock, exercise, time, interest, _
    divyield, sigma), 1)
End Function

Function Vanna(stock, exercise, time, _
interest, divyield, sigma)
    Vanna = stock * Sqr(time) * _
    Application.Norm_S_Dist(dOne(stock, exercise, _
    time, interest, divyield, sigma), 0) * _
    Exp(-divyield * time) / stock * (1 - dOne _
    (stock, exercise, time, interest, divyield, _
    sigma) / (sigma * Sqr(time)))
End Function

Function CharmCall(stock, exercise, time, interest, _
divyield, sigma)
    CharmCall = divyield * Exp(-divyield * time) * _
    Application.Norm_S_Dist(dOne(stock, exercise, _
    time, interest, divyield, sigma), 1) - _
    Exp(-divyield * time) * _
    Application.Norm_S_Dist(dOne(stock, exercise, _
    time, interest, divyield, sigma), 0) _
    * (2 * (interest - divyield) * time - dTwo _
    (stock, exercise, time, interest, divyield, _
    sigma) * sigma * Sqr(time)) / (2 * time _
    * sigma * Sqr(time))
End Function

Function CharmPut(stock, exercise, time, interest, _
```

```
divyield, sigma)
    CharmPut = -divyield * Exp(-divyield * time) _
    * Application.Norm_S_Dist(-dOne(stock, _
    exercise, time, interest, divyield, sigma), 1) _
    - Exp(-divyield * time) * Application.Norm_S_ Dist _
    (dOne(stock, exercise, time, interest, divyield, _
    sigma), 0) * (2 * (interest - divyield) * _
    time - dTwo(stock, exercise, time, interest, _
    divyield, sigma) * sigma * Sqr(time)) _
    / (2 * time * sigma * Sqr(time))
End Function

Function Vomma(stock, exercise, time, _
interest, divyield, sigma)
    Vomma = stock * Sqr(time) * _
    Application.Norm_S_Dist(dOne(stock, exercise, _
    time, interest, divyield, sigma), 0) * _
    Exp(-divyield * time) * dOne(stock, exercise, _
    time, interest, divyield, sigma) * dTwo(stock, _
    exercise, time, interest, divyield, sigma) _
    / sigma
End Function

Function Veta(stock, exercise, time, _
interest, divyield, sigma)
    Veta = -stock * Exp(-divyield * time) _
    * Application.Norm_S_Dist(dOne(stock, exercise, _
    time, interest, divyield, sigma), 0) * Sqr(time) _
    * (divyield + (interest - divyield) * _
    dOne(stock, exercise, time, interest, _
    divyield, sigma) / (sigma * Sqr(time)) _
    - (1 + dOne(stock, exercise, time, interest, _
    divyield, sigma) * dTwo(stock, exercise, _
    time, interest, divyield, sigma)) / (2 * time))
End Function

Function Speed(stock, exercise, time, _
interest, divyield, sigma)
    Speed = -Exp(-divyield * time) * _
    Application.Norm_S_Dist(dOne(stock, exercise, _
    time, interest, divyield, sigma), 0) / _
    (stock * sigma * Sqr(time)) / stock * _
    (dOne(stock, exercise, time, interest, _
    divyield, sigma) / (sigma * Sqr(time)) + 1)
End Function

Function Zomma(stock, exercise, time, _
interest, divyield, sigma)
    Zomma = Exp(-divyield * time) * _
    Application.Norm_S_Dist(dOne(stock, exercise, _
    time, interest, divyield, sigma), 0) / _
    (stock * sigma * Sqr(time)) * (dOne(stock, _
```

```
    exercise, time, interest, divyield, sigma) _
    * dTwo(stock, exercise, time, interest, _
    divyield, sigma) - 1) / sigma
End Function

Function Color(stock, exercise, time, _
interest, divyield, sigma)
    Color = -Exp(-divyield * time) * _
    Application.Norm_S_Dist(dOne(stock, exercise, _
    time, interest, divyield, sigma), 0) / _
    (stock * sigma * Sqr(time)) / (2 * time) * _
    (1 + 2 * divyield * time + 2 * dOne(stock, _
    exercise, time, interest, divyield, sigma) _
    * (interest - divyield) * time / (sigma * _
    Sqr(time)) - dOne(stock, exercise, time, _
    interest, divyield, sigma) * _
    dTwo(stock, exercise, _
    time, interest, divyield, sigma))
End Function
```

Appendix 19.2: R Code for Greeks

Calculating the Greeks in R looks very similar to the VBA. Here is the R code:

```
# we thank Sagi Haim for developing this script

##############################
# CHAPTER 19 - THE GREEKS
##############################

S <- 100
X <- 90
r <- 0.05
sigma <- 0.35
t <- 0.5
k <- 0.015

# d1
fm5_d1 <- function(S, X, r, sigma, t, k = 0){
    return( (log(S/X) + (r - k + 0.5 * sigma^2) * t) / (sigma * sqrt(t)) )
    }
fm5_d1(S, X, r, sigma, t, k)

# d2
fm5_d2 <- function(S, X, r, sigma, t, k = 0){
    return ( (log(S/X) + (r - k + 0.5 * sigma^2) * t) /
               (sigma * sqrt(t)) - sigma * sqrt(t) )
    }
fm5_d2(S, X, r, sigma, t, k)

# N(d1)
fm5_N_d1 <- function(S, X, r, sigma, t, k = 0){
    d1 <- fm5_d1(S, X, r, sigma, t, k)
    return(pnorm(d1, mean = 0, sd = 1))
    }
fm5_N_d1(S, X, r, sigma, t, k)
```

```r
# N(d2)
fm5_N_d2 <- function(S, X, r, sigma, t, k = 0){
    d2 <- fm5_d2(S, X, r, sigma, t, k)
    return(pnorm(d2, mean = 0, sd = 1))
    }
fm5_N_d2(S, X, r, sigma, t, k)

# N'(d1) and N'(d2)
fm5_N_tag_d1 <- function(S, X, r, sigma, t, k = 0){
    d1 <- fm5_d1(S, X, r, sigma, t, k)
    return(dnorm(d1, mean = 0, sd = 1))
    }
fm5_N_tag_d1(S, X, r, sigma, t, k)

fm5_N_tag_d2 <- function(S, X, r, sigma, t, k = 0){
    d2 <- fm5_d2(S, X, r, sigma, t, k)
    return(dnorm(d2, mean = 0, sd = 1))
    }
fm5_N_tag_d2(S, X, r, sigma, t, k)

## First order Greeks
# Delta Call
fm5_delta_call <- function(S, X, r, sigma, t, k = 0){
    N_d1 <- fm5_N_d1(S, X, r, sigma, t, k)
    return( exp(-k*t) * N_d1)
    }
fm5_delta_call(S, X, r, sigma, t, k)

# Delta Put
fm5_delta_put <- function(S, X, r, sigma, t, k = 0){
    N_d1 <- fm5_N_d1(S, X, r, sigma, t, k)
    return( -exp(-k*t) *(1 - N_d1))
    }
fm5_delta_put(S, X, r, sigma, t, k)

# Vega (same for put and call)
fm5_vega <- function(S, X, r, sigma, t, k = 0){
    N_tag_d1 <- fm5_N_tag_d1(S, X, r, sigma, t, k)
    return( S * exp(- k * t) * sqrt(t) * N_tag_d1)
    }
fm5_vega(S, X, r, sigma, t, k)

```

```r
# Theta call
fm5_theta_call <- function(S, X, r, sigma, t, k = 0){
    N_d1 = fm5_N_d1(S, X, r, sigma, t, k)
    N_d2 = fm5_N_d2(S, X, r, sigma, t, k)
    N_tag_d1 =  fm5_N_tag_d1(S, X, r, sigma, t, k)
    return( -(S * exp(-k * t) * N_tag_d1 * sigma) / (2 * sqrt(t)) +
              k * S * exp(-k * t) * N_d1 - r * X * exp(-r * t) * N_d2 )
    }
fm5_theta_call(S, X, r, sigma, t, k)

# Theta Put
fm5_theta_put <- function(S, X, r, sigma, t, k = 0){
    d1 = fm5_d1(S, X, r, sigma, t, k)
    d2 = fm5_d2(S, X, r, sigma, t, k)
    N_tag_d1 =  fm5_N_tag_d1(S, X, r, sigma, t, k)
    N_minus_d1 = pnorm(-d1, mean = 0, sd = 1)
    N_minus_d2 = pnorm(-d2, mean = 0, sd = 1)
    return(-(S * exp(-k * t) * N_tag_d1 * sigma) / (2 * sqrt(t)) -
             k * S * exp(-k * t) * N_minus_d1 +
             r * X * exp(-r * t) * N_minus_d2)
    }
fm5_theta_put(S, X, r, sigma, t, k)

# Rho call
fm5_rho_call <- function(S, X, r, sigma, t, k = 0){
    N_d2 = fm5_N_d2(S, X, r, sigma, t, k)
    return( X * t * exp(-r * t) * N_d2 )
    }
fm5_rho_call(S, X, r, sigma, t, k)

# Rho Put
fm5_rho_put <- function(S, X, r, sigma, t, k = 0){
    d2 = fm5_d2(S, X, r, sigma, t, k)
    N_minus_d2 = pnorm(-d2, mean = 0, sd = 1)
    return(-X * t * exp(-r * t) * N_minus_d2)
    }
fm5_rho_put(S, X, r, sigma, t, k)

## Second order Greeks
# Gamma
fm5_opt_gamma <- function(S, X, r, sigma, t, k = 0){
    N_tag_d1 = fm5_N_tag_d1(S, X, r, sigma, t, k)
    return( (exp(-k * t) * N_tag_d1) /
              (S * sigma * sqrt(t) ))
    }
fm5_opt_gamma(S, X, r, sigma, t, k)

# Vanna
fm5_vanna <- function(S, X, r, sigma, t, k = 0){
    d2 = fm5_d2(S, X, r, sigma, t, k)
    N_tag_d1 =  fm5_N_tag_d1(S, X, r, sigma, t, k)
    return( -exp(-k * t) * N_tag_d1 * d2 / sigma)
    }
fm5_vanna(S, X, r, sigma, t, k)
```

```r
# Charm
fm5_charm_call <- function(S, X, r, sigma, t, k = 0){
    N_d1 = fm5_N_d1(S, X, r, sigma, t, k)
    d2 = fm5_d2(S, X, r, sigma, t, k)
    N_tag_d1 =  fm5_N_tag_d1(S, X, r, sigma, t, k)
    return( k * exp(-k * t) * N_d1 - exp(-k * t) * N_tag_d1 *
              (2 * (r - k) * t - d2 * sigma * sqrt(t)) / (2 * t * sigma * sqrt(t)))
    }
fm5_charm_call(S, X, r, sigma, t, k)

fm5_charm_put <- function(S, X, r, sigma, t, k = 0){
    d1 =  fm5_d1(S, X, r, sigma, t, k)
    N_minus_d1 = pnorm(-d1, mean = 0, sd = 1)
    N_tag_d1 =  fm5_N_tag_d1(S, X, r, sigma, t, k)
    d2 = fm5_d2(S, X, r, sigma, t, k)
    return( -k * exp(-k * t) * N_minus_d1 - exp(-k * t) * N_tag_d1 *
                 (2 * (r - k) * t - d2 * sigma * sqrt(t)) / (2 * t * sigma * sqrt(t)))
    }
fm5_charm_put(S, X, r, sigma, t, k)

# Vomma
fm5_vomma <- function(S, X, r, sigma, t, k = 0){
    d1 = fm5_d1(S, X, r, sigma, t, k)
    N_tag_d1 =  fm5_N_tag_d1(S, X, r, sigma, t, k)
    d2 = fm5_d2(S, X, r, sigma, t, k)
    return( S * exp(-k * t) * N_tag_d1 * d1 * d2 / sigma * sqrt(t))
    }
fm5_vomma(S, X, r, sigma, t, k)

# Veta
fm5_veta <- function(S, X, r, sigma, t, k = 0){
    d1 = fm5_d1(S, X, r, sigma, t, k)
    N_tag_d1 =  fm5_N_tag_d1(S, X, r, sigma, t, k)
    d2 = fm5_d2(S, X, r, sigma, t, k)
    return( -S * exp(-k * t) * N_tag_d1 * sqrt(t) *
                 (k + ((r - k) * d1) / (sigma * sqrt(t)) - ((1 + d1 * d2) / (2 * t))))
    }
fm5_veta(S, X, r, sigma, t, k)
## Third Order Greeks
# Speed
fm5_speed <- function(S, X, r, sigma, t, k = 0){
    d1 = fm5_d1(S, X, r, sigma, t, k)
    N_tag_d1 = fm5_N_tag_d1(S, X, r, sigma, t, k)
    return( -exp(-k * t) * N_tag_d1 / (S^2 * sigma * sqrt(t)) *
                 (d1 / (sigma * sqrt(t)) + 1))
    }
fm5_speed(S, X, r, sigma, t, k)

# Zomma
fm5_zomma <- function(S, X, r, sigma, t, k = 0){
    d1 = fm5_d1(S, X, r, sigma, t, k)
    N_tag_d1 = fm5_N_tag_d1(S, X, r, sigma, t, k)
    d2 = fm5_d2(S, X, r, sigma, t, k)
    return(exp(-k * t) * (N_tag_d1 * (d1 * d2 - 1)) / (S * sigma^2 * sqrt(t)))
    }
fm5_zomma(S, X, r, sigma, t, k)
```

```r

# Color
fm5_color <- function(S, X, r, sigma, t, k = 0){
    d1 = fm5_d1(S, X, r, sigma, t, k)
    N_tag_d1 = fm5_N_tag_d1(S, X, r, sigma, t, k)
    d2 = fm5_d2(S, X, r, sigma, t, k)
    return( -exp(-k * t) * N_tag_d1 / (2 * t * S * sigma * sqrt(t)) *
                ( 1 + (2 * k * t) + (2 * d1 * (r - k) * t) /
                        (sigma * sqrt(t)) - (d1 * d2)))
    }
fm5_color(S, X, r, sigma, t, k)
```

---

1. 1. Remember that $f'(x) = \lim_{h \to 0} \dfrac{f(x+h) - f(x)}{h}$.
2. 2. This topic is discussed again in Chapter 26.

3. [3]. In Excel, instead of **False** or **True** we can also use 0 or 1. When using these functions in VBA, 0 or 1 is mandatory.

20 Real Options

20.1 Overview

The standard net present value (NPV) analysis of capital budgeting values a project by discounting its expected cash flows at a risk-adjusted cost of capital. This *discounted cash flow* (DCF) technique is by far the most widely used practice for evaluating capital projects, be they acquisitions of companies or the purchases of machines. However, standard NPV analysis does not take account of the *flexibility* inherent in the capital budgeting process. Part of the complexity of the capital budgeting process is that the firm can change its decisions dynamically, depending on the circumstances.

Here are two examples:

1. A firm is considering replacing some of its machines with a new type of machine. Instead of replacing all the machines together, it can first replace one machine. Based on the performance of the first machine replaced, the firm can then decide whether to replace the rest of the machines. This "option to wait" (or perhaps the "option to expand") is not valued in the standard NPV process. It is essentially a call option.
2. A firm is considering investing in a project that will produce (uncertain) cash flows over time. One option—not valued in the standard NPV framework—is to *abandon* the project if its performance is not satisfactory. The *abandonment option* is, as we see below, a put option that is implicit in many projects. It is also sometimes called the "option to contract scale."

There are many other real options. In the leading book on the valuation of real options, Trigeorgis (1996) lists the following common real options:

- • The option to defer or to wait when developing a natural resource or building a plant.
- • The time-to-build option (staged investment)—at each stage the investment can be reevaluated and (possibly) abandoned or expanded.
- • The option to alter operating scale (expand, contract, shut down, or restart).
- • The option to abandon.
- • The option to switch inputs or outputs.
- • The growth option—an early investment in a project constitutes an option to "get into the market" at a later date.

The recognition of real options is an important extension of the NPV techniques. However, modeling and valuing real options is more difficult than modeling and valuing standard cash flows by the DCF method. Our examples below illustrate these difficulties. Often it is best to implement real options by recognizing that the DCF technique misjudges the value of a project because it ignores the project's real options. Our usual conclusion will be that real options add to the value of a project and that the NPV thus underestimates the true value.

20.2 A Simple Example of the Option to Expand

In this section we give a simple example of the option to expand. Consider ABC Corp., which has six widget machines. ABC is considering replacing each of the old machines with a new machine that costs $1,000. The new machines have a five-year life. The anticipated cash flows for the new machine are given below.[1]

	A	B	C	D	E	F	G
1	**THE OPTION TO EXPAND**						
2	**Year**	**0**	**1**	**2**	**3**	**4**	**5**
3	CF of single machine	-1,000	220	300	400	200	150
4							
5	Riskless discount rate	6%					
6	Discount rate for machine cash flows (risk-adjusted)	12%					
7	Present value of machine's future cash flows	932.52	<--=NPV(B6,C3:G3)				
8	NPV of single machine	-67.48	<--=NPV(B6,C3:G3)+B3				

The financial analyst working on the replacement project has estimated a cost of capital for the project of 12%. Using these anticipated cash flows and the 12% cost of capital, the analyst has concluded that the replacement of a single old machine by a new machine is unprofitable because the NPV is negative:

$$-1000+\underbrace{\frac{220}{1.12}+\frac{300}{(1.12)^2}+\frac{400}{(1.12)^3}+\frac{200}{(1.12)^4}+\frac{150}{(1.12)^5}}_{\text{The present value of the machine's future cash flow is \$932.52}}=-67.48$$

Now comes the (real options) twist. The line manager in charge of the widget line says, "I want to try one of the new machines for a year. At the end of the year, if the experiment is successful, I want to replace five other similar machines on the line with the new machines."

Does this change our previously negative conclusion about replacing a single machine? The answer is yes. To see this, we now realize that what we have is a package:

- • Replacing a single machine today. This option has an NPV of −67.48.
- • The *option* of replacing five more machines in one year. Suppose that the risk-free rate is 6%. Then we view each such option as a call option on an asset that

has a current value S equal to the present value of the machine's future cash flows. As can be seen in cell B7 above, this present value is S = 932.52. The exercise price of this option is X = 1,000. Of course, these call options can be exercised only if we purchase the first machine now.[2]

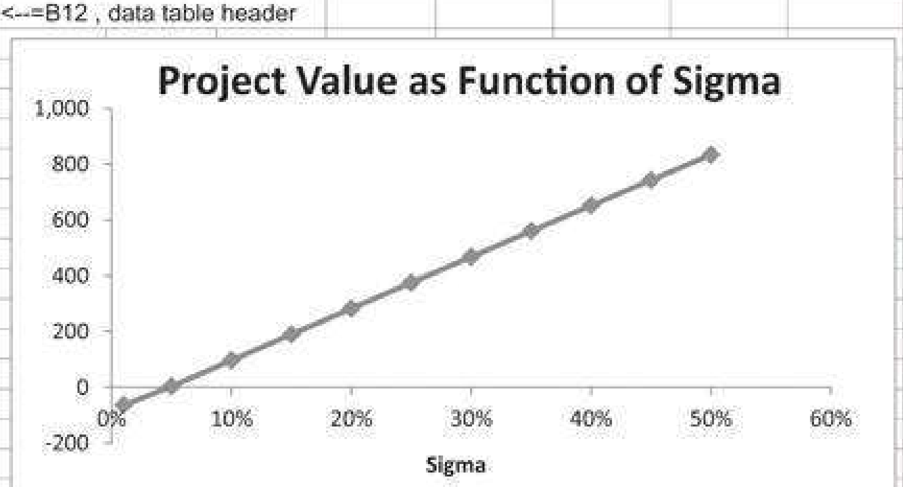
Suppose we assume that the Black-Scholes option pricing model can price this option. In this case we have:

	A	B	C	D	E	F	G
1	**THE OPTION TO EXPAND**						
2	**Year**	**0**	**1**	**2**	**3**	**4**	**5**
3	CF of single machine	-1,000	220	300	400	200	150
4							
5	Riskless discount rate	6%					
6	Discount rate for machine cash flows (risk-adjusted)	12%					
7	Present value of machine's future cash flows	932.52	<--=NPV(B6,C3:G3)				
8	NPV of single machine	-67.48	<--=NPV(B6,C3:G3)+B3				
9							
10	Number of machines bought next year	5					
11	Option value today of single machine	70.12	<--=B24				
12	NPV of total project	283.09	<--=B8+B10*B11				
13							
14	**Black-Scholes Option Pricing Formula**						
15	S	932.52	PV of machine CFs				
16	X	1000	Exercise price = Machine cost				
17	r	6%	Risk-free rate of interest				
18	T	1	Time to maturity of option (in years)				
19	Sigma	20%	<-- Volatility				
20	d_1	0.0507	<--=(LN(B15/B16)+(B17+0.5*B19^2)*B18)/(B19*SQRT(B18))				
21	d_2	-0.1493	<--=B20-SQRT(B18)*B19				
22	$N(d_1)$	0.5202	<--=NORM.S.DIST(B20,1)				
23	$N(d_2)$	0.4406	<--=NORM.S.DIST(B21,1)				
24	Option value = BS call price	70.12	<--=B15*B22-B16*EXP(-B17*B18)*B23				

As cell B12 shows, the value of the whole project is 283.09.

Our conclusion: Buying one machine today and knowing that we have the option to purchase five more machines in one year is a worthwhile project. One critical element here is the volatility. The lower the volatility (i.e., the lower the uncertainty), the less worthwhile this project is.

	B	C	D
27	**Data Table**		
28	σ	283	<--=B12 , data table header
29	1%	-63	
30	5%	5	
31	10%	97	
32	15%	190	
33	20%	283	
34	25%	376	
35	30%	468	
36	35%	561	
37	40%	652	
38	45%	744	
39	50%	835	



This is not very surprising. The value of the project as a whole comes from our uncertainty about the actual cash flows a year from today. The less this uncertainty is (measured by σ), the less valuable the project will be.

Is Black-Scholes the Appropriate Valuation Tool for Real Options?

The answer is almost certainly no: Black-Scholes is not the appropriate tool. However, the Black-Scholes model is by far the most numerically tractable (i.e., easiest) model we have for valuing options of any kind. In valuing real options, we often use the Black-Scholes model, realizing that at best it can give an approximation to the actual option value. Such is life.

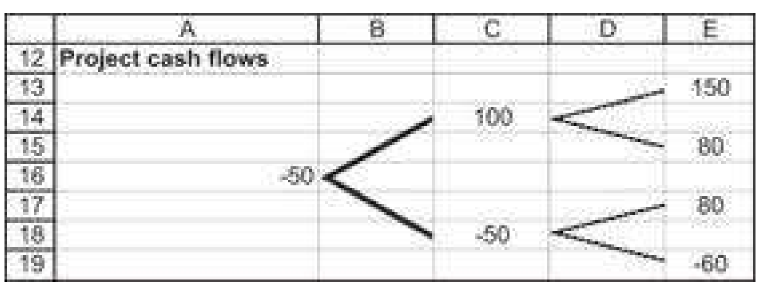
Having said this, you should realize that the assumptions of the Black-Scholes option valuation model—continuous trading, constant interest rate, the ability to trade any fraction of securities, no exercise before final option maturity—are not really appropriate to the real options considered in this chapter. In many cases real options involve what, in a securities option context, would be considered dividend-paying securities and/or early exercise. Two examples:

- • The staged-investment real option, when we have the opportunity to expand or contract the investment over time, is intrinsically an option with early exercise.
- • When an option to abandon an investment exists, as long as the investment is still in place and not abandoned, it continues to pay "dividends" in the form of cash flows.

We can only hope that the Black-Scholes model gives an *approximation* to the option value intrinsic in the real options.

20.3 The Abandonment Option

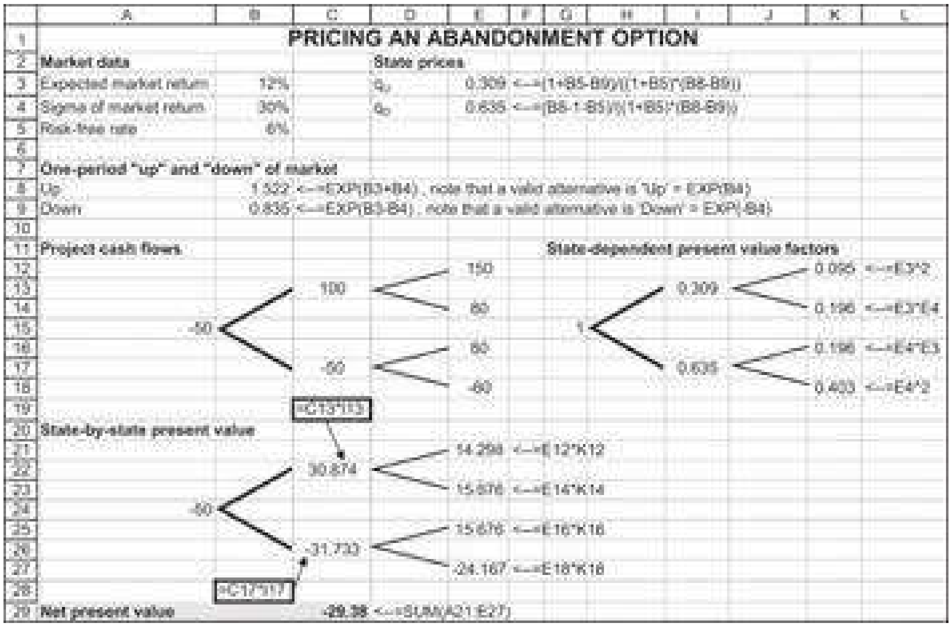
Consider the following capital budgeting project:



The initial cost of this project is 50. In one period the project will produce cash flows of either \$100 or −\$50; that is, on average, it will lose money. Two periods hence the project again has chances of either losing money (in the worst case) or making money.

Valuing the Project

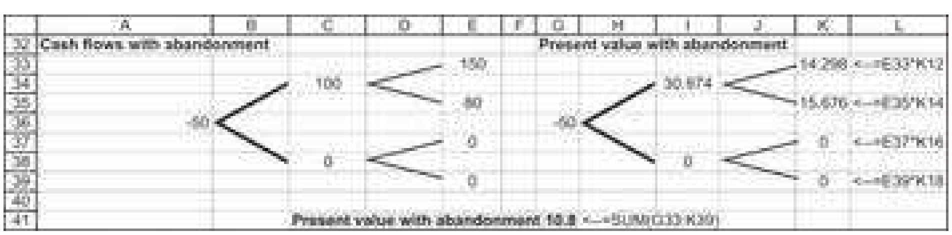
In order to value the project, we use the state prices from option pricing.[3] The state price q_u is the price today of \$1 to be paid in the succeeding period in the "Up" state, and the price q_d is the price today of \$1 to be paid in the "Down" state. The spreadsheet below shows all the relevant details, leading to a project valuation of −\$29.38 (implying rejection of the project):



The methodology is to calculate state-dependent present value factors (how this is done is discussed below) and to multiply these factors times the individual state-dependent cash flows. Each node of the tree is discounted by the relevant state price for the node; for example, the cash flow of 80 that occurs at date 2 is discounted by $q_U q_D$. The NPV of the project is the sum of all the discounted cash flows plus the initial cost (cell B29).

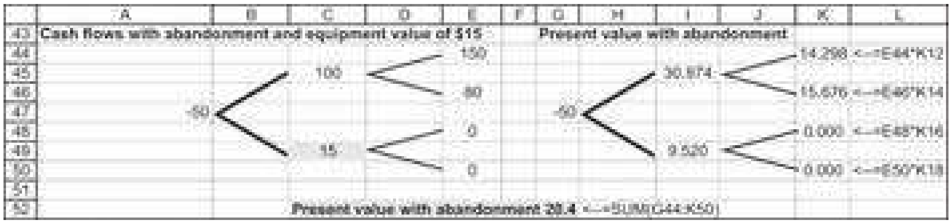
The Abandonment Option Can Enhance Value

Now suppose that we can abandon the project at date 1 if its cash flow "threatens" to be −50. Suppose, furthermore, that this abandonment means that all subsequent cash flows will also be zero. As the spreadsheet below shows, this *option to abandon the project* enhances its value:



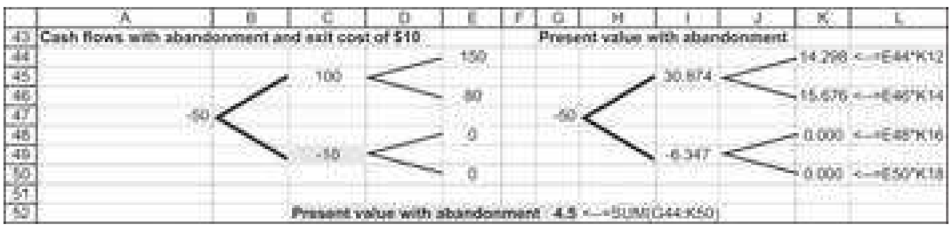
Abandonment When We Sell the Equipment

Another possibility is, of course, that "abandonment" means selling the equipment. In this case there might even be a positive cash flow from abandonment. As an example, suppose that we can sell the asset for $15:



Abandonment When We Pay for Abandon (Exit Cost)

Thinking about this further, it is clear that it might even be worthwhile to *pay to abandon the project*—this payment is also called *exit cost*. Here's what the project looks like when we pay $10 to abandon it in the troublesome state (this payment can be thought of as representing the cost of closing down a facility):



Determining the State Prices

The method we have used above to determine the state prices was explained in greater detail in Chapter 17. We assume that in each period the market portfolio (by which we mean some large, diversified stock market portfolio such as the Standard & Poor's 500) either moves "Up" or "Down," and that the size of these moves is determined by the mean return μ of the market portfolio and by the standard deviation σ of the market portfolio's returns. Assuming that the returns on the market portfolio have mean μ = 12% and standard deviation of returns σ = 30%, we have—in the above examples—calculated the following:

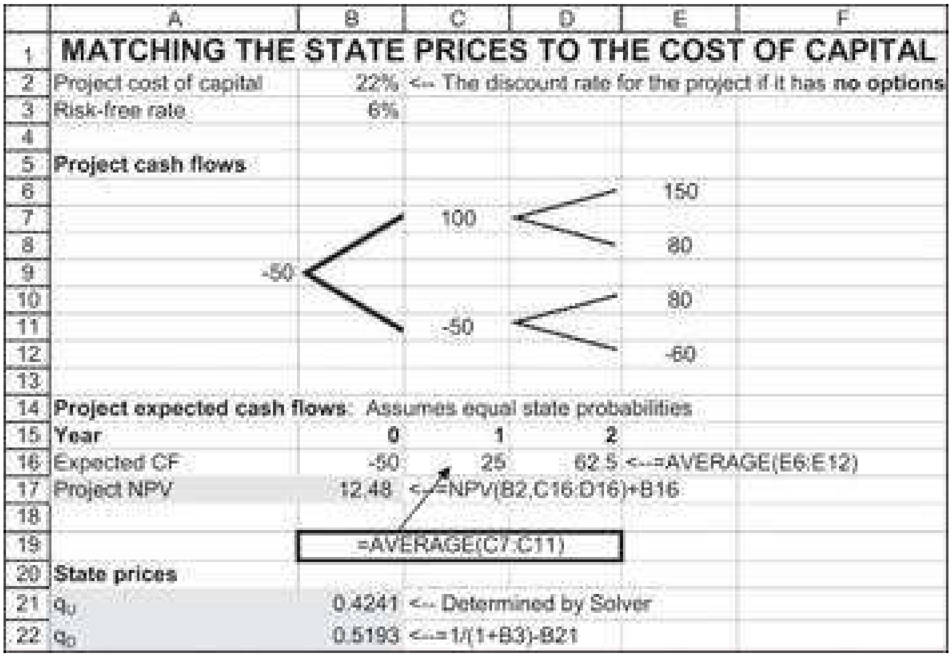
$Up = e^{\mu + \sigma} = 1.52$, and $Down = e^{\mu - \sigma} = 0.84$

Denote by q_U the price today for $1 in the "Up" state in one period and denote by q_D the price today for $1 in the "Down" state in one period. Also, R represent the risk-free rate plus one ($R = 1 + r_f$). Then—as explained in Chapter 17—the state prices are calculated by solving the system of linear equations:

$$q_U = \frac{R - Down}{R * (Up - Down)}, q_D = \frac{Up - R}{R * (Up - Down)}$$

Alternative State Price Determinations

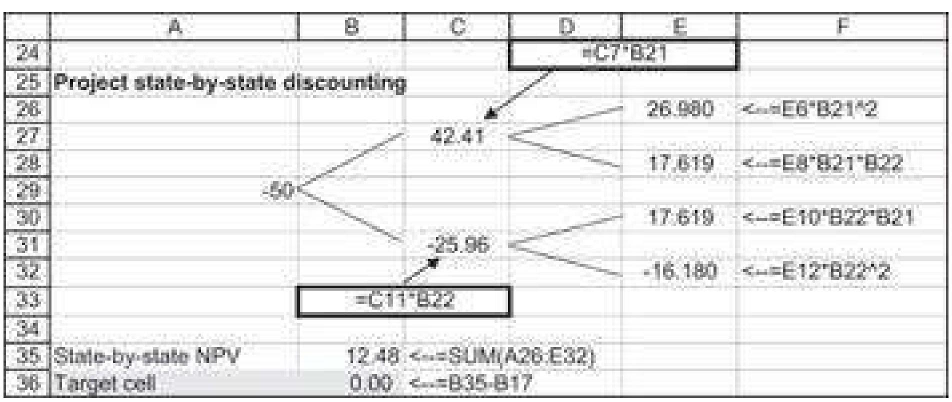
An alternative method of calculating the state prices is to try to match them to the project's cost of capital. Reconsider the project discussed above and suppose that the actual probability of each state's occurrence is 50%. Furthermore, suppose that the risk-free rate is 6%. Finally, assume that the project's discount rate—if it has no options whatsoever—is 22%. Then we can calculate the project's NPV without real options as 12.48:



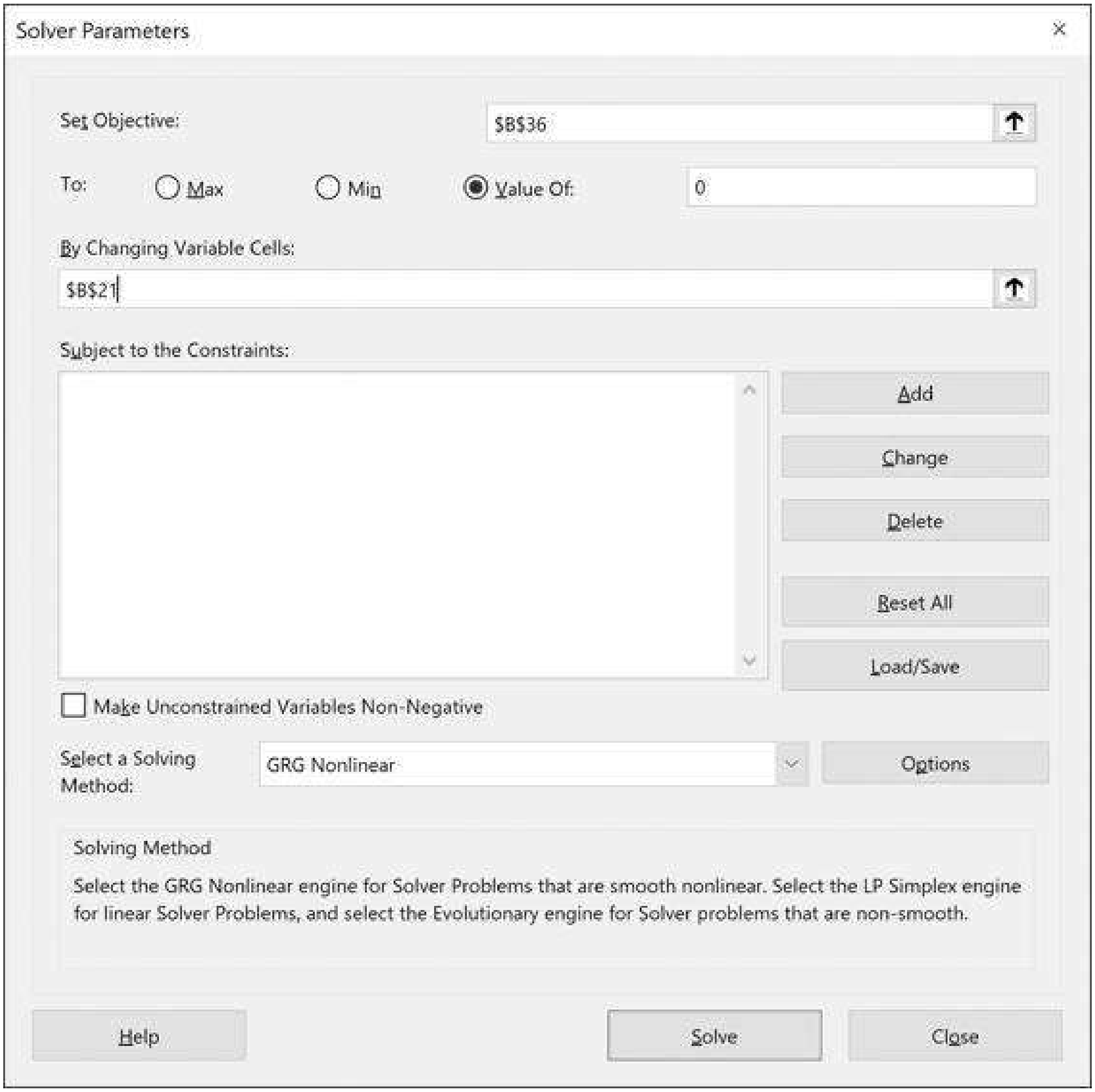
In cells B21 and B22 we look for state prices q_U and q_D, which have two properties:

1. They are consistent with the risk-free interest rate. This means that $q_U + q_D = \frac{1}{R} = \frac{1}{1.06}$.
2. The state prices give the same NPV for the project as that calculated by the cost of capital.

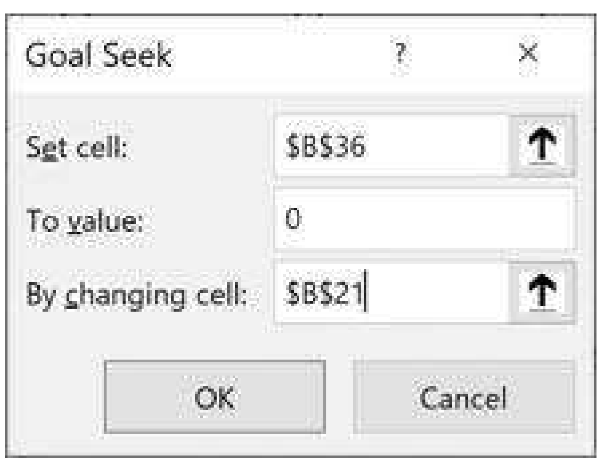
The second requirement means that we have to use the Excel **Solver** to determine the state prices. Here's what the solution looks like (the discussion of how Solver was used follows the next spreadsheet):



To determine the state prices, we use the Solver (**Data|Solver**):



One more note: You can also use Goal Seek (**Data|What-If Analysis|Goal Seek**) to get the same result. However, Excel's Goal Seek does not remember its previous settings, which means that each time you repeat this calculation you will have to reset the cell references. Here's what the Goal Seek dialog box looks like:



20.4 Valuing the Abandonment Option as a Series of Puts

The example above shows how and why the abandonment option can have value. It also illustrates another, more troublesome feature of the abandonment option—namely, that it may be very difficult to value. While it is difficult enough to project expected cash flows, it is even more difficult to project state-by-state cash flows and state prices for a complex project.

A possible compromise in the valuation of an abandonment option is to value a project as a series of cash flows *plus* a series of Black-Scholes put options. Consider the following example: You are valuing a 4-year project with the expected cash flows given below and with a risk-adjusted discount rate of 12%. As you can see, the project has a negative NPV:

	A	B	C	D	E	F
1	**STANDARD DCF PROJECT VALUATION**					
2	**Project cash flows**					
3	Year	0	1	2	3	4
4	Cash flow	-750	100	200	300	400
5						
6	Risk-adjusted discount rate	12%	The project's cost of capital			
7	NPV without options	-33.53	<-- =B4+NPV(B6,C4:F4)			

Suppose that we can abandon the project at the end of any of the next four years, selling the equipment for 300. Although this abandonment option is an American option and not a Black-Scholes option, we value it as a series of Black-Scholes put options. In each case we suppose that we first get the year-end cash flow; we then value the abandonment option on the remaining project value.

- • End of year 1: The asset's expected value at the end of year 1 will be the discounted value of its future expected cash flows:

$$702.44 = \frac{200}{(1.12)^1} + \frac{300}{(1.12)^2} + \frac{400}{(1.12)^3}.$$

- The abandonment option means that we can get $300 for the asset during the next three years. Suppose that the value has a volatility of 50%; then valuing this option as a Black-Scholes put with one year to maturity gives its value as 19.53. The following spreadsheet uses the VBA function **BSPut** defined in Chapter 18:

	A	B	C	D	E	F
1	**ABANDONMENT VALUE – DETAILS OF YEAR 1 CALCULATION**					
2	**Project cash flows**					
3	Year	**0**	**1**	**2**	**3**	**4**
4	Cash flow	-750	100	200	300	400
5						
6	Risk-adjusted discount rate	12%	The project's cost of capital			
7	NPV without options	-33.53	<-- =B4+NPV(B6,C4:F4)			
8						
9	**Valuing the year-1 abandonment put**					
10	Value of project, end year 1	702.44	<--=NPV(B6,D4:F4)			
11	Abandonment value	300	Like strike price in put formula			
12	Time to option maturity (years)	3	<--=F3-1			
13	Risk-free rate	6%				
14	Sigma	50%				
15	Put value	19.53	<--=bsput(B10,B11,B12,B13,B14)			

- • End of year 2: We have a put option with exercise price \$300 on an asset worth $586.73 = \frac{300}{1.12} + \frac{400}{(1.12)^2}$. Valuing the abandonment option as a Black-Scholes put option with two years to exercise gives its value (when $\sigma = 50\%$) as 17.74.
- • End of year 3: We have a put option with exercise price \$300 on an asset worth $357.14 = \frac{400}{1.12}$. The option has one more year remaining to its life and is worth 32.47.
- • End of year 4: The asset is worthless in terms of future anticipated cash flows but can be abandoned for \$300 (this is thus its scrap or salvage value). The abandonment option is worth \$300.

In the spreadsheet below the asset has been valued as the sum of:

- • The present value of the future expected cash flows. As we showed above, this is −\$33.53.
- • The present value (at the risk-free rate) of a series of Black-Scholes puts. This value is \$299.10.

The total value of the project is −\$33.53 + \$299.10 = \$265.57.

	A	B	C	D	E	F	G
1	PRICING AN ABANDONMENT OPTION AS A SERIES OF PUTS						
2	Project cash flows						
3	Year	0	1	2	3	4	
4	Cash flow	-750	100	200	300	400	
5							
6	Risk-adjusted discount rate	12%	The project's cost of capital				
7	NPV without options	-33.53	<--=NPV(B6,C4:F4)+B4				
8							
9	Sigma	50%					
10	Risk-free rate	6%					
11	Abandonment value	300	Abandonment value at end of any year				
12							
13	NPV of cash flows at RADR	-33.53	<--=B7				
14	Value of abandonment option	299.10	<--=NPV(B10,C19:F19)				
15	Adjusted present value	265.57	<--=B14+B13				
16					Function in cell D18: =NPV($B$6,E4:$F$4)		
17	Year	0	1	2	3	4	
18	End-year value of remaining cash flows		702.44	586.73	357.14	0.00	
19	Put option value		19.53	17.74	32.47	300.00	
20							
21		Function in cell D19: =bsput(D18,$B$11,$F$3-D3,$B$10,$B$9)					

20.5 Valuing a Biotechnology Project[4]

One of the interesting features of the biotech industry is the existence of highly valued firms that have no revenues. It is common understanding that the value of those firms is in their future cash flow opportunities. Therefore, understanding the translation of qualitative investment opportunities into quantitative valuation is of great importance when valuing those firms. In this section we use the real option method to value a biotechnology project and to illustrate the application of the real option approaches.

Consider the following story.[5] A firm is considering the initiation of research into a new drug. It knows that there are three stages to the drug's development:

- • In the **discovery phase**, the firm does preliminary research about the viability of the idea. This research takes one year and costs $1,000 at the beginning of the year. With 50% probability, the results will be positive enough to proceed to the next stage of research.
- • If the discovery phase yields success, then the drug goes into the **clinical phase**, in which the drug is tested. This stage lasts one year and costs $2,000 at the beginning of the year, and with a probability of 30% that it yields enough positive results to proceed to the next stage.
- • If the drug passes the clinical phase successfully, then it goes into the **market stage**, in which it is sold. This phase costs $15,000 per year (at the beginning of each year) and on average lasts five years. On average, a successful drug can be expected to start the marketing phase with income of $20,000. This income grows with annual mean 10% and standard deviation σ = 100%.

The expected return on project of this type is 25%. We assume that this is the cost of capital of the project in the case of a DCF valuation.

The Expected Value of the Project Using Traditional DCF Analysis

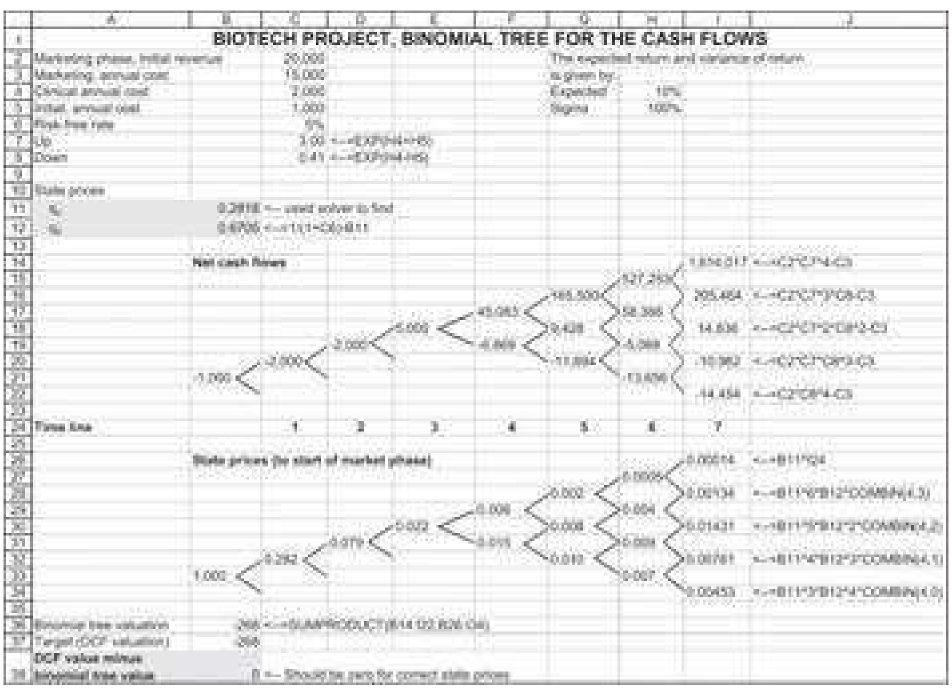
If we estimate the value of this project using traditional discounted cash flow analysis, we get a negative net present value for the project:

	A	B	C	D	E	F	G	H
1	**BIOTECH PROJECT EXPECTED CASH FLOWS**							
2	Discount rate		25%					
3	Growth (Marketing)		10%					
4								
5	**Year**	**Stage**	**Cost**	**Income**	**Net**	**Probability**	**Expected cash flow**	
6	0	Discovery	-1,000	0	-1,000	100%	-1,000	<--=F6*E6
7	1	Clinical	-2,000	0	-2,000	50%	-1,000	<--=F7*E7
8	2	Clinical	-2,000	0	-2,000	50%	-1,000	
9	3	Marketing	-15,000	20,000	5,000	15%	750	
10	4	Marketing	-15,000	22,000	7,000	15%	1,050	
11	5	Marketing	-15,000	24,200	9,200	15%	1,380	
12	6	Marketing	-15,000	26,620	11,620	15%	1,743	
13	7	Marketing	-15,000	29,282	14,282	15%	2,142	
14								
15	**Project NPV**	**-268**	<--=G6+NPV(C2,G7:G13)					

Since the project's net present value is negative, the DCF approach indicates that it should not be undertaken.

Using a Real Options Approach

An alternative method for estimating the present value of proceeds is to plot the project's cash flows on a binomial tree. This is done below:



We use the Excel function **Sumproduct** to do this computation.

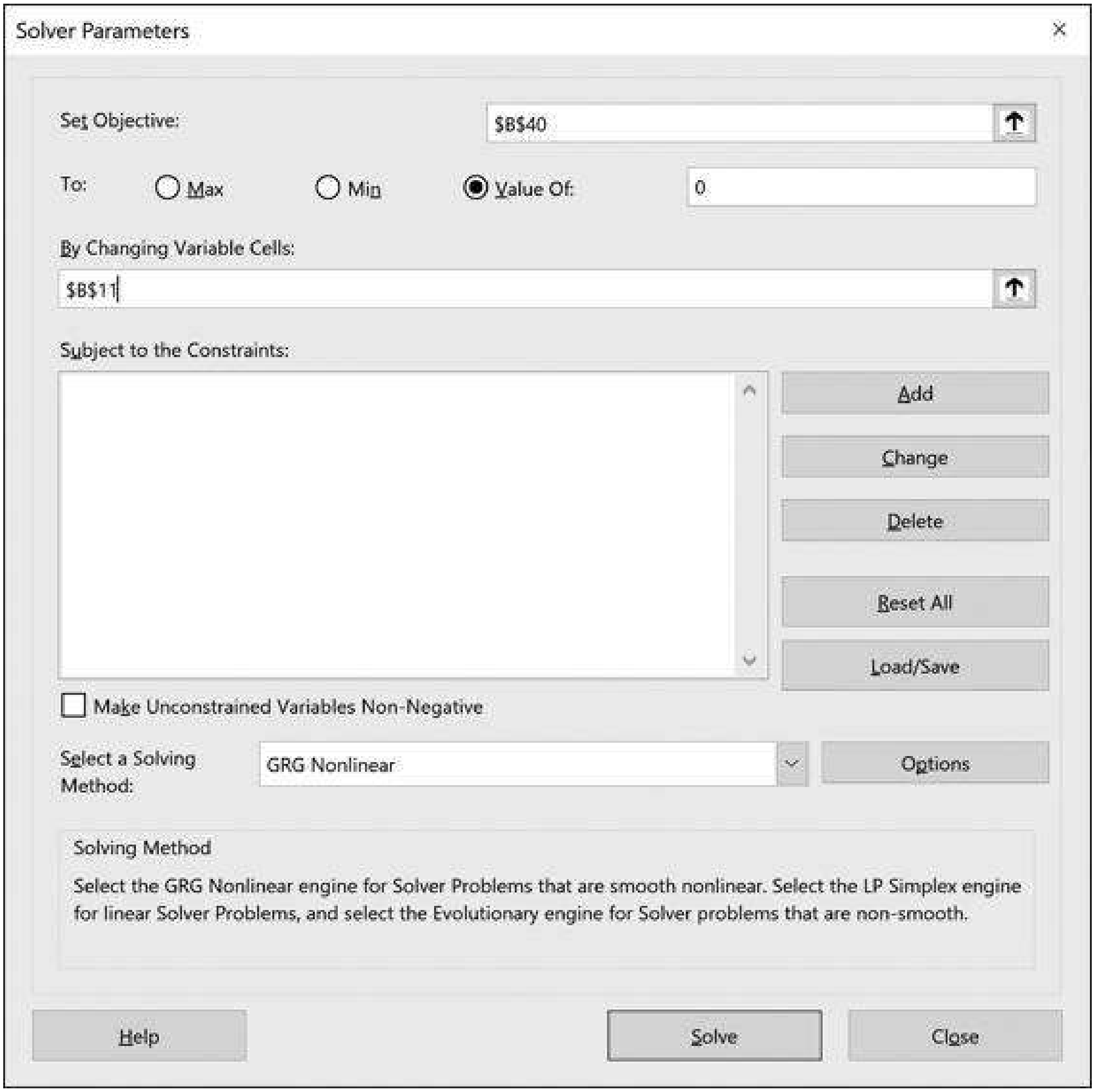
A Note about the State Prices

The net present value of the proceeds from project is the product of the net cash flows and the appropriate state prices:

$$NPV = \sum_{t=0}^{7} \sum_{j=0}^{t} CF_{jt} * (q_U)^j * (q_D)^{t-j} * (Number\ of\ paths\ to\ node)$$

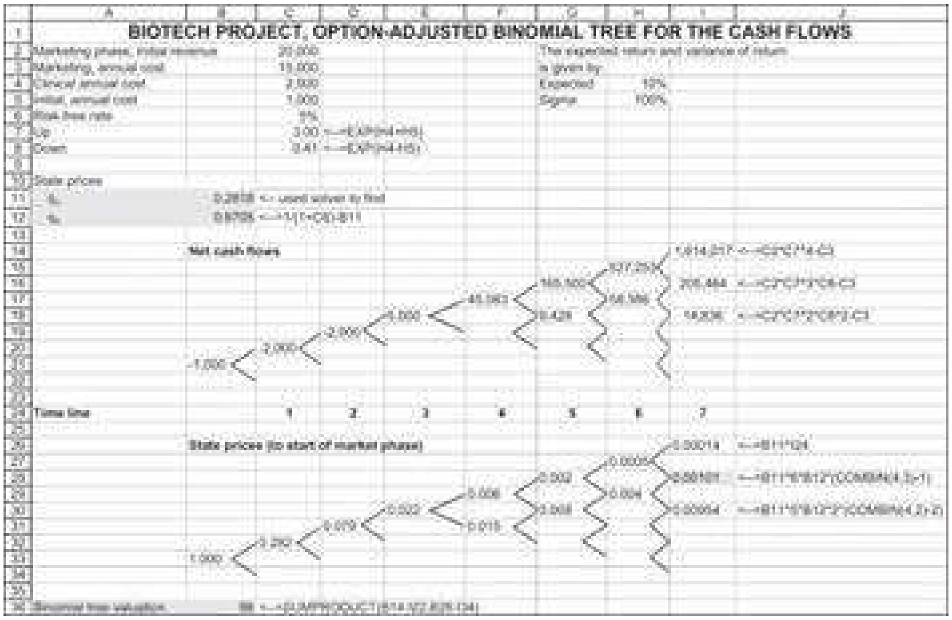
Here, $CF_{j,t}$ denotes the proceeds from the project at date t and state j, where j is the number of Up moves. As explained in Chapter 17, in the standard binomial model, the state price for a node is $(q_U)^j * (q_D)^{n-j} \binom{t}{j}$, where n is the time at which the node occurs, j is the number of Up steps needed to get to the node, and $\binom{t}{j}$ is the number of paths to reach the node. The latter expression is computed in Excel by using the function **Combin(n,j)**. However, for the real options model above, the number of paths to each node is slightly different, since the beginning of the tree (the initial and clinical states) is accessible via only one path.

In the spreadsheet above, the prices q_U and q_D were computed (using **Solver**) so that the present value of the project on the binomial tree equals that of the DCF valuation and that the equilibrium condition $q_U + q_D = \frac{1}{1.05}$ holds. Here's the **Solver** screen:



The Real Options Approach

The real options approach to research and development (R&D) recognizes that at each stage in the project, the managers can choose whether to continue the project or not. They do this by comparing the value and costs of continuation. In option terminology, at each stage, the managers exercise their continuation option if the value from exercising the option exceeds the exercise price. In the spreadsheet below, we have eliminated obvious negative cash flows from the marketing stage, taking care to also eliminate the subsequent cash flows and to make an adjustment to the state prices (more on that below):



Another Note about State Prices

When we eliminate states in the real option approach, we must also adjust the number of paths to each node to account for the fact that some states are no longer reachable. This has been done in the above spreadsheet. For example, the state in cell I28 (highlighted) is now reachable by one fewer path.

20.6 Summary

Recognizing that capital budgeting should include option aspects of projects is clear and obvious. Valuing these options is often difficult. In this chapter we have tried to emphasize the intuitions and—insofar as is possible—to give some implementation of the valuation.

Exercises

1. Your company is considering purchasing 10 machines, each of which has the following expected cash flows (the entry in B3 of −$550 is the cost of the machine):

	A	B	C	D	E	F
1	MACHINE REPLACEMENT					
2	Year	0	1	2	3	4
3	CF of single machine	-550	100	200	300	400

You estimate the appropriate discount rate for the machines as 25%.

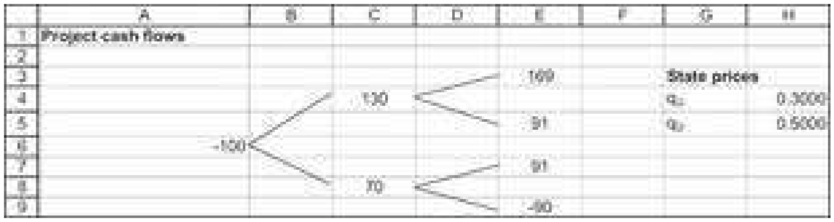
1. Would you recommend buying just one machine, if there are no options effects?
2. Your purchase manager recommends buying one machine today and then—after seeing how the machine operates—reconsidering the purchase of the other nine machines in six months. Assuming that the cash flows from the machines have a standard deviation of 30% and that the risk-free rate is 5%, value this strategy.

2. Your company is considering the purchase of a new piece of equipment. The equipment costs $50,000 and your analysis indicates that the PV of the future cash flows from the equipment is $45,000. Thus, the NPV of the

equipment is −$5,000. This estimated NPV is based on some initial numbers provided by the manufacturer plus some creative thinking on the part of your financial analyst.

The seller of the new piece of equipment is offering a course on how it works. The course costs $1,500. You estimate that the σ of the equipment's cash flows is 30%, the risk-free rate is 6%, and you will have another half year after the course to purchase the equipment at the price of $50,000. Is it worth taking the course?

3. Consider the project whose cash flows are given below:

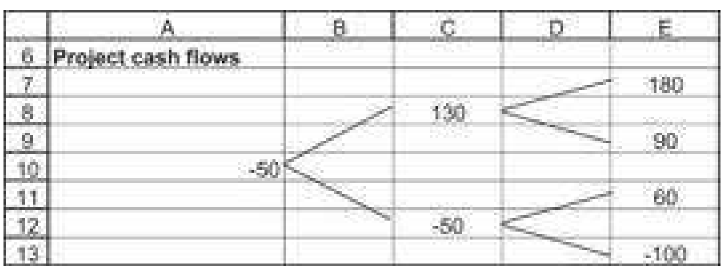


1. Using the state prices, value the project.
2. Suppose that at date 2 the project can be abandoned at no cost. How is this affecting the value of the project?
3. Suppose that at any time the project can be sold for 100. Show the tree of cash flows and value the project.

4. Suppose that the market portfolio has mean $\mu = 15\%$ and standard deviation $\sigma = 20\%$.

1. If the risk-free rate of interest is 4%, calculate the 1-period state prices for an "Up" and a "Down" state.
2. Show the effect (in a data table) of the risk-free rate on the state prices.
3. Show the effect of the σ on the state prices.

5. Consider the cash flows below:



1. If the cost of capital is 30% and the risk-free rate is 5%, find the state prices that match the project's NPV.
2. If there exists an abandonment option so that we can change all negative cash flows to zero, value the project.

---

1. 1. These cash flows are the incremental cash flow of replacing a single old machine with a new machine. The computations include taxes, incremental depreciation, and the sale of the old machine.

2. 2. What we're really doing is pricing the cost of learning!
3. 3. A detailed discussion is provided in section 17.3. See the short discussion below on how to calculate these state prices.
4. 4. A version of this example originally appeared in Benninga and Tolkowsky (2002).
5. 5. We have made the story simple enough to fit an understandable spreadsheet. For a somewhat more complicated story in the same spirit, see Kellogg and Charnes (2000).

V MONTE CARLO METHODS

21 Generating and Using Random Numbers

21.1 Overview

In this chapter we discuss techniques for computing random numbers. We use random numbers extensively in this section to simulate stock prices, investment strategies, and option strategies. In this chapter we show how to produce both uniformly distributed and normally distributed random numbers.

A random-number generator on a computer is a function that produces a seemingly unrelated set of numbers. The question of *what is* a random number is a philosophical one.[1] In this chapter we will ignore philosophy and concentrate on some simple random-number generators—primarily the Excel random-number generator **Rand()** and the VBA random-number generator **Rnd**.[2] We will show how to use these generators to produce uniform random numbers and, subsequently, random numbers that are normally distributed. At the end of the chapter we use the Cholesky decomposition to produce correlated random numbers.

To imagine a set of uniformly distributed random numbers, think of an urn filled with 1,000 little balls numbered 000, 001, 002, …, 999. Suppose we perform the following experiment: Having shaken the urn to mix up the balls, we draw one ball out of the urn and record the ball's number. Next, we put the ball back into the urn, shake the urn thoroughly so that the balls are mixed up again, and then draw out a new ball. The series of numbers produced by repeating this procedure many times should be *uniformly distributed* between 000 and 999.

A random-number generator on a computer is a function that imitates this procedure. The random-number generators considered in this chapter are sometimes termed *pseudo-random-number generators*, since they are actually deterministic functions whose values are indistinguishable from random numbers. All pseudo-random-number generators have cycles (i.e., they eventually start to repeat themselves). The trick is to find a random-number generator with a long cycle. The Excel **Rand()** function has very long cycles and is a respectable random-number generator.

If you have never used a random-number generator, open an Excel spreadsheet and type **=Rand()** in any cell. You will see a 15-digit number between 0.000000000000000 and 0.999999999999999. Every time you recalculate the spreadsheet (e.g., by pressing the **F9** button), the number changes. We leave the technical details of how **Rand()** works for the exercises to this chapter, where we show you how to design your own random-number generator. Suffice it to say, however, that the series of numbers produced by the function should be “unpredictable to the uninitiated” (to use Lehmer’s terminology from footnote 1).

In this chapter we shall deal with several kinds of random-number generators: We first examine the uniform random-number generators that come with Excel and VBA. Subsequently, we generate normally distributed random numbers.[3] Finally, we generate correlated random numbers using the Cholesky decomposition.

21.2 Rand() and Rnd: The Excel and VBA Random-Number Generators

Suppose you simply wanted to generate a list of random numbers. One way to do this would be to copy the Excel function **Rand()** to a range of cells.

	A	B	C	D	E
1	USING EXCEL'S RAND() FUNCTION				
2	0.5673	0.9469	0.5465	0.3589	<--=RAND()
3	0.6945	0.6012	0.6434	0.8983	
4	0.5017	0.4066	0.5307	0.9123	
5	0.3116	0.2463	0.8787	0.4516	
6	0.7713	0.8141	0.3733	0.7815	
7	0.5618	0.5266	0.6794	0.2371	
8	0.1857	0.0315	0.3826	0.9804	
9	0.5610	0.6642	0.3861	0.4782	
10	0.4040	0.8802	0.6569	0.3516	
11					
12	Each cell contains the function **Rand()**. Each time you update the spreadsheet or press F9 the block of cells will produce a new set of random numbers.				

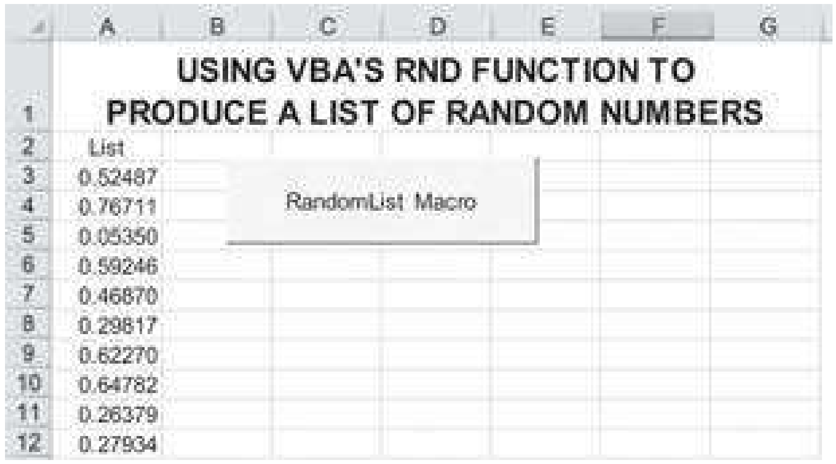
In section 21.2 we will develop a crude test of how well **Rand()** works.

Using VBA’s Rnd Function

VBA contains its own function **Rnd,** which is equivalent to the Excel **Rand** function.[4] Here is a small VBA program that illustrates a basic use of the **Rnd** function:

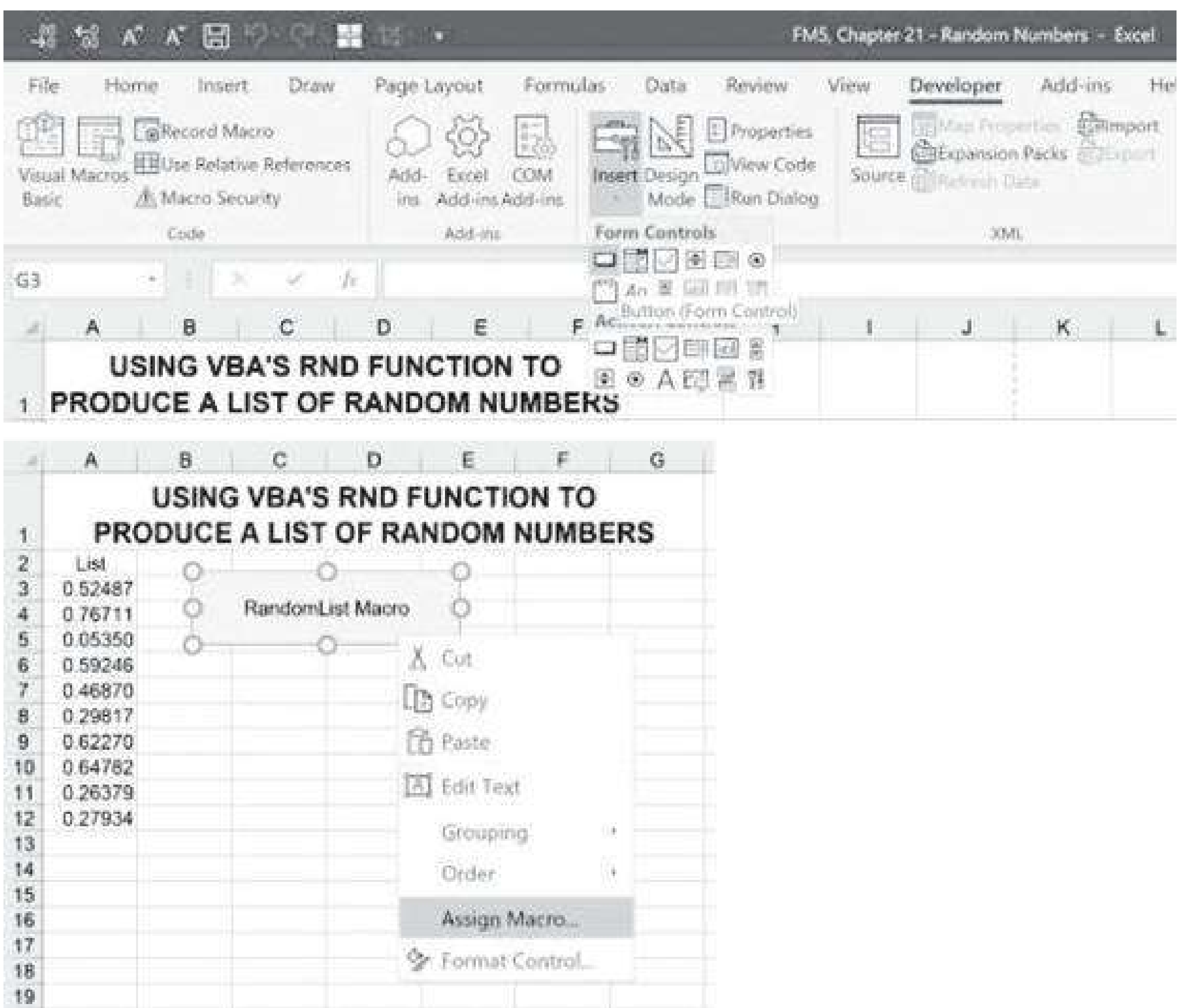
```
Sub RandomList()
'Produces a simple list of random numbers
    For Index = 1 To 10
        Range("A4").Cells(Index, 1) = Rnd
    Next Index
End Sub
```

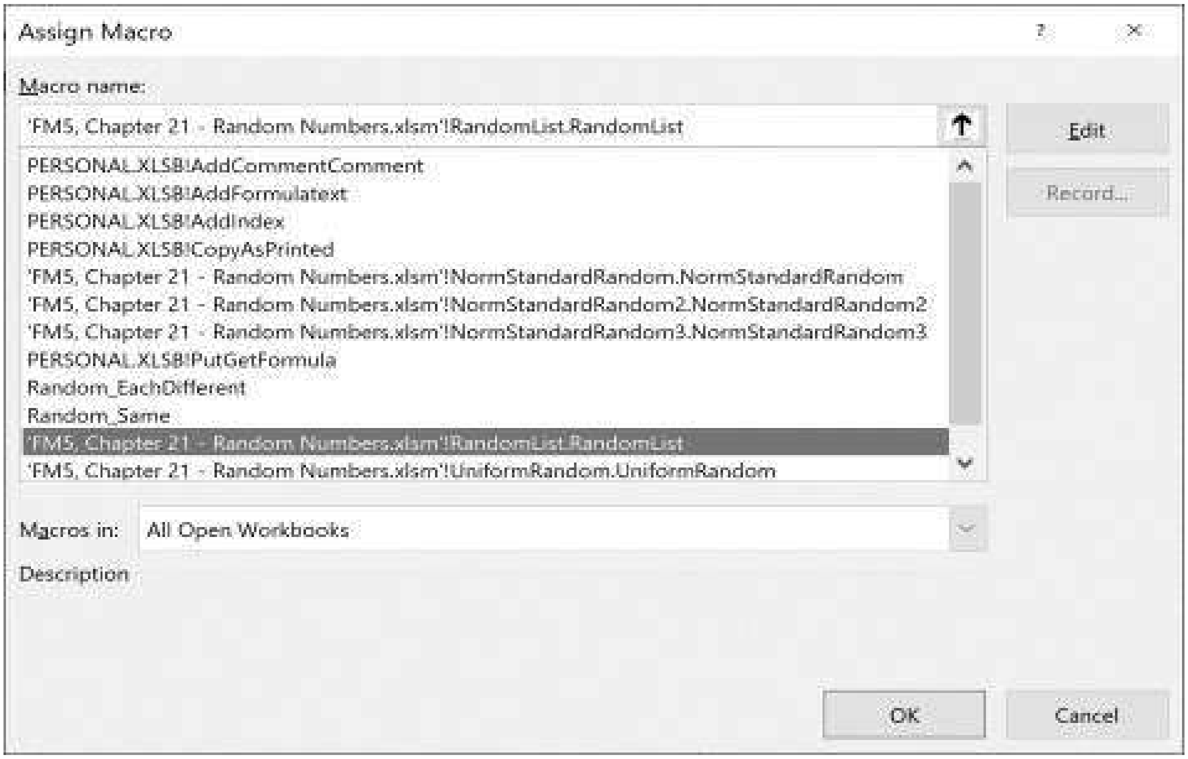
In the spreadsheet below, the VBA program has been assigned to a button so that each time we click on the button, it runs a VBA program that produces 10 random numbers:



Assigning a Macro to a Button or to a Control Sequence

In the spreadsheet above, we have assigned the macro **RandomList** to the button marked "RandomList Macro." Any drawing shape in Excel can be assigned a VBA program. In this case we have created a Button; right-clicking on this rectangle, we have assigned it a macro:





Testing Random-Number Generators

Producing lists of random numbers is interesting, though a bit uninformative. Is the list of numbers thus produced really uniformly distributed? A simple test is to generate each number and determine whether it falls into the interval $[0, 0.1)$, $[0.1, 0.2)$, …, $[0.9, 1)$. The notation $[a, b)$ denotes the *half-open* interval between a and b; a number x is in this interval if $a \le x < b$. If the list of numbers is really uniformly distributed, we would expect roughly an even number of the "random" numbers to be in each of the 10 intervals.

One way to test this is to generate a list of random numbers on the spreadsheet by copying **Rand()** to many cells and then using the Excel array function **Frequency(data_array,bins_array)**.[5] This is illustrated in the following spreadsheet picture:

	A	B	C	D	E
1	USING EXCEL'S FREQUENCY FUNCTION TO TEST THE DISTRIBUTION OF RAND()				
2	Random numbers		Bin	Frequency	
3	0.7012	<--=RAND()	0.1	1	
4	0.8866		0.2	1	
5	0.6692		0.3	0	
6	0.7827		0.4	1	
7	0.0023		0.5	0	<--{=FREQUENCY(A3:A12,C3:C12)}
8	0.9184		0.6	0	
9	0.7687		0.7	2	
10	0.6285		0.8	3	
11	0.1299		0.9	1	
12	0.3432		1	1	
13					
14	Each cell in the range A3:A12 contains the formula **Rand()**. Pressing F9 will produce a new set of random numbers and frequencies.				

Writing **Frequency(A:A,D3:D12)** refers to all non-empty cells in column A:

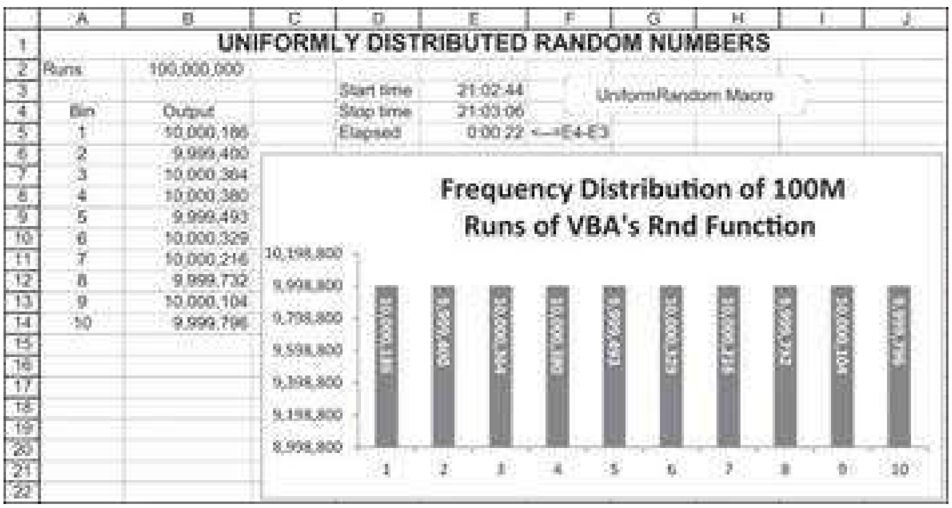
	A	B	C	D	E
1	USING EXCEL'S FREQUENCY FUNCTION TO TEST THE DISTRIBUTION OF RAND()				
2	0.018963	<-- =RAND()	Bin	Frequency	
3	0.308998		0.1	136	
4	0.077996		0.2	172	
5	0.942367		0.3	148	
6	0.295612		0.4	151	
7	0.988988		0.5	140	<--{=FREQUENCY(A:A,C3:C12)}
8	0.446434		0.6	168	
9	0.787454		0.7	131	
10	0.019319		0.8	159	
11	0.323767		0.9	145	
12	0.181345		1	145	
13	0.230781				
14	0.214178		Total	1495	<--=SUM(D3:D12)
1494	0.215724				
1495	0.61205	<--=RAND()			
1496	0.750147	<--=RAND()			

This method is obviously not efficient (or even feasible) when we want to test the random-number generator for large numbers of random draws. The following program uses VBA to generate many random numbers and puts them into the bins in range A3:A12:

```
Sub UniformRandom()
'Puts random numbers into bins
    Range("E3") = Time
    N = Range("B2").Value 'the number of random draws
    Dim distribution(10) As Long 'bins
    For k = 1 To N
        draw = Rnd
        distribution(Int(draw * 10) + 1) = _
```

```
                distribution(Int(draw * 10) + 1) + 1
        Next k
        For Index = 1 To 10
            Range("B5").Cells(Index, 1) = distribution(Index)
        Next Index
        Range("E4") = Time
    End Sub
```

In the spreadsheet below we have generated 100 million random numbers in 25 seconds:



Here are some things to note about **UniformRandom**:

- • The program has a "clock" to measure the amount of time it takes to run. At the start of the program, we **use Range("E3")=Time** to put the starting time into cell E3. At the end of the program, **Range("E4")=Time** puts in the ending time. The cell **elapsed** calculates the difference between the starting time and the ending time. Note that for the cells to read correctly, you have to use the command **Format Cells|Number|Time** on the relevant cells.
- • The heart of the program uses the function `Int(draw * 10) + 1`. Multiplying the random draw by 10 produces a number whose first digit is 0, 1, …, or 9. The VBA function **Int** gives this integer. **Distribution** is a VBA array numbered 1 to 10, with **Distribution(1)** being the number of random numbers in [0, 0.1), **Distribution(2)** the number of random numbers in [0.1, 0.2), etc. Thus, `Int(draw * 10) + 1` is the proper place for the current random draw in **Distribution**.

Using Randomize to Produce the Same List (or Not) of Random Numbers

Most random-number generators use the last generated “random” number to produce the next.[6] The first number used in a particular sequence is controlled by the “seed,” which is typically taken from the computer’s clock. VBA’s **Rnd** is no exception, but it allows you to control the seed by using the command **Randomize**. The two small programs below illustrate two uses of this command.

- • Using **Randomize** without any numeric argument resets the seed (meaning—it breaks the connection between the next random number and the current random number). This is illustrated in the macro **Random_EachDifferent**, though it is difficult to see the effect.

```
Sub Random_EachDifferent()
'Produces a list of random numbers
Randomize 'Initializes the VBA random number generator
    For Index = 1 To 10
        Range("A5").Cells(Index, 1) = Rnd()
    Next Index
End Sub
```

• Using **Randomize(seed)** uses a particular number as the seed.

- • Using the sequence of commands **Rnd(negative number)** and **Randomize(seed)** guarantees the same sequence of random numbers. This is illustrated in the macro **Random_Same**.

```
Sub Random_Same()
'Produces the same list of random numbers which is always the
same
Rnd (-4)
Randomize (Range("seed")) 'Initializes the VBA random-number
generator
    For Index = 1 To 10
        Range("B5").Cells(Index, 1) = Rnd()
    Next Index
End Sub
```

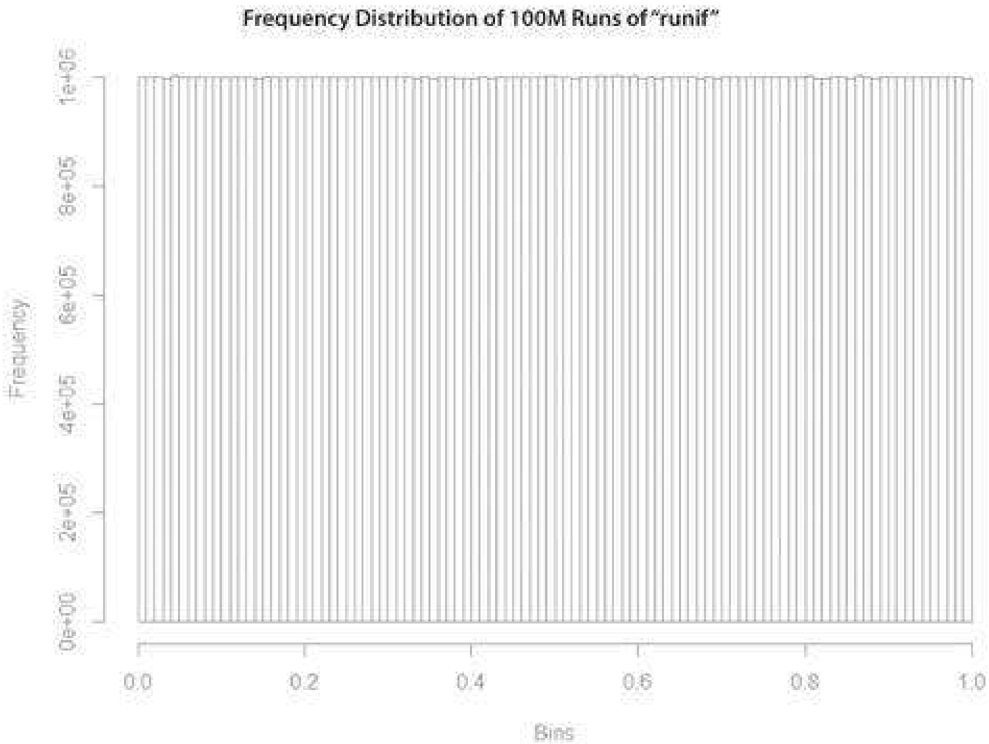
In the spreadsheet below, pushing the top button produces a random set of random numbers. Pushing the bottom button activates the macro **Random_Same** and produces the same set of random numbers each time—provided the **Seed** (cell B2) isn’t changed.

	A	B	C
1	**PRODUCING LISTS OF RANDOM NUMBERS**		
2	Seed	334	
3			
4	**Each run different**	**Output: Each run same**	Run Random_EachDifferent
5	0.78866	0.29708	
6	0.83370	0.70653	
7	0.84234	0.65463	
8	0.57270	0.96848	Run Random_Same
9	0.55024	0.48999	
10	0.79921	0.72373	
11	0.44507	0.06518	
12	0.51669	0.60034	
13	0.71168	0.25382	
14	0.58467	0.70398	
15			
16	**Note**: To see the effect of the "Run Random_Same" button, erase the cells B5:B14. Changing the seed in cell B2 changes the output in column B.		

Using "runif" in R to Produce Uniform Deviates

The R command "runif(n, min = 0, max = 1)" is equivalent to the Rand() function in Excel. Below, please find the equivalent code in R to the 100 million (M) runs:

```
# We thank Sagi Haim for developing this script
################################
# CHAPTER 21 - RANDOM NUMBERS
################################

## 21.2  Rand() and Rnd - the Excel and VBA Random-Number Generators
# 100M Trials
uni_rand_var <- runif(n = 100000000, min = 0, max = 1)

# Histogram
hist(uni_rand_var, breaks = 100,
     main="Frequency Distribution of 100M runs of 'runif'", xlab="Bins")
```



21.3 Scaling Uniformly Distributed Numbers

In many cases we are interested in uniformly distributed random numbers that are not distributed between 0 and 1. This scaling is relatively easy. If we are interested in a uniformly distributed number that ranges between X and Y (i.e., $U \sim U\,[X, Y]$) we should use the following scaling formula:

$U = X + (Y - X) * Rand()$

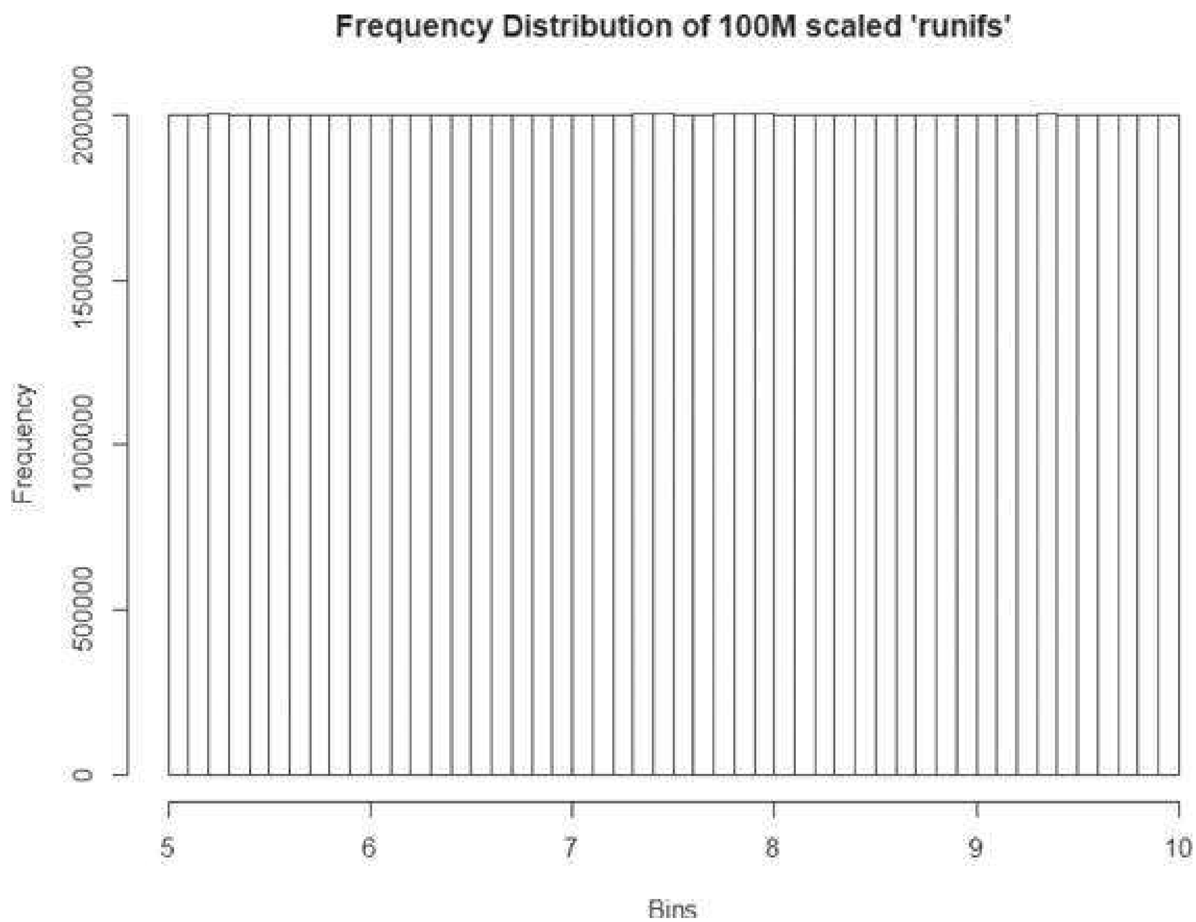
where $Rand() \sim [0, 1]$.

The following Excel clip uses this procedure:

	A	B	C	D
1	SCALING UNIFORMLY RANDOM NUMBERS			
2	Lower bound (X)	5		
3	Upper bound (Y)	10		
4				
5	U_1~U[0,1]	0.2856	<--=RAND()	
6	U_2~U[X,Y]	6.4281	<--=$B$2+($B$3-$B$2)*B5	

Implementing this approach in R looks like this:

```
x <- 5 # Lower bound
y <- 10 #Upper bound

scaled_uni_rand_var <- x + (y-x) * uni_rand_var

# Histogram
hist(scaled_uni_rand_var, breaks = 50,
     main="Frequency Distribution of 100M scaled 'runifs'", xlab="Bins")
```

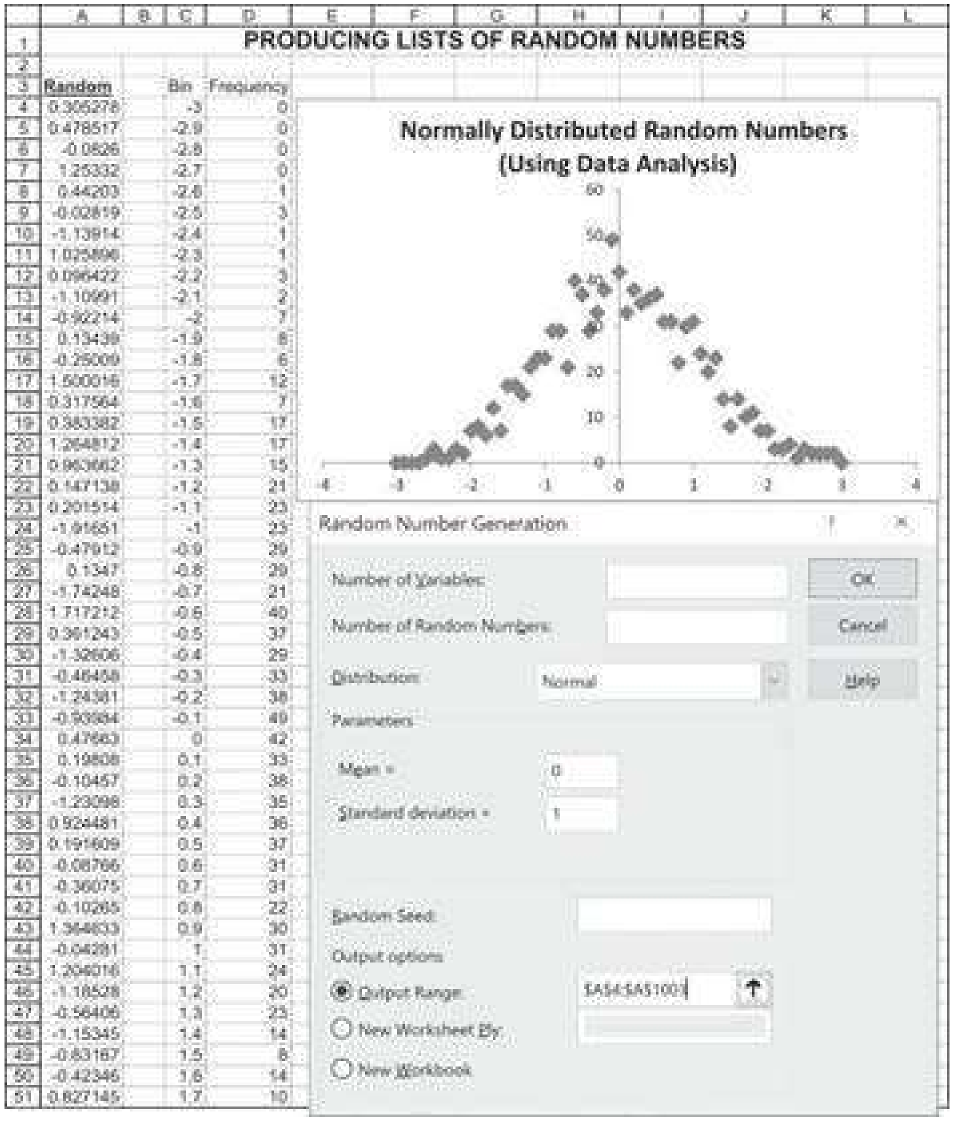


21.4 Generating Normally Distributed Random Numbers

In the sections above we have generated numbers that are uniformly distributed. In this section we explore four ways to produce normally distributed random numbers using Excel.

Method 1: Normally Distributed Numbers Using Data|Data Analysis|Random Number Generation

One way to do this is to use the Excel command **Data|Data Analysis|Random Number Generation**. Here's how we get Excel to produce 1,000 random numbers that are normally distributed (with $\mu = 0$ and $\sigma = 1$) in column G of the spreadsheet:



If we want to see whether the output is distributed normally, we can have Excel do a frequency distribution (either by using the array function **Frequency** or by using **Data|Data Analysis|Histogram**). As the graph above shows, the output appears to be normally distributed.

Method 2: Normally Distributed Numbers Using Norm.S.Inv(Rand())

The Excel function **Norm.S.Inv(Rand())** can be used to generate normally distributed random numbers. In order to understand the use of this function, we start with an explanation of the function **Norm.S.Dist**. The Excel function **Norm.S.Dist** computes values of the standard normal distribution. In the spreadsheet below, for example, we use **Norm.S.Dist**(0.5,1) to compute the probability that a normally distributed random number will be below or equal to 0.5. This is denoted as $N(0.5)$ and sometimes as

$\Phi(0.5)$. We also use **Norm.S.Dist(1,1)—Norm.S.Dist(-1,1)** to compute the percentage of the standard normal distribution that is between −1 and +1:

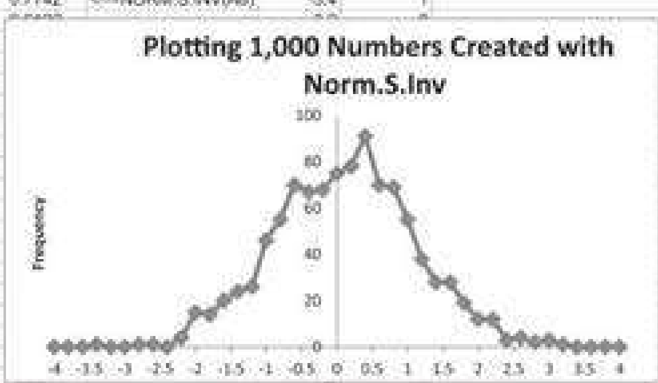
	A	B	C
1	USING NORM.S.DIST		
2	x	0.500	
3	N(x)	0.6915	<--=NORM.S.DIST(B2,1)
4			
5	x_1	1	
6	x_2	-1	
7	$N(x_2)$-$N(x_1)$	0.6827	<--=NORM.S.DIST(B5,1)-NORM.S.DIST(B6,1)

Excel's **Norm.S.Inv()** function inverts the function **Norm.S.Dist**. Given a number **x** between 0 and 1, **Norm.S.Inv(x)** produces number **y** such that **Norm.S.Dist(y,1)** = **x**. The function **NormSDist(Rand())** should produce a set of random numbers that is distributed standard normal:

	A	B	C
1	NORMALLY DISTRIBUTED RANDOM NUMBERS USING NORM.S.INV()		
2	Any number between 0 and 1	0.6000	
3	Normal number	0.2533	<--=NORM.S.INV(B2)
4	Check:	0.6000	<--=NORM.S.DIST(B3,1)
5			
6	Random normal number	-0.1655	<--=NORM.S.INV(RAND())

In the spreadsheet below, we produce 1,000 iterations of **Norm.S.Inv(Rand())** and graph the resulting frequencies. They look normally distributed:

	A	B	C	D	E	F
1	NORMALLY DISTRIBUTED RANDOM NUMBERS USING NORM.S.INV()					
2	Random numbers (produced with Rand())	Norm.S.Inv		Bin	Frequency	
3	0.9877	2.2483	<--=NORM.S.INV(A3)	-4	0	
4	0.1827	-0.9050	<--=NORM.S.INV(A4)	-3.8	0	
5	0.6897	0.4950	<--=NORM.S.INV(A5)	-3.6	0	<--{=FREQUENCY(B:B,D3:D43)}
6	0.7806	0.7742	<--=NORM.S.INV(A6)	-3.4	1	
7	0.7298					
8	0.1571					
9	0.4828					
10	0.6029					
11	0.5143					
12	0.0454					
13	0.8118					
14	0.7083					
15	0.0784					
16	0.8559					
17	0.4914					
18	0.1353					
19	0.2834					
20	0.7876					
21	0.4389					
22	0.3718					
23	0.3617					
24	0.2772					

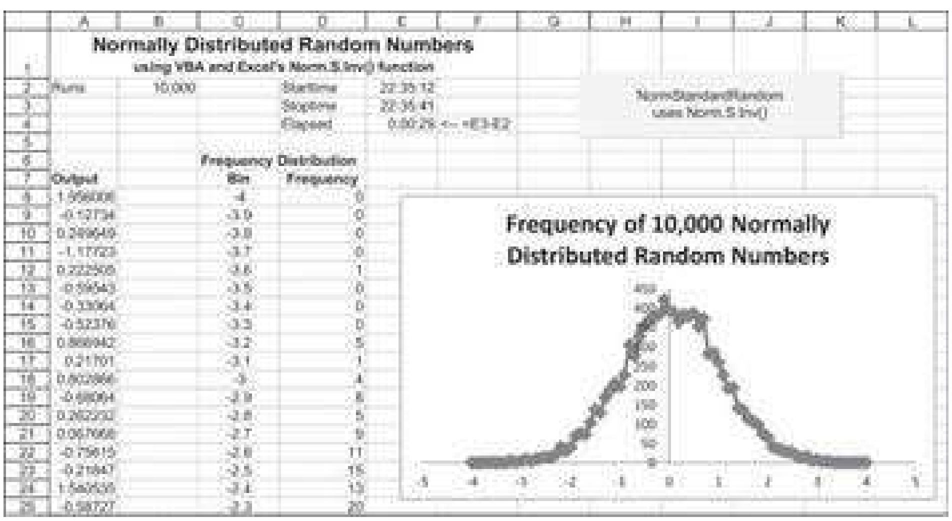


Method 3: Incorporating Norm.S.Inv() into VBA

The VBA program **NormStandardRandom** uses **Norm.S.Inv** to produce random deviates. Here is the program and below is the output it produces. Note that in VBA we replace the periods in the function with underscores, writing **Norm_S_Inv**.

```
Sub NormStandardRandom()
'Produces a list of normally distributed random numbers
Randomize 'Initializes the VBA random-number generator
Application.ScreenUpdating=False
Range("E2") = Time
    Range("A8").Range(Cells(1, 1), Cells(64000, 1)).Clear
    N = Range("B2").Value
    For Index = 1 To N
        Range("A8").Cells(Index, 1) = Application.Norm_S_Inv(Rnd)
    Next Index
Range("E3") = Time
End Sub
```

The program **NormStandardRandom** includes two lines that measure the time taken for the whole simulation to run. The program is very slow, largely because of repeated calls on the spreadsheet function. As you can see below, 10,000 runs of the program take about 29 seconds on the author's Lenovo Yoga 920. Here is a sample screen (the button operates the macro):



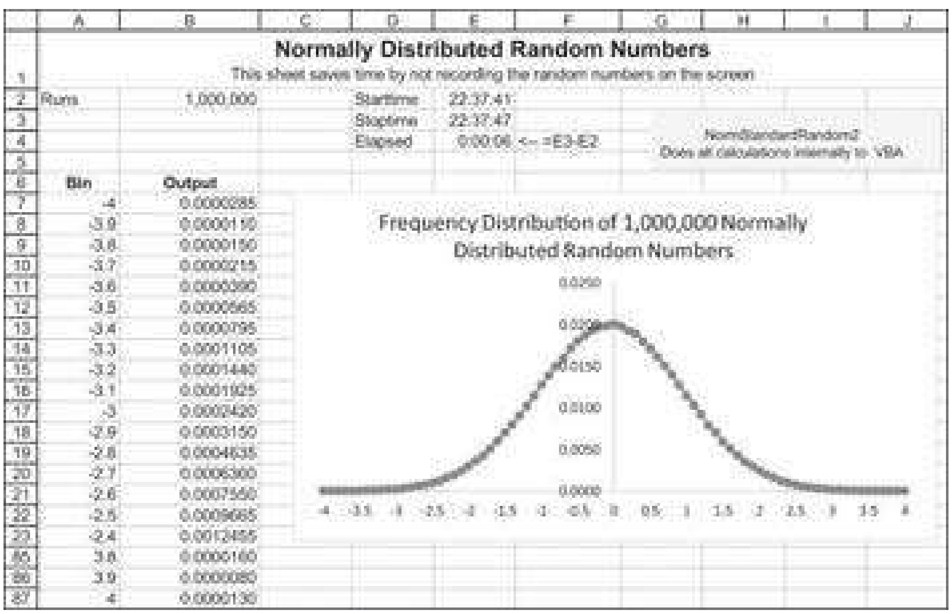
A Faster Version of Method 3

We can make method 3 much faster by storing all the data in VBA and only writing the final frequency distribution on the screen:

```
Sub NormStandardRandom2()
Randomize 'Initializes the VBA random number generator
Dim distribution(-40 To 40) As Double
```

```
Application.ScreenUpdating=False
Range("E2") = Time
    N = Range("B2").Value
    For Index = 1 To N
        X=Application.Norm_S_Inv(Rnd())
    If X < -4 Then
        distribution(-40) = distribution(-40) + 1
    ElseIf X > 4 Then
        distribution(40) = distribution(40) + 1
    Else: distribution(Int(X / 0.1)) = distribution(Int(X / 0.1))
    + 1
    End If
    Next Index
For Index = -40 To 40
    Range("B7").Cells(Index + 41, 1) = distribution(Index) / (2 *
    N)
Next Index
Range("E3") = Time
End Sub
```

Here's the output for 100,000 iterations. Note the time in cell E4:



There are a few things to note about this program:

- • Most of the results of the normal distribution are between −4 and +4. When, in **NormStandardRandom2**, we classify the output into bins, we want these bins to be (−∞, −3.9], (−2.9, −2.8], …, (−3.9, ∞).To do this we first define an array `distribution(-40 To 40)`; this array has 81 indices. To classify a particular random number (say, X) into the bins of this array, we use the function:

```
If X < -4 Then
    distribution(-40) = distribution(-40) + 1
    ElseIf X > 4 Then
    distribution(40) = distribution(40) + 1
    Else: distribution(Int(X / 0.1)) = distribution(Int(X / 0.1))
    + 1
End If
```

• **NormStandardRandom2** produces not a histogram (which is a *count* of how many times a number falls into a particular bin) but a *frequency distribution*. We do this by dividing by twice the number of runs (remember that each successful run produces two random numbers), $2N$, before we output the data to the spreadsheet:

```
For Index = -40 To 40
    Range("output").Cells(Index + 41, 1) = _
    distribution(Index) / (2 * N)
Next Index
```

• Finally, note that the command `Application.ScreenUpdating=False` makes a big difference! This command prevents both the updating of the output in the cells and the Excel chart. Try running the program with and without this command to see the effect.

Method 4: The Box-Muller Method

The Box-Muller method for creating randomly distributed normal deviates is the fastest method of the four.[7] The eight lines that follow `Start` in the VBA program below define a routine that, in each successful iteration, creates two numbers drawn from a standard normal distribution. The routine creates two random numbers, $rand_1$ and $rand_2$, between -1 and $+1$. If the sum of the squares of these numbers is within the unit circle, then the two normal deviates are defined by:

$$\{X_1, X_2\} = \left\{rand_1 * \sqrt{\frac{-2ln(S_1)}{S_1}}, rand_2 * \sqrt{\frac{-2ln(S_1)}{S_1}}\right\}$$

where

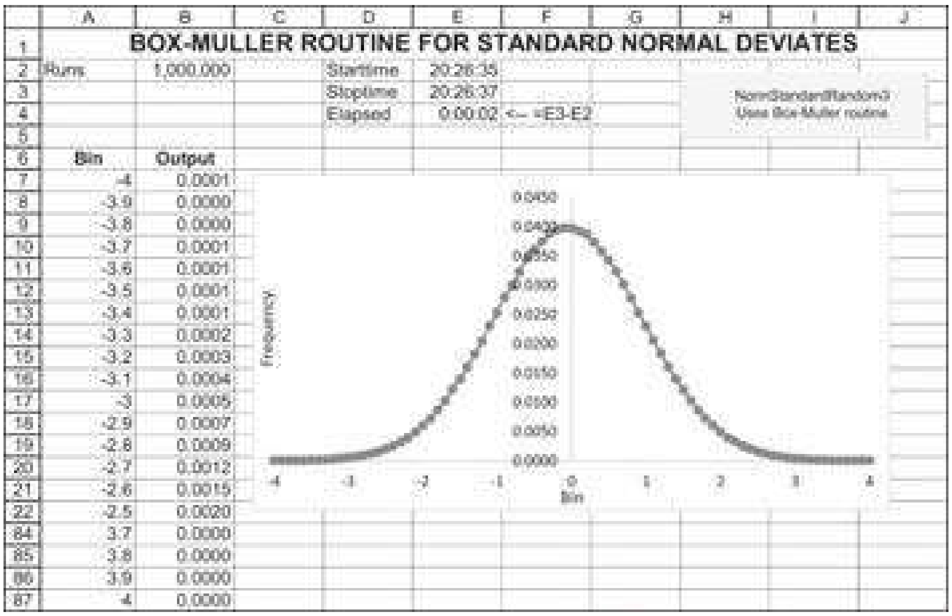
$$S_1 = rand_1^2 + rand_2^2$$

Here’s the VBA program:

```
Sub NormStandardRandom3()
'Box-Muller for producing standard normal deviates
Dim distribution(-40 To 40) As Long
Range("E2") = Time
N = Range("B2").Value
Application.ScreenUpdating=False
For Index = 1 To N
start:
    Static rand1, rand2, S1, S2, X1, X2
    rand1 = 2 * Rnd - 1
    rand2 = 2 * Rnd - 1
    S1 = rand1 ^ 2 + rand2 ^ 2
    If S1 > 1 Then GoTo start
    S2 = Sqr(-2 * Log(S1) / S1)
    X1 = rand1 * S2
    X2 = rand2 * S2
    If X1 < -4 Then
        distribution(-40) = distribution(-40) + 1
    ElseIf X1 > 4 Then
        distribution(40) = distribution(40) + 1
    Else: distribution(Int(X1 / 0.1)) = distribution(Int(X1 /
    0.1)) + 1
    End If
    If X2 < -4 Then
        distribution(-40) = distribution(-40) + 1
    ElseIf X2 > 4 Then
        distribution(40) = distribution(40) + 1
    Else: distribution(Int(X2 / 0.1)) = distribution(Int(X2 /
    0.1)) + 1
    End If
Next Index
For Index = -40 To 40
```

```
        Range("B7").Cells(Index + 41, 1) = distribution(Index) / (2 *
        N)
    Next Index
    Range("E3") = Time
    End Sub
```

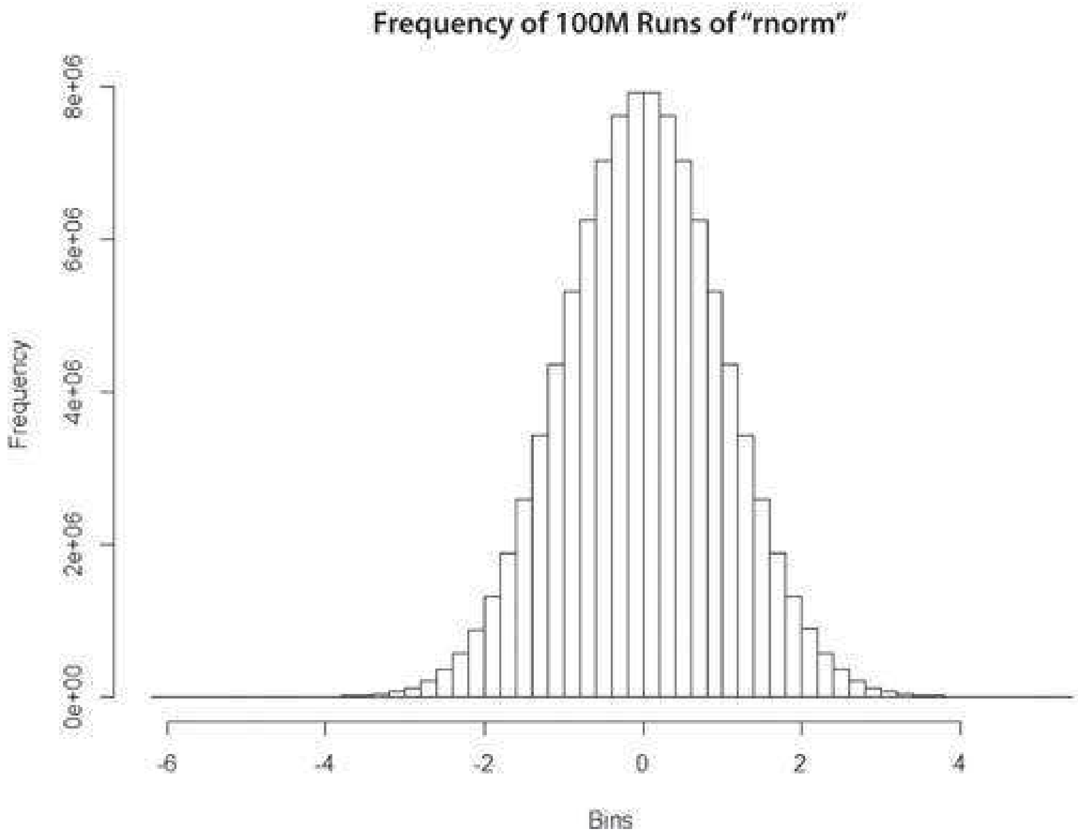
This routine is very fast; in the spreadsheet below we produce one million normal variates in one second:



Using R to Generate Standard Normal Deviates

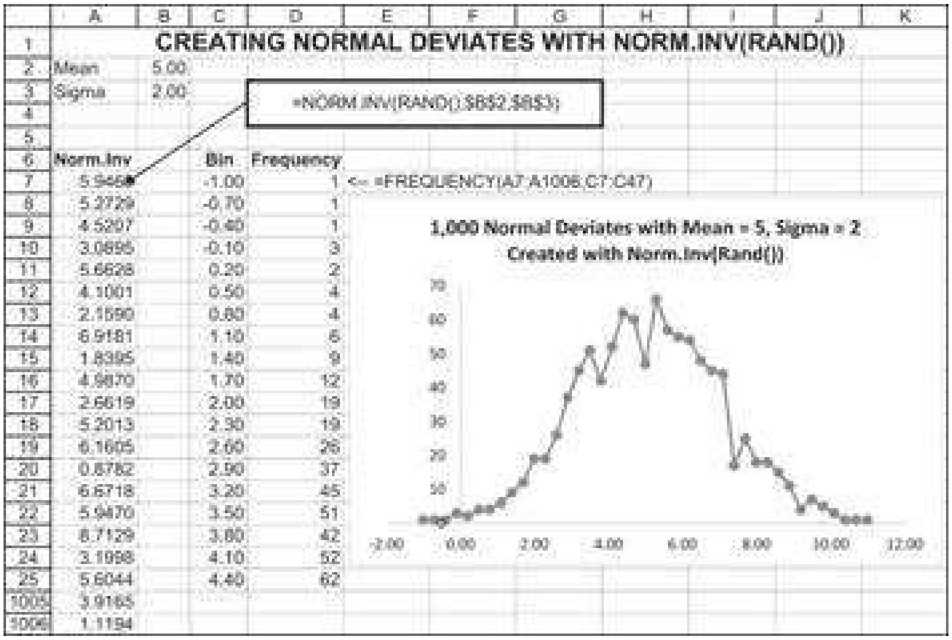
R is very useful in generating standard normal deviates. The function is "rnorm." The code to do so is very efficient and looks like this:

```
### 21.4 Generating a (standard) normally distributed random numbers
norm_rand_var <- rnorm(n = 100000000, mean = 0, sd = 1)

# Histogram
hist(norm_rand_var, breaks = 50,
     main="Frequency of 100M runs of 'rnorm'", xlab="Bins")
```



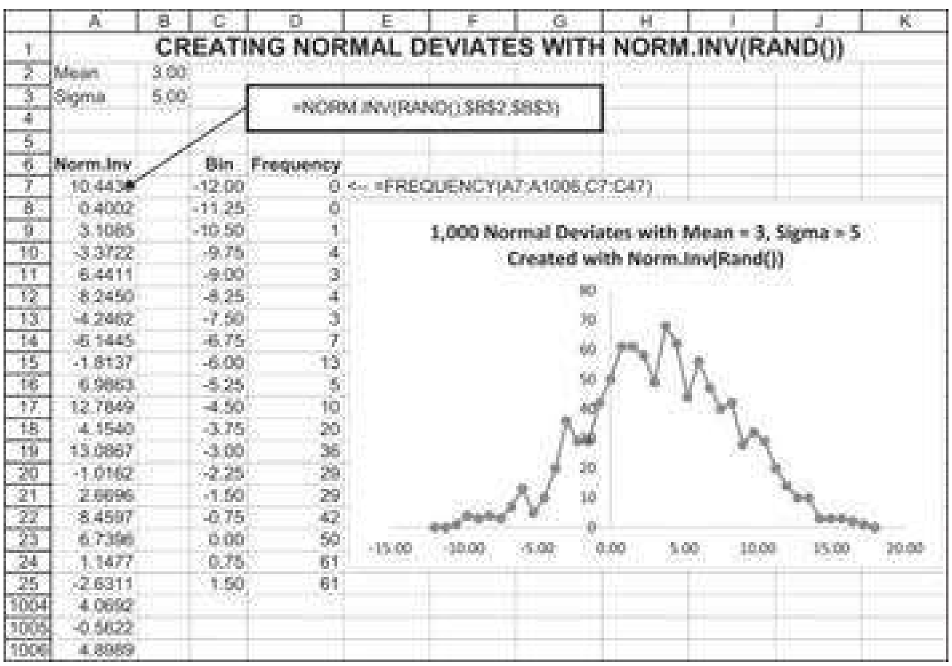
21.5 Norm.Inv: Another Way to Generate Normal Deviates

The Excel function **Norm.Inv(rand(),mean,sigma)** can also generate normal deviates. Whereas **Norm.S.Inv(rand())** generates only standard normal deviates, we can use **Norm.Inv** to change the mean and standard deviation of our deviates. In Chapter 23, we will sometimes use this function to generate normally distributed stock returns.



Norm.Inv as Alternative to Norm.S.Inv

Norm.S.Inv(Rand()) inverts the standard normal distribution and can be used to create standard normal deviates. The Excel function **Norm.Inv(Rand(), mean,sigma)** can be used to create normal deviates that are normally distributed with any normal distribution. The example below illustrates:



Getting Ahead of Ourselves: Do We Prefer Norm.S.Inv or Norm.Inv?

Ultimately, our goal is to simulate stock returns; these (as we discuss in Chapter 23) are generally assumed to be normally distributed. If we are simulating the returns of a stock based on annual mean μ and standard deviation σ, then it is obviously more convenient to use Norm.S.Inv(rand(),μ,σ). As we discuss in Chapter 23, if (μ, σ) are the annual statistics of the stock return, and if we divide the year into n subperiods, then the periodic stock returns can be simulated with $Norm.Inv(Rand(), \mu \cdot \Delta t, \sqrt{\Delta t} \cdot \sigma)$, where $\Delta t = 1/n$.

On the other hand, much financial theory is phrased in terms of lognormal price processes. These are generally written as $r = \mu \cdot \Delta t + \sigma \cdot \sqrt{\Delta t} \cdot Z$, where Z is a standard normal deviate. To conform to this writing, it is often more convenient (and theoretically equivalent) to simulate returns by using:

$$r = \mu \cdot \Delta t + \sigma \cdot \sqrt{\Delta t} \cdot Norm.S.Inv(Rand())$$

In this book, we interchangeably use both writings.

21.6 Scaling Normally Distributed Numbers

In many cases we are interested in normally distributed random numbers that are not standard normal distribution (with mean not equal to 0 and standard-deviation other than 1). This scaling is relatively easy. If we are interested in a normally distributed number that has mean of μ and a standard deviation of σ (i.e., $N \sim N(\mu, \sigma)$), we should use the following scaling formula:

$$N = \mu + \sigma * \underbrace{Norm.S.Dist(Rand())}_{\sim N(0,1)}$$

The following Excel clip uses this procedure:

	A	B	C
1	SCALING NORMAL DEVIATES		
2	Mean (μ)	3	
3	Sigma (σ)	5	
4			
5	N_1~N(0,1)	0.2639	<--=NORM.S.INV(RAND())
6	N_2~N(μ,σ)	4.3196	<--=$B$2+$B$3*B5

21.7 Generating Correlated Random Numbers

In this section we show how to produce correlated pseudo-random numbers. We start by showing how to generate two correlated normal and uniform random numbers and then go on to the multivariate case.

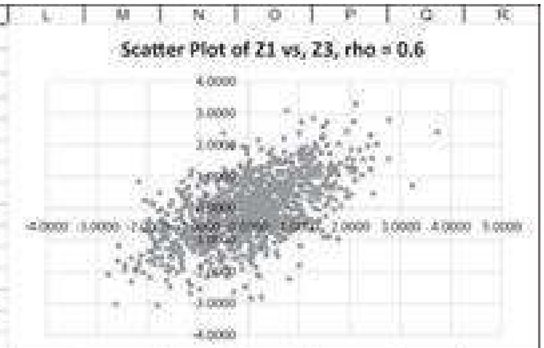
Example 1: Two Correlated Standard Normal Variables

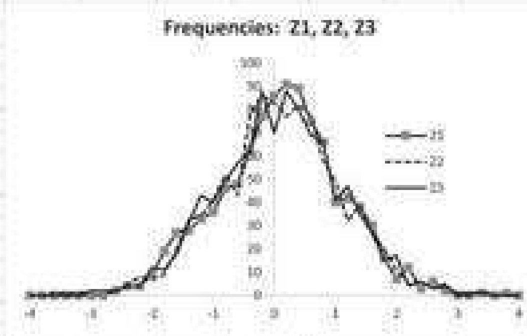
Suppose that Z_1 and Z_2 are standard normal deviates (created in the spreadsheet below with **Norm.S.Inv(Rand())**. Defining $z_3 = \rho z_1 + z_2\sqrt{1-\rho^2}$, we create a set of simulates (Z_3) that has the desired correlation with Z_1:

	A	B	C	D	E
1	1,000 CORRELATED STANDARD NORMAL DEVIATES				
2	Desired correlation (ρ)	0.6			
3		Z_1	Z_2	Z_3	
4	Mean	0.0421	0.0155	0.0376	<--=AVERAGE(D12:D1011)
5	Sigma	1.0129	1.0137	1.0292	<--=STDEV.S(D12:D1011)
6	Skewness	0.0260	0.0591	0.1419	<--=SKEW(D12:D1011)
7	Kurtosis	-0.1475	-0.2036	-0.2256	<--=KURT(D12:D1011)
8	Count	1000	1000	1000	<--=COUNT(D12:D1011)
9	Correlation coefficient (z_1,z_3)	0.616	<--=CORREL(B12:B1011,D12:D1011)		
10					
11		Z_1	Z_2	Z_3	
12	=NORM.S.INV(RAND()) -->	0.8083	0.0504	0.5252	<--=$B$2*B12+SQRT(1-$B$2^2)*C12
13	=NORM.S.INV(RAND()) -->	-0.8004	0.2876	-0.2502	<--=$B$2*B13+SQRT(1-$B$2^2)*C13
14	=NORM.S.INV(RAND()) -->	-0.9457	0.6573	-0.0416	<--=$B$2*B14+SQRT(1-$B$2^2)*C14
15		-0.0813	2.5937	2.0262	
1010		0.2184	0.7184	0.7057	
1011		-0.1088	-0.2547	-0.2690	

The frequencies and distributions are shown below:

	G	H	I	J
11				
12	Frequency distribution, Z_1, Z_2, Z_3			
13		Z_1	Z_2	Z_3
14	-4	0	0	0
15	-3.8	0	0	0
16	-3.6	0	1	0
17	-3.4	0	1	0
18	-3.2	0	1	0
19	-3	0	2	2
20	-2.8	0	2	2
21	-2.6	2	1	2
22	-2.4	4	5	4
23	-2.2	4	8	4
24	-2	9	7	12
25	-1.8	19	9	11
26	-1.6	27	20	18
27	-1.4	28	33	31
28	-1.2	33	34	43
29	-1	36	41	40
30	-0.8	46	52	49
31	-0.6	48	43	57
32	-0.4	62	78	62
33	-0.2	77	86	88
34	0	86	83	70
35	0.2	91	77	88
36	0.4	89	82	80
37	0.6	75	77	69
38	0.8	66	58	66
39	1	40	49	41
40	1.2	42	32	47
41	1.4	38	39	35
42	1.6	30	27	27
43	1.8	16	20	15





The equivalent code in R looks like this:

```
## Two correlated Standard Normal Variables
# Start by generating two uncorrelated normal deviates:
Z1 <- rnorm(n = 1000000, mean = 0, sd = 1)
Z2 <- rnorm(n = 1000000, mean = 0, sd = 1)

# Input:
Rho <- 0.6

# Generate Z3 that is a normal deviate and is correlated with Z1
Z3 <- Z1 * Rho + Z2 * sqrt(1 - Rho^2)
cor(Z1,Z3) # Checking the correlation of Z1 and Z3
```

This code, in the simulation run on the authors' machines, generated a correlation of 0.582—very close to the 0.6 desired correlation.

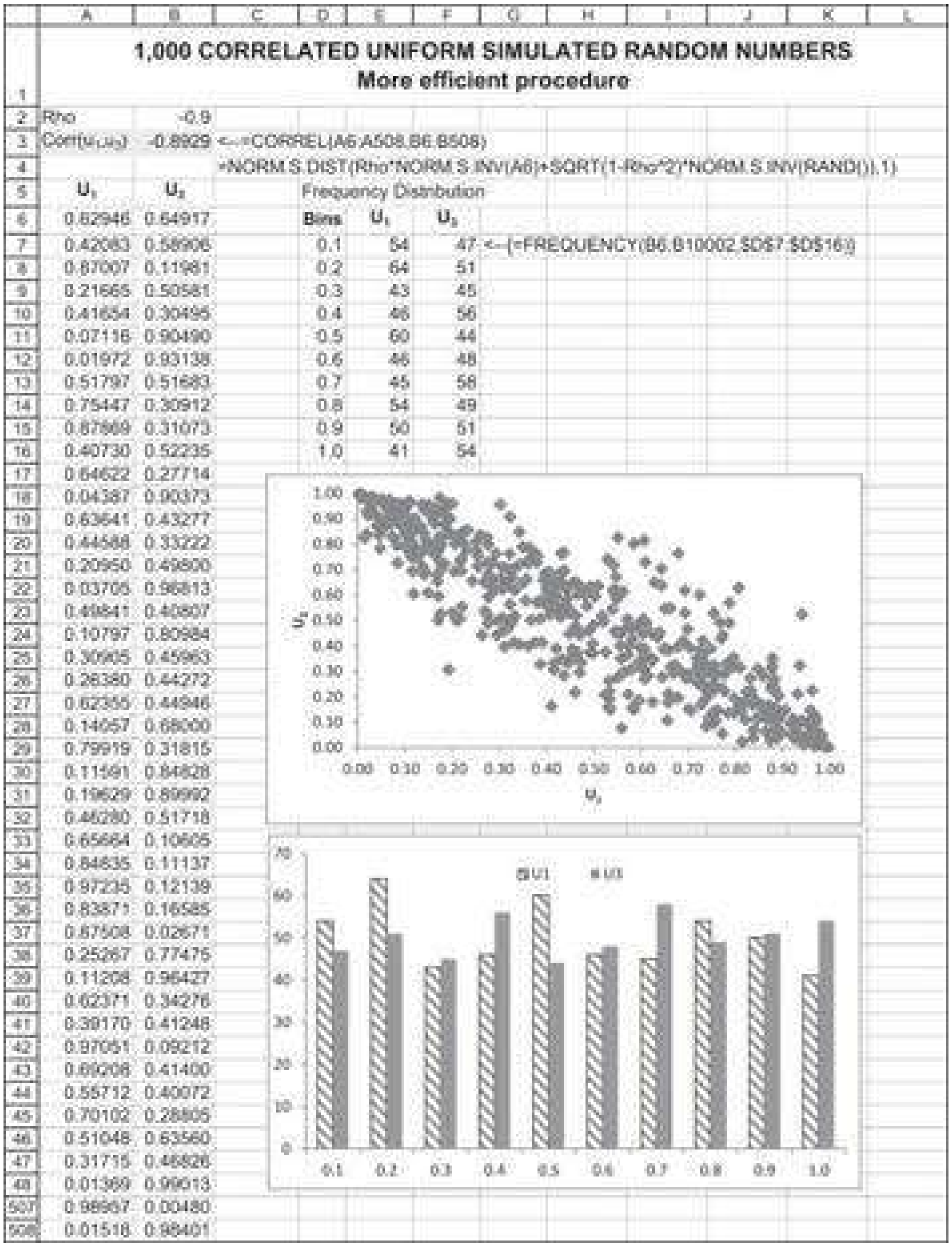
Example 2: Two Correlated Uniform Variables

To create two uniform variables (u_1 and u_3) that are correlated, we first produce correlated standard normal variables, then use **Norm.S.Dist** to produce the uniform variables. Here is the procedure:

- • Use **Norm.S.Inv(Rand())** to produce two random numbers that are normally distributed. Call these numbers z_1 and z_2.
- • Let $z_3 = \rho z_1 + z_2\sqrt{1-\rho^2}$. As we showed above, z_3 and z_1 are correlated with correlation ρ.
- • Now define u_1=**Norm.S.Dist(z$_1$,1)** and u_3=**Norm.S.Dist(z$_3$,1)**. Then u_1 and u_3 are uniformly distributed and correlated with correlation ρ.

We show this routine twice. The first time is in an extensive spreadsheet (see below). Note the "diagonal aspect" of the scatter diagram of u_3 versus u_1 (this shows the correlation).

We can make this process more efficient (and also more obscure) by combining functions, as shown in the next spreadsheet. Column A cells are defined with **Rand()**. Column B contains the formula **=NORM.S.DIST(Rho*NORM.S.INV(A7)+SQRT(1-Rho^2)*NORM.S.INV(RAND()),1)**.



1,000 CORRELATED UNIFORM SIMULATED RANDOM NUMBERS
More efficient procedure

Row	A	B	C	D	E	F	G
2	Rho	-0.9					
3	Corr(u_1,u_2)	-0.8929	<--=CORREL(A6:A508,B6:B508)				
4			=NORM.S.DIST(Rho*NORM.S.INV(A6)+SQRT(1-Rho^2)*NORM.S.INV(RAND()),1)				
5	U_1	U_2		Frequency Distribution			
6	0.62946	0.64917		Bins	U_1	U_2	
7	0.42083	0.58906		0.1	54	47	<--{=FREQUENCY(B6:B10002,$D$7:$D$16)}
8	0.87007	0.11981		0.2	64	51	
9	0.21665	0.50581		0.3	43	45	
10	0.41654	0.30495		0.4	46	56	
11	0.07116	0.90490		0.5	60	44	
12	0.01972	0.93138		0.6	46	48	
13	0.51797	0.51683		0.7	45	58	
14	0.75447	0.30912		0.8	54	49	
15	0.87869	0.31073		0.9	50	51	
16	0.40730	0.52235		1.0	41	54	
17	0.64622	0.27714					
18	0.04387	0.90373					
19	0.63641	0.43277					
20	0.44588	0.33222					
21	0.20950	0.49800					
22	0.03705	0.96813					
23	0.49841	0.40807					
24	0.10797	0.80984					
25	0.30905	0.45963					
26	0.26380	0.44272					
27	0.62355	0.44946					
28	0.14057	0.68000					
29	0.79919	0.31815					
30	0.11591	0.84828					
31	0.19629	0.89992					
32	0.46280	0.51718					
33	0.65664	0.10605					
34	0.84835	0.11137					
35	0.97235	0.12139					
36	0.83871	0.16585					
37	0.87508	0.02671					
38	0.25267	0.77475					
39	0.11208	0.96427					
40	0.62371	0.34276					
41	0.39170	0.41248					
42	0.97051	0.09212					
43	0.69208	0.41400					
44	0.55712	0.40072					
45	0.70102	0.28805					
46	0.51048	0.63560					
47	0.31715	0.46826					
48	0.01369	0.99013					
507	0.98957	0.00480					
508	0.01518	0.98401					

Implementing this in R looks like this:

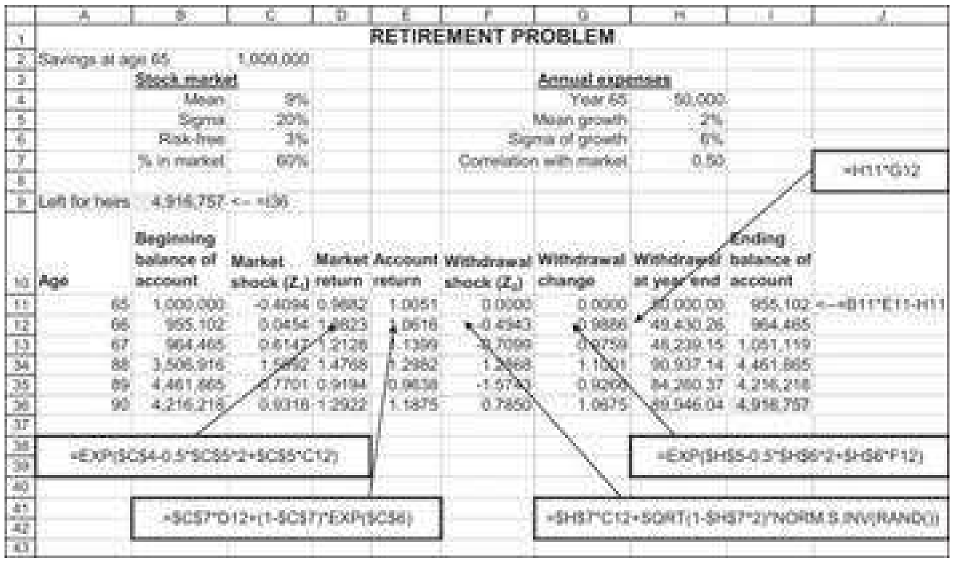
```
# Convert to two correlated uniform variables
U1 <- pnorm(Z1, mean = 0, sd = 1, lower.tail = TRUE)
U3 <- pnorm(Z3, mean = 0, sd = 1, lower.tail = TRUE)

# Checking the correlation of U1 and U3
cor(U1,U3)
```

21.8 What's Our Interest in Correlation? A Small Case[8]

Jacob has just retired at age 65 with savings of $1 million. He intends to invest 60% of this in a market index fund and the remaining 40% in a risk-free asset. He estimates the return on the risk-free asset as r_f = 3% annually, and he estimates that the market portfolio will have normally distributed returns with mean return μ = 9% and σ = 20%.

Jacob intends to withdraw $50,000 from his account at the end of this year. He thinks that this amount will, on average grow at 3% per year, with a standard deviation of 6%. Furthermore, he thinks that the growth rate of his annual spending with be correlated ρ = 0.5 with the stock market.[9]

Our question: If Jacob's life expectancy is age 90, how much will he leave to his beloved son Simon? Here is a simulation that answers this question (some rows are hidden):



Some details of the simulation:

- • Whatever savings are left at the end of a particular year constitute the initial savings for the next year.
- • Z_1 and Z_2 are correlated standard normal variables. Note the formulas:

Z_1 = ***Norm.S.Inv***(***Rand***()),

$Z_2 = \rho * Z_1 +$ ***Sqrt***$(1 - \rho^2) *$ ***Norm.S.Inv***(***Rand***())

In this particular simulation, Jacob leaves a $4,916,757 for his heirs. Using the technique of running a data table on a blank cell (see Chapter 28), we can run many simulations of this same problem:

	L	M	N
11	**Data table:**	**20 simulations of bequest**	
12	Simulation		
13	1	4,916,757	<--=$B$9, data table header
14	2	184,456	
15	3	454,990	
16	4	221,263	
17	5	715,299	
18	6	209,259	
19	7	1,154,062	
20	8	-819,960	
21	9	4,425,121	
22	10	2,349	
23	11	1,139,100	
24	12	1,301,247	
25	13	382,779	
26	14	4,485,559	
27	15	458,402	
28	16	1,479,179	
29	17	126,445	
30	18	413,479	
31	19	291,809	
32	20	7,112,280	
33			
34	Average	1,432,694	<-- =AVERAGE(M13:M32)
35	Sigma	2,078,282	<-- =STDEV.S(M13:M32)
36	Min	-819,960	<-- =MIN(M13:M32)
37	Max	7,112,280	<-- =MAX(M13:M32)
38	Negative	1	<-- =COUNTIF(M13:M32,"<0")

On average, Jacob's heirs should do pretty well. But in one out of the 20 simulations, his heirs will be left intestate. They might also use **Data Table** to check whether the correlation affects their inheritance.

	P	Q	R	S	T	U	V
10			=$B$9, data table header				
11	Data table: 20 simulations of bequest for different correlations						
12		-0.80	-0.40	0.00	0.40	0.80	<-- correlation
13	1	68,297	619,183	2,321,331	5,346,583	-316,552	
14	2	-192,765	812,620	933,513	494,140	467,472	
15	3	1,138,587	7,247,582	-556,280	84,664	5,729,569	
16	4	7,245,074	-710,652	206,418	691,018	3,215,670	
17	5	163,382	2,302,918	1,626,398	1,302,367	1,972,439	
18	6	2,802,985	2,537,870	2,605,451	3,179,629	3,845,860	
19	7	1,877,881	1,713,618	5,381,971	943,328	448,468	
20	8	2,842,405	-450,402	-149,991	3,653,142	341,838	
21	9	830,500	717,112	-748,786	1,060,614	1,007,724	
22	10	730,267	1,351,204	708,875	-644,918	2,502,479	
23	11	-125,901	-279,274	-456,514	3,349,785	665,521	
24	12	5,387,932	-728,740	3,436,911	1,484,252	190,683	
25	13	5,685,703	11,231,580	2,073,198	-224,699	-197,962	
26	14	-996,071	121,732	-421,460	884,773	2,185,810	
27	15	5,831,225	2,202,042	10,670,635	4,687,831	1,882,234	
28	16	-506,170	1,011,221	-1,166,483	1,691,880	1,780,238	
29	17	2,765,523	960,751	128,149	9,450,000	2,741,953	
30	18	115,496	162,329	2,049,192	-442,311	2,277,788	
31	19	1,419,299	6,300,926	3,622,498	1,747,645	329,853	
32	20	1,460,223	-753,556	2,177,613	4,171,357	2,058,757	
33							
34	Average	1,927,194	1,818,503	1,717,081	2,145,554	1,656,492	<--=AVERAGE(U13:U32)
35	Sigma	2,391,790	3,069,574	2,724,391	2,448,171	1,520,857	<--=STDEV.S(U13:U32)
36	Min	-996,071	-753,556	-1,166,483	-644,918	-316,552	<--=MIN(U13:U32)
37	Max	7,245,074	11,231,580	10,670,635	9,450,000	5,729,569	<--=MAX(U13:U32)
38	Negative	4	5	6	3	2	<--=COUNTIF(U13:U32,"<0")

As might be expected, in general the higher the negative correlation between the stock market and Jacob's expenses, the lower will be the inheritance. This makes sense: With negative correlation, when the stock market goes down, expenses go up (and vice versa).

21.9 Multiple Random Variables with Correlation: The Cholesky Decomposition

We can also create multiple correlated random deviates by using the Cholesky decomposition. A bit of background: A square matrix is called *positive definite* if, for any row vector x, the product $x \cdot S \cdot x^T > 0$. The variance-covariance matrix for security returns discussed in Chapters 10–14,

$$S = \begin{bmatrix} \sigma_{11} & \sigma_{12} & \cdots & \sigma_{1N} \\ \sigma_{21} & \sigma_{22} & & \sigma_{2N} \\ \vdots & & \ddots & \vdots \\ \sigma_{N1} & \sigma_{N2} & \cdots & \sigma_{NN} \end{bmatrix},$$

is positive definite and symmetric (since $\sigma_{ij} = \sigma_{ji}$). A square matrix is called *upper triangular* if it has non-zero entries only on the diagonal and elements above the diagonal. A square matrix is called *lower triangular* if (you can fill in the blanks).

The French mathematician André-Louis Cholesky (1875–1918) proved that any symmetric positive-definite matrix S can be written as the product of a lower-triangular matrix L and its transpose L^T. This is the *Cholesky decomposition*.

Example

In the example below, cells A3:D6 contain a 4 × 4 variance-covariance matrix. Cells B8:D11 contain the Cholesky decomposition of this matrix—a lower-triangular matrix

L. Cells A15:D18 multiply L times its transpose. As you can see, the result is to give back the original variance-covariance matrix.

	A	B	C	D	E
1	CHOLESKY DECOMPOSITION OF SYMMETRIC POSITIVE-DEFINITE MATRIX				
2	Variance-covariance matrix, S				
3	0.400	0.030	0.020	0.000	
4	0.030	0.200	0.000	-0.060	
5	0.020	0.000	0.300	0.030	
6	0.000	-0.060	0.030	0.100	
7					
8	Cholesky decomposition, L				
9	0.632	0.000	0.000	0.000	<--{=cholesky(A3:D6)}
10	0.047	0.445	0.000	0.000	
11	0.032	-0.003	0.547	0.000	
12	0.000	-0.135	0.054	0.281	
13					
14	Check: Multiply above matrix by its transpose				
15	0.400	0.030	0.020	0.000	<--{=MMULT(A9:D12,TRANSPOSE(A9:D12))}
16	0.030	0.200	0.000	-0.060	
17	0.020	0.000	0.300	0.030	
18	0.000	-0.060	0.030	0.100	

The VBA function **Cholesky** can be found on the companion website for *Financial Modeling*.[10]

R has a prebuilt function to compose the Cholesky decomposition. The code below shows how to apply it in our example:

```
## 21.9  Multiple Random Variables with Correlation: The Cholesky Decomposition
# Example
cov_mat <- matrix(c(0.40, 0.03, 0.02, 0.01,
                    0.03, 0.30, 0.00, -0.06,
                    0.02, 0.00, 0.20, 0.03,
                    0.01, -0.06, 0.03, 0.10),
                    nrow = 4)

chol_decompos <- t(chol(cov_mat)) # Cholesky decomposition

# Check
chol_decompos %*% t(chol_decompos)
```

Running this code results in:

```
> chol_decompos
           [,1]         [,2]       [,3]      [,4]
[1,] 0.63245553  0.000000000 0.00000000 0.0000000
[2,] 0.04743416  0.545664732 0.00000000 0.0000000
[3,] 0.03162278 -0.002748941 0.44608569 0.0000000
[4,] 0.01581139 -0.111332099 0.06544472 0.2882224
> # Check
> chol_decompos %*% t(chol_decompos)
     [,1]  [,2] [,3]  [,4]
[1,] 0.40  0.03 0.02  0.01
[2,] 0.03  0.30 0.00 -0.06
[3,] 0.02  0.00 0.20  0.03
[4,] 0.01 -0.06 0.03  0.10
> 
```

Using the Cholesky Decomposition to Produce Correlated Normal Simulate

We start with a var-cov matrix $S = \begin{bmatrix} \sigma_{11} & \sigma_{12} & \sigma_{13} & \sigma_{14} \\ \sigma_{21} & \sigma_{22} & \sigma_{23} & \sigma_{24} \\ \sigma_{31} & \sigma_{32} & \sigma_{33} & \sigma_{34} \\ \sigma_{41} & \sigma_{42} & \sigma_{43} & \sigma_{44} \end{bmatrix}.$ We want to run a simulation which in each iteration produces a vector $\begin{Bmatrix} z_{1t} \\ z_{2t} \\ z_{3t} \\ z_{4t} \end{Bmatrix}$ of four random numbers so that these numbers have the following properties:

- • z_i is distributed normally.
- • The mean of each of the numbers is zero: $\frac{1}{n}\sum_{t=1}^{n} z_{it} \approx 0.$ (Because of randomness, we can never demand exact equality to zero, only an approximation.)
- • The variance of each series of numbers matches the var-cov matrix:

$Var\{z_{11}, z_{12}, z_{13}, \ldots\} \approx \sigma_{11}$
$Var\{z_{21}, z_{22}, z_{23}, \ldots\} \approx \sigma_{22}$
…

- • The covariance of any two of the series of numbers matches the covariance in the var-cov matrix:

$Cov\{(z_{11}, z_{12}, z_{13}, \ldots), (z_{21}, z_{22}, z_{23}, \ldots)\} \approx \sigma_{21} = \sigma_{12}$ etc.

The way to do this is in two steps:

1. Produce a set of normally distributed numbers between 0 and 1.
2. Pre-multiply each vector of such numbers by the Cholesky decomposition L.

In the illustration below, we show one step of this simulation:

	A	B	C	D	E	F
1	**MULTIVARIATE NORMAL SIMULATION**					
2	**Variance-covariance matrix**					
3	0.40	0.03	0.02	0.01		
4	0.03	0.30	0.00	-0.06		
5	0.02	0.00	0.20	0.03		
6	0.01	-0.06	0.03	0.10		
7						
8	**Cholesky decomposition**					
9	0.6325	0.0000	0.0000	0.0000	<--{=cholesky(varcov)}	
10	0.0474	0.5457	0.0000	0.0000		
11	0.0316	-0.0027	0.4461	0.0000		
12	0.0158	-0.1113	0.0654	0.2882		
13						
14	**Generating four random normal variates**					
15		-0.9351	<--=NORM.S.INV(RAND())			
16		0.8520	<--=NORM.S.INV(RAND())			
17		-0.3316	<--=NORM.S.INV(RAND())			
18		-1.0994	<--=NORM.S.INV(RAND())			
19						
20	**Generating multinomial normal output**					
21	Z_1	-0.5914	<--{=MMULT($A$9:$D$12,B15:B18)}			
22	Z_2	0.4206				
23	Z_3	-0.1798				
24	Z_4	-0.4482				

In cells B15:B18, we use **Norm.S.Inv(Rand())** to produce four random numbers, each of which is normally distributed with mean 0 and standard deviation 1.[11] In cells B21:B24, we pre-multiply this random vector by the Cholesky matrix in cells A9:D12. The claim is that the numbers in B21:B24 are normally distributed with mean 0 and the variance-covariance structure given by the var-cov matrix. We cannot, of course, prove this claim from a single simulation. In the spreadsheet below, we replicate the above procedure 400 times:

	A	B	C	D	E	F	OJ	OK	OL
26	Trial	1	2	3	4	5	399	400	
27	Non-correlated normal deviates (using Norm.S.Inv(rand()))								
28	Z_1	-1.4170	0.1287	0.3837	1.0446	-0.3449	-0.6952	-1.2292	<--=NORM.S.INV(RAND())
29	Z_2	1.5456	-0.5539	-1.0315	0.0894	1.6447	0.6925	-0.1355	<--=NORM.S.INV(RAND())
30	Z_3	0.5854	0.2363	1.0868	-0.7650	0.9927	-0.5196	-0.3432	
31	Z_4	0.5011	-0.2640	-0.3857	-2.3177	0.9521	0.6830	-0.7535	
32									
33	Correlated normal deviates (formula in cells $B$34:$B$37 is: {=MMULT($A$9:$D$12,B28:B31)})								
34	Z_1	-0.8962	0.0891	0.2427	0.6608	-0.2180	-0.4397	-0.7774	<--{=MMULT($A$9:$D$12,OK28:OK31)}
35	Z_2	0.7761	-0.2962	-0.5444	0.0963	0.8811	0.2963	-0.1322	<--{=MMULT($A$9:$D$12,OK28:OK31)}
36	Z_3	0.2121	0.1110	0.4998	-0.3084	0.4274	-0.2581	-0.1916	
37	Z_4	-0.0117	-0.0027	0.0808	-0.7115	0.1502	0.1436	-0.2440	
38									
39		Checking							
40	Number of simulations	400	<--=COUNT(28:28)						
41									
42	Mean1	-0.03313	<--=AVERAGE(34:34)						
43	Mean2	-0.02926	<--=AVERAGE(35:35)						
44	Mean3	0.04381	<--=AVERAGE(36:36)						
45	Mean4	0.02656	<--=AVERAGE(37:37)						
46									
47	Var-cov matrix from simulation								
48		Z_1	Z_2	Z_3	Z_4				
49	Z_1	0.450	0.028	0.016	0.017	<--=COVARIANCE.P($B37:$37,34:34)			
50	Z_2	0.028	0.324	-0.006	-0.066	<--=COVARIANCE.P($B37:$37,35:35)			
51	Z_3	0.016	-0.006	0.206	0.029	<--=COVARIANCE.P($B37:$37,36:36)			
52	Z_4	0.017	-0.066	0.029	0.095	<--=COVARIANCE.P($B37:$37,37:37)			
53									
54	Variance-covariance matrix								
55		1	2	3	4				
56	1	0.400	0.030	0.020	0.010				
57	2	0.030	0.300	0.001	-0.060				
58	3	0.020	0.000	0.200	0.030				
59	4	0.010	-0.060	0.030	0.100				

In rows 42–59, we check the mean, variance, and covariance of each of the variables, comparing the means to zero and the variances and covariances to the data in the var-cov matrix. By pressing **F9**, you can repeat the simulations and convince yourself that we have indeed produced a set of multivariate normal simulations corresponding to the target variance-covariance structure.

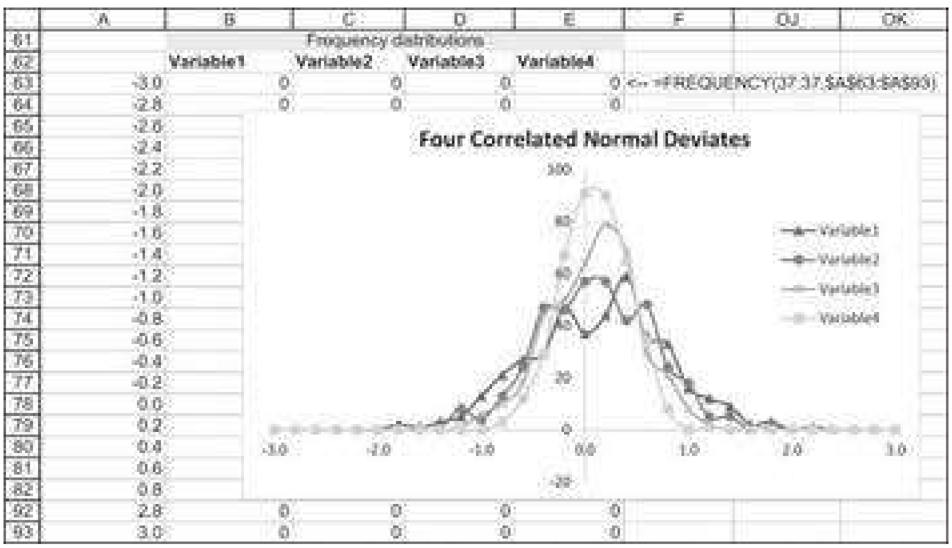
Using R to Generate Multiple Normal Deviates

Implementing the Cholesky decomposition approach in R is very easy and efficient. The code for the above example is:

```
## Using the Cholesky Decomposition to Produce Correlated Normal Deviates
# Step 1: Non-corelated normal deviates:
norm_var <- matrix(rnorm(n = 1000000 * 4, mean = 0, sd = 1), nrow = 4)
# Step 2: Corelated normal deviates:
corr_norm_var <- t(chol_decompos %*% norm_var)

# Check
colMeans(corr_norm_var)
cov(corr_norm_var)

# Density plot
plot(density(corr_norm_var[,1]), col = 1, ylim = c(0,1.5),
     main = "Four correlated normal deviates")
    lines(density(corr_norm_var[,2]), col = 2)
    lines(density(corr_norm_var[,3]), col = 3)
    lines(density(corr_norm_var[,4]), col = 4)
```

Further Convincing

Another way to convince yourself that we have done something sensible is to plot the frequency distribution of each of the variables, using the Excel **Frequency** function:



Note: Simulations are tricky! Press **F9** multiple times to get different frequencies and statistics.

21.10 Multivariate Uniform Simulations

Once we have simulated, correlated normal distributions, we can easily simulate uniform distributions by using **Norm.S.Dist**. In the example below, we first create four series of correlated standard normal deviates (in rows 34–37). Next, we generate uniform variates by applying **Norm.S.Dist** to the normal variates as shown below:

	A	B	C	D	E	F	OJ	OK	OL
1	MULTIVARIATE UNIFORM SIMULATION								
2	Variance-covariance matrix								
3	0.40	0.03	0.02	0.01					
4	0.03	0.30	0.00	-0.06					
5	0.02	0.00	0.20	0.03					
6	0.01	-0.06	0.03	0.10					
7									
8	Cholesky decomposition								
9	0.6325	0.0000	0.0000	0.0000	<--{=cholesky(varcov)}				
10	0.0474	0.5457	0.0000	0.0000					
11	0.0316	-0.0027	0.4461	0.0000					
12	0.0158	-0.1113	0.0654	0.2682					
13									
14	Generating four random normal variates								
15		-0.9312	<--=NORM.S.INV(RAND())						
16		-0.3086	<--=NORM.S.INV(RAND())						
17		0.2259	<--=NORM.S.INV(RAND())						
18		-1.1496	<--=NORM.S.INV(RAND())						
19									
20	Generating multinomial normal output								
21	Z_1	-0.5860	<--{=MMULT($A$9:$D$12,B15:B18)}						
22	Z_2	-0.2126							
23	Z_3	0.0722							
24	Z_4	-0.2969							
25									
26	Trial	1	2	3	4	5	399	400	
27	Non-correlated normal deviates (using Norm.S.Inv(rand()))								
28	Z_1	-0.4040	-1.3845	-2.1923	-1.7359	0.4976	-0.7783	0.1762	<--=NORM.S.INV(RAND())
29	Z_2	-1.5526	0.7755	0.0696	-0.5805	1.2085	0.1440	0.8745	<--=NORM.S.INV(RAND())
30	Z_3	0.8180	-1.0582	0.1378	-0.4396	-1.1187	-0.4904	0.6056	
31	Z_4	0.9366	0.1928	-0.2011	0.8313	-0.1711	1.6068	1.0134	
32									
33	Correlated normal deviates (formula in cells $B$34:$B$37 is: {=MMULT($A$9:$D$12,B28:B31)})								
34	Z_1	-0.2567	-0.8756	-1.3865	-1.0979	0.3147	-0.4923	0.1114	<--{=MMULT($A$9:$D$12,OK28:OK31)}
35	Z_2	-0.8664	0.3575	-0.0552	-0.3882	0.6830	0.0417	0.4856	<--{=MMULT($A$9:$D$12,OK28:OK31)}
36	Z_3	0.3564	-0.5180	-0.0081	-0.2494	-0.4866	-0.2437	0.2733	
37	Z_4	0.4899	-0.1219	-0.0936	0.2456	-0.2492	0.4027	0.2371	
38									
39	Correlated uniformly deviates								
40	U_1	0.3991	0.19061	0.08279	0.13613	0.6235	0.31127	0.544361046	<--=NORM.S.DIST(OK34,1)
41	U_2	0.19314	0.63964	0.478	0.34893	0.75271	0.51662	0.68635119	<--=NORM.S.DIST(OK35,1)
42	U_3	0.63922	0.30224	0.49676	0.40152	0.31327	0.40371	0.607695588	<--=NORM.S.DIST(OK36,1)
43	U_4	0.68791	0.45146	0.46273	0.50707	0.4016	0.65641	0.590724977	<--=NORM.S.DIST(OK37,1)

The statistics for the simulated uniform variates are given below:

	A	B	C	D	E	F	OJ	OK
45		Checking						
46	Number of simulations	400	<--=COUNT(40:40)					
47								
48	Mean1	0.51360	<--=AVERAGE(40:40)					
49	Mean2	0.50993	<--=AVERAGE(41:41)					
50	Mean3	0.50447	<--=AVERAGE(42:42)					
51	Mean4	0.49767	<--=AVERAGE(43:43)					
52								
53	Correlation matrix from simulation							
54		U1	U2	U3	U4			
55	U1	1.000	0.131	-0.073	-0.021	<--=CORREL($43:$43,40:40)		
56	U2	0.131	1.000	0.015	-0.299	<--=CORREL($43:$43,41:41)		
57	U3	-0.073	0.015	1.000	0.143	<--=CORREL($43:$43,42:42)		
58	U4	-0.021	-0.299	0.143	1.000	<--=CORREL($43:$43,43:43)		
59				=B3/($OJ$63*OJ62)				
60	Correlation matrix							
61		1	2	3	4		Sigma	
62	1	1.000	0.087	0.071	0.050		0.63	<--=SQRT(A3)
63	2	0.087	1.000	0.004	-0.346		0.55	<--=SQRT(B4)
64	3	0.071	0.000	1.000	0.212		0.45	<--=SQRT(C5)
65	4	0.050	-0.346	0.212	1.000		0.32	<--=SQRT(D6)

Note that without very large samples, it is extremely difficult to match covariances. But a number of hits on **F9** might convince you that we have indeed matched the desired correlation structure.

Using R to Generate Multivariate Uniform Deviates

As before, R is very efficient in generating uniform deviates. The code to implement this approach is:

```
## Multivariate Correlated Uniform Simulations
corr_uni_var <- pnorm(corr_norm_var, mean = 0, sd = 1, lower.tail = TRUE)

# Simulation correlation matrix
cor(corr_uni_var)

# Expected correlation matrix from assumption
sigma <- matrix(sqrt(diag(cov_mat)))
cov_mat / sigma %*% t(sigma)
```

And the results are very close to the assumed correlations:

```
> # Simulation correlation matrix
> cor(corr_uni_var)
           [,1]         [,2]         [,3]        [,4]
[1,] 1.00000000  0.084798096  0.069422167  0.04872507
[2,] 0.08479810  1.000000000 -0.002566323 -0.34649706
[3,] 0.06942217 -0.002566323  1.000000000  0.21288522
[4,] 0.04872507 -0.346497056  0.212885221  1.00000000
> # Expected correlation matrix from assumption
> sigma <- matrix(sqrt(diag(cov_mat)))
> cov_mat / sigma %*% t(sigma)
           [,1]        [,2]       [,3]       [,4]
[1,] 1.00000000  0.08660254 0.07071068  0.0500000
[2,] 0.08660254  1.00000000 0.00000000 -0.3464102
[3,] 0.07071068  0.00000000 1.00000000  0.2121320
[4,] 0.05000000 -0.34641016 0.21213203  1.0000000
> |
```

21.11 Summary

Random numbers are widely used in financial engineering, especially in option pricing. This chapter has introduced you to the Excel and VBA random-number generators and has shown a number of techniques for producing normally distributed random numbers.

Exercises

1. Here is a random-number generator you can make yourself:

   - Start with some number, *Seed*.
   - Let $X_1 = Seed + \pi$. Let $X_2 = e^{5+\ln(X_1)}$.
   - The first random number is $Random = X_2 - Integer(X_2)$, where $Integer(X_2)$ is the integer part of X_2.
   - Repeat the process, letting $Seed = Random$.

   Run 1,000 of these random numbers and use **Frequency** to produce a frequency distribution in the intervals 0, 0.1, 0.2, …, 1.

2. Write a VBA **Exercise1(seed)** that produces a random number based on a **Seed** and the rule of the previous exercise.
3. Define *AmodB* as the *remainder* when *A* is divided by *B*. For example, 36mod25 = 11. Excel has this function; it is written **Mod(A,B)**. Now here is another random-number generator:

   - Let X_0 = **seed**.

- Let $X_{n+1} = (7 * X_n)\text{mod}10^8$.
- Let $U_{n+1} = X_{n+1}/10^8$.

The list of numbers U_1, U_2, … are the pseudo-random numbers generated by this random-number generator. This is one of the many uniform random-number generators given in Abramowitz and Stegun (1972).

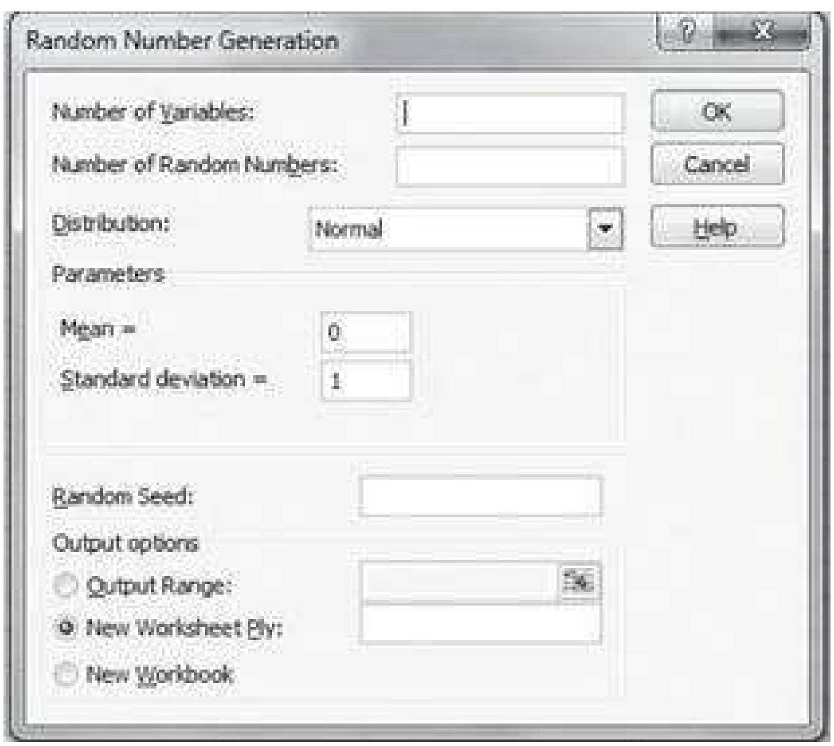
4. Many states have daily lotteries, which are played as follows: Sometime during the day, you buy a lottery ticket, on which the seller inscribes a number you choose, between 000 and 999. That night there is a drawing on television. A three-digit number is drawn. If the number on your ticket matches the number drawn, you win and collect $500. If you lose, you get nothing.

   1. Write an Excel function that produces a random number between 000 and 999. (Hint: Use **Rand()** and **Int()**.)
   2. Assume that you bet $1 every day of the year on the same number. Show your cumulative winnings throughout the year.

5. Tareq and Jamillah are playing with a single die for money. According to the rules of their game, Tareq pays Jamillah $0.50 at the start of every round, before they throw a die. They then throw the die; if it falls on an even number, Jamillah pays that amount in dollars to Tareq, and if it falls on an odd number, Tareq pays that amount to Jamillah.

   - Simulate Jamillah's cumulative winnings after 25 rounds of the game.
   - Simulate 50 games, and graph the results.

6. Stock price simulation: A stock's price is lognormally distributed with mean $\mu = 15\%$. The current stock price is $S_0 = 35$. Following the template on the spreadsheet, create 60 *static* standard normal deviates using **Data|Data Analysis|Random number generation**. Use these random numbers to simulate the stock price path over 60 months. Create price paths for $\sigma = 15\%$, 30%, and 60% and graph these three paths on the same axes.



7. Stock price simulation: A stock's price is lognormally distributed with mean μ = 15% and σ = 50%. The current stock price is S_0 = 35. Following the template on the spreadsheet, create 60 *dynamic* standard normal deviates using **NormSInv(Rand())**. Use these random numbers to simulate the stock price path over 60 months and graph.

8. Marcus is 25 years old. He has a new job and intends to save $10,000 today and in each of the next 34 years (35 deposits altogether). He has decided on an investment policy in which he invests 30% of his assets in a risk-free bond with 3% continuously compounded annual interest and the remainder in the market portfolio that has lognormal returns μ = 12% and σ = 35%. Write a spreadsheet showing Marcus's accumulation by the time he is 60. A sample output is given below:

	A	B	C	D	E
1	**MARCUS'S INVESTMENT/SAVINGS DECISION**				
2	Annual deposit	10,000			
3	Risk-free rate	3%			
4	Market portfolio mean	12%			
5	Market portfolio sigma	35%			
6	Proportion in market portfolio	70%			
7	Accumulation at age 60	12,048,869	<-- =B45		
8					
9	**Age**	**Total investment at beginning of period**	**New investment**	**Total investment at end of period**	
10	25	0	10,000	13,065.44	<--
11	26	13,065.44	10,000	27,795.31	=(B10+C10)*($B$6*EXP($B$4+
12	27	27,795.31	10,000	52,906.64	$B$5*NORM.S.INV(RAND()))+(1-
13	28	52,906.64	10,000	58,590.40	$B$6)*EXP($B$3))
14	29	58,590.40	10,000	99,352.45	

9. Marcus decides that he needs at least $2 million by the time he hits 60.

- Run 100 simulations in order to determine the approximate probability of achieving this goal.
- Compute the average and standard deviation of the terminal wealth.
- Create a **Data Table** to determine the relation between the proportion invested in the risky asset and the probability of achieving the minimum at age 60. Set the proportions in the risky to 0%, 10%, …, 100%.

10. Martha is playing a coin-toss game in which she tosses two coins. The probability of heads on the second coin is correlated with correlation ρ = 0.6 with the probability of heads on the first coin. In this particular game, Martha wins $1 for each heads that she tosses.

- Model one round of two-coin tosses.
- If she plays 10 rounds of two-coin tosses each, how much will she win?
- Use **Data Table** on a blank cell to model 25 cycles of this 10-round game.

---

1. 1. Philosophical? Perhaps theological. Knuth (1981, 142) gives the following quote: "A random sequence is a vague notion embodying the idea of a sequence in which each term is unpredictable to the uninitiated and whose digits pass a certain number of tests, traditional with statisticians and depending somewhat on the uses to which the sequence is to be put." This quote is attributed to Lehmer (1951).
2. 2. In this book we usually write Excel functions in boldface without the parentheses. In this chapter we generally write **Rand()** with the parentheses to emphasize that (i) the parentheses are necessary and (ii) that they are empty.

3. 3. A common nomenclature speaks of "random deviates." Only in financial engineering can one find "normal deviates"!
4. 4. Confusing, no? There are two different functions in the same computer package that do the same thing.
5. 5. Array functions are explained in Chapter 31.
6. 6. There are more examples in the exercises at the end of the chapter.
7. 7. See Box and Muller (1958) or Knuth (1981).
8. 8. This section uses some materials from Chapter 24 and may be skipped on first reading.
9. 9. Jacob's story: When the market goes up, everyone spends more!
10. 10. Our thanks to Antoine Jacquier, who posted this on Wilmott.com and has allowed us to use it in this book.
11. 11. For details of why this works, see section 21.4.

22 An Introduction to Monte Carlo Methods

22.1 Overview

"Monte Carlo" (MC) methods refer to a variety of random simulations used to determine the values of parameters. In this introductory chapter to MC methods, we use MC to determine the value of π. In subsequent chapters we use MC to gain insight into investment and option strategies. The Monte Carlo method has its source in physics, where it is often used to determine model values for which there is no analytical solution.[1] One use of Monte Carlo in finance is similar: Monte Carlo methods use simulation to price assets whose prices are not readily determined by analytical means. In short, if there isn't a formula for computing the value of an asset, maybe we can determine its value with a simulation.

In Chapters 23–27 we also use Monte Carlo methods to give a feel for the uncertainty surrounding various investment and option strategies. When we do this, we are not necessarily interested in the pricing implications of uncertainty—we want to illustrate what kinds of results can occur from a given investment strategy in assets whose returns are uncertain.

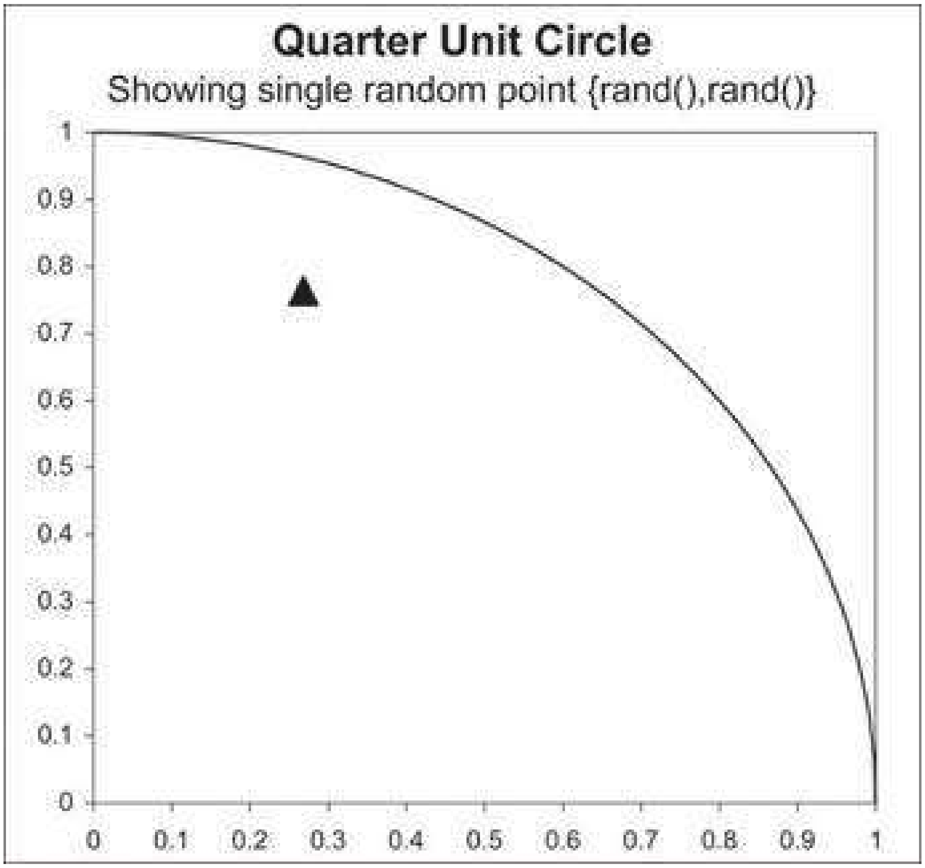
In this chapter we give a layperson's introduction to Monte Carlo pricing.[2] We assume that you have read Chapter 21, which gives an introduction to random numbers.

22.2 Computing π Using Monte Carlo

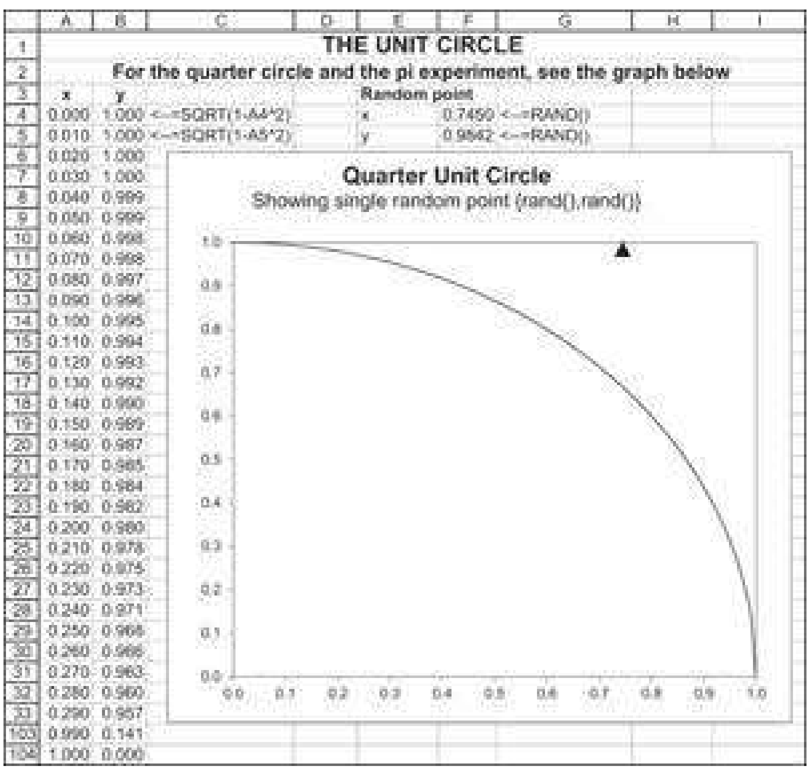
All Monte Carlo methods involve random simulation. We illustrate how to use Monte Carlo to calculate the value of π, a number with which you are presumably familiar.

Here's our method: We know that the area of the unit circle (circle with radius 1) is π. It follows that the area of a quarter circle is π /4. We inscribe a quarter circle into a unit square, as illustrated below. We then proceed to "shoot" random points at the unit square. Each random point has an x component and a y component. We generate such points by using the Excel function **Rand**.

The picture below shows the quarter circle inscribed in the unit square and a single random point, which happened to land inside the circle.[3]



By pressing **F9** in the spreadsheet that accompanies this chapter, you can generate different values of the random point. In some cases, the point will be outside the unit circle:



We can easily do a calculation of the probability that the point will be inside the circle:

- • The area of the whole unit circle is $\pi * r^2 = \pi$. Thus, the area of the quarter unit circle is $\pi/4$.
- • The random point generated by **{Rand(), Rand()}** will always be inside the unit square, whose area is 1.
- • Thus, the probability of the random point being inside the unit circle is:

$$\frac{\text{Area unit circle}}{\text{Area unit square}} = \frac{\pi/4}{1} = \pi/4$$

Monte Carlo Computation of π

If we count the relative number of points that fall inside the unit circle, we should approximate $\pi/4$. Thus:

$$\text{Monte Carlo approximation of } \pi = 4 * \frac{\text{Number points inside unit circle}}{\text{Total number of points}}$$

In the spreadsheet below we generate a list of random numbers (columns B and C) and then use a Boolean function to test whether the numbers are in or out of the unit circle. In row 8 below, for example, the function **=(B8∧2+C8∧2 <= 1)** returns **TRUE** if the sum of the squares of B8 and C8 are less than or equal to 1 and 0 otherwise. In cell B2, we use **Count** to count the number of total points generated and **Countif** to determine the number of data points that fall inside the unit circle.[4]

	A	B	C	D	E
1	COMPUTING PI USING MONTE CARLO METHODS INITIAL EXPERIMENT				
2	Number of data points	4400	<--=COUNT(A:A)		
3	Inside circle	3445	<--=COUNTIF(D:D,TRUE)		
4	Estimated Pi	3.131818182	<--=B3/B2*4		
5	Actual Pi	3.141592654	<--=PI()		
6	How far are we?	-0.311%	<--=B4/B5-1		
7		Each cell in these columns contains the Excel function =Rand()			
8	Experiment	Random1	Random2	In unit circle?	
9	1	0.66424	0.00916	TRUE	<--=(B9^2+C9^2<=1)
10	2	0.33880	0.88906	TRUE	<--=(B10^2+C10^2<=1)
11	3	0.78279	0.43630	TRUE	
12	4	0.64754	0.09633	TRUE	
4406	4398	0.99310	0.16856	FALSE	
4407	4399	0.29326	0.40386	TRUE	
4408	4400	0.97433	0.23914	FALSE	

Each time we press **F9** to recalculate the spreadsheet, we get different values for the **=Rand**, and hence a different Monte Carlo estimated value for π. Here are the results from 10 other examples:

	H	I	J
7	Trial	Estimated Pi	How far are we?
8			
9	1	3.1364	-0.17%
10	2	3.1809	1.25%
11	3	3.1209	-0.66%
12	4	3.1827	1.31%
13	5	3.1400	-0.05%
14	6	3.1564	0.47%
15	7	3.1555	0.44%
16	8	3.1200	-0.69%
17	9	3.1382	-0.11%
18	10	3.1318	-0.31%
19			
20	**Average**	**3.1463**	**0.149%**

Now it's clear that the values we get for π are experimental, but if we did this for a lot of points we would get closer to the actual value of π. In the example below we run our experiment 65,000 times:

	A	B	C	D	E
1	**COMPUTING PI USING MONTE CARLO METHODS 65,000 Iterations**				
2	Number of data points	65,000	<--=COUNT(A:A)		
3	Inside circle	51,046	<--=COUNTIF(D:D,TRUE)		
4	Estimated Pi	3.141292308	<--=B3/B2*4		
5	Actual Pi	3.141592654	<--=PI()		
6	How far are we?	-0.010%	<--=B4/B5-1		
7	**Experiment**	**Random1**	**Random2**	**In unit circle?**	
8	1	0.19857	0.15981	TRUE	<--=(B8^2+C8^2<=1)
9	2	0.12376	0.29234	TRUE	
10	3	0.68012	0.91166	FALSE	
11	4	0.60882	0.55377	TRUE	
65005	64998	0.17127	0.34011	TRUE	
65006	64999	0.44529	0.70434	TRUE	
65007	65000	0.55509	0.63239	TRUE	

Pressing **F9** still gives much more accurate values for π. A few more presses of **F9** produced the following Monte Carlo values for π:

	H	I	J
7	Trial	Estimated Pi	How far are we?
8			
9	1	3.14517	0.11%
10	2	3.13951	-0.07%
11	3	3.14185	0.01%
12	4	3.13237	-0.29%
13	5	3.13865	-0.09%

Using more data points makes our MC value for π more accurate, though none of these values is exactly the value of π.[5]

22.3 Programming the Monte Carlo Approach to Estimate π

VBA Program to Estimate π

This Monte Carlo business works much better and faster with programming (versus running it in the Excel spreadsheet). We start by introducing the VBA approach and in the next subsection we show a similar implementation in R. Here's the VBA program:

```
Sub MonteCarlo()
n = Worksheets(“MC”).Range(“Number”)
Hits = 0
For Index = 1 To n
    If Rnd ∧ 2 + Rnd ∧ 2 < 1 Then Hits = Hits + 1
Next Index
Range(“Estimate”) = 4 * Hits / n
End Sub
```

The spreadsheet below shows this program and two other VBA program variants. The program **MonteCarloTimer** computes both the **StartTime** and the **StopTime** so that we can compute the elapsed time for the computations. Below, you can see that 10 million iterations of the program took three seconds on the author's Lenovo Yoga 920 laptop.

The program **MonteCarloTimeRecord** records each iteration of the program on the screen. This VBA routine allows you to see how the values of π in cell B3 develop. You can stop this or any VBA routine in midstream by pressing [Ctrl] + [Break]. This macro is incredibly time wasteful. It took us 64 seconds to run 5,000 iterations of the routine (compare this to running the 10 million iterations without the screen updating).

	A	B	C
1	**COMPUTING PI USING VBA**		
2	Number of data points	10,000,000	<-- This cell is called "Number"
3	Estimated Pi	3.141448800	<-- This cell is called "Estimate"
4	Actual Pi	3.141592654	<--=PI()
5	How far are we?	-0.00458%	<--=Estimate/B4-1
6			
7	StartTime	22:42:37	<-- This cell called "StartTime"
8	StopTime	22:42:40	<-- This cell called "StopTime"
9	Elapsed	0:00:03	<-- =Stoptime-StartTime

```
Sub MonteCarloTime()
'Includes timer
    n = Range(“Number”)
    Range(“StartTime”) = Time
    n = Range(“Number”)
    Hits = 0
    For Index = 1 To n
    If Rnd ∧ 2 + Rnd ∧ 2 < 1 Then Hits = Hits + 1
    Next Index
```

```
        Range(“Estimate”) = 4 * Hits / n
        Range(“StopTime”) = Time
    End Sub
    Sub MonteCarloTimeRecord()
    ‘Records everything (takes a long time)
        n = Range(“Number”)
        Range(“StartTime”) = Time
        n = Range(“Number”)
        Hits = 0
        For Index = 1 Ton
            Range(“Number”) = Index
            If Rnd ∧ 2 + Rnd ∧ 2 < 1 Then Hits = Hits + 1
            Range(“Estimate”) = 4 * Hits / Index
            Range(“StopTime”) = Time
        Next Index
    End Sub
```

R Program to Estimate π

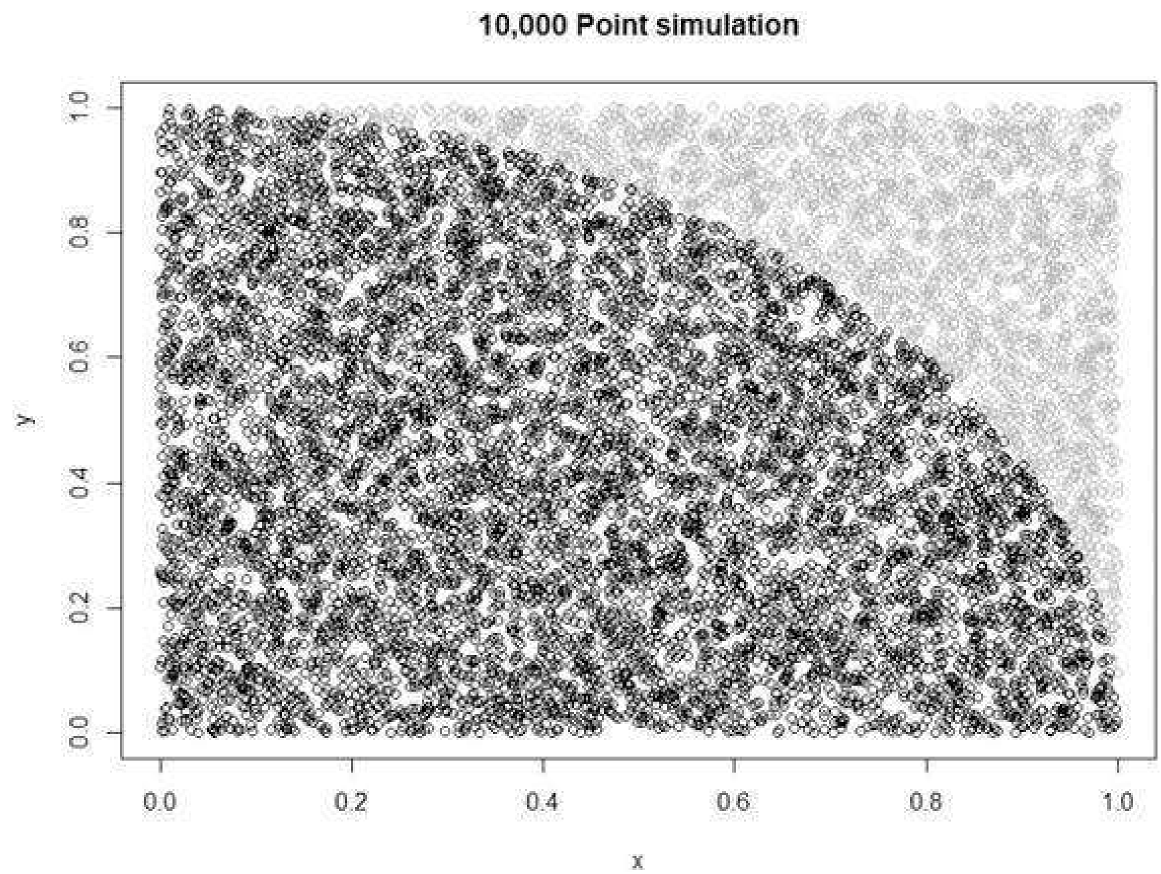
Here is the code we ran in R to estimate π.

```
# Random Points
n <- 10000 # number of trials
z1 <- runif(n, 0, 1) # random uniform variable #1
z2 <- runif(n, 0, 1) # random uniform variable #2

# Check if the point is inside or outside the circle)
unit_cir_est <- z1 ^ 2 + z2 ^ 2 <= 1

# plot
plot(z1, z2, type = "p", col = unit_cir_est+8, xlab="x", ylab="y",
     main="10,000 Point simulation", )

# Calculate Pi
mc_pi <- length(which(unit_cir_est)) / n * 4 # simulated Pi

# Compare the estimated Pi to the actual one
mc_pi
pi # actual Pi
```

The results are very close:

```
> # Compare the estimated Pi to the actual one
> mc_pi
[1] 3.14184
> pi # actual Pi
[1] 3.141593
> |
```

And the chart looks like this:



22.4 Another Monte Carlo Problem: Investment and Retirement[6]

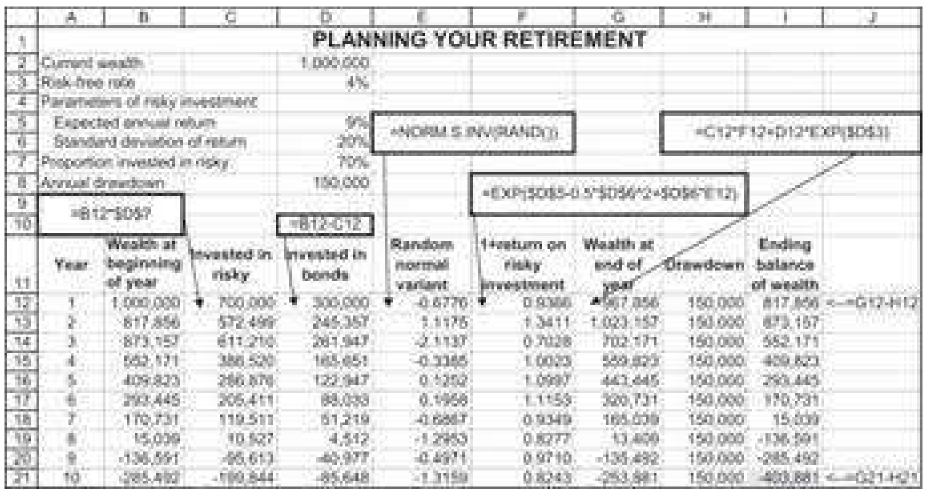
The problem: You are 65 years old and you have \$1,000,000. You are trying to decide on a mix of investments. There is a riskless bond with an annual return of 4% and a risky stock portfolio with an expected log return of 9% and a standard deviation of return of 20%. Your limitations: You want to take \$150,000 out of the account every year and have something left over at age 75.

To get a better handle on this situation, you plot out a spreadsheet:

	A	B	C	D	E	F	G	H	I	J
1				PLANNING YOUR RETIREMENT						
2	Current wealth			1,000,000						
3	Risk-free rate			4%						
4	Parameters of risky investment									
5	Expected annual return			9%	=NORM.S.INV(RAND())			=C12*F12+D12*EXP($D$3)		
6	Standard deviation of return			20%						
7	Proportion invested in risky			70%						
8	Annual drawdown			150,000		=EXP($D$5-0.5*$D$6^2+$D$6*E12)				
9	=B12*$D$7									
10				=B12-C12						
11	Year	Wealth at beginning of year	Invested in risky	Invested in bonds	Random normal variant	1+return on risky investment	Wealth at end of year	Drawdown	Ending balance of wealth	
12	1	1,000,000	700,000	300,000	-1.2855	0.8294	892,797	150,000	742,797	<--=G12-H12
13	2	742,797	519,958	222,839	0.8888	1.2760	895,424	150,000	745,424	
14	3	745,424	521,797	223,627	1.6809	1.5011	1,016,015	150,000	866,015	
15	4	866,015	606,210	259,804	-0.0688	1.0579	911,690	150,000	761,690	
16	5	761,690	533,183	228,507	1.6003	1.4771	1,025,385	150,000	875,385	
17	6	875,385	612,770	262,616	0.2674	1.1314	966,643	150,000	816,643	
18	7	816,643	571,650	244,993	-0.0861	1.0542	857,624	150,000	707,624	
19	8	707,624	495,337	212,287	-0.1723	1.0362	734,208	150,000	584,208	
20	9	584,208	408,946	175,263	0.6887	1.2309	685,783	150,000	535,783	
21	10	535,783	375,048	160,735	0.0605	1.0856	574,434	150,000	424,434	<--=G21-H21

In this spreadsheet, column B shows the wealth at the beginning of every year. The wealth is divided between risky and riskless investments according to the proportions in cell D7. The riskless investment earns a continuously compounded return of 4% (meaning $100 invested in the riskless grows to 100 * $e^{4\%}$ = 104.08 at the end of year). The risky part of the investment grows by a factor of $e^{(\mu-0.5*\sigma^2)+\sigma*Z} = e^{(9\%-0.5*20\%)+20\%*Z}$, where Z is a random number that is normally distributed with mean 0 and standard deviation 1. As explained in Chapter 21, one way of generating these numbers is to use the Excel function **Norm.S.Inv(Rand())**. With each press of **F9**, this function recomputes and produces another set of normally distributed random numbers.

In the above simulation the investor has money left at the end of the 10-year period. But it is clear that not every simulation will leave the investor with spare cash at the end of year 10. A few presses of **F9** to recalculate the spreadsheet will produce something like this:



	A	B	C	D	E	F	G	H	I	J
1				PLANNING YOUR RETIREMENT						
2	Current wealth			1,000,000						
3	Risk-free rate			4%						
4	Parameters of risky investment									
5	Expected annual return			9%	=NORM.S.INV(RAND())			=C12*F12+D12*EXP($D$3)		
6	Standard deviation of return			20%						
7	Proportion invested in risky			70%						
8	Annual drawdown			150,000		=EXP($D$5-0.5*$D$6^2+$D$6*E12)				
9	=B12*$D$7									
10				=B12-C12						
11	Year	Wealth at beginning of year	Invested in risky	Invested in bonds	Random normal variant	1+return on risky investment	Wealth at end of year	Drawdown	Ending balance of wealth	
12	1	1,000,000	700,000	300,000	-0.6776	0.9366	967,856	150,000	817,856	<-- =G12-H12
13	2	817,856	572,499	245,357	1.1176	1.3411	1,023,157	150,000	873,157	
14	3	873,157	611,210	261,947	-2.1137	0.7028	702,171	150,000	552,171	
15	4	552,171	386,520	165,651	-0.3385	1.0023	559,823	150,000	409,823	
16	5	409,823	286,876	122,947	0.1252	1.0997	443,445	150,000	293,445	
17	6	293,445	205,411	88,033	0.1958	1.1153	320,731	150,000	170,731	
18	7	170,731	119,511	51,219	-0.6867	0.9349	165,039	150,000	15,039	
19	8	15,039	10,527	4,512	-1.2953	0.8277	13,409	150,000	-136,591	
20	9	-136,591	-95,613	-40,977	-0.4971	0.9710	-135,492	150,000	-285,492	
21	10	-285,492	-199,844	-85,648	-1.3159	0.8243	-253,881	150,000	-403,881	<-- =G21-H21

What interests us is *what percentage of the investment-consumption paths will end with a positive remainder*? We will use Monte Carlo techniques to answer this question. But before we do, we consider an economic question.

Should We Withdraw $150,000 Blindly?

In the above simulations we withdrew a fixed amount ($150,000) no matter what the account balance is. This is done irrespective of whether there are funds in the account.

In the spreadsheet below we correct this. We assume that the investor defines a "safety cushion." The cushion in cell B8 below is 3, which is taken to mean that the annual drawdown is $150,000 if the investor's portfolio is worth at least 3 * $150,000 at the end of the year. If this is not true, then the investor takes one-third of their portfolio value as a drawdown:

	A	B	C	D	E	F	G	H	I	J
1	PLANNING YOUR RETIREMENT Using a safety cushion of 3 Investor takes 150,000 if end-year wealth > 3*150,000, else takes end-year wealth/3									
2	Current wealth			1,000,000						
3	Risk-free rate			4%						
4	Parameters of risky investment									
5	Expected annual return			9%						
6	Standard deviation of return			20%						
7	Proportion invested in risky			70%						
8	Safety cushion factor			3						
9	Annual drawdown			150,000						
10										
11										
12	Year	Wealth at beginning of year	Invested in risky	Invested in bonds	Random normal variant	1+return on risky investment	Wealth at end of year	Drawdown	Ending balance of wealth	
13	1	1,000,000	700,000	300,000	-2.3741	0.6671	779,215	150,000	629,215	<--=G13-H13
14	2	629,215	440,450	188,764	0.5082	1.1873	719,395	150,000	569,395	
15	3	569,395	398,576	170,818	0.5866	1.2012	656,556	150,000	506,556	
16	4	506,556	354,589	151,967	-0.4431	0.9816	506,217	150,000	356,217	
17	5	356,217	249,352	106,865	-0.2412	1.0220	366,062	122,021	244,041	
18	6	244,041	170,829	73,212	-2.2900	0.6784	192,092	64,031	128,062	
19	7	128,062	89,643	38,418	-0.6068	0.9499	125,142	41,714	83,428	
20	8	83,428	58,400	25,028	0.1464	1.1044	90,545	30,182	60,363	
21	9	60,363	42,254	18,109	0.4316	1.1692	68,252	22,751	45,501	
22	10	45,501	31,851	13,650	-0.0681	1.0580	47,905	15,968	31,937	<--=G22-H22

=B13*$D$7
=B13-C13
=NORM.S.INV(RAND())
=EXP($D$5-0.5*$D$6^2+$D$6*E13)
=C13*F13+D13*EXP($D$3)
=MIN($D$9,G13/$D$8)

Many rules like this can be devised. The problem here—phrased as investment and payouts over retirement—is substantially the same as that faced by any endowment manager struggling with the problem of how to determine simultaneously the investment and the drawdown policy of the endowment. As far as we know, there is no analytical solution to this problem, though as can be seen, it is not difficult to simulate the problem.

22.5 A Monte Carlo Simulation of the Investment Problem

In this section we run multiple simulations of the investment problem. The first set of simulations is run in the Excel spreadsheet itself, using the technique of “Data table on a blank cell” explained in section 28.7. The second set of simulations is run in VBA.

Data Table on a Blank Cell

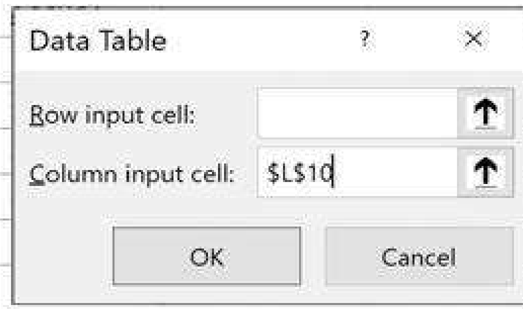
We set up a **Data Table** on the investment simulation spreadsheet. Note that the data table header refers to the bequest in cell I20 and that the **Column input cell** in the Data Table dialog box refers to an empty cell. This is the technique explained in section 28.7.

	I	J	K	L	M	N
10						
11	Ending balance of wealth			Simulation	Ending balance of wealth	
12	991,591	<--=G12-H12		1	122,470	<--=I21 data table header
13	981,037			2	172,878	
14	1,010,965			3	719,146	
15	864,036			4	199,573	
16	838,234			5	-95,771	
17	739,976			6	75,834	
18	604,487			7	-632,498	
19	455,578			8	-263,358	
20	218,909			9	41,563	
21	122,470	<--=G21-H21		10	-250,012	
22				11	68,036	
23				12	-107,679	

Data Table ? ×
Row input cell:
Column input cell: $L$10
OK Cancel

There are 20 rows in the **Data Table** above though only 12 are shown. When we select OK, we run the simulation 20 times. Here is one run and some statistics. Pressing **F9** will run the simulation again (now with all 20 lines presented):

	L	M	N
6	Average	-142,560	<--=AVERAGE(M12:M31)
7	Sigma	349,123	<--=STDEV.S(M12:M31)
8	Negative bequest	13	<--=COUNTIF(M12:M31,"<0")
9			
10			
11	**Simulation**	**Ending balance of wealth**	
12	1	122,470	<--=I21 data table header
13	2	172,878	
14	3	719,146	
15	4	199,573	
16	5	-95,771	
17	6	75,834	
18	7	-632,498	
19	8	-263,358	
20	9	41,563	
21	10	-250,012	
22	11	68,036	
23	12	-107,679	
24	13	-5,727	
25	14	-359,396	
26	15	-549,746	
27	16	-755,889	
28	17	-326,207	
29	18	-108,518	
30	19	-155,110	
31	20	-640,790	



As can be seen, many of the outcomes are negative (13 out of 20). It's not looking good for our retirement.

Using VBA to Analyze the Impact of Percentage in a Risky Asset

In the above simulations we have split our investment between the risky and the riskless investments mechanically. We write a VBA function **SuccessfulRuns**, which simulates the above problem. Given an investment policy, the function **SuccessfulRuns** determines the percentage of investment/drawdown trajectories that will leave the retiree with positive wealth at the end of his investment horizon.

Here is some output from this function:

	A	B	C	D	E
1	**HOW WELL DO WE DO? PERCENTAGE OF POSITIVE OUTCOMES**				
2	Current wealth	1,000,000			
3	Risk-free rate	4%			
4	Parameters of risky investment				
5	Expected annual return	9%			
6	Standard deviation of return	20%			
7	Proportion invested in risky	40%			
8	Annual drawdown	100,000			
9	Years of investment	10			
10					
11	Runs	1,000			
12					
13	Successful runs	91.50%	<--=successfulruns(B2,B8,B3,B5,B6,B7,B9,B11)		

The function result in cell B13 considers a retiree starting with \$1 million, making an investment decision between a risk-free asset with return 4% and a risky asset with stochastic returns $\mu = 9\%$ and $\sigma = 20\%$, and desiring to draw down \$100,000 per year. In this analysis, the investor is allocating 40% to risky assets and 60% to the risk-free asset. Simulating 1,000 returns, we determine that in 91.5% of the cases, the investor will finish with positive wealth at the end of 10 years. Running the function again will, of course, produce different results.

Here Is the VBA Code for SuccessfulRuns

```
Function SuccessfulRuns(Initial, Drawdown, Interest, Mean, Sigma,
PercentRisky, Years, Runs)
    Dim PortfolioValue() As Double
    ReDim PortfolioValue(Years + 1)
    Dim Success As Integer
    Up = Exp(Mean + Sigma)
    Down = Exp(Mean - Sigma)
    PiUp = (Exp(Interest) - Down) / (Up - Down)
    PiDown = 1 - PiUp
    For Index = 1 To Runs
    For j = 1 To Years
        Randomize
        PortfolioValue(0) = Initial
        If Rnd > PiDown Then
        PortfolioValue(j) = PortfolioValue(j - 1) * PercentRisky *
        Up + _ PortfolioValue (j - 1) * (1 - PercentRisky) *
        Exp(Interest) - Drawdown
        Else:
        PortfolioValue(j) = PortfolioValue(j - 1) * PercentRisky *
        Down + _PortfolioValue (j - 1) * (1 - PercentRisky) *
        Exp(Interest) - Drawdown
```

```
End If
Next j
    If PortfolioValue(Years) > 0 Then Success = Success + 1
Next Index
SuccessfulRuns = Success / Runs
End Function
```

Using R to Simulate the Investment Problem

Problems like the investment problem presented above would usually be modeled in Excel. This is mainly because it is not data intensive and probably much more intuitive to model in Excel. However, we can implement the same approach in R, as shown below.

We start by modeling a single try, which should look like this:

```
# Inputs
savings <- 1000000
riskless_r <- 0.04
risky_r <- 0.09
risky_sigma <- 0.2
allowance <- 150000
m <- 10
risky_prop <- 0.4

# Run
balance <- savings
table <- c()

# single trial
for (i in 1:m){
  risky_begin <- balance * risky_prop
  risky_ret <- exp((risky_r - 0.5 * risky_sigma^2) +
                    risky_sigma * rnorm(1, mean = 0, sd = 1)) # risky asset
  risky_end <- risky_begin * risky_ret # risk free asset
  bond_begin <- balance * (1 - risky_prop)
  bond_end <- bond_begin * (1 + riskless_r)
  balance <- sum(risky_end, bond_end, -allowance) # total wealth
  table <- rbind(table, c(risky_begin = risky_begin, risky_ret = risky_ret,
                          risky_end = risky_end, bond_begin = bond_begin,
                          bond_end = bond_end, balance = balance))
               }
table # result
```

The result looks like this:

```
> table # result
      risky_begin risky_ret risky_end bond_begin bond_end   balance
 [1,]    400000.0 1.5124566  604982.6   600000.0 624000.0 1078982.6
 [2,]    431593.0 1.3468234  581279.6   647389.6 673285.2 1104564.8
 [3,]    441825.9 1.3324119  588694.1   662738.9 689248.4 1127942.5
 [4,]    451177.0 1.0857646  489872.0   676765.5 703836.1 1043708.2
 [5,]    417483.3 0.9788526  408654.6   626224.9 651273.9  909928.5
 [6,]    363971.4 1.3941241  507421.3   545957.1 567795.4  925216.7
 [7,]    370086.7 1.2831153  474863.9   555130.0 577335.2  902199.1
 [8,]    360879.6 1.5610043  563334.7   541319.4 562972.2  976306.9
 [9,]    390522.8 1.0437134  407593.8   585784.1 609215.5  866809.3
[10,]    346723.7 1.1204343  388481.2   520085.6 540889.0  779370.2
>
```

In the following step we use the same approach form m trials. Note the difference in the function generating the random normal deviates: rnorm(m, mean = 0, sd = 1)) instead of rnorm(1, mean = 0, sd = 1)) in the single trial.

```
## m trials
m <- 1000

# Run
balance <- savings
for(i in 1:N){
  # risky asset
  risky_invest <- balance * risky_prop
  risky_ret <- exp(risky_r - 0.5 * risky_sigma^2 +
                   risky_sigma * rnorm(m, mean = 0, sd = 1 ))
  risky_end <- risky_invest * risky_ret
  bond_invest <- balance - risky_invest
  bond_end <- bond_invest * exp(riskless_r)
  balance <- risky_end + bond_end - allowance
            }

mean(balance)
sd(balance)
length(which(balance>=0))
```

And the results are:

```
> mean(balance)
[1] -160850.6
> sd(balance)
[1] 275232.4
> length(which(balance>=0))
[1] 228
> |
```

And if we plug in an annual allowance of 100,000 instead of the 150,000 we get:

```
> mean(balance)
[1] 503786.1
> sd(balance)
[1] 337964.5
> length(which(balance>=0))
[1] 981
> |
```

22.6 Summary

Monte Carlo methods are experimental techniques for determining the numerical value of a function or a procedure. In this chapter we have used the example of π to illustrate how Monte Carlo might be applied. As a valuation tool, MC methods are to be avoided when there is another, closed-form way of determining the value. As we illustrate in the exercises to this chapter, MC is not a good way to compute the value of π, given that many excellent formulas exist that approximate π with great accuracy. In cases where this is not true, however, you can use Monte Carlo to approximate the value.

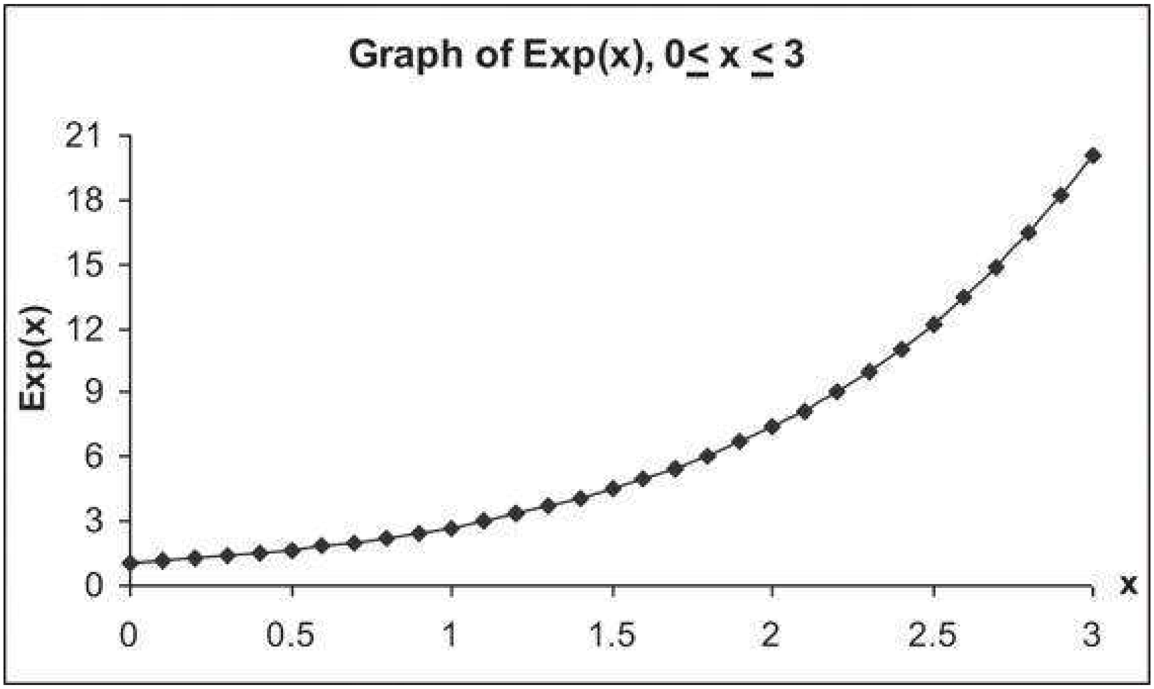
Monte Carlo methods can also be used to simulate investment problems in order to gain insight into the meaning of asset return uncertainty. In this chapter we illustrated this with some investment examples.

Exercises

1. In section 22.2 we designed a macro that calculates the value of π using Monte Carlo and updates the screen every iteration. Modify the macro so that it updates the screen only every 1,000 iterations.
   Hints:
   - Use the VBA function **Mod**. From the VBA help menu, note that this function has syntax **a Mod b.** (A similar Excel function has syntax **Mod(a,b)**, but cannot be used in VBA.)

- Use the VBA commands Application.ScreenUpdating=True and Application.ScreenUpdating=False to control the updating of the screen.

2. In the previous exercise, put a "switch" on the spreadsheet itself, which controls the updating of the macro (whether to update, yes or no, and how often to update).
3. Use Monte Carlo to calculate the integral of the function Exp(x) for 0 < x < 3. Here's the graph of this function:



4. One of the messages of this chapter is that while Monte Carlo is a clever method of calculation, it shouldn't be used when some better method exists. The MC valuation of π in section 22.2, for example, converges very slowly. It's a crummy way to compute π, since there are well-known methods for doing this computation. To see this, answer the following questions:

- Approximately how many runs does it require until you get four-decimal accuracy for π using our MC simulation?
- Approximately how many runs does it require until you get eight-decimal accuracy for π using our MC simulation?
- Approximately how many runs does it require until you get 15-decimal accuracy for π using our MC simulation?

5. Here in exercise 5, we show you two alternative ways to compute the value of π. All are substantially better than Monte Carlo!

   a. The first method for computing π is:

$$\frac{\pi^2}{6} = 1 + \frac{1}{2^2} + \frac{1}{3^2} + \cdots$$

Use this formula to approximate π.

   b. The great Indian mathematical genius Ramanujan showed that:

$$\frac{1}{\pi} = \frac{\sqrt{8}}{9801} \sum_{n=0}^{\infty} \frac{(4n)!}{(n!)^4} \frac{(1103 + 26390n)}{396^{4n}}$$

The $n!$ indicates the factorial:

$n! = n * (n-1) * (n-2) * \ldots * 2 * 1$
$0! = 1$

Excel's function **Fact** computes the factorial. R's "factorial(x)" computes the factorial of x.

Use this series to construct a program (in R or a VBA function) to value π, where n is the number of terms in the series. Show that two iterations give you over 15 digits of accuracy.

Appendix: Some Comments on the Value of π

Exercise 5 is based on Borwein, Borwein, and Bailey (1989). Ramanujan's method in exercise 5 adds roughly eight digits with each iteration. Ramanujan developed an even faster method, where only 13 iterations provide more than *1 billion* digits of π. (Excel's maximal accuracy is only 15 digits, but there is a nice Excel add-in that extends the precision to 32,767 digits; it is available at http://precisioncalc.com/.)

Srinivasa Ramanujan (1887–1920) was one of the great mathematical geniuses of all time. Read a biography of this unique individual, *The Man Who Knew Infinity: A Life of the Genius Ramanujan* (Kanigel 1991).

Finally: So what's the value of π ? Mathematica—a very sophisticated mathematical programming language (https://www.wolframalpha.com/input/?i=pi)—gives the following values, which are computed using one of Ramanujan's formulas.

Number of Significant Digits	Value of π
25	3.141592653589793238462643
50	3.1415926535897932384626433832795028841971693993751
75	3.1415926535897932384626433832795028841971693993751058 2097494459230781640629
500	3.1415926535897932384626433832795028841971693993751058 2097494459230781640628620899862803482534211706798214808 6513282306647093844609550582231725359408128481117450284 1027019385211055596446229489549303819644288109756659 3344612847564823378678316527120190914564856692346034861 0454326648213393607260249141273724587006606315588174881 5209209628292540917153643678925903600113305305488204 6652138414695194151160943305727036575959195309218611738 1932611793105118548074462379962749567351885752724891227 9381830119491

1. 1. Two good websites with introductions to Monte Carlo methods are www.ornl.gov/~pk7/thesis/pkthnode19.html and http://marcoagd.usuarios.rdc.puc-rio.br/monte-carlo.html.
2. 2. An excellent nonintroductory text is Glasserman (2004).
3. 3. Remember that a circle with a center at (0,0) can be described by $x^2 + y^2 = r^2$, and in the case of a unit square by $x^2 + y^2 = 1$ or $y = \sqrt{1 - x^2}$.
4. 4. All of the functions in this paragraph—Boolean functions, **Count,** and **Countif**—are discussed in Chapter 30.
5. 5. The actual value of π, correct to 50 digits, is
6. 3.14159265358979323846264338327950288419716939937510. One of the end-of-chapter problems shows you a quick way of computing this value using several remarkable functions created by the Indian mathematician Srinivasa Ramanujan (1887–1920).
7. 6. The material in the remainder of the chapter anticipates some results of the next several chapters and can be skipped on first reading.

23 Simulating Stock Prices

23.1 Overview

In Chapter 17 we discussed the pricing of options using the binomial option pricing model. The binomial model—besides being an attractive and intuitive way to price options and other derivative securities—also has a deeper message for derivative asset pricing: It shows us that, given some assumptions about the uncertainty governing the stock price and given a risk-free interest rate, we can price options and other assets whose prices are dependent on the price of an underlying stock.

A problem with the binomial option pricing approach is that we were not able to give simple formulas for the pricing of options. The pricing approach developed in Chapter 17 is computational, not analytic. In order to develop a *formula* for the pricing of options (such as the Black-Scholes formula, discussed in Chapter 18), we need to make some assumptions about the statistical properties of the underlying stock price.

A central assumption of the Black-Scholes pricing model is that stock prices are distributed lognormally. An alternative statement of this assumption is that stock returns are distributed normally. In this chapter we attempt to give this assumption enough content so that you will be happy using it. Our method is as follows: We shall not, in this book, prove the Black-Scholes option pricing formula. Instead, we shall try to convince you in this chapter that the basic assumption made by the Black-Scholes model with regard to stock prices—the lognormality of stock prices—is reasonable. If we can convince you of this, then we leave the technical details of the Black-Scholes proof to other, more advanced, texts.

The structure of this chapter is as follows:

- • We start with a discussion of what constitutes “reasonable” assumptions about stock prices.
- • We then discuss why the lognormal distribution is a reasonable distribution for stock prices.
- • Next, we show how to simulate lognormal price paths.
- • Finally, we show you how to derive the parameters of the lognormal distribution —the mean and standard deviation of the stock returns—from historical stock price data.

23.2 What Do Stock Prices Look Like?

What are reasonable assumptions about the way stock prices behave over time? Clearly, the price of a stock (or any other risky financial asset) is uncertain. What is its distribution? This is a perplexing question. One way to answer this question is to ask what reasonable statistical properties of a stock price are. Here are five reasonable properties:

1. The stock price is uncertain. Given the price today, we do not know the price tomorrow.
2. Changes in the stock's price are continuous. Over short periods of time, changes in a stock's price are very small, and the change goes to zero as the time span goes to zero.[1]
3. The stock price is never zero or negative. This property means that we exclude the stocks of "dead" companies.
4. The return from holding a stock tends to increase over time on average. Notice the word "tends." We do not *know* that holding a stock for a longer time will lead to higher return; however, we *expect* that holding a risky asset over a longer term will lead to a higher return *on average*.
5. The *uncertainty* associated with the return from holding a stock also tends to increase the longer the stock is held. Thus, given the stock's price today, the variance of the stock price tomorrow is small; however, the price variance in one month is larger and the variance in one year is larger still.

Reasonable Stock Properties and Stock Price Paths

One way of viewing these five "reasonable properties" of stock prices is to think about *price paths*. A stock price path is a graph of a stock price over a period of time. Here, for example, is the price of paths of three stocks:

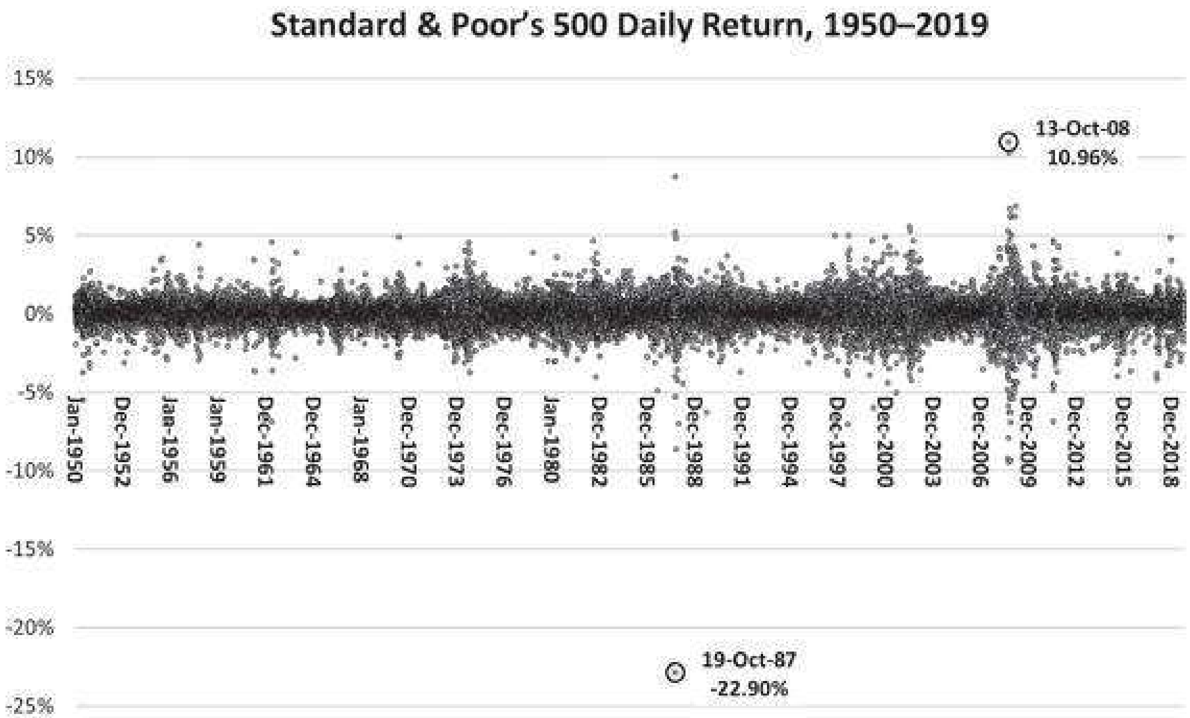
If we simulated stock price paths (something we will do using the lognormal model later in this chapter), how would we expect them to look? Our five properties imply that we would expect:

1. Wiggly lines.
2. Lines that are continuous (solid), with no jumps.
3. Lines that are always positive and never cross zero, no matter how low they get.
4. That at a given point in time, the average over all plausible lines is greater than the initial price of the stock. The farther out we go, the higher this average becomes.
5. That the uncertainty over all plausible lines is greater the farther out we go.

Here's another way of thinking about stock prices. Suppose we take the daily returns on the Standard & Poor's 500 Index (SP500). (In our examples, we only show the start and the end of the data—many lines are hidden.)

	A	B	C	D
1	**SP500 DAILY PRICES Jan 1950–Oct 2019**			
2	**Date**	**Close**	**Return**	
3	03-Jan-50	16.66		
4	04-Jan-50	16.85	1.13%	<--=LN(B4/B3)
5	05-Jan-50	16.93	0.47%	<--=LN(B5/B4)
6	06-Jan-50	16.98	0.29%	<--=LN(B6/B5)
7	09-Jan-50	17.08	0.59%	
17569	24-Oct-19	3,010.29	0.19%	
17570	25-Oct-19	3,022.55	0.41%	
17571	28-Oct-19	3,039.42	0.56%	
17572	29-Oct-19	3,036.89	-0.08%	
17573	30-Oct-19	3,046.77	0.32%	

If we graph these returns over any given period, we get a mess of dots, which is difficult to interpret:

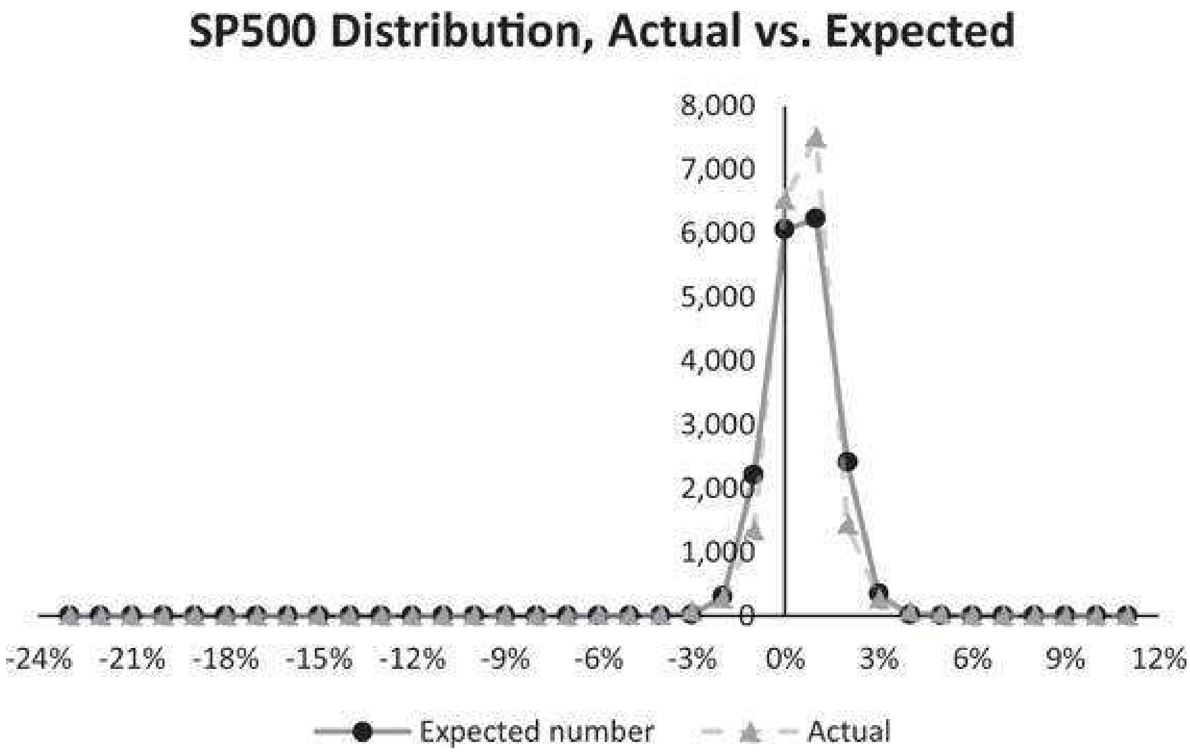


We have marked the largest daily decline and increase with circles: −22.9% on October 19, 1987, and +10.96% on October 13, 2008.

This smear of dots on a graph is difficult to interpret. Excel can help us make some sense of the data:

	F	G	H
3	Number of daily returns	17,570	<--=COUNT(C:C)
4	Period in years	69.82	<--=YEARFRAC(A3,A17573,1)
5	Trading days in a year	252	<--=G3/G4
6			
7	Max	10.96%	13-Oct-08
8	Min	-22.90%	19-Oct-87
9			
10	Number of returns between -1% and +1%	14,042	<--=COUNTIFS(C:C,"<=1%",C:C,">=-1%")
11	Average daily return	0.0296%	<--=AVERAGE(C:C)
12	Standard deviation of daily return	0.9648%	<--=STDEV.S(C:C)
13	Average annual return	7.46%	<--=G5*G11
14	Standard deviation of annual return	15.31%	<--=SQRT(G5)*G12
15			
16	Skewness	-1.00449	<--=SKEW(C:C)
17	Standard error of skewness (SE)	0.018479	<--=SQRT(6/G3)
18	Skewness/SE	-54.357	<--=G16/G17
19			
20	Kurtosis	26.74501	<--=KURT(C:C)
21	Standard error of kurtosis	0.036959	<--=SQRT(24/G3)
22	Kurtosis/SE	723.6407	<--=G20/G21

Another way to look at the data is to plot the frequency of the returns using Excel's **Frequency** function. The details can be found on this book's companion website. The plot looks approximately normally distributed, though perhaps with some left skewness:



Computing Returns and Their Distribution for a Continuous Return-Generating Process

We can compute the stock return over a period by taking the natural logarithm of the price relatives, defined as $Ln\left(\frac{Price_t}{Price_{t-1}}\right)$. Furthermore, if $\{r_1, r_2, \ldots, r_M\}$ is a series of periodic returns, we can compute the mean, variance, and standard deviation of the returns as:

$$Periodic\ mean = \mu_{Periodic} = \underbrace{\frac{1}{M}\sum_{t=1}^{M} r_t}_{\substack{\uparrow \\ \text{Use Excel's} \\ \text{Average function}}},$$

$$Periodic\ variance = \sigma^2_{Periodic} = \underbrace{\frac{1}{M-1}\sum_{t=1}^{M}(r_t - \mu_{Periodic})^2}_{\substack{\uparrow \\ \text{Use Excel's} \\ \text{Stdev.s function}}}$$

Assuming that there are n periods per year, we can compute the annualized returns by:

$$Annualized\ mean = n * \mu_{Periodic}$$

$$Annualized\ variance = n * \sigma^2_{Periodic}$$

$$Annualized\ standard\ deviation = \sigma\sqrt{n}$$

23.3 Lognormal Price Distributions and Geometric Diffusions

Now we can get a bit more formal and describe what we mean by a lognormal price distribution. We then relate the lognormal price process to a geometric diffusion.

Suppose we denote by S_t the price at time t of a share of stock. The lognormal distribution assumes that *the natural logarithm of one-plus-the-return* from holding a share of stock between time t and time $t + \Delta t$ is normally distributed with mean μ and standard deviation σ. Denote the (uncertain) rate of return over an interval Δt by $\tilde{r}_{\Delta t}$. Then we can write $S_{t+\Delta t} = S_t exp(\tilde{r}_{\Delta t}\Delta t)$. In the lognormal distribution, we assume that the rate of return $\tilde{r}_{\Delta t}$ over a short period Δt is normally distributed with mean $\mu \Delta t$ and variance $\sigma^2 \Delta t$.

A more accurate way of writing this relation is to write the stock price $S_{t+\Delta t}$ at time $t + \Delta t$ in the following way (after applying Itô's lemma):

$$ln\left(\frac{S_{t+\Delta t}}{S_t}\right) = (\mu - 0.5 \cdot \sigma^2) \cdot \Delta t + \sigma \cdot \sqrt{\Delta t} \cdot Z$$

Here, Z is a standard normal variable (mean = 0, standard deviation = 1).[2] To see what this assumption means, suppose first that $\sigma = 0$. In this case we have:

$$S_{t+\Delta t} = S_t\, exp(\mu \cdot \Delta t)$$

This simply says that the stock price grows at an exponential rate with certainty. In this case, the stock is like a riskless bond that bears interest rate μ, continuously compounded.

Now suppose that $\sigma > 0$. In this case, the lognormal assumption says that, although the tendency is for the stock price to increase, there is an uncertain element (normally distributed) that must be taken into account. The best way to think about this is in terms of a simulation. Suppose, for example, that we are trying to simulate a lognormal price process in which $\mu = 15\%$, $\sigma = 30\%$, and $\Delta t = 0.004$ (one trading day).[3] Suppose the price at time 0 is $S_0 = 35$. To simulate the possible stock prices at time Δt, we first have to pick (at random) a number Z from a standard normal distribution.[4] Suppose that this number is 0.1165. Then the stock price $S_{\Delta t}$ at time Δt will be:

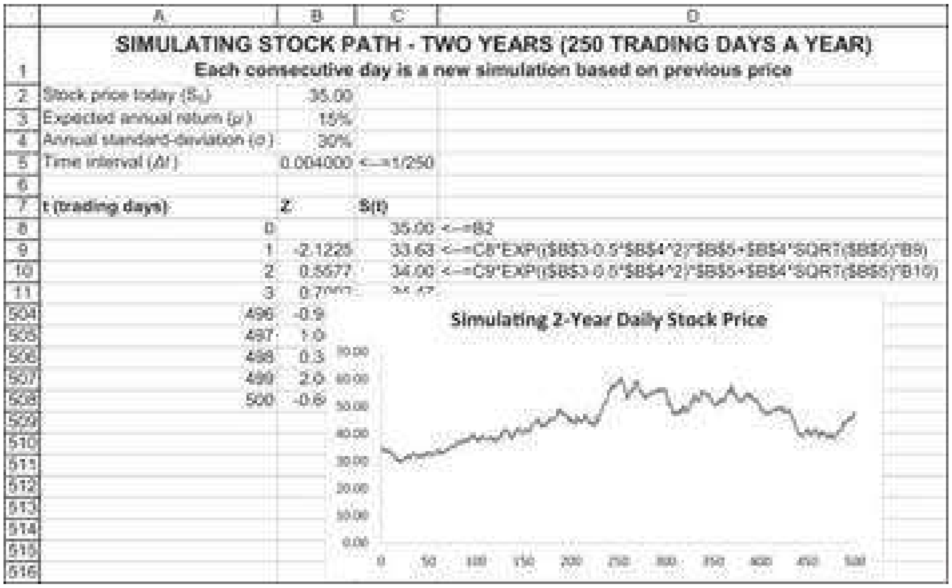
$$S_{\Delta t} = S_0 * exp[(\mu - 0.5\sigma^2)\Delta t + \sigma Z\sqrt{\Delta t}]$$
$$= 35 * e^{[(0.15 - 0.5*0.3^2)*0.004 + 0.3*0.1165*\sqrt{0.004}]} = 35.0922$$

	A	B	C
1	SIMULATING STOCK PRICE		
2	Stock price today (S_0)	35.00	
3	Expected annual return (μ)	15%	
4	Annual standard-deviation (σ)	30%	
5	Time interval (Δt)	0.004000	<--=1/250
6	Normal variant (Z)	0.1165	"<-- should be using Norm.S.Inv(rand())
7			
8	Stock price in two days ($S_{t+\Delta t}$)	35.0922	<--=B2*EXP((B3-0.5*B4^2)*B5+B4*SQRT(B5)*B6)

Of course, we could have drawn a different random number. This process is illustrated in the spreadsheet below, where we generated a list of 1,000 numbers picked from a standard normal distribution (the technical nomenclature is "standard normal deviates"). Each is an equally likely potential candidate to be Z (note that many lines are hidden).

	A	B	C
1	SIMULATING STOCK PRICE		
2	Stock price today (S_0)	35.00	
3	Expected annual return (μ)	15%	
4	Annual standard-deviation (σ)	30%	
5	Time interval (Δt)	0.004000	<--=1/250
6	Normal variant (Z)	0.1165	"<-- should be using Norm.S.Inv(rand())
7			
8	Stock price in two days ($S_{t+\Delta t}$)	35.0922	<--=B2*EXP((B3-0.5*B4^2)*B5+B4*SQRT(B5)*B6)
9			
10	**Z**	**S(t)**	
11	-0.6195	34.6055	<--=$B$2*EXP(($B$3-0.5*$B$4^2)*$B$5+$B$4*SQRT($B$5)*A11)
12	-0.2986	34.8169	<--=$B$2*EXP(($B$3-0.5*$B$4^2)*$B$5+$B$4*SQRT($B$5)*A12)
13	-1.0059	34.3528	
14	1.3151	3[illegible]	
1007	-2.0201		
1008	0.6340	3	
1009	-0.3363	3	
1010	-0.3159	3	
1011			
1012			
1013			
1014			
1015			
1016			
1017			
1018			
1019			
1020			
1021			

Having picked Z for a particular time interval Δt, the price $S_{t+\Delta t}$ follows:



To summarize: In order to simulate *the growth of the stock price*, when the price follows a lognormal price distribution:

- • Multiply μ (the average rate of growth) minus half a variance by Δt (the elapsed time interval). This gives the certain portion of the return in a log normal distribution.
- • Take a draw Z from a random variable which is standard normal and then multiply this draw by $\sigma\sqrt{\Delta t}$. This gives the uncertain portion of the return. (The square root implies that the variance of the stock's return is linear in time. See below.)
- • Add the two results and exponentiate. The daily return is $exp[(\mu - 0.5\sigma^2)\Delta t + \sigma Z\sqrt{\Delta t}]$. If the price on date t is S_t , then the price on date $t + 1$ is $S_{t+1} = S_t \cdot exp[(\mu - 0.5 \cdot \sigma^2)\Delta t + \sigma \cdot Z \cdot \sqrt{\Delta t}]$.

Why Do We Assume That the Stock's Mean Return Is $r - 0.5 * \sigma^2$?

Suppose that the stock returns are normal with mean μ and standard deviation σ. As discussed above, this means that the expected stock price is $S_0\ Exp[(\mu + 0.5 \cdot \sigma^2)t]$. In our first meaning of risk-neutrality, we want to transform the return distribution so that the expected price has mean r. We do this by replacing μ by $-0.5 \cdot \sigma^2$. This guarantees that the expected future price of the stock grows at rate r, which is the basic condition for the risk-neutral pricing:

$$S_0Exp[(\mu + 0.5 \cdot \sigma^2)t^] = S_0Exp[((r - 0.5 \cdot \sigma^2) + 0.5 \cdot \sigma^2)t] = S_0\ exp[rt].$$

23.4 What Does the Lognormal Distribution Look Like?

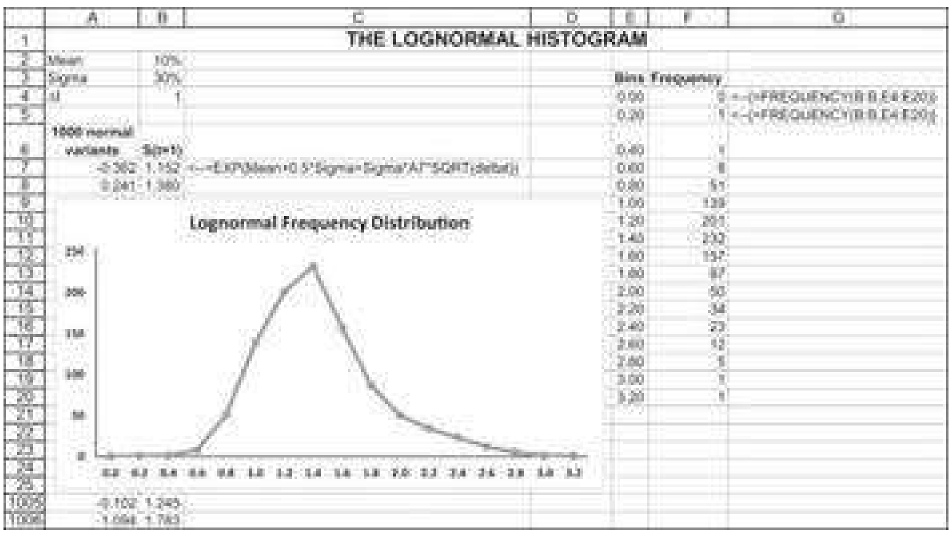
We know that the normal distribution produces a "bell curve." What about the lognormal distribution? In the following experiment, we simulate 1,000 random end-of-year stock prices. The experiment is a continuation of the experiment performed in the previous section; since we are simulating end-of-year prices, we set $\Delta t = 1$. To perform this experiment:

- • We produce a list of 1,000 normal deviates.
- • We use each normal deviate to produce an end-of-period stock price:

$$
\begin{aligned}
S_1 &= S_0 * exp[(\mu - 0.5 * \sigma^2)\Delta t + \sigma Z\sqrt{\Delta t}\,] \\
&= S_0 * exp[(\mu - 0.5 * \sigma^2) + \sigma * Z],\ since\ \Delta t = 1
\end{aligned}
$$

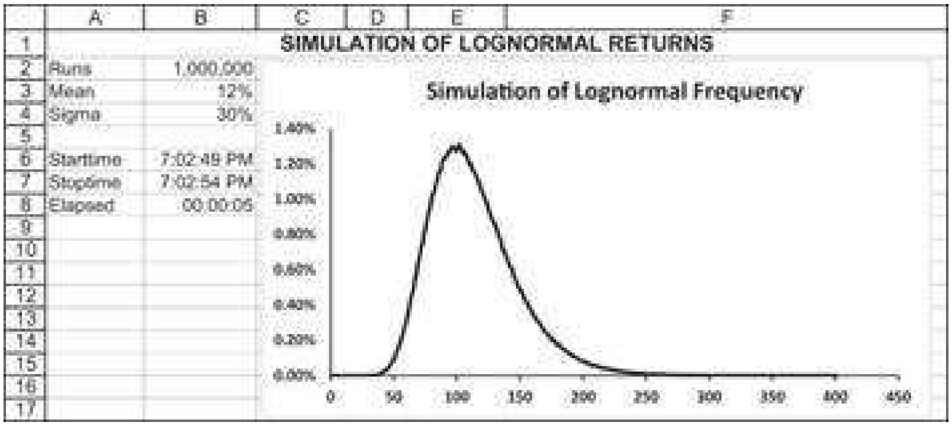
- • We put the stock prices into bins and produce a histogram.

Here's what the spreadsheet for this experiment looks like:



Having produced 1,000 lognormal price relatives $exp[(\mu - 0.5\sigma^2)\Delta t + \sigma Z\sqrt{\Delta t}\,]$, we can use the array function **Frequency()** (this function is discussed in Chapter 31) to put them into bins.

When we do this simulation for a large number of points, the resulting density curve becomes smooth. Here, for example, is the frequency distribution of 1,000,000 trials with $\mu = 12\%$, $\sigma = 30\%$, and $\Delta t = 1$:



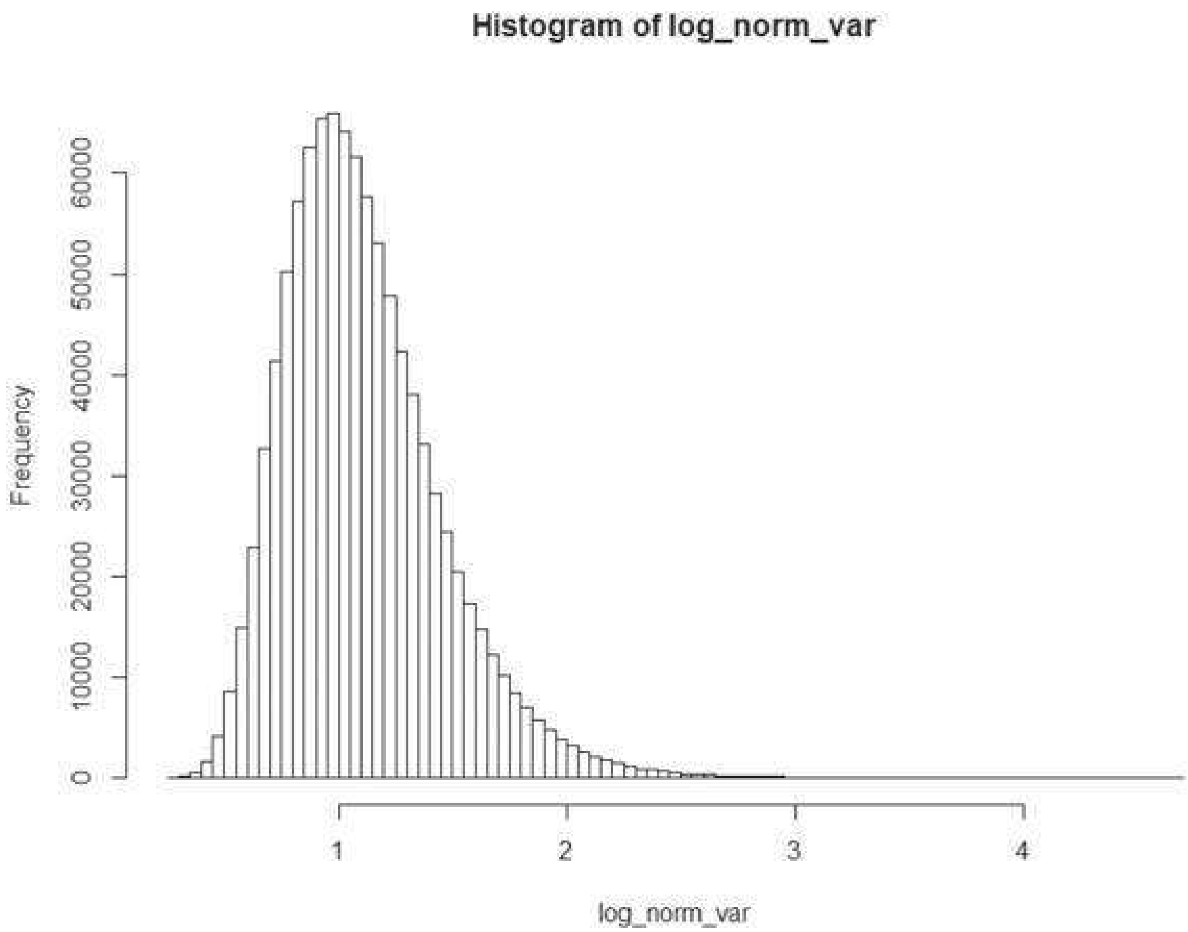
The VBA program that produced this output is shown below:

```
Sub RandomNumberSimulation()
'Simulating the lognormal distribution
'Note that we take delta = 1!
Application.ScreenUpdating=False
Range(“starttime”) = Time
N = Range(“runs”).Value
mean = Range(“mean”)
sigma = Range(“sigma”)
ReDim Frequency(0 To 1000) As Integer
For Index = 1 To N
    Return1 = Exp((mean - 0.5 * (sigma) ∧ 2) _
    + sigma * WorksheetFunction.Norm_S_Inv(Rnd))
    Frequency(Int(Return1 / 0.01)) = _
    Frequency(Int(Return1 / 0.01)) + 1
Next Index
For Index = 0 To 400
    Range(“simuloutput”).Cells(Index + 1, 1) = _
    Frequency(Index) / N
Next Index
Range(“stoptime”) = Time
Range(“elapsed”) = Range(“stoptime”) - _
Range(“starttime”)
Range(“elapsed”).NumberFormat = “hh:mm:ss”
End Sub
```

The routine that produces randomly distributed standard normal deviates is contained in the eight lines following the word start; this routine is further explained in Chapter 24.

Simulating the lognormal distribution in R looks like this:

```
# Inputs:
mean <- 0.1
sigma <- 0.3
delta_t <- 1
Z <- rnorm(1000000, 0,1) # 1M normally distributed variables

# Calculate returns based on 1M Z's
log_norm_var <- exp( (mean - 0.5 * sigma ^ 2) * delta_t +
                        sigma * sqrt(delta_t) * Z)

# Lognormal Frequenc Distribution
hist(log_norm_var, breaks = 100)
```



23.5 Simulating Lognormal Price Paths

We now return to the problem of simulating lognormal price paths that we started to discuss in section 23.3. We shall try to understand, through a simulation written in VBA, the meaning of the following sentences: “The price of a stock today is $25. The price of the stock is distributed lognormally, with an annual log mean return of 10% and an annual log standard deviation of 20%.” We want to know how the price of the stock might behave on a daily basis throughout the next year. There are an infinite potential number of price paths for the stock. What we will do is simulate (randomly) one of these paths. If we want another price path, we can merely rerun the simulation.

There are about 250 business days in a year. Therefore, the daily price movement of the stock between day t and day $t + 1$ can be simulation by setting $\Delta t = 1/250 = 0.004$,

$\mu = 10\%$, and $\sigma = 20\%$. If the initial price of the stock $S_0 = \$25$, then the price after one day will be:

$$S_{\Delta t} = S_0 * exp[(\mu - 0.5\sigma^2)\Delta t + \sigma Z\sqrt{\Delta t}]$$
$$= 25 * exp[(0.15 - 0.5 \cdot 0.20^2) \cdot 0.004 + 0.20 \cdot Z\sqrt{0.004}]$$

And the price after two days will be:

$$S_{0.008} = S_{0.004} * exp[(0.15 - 0.5 \cdot 0.20^2) \cdot 0.004 + 0.20 \cdot Z\sqrt{0.004}]$$

And so on. At each step, the random normal deviate Z is the uncertain factor in the price return. Because of this uncertainty, all paths produced will be different.

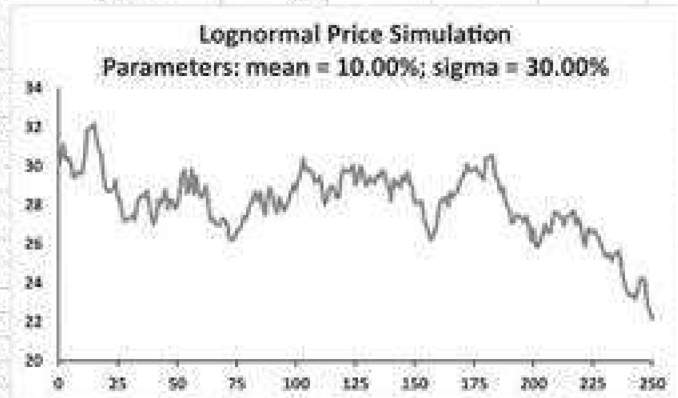
Here is a VBA program **PricePathSimulation** that reproduces a typical price path:

```
Sub PricePathSimulation()
Range("starttime") = Time
N = Range("runs").Value
mean = Range("mean")
sigma = Range("sigma")
delta_t = 1 / (2 * N)
ReDim price(0 To 2 * N) As Double
price(0) = Range("initial_price")
For Index = 1 To N
start:
    Static rand1, rand2, S1, S2, X1, X2
    rand1 = 2 * Rnd - 1
    rand2 = 2 * Rnd - 1
    S1 = rand1 ∧ 2 + rand2 ∧ 2
    If S1 > 1 Then GoTo start
    S2 = Sqr(-2 * Log(S1) / S1)
    X1 = rand1 * S2
    X2 = rand2 * S2
price(2 * Index - 1) = price(2 * Index - 2) * Exp(mean * delta_t +
_
        sigma * Sqr(delta_t) * X1)
    price(2 * Index) = price(2 * Index - 1) * Exp(mean * delta_t +
    _
        sigma * Sqr(delta_t) * X2)
Next Index
For Index = 0 To 2 * N
    Range("output").Cells(Index + 1, 1) = Index
    Range("output").Cells(Index + 1, 2) = price(Index)
```

```
Next Index
Range("stoptime") = Time
Range("elapsed") = Range("stoptime")- Range("starttime")
Range("elapsed").NumberFormat = "hh:mm:ss"
End Sub
```

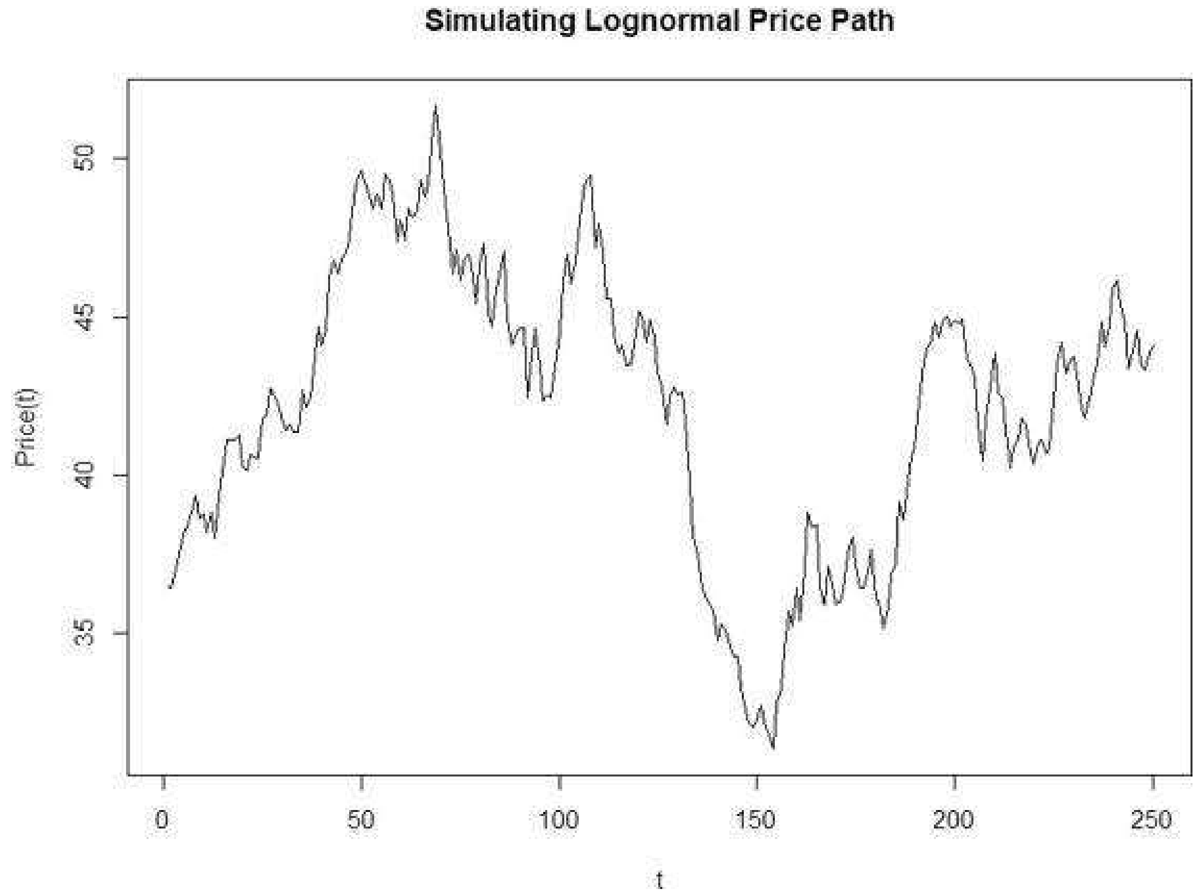
The output from this program looks like the graph on the spreadsheet:

	A	B	C	D	E	F	G	H	I
1	SIMULATING LOGNORMAL PRICE PATHS WITH VBA								
2	Day	S(t)							
3	0	30.00		Trading days	250		Start time	7:28:20 PM	
4	1	30.48		Initial price	30		Stop time	7:28:21 PM	
5	2	31.11		Mean	10%		Elapsed	00:00:01	
6	3	30.34		Sigma	30%				
7	4	30.41							
8	5	30.29							
9	6	29.48							
10	7	29.42							
11	8	29.72							
12	9	29.72							
13	10	29.60							
14	11	30.51							
15	12	31.80							
16	13	31.94							
17	14	32.00							
18	15	32.19							
19	16	31.38							
20	17	30.86							
21	18	30.53							
22	19	29.36							
23	20	28.80							
251	248	22.88							
252	249	22.60							
253	250	22.18							



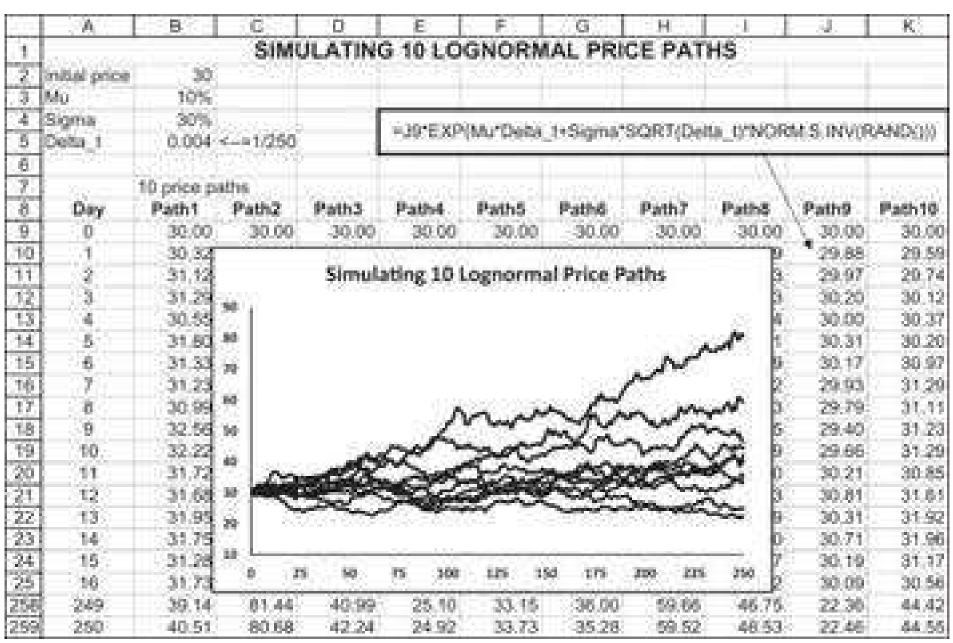
And when implemented in R, it will look like this:

```
## Single Trial
# Input:
mean <- 0.1
sigma <- 0.3
delta_t <- 1/250
s_zero <- 35

# 250 normally distributed varibles
Z <- rnorm(250, 0, 1)

# Convert to 250 LOG-normally distributed varibles
log_norm_var <- exp( (mean - 0.5 * sigma ^ 2) * delta_t +
                     sigma * sqrt(delta_t) * Z)

# Compute compounded returns
comp_ret <- cumprod(log_norm_var)

# Initial price * compounded returns
sim_price <- s_zero * comp_ret

# Plot
plot(1:250, sim_price, type = "l", xlab = "t", ylab = "Price(t)",
     main = "Simulating Lognormal Price Path")
```



Ten Lognormal Price Paths

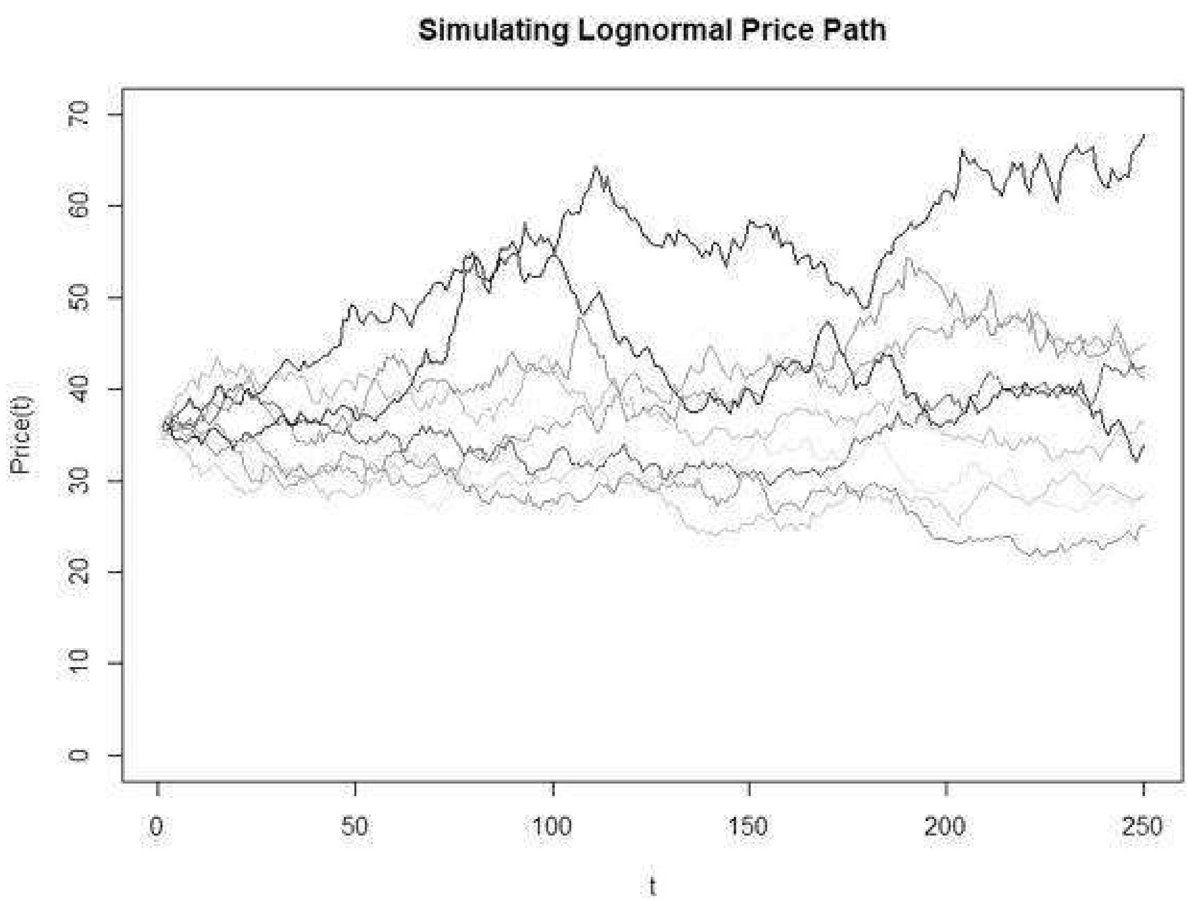
In the spreadsheet below we use a slightly different technique to simulate 10 lognormal price paths with the same statistical parameters. In each cell we use **Norm.S.Inv(rand())** to draw a number from the standard normal distribution. This normal deviate is then used to simulate the stock price, as illustrated below:



As you can see, on average the price of the asset increases over time, as does the variance of the returns. This accords with properties 4 and 5 of stock prices in section 23.2—we expect both the return on an asset and the uncertainty associated with this return to increase over time.

Implementing this in R should look like this:

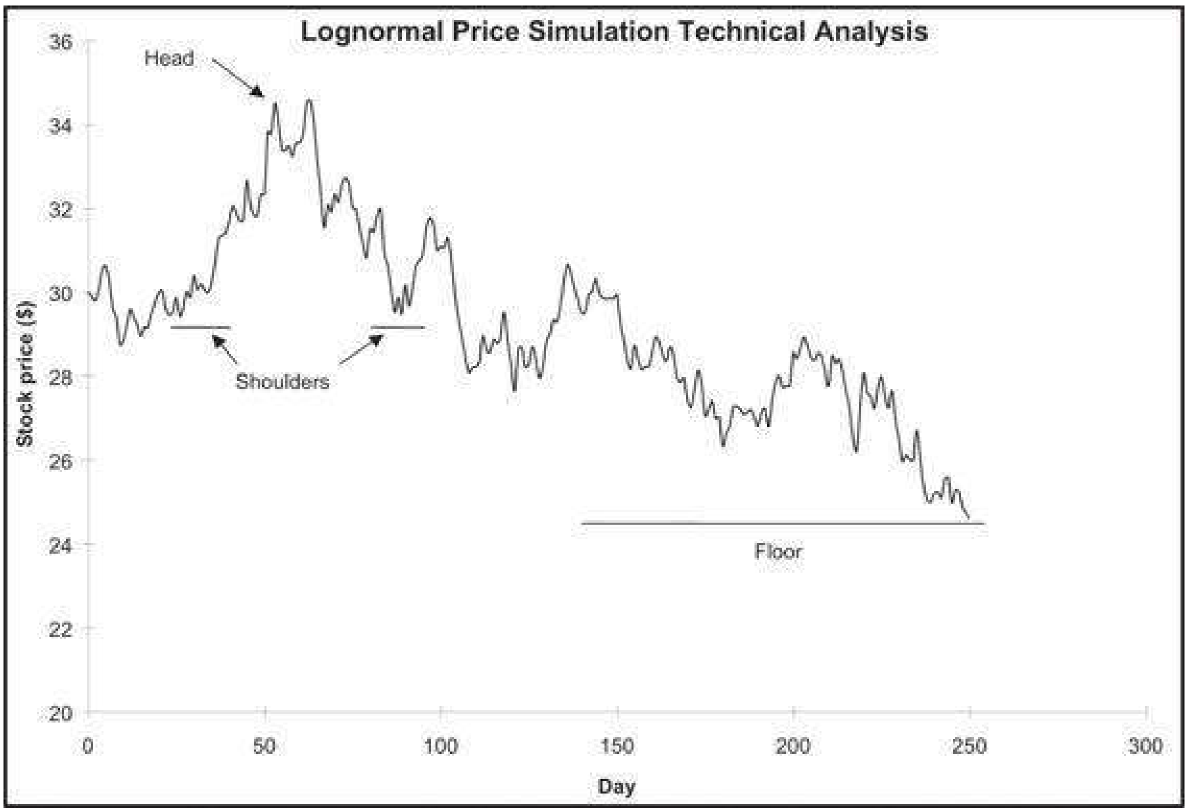
```r
## n-Trials Approach (10 trials example)
n <- 10

# n x 250 normally distributed varibles
z <- matrix(rnorm(250 * n, 0, 1),
            nrow = n)

# Convert to n x 250 LOG-normally distributed varibles
log_norm_var <- exp( (mean - 0.5 * sigma ^ 2) *
                     delta_t + sigma * sqrt(delta_t) * z)

# Compute compounded returns
comp_ret <- apply(log_norm_var, 1, cumprod)

# Initial price * the compounded return
sim_price <- s_zero * comp_ret

# Plot
plot(1:250, sim_price[,1],col=1, type = "l", ylim = c(0,70),
     xlab = "t", ylab = "Price(t)", main = "Simulating Lognormal Price Path")
for(i in 1:n){lines(1:250, sim_price[,i], col=i)}
```



23.6 Technical Analysis

Security analysts are divided into “fundamentalists” and “technicians.” This division has nothing to do with their outlook on the Creator of the Universe, but rather with the way they regard stock prices. Fundamentalists believe that the value of a stock is ultimately determined by underlying economic variables. Thus, when fundamentalists analyze a company, they will look at its earnings, its debt-to-equity ratio, its markets, and so forth.

Technicians, in contrast, think that stock prices are determined by patterns. They believe that, by examining the pattern of past prices of a stock, they can predict (or at least make sensible statements about) the stock's future prices. A technician may tell you that "we're currently in a head-and-shoulders pattern," meaning that a graph of the stock price looks like the figure below:



Other terms used by technicians include "floors" (there's one in the graph), "rebound levels," and "pennants."

The academic (some would say ivory tower) view of technical analysis is that it is worthless. A basic theory of financial theory says that markets efficiently incorporate the information known about the securities traded on them. There are several versions of this theory. One of them, the *weak efficient markets hypothesis*, says that at the very least, all information about past prices is incorporated into the current price. The weak efficient markets hypothesis means that technical analysis cannot make predictions about futures prices because technical analysis is based solely on past price information.[5]

Nevertheless, a lot of people believe in technical analysis (this in itself may give technical analysis some validity). The simulations we are running in this chapter will allow us to generate a myriad of patterns which, when analyzed, will yield "good" predictions of future prices. For example, in the figure above it appears that \$24 is a floor for the stock price, since it never goes any lower. A perspicacious analyst can detect a clear head-and-shoulders pattern between days 40 and 100. There appears to be a ceiling of \$35. Thus, a technician might predict that the stock price will stay below \$37 unless it rises above that level. (If you are going to be a technician, you have to learn to say these things with a straight face.)

23.7 Calculating the Parameters of the Lognormal Distribution from Stock Prices

The main purpose of this section is to show you how stock price data can be used to compute the annual mean return μ and the standard deviation of the annual return σ needed in the lognormal simulations (and—in Chapter 18—the σ needed as an input to the Black-Scholes formula). Before doing this, note that the mean and variance of the logarithm of the stock return over an interval Δt are as follows:

$$E\left[ln\left(\frac{S_{t+\Delta t}}{S_t}\right)\right] = E[(\mu - 0.5\sigma^2)\Delta t + \sigma Z\sqrt{\Delta t}] = \mu\Delta t$$

$$var\left[ln\left(\frac{S_{t+\Delta t}}{S_t}\right)\right] = var[(\mu - 0.5\sigma^2)\Delta t + \sigma Z\sqrt{\Delta t}] = \sigma^2\Delta t$$

This means that both the expected log return and the variance of the log return are linear in time.

Now suppose we want to estimate the lognormal μ and σ from data on historical prices. It follows that:

$$\mu = \frac{mean\left[ln\left(\frac{S_{t+\Delta t}}{S_t}\right)\right]}{\Delta t}, \quad \sigma^2 = \frac{var\left[ln\left(\frac{S_{t+\Delta t}}{S_t}\right)\right]}{\Delta t}$$

To make things specific, the following spreadsheet gives monthly prices for a particular stock. From these prices we calculate the log returns and the *annualized* mean and standard deviation. Note that we have used the function **Stdev.s** to calculate σ. This assumes that the data is a good sample to the actual distribution.

	A	B	C	D
1	**ESTIMATING THE ANNUAL MEAN AND SIGMA OF RETURNS FROM MONTHLY PRICE DATA FOR APPLE INC.** Nov. 2014–Nov. 2019			
2	Monthly average	1.31%	<--=AVERAGE(C:C)	
3	Monthly standard deviation	8.51%	<--=STDEV.S(C:C)	
4				
5	Annual average, μ	15.78%	<--=12*B2	
6	Annual standard deviation, σ	29.49%	<--=SQRT(12)*B3	
7				
8	**Date**	**Adj. close price**	**Monthly return**	
9	01-Nov-19	261.78	-0.38%	<--=LN(B9/B10)
10	01-Oct-19	262.78	-0.38%	<--=LN(B10/B11)
11	01-Sep-19	263.78	-0.38%	
12	01-Aug-19	264.78	-0.38%	
66	01-Feb-15	128.46	9.21%	
67	01-Jan-15	117.16	5.96%	
68	01-Dec-14	110.38	-7.46%	<--=LN(B68/B69)
69	01-Nov-14	118.93		

Note that the annual average log return is 12 times the monthly average log return, whereas the annual standard deviation is $\sqrt{12}$ the monthly standard deviation. In general, if the return data are generated for n periods per year, then we get the following:

$$mean_{annual\ return} = n \cdot mean_{periodic\ return},$$

$$\sigma_{annual\ return} = \sqrt{n} \cdot \sigma_{periodic\ return}$$

Of course, this is not the only way to calculate the parameters of the lognormal distribution. We should mention at least two other methods:

- • We can use some other procedure to extrapolate the mean and standard deviation of *future* returns from the past history of returns. One example would be to use a moving average.
- • We can use forward-looking data such as the Black-Scholes formula to find the *implied volatility* (the σ of the stock's log returns that fits the price of an option on the stock) and the Black-Litterman model to find the expected return. This is illustrated in Chapter 18 (section 18.4, "Implied Volatility") and in Chapter 15.

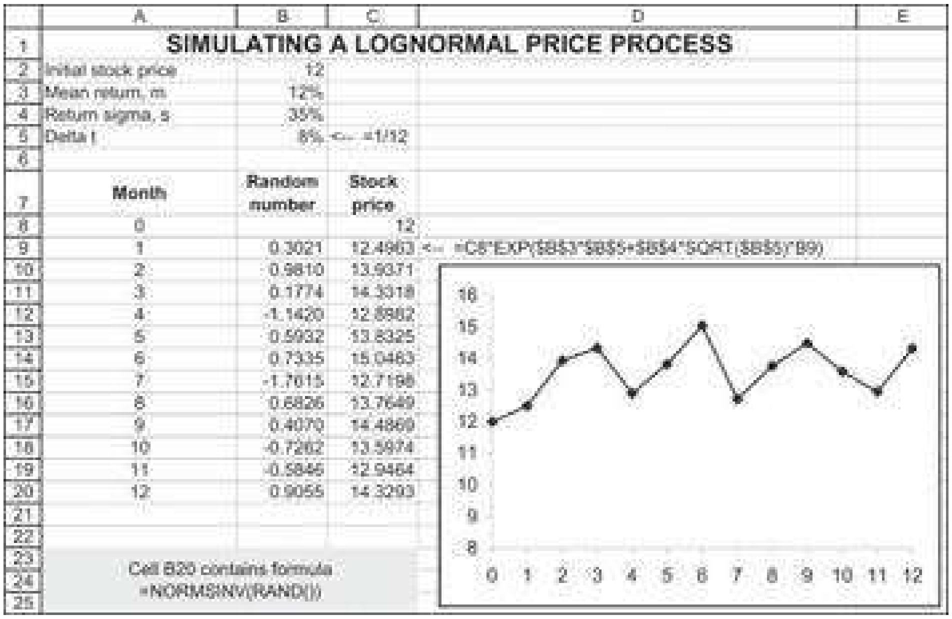
23.8 Summary

The lognormal distribution is one of the foundations of the Black-Scholes formula for option pricing discussed in Chapter 18. In this chapter we have explored the meaning of lognormality for stock prices. We have shown how lognormality—the assumption that the returns on an asset are normally distributed—can be justified visually for the SP500 portfolio. We have also shown how to simulate price paths that are lognormally distributed. Finally, we have shown how to compute the mean and the standard deviation of a lognormal distribution from the historic returns of an asset.

Exercises

1. Assume that a stock is traded at $12. The stock has an expected annual return of 12% and standard deviation of 35% per year.

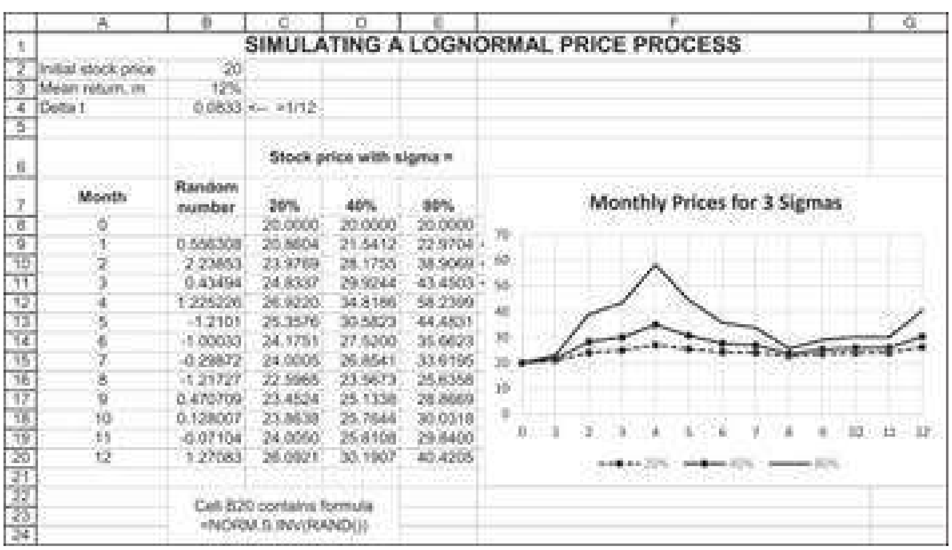
    a. Use **Norm.S.Inv(Rand())** to produce a simulation of monthly stock prices, as illustrated below.[6]



    b. Repeat section a. and implement the same exercise in R—remembering that the R equivalent to Excel's **Norm.S.Inv(Rand())** is **rnorm()**.

2. Expand the previous exercise to produce a simulation of daily stock prices for 250 days (approximately one year of trading days).

3. Recreate the spreadsheet below. Play with the spreadsheet (each press of **F9** will recompute the numbers) to convince yourself that higher σ means a more volatile price path for the stock.



4. Repeat question 3 but in R and not Excel.

5. Write a program (either in VBA or R) that reproduces the lognormal frequency distribution for an arbitrary number of runs. That is, this program should:

- Produce N normal random deviates.
- For each deviate produce a lognormal price relative $\exp[(\mu - 0.5 \cdot \sigma^2) \cdot \Delta t + \sigma \cdot \sqrt{\Delta t} \cdot z$.
- Classify each price relative into a set of bins running from 0, 0.1, …, 3.
- Put the frequencies on the spreadsheet and produce a frequency graph such as the one in section 23.4.

6. Run a few of the lognormal price path simulations. Examine the price pattern for trends. Find one or more of the following technical patterns:

   support area
   resistance area
   uptrend/downtrend
   head and shoulders
   inverted head and shoulders
   double top/bottom
   rounded top/bottom
   triangle (ascending, symmetrical, descending)
   flag

7. The exercise file for this chapter contains daily price data for the SP500 and for Abbott Laboratories for the three months from April to June 2020. Use this data to compute the annual average, variance, and standard deviation of the logarithmic returns for the SP500 and for Abbott. What is the correlation between the returns of the SP500 and Abbott?
8. The exercise file for this chapter gives daily returns from 1987 to 2020 for the Vanguard Index 500 Fund (VFINX). This is a fund that tracks the SP500, where the returns include dividends.

- Compute the overall daily return statistics: average and standard deviation.
- Annualize these statistics, assuming that there are 250 days per year.
- Compute the daily and annualized return statistics by year.

Hint: Take a look at the functions **DAverage** and **DStdev** discussed in Chapter 30.

Appendix 23.1: Itô's Lemma

Itô's lemma is an identity used to find the differential of a time-dependent function of a stochastic process. Intuitively, it can be derived from the Taylor series expansion of the function up to its second derivatives, retaining terms up to first order in the time increment and second order in the Wiener process increment. Itô's lemma, which is named after Kiyosi Itô, is occasionally referred to as the Itô–Doeblin theorem in recognition of posthumously discovered work of Wolfgang Doeblin.[7] The formal discussion about the lemma extends beyond the scope of this book.[8]

We first set the the lognormal price process is a *geometric diffusion*, $\frac{dS}{S} = \mu dt + \sigma dB$, where dB is a Wiener process ("white noise"); $dB = Z\sqrt{dt}$, where Z is a standard random variable. Note that:

$$\frac{dS}{S} = d(ln(S_t)) = ln(S_t) - ln(S_{t-1}) = ln\left(\frac{S_t}{S_{t-1}}\right) \Rightarrow ln\left(\frac{S_t}{S_{t-1}}\right) = \mu dt + \sigma dB$$

Itô's lemma has the following general form: For any function $G(X, t)$ of two variables X and t, where X satisfies the following stochastic differential equation $dX = a \cdot dt + b \cdot dB_t$, for any constants a and b, we get:

$$dG = \left(a\frac{\partial G}{\partial X} + \frac{\partial G}{\partial t} + 0.5 \cdot b^2 \cdot \frac{\partial^2 G}{\partial X^2}\right)dt + \frac{\partial G}{\partial X}db$$

And if we set $G = ln(S_t)$, where $dSt = \mu \cdot S_t \cdot dt + \sigma \cdot St \cdot dB$, we get:

$$\frac{\partial G}{\partial S} = \frac{1}{S_t}$$

$$\frac{\partial^2 G}{\partial S^2} = -\frac{1}{S_t^2}$$

$$\frac{\partial G}{\partial t} = 0$$

Substituting this is the Itô's formula, we get:

$$d(ln(S_t)) = \left(\mu \cdot S_t \frac{1}{S_t} + 0 + 0.5 \cdot \sigma^2 \cdot S_t^2 \left(\frac{-1}{S_t^2}\right)\right)dt + \sigma \cdot S_t \cdot \frac{1}{S_t} dB_t$$

So:

$$ln\left(\frac{S_t}{S_{t-1}}\right) = (\mu - 0.5 \cdot \sigma^2)dt + \sigma \cdot dB_t$$

And we finally get:

$$S_t = S_{t-1} \cdot e^{(\mu - 0.5 \cdot \sigma^2)dt + \sigma \cdot \sqrt{t} \cdot Z}; \text{ where } Z \sim N(0,1)$$

---

1. 1. If you have watched stock prices, you know that continuity is usually not a bad assumption. Sometimes, however, it can be disastrous (look at the way stock market prices behaved in October 1987, for a dramatic example of price *discontinuities*). It is possible to build a stock price model that assumes that prices are usually continuous but have occasional (and random) jumps. See Cox and Ross (1976), Jarrow and Rudd (1983), and Merton (1976).
2. 2. If you know about diffusion processes, then the lognormal price process is a *geometric diffusion*, $\frac{dS}{S} = \mu dt + \sigma dB$, where dB is a Wiener process ("white noise"); $dB = Z\sqrt{dt}$, where Z is a standard random variable. See appendix 23.1 for implementation of Ito's lemma.
3. 3. The number of business days in a year is approximately 250. Thus, when we define $\Delta\ t = 1/250 = 0.004$, we are simulating the stock price on a daily basis over the course of a year.
4. 4. See Chapter 21 for techniques (using Excel, VBA, and R) for generating random numbers.
5. 5. For a discussion of this point, see Brealey, Myers, and Allen (2013), chap. 13; for a more advanced treatment, see Copeland, Weston, and Shastri (2003), chaps. 10–11.
6. 6. The use of **Norm.S.Inv** and **Rand()** in Excel, and the use of **rnorm()** in R, for this purpose is discussed in section 21.4.
7. 7. Itô (1944, 1951).
8. 8. For a good reference, see Protter (2005).

24 Monte Carlo Simulations for Investments

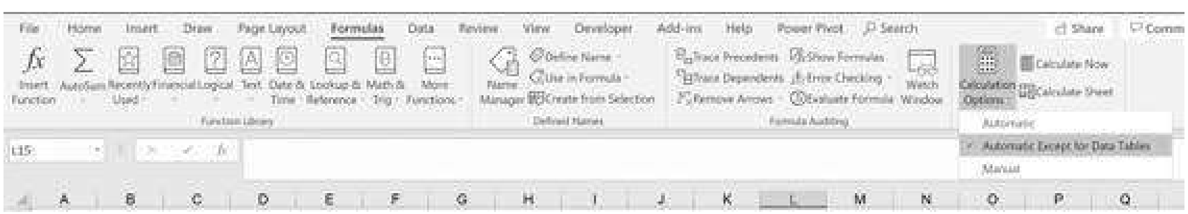
24.1 Overview

In this chapter we simulate the performance of portfolio investments of one or more stocks. We start with simulations of a single stock (a slight repetition of materials covered in Chapter 23). We then deal with correlation, first discussing the case of two correlated stock returns, then generalizing to portfolios of multiple stocks, using the Cholesky decomposition (Chapter 21). We go on to discuss simulations of pension problems, where portfolio investment is used to finance future withdrawals. Finally, we discuss the simulation of β, showing that low β stocks have higher α.

Throughout this chapter we use the technique of running a **Data Table** on a blank cell to do sensitivity analysis. This technique, which allows us to run multiple random simulations, is described in Chapter 28.

A Computational Note

The spreadsheets for this chapter are very computationally intensive. We advise turning off the automatic computation of **Data Tables** (**File|Options|Calculation Options|Automatic except for Data Tables**).



24.2 Simulating Price and Returns for a Single Stock

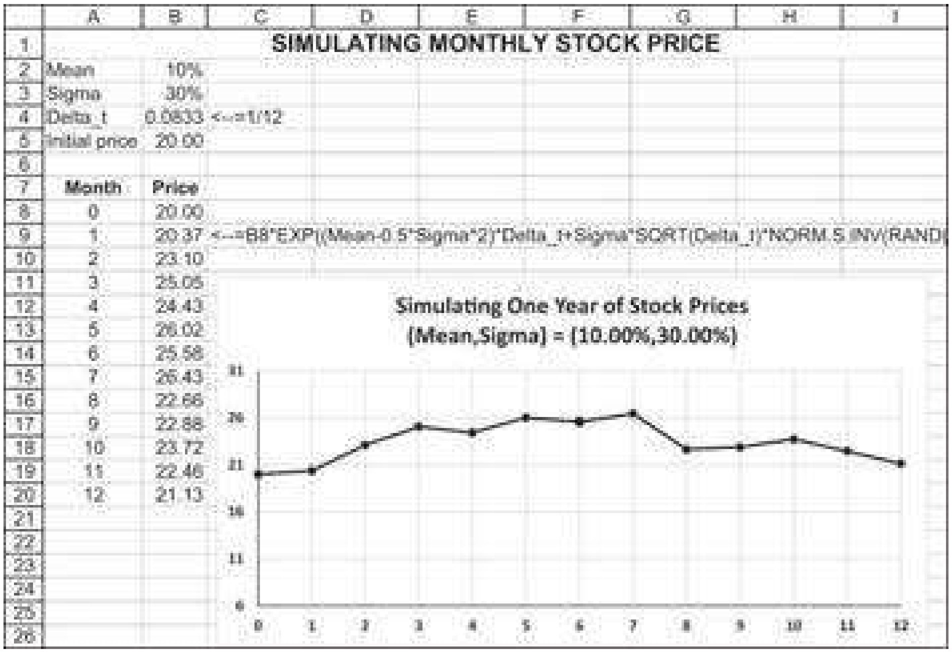
In this section we simulate portfolio investments. We start with an exercise we have already done in Chapter 23—namely, simulating stock returns over time. Suppose we start with one stock and consider a possible future price path for this stock. In the spreadsheet below we simulate such a price path: At each point t, we generate a random return for the stock:

$$1+\tilde{r}_t = e^{(\mu-0.5\sigma^2)\Delta t+\sigma\sqrt{\Delta t}Z}$$

Here, Z is a standard normal deviate generated by **Norm.S.Inv(rand())**.[1] The stock price at time t is given by:

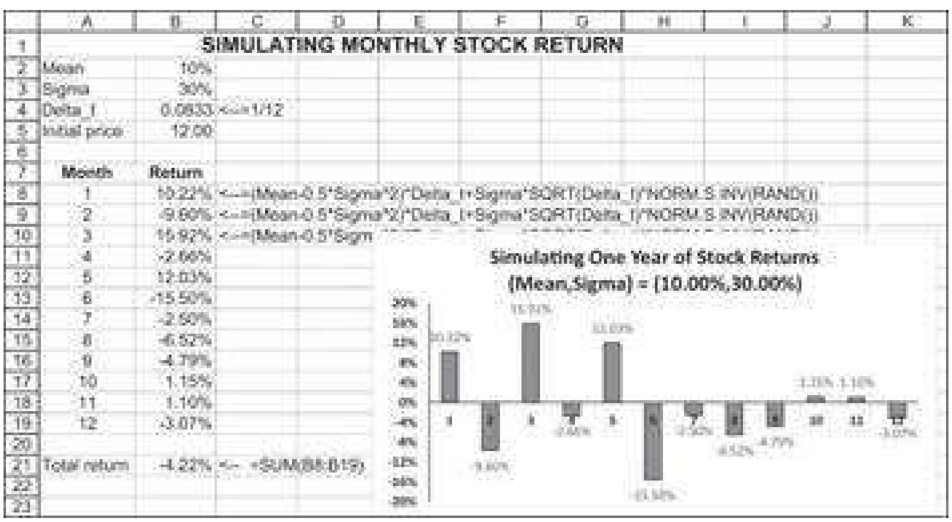
$$S_{t+\Delta t} = S_t exp[(\mu-0.5\sigma^2)\Delta t+\sigma\sqrt{\Delta t}Z]$$

Below is a simulation example for monthly stock prices:



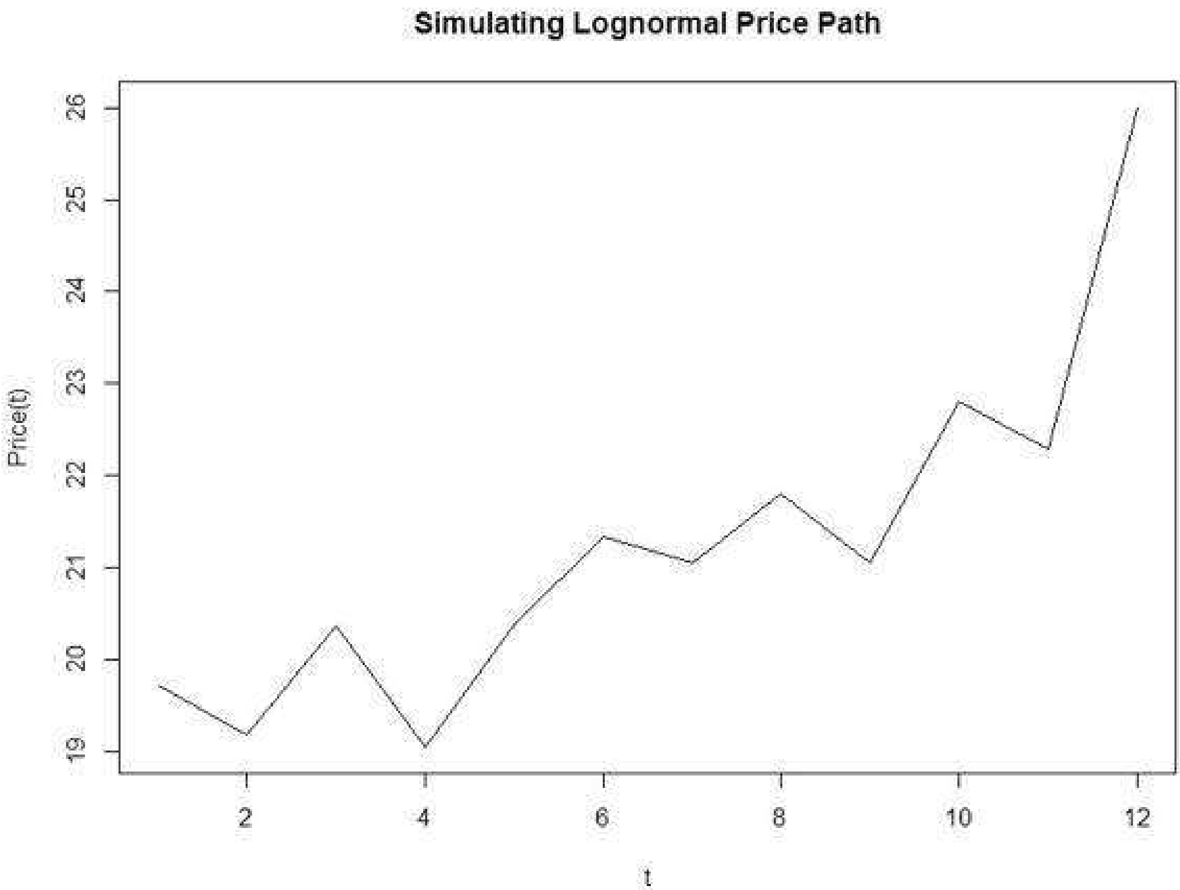
Working with Returns Instead of Prices

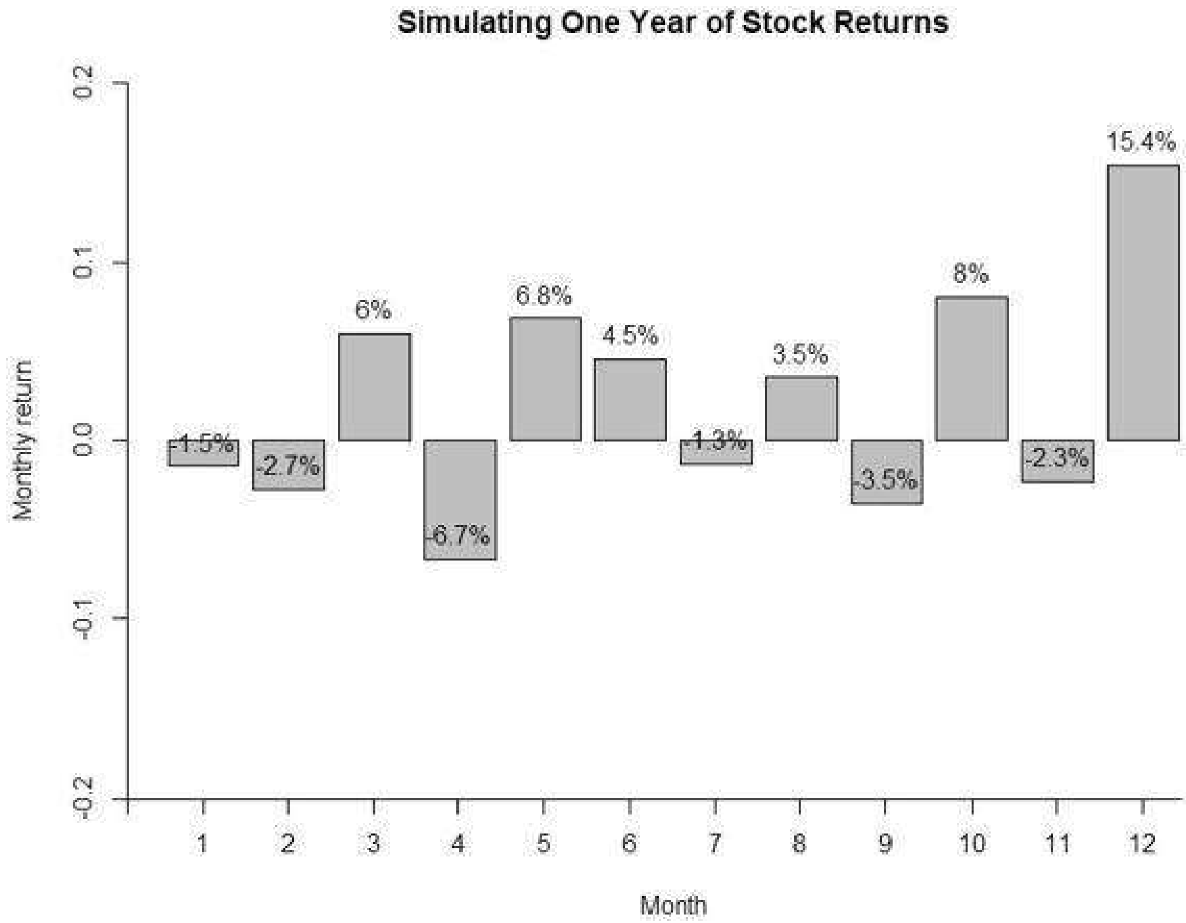
If we do this simulation with stock returns, we get the following:



Implementing this approach in R looks like this:

```r
# Inputs:
mean <- 0.1
sigma <- 0.3
delta_t <- 1/12
s_zero <- 20

Z <- rnorm(12, 0, 1) # Generate 12 normally distributed varibles

# convert to 12 log-normally distributed varibles
log_norm_var <- exp( (mean - 0.5 * sigma ^ 2) * delta_t +
                     sigma * sqrt(delta_t) * Z)

# Compute compounded returns
comp_ret <- cumprod(log_norm_var) #cumprod is like sumproduct() in excel

# End price = initial price * compounded returns
sim_price <- s_zero * comp_ret

# Plotting Stock Price
plot(1:12, sim_price, type = "l", xlab = "t", ylab = "Price(t)",
     main = "Simulating Lognormal Price Path")

# plot returns
monthly_returns <- as.table(log(log_norm_var))
row.names(monthly_returns) <- as.numeric(1:12)

bp <- barplot(monthly_returns,  ylab="Monthly return", xlab="Month", ylim=c(-0.2,0.2),
              main = "Simulating One Year of Stock Returns")
bp_lables <- paste(round(log(log_norm_var),3) * 100, "%", sep = "")
text(x = bp, y = log(log_norm_var),labels = bp_lables, pos = 3)
```





24.3 Portfolio of Two Stocks

We extend the exercise from the previous section to two stocks with correlation ρ. To simplify matters, we will work only in returns, not in prices.

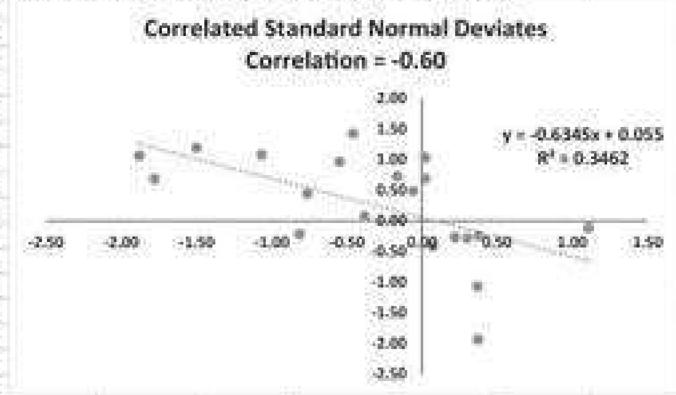
Theory

We recall from Chapter 21 that two standard normal deviates Z_1 and Z_2 are correlated with correlation ρ if

$$Z_2 = \rho Z_1 + Z_3\sqrt{1-\rho^2},$$

where Z_1 and Z_3 are both standard uncorrelated normal deviates (produced with the Excel function **Norm.S.Inv(rand())**. We simulate this below:

	A	B	C
1	SIMULATING TWO CORRELATED NORMAL DEVIATES		
2	Correlation	-0.6	
3			
4	Z_1	Z_2	
5	-1.8798	1.0618	<--=$B$2*A5+SQRT(1-$B$2^2)*NORM.S.INV(RAND())
6	-1.5015	1.1861	
7	0.0212	1.0284	
8	-0.5472	0.9640	
9	-0.8144	-0.2236	
10	0.3682	-1.0699	
11	0.0737	-0.4209	
12	-0.3801	0.0717	
13	1.1096	-0.1173	
14	0.3015	-0.2848	
15	0.0208	0.6868	
16	-1.0677	1.0769	
17	-0.0579	0.4853	
18	0.3720	-1.9384	
19	-0.7620	0.4423	
20	-0.1633	0.7248	
21	0.2180	-0.2698	
22	-0.4592	1.4206	
23	0.3776	-0.2497	
24	-1.7813	0.6825	

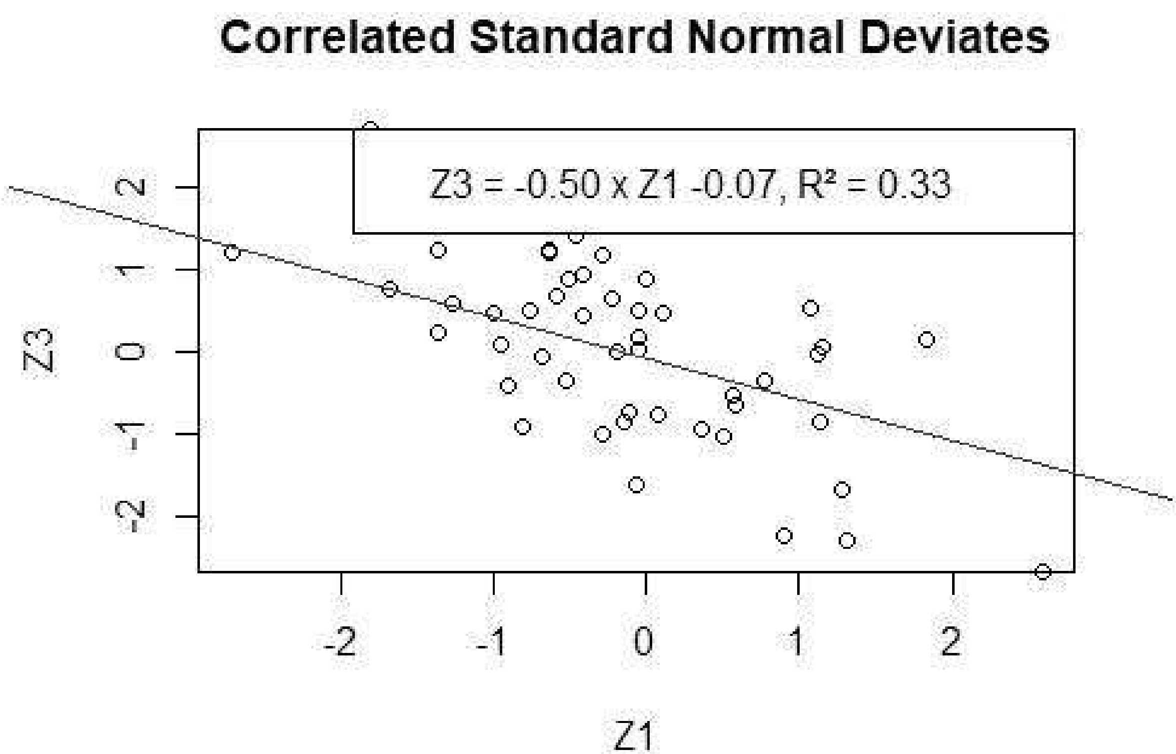


The simulation shows the regression of Z_2 on Z_1. The anticipated regression intercept should be 0, the anticipated regression slope should be the correlation ρ, and the R^2 should be ρ^2. But because we are simulating random numbers, this will never exactly happen. However, if we run this experiment many times (using **Data Table** on blank cells, section 28.7), we see that we get the approximate result:

	J	K	L	M	N	O
2		Data table statistics				
3		Correlation	Intercept	Slope	R-squared	
4	Average	-0.597	-0.018	-0.599	0.378	<--=AVERAGE(N12:N211)
5	Anticipated	-0.600	0.000	-0.600	0.360	<--=B2^2
6	Max	-0.139	0.553	-0.154	0.749	<--=MAX(N12:N211)
7	Min	-0.866	-0.528	-1.180	0.019	<--=MIN(N12:N211)
8						
9		Data table: 200 simulations				
10	Simulation	Correlation	Intercept	Slope	R-squared	
11		-0.6266	0.3236	-0.4858	0.3926	<--=RSQ(B5:B24,A5:A24)
12	1	-0.7108	-0.2080	-0.5656	0.5053	=SLOPE(B5:B24,A5:A24)
13	2	-0.5064	-0.1525	-0.8115	0.2565	=INTERCEPT(B5:B24,A5:A24)
14	3	-0.7427	0.4290	-0.9560	0.5517	=CORREL(A5:A24,B5:B24)
15	4	-0.7410	0.1736	-0.7220	0.5491	
16	5	-0.6020	-0.1364	-0.6815	0.3624	
207	196	-0.4971	-0.0528	-0.4495	0.2471	
208	197	-0.1924	-0.1772	-0.1998	0.0370	
209	198	-0.3298	0.0977	-0.1702	0.1088	
210	199	-0.5936	0.0160	-0.5233	0.3523	
211	200	-0.6923	-0.1607	-0.8104	0.4793	

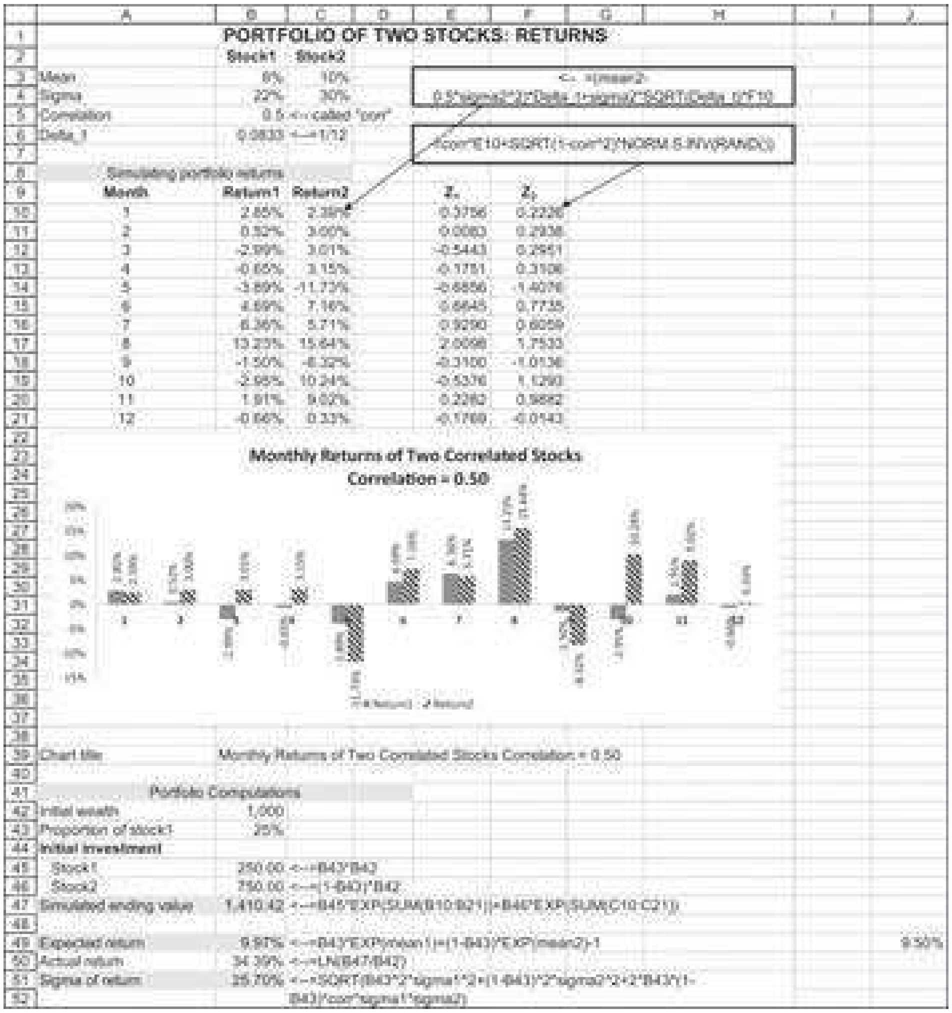
Performing the same in R should look like this:

```r
# Two Correlated Standard Normal Variables
n <- 50
Z1 <- rnorm(n = n, mean = 0, sd = 1)
Z2 <- rnorm(n = n, mean = 0, sd = 1)

# Input desired correlation:
Rho <- -0.6

# Two correlated numbers
Z3 <- Z1 * Rho + Z2 * sqrt(1 - Rho^2)

# Plot Envelope
plot(Z1, Z3, main="Correlated Standard Normal Deviates",
     type="p", xlim=c(min(Z1),max(Z1)))
regression <- lm(Z1 ~ Z3)
abline(regression, col="blue", lwd=1)
coeff <- round(coef(regression), 4)
rsquared <- summary(regression)$r.squared
legend('topright', legend = sprintf("Z3 = %3.2f x Z1 %+ 3.2f, R\UB2 = %3.2f",
                                   coeff[2], coeff[1], rsquared))
```



Simulating Correlated Stock Returns

Below we simulate two correlated normal standard deviates Z_1 and Z_2. We then use these deviates to compute the returns of the two stocks in months 1, 2, …, 12.



	A	B	C	D	E	F	G	H	I	J
1		PORTFOLIO OF TWO STOCKS: RETURNS								
2		Stock1	Stock2							
3	Mean	6%	10%		<-- =(mean2-					
4	Sigma	22%	30%		0.5*sigma2^2)*Delta_t+sigma2*SQRT(Delta_t)*F10					
5	Correlation	0.5	<-- called "corr"							
6	Delta_t	0.0833	<-- =1/12		=corr*E10+SQRT(1-corr^2)*NORM.S.INV(RAND())					
7										
8	Simulating portfolio returns									
9	Month	Return1	Return2		Z_1	Z_2				
10	1	2.65%	2.38%		0.3756	0.2226				
11	2	0.52%	3.00%		0.0083	0.2938				
12	3	-2.99%	3.01%		-0.5443	0.2951				
13	4	-0.65%	3.15%		-0.1751	0.3106				
14	5	-3.89%	-11.73%		-0.6856	-1.4076				
15	6	4.69%	7.16%		0.6645	0.7735				
16	7	6.36%	5.71%		0.9290	0.6059				
17	8	13.23%	15.64%		2.0098	1.7533				
18	9	-1.50%	-8.32%		-0.3100	-1.0136				
19	10	-2.95%	10.24%		-0.5376	1.1293				
20	11	1.91%	9.02%		0.2262	0.9882				
21	12	-0.66%	0.33%		-0.1769	-0.0143				
39	Chart title	Monthly Returns of Two Correlated Stocks Correlation = 0.50								
41	Portfolio Computations									
42	Initial wealth	1,000								
43	Proportion of stock1	25%								
44	**Initial investment**									
45	Stock1	250.00	<-- =B43*B42							
46	Stock2	750.00	<-- =(1-B43)*B42							
47	Simulated ending value	1,410.42	<-- =B45*EXP(SUM(B10:B21))+B46*EXP(SUM(C10:C21))							
49	Expected return	9.97%	<-- =B43*EXP(mean1)+(1-B43)*EXP(mean2)-1							9.50%
50	Actual return	34.39%	<-- =LN(B47/B42)							
51	Sigma of return	25.70%	<-- =SQRT(B43^2*sigma1^2+(1-B43)^2*sigma2^2+2*B43*(1-							
52			B43)*corr*sigma1*sigma2)							

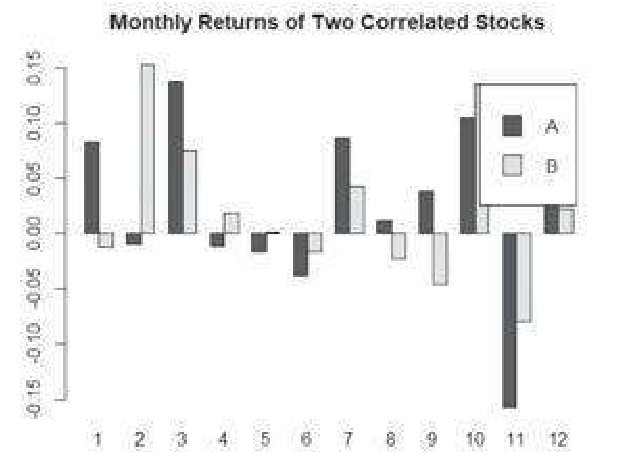
Implementing the same approach in R looks like this:

```r
## two correlated stocks
# Input
mean <- 0.1
sigma <- 0.3
n <- 12
t <- 1
delta_t <- t/n
S0 <- 20

# Two correlated numbers
Z1 <- rnorm(n = n, mean = 0, sd = 1)
Z2 <- rnorm(n = n, mean = 0, sd = 1)
Rho <- 0.5
Z3 <- Z1 * Rho + Z2 * sqrt(1 - Rho^2)

# Calculate St based on Z1 and Z3
log_norm_ret <- sapply(list(Z1, Z3) , function(Z) exp( (mean - 0.5 * sigma ^ 2) *
                          delta_t + sigma * sqrt(delta_t) * Z))

# Compute compounded returns
comp_ret <- apply(log_norm_ret, 2, cumprod)

# Initial price * compounded returns
sim_price <- S0 * comp_ret

# Returns table
log_norm_ret_Tables <- rep(c(1,2),12)

# Plot
monthly_returns <- as.table(log(log_norm_ret))
row.names(monthly_returns) <- as.numeric(1:12)

bp <- barplot(t(monthly_returns),beside = TRUE,
        main = 'Monthly Returns of Two Correlated Stocks',
        legend = colnames(monthly_returns))
```

Results:

```
> monthly_returns
              A            B
1   0.083096708 -0.012264143
2  -0.009426278  0.152776440
3   0.136817200  0.074355347
4  -0.012232044  0.017912154
5  -0.016596626  0.001106273
6  -0.038442610 -0.017079708
7   0.087141214  0.042028206
8   0.011433479 -0.022865491
9   0.038539706 -0.045438211
10  0.104381756  0.134807160
11 -0.156969585 -0.080178413
12  0.059421752  0.022253362
> |
```

Plot:



24.4 Adding a Risk-Free Asset

We add a risk-free asset to the exercise in the previous section, simulating the performance of an investment in a portfolio of the two risky stocks and the risk-free asset. Cells B13 and B14 below give the risky portfolio expected annual return and annual sigma. Cells B19 and B20 show the expected return and sigma for a portfolio invested 40% in the risky stocks and 60% in the risk-free asset:

	A	B	C	D
1	PORTFOLIO OF TWO STOCKS AND RISK-FREE			
2		Stock1	Stock2	
3	Mean	8%	10%	<--cell names: mean1, mean2
4	Sigma	15%	20%	<--cell names: sigma1, sigma2
5	Correlation	0.2	<--cell name: corr	
6	Delta_t	0.0833	<--=1/12, cell name: delta_t	
7				
8	Risk-free rate, r_f	2%	<--cell name: rf	
9				
10	**Risky portfolio**			
11	Stock1	30%	<--cell name: prop1	
12	Stock2	70%	<--=1-prop1	
13	Expected annual return	9.86%	<--=prop1*EXP(mean1)+(1-prop1)*EXP(mean2)-1	
14	Sigma annual return	15.54%	<--=SQRT(prop1^2*sigma1^2+(1-	
15			prop1)^2*sigma2^2+2*prop1*(1-prop1)*corr*sigma1*sigma2)	
16	**Investment**			
17	In risky portfolio	40%	<--cell name: prop	
18	In risk free	60%	<--=1-prop	
19	Expected return	5.36%	<--=prop*EXP(B13)+(1-prop)*EXP(rf)-1	
20	Sigma return	6.22%	<--=prop*B14	
21				
22	**Annual return**			
23	Simulated	14.09%	<--=SUM(D36:D47)	
24	Expected	5.36%	<--=B19	
25				
26	**Sigma of annualized return**			
27	Simulated	5.90%	<--=SQRT(12)*STDEV.S(D36:D47)	
28	Expected	6.22%	<--=B20	
29				
30	=prop*(prop1*(EXP((mean1-0.5*sigma1^2)*delta_t+sigma1*SQRT(delta_t)*B36))+(1-			
31	prop1)*EXP(((mean2-0.5*sigma2^2)*delta_t)+sigma2*SQRT(delta_t)*C36))+(1-			
32				
33		=NORM.S.INV(RAND())		
34				
35	**Month**	**Z_1**	**Z_2**	**Portfolio return**
36	1	1.02029	0.49823	1.72%
37	2	-1.32364	-0.1534	-0.56%
38	3	1.60628	1.8338	4.37%
39	4	0.77866	0.6399	1.83%
40	5	-0.54627	-0.30933	-0.42%
41	6	-0.32802	-1.57877	-2.27%
42	7	0.9791	0.33394	1.43%
43	8	0.60101	0.63958	1.73%
44	9	0.09366	1.07965	2.22%
45	10	-0.11854	0.88928	1.78%
46	11	-0.3992	0.03615	0.21%
47	12	0.59715	0.82629	2.05%
48				
49	=corr*B36+SQRT(1-corr^2)*NORM.S.INV(RAND())			
50				

24.5 Multiple Stock Portfolios

So far, we have simulated the performance of a portfolio of two stocks. When we turn to multiple stocks, we need to make use of the Cholesky decomposition (see section 21.9). We remind you how the Cholesky decomposition works: We want to create a set of normal deviates that has variance-covariance structure S:

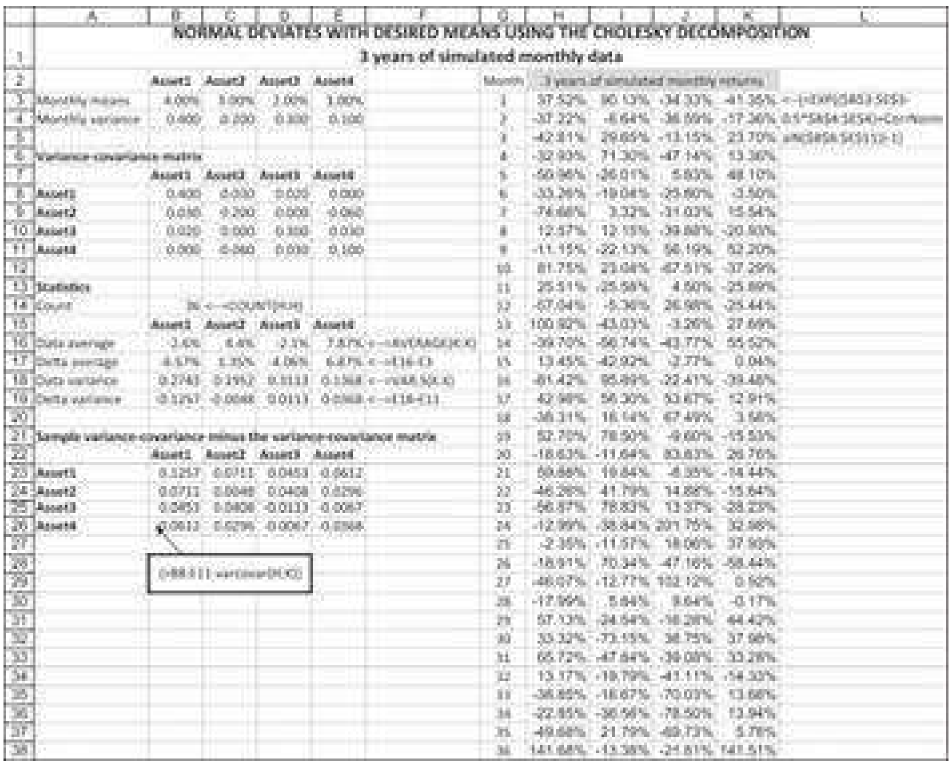
$$S = \begin{bmatrix} \sigma_{11} & \sigma_{12} & \cdots & \sigma_{1N} \\ \sigma_{21} & \sigma_{22} & \cdots & \sigma_{2N} \\ \vdots & \vdots & \ddots & \vdots \\ \sigma_{N1} & \sigma_{N2} & \cdots & \sigma_{NN} \end{bmatrix}$$

As explained in Chapter 21, the steps in doing so are as follows:

1. Create a lower-triangular Cholesky decomposition of S. We denote this matrix by L and use the VBA function **Cholesky** (explained in chapter 21) to compute the matrix.
2. Create a column vector of N standard normal deviates. We use the Excel function **Norm.S.Inv(Rand())**.
3. Multiply L times the column vector of standard normal deviates.
4. The result is a set of correlated standard normal variates.

Example: Standard Normal Deviates with Given Correlation Structure

Here's an example. We define a VBA function **CorrNormal** that takes as its argument the desired variance-covariance matrix. Below we use this function to simulate three years of monthly returns with means specified in cells B3:E3 and variance-covariance in the cells below the means.



NORMAL DEVIATES WITH DESIRED MEANS USING THE CHOLESKY DECOMPOSITION

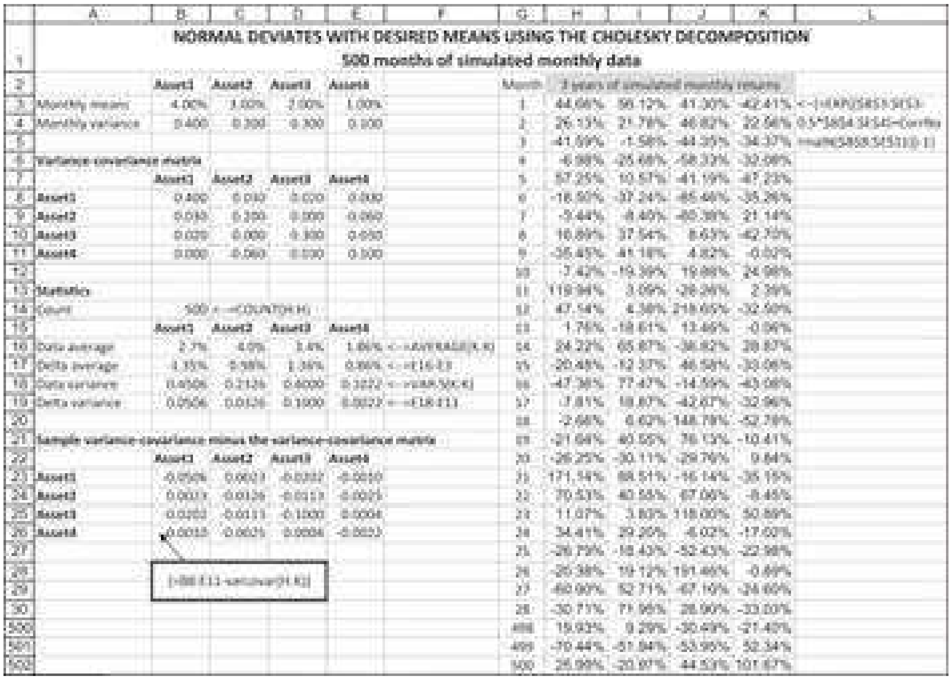
3 years of simulated monthly data

	Asset1	Asset2	Asset3	Asset4	
Monthly means	4.00%	5.00%	2.00%	3.00%	
Monthly variance	0.400	0.200	0.300	0.100	
Variance-covariance matrix					
	Asset1	Asset2	Asset3	Asset4	
Asset1	0.400	0.030	0.020	0.000	
Asset2	0.030	0.200	0.000	-0.060	
Asset3	0.020	0.000	0.300	0.030	
Asset4	0.000	-0.060	0.030	0.100	
Statistics					
Count		36 <-- =COUNT(H:H)			
	Asset1	Asset2	Asset3	Asset4	
Data average	-2.4%	6.4%	-2.1%	7.87%	<-- =AVERAGE(K:K)
Delta average	-6.57%	1.35%	-4.06%	6.87%	<-- =E16-E3
Data variance	0.2743	0.1952	0.3113	0.1368	<-- =VAR.S(K:K)
Delta variance	-0.1257	-0.0048	0.0113	0.0368	<-- =E18-E11
Sample variance-covariance minus the variance-covariance matrix					
	Asset1	Asset2	Asset3	Asset4	
Asset1	-0.1257	0.0711	0.0453	-0.0612	
Asset2	0.0711	-0.0048	0.0408	0.0296	
Asset3	0.0453	0.0408	-0.0113	-0.0067	
Asset4	-0.0612	0.0296	-0.0067	0.0368	

{=B8:E11-varcovar(H:K)}

Month	3 years of simulated monthly returns				
1	37.52%	90.13%	-34.33%	-41.35%	<-- {=EXP($B$3:$E$3-0.5*$B$4:$E$4+CorrNormal($B$8:$E$11))-1}
2	-37.22%	-6.64%	-36.59%	-17.36%	
3	-42.81%	29.65%	-13.15%	23.70%	
4	-32.93%	71.30%	-47.14%	13.36%	
5	-50.96%	-26.01%	5.83%	48.10%	
6	-33.26%	-19.04%	-25.60%	-3.50%	
7	-74.68%	3.32%	-31.03%	15.54%	
8	12.57%	12.15%	-39.88%	-20.93%	
9	-11.15%	-22.13%	56.19%	52.20%	
10	81.75%	23.08%	-67.51%	-37.29%	
11	25.51%	-25.58%	4.50%	-25.89%	
12	-57.04%	-5.36%	26.98%	-25.44%	
13	100.92%	-43.03%	-3.26%	27.69%	
14	-39.70%	-56.74%	-43.77%	55.52%	
15	13.45%	-42.92%	-2.77%	0.04%	
16	-81.42%	95.89%	-22.41%	-39.48%	
17	42.98%	56.30%	53.67%	12.91%	
18	-38.31%	16.14%	67.49%	3.58%	
19	52.70%	78.50%	-9.60%	-15.53%	
20	-18.63%	-11.64%	83.83%	26.76%	
21	59.88%	19.84%	-8.35%	-14.44%	
22	-46.28%	41.79%	14.88%	-15.64%	
23	-56.87%	78.83%	13.37%	-28.23%	
24	-12.99%	-38.84%	201.75%	32.98%	
25	-2.35%	-11.57%	18.06%	37.93%	
26	-18.91%	70.34%	-47.16%	-58.44%	
27	-46.07%	-12.77%	102.12%	0.92%	
28	-17.99%	5.84%	9.64%	-0.17%	
29	57.13%	-24.54%	-16.28%	44.42%	
30	33.32%	-73.15%	38.75%	37.98%	
31	65.72%	-47.84%	-39.08%	33.28%	
32	13.17%	-19.79%	-41.11%	-54.33%	
33	-36.85%	-16.67%	-70.03%	13.68%	
34	-22.85%	-38.58%	-78.50%	13.94%	
35	-49.68%	21.79%	-69.73%	5.78%	
36	141.68%	-13.38%	-21.81%	141.51%	

If we rerun this simulation for 500 months, the simulation more closely corresponds to the priors:



The VBA Function CorrNormalN

The VBA function that creates the correlated normal deviates is a combination of two functions. The second of these functions **URandomlist** creates a column vector of standard normal deviates. These deviates are then multiplied times the Cholesky matrix.

```
Function CorrNormalN(mat As Range) As Variant
CorrNormalN =
Application.Transpose(Application.MMult(Cholesky(mat), _
    urandomlist(mat)))
End Function
Function urandomlist(mat As Range) As Variant
    Application.Volatile
    Dim vector() As Double
    numCols = mat.Columns.Count
    ReDim vector(numCols - 1, 0)
    For i = 1 To numCols
        vector(i - 1, 0) = Application.Norm_S_Inv(Rnd)
    Next i
urandomlist = vector
End Function
```

Implementing this example in R should look like this:

```
# Input: Variance-Covariance Matrix
cov_mat <- matrix(c(0.400,  0.030,  0.020,  0.000,
                    0.030,  0.200,  0.000,  -0.060,
                    0.020,  0.000,  0.300,  0.030,
                    0.000,  -0.060, 0.030,  0.100),
                  nrow = 4)

# Lower-Triangular Cholesky Decomposition
chol_decompos <- t(chol(cov_mat))

# Simulate Monthly Returns Of n Correlated Stocks
month_ret <- sapply(1:500000, function(x)
    chol_decompos %*% rnorm(n = 4, mean = 0, sd = 1))

# Check: Is The Sample Varince Close To The Input Variance?
cov_mat - cov(t(month_ret))
```

The “check” result is:

```
> cov_mat - cov(t(month_ret))
              [,1]          [,2]          [,3]          [,4]
[1,] -1.236769e-04 -0.0008887235  9.826772e-06  7.154813e-04
[2,] -8.887235e-04 -0.0006424022  8.960800e-05  1.950085e-04
[3,]  9.826772e-06  0.0000896080 -3.990227e-04 -4.030114e-05
[4,]  7.154813e-04  0.0001950085 -4.030114e-05 -1.474139e-04
> |
```

That is, it is very small and, for all practical use, can be considered to be 0.

24.6 Simulating Savings for Pensions

We return to a problem that we discussed in Chapter 1: the case of a potential pensioner who wants to save for five years in order to enable eight subsequent withdrawals of 30,000. The question we discussed in section 1.6 is the calibration of the annual deposit so that the pension fund (with accumulated 8% annual interest) would be completely depleted after eight years.

Here’s the answer to this problem:

	A	B	C	D	E	F
1	A RETIREMENT PROBLEM, Section 1.6					
2	Interest	8%				
3	Annual deposit	29,386.55				=$B$2*(C7+B7)
4	Annual retirement withdrawal	30,000.00				
5						
6	Year	Account balance, beginning of year	Deposit at beginning of year	Interest earned during year	Total in account, end year	
7	1	0.00	29,386.55	2,350.92	31,737.48	<--=D7+C7+B7
8	2	31,737.48	29,386.55	4,889.92	66,013.95	
9	3	66,013.95	29,386.55	7,632.04	103,032.54	
10	4	103,032.54	29,386.55	10,593.53	143,012.62	
11	5	143,012.62	29,386.55	13,791.93	186,191.10	
12	6	186,191.10	-30,000.00	12,495.29	168,686.39	
13	7	168,686.39	-30,000.00	11,094.91	149,781.30	
14	8	149,781.30	-30,000.00	9,582.50	129,363.81	
15	9	129,363.81	-30,000.00	7,949.10	107,312.91	
16	10	107,312.91	-30,000.00	6,185.03	83,497.94	
17	11	83,497.94	-30,000.00	4,279.84	57,777.78	
18	12	57,777.78	-30,000.00	2,222.22	30,000.00	
19	13	30,000.00	-30,000.00	0.00	0.00	
20						
21	**Note**: This problem has 5 deposits and 8 annual withdrawals, all made at the beginning of the year. The beginning of year 13 is the last year of the retirement plan; if the annual deposit is correctly computed, the balance at the beginning of year 13 after the withdrawal should be zero.					

In this section we discuss a Monte Carlo variation of this problem. As in Chapter 1, the future pensioner has no pension savings and desires to make deposits to fund eight subsequent withdrawals. In this version, however, the savings are invested in a risky portfolio having mean return of 8% and return sigma of 18%. We are curious to see whether a specific level of annual deposits can fund the future planned withdrawals of 30,000 per year. To do this, we examine the bequest—the amount left in the pension plan at the end of year 13. Several things are clear:

- Except for the case where the investment is riskless, there is no longer any certainty about the savings or the bequest.
- On average, the larger the variance in the risky asset, the greater will be the average bequest.

Here is one simulation:

	A	B	C	D	E
1	A RETIREMENT PROBLEM - REVISED TO INCLUDE STOCHASTIC RETURN				
2	Risky asset				
3	Mean	8%			
4	Sigma	18%			
5	Annual deposit	29,386.55	<-- Years 1-5		
6	Retirement withdrawal	30,000	<-- Years 6-13		
7					
8	Year	Account balance, beginning of year	Deposit at beginning of year	In account at end of year	
9	1	0	29,387	35,603	<--=SUM(B9:C9)*EXP(($B$3-0.5*$B$4^2)*1+$B$4*SQRT(1)*NORM.S.INV(RAND()))
10	2	35,603	29,387	58,392	
11	3	58,392	29,387	120,929	
12	4	120,929	29,387	157,900	
13	5	157,900	29,387	200,893	
14	6	200,893	-30,000	191,827	
15	7	191,827	-30,000	192,748	
16	8	192,748	-30,000	163,159	
17	9	163,159	-30,000	130,764	
18	10	130,764	-30,000	134,860	
19	11	134,860	-30,000	111,598	
20	12	111,598	-30,000	105,539	
21	13	105,539	-30,000	91,120	

To get a feel for the uncertainty, we run our standard **Data Table** on an empty cell:

	G	H	I
2		Statistics for data table	
3	Average	-77,318	<--=AVERAGE(H12:H21)
4	Max	43,042	<-- =MAX(H12:H21)
5	Min	-181,730	<-- =MIN(H12:H21)
6	Sigma	80,024	<-- =STDEV.S(H12:H21)
7	% positive bequests	20%	<-- =COUNTIF(H12:H21,">0")/10
8			
9		Data table:	
10	Simulation	Bequest	
11		-155,714	<-- =D21, data table header
12	1	18,456	
13	2	-110,970	
14	3	43,042	
15	4	-57,014	
16	5	-26,736	
17	6	-122,816	
18	7	-18,370	
19	8	-181,730	
20	9	-147,626	
21	10	-169,420	

The average bequest for this series of 10 simulations is −77,318, but with considerable variation. In only 20% of the cases can we count on full funding of the pension (in the sense that the bequest is positive).

The above example, implemented using R, looks like this:

```
##  24.6 Simulating Savings for Pensions
mean_ret <- 0.08
sigma_ret <- 0.18
annual_deposit <- 29386.55
retirement_withdrawl <- 30000
balance <- 0
trials <- 20

for (i in 1:13){
  if (i > 5){
    change <- -retirement_withdrawl
          } else {change <- annual_deposit}
  balance <- (balance + change) * exp(mean_ret - 0.5 * sigma_ret^2
                         + sigma_ret * rnorm(trials))
          }
summary(balance)
sd(balance)
```

The results are:

```
> summary(balance)
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
-329084 -187564   -4978  305327  193279 2913190
> sd(balance)
[1] 905131.9
> 
```

24.7 Beta and Return

In this section we simulate a typical beta calculation. We assume that we know the "actual" beta, in the sense that we know the σ_M, σ_i, and the correlation ρ between stock i and the market. We then simulate this beta by creating returns drawn from an appropriate distribution. We use this example to illustrate how far the actual beta, derived from data, can be from the theoretical beta.

Recall from Chapter 11 that for an asset i, β_i is defined as follows:

$$\beta_i = \frac{Cov(r_i, r_M)}{Var(r_M)}$$

In the simulations below, we use two equivalent expressions for β_i. First, using the correlation ρ between i and M, we can write β_i as:

$$\beta_i = \frac{Cov(r_i, r_M)}{Var(r_M)} = \frac{\sigma_i \sigma_M \rho}{\sigma_M^2} = \frac{\sigma_i}{\sigma_M}\rho$$

Second, if we have time-series data $\{r_{it}, r_{Mt}\}$ for the returns on the stock and the market, then we can estimate β_i by running the regression:

$$r_{it} = \alpha_i + \beta_i r_{Mt}$$

If the data is correlated with correlation ρ, then we expect:

$$\alpha_i = E(r_M) - \beta_i E(r_i), \text{ where } \beta_i = \frac{\sigma_i}{\sigma_M}\rho, \text{ and } R^2 = \rho^2$$

In capital market calculations β_i is typically calculated for monthly returns over a period of three to five years. We replicate below this procedure by simulating the 60 correlated returns of two assets. We will call the first asset "i " and the second asset "M." We start with some basic assumptions for i and M:

	A	B	C	D	E	F
1	SIMULATING BETA AND ALPHA					
2		Mean	Sigma			
3	Stock i	6%	22%			
4	Market (M)	10%	15%			
5	Correlation(i,M)	0.3000				
6						
7	Anticipated α_i	0.0160	<--=mu_i-B8*mu_m			
8	Anticipated β_i	0.4400	<--=rho*sigma_i/sigma_m			
9	Anticipated R^2	0.0900	<--=rho^2			

Next we simulate our data:

	A	B	C	D	E	F
1	SIMULATING BETA AND ALPHA					
2		Mean	Sigma			
3	Stock i	6%	22%			
4	Market (M)	10%	15%			
5	Correlation(i,M)	0.3000				
6						
7	Anticipated α_i	0.0160	<--=mu_i-B8*mu_m			
8	Anticipated β_i	0.4400	<--=rho*sigma_i/sigma_m			
9	Anticipated R^2	0.0900	<--=rho^2			
10						
11	Regress r_i on r_M					
12	Alpha	0.0077	<--=INTERCEPT(E19:E78,F19:F78)			
13	Slope	0.4827	<--=SLOPE(E19:E78,F19:F78)			
14	R-squared	0.1236	<--=RSQ(E19:E78,F19:F78)			
15						
16	Simulation					
17	Month	Corr. normal deviates			Returns	
18		Z_1	Z_2		Stock	Market
19	1	-0.7011	-0.3937		-4.15%	-0.97%
20	2	-1.5012	-1.2862		-9.24%	-4.83%
21	3	-0.8703	-0.6137		-5.23%	-1.92%
22	4	0.4765	-0.2016		3.32%	-0.13%
23	5	-0.9787	-0.2497		-5.92%	-0.34%
76	58	1.3179	-1.5871		8.67%	-6.13%
77	59	-0.1241	-0.7526		-0.49%	-2.52%
78	60	-1.5815	0.2181		-9.75%	1.68%
79						
80	Formulas					
81	Cell B19:=NORM.S.INV(RAND())					
82	Cell C19:=rho*B19+SQRT(1-rho^2)*NORM.S.INV(RAND())					
83	Cell E19:=(mu_i-0.5*sigma_i^2)*(1/12)+sigma_i*SQRT(1/12)*B19					
84	Cell F19:=(mu_m-0.5*sigma_m^2)*(1/12)+sigma_m*SQRT(1/12)*C19					

The formulas used in the spreadsheet are the following:

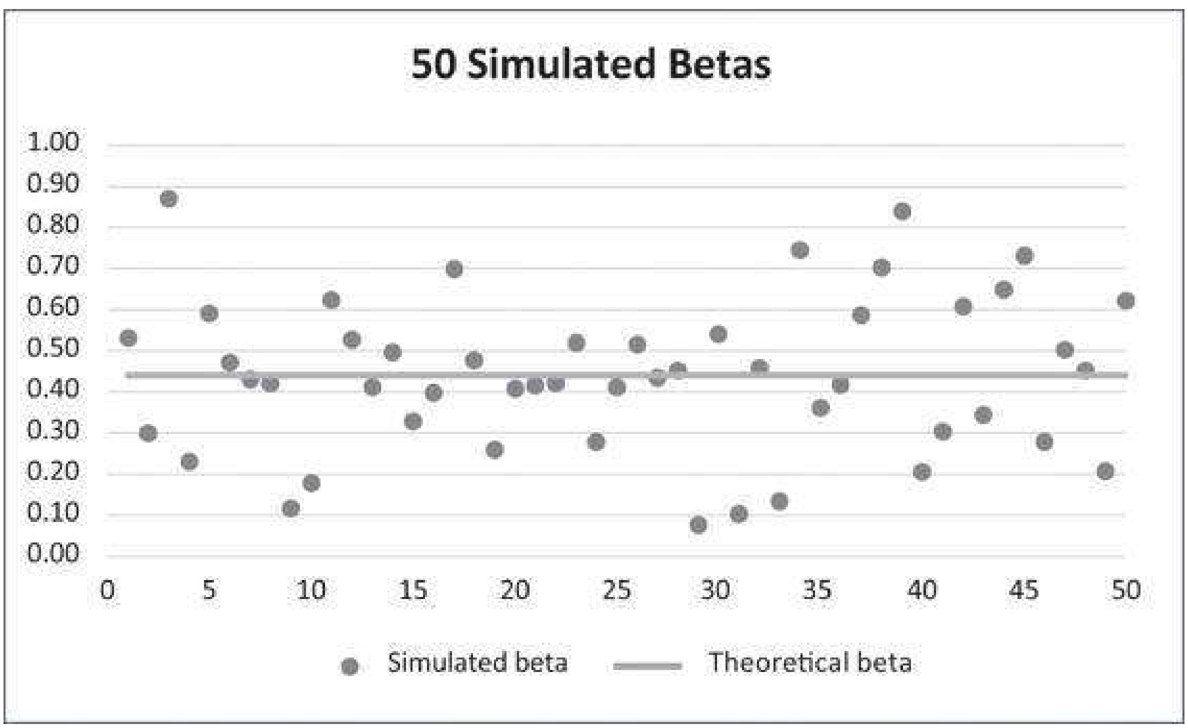
- • $Z_1 = Norm.S.Inv\ (Rand())$. As discussed in Chapter 21, this creates a standard normal deviate.
- • $Z_2 = \rho * Z_1 + Norm.S.Inv(Rand()) * \sqrt{1-\rho^2}$. As discussed in Chapter 21, the standard normal deviate Z_2 has correlation ρ with Z_1.
- • The column for the stock returns is created by $r_{stock} = (\mu_{stock} - 0.5 * \sigma^2_{stock}) * (1/12) + \sigma_{stock} * \sqrt{1/12} * Z_1$.
- • The column for the market returns is created by $r_{market} = (\mu_{market} - 0.5 * \sigma^2_{market}) + \sigma_{market} * \sqrt{1/12} * Z_2$.

The result is that the market and stock returns are correlated with correlation ρ. We now run a standard first-pass regress of r_i on r_M. For the simulations discussed above, the results of the Monte Carlo simulation are not far from the theoretical results:

	A	B	C	D	E	F
1	SIMULATING BETA AND ALPHA					
2		Mean	Sigma			
3	Stock i	6%	22%			
4	Market (M)	10%	15%			
5	Correlation(i,M)	0.3000				
6						
7	Anticipated α_i	0.0160	<--=mu_i-B8*mu_m			
8	Anticipated β_i	0.4400	<--=rho*sigma_i/sigma_m			
9	Anticipated R^2	0.0900	<--=rho^2			
10						
11	Regress r_i on r_M					
12	Alpha	0.0077	<--=INTERCEPT(E19:E78,F19:F78)			
13	Slope	0.4827	<--=SLOPE(E19:E78,F19:F78)			
14	R-squared	0.1236	<--=RSQ(E19:E78,F19:F78)			

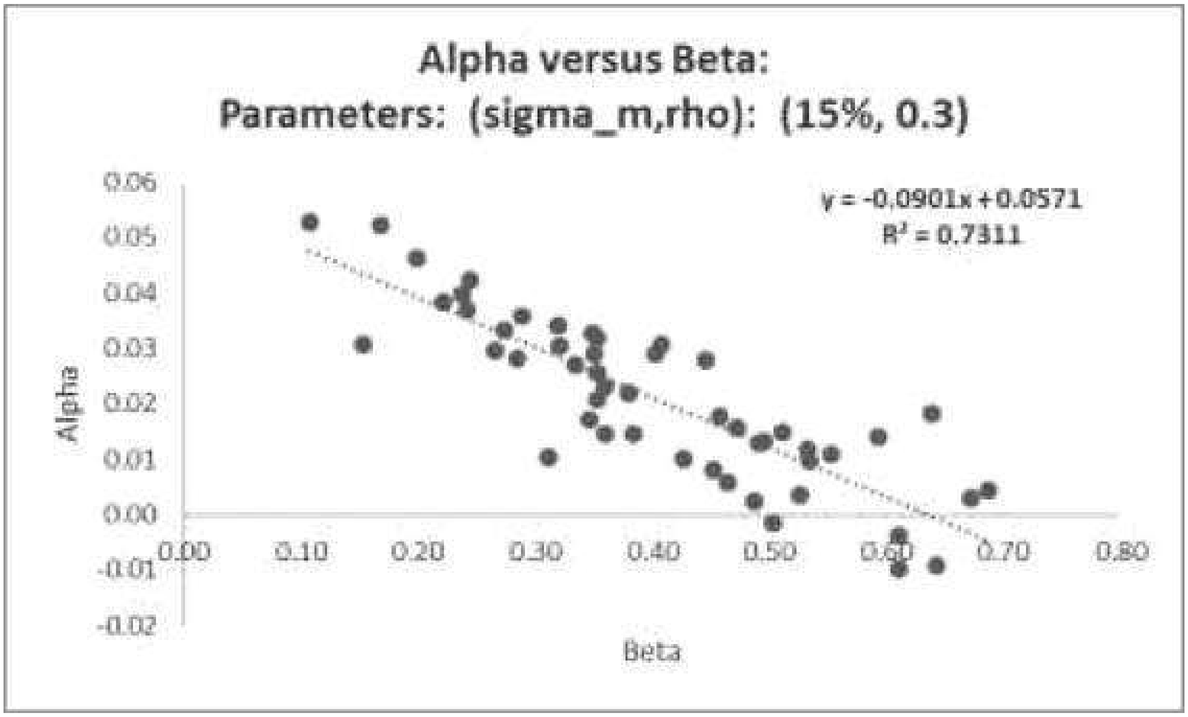
Rerunning the simulation shows that not all results will be close to the desired. If we repeat our experiment 50 times, we see that the computed β_i exhibits considerably variability:

	H	I	J	K	L
3		Statistics for 50 simulations			
4		theoretical beta = 0.44			
5		Average	St. dev	Max	Min
6	Alpha	0.0208	0.0150	0.0529	-0.0097
7	Slope	0.4035	0.1422	0.6872	0.1070
8	R-squared	0.0802	0.0524	0.2206	0.0047



Are Beta and Alpha Related?

There is a growing literature showing that low beta stocks have high alphas and vice versa.[2] We show that this is true in our Monte Carlo simulation. In the spreadsheet below we run a **Data Table** showing beta and alpha as functions of σ_i.[3]



24.8 Summary

When examining investment problems, Monte Carlo techniques give insight that goes beyond the standard calculations of mean and standard deviation of returns. In this chapter we examine some common cases of asset management: the return on a single stock, on a portfolio of risky assets with a given correlations structure, a standard savings/pension problem with an uncertain investment component, and the computation of asset betas.

Exercises

1. Consider a portfolio of two stocks whose statistical parameters are given below.

- Stock A: annual mean return = 15%, annual standard deviation of return = 30%.
- Stock B: μ = 8%, σ = 15%.
- Correlation (A,B) = $\rho_{A,B}$ = 0.3.

An investor with a buy-and-hold strategy buys a portfolio composed of 60% A and 40% B and holds it for 20 years. Simulate the annual returns on the portfolio. A suggested template is given below.

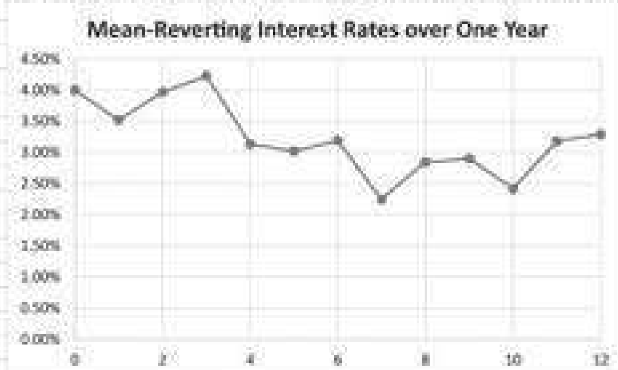
	A	B	C	D	E	F	G	H
1	INVESTING IN A PORTFOLIO OF TWO STOCKS							
2		StockA	StockB					
3	Mean	15%	8%					
4	Sigma	30%	15%					
5	Correlation	0.3						
6	Proportion of A	60%						
7								
8	Summary of portfolio returns							
9		Theoretical	Actual					
10	Mean							
11	Sigma							
12								
13		Simulated returns				Normal deviates		
14	Year	A	B					
15	1							
16	2							

2. Reconsider the problem above. Assume that the risk-free rate is 4% and that the investor (still "buy and hold") invests in a portfolio composed of 50% risk-free and 50% invested in the 60/40 portfolio of A and B. Compare the theoretical to the simulated returns.
3. The previous example assumes that the risk-free rate is constant. An alternative, perhaps more plausible model might be to assume that the risk-free rate is mean reverting, with a long-run mean. Under this assumption, if the current rate is above the long-run mean, the next period rate will trend downward, and vice versa. One such model is the Ornstein-Uhlenbeck process (see Ornstein and Uhlenbeck 1930):

$$r_t = r_{t-1} + \underbrace{\varphi(\mu - r_{t-1})\Delta t + \sigma\sqrt{\Delta t}Z}_{\text{Innovation in the rate}}$$

Simulate this process over 12 months:

	A	B	C
1	ORNSTEIN-UHLENBECK PROCESS FOR INTEREST RATES		
2	Current rate	4%	
3	Mean, μ	3%	$r_t = r_{t-1} + \underbrace{\varphi(\mu - r_{t-1})\Delta t + \sigma\sqrt{\Delta t}\,Z}_{\text{Innovation in the rate}}$
4	"Pressure", φ	0.10	
5	Sigma, σ	2%	
6	Δt	0.0833	<-- =1/12
7			
8	Month	Rate	
9	0	4.00%	
10	1	3.52%	<-- =B9+$B$4*($B$3-B9)*$B$6+$B$5*SQRT($B$6)*NORM.S.INV(RAND())
11	2	3.97%	
12	3	4.23%	
13	4	3.13%	
14	5	3.02%	
15	6	3.19%	
16	7	2.25%	
17	8	2.84%	
18	9	2.90%	
19	10	2.42%	
20	11	3.18%	
21	12	3.29%	



4. The website that accompanies this book gives five years of monthly price data for five U.S. stocks.

- Compute the monthly returns for the stocks.
- Compute the stocks' average monthly returns and standard deviations.
- Compute the variance-covariance matrix for the stock returns.
- Compute the correlation matrix of returns for the stocks.

- Compute the lower-Cholesky matrix for the variance-covariance structure.

Using the data from this file, simulate 36 months of stock returns assuming the same variance-covariance structure as the historical returns. Notice that it doesn't make sense to assume that the forward-looking expected monthly returns are the same as the historical returns. Instead, use the following values:

	A	B	C	D	E	F
2		Monthly mean and sigma				
3		JCP	AAPL	C	F	K
4	Historical mean	-1.08%	1.46%	-2.02%	2.02%	0.75%
5	Anticipated future mean	2.00%	1.50%	1.00%	2.00%	0.60%

---

1. 1. In case you are using Excel 2007 or earlier versions, replace this function with **NormSInv(Rand())**.
2. 2. For references, see Baker and Haugen (2012); Cremers, Petajisto, and Zitzewitz (2010); Frazzini and Pedersen (2012); and Hong and Sraer (2012).
3. 3. Of course, the higher the σ_I, the higher will be β_I.

25 Value at Risk (VaR)

25.1 Overview

Value at risk (VaR) measures the worst expected loss under normal market conditions over a specific time interval at a given confidence level. As one of our references states: "VaR answers the question: How much can I lose with x% probability over a preset horizon?" (Finger et al. 2002). Another way of expressing this is that VaR is the lowest percentile of the potential losses that can occur within a given portfolio during a specified period. The basic period "T" and the confidence level (the percentile) "p" are the two major parameters that should be chosen in a way appropriate to the overall goal of risk measurement. The time horizon can differ from a few hours for an active trading desk to a year for a pension fund. When the primary goal is to satisfy external regulatory requirements, such as bank capital requirements, the percentile is typically very small (e.g., 1% of worst outcomes).

However, for an internal risk management model used by a company to control the risk exposure, the typical number is around 5%. A general introduction to VaR can be found in Linsmeier and Pearson (2000) and in Jorion (1997).

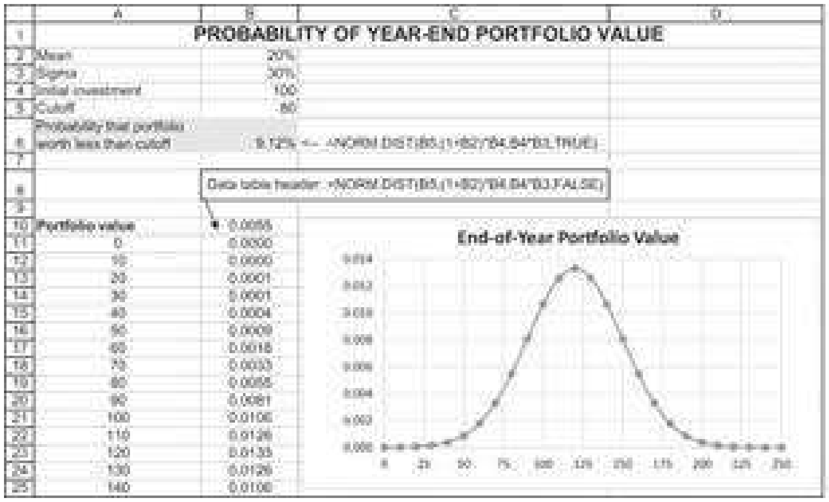
In the jargon of VaR, suppose that a portfolio manager has a daily VaR equal to $1 million at 1%. This statement means that there is only one chance in 100 that a daily loss bigger than $1 million occurs under normal market conditions.

A Simple Example

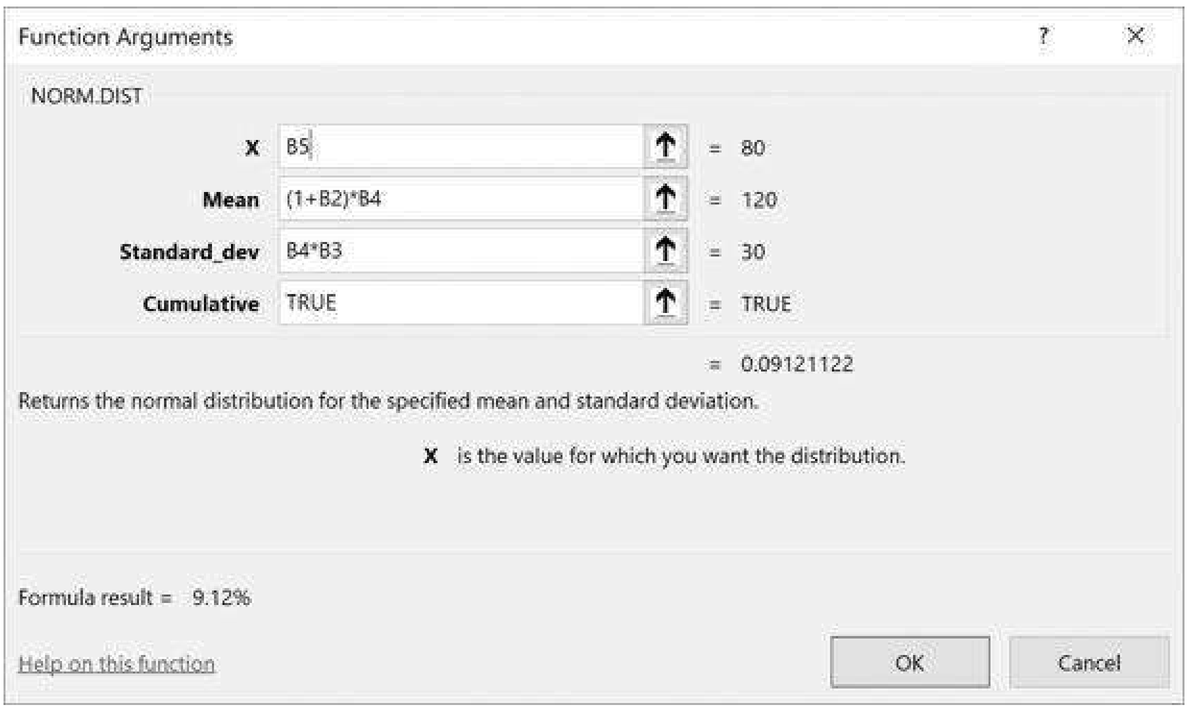
Suppose a manager has a portfolio that consists of a single asset. The return of the asset is normally distributed with mean return of 20% and a standard deviation of 30%. The value of the portfolio today is $100 million. We want to answer various simple questions about the end-of-year distribution of portfolio value:

1. What is the distribution of the end-of-year portfolio value?
2. What is the probability of a loss of more than $20 million by year-end (i.e., what is the probability that the end-of-year value is less than $80 million)?
3. With 1% probability, what is the maximum loss at the end of the year? This is the VaR at 1%.

The probability that the end-of-year portfolio value is less than $80 million is about 9%. Note that the 9.12% in the following spreadsheet is the area under the normal distribution to the left of $X = 80$:



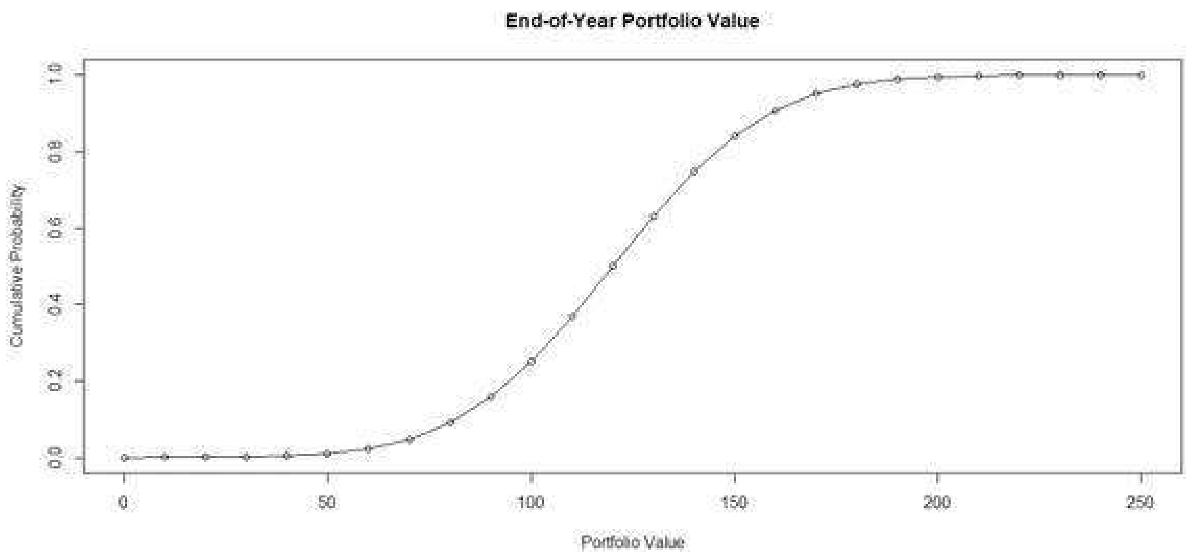
Excel's **Norm.Dist** function can return both the cumulative distribution and the probability mass function. Here's the way the screen looks when we apply the **Norm.Dist** function in cell B6:



The spreadsheet uses two versions of **Norm.Dist**: First, we use the function in cell B6 to determine the probability that the year-end value of the portfolio is less than 80. In this version of the function, we use the value TRUE for the last entry in **Norm.Dist**; when we write =NORM.DIST(B5,(1+B2) * B4,B4 * B3,TRUE), **Norm.Dist** returns values of the cumulative normal distribution. In the data table we set this value to FALSE to plot the probability mass function of the year-end portfolio value.

Implementing the above example in R looks like this:

```r
# Inputs:
mean <- 0.20
sigma <- 0.30
init_invst <- 100
cutoff <- 80

# (Cumulative) Probability that portfolio worth less than cutoff
pnorm(cutoff, mean = init_invst * (1 + mean),
      sd = sigma * init_invst, log = FALSE) # similar to Excel's NORM.DIST

# Probability distribution
port_value <- c(0:25)*10  #bins
prob <- sapply(port_value, function(x) pnorm(x, mean = init_invst * (1 + mean),
                                       sd = sigma * init_invst, log = FALSE))

# Plot (cumulative distribution)
plot(port_value, prob, type = "l", main = "End-of-Year Portfolio Value",
     xlab = "Portfolio Value", ylab = "Cumulative Probability")
lines(port_value,prob, type = "p")
```



The Three Steps to Calculate VaR

VaR is calculated in three steps:

- **Step 1**. Identify the risk factors for the target portfolio. If, for example, we are trying to estimate the VaR of a long position on 10 million euros, then the risk factor will be the U.S. dollar (USD) to euro exchange rate, and if we are trying to estimate the VaR of options on a stock portfolio, the VaR will probably be the stock price.
- **Step 2**. Estimate the change in the risk factors. If, for example, the risk factor is the USD/euro exchange rate, then we would like to model the movement of the exchange rate. The same goes for stocks and other risk factors.
- **Step 3**. Estimate the effect of changes in the risk factor on the portfolio. In some cases this effect is direct, like in the example of a long position on 10 million euros (simply 10M * exchange rate). In other cases it may require more complicated analysis (like in the options example because the options should probably be priced using the Black-Scholes model; see chapter 18).

25.2 The Three Types of VaR Models

There are basically three types of VaR models. The three types differ by the assumptions each makes on the distribution of the risk factors. Below we use a simple and illustrative example of VaR for a long position on 10 million euros (€10M) and in later sections we extend the discussion to more complex examples.

Analytic VaR

In the analytic VaR, we assume a mathematical distribution of the risk factor. We then use the mathematical distribution to estimate the VaR of the portfolio.

Below is a simple example of how to implement the VaR on a long position on €10M; we calculated weekly VaR in this example (note that as explained in appendix 10.1 of Chapter 10, $\sigma_{week} = \sigma_{year} * \sqrt{\frac{1}{52}}$):

	A	B	C	D
1	ANALYTIC VaR OF €10M (WEEKLY)			
2	Mean (weekly)	0.04%	<-- =2%/52	
3	Sigma (annual)	10%		
4	Sigma (week)	1.39%	<-- =B3*SQRT(1/52)	
5	Portfolio size	10,000,000	Euros	
6	Spot rate	1.15	$/€	=NORM.S.INV(A9)
7				
8	Probability	Standard- deviations	VaR (1 week, $)	
9	90%	1.2816	204,377	<-- =B9*$B$4*$B$5*$B$6
10	95%	1.6449	262,315	<-- =B10*$B$4*$B$5*$B$6
11	99%	2.3263	370,997	<-- =B11*$B$4*$B$5*$B$6

In this example, the risk factor is the USD/euro exchange rate. The weekly distribution of the exchange rate is assumed to be normal with mean of 0.04% and sigma of 1.39%—that is, $S_{week} \sim N(0.04\%, 1.39\%^2)$.

A probability of 90% is 1.2816 standard deviations from the mean, so there is 90% probability that the portfolio will not lose more than $204,377.

In the same way, there is a 99% probability that the portfolio value will not lose more than $370,997.

Implementing the analytic VaR approach in R looks like this:

```
# Inputs:
mean_annual <- 0.02
mean_weekly <- mean_annual*1/52
sigma_annual  <- 0.10
sigma_week    <- sigma_annual * sqrt(1/52)
port_size     <- 10000000
spot_rate  <- 1.15 #(Dollar/Euro spot rate)

# Standard deviations
prob <- c(0.9, 0.95, 0.99)
st_dev <- qnorm(prob, mean = 0, sd = 1, lower.tail = TRUE, log.p = FALSE)
names(st_dev) <- prob

# VaR (1 week, $)
var_analytic <- st_dev * sigma_week * port_size * spot_rate
names(var_analytic) <- prob
var_analytic
```

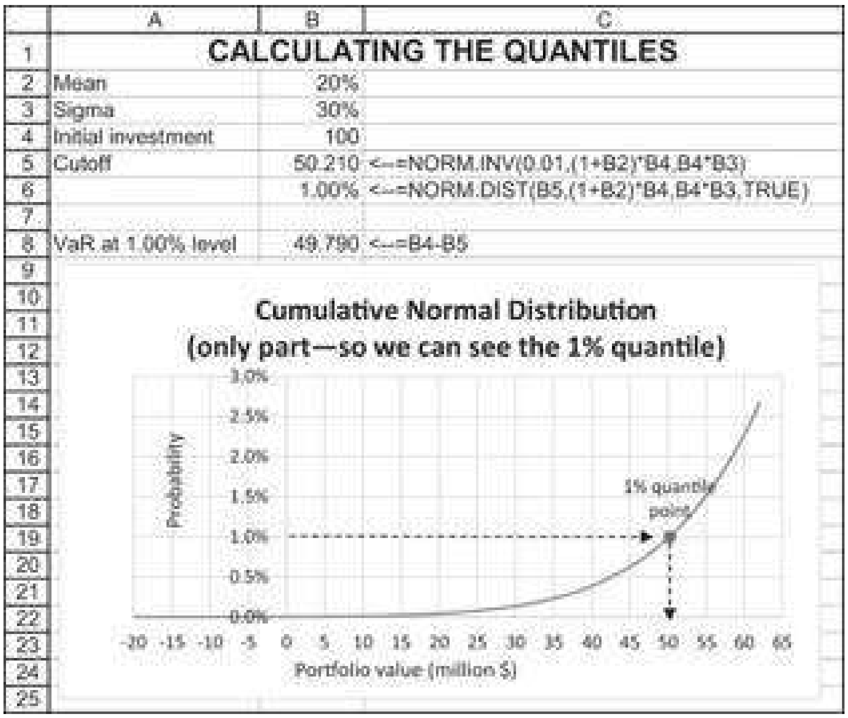
The results are as follows:

```
> var_analytic
     0.9     0.95     0.99
204377.1 262315.2 370997.4
> |
```

Defining Quantiles in Excel

The VaR (or cutoff) is known as "the quantile of the distribution." In Excel, it can be determined by using **Solver**. For two distributions that are widely used by practitioners—the normal and the lognormal distributions—Excel has built-in functions that find the quantile. These functions—**Norm.Inv**, **Norm.S.Inv**, and **Lognormal.inv**—find the inverse for the normal, standard normal, and lognormal distributions.

Here is an example: The value of the asset today is $100 million, and the asset is distributed with a mean return of 20% and a standard deviation of 30%. We are interested in the 1-year VaR of 1%. This time we have written the function**=NORM.INV(0.01,(1+B3)*B5,B5*B4)** in cell B6. This function finds the cutoff point for which the normal distribution with mean = 120 [100 * (1 + 20%) = 120] and standard deviation = 30 [30% * 100 = 30] has probability of 1%. You can see this point on the graph below, which shows part of the cumulative distribution:



And the implementation in R looks like this:

```r
mean <- 0.20
sigma <- 0.30
init_invst <- 100
cutoff <- qnorm(0.01, mean = init_invst * (1 + mean),
      sd = sigma * init_invst, lower.tail = TRUE, log.p = FALSE)

# VaR at 1.00% level
var_analytic <- init_invst - cutoff
```

The lognormal distribution is a more reasonable distribution for many asset prices (which cannot become negative) than the normal distribution. Suppose that price is $V_T = V_0 * e^{r * T}$, or that the return [return = Log(price relative)] on the portfolio is normally distributed with annual mean μ and annual standard deviation σ. Furthermore, suppose that the current value of the portfolio is given by V_0. Then it follows that the logarithm of the portfolio value at time T, V_T, is normally distributed.[1] This means:

$$ln(V_T) \sim Normal\left[\underbrace{ln(V_0) + (\mu - 0.5 * \sigma^2) * T}_{mean},\ \underbrace{\sigma * \sqrt{T}}_{sigma}\right]$$

Suppose, for example, that $V_0 = 100$, $\mu = 10\%$, and $\sigma = 30\%$. Thus, the end-of-year log of the portfolio value is distributed normally:

$$ln(V_T) \sim Normal\left[\underbrace{ln(100) + (0.1 - 0.5 * 0.3^2) * 1}_{mean}, \underbrace{0.3 * \sqrt{1}}_{sigma}\right] = Normal[4.6602, 0.3]$$

Thus, a portfolio whose initial value is \$100 million and whose annual returns are lognormally distributed with parameters $\mu = 10\%$ and $\sigma = 30\%$, has an annual VaR equal to \$47.42 million at 1%:

	A	B	C
1	QUANTILES FOR LOGNORMAL DISTRIBUTION		
2	Initial value, V_0	100	
3	Mean, μ	10%	
4	Sigma, σ	30%	
5	Time period, T	1	<-- in years
6			
7	Parameters of normal distribution of ln(V_T)		
8	Mean	4.6602	<--=LN(B2)+(B3-B4^2/2)*B5
9	Sigma	0.3000	<--=B4*SQRT(B5)
10			
11	Cutoff	52.576	<--=LOGNORM.INV(0.01,B8,B9)
12	VaR at 1% level	47.424	<--=B2-B11

The implementation in R would appear as follows:

```
init_invst <- 100
mean <- 0.10
sigma <- 0.30
t <- 1.0

#Parameters of normal distribution of ln(VT)
mean <- log(init_invst) + (mean - 0.5*sigma^2) * t
sigma <- sigma * sqrt(t)
cutoff <- qlnorm(0.01, mean = mean, sd = sigma)

# VaR at 1.00% Level
var_analytic <- init_invst - cutoff
```

Most VaR calculations are not concerned with annual value at risk. The main regulatory and management concern is with loss of portfolio value over a much shorter time period (typically several days or perhaps weeks). It is clear that the distribution formula $ln(V_T) \sim Normal\left[\underbrace{ln(V_0) + (\mu - 0.5 * \sigma^2) * T}_{mean}, \underbrace{\sigma * \sqrt{T}}_{sigma}\right]$ can be used to calculate the VaR over any horizon. Recall that T is measured in annual terms; if there are 250 business days in a year, then the daily VaR corresponds to $T = 1/250$ (for many fixed income instruments, one should use 1/360, 1/365, or 1/365.25, depending on the market convention).

Monte Carlo VaR

In the Monte Carlo VaR, we usually assume a mathematical distribution of the risk factor. We then use simulations to estimate the VaR of the portfolio.

Below is a simple example on how to implement the Monte Carlo VaR on a long position on €10M. As before, we calculated weekly VaR in this example (note that many lines are hidden):

	A	B	C	D	E	F	G	H
1	MONTE CARLO VaR OF €10M (WEEKLY)							
2	Mean (weekly)	0.04%	<-- =2%/52					
3	Sigma (annual)	10%						
4	Sigma (week)	1.39%	<-- =B3*SQRT(1/52)					
5	Portfolio size	10,000,000	Euros			=PERCENTILE.INC($C$9:$C$2008,1-E9)		
6	Spot rate	1.15	$/€					
7								
8	Trial	Z	$ln(S_t/S_0)$		Probability	Percentile	Var (1-week)	
9	1	1.0976	1.55%		90%	-1.72%	-197,322	<-- =F9*$B$5*$B$6
10	2	-0.4356	-0.58%		95%	-2.28%	-261,857	<-- =F10*$B$5*$B$6
11	3	-0.0593	-0.05%		99%	-3.20%	-367,683	<-- =F11*$B$5*$B$6
12	4	-0.0012	0.03%					
13	5	-0.8477	-1.15%	<-- =($B$2-0.5*$B$4^2)*1+$B$4*B13				
14	6	-1.3537	-1.85%	<-- =($B$2-0.5*$B$4^2)*1+$B$4*B14				
15	7	0.6662	0.95%					
2007	1999	0.2844	0.42%					
2008	2000	1.5346	2.16%					

In this example we used a simulation of 2,000 trials. Practitioners usually use at least 10,000 trials. We have used the lognormal distribution assumption (see Chapter 23) to model the expected change in exchange rate:

$$S_t = S_0 * e^{(r_\$ - r_€ - 0.5*\sigma)*t + \sigma*\sqrt{T}*Z}, \text{ where } Z \sim N(0,1).$$

In every row "i" we used

$$\underbrace{ln\left(\frac{S_t}{S_0}\right)}_{\%\ change} = \left(\underbrace{r_\$ - r_€}_{0.04\%} - 0.5 * \underbrace{\sigma}_{1.39\%}\right) * \underbrace{t}_{1} + \underbrace{\sigma}_{1.39\%} * \underbrace{\sqrt{T}}_{1} * Z_i.$$

In our sample of 2,000 simulations, 10% of the results were below −1.81% while only 1% of the results were below −3.26%. We used the function PERCENTILE.INC() to find the threshold simulated return. We then used this change to deduce that based on our simulation, there is a 90% probability that the loss will not be above $197,322 (cell G9) and a 99% probability that the loss will not be more than $367,683 (cell G11).

Implementing this approach in R looks like this:

```
# Simulate weekly returns
n <- 10000 #number of simulations
Z <- rnorm(n, mean = 0, sd = 1)
week_ret <- (mean_weekly - 0.5 * sigma_week ^ 2) + sigma_week * Z

# Calculate Percentiles
percentile <- quantile(week_ret, probs = 1-prob) # same as PERCENTILE.INC in Excel

# Calculate Value at Risk
var_mc <- percentile * port_size * spot_rate
```

Historical VaR

The historical VaR is using the historical observations within a sample to generate the distribution of movements in the risk factor. The benefit of this method, over the previous two, is that it does not assume any distribution to the risk factors. Instead, it only assumes a stationary distribution (i.e., that the same distribution in the past is expected to occur in the VaR period). In other words, the only assumption is that the historical distribution within the sample is valid for our VaR period.

Below is a simple example on how to implement the historical VaR on a long position on €10M. As before, we calculated weekly VaR in this example:

	A	B	C	D	E	F	G	H
1	HISTORICAL VaR OF €10M (WEEKLY)							
2	Mean (weekly)	0.04% <--=2%/52						
3	Sigma (annual)	10%						
4	Sigma (week)	1.39% <--=B3*SQRT(1/52)						
5	Portfolio size	10,000,000 Euros						
6	Spot rate	1.15 $/€						
7								
8	Date	Exchange rate	$\ln(S_t/S_{t-1})$	Portfolio change		Probability	VaR (1-week)	
9	27-Apr-18	1.2108	-1.42684%	-141,671		90%	-115,688	<--=PERCENTILE.INC(D9:D111,1-F9)
10	20-Apr-18	1.2282	-0.33%	-32,462		95%	-140,662	<--=PERCENTILE.INC(D9:D111,1-F10)
11	13-Apr-18	1.2322	0.39%	39,107		99%	-214,146	<--=PERCENTILE.INC(D9:D111,1-F11)
12	06-Apr-18	1.2274	-0.37%	-37,336				
13	30-Mar-18	1.2320	-0.32%	-32,362 <--=$B$5*EXP(C13)-$B$5				
14	23-Mar-18	1.2360	0.65%	65,147 <--=$B$5*EXP(C14)-$B$5				
15	16-Mar-18	1.2280	-0.37%	-37,319				
16	09-Mar-18	1.2326	0.10%	9,745				
110	20-May-16	1.1207	-0.77%	-77,032				
111	13-May-16	1.1294	-1.12%	-111,199				
112	06-May-16	1.1421						

In this approach, we used a historical window (observation period) of two years (104 weeks). Choosing the observation period involves a trade-off between using a long period (and hence covering more extreme cases) and using a short period (and including more relevant data).

For each week in our observation period we have calculated the weekly changes using continuous compounding and the portfolio value change, assuming the weekly changes were implemented on the portfolio.

We then used the function PERCENTILE.INC() to find the threshold portfolio loss for three VaR calculations. From that, we learn that based on our 2-year sample, there is a 90% chance that the loss will not be above \$115,688 and a 99% chance that the loss will not be more than \$214,146.

Implementing the same approach in R looks like this:

```
# Set working directory
workdir <- readline(prompt="working directory?")
setwd(workdir)

# Read data
data <- read.csv("chapter_25_data_1.csv", row.names = 1)

# Sort the data by date
data <- data[order(as.Date(row.names(data), format="%d-%m-%y"),decreasing = FALSE),]

# Calculate returns
week_ret <- diff(log(data))
port_change <- port_size * exp(week_ret) - port_size

# Calculate Value at Risk
Var_his <- quantile(port_change, probs = 1 - prob) # same as PERCENTILE.INC in Excel
```

25.3 VaR of an *N*-Asset Portfolio

In this section we show how to implement the VaR in a case of an *n*-asset portfolio. For simplicity, we use an example of a three-asset portfolio. This example can easily be extended for any number of securities.

As can be seen from the preceding examples, VaR is not—in principle, at least—a very complicated concept. In the implementation of VaR, however, there are two big problems (both problems are discussed in much greater detail in the material available in Holton 2014, sec. 1.9.5).

1. The first problem is the estimation of the parameters of asset return distributions. In "real world" applications of VaR, it is necessary to estimate means, variances, and correlations of returns. This is a not inconsiderable problem! In this section we illustrate the importance of the correlations between asset returns. In the following section we give a highly simplified example of the estimation of return distributions from market data. For example, you can imagine that a long position in euros and a short position in U.S. dollars are less risky than a position in only one of the currencies, because of a high probability that profits of one position will be mainly offset by losses of another.
2. The second problem is the actual calculation of position sizes. A large financial institution may have thousands of loans outstanding. The database of these loans may not classify them by their riskiness or even by their term to maturity. Or—to give a second example—a bank may have offsetting positions in foreign currencies at different branches in different locations. A long position in euros in New York may be offset by a short position in euros in Geneva; the bank's risk—which we intend to measure by VaR—is based on the *net position*.

Analytic VaR of a Three-Asset Portfolio

We start with the problem of correlations between asset returns. We continue the previous example but assume that there are three risky assets. As before, the parameters of the distributions of the asset returns are known: all the means, μ_1, μ_2, μ_3, as well as the variance-covariance matrix of the returns:

$$S = \begin{pmatrix} \sigma_{1,1} & \sigma_{1,2} & \sigma_{1,3} \\ \sigma_{2,1} & \sigma_{2,2} & \sigma_{2,3} \\ \sigma_{3,1} & \sigma_{3,2} & \sigma_{3,3} \end{pmatrix}$$

The matrix S is of course symmetric; $\sigma_{i,i}$ is the variance of the i^{th} asset's return, and $\sigma_{i,j}$ is the covariance of the returns of assets i and j (in the case where $i = j$, $\sigma_{i,j}$ is the variance of asset i's return).

Suppose that the total portfolio value today is \$100 million, with \$30 million invested in asset 1, \$25 million in asset 2, and \$45 million in asset 3. Then the return distribution of the portfolio is given as follows:

Mean return $= x_1 \cdot \mu_1 + x_2 \cdot \mu_2 + x_3 \cdot \mu_3$ *Variance of return* $= \{x_1, x_2, x_3\} \cdot S \cdot \{x_1, x_2, x_3\}^T$

Here, $x = \{x_1, x_2, x_3\} = \{0.3, 0.25, 0.45\}$ is the vector of proportions invested in each of the three assets. Assuming that the returns are normally distributed (meaning that prices are lognormally distributed), we may calculate the VaR as in the following spreadsheet:

	A	B	C	D	E	F	G	H
1	ANALYTIC VaR FOR 3-ASSET PROBLEM							
2		Mean returns		Variance-covariance matrix				Portfolio proportions
3	Asset 1	10%		0.10	0.04	0.03		0.30
4	Asset 2	12%		0.04	0.20	-0.04		0.25
5	Asset 3	13%		0.03	-0.04	0.60		0.45
6								
7	Initial investment	100						
8	Mean return	11.85%	<--{=MMULT(TRANSPOSE(B3:B5),H3:H5)}					
9	Portfolio sigma	38.48%	<--{=SQRT(MMULT(MMULT(TRANSPOSE(H3:H5),D3:F5),H3:H5))}					
10								
11	Mean investment value	111.85	<--=(1+B8)*B7					
12	Sigma of investment value	38.48	<--=B9*B7					
13								
14	Cutoff	22.3234	<--=NORM.INV(0.01,(1+B8)*B7,B9*B7)					
15	Cumulative PDF (check)	0.01	<--=NORM.DIST(B14,B11,B12,TRUE)					
16	VaR at 1.00% level	-77.6766	<--=B14-B7					
17								
18	Note that the functions in cells B8 and B9 are array functions. You must press [Ctrl]+[Shift]+[Enter] after you							
19	write the function in the cell. The curly brackets {} are not written—they appear automatically.							

Implementing the above approach in R looks like this:

```
# Inputs:
mean_ret <- c(0.1, 0.12, 0.13)
cov_mat <- matrix(c(0.10, 0.04, 0.03,
                    0.04, 0.20, -0.04,
                    0.03, -0.04,  0.60),
                  nrow = 3)
port_prop <- c(0.3, 0.25, 0.45)
Init_invst  <- 100

# Portfolio Stats
port_mean_ret <- mean_ret %*% port_prop
port_sigma   <- sqrt(port_prop %*% cov_mat %*% port_prop)
mean_invs_value <- Init_invst * ( 1 + port_mean_ret )
invs_value_sigma <- port_sigma * Init_invst

# Analytic VaR
sig_level <- 0.01
cutoff <- qnorm(p = sig_level, mean = mean_invs_value, sd = invs_value_sigma)

# VaR at 1.00%  level
var_analytic <- cutoff - Init_invst
```

Monte Carlo Simulation VaR of a Three-Asset Portfolio

In the Monte Carlo simulation approach, we will simulate returns for our three assets. We use the Cholesky decomposition explained in Chapter 21 to generate the normal correlated random numbers. This is done by implementing the following three steps:

1. Create a lower-triangular Cholesky decomposition of S. We denote this matrix by L and use the VBA function **Cholesky** on the companion website of this book to compute the matrix.
2. Create a column vector of N standard normal deviates. We use the Excel function **Norm.S.Inv(Rand())**.
3. Multiply L times the column vector of standard normal deviates.

The result is a set of correlated standard normal variates. We then use these simulated variates to model the changes in the assets-values. Below is a 2,000 trials simulation of a 1% VaR for a 1-year portfolio.

	A	B	C	D	E	F	G	H	I	J	K
1	MONTE CARLO VaR OF 3-ASSET PORTFOLIO										
2		Mean returns		Variance-covariance matrix				Portfolio proportions			
3	Asset 1	10%		0.10	0.04	0.03		0.30			
4	Asset 2	12%		0.04	0.20	-0.04		0.25			
5	Asset 3	13%		0.03	-0.04	0.60		0.45			
6											
7	Initial investment	100									
8	Probability	1.00%						=NORM.S.INV(RAND())			
9	Percentile	-70.29%	<-- =PERCENTILE.INC(K20:K2019,B8)								
10	VaR	-70.2893	<-- =B9*B7					{=TRANSPOSE(MMULT($B$14:$D$16,TRANSPOSE(B20:D20)))}			
11											
12	Cholesky decomposition										
13		Asset 1	Asset 2	Asset 3				=$B$3-0.5*$D$3^1+SQRT($D$3)^1*E20			
14	Asset 1	0.316228	0	0	<-- {=cholesky(D3:F5)}						
15	Asset 2	0.126491	0.428952	0					=MMULT(H20:J20,$H$3:$H$5)		
16	Asset 3	0.094868	-0.12123	0.756147							
17											
18		Non-correlated random numbers			Correlated random numbers			Simulated asset returns			Portfolio
19	Trial	Z1	Z1	Z1	Z(Asset 1)	Z(Asset 2)	Z(Asset 3)	Asset 1	Asset 2	Asset 3	
20	1	-0.6386	-1.9094	2.6319	-0.2019	-0.8998	2.3207	-1.30%	-38.24%	162.76%	63.27%
21	2	-1.1307	-0.3610	0.9876	-0.3576	-0.2979	0.6862	-6.31%	-11.32%	36.15%	11.55%
22	3	-1.4879	-1.2735	0.3854	-0.4705	-0.7544	0.3058	-9.88%	-30.84%	6.88%	-7.67%
2016	1997	0.6520	0.2509	-0.5233	0.2062	0.1901	-0.3658	11.52%	10.50%	-45.34%	-14.32%
2017	1998	-0.9998	-0.2454	-1.2736	-0.3162	-0.2317	-1.0319	-5.00%	-6.36%	-96.93%	-47.21%
2018	1999	-1.6224	-0.5232	-0.2500	-0.5130	-0.4297	-0.2803	-11.22%	-17.22%	-38.71%	-25.09%
2019	2000	-0.5823	0.2509	0.6811	-0.1841	0.0340	0.4390	-0.82%	3.52%	17.01%	8.29%

In columns E, F, and G we used the Cholesky decomposition in order to generate correlated standard normally distributed numbers. We then used these numbers to model the change in asset prices, using lognormal distribution (see columns H, I, and J).[2] Once the assets-returns are modeled, it is straightforward to model the portfolio return (see Chapter 10). We then used the Percentile.Inc() formula to estimate the 1% VaR (cell B10).

Implementing this approach in R looks like this:

```
## Monte-Carlo simulation VaR of the 3-asset portfolio

# Assets Variance
asset_var <- diag(cov_mat)

# Lower-Triangular Cholesky Decomposition
chol_decompos <- t(chol(cov_mat))

n <- 10000 # number of simulations

# Simulate correlated random variables
z <- t( sapply(1:n, function(x) chol_decompos %*% rnorm(n = 3, mean = 0, sd = 1)))

# Simulate Monthly Returns Of 3 Correlated Stocks
sim_asset_ret <- apply(z, 1, function(z) { (mean_ret - 0.5 * asset_var)
                                          + sqrt(asset_var) * z) }

# Simulate Portfolio Return
port_ret <- t(sim_asset_ret) %*% port_prop

# Calculate the 1% Percentile Return
percentile <- quantile(port_ret, probs = 0.01)

# VaR at 1.00%  level
var_monte_carlo <- percentile * init_invst
```

Historical VaR of a Three-Asset Portfolio

The historical VaR of a three-asset portfolio is easy to calculate. We simply observe the return of each asset in each year in the past sample period (50 years in our example below). We then calculate the portfolio return in each period assuming the proportions are constant. Finally, we use the Percentile.Inc() function to estimate the 1% VaR of our portfolio (cell B10).

	A	B	C	D	E	F
1		HISTORICAL VaR FOR 3-ASSET PROBLEM				
2		Portfolio proportions				
3	Asset 1	0.30				
4	Asset 2	0.25				
5	Asset 3	0.45				
6						
7	Initial investment	100				
8	Probability	1.00%				
9	Percentile	-59.30%	<-=PERCENTILE.INC(E14:E63,B8)			
10	VaR	-59.29806	<-=B9*B7			
11						
12		Actual assets return			Portfolio	
13	Trial	Asset 1	Asset 2	Asset 3		
14	1	-3.15%	-15.77%	-22.12%	-14.84%	<-=MMULT(B14:D14,$B$3:$B$5)
15	2	6.32%	-19.20%	-50.93%	-25.83%	
16	3	3.97%	11.45%	-13.30%	-1.93%	
62	49	2.31%	-26.09%	176.67%	73.67%	
63	50	-10.60%	32.01%	-75.94%	-29.35%	

And the R implementation is as follows:

```
## Historical VaR of the 3-asset portfolio
# Read the return data from the csv file
asset_ret <- as.matrix(read.csv("Chapter_25_Data_2.csv"))
port_ret <- asset_ret %*% port_prop

# calculate the 1% Percentile Return
percentile <- quantile(port_ret, probs = 0.01)

# VaR at 1.00%  level
Var_historical <- percentile * Init_invst
```

25.4 Backtesting

Any empirical work should be assessed based on the usefulness of its predictions. Backtesting is a process of assessing the usefulness of a VaR. Once sufficient data has been collected, we can then use a hindsight approach to test, statistically, how well the VaR measurements reflect the riskiness of the portfolio. Generally speaking, there are two backtesting methodologies:

1. **Coverage tests** are basically statistical tests where the null hypothesis (H_0) is that each observed period is drawn from a binomial distribution. This is also called hypothesis testing. These test methods are checking whether the frequency of observed values is consistent with the quantile of loss a VaR measure is intended to reflect.
2. **Whole distribution tests** are checking whether the distribution assumed in the VaR is observed in the sample. Generally speaking, these tests are goodness-of-fit tests applied to the overall distribution forecast by the VaR measures.

A Standard Coverage Test

In the standard coverage test, we start by formulating our null hypothesis: The observed distribution is drawn from a binomial distribution. For example, if our VaR is a 1% VaR and we are testing it on 750 observations, $H_0 = Observation_i \sim B(750, 99\%)$, we can then say, with a significance level of 5%, that more than 13 observations, or less than three that exceed the VaR, will reject the VaR measure.

	A	B	C	D	E	F	G
1	BACKTESTING - THE STANDARD COVERAGE TEST						
2	VaR level	1%					
3	n	750					
4	Significance level	5%					
5							
6	**Confidence interval**						
7	Upper bound	13	<--=B3-BINOM.INV(B3,1-B2,B4/2)				
8	Lower bound	3	<--=B3-BINOM.INV(B3,1-B2,1-B4/2)				
9	Non-rejection values	3-13	<--=B8&"-"&B7				

Note that we have used the function Binom.inv() with a symmetric distribution assumption to test out the VaR.

And in R, the implementation looks like this:

```
## The Standard Coverage Test
# Inputs
VaR_level <- 0.01
n <- 750
Signif_level <- 0.05

# Confidence interval
upper_bound <- qbinom(1-Signif_level/2, n, VaR_level) # Similar to Excel's BINOM.INV
lower_bound <- n - qbinom(1-Signif_level/2, n, 1-VaR_level)
non_rejection_values <- paste(lower_bound, "-", upper_bound, sep = "")
```

We can use this result to formulate the confidence interval (our non-rejection values) for various VaRs and sample size:

	A	B	C	D	E	F	G
1	BACKTESTING - THE STANDARD COVERAGE TEST						
2	VaR level	1%					
3	n	750					
4	Significance level	5%					
5							
6	**Confidence interval**						
7	Upper bound	13	<--=B3-BINOM.INV(B3,1-B2,B4/2)				
8	Lower bound	3	<--=B3-BINOM.INV(B3,1-B2,1-B4/2)				
9	Non-rejection values	3-13	<--=B8&"-"&B7				
10							
11	**Non-rejection values (5% significance level)**				Data-table header: =B9		
12	↓ Sample size: → VaR level						
13	3-13	**0.50%**	**1%**	**2.50%**	**5%**	**10%**	
14	**50**	0-2	0-2	0-4	0-6	1-9	<--{=TABLE(B2,B3)}
15	**100**	0-2	0-3	0-6	1-10	5-16	
16	**200**	0-3	0-5	1-10	4-16	12-29	
17	**250**	0-4	0-6	2-11	6-20	16-35	
18	**500**	0-6	1-10	6-20	16-35	37-64	
19	**750**	1-8	3-13	11-28	26-50	59-91	
20	**1000**	1-10	4-17	16-35	37-64	82-119	
21	**1250**	2-12	6-20	21-42	48-78	105-146	
22	**1500**	3-13	8-23	26-50	59-92	128-173	

Whole Distribution Test

Distribution tests are tests that do more than test a specific VaR quantile-of-loss measure. They examine the whole distribution that the VaR characterizes. For example, a simple distribution test can be implemented by performing multiple coverage tests for many different VaR quantiles. The basic coverage test is applied to assess how well the VaR measure estimates at 1%, 2.5%, 5%, 10%, 15%, …, 80%. Collectively, these analyses provide a rough goodness-of-fit test for how well the VaR measure characterized the overall distribution of the portfolio. We use our previous example to illustrate how to test the whole distribution:

	A	B	C
1	**BACKTESTING - WHOLE DISTRIBUTION, SIMPLE APPROACH**		
2	VaR level	1%	
3	n	750	
4	Significance level	5%	
5			
6	**Confidence interval**		
7	Upper bound	13	<--=B3-BINOM.INV(B3,1-B2,B4/2)
8	Lower bound	3	<--=B3-BINOM.INV(B3,1-B2,1-B4/2)
9	Non-rejection values	3-13	<--=B8&"-"&B7
10			
11	**Non-rejection values (5% significance level)**		
12	**VaR level**	**Non-rejection values**	
13		3-13	Data-table header: =B9
14	0.50%	1-8	<--{=TABLE(,B2)}
15	1.00%	3-13	<--{=TABLE(,B2)}
16	2.50%	11-28	
17	5.00%	26-50	
18	10.00%	59-91	
19	15.00%	94-132	
20	20.00%	129-172	
21	35.00%	237-288	
22	50.00%	348-402	
23	80.00%	578-621	

The implementation in R is shown as follows:

```
# Data table variables
VaR_levels <- c(0.005, 0.01, 0.025, 0.05, 0.1)
n <- c(50, 100, 200, 250, 500, 750, 1000, 1250, 1500) #sample size

#  define function for the non-rejection values
non_rej_val <- function(VaR_level, n){
  upper_bound <- qbinom(1-signif_level/2, n, VaR_level) # similar to Excel's BINOM.INV
  lower_bound <- n - qbinom(1-signif_level/2, n, 1-VaR_level)
  non_rejection_values <-  paste(lower_bound, "-", upper_bound, sep = "")
  return(non_rejection_values)
}

# run function for each combination
results <- outer(VaR_levels, n, FUN = "non_rej_val")
colnames(results) <- n
rownames(results) <- VaR_levels
results
```

And the results are:

```
> results
      50    100    200     250     500     750     1000     1250      1500
0.005 "0-2" "0-2"  "0-3"   "0-4"   "0-6"   "1-8"   "1-10"   "2-12"    "3-13"
0.01  "0-2" "0-3"  "0-5"   "0-6"   "1-10"  "3-13"  "4-17"   "6-20"    "8-23"
0.025 "0-4" "0-6"  "1-10"  "2-11"  "6-20"  "11-28" "16-35"  "21-42"   "26-50"
0.05  "0-6" "1-10" "4-16"  "6-20"  "16-35" "26-50" "37-64"  "48-78"   "59-92"
0.1   "1-9" "5-16" "12-29" "16-35" "37-64" "59-91" "82-119" "105-146" "128-173"
> 
```

---

1. 1. See section 23.3 of this volume and Hull (2015).
2. 2. For more details, see Chapter 23.

26 Replicating Options and Option Strategies

26.1 Overview

In this chapter we simulate options and option strategies. We start by showing how an option on a stock can be replicated by a dynamic portfolio of the stock and a bond. We go on to apply this approach to portfolio insurance (a combination of a put and stock) and to a butterfly. Our approach has its roots in the Black-Scholes formula. This formula can be interpreted as showing that an option is a portfolio of a position in the underlying asset and a position in the risk-free asset, where both positions are adjusted dynamically over time. Remarkably, the Black-Scholes formula shows that if the adjustment process is continuous, then the dynamic strategy is self-financing: It requires no additional cash flows after the establishment of the initial portfolio.

In a more realistic situation, it is of course impossible to replicate the option strategy portfolio continuously. We have to compromise by making only periodic adjustments, and this forces us to consider (1) how to define the strategy over time and (2) how well the compromise strategy performs in replicating the option strategy. We will show several approaches to this problem.

Throughout this chapter we assume the replication is done for pricing purposes, thus we employ a risk-neutral approach (i.e., performing the underlying asset simulation with r_f and not with the expected (risky) return).

Background: Price Simulations and the Black-Scholes Formula

Recall from Chapter 23 that the way to simulate a stock price is to simulate $S_{t+\Delta t} = S_t e^{(\mu - 0.5\sigma^2)\Delta t + \sigma\sqrt{\Delta t} Z}$, where Z is a draw from a standard normal distribution. In Excel and R, this formula becomes:

$$S_{t+\Delta t} = S_t * exp[(\mu - 0.5\sigma^2)\Delta t + \sigma\sqrt{\Delta t} * \text{Norm.S.Inv(Rand())}]$$

We will use this formula throughout the chapter to simulate stock prices.

Recall also that the Black-Scholes formula, explained in Chapter 18, states that the prices of a European call and put are given by:

$Call = SN(d_1) - Xe^{-rT} N(d_2)$

$Put = Xe^{-rT} N(-d_2) - SN(-d_1)$

where

$$d_1 = \frac{ln\left(\frac{S}{X}\right) + (r + 0.5 \cdot \sigma^2)T}{\sigma\sqrt{T}}, \quad d_2 = d_1 - \sigma\sqrt{T}$$

In this chapter our interpretation of this formula is that an option can be replicated by a long or short position in the underlying stock and a long or short position in the risk-free asset (i.e., "the bond").

	Stock position	**Bond position**
$Call = SN(d_1) - Xe^{-r\,T} N(d_2)$	$S_t N(d_1)$	$-Xe^{-r * T(d_2)}$
	Long position in stock	Short position in bond
$Put = -SN(-d_1) + Xe^{-r\,T} N(-d_2)$	$-S_t N(-d_1)$	$Xe^{-r * T(-d_2)}$
	Short position in stock	Long position in bond

It follows that if we replicate an option with a portfolio of stocks and bonds, this portfolio needs to be rebalanced often. A remarkable fact, proved by Black and Scholes in their pathbreaking 1973 paper, is that if the rebalancing is continuous, then the changes in the stock and bond positions exactly offset each other. This zero-investment property is also termed "self-financing" and is a hallmark of the Black-Scholes replicating portfolio.

While we cannot prove this fact in this framework, we can give some intuition. In the spreadsheet below, we look at two very close times. At time t = 0, we price a call option on a stock whose current price is S_0 = 50. The option has exercise price X = 50 and expires at T = 0.5. The replicating portfolio for the option is 28.9698 in the stock and −24.2745 in the bond, giving a call option value of 4.6952 (cell B19).

At time $\Delta\ t$ = 1/250 (roughly one day later), a randomly generated stock price is 49.8708. The replicating portfolio is now 28.6401 in the stock and −24.0399 in the bond. To compute the investment required to reach this portfolio, we have to revalue the previous investments in the stock and the bond. The previous stock position has changed in value to $28.9698 * \frac{49.8708}{50} = 28.8949$ and the previous bond position now has value of −24.2745 * $e^{0.04 * 1/250}$ = −24.2707 (one day's interest has been added). This means that we must sell off some of the stock: (28.6401 − 28.8949) = 0.2548. Thus, the cash flow from the change in the stock position is positive. In the bond position we want to change the debt from −24.2784 to −24.0399; this means that we require cash of 0.2385 to pay off some of the debt. The net cash flow from these changes is +0.2549 − 0.2386 = 0.0163. In a spreadsheet:

	A	B	C	D
1		**BLACK-SCHOLES AS A PORTFOLIO**		
2	Δt	0.0040	<-- =1/250	
3				
4		**Time = 0**	**Time = Δt**	
5	S	50.0000	49.8708	<-- 49.8708
6	X	50.0000	50.0000	
7	r	4.00%	4.00%	
8	T, time to option maturity	0.5000	0.4960	<-- =B8-B2
9	Sigma	30%	30%	
10				
11	d_1	0.2003	0.1873	<-- =(LN(C5/C6)+(C7+0.5*C9^2)*C8)/(C9*SQRT(C8))
12	d_2	-0.0118	-0.0240	<-- =C11-SQRT(C8)*C9
13				
14	$N(d_1)$	0.5794	0.5743	<-- =NORM.S.DIST(C11,1)
15	$N(d_2)$	0.4953	0.4904	<-- =NORM.S.DIST(C12,1)
16				
17	Stock position, $S*N(d_1)$	28.9698	28.6401	<-- =C5*C14
18	Bond position, $-X*N(d_2)$	-24.2745	-24.0399	<-- =-C6*EXP(-C8*C7)*C15
19	**Value of call option**	**4.6952**	**4.6002**	<-- =SUM(C17:C18)
20				
21	**Change in position value:**			
22		**Value at Δt of previous position**	**Cash from position change at Δt**	
23	In stocks	28.8949	0.2548	<-- =B23-C17
24	In bonds	-24.2764	-0.2385	<-- =B24-C18
25	**Total**		**0.0163**	<-- =SUM(C23:C24)
26				
27	=B17*C5/B5			=B18*EXP(B7*B2)
28				

The Excel notebook with this chapter gives a dynamic spreadsheet for the above example that changes values with each press of **F9**. Using this spreadsheet you can confirm:

- • The position changes rarely completely net out.
- • The smaller the Δt, the smaller the net cash flow will be from the position change. Black-Scholes proved that in the limit, this net cash flow is always zero and that the strategy is thus completely self-financing.

One-Week Simulation Using R

Implementing the same approach will be done in two steps. At the first step we define the Black-Scholes (Merton) option pricing formulas. This should look like this:

```r
# B&S d1
fm5_bs_d1 <- function(S, X, t, r, sigma_S, k=0){
    return( ( log(S/X) + (r -k + 0.5 * sigma_S ^ 2) * t ) / ( sigma_S * sqrt(t)) )
                                                            }

# Function: d2
fm5_bs_d2 <- function(S, X, t, r, sigma_S, k=0){
    return(fm5_bs_d1(S, X, t, r, sigma_S, k) - sigma_S * sqrt(t))
                                                      }

# B&S Call option
fm5_bs_call <- function(S, X, t, r, sigma_S, k = 0){
    d_1 <- fm5_bs_d1 (S, X, t, r, sigma_S, k=0)
    d_2 <- fm5_bs_d2 (S, X, t, r, sigma_S, k=0)
    return(S * pnorm(d_1, mean = 0, sd = 1) * exp(-k * t) -
              X * exp(-r*t) * pnorm(d_2, mean = 0, sd = 1))
                                                              }
# B&S Put option
fm5_bs_put <- function(S, X, t, r, sigma_S, k = 0){
    d_1 <- fm5_bs_d1 (S, X, t, r, sigma_S, k=0)
    d_2 <- fm5_bs_d2 (S, X, t, r, sigma_S, k=0)
    return(X * exp(-r*t) * pnorm(-d_2, mean = 0, sd = 1) -
              S * pnorm(-d_1, mean = 0, sd = 1) * exp(-k * t))
                                                              }

# Option pricing example
fm5_bs_call(S = 50, X = 50, r = 0.04, t = 0.5, sigma_S = 0.3)
fm5_bs_put(S = 50, X = 50, r = 0.04, t = 0.5, sigma_S = 0.3)
```

In the second step we define the self-financing portfolio and implement it in the previous example, as follows:

```r
# Self-Financing Call position
fm5_sf_call_pos <- function(S, X, t, r, sigma_S, k=0){
    # B&S d1
    d_1 <- fm5_bs_d1 (S, X, t, r, sigma_S, k=0)
    # B&S d2
    d_2 <- fm5_bs_d2 (S, X, t, r, sigma_S, k=0)
    # returns position in (S, X)
    return(c( S = S * pnorm(d_1, mean = 0, sd = 1),
              X = - X * exp(-r*t) * pnorm(d_2, mean = 0, sd = 1)))
                                                              }

# Example 1: Option Replication
S0 <- 50
X <- 50
r <- 0.04
t0 <- 0.5
sigma_S <- 0.3

# T = 0
pos_t0 <-fm5_sf_call_pos(S = S0, X = X, t = t0, r = r, sigma_S = sigma_S)

# Delta t = 1/250
del_t <- 1/250
t <- 0.5 - del_t
St <- 49.8708

# Calculate new position
pos_t0.496 <- fm5_sf_call_pos(S = St, X = X, t = t, r = r, sigma_S = sigma_S)

# Value at time t of the initial position
val_del_t <- c(St/S0, exp(r*del_t)) * pos_t0

# Change in cash
sum(val_del_t - pos_t0.496)
```

The result is:

```
> # Change in cash
> sum(val_del_t - pos_t0.496)
[1] 0.01627404
>
```

26.2 Imperfect but Cashless Replication of a Call Option

To perfectly replicate a call option using the Black-Scholes formula will require continuous trading. Suppose we compromise a bit on strictly following the Black-

Scholes formula. In what follows we use the strategy of replication but force it to be self-financing. We do this as follows:

- • At time 0, we set the initial portfolio equal to that prescribed by the Black-Scholes formula:

$Stock_0 = S_0 N(d_1)$, $Bond_0 = -X \cdot e^{-rT} N(d_2)$

- • At times $t > 0$, we set the stock position to $S_t N(d_1)$, but we set the bond position equal to the cash flow from the change in the stock position:

$$Bond_{t+\Delta t} = \underbrace{Bond_t * e^{r\Delta t}}_{\substack{t+\Delta t \text{ value} \\ \text{of time } t \text{ bond} \\ \text{position}}} + \underbrace{S_t N(d_{1,t}) * \frac{S_{t+\Delta t}}{S_t}}_{\substack{t+\Delta t \text{ value} \\ \text{of time } t \\ \text{stock position}}} - \underbrace{S_{t+\Delta t} N(d_{1,t+\Delta t})}_{\substack{t+\Delta t \text{ desired} \\ \text{stock position}}}$$

$$= Bond_t * e^{r\Delta t} + S_{t+\Delta t}[N(d_{1,t}) - N(d_{1,t+\Delta t})]$$

Here's how this looks. At $t = 0$, the replicating portfolio is exactly equivalent to the Black-Scholes (cells H13:I13), but at $t > 0$, the Black-Scholes value is different from the portfolio value (though quite close).

	A	B	C	D	E	F	G	H	I
1	**IMPERFECT BUT CASHLESS REPLICATION OF CALL OPTION WITH PORTFOLIO** Requires no investment over time but doesn't perfectly match Black-Scholes								
2	S_0, stock price	50							
3	X, exercise price	50		=E12*EXP(interest*Delta_t)+B13*(NORM.S.DIST(done(B12,exercise,C12,interest,Sigma),1)-NORM.S.DIST(done(B13,exercise,C13,interest,Sigma),1))					
4	T, strike date	0.5							
5	Sigma of stock return	30%							
6	r, interest rate	4%							
7	Delta_t	0.0192	<--=1/52	=B13*NORM.S.DIST(done(B13,exercise,C13,interest,Sigma),1)					
8									
9									
10				**Replicating portfolio**			**Comparing BS to portfolio value**		
11	**Week**	**Stock price**	**Time remaining**	**Stock**	**Bond**		**Portfolio**	**BS**	**Portfolio Minus BS**
12	0	50.0000	0.5000	28.9698	-24.2745		4.6952	4.6952	0.0000
13	1	48.6044	0.4808	25.4719	-21.6040		3.8679	3.8267	0.0412
14	2	46.6895	0.4615	20.6785	-17.8308		2.8478	2.8051	0.0427
15	3	44.3105	0.4423	14.9915	-13.2111		1.7804	1.7888	-0.0084
16	4	44.7564	0.4231	15.6955	-13.7744		1.9211	1.8601	0.0610
17	5	45.2698	0.4038	16.5897	-14.4991		2.0905	1.9578	0.1327
18	6	43.2159	0.3846	11.7370	-10.4103		1.3267	1.2211	0.1056
19	7	45.0803	0.3654	15.5353	-13.7103		1.8251	1.7119	0.1131
20	8	47.8504	0.3462	22.3564	-19.5873		2.7691	2.7348	0.0343
21	9	48.5125	0.3269	23.9839	-20.9206		3.0634	2.9420	0.1214
22	10	50.5158	0.3077	29.6333	-25.5955		4.0377	3.9065	0.1312
23	11	50.3070	0.2885	28.9471	-25.0516		3.8955	3.6614	0.2341
24	12	51.2458	0.2692	31.7653	-27.3489		4.4164	4.0948	0.3217
25	13	51.1871	0.2500	31.5918	-27.2328		4.3590	3.9266	0.4324
26	14	51.6998	0.2308	33.2735	-28.6191		4.6544	4.1136	0.5408
27	15	53.9427	0.2115	40.5580	-34.4820		6.0760	5.5439	0.5321
28	16	53.8217	0.1923	40.5301	-34.5717		5.9584	5.3165	0.6420
29	17	60.3168	0.1731	57.1525	-46.3280		10.8244	10.8383	-0.0139
30	18	61.6999	0.1538	59.9200	-47.8225		12.0974	12.0908	0.0066
31	19	62.0409	0.1346	60.8313	-48.4396		12.3918	12.3600	0.0318
32	20	61.9402	0.1154	61.0739	-48.8181		12.2557	12.2025	0.0532
33	21	60.5435	0.0962	59.6755	-48.7344		10.8410	10.7686	0.0724
34	22	59.7946	0.0769	59.0221	-48.9555		10.0666	9.9713	0.0954
35	23	59.0684	0.0577	58.5599	-49.2468		9.3131	9.1972	0.1159
36	24	59.2384	0.0385	59.1404	-49.6976		9.4428	9.3170	0.1258
37	25	60.8720	0.0192	60.8720	-49.8366		11.0354	10.9105	0.1249
38	26	63.9895	0.0000						
39									
40	**Final payoff of portfolio**								
41	Perfectly replicated	13.9895	<--=MAX(B38-exercise,0)						
42	Investment payoff	14.1145	<--=D37*B38/B37+EXP(interest*Delta_t)*E37						

Implementing this in R looks like this:

```r
# Inputs:
S0  <- 50.00
X   <- 50.00
t <- 0.5 # strike date
sigma <- 0.30
r <- 0.04
delta_t <- 1/52

# Generate a Random Stock Price Path
stock_ret <- sapply(rnorm(t/delta_t), function(z)
    exp((r - 0.5 * sigma ^ 2) * delta_t + sqrt(delta_t) * sigma * z) )
St <- c(S0, cumprod(stock_ret) * S0)

# Time remaining
time <- t - delta_t * 0:26

# Calculate Nd1
Nd1 <- pnorm(mapply(fm5_bs_d1, St, X, time, r,sigma))

# Stock Position
stock_pos <- Nd1 * St
stock_pos[27] <- St[27]/St[26] * stock_pos[26]

# Bond Initial position
bond_pos <- -exp(-r * t) * X * pnorm( fm5_bs_d2(S0, X, t, r,sigma) )

# Bond Cash Flow
cf <- c(bond_pos, -diff(Nd1) * St[-1])

# Bond position t = 1 to t = 25
for (i in 2:26){
    bond_pos[i] <- bond_pos[i-1] * exp(r*delta_t) + cf[i]
               }

# Bond position t = 26
bond_pos[27] <- bond_pos[26] * exp(r*delta_t)

# Investment payoff
investment_payoff <- stock_pos[27] + bond_pos[27]

# Perfectly replicated payoff
final_payoff <- max(0, St[27]-X)
```

We run this simulation 50 times and compare the payoffs on the perfect replication with those of the investment portfolio. We use the technique from Chapter 28, "**Data Table** on a blank cell," to run the simulation. Our conclusion: By and large, the imperfect replication strategy works okay.[1]

Data table: running 50 replications

Simulation	Perfect replication	Portfolio	Difference: Perfect - portfolio
1	0.0000	1.0317	-1.0317
2	0.0000	0.6299	-0.6299
3	7.0426	6.7303	0.3122
4	0.0000	0.5086	-0.5086
5	4.9102	4.6482	0.2620
6	2.4965	2.2094	0.2871
7	14.5895	14.6526	-0.0631
8	2.0490	2.6650	-0.6160
9	12.2000	12.7066	-0.5066
10	0.0000	0.0427	-0.0427
11	9.7744	9.9166	-0.1422
12	10.0207	10.4603	-0.4396
13	6.2331	5.6813	0.5518
14	0.0000	0.3818	-0.3818
15	0.0000	0.5602	-0.5602
16	16.8316	16.3865	0.4451
17	0.0000	1.3585	-1.3585
18	0.0000	0.4205	-0.4205
19	0.0000	1.4843	-1.4843
20	0.2534	0.7866	-0.5332
21	0.0000	0.1260	-0.1260
48	0.0000	0.0890	-0.0890
49	9.0218	9.2419	-0.2201
50	40.4557	40.7008	-0.2452

Statistics for 50 simulations

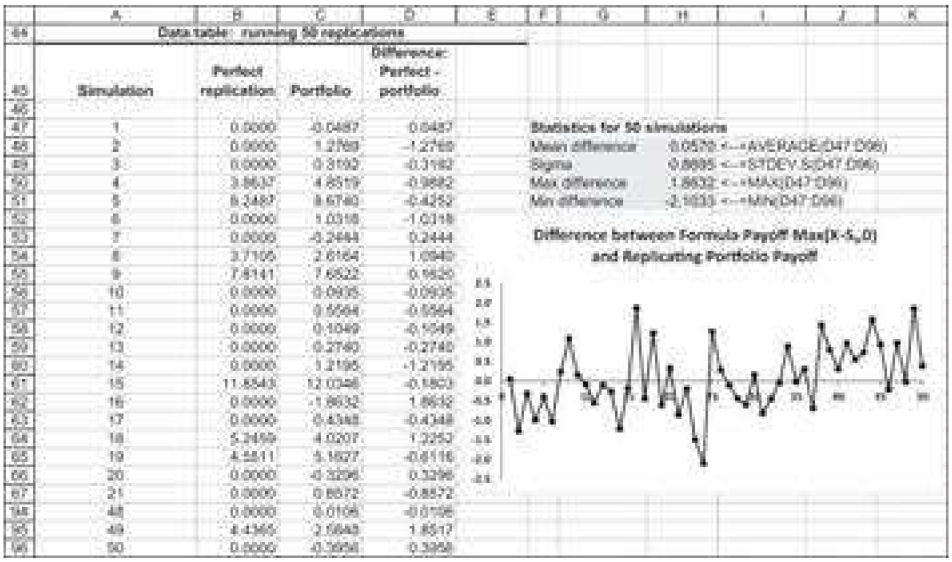
Mean difference	-0.0536	<-- =AVERAGE(D47:D96)
Sigma	0.6725	<-- =STDEV.S(D47:D96)
Max difference	1.6468	<-- =MAX(D47:D96)
Min difference	-1.4843	<-- =MIN(D47:D96)

With a straightforward analysis, we can do the same for a put. Redoing the mathematics for a put gives the following:[2]

$$Bond_{t+\Delta t} = \underbrace{Bond_t * e^{r\Delta t}}_{\substack{\uparrow \\ t+\Delta t \text{ value} \\ \text{of time } t \text{ bond} \\ \text{position}}} - \underbrace{S_t N(-d_{1,t}) * \frac{S_{t+\Delta t}}{S_t}}_{\substack{\uparrow \\ t+\Delta t \text{ value} \\ \text{of time } t \\ \text{stock position}}} + \underbrace{S_{t+\Delta t} N(-d_{1,t+\Delta t})}_{\substack{\uparrow \\ t+\Delta t \text{ desired} \\ \text{stock position}}}$$

$$= Bond_t * e^{r\Delta t} + S_{t+\Delta t}[-N(-d_{1,t}) + N(-d_{1,t+\Delta t})]$$

Apply this in a simulation and run the simulation 50 times:



	A	B	C	D
44	Data table: running 50 replications			
45	Simulation	Perfect replication	Portfolio	Difference: Perfect - portfolio
46				
47	1	0.0000	-0.0487	0.0487
48	2	0.0000	1.2769	-1.2769
49	3	0.0000	0.3192	-0.3192
50	4	3.8637	4.8519	-0.9882
51	5	8.2487	8.6740	-0.4252
52	6	0.0000	1.0318	-1.0318
53	7	0.0000	-0.2444	0.2444
54	8	3.7105	2.6164	1.0940
55	9	7.8141	7.6522	0.1620
56	10	0.0000	0.0935	-0.0935
57	11	0.0000	0.5564	-0.5564
58	12	0.0000	0.1049	-0.1049
59	13	0.0000	0.2740	-0.2740
60	14	0.0000	1.2195	-1.2195
61	15	11.8543	12.0346	-0.1803
62	16	0.0000	-1.8632	1.8632
63	17	0.0000	0.4348	-0.4348
64	18	5.2459	4.0207	1.2252
65	19	4.5511	5.1627	-0.6116
66	20	0.0000	-0.3296	0.3296
67	21	0.0000	0.8572	-0.8572
94	48	0.0000	0.0106	-0.0106
95	49	4.4365	2.5848	1.8517
96	50	0.0000	-0.3958	0.3958

26.3 Simulating Portfolio Insurance

Options can be used to guarantee minimum returns from stock investments. As we showed in our discussion of option strategies in section 16.5, when you purchase a stock (or a portfolio of stocks) and simultaneously purchase a put on the stock (on the portfolio), you are assured that the strategy payoff will never be lower than the exercise price on the put:

$$Stock + Put = S_T + Max(X - S_T, 0) = \begin{cases} S_T & if\ S_T > X \\ X & if\ S_T \le X \end{cases}$$

It is not always possible to find marketed puts in all portfolios; in this case the Black-Scholes option pricing formula can show us how to replicate a put by a dynamic strategy in which the investment in a risky asset (be it a single stock or a portfolio) and the investment in riskless bonds changes over time to mimic the returns of a put option. Such replication strategies are at the heart of the portfolio insurance strategies discussed here.

We start by considering the following simple example: You decide to invest in one share of General Pills (GP) stock, which currently costs \$56. The stock pays no dividends. You hope for a large capital gain at the end of the year, but you worry that the stock's price may decline. To guard against a decline in the stock's price, you decide to purchase a European put on the stock. The put you purchase allows you to sell the stock at the end of one year for $50. The cost of the put, $2.38, is derived from the Black-Scholes model (see Chapter 18) using the following data: S_0 = $56, X = $50, σ = 30%, and r = 4%:

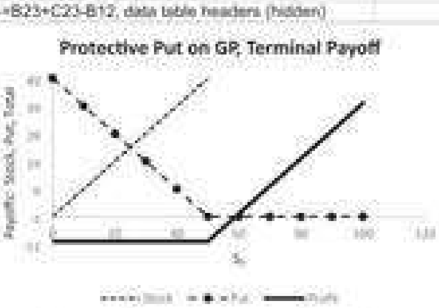
	A	B	C	D	E
2	S_0	56.00			
3	X	50.00			
4	T	1			
5	r	4.00%			
6	Sigma	30%			
7	Put price	3.01	<--=bsput(B2,B3,B4,B5,B6)		

This *protective put* or *portfolio insurance* strategy guarantees that you will lose no more than $6 on your share of GP stock (56 − 50 = 6). If the stock's price at the end of the year is more than $50, you will simply let the put expire without exercising it. However, if the stock's price at the end of the year is less than $50, you will exercise the put and collect $50. It is as if you had purchased an insurance policy on the stock with a $6 deductible.

Of course, this protection doesn't come for free: Instead of investing $56 in your single share of stock, you have invested $59.01. You could have deposited the additional $3.01 in the bank and earned interest of 4% * $3.01 = $0.1203 in the course of the year; alternatively, you could have used the $3.01 to buy more shares.

To see how this strategy works, we perform sensitivity analysis of the strategy profit as a function of the terminal price of the stock S_T:

	A	B	C	D	E
9	PORTFOLIO INSURANCE STRATEGY				
10	Buy stock	56.00	<--=B2		
11	Buy put	3.01	<--=B7		
12	Total cost	59.01	<--=B10+B11		
13					
14	Payoff at date T				
15	S_T	35.00			
16	Put payoff	15.00	<-- =MAX(B3-B15,0)		
17	Total payoff	50.00	<-- =SUM(B15:B16)		
18	Profit	-9.01	<-- =B17-B12		
19					
20	Data table: Total strategy profit as function of S_T				
21					
22	S_T	Stock	Put	Profit	
23					<--=B23+C23-B12, data table headers (hidden)
24	0	0	50	-9.007	
25	10	10	40	-9.007	
26	20	20	30	-9.007	
27	30	30	20	-9.007	
28	40	40	10	-9.007	
29	50	50	0	-9.007	
30	60	60	0	0.993	
31	70	70	0	10.993	
32	80	80	0	20.993	
33	90	90	0	30.993	
34	100	100	0	40.993	



Portfolio Insurance When There Are No Traded Puts

In the example above, we have implemented a portfolio insurance strategy by purchasing a put whose underlying asset exactly corresponds to our share portfolio. But this technique may not always be possible:

- • It could be that there is no traded put option on the shares we wish to insure.
- • It could also be that we want to purchase portfolio insurance on a more complicated basket of assets, such as a portfolio of shares. Puts on portfolios do exist (for example, there are traded puts on the Standard & Poor's 100 and the Standard & Poor's 500 portfolios), but there are no traded puts on most portfolios.

It is here that the Black-Scholes option pricing model comes to our aid. From this formula it follows that a put option on a stock (from here on, "stock" will be used to refer to a portfolio of stocks as well as a single stock) is simply a portfolio consisting of a short position in the stock and a long position in the risk-free asset, with both positions being adjusted continuously. For example, consider the Black-Scholes formula for a put with expiration date $T = 1$ and exercise price X. At time t, $0 \leq t < 1$, the put has the following value:

$$P_t = Xe^{-r(1-t)}N(-d_2) - S_tN(-d_1)$$

$$d_1 = \frac{ln\left(\frac{S_t}{X}\right) + (r + 0.5 * \sigma^2)(1-t)}{\sigma\sqrt{1-t}}, d_2 = d_1 - \sigma\sqrt{1-t}$$

where

$1 - t$ is time remaining to maturity S_t is the price of the stock at time t

Thus, buying a put is equivalent to investing $Xe^{-r(1-t)} N(-d_2)$ in a risk-free bond that matures at time 1 and investing $-S_tN(-d_1)$ in the stock. Since the investment in the stock is negative, this means that a put is equivalent to a short position in the stock and a long position in the risk-free asset.

The total investment required to buy one share of the stock plus a put on the stock is $S_t + P_t$. Writing this out and substituting in the Black-Scholes put formula gives:

$$\begin{aligned} \textit{Total investment, protective put} &= S_t + P_t \\ &= S_t + [Xe^{-r(1-t)} N(-d_2) - S_tN(-d_1)] \\ &= S_t(1 - N(-d_1)) + Xe^{-r(1-t)} N(-d_2) \\ &= S_tN(d_1) + Xe^{-r(1-t)} N(-d_2) \end{aligned}$$

Here the last equality uses the fact that for the standard normal distribution $N(-x) = 1 - N(x)$. Another way of looking at this problem is to regard the total investment $S_t + P_t$ at time t as a portfolio of a stock and a bond; we can then ask, What is the proportion ω_t of this portfolio invested in the stock at time t? Rewriting the formula above in terms of portfolio proportions gives:

$$\textit{Proportion invested in stock } (\omega_t) = \frac{S_t N(d_1)}{S_t N(d_1) + Xe^{-r(1-t)} N(-d_2)}$$

$$\textit{Proportion in risk-free asset } (1-\omega_t) = \frac{Xe^{-r(1-t)} N(-d_2)}{S_t N(d_1) + Xe^{-r(1-t)} N(-d_2)}$$

To sum up: If you want to buy a specific portfolio of assets *and* an insurance policy guaranteeing that at $t = 1$ your total investment will not be worth less than X, then at each point in time (t), you should invest a proportion ω_t of your wealth in the specific portfolio you have chosen and a proportion $1 - \omega_t$ in riskless, pure discount bonds that mature at $t = 1$. The Black-Scholes put pricing formula can be used to determine these proportions.

An Example

Suppose you decide to invest $1,000 in GP stock (currently selling at $56) and in protective puts on the shares with an exercise price of $50 and an expiration date one year from now. This ensures that your dollar value per share at the end of one year will be no less than $50. Suppose that there is no traded put on GP, so that you will have to create your own put by investing in the share and in riskless discount bonds. The riskless rate of interest is 4%, and the standard deviation of the GP log return is 30%.

We will construct a series of portfolios that implements this strategy on a week-by-week basis. Based on our discussion from the previous section, we know that this replication strategy will not be perfect. We simulate it to see how it works.

Week 0: At the beginning of this week, the initial investment in shares of GP should be:

$$\omega_0 = \frac{S_0 N(d_1)}{S_0 + P_0} = \frac{56 * 0.7457}{56 + 3.01} = 70.77\%$$

The remaining proportion, $1 - \omega_0 = 29.23\%$, is invested in riskless discount bonds maturing in one year. If traded European puts on GP existed and if these puts had an exercise price of 50 with an exercise date one year from now, these would be trading at $3.01. Your strategy would consist of buying 16.95 shares of GP (cost = $949.03) and the same number of puts (total cost = 16.95 * 3.01 = $50.97). The number of shares and puts can easily be calculated by: 1,000/ (56 + 3.01) = 16.95. Buying $707.72 worth of shares and $292.28 worth of bonds exactly duplicates the initial investment in 16.95

shares and puts. This equivalence is guaranteed by Black-Scholes. The calculations of the option price and the appropriate portfolio proportions are shown in the spreadsheet below:

	A	B	C
1	**BLACK-SCHOLES OPTION PRICING FORMULA APPLIED TO GENERAL PILLS PUT**		
2	S	56.00	Stock price
3	X	50.00	Exercise price
4	T	1.00	Time remaining
5	r	4%	Risk-free rate of interest
6	Sigma	30%	Stock volatility
7	Put price	3.01	<-- call price - S + X*Exp(-r*T): by Put-Call parity
8			
9	**Calculating the portfolio insurance proportions**		
10	Omega	70.77%	<-- =B2*NORM.S.DIST(done(B2,B3,B4,B5,B6),1)/(B2+B7), proportion in shares
11	1-omega	29.23%	<-- =1-B10, proportion in bonds
12			
13	Initial investment	1,000.00	
14	In stocks	707.72	<--=B13*B10
15	Number of stocks	12.64	<--=B14/B2
16	In risk-free	292.28	<--=B13*B11

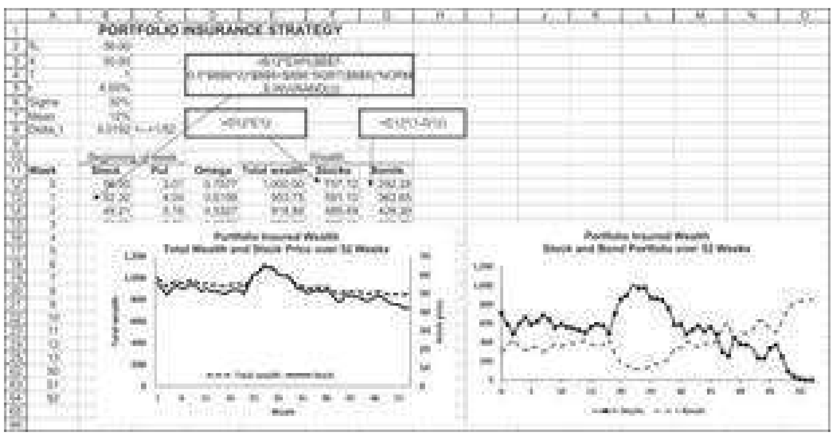
You started at t = 0 with an initial investment of $1,000; now suppose that by the beginning of the next week (t = 1/52 = 0.0192) the price of GP shares increased to $60. Here is your updated portfolio at the beginning of week 1. You can see that when the stock price increases, the portfolio proportion of the stock increases:

	A	B	C
1	**UPDATING THE PORTFOLIO INSURANCE PROPORTIONS ($S_{t+\Delta t}$=60)**		
2	Previous stock price S_t	56.00	
3	Previous time to maturity	1.00	
4	Time interval, Δt	0.0192	<--=1/52
5	New time to maturity	0.9808	<--=B3-B4
6	r	4.00%	Risk-free rate of interest
7	X	50	Exercise price
8	Sigma	30%	Stock volatility
9	Current stock price $S_{t+\Delta t}$	60.00	New stock price
10	Current put price $P_{t+\Delta t}$	2.10	<--=bsput(B9,B7,B5,B6,B8)
11			
12	Previous portfolio		
13	Stock	707.72	
14	Bonds	292.28	
15			
16	**Current portfolio before readjustment**		
17	Stock	758.27	<--=B13*B9/B2
18	Bonds	292.51	<--=B14*EXP(B6*B4)
19	**Total**	**1,050.78**	<--=SUM(B17:B18)
20			
21	**Calculating the portfolio insurance proportions**		
22	Proportion of stock omega, ω	78.69%	<--=B9*NORM.S.DIST(done(B9,B7,B5,B6,B8),1)/(B9+B10)
23	1-omega	21.31%	<--=1-B22

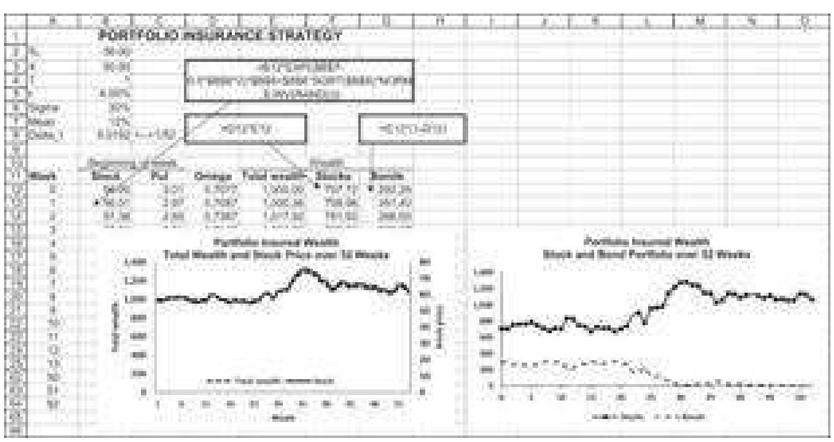
If, conversely the stock price at time Δ t is 52, the proportion of the stock decreases and the proportion of the risk-free increases:

	A	B	C
1	**UPDATING THE PORTFOLIO INSURANCE PROPORTIONS ($S_{t+\Delta t}$=50)**		
2	Previous stock price S_t	56.00	
3	Previous time to maturity	1.00	
4	Time interval, Δt	0.0192	<--=1/52
5	New time to maturity	0.9808	<--=B3-B4
6	r	4.00%	Risk-free rate of interest
7	X	50	Exercise price
8	Sigma	30%	Stock volatility
9	Current stock price $S_{t+\Delta t}$	50.00	New stock price
10	Current put price $P_{t+\Delta t}$	4.88	<--=bsput(B9,B7,B5,B6,B8)
11			
12	Previous portfolio		
13	Stock	707.72	
14	Bonds	292.28	
15			
16	**Current portfolio before readjustment**		
17	Stock	631.89	<--=B13*B9/B2
18	Bonds	292.51	<--=B14*EXP(B6*B4)
19	**Total**	**924.40**	<--=SUM(B17:B18)
20			
21	**Calculating the portfolio insurance proportions**		
22	Proportion of stock omega, ω	55.62%	<--=B9*NORM.S.DIST(dOne(B9,B7,B5,B6,B8),1)/(B9+B10)
23	1-omega	44.38%	<--=1-B22

We now simulate this strategy over time:



In the above simulation, the stock price decreased during the year, and the final proportions of the portfolio insurance strategy will be wholly in the risk-free asset. Below we present another possibility: The stock price increases over the year, and the portfolio proportions are increasingly in the stock, with the bond going to zero.



On the companion website to this book you'll find the R code that replicates the above analysis; we believe that it is too technical and long to be printed in the book.

26.4 Some Properties of Portfolio Insurance

The preceding example illustrates some of the typical properties of portfolio insurance. Three important properties are the following:

PROPERTY 1 When the stock price is above the exercise price X, then the proportion ω invested in the risky asset is greater than 50%.

Proof The proof of this property requires a little manipulation of our formula for ω. Rewrite ω as:

$$\omega = \frac{S \cdot N(d_1)}{S \cdot N(d_1) + Xe^{-r(1-t)}N(-d_2)} = \frac{1}{1 + \left(\dfrac{Xe^{-r(1-t)}N(-d_2)}{SN(d_1)} \right)}$$

We will show that when $S \geq X$, the denominator of ω is < 2, which will prove the proposition. First, note that when $S \geq X$, $X/S < 1$. Next, note that $e^{-r(1-t)} < 1$ for all $0 \leq t \leq 1$. Finally, examine the following expression:

$$\frac{N(-d_2)}{N(d_1)} = \frac{N(\sigma\sqrt{1-t} - d_1)}{N(d_1)} = \frac{N\left(0.5\sigma\sqrt{1-t} - \left[ln\left(\dfrac{S}{X}\right) + r(1-t) \right] / \sigma(1-t)\right)}{N\left(0.5\sigma\sqrt{1-t} + \left[ln\left(\dfrac{S}{X}\right) + r(1-t) \right] / \sigma(1-t)\right)} < 1$$

This proves the property.

PROPERTY 2 When the stock's price increases, the proportion ω invested in the stock increases, and vice versa.

Proof To see this property, it is enough to see that when S increases, the value of the put decreases and $N(-d_1)$ decreases. Rewrite the original definition of ω as:

$$\omega = \frac{S[1 - N(-d_1)]}{S + P} = \frac{[1 - N(-d_1)]}{1 + P/S}$$

Thus, when S increases, the denominator of ω decreases and the numerator increases, which proves Property 2.

PROPERTY 3: As $t \to 1$, one of two things happens: If $S_t > X$, then $\omega_t \to 1$. If $S_t < X$, then $\omega_t \to 0$.

Proof To see this, note that when $S_t > X$ and $t \to 1$, $N(d_1) \to 1$ and $N(-d_1) \to 0$. Thus, for this case $\omega_t \to 1$. Conversely, when $S_t < X$ and $t \to 1$, $N(d_1) \to 0$ and $N(-d_1) \to 1$ and thus $\omega_t \to 0$. Strictly speaking, these statements are only true as "probability limits”; see Billingsley (1968). What about the case when, as $t \to 1$, $S_t / X \to 1$? In this case $\omega_t \to ½$. However, the probability of this occurring is zero.

26.5 Digression: Insuring Total Portfolio Returns

We digress slightly to consider an interesting portfolio insurance problem. So far, we have considered only the problem of constructing artificial puts, one per share. A slightly different version of this problem involves constructing a portfolio of puts and shares that guarantees the *total* dollar returns on the *total* initial investment. A typical story goes like this:

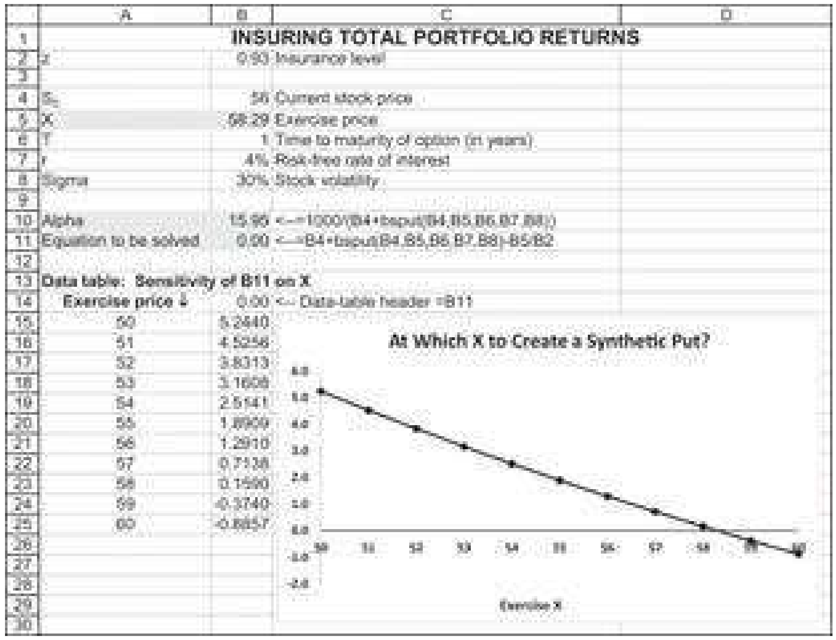
You have \$1,000 to invest, and you want to guarantee that a year from now you will have at least \$1,000 × z. Here z is some number, generally between 0 and 1; for example, if $z = 0.93$, this means that you want your final wealth to be at least \$930.[3] You want to invest in a stock whose current price is S_0 and in a put on the stock with an exercise price X. You want the number of puts to be equal to the number of shares so that each “package” of share + put costs you today $S_0 + P(S_0, X)$. To implement the strategy, you must therefore buy α shares, where:

$$\alpha = \frac{1{,}000}{S_0 + P(S_0, X)}$$

Since you have bought α shares and α puts with an exercise price of X, the minimum dollar return from your portfolio is αX. You want this to be equal to 1,000 * z, and therefore you solve to get $\alpha = 1{,}000 \cdot z/X$. Thus, you can guarantee your minimum return if:

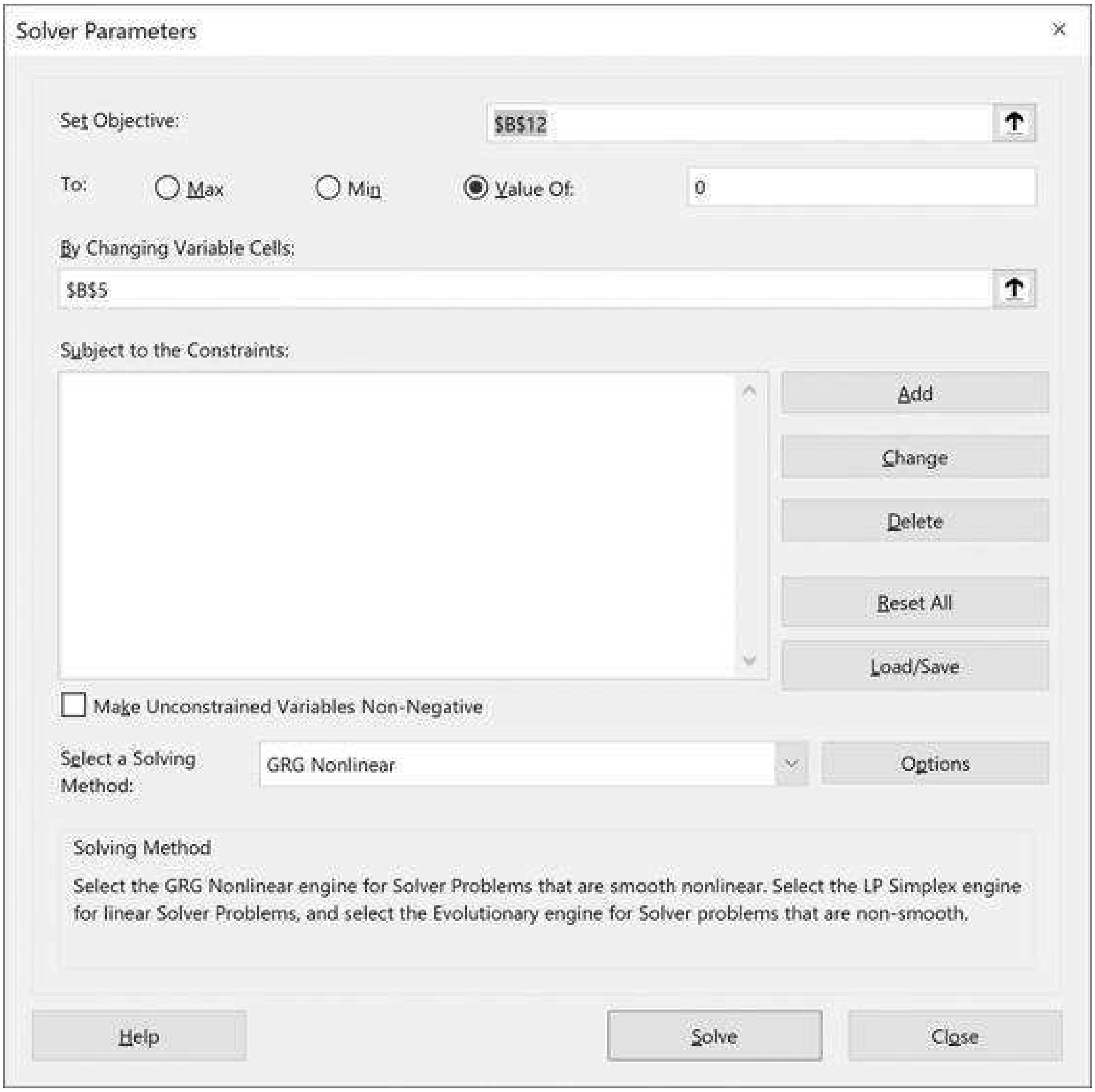
$$S_0 + P(S_0, X) = X/z$$

Here is a spreadsheet implementation of this equation. The data table shows the graph of $S_0 + P(S_0, X) - X/z$; where this graph crosses the x-axis is the solution for the put exercise price X when $S_0 = 56$, $\sigma = 30\%$, $r = 3\%$, $T = 1$, and $z = 93\%$.



	A	B	C	D
1		INSURING TOTAL PORTFOLIO RETURNS		
2	z	0.93	Insurance level	
3				
4	S_0	56	Current stock price	
5	X	58.29	Exercise price	
6	T	1	Time to maturity of option (in years)	
7	r	4%	Risk-free rate of interest	
8	Sigma	30%	Stock volatility	
9				
10	Alpha	15.96	<--=1000/(B4+bsput(B4,B5,B6,B7,B8))	
11	Equation to be solved	0.00	<--=B4+bsput(B4,B5,B6,B7,B8)-B5/B2	
12				
13	**Data table: Sensitivity of B11 on X**			
14	**Exercise price ↓**	0.00	<-- Data-table header =B11	
15	50	5.2440		
16	51	4.5256		
17	52	3.8313		
18	53	3.1608		
19	54	2.5141		
20	55	1.8909		
21	56	1.2910		
22	57	0.7138		
23	58	0.1590		
24	59	-0.3740		
25	60	-0.8857		

As you can see, the X for which the equation in cell B11 equals zero is between 58 and 59. We can use **Solver** to find the exact value, $X = 58.29$:



Solver gives the solution to the equation $S_0 + Put(S_0, X) - X/z = 0$.

The solution is to buy α = 15.9536 puts and shares (the cost of which is $1,000, as you can see in cell B16 below). The minimum return of this portfolio is $\alpha * X$ = 15.9536 * 58.2941 = $930 (cell B19).

	A	B	C
1	**INSURING TOTAL PORTFOLIO RETURNS**		
2	z	0.9300	Insurance level
3			
4	S_0	56.0000	Current stock price
5	X	58.2942	Exercise price
6	T	1	Time to maturity of option (in years)
7	r	4.00%	Risk-free rate of interest
8	Sigma	30%	Stock volatility
9			
10	Alpha	15.9536	<--=1000/(B4+bsput(B4,B5,B6,B7,B8))
11	Equation to be solved	0.00	<--=B4+bsput(B4,B5,B6,B7,B8)-B5/B2
12			
13	**Check:**		
14	Cost of shares	893.40	<--=B10*B4
15	Cost of puts	106.60	<--=B10*bsput(B4,B5,B6,B7,B8)
16	**Total cost**	**1,000.00**	<--=B14+B15
17			
18	Minimum portfolio return	930.00	<--=B10*B5

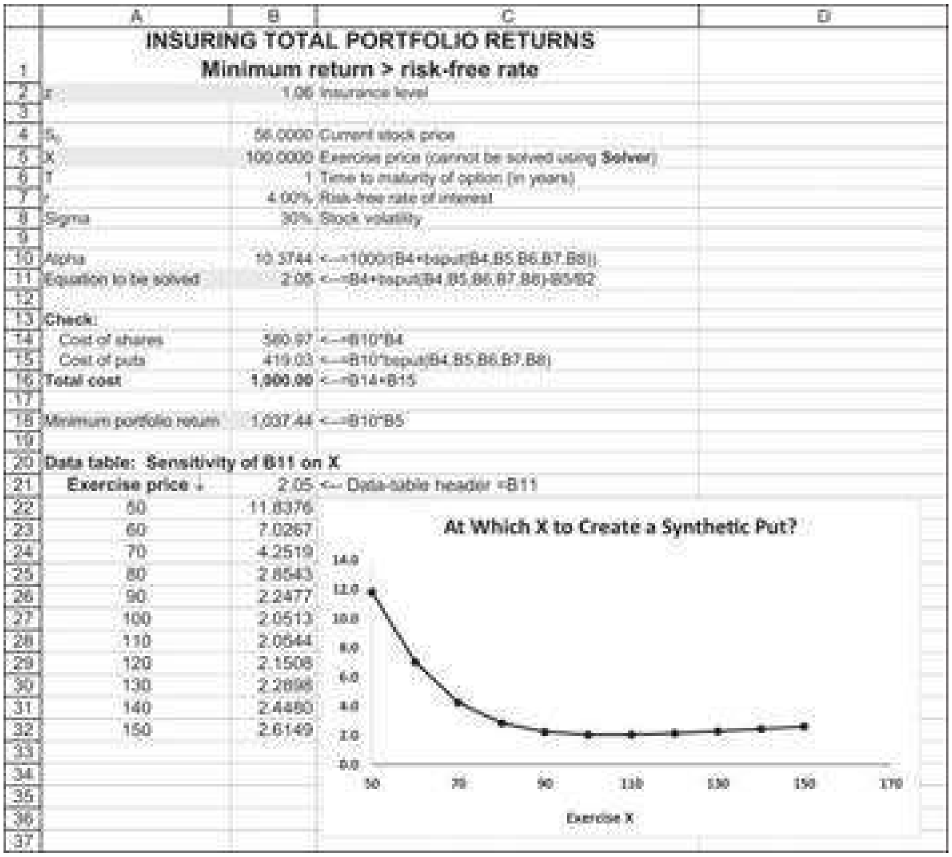
Can You Insure for More Than Your Initial Investment?

When we raise the insurance level, the implied exercise price of the put must go up. This means that as we buy more insurance, we spend relatively more of our $1,000 on puts (insurance) and relatively less on the stocks (which have the upside potential).

Can we insure for more than our current level of investment? To put it another way, can we set $z > 1$? This means that we are picking an insurance level that guarantees that we end up with *more* than our initial investment. A little thought and some calculations reveal that we can indeed choose $z > 1$ as long as $z < (1 + r)$. That is, we cannot guarantee ourselves a return greater than the riskless interest rate! To see this, we offer two examples. In the first example below, we solve for $z = 1.04 = 1 + r$. This has a solution (note that the value of cell B11 is zero):

	A	B	C
1	**INSURING TOTAL PORTFOLIO RETURNS Minimum return = risk-free rate**		
2	z	1.0400	Insurance level
3			
4	S_0	56.0000	Current stock price
5	X	116.0771	Exercise price (solved using **Solver**)
6	T	1	Time to maturity of option (in years)
7	r	4.00%	Risk-free rate of interest
8	Sigma	30%	Stock volatility
9			
10	Alpha	8.9596	<--=1000/(B4+bsput(B4,B5,B6,B7,B8))
11	Equation to be solved	0.00	<--=B4+bsput(B4,B5,B6,B7,B8)-B5/B2
12			
13	**Check:**		
14	Cost of shares	501.74	<--=B10*B4
15	Cost of puts	498.26	<--=B10*bsput(B4,B5,B6,B7,B8)
16	**Total cost**	**1,000.00**	<--=B14+B15
17			
18	Minimum portfolio return	1,040.00	<--=B10*B5

When $z > 1.04$, however, there is no solution. Returning to the first chart of this section, we can see that there is no solution that insures for 6% ($z = 1.06$) that is greater than the interest rate of 4%:



	A	B	C	D
1	**INSURING TOTAL PORTFOLIO RETURNS** **Minimum return > risk-free rate**			
2	z	1.06	Insurance level	
3				
4	S_0	56.0000	Current stock price	
5	X	100.0000	Exercise price (cannot be solved using **Solver**)	
6	T	1	Time to maturity of option (in years)	
7	r	4.00%	Risk-free rate of interest	
8	Sigma	30%	Stock volatility	
9				
10	Alpha	10.3744	<--=1000/(B4+bsput(B4,B5,B6,B7,B8))	
11	Equation to be solved	2.05	<--=B4+bsput(B4,B5,B6,B7,B8)-B5/B2	
12				
13	**Check:**			
14	Cost of shares	580.97	<--=B10*B4	
15	Cost of puts	419.03	<--=B10*bsput(B4,B5,B6,B7,B8)	
16	**Total cost**	**1,000.00**	<--=B14+B15	
17				
18	Minimum portfolio return	1,037.44	<--=B10*B5	
19				
20	**Data table: Sensitivity of B11 on X**			
21	**Exercise price ↓**	2.05	<-- Data-table header =B11	
22	50	11.8376		
23	60	7.0267		
24	70	4.2519		
25	80	2.8543		
26	90	2.2477		
27	100	2.0513		
28	110	2.0544		
29	120	2.1508		
30	130	2.2898		
31	140	2.4480		
32	150	2.6149		

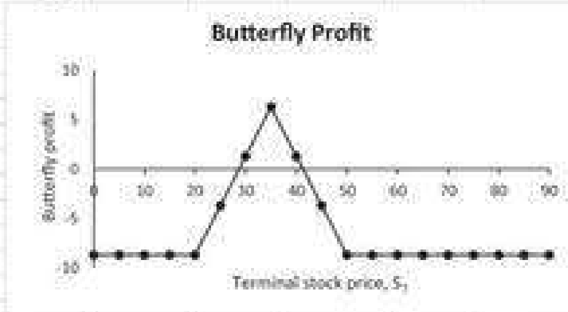
26.6 Simulating a Butterfly

For our last exercise in this chapter, we simulate a butterfly strategy. A butterfly consists of three options. In this section we simulate a butterfly over one month (22 trading days), rebalancing the position daily. Our butterfly consists of three calls written on a stock with current price $S_0 = 35$. We assume that the stock price is lognormal with $\sigma = 35\%$ and that $r = 2\%$. The calls are as follows:

- • Call 1: $X = 20$, position: 1 call purchased
- • Call 2: $X = 35$, position: 2 calls written
- • Call 3: $X = 50$, position: 1 call purchased

The payoff/profit pattern of the butterfly is described below:

	A	B	C	D
1	**BUTTERFLY**			
2	S_0	35		
3	T	0.087302	<--=22/252	
4	Sigma	35%		
5	r	2%		
6				
7		**X**	**Position**	**Cost**
8	Call1	20	1	15.03 <--=bscall($B$2,B8,$B$3,$B$5,$B$4)
9	Call2	35	-2	1.47 <--=bscall($B$2,B9,$B$3,$B$5,$B$4)
10	Call3	50	1	0.00 <--=bscall($B$2,B10,$B$3,$B$5,$B$4)
11	Initial cost	12.09	<--=SUMPRODUCT(C8:C10,D8:D10)	
12				
13	**One profit example (used in data table)**			
14	S_T	47		
15	Profit1	11.97	<--=C8*(MAX($B$14-B8,0)-D8)	
16	Profit2	-21.05	<--=C9*(MAX($B$14-B9,0)-D9)	
17	Profit3	0.00	<--=C10*(MAX($B$14-B10,0)-D10)	
18	Total	-9.09	<--=SUM(B15:B17)	
19				
20	**Data table: profit as function of S_T**			
21	**S_T**	**Payoff**		
22		-9.09	<--=B18	
23	0	-8.70		
24	5	-8.70		
25	10	-8.70		
26	15	-8.70		
27	20	-8.70		
28	25	-3.70		
29	30	1.30		
30	35	6.30		
31	40	1.30		
32	45	-3.70		
33	50	-8.70		
34	55	-8.70		
35	60	-8.70		
41	90	-8.70		



We recall the Black-Scholes formula and denote the time remaining to option maturity by T:

$$Call(X) = S \cdot N(d_1, T) - X \cdot e^{-rT} \cdot N(d_2, T)$$

$$d_1 = \frac{Ln\left(\frac{S}{X}\right) + (r + 0.5 \cdot \sigma^2) \cdot T}{\sigma\sqrt{T}}, d_2 = d_1 - \sigma\sqrt{T}$$

In a table for the butterfly:

N_{Low} calls $\quad Call(X_{Low}, t) = N_{Low}\ [S \cdot N(d_1(X_{Low}, T)) - X_{Low} \cdot e^{-rT} N(d_2(X_{Low}, T))]$

$$d_1(X_{Low}, T) = \frac{Ln\left(\frac{S}{X_{Low}}\right) + (r + 0.5 \cdot \sigma^2)T}{\sigma\sqrt{T}},\ d_2(X_{Low}, T) = d_1(X_{Low}, T) - \sigma\sqrt{T}$$

N_{Mid} calls $\quad Call(X_{Mid}, t) = N_{Mid}\ [S \cdot N(d_1(X_{Mid}, T)) - X_{Mid} \cdot e^{-rT} N(d_2(X_{Mid}, T))]$

$$d_1(X_{Mid}, T) = \frac{Ln\left(\frac{S}{X_{Mid}}\right) + (r + 0.5 \cdot \sigma^2)T}{\sigma\sqrt{T}},\ d_2(X_{Mid}, T) = d_1(X_{Mid}, T) - \sigma\sqrt{T}$$

N_{High} calls $\quad Call(X_{High}, t) = N_{High} [S \cdot N(d_1(X_{High}, T)) - X_{High} \cdot e^{-rT} N(d_2(X_{High}, T))]$

$$d_1(X_{High}, T) = \frac{Ln\left(\frac{S}{X_{High}}\right) + (r + 0.5 \cdot \sigma^2)T}{\sigma\sqrt{T}},\ d_2(X_{High}, T) = d_1(X_{High}, T) - \sigma\sqrt{T}$$

Adding these together, we get:

$$\begin{aligned} Butterfly(t) = S_t\{&N_{Low} \cdot N(d_1(X_{Low}, T)) + N_{Mid} \cdot N(d_1(X_{Mid}, T)) \\ &+ N_{High} \cdot N(d_1(X_{High}, T))\} + -e^{-rT}\{N_{Low} \cdot X_{Low} N(d_2(X_{Low}, T)) \\ &+ N_{Mid} \cdot X_{Mid} N(d_2(X_{Mid}, T)) + N_{High} \cdot X_{High} N(d_2(X_{High}, T))\} \end{aligned}$$

Newly defined VBA functions are:

$$\begin{aligned} Butterfly(t) = S_t * butterflyNd_1(&X_{Low}, X_{Mid}, X_{High}, N_{Low}, N_{Mid}, N_{High}, S_t, X, t, \sigma, r) \\ -e^{-rT} butterflyNd_2(&X_{Low}, X_{Mid}, X_{High}, N_{Low}, N_{Mid}, N_{High}, S_t, X, t, \sigma, r) \end{aligned}$$

The VBA function **butterflyNd1** is defined below (the function **butterflyNd2** is similar):

```
Function butterflyNd1(XLow, XMid, XHigh, NumberLow, NumberMid,
NumberHigh, Stock, Time, Interest, sigma)
butterflyNd1 = Stock * _
    (NumberLow * Application.Norm_S_Dist (dOne(Stock, XLow, Time,
    Interest, sigma), 1) _
    + NumberMid * Application.Norm_S_Dist (dOne(Stock, XMid,
    Time, Interest, sigma), 1) _
    + NumberHigh * Application.Norm_S_Dist (dOne(Stock, XHigh,
    Time, Interest, sigma), 1))
```

```
End Function
```

Self-Financing Butterfly Portfolio

We attempt to replicate this position dynamically, maintaining the self-financing requirement (i.e., no cash flows after time $t = 0$).

At time $t = 0$, we define the investment in the stock and the bond as:

$$Stock(0) = S_0 \begin{Bmatrix} N_{Low} * N(d_1(X_{Low}, t = 0)) + N_{Mid} * N(d_1(X_{Mid}, t = 0)) \\ + N_{High} * N(d_1(X_{High}, t = 0)) \end{Bmatrix}$$

Afterward, at time $t + \Delta t$, we define the stock investment as above and adjust the bond investment to net out the stock:

$$Stock(t + \Delta t) = S_{t+\Delta t} \begin{Bmatrix} N_{Low} * N(d_1(X_{Low}, t + \Delta t)) + N_{Mid} * (d_1(X_{Mid}, t + \Delta t)) \\ + N_{High} * N(d_1(X_{High}, t + \Delta t)) \end{Bmatrix}$$

Cash Flow from the Stock Position

The cash flow from the stock position is the value of the time t position minus the value of the time $t + \Delta t$ position:

$$\text{Stock cash flow}(t + \Delta t) = Stock(t) * \frac{S_{t+\Delta t}}{S_t} - Stock(t + \Delta t) =$$

$$\frac{S_{t+\Delta t}}{S_t}\{N_{Low} * N(d_1(X_{Low}, S_t, t)) + N_{Mid} * N(d_1(X_{Mid}, S_t, t))$$

$$+ N_{High} * N(d_1(X_{High}, S_t, t))\} -$$

$$S_{t+\Delta t} \begin{Bmatrix} N_{Low} * N(d_1(X_{Low}, S_{t+\Delta t}, t + \Delta t)) + N_{Mid} * (d_1(X_{Mid}, S_{t+\Delta t}, t + \Delta t)) \\ + N_{High} * (d_1(X_{High}, S_{t+\Delta t}, t + \Delta t)) \end{Bmatrix}$$

In Excel:

$$\frac{S_{t+\Delta t}}{S_t} * butterflyNd_1(X_{Low}, X_{Mid}, X_{High}, N_{Low}, N_{Mid}, N_{High}, S_t, X, t, r, \sigma)$$

$$- butterflyNd_1(X_{Low}, X_{Mid}, X_{High}, N_{Low}, N_{Mid}, N_{High}, S_{t+\Delta t}, X, t + \Delta t, r, \sigma)$$

Zero-Investment Cash Flow from the Bond Position

In order to define a self-financing strategy, we let the bond position at each time t "absorb" the change in the stock position. At time $t = 0$, the bond position is:

$$Bond(0) = -e^{-rT}\begin{Bmatrix} N_{Low} * X_{Low} N(d_2(X_{Low})) + N_{Mid} * X_{Mid} N(d_2(X_{Mid})) \\ + N_{High} * X_{High} N(d_2(X_{High})) \end{Bmatrix}$$

At time $t + \Delta\, t$, the bond position is the previous position minus the cash flow on the stock position:

$$Bond(t + \Delta) = Bond(t) * e^{r\Delta t}$$
$$-\begin{Bmatrix} \dfrac{S_{t+\Delta t}}{S_t} * butterflyNd_1(X_{Low}, X_{Mid}, X_{High}, N_{Low}, N_{Mid}, N_{High}, S_t, X, t, r\sigma) \\ -butterflyNd_1(X_{Low}, X_{Mid}, X_{High}, N_{Low}, N_{Mid}, N_{High}, S_{t+\Delta t}, X, t + \Delta t, r, \sigma) \end{Bmatrix}$$

The cash flow on the bond position is:

$$\begin{aligned} \text{Bond cash flow}(t + \Delta t) &= Bond(t) * e^{r\Delta t} = Bond(t + \Delta t) \\ &= Bond(t) * e^{r\Delta t} - Bond(t) * e^{r\Delta t} \end{aligned}$$
$$+\begin{Bmatrix} \dfrac{S_{t+\Delta t}}{S_t} * butterflyNd_1(X_{Low}, X_{Mid}, X_{High}, N_{Low}, N_{Mid}, N_{High}, S_t, X, t, r\sigma) \\ - butterflyNd_1(X_{Low}, X_{Mid}, X_{High}, N_{Low}, N_{Mid}, N_{High}, S_{t+\Delta t}, X, t + \Delta t, r, \sigma) \end{Bmatrix}$$

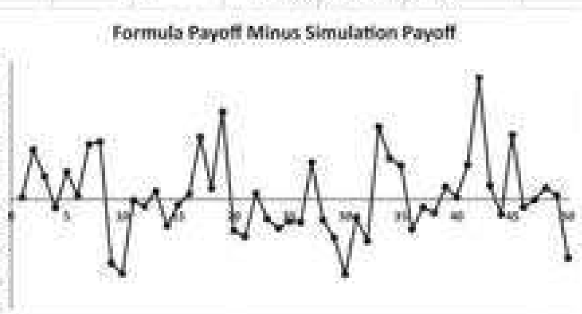
Running the Simulation

We run the simulation as shown in the next spreadsheet:

	A	B	C	D	E	F	G	H
1				SIMULATING A BUTTERFLY				
2	S_0	35			Butterfly	X	Number	
3	T	0.087302	<--=22/252		Xlow	20	-1	
4	Sigma	35%			Xmid	35	2	
5	r	2%			Xhigh	50	-1	
6	Delta_t	0.0040	<--=1/252					
7								
8	Butterfly simulation							
9	Time to maturity	Stock	Stock position	Bond position				
10	0.0873	35.0000	1.9013	-13.9909				
11	0.0833	34.0505	-5.5392	-6.6031	<--			
12	0.0794	34.2071	-4.5392	-7.6291	=D10*EXP($B$5*$B$6)+B11/B10*b			
13	0.0754	33.8898	-7.2575	-4.8693	utterflyNd1($F$3,$F$4,$F$5,$G$3,$			
14	0.0714	33.2002	-12.7681	0.7886	G$4,$G$5,B10,A10,$B$5,$B$4)-			
15	0.0675	32.8229	-15.8152	3.9808	butterflyNd1($F$3,$F$4,$F$5,$G$3,			
16	0.0635	30.6268	-26.1650	15.3891	$G$4,$G$5,B11,A11,$B$5,$B$4)			
17	0.0595	31.3807	-24.4194	13.0007				
18	0.0556	31.6408	-23.9631	12.3429				
19	0.0516	32.5731	-19.7188	7.3936				
20	0.0476	32.8801	-18.3194	5.8089				
21	0.0437	33.4902	-14.0921	1.2422				
22	0.0397	34.0205	-9.5873	-3.4856				
23	0.0357	34.4735	-5.0679	-8.1330				
24	0.0317	34.7321	-2.2657	-10.9738				
25	0.0278	34.3782	-7.2801	-5.9373				
26	0.0238	33.9446	-13.7363	0.6103				
27	0.0198	35.2047	4.2256	-17.8615				
28	0.0159	35.4341	8.5939	-22.2036				
29	0.0119	35.6336	13.5231	-27.0862				
30	0.0079	35.6983	21.3734	-34.8382				
31	0.0040	36.9223	36.3797	-49.2376				
32	0.0000	37.3099	36.7615	-49.2415				
33								
34	Comparison of formula payoff to simulation payoff							
35	Formula	-12.6901	<--=G3*MAX(B32-F3,0)+G4*MAX(B32-F4,0)+G5*MAX(B32-F5,0)					
36	Simulation	-12.4800	<--=SUM(C32:D32)					

Now, repeat this simulation 50 times and compare the formula payoff (B35) to the simulation payoff (B36):

	A	B	C	D	E	F	G	H	I
38	Data table: 50 runs of the simulation								
39	Run	0.6037	<-- =B35-B36, data table header						
40	1	0.0675							
41	2	1.8097			Statistics for 50 runs: Formula payoff-simulation				
42	3	0.8366			Average	0.1175	<--=AVERAGE(B40:B89)		
43	4	-0.3115			Sigma	1.4897	<--=STDEV.S(B40:B89)		
44	5	0.9994			Max	4.4473	<--=MAX(B40:B89)		
45	6	0.1257			Min	-2.7567	<--=MIN(B40:B89)		
46	7	2.0113							
47	8	2.1040							
48	9	-2.3427							
49	10	-2.7383							
50	11	-0.0528							
51	12	-0.2782							
52	13	0.2862							
53	14	-0.9759							
54	15	-0.2089							
55	16	0.1889							
56	17	2.2879							
57	18	0.3892							
58	19	3.1930							
59	20	-1.1574							
60	21	-1.3782							
61	22	0.2152							
62	23	-0.7180							
88	49	0.1495							
89	50	-2.1676							



The R code to simulate the butterfly looks like this:

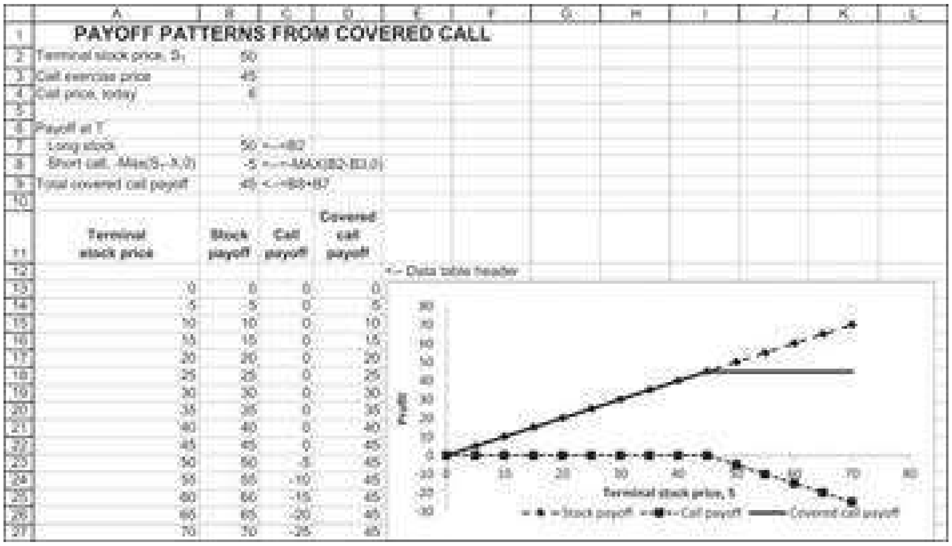
```r
# Inputs:
S <- 35
sigma_S <- 0.35
r <- 0.02
t <- 22/252
X1 <- 20
X2 <- 35
X3 <- 50

# Price of a Butterfly
fm5_bs_call(S = S, X = X1, r = r , t = t, sigma_S = sigma_S) -
    2 * fm5_bs_call(S = S, X = X2, r = r , t = t, sigma_S = sigma_S) +
    fm5_bs_call(S = S, X = X3, r = r , t = t, sigma_S = sigma_S)

## Self Financed Butterfly
# Define a function for a butterfly strategy
pos1 <- 1 * fm5_sf_call_pos(S = S, X = 20, r = r, t = t, sigma_S = sigma_S)
pos2 <- -2 * fm5_sf_call_pos(S = S, X = 35, r = r , t = t, sigma_S = sigma_S)
pos3 <- 1 * fm5_sf_call_pos(S = S, X = 50, r = r , t = t, sigma_S = sigma_S)

# summarize position
positions <- c(stock_position = sum(pos1["S"], pos2["S"], pos3["S"]),
               X1_position = pos1["X"],
               X2_position = pos2["X"],
               X3_position = pos3["X"])

# Calulate price
butterfly_price <- sum(positions)
```

26.7 Summary

This chapter concentrated on simulating option replication strategies. All the replication strategies are based on the idea that the Black-Scholes formula gives the dynamic investment over time in the risky and riskless asset. When implementing this idea, we had to adjust this dynamic formula so that the net investment over time is zero (the so-called self-financing or zero-investment strategies). The replication strategies then become somewhat less than perfect so that the terminal payoff of these strategies matches the formula payoff only approximately. For more complicated strategies (we illustrated for a butterfly), the mismatch between the formula and the strategy payoff can be significant.

Exercises

1. You are a portfolio manager, and you want to invest in an asset having $\sigma = 40\%$. You want to create a put on the investment so that at the end of the year you have losses no greater than 5%. Since there is no put on this specific asset, you plan to create a synthetic put by engaging in a dynamic investment strategy—purchasing a portfolio composed of dynamically changing proportions of the risky asset and riskless bonds. If the interest rate is 6%, how much should your initial investment be in the portfolio and in the riskless bond?
2. Simulate the above strategy, assuming weekly rebalancing of the portfolio.
3. Go back to the numerical example of section 26.5. Write a VBA function that solves for the implied asset value V_a. (Hint: Use the bisection method.) Then use this function to create a graph showing the trade-off between the implied asset value and the asset volatility.
4. You have been offered the chance to purchase stock in a firm. The seller wants \$55 per share but offers to repurchase the stock at the end of one-half year for \$50 per share. If the σ of the share's log returns is 80%, determine the true value per share. Assume that the interest rate is 10%.
5. A covered call is a long stock and short call. The pattern of payoffs is given below:

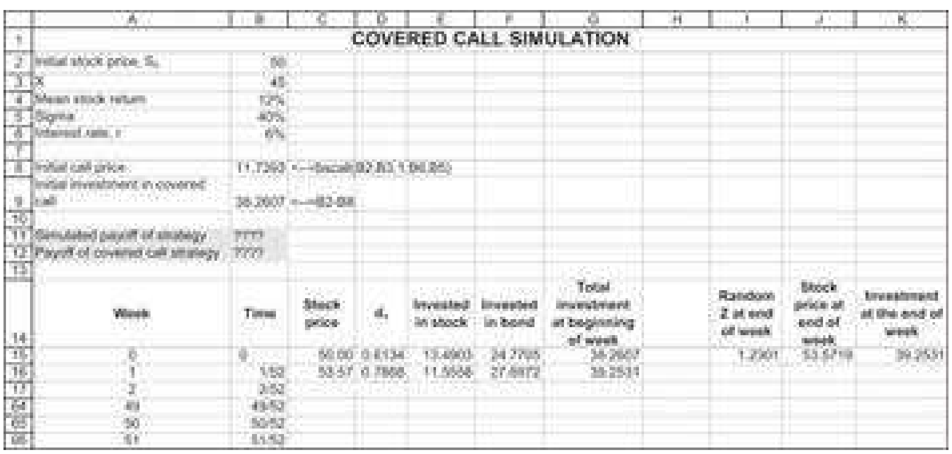


	A	B	C	D	E
1	**PAYOFF PATTERNS FROM COVERED CALL**				
2	Terminal stock price, S_T	50			
3	Call exercise price	45			
4	Call price, today	6			
5					
6	Payoff at T				
7	Long stock	50	<-- =B2		
8	Short call, -Max(S_T-X,0)	-5	<-- =-MAX(B2-B3,0)		
9	Total covered call payoff	45	<-- =B8+B7		
10					
11	Terminal stock price	Stock payoff	Call payoff	Covered call payoff	
12					<-- Data table header
13	0	0	0	0	
14	5	5	0	5	
15	10	10	0	10	
16	15	15	0	15	
17	20	20	0	20	
18	25	25	0	25	
19	30	30	0	30	
20	35	35	0	35	
21	40	40	0	40	
22	45	45	0	45	
23	50	50	-5	45	
24	55	55	-10	45	
25	60	60	-15	45	
26	65	65	-20	45	
27	70	70	-25	45	

In this problem, you are asked to simulate the payoffs of a covered call over 52 weeks, with weekly updating of the positions. Start by deriving the formula for the covered call. Add together the Black-Scholes price and the stock price:

$$\underbrace{S_0}_{\substack{\uparrow \\ \text{Long stock}}} \quad \underbrace{-S_0N(d_1)+Xe^{-rT}N(d_2)}_{\substack{\uparrow \\ \text{Short call}}} = \underbrace{S_0(1-N(d_1))}_{\substack{\uparrow \\ \text{Long position} \\ \text{in stock}}} + \underbrace{Xe^{-rT}N(d_2)}_{\substack{\uparrow \\ \text{Long position} \\ \text{in bond}}}$$

Thus, we see that a covered call is a long position in the stock and a long position in the bond. Now implement the following spreadsheet to test the effectiveness of a simulated covered call strategy.



	A	B	C	D	E	F	G	H	I	J	K
1					COVERED CALL SIMULATION						
2	Initial stock price, S_0	50									
3	X	45									
4	Mean stock return	12%									
5	Sigma	40%									
6	Interest rate, r	6%									
7											
8	Initial call price	11.7393	<-- =bscall(B2,B3,1,B6,B5)								
9	Initial investment in covered call	38.2607	<-- =B2-B8								
10											
11	Simulated payoff of strategy	????									
12	Payoff of covered call strategy	????									
13											
14	Week	Time	Stock price	d_1	Invested in stock	Invested in bond	Total investment at beginning of week		Random Z at end of week	Stock price at end of week	Investment at the end of week
15	0	0	50.00	0.6134	13.4903	24.7705	38.2607		1.2901	53.5719	39.2531
16	1	1/52	53.57	0.7868	11.5558	27.6972	39.2531				
17	2	2/52									
64	49	49/52									
65	50	50/52									
66	51	51/52									

6. Section 26.2 discusses the cashless replication of a call. Use the same logic to program in a spreadsheet the replication of a put.

---

1. 1. On the companion website for this book you can find the 50 trials performed in R.
2. 2. On the companion website for this book you can find the model for a put and the 50 trials in R.

3. 3. As we show below, it is possible to insure (up to a point) even with $z > 1$.

27 Using Monte Carlo Methods for Option Pricing

27.1 Overview

This chapter continues the discussion of the previous chapter and shows how to implement Monte Carlo methods for pricing options. The main objective of the chapter is to show how to price the most common exotic options like Asian options and barrier options. Many of the exotic options are *path-dependent*: Their payoffs depend not just on the terminal price of an underlying asset but also on the intermediate asset prices before the terminal price. Whenever possible, we'll present the closed-form formula for pricing; in other cases we show how to implement the Monte Carlo pricing approach.

What Does Risk-Neutrality Mean?

We continue, as in Chapter 10, to think about a situation in which there are two basic securities: a riskless bond and a stock. The interest rate on the bond is called the risk-free rate and is usually denoted by *r*. In the context of this situation of two basic assets, "risk-neutral pricing" can have two meanings. In both meanings of risk-neutrality derivative (i.e., nonbasic) a security price is the discounted expected payoffs (at the risk-free rate). The two meanings of risk-neutrality differ in the procedure used to arrive at the payoffs of the risky basic security (the stock).

- • In the first meaning of risk-neutrality, we change the underlying distribution of the stock's returns so that the expected stock price is one-plus the risk-free rate. To price an option, we apply the basic risk-neutral pricing principle of discounting the expected option payoffs by the risk-free rate. In this use of risk-neutrality, we transform the stock's returns but compute the expected payoffs using the actual return probabilities. We illustrate this procedure in section 27.2.
- • The second meaning of risk-neutrality uses the binomial model. We transform the state prices into risk-neutral probabilities. In this use of risk-neutrality, we do not transform the stock's payoffs but instead replace the actual state probabilities by the equivalent risk-neutral probabilities. We then price derivative securities by discounting their expected payoffs at the risk-free rate.

The first method of pricing is ideal for pricing options whose payoffs depend only on the terminal price of the stock. We call such options *path-independent*. The second method is more general and can in principle be used to price any option, even if the payoffs are path-dependent. Examples of such options explored in this chapter are Asian and barrier options. Both are path-dependent options—options whose price depends not only on the terminal price of the asset, but also on the path of the prices by which the terminal price was reached. In general, a path-dependent option does not have an analytic price solution. Monte Carlo provides us with a handy numerical tool for pricing such options. Monte Carlo pricing of path-dependent options depends on a simulation of the price path of the underlying asset.

The Structure of This Chapter

In section 27.2 we show how the first risk-neutrality pricing principle (transforming the stock's returns) can be used to Monte Carlo price a standard call and a put. We then continue with the second meaning of risk-neutrality and show how it can be used to price exotic options.

In order to make this chapter more self-contained, we include a brief review of state prices and risk-neutrality (section 27.3). We then show, in section 27.4, how to price a plain-vanilla option with a Monte Carlo algorithm. Since plain-vanilla options—jargon by which we simply mean European calls and puts on a stock whose price process is lognormal—are accurately priced using the Black-Scholes formula, this exercise allows us to check our pricing method against a known result and also allows us to develop the proper intuitions about the Monte Carlo pricing of more complicated options. In later sections we show how to implement the methods to price the most common exotic options, including Asian options (section 27.5), barrier options (section 27.6), basket options (section 27.7), rainbow options (section 27.8), binary options (section 27.9), chooser options (section 27.10), and lookback options (section 27.11).

27.2 Pricing Plain-Vanilla Options Using Monte Carlo Methods

In this section we explore the pricing of standard European calls and puts using Monte Carlo. We employ the first meaning of risk-neutrality discussed previously—that is, transforming the returns on the stock so that the stock's expected return is the risk-free rate and then discounting the expected option payoffs at the risk-free rate.

To use this complicated model to price European calls and puts with Monte Carlo methods might be viewed as an immense waste of time—the Black-Scholes formula (Chapter 18) gives a wonderful pricing solution for European calls and puts. However, like the example of estimating the value of π discussed in Chapter 22, the exercise of pricing a plain-vanilla call using Monte Carlo methods gives us considerable insight into the application of Monte Carlo methods.

The Procedure

Consider a call on a stock with the parameters given below. This call can be priced using the Black-Scholes formula. We will show how our first risk-neutral procedure of transforming the returns can be used to price this call using Monte Carlo.

	A	B	C
1	BLACK-SCHOLES CALL PRICE		
2	Stock price today, S_0	50	
3	Exercise price, X	45	
4	Time to maturity, T	1	
5	Risk-free interest rate, r	3%	
6	Annual volatility, Sigma	25%	
7			
8	Black-Scholes call price	8.4859	<--=BSCall(B2,B3,B4,B5,B6)

We now simulate the option price as follows:

- • Step 1: We simulate a set of stock prices at time T. The stock prices have the property that they are lognormally distributed with mean r and standard deviation σ.
- • Step 2: We compute the terminal payoffs of our option for these stock prices: $Max(S_T - X, 0)$.
- • Step 3: We compute the average discounted payoff:

$exp[-r * T] * Average[Max(S_T - X, 0)]$.

This should be the same as the Black-Scholes price (and approximately it is).

Step 1: Generating the Price Data

Having cleared up this theoretical issue, we now return to our computations. We first generate 1,000 future stock prices. Each price is generated using the formula:

$$S_T = S_0 \cdot exp[(r - 0.5 \cdot \sigma^2) \cdot T + \sigma \cdot \sqrt{T} \cdot Z]$$

Here, Z is the standard normal deviate produced (i.e., $Z \sim N(0, 1)$), as illustrated in Chapter 22, by the Excel function **Norm.S.Inv(Rand()).** Here's some sample data:

	A	B	C	D	E	F	G
1	**SIMULATING RISK-NEUTRAL STOCK PRICES**						
2	Stock price today, S_0		50				
3	Exercise price, X		45				
4	Time to maturity, T		1				
5	Risk-free interest rate, r		3%				
6	Annual volatility, Sigma		25%				
7							
8	Black-Scholes call price		8.4859	<--=BSCall(C2,C3,C4,C5,C6)			
9							
10	Simulated time T stock prices						
11	Trial	S(T)					
12	1	50.9034	<--=$C$2*EXP(($C$5-0.5*$C$6^2)*$C$4+$C$6*SQRT($C$4)*NORM.S.INV(RAND()))				
13	2	45.3842					
14	3	49.6520		Stock prices			
15	4	62.1596		Count	1,000	<--=COUNT(B:B)	
16	5	45.7052		Max	135.8605	<--=MAX(B:B)	
17	6	60.2516		Min	22.4292	<--=MIN(B:B)	
18	7	55.9429		Average	52.4072	<--=AVERAGE(B:B)	
19	8	51.4079		Sigma	13.9802	<--=STDEV.S(B:B)	
20	9	55.7568					
21	10	92.6016		Data statistics			
22	11	53.5209		Average return	4.70%	<--=LN(E18/C2)	
23	12	44.8678		Sigma of return	26.68%	<--=E19/E18	
24	13	36.1275					
25	14	63.2798		Theoretical statistics			
26	15	40.3161		Return	3.05%	<--=EXP(C5*C4)-1	
27	16	66.8012		Sigma of return	25.00%	<--=C6	
1010	999	53.8970					
1011	1000	43.8951					

Steps 2 and 3: Terminal Payoffs and Option Prices

We now use the sample data to compute the payoffs of our option. Below we compute the call payoff for each simulated stock price (column E), and in cell B10 we computed the discounted average (at the risk-free rate) of these payoffs $exp[-r * T] * Average[Max(S_T - X, 0)]$. As predicted, this discounted average closely matches the Black-Scholes price.

	A	B	C	D	E	F
1	**PRICING A CALL WITH RISK-NEUTRAL STOCK PRICES**					
2	Stock price today, S_0	50				
3	Exercise price, X	45		Call option valuation		
4	Time to maturity, T	1		Average payoff	9.3497	<--=AVERAGE(C:C)
5	Risk-free interest rate, r	3%		Discounted avg(payoff)	9.0734	<--=E4*EXP(-B5*B4)
6	Annual volatility, Sigma	25%				
7						
8	Black-Scholes call price	8.4859	<--=BSCall(B2,B3,B4,B5,B6)			
9						
10	Simulated time T stock prices					
11	Trial	S(T)	Call payoff at T			
12	1	54.1827	9.1827	<--=MAX(0,B12-$B$3)		
13	2	54.6826	9.6826			
14	3	77.1680	32.1680			
1010	999	55.8650	10.8650			
1011	1000	40.8390	0.0000			

The same procedure works for puts:

	A	B	C	D	E	F
1	**PRICING A PUT WITH RISK-NEUTRAL STOCK PRICES**					
2	Stock price today, S_0	50				
3	Exercise price, X	45		**Put option valuation**		
4	Time to maturity, T	1		Average payoff	2.0717	<--=AVERAGE(C:C)
5	Risk-free interest rate, r	3%		Discounted avg(payoff)	2.0104	<--=E4*EXP(-B5*B4)
6	Annual volatility, Sigma	25%				
7						
8	Black-Scholes put price	2.1560	<--=bsput(B2,B3,B4,B5,B6)			
9						
10	**Simulated time T stock prices**					
11	**Trial**	**S(T)**	**Put payoff at T**			
12	1	45.7947	0.0000	<--=MAX(0,$B$3-B12)		
13	2	65.3002	0.0000			
14	3	71.2007	0.0000			
1010	999	54.3496	0.0000			
1011	1000	40.0642	4.9358			

Using Monte Carlo to Price Plain-Vanilla Options with R

Applying the same approach in R should look like this:
Initially, we define the Black-Scholes option pricing formulas:

```
# Inputs
S0 <- 50
X <- 45
t <- 1
r <- 0.03
sigma_s <- 0.25

# B&S call option (from Chapter 15)
fmS_bs_call <- function(S, X, t, r, sigma_s, k = 0){
    d_1 = ( log(S/X) + (r - k + 0.5 * sigma_s ^ 2) * t ) /  ( sigma_s * sqrt(t) )
    d_2 = d_1 - sigma_s * sqrt(t)
    return(S * pnorm(d_1, mean = 0, sd = 1) * exp(-k * t) -
                 X * exp(-r*t) * pnorm(d_2, mean = 0, sd = 1))
                                              }

# Function: B&S Put option
fmS_bs_put <- function(S, X, t, r, sigma_s, k = 0){
    d_1 = ( log(S/X) + (r - k + 0.5 * sigma_s ^ 2) * t ) /  ( sigma_s * sqrt(t) )
    d_2 = d_1 - sigma_s * sqrt(t)
    return(X * exp(-r*t) * pnorm(-d_2, mean = 0, sd = 1) -
                 S * pnorm(-d_1, mean = 0, sd = 1) * exp(-k * t))
                                              }
```

We then employ the three-step procedure as described above. The first step is to generate price data for our stock (note that since R is very efficient, we used 100,000 trials and not 1,000 as in the implementation in Excel):

```
## Step 1:Generating the price data (Stock Prices at Time t)
# Number of runs (100K)
n <- 100000

# n Stock Prices
stock_prices <- sapply(rnorm(n), function(z)
    S0 * exp((r - 0.5 * sigma_s ^ 2) * t + sqrt(t) * sigma_s * z) )

# Stock price statistics
summary(stock_prices) # Min, Max, Median, 1st and 3rd quartiles
sd(stock_prices) # standard deviation
length(stock_prices) # count
log(mean(stock_prices) / S0) # mean return
sd(stock_prices) / mean(stock_prices) # return's sigma

# Theoretical statistics
exp(r*t) - 1 # mean return
sigma_s # return's sigma
```

The results are as follows:

```
> # stock price statistics
> summary(stock_prices) # Min, Max, Median, 1st and 3rd quartiles
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
  16.08   42.21   49.93   51.53   59.10  147.55
> sd(stock_prices) # standard deviation
[1] 13.06021
> length(stock_prices) # count
[1] 100000
> log(mean(stock_prices) / S0) # mean return
[1] 0.03017472
> sd(stock_prices) / mean(stock_prices) # return's sigma
[1] 0.2534402
>
> # Theoretical statistics
> exp(r*t) - 1 # mean return
[1] 0.03045453
> sigma_S # return's sigma
[1] 0.25
>
```

The second step (calculate the terminal payoff) and the third step (calculate the discounted average payoff) look like this:

```
## Steps 2 and 3
# Call payoff at time t
payoff <- pmax(stock_prices - X, 0)
avg_payoff <- mean(payoff)
dscnt_avg_payoff <- avg_payoff * exp(-r*t)

# Compare MC result to B&S call price
dscnt_avg_payoff
fm5_bs_call(S = S0, X = X, t = t, r = r, sigma_S = sigma_S)

# Put payoff at time t
payoff <- pmax(X - stock_prices, 0)
avg_payoff <- mean(payoff)
dscnt_avg_payoff <- avg_payoff * exp(-r*t)

# Compare MC result to B&S put price
dscnt_avg_payoff
fm5_bs_put(S = S0, X = X, t = t, r = r, sigma_S = sigma_S)
```

And the results are:

Call option	`> # Compare MC result to B&S call price`
	`> dscnt_avg_payoff`
	`[1] 8.484912`
	`> fm5_bs_call(S = S0, X = X, t = t, r = r, sigma_S = sigma_S)`
	`[1] 8.485938`
	`>`

Put option	`> # Compare MC result to B&S put price`
	`> dscnt_avg_payoff`
	`[1] 2.168259`
	`> fm5_bs_put(S = S0, X = X, t = t, r = r, sigma_S = sigma_S)`
	`[1] 2.155987`
	`>`

Example: Second Digit Option

To show the power of this method, suppose we wish to price a security that pays off the second digit of the terminal stock price. As an example: If the $S_T = 43.5323$, our "digital security" payoff is 3; if $S_T = 50.5323$, the payoff is 0. (Why anyone would want to buy such a security is a separate question.)

The spreadsheet below shows that this security can easily be priced using the Monte Carlo method explained in this section. We simulate 1,000 stock prices and use the

Excel function **Int(Mod(S_T,10))** to determine the second digit of the terminal stock price:

	A	B	C	D	E	F
1	USING MONTE CARLO FOR PRICING A SECOND DIGIT OPTION					
2	Stock price today, S_0	50				
3	Exercise price, X	45		Put option valuation		
4	Time to maturity, T	1		Average payoff	4.4780	<-- =AVERAGE(C:C)
5	Risk-free interest rate, r	3%		Discounted avg(payoff)	4.3457	<-- =E4*EXP(-B5*B4)
6	Annual volatility, Sigma	25%				
7						
8	Simulated time T stock prices					
9	Trial	S(T)	2nd digit payoff at T			
10	1	73.4811	3.0	<-- =INT(MOD(B10,10))		
11	2	60.4104	0.0			
12	3	104.3450	4.0			
1008	999	62.7548	2.0			
1009	1000	66.8287	6.0			

27.3 State Prices, Probabilities, and Risk-Neutrality

In the remainder of this chapter we use the second meaning of risk-neutrality discussed at the beginning of this chapter: In the framework of a binomial model, we compute risk-neutral state probabilities that are then used to compute the expected payoffs of derivative securities. Our main focus will be on the pricing of barrier and Asian options, but the principles are general and can be applied to most derivatives.

In this section we start with a brief recapitulation of the basic facts about state prices and risk-neutral pricing discussed in Chapter 17.3. Suppose we have a binomial framework with stock price S that grows at U or D in every period. Suppose the interest rate is R.[1] Risk-neutrality is a property of every set of state prices: Given state prices $\left\{q_U = \frac{R-D}{R(U-D)}, q_D = \frac{U-R}{R(U-D)}\right\}$, the risk-neutral probabilities are defined by $\{\pi_U = R \cdot q_U, \pi_D = R \cdot q_D\}$. The risk-neutral probabilities can be used to price assets by taking the *discounted expected value* of the asset payoffs.[2] An asset with payoffs one period hence of $\{Payoff_U, Payoff_D\}$ in states U and D, respectively, has a value today of:

$$Asset\ value\ today = q_U Payoff_U + q_D Payoff_D = \underbrace{\frac{\pi_U Payoff_U + \pi_D Payoff_D}{R}}_{\substack{\uparrow \\ \text{The risk-neutral expected payoff} \\ \text{(expectation computed} \\ \text{with the risk-neutral probabilities)} \\ \text{discounted at the risk-free interest rate}}}.$$

The $\{Payoff_U, Payoff_D\}$ are usually functions of the underlying asset price. For an ordinary call, for example, $\{Payoff_U = Max(S * U - X, 0), Payoff_D = Max(S * D - X, 0)\}$.

We can extend the risk-neutral pricing scheme to multi-period frameworks. Consider a multi-period binomial setting where U and D do not change over time and indicate the date-n state payoffs by $Payoff_{n,j}$, $j = 0, \ldots, n$. This notation $Payoff_{n,j}$ indicates the date-n payoff of the asset in a state where there are j Up moves on the binomial tree (and hence

$n - j$ Down moves); for a call in a binomial framework, $Payoff_{n,j} = Max(S * U^j D^{n-j} - X, 0)$. Then the value of this asset is given as follows:

$$\text{Asset value today} = \sum_{j=0}^{n} \binom{n}{j} q_U^j q_D^{n-j} Payoff_{n,j} = \underbrace{\frac{1}{R^n} \sum_{j=0}^{n} \binom{n}{j} \pi_U^j \pi_D^{n-j} Payoff_{n,j}}_{\text{The risk-neutral expected discounted value}}$$

This particular notation assumes that the tree is *recombining*. It assumes, in other words, that the date-n payoffs are *path-independent*—the option payoff is a function only of the terminal stock price and does not depend on the path by which this price is reached. We return to this topic later in this chapter. Here is the Excel for a four-period model:

	A	B	C	D	E	F	G	H
1	BINOMIAL OPTION PRICING WITH STATE PRICES IN A FOUR-PERIOD (FIVE-DATE) MODEL							
2	Up, U	1.10						
3	Down, D	0.97		State prices				
4	Interest rate, R	1.06		q_U	0.6531	<--=(B4-B3)/(B4*(B2-B3))		
5	Initial stock price, S	50.00		q_D	0.2903	<--=(B2-B4)/(B4*(B2-B3))		
6	Option exercise price, X	50.00						
7					=$E$4^A12*$E$5^B12			
8	=$B$5*$B$2^A12*$B$3^B12							
9				=MAX(C12-$B$6,0)			=COMBIN(4,B12)	
10	Number of "up" steps at terminal date	Number of "down" steps at terminal date	Terminal stock price	Option payoff at terminal state	State price for terminal date	Number of paths to terminal state	Value	
11	4	0	73.2050	23.2050	0.1820	1	4.2224	
12	3	1	64.5535	14.5535	0.0808	4	4.7078	<--=D12*E12*F12
13	2	2	56.9245	6.9245	0.0358	6	1.4833	
14	1	3	50.1970	0.1970	0.0160	4	0.0126	
15	0	4	44.2646	0.0000	0.0071	1	0.0000	
16						Call price	10.4360	<--=SUM(G11:G15)
17						Put price	0.0407	<--=G16-B6/B4^4+B5
18	Notes							
19	There are 5 dates in this model (0, 1, ..., 5) but only 4 periods and thus only 4 possible "up" or "down" steps.							
20	The put price in cell G17 is computed using put-call parity: put = call + PV(X) - stock							

In figure 27.1, on the left we present a recombining binomial model, and on the right we present non-recombining binomial models.

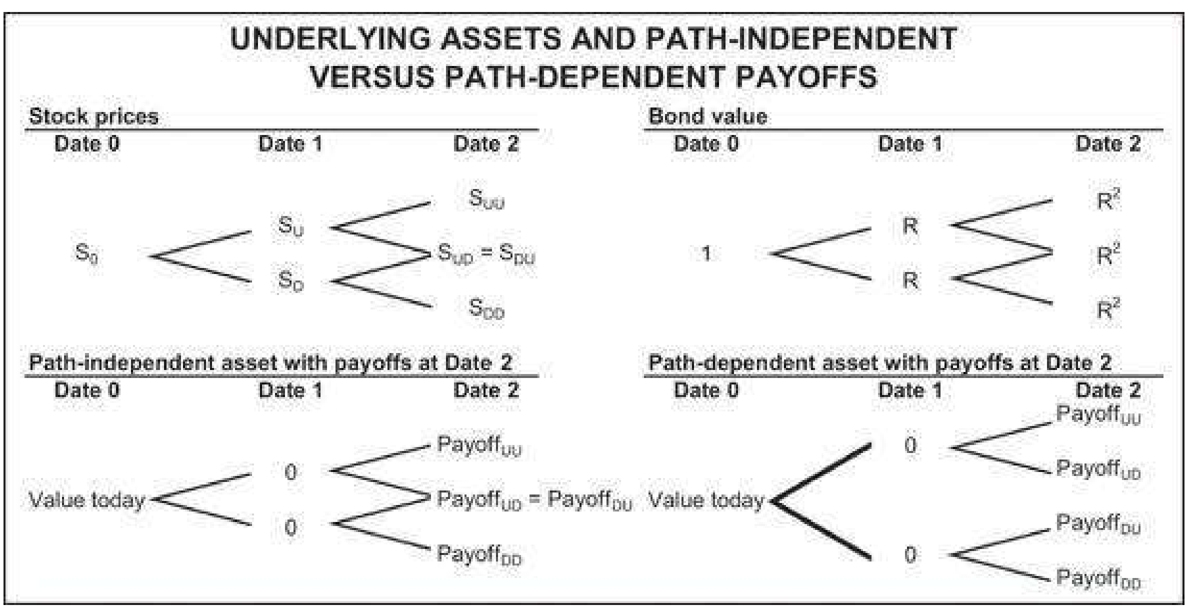


Figure 27.1
Recombining versus non-recombining binomial models.

Using R to Implement the Binomial Model

Applying the same approach in R would look like this:

```
## Binomial Option Pricing with State Prices in a Four-Period Model
# Inputs:
U <- 1.1 # Up
D <- 0.97 # Down
r <- 1.06 # 1 + Interest Rate
S0 <- 50 # Initial Stock Price
X <- 50 # Option Exercise Price
periods <- 4

# State Prices
qU <- (r - D) / (r * (U - D))
qD <- (U - r) / (r * (U - D))

ups <- c(0:periods) # number of "up" steps at terminal date
downs <-  periods-ups

# Terminal stock price
St <- S0 * U^ups * D^downs

# Option payoff at T
payoff <- pmax(St - X, 0)

# State price for terminal date
state_price <- qU^ups * qD^downs

# Number of paths to terminal state
n_paths <- choose(periods, downs)

# Call value
call_price <- sum( payoff * state_price * n_paths)

# Put value
put_price <- call_price + X / r^periods - S0
```

27.4 Pricing Plain-Vanilla Options—Monte Carlo Binomial Model Approach

We start with a ridiculously simple example. We use Monte Carlo to price a European call in a two-period setting in which the stock price goes either up or down each period. In the spreadsheet below we use Monte Carlo methods to price an at-the-money option on a stock whose price today is S_0 = 50. There are two periods, and in each period the stock price goes either *Up* = 1.4 or *Down* = 0.9; the interest rate is *R* = 1.05. Given Up and Down, the stock price tree is shown in figure 27.2.

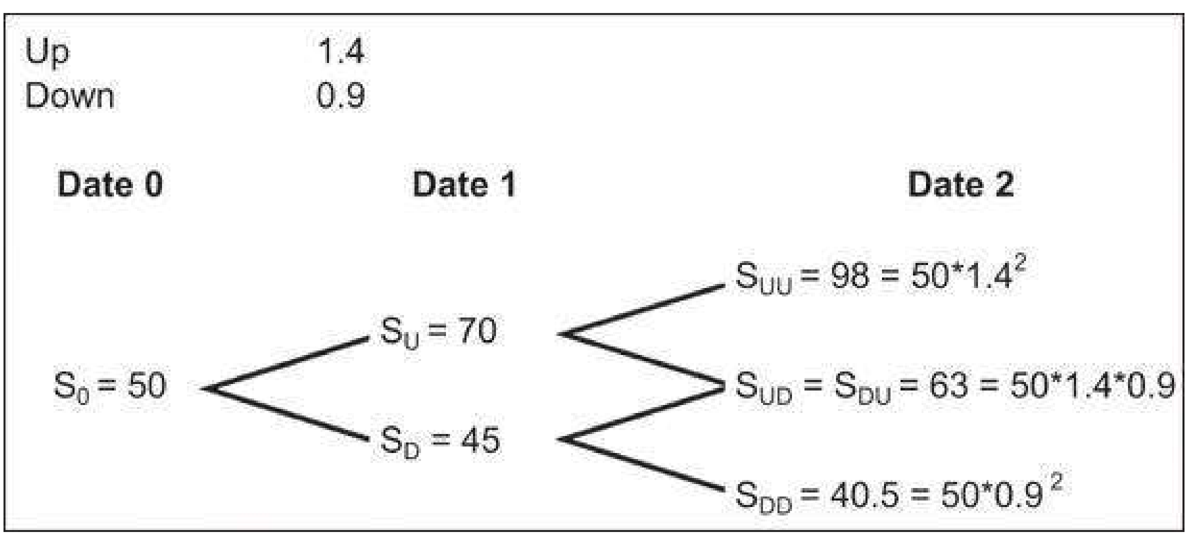


Figure 27.2
Stock price tree in a standard recombining binomial model.

The spreadsheet below shows two random price paths and their pricing:

	A	B	C	D
1	**SIMPLE SIMULATION: TWO PATHS IN A TWO-DATE MODEL**			
2	Initial stock price, S_0	50		
3	Exercise (strike) price, X	50		
4				
5	Up, U	1.4		
6	Down, D	0.9		
7	Risk-free rate, R	1.05		
8				
9	**State prices**			
10	q_U	0.2857	<--=(B7-B6)/(B7*(B5-B6))	
11	q_D	0.6667	<--=(B5-B7)/(B7*(B5-B6))	
12				
13	**Risk-neutral probabilities**			
14	π_U	0.3000	<--=B10*$B$7	
15	π_D	0.7000	<--=B11*$B$7	
16				
17	**Random paths and the Monte Carlo price**			
18		Path 1	Path 2	
19	First period, up (1) or down (0)?	0	0	<--=IF(RAND()>$B$15,1,0)
20	Second period, up (1) or down (0)?	1	0	<--=IF(RAND()>$B$15,1,0)
21				
22	Total ups	1	0	<--=SUM(C19:C20)
23	Terminal stock price	63	40.5	<--=$B$2*Up^C22*Down^(2-C22)
24	Option payoff	13	0	<--=MAX(C23-$B$3,0)
25				
26	**Average discounted payoff**	5.8957	<--=AVERAGE(24:24)/R_^2	
27				
28	**Computing the actual option price with state prices**			
29	**Payoffs**			
30	top	48	<--=MAX(B2*Up^2-B3,0)	
31	middle	13	<--=MAX(B2*Up*Down-B3,0)	
32	bottom	0	<--=MAX(B2*Down^2-B3,0)	
33	Actual option price	8.8707	<--=q_up^2*B30+2*q_up*q_down*B31+q_down^2*B32	

The state prices and risk-neutral probabilities are computed in cells B10:B11 and B14:B15. Cells B19:B20 and C19:C20 show two random price paths. In each period we use a random number from Excel generated by the function **Rand**. If ***Rand***() $\leq \pi_D$ then the stock price goes up, and if ***Rand***() $> \pi_D$, then the stock price goes down.

- • In the first price path (cells B19:B20), the stock price goes down in the first period and up in the second period. The terminal stock price is 63 and the option's payoff is 13 (cell B24).
- • In the second price path (cells C19:C20), the stock price goes down in both periods. The terminal stock price is 40.5 and the option's payoff is 0 (cell C24).

If these were the only two random price paths, the Monte Carlo option price would be the discounted average 5.8957 (cell B26). Notice that we have also computed the *actual call price* using the state prices; cell B33 shows this price to be 8.8707.

Risk-Neutral Probabilities in Monte Carlo

Notice the role of the risk-neutral probabilities in the Monte Carlo simulation: The price path is determined *not by the actual probabilities* but by the risk-neutral probabilities π_U and π_D. There is no role for the actual probabilities in the Monte Carlo risk-neutral pricing.

Implementing the above example in R looks like this:

```r
# Simple example (2 period example)
U <- 1.4 # Up
D <- 0.9 # Down
r <- 1.05 # 1 + Interest Rate
S0 <- 50 # initial Stock Price
X <- 50 # option exercise price

# state prices
qU <- (r - D) / (r * (U - D))
qD <- (U - r) / (r * (U - D))

# Risk neutral probabilities
pi_U <- qU * r
pi_D <- qD * r

# Random path (2 runs example)
periods <- 2 # number of periods
paths <- 2 # number of paths

# Total Ups
total_ups <- rbinom(n = paths, size = periods, prob = pi_U)

# Terminal Stock Price
St <- S0 * U^total_ups * D^(periods - total_ups)

call_payoff <- pmax(St - X, 0)
```

And:

```r
# Computing the actual option price with state prices
ups <- c(0:periods) # Number of "up" steps at terminal date
downs <- periods-ups # Number of "down" steps at terminal date
St <- S0 * U^ups * D^downs # Terminal stock price
payoff <- pmax(St - X, 0) # Option payoff at X
state_price <- qU^ups * qD^downs # State price for terminal date
n_paths <- choose(periods, downs) # Number of paths to terminal state
call_actual_price <- sum( payoff * state_price * n_paths) # Value
```

Extending the Two-Period Model

You can’t, of course, run a Monte Carlo simulation by using only two price paths. In the spreadsheet below we have extended our price paths to all the columns in the spreadsheet. Excel’s spreadsheet is 16,384 columns wide; this means that the computation in cell B25 is the average of the option value over less than 16,384 simulated price paths; in our example we used 1,000 (note the hidden columns).

	A	B	C	D	ALL	ALM	ALN
1	**SIMPLE SIMULATION: 1,000 PATHS IN A TWO-DATE MODEL**						
2	Initial stock price, S_0	50					
3	Exercise (strike) price, X	50					
4							
5	Up, U	1.4					
6	Down, D	0.9					
7	Risk-free rate, R	1.05					
8							
9	**State prices**						
10	q_U	0.2857	<--=(B7-B6)/(B7*(B5-B6))				
11	q_D	0.6667	<--=(B5-B7)/(B7*(B5-B6))				
12							
13	**Risk-neutral probabilities**						
14	π_U	0.3000	<--=B10*$B$7				
15	π_D	0.7000	<--=B11*$B$7				
16							
17	**Random paths and the Monte Carlo price**						
18		**Path 1**	**Path 2**	**Path 3**	**Path 999**	**Path 1000**	
19	First period, up (1) or down (0)?	0	0	1	0	0	<--=IF(RAND()>$B$15,1,0)
20	Second period, up (1) or down (0)?	1	0	1	1	1	<--=IF(RAND()>$B$15,1,0)
21							
22	Total ups	1	0	2	1	1	<--=SUM(ALM19:ALM20)
23	Terminal stock price	63	40.5	98	63	63	<--=$B$2*Up^ALM22*Down^(2-ALM22)
24	Option payoff	13	0	48	13	13	<--=MAX(ALM23-$B$3,0)
25							
26	**Average discounted payoff**	8.8163	<--=AVERAGE(24:24)/R_^2				
27							
28	**Computing the actual option price with state prices**						
29	**Payoffs**						
30	top	48	<--=MAX(B2*Up^2-B3,0)				
31	middle	13	<--=MAX(B2*Up*Down-B3,0)				
32	bottom	0	<--=MAX(B2*Down^2-B3,0)				
33	Actual option price	8.8707	<--=q_up^2*B30+2*q_up*q_down*B31+q_down^2*B32				

The average discounted payoff (cell B26) is 8.8163. This value is random, meaning it will change each time we press F9 to produce a new set of random paths. Monte Carlo methods imply that for even more paths we would converge to the actual option price of 8.8707. In the next section we show that the Monte Carlo method eventually produces convergence to this price.

Implementing the above approach in R is relatively easy and efficient. In the implementation below we used a 10,000-paths simulation to model the previous example:

```
# Simple simulation: 10,000 paths in a two-date model
S0 <- 50 # initial stock Price
X <- 50 # option exercise price
U <- 1.4 # Up
D <- 0.9 # Down
r <- 1.05 # 1 + interest Rate
paths <- 10000 # number of paths
periods <- 2 # number of periods

# Total Ups
total_ups <- rbinom(n = paths, size = periods, prob = pi_U)

# Terminal Stock Price
St <- S0 * U^total_ups * D^(periods - total_ups)

# Option payoff
call_payoff <- pmax(St - X, 0)

# Average discounted payoff
call_price <- mean(call_payoff / r^periods)
```

In our simulation the estimated call price is 8.852971, where the actual call price should be 8.870748, which is a 0.2% difference.

Monte Carlo Plain-Vanilla Call Pricing Converges to Black-Scholes

Now that we understand the principles, we extend our logic. We write a VBA routine that prices a plain-vanilla call using Monte Carlo methods under conditions that converge to Black-Scholes pricing.

Our basic setup is as follows: We price a European call on a stock whose current price is S_0. The option's exercise price is X, and the time to maturity of the option is T. We assume that the stock price is lognormally distributed with mean μ and standard deviation σ.

To price the call using Monte Carlo:

- • We divide the unit time interval into n divisions. This means that $\Delta t = 1/n$.
- • For each Δt, we define $Up_{\Delta t} = exp[(r - 0.5 * \sigma^2)\Delta t + \sigma\sqrt{\Delta t}]$ and $Down_{\Delta t} = exp[(r - 0.5 * \sigma^2)\Delta t - \sigma\sqrt{\Delta t}]$. The interest rate on the interval Δt is $R_{\Delta t} = exp[r\Delta t]$.
- • This means that the state prices and risk-neutral probabilities are given as follows:

$$q_u = \frac{R_{\Delta t} - Down_{\Delta t}}{R_{\Delta t}(Up_{\Delta t} - Down_{\Delta t})}, \quad q_d = \frac{Up_{\Delta t} - R_{\Delta t}}{R_{\Delta t}(Up_{\Delta t} - Down_{\Delta t})}$$

$$\pi_u = \frac{R_{\Delta t} - Down_{\Delta t}}{Up_{\Delta t} - Down_{\Delta t}}, \qquad \pi_d = \frac{Up_{\Delta t} - R_{\Delta t}}{Up_{\Delta t} - Down_{\Delta t}} = 1 - \pi_u$$

- • Since the time to maturity of the option is T, the price path to T requires $m = \frac{T}{\Delta t}$ periods. A price path of length m is created by determining the up or down move of the stock as a function of a random number between 0 and 1 and the risk-neutral probability π_d. As discussed in the example of section 27.3, if the random number is greater than π_d, the stock makes an Up move; otherwise it makes a Down move.

A VBA Routine

The VBA routine below defines a function **VanillaCall**. This function requires as inputs the variables mentioned above. The variable **Runs** is the number of random price paths created; these paths are averaged to determine the Monte Carlo value of the call:

```
Function VanillaCall(S0, Exercise, sigma, _
Interest, Time, Divisions, Runs)
    deltat = 1 / Divisions
    interestdelta = Exp(Interest * deltat)
    Up = Exp((Interest - 0.5 * (sigma) ^ 2) * deltat + _
    sigma * Sqr(deltat))
```

```
    Down = Exp((Interest - 0.5 * (sigma) ∧ 2) * deltat - _
    sigma * Sqr(deltat))
    pathlength = Int(Time / deltat)
'Risk-neutral probabilities
piup = (interestdelta - Down) / (Up - Down)
pidown = 1 - piup
Temp = 0
For Index = 1 To Runs
    Upcounter = 0
    'Generate terminal price
    For j = 1 To pathlength
        If Rnd > pidown Then Upcounter = _
        Upcounter + 1
    Next j
    callvalue = Application.Max(S0 * _
    (Up ∧ Upcounter) * (Down ∧ (pathlength - _
    Upcounter)) - Exercise, 0) _
    / (interestdelta ∧ pathlength)
    Temp = Temp + callvalue
Next Index
VanillaCall = Temp / Runs
End Function
```

The number of Up moves is stored in a counter called **Upcounter** and the value of the call for each **Run** is the discounted value of the call payoff for a particular terminal price $S_0 * Up^{Upcounter}\ Down^{pathlength-Upcounter}$, where `pathlength = Int(Time / deltat)` is the integer part of $T/\Delta t$:

```
callvalue = Application.Max(S0 * (up ∧ Upcounter) * _
(down ∧ (pathlength - Upcounter)) - Exercise, 0) / (interestdelta
∧ pathlength)
```

The Monte Carlo value of the call is given by: VanillaCall = Temp/Runs.

Understanding the Principles of the Monte Carlo Simulation

For future reference, we state the principles of the Monte Carlo simulation. These principles hold not only for the plain-vanilla options of this section, but also for the Asian options treated later in this chapter.

- • Price paths are generated by using the risk-neutral probabilities. In the program **VanillaCall**, for example, the price of the stock moves up if the random number generator $>\pi_D$ and moves down if the random number generator $\leq \pi_D$. Effectively this means that the risk-neutral probabilities $\{\pi_U = 1 - \pi_D, \pi_D\}$ of each price path are incorporated into the price path itself.
- • The value of the option using Monte Carlo is determined by the discounted value of the simple average of all results over the price paths generated.

Implementing the Monte Carlo Function VanillaCall in a Spreadsheet

The next spreadsheet shows the implementation of **VanillaCall**. The value in cell B13 is the option value as computed by the Black-Scholes formula; the function **BSCall** was defined in Chapter 18.

	A	B	C
1	MONTE CARLO PRICING OF PLAIN-VANILLA CALLS		
2	S_0 , current stock price	50	
3	X, exercise price	50	
4	r, interest rate	5%	
5	T, time	0.8	
6	σ, sigma	30%	<--standard deviation of stock return
7			
8	n, divisions of unit time	200	
9	Runs	3,000	
10			
11	Vanilla call price using Monte Carlo	6.5106	<--=vanillacall(B2,B3,B6,B4,B5,B8,B9)
12			
13	Vanilla call price using Black-Scholes	6.2697	<--=BSCall(B2,B3,B5,B4,B6)

The function divides the time to option expiration $T = 0.8$ years (cell B5) into 200 divisions (cell B9), so that $\Delta\ t = 1/200$. Each time the function is called, it runs 3,000 price paths (cell B9). A particular call of the function shown above produced the value 6.2697 (cell B13), whereas the Black-Scholes call value—computed with the function **BSCall** defined in Chapter 18—is 6.5106 (cell B11).

How good is this Monte Carlo routine? One way to test it is to run it many times. In the spreadsheet below, we've run 40 instances of the function **VanillaCall**.

	A	B	C	D	E
1	**RUNNING THE MONTE CARLO FUNCTION MANY TIMES**				
2	S_0 , current stock price	50			
3	X, exercise price	50			
4	r, interest rate	5%			
5	T, time	0.8			
6	σ, sigma	30%	<--standard deviation of stock return		
7					
8	n, divisions of unit time	100			
9	Runs	3,000			
10					
11	Vanilla call price using Monte Carlo	6.1686	<--=vanillacall(B2,B3,B6,B4,B5,B8,B9)		
12					
13	Vanilla call price using Black-Scholes	6.2697	<--=BSCall(B2,B3,B5,B4,B6)		
14					
15	**Multiple runs of the function**				
16	6.4021	6.0120	6.0779	6.0311	<--
17	6.3201	6.3159	6.4282	6.4124	=vanillacall(B2,B3,B6,
18	5.8646	6.6522	6.4013	6.2044	B4,B5,B8,B9)
19	6.3612	6.4910	6.5033	6.3057	
20	6.0024	5.8793	6.0839	6.1823	
21	6.1658	6.0912	6.2631	6.3815	
22	5.9513	6.1238	6.2364	6.1607	
23	5.9291	6.2657	6.2007	6.2288	
24	6.0843	6.2784	6.1989	6.4422	
25	6.0536	6.7148	6.2580	6.1837	
26					
27	Average	6.2286	<-- =AVERAGE(A16:D25)		
28	Standard-deviation	0.1965	<-- =STDEV.S(A16:D25)		

The average value (cell B27) has a relatively low standard deviation (cell B28). The Monte Carlo routine works pretty well.

Improving the Efficiency of the Monte Carlo Routine

Monte Carlo routines are inherently very wasteful—you must run them many times to get a reasonable approximation to the true value. Thus there is a lot of mileage in making a particular routine more efficient. Continuing with our **VanillaCall** example, we show one example of such an efficiency gain.

Suppose that after j random numbers the random price is such that there is no chance that the call option will be in the money. Denote the number of Ups after j random coin tosses by *Upcounter(j)*. Then the call option cannot be in the money after n random numbers if $S_0 Up^{Upcounter(j)\,+\,(n\,-\,j)}\,Down^{j\,-\,Upcounter(j)} < X$. This formula assumes that all the remaining random numbers (there will be $n - j$ such numbers) will give an Up stock price movement.

For this case, we should stop choosing random numbers after j and let the call value be zero. The VBA routine below implements this logic:

```
Function BetterVanillaCall(S0, Exercise, Mean, _
sigma, Interest, Time, Divisions, Runs)
    deltat = Time / Divisions
    interestdelta = Exp(Interest * deltat)
Up = Exp((Interest - 0.5 * (sigma) ^ 2) * deltat + sigma *
Sqr(deltat))
Down = Exp((Interest - 0.5 * (sigma) ^ 2) * deltat - sigma *
Sqr(deltat))
```

```
pathlength = Int(Time / deltat)
'Risk-neutral probabilities
piup = (interestdelta - Down) / (Up - Down)
pidown = 1 - piup
Temp = 0
For Index = 1 To Runs
    Upcounter = 0
    'Generate terminal price
    For j = 1 To pathlength
    If Rnd > pidown Then Upcounter = _
    Upcounter + 1
    If S0 * Up ^ (Upcounter + pathlength - j) _
    * Down ^ (j - Upcounter) < X Then GoTo Compute
    Next j
Compute:
    callvalue = Application.Max(S0 * _
    (Up ^ Upcounter) * (Down ^ _
    (pathlength - Upcounter)) _
    - Exercise, 0) / (interestdelta _
    ^ pathlength)
    Temp = Temp + callvalue
Next Index
BetterVanillaCall = Temp / Runs
End Function
```

The highlighted portions (in **bold**) of the code show the changes. The lines called **Compute** simply calculate the **Callvalue**.

The spreadsheet below shows the implementation:

	A	B	C
1	**MONTE CARLO PRICING OF PLAIN-VANILLA CALLS** BetterVanillaCall: A somewhat more efficient function: If, after *j* random numbers which produce *k* Up moves, S_0*$Up^{(k+n-j)}$*$Down^{(j-k)}$<X, then we abort the random price path and let the call value = 0		
2	S_0, current stock price	50	
3	X, exercise price	50	
4	r, interest rate	5%	
5	T, time	0.8	
6	σ, sigma	30%	<--standard deviation of stock return
7			
8	n, divisions of unit time	100	
9	Runs	3,000	
10			
11	Vanilla call price using Monte Carlo (improved)	6.5511	<--=bettervanillacall(B2,B3,B6,B4,B5,B8,B9)
12			
13	Vanilla call price using Black-Scholes	6.2697	<--=BSCall(B2,B3,B5,B4,B6)

Performing the Routine in R

Implementing the same approach in R for 10,000 is presented below:

```
# Inputs:
S0 <- 50 # initial stock price
X <- 50 # exercise price
r <- 0.1 # interest rate
t <- 0.8 # time
sigma_S <- 0.3 # standard deviation of the stock return
n <- 200 # divisions of unit time
runs <- 10000 # number of simulations

# Define time interval and interest rate
delta_t <- 1 / n
m <- t / delta_t
r_delta_t <- exp(r * delta_t)

# Calculate Up and Down returns
U <- exp((r - 0.5 * sigma_S^2) * delta_t + sigma_S * sqrt(delta_t)) # Up
D <- exp((r - 0.5 * sigma_S^2) * delta_t - sigma_S * sqrt(delta_t)) # Down

# State prices
qU <- (r_delta_t - D) / (r_delta_t * (U - D))
qD <- (U - r_delta_t) / (r_delta_t * (U - D))

# Risk neutral probabilities
pi_U <- (r_delta_t - D) / (U - D)
pi_D <- 1 - pi_U

# Pricing the option:
total_ups <- rbinom(n = runs, size = m, prob = pi_U) # Total ups
ST <- S0 * U^total_ups * D^(m - total_ups) # Terminal Stock Prices
call_payoff <- pmax(ST - X, 0) # Option payoffs
call_price <- mean(call_payoff / exp(r*t)) # Average discounted payoff

# Compare to B&S call price
fm5_bs_call(S = S0, X = X, T = t, r = r, sigma_S = sigma_S)
```

The results are about 0.1% off the Black-Scholes price. Quite impressive, isn't it?

```
> # Compare to B&S call price
> fm5_bs_call(S = S0, X = X, t = t, r = r, sigma_S = sigma_S)
[1] 7.278185
> call_price
[1] 7.271232
```

Where Do We Go from Here?

Now that we understand the Monte Carlo technology and its implementation in VBA, we can extend our examples in two directions. In the next section we discuss the pricing of Asian options in which the option's terminal payoff depends on the average price over the path. In section 27.6 we discuss the pricing of barrier options.

27.5 Pricing Asian Options

An Asian option is an option whose payoff depends in some way on the average price of the asset over a period of time prior to option expiration.[3] Asian options are sometimes called "average price options." There are two common kinds of Asian options (presented for call options, the equivalence Asian puts are straightforward):

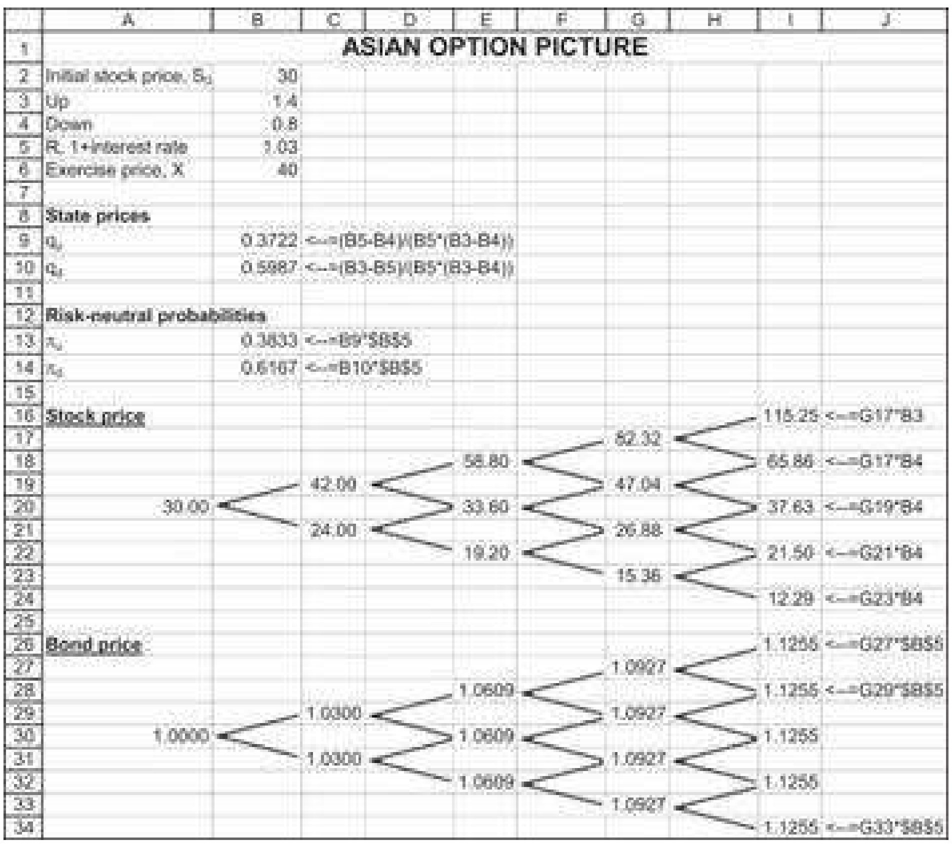
- • In the first kind of Asian option, the option's payoff is based on the difference between the average price of the underlying asset and the strike price: Max[Average underlying − Strike, 0]. Such options are common for commodity prices like crude oil and others.
- • In the second kind of Asian option, the option's exercise price is the average of the underlying's price over a period preceding option maturity: *Max*[*Terminal*

underlying—*averageunderlying*, 0]. Such *average strike* options are common in markets for electric energy. They assist hedgers whose primary risks are related to the average price of the underlying.

Asian options are particularly useful when the user sells the underlying during the period and is therefore exposed to the average price and when there is danger of price manipulation in the underlying. The Asian option mitigates the effect of manipulation because it is based not on a single price but on a sequence of prices.

An Initial Example of an Asian Option

We start by considering an Asian option on a stock whose price increases by either 40% or decreases by 20% each period. We look at five dates, starting with date 0:



To compute the value of the option, we first compute each *price path*. There are 16 such paths. The spreadsheet below shows each path, the average stock price over the path, the option payoff, and the path's risk-neutral probability:

PRICING AN ASIAN OPTION BY PRICING ALL THE PATHS		
Initial stock price, S_0	30	
Up	1.4	
Down	0.8	
R, 1+interest rate	1.03	
Exercise price, X	40	
State price, Up: q_u	0.3722	<-- =(B5-B4)/(B5*(B3-B4))
State price, Down, q_d	0.5987	<-- =(B3-B5)/(B5*(B3-B4))
Risk-neutral prob., Up	0.3833	<-- =B8*B5
Risk-neutral prob., Down	0.6167	<-- =B9*B5

Formula in cell O16: =$B$11^4

Formula in cell O18: =$B$11^3*$B$12

Formula in cell M16: =AVERAGE(G16:K16)

Formula in cell N16: =MAX(M16-$B$6,0)

					STOCK PRICE							
Paths	Period 1	Period 2	Period 3	Period 4	Period 0	Period 1	Period 2	Period 3	Period 4	Average stock price	Option payoff	Path risk-neutral probability
All up (1 path)	up	up	up	up	30.00	42.00	58.80	82.32	115.25	65.67	25.67	0.0216
One down (4 paths)	down	up	up	up	30.00	24.00	33.60	47.04	65.86	40.10	0.10	0.0347
	up	down	up	up	30.00	42.00	33.60	47.04	65.86	43.70	3.70	0.0347
	up	up	down	up	30.00	42.00	58.80	47.04	65.86	48.74	8.74	0.0347
	up	up	up	down	30.00	42.00	58.80	82.32	65.86	55.80	15.80	0.0347
Two down (6 paths)	down	down	up	up	30.00	24.00	19.20	26.88	37.63	27.54	0.00	0.0559
	down	up	down	up	30.00	24.00	33.60	26.88	37.63	30.42	0.00	0.0559
	down	up	up	down	30.00	24.00	33.60	47.04	37.63	34.45	0.00	0.0559
	up	down	down	up	30.00	42.00	33.60	26.88	37.63	34.02	0.00	0.0559
	up	up	down	down	30.00	42.00	58.80	47.04	37.63	43.09	3.09	0.0559
	up	down	up	down	30.00	42.00	33.60	47.04	37.63	38.05	0.00	0.0559
Three down (4 paths)	up	down	down	down	30.00	42.00	33.60	26.88	21.50	30.80	0.00	0.0899
	down	up	down	down	30.00	24.00	33.60	26.88	21.50	27.20	0.00	0.0899
	down	down	up	down	30.00	24.00	19.20	26.88	21.50	24.32	0.00	0.0899
	down	down	down	up	30.00	24.00	19.20	15.36	21.50	22.01	0.00	0.0899
Four down (1 path)	down	down	down	down	30.00	24.00	19.20	15.36	12.29	20.17	0.00	0.1446
										Option value	1.5206	

Formula in cell N37: =SUMPRODUCT(N16:N35,O16:O35)/B5^4

To price the Asian option, we attach a risk-neutral probability to each price path (column O). The option price is the discounted expected payoff value, where the expectations are computed with the risk-neutral probabilities:

$$Price = \frac{\sum_{all\ paths} \pi_{path} * option\ payoff\ on\ path}{R^n} = 1.5206$$

The paths are determined by the number and the sequence of the *Up* and the *Down* movements of the stock. For explanatory purposes we have highlighted two price paths:

- • Along the path {*up, down, up, up*} the terminal stock price is 65.856, the average price is 43.70, the option payoff is 3.70, and the discounted expected value using the risk-neutral prices is 0.0347 * *max*(0, 43.70 − 40) = 0.114:

PATH PRICE EXAMPLE: (Up, Down, Up, Up)		
Initial stock price, S_0	30	
Up	1.4	
Down	0.8	
R, 1+interest rate	1.03	
Exercise price, X	40	
State price, Up: q_u	0.3722	<-- =(B5-B4)/(B5*(B3-B4))
State price, Down, q_d	0.5987	<-- =(B3-B5)/(B5*(B3-B4))
Risk-neutral prob., Up	0.3833	<-- =B8*B5
Risk-neutral prob., Down	0.6167	<-- =B9*B5
Date	**Price at beginning of period**	**Price movement: Up or Down**
0	30.000	
1	42.000	Up
2	33.600	Down
3	47.040	Up
4	65.856	Up
Average price along path	43.70	<-- =AVERAGE(B15:B19)
Option payoff at path end	3.70	<-- =MAX(B20-$B$6,0)
Path risk-neutral price	0.0347	<-- =B11^COUNTIF(C16:C19,"Up")*B12^COUNTIF(C16:C19,"Down")
Value of path:	**0.114**	<-- =B21*B22/B5^4

- • Along the path {*up, up, down, up*} the terminal stock price is the same as before: 65.856. However, the average price and thus the option payoff and value are different:

	A	B	C
1	PATH PRICE EXAMPLE: (Up, Up, Down, Up)		
2	Initial stock price, S_0	30	
3	Up	1.4	
4	Down	0.8	
5	R, 1+interest rate	1.03	
6	Exercise price, X	40	
7			
8	State price, Up: q_U	0.3722	<--=(B5-B4)/(B5*(B3-B4))
9	State price, Down, q_D	0.5987	<--=(B3-B5)/(B5*(B3-B4))
10			
11	Risk-neutral prob., Up	0.3833	<--=B8*B5
12	Risk-neutral prob., Down	0.6167	<--=B9*B5
13			
14	Date	Price at beginning of period	Price movement: Up or Down
15	0	30.000	
16	1	42.000	Up
17	2	58.800	Up
18	3	47.040	Down
19	4	65.856	Up
20	Average price along path	48.74	<--=AVERAGE(B15:B19)
21	Option payoff at path end	8.74	<--=MAX(B20-$B$6,0)
22	Path risk-neutral price	0.0347	<--=B11^COUNTIF(C16:C19,"Up")*B12^COUNTIF(C16:C19,"Down")
23	**Value of path:**	**0.270**	<--=B21*B22/B5^4

These two paths illustrate what we mean when we say that an Asian option price is *path-dependent*: Two paths—both starting at the initial stock price of 30 and ending at 65.856—have different option payoffs because the stock price average along the path is different.

This example, which we've not yet concluded, also illustrates the difficulty of pricing Asian options: Each single path must be dealt with. So, for example, 16 separate paths ($2^4 = 16$). This distinguishes Asian options from the case of ordinary options ("plain vanilla," in the jargon of option traders); for the particular example we are considering here, a plain-vanilla option requires dealing only with five ending prices.

Risk-Neutral Probabilities—Again

We repeat our earlier comments (see section 27.1 of this volume) about the role of risk-neutral probabilities: Each path is priced by its discounted risk-neutral probability, which is a function of the Up, Down, and R. The actual state probabilities are not relevant.

Pricing Asian Options with R—Modeling the Full Binomial Tree

We can also implement the above approach in R. In the first step, we model the whole binomial tree. In our example this is not a fearsome task since there are, in the general case, 2^m paths and in our example, since $m = 4$, we get 16 paths. This is the code to implement the procedure:

```
# Inputs
S0 <- 30
X <- 40
U <- 1.4
D <- 0.8
r <- 1.03
m <- 4 # number of periods

# State prices
qU <- (r - D) / (r * (U - D))
qD <- (U - r) / (r * (U - D))

# Risk neutral probabilities
pi_U <- qU * r
pi_D <- qD * r

# A matrix of all binomial tree paths (first approach - good for small samples)
paths <- sapply(1:m, function(x){
    rep(c(rep(U, (2^x) / 2), rep(D, (2^x) / 2)), 2^(m-x))
                    })
# Stock price paths
stock_paths <- cbind(S0, t(S0 * apply(paths, 1, cumprod)))

# Average stock price
avg_St <- rowMeans(stock_paths)

# Option payoff
asian_call_payoff <- pmax(avg_St - X, 0)

# Path risk-neutral probability
prob_mat <- ifelse(paths == U, pi_U, pi_D)
prob_vec <- apply(prob_mat, 1, prod)

# Average discounted payoff
asian_call_value <- asian_call_payoff %*% prob_vec / r^m
```

Pricing Asian Options with a VBA Program—the Monte Carlo Approach

Our spreadsheet example in the previous section illustrates both the principle of Monte Carlo pricing of options and the problematics of pricing the options directly in a spreadsheet. For four periods, we require the computation of $2^4 = 16$ paths. For a more general problem with n periods, there are 2^n paths to consider; this rapidly becomes too large for even a very powerful computer. In order to accurately price the options, you would need hundreds or thousands of simulations. Doing this in a spreadsheet directly would be cumbersome. The obvious answer is to write some VBA code to automate the process, which would allow us to run an arbitrarily large number of simulations.

In this section we write VBA code to do a Monte Carlo simulation of Asian pricing options. We generate price paths by simulating a sequence of *Up* and *Down* movements of the underlying stock price; the probability of *Up* or *Down* depends on the risk-neutral probabilities—in this sense, our Monte Carlo simulation for Asian options is similar to that illustrated in section 27.4 for plain-vanilla options. For each price path generated, we calculate the option payoff, and after generating a large number of price paths, we compute the option price by discounting and averaging these payoffs. The VBA function **MCAsianCall**, which prices the Asian options, is given below.

VBA Asian Call Option Pricing Function

```
Function MCAsianCall(initial, Exercise, Up, Down, _
    Interest, Periods, Runs)
    Dim PricePath() As Double
    ReDim PricePath(Periods + 1)
    'Risk-neutral probabilities
    piup = (Interest - Down) / (Up - Down)
    pidown = 1 - piup
```

```
        Temp = 0
        For Index = 1 To Runs
            `Generate path
            For i = 1 To Periods
                PricePath(0) = initial
                pathprob = 1
                If Rnd > pidown Then
                    PricePath(i) = PricePath(i - 1) * Up
                    Else:
                    PricePath(i) = PricePath(i - 1) * Down
                End If
            Next i
            PriceAverage = Application.Sum (PricePath) / (Periods + 1)
            callpayoff=Application.Max(PriceAverage - Exercise, 0)
            Temp = Temp + callpayoff
        Next Index
        MCAsianCall = (Temp / Interest ^ Periods) / Runs
    End Function
```

Here is the implementation of the function in a spreadsheet for a 20-period option:

	A	B	C
1	**PRICING AN ASIAN OPTION BY MONTE CARLO**		
2	Initial stock price, S_0	30	
3	Up	1.4	
4	Down	0.8	
5	R, 1+interest rate	1.03	
6	Exercise price, X	40	
7	Periods	4	
8	Runs	500	
9	Asian call value	1.5838	<--=MCAsianCall(B2,B6,B3,B4,B5,B7,B8)

The function in cell B9 is our Monte Carlo valuation of the option—the simulated value. Any recalculation of the spreadsheet will cause the function to rerun and recompute the option value.

Below we show a block of replications of the function for the four-period option presented in section 27.5. Each of the cells A11:D20 contains the formula **=MCAsianCall(Initialprice,exercise,Up,Down,Interest,Periods,Runs)**, so that we calculate 40 simulations of the option value.

	A	B	C	D	E
1	PRICING AN ASIAN OPTION--VBA FUNCTION Prices an Asian option with 4 periods and 500 runs for each simulation				
2	Initial stock price, S_0	30			
3	Up	1.4			
4	Down	0.8			
5	R, 1+interest rate	1.03			
6	Exercise price, X	40			
7	Periods	4			
8	Runs	500			
9	Asian call value	1.5966	<--=MCAsianCall(B2,B6,B3,B4,B5,B7,B8)		
10					
11	1.4746	1.3457	1.3395	1.8640	<--=MCAsianCall($B$2,$B$6,$B$3,$B$4,$B$5,$B$7,$B$8)
12	1.7995	1.7303	1.3741	1.4809	
13	1.5091	1.4318	1.1891	1.3472	
14	1.2877	1.5226	1.5246	1.3527	
15	1.7662	1.6069	1.4062	1.6686	
16	1.5454	1.1928	1.1853	1.2530	
17	1.2186	1.3843	1.6770	1.4759	
18	1.5916	1.5250	1.6145	1.8181	
19	1.4840	1.4129	1.6659	1.4618	
20	1.5132	1.1834	1.6382	1.6184	
21					
22	Average of MC simulations	1.4870	<--=AVERAGE(A11:D20)		
23	True value	1.5206	<-- From pricing the full tree		
24	Stdev of MC simulations	0.184	<--=STDEV.S(A11:D20)		

We have deliberately priced the Asian option for which we know the true value. As we illustrated above, the value of an Asian option in a four-period model with the parameters for *Up*, *Down*, and *Interest* is 1.5206 (cell B23). The average of our Monte Carlo simulations is 1.4870, with a standard deviation of 0.184.

When we increase the number of runs (cell B8), we will usually decrease the standard deviation of our estimates (cell B24), which is equivalent to increasing the accuracy of the simulation. In the example below, we have run the 40 simulations 5,000 runs each:

	A	B	C	D	E
1	PRICING AN ASIAN OPTION—VBA FUNCTION Asian option with 4 periods and 5000 runs for each simulation				
2	Initial stock price, S_0	30			
3	Up	1.4			
4	Down	0.8			
5	R, 1+interest rate	1.03			
6	Exercise price, X	40			
7	Periods	4			
8	Runs	5000			
9	Asian call value	1.4119	<--=MCAsianCall(B2,B6,B3,B4,B5,B7,B8)		
10					
11	1.4045	1.4877	1.5087	1.4637	<--=MCAsianCall($B$2,$B$6,$B$3,$B$4,$B$5,$B$7,$B$8)
12	1.5452	1.4973	1.5168	1.5604	
13	1.5376	1.5483	1.6025	1.4479	
14	1.5201	1.5755	1.4378	1.5408	
15	1.5472	1.4996	1.5487	1.3978	
16	1.5003	1.4038	1.4663	1.5758	
17	1.4071	1.4735	1.5811	1.5265	
18	1.5460	1.5904	1.5870	1.5732	
19	1.5461	1.5192	1.4655	1.5501	
20	1.5105	1.5648	1.6321	1.4655	
21					
22	Average of MC simulations	1.5168	<--=AVERAGE(A11:D20)		
23	True value	1.5206	<-- From pricing the full tree		
24	Stdev of MC simulations	0.059	<--=STDEV.S(A11:D20)		

As you can see, the standard deviation is greatly reduced—less than half the standard deviation with 500 runs.

Pricing Asian Options with R—the Monte Carlo Approach

The same approach as presented in VBA when implement in R looks like this (note that we have used 100,000 simulations and R is running this procedure in seconds):

```
# Inputs:
S0 <- 30 # initial stock price
U <- 1.4
D <- 0.8
r <- 1.03
X <- 40
m <- 4 # number of periods
runs <- 100000 # number of simulations

# Risk neutral probabilities
pi_U <- qU * r

# Create a matrix of [runs,periods]
paths <- matrix( rbinom(n = runs * m, 1, prob = pi_U), nrow = runs)

# Convert matrix to Up/Down
paths <- ifelse(paths == 1, U, D)

# Stock price paths
stock_paths <- cbind(S0, t(S0 * apply(paths, 1, cumprod)))

# Average stock price
avg_St <- rowMeans(stock_paths)

# Option payoff
asian_call_payoff <- pmax(avg_St - X, 0)
```

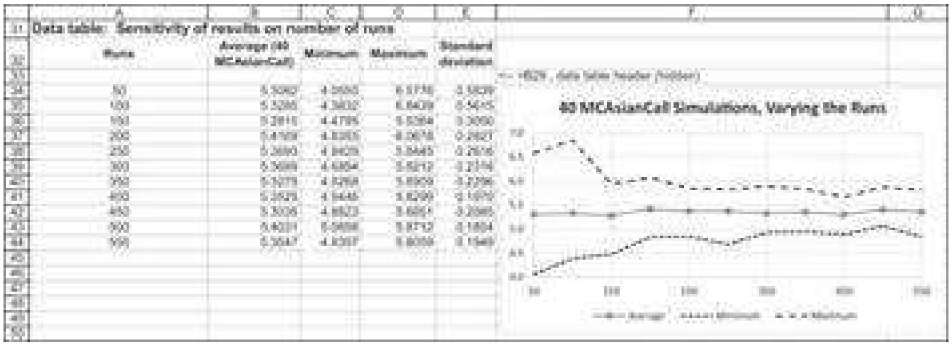
Asian Options with More Periods

In the following spreadsheet we divide the unit time interval into n subperiods. We follow the procedure of section 27.4 for defining the returns, state prices, and risk-neutral probabilities over the subperiod Δt. Here is the resulting spreadsheet:

	A	B	C	D	E	F
1	**PRICING AN ASIAN OPTION—VBA FUNCTION** **Each time interval is divided into n subintervals. In this simulation the maturity date is 0.50 years and the unit time interval is divided into 80 subintervals. There are 100 runs in each Monte Carlo simulation.**					
2	S_0, current stock price	50				
3	K, exercise price	45				
4	T, time to option exercise	0.5				
5	r, interest rate	3%				
6	σ, standard deviation of stock	30%				
7						
8	n, number of sub-intervals of T	80				
9	Delta t	0.0063	<-- =B4/B8			
10						
11	Up over 1 sub-interval	1.0239	<-- =EXP((B5-0.5*B6^2)*B9+B6*SQRT(B9))			
12	Down over 1 sub-interval	0.9765	<-- =EXP((B5-0.5*B6^2)*B9-B6*SQRT(B9))			
13	Interest over 1 sub-interval	1.0002	<-- =EXP(B5*B9)			
14						
15	Runs	100				
16						
17	5.6114	5.3648	5.6692	6.2083	5.5143	<-- =MCAsianCall($B$2,$B$3,$B$11,$B$12,$B$13,$B$4*$B$8,$B$15)
18	5.9405	5.7131	5.4355	6.1569	5.4323	
19	4.9257	5.7651	5.4356	5.1464	4.6015	
20	4.9599	5.1827	5.1118	5.8008	5.7492	
21	5.3142	4.7012	5.1756	4.8406	5.4588	
22	5.0335	5.2156	5.0096	4.7831	4.9774	
23	4.9069	5.9379	4.8100	5.5290	4.9667	
24	4.9258	5.7244	5.4049	5.3084	5.2161	
25						
26	Average of above	5.3314	<-- =AVERAGE(A17:E24)			
27	Minimum	4.6015	<-- =MIN(A17:E24)			
28	Maximum	6.2083	<-- =MAX(A17:E24)			
29	Standard deviation	0.4056	<-- =STDEV.S(A17:E24)			

The block of results in cells A17:E24 gives 40 results for running the function **MCAsianCall**. Below this block we give the statistics for these simulations.

The above simulations use 100 price paths per iteration of the function **MCAsianCall**; this is the number contained in cell B15. We can use **Data Table** to see the effect of changing the number of runs:



It is clear that increasing the number of runs narrows the bounds on the simulation. Implementing the above approach in R should look like this:

```
# Inputs:
S0 <- 50 # initial stock price
X <- 45 # exercise price
t <- 0.5 # Time to exercise
r <- 0.03 # interest rate
sigma_S <- 0.3 # standard deviation of the stock return

# Define time interval
n <- 80 # divisions of unit time
delta_t <- t / n

# Calculate Up, Down and interest rate returns over 1 sub-interval
U <- exp((r - 0.5*sigma_S^2) * delta_t + sigma_S * sqrt(delta_t)) # Up
D <- exp((r - 0.5*sigma_S^2) * delta_t - sigma_S * sqrt(delta_t)) # Down
r_delta_t <- exp(r * delta_t)

# Risk neutral probabilities
pi_U <- (r_delta_t - D) / (U - D)

# Create a matrix of [runs,periods]
runs <- 100000
paths <- matrix( rbinom(n = runs * n, 1, prob = pi_U), nrow = runs)

# Convert matrix to Up/Down
paths <- ifelse(paths == 1, U, D)

# Stock price paths
stock_paths <- cbind(S0, t(S0 * apply(paths, 1, cumprod)))

# Average stock price
avg_St <- rowMeans(stock_paths)

# Option payoff
asian_call_payoff <- pmax(avg_St - X, 0)

# Average of MC simulations
asian_call_value_mc <- mean(asian_call_payoff) * exp(-r*t)
```

27.6 Barrier Options

A barrier option's payoff depends on whether the price reaches a specific level during the life of the option:

- • A *knock-in* barrier call option has payoff $Max(S_T - X, 0)$ only if at some time $t < T$, S_t crossed the barrier K. A knock-in put has the same condition but pays off $Max(X - S_T, 0)$.

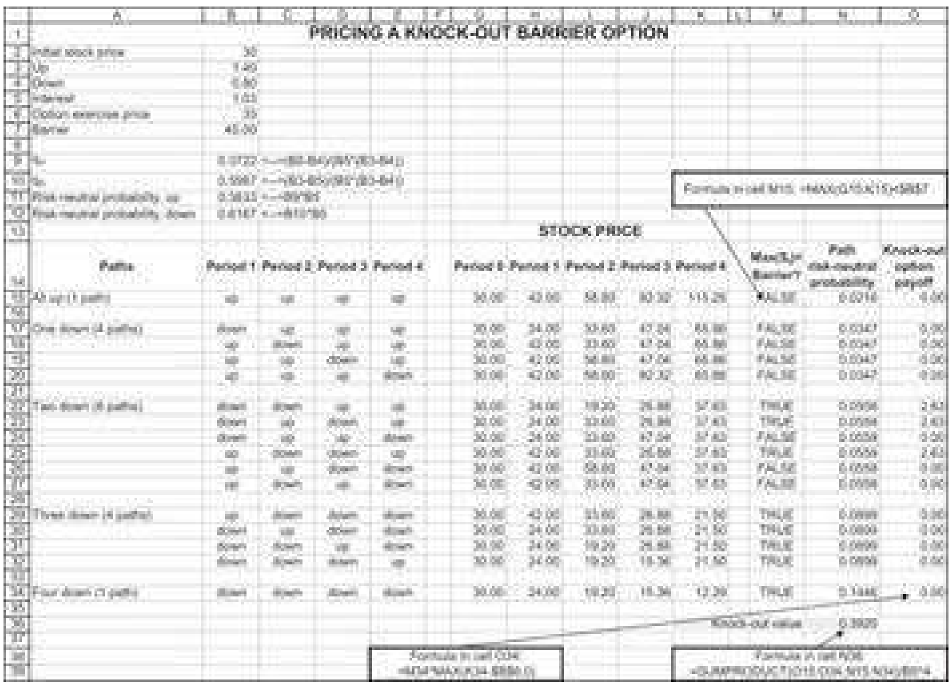
- • A *knock-out* barrier call or put option has these payoffs provided that at no time before T does the stock price reach the barrier.

Imposing a barrier makes it more difficult for an option to be in the money at expiration; thus, barrier options have lower value than regular options. There are four types of barrier options: (1) Down and in, (2) Down and out, (3) Up and in, (4) Up and out.

Pricing barrier options with Monte Carlo isn't necessarily a good idea, but it's a good exercise.[4]

A Simple Example of a Knock-Out Barrier Call Option

Below, we show an extended example for a knock-out barrier option that is similar to the example for an Asian option given in section 27.5.



In this example, we model a five-date, four-period barrier call option. The barrier is 45 (cell B7). A knock-out option pays off only if the price never goes through this barrier, and a knock-in option pays off only if the stock price goes through the barrier. In an equation:

$$Barrier\ Knock\text{-}In\ call\ payoff = \begin{cases} Max[S_T - X, 0] & if\ S_t > Barrier\ for\ t < T \\ 0 & Otherwise \end{cases}$$

$$Barrier\ Knock\text{-}Out\ call\ payoff = \begin{cases} Max[S_T - X, 0] & if\ S_t < Barrier\ for\ t < T \\ 0 & Otherwise \end{cases}$$

The knock-out barrier call illustrated above pays off only when two things happen simultaneously:

- • The stock price did not exceed the barrier. This happens for all the paths labeled "TRUE" in column M. To check this condition, in cell M15 we use the Boolean function (=MAX(G15:K15)<$B$7).[5] This function evaluates to TRUE or FALSE, depending on whether the condition is met. Other cells in column M use a similar condition. When used in a formula (as in the bullet below), the Boolean function evaluates to 1 if TRUE and to 0 if FALSE.
- • The terminal stock price, S_T, is 115.25 (cell K15), which is higher than the option exercise price X of 35 (cell B6). In cell O15 we use the condition M15 * MAX(K15 − $B$6,0) to evaluate the option payoff.
    - ◦ If $M15 = 0$ (meaning that $S_t > 50$ somewhere along the path and the option was "knocked out"), then the option doesn't pay off.
    - ◦ If $M15 = 1$ (so that $S_t < 50$ throughout the path), then the option has a standard call payoff of $Max(S_T - X, 0)$.

As in all previous cases discussed in this chapter, the barrier call's value is the discounted expected payoff of the option, where the probabilities are the risk-neutral probabilities:

$$Option\ value = \frac{\sum_{all\ states} \pi_j\, Payoff}{R^4} = 0.3920$$

Implementing this approach in R looks like this:

```
## Pricing Knock-Out Barrier Option with Monte Carlo
# Inputs:
S0 <- 30
U <- 1.4
D <- 0.8
r <- 1.03
X <- 35
barrier <- 45
m <- 4 # number of periods

# State prices
qU <- (r - D) / (r * (U - D))
qD <- (U - r) / (r * (U - D))

# Risk neutral probabilities
pi_U <- qU * r
pi_D <- qD * r

# A matrix of all binomial tree paths
paths <- sapply(1:m, function(x){
    rep(c(rep(U, (2^x) / 2), rep(D, (2^x) / 2)), 2^(m-x))
                            })

# Stock price paths
stock_paths <- cbind(S0, t(S0 * apply(paths, 1, cumprod)))

# Knocked out if maximum price in the path > 45
knocked_out <- apply(stock_paths, 1, max) > barrier

# Option payoff
ko_call_payoff <-  pmax(stock_paths[,m+1] - X, 0)*(1-knocked_out)

# Path risk-neutral probability
prob_mat <- ifelse(paths == U, pi_U, pi_D)
prob_vec <- apply(prob_mat, 1, prod)

# Average discounted payoff
ko_call_value <- sum(ko_call_payoff * prob_vec) / r^m
```

The Knock-In Barrier Call

By changing the condition in column O, we can price the knock-in barrier call. This time we write (in cell O15, for example) the function $= (1 - M15) * MAX(K15 - \$B\$6, 0)$. The value in M15 tests whether the barrier has ever been passed; if this is FALSE (i.e., has a value of zero), then the option is "knocked in" and the payoff is like that of a regular call. If M15 is TRUE, then the barrier has not been passed and the option does not pay off:

PRICING A KNOCK-IN BARRIER OPTION

Initial stock price	30	
Up	1.40	
Down	0.80	
Interest	1.03	
Option exercise price	35	
Barrier	45.00	
q_u	0.3722	<-- =(B5-B4)/(B5*(B3-B4))
q_d	0.5987	<-- =(B3-B5)/(B5*(B3-B4))
Risk-neutral probability, up	0.3833	<-- =B9*B5
Risk-neutral probability, down	0.6167	<-- =B10*B5

Formula in cell M15: =MAX(G15:K15)>$B$7

Paths	Period 1	Period 2	Period 3	Period 4	STOCK PRICE Period 0	Period 1	Period 2	Period 3	Period 4	Max(S_t)> Barrier?	Path risk-neutral probability	Knock-in option payoff
All up (1 path)	up	up	up	up	30.00	42.00	58.80	82.32	115.25	FALSE	0.0216	80.25
One down (4 paths)	down	up	up	up	30.00	24.00	33.60	47.04	65.86	FALSE	0.0347	30.86
	up	down	up	up	30.00	42.00	33.60	47.04	65.86	FALSE	0.0347	30.86
	up	up	down	up	30.00	42.00	58.80	47.04	65.86	FALSE	0.0347	30.86
	up	up	up	down	30.00	42.00	58.80	82.32	65.86	FALSE	0.0347	30.86
Two down (6 paths)	down	down	up	up	30.00	24.00	19.20	26.88	37.63	TRUE	0.0559	0.00
	down	up	down	up	30.00	24.00	33.60	26.88	37.63	TRUE	0.0559	0.00
	down	up	up	down	30.00	24.00	33.60	47.04	37.63	FALSE	0.0559	2.63
	up	down	down	up	30.00	42.00	33.60	26.88	37.63	TRUE	0.0559	0.00
	up	up	down	down	30.00	42.00	58.80	47.04	37.63	FALSE	0.0559	2.63
	up	down	up	down	30.00	42.00	33.60	47.04	37.63	FALSE	0.0559	2.63
Three down (4 paths)	up	down	down	down	30.00	42.00	33.60	26.88	21.50	TRUE	0.0899	0.00
	down	up	down	down	30.00	24.00	33.60	26.88	21.50	TRUE	0.0899	0.00
	down	down	up	down	30.00	24.00	19.20	26.88	21.50	TRUE	0.0899	0.00
	down	down	down	up	30.00	24.00	19.20	15.36	21.50	TRUE	0.0899	0.00
Four down (1 path)	down	down	down	down	30.00	24.00	19.20	15.36	12.29	TRUE	0.1446	0.00
										Knock-in value	5.7407	

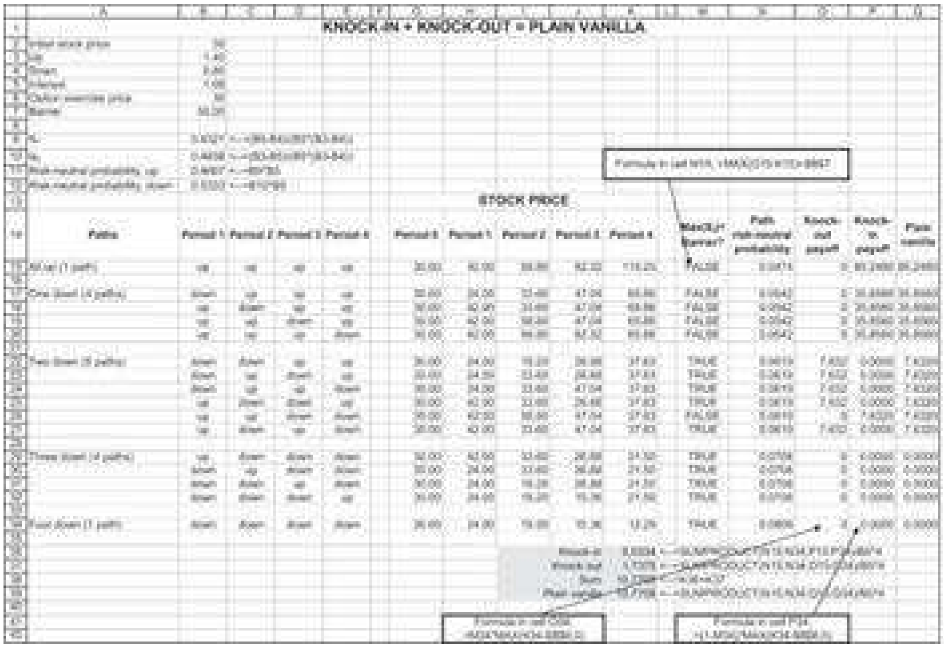
Formula in cell O34: =(1-M34)*MAX(K34-$B$6,0)

Formula in cell N36: =SUMPRODUCT(O15:O34,N15:N34)/B5^4

Implementing this in R is very straightforward. We just need to reverse the condition and check where we "knocked in"—which in our example is when $Max(S_t) > 45$:

```r
## Pricing Knock-In Barrier Option
# Option payoff
knocked_in <- apply(stock_paths, 1, max) > barrier
ki_payoff <- pmax(stock_paths[,m+1] - X, 0) * knocked_in

# Path risk-neutral probability
prob_mat <- ifelse(paths == U, pi_U, pi_D)
prob_vec <- apply(prob_mat, 1, prod)

# Average discounted payoff
ki_call_value <- sum(ki_payoff * prob_vec) / r^m
```

The spreadsheets for the knock-out and knock-in barriers illustrate another principle of pricing barrier options: The price of a knock-in plus a knock-out call equals the price of a regular plain-vanilla call:



Using VBA and Monte Carlo to Price Knock-In Barrier Option

We write two VBA functions to price knock-in and knock-out barrier options. Here is the function for knock-out options:

```
Function MCBarrierOut(Initial, Exercise, Barrier, Up, _
    Down, Interest, Periods, Runs)
    Dim PricePath() As Double
    ReDim PricePath(Periods + 1)
'Risk-neutral probabilities
    piup = (Interest - Down) / (Up - Down)
    pidown = 1 - piup
    Temp = 0
    For Index = 1 To Runs
        'Generate path
        For i = 1 To Periods
            PricePath(0) = Initial
            pathprob = 1
            If Rnd > pidown Then
                PricePath(i) = PricePath(i - 1) * Up
                Else:
                PricePath(i) = PricePath(i - 1) * Down
            End If
        Next i
            If Application.Max(PricePath) < Barrier Then
            Callpayoff = _
                Application.Max(PricePath(Periods) - Exercise, 0)
                _
```

```
                Else Callpayoff = 0
                Temp = Temp + Callpayoff
        Next Index
        MCBarrierOut = (Temp / Interest ^ Periods) / Runs
    End Function
```

Since this function is very similar to the function **MCAsianCall** of section 27.5, we will not discuss it, except to point out that the operative part for the "knock in" option is contained in the following lines (note the use of Excel's **Max** function in the form of **Application.Max**; VBA does not have its own maximum function):

```
    If Application.Max(PricePath) < Barrier Then _
        Callpayoff = Application.Max(PricePath (Periods) - Exercise,
        0) _
        Else Callpayoff = 0
```

In the spreadsheet below we use this function and its associated function **MCBarrierOut** to price the options previously priced in our extensive example:

	A	B	C	D	E	F
1	PRICING BARRIER OPTIONS BY MONTE CARLO					
2	Up	1.4				
3	Down	0.8				
4	Interest	1.03				
5						
6	Initial price	30				
7	Periods	4				
8	Exercise	35				
9	Barrier	45				
10						
11	Runs	500				
12						
13	Knock-in option value	5.4419	<--=mcbarrierin(B6,B8,B9,B2,B3,B4,B7,B11)			
14	Actual value	5.7407	<-- Determined from fully worked out example			
15						
16	Knock-out option value	0.3882	<-- =mcbarrierout(B6,B8,B9,B2,B3,B4,B7,B11)			
17	Actual value	0.3920	<-- Determined from fully worked out example			
18						
19	**40 iterations of MCBarrierIn**					
20	6.1819	6.2790	6.3134	6.1708	5.8638	
21	5.4111	6.6690	4.3880	4.7686	6.2739	
22	4.9627	5.0649	5.0900	5.7056	5.4090	
23	6.1853	7.1877	5.3405	5.2549	5.1337	
24	4.8770	5.8871	5.7041	5.4375	5.2655	
25	5.7594	5.0817	5.0458	6.8335	4.4206	
26	5.2430	6.7304	5.3516	6.3417	5.2953	
27	5.6383	5.0755	6.2745	5.9626	5.3669	
28						
29	Average of simulations	5.6311	<--=AVERAGE(A20:E27)			
30	True value	5.7407	<--=B14			
31						
32	Minimum value	4.3880	<--=MIN(A20:F27)			
33	Maximum value	7.1877	<--=MAX(A20:F27)			
34	Standard deviation	0.6606	<--=STDEV(A20:F27)			

Finally, we can also show the implementation of the functions **MCBarrierIn** and **MCBarrierOut** for the case where the unit period is divided into *n* subperiods:

	A	B	C	D	E	F	G	H
1	**PRICING BARRIER OPTIONS—VBA FUNCTION** **Each time interval is divided into n subintervals. In this simulation the maturity date is 0.33 years and the unit time interval is divided into 60 subintervals. There are 100 runs in each Monte Carlo simulation.**							
2	S_0, current stock price	50						
3	X, exercise price	45						
4	Barrier	55						
5	T, time to option exercise	0.33						
6	r, interest rate	3%						
7	σ, standard deviation of stock return	35%						
8								
9	n, number of sub-intervals of 1 period	60						
10	Delta t	0.0167	<--=1/B9					
11								
12	Up over 1 sub-interval	1.0457	<--=EXP((B6-0.5*B7^2)*B10+B7*SQRT(B10))					
13	Down over 1 sub-interval	0.9553	<--=EXP((B6-0.5*B7^2)*B10-B7*SQRT(B10))					
14	Interest over 1 sub-interval	1.0005	<--=EXP(B6*B10)					
15								
16	Runs	100						
17								
18	6.6719	6.1465	7.0145	6.4619	5.2856	<--		
19	6.7065	6.6547	5.5532	5.7767	5.6409	=mcbarrierin($B$2,$B$3,$B$4,$B$12,$B$13,$B$14,		
20	6.5283	5.6787	4.9424	5.3696	6.7397	INT($B$9*$B$5),$B$16)		
21	6.3629	6.9002	4.5521	7.5493	7.3908			
22	7.3728	6.7511	6.0093	6.0384	6.9754			
23	5.9851	4.7504	6.9384	5.5693	5.3780			
24	7.2473	5.9356	6.7720	5.3887	6.7156			
25	7.0990	4.9855	7.3163	5.9134	6.1064			
26								
27	Average of above	6.2308	<--=AVERAGE(A18:F25)					
28	Minimum	4.5521	<--=MIN(A18:F25)					
29	Maximum	7.5493	<--=MAX(A18:F25)					
30	Standard deviation	0.7951	<--=STDEV(A18:F25)					

As for the case discussed in section 27.5 for Asian options, as the number of runs (cell B16) gets larger, the approximations become better, although the improvement is not dramatic:

	A	B	C	D	E	F
32	**Data table: Sensitivity of results on number of runs**					
33	**Runs**	**Avg of 40 MCBarrierIn**	**Minimum**	**Maximum**	**Standard deviation**	
34						<--=B30 , data table header (hidden)
35	50	6.0025	3.8121	8.0057	1.1237	
36	100	6.2571	4.8644	8.2864	0.8455	
37	150	6.1306	5.0046	8.1995	0.6800	
38	200	6.4206	5.0307	7.4681	0.6176	
39	250	6.2210	5.2055	7.6223	0.6152	
40	300	6.3269	5.3869	7.4253	0.4775	
41	350	6.1349	5.2299	6.8781	0.3260	
42	400	6.1945	5.3085	7.0076	0.3924	
43	450	6.3132	5.3452	7.5765	0.4055	
44	500	6.2106	5.2726	6.9252	0.3640	
45	550	6.3276	5.6096	7.4052	0.4056	

Finally, we can show that the sum of the knock-in plus knock-out is approximately equal to the Black-Scholes call when *n*, the number of divisions of the unit time interval, is very large:

	A	B	C
1	**KNOCK-IN + KNOCK-OUT = CALL** **The almost-continuous case**		
2	S_0, current stock price	30	
3	X, exercise price	35	
4	Barrier	45	
5	T, time to option exercise	0.33	
6	r, interest rate	3%	
7	σ, standard deviation of stock return	35%	
8			
9	n, number of sub-intervals of 1 period	200	
10	Delta t	0.0017	<--=B5/B9
11			
12	Up over 1 sub-interval	1.0143	<--=EXP((B6-0.5*B7^2)*B10+B7*SQRT(B10))
13	Down over 1 sub-interval	0.9858	<--=EXP((B6-0.5*B7^2)*B10-B7*SQRT(B10))
14	Interest over 1 sub-interval	1.0000	<--=EXP(B6*B10)
15			
16	Runs	1000	
17			
18	Knock-out barrier	0.501811	<--=mcbarrierout(B2,B3,B4,B12,B13,B14,B9,B16)
19	Knock-in barrier	0.382277	<--=mcbarrierin(B2,B3,B4,B12,B13,B14,B9,B16)
20	Sum of knock-out + knock-in	0.884088	<--=B18+B19
21	Black-Scholes call price	0.896622	<--=BSCall(B2,B3,B5,B6,B7)

Using R and Monte Carlo to Price Barrier Options

Implementing the same approach in R would look like this:

```
## Barrier option with MC (Knock-in and Knock-out Example)
# Inputs:
S0 <- 30 # initial stock price
X <- 35
Barrier <- 45
r <- 0.03 # risk-free rate
t <- 0.33 # time to option exercise
n <- 200 # number of sub-intervals of t
sigma <- 0.35 # annual volatility
runs <- 10000 # number of simulations
delta_t <- t/n

# Up Down and periodic interest rate
U <- exp((r-0.5*sigma^2)*delta_t + sigma*sqrt(delta_t))
D <- exp((r-0.5*sigma^2)*delta_t - sigma*sqrt(delta_t))
interest <- exp(r*delta_t)

# Risk neutral probabilities
pi_U <- (interest - D) / (U - D)
pi_D <- (U - interest) / (U - D)

# Create a matrix of [runs,periods]
paths <- matrix(rbinom(n = runs * n, 1, prob = pi_U), nrow = runs)
paths <- ifelse(paths == 1, U, D) # convert matrix to Up/Down

# Stock price paths
stock_paths <- cbind(S0, t(S0 * apply(paths, 1, cumprod)))

# Maximum price in the path > 45?
p_crossed_barrier <- apply(stock_paths, 1, max) > barrier

# Option payoff
vanilla_payoff <-  pmax(stock_paths[,n+1] - X, 0)
mc_ki_payoff <- vanilla_payoff*p_crossed_barrier
mc_ko_payoff <- vanilla_payoff*(1-p_crossed_barrier)
```

27.7 Basket Options

Basket options are based on several underlying assets. The payoff of a basket option is similar to a plain-vanilla option (call: Max($S_T - X$, 0), and put: Max($X - S_T$, 0)). The difference here is that the underlying asset (the basket) is a weighted average of all specific underlying assets. Note that the weights of the underlying assets are not always equal.

An Example of Basket Options

Let us assume that we are pricing a call basket option on the FAAMG stocks (Facebook-FB, Apple-AAPL, Amazon-AMZN, Microsoft-MSFT, and Google-GOOG).

The basket has the following characteristics:

	A	B	C	D	E	F
1	**PRICING A CALL BASKET OPTION**					
2		**FB**	**AAPL**	**AMZN**	**MSFT**	**GOOG**
3	Asset price	222.14	318.73	1864.72	167.1	1480.39
4	Weights	10%	25%	20%	25%	20%
5						
6	Basket price (t=0)	812.69	<--=SUMPRODUCT(B4:F4,B3:F3)			
7	r	2%				
8	T (years)	0.5				
9	X	800				
10						
11	**Var-Cov matrix (weekly)**					
12		**FB**	**AAPL**	**AMZN**	**MSFT**	**GOOG**
13	**FB**	0.002076	0.000498	0.000817	0.000546	0.000748
14	**AAPL**	0.000498	0.001368	0.000556	0.000435	0.000508
15	**AMZN**	0.000817	0.000556	0.001731	0.000623	0.000804
16	**MSFT**	0.000546	0.000435	0.000623	0.000967	0.000623
17	**GOOG**	0.000748	0.000508	0.000804	0.000623	0.001071

Pricing such options required the use of several tools described throughout this book; we use Cholesky decomposition (Chapter 21) to simulate the individual asset prices in the basket (cells I8:M34 in the next spreadsheet). We then use this one simulation to price the basket option (cell B24). At the final step, we use a data table on blank cell to reiterate this simulation 1,000 times[6] (cells A31:B1031) to get the expected option value of 60.27 (cell B27). Here is the spreadsheet:

Pricing Basket Options with R

Implementing the above approach in R (starting with the inputs and data preparation) looks like this:

```
# Inputs:
r <- 0.02 # annual risk-free rate
t <- 0.5
X <- 800
n <- 52*t # weekly periods
asset_price <- c("FB" = 222.14, "AAPL" = 318.73, "AMZN" = 1864.72, "MSFT" = 167.1, "GOOG" = 1480.39)
weights <-  c("FB" = 0.10,  "AAPL" = 0.25, "AMZN" = 0.2 , "MSFT" = 0.25, "GOOG" = 0.20)
var_cov_mat <- matrix(c(0.002075976,    0.000498403, 0.00081678, 0.000545603, 0.000748417,
                        0.000498403, 0.001367649, 0.000556148, 0.000435025, 0.000508491,
                        0.00081678, 0.000556148, 0.001731161, 0.000623072, 0.000804151,
                        0.000545603, 0.000435025, 0.000623072, 0.000966882, 0.000623484,
                        0.000748417, 0.000508491, 0.000804151, 0.000623484, 0.001071263), nrow = 5)

# Assets Variance
variance <- diag(var_cov_mat)

# Basket Initial Values
initial_values <- asset_price * weights

# Basket price (t=0)
basket_value_today <- sum(initial_values)
```

We then implement and price one simulation:

```
# Lower-Triangular Cholesky Decomposition
chol_decompos <- t(chol(var_cov_mat))

# Simulate weekly Returns of n Correlated Stocks
weekly_ret <- sapply(1:n, function(x)
    exp(chol_decompos %*% rnorm(n = 5, mean = 0, sd = 1) + r/52 - 0.5 * variance))

# Written as a concise function
basket_ret <- function(var_cov_mat, r, t, n){
    return( exp(t(chol(var_cov_mat)) %*%
                  matrix(rnorm(n = ncol(var_cov_mat) * n), nrow = ncol(var_cov_mat))+
                  r/(n/t) - 0.5 * diag(var_cov_mat)))
                                       }
weekly_ret <- basket_ret(var_cov_mat, r, t, n)

# Total Simulated Compounded return
total_ret <- apply(month_ret, 1, prod)

# Simulated Basket Value
basket_value_end <- (asset_price * total_ret) %*% weights

# Call payoff
call_payoff <- max(0,basket_value_end - X)

# Call Value
call_value <- call_payoff * exp(-r*t)
```

At the final stage, we run the simulation 10,000 times:

```
# Run 10000 simulations
m <- 10000 # number of simulations
basket_prices <- sapply(1:m, function(x){
    returns = basket_ret(var_cov_mat, r, t, n)
    total_ret = apply(returns , 1, prod)
    return ((asset_price * total_ret) %*% weights)
                                         })

# Call values
call_value_end <- pmax(0,basket_prices - X) * exp(-r*t)
call_price <- mean(call_value_end)
```

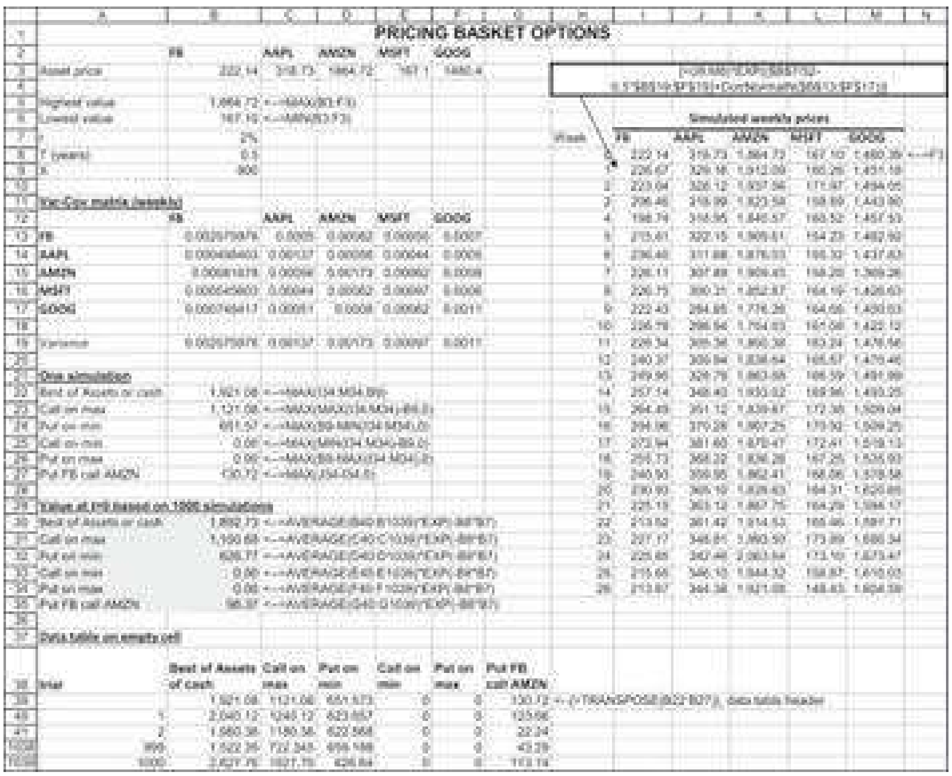
27.8 Rainbow Options

Rainbow options are those whose payoff depends on more than one underlying uncertain asset. There are a few types of rainbow options that differ by their payoff:

- • **Best of assets or cash option**—delivers the maximum of two risky assets and cash at expiry, payoff: Max(S_1, S_2, …, S_n, X).
- • **Call on max option**—gives the holder the right to purchase the highest-value asset for the strike price at expiry date, payoff: Max[$Max(S_1, S_2, \ldots, S_n) - X$, 0].
- • **Put on min option**—gives the holder the right to sell the lowest-value asset for the strike price at expiry, payoff: Max[$X - Min(S_1, S_2, \ldots, S_n)$, 0].

- • **Call on min option**—gives the holder the right to purchase the lowest-value asset for the strike price at expiry date, payoff: Max[$Min\ (S_1, S_2, \ldots, S_n) - X$, 0].
- • **Put on max option**—gives the holder the right to sell the highest-value asset for the strike price at expiry, payoff: Max[$X - Max(S_1, S_2, \ldots, S_n)$, 0].
- • **Put 2 and call 1**, also called **an exchange option**—gives the holder the right to put a predefined risky asset and call the other risky asset. Thus, asset 1 is called with the "strike" being asset 2, payoff: Max($S_1 - S_2$, 0).

We use our FAAMG stocks to implement the pricing approach for the different types of rainbow options:



Pricing Basket Options with R

Implementing the above approach in R is relatively simple and very efficient. Here is the way we implemented it:

```
### 27.7. Rainbow option
m <- 10000 # number of simulations
asset_prices_t <- sapply(1:m, function(x){
    returns = basket_ret(var_cov_mat, r, t, n)
    total_ret = apply(returns , 1, prod)
    return (asset_price * total_ret)
                                        })

# best of assets of cash
option_payoffs <- apply( rbind(asset_prices_t, X), 2, max)
best_of_a_or_c <- mean(option_payoffs) * exp(-r*t)

# Call on max
option_payoffs <- pmax( apply( asset_prices_t, 2, max) - X, 0)
call_max <- mean(option_payoffs)* exp(-r*t)

# Put on min
option_payoffs <- pmax( X - apply( asset_prices_t, 2, min), 0)
put_min <-  mean(option_payoffs)* exp(-r*t)

# Call on min
option_payoffs <- pmax( apply( asset_prices_t, 2, min) - X, 0)
call_min <-  mean(option_payoffs)* exp(-r*t)

# Put on max
option_payoffs <- pmax( X - apply( asset_prices_t, 2, max), 0)
put_max <-  mean(option_payoffs)* exp(-r*t)

# Put FB call AMZN
option_payoffs <- pmax(asset_prices_t["AAPL",] - asset_prices_t["FB",], 0)
put_FB_call_AMZN <-  mean(option_payoffs)* exp(-r*t)
```

27.9 Binary Options

Binary options are also known as **digital options**. They guarantee a payoff in the form of a fixed amount or a predetermined asset in case an event has occurred. Conversely, in case the event has not occurred, the payoff is nothing. In other words, binary options provide all-or-nothing payoffs. The following common types of binary options are:

- • **Cash-or-nothing call**—pays out N units of cash (N) if the underlying is above the strike at maturity, payoff: $\{N \; if \; S_T \geq X$, and $0 \; if \; S_T < X\}$.
- • **Cash-or-nothing put**—pays out N units of cash (M) if the underlying is below the strike at maturity, payoff: Cash-or-nothing call—pays out M units of cash (M) if the underlying is above the strike at maturity, payoff: $\{M \; if \; S_T \leq X$, and $0 \; if \; S_T > X\}$.
- • **Asset-or-nothing call**—pays out one unit of asset if the underlying is above the strike at maturity, payoff: $\{S_T \; if \; S_T \geq X$, and $0 \; if \; S_T < X\}$.
- • **Asset-or-nothing put**—pays out one unit of asset if the underlying is below the strike at maturity, payoff: $\{S_T \; if \; S_T \leq X$, and $0 \; if \; S_T > X\}$.

There is actually a closed-form formula for the binary options, we use it to check the results from our Monte Carlo approach:

Option	**Price**
Cash-or-nothing call	$C = M * e^{-r \cdot T} * N(d_2)$
Cash-or-nothing put	$P = M * e^{-r \cdot T} * N(-d_2)$
Asset-or-nothing call	$C = S_0 * e^{-k \cdot T} * N(d_1)$
Asset-or-nothing put	$P = S_0 * e^{-k \cdot T} * N(-d_1)$

Where:

k is the continuous dividend yield

$$d_1 = \frac{ln(S/X) + \left(r - k + \frac{\sigma^2}{2}\right) \cdot T}{\sigma\sqrt{T}}; \quad d_2 = d_1 - \sigma\sqrt{T}$$

The spreadsheet below shows the implementation:

	A	B	C
1	**PRICING A BINARY OPTION—CLOSED FORM**		
2	Initial stock price, S_0	30	
3	Interest rate, r	3%	
4	Dividend rate, k	0%	
5	Strike price, X	32	
6	Time to exercise, T	0.01923	<--=1/52
7	Sigma, σ	30%	
8	Cash payment, M	50	
9			
10	d1	-1.517	<--=(LN(B2/B5)+(B3-B4+0.5*B7^2)*B6)/(B7*SQRT(B6))
11	d2	-1.558	<--=B10-B7*SQRT(B6)
12			
13	Cash-or-nothing call	2.978	<--=B8*EXP(-B3*B6)*NORM.S.DIST(B11,1)
14	Cash-or-nothing put	46.994	<--=B8*EXP(-B3*B6)*NORM.S.DIST(-B11,1)
15	Asset-or-nothing call	1.9403	<--=B2*EXP(-B4*B6)*NORM.S.DIST(B10,1)
16	Asset-or-nothing put	28.06	<--=B2*EXP(-B4*B6)*NORM.S.DIST(-B10,1)

Implementing the above in R looks like this:

```
# Inputs:
S0 <- 30
r <- 0.03
k <- 0
X <- 32
t<- 1/52
sigma_S <- 0.30
M <- 50 #cash payments

d1 <- (log(S0/X) + (r - k + 0.5 * sigma_S ^ 2) * t) / (sigma_S * sqrt(t))
d2 <- d1 - sigma_S * sqrt(t)

# Cash-or-nothing call
cash_nothing_call <- M * exp(-r*t) * pnorm(d2)

# Cash-or-nothing put
cash_nothing_put <- M * exp(-r*t) * pnorm(-d2)

# Asset-or-nothing call
asset_nothing_call <- S0 * exp(-k*t) * pnorm(d1)

# Asset-or-nothing put
asset_nothing_put <- S0 * exp(-k*t) * pnorm(-d1)
```

27.10 Chooser Options

Chooser exotic options provide the holder with the right to decide whether the option is a call or a put. The decision can be made only at a fixed date (T_1) prior to the expiration of the contracts (T_2). There is a closed-form solution to this option.

Note that at time T_1, the option value is $Max[C(X, T_2 - T_1), P(X, T_2 - T_1)]$. Using put-call-parity (PCP) we get:

$$Chooser = Max[C, C + Xe^{-r(T_2-T_1)} - S_{T_1}] = C + Max[0, Xe^{-r(T_2-T_1)} - S_1]$$

$$Chooser = C(X, T_2) + P(X \cdot e^{-r(T_2-T_1)}, T_1)$$

The spreadsheet below shows the implementation of this:

	A	B	C
1	PRICING A CHOOSER OPTION - CLOSED FORM APPROACH		
2	Initial stock price, S_0	70	
3	Interest rate, r	3%	
4	Strike price, X	75	
5	Time to exercise, T_1	0.167	<-- =2/12
6	Time to exercise, T_2	0.250	<-- =3/12
7	Sigma, σ	30%	
8			
9	Chooser option	8.64	<-- =BSCall(B2,B4,B6,B3,B7)+BSput(B2,B4*EXP(-B3*(B6-B5)),B5,B3,B7)

The implementation of the R version would look like this:

```
# Inputs:
S0 <- 70
r <- 0.03
X <- 75
t1 <- 2/12
t2 <- 3/12
sigma_s <- 0.30

# chooser option
chooser_option <- (fm5_bs_call(S0, X, t2, r, sigma_s) +
    fm5_bs_put(S0, 75 * exp(-r*(t2-t1)), t1, r, sigma_s))
```

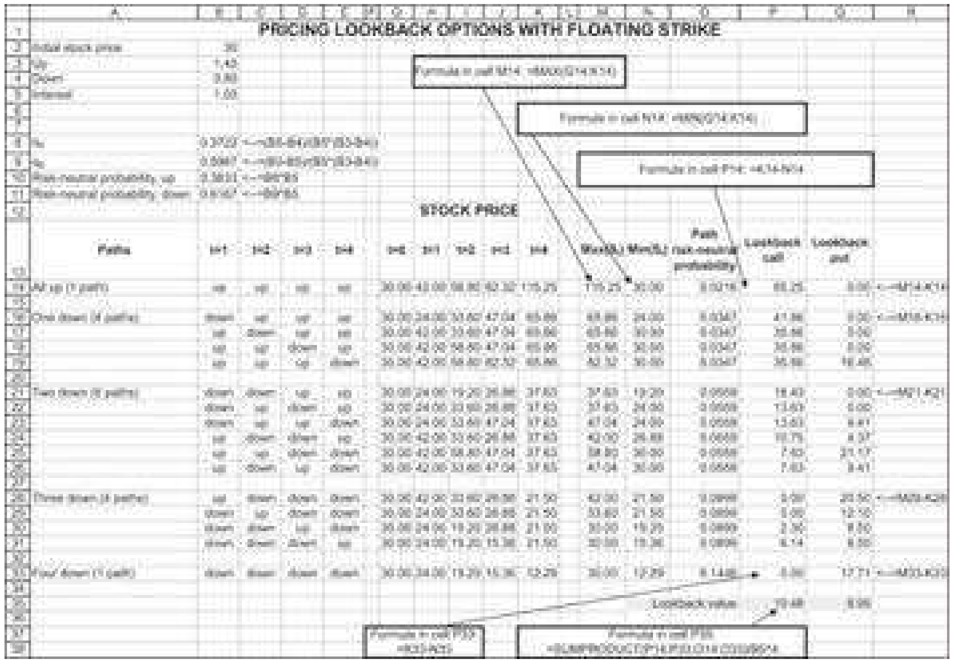
27.11 Lookback Options

A lookback option allows the holder to choose the most favorable strike price or asset price among the prices that have occurred during the lifetime of the options. There are two types of lookback options:

- • **Lookback option with floating strike**—where the strike price is determined at maturity as the optimal value of the underlying asset's price during the option life. For the call, the strike price is the asset's lowest price during the option's life; for the put, it is the asset's highest price. The payoff functions are given by:

- **Lookback call with floating strike**: $payoff = S_T - \min(S_t)$
- **Lookback put with floating strike**: $payoff = \max(S_t) - S_T$

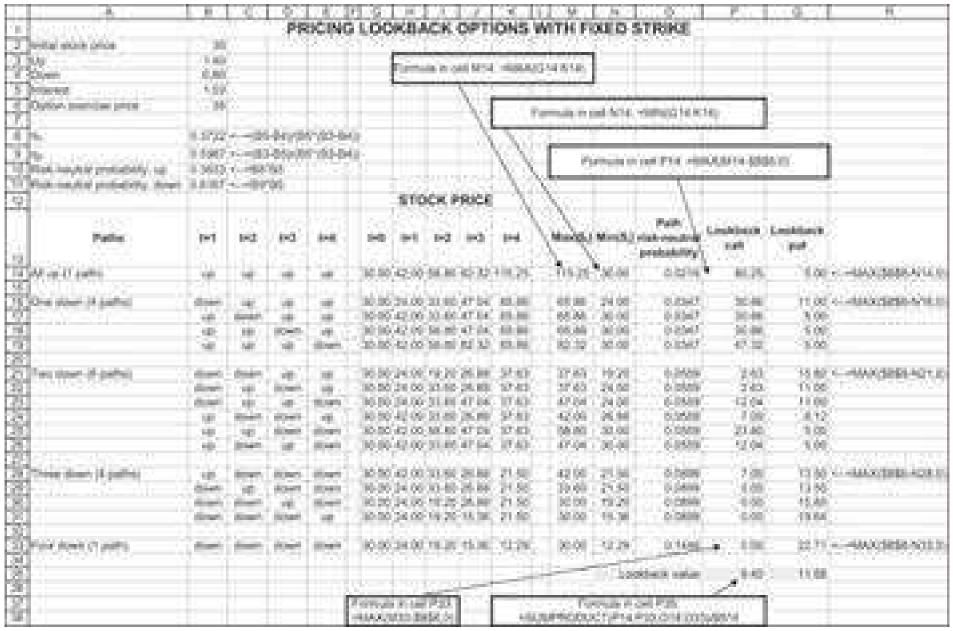
The spreadsheet below shows the implementation:



- • **Lookback option with fixed strike**—in this case the option's strike price is fixed. The option is exercised at the best price at maturity. The payoff is the maximum difference between the optimal underlying asset price and the strike. For the call option, the underlying asset price is at its highest level during the option's life. For the put option, the underlying price is at its lowest level during the option's life. The payoff functions are given by:

- **Lookback call with fixed strike:** $payoff = max(S_t) - X$
- **Lookback put with floating strike:** $payoff = X - \min(S_t)$

The spreadsheet below shows the implementation:



We leave it to the reader and the homework assignment at the end of this chapter to create a VBA function for the lookback option.

The R implementation is presented below. For the lookback options with floating strike, we use the following:

```
# Lookback option with floating strike
# Inputs:
S0 <- 30
U <- 1.4
D <- 0.8
r <- 1.03
m <- 4 # number of periods

# State prices
qU <- (r - D) / (r * (U - D))
qD <- (U - r) / (r * (U - D))

# Risk neutral probabilities
pi_U <- qU * r
pi_D <- qD * r

# A matrix of all binomial tree options
paths <- sapply(1:m, function(x){
    rep(c(rep("U", (2^x) / 2), rep("D", (2^x) / 2)), 2^(m-x))
                              })

# A matrix of Stock Prices
St <- cbind(S0 ,t(S0 * apply(ifelse(paths == "U", U, D), 1, cumprod)))

# Path risk-neutral probability
prob_mat <- apply(ifelse(paths == "U", pi_U, pi_D), 1, prod)

# Lookback Call Payoffs
lb_call_floating <- ((St[,m+1] - apply(St, 1, min)) %*% prob_mat) / r^m

# Lookback Put Payoffs
lb_put_floating <- ((apply(St, 1, max) - St[,m+1]) %*% prob_mat) / r^m
```

And for the lookback options with fixed price we use:

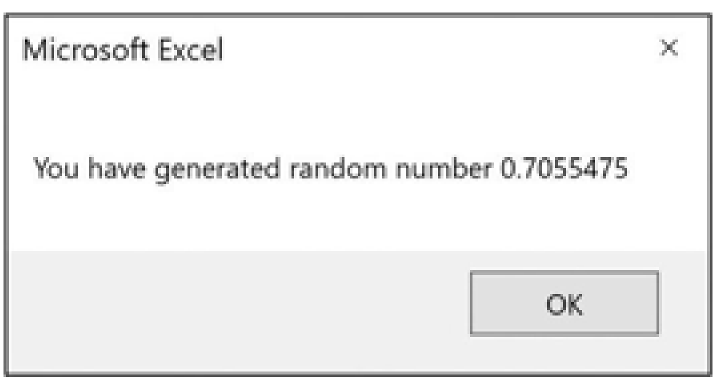
```r
# Input:
s0 <- 30
U <- 1.4
D <- 0.8
r <- 1.01
X <- 35
m <- 4 # number of periods

# State prices
qU <- (r - D) / (r * (U - D))
qD <- (U - r) / (r * (U - D))

# Risk neutral probabilities
pi_U <- qU * r
pi_D <- qD * r

# A matrix of all binomial tree options
paths <- sapply(1:m, function(x){
    rep(c(rep("U", (2^x) / 2), rep("D", (2^x) / 2)), 2^(m-x))
                              })

# A matrix of Stock Prices
St <- cbind(s0 ,t(s0 * apply(ifelse(paths == "U", U, D), 1, cumprod)))

# Path risk-neutral probability
prob_mat <- apply(ifelse(paths == "U", pi_U, pi_D), 1, prod)

# Lookback Put Payoffs
lb_put_fixed <- (pmax(X - apply(St, 1, min),0) %*% prob_mat) / r^m

# Lookback Call Payoffs
lb_call_fixed <- (pmax(apply(St, 1, max) - X,0) %*% prob_mat) / r^m
```

27.12 Summary

Monte Carlo methods—simulations of option pricing performed by tracing out many paths of the stock price—are at best a "second best" method of pricing. But in cases where no analytical formulas are available, Monte Carlo is easy to program in VBA and easy to see in Excel. In this chapter we have illustrated Monte Carlo methods for plain-vanilla options, Asian options, barrier options, basket options, rainbow options, binary options, chooser options, and lookback options. Other variations of path-dependent options and their Monte Carlo solutions are considered in the exercises.

Exercises

1. Create a VBA subroutine (call it **Exercise1()**) that generates a random number and prints it on the screen in a message box, which looks like this:



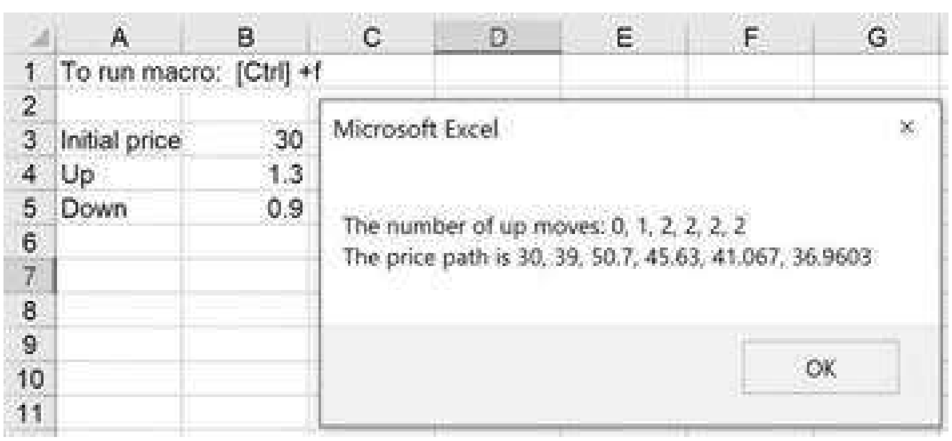
   Note: Use the VBA keyword **Rnd**.

2. Create a VBA subroutine (call it **Exercise2()**) that generates five random numbers and prints them out on the screen in a message box like the following:

Note: Use **FormatNumber(***Expression*,*NumDigitsAfterDecimal*) to print out only four digits.

3. Create a VBA subroutine (call it **Exercise3()**) that generates five random digits of either 1 or 0 and prints them out on the screen in a message box like the following:

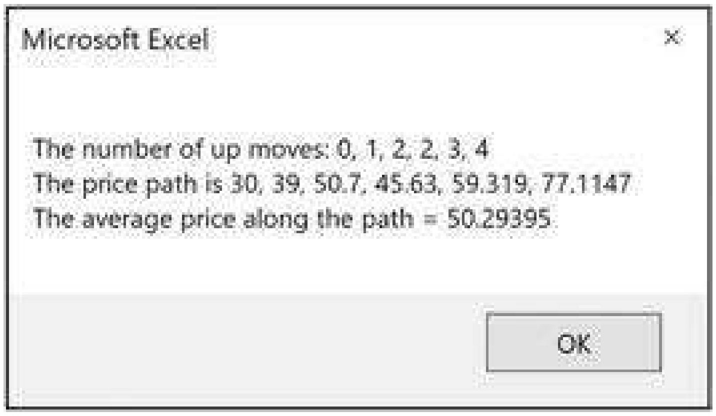
4. Suppose a stock price follows a binomial distribution. We want to create a *random price path* for the stock in a VBA macro. Here is the input with some sample output:
   Write a VBA an appropriate subroutine.



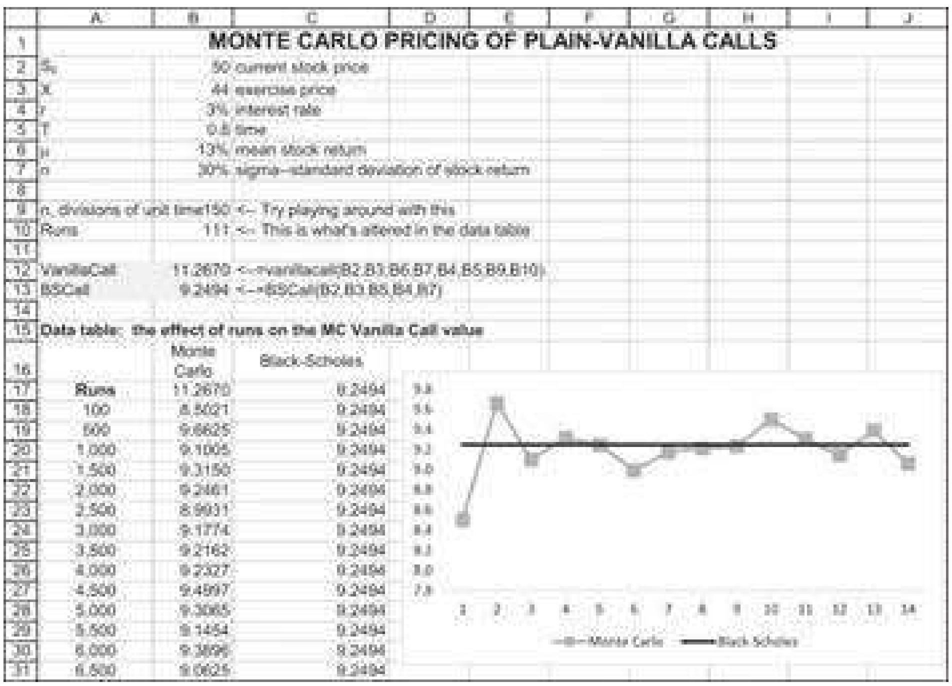
Notes:

- In the VBA message box, use **Chr(13)** to start a new line.
- Use range names in the spreadsheet to transfer values from the spreadsheet to the VBA routine.

5. Repeat the last exercise. This time compute the average price of a path:



6. Use the function **VanillaCall** defined in the chapter to create a **Data Table** in which you can see the relation between the number of runs incorporated in the function and the Black-Scholes value of a call. Your result should look something like the following:



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1. [1]. Properly speaking, *U*, *D*, and *R* are *one-plus* the growth and interest rates. For the sake of linguistic parsimony, we will use the terms "up growth," "down growth," and "interest rate," even though we mean something slightly different.
2. [2]. As discussed in Chapter 17, the risk-neutral probabilities are not the actual probabilities of the state occurrences. They are in fact "pseudo probabilities," which derive from the state prices.
3. [3]. See https://en.wikipedia.org/wiki/Asian_option for some definitions and a discussion of the literature. A list of references is also given at this link.
4. [4]. See Broadie, Glasserman, and Kou (1997) and Kou (2003) for a complete discussion.
5. [5]. Boolean functions are discussed in Chapter 30.

6. 6. We used a relatively small number of simulations (1,000) for simplicity. In real-life calculations, we suggest you use more than 10,000 simulations.

VI TECHNICAL

28 Data Tables

28.1 Overview

Data table commands are powerful commands that make it possible to do complex sensitivity analyses. Excel offers the opportunity to build a table in which only one variable is changed or one in which two variables are changed. Excel data tables are array functions and thus change dynamically when related spreadsheet cells are changed.

In this chapter you will learn how to build both one-dimensional and two-dimensional Excel data tables. At the end of this chapter we show how to hack Excel's data tables capabilities to perform multiple simulations for us—this is, using data tables on blank cells.

28.2 An Example

Consider a project that has an initial cost of \$1,000 and seven subsequent cash flows. The cash flows in years 1 to 5 grow at rate g, so that the cash flow in year t is $CF_t = CF_{t-1} * (1 + g)$. Given a discount rate r, the net present value (NPV) of the project is

$$NPV = -1{,}150 + \frac{CF_1}{(1+r)^1} + \frac{CF_1(1+g)}{(1+r)^2} + \frac{CF_1(1+g)^2}{(1+r)^3} + \cdots + \frac{CF_1(1+g)^4}{(1+r)^5}$$

The internal rate of return (IRR) is the rate at which the NPV equals zero:

$$0 = 1{,}150 + \frac{CF_1}{(1+IRR)^1} + \frac{CF_1(1+g)}{(1+IRR)^2} + \frac{CF_1(1+g)^2}{(1+IRR)^3} + \cdots + \frac{CF_1(1+g)^4}{(1+IRR)^5}$$

These calculations are easily done in Excel. In the following example the initial cash flow is 300, the growth rate g = 5%, and the discount rate r = 10%:

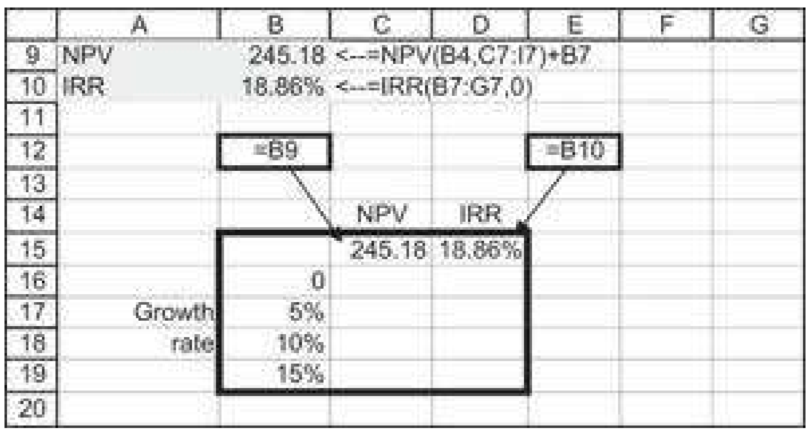
	A	B	C	D	E	F	G	H
1			DATA TABLE EXAMPLE					
2	CF_1	300						
3	Growth rate	5%						
4	Discount rate	10%						
5								
6	Year	0	1	2	3	4	5	
7	Cash flow	-1000.00	300.00	315.00	330.75	347.29	364.65	<--=F7*(1+$B$3)
8								
9	NPV	245.18	<--=NPV(B4,C7:I7)+B7					
10	IRR	18.86%	<--=IRR(B7:G7,0)					

Note the cell addresses for the growth rate, the discount rate, the NPV, and the IRR. They will be needed below.

28.3 Creating a One-Dimensional Data Table

We refer to a data table that does sensitivity analysis by varying one parameter as a *one-dimensional data table*. This section discusses such tables, and the next section discusses two-dimensional data tables.

Suppose we want to know how the NPV and IRR are affected by a change in the growth rate. The command **Data Table** allows us to do this simply. The first step is to set up the table's structure. In the example below, we put the formulas for the NPV and IRR on the top row and we put the variable we wish to vary (in this case the growth rate) in the first column. At this point the table looks like this:



The actual table (as opposed to the labels for the columns and the rows) is outlined in the dark border. The numbers directly under the labels "NPV" and "IRR" refer to the corresponding formulas in the previous picture. Thus, if the cell B9 contains the calculation for the NPV, then the cell under the letters "NPV" contains the formula "=B9." Similarly, if the cell B10 contains the original calculation for the IRR, then the cell under "IRR" in the table contains the formula "=B9."

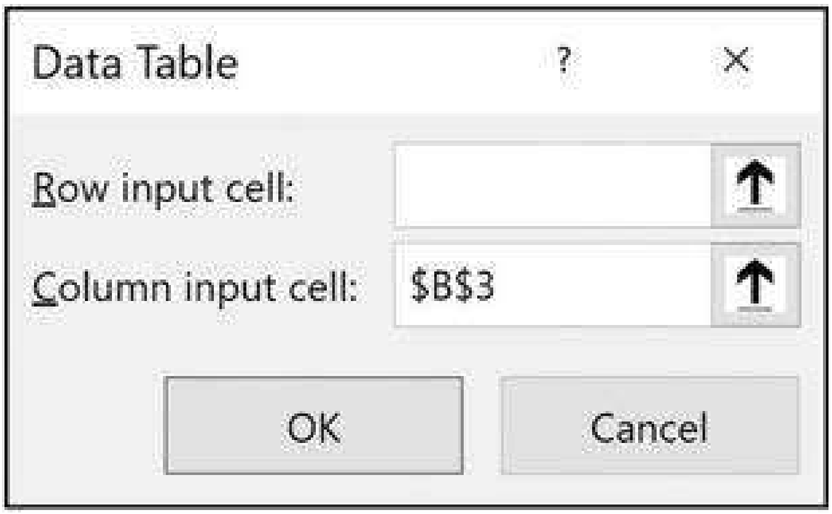
We like to think of a data table spreadsheet as having two parts:

- A basic example.
- A table that does a sensitivity analysis on the basic example. In our example, the first row of the table contains references to calculations done in our basic example. While there are other ways to do data tables, this structure is both typical and easy to understand.

Now do the following:

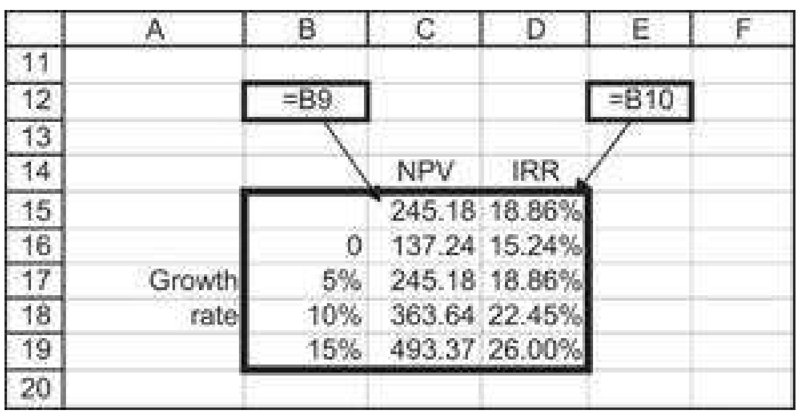
- • Highlight the table area (outlined in the dark border).

- • Activate the command **Data|What-if Analysis|Data Table**. You will get a dialog box that asks you to indicate a **Row input cell** and/or a **Column input cell**.



In this case, the variable we wish to change is in the left-hand column of our table, so we leave the **Row input cell** blank and indicate the cell B2 (this cell contains the growth rate in our basic example) in the **Column input cell** box.

Here's the result:



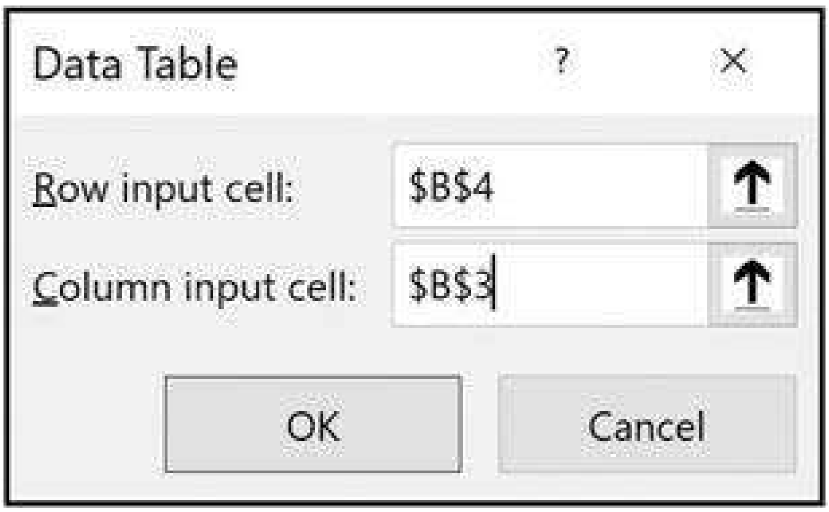
28.4 Creating a Two-Dimensional Data Table

We can also use the **Data Table** command to vary *one* formula while changing *two* parameters. Suppose, for example, that we want to calculate the NPV of the cash flows for different growth rates and different discount rates. We create a new table that looks like this:

	A	B	C	D	E	F
20						
21	=B9					
22			Discount rate ↓			
23		245.18	8%	10%	12%	
24	Growth	0				
25	rate -->	5%				
26		10%				
27		15%				
28						

The upper left-hand corner of the table contains the formula “=B9” as a reference to the NPV in our example.

We now use the **Data Table** command again. This time we fill in both the **Row input cell** (indicating cell B4, the site of the discount rate in our basic example) and the **Column input cell** (indicating B3).

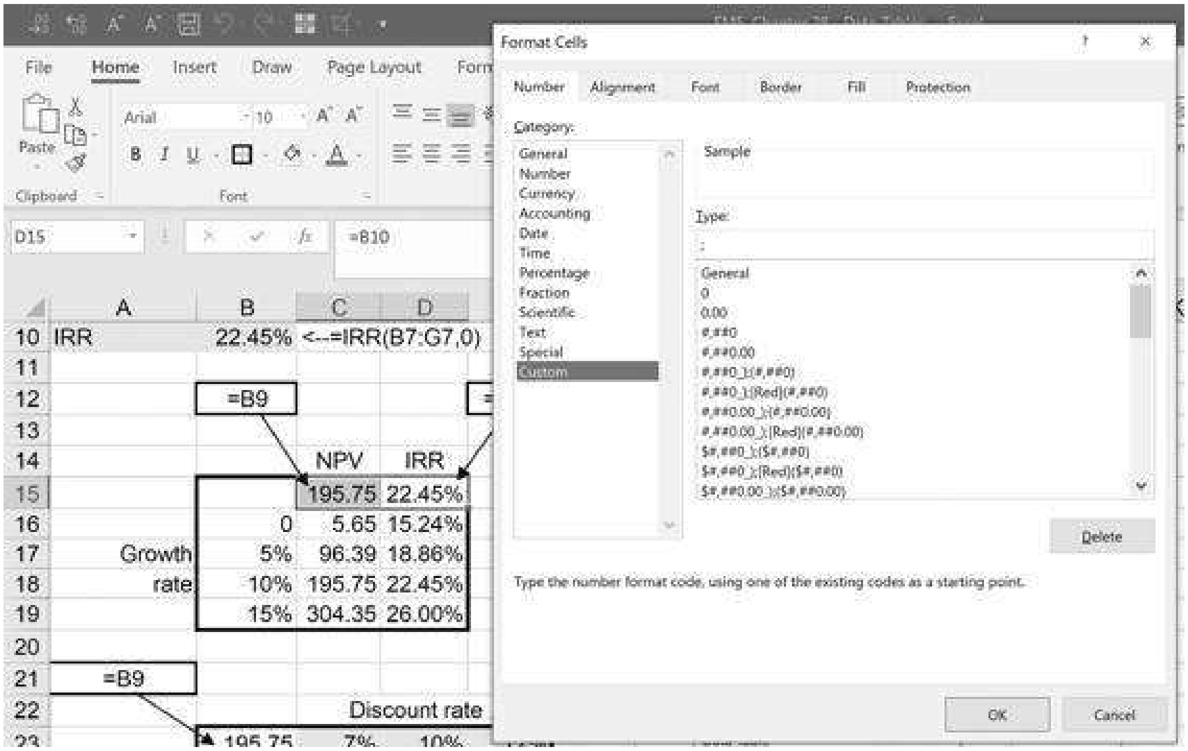


Here’s the result:

	A	B	C	D	E	F
20						
21	=B9					
22			Discount rate ↓			
23		245.18	8%	10%	12%	
24	Growth	0	197.81	137.24	81.43	
25	rate -->	5%	313.84	245.18	182.02	
26		10%	441.29	363.64	292.30	
27		15%	580.98	493.37	412.98	
28						

28.5 An Aesthetic Note: Hiding the Formula Cells

Data tables tend to look a bit strange because the formula being calculated shows up in the data table (in our examples, in the top row of the first data table and in the top left-hand corner of the second data table). You can make your tables look nicer by *hiding* the formula cells. To do this, mark the offending cells and use the **Format Cells** command (or press the right mouse button and click **Format Cells** and then go to the **Number|Custom**). In the dialog box, go to the box marked **Type** and insert a semicolon into the box. Here is the way this looks for the previous example:



The cell contents will now be hidden. This gives the following result:

	A	B	C	D	E	F
11						
12		=B9			=B10	
13						
14			NPV	IRR		
15						
16		0	5.65	15.24%		
17	Growth	5%	96.39	18.86%		
18	rate	10%	195.75	22.45%		
19		15%	304.35	26.00%		
20						

28.6 Excel Data Tables Are Arrays

This means that Excel data tables are dynamically linked to your initial example. When you change a parameter in the original example, the corresponding column or row of the data table changes. For example, if we change the initial cash flow from 300 to 250, here is what will happen in the data table pictured above:

	A	B	C	D	E	F	G	H
1			DATA TABLE EXAMPLE					
2	CF_1	250						
3	Growth rate	5%						
4	Discount rate	10%						
5								
6	**Year**	**0**	**1**	**2**	**3**	**4**	**5**	
7	Cash flow	-1000.00	250.00	262.50	275.63	289.41	303.88	<--=F7*(1+$B$3)
8								
9	NPV	37.65	<--=NPV(B4,C7:I7)+B7					
10	IRR	11.41%	<--=IRR(B7:G7,0)					
11								
12		=B9			=B10			
13								
14			NPV	IRR				
15			37.65	11.41%				
16		0	-52.30	7.93%				
17	Growth	5%	37.65	11.41%				
18	rate	10%	136.36	14.86%				
19		15%	244.47	18.27%				
20								

28.7 Data Tables on Blank Cells (Advanced)

An exciting use of data tables is to run multiple iterations of random simulations. As an example, suppose you use **Rand()** to create 10 random numbers between 0 and 1. This Excel random number generator is discussed in detail in Chapter 21. The simulation we are describing might look like this:

	A	B	C	D	E	F
1			**10 RANDOM NUMBERS**			
2	0.1275	<--=RAND()			**Statistics**	
3	0.1966	<--=RAND()		Mean	0.4205	<--=AVERAGE(A2:A11)
4	0.3828	<--=RAND()		Variance	0.0484	<--=VAR.P(A2:A11)
5	0.4432	<--=RAND()		Sigma	0.2201	<--=SQRT(E4)
6	0.6830	<--=RAND()				
7	0.6426	<--=RAND()				
8	0.5249	<--=RAND()				
9	0.6764	<--=RAND()				
10	0.0374	<--=RAND()				
11	0.4910	<--=RAND()				

If you want to repeat this experiment 10 times, your spreadsheet might look like this:

	A	B	C	D	E	F	G	H	I	J	K	L
1					**10 RANDOM NUMBERS X 10 TRIALS**							
2	0.0321	<--=RAND()			**Statistics**							
3	0.7751	<--=RAND()		Mean	0.2704	<--=AVERAGE(A2:A11)						
4	0.4597	<--=RAND()		Variance	0.0535	<--=VAR.P(A2:A11)						
5	0.0563	<--=RAND()		Sigma	0.2314	<--=SQRT(E4)						
6	0.0214	<--=RAND()										
7	0.0589	<--=RAND()										
8	0.3853	<--=RAND()										
9	0.4017	<--=RAND()										
10	0.2086	<--=RAND()										
11	0.3048	<--=RAND()										
12												
13						10 experiments: each cell contains **Rand()**						
14		**1**	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**	
15		0.0004	0.8487	0.5822	0.4819	0.2723	0.6480	0.1461	0.6003	0.2497	0.3836	
16		0.4091	0.3537	0.8759	0.9718	0.7690	0.1508	0.2380	0.1996	0.7986	0.1436	
17		0.0090	0.2638	0.7985	0.7628	0.4136	0.2802	0.1616	0.7620	0.6106	0.3422	
18		0.4860	0.7229	0.1836	0.1829	0.5291	0.3738	0.3736	0.9986	0.9144	0.1241	
19		0.4999	0.9571	0.3133	0.4363	0.3940	0.5596	0.2879	0.0023	0.0298	0.4717	
20		0.9928	0.7597	0.7331	0.2678	0.8915	0.3303	0.5704	0.6525	0.4376	0.1150	
21		0.3488	0.1388	0.6333	0.7346	0.8820	0.1777	0.9409	0.2371	0.5574	0.9611	
22		0.5326	0.0779	0.5316	0.9438	0.5237	0.6016	0.0497	0.8726	0.3886	0.0352	
23		0.3301	0.1572	0.2483	0.9835	0.3126	0.3475	0.8237	0.4885	0.0927	0.7312	
24		0.5442	0.7408	0.7728	0.8511	0.8275	0.9412	0.9067	0.2705	0.0237	0.8128	
25												
26	**Statistics for the 10 experiments**											
27	Mean	0.4153	0.5021	0.5673	0.6616	0.5815	0.4411	0.4529	0.5084	0.4103	0.4121	<--=AVERAGE(K15:K24)
28	Variance	0.0725	0.1009	0.0538	0.0798	0.0519	0.0538	0.1045	0.0949	0.0889	0.0958	<--=VAR.P(K15:K24)
29	Sigma	0.2693	0.3176	0.2319	0.2825	0.2279	0.2319	0.3232	0.3080	0.2982	0.3096	<--=SQRT(K28)

Using Data Table with Blank Cell Reference

There's a more efficient way of running this experiment, as illustrated below. Note that the cell referenced in the **Data Table** dialog box is **empty**.

	A	B	C	D	E	F
4	0.3038	<--=RAND()		Variance	0.0765	<--=VAR.P(A2:A11)
5	0.4873	<--=RAND()		Sigma	0.2766	<--=SQRT(E4)
6	0.8441	<--=RAND()				
7	0.3890	<--=RAND()				
8	0.7153	<--=RAND()				
9	0.2473	<--=RAND()				
10	0.4344	<--=RAND()				
11	0.9943	<--=RAND()				
12						
13		Data table				
14	**Experiment**	**Mean**	**Variance**	**Sigma**		
15		0.4953	0.0765	0.2766	<-- =E5, data table header	
16	**1**					
17	**2**					
18	**3**					
19	**4**					
20	**5**					
21	**6**					
22	**7**					
23	**8**					
24	**9**					
25	**10**					

Data Table ? ×

Row input cell:

Column input cell: $D$7

OK Cancel

The result looks something like this ("something" means it is an approximation because by the nature of this experiment, the results are random, so each recalculation, like pressing **F9**, will produce a different set of numbers):

	A	B	C	D	E	F
1	**BETTER WAY OF RUNNING THE EXPERIMENT**					
2	0.5888	<--=RAND()			**Statistics**	
3	0.6716	<--=RAND()		Mean	0.5170	<--=AVERAGE(A2:A11)
4	0.4516	<--=RAND()		Variance	0.0718	<--=VAR.P(A2:A11)
5	0.8478	<--=RAND()		Sigma	0.2679	<--=SQRT(E4)
6	0.1055	<--=RAND()				
7	0.2465	<--=RAND()				
8	0.7584	<--=RAND()				
9	0.7914	<--=RAND()				
10	0.6240	<--=RAND()				
11	0.0839	<--=RAND()				
12						
13		Data table				
14	**Experiment**	**Mean**	**Variance**	**Sigma**		
15		0.5170	0.0718	0.2679	<-- =E5, data table header	
16	**1**	0.4458	0.0759	0.2755		
17	**2**	0.3420	0.0346	0.1859		
18	**3**	0.6024	0.0701	0.2648		
19	**4**	0.4631	0.0785	0.2802		
20	**5**	0.5159	0.0742	0.2723		
21	**6**	0.3904	0.0385	0.1962		
22	**7**	0.5215	0.0937	0.3061		
23	**8**	0.3715	0.0892	0.2986		
24	**9**	0.4442	0.0696	0.2638		
25	**10**	0.6146	0.0806	0.2840		

A Slightly More Realistic Example (Also More Advanced)

Why use this technique? Go back to the simulated pension problems discussed in Chapter 24, section 24.6. Suppose you retire at age 75 with 1 million euros in savings. You want to invest this 60% in the risky asset that has a mean annual return $\mu = 11\%$ and a standard deviation of return $\sigma = 30\%$. You plan to withdraw 100,000 euros at the end of this year and each of the next nine years, for a total of 10 withdrawals.

The amount you'll have left at the end of 10 years depends on the random annual return on the risky asset. As explained in Chapter 24, this can be modeled by using ***Exp*[(*μ* − 0.5 * *σ*** ^{2}**) + *σ* * *Norm.S.Inv*(*rand*())]**. We embed this function and the assumptions in the spreadsheet below. The amount left is cell B24. In the example shown below, this number is positive, but it can also be negative—depending on the parameters.[1]

	A	B	C	D	E	F
1	PENSION PROBLEM					
2	Current savings	1,000,000				
3	Invested in					
4	Risky asset	60%				
5	Risk free asset	40%	<--=1-B4			
6	Annual withdrawal	100,000				
7						
8	Return paramets					
9	Risk free rate	4%				
10	Risky asset mean return μ	11%				
11	Risky asset sigma σ	30%				
12						
13	Age	Savings at beginning of year	Savings at end of year	Withdrawal at end of year	Net after withdrawal	
14	75	1,000,000	696,040	100,000	596,040	<--=C14-D14
15	76	596,040	594,241	100,000	494,241	
16	77	494,241	628,314	100,000	528,314	
17	78	528,314	424,906	100,000	324,906	
18	79	324,906	708,107	100,000	608,107	
19	80	608,107	525,396	100,000	425,396	
20	81	425,396	561,776	100,000	461,776	
21	82	461,776	589,250	100,000	489,250	
22	83	489,250	626,554	100,000	526,554	
23	84	526,554	382,577	100,000	282,577	
24	85	282,577				

=B14*$B$4*EXP((mu-sigma^2)+sigma*NORM.S.INV(RAND()))+$B$5*EXP(riskfree))

Now we use our new "data table on blank cell" technique to run 20 simulations to see what the final (cell B24) amount might be:

	A	B	C	D	E	F	G	H	I
1	PENSION PROBLEM—USING DATA TABLE ON BLANK CELL TO RUN 20 SIMULATIONS								
2	Current savings	1,000,000					Data table		
3	Invested in						Simulation	Ending amount	
4	Risky asset	60%						444,404	<--=B24, data table header
5	Risk free asset	40%	<--=1-B4				1	-78,779	<--{=TABLE(,E2)}
6	Annual withdrawal	100,000					2	290,880	
7							3	-403,106	
8	Return paramets						4	-45,091	
9	Risk free rate	4%					5	-339,303	
10	Risky asset mean return μ	11%					6	568,486	
11	Risky asset sigma σ	30%					7	5,640,588	
12							8	-62,341	
13	Age	Savings at beginning of year	Savings at end of year	Withdrawal at end of year	Net after withdrawal		9	-250,531	
14	75	1,000,000	942,023	100,000	842,023	<--=C14-D14	10	-343,578	
15	76	842,023	828,215	100,000	728,215		11	336,507	
16	77	728,215	917,842	100,000	817,842		12	-269,255	
17	78	817,842	1,209,828	100,000	1,109,828		13	-362,088	
18	79	1,109,828	1,080,695	100,000	980,695		14	-511,327	
19	80	980,695	804,533	100,000	704,533		15	-134,986	
20	81	704,533	659,213	100,000	559,213		16	2,910,208	
21	82	559,213	700,163	100,000	600,163		17	965,653	
22	83	600,163	851,260	100,000	751,260		18	-136,574	
23	84	751,260	544,404	100,000	444,404		19	870,565	
24	85	444,404					20	-486,760	

=B14*$B$4*EXP((mu-sigma^2)+sigma*NORM.S.INV(RAND()))+$B$5*EXP(riskfree))

Even Better

We can use a two-dimensional data table, combined with a data table on empty cells, to run many simulations for different annual amounts withdrawn:

	A	B	C	D	E	F	G	H
13	Age	Savings at beginning of year	Savings at end of year	Withdrawal at end of year	Net after withdrawal			
14	75	1,000,000	1,177,007	100,000	1,077,007	<--=C14-D14		
15	76	1,077,007	1,320,089	100,000	1,220,089			
16	77	1,220,089	922,158	100,000	822,158			
17	78	822,158	783,819	100,000	683,819			
18	79	683,819	680,089	100,000	580,089			
19	80	580,089	603,004	100,000	503,004			
20	81	503,004	602,318	100,000	502,318			
21	82	502,318	636,608	100,000	536,608			
22	83	536,608	551,380	100,000	451,380			
23	84	451,380	495,081	100,000	395,081			
24	85	395,081						
25								
26	Data table: Terminal legacy as function of withdrawal							
27	↓Simulation: → Annual withdrawal							
28	395,081	50,000	75,000	100,000	125,000	150,000	175,000	200,000
29	1							
30	2							
31	3							
32	4							
33	5							
34	6							
35	7							
36	8							
37	9							
38	10							
39	11							
40	12							
41	13							
42	14							
43	15							
44	16							
45	17							
46	18							
47	19							
48	20							

Data Table ? ×

Row input cell: $B$6

Column input cell: $F$25

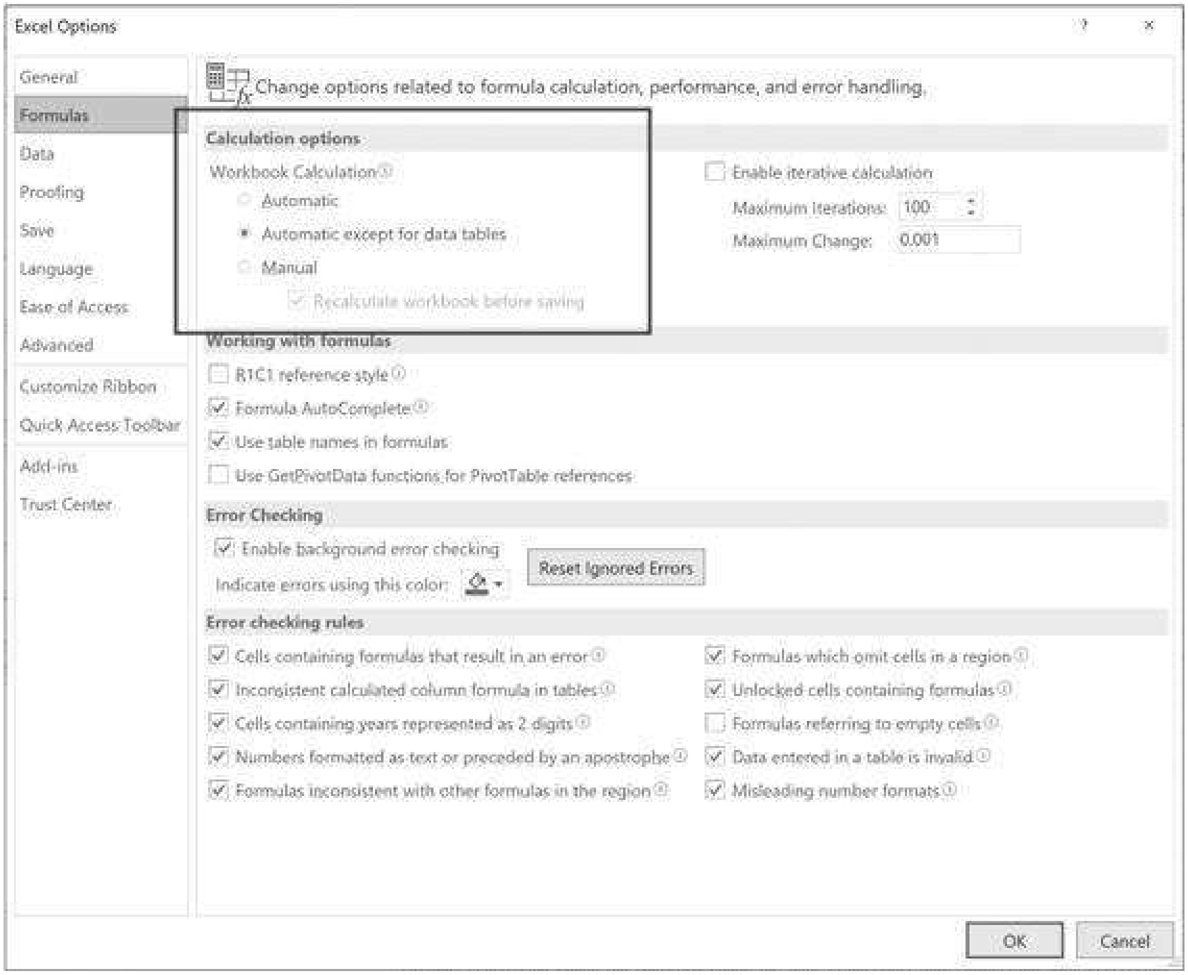
OK Cancel

Here are some results, showing 20 simulations for seven different annual withdrawal amounts. For fun and for a sanity check, we've also run some statistics in rows 50 and 51.

	A	B	C	D	E	F	G	H
26	Data table: Terminal legacy as function of withdrawal							
27	↓Simulation: → Annual withdrawal							
28	396,898	50,000	75,000	100,000	125,000	150,000	175,000	200,000
29	1	3,111,612	2,031,251	13,250	-138,863	-637,788	324,585	-564,845
30	2	2,884,438	259,987	-113,450	-724,760	-2,750,049	4,240,285	-524,119
31	3	1,056,579	3,138,210	-434,600	1,122,369	2,978,810	-798,117	-1,319,325
32	4	421,733	458,961	-358,437	2,368,624	-834,680	-625,589	-1,232,510
33	5	1,038,593	1,836,876	-402,542	-692,606	-1,015,306	-1,416,864	-676,858
34	6	1,031,642	422,165	-66,701	437,456	-2,706,315	853,145	-1,581,682
35	7	-53,562	-59,350	-353,067	293,204	-1,500,208	-471,495	-859,106
36	8	562,419	-88,265	1,747,466	-159,823	421,377	-914,636	-1,147,019
37	9	2,737,063	-173,716	-687,531	-3,120	-507,777	-1,607,479	-121,020
38	10	987,040	948,859	2,100,335	1,063,334	-152,207	-272,137	-722,380
39	11	1,145,094	-277,259	6,135,849	295,631	-1,284,026	81,735	-780,430
40	12	144,817	2,292,042	4,015,015	-617,300	1,860,748	-465,980	746,413
41	13	1,412,247	-138,034	-57,044	-925,456	-87,106	-1,385,487	518,009
42	14	1,419,055	-271,428	-281,127	-130,956	-606,542	221,781	-1,208,965
43	15	623,530	827,330	-674,147	-85,204	-996,976	883,534	-894,940
44	16	937,752	477,680	208,955	376,453	-747,478	-638,608	-1,490,481
45	17	-85,767	3,269,128	443,480	1,378,234	-694,229	-482,483	-1,041,159
46	18	1,343,319	-346,481	148,814	-613,199	-765,109	-835,999	-871,982
47	19	-91,966	68,282	477,676	-624,192	503,679	-1,139,468	-462,552
48	20	7,660,635	738,692	-80,659	-1,382,785	-1,044,675	-2,021,826	-1,374,778
49								
50	Average	1,424,314	770,741	594,077	61,852	-528,243	-333,605	-779,487
51	Sigma (Stdev.s)	1,775,110	1,136,917	1,719,488	897,942	1,307,319	1,312,672	610,764

28.8 Data Tables Can Stop Your Computer

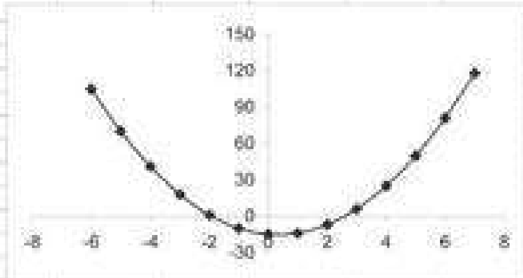
Data tables are amazing, but they can be an incredibly wasteful use of computer resources! A few juicy data tables can slow your spreadsheet to a crawl. One way to get around this is to set the recalculation to manual (**File|Options**):



Exercises

1. a. Use **Data|Table** to graph the function $f(x) = 3x^2 - 2x - 15$, as illustrated below:

	A	B	C	D
1	**USING DATA TABLE TO GRAPH A FUNCTION**			
2	x		3	
3	f(x)		6	<-- =3*B2^2-2*B2-15
4				
5				
6		x	6	<-- =B3, data table header
7		-6	105	
8		-5	70	
9		-4	41	
10		-3	18	
11		-2	1	
12		-1	-10	
13		0	-15	
14		1	-14	
15		2	-7	
16		3	6	
17		4	25	
18		5	50	
19		6	81	
20		7	118	



b. Use **Solver** or **Goal|Seek** to find two values of x for which $f(x) = 0$.

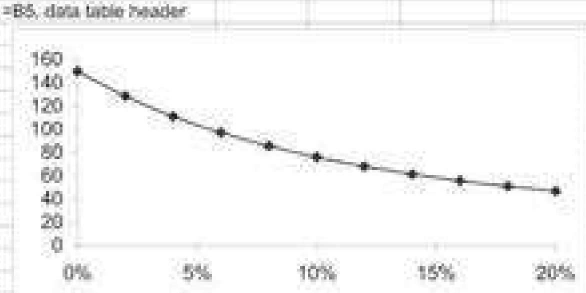
2. The Excel function **PV(rate, number_periods, payment)** calculates the present value of a constant payment. In the spreadsheet example below, for example:

$$PV(15\%, 15, -10) = \sum_{t=1}^{15} \frac{10}{(1+15\%)^t} = 58.47$$

(Note that we have put the payment as a negative number, since otherwise Excel returns a negative value! This little irritation is discussed in Chapters 1 and 30.)

Use **Data Table** to graph the present value as a function of the discount rate, as illustrated below:

	A	B	C
1	**DATA TABLE AND PV**		
2	Rate	15%	
3	Number of periods	15	
4	Payment	-10	To get a positive PV, we let the payment be negative (see Chapters 1 & 34)
5	Present value	$58.47	<-- =PV(B2,B3,B4)
6			
7			
8	**Rate**	$58.47	<-- =B5, data table header
9	0%	150.00	
10	2%	128.49	
11	4%	111.18	
12	6%	97.12	
13	8%	85.59	
14	10%	76.06	
15	12%	68.11	
16	14%	61.42	
17	16%	55.75	
18	18%	50.92	
19	20%	46.75	
20			
21			



3. The spreadsheet fragment below shows a net present value and internal rate of return calculation for a project:

	A	B	C	D	E	F	G	H
1		NPV, DISCOUNT AND GROWTH RATES						
2	Growth rate	10%						
3	Discount rate	15%						
4	Cost	500						
5	Year 1 cash flow	100						
6								
7	Year	0	1	2	3	4	5	
8	Cash flow	-500.00	100.00	110.00	121.00	133.10	146.41	
9								
10	NPV	(101.42)	<-- =NPV(B3,C8:G8)+B8					
11	IRR	6.60%	<-- =IRR(B8:G8)			Cell B15 contains the data table		
12						function =B10		
13								
14			Growth					
15		($101.42)	0%	3%	6%	9%	12%	
16		0%	0.00	30.91	63.71	98.47	135.28	
17	Discount rate	3%	-42.03	-14.56	14.55	45.38	78.01	
18		6%	-78.76	-54.26	-28.30	-0.84	28.21	
19		9%	-111.03	-89.08	-65.85	-41.28	-15.33	
20		12%	-139.52	-119.76	-98.91	-76.86	-53.57	
21		15%	-164.78	-146.97	-128.15	-108.28	-87.32	
22		18%	-187.28	-171.15	-154.13	-136.16	-117.23	
23		21%	-207.40	-192.75	-177.30	-161.01	-143.84	
24		24%	-225.46	-212.11	-198.04	-183.22	-167.62	

Use **Data Table** to do a sensitivity analysis on the NPV of the project, varying the discount rates from 0%, 3%, 6%, …, 21% and varying the growth rates from 0%, 3%, …, 12%.

4. Using **Data Table**, graph the function $sin(x * y)$ for $x = 0, 0.2, 0.4, \ldots, 1.8, 2$ and $y = 0, 0.2, 0.4, \ldots, 1.8, 2$. Use the "Surface" graph option to make a three-dimensional graph of the function.
5. Boris and Tareq are tossing coins. For each toss, if the coin falls on heads, Tareq wins $1. If the coin falls on tails, Tareq pays Boris $1.

- Simulate 10 rounds of this game, showing Tareq's cumulative winnings.
- Use **Data Table** on a blank cell to simulate 25 games of 10 rounds each, showing Tareq's cumulative winnings.

6. Maria and Shavit are tossing coins. Their game works as follows:

- On the first toss, if the coins falls on heads, Shavit pays Maria $1 (and vice versa).
- On each successive toss:
  - If the coin falls on heads and Maria is ahead, Shavit pays her the square of her previous winnings.
  - If the coin falls on heads and Shavit is ahead, this cancels all of Maria's debt to Shavit.

Simulate Maria's winnings after 10 tosses.

---

1. 1. This negative number is, of course, problematic! Maybe you have other resources, maybe you change your spending policy during the 10 years, or maybe you die sooner (heaven forfend!). We ignore all these issues.

29 Matrices

29.1 Overview

The portfolio optimization chapters of *Financial Modeling* (Chapters 11–12) make extensive use of matrices to find efficient portfolios. This chapter contains enough information about matrices to make it possible for you to do the calculations required for portfolio mathematics.

A matrix is a rectangular array of numbers. All of the following are matrices:

	A	B	C	D	E	F	G	H	I
1	MATRICES IN EXCEL								
2	Matrix A (a row vector)				Matrix B (square 3 x 3 matrix)				Matrix C (column vector)
3	2	3	4		13	-8	-3		13
4					-8	10	-1		-8
5					-3	-1	11		-3
6									
7	Matrix D (a 4 x 3 matrix)								
8	13	-8	-3						
9	-8	10	-1						
10	-3	-1	11						
11	0	13	3						

A matrix with only one row is also called a *row vector*; a matrix with only one column is also called a *column vector*. A matrix with an equal number of rows and columns is called a *square matrix*.

A single letter is often used to denote a matrix or a vector. In this case we often write, for example, $B = [b_{ij}]$, where b_{ij} stands for the entry in row i and column j of the matrix. For a vector we might write $A = [a_i]$ or $C = [c_i]$. Thus, for the examples given above:

$$a_3 = 4,\ b_{22} = 10,\ c_1 = 13,\ d_{41} = 0$$

The matrix B above is *symmetric*, meaning that $b_{ij} = b_{ji}$. (The variance-covariance matrices used in the portfolio discussion of Chapters 11–12 are symmetric.)

29.2 Matrix Operations

In this section we briefly review the basic operations on a matrix: multiplying a matrix by a scalar, adding matrices, transposition of matrices, and matrix multiplication.

Multiplication by a Scalar

Multiplying a matrix by a scalar multiplies every entry in the matrix by the scalar. Thus, for example:

	A	B	C	D	E
1	MULTIPLYING A MATRIX BY A SCALAR				
2	Scalar	6			
3					
4	Matrix B				
5	13	-8	-3		
6	-8	10	-1		
7	-3	-1	11		
8					
9	**Scalar * Matrix B**				
10	78	-48	-18	<--=C5*$B$2	
11	-48	60	-6		
12	-18	-6	66		

Matrix Addition

Matrices may be added together provided they have the same number of rows and columns. Adding two vectors or matrices is accomplished by adding their corresponding entries. Thus, if $A = [a_{ij}]$ and $B = [b_{ij}]$, $A + B = [a_{ij} + b_{ij}]$:

	A	B	C	D	E	F	G	H	I
1	ADDITION OF MATRICES								
2	Matrix A			Matrix B			Sum of A + B		
3	1	3		2	3		3	6	<--=B3+E3
4	3	0		23	5		26	5	
5	6	-9		8	6		14	-3	
6	5	11		-15	1		-10	12	
7	7	12		4	-1		11	11	

Matrix Transposition

Transposition is an operation by which the rows of a matrix are turned into columns and vice versa. Thus, for the matrix *E*:

	A	B	C	D	E	F	G	H	I
1	TRANSPOSITION OF MATRICES								
2	Matrix E					Transpose of E: E^T			
3	1	2	3	4		1	0	16	<--{=TRANSPOSE(A3:D5)}
4	0	3	77	-9		2	3	7	
5	16	7	7	2		3	77	7	
6						4	-9	2	
7									
8	Cells F3:H6 are generated with the array function Transpose(A3:D5). This function is inserted by marking off the target area, typing the formula, and then finishing by pressing [Ctrl]+[Shift]+[Enter] . See Chapter 31 for more details.								

The illustration above uses the array function **Transpose**. More details on the use of array functions are given in Chapter 31.

Multiplication of Matrices

You can multiply matrix A by matrix B to get product AB. However, you can only do this if the number of columns in A equals the number of rows in B. The resulting product AB is a matrix with the number of rows as A and the number of columns of B.

Confused? A couple of examples will help. Suppose that X is a row vector and that Y is a column vector, both with n coordinates:

$$X = [x_1 \quad \dots \quad x_n], Y = \begin{bmatrix} y_1 \\ \vdots \\ y_n \end{bmatrix}$$

Then the *product of X and the product of X and Y* are defined as follows:

$$XY = [x_1 \quad \dots \quad x_n] \begin{bmatrix} y_1 \\ \vdots \\ y_n \end{bmatrix} = \sum_{i=1}^{n} x_i y_i$$

Now suppose that A and B are two matrices, and that A has n columns and p rows and B has n rows and m columns:

$$A = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ \vdots & & \ddots & \vdots \\ a_{p1} & a_{p2} & \cdots & a_{pn} \end{bmatrix} B = \begin{bmatrix} b_{11} & \cdots & b_{1m} \\ b_{21} & \cdots & b_{2m} \\ \vdots & \ddots & \vdots \\ b_{n1} & \cdots & b_{nm} \end{bmatrix}$$

Then the product of A and B, written AB, is defined by the matrix:

$$AB = \begin{bmatrix} \sum_{h=1}^{n} a_{1h} b_{h1} & \sum_{h=1}^{n} a_{1h} b_{h2} & \cdots & \sum_{h=1}^{n} a_{1h} b_{hm} \\ \sum_{h=1}^{n} a_{2h} b_{h1} & \sum_{h=1}^{n} a_{2h} b_{h2} & \cdots & \sum_{h=1}^{n} a_{2h} b_{hm} \\ \vdots & \vdots & & \vdots \\ \sum_{h=1}^{n} a_{ph} b_{h1} & \sum_{h=1}^{n} a_{ph} b_{h2} & \cdots & \sum_{h=1}^{n} a_{ph} b_{hm} \end{bmatrix}, \textit{with } ij^{th} \textit{ element} = \sum_{h=1}^{n} a_{ih} b_{hj}$$

Note that the ij^{th} coordinate of AB is the product of the i^{th} row of A times the j^{th} column of B. Now, in this example, if

$$A=\begin{bmatrix} 2 & -6 \\ -9 & 3 \end{bmatrix},\ B=\begin{bmatrix} 6 & 9 & -12 \\ -5 & 2 & 4 \end{bmatrix} \text{ then } AB=\begin{bmatrix} 42 & 6 & -48 \\ -69 & -75 & 120 \end{bmatrix},$$

the order of matrix multiplication is critical. Multiplication of matrices is not commutative; that is, $AB \neq BA$. As the above example shows, the fact that it is possible to multiply A times B does not always imply that the multiplication BA is even defined.

To multiply matrices in Excel, we use the array function **MMult**:

	A	B	C	D	E	F
1	MULTIPLYING MATRICES					
2	Matrix A			Matrix B		
3	2	-6		6	9	-12
4	-9	3		-5	2	4
5						
6	Product AB					
7	42	6	-48	<--{=MMULT(A3:B4,D3:F4)}		
8	-69	-75	120			

To multiply two matrices together, the number of columns in the first matrix must equal the number of rows in the second. Thus, we can multiply A times B, but we cannot multiply B times A. If you try this in Excel, the function **MMult** will give you an error message:

	A	B	C	D	E	F
1	MATRIX MULTIPLICATION Number of columns of first matrix must equal number of rows of second matrix					
2	Matrix A			Matrix B		
3	2	-7		6	9	-12
4	0	3		-5	2	4
5						
6	Product BA					
7	#VALUE!	#VALUE!	#VALUE!	<--{=MMULT(D3:F4,A3:B4)}		
8	#VALUE!	#VALUE!	#VALUE!			

29.3 Matrix Inverses

A square matrix I is called the *identity matrix* if all its off-diagonal entries are 0 and all its diagonal entries are 1. Thus:

$$I=\begin{bmatrix} 1 & 0 & \cdots & 0 & 0 \\ 0 & 1 & \cdots & 0 & 0 \\ \vdots & \vdots & & \vdots & \vdots \\ 0 & 0 & & 1 & 0 \\ 0 & 0 & \cdots & 0 & 1 \end{bmatrix}$$

It is easy to confirm that multiplying any matrix A by the identity matrix of the proper dimension leaves that A unchanged. Thus, if I_n is an $n \times n$ identity matrix and A is an $n \times m$ matrix, $IA = A$. Similarly, if I_m is an $m \times m$ identity matrix, $AI = A$.

Now suppose we are given a *square* matrix A of dimension n. The $n \times n$ matrix A^{-1} is called the *inverse* of A if $A^{-1} A = AA^{-1} = I$. The computation of an inverse matrix can be a lot of work; fortunately, Excel has the array function **MInverse** to do the calculations for us. Here's an example:

	A	B	C	D	E	F	G	H	I	J
1				MATRIX INVERSE Use array function Minverse to compute the inverse of a square matrix						
2		Matrix A					Inverse of A			
3	1	-9	16	1		-0.0217	1.8913	0.5362	-1.1449	<--{=MINVERSE(A3:D6)}
4	3	3	2	3		0.0000	-1.0000	-0.1667	0.6667	
5	2	4	0	-2		0.0652	-0.6739	-0.1087	0.4348	
6	5	7	3	4		-0.0217	-0.1087	-0.2971	0.1884	
7										
8	Verifying the inverse									
9	We multiply A*Inverse(A):									
10	1.00	0.00	0.00	0.00	<--{=MMULT(A3:D6,F3:I6)}					
11	0.00	1.00	0.00	0.00						
12	0.00	0.00	1.00	0.00						
13	0.00	0.00	0.00	1.00						

As illustrated above, you can use **MMult** to verify that the product of the above matrix and its inverse indeed give the identity matrix. In the above example we used **Format|Cells|Number** to specify two decimal places in order to get rid of expressions like 1.07E-15 (this expression means $1.07 * 10^{-15}$ and thus essentially zero).

A square matrix that has an inverse is called a *nonsingular matrix*. The conditions for a matrix to be nonsingular are the following: Consider a square matrix A of dimension n. It can be shown that $A = [a_{ij}]$ is nonsingular if and only if the only solution to the n equations

$$\sum_i a_{ij} x_i = 0, \; j = 1, \dots, n$$

is $x_i = 0$, $i = 1, \dots, n$. Matrix inversion is a tricky business. If there exists a vector X whose components are almost zero and which solves the above system, then the matrix is *ill-conditioned*, and it may be very difficult to find an accurate inverse.

29.4 Solving Systems of Simultaneous Linear Equations

A system of n linear equations in m unknown is written as:

$$\begin{array}{ccccccccc}
a_{11}x_1 & + & a_{12}x_2 & + & \cdots & + & a_{1n}x_n & = & y_1 \\
a_{21}x_1 & + & a_{22}x_1 & + & \cdots & + & a_{2n}x_1 & = & y_2 \\
\vdots & & \vdots & & & & \vdots & & \vdots \\
a_{n1}x_1 & + & a_{n2}x_1 & + & \cdots & + & a_{nn}x_1 & = & y_n
\end{array}$$

Writing the matrix of coefficients as $A = [a_{ij}]$, the column vector of unknowns as $X = [x_j]$, and the column vector of constants as $Y = [y_j]$, we may write the above system in matrix notation as $AX = Y$.

Not every system of linear equations has a solution, and not every solution of such a system is unique. The system $AX = Y$ *always* has a unique solution, however, if the matrix A is square and nonsingular. In this case the solution is found by pre-multiplying both sides of the equation $AX = Y$ by the inverse of A:

since $AX = Y \Rightarrow A^{-1}AX = A^{-1}Y \Rightarrow X = A^{-1}Y$

Here is an example. Suppose we want to solve the 3 × 3 system of equations:

$$\begin{array}{ccccccc} 3x_1 & + & 4x_2 & + & 66x_3 & = & 16 \\ 0 & + & -33x_2 & + & x_3 & = & 77 \\ 42x_1 & + & 3x_2 & + & 2x_3 & = & 12 \end{array}$$

We set this up and solve it in Excel as follows:

	A	B	C	D	E	F	G	H
1			SOLVING SIMULTANEOUS EQUATIONS					
2		Matrix A of coefficients			Column vector y		Solution $A^{-1}Y$	
3	3	4	66		16		0.4343	
4	0	-33	1		77		-2.3223	<--{=MMULT(MINVERSE(A3:C5),E3:E5)}
5	42	3	2		12		0.3634	
6								
7	Checking that the solution works							
8					16			
9	A	x	B	=	77	<--{=MMULT(A3:C5,G3:G5)}		
10					12			

In cells E8:E10 we check that the solution indeed solves the system by multiplying the matrix A times the column vector G3:G5.

Exercises

1. Use Excel to perform the following matrix operations:

   a. $\begin{bmatrix} 2 & 12 & 6 \\ 4 & 8 & 7 \\ 1 & 0 & -9 \end{bmatrix} + \begin{bmatrix} 1 & 1 & 2 \\ 8 & 0 & -23 \\ 1 & 7 & 3 \end{bmatrix}$

   b. $\begin{bmatrix} 2 & -9 \\ 5 & 0 \\ 6 & -6 \end{bmatrix} \begin{bmatrix} 3 & 1 & 1 \\ 2 & 3 & 2 \end{bmatrix}$

   c. $\begin{bmatrix} 2 & 0 & 6 \\ 4 & 8 & 7 \\ 1 & 0 & -9 \end{bmatrix} \begin{bmatrix} 1 & 1 & 2 \\ 8 & 0 & -2 \\ 1 & 7 & 3 \end{bmatrix}$

2. Find the inverses of the following matrices:

a. $\begin{bmatrix} 1 & 2 & 8 & 9 \\ 2 & 5 & 3 & 0 \\ 4 & 4 & 2 & 7 \\ 5 & -2 & 1 & 6 \end{bmatrix}$

b. $\begin{bmatrix} 3 & 2 & 1 \\ 6 & -1 & 3 \\ 7 & 4 & 3 \end{bmatrix}$

c. $\begin{bmatrix} 20 & 2 & 3 & -3 \\ 2 & 10 & 2 & -2 \\ 3 & 2 & 40 & 9 \\ -3 & -2 & 9 & 33 \end{bmatrix}$

3. Transpose the following matrices using the Excel array function **Transpose**.

a. $A = \begin{bmatrix} 3 & 2 & 1 \\ -15 & 4 & 1 \\ 6 & -9 & 1 \end{bmatrix}$

b. $B = \begin{bmatrix} 1 & 2 & 3 & 4 & 5 \\ -2 & 7 & -9 & 0 & 0 \\ 3 & -3 & 11 & 12 & 1 \end{bmatrix}$

4. Solve the following system of equations by using matrices:

$$\begin{aligned} 3x \ +4y \ -6z \ -9w &= 15 \\ 2x \ -y \ \qquad +w &= 2 \\ y \ +z \ +w &= 3 \\ x \ +y \ -z \qquad &= 1 \end{aligned}$$

5. Solve the equations $AX = Y$, where:

$$A = \begin{bmatrix} 13 & -8 & -3 \\ -8 & 10 & -1 \\ -3 & -1 & 11 \end{bmatrix}, \ Y = \begin{bmatrix} 20 \\ -5 \\ 0 \end{bmatrix}, \ X = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

6. An ill-conditioned matrix is a matrix that "almost doesn't have" an inverse. A set of examples of such matrices are Hilbert matrices. An n-dimensional Hilbert matrix looks like this:

$$H_n = \begin{bmatrix} 1 & 1/2 & \cdots & 1/n \\ 1/2 & 1/3 & \cdots & 1/(n+1) \\ \vdots & & & \\ 1/n & 1/(n+1) & & 1/(2n-1) \end{bmatrix}$$

a. Calculate the inverses of H_2, H_3, and H_8.

b. Consider the following system of equations:

$$H_n \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} = \begin{bmatrix} 1 + 1/2 + \cdots + 1/n \\ 1/2 + 1/3 + \cdots 1/(n+1) \\ \vdots \\ 1/n + 1/(n+1) + \cdots + 1/(2n-1) \end{bmatrix}$$

Find the answers to these problems by inspection.

c. Now solve $H_n * X = Y$ for $n = 2, 8, 14$. How do you explain the differences?

30 Excel Functions

30.1 Overview

Excel contains several hundred functions. This chapter surveys only the main functions used in this book. The functions discussed are the following:

- • Financial functions: **NPV**, **IRR**, **PV**, **PMT**, **XIRR**, **XNPV**, **IPMT**, and **PPMT**
- • Date functions: **Now**, **Today**, **Date**, **Weekday**, **Month**, **Datedif**
- • Statistical functions: **Average**, **Var**, **Varp**, **Stdev**, **Stdevp**, **Correl**, **Covar**
- • Regression functions: **Slope**, **Intercept**, **Rsq**, **Linest**
- • Conditional functions: **If**, **Countif**, **Countifs**, **Averageif**, **Averageifs**, **Countif**
- • Reference function: **VLookup**, **HLookup**, **XLookup**, **Offset**
- • **Large**, **Rank**, **Percentile**, **Percentrank**
- • **Count**, **CountA**

A separate chapter, Chapter 31, is devoted to the important topic of array functions.

30.2 Financial Functions

Excel has a large number of financial functions. The main functions used in this book are explored in this section.

NPV

The Excel definition of net present value (NPV) differs somewhat from the standard finance definition. In the finance literature, the NPV of a sequence of cash flows C_0, C_1, C_2, …, C_n at a discount rate r refers to the following expression:

$$NPV = \sum_{t=0}^{n} \frac{C_t}{(1+r)^t} \text{ or } NPV = C_0 + \sum_{t=1}^{n} \frac{C_t}{(1+r)^t}$$

In many cases C_0 represents the cost of the asset purchased and is therefore negative.

The Excel definition of **NPV** always assumes that the first cash flow occurs after one period. The user who wants the standard finance expression must therefore calculate **NPV(r,(C_1, …, C_n)) + C_0**. Here is an example:

	A	B	C	D
1	EXCEL'S NPV FUNCTION			
2	Discount rate	5%		
3				
4	Year	Cash flow	Present value	
5	0	-100.00	-100.00	<--=B5/(1+$B$2)^A5
6	1	35.00	33.33	
7	2	33.00	29.93	
8	3	34.00	29.37	
9	4	25.00	20.57	
10	5	16.00	12.54	
11				
12	PV(future cash flows)	125.74	<--=SUM(C6:C10)	
13	PV(future cash flows)	125.74	<--=NPV(B2,B6:B10)	
14	Net present value	25.74	<--=SUM(C5:C10)	
15	Net present value	25.74	<--=B5+NPV(B2,B6:B10)	

It is important to note that the **NPV** function differentiates between blank cells and cells containing zeros. This can cause some confusion, as can be seen in the next example. The present value of the cash flows in B5:B7 is 86.38, which corresponds to $\frac{100}{1.05^3}$. But in the otherwise similar example of the cash flows in B11:B13, Excel's NPV function regards the first cash flow as being 100 and returns the answer $\frac{100}{1.05} = 95.24$. So, in using **NPV**, you have to be explicit in putting in zeros for zero cash flows.

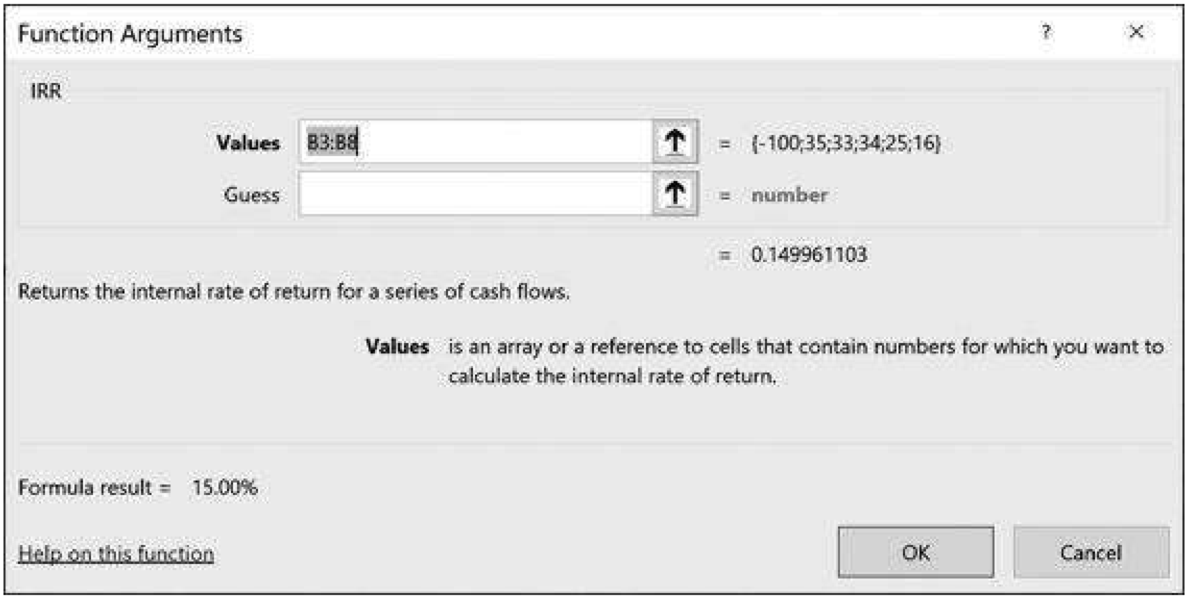
	A	B	C
1	NPV IGNORES BLANK CELLS!		
2	Discount rate, r	5%	
3			
4	Year	Cash flow	
5	1	0.00	
6	2	0.00	
7	3	100.00	
8	Present value	86.38	<--=NPV(B2,B5:B7)
9			
10	Year	Cash flow	
11	1		
12	2		
13	3	100.00	
14	Present value	95.24	<--=NPV(B2,B11:B13)

IRR

The internal rate of return (IRR) of a sequence of cash flows C_0, C_1, C_2, …, C_n is an interest rate r such that the net present value of the cash flows is zero:

$$\sum_{t=0}^{n} \frac{C_t}{(1+IRR)^t} = 0$$

The Excel syntax for the **IRR** function is **IRR(cash flows**, **guess)**. Here, **cash flows** represent the whole sequence of cash flows, including the first cash flow C_0, and **guess** is an optional starting point for the algorithm that calculates the IRR.

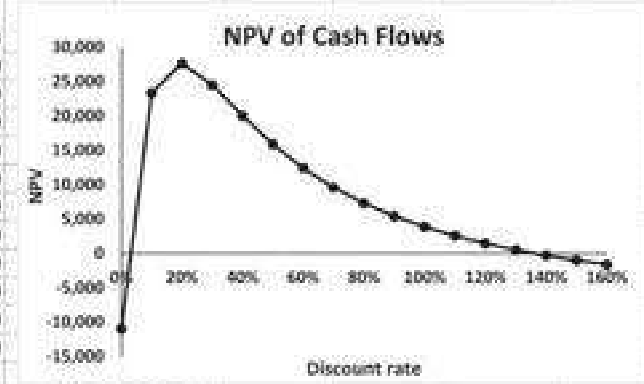


Guess is not required, and when there is only one IRR it is usually irrelevant.

	A	B	C
1	EXCEL'S IRR FUNCTION		
2	Year	Cash flow	
3	0	-100	
4	1	35	
5	2	33	
6	3	34	
7	4	25	
8	5	16	
9			
10	IRR	15.00%	<--=IRR(B3:B8)
11	IRR	15.00%	<--=IRR(B3:B8), IRR Guess = 5%

If there are multiple IRRs, then the choice of **guess** can make a difference. Consider, for example, the following cash flows:

	A	B	C
1			MULTIPLE IRRs
2	Year	Cash flows	
3	0	-11,000	
4	1	15,000	
5	2	15,000	
6	3	15,000	
7	4	15,000	
8	5	15,000	
9	6	15,000	
10	7	15,000	
11	8	15,000	
12	9	15,000	
13	10	-135,000	
14			
15	IRR(1)	1.86%	<--=IRR(B3:B13,0)
16	IRR(2)	135.99%	<--=IRR(B3:B13,2)



The graph (created from a **Data Table** that is not shown) shows that there are two IRRs, since the NPV curve crosses the *x*-axis twice. To find both these IRRs, we must change the **guess** (though the precise value of **guess** is still not critical). In the example below we have changed both guesses but still get the same answer:

	A	B	C
15	IRR(1)	1.86%	<--=IRR(B3:B13,0)
16	IRR(2)	135.99%	<--=IRR(B3:B13,3)

Note: A given set of cash flows typically has more than one IRR if there is more than one change of sign in the cash flows. In the above example, the initial cash flow is negative, and CF_1 to CF_9 are positive (this accounts for one change of sign); but then CF_{10} is negative, making a second change of sign. If you suspect that a set of cash flows has more than one IRR, the first thing to do is to use Excel to make a graph of the NPVs, as we did above. The number of times that the NPV graph crosses the *x*-axis identifies the number of IRRs (and also their approximate values).

PV

This function calculates the present value (PV) of an annuity, which is a series of fixed periodic payments. For example:

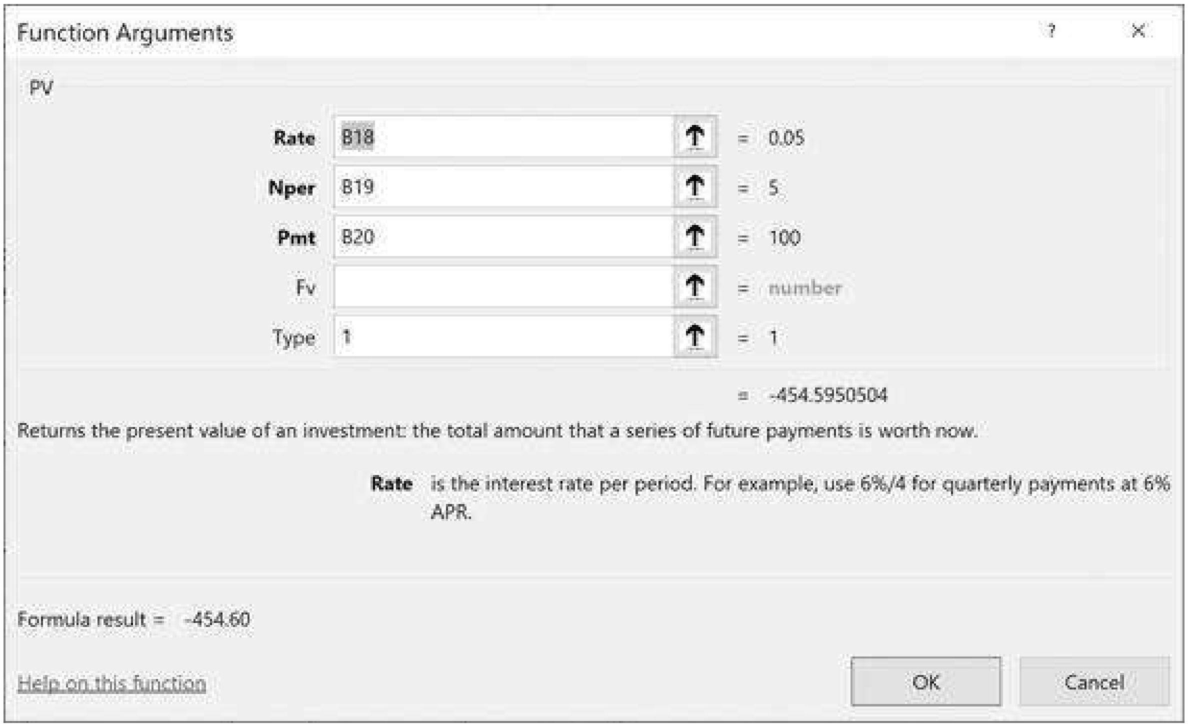
	A	B	C	D
1	THE PV FUNCTION			
2	Payments made at the end of the period			
3	Rate	5%		
4	Number of periods	5		
5	Periodic payment	100.00		
6	Present value	-432.95	<--=PV(B3,B4,B5)	
7	Present value	432.95	<--=SUM(C11:C15)	
8				
9	Year	Cash flows	Present value	
10	0			
11	1	100	95.24	<--=B11/(1+$B$3)^A11
12	2	100	90.70	
13	3	100	86.38	
14	4	100	82.27	
15	5	100	78.35	<--=B15/(1+$B$3)^A15

Thus, $\$432.95 = \sum_{t=1}^{5} \frac{100}{(1.05)^t}$. Here are two things to note about the **PV** function:

- • Writing **PV(B3,B4,B5)** assumes that payments are made at dates 1, 2, …, 5. If the payments are made at dates 0, 1, 2, 3, 4, the spreadsheet below shows what you should write:

	A	B	C	D
17	Payments made at the beginning of the period			
18	Rate	5%		
19	Number of periods	5		
20	Periodic payment	100.00		
21	Present value	-454.60	<--=PV(B18,B19,B20,,1)	
22	Present value	454.60	<--=SUM(C25:C29)	
23				
24	Year	Cash flows	Present value	
25	0	100	100.00	
26	1	100	95.24	<--=B26/(1+$B$18)^A26
27	2	100	90.70	
28	3	100	86.38	
29	4	100	82.27	<--=B29/(1+$B$18)^A29
30	5			

The formula **PV(B18,B19,B20,,1)** can also be generated from the dialog box:



- • Irritatingly, the **PV** function (and the **PMT**, **IPMT**, and **PPMT** functions—see below) produces a negative answer when the payment or future value is positive (there is a logic here, but it's not worth explaining). The solution is obvious: Either write **-PV(B3,B4,B5)** or let the payment be negative by writing **PV(B3,B4,-B5)**.

Excel's approach to calculate the annuity is by solving the following formula:

$$0 = PV + \frac{PMT}{RATE} * \left[1 - \left(\frac{1}{(1+RATE)^{NPER}}\right)\right] + \frac{FV}{(1+RATE)^{NPER}}$$

Looking at this representation we understand why the PV is negative. We can also understand the relationship between the functions **PV(), PMT(), RATE(), NPER(),** and **FV()** in Excel.

PMT

This function calculates the payment necessary to pay off a loan with equal payments over a fixed number of periods. For example, the first calculation below shows that a loan of $1,000, which is to be paid off over 10 years at an interest rate of 8%, will require equal annual payments of interest and principal of $149.03. The calculation performed is the solution of the following equation:

$$Initial\ loan\ principal = \sum_{t=1}^{n} \frac{X}{(1+r)^t}$$

	A	B	C
1	**THE PMT FUNCTION**		
2	**Payments made at the end of the period**		
3	Rate	8%	
4	Number of periods	10.00	
5	Principal	1,000.00	
6	Payment	-149.03	<--=PMT(B3,B4,B5)
7			
8	**Payments made at the beginning of the period**		
9	Rate	8%	
10	Number of periods	10.00	
11	Principal	1,000.00	
12	Payment	-137.99	<--=PMT(B9,B10,B11,,1)

Mortgage loan amortization tables can be calculated using the **PMT** function. These tables—explained in detail in Chapter 1—show the split between interest and principal of each payment. In each period, the payment on the loan (calculated with **PMT**) is split:

- • We first calculate the interest owing for that period on the principal outstanding at the beginning of the period. In the table below, at the end of year 1, we owe $80 (= 8% * $1,000) of interest on the loan principal outstanding at the beginning of the year.
- • The remainder of the payment (for year 1, $69.03) goes to reduce the principal outstanding.

	A	B	C	D	E	F
1	**LOAN AMORTIZATION TABLE (MORTGAGE)**					
2	Interest	8%				
3	Number of periods	10				
4	Principal	1,000				
5	Annual payment	149.03	<--=-PMT(B2,B3,B4)			
6					=$B$2*B10	
7	=B10-E10		=$B$5			
8				Split payment into		
9	**Year**	**Principal at beginning of year**	**Payment**	**Interest**	**Repayment of principal**	
10	1	1,000.00	149.03	80.00	69.03	<--=C10-D10
11	2	930.97	149.03	74.48	74.55	
12	3	856.42	149.03	68.51	80.52	
13	4	775.90	149.03	62.07	86.96	
14	5	688.95	149.03	55.12	93.91	
15	6	595.03	149.03	47.60	101.43	
16	7	493.60	149.03	39.49	109.54	
17	8	384.06	149.03	30.73	118.30	
18	9	265.76	149.03	21.26	127.77	
19	10	137.99	149.03	11.04	137.99	

Note that at the end of the 10 years, the repayment of principal is exactly equal to the principal outstanding at the beginning of the year (i.e., the loan has been paid off).

The Functions IPMT and PPMT

As you have seen above, a loan table shows the split of a loan's flat payments (computed with **PMT**) between interest and principal. In the loan table of the previous subsection, we computed this split by first computing the flat payment per period (column C), then taking the interest on the principal at the beginning of the period (column D), and finally subtracting this interest from the periods total payment (column E).

IPMT and **PPMT** perform this calculation without the necessity of relying on the total payment. Here's an example:

	A	B	C	D	E
1			IPMT AND PPMT		
2	Interest	8%			
3	Number of periods	10			
4	Principal	1,000			
5					
6	Year	Principal payment at end year		Interest payment at end year	
7	1	69.03	<--=PPMT($B$2,A7,$B$3,-$B$4)	80.00	<--=IPMT($B$2,A7,$B$3,-$B$4)
8	2	74.56	<--=PPMT($B$2,A8,$B$3,-$B$4)	74.48	<--=IPMT($B$2,A8,$B$3,-$B$4)
9	3	80.52		68.51	
10	4	86.96		62.07	
11	5	93.91		55.12	
12	6	101.43		47.60	
13	7	109.54		39.49	
14	8	118.30		30.73	
15	9	127.77		21.26	
16	10	137.99		11.04	

As you can see, the payments computed are the same as in the loan table of the previous subsection.

The Functions XIRR and XNPV

The functions **XIRR** and **XNPV** calculate the internal rate of return and the net present value for a series of cash flows received on specific dates. They are especially useful for calculating IRR and NPV when the dates are unevenly spaced.[1]

XIRR

Here's an example: You pay $600 on February 16, 2021, for an asset that repays $100 on April 5; $100 on July 15, 2021; and then $100 on every September 22 from 2021 until 2029. The dates are not evenly spaced, so you cannot use **IRR**. With **XIRR** (cell B16 below), you can compute the *annualized IRR*, the effective annual interest rate (EAIR), as defined in Chapter 1.

	A	B	C
1	THE EXCEL XIRR FUNCTION		
2	Date	Payment	
3	16-Feb-21	-600	
4	05-Apr-21	100	
5	15-Jul-21	100	
6	22-Sep-21	100	
7	22-Sep-22	100	
8	22-Sep-23	100	
9	22-Sep-24	100	
10	22-Sep-25	100	
11	22-Sep-26	100	
12	22-Sep-27	100	
13	22-Sep-28	100	
14	22-Sep-29	100	
15			
16	XIRR	21.97%	<--=XIRR(B3:B14,A3:A14)

The **XIRR** function works by discounting each cash flow at the daily rate. In our example, the first cash flow of $100 occurs 48 days from now, the second in 149 days, and so on. The **XIRR** transforms 21.97% to a daily rate and uses it to discount the cash flows:

$$-600 + \frac{100}{(1.2197)^{48/365}} + \frac{100}{(1.2197)^{149/365}} + \cdots + \frac{100}{(1.2197)^{3140/365}} = 0$$

	A	B	C	D	E
1	HOW DOES XIRR WORK? XIRR computes the daily internal rate of return				
2	Date	Payment	Days from initial date	Present value	
3	16-Feb-21	-600		-600.00	<--=B3
4	05-Apr-21	100	48	97.42	<--=B4/(1+$B$16)^(C4/365)
5	15-Jul-21	100	149	92.21	<--=B5/(1+$B$16)^(C5/365)
6	22-Sep-21	100	218	88.81	<--=B6/(1+$B$16)^(C6/365)
7	22-Sep-22	100	583	72.81	<--=B7/(1+$B$16)^(C7/365)
8	22-Sep-23	100	948	59.70	<--=B8/(1+$B$16)^(C8/365)
9	22-Sep-24	100	1,314	48.91	
10	22-Sep-25	100	1,679	40.10	Cell C4 contains the formula
11	22-Sep-26	100	2,044	32.88	=A4-$A$3
12	22-Sep-27	100	2,409	26.96	
13	22-Sep-28	100	2,775	22.09	
14	22-Sep-29	100	3,140	18.11	
15					
16	XIRR	21.97%	<--=XIRR(B3:B14,A3:A14)	0.00	<--=SUM(D3:D14)

XNPV

The **XNPV** function computes the NPV for unevenly spaced cash flows. In the example below, we use the function to compute the NPV on the same example we used for **XIRR**.

	A	B	C
1	THE EXCEL XNPV FUNCTION		
2	Date	Payment	
3	16-Feb-21	-600	
4	05-Apr-21	100	
5	15-Jul-21	100	
6	22-Sep-21	100	
7	22-Sep-22	100	
8	22-Sep-23	100	
9	22-Sep-24	100	
10	22-Sep-25	100	
11	22-Sep-26	100	
12	22-Sep-27	100	
13	22-Sep-28	100	
14	22-Sep-29	100	
15			
16	Discount rate	12%	
17	XNPV	151.59	<--=XNPV(B16,B3:B14,A3:A14)

Notice that **XNPV** requires you to indicate all the cash flows (starting with the initial cash flow), as opposed to **NPV**, which starts from the first cash flow.

Bugs in XIRR and XNPV

XIRR and **XNPV** address the problems of discounting when cash flows are time-dated, but they have two bugs: XNPV does not work with zero discount rates, and XIRR does not solve multiple internal rates of return. Below we discuss these problems. We then define two new functions, **NXNPV** and **NXIRR**, that solve these problems. These two functions are included as additional functions in the spreadsheet that accompanies this chapter.[2]

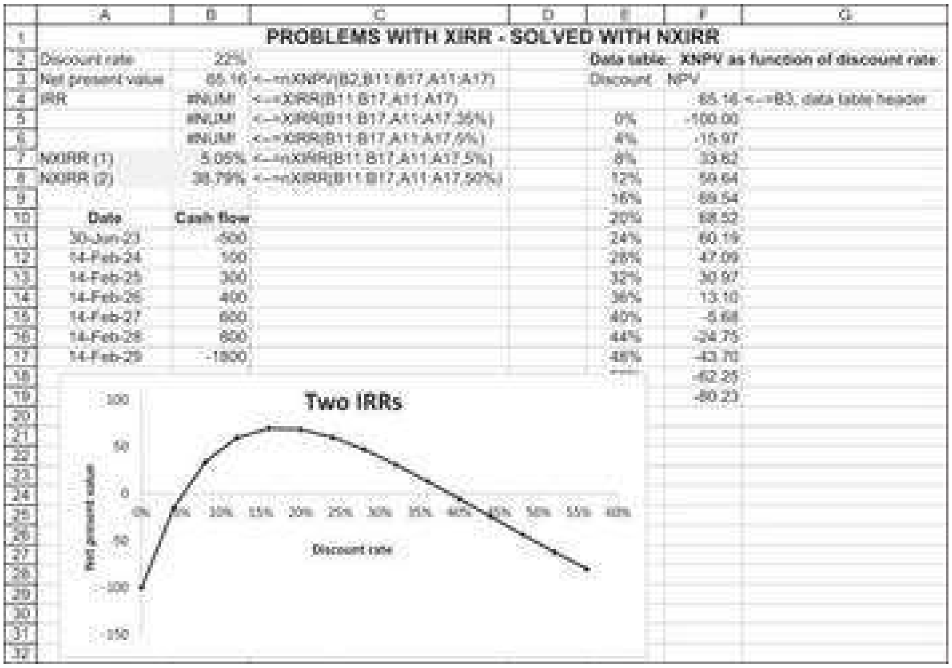
Problem 1: XNPV Doesn't Work with Zero Discount Rates

XNPV works fine with positive discount rates but fails when discount rates are zero.[3] We illustrate this below (cell E3). Our new function **NXNPV** solves this problem (cell E4):

	A	B	C	D	E	F
1	XNPV DOES NOT WORK WITH ZERO DISCOUNT RATES - SOLVED WITH NXNPV					
2	Discount rate	3%		Discount rate	0%	
3	XNPV	579.05	<--=XNPV(B2,B6:B11,A6:A11)	XNPV	#NUM!	<--=XNPV(E2,E6:E11,D6:D11)
4				NXNPV	680	<--=nXNPV(E2,E6:E11,D6:D11)
5	Date	Cash flow		Date	Cash flow	
6	13-Jan-22	-1,000		13-Jan-22	-1,000	
7	18-Aug-22	115		18-Aug-22	115	
8	20-Jan-23	121		20-Jan-23	121	
9	15-Jul-23	100		15-Jul-23	100	
10	01-Jan-24	333		01-Jan-24	333	
11	16-Jul-24	1,011		16-Jul-24	1,011	

Problem 2: XIRR Doesn't Work with Two IRRs

The **Guess** switch on **XIRR** doesn't work. As shown below, this means that **XIRR** is unable to compute the IRR for cash flows having multiple rates of return:



	A	B	C	D	E	F	G
1			PROBLEMS WITH XIRR - SOLVED WITH NXIRR				
2	Discount rate	22%			Data table:	XNPV as function of discount rate	
3	Net present value	65.16	<--=nXNPV(B2,B11:B17,A11:A17)		Discount	NPV	
4	IRR	#NUM!	<--=XIRR(B11:B17,A11:A17)			65.16	<--=B3, data table header
5		#NUM!	<--=XIRR(B11:B17,A11:A17,35%)		0%	-100.00	
6		#NUM!	<--=XIRR(B11:B17,A11:A17,5%)		4%	-15.97	
7	NXIRR (1)	5.05%	<--=nXIRR(B11:B17,A11:A17,5%)		8%	33.62	
8	NXIRR (2)	38.79%	<--=nXIRR(B11:B17,A11:A17,50%)		12%	59.64	
9					16%	69.54	
10	Date	Cash flow			20%	68.52	
11	30-Jun-23	-500			24%	60.19	
12	14-Feb-24	100			28%	47.09	
13	14-Feb-25	300			32%	30.97	
14	14-Feb-26	400			36%	13.10	
15	14-Feb-27	600			40%	-5.68	
16	14-Feb-28	800			44%	-24.75	
17	14-Feb-29	-1800			48%	-43.70	
18						-62.25	
19						-80.23	

From the data table, it is evident that there are two internal rates of return (around 5% and around 40%). But the **XIRR** function does not identify either (see cells B4:B6). The additional function **NXIRR** solves this problem (cells B7:B8).

30.3 Dates and Date Functions

The following statement, taken from Excel help, tells you almost everything you need to know about entering dates into your spreadsheet.

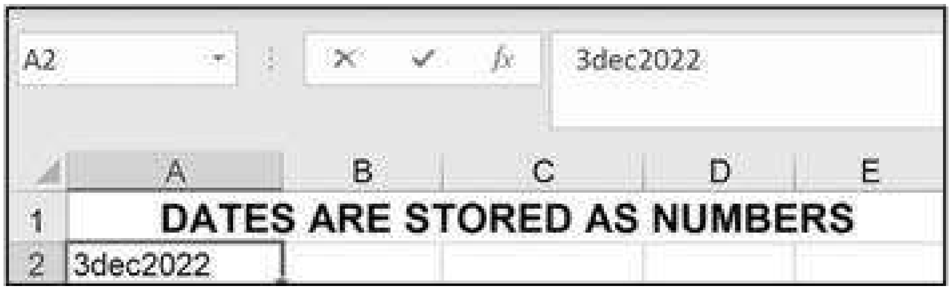
EXCEL HELP: learn about date calculations and formats

Excel stores dates as sequential numbers that are called serial values. For example, in Excel for Windows, January 1, 1900 is serial number 1, and January 1, 2008 is serial number 39448 because it is 39,448 days after January 1, 1900.

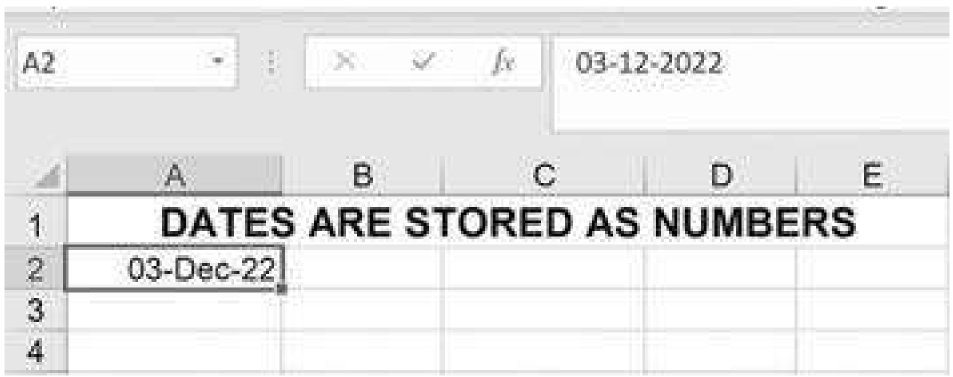
Excel stores times as decimal fractions because time is considered a portion of a day. The decimal number is a value ranging from 0 (zero) to 0.99999999, representing the times from 0:00:00 (12:00:00 A.M.) to 23:59:59 (11:59:59 P.M.).

Because dates and times are values, they can be added, subtracted, and included in other calculations. You can view a date as a serial value and a time as a decimal fraction by changing the format of the cell that contains the date or time to General format.

The basic fact you need to know is that Excel translates dates into a number. As an example, suppose you decide to type a date into a cell:



When you hit **Enter**, Excel decides that you've entered a date. Here's the way it appears:



Note that in the formula bar (indicated by the arrow above), Excel interprets the date entered as **03-12-2022**.[4] When you reformat the cell as **Format Cells|Number|General**, you see that Excel interprets this date as the number 44898, the number 1 being 1 January 1900.

Spreadsheet dates can be subtracted: In the spreadsheet below, we have entered two dates and subtracted them to find the number of days between the dates:

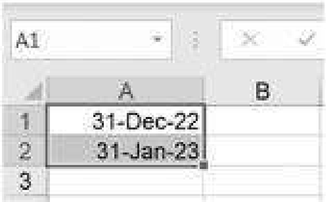
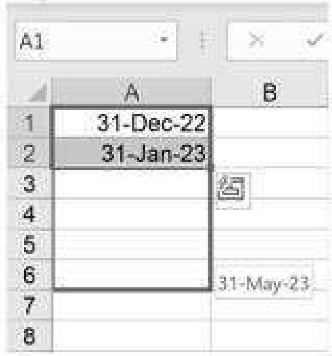
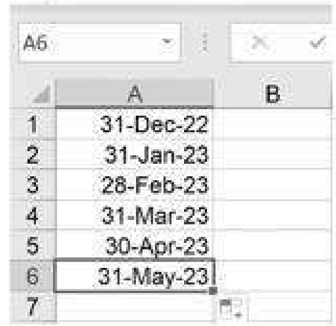
	A	B	C
4	Date1	07-Sep-07	
5	Date2	06-Jul-09	
6	Days between	668	<--=B5-B4

You can also add a number to a date to find another date. What, for example, was the date 165 days after 16 November 1947?

	B	C
8	16-Nov-47	
9	29-Apr-48	<--=B8+165

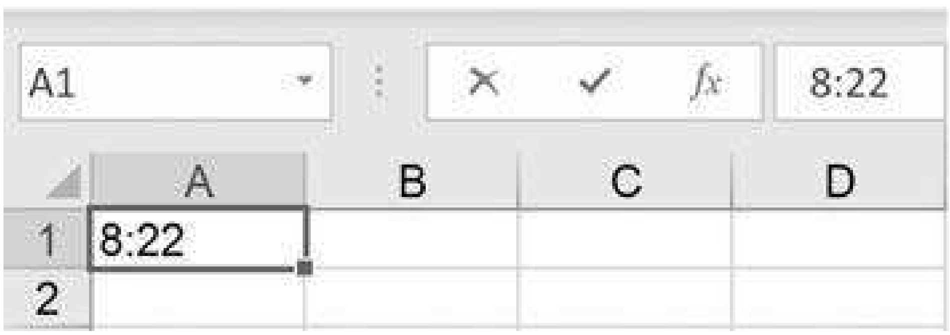
Stretching Out Dates

In the cells below we've put in two dates and then "stretched" the cells out to add more dates with the same difference between them:

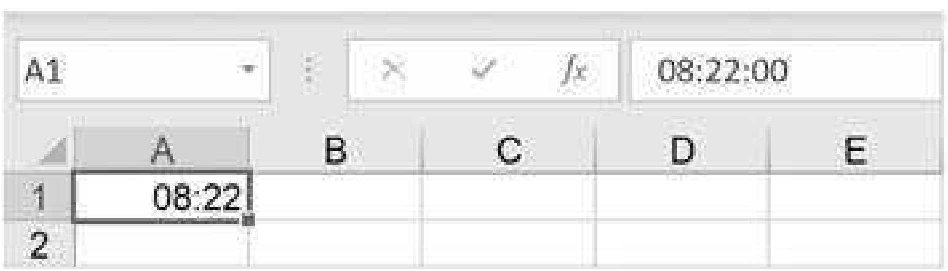
Write in two dates; mark both cells. 	Grab the handle (arrow on previous drawing) and pull down. 	The result: More dates added with same spacing (in this case, 1 month): 

Times in a Spreadsheet

Hours, minutes, and seconds can also be typed into a cell. In the cell below, we've typed in 8:22:



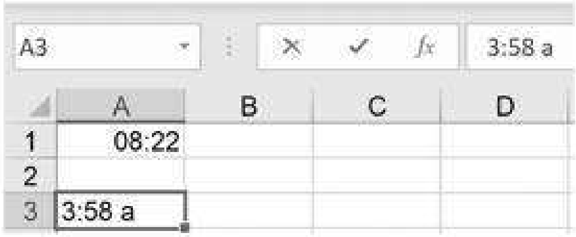
When we hit **Enter**, Excel interprets this as 8:22 A.M.:



Excel recognizes 24-hour times and also recognizes the symbol **a** for A.M. and **p** for P.M.:

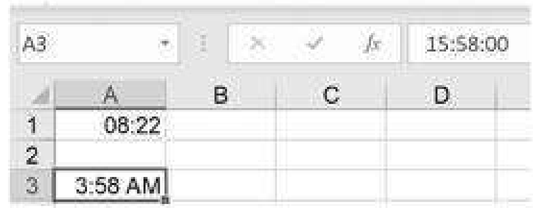
EXCEL RECOGNIZES A.M./P.M. ENTRIES

As entered:



Note that the **a** is separated from the time by a space. (Of course, A.M. is represented by an **a**, and P.M. is represented by a **p**.)

When you hit **Enter**:



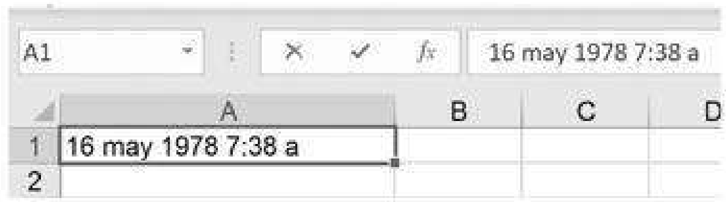
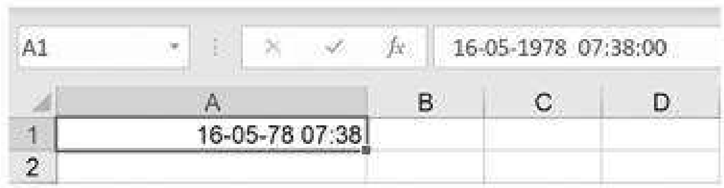
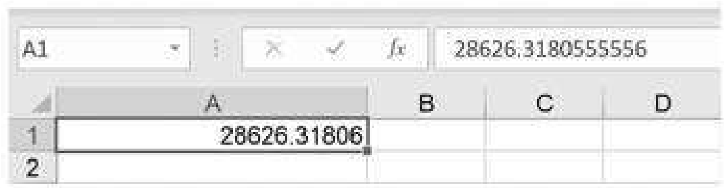
You can subtract times just like you subtract dates; cell D4 below tells you that 4 hours and 13 minutes have elapsed between the two times on 3-Feb-2023:

	A	B	C	D	E
1	Total hours	09:49	<--=SUM(D4:D5)		
2					
3	**Date**	**Start**	**Stop**	**Elapsed**	
4	03-Feb-23	08:32	12:45	04:13	<--=C4-B4
5	05-Apr-23	16:24	22:00	05:36	<--=C5-B5

When you reformat the cells above with **Format|Cells|Number|General**, you can see that times are represented in Excel as fractions of a day:

	A	B	C	D	E
1	Total hours	0.409	<--=SUM(D4:D5)		
2					
3	**Date**	**Start**	**Stop**	**Elapsed**	
4	44960	0.356	0.531	0.17569	<--=C4-B4
5	45021	0.683	0.917	0.23333	<--=C5-B5

If you type in a date and a time and reformat, you can also see this:

Here's what you typed:	
Here's how it appears:	
If you reformat this to **General**:	

Time and Date Functions in Excel

Excel has a whole set of time and date functions. Here are several functions that we find useful:

- • **Now** reads the computer clock and represents the date and the time. **Now** takes no arguments and is written with empty parentheses: **Now()**.
- • **Today** reads the computer's clock and prints the date. This function, like **Now**, is written with empty parentheses: **Today()**.
- • **Date(year,month,day)** gives the date entered.
- • **Weekday** gives the day of the week.
- • **Month, Week, Day** give the month, week, day of a date.

Here are the some of these functions in a spreadsheet:

	A	B	C
1	Serial representation	Date/Time Format	
2	43855.43354	25-Jan-20 10:24	<--=NOW()
3	43855	25-Jan-20	<--=TODAY()
4	43924	03-Apr-22	<--=DATE(2022,4,3)
5			
6	Different formatting of Now()		
7		25-Jan-20	<--=NOW()
8		01-25-20 10:24	<--=NOW()
9		10:24 AM	<--=NOW()
10			
11	Using Weekday, Month, Weeknum, Day		
12		7	<--=WEEKDAY(NOW())
13		5	<--=WEEKDAY("3apr1947")
14		4	<--=MONTH(B4)
15		1	<--=MONTH(NOW())
16		4	<--=WEEKNUM(NOW())
17		25	<--=DAY(NOW())

Calculating the Difference between Two Dates: The Functions Yearfrac and Datedif

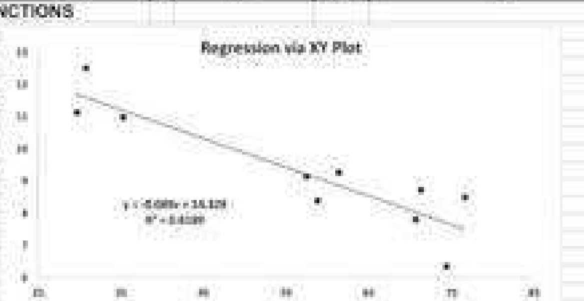
The Excel functions **Yearfrac** and **Datedif** computes the difference between two dates in various useful ways:

	A	B	C	D
1	DATEDIF AND YEARFRAC COMPUTES THE DIFFERENCE BETWEEN TWO DATES			
2	Date1	07-Apr-77		
3	Date2	31-Mar-20		
4				
5		42.98	<--=YEARFRAC(B2,B3,0)	Years between dates, 30 days per month, 360 days a year
6		42.98	<--=YEARFRAC(B2,B3,1)	Years between dates, actual days per month, actual days a year
7		43.61	<--=YEARFRAC(B2,B3,2)	Years between dates, actual days per month, 360 days a year
8		43.01	<--=YEARFRAC(B2,B3,3)	Years between dates, actual days per month, 365 days a year
9		42.98	<--=YEARFRAC(B2,B3,4)	Years between dates, 30 days per month, 360 days a year (european)
10				
11		42	<--=DATEDIF(B2,B3,"y")	Number of years between dates
12		515	<--=DATEDIF(B2,B3,"m")	Number of months between dates
13		15699	<--=DATEDIF(B2,B3,"d")	Number of days between dates
14		24	<--=DATEDIF(B2,B3,"md")	Number of days in excess of full number of months
15		11	<--=DATEDIF(B2,B3,"ym")	Number of months in excess of full number of years
16		358	<--=DATEDIF(B2,B3,"yd")	Number of days in excess of full number of years

30.4 Statistical Functions

Excel contains a number of statistical functions. We illustrate these functions using the following example, and assuming you are using Excel version 2013 or older.

	A	B	C	D	E	F	G	H
1	BASIC STATISTICAL FUNCTIONS							
2	Observation	X	Y					
3	1	35.30	10.98					
4	2	29.70	11.13					
5	3	30.80	12.51					
6	4	58.80	8.40					
7	5	61.40	9.27					
8	6	71.30	8.73					
9	7	74.40	6.36					
10	8	76.70	8.50					
11	9	70.70	7.82					
12	10	57.50	9.14					
13								
14								
15								
16								
17								
18								
19	Average	56.66	9.28	<--=AVERAGE(C3:C12)		Correlation	-0.9049	<--=CORREL(B2:B12,C2:C12)
20	Sample variance	334.15	3.23	<--=VAR.S(C3:C12)		Covariance	26.9988	<--=COVARIANCE.S(B19:B25,C19:C25)
21	Population variance	300.73	2.91	<--=VAR.P(C3:C12)			23.1417	<--=COVARIANCE.P(B19:B25,C19:C25)
22	Sample standard deviation	18.28	1.80	<--=SQRT(C20)				
23		18.28	1.80	<--=STDEV.S(C3:C12)		Regression intercept	14.3285	<--=INTERCEPT(C3:C12,B3:B12)
24	Population standard deviation	17.34	1.71	<--=SQRT(C21)		Regression slope	-0.0890	<--=SLOPE(C3:C12,B3:B12)
25		17.34	1.71	<--=STDEV.P(C3:C12)		Regression r-squared	0.6189	<--=RSQ(C3:C12,B3:B12)



The functions **Var.p**, and **Stdev.p** calculate the population variance and standard deviation, whereas the functions **Var.s**, and **Stdev.s** compute the sample variance and standard deviation. The difference between these two functions is that **Var.p** assumes that your data are the whole population and thus divides by the number of data points, whereas **Var.s** assumes that the data are samples from the distribution:

$$Var.p(x_1, \ldots, x_N) = \frac{1}{N}\sum_{i=1}^{N}(x_i - Average(x_1, \ldots, x_N))^2$$
$$Stdev.p(x_1, \ldots, x_N) = \sqrt{Varp(x_1, \ldots, x_N)}$$
$$Var.s(x_1, \ldots, x_N) = \frac{1}{N-1}\sum_{i=1}^{N}(x_i - Average(x_1, \ldots, x_N))^2$$
$$Stdev.s(x_1, \ldots, x_N) = \sqrt{Var(x_1, \ldots, x_N)}$$

Covariance.s, Covariance.p, and Correl

These functions—used extensively in Chapters 10–15 on portfolios—are used to compute the covariance and correlation of two series of numbers. For the definitions, we refer you to section 10.2. The function **Covarianc.p** is the population covariance (that is, it divides by the population size *M)* where the **Covariance.s** is the sample covariance (divides by $M - 1$). Below is an example in which we compute the covariance and correlation for returns on McDonald's stock and Wendy's stock. Note the two computations for the correlation. In cell E7, we use the Excel **Correl** function; in cells E8 and E9 we show alternative calculations by using the definition $Correlation(MCD, WEN) = \frac{Covar(MCD, WEN)}{(\sigma_{MCD} * \sigma_{WEN})}$.

	A	B	C	D	E	F
1	COVARIANCE AND CORRELATION: MCDONALD'S (MCD) AND WENDY'S (WEN)					
2	Date	MCD	WEN		Covariance	
3	01-Dec-19	2.24%	4.08%		0.00092	<--=COVARIANCE.P(B:B,C:C)
4	01-Nov-19	-1.14%	1.22%		0.00094	<--=COVARIANCE.S(B:B,C:C)
5	01-Oct-19	-8.76%	6.80%			
6	01-Sep-19	-0.98%	-9.18%		Correlation	
7	01-Aug-19	3.38%	19.02%		0.33296	<--=CORREL(B:B,C:C)
8	01-Jul-19	1.46%	-7.36%		0.33296	<--=COVARIANCE.P(B:B,C:C)/(STDEV.P(B:B)*STDEV.P(C:C))
9	01-Jun-19	5.21%	6.81%		0.33296	<--=COVARIANCE.S(B:B,C:C)/(STDEV.S(B:B)*STDEV.S(C:C))
10	01-May-19	0.35%	-1.19%			
11	01-Apr-19	3.96%	3.98%			
58	01-May-15	-0.64%	10.50%			
59	01-Apr-15	-0.92%	-7.42%			
60	01-Mar-15	-0.62%	-1.24%			
61	01-Feb-15	6.75%	5.09%			

Computing Statistics for a Database

Excel has a whole family of statistical functions designed to work on databases. These functions all start with the letter "D" (for "data," get it?). The list includes **DAverage**, **DCount**, **DMin**, **DVar** (the sample variance), **DVarP** (the population variance), **DStdev**, and **DStevP**. There are more, but we'll leave them to you to explore.

To see how these functions work, consider the following example: We have a set of monthly stock prices and returns for Apple. Suppose we want to calculate the average return of all the returns that are greater than 10%:

	A	B	C	D	E	F	G	H	I
1	MANIPULATING APPLE STOCK RETURNS USING DAVERAGE, DVAR, etc.								
2						Criterion range (2 rows)			
3	Date	AAPL	Return			Date	AAPL	Return	
4	01-Dec-19	293.65	9.72%	<--=LN(B4/B5)				>10%	
5	01-Nov-19	266.45	7.17%	<--=LN(B5/B6)					
6	01-Oct-19	248.02	10.50%			Average	15.78%	<--=DAVERAGE(A3:C242,3,F3:H4)	
7	01-Sep-19	223.30	7.42%			Variance	0.0028	<--=DVAR(A3:C242,3,F3:H4)	
8	01-Aug-19	207.33	-2.04%			Sigma	5.24%	<--=DSTDEV(A3:C242,3,F3:H4)	
9	01-Jul-19	211.60	7.36%						
241	01-Mar-00	4.22	16.96%						
242	01-Feb-00	3.56	9.97%						
243	01-Jan-00	3.22							

Looking at the above example, we see that there are three parts:

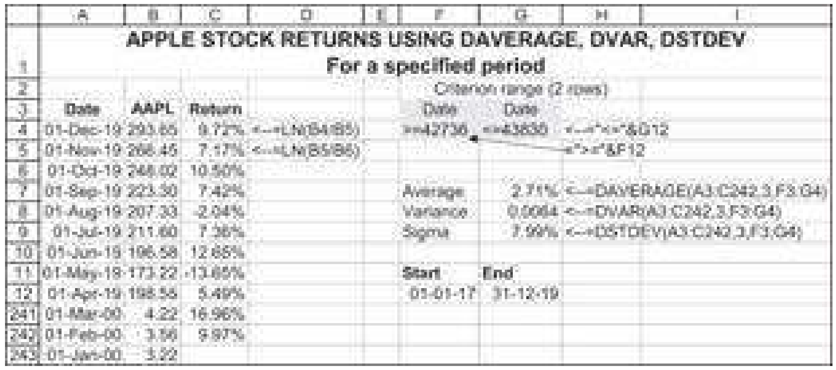
- • The database is in cells A3:C242 (note that many rows are hidden). It has headers **Date**, **AAPL**, **Return**.
- • There is a two-row **Criterion range**: The top row matches the headers of the database, and the bottom row has “>10%” under **Return**.
- • Now the functions: **DAverage(A3:C242,3,F3:H4)** computes the average for all returns that are greater than 10%. **DVar** with the arguments computes the sample variance, and **DStdev** computes the sample standard deviation (where the sample is the month in which Apple had a return higher than 10%).

If you change the criterion range, you get a different answer:

	F	G	H	I
2	Criterion range (2 rows)			
3	Date	AAPL	Return	
4			<8%	
5				
6	Average	-3.05%	<--=DAVERAGE(A3:C242,3,F3:H4)	
7	Variance	0.0121	<--=DVAR(A3:C242,3,F3:H4)	
8	Sigma	11.02%	<--=DSTDEV(A3:C242,3,F3:H4)	

Advanced Use of the “D” Functions

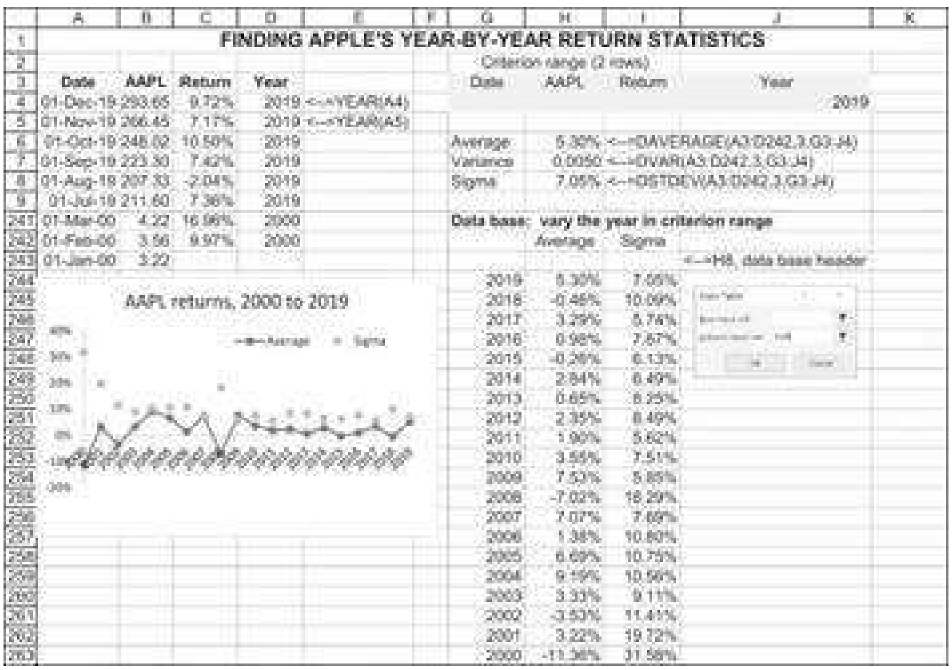
Suppose we want to know the statistics for Apple’s returns for a specific year. Here’s the trick:



	A	B	C	D	E	F	G	H	I
1	APPLE STOCK RETURNS USING DAVERAGE, DVAR, DSTDEV For a specified period								
2						Criterion range (2 rows)			
3	Date	AAPL	Return			Date	Date		
4	01-Dec-19	293.65	9.72%	<--=LN(B4/B5)		>=42736	<=43830	<--="<="&G12	
5	01-Nov-19	266.45	7.17%	<--=LN(B5/B6)				=">="&F12	
6	01-Oct-19	248.02	10.50%						
7	01-Sep-19	223.30	7.42%			Average	2.71%	<--=DAVERAGE(A3:C242,3,F3:G4)	
8	01-Aug-19	207.33	-2.04%			Variance	0.0064	<--=DVAR(A3:C242,3,F3:G4)	
9	01-Jul-19	211.60	7.36%			Sigma	7.99%	<--=DSTDEV(A3:C242,3,F3:G4)	
10	01-Jun-19	196.58	12.65%						
11	01-May-19	173.22	-13.65%			Start	End		
12	01-Apr-19	198.56	5.49%			01-01-17	31-12-19		
241	01-Mar-00	4.22	16.96%						
242	01-Feb-00	3.56	9.97%						
243	01-Jan-00	3.22							

We've put the date in cells F12:G12 and used the **Text** function to create an appropriate criterion under the **Date** header.

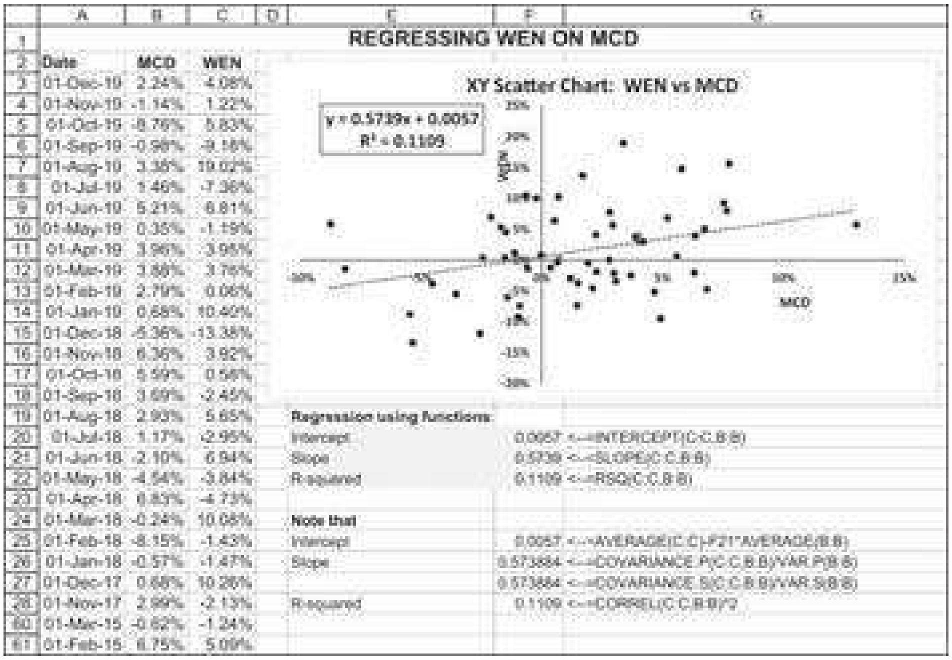
One more example and we leave the rest to you. We want to find Apple's return statistics for each of the years. We expand the database and the criterion range by adding in the year. A single example for 2019 shows the average return and sigma for that year to be 5.3% and 7.05%. Running a data table with **Year** as the parameter to be varied gives the results below:



	A	B	C	D	E	F	G	H	I	J
1	FINDING APPLE'S YEAR-BY-YEAR RETURN STATISTICS									
2							Criterion range (2 rows)			
3	Date	AAPL	Return	Year			Date	AAPL	Return	Year
4	01-Dec-19	293.65	9.72%	2019	<--=YEAR(A4)					2019
5	01-Nov-19	266.45	7.17%	2019	<--=YEAR(A5)					
6	01-Oct-19	248.02	10.50%	2019			Average	5.30%	<--=DAVERAGE(A3:D242,3,G3:J4)	
7	01-Sep-19	223.30	7.42%	2019			Variance	0.0050	<--=DVAR(A3:D242,3,G3:J4)	
8	01-Aug-19	207.33	-2.04%	2019			Sigma	7.05%	<--=DSTDEV(A3:D242,3,G3:J4)	
9	01-Jul-19	211.60	7.36%	2019						
241	01-Mar-00	4.22	16.96%	2000			Data base: vary the year in criterion range			
242	01-Feb-00	3.56	9.97%	2000				Average	Sigma	
243	01-Jan-00	3.22								<--=H8, data base header
244							2019	5.30%	7.05%	
245							2018	-0.46%	10.09%	
246							2017	3.29%	5.74%	
247							2016	0.98%	7.67%	
248							2015	-0.26%	6.13%	
249							2014	2.84%	6.49%	
250							2013	0.65%	8.25%	
251							2012	2.35%	8.49%	
252							2011	1.90%	5.62%	
253							2010	3.55%	7.51%	
254							2009	7.53%	5.85%	
255							2008	-7.02%	18.29%	
256							2007	7.07%	7.69%	
257							2006	1.38%	10.80%	
258							2005	6.69%	10.75%	
259							2004	9.19%	10.56%	
260							2003	3.33%	9.11%	
261							2002	-3.53%	11.41%	
262							2001	3.22%	19.72%	
263							2000	-11.36%	31.58%	

30.5 Doing Regressions with Excel

There are several techniques to produce an ordinary least-squares regression with Excel. We illustrate three techniques using the McDonald's and Wendy's data just discussed.



REGRESSING WEN ON MCD

Date	MCD	WEN
01-Dec-19	2.24%	4.08%
01-Nov-19	-1.14%	1.22%
01-Oct-19	-8.76%	5.83%
01-Sep-19	-0.98%	-9.16%
01-Aug-19	3.38%	19.02%
01-Jul-19	1.46%	-7.36%
01-Jun-19	5.21%	6.81%
01-May-19	0.35%	-1.19%
01-Apr-19	3.96%	3.95%
01-Mar-19	3.88%	3.76%
01-Feb-19	2.79%	0.06%
01-Jan-19	0.68%	10.40%
01-Dec-18	-5.36%	-13.38%
01-Nov-18	6.36%	3.92%
01-Oct-18	5.59%	0.58%
01-Sep-18	3.69%	-2.45%
01-Aug-18	2.93%	5.65%
01-Jul-18	1.17%	-2.95%
01-Jun-18	-2.10%	6.94%
01-May-18	-4.54%	-3.84%
01-Apr-18	6.83%	-4.73%
01-Mar-18	-0.24%	10.08%
01-Feb-18	-8.15%	-1.43%
01-Jan-18	-0.57%	-1.47%
01-Dec-17	0.68%	10.28%
01-Nov-17	2.99%	-2.13%
01-Mar-15	-0.62%	-1.24%
01-Feb-15	6.75%	5.09%

Regression using functions		
Intercept	0.0057	<--=INTERCEPT(C:C,B:B)
Slope	0.5739	<--=SLOPE(C:C,B:B)
R-squared	0.1109	<--=RSQ(C:C,B:B)
Note that		
Intercept	0.0057	<--=AVERAGE(C:C)-F21*AVERAGE(B:B)
Slope	0.573884	<--=COVARIANCE.P(C:C,B:B)/VAR.P(B:B)
	0.573884	<--=COVARIANCE.S(C:C,B:B)/VAR.S(B:B)
R-squared	0.1109	<--=CORREL(C:C,B:B)^2

Using Excel Functions

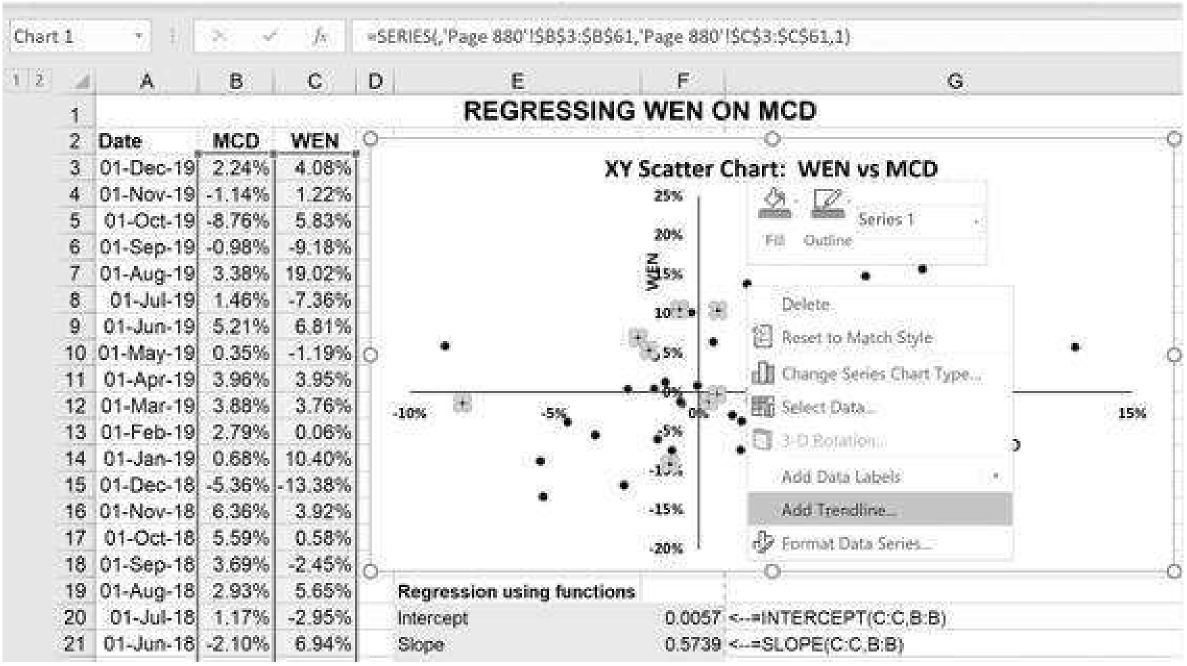
The first technique involves the functions **Slope**, **Intercept**, and **Rsq**; these functions give the parameters for a simple regression of the data in column B on column C. Using these numbers, the best linear explanation of the relation between the returns on WEN and MCD is:

$r_{WEN} = 0.0057 + 0.573884 \cdot r_{MCD}, \quad R^2 = 11.09\%$

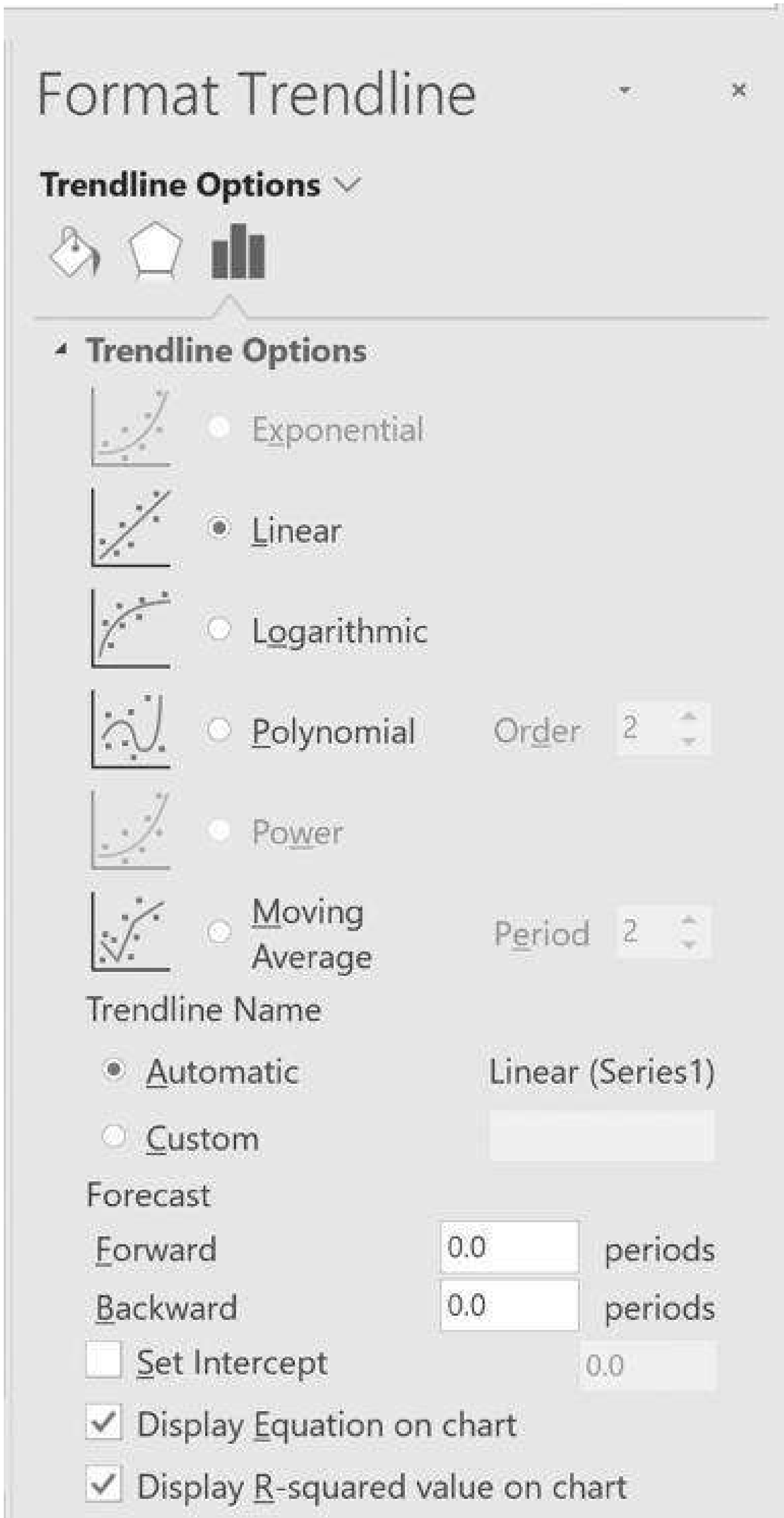
Using a Scatter Plot and Trendline

Another way that we can produce a simple regression is to graph the data in a scatter chart and then use the **Trendline** function to compute the regression:

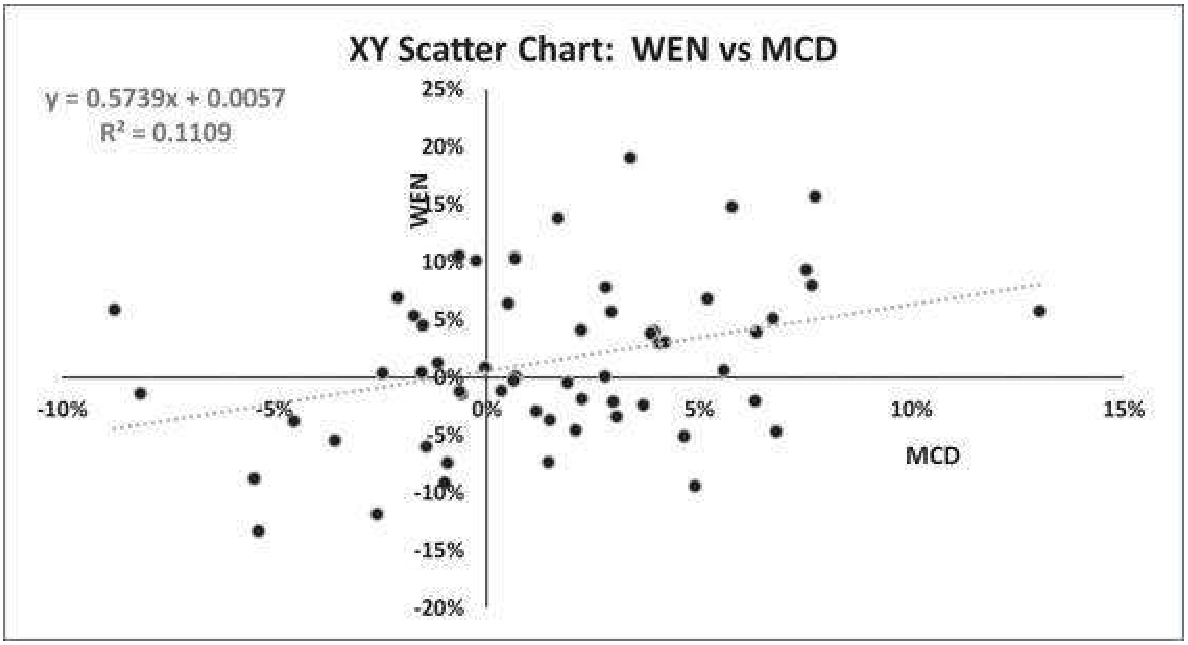
- • First, plot the data using an **XY Scatter Chart**.
- • Click the data and then go to **Add Trendline**. The menu of regressions is given below the picture that follows this bullet.



Here's the menu. We've indicated that we want a linear regression and clicked on two boxes to display the equation and the R^2 in the chart.

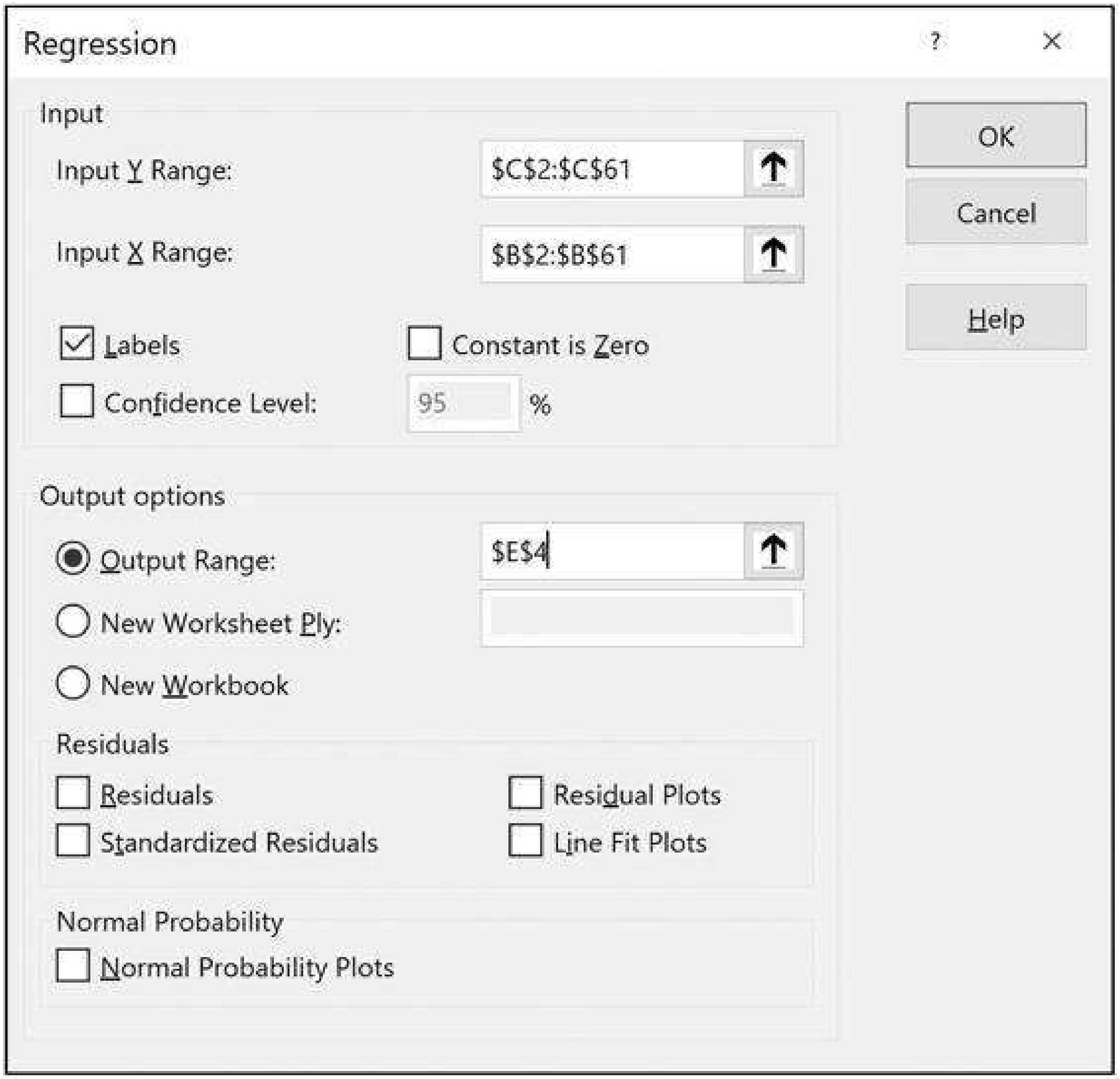

Format Trendline
Trendline Options
Trendline Options
Exponential
Linear
Logarithmic
Polynomial
Order
2
Power
Moving Average
Period
2
Trendline Name
Automatic
Linear (Series1)
Custom
Forecast
Forward
0.0
periods
Backward
0.0
periods
Set Intercept
0.0
Display Equation on chart
Display R-squared value on chart

Here's the final result:

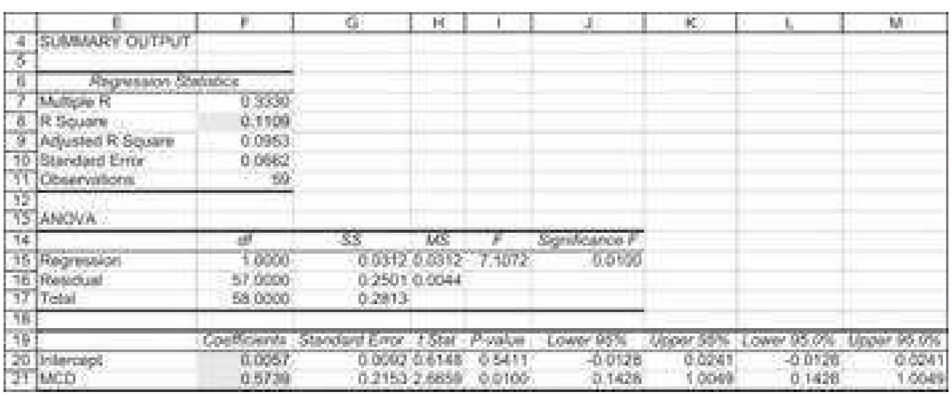


Using Data Analysis for Regressions

Using the same data as before, we press **Data|Data Analysis|Regression**.[5] In the picture below, we've asked that the output be put in cell E4:



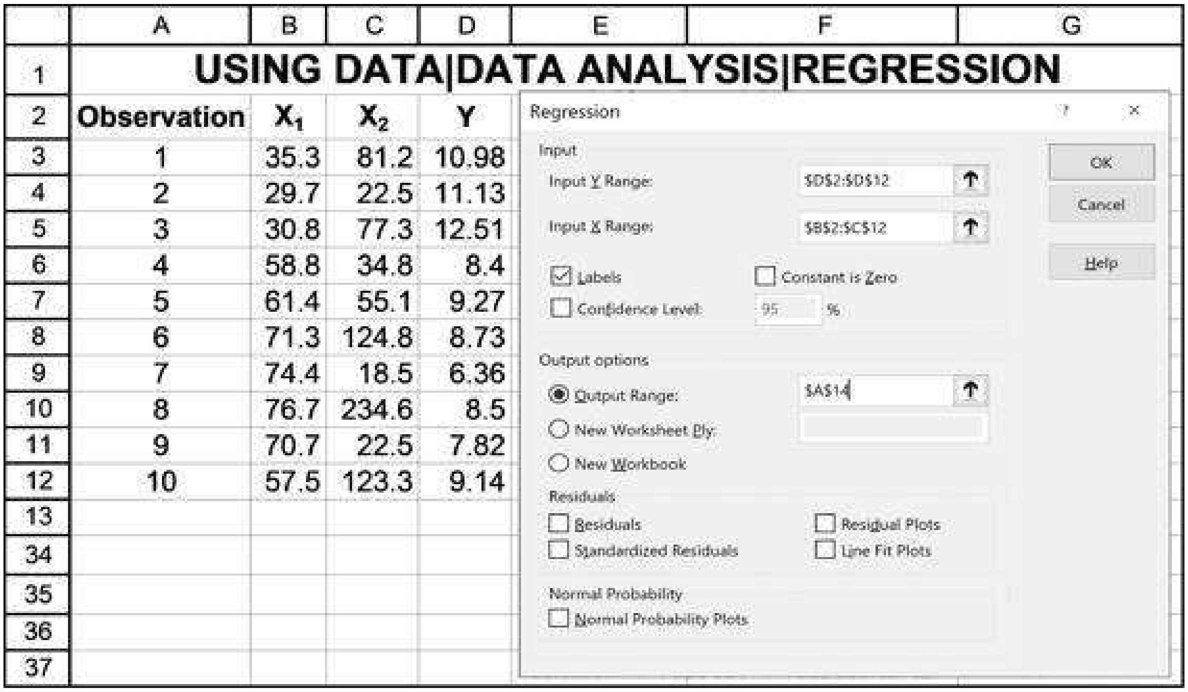
Here's the output. We have highlighted cells corresponding to the intercept (cell F20), slope (cell F21), and R^2 (cell F8) as computed by the relevant Excel functions:



	E	F	G	H	I	J	K	L	M
4	SUMMARY OUTPUT								
5									
6	*Regression Statistics*								
7	Multiple R	0.3330							
8	R Square	0.1109							
9	Adjusted R Square	0.0953							
10	Standard Error	0.0662							
11	Observations	59							
12									
13	ANOVA								
14		*df*	*SS*	*MS*	*F*	*Significance F*			
15	Regression	1.0000	0.0312	0.0312	7.1072	0.0100			
16	Residual	57.0000	0.2501	0.0044					
17	Total	58.0000	0.2813						
18									
19		*Coefficients*	*Standard Error*	*t Stat*	*P-value*	*Lower 95%*	*Upper 95%*	*Lower 95.0%*	*Upper 95.0%*
20	Intercept	0.0057	0.0092	0.6148	0.5411	-0.0126	0.0241	-0.0126	0.0241
21	MCD	0.5738	0.2153	2.6659	0.0100	0.1426	1.0049	0.1426	1.0049

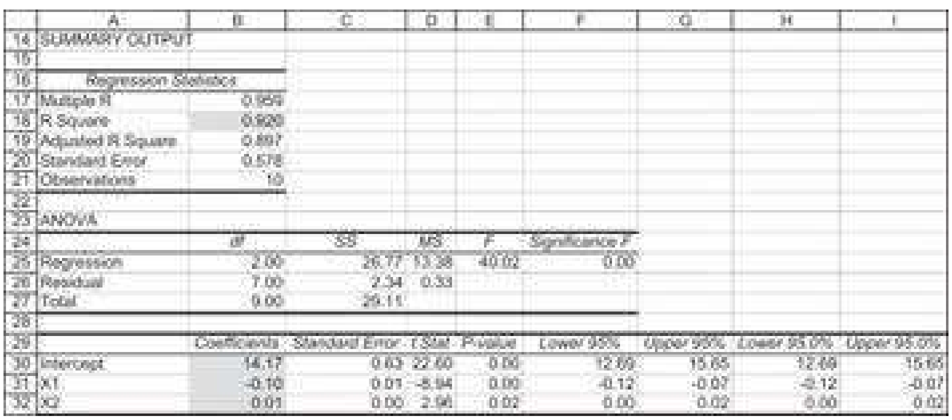
Using Data Analysis|Regression for Multiple Explanatory Variables

The same module can be used for multiple explanatory variables. In the picture below, we've asked that the output be placed in cell F3:



	A	B	C	D
1	USING DATA|DATA ANALYSIS|REGRESSION			
2	Observation	X_1	X_2	Y
3	1	35.3	81.2	10.98
4	2	29.7	22.5	11.13
5	3	30.8	77.3	12.51
6	4	58.8	34.8	8.4
7	5	61.4	55.1	9.27
8	6	71.3	124.8	8.73
9	7	74.4	18.5	6.36
10	8	76.7	234.6	8.5
11	9	70.7	22.5	7.82
12	10	57.5	123.3	9.14

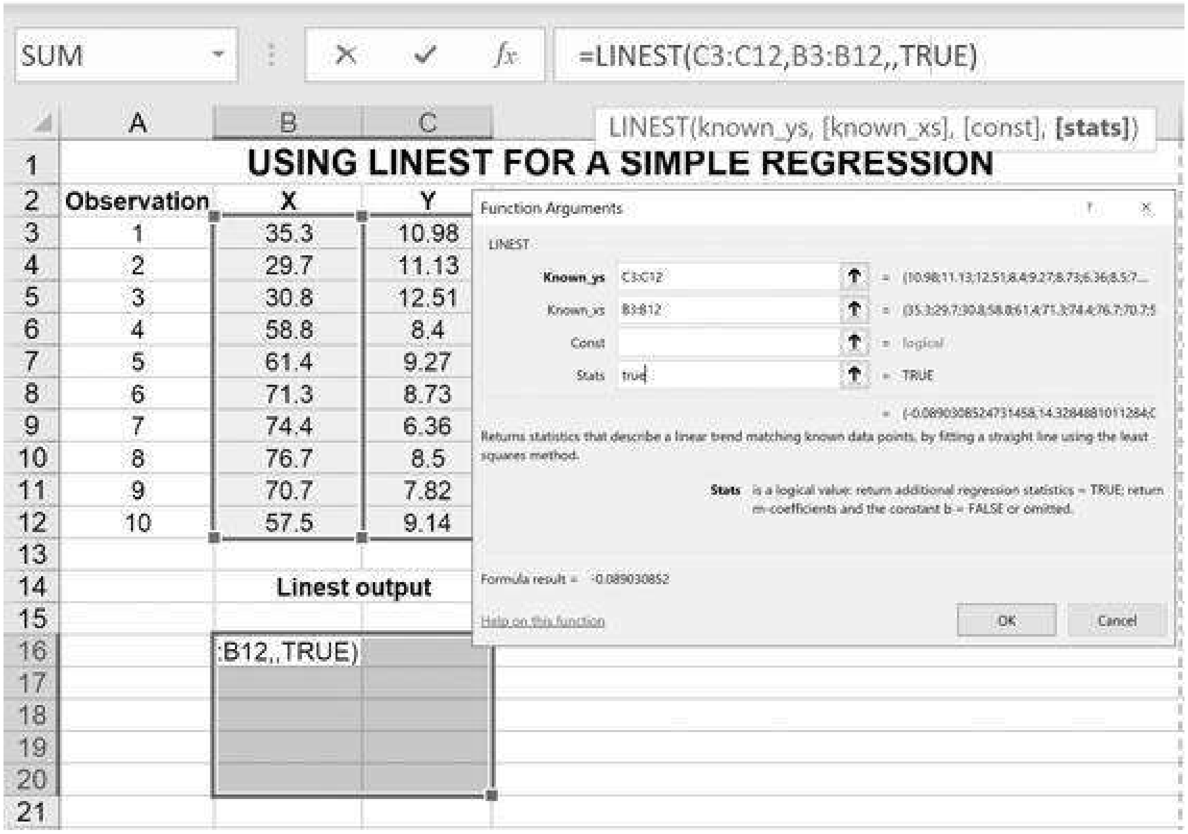
Here's the output. Highlighted cells correspond to the most important values produced in the previous example.



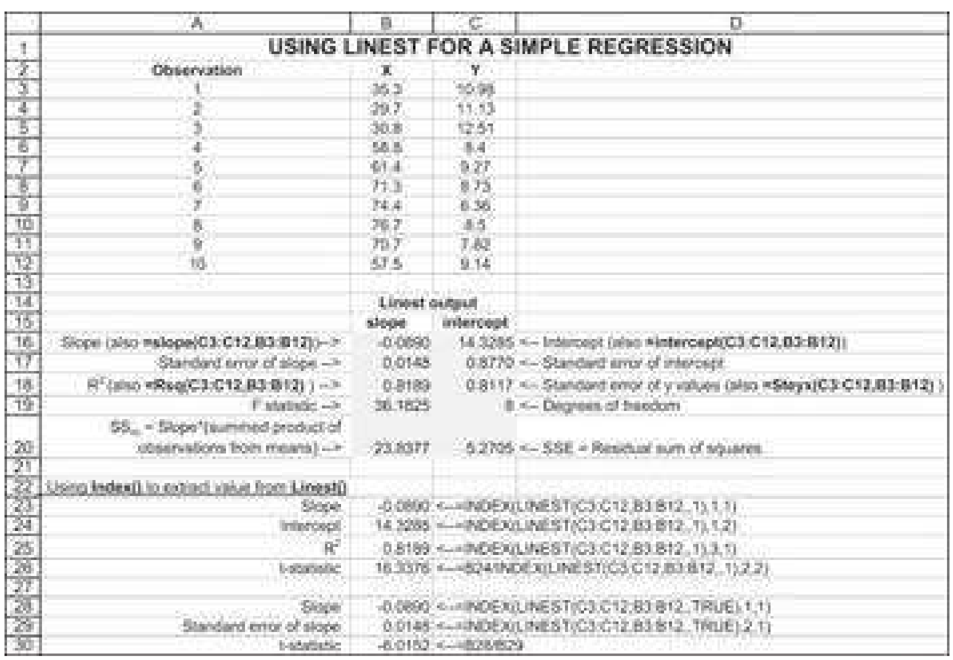
	A	B	C	D	E	F	G	H	I
14	SUMMARY OUTPUT								
16	Regression Statistics								
17	Multiple R	0.959							
18	R Square	0.920							
19	Adjusted R Square	0.897							
20	Standard Error	0.578							
21	Observations	10							
23	ANOVA								
24		df	SS	MS	F	Significance F			
25	Regression	2.00	26.77	13.38	40.02	0.00			
26	Residual	7.00	2.34	0.33					
27	Total	9.00	29.11						
29		Coefficients	Standard Error	t Stat	P-value	Lower 95%	Upper 95%	Lower 95.0%	Upper 95.0%
30	Intercept	14.17	0.63	22.60	0.00	12.69	15.65	12.69	15.65
31	X1	-0.10	0.01	-8.94	0.00	-0.12	-0.07	-0.12	-0.07
32	X2	0.01	0.00	2.96	0.02	0.00	0.02	0.00	0.02

Using Linest

Excel has an array function **Linest** whose output consists of a number of regression statistics for an ordinary least-squares regression.[6] Here is a picture of the spreadsheet and the **Linest** dialog box:



Linest is an array function (see Chapter 31), which means that instead of **[Enter]**, we press **[Control]** + **[Shift]** + **[Enter]** at the same time. With the data from the example above, we can use **Linest** to produce the following output:



Linest produces a block of output without column headers or row labels which identify the output. Excel's Help provides a good explanation of the meaning of the output; in the above picture, we have added in the explanations.

Note the syntax of this function: **Linest(y-range,x-range,constant,statistics)**. The **y-range** is the range of dependent variables and the **x-range** is the range of the independent variables. If **constant** is omitted (as in the case above) or set to **True**, then the regression is calculated normally; if **constant** is set to **False**, then the intercept is forced to be zero. If **statistics** is set to **True** (as in the case above), then the range of statistics is calculated; otherwise only the slope and intercept are calculated.

Individual items of this output can be called by using the function **Index** (discussed below). Suppose, for example, that we want to do a simple *t*-test on the slope; doing this requires us to divide the slope value by its standard error.

Index

The discussion of the **Index** function would normally belong in section 30.7 on reference functions. It is presented here only because we used it in the previous subsection. We sometimes want to pick an individual value out of an array. In the example below, the range of cells A2:C4 contains a mixture of numbers and names. To pick out an individual item from this range, we use **=Index(A2:C4,row,column)**, where **row** and **column** are relative to the range itself. Thus "Howie" appears in row 2 and column 3 of the range A2:C4.

	A	B	C
1	**THE INDEX FUNCTION**		
2	a	Terry	3
3	Simon	6	Howie
4	Lizzie	7	Tal
5			
6	Howie	<--=INDEX(A2:C4,2,3)	

In the next subsection we use the **Index** function to pick out a single item in the **Linest** array.

Multiple Regressions with Linest

Linest can also be used to do a multiple regression, as illustrated below:

	A	B	C	D	E	F
1	USING LINEST TO DO A MULTIPLE REGRESSION					
2		Observation	X_1	X_2	Y	
3		1	35.3	81.2	10.98	
4		2	29.7	22.5	11.13	
5		3	30.8	77.3	12.51	
6		4	58.8	34.8	8.4	
7		5	61.4	55.1	9.27	
8		6	71.3	124.8	8.73	
9		7	74.4	18.5	6.36	
10		8	76.7	234.6	8.5	
11		9	70.7	22.5	7.82	
12		10	57.5	123.3	9.14	
13						
14			x_2 coeff.	x_1 coeff.	Intercept	
15		Slope -->	0.0089	-0.0987	14.1705	<-- Intercept
16		Standard error -->	0.0030	0.0110	0.6271	
17		R^2 -->	0.9196	0.5783	#N/A	
18		F statistic -->	40.0228	7.0000	#N/A	
19		SS_{xy} -->	26.7674	2.3408	#N/A	
20						
21		{=LINEST(E3:E12,C3:D12,,TRUE)}				
22						

30.6 Conditional Functions

If is the most common function that allows you to put in conditional statements.

The syntax of Excel's **If** statement is **If(condition, output if condition is true, output if condition is false)**. In the example below, if the initial number is B3 $\leq$ 3, then the desired output is 15. If B3 > 3, then the output is 0:

	A	B	C
1	THE IF FUNCTION		
2	Initial number	2	
3	If statement	15	<--=IF(B2<=3,15,0)
4			
5	Initial number	2	
6	If statement	Less than or equal to 3	<--=IF(B5<=3,"Less than or equal to 3","More than 3")

As you can see in row 6, you can make **If** print text also, by enclosing the desired text in double quotes.

Boolean Functions

A Boolean function is similar to an if statement. When you include a question in parentheses, you are setting up a *Boolean function*:

	A	B	C
1	BASIC BOOLEAN FUNCTIONS		
2	x	22	
3	y	-15	
4			
5	Number	25	
6	Is number <= x?	FALSE	<--=(B5<=B2)
7	Is number > y?	TRUE	<--=(B5>B3)
8			
9	Multiplying	0	<--=B6*B7

In cell B6, we have written **=(B5<=B2)**; this asks whether B5 is less than or equal to B2. If the answer is positive, Excel returns **False**, otherwise it returns **True**. Multiplying **False*True** or **False*False** gives 0 (see cell B9) and multiplying **True*True** gives 1.

	A	B	C
1	**BASIC BOOLEAN FUNCTIONS**		
2	x	22	
3	y	-15	
4			
5	Number	20	
6	Is number <= x?	TRUE	<--=(B5<=B2)
7	Is number > y?	TRUE	<--=(B5>B3)
8			
9	Multiplying	1	<--=B6*B7

Boolean functions can be useful in the most unexpected places. In the spreadsheet below, the first two columns contain the monthly returns for Marriott for a 2-year period. The problem we encounter is counting the number of returns that are between two arbitrary bounds and taking the average of the returns that are between these bounds:

	A	B	C	D	E	F	G	H
1						**USING BOOLEAN FUNCTIONS**		
2	Date	Marriott price	Return					
3	01-Dec-19	151.43	7.95%	<--=LN(B3/B4)		How many data points?	239	<--=COUNT(C:C)
4	01-Nov-19	139.86	10.36%	<--=LN(B4/B5)		Maximal return	36.99%	<--=MAX(C:C)
5	01-Oct-19	126.10	1.74%			Minimal return	-27.22%	<--=MIN(C:C)
6	01-Sep-19	123.93	-0.98%					
7	01-Aug-19	125.15	-9.81%			Upper bound	5%	
8	01-Jul-19	138.06	-0.88%			Lower bound	-2%	
9	01-Jun-19	139.28	12.04%					
10	01-May-19	123.48	-8.87%			How many < upper bound?	164	<--=COUNTIF(C:C,"<"&G7)
11	01-Apr-19	134.93	8.67%			How many > lower bound?	159	<--=COUNTIF(C:C,">"&G8)
12	01-Mar-19	123.72	0.18%					
13	01-Feb-19	123.50	8.96%			How many between upper and lower bounds?	84	<--=SUMPRODUCT((C3:C241>G8)*(C3:C241<G7),(C3:C241>G8)*(C3:C241<G7))
14	01-Jan-19	112.91	5.35%					
15	01-Dec-18	107.03	-5.44%			Average of returns between the bounds	1.32%	<--=SUMPRODUCT((C3:C241>G8)*(C3:C241<G7),(C3:C241>G8)*(C3:C241<G7),C3:C241)/SUMPRODUCT((C3:C241>G8)*(C3:C241<G7),(C3:C241>G8)*(C3:C241<G7))
236	01-Jul-00	15.22	9.91%					
237	01-Jun-00	13.78	-0.52%					
238	01-May-00	13.86	12.47%					
239	01-Apr-00	12.23	1.76%					
240	01-Mar-00	12.02	13.35%					
241	01-Feb-00	10.52	-11.95%					
242	01-Jan-00	11.85						

In cell G3, we use **Count** to determine the number of returns. Cells G10 and G11 use **Countif** to determine how many returns are below the upper bound in cell G7 and how many are above the lower bound in cell G8. But how many of the returns are between the two bounds? You can't do it with **Countif**, but we can do it (cell G14) with a trick involving Boolean functions:

$$= \underbrace{\text{SUMPRODUCT}(\underbrace{(C3{:}C241 > G8) * (C3{:}C241 < G7)}_{\substack{\text{Creates a vector of 1's where}\\ \text{the returns are > lower bound (G7)}\\ \text{and 0's where}\\ \text{the returns are < upper bound (G8)}}},\ \underbrace{(C3{:}C241 > G8) * (C3{:}C241 < G7))}_{\text{Ditto}}}_{\substack{\text{Sumproduct multiplies the two vectors and sums the results.}\\ \text{This gives the number of data points between the two bounds.}}}$$

A similar trick is employed in cell G15 to find the average of the returns that are between the two bounds:

$$= \frac{\text{SUMPRODUCT}((C3{:}C241 > G8) * (C3{:}C241 < G7), C3{:}C241)}{\text{SUMPRODUCT}((C3{:}C241 > G8) * (C3{:}C241 < G7), (C3{:}C241 > G8) * (C3{:}C241 < G7))}$$

$$= \frac{\substack{\text{Numerator: Multiplies vector of 1's and 0's times the returns,}\\ \text{thus sums the returns which are between the bounds}}}{\text{Counts the returns which are between the bounds}}$$

This is all very tricky, but it is also very useful!

30.7 Reference Functions

VLookup and Hlookup

Since **VLookup**, and **HLookup** both have the same structure, we will concentrate on **VLookup** and leave you to figure out **HLookup** for yourself. **VLookup** is a way to introduce a table search in your spreadsheet. For example, suppose the marginal tax rates on income are given by the table below (i.e., for income less than \$8,000, the marginal tax rate is 0%; for income above \$8,000, the marginal tax rate is 15%). Cell B9 illustrates how the function **VLookup** is used to look up the marginal tax rate.

	A	B	C
1	VLOOKUP FUNCTION		
2	Income	Tax rate	
3	0	0%	
4	8,000	15%	
5	14,000	25%	
6	25,000	38%	
7			
8	Income	15,000	
9	Tax rate	25%	<--=VLOOKUP(B8,A3:B6,2)

The syntax of this function is **VLookup(lookup_value,table,column)**. The first column of the lookup table, A3:B6, must be arranged in ascending (increasing) order.

The **lookup_value**, in this case the income of 15,000, is used to determine the applicable row of the **table**. The row is the first row whose value is ≤ the **lookup_value**; in this case, this is the row that starts with 14,000. The **column** entry determines from which column of the applicable row the answer is taken; in this case the marginal tax rates are in column 2.

Dget

Dget is a useful database function. It can fetch a specific value from a table with multiple values. The structure of the function is **Dget(Database, field, criteria)**. Here is a simple example.

	A	B	C
1	**DGET FUNCTION**		
2	**Product**	**Sales (M)**	
3	A	1.50	
4	B	2.30	
5	C	2.60	
6	D	4.30	
7			
8	**Product**	**Sales (M)**	
9	A	1.50	<--=DGET(A2:B6,B2,A8:A9)

Offset

The function **Offset** allows us to specify a cell or a block of cells in an array. It cannot be used by itself—rather, it must be part of another Excel function. The example below shows a large array of numbers. We want to sum a four-row, five-column array of the larger array (these numbers are specified in cells B6 and B7); as specified in cells B3 and B4, we want this summed array to start to the right and below the third row and the second column of the large block of numbers:

	A	B	C	D	E	F	G	H
1		**USING OFFSET**						
2	**Starting corner**							
3	Rows down	3						
4	Columns over	2						
5	**Range to be summed**							
6	Number of rows	4						
7	Number of columns	5						
8	Sum	811	<--=SUM(OFFSET(A11:H20,B3,B4,B6,B7))					
9	**Check**	811	<--=SUM(C14:G17)					
10								
11	89	34	72	42	41	89	75	41
12	33	6	49	7	62	50	38	17
13	71	69	42	68	39	75	32	77
14	2	69	8	79	40	8	67	46
15	70	12	44	48	88	27	38	51
16	85	0	23	35	83	30	17	52
17	30	50	16	28	73	4	55	68
18	35	56	31	24	15	47	89	88
19	99	31	55	60	45	24	28	3
20	93	72	7	75	90	81	52	71

The function **OFFSET(A11:H20,B3,B4,B6,B7)** in cell B8 specifies a block of cells within the range A11:H20. This range starts *three rows below* (cell B3) and *two columns to the right* (cell B4) of the top left-hand cell of the range A11:H20. The range itself is four rows deep (cell B6) and five columns wide (cell B7).

The values in cells B6 and B7 always have to be positive, but the values in B3 and B4 can be negative as well as positive.

For an innovative use of **Offset**, see section 14.6.

30.8 Large, Rank, Percentile, and Percentrank

Large(array, k) returns the *k*th largest number of the **array**, and **Rank(number, array)** returns the rank in **array** of **number**.

Here is an example of each function:

	A	B	C
1	LARGE, RANK, PERCENTILE, PERCENTRANK		
2	Data		
3	10.98	8.73	
4	11.13	6.36	
5	12.51	8.50	
6	8.40	7.82	
7	9.27	9.14	
8			
9	Ranking, k	3	
10	K-th largest	10.98	<--=LARGE(A3:B7,B9)
11			
12	Specific number	9.27	
13	Rank from top	4	<--=RANK(B12,A3:B7)
14	Rank from bottom	7	<--=RANK(B12,A3:B7,1)
15			
16	Percentile rank	0.8	
17	Percentile	11.01	<--=PERCENTILE(A3:B7,B16)
18			
19	Specific number	9.27	
20	Percentile ranking	0.666	<--=PERCENTRANK(A3:B7,B19)

Thus, the third-largest number in the range A3:B7 is 10.98 and 9.27 is the fourth-largest number in the range A3:B7. If, as in cell B19, you specify an additional parameter in the function **Rank**, you will see that 9.27 is the seventh-ranking number from the bottom of the range A3:B7.

As illustrated, Excel has similar functions for percentiles: **Percentile** and **PercentRank**.

30.9 Count, CountA, Countif, Countifs, Averageif, Averageifs

As their names suggest, all six of these functions operate as follows:

- • **Count**: Counts the number of numeric entries in a range of cells.

- • **CountA**: Counts all the non-blank cells in a range.
- • **Countif**: Counts cells that fulfill a specific condition.
- • **Countifs**: Counts cells depending on multiple conditions.
- • **Averageif** and **Averageifs**: Obvious.

Examples of **Count** and **CountA** are given below:

	A	B	C
1	**COUNT AND COUNTA**		
2	**Count**: Count only numerical values	5	<--=COUNT(A6:C8)
3	**CountA**: Count all non-blank cells	8	<--=COUNTA(A6:C8)
4			
5	**Data:**		
6	1	two	3
7	4		six
8	seven	8	9

To use **Countif**, we have to specify a condition. The spreadsheet below gives a year of weekly stock returns for Merck (many rows have been hidden):

	A	B	C	D
1	**USING COUNTIF ON MERCK'S WEEKLY STOCK RETURNS**			
2	Number of returns	1043	<--=COUNT(C:C)	
3	Returns over 2%	266	<--=COUNTIF(C:C,">2%")	
4				
5	Start	01-Jan-10		
6	End	31-Dec-19		
7	Observations	522	<--=COUNTIFS(A11:A1054,">="&B5,A11:A1054,"<="&B6)	
8	Average return	0.24%	<--=AVERAGEIFS(C11:C1053,A11:A1053,">="&B5,A11:A1053,"<="&B6)	
9				
10	**Date**	**MRK**	**Return**	
11	28-Dec-19	90.95	-0.60%	<--=LN(B11/B12)
12	21-Dec-19	91.50	-0.09%	<--=LN(B12/B13)
13	14-Dec-19	91.58	3.33%	
14	07-Dec-19	88.58	0.38%	
1050	29-Jan-00	35.47	-1.73%	
1051	22-Jan-00	36.09	5.29%	
1052	15-Jan-00	34.22	-2.30%	
1053	08-Jan-00	35.02	-1.09%	
1054	01-Jan-00	35.41		

In cell B3 we count all stock returns that are over 2%. In cells B7:B8 we illustrate a different technique. The number of observations for a specific time frame can be calculated by **Countifs** where the two criteria are a date greater than or equal to the start date **and also** lower than or equal to the end date. Changing the entries in cells B5:B6 allows us to count and calculate the average of returns for different time frames.

---

1. 1. Excel's **IRR** function assumes that the first cash flow occurs today, the next cash flow occurs one period hence, the following cash flow two periods hence, and so on. Excel's **NPV** function assumes that the first cash

flow occurs one period from now, the next cash flow in two periods, etc. We call this “even spacing of cash flows.” When this is not the case, you’ll need the **XIRR** and **XNPV** functions.

2. 2. The two additional functions **NXIRR** and **NXNPV** were developed by Benjamin Czaczkes. Read “Adding Getformula to Your Spreadsheet” on the website that accompanies this book to see how you can copy these functions into your own spreadsheets.
3. 3. Recall that when the discount rate is zero, the net present value is just the sum of all the cash flows.
4. 4. The way this appears and is interpreted depends on the Regional Settings entered in the Windows Control Panel.
5. 5. This procedure is available only if you’ve installed the **Data Analysis** add-in. If you don’t see **Data Analysis** on your Excel **Data** menu, go to **File|Options|Add-ins**. At the bottom of the resulting screen, indicate **Manage Add-ins**. On the next screen, put a checkmark next to **Analysis ToolPak**.
6. 6. There is also an Excel function **Logest**, whose syntax is exactly the same as that of **Linest**. **Logest** calculates the parameters to fit an exponential curve.

31 Array Functions

31.1 Overview

An Excel array function or formula performs an operation on a rectangular block of cells. In the simplest cases, built-in Excel array functions such as **Transpose** or **MMult** take an array and transpose it or take two matrices and multiply them. Once you get the hang of array functions, you can design your own array formulas. In this chapter, for example, we show how to use array formulas to find the minimum or maximum of the off-diagonal elements of a matrix or to pick out the diagonal of a matrix—all useful tricks to know when doing portfolio calculations such as those discussed in Chapters 10–15.

The one critical thing to remember about array functions or formulas in that they are entered into a spreadsheet by pressing **[Ctrl] + [Shift] + [Enter]** keys simultaneously; this contrasts with the usual procedure whereby we enter a function or formula by pressing **[Enter]**.

If you are using excel version 2019 or later, you can achieve the same goals as with the automatic "Spill feature" by simply pressing [Enter]. The main differences between the classic array functions and the "spill" feature are:

1. There is no need to select the range where the results are to be populated. Excel will determine the size of the output range automatically.
2. The output range starts from the formula cell, and "spills" the results by populating the cells through right and bottom, based on the return array.
3. Use the first cell (top-left) in the spill range to edit the formula. Other cells in the spill range will show the formula as well, but the formula will be "grayed" and cannot be updated.
4. When you select a cell within the spill area, the range will be highlighted with a blue border.
5. A spilled range can be referenced using the dash character (#) after the top-left cell. For example, you can use **B2#** instead of **B2:B6.** This is called "spilled range" operator.
6. If there are any non-empty cells in the spill area, the formula returns a **#SPILL!** error. Excel will automatically populate the cells when the occupied cells are cleared.

31.2 Some Built-In Excel Array Functions

In this section we discuss some built-in Excel array functions: **Transpose**, **MMult**, **MInverse**, and **Frequency**. Other functions are discussed elsewhere in this book—for example, the function **Linest** is discussed in Chapter 30.

Transpose

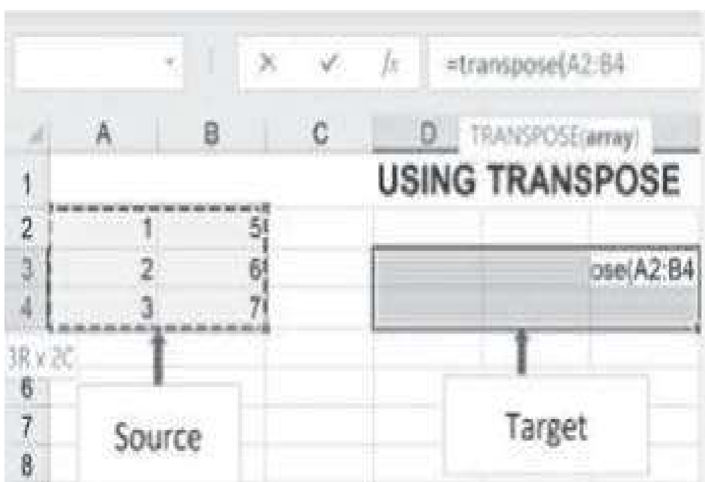
Suppose we are trying to calculate the transpose of a 3 × 2 (three rows, two columns) matrix that is in cells A2:B4 of the spreadsheet.

	A	B
2	1	5
3	2	6
4	3	7

Excel has a function called **Transpose()**. Like all array functions, its use requires care:

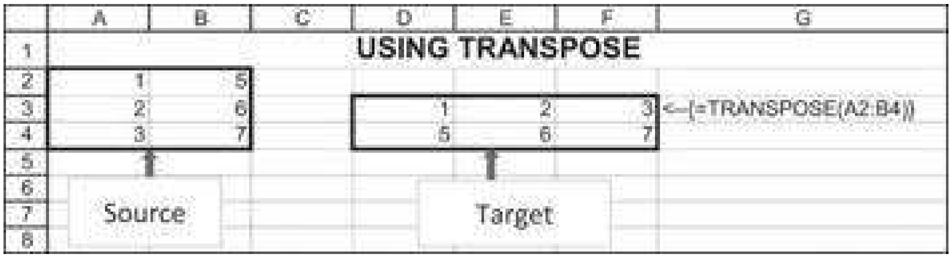
- • **Mark the target**: Block off the cells D3:F4 into which you intend to put the transposed matrix.
- • **Type the array function**: Now type **=Transpose(A2:B4)**. This will appear in the top left-hand corner of the blocked-off cells. Of course, you can use the usual tricks to show Excel which cells you want (e.g., pointing or using named ranges).

At this point your spreadsheet looks like this:



- • **[Ctrl] + [Shift] + [Enter]**: When you have finished typing the formula. This will put the array function into all of the blocked-off cells.

Here’s what the final product will look like:



Notice that the array function is surrounded by curly brackets {}. You don't type these in—Excel puts them in automatically.

Paste|Paste Special|Transpose

There is, of course, another way to transpose an array: You can copy the original array and then use **Paste|Paste Special** to transpose the range, clicking on **Transpose**:

This will transpose the range, but it will not link the original with the target range. When you change something in the original range, the target range is unchanged. The neat thing about the array function **Transpose** is that it is a *dynamic function*, as are all the array functions and formulas: When you change one of the initial set of cells, the transposed array also changes.

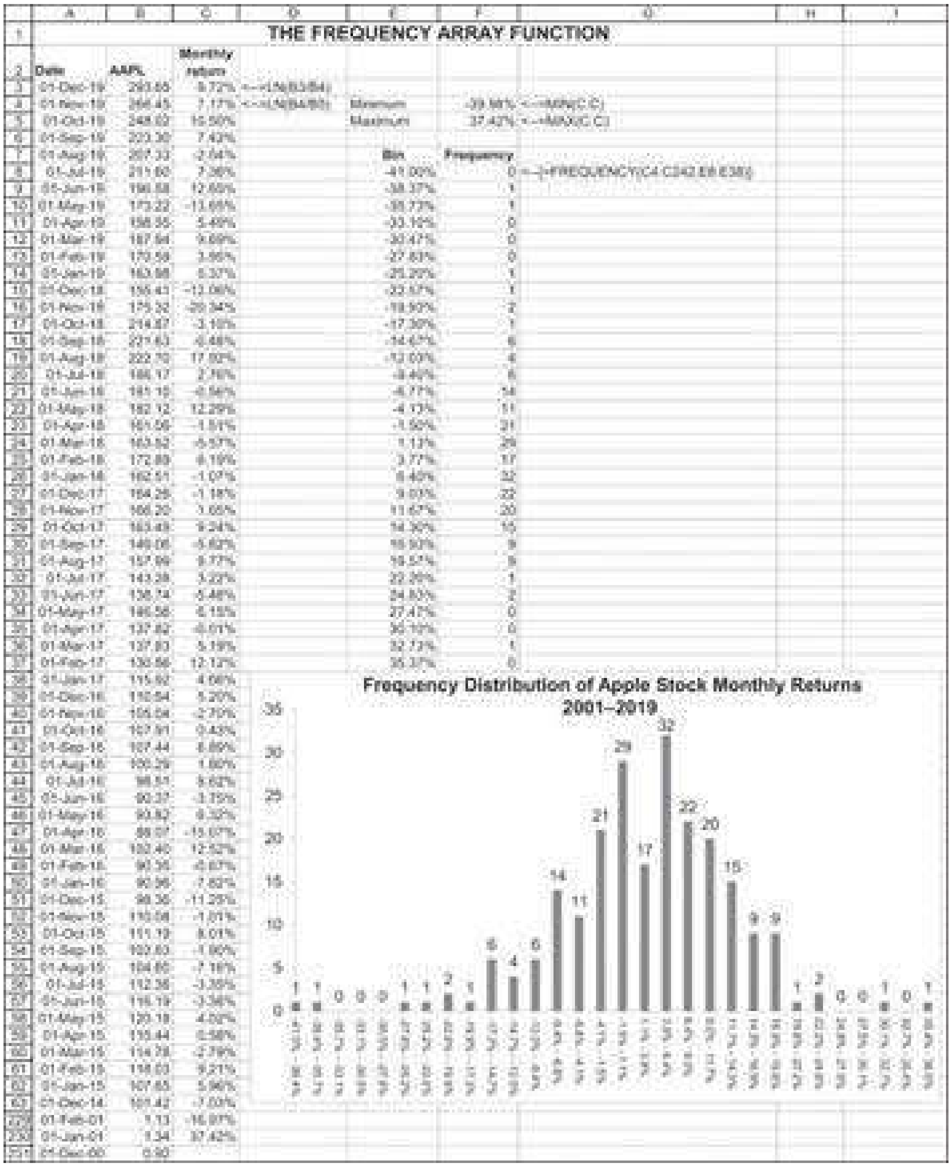
MMult and MInverse—Multiplying and Inverting Matrices

These two functions are used and explained in Chapters 10–15, on portfolio calculations, so we only recapitulate briefly:

- • **MMult(range1,range2)** multiplies the matrix in **range1** times that in **range2**. Of course, this is only possible if the number of columns in **range1** equals the number of rows in **range2**.
- • **MInverse(range)** calculates the inverse of the matrix in **range**. Note that **range** must be rectangular.

Frequency

The Excel array function **Frequency(data_array,bins_array)** calculates the frequency distribution of a data set. The spreadsheet below shows monthly return data for Apple (AAPL) stock over the period January 2001 to December 2019. In column E we have put the bins, taking care that the first bin will be *below* the minimum monthly return over the period and that the last bin will be *above* the maximum monthly return. The range F8:F38 contains the array function **Frequency (C4:C363,E8:E38)**. From the output we can, for example, deduce that in the 19-year period there were two monthly returns between 22.2% and 24.8%, and 20 monthly returns between 9% and 11.7%.



31.3 Homemade Array Functions

In our experience, array functions often arise out of situations where you are called on to do long, repetitive calculations. You then discover that the same calculation can be done in a single array function. In many cases it is not straightforward to understand why a particular array technique should work. For example, in this chapter we will use the fact that A3+B6:B8 adds the contents of cell A3 to each of cells B6:B8. Why? Heaven only knows! We also use the (undocumented, as far as we are aware) trick that B3:B7∧A3:A7 raises cell B3 to the power A3, cell B4 to the power A4, …, and so on.

In this section we illustrate homemade array functions with two examples having to do with investment returns.

Computing the Compound Annual Return from 10 Years of Return Data

The table below gives the annual returns for the Harvard University endowment. You are asked to compute the compound annual return over the 10-year period. Assuming that the returns have been discretely computed (meaning that

$r_t = \dfrac{Endowment\ value_t}{Endowment\ value_{t-1}}, t = 1, \ldots, 10$), you realize that the compounded annual return is $r = ((1 + r_{2002}) * (1 + r_{2003}) \ldots * (1 + r_{2011}))^{1/10} - 1$. In cell B14 below, we do this calculation with a single array function:

	A	B	C
1	**HARVARD UNIVERSITY ENDOWMENT RETURNS** Years ending June 30		
2	**Year**	**Return**	
3	2010	10.98%	
4	2011	21.36%	
5	2012	-0.05%	
6	2013	11.31%	
7	2014	15.40%	
8	2015	5.78%	
9	2016	-2.04%	
10	2017	8.06%	
11	2018	9.96%	
12	2019	6.50%	
13			
14	Compound annual return	8.53%	<--{=PRODUCT(1+B3:B12)^(1/10)-1}

The Excel function **Product** multiplies the entries in a range of cells. The entry in cell B14 adds 1 to each cell in B3:B12, multiplies the cells, takes the tenth root, and subtracts 1 from the result—all in one cell (entered, of course, with [Ctrl] + [Shift] + [Enter]).[1]

Computing the Compound Annual Continuous Return

Column B of the spreadsheet below gives the amounts that accumulated in a customer account of the Youngtalk Investment Fund. The annual continuously compounded return is computed by $r_t = Ln\left(\dfrac{Account_t}{Account_{t-1}}\right)$, and the average return over the period is $\dfrac{1}{10}\sum_{t=1}^{10} r_t$. In cell B15 below we do this calculation by averaging the annual returns, and in cell B17 we show an array function that does the whole calculation in a single cell. Pretty neat!

	A	B	C	D
1	**YOUNGTALK INVESTMENT FUND**			
2	**Year**	**Investment beginning of**	**Continuous return for year**	
3	2012	100.00		
4	2013	121.51	19.48%	<--=LN(B4/B3)
5	2014	132.22	8.45%	
6	2015	98.63	-29.31%	
7	2016	75.65	-26.53%	
8	2017	140.48	61.90%	
9	2018	221.40	45.49%	
10	2019	243.46	9.50%	
11	2020	280.11	14.02%	
12	2021	398.72	35.31%	
13	2022	543.58	30.99%	
14				
15	Compound annual return		16.93%	<--=AVERAGE(C4:C13)
16			16.93%	<--=LN(B13/B3)/10
17	Same calculation with array function		16.93%	<--{=AVERAGE(LN(B4:B13/B3:B12))}

A final note: Look at cell C16. If you know some continuous-time mathematics, you will know that **LN(B13/B3)/10** produces the same result. Simpler yet!

Computing Discount Factors with an Array Function

We are given a set of interest rates $r_1, r_2, \ldots, r_n$ and we want to compute the formula $\sum_{t=1}^{n} \frac{1}{(1+r_t)^t}$. Excel's **NPV** function won't work for this, so we'll have to build our own function. The spreadsheet below shows two methods:

	A	B	C	D
1	COMPUTING PRESENT VALUE FACTORS WITH AN ARRAY FUNCTION			
2	Year	Interest rate		
3	1	6.23%		
4	2	4.00%		
5	3	4.20%		
6	4	4.65%		
7	5	4.80%		
8				
9	Present value	4.3746	<-- {=SUM(1/((1+B3:B7)^A3:A7))}	
10				
11	Checking the formula with a recursive formula			
12	Year	Interest rate	Sum of $1/(1+r_t)^t$	
13	1	6.23%	0.9414	<--=1/(1+B13)^A13
14	2	4.00%	1.8659	<--=C13+(1/(1+B14)^A14)
15	3	4.20%	2.7498	<--=C14+(1/(1+B15)^A15)
16	4	4.65%	3.5836	<--=C15+(1/(1+B16)^A16)
17	5	4.80%	4.3746	<--=C16+(1/(1+B17)^A17)

In cell B9 we use an array function {=SUM(1/((1+B3:B7)∧A3:A7))}. Writing (1+B3:B7)∧A3:A7) adds 1 to each of cells B3:B7 and raises the result to the power in cells A3:A7. Applying **Sum** gives the result (of course, this being an array function, you have to enter it with [Ctrl] + [Shift] + [Enter]).

An alternative, given in rows 13–17, is to build the result recursively. This gives the same result, but with more work.

31.4 Array Formulas with Matrices

In this section we create some array functions that have to do with matrices.

Subtracting a Constant from a Matrix

In the portfolio computations of Chapters 10–15, we often have to subtract a constant from a matrix. This is easy to do with an array formula, entered with [Ctrl] + [Shift] + [Enter].

	A	B	C	D	E	F	G	H
1	SUBTRACTING A CONSTANT FROM A MATRIX							
2	Matrix			Constant		Matrix minus		
3	1	6		3		-2	3	
4	2	6				-1	3	<--{=A3:B7-D3}
5	3	8				0	5	
6	4	9				1	6	
7	5	10				2	7	

Creating a Matrix with Ones on the Diagonal and Zeros Elsewhere

This is a problem that comes up in Chapter 9: We want a matrix that has a diagonal of entries that are 1 but has 0 in the off-diagonal elements. The spreadsheet below shows two ways of doing this:

	A	B	C	D	E	F
1	CREATING A MATRIX OF 1'S AND 0'S We want 1 on diagonal, and 0 elsewhere					
2	The cells below contain the array formula {=IF(B3:E3=A4:A7,1,0)}					
3		A	B	C	D	
4	A	1	0	0	0	
5	B	0	1	0	0	
6	C	0	0	1	0	
7	D	0	0	0	1	
8						
9	The cells below contain the formula =IF(B$10=$A11,1,0)					
10		A	B	C	D	
11	A	1	0	0	0	
12	B	0	1	0	0	
13	C	0	0	1	0	
14	D	0	0	0	1	

The first and second methods rely on the labeling of the rows and the columns. In the first example, the formula =IF(B3:E3=A4:A7,1,0) tests whether the row label equals the column heading; if this is true, we put a 1 in the cell, and otherwise we put in a 0. In the second example we use an **If** function with mixed absolute and relative references to create the same effect.

Finding the Maximum and Minimum Off-Diagonal Elements of a Matrix

We want to find the maximal and minimal elements of the off-diagonal elements of a matrix. Two ways to do this are illustrated below:

	A	B	C	D	E	F
1	FINDING THE MAX,MIN OF OFF-DIAGONAL ELEMENTS The long way					
2		The source matrix				
3		A	B	C	D	
4	A	10	2	3	4	
5	B	-3	20	4	-3	
6	C	1	5	60	6	
7	D	4	2	-10	25	
8						
9	Range below contains array formula =IF(B3:E3=A4:A7,"",B4:E7)					
10		A	B	C	D	
11	A		2	3	4	<--{=IF(B3:E3=A4:A7,"",B4:E7)}
12	B	-3		4	-3	
13	C	1	5		6	
14	D	4	2	-10		
15						
16	Max of off-diagonals	6	<--=MAX(B11:E14)			
17	Min of off-diagonals	-10	<--=MIN(B11:E14)			
18						
19						
20	Using only non-array formulas					
21	Range below contains the non-array formula =IF(B$3=$A4,"",B4)					
22		A	B	C	D	
23	A		2	3	4	<--=IF(E$3=$A4,"",E4)
24	B	-3		4	-3	
25	C	1	5		6	
26	D	4	2	-10		
27						
28	Max of off-diagonals	6	<--=MAX(B23:E26)			
29	Min of off-diagonals	-10	<--=MIN(B23:E26)			

In the example above, we first use an array function to replace all the diagonal elements with a blank cell. We can then use **Max** and **Min** to determine the extreme off-diagonal elements. As shown in rows 20–29, we could also have used the non-array formula = IF(B$3=$A4, "", B4) in cell B11 and copied it to the rest of the matrix.

We can also find the maximum and minimum by incorporating the array formula directly into the **Max** and **Min**, as shown in the next spreadsheet:

	A	B	C	D	E	F
1	FINDING THE MAX,MIN OF OFF-DIAGONAL ELEMENTS In one step					
2		The source matrix				
3		A	B	C	D	
4	A	10	2	3	4	
5	B	-3	20	4	-3	
6	C	1	5	60	6	
7	D	4	2	-10	25	
8						
9	Max of off-diagonals	6	<--{=MAX(IF(B3:E3=A4:A7,"",B4:E7))}			
10	Min of off-diagonals	-10	<--{=MIN(IF(B3:E3=A4:A7,"",B4:E7))}			

Replacing the Off-Diagonals Using VLookup

Now suppose that we want to replace the off-diagonal elements using a lookup table, as illustrated below:

	A	B	C	D	E	F
1	REPLACING THE MAX,MIN OF OFF-DIAGONAL ELEMENTS OF A MATRIX Long way, non-array formulas					
2		The source matrix				
3		A	B	C	D	
4	A	10	2	3	4	
5	B	-3	20	4	-3	
6	C	1	5	60	6	
7	D	4	2	-10	25	
8						
9	Lookup table for replacements					
10	-10	i				
11	-6	ii				
12	-2	iii				
13	2	iv				
14	6	v				
15	10	vi				
16						
17	Range below contains non-array formula =IF(B$3=$A4,B4,VLOOKUP(B4,$A$10:$B$15,2)), which has been copied to all the cells					
18		A	B	C	D	
19	A	10	iv	iv	iv	<--=IF(E$3=$A4,E4,VLOOKUP(E4,$A$10:$B$15,2))
20	B	ii	20	iv	ii	
21	C	iii	iv	60	v	
22	D	iv	iv	i	25	

Array formulas can simplify this procedure:

	A	B	C	D	E	F
1	REPLACING THE MAX,MIN OF OFF-DIAGONAL ELEMENTS OF A MATRIX Using array formula					
2		The source matrix				
3		A	B	C	D	
4	A	10	2	3	4	
5	B	-3	20	4	-3	
6	C	1	5	60	6	
7	D	4	2	-10	25	
8						
9	Lookup table for replacements					
10	-10	i				
11	-6	ii				
12	-2	iii				
13	2	iv				
14	6	v				
15	10	vi				
16						
17	Range below contains array formula =IF(B3:E3=A4:A7,B4:E7,VLOOKUP(B4:E7,A10:B15,2))					
18		A	B	C	D	
19	A	10	iv	iv	iv	<--{=IF(B3:E3=A4:A7,B4:E7,VLOOKUP(B4:E7,A10:B15,2))}
20	B	ii	20	iv	ii	
21	C	iii	iv	60	v	
22	D	iv	iv	i	25	

Exercises

1. Use a homemade array function to multiply the vector {1, 2, 3, 4, 5} times the constant 3.
2. Use the array functions **Transpose** and **MMult** to multiply the row vector {1, 2, 3, 4, 5} times the column vector $\left\{\begin{bmatrix} -8 \\ -9 \\ 7 \\ 6 \\ 5 \end{bmatrix}\right\}$.
3. Below you will find the variance-covariance matrix of six stocks. Use an array function to create a matrix with only variances on the diagonal and with zeros elsewhere.

	A	B	C	D	E	F	G
1		GE	MSFT	JNJ	K	BA	IBM
2	GE	**0.1035**	0.0758	0.0222	-0.0043	0.0857	0.1414
3	MSFT	0.0758	**0.1657**	0.0412	-0.0052	0.0379	0.1400
4	JNJ	0.0222	0.0412	**0.0360**	0.0181	0.0101	0.0455
5	K	-0.0043	-0.0052	0.0181	**0.0570**	-0.0076	0.0122
6	BA	0.0857	0.0379	0.0101	-0.0076	**0.0896**	0.0856
7	IBM	0.1414	0.1400	0.0455	0.0122	0.0856	**0.2993**

4. For the problem above: Use an array function to create a matrix with zeros on the diagonal and the covariances off-diagonal.
5. In the companion website to this book you'll find the exercise Excel file for this chapter. This file gives data for three mutual funds. Compute the discrete annual returns for each fund and then use an array function to compute the compound annual return over the period. Recall that discretely compounded, the return in year t is $(Fund\ value_t/Fund\ value_{t-1}) - 1$. If the returns were continuously compounded, then the year t return would be $Ln(Fund\ value_t/Fund\ value_{t-1})$.

---

1. 1. There is nothing in Excel documentation that indicates why this marvelous feature should work. But it does.

32 Some Excel Hints

32.1 Overview

This chapter covers a grab bag of Excel hints dealing with problems and needs that we sometimes run into. The chapter makes no pretense at uniformity or extensiveness of coverage. Topics covered include:

- • Fast fills and copy
- • Graph titles that change when data changes
- • Creating multi-line cells (useful for putting line breaks in cells and linked graph titles)
- • Typing Greek symbols
- • Typing subscripts and superscripts (but not both)
- • Naming cells
- • Hiding cells
- • Formula auditing
- • Writing on multiple spreadsheets
- • Using Excel's personal notebook to copy and paste and format quickly

32.2 Fast Copy: Filling in Data Next to Filled-In Column

Usually, we copy cells by dragging on the fill handle of the cell with the formula. There is sometimes an easier method. Consider the following situation:

	A	B	C
1	AUTO FILL/COPY		
2	1	2	
3	2	5	<-- =B2+3
4	3		
5	4		
6	5		
7	6		
8	7		
9	8		

Double-click on "fill handle" (shown below with the cross). After double-clicking, the range B2:B9 will automatically fill with the formula in B3.

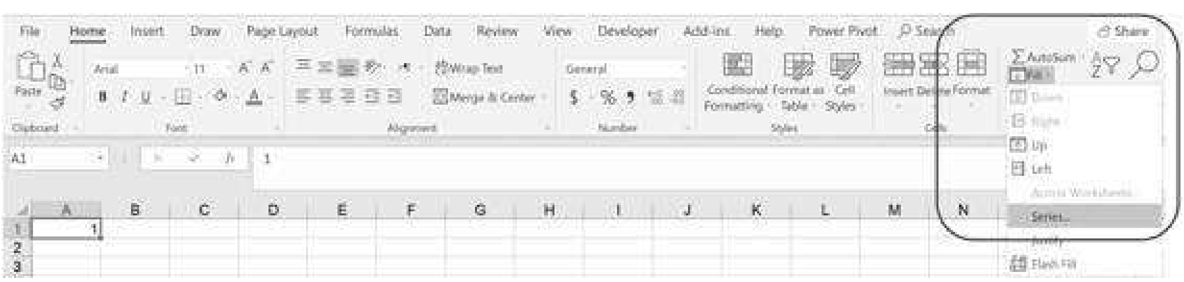
	A	B	C
1	AUTO FILL/COPY		
2	1	2	
3	2	5	<-- =B2+3
4	3		
5	4		
6	5		
7	6		
8	7		
9	8		

Here's the result:

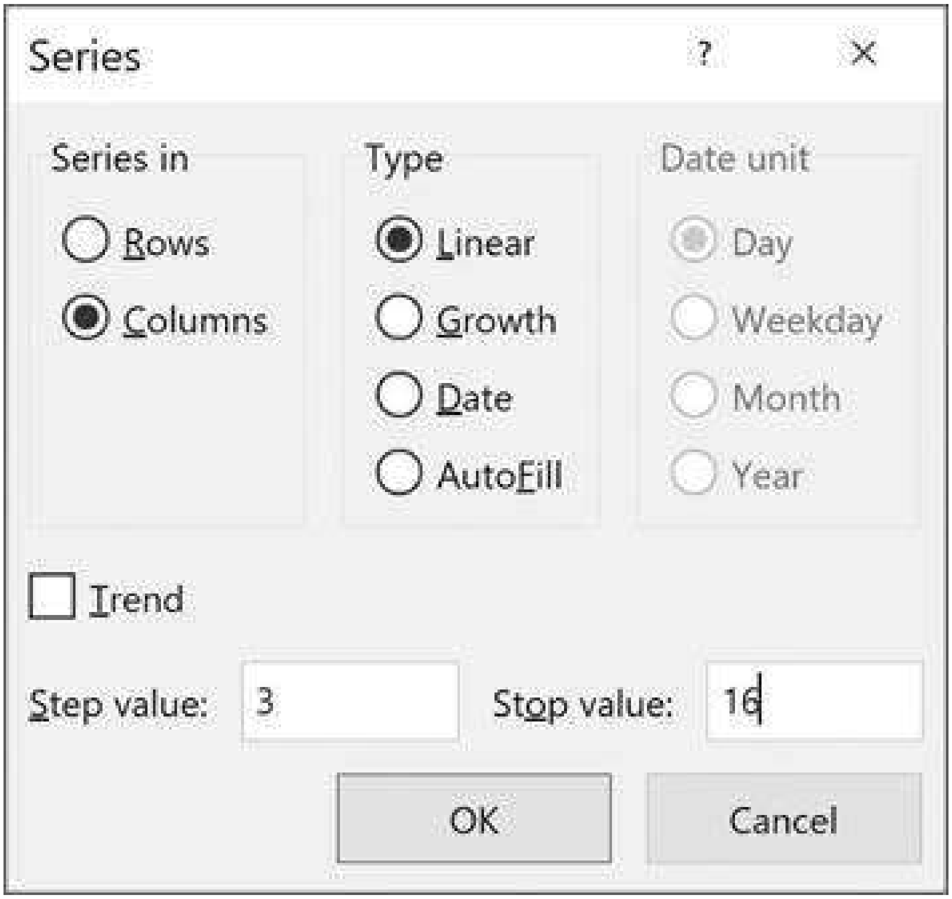
	A	B	C
1	AUTO FILL/COPY		
2	1	2	
3	2	5	<-- =B2+3
4	3	8	
5	4	11	
6	5	14	
7	6	17	
8	7	20	
9	8	23	
10			
11	Double-clicking on the "fill handle" of a cell will fill in the rest of the column provided there's a filled cell next to it.		

32.3 Filling Cells with a Series

Sometimes we want to fill a set of cells with a series. This can be accomplished by going to **Editing|Fill|Series** on the **Home** tab:



Here's an example. Starting with cell A1, we want to fill a column with cells increasing by 3 until we get to 16:



Clicking **OK** gives the following:

	A
1	1
2	4
3	7
4	10
5	13
6	16

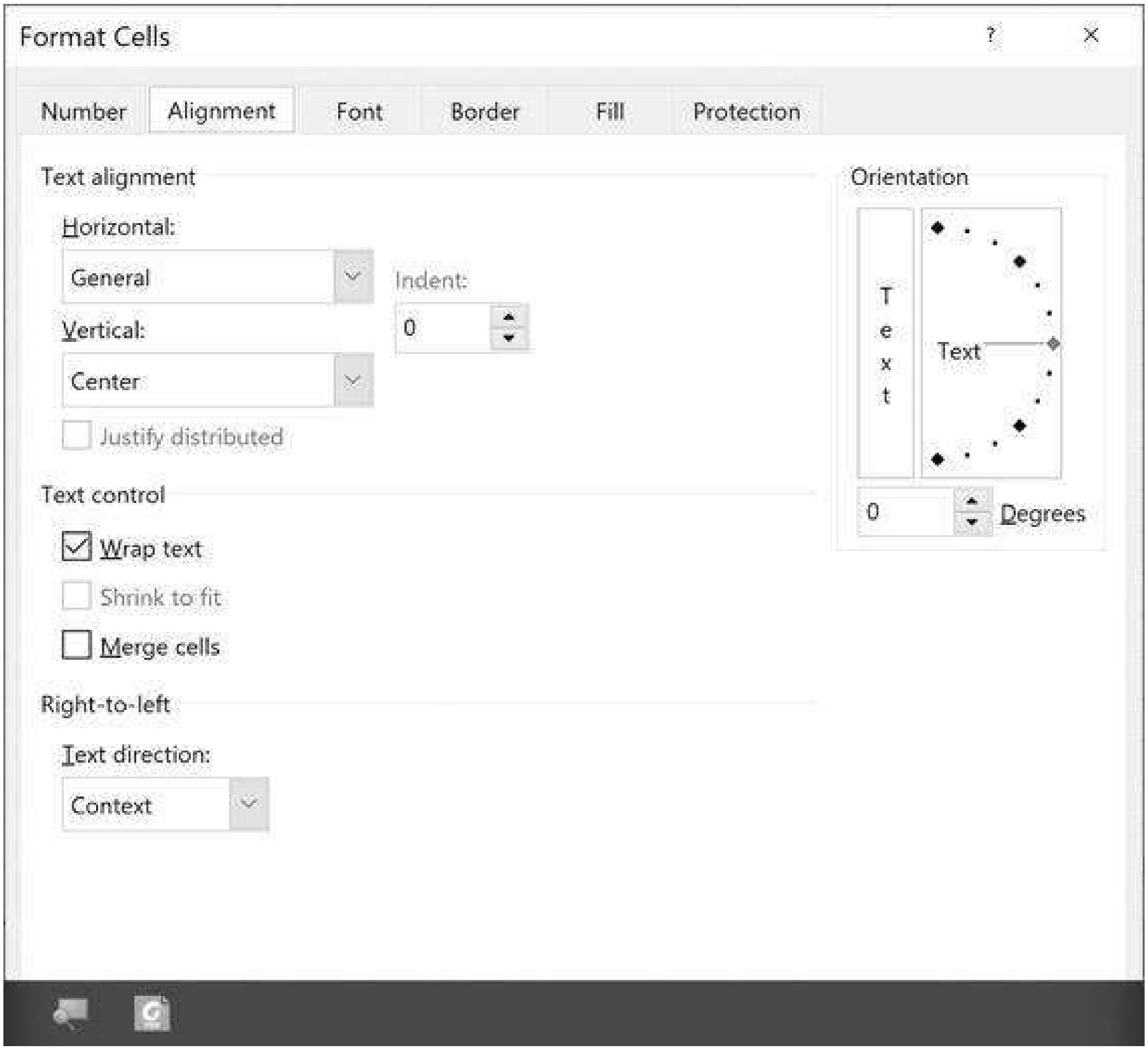
This interesting command has several other options, which we leave for you to try.

32.4 Multi-line Cells

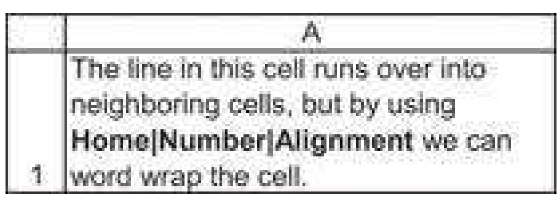
It is sometimes useful to put a line break in a cell, thus creating a multi-line cell. Do this with **[Alt]** + **[Enter]** where you want a line break.

	A
1	**PUTTING LINE BREAKS IN CELLS**
2	This is a multi-line cell. The break was entered by inserting [Alt]+[Enter] at the desired break points.

There are, of course other ways to make a cell multi-lined. The most obvious is to use the **Wrap text** box in the **Format Cells|Alignment** command:



Here's the cell after we selected to wrap text. (Note that in the dialog box above, we have also set the vertical alignment of the cell to **Center**.)



32.5 Multi-line Cells with Text Formulas

Sometimes you want to put a line break in a cell that has text formulas in it. In the example below, the text formula in cell A5 combines the text in cells A2 and A3.

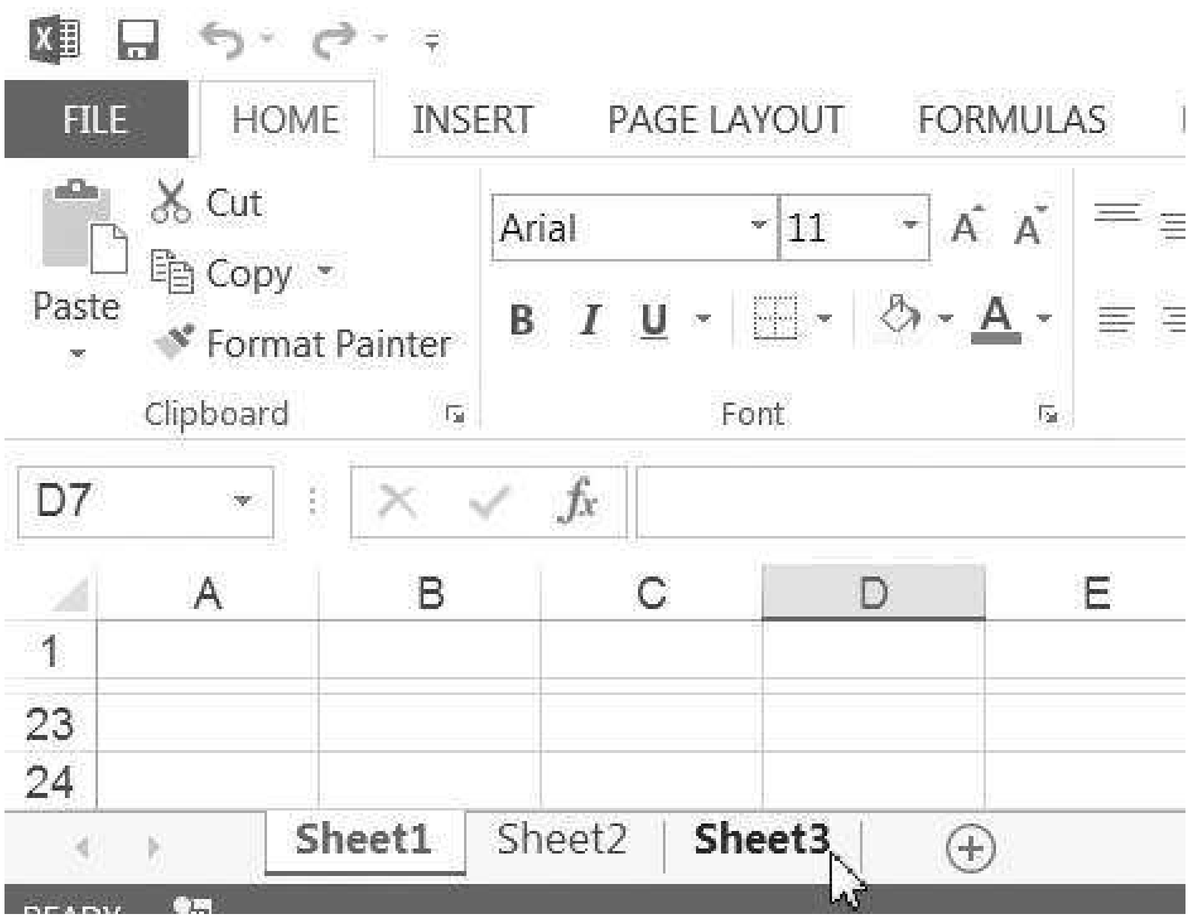
	A	B
1	ADDING A BREAK TO CONCATENATED TEXT	
2	Simon	
3	Jack	
4		
5	SimonJack	<--=A2&A3
6	SimonJack	<--=A2&CHAR(10)&A3, not correctly formatted
7	Simon Jack	<--=A2&CHAR(10)&A3, pushed the Wrap Text on the Home tab

We can put a line break into the text formula by doing two things:

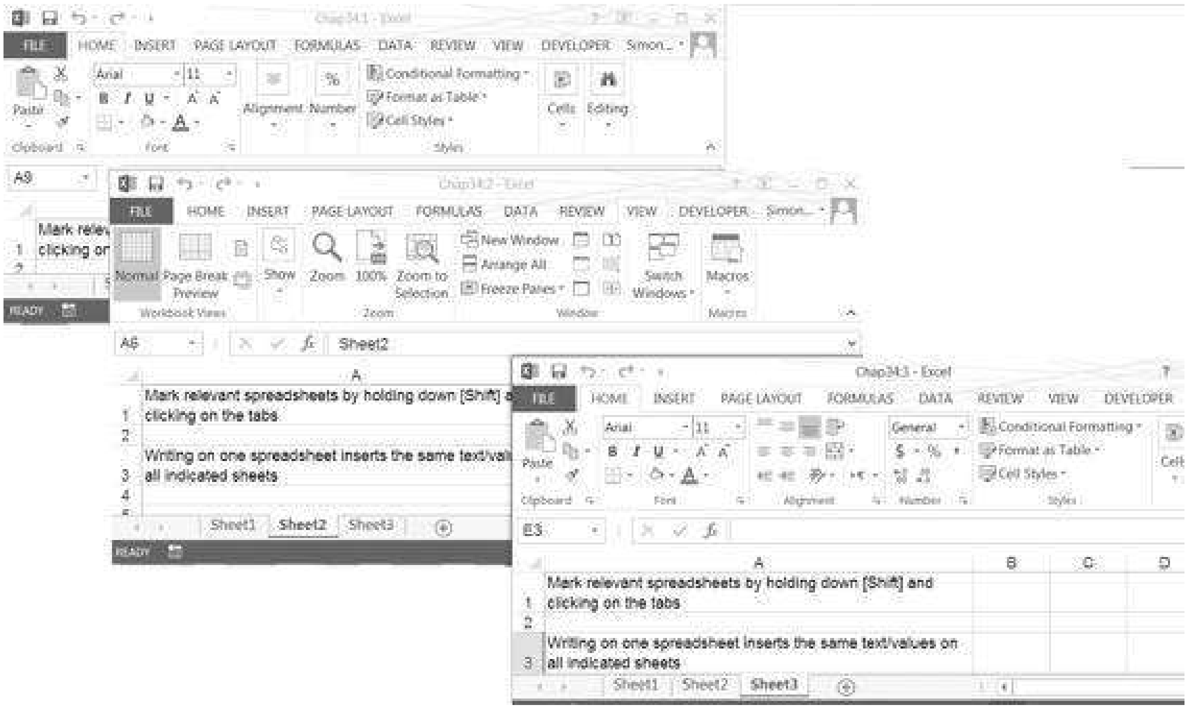
- • Put **Char(10)** between A2 and A3—that is, write the formula =**A2&Char(10)&A3** into the cell. **Char(10)** is the code for a hard line return.
- • Push **Word Wrap** on the **Home** tab. Now you will have a break between the contents of the two cells at the point you entered **Char(10)**.

32.6 Writing on Multiple Spreadsheets

This Excel trick allows you to write in multiple spreadsheets at the same time. First, hold down [Shift] and indicate several spreadsheet tabs by clicking on them. In the example below, we've clicked on the tabs labeled **Sheet1**, **Sheet2**, **Sheet3**.



Now, anything we write in one of the sheets is also written in the same cells of all the others, so that we can produce three identical spreadsheets:



32.7 Moving Multiple Sheets of an Excel Notebook

We write on multiple sheets by holding down [Shift] and marking the tabs of the relevant spreadsheets. A similar trick works to move multiple sheets of the same Excel notebook:

- • Mark the multiple sheets by holding down [Shift] and clicking on the appropriate sheets.
- • Now use **Edit|Move** or **copy sheet** to move or copy the sheets to another location on the same spreadsheet or to a different spreadsheet.

32.8 Text Functions in Excel

The **Text** function lets you change numbers to text. Here are some examples:

	A	B	C
1	**TEXT FUNCTIONS**		
2	Income	15,000	
3	Tax rate	35%	
4	Taxes owed	5,250	<--=B2*B3
5			
6	Tax rate as text	35.00%	<--=TEXT(B3,"0.00%")
7		0.4	<--=TEXT(B3,"0.0")
8			
9	Income as date	Jan.24,1941	<--=TEXT(B2,"mmm.dd,yyyy")

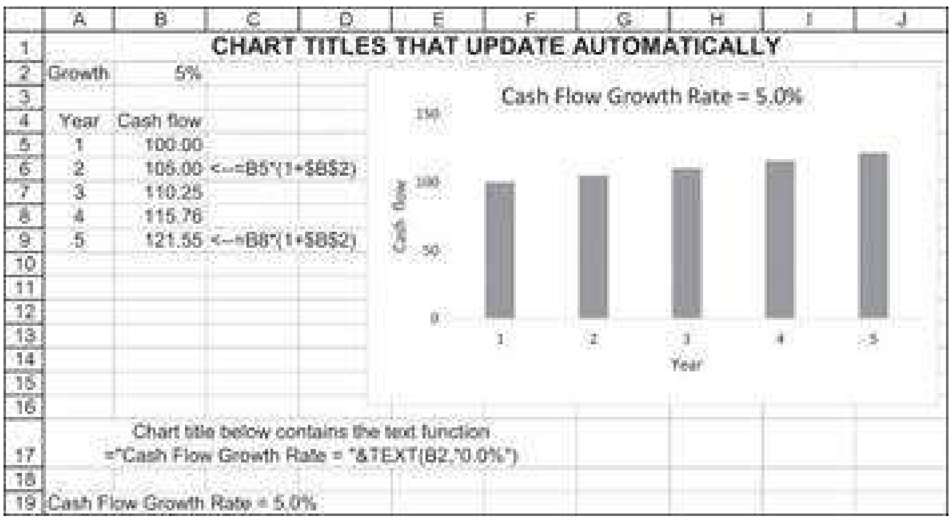
Note that you can choose different ways of formatting the cell B3 in text form—in cell B6 we have formatted the tax rate as a percentage with two decimal points, whereas in cell B7 we have formatted the tax rate as one decimal, causing it to be rounded off.

Note also the somewhat stupid example in cell B9: Since dates in Excel are just numbers that express the number of days from January 1, 1900, we can express the income of $15,000 in cell B2 as a date.

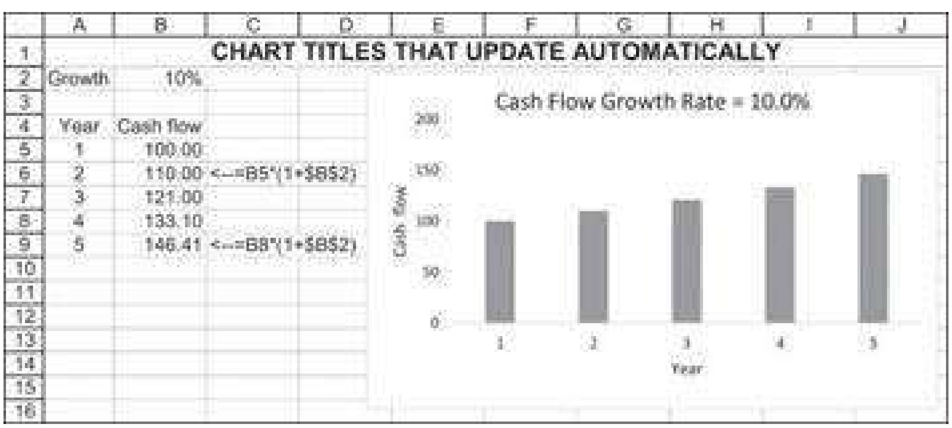
In the next section we use text functions to create chart titles that update themselves.

32.9 Chart Titles That Update

You want to have the chart title change when a parameter on the spreadsheet changes. For example, in the next spreadsheet, you want the chart title to indicate the growth rate.

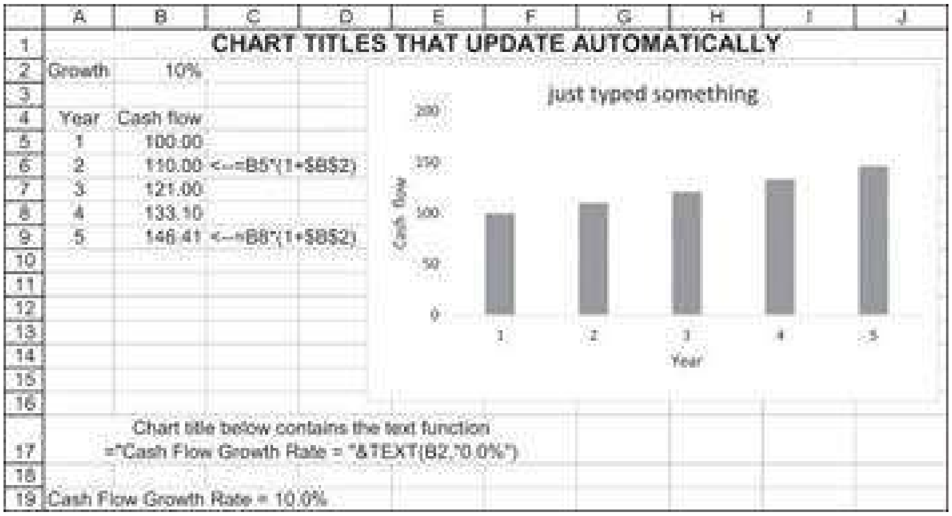


Once we have completed the necessary steps, changing the growth rate will change both the graph *and* its title:



To make graph titles update automatically, carry out the following steps:

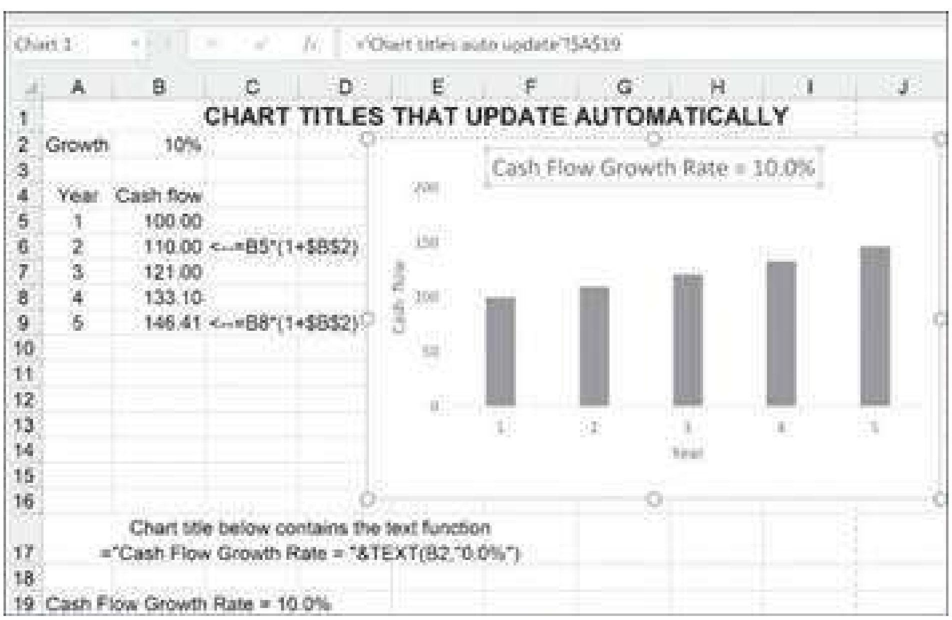
- • Create the graph you want in the format you want it. Give the graph a "proxy title." (It makes no difference what it is; you are going to eliminate it soon.) At this stage, your graph might look like this:



- • Create the title you want in a cell. In the example above, cell A19 contains the formula:

=“Cash Flow Graph when Growth = “&TEXT(B2,”0.0%”)

- • Click on the graph title to mark it, and then go to the formula bar and insert an equal sign to indicate a formula. Then **point** at cell A19 with the formula and click **[Enter]**. In the picture below, you see the chart title highlighted and in the formula bar “=‘Chart titles auto update’!$A$19” indicating the title of the graph.

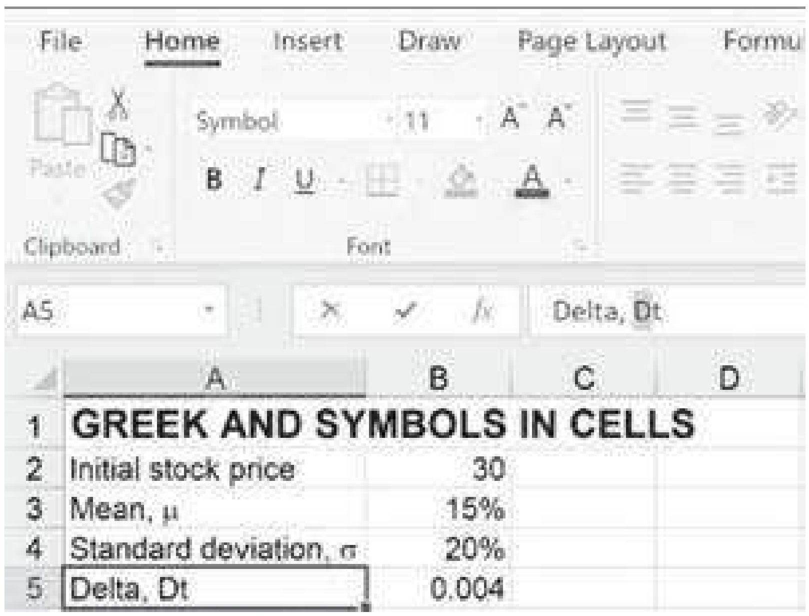


32.10 Putting Greek Symbols in Cells

How do we type Greek letters in a spreadsheet?

	A	B
1	**GREEK AND SYMBOLS IN CELLS**	
2	Initial stock price	30
3	Mean, μ	15%
4	Standard deviation, σ	20%
5	Delta, Δt	0.004

This is fairly simple, if you know the Greek equivalents for the Greek letters (for example, μ and σ are lowercase “m” and “s,” respectively, while Σ and Δ are uppercase “S” and “D”). For example, we first typed “Delta, Dt” into cell A5 and then marked the “D” in the formula bar and change the font type to Symbol:



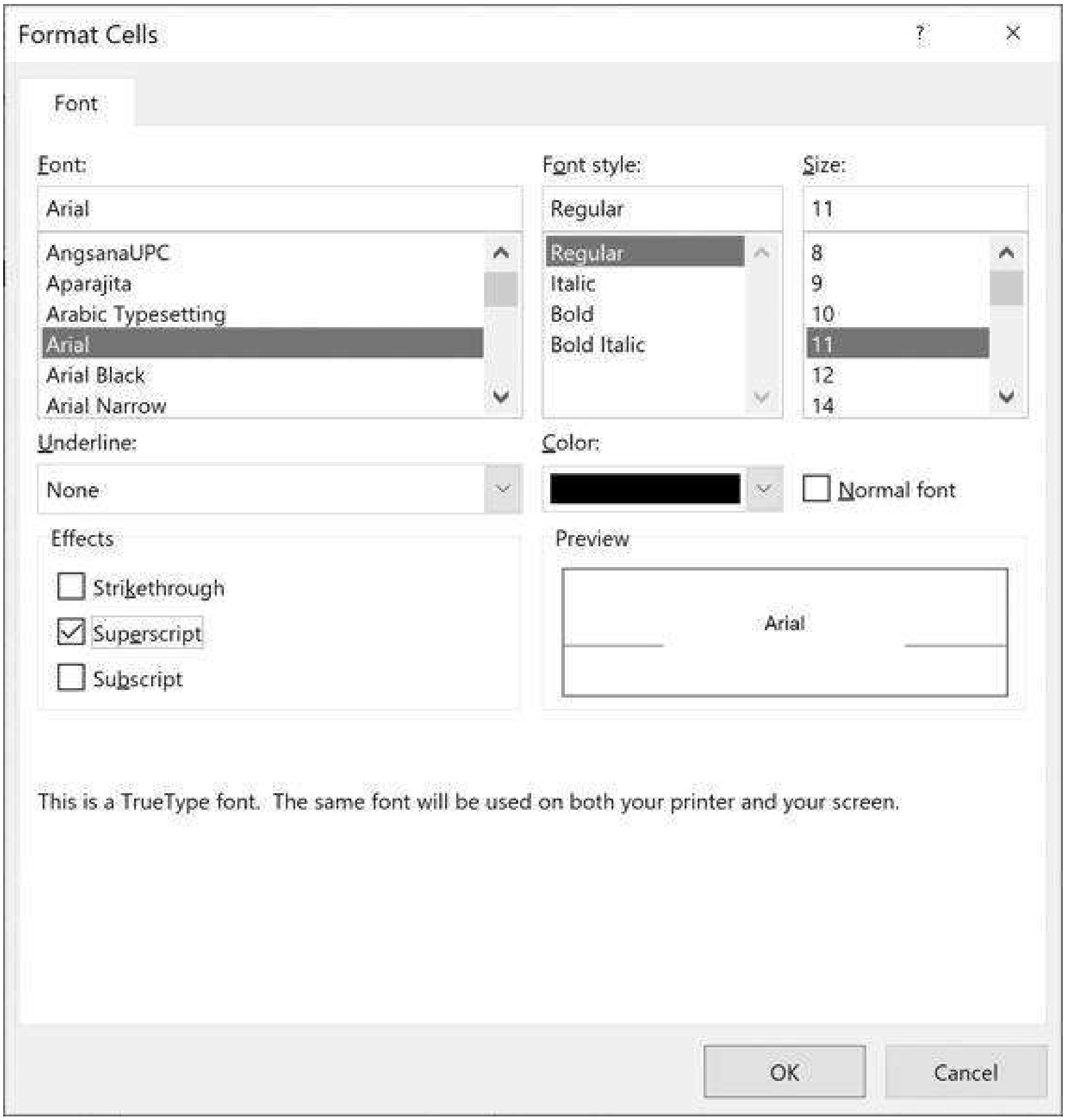
Pressing **[Enter]** produces the desired result.

	A	B	C	D
1	**GREEK AND SYMBOLS IN CELLS**			
2	Initial stock price	30		
3	Mean, μ	15%		
4	Standard deviation, σ	20%		
5	Delta, Δt	0.004		

32.11 Superscripts and Subscripts

It is not a problem to type subscripts or superscripts in Excel. Enter text into a cell, and then mark the letters you want to turn into a subscript or superscript:

Now, go to **Format Cells** and check the **Superscript** box:



Here’s the result:

	A	B	C	D	E	F
1	SUPERSCRIPT AND SUBSCRIPT IN CELLS					
2	x^2					
3						
4	x^2_i		Cannot put superscript and			
5			subscript one above the other			

As you can see in cell A4 above, you cannot put a subscript and a superscript on the same letter. That is, you cannot create x_i^2.

32.12 Named Cells

It is sometimes useful to give a name to a cell. Here's an example:

	A	B	C
1	NAMED CELLS		
2	Income	15,000	
3	Tax rate	33%	
4	Tax paid	4,950	<-- =B3*B2

We want to refer to cell B3 by the name "tax." To do this, we mark the cell and then go to the name tab on the toolbar:

	A	B	C	D
1	NAMED CELLS			
2	Income	15,000		
3	Tax rate	33%		
4	Tax paid	4,950	<--=B3*B2	

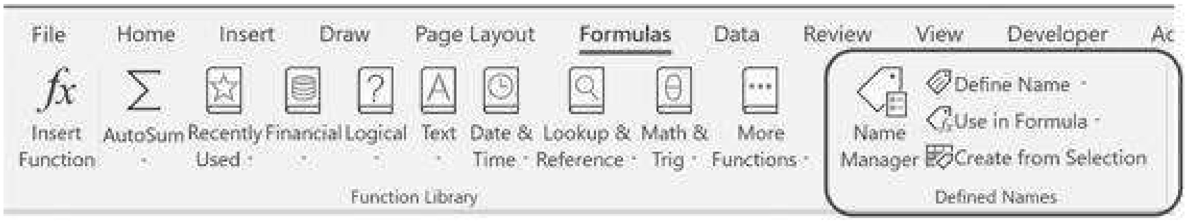
Typing in the word "tax" on the highlighted B3 allows us to reference B3 by this name anywhere in the Excel notebook:

	A	B	C
1	NAMED CELLS		
2	Income	15,000	
3	Tax rate	33%	
4	Tax paid	4,950	<--=tax*B2

Sometimes Excel lets us use cell names without ever actually going through the procedure just described. In the next example, Excel lets us use the column headers as cell names:

	A	B	C	D
7	Sales	Margin	Profit	
8	1000	20%	200	<-- =Sales*Margin
9	5000	30%	1500	<-- =Sales*Margin

To manage the named cells, go to the **Name Manager** on the **Formulas** tab:

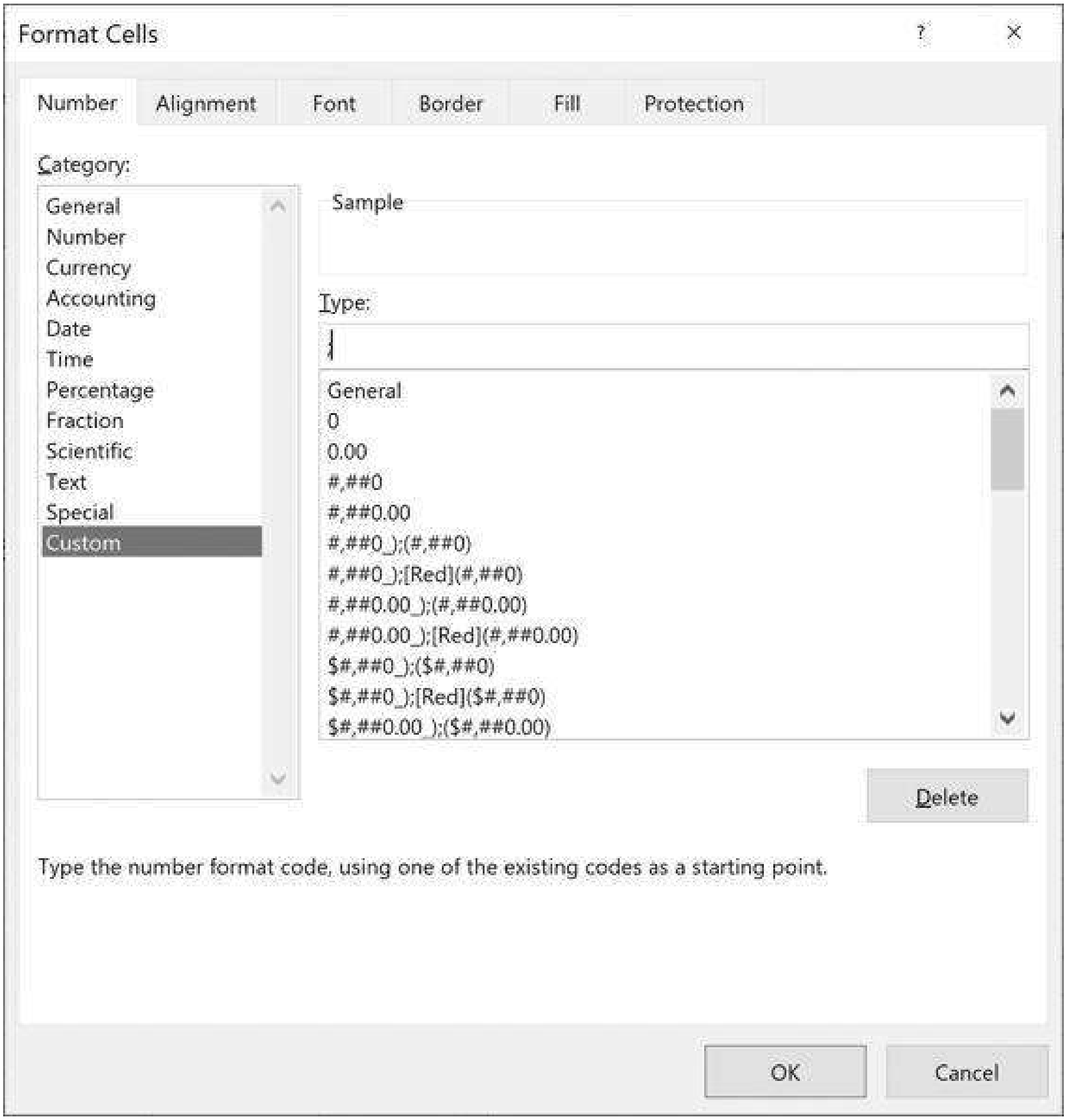


32.13 Hiding Cells (in Data Tables and Other Places)

In this text, we have often hidden the cell contents of data table headers. Here's a simple data table (this topic is discussed in detail in Chapter 28):

	A	B	C	D
1			**HIDING CELLS**	
2	Payment	100		
3	Number of payments	15		
4	Discount rate	15%		
5	Present value	$584.74		
6				
7			**PV of payments**	
8	**Data table**		584.74	<--=B5 data table header
9		0%	1,500.00	
10		3%	1,193.79	
11		6%	971.22	
12		9%	806.07	
13		12%	681.09	
14		15%	584.74	
15		18%	509.16	
16		21%	448.90	

The data table header in cell C8 is necessary for the table to work, but it is ugly and may be confusing if the table is copied into other documents. To hide the contents of C8, mark the cell and go to the **Format Cells** menu (or click on the right mouse button):



In the **Number|Custom|Type** box we have put in a semicolon. This preserves the cell contents but prevents them from being seen. Now when you copy the cells, this is the way they will appear:

	A	B	C	D
1	HIDING CELLS			
2	Payment	100		
3	Number of payments	15		
4	Discount rate	15%		
5	Present value	$584.74		
6				
7			PV of payments	
8	Data table			<--=B5 data table header
9		0%	1,500.00	
10		3%	1,193.79	
11		6%	971.22	
12		9%	806.07	
13		12%	681.09	
14		15%	584.74	
15		18%	509.16	
16		21%	448.90	

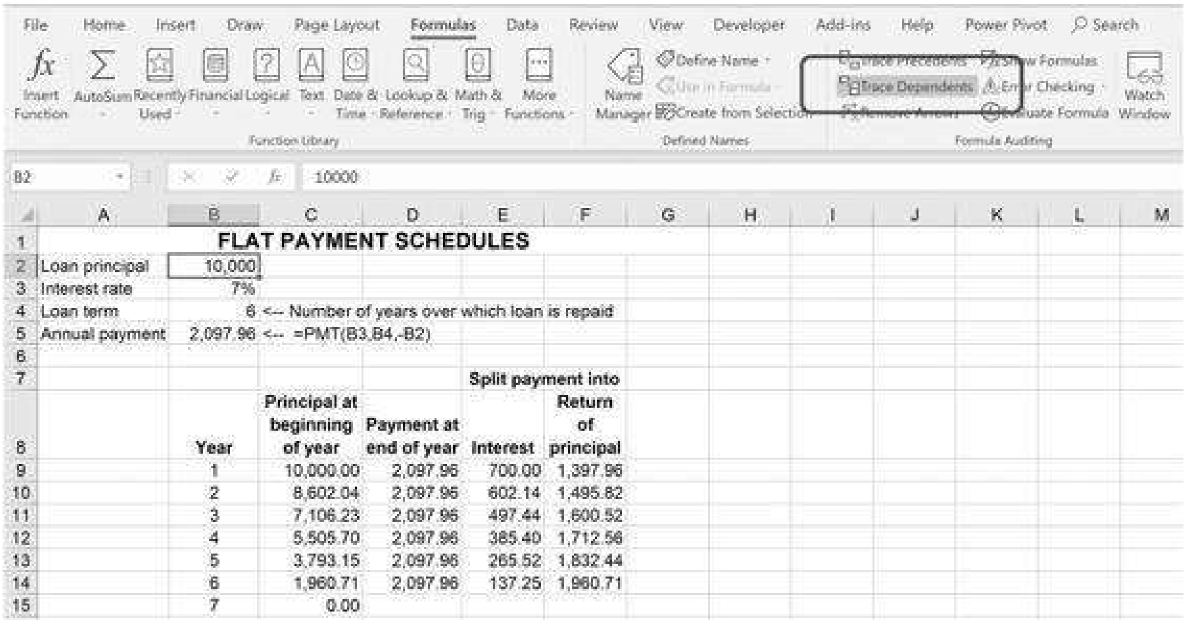
Note the comment in cell D8: We advise you always to *annotate* your spreadsheet so that when you come back to it after a few weeks/months, you will know that cell C8 really does have something in it!

One final note: To hide a cell that contains a reference to another cell containing a formula, three semicolons (;;;) may be necessary. In the spreadsheet below, cell B4 contains the function **IF**. Cell B6 refers to this cell. To hide B6, we use three semicolons instead of one in the **Format Cells|Number|Custom|Type.**

	A	B	C
1	HIDING A CELL WHICH REFERS TO A FORMULA		
2	a	33	
3	b	8	
4	c	bbb	<-- =IF(B2+B3<15,"aaa","bbb")
5			
6	Cell to be hidden -->		<-- =B4

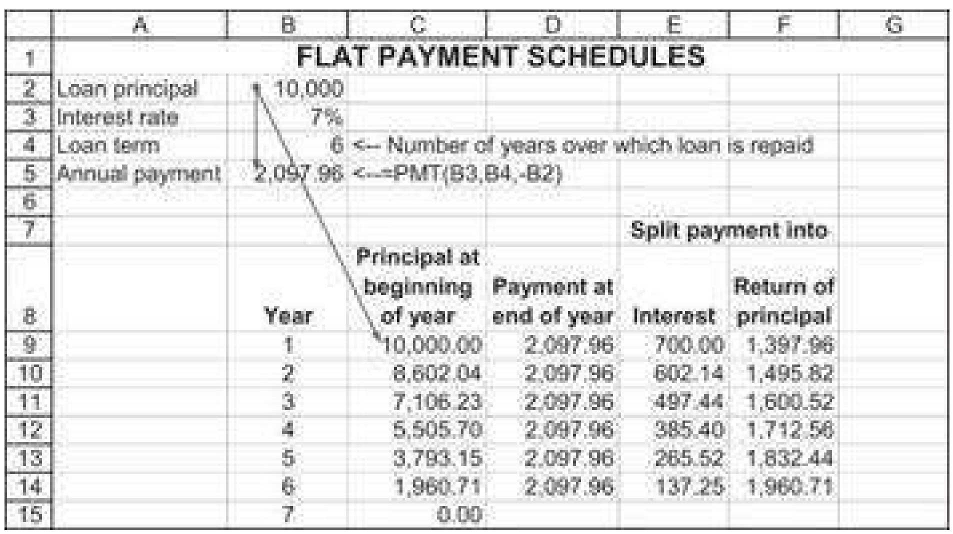
32.14 Formula Auditing

Excel can tell you where you have used a cell in your formulas and which cells a particular formula depends on. Clicking on **Formula| Trace Dependents** allows you to do this:



	A	B	C	D	E	F
1		FLAT PAYMENT SCHEDULES				
2	Loan principal	10,000				
3	Interest rate	7%				
4	Loan term	6	<-- Number of years over which loan is repaid			
5	Annual payment	2,097.96	<-- =PMT(B3,B4,-B2)			
6						
7					Split payment into	
8		Year	Principal at beginning of year	Payment at end of year	Interest	Return of principal
9		1	10,000.00	2,097.96	700.00	1,397.96
10		2	8,602.04	2,097.96	602.14	1,495.82
11		3	7,106.23	2,097.96	497.44	1,600.52
12		4	5,505.70	2,097.96	385.40	1,712.56
13		5	3,793.15	2,097.96	265.52	1,832.44
14		6	1,960.71	2,097.96	137.25	1,960.71
15		7	0.00			

Here’s the result:



	A	B	C	D	E	F
1		FLAT PAYMENT SCHEDULES				
2	Loan principal	10,000				
3	Interest rate	7%				
4	Loan term	6	<-- Number of years over which loan is repaid			
5	Annual payment	2,097.96	<--=PMT(B3,B4,-B2)			
6						
7					Split payment into	
8		Year	Principal at beginning of year	Payment at end of year	Interest	Return of principal
9		1	10,000.00	2,097.96	700.00	1,397.96
10		2	8,602.04	2,097.96	602.14	1,495.82
11		3	7,106.23	2,097.96	497.44	1,600.52
12		4	5,505.70	2,097.96	385.40	1,712.56
13		5	3,793.15	2,097.96	265.52	1,832.44
14		6	1,960.71	2,097.96	137.25	1,960.71
15		7	0.00			

In a similar way we can check to see which cells are precedents of a particular cell:

	A	B	C	D	E	F	G
1	FLAT PAYMENT SCHEDULES						
2	Loan principal	10,000					
3	Interest rate	7%					
4	Loan term	6	<-- Number of years over which loan is repaid				
5	Annual payment	2,097.96	<--=PMT(B3,B4,-B2)				
6							
7					Split payment into		
8		Year	Principal at beginning of year	Payment at end of year	Interest	Return of principal	
9		1	10,000.00	2,097.96	700.00	1,397.96	
10		2	8,602.04	2,097.96	602.14	1,495.82	
11		3	7,106.23	2,097.96	497.44	1,600.52	
12		4	5,505.70	2,097.96	385.40	1,712.56	
13		5	3,793.15	2,097.96	265.52	1,832.44	
14		6	1,960.71	2,097.96	137.25	1,960.71	
15		7	0.00				

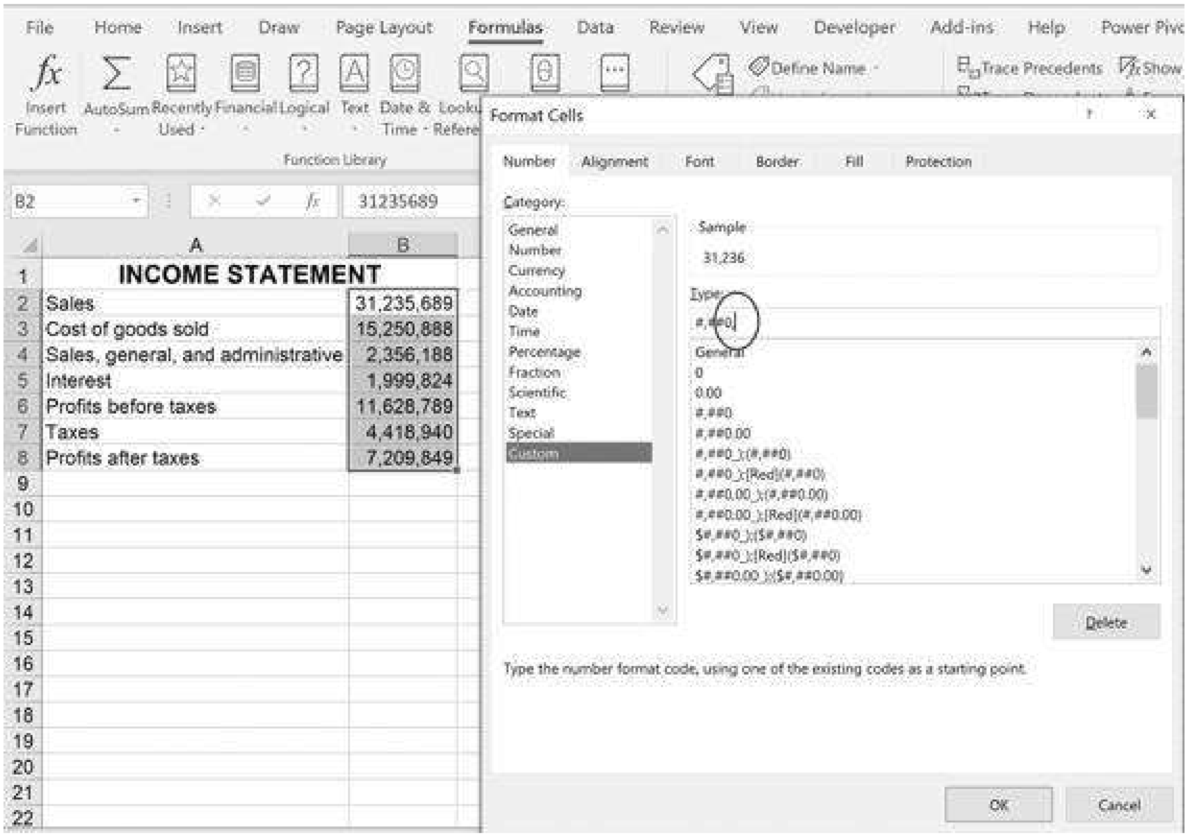
Formula auditing can help you implement a general rule of good spreadsheet writing: You should try to avoid cells that don't have either precedents or dependents.

32.15 Formatting Millions as Thousands

By using **Format Cells|Custom** you can change millions into thousands. To see where this is handy, consider the following income statement:

	A	B
1	INCOME STATEMENT	
2	Sales	31,235,689
3	Cost of goods sold	15,250,888
4	Sales, general, and administrative	2,356,188
5	Interest	1,999,824
6	Profits before taxes	11,628,789
7	Taxes	4,418,940
8	Profits after taxes	7,209,849

We want to make the income statement appear in thousands (in other words, instead of 31,235,689 we will see 31,236). Here's how this can be done:



In the **Type** box we have indicated **#,##0,**. The comma at the end indicates that we want Excel to drop the last three digits in the number (and round the number). This is merely a formatting change—the actual numbers are not changed: In cell B10 in the output below, we've multiplied the Sales by 2; the result is 62,471,378.

	A	B	C
1	INCOME STATEMENT		
2	Sales	31,236	
3	Cost of goods sold	15,251	
4	Sales, general, and administrative	2,356	
5	Interest	2,000	
6	Profits before taxes	11,629	
7	Taxes	4,419	
8	Profits after taxes	7,210	
9			
10	The cells retain their values	62,471,378	<--=B2*2

Adding another comma to the **Type** box (that is, **#,##0,,**) will drop another three digits.

32.16 Excel's Personal Notebook: Automating Frequent Procedures

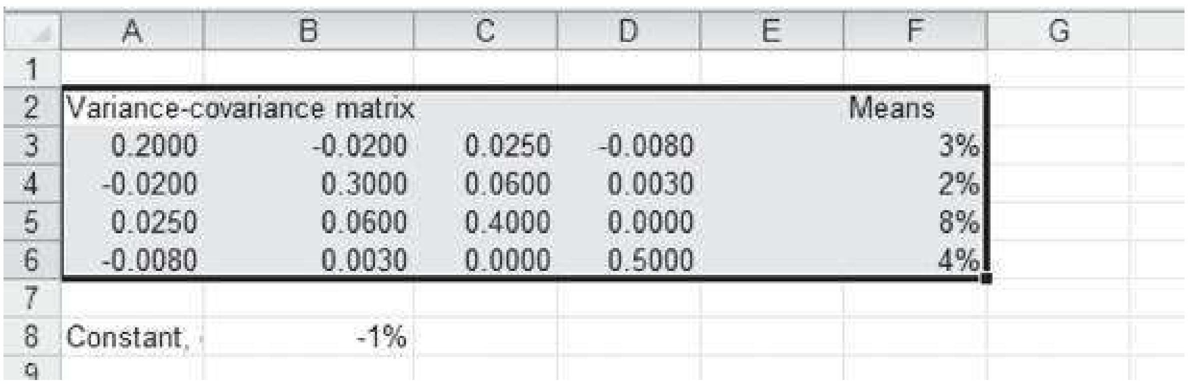
Excel's personal notebook allows you to save macros and procedures that only you can access. We give two examples of such procedures:

- • We explain Excel's **Copy as Picture** feature and show how to attach this to a macro stored in your personal notebook. This greatly simplifies copying from Excel and pasting into Microsoft Word (all the copy/pastes in this book were made this way).
- • We explain how to save number formatting in the personal notebook.

Using the Copy as Picture Feature of Excel

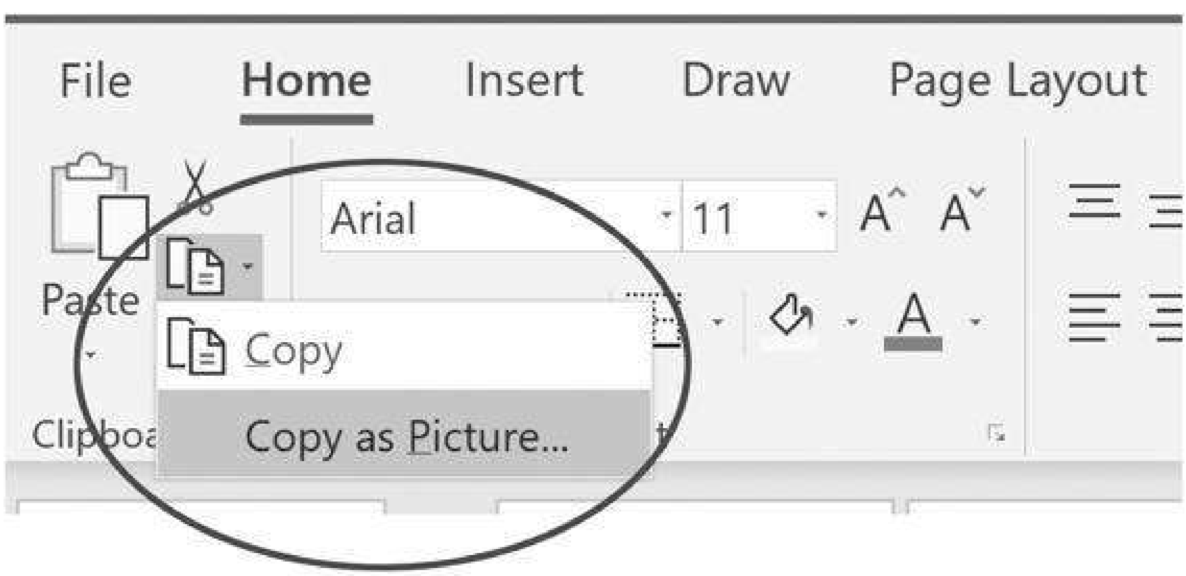
Excel contains a very nice way to copy from Excel as a picture. This is useful for embedding pictures of Excel spreadsheets into Word without a link. Here's the way it works:

1. In Excel, indicate the selection you want to copy:

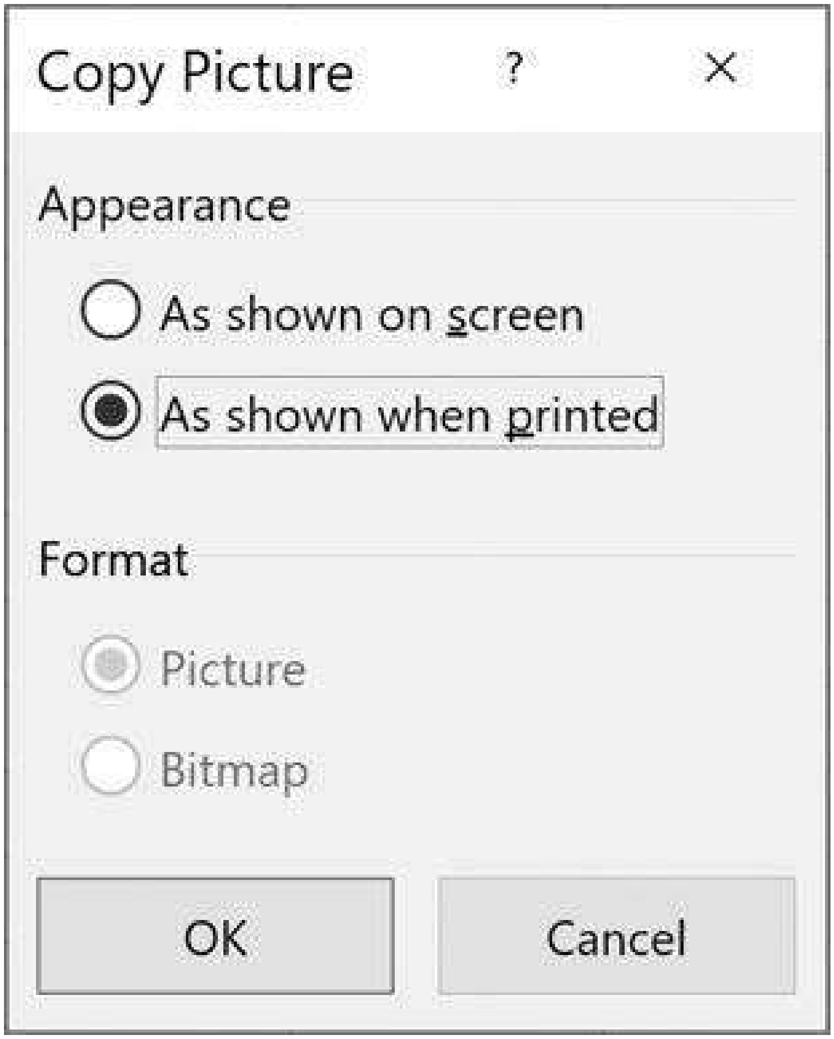


	A	B	C	D	E	F	G
1							
2	Variance-covariance matrix					Means	
3	0.2000	-0.0200	0.0250	-0.0080		3%	
4	-0.0200	0.3000	0.0600	0.0030		2%	
5	0.0250	0.0600	0.4000	0.0000		8%	
6	-0.0080	0.0030	0.0000	0.5000		4%	
7							
8	Constant,	-1%					

2. On the **Home** tab, go to **Copy|Copy as Picture**:



3. Indicate **As shown when printed**:



4. You can now go to any other program (Microsoft Word/ PowerPoint/ Outlook) and select Paste. The result: You get a picture of the spreadsheet.

	A	B	C	D	E	F
2	Variance-covariance matrix					Means
3	0.2000	-0.0200	0.0250	-0.0080		3%
4	-0.0200	0.3000	0.0600	0.0030		2%
5	0.0250	0.0600	0.4000	0.0000		8%
6	-0.0080	0.0030	0.0000	0.5000		4%

Automating the Procedure

We want to automate this procedure:

- • Turn it into a macro.
- • Attach a key sequence (in this case, **[Ctrl] + w**) to the macro.
- • Make the macro and key sequence available in all your Excel spreadsheets.

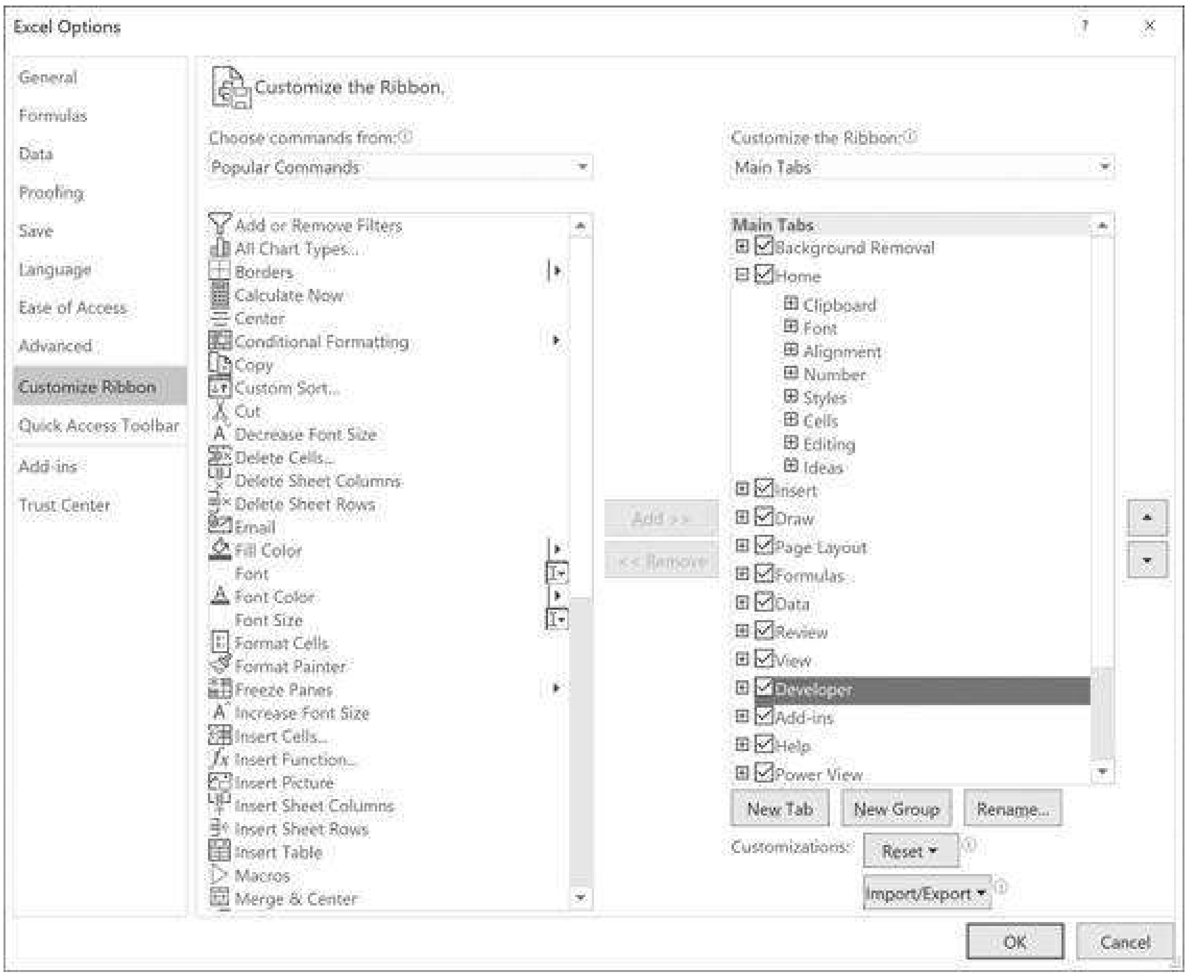
To do this, you have to create a **Personal.xlsb** file. This file is hidden but activates each time you start Excel. It's yours only—other readers of your spreadsheets won't see it.

Here are the steps:

- • Activate the **Developer** tab on the menu bar.
- • Use **Record Macro** to save a macro as a personal notebook.
- • Edit the personal notebook with what you want.

Activate the Developer Tab

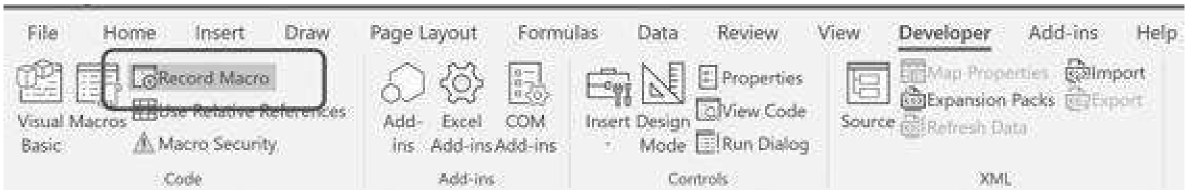
Go to **File|Options|Customize Ribbon** and activate the **Developer** tab as shown below:



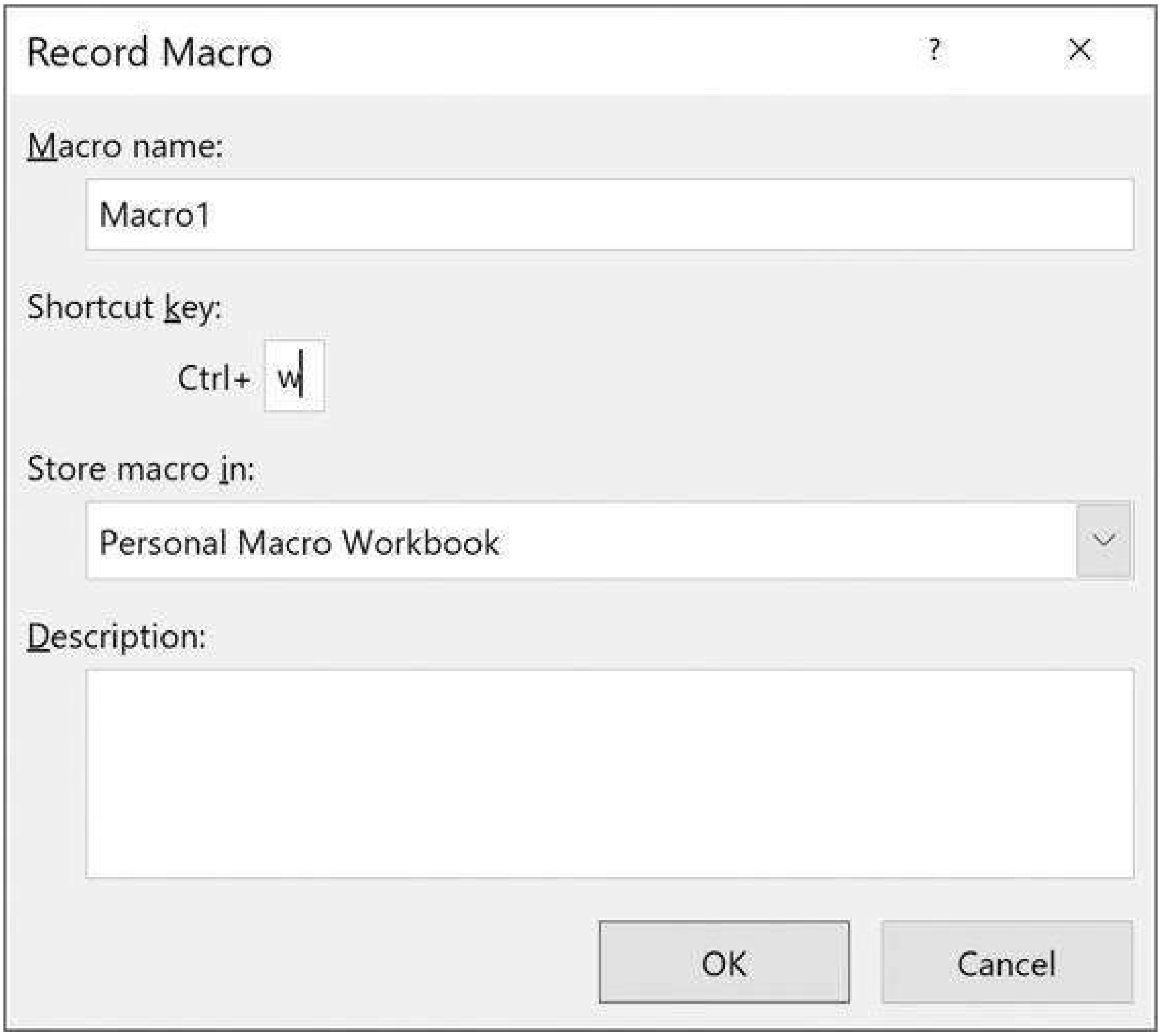
Use Record Macro

The **Developer** tab allows you to record a macro and save it as part of the **Personal.xlsb** notebook. We will illustrate with the **Copy as Picture** feature.

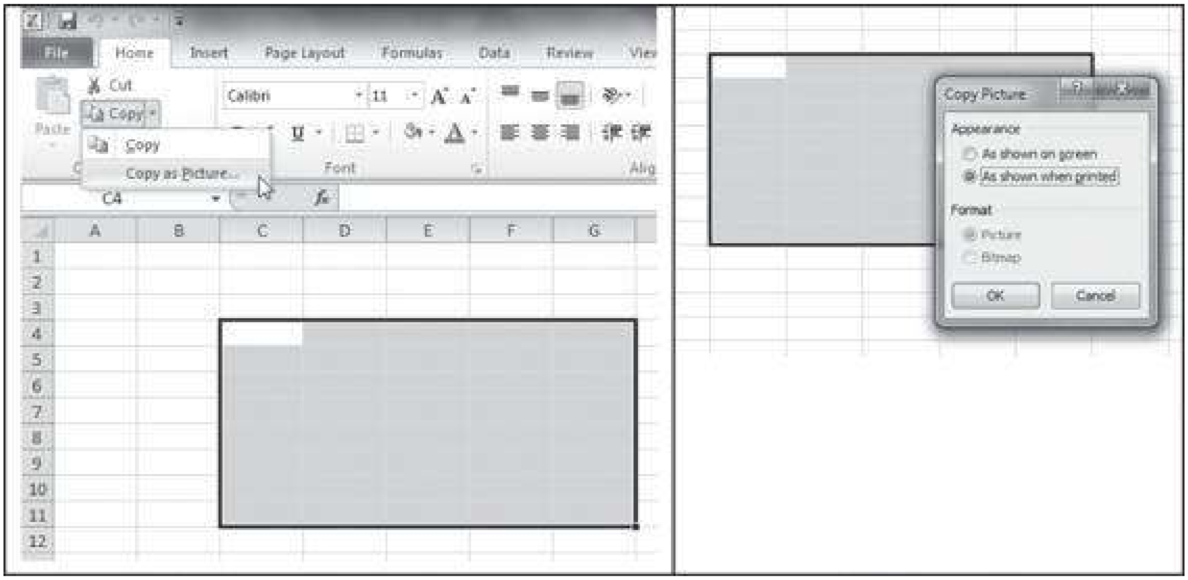
1. Open a blank Excel notebook and click on the **Developer** tab and then on **Record Macro**, as shown:



Excel will ask for details of the recording. Here's what we wrote. Note that we are saving this as a **Personal Macro Workbook** and that we are using the shortcut keystrokes **[Ctrl] + w**:



2. Now go to the **Home** tab, mark an area of the spreadsheet, and go through the whole **Copy as Picture** feature:



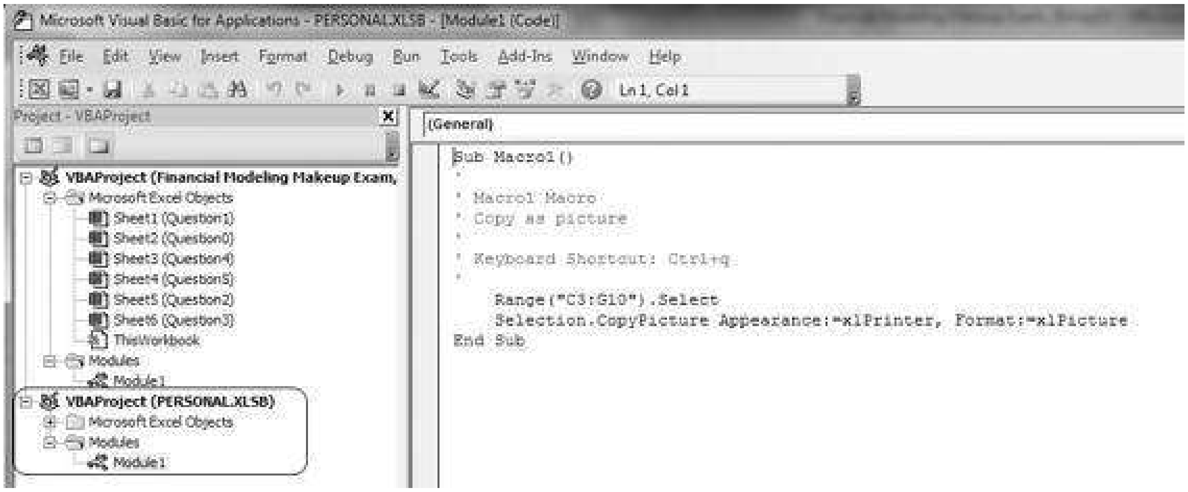
3. Go back to the **Developer** tab and stop the recording:

4. Close down Excel. Excel will ask you if you want to save the Personal notebook. The answer is, of course, yes.

This creates a PERSONAL.XLSB in your Excel folder (usually under: C:\Users\ … \AppData\Roaming\Microsoft\Excel\XLSTART).

Edit the Personal Notebook

Now that you have saved the **Personal.xlsb** file, you may want to edit it. Open any Excel file, hit **[Alt] + [F11]** to go the VBA editor. Note that the Personal file is also there:



We have edited this by deleting the line beginning with **`Range("C3:G10")`**:

```
Sub Macro1()
'
' Macro1 Macro
' Copy as picture
'
' Keyboard Shortcut: Ctrl+q
    Selection.CopyPicture Appearance:=xlPrinter,
    Format:=xlPicture
End Sub
```

Using the Macro

From now on, whenever you open a file on *your computer*, you can use [**Ctrl**] + **q** to copy a region as a picture. Note that this feature works only on your machine because it's in your personal notebook.

32.17 Quick Number Formatting

We often want to format numbers with comma separators and without decimals. For example:

	A	B
1	1356.001	1987.398
2	3387.3	4458.98

To make the formatting of these numbers uniform:

- • Mark the numbers.

- • Press **[CTRL] + [SHIFT] + [1].** This will format the cells with a thousand separator and two decimal digits.

The result is:

	A	B
1	1,356.00	1,987.40
2	3,387.30	4,458.98

You can use the same trick to format numbers as time by pressing **[CTRL] + [SHIFT] + [2]**; as dates by pressing **[CTRL] + [SHIFT] + [3]**; as currency by pressing **[CTRL] + [SHIFT] + [4]**; and as percentages by pressing **[CTRL] + [SHIFT] + [5]**. The spreadsheet below illustrates the results.

	A	B	C
1	**QUICK FORMAT CELLS**		
2	Cell value	853.678543	
3			
4	**Different formatting results by pressing:**		
5	[CTRL]+[SHIFT]+[1]	853.68	<--=$B$2
6	[CTRL]+[SHIFT]+[2]	16:17	<--=$B$2
7	[CTRL]+[SHIFT]+[3]	02-May-1902	<--=$B$2
8	[CTRL]+[SHIFT]+[4]	$853.68	<--=$B$2
9	[CTRL]+[SHIFT]+[5]	85368%	<--=$B$2

33 Essentials of R Programming

This chapter does not try in any way to be a comprehensive discussion about how to work with R. Since R is an open code program, and since it is so common, there are excellent sources on the web to do so.[1]

In this book, we use RStudio to do the programming in R. We highly recommend you get used to working with it because RStudio is also the industry standard.[2]

33.1 Rule #1: Use the Provided Help for R Functions

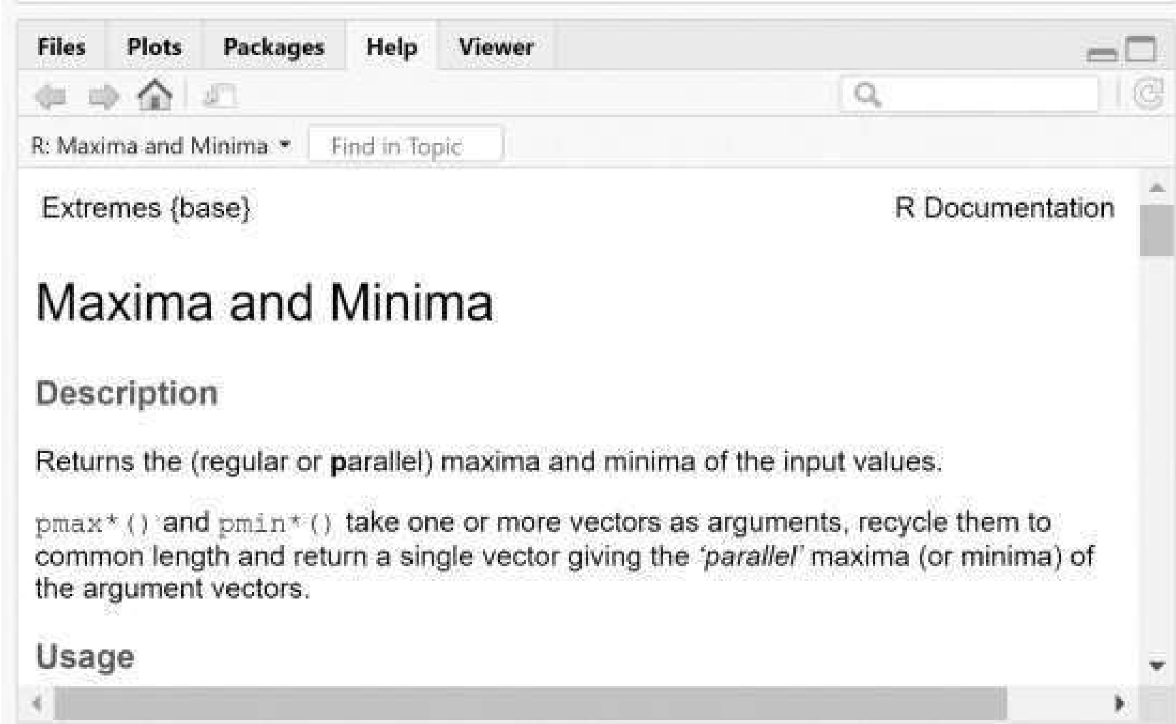
Many times, while using R you will have to use new and unfamiliar functions. Simply by adding a “?” before each function, R will return a detailed description of the function and how to use it. For example, look at the max function below:

```
5  #33.1  using ? to get help on formula
6  max(10,12) # stand on this line and click [ctrl]+[enter] to run
7  ?max # stand on this line and click [ctrl]+[enter] to get help
```

When we run line 6 we get:

```
> max(10,12)
[1] 12
```

And when we run line 7 we get:



This functionality of R is extremely popular and is highly recommended for everyone.

33.2 Installing a Package

Probably the best part of using R is that it is open source, so there are many contributors to its content. This means that we can use functions other people wrote. R offers an enormous number of such packages and all you need to do in order to use them is to activate them. Throughout this book we have tried, to the best of our ability, not to use external packages because we had a didactic goal in mind. However, practitioners use them all the time.

If you want to activate a package and load it wherever you want to use it, simply type:

- • install.packages("package-name")—to install a package in R. if you know the package name, just put it in quotation marks within parentheses.
- • library(package-name)—to load the package whenever you want to use it.

So, if we would like to install and use the very famous plotting package (called ggplot), the code will look like this:

```
#Installing and using a package (ggplot)
install.packages("ggplot2") # click [ctrl]+[enter] to install
library("ggplot2") # click [ctrl]+[enter] to load
```

33.3 Setting a Default Folder (Working Directory)

When working with external data, it is important to define the working directory first. Setting a default folder in R is called setting a *working directory*, and it is done by using:

```
setwd("C:\\Users\\mofka")
```

```
13  # setting a working directory
14  setwd("C:\\Users\\Mofka") #  "\\" is needed to specify sub-directory
```

Although this approach would work, it is extremely annoying since we need to add two "\" lines to specify sub-directories.

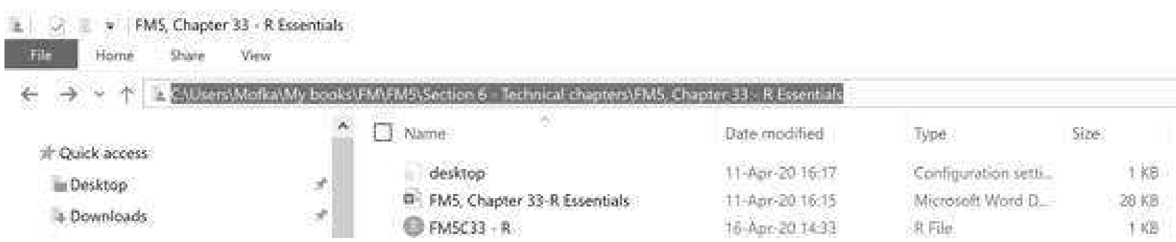
In this book we use a different and more flexible approach. We are prompting the user to enter the file path using **`readline(prompt="working directory?")`**. It looks like this:

```
16  # Set working directory (better approach)
17  workdir <-readline(prompt="working directory?") # paste path below
18  setwd(workdir) #click [ctrl]+[enter] to run
```

We then get the following prompt in the console below:

```
> # Set working directory (better approach)
> workdir <-readline(prompt="working directory?")
working directory?
```

All we need to do is to copy the folder address (using file explorer):



Then we paste the folder address next to the prompt question:

```
working directory?C:\Users\Mofka\My books\FM\FM5\Section 6 - Technical chapters\FM5, Chapter 33 - R
                Essentials
```

And click [Enter]. R automatically replaces the "\" with "\\" as it sets the workdir variable. Then we would simply click [CTRL] + [ENTER] on line 18 in our example to set the working directory.

33.4 Understanding Data Types in R

Here are the basic types of data that R allows you to store:[3]

- • **Numeric**: a decimal number, which is the default data type for numbers
- • **Integer**: a round number (with no leading zeros)
- • **Complex**: a complex number defined by the value i
- • **Logical**: a Boolean object of either TRUE (also T and 1) or FALSE (also F and 0)
- • **Character:** a text object

R allows you to store data in various structures. Here are the basic structures, in a nutshell:

- • **Vector:** an indexed array of a single data type (could be any data type)—defined by c()
- • **List**: an indexed array of all data types and data structures (including tables or even other lists)—defined by list()
- • **Matrix**: a table of a single data type—defined by matrix()
- • **Data frame**: a table of all data types (unconstrained)—defined by data.frame()
- • **Factors**: an array of categorized textual data—defined by factor()

33.5 How to Read a Table from a CSV File

To read a csv file, we use the **read.csv()** function. Here are some of the function's basic elements:

- **`file`**—the file's name. The file must be in the working directory (see section 33.3).
- **`header`**—indicates whether the table has headers or not (True or False).
- **`stringsAsFactors`**—should be True when the table stores text as a Factors data type. Factors data types are good for memory efficiency, but they can cause some errors if you are not familiar with how to treat them.
- **`colClasses`**—define the data type for each column.

Here is the example we used in Chapter 10 to read stocks data:

```
# read data from a csv file (taken from Chapter 10)
monthly_prices <- read.csv("Chapt_10_data.csv", row.names = 1, stringsAsFactors = FALSE,
                           colClasses=c("character", "double", "double"))
head (monthly_prices) # head presents the first part of the table
```

In this example we have defined a table (data frame) called monthly_prices. This table loads all the price data from the table in the csv file. This is how the results look:

```
> # read data from a csv file (taken from chapter 10)
> monthly_prices <- read.csv("chapt_10_data.csv", row.names = 1, stringsAsFactors = FALSE,
+                            colClasses=c("character", "double", "double"))
> head (monthly_prices) # head presents the first part of the table
            AAPL     K
01-Dec-18 157.07 56.45
01-Nov-18 177.20 62.46
01-Oct-18 217.17 64.26
01-Sep-18 223.99 68.72
01-Aug-18 225.08 69.90
01-Jul-18 188.16 69.16
> |
```

33.6 How to Directly Import Stock Price Data to R

Stock data can be obtained from a number of sources, including Yahoo (https://finance.yahoo.com/) or Google (https://www.google.com/finance/). The **quantmod** package provides easy access to those sources. **quantmod** also provides a number of other useful features for financial modeling, so we strongly advise you to play with it and explore it further. In the following example, we show how to extract the data that was saved as a csv file in Chapter 10.

```
##33.6  Import stock data from finance.yahoo directly
# Install quantmod package
  install.packages("quantmod")
  library(quantmod)

start_date <- as.Date("2008-12-01") # yyyy-mm-dd
end_date <- as.Date("2018-12-01") # yyyy-mm-dd

# getSymbols load data directly into the global environment.
price_Data <- new.env()
ticker_symbol <- c("AAPL","K")
getSymbols(ticker_symbol, src = "yahoo", from = start_date, to = end_date,
           periodicity = 'monthly', env=price_Data )

# collect the adjusted prices of all stocks to one table
Stocks_Adj_Data <- do.call(merge, eapply(price_Data, Ad))
```

33.7 Defining a Function

In many cases it is efficient and comfortable to create a function instead of rewriting the procedure many times. We used this approach a lot when simulating for option pricing in the Monte Carlo chapters. The structure of functions is shown below (note that the line breaks are needed):

```
function_name <- function(inputs){
    result <- f(inputs)
    return(result)
        }
```

Once the function is defined, we can call it simply by using function_name (inputs).

In the following two examples, we show how to implement the functions in R. We start with a simple example of how to convert temperatures from Fahrenheit to Celsius. Say we want to convert 100 degrees Fahrenheit. We would implement this conversion with the following equation: $celsius = (fahrenheit - 32) * \frac{5}{9}$.

```
# First example: Convert temperatures from Fahrenheit to Celsius
fahr_to_celsius <- function(temp) {
  celsius <- ((temp - 32) * (5 / 9))
  return(celsius)
          }
fahr_to_celsius(100)
```

And the result is as follows:

```
> fahr_to_celsius(100)
[1] 37.77778
>
```

In the second example, we want to calculate the present value perpetuity factor. To implement, we use the following equation: $pv_{factor} = \frac{1}{r-g}$.

```
# Second example: Calculate the PV perpetuity factor
pv_factor <- function(r, g) {
  pv_factor <- 1/(r-g)
  return(pv_factor)
          }
pv_factor(0.1, 0.02)
```

And the result is as follows:

```
> pv_factor(0.1, 0.02)
[1] 12.5
>
```

33.8 Plotting Data in R

Practitioners use the “ggplot2” package to plot data in R. However, the basic charts in R are quite good and sufficient for the examples in this book (again, we have tried to avoid using external packages as much as possible). However, we highly recommend using the ggplot2 package to plot data in R.

The basic (two-dimensional) plot in R is comprised of X values and Y values:

```
plot(x, y)
```

Additional parameters could be:

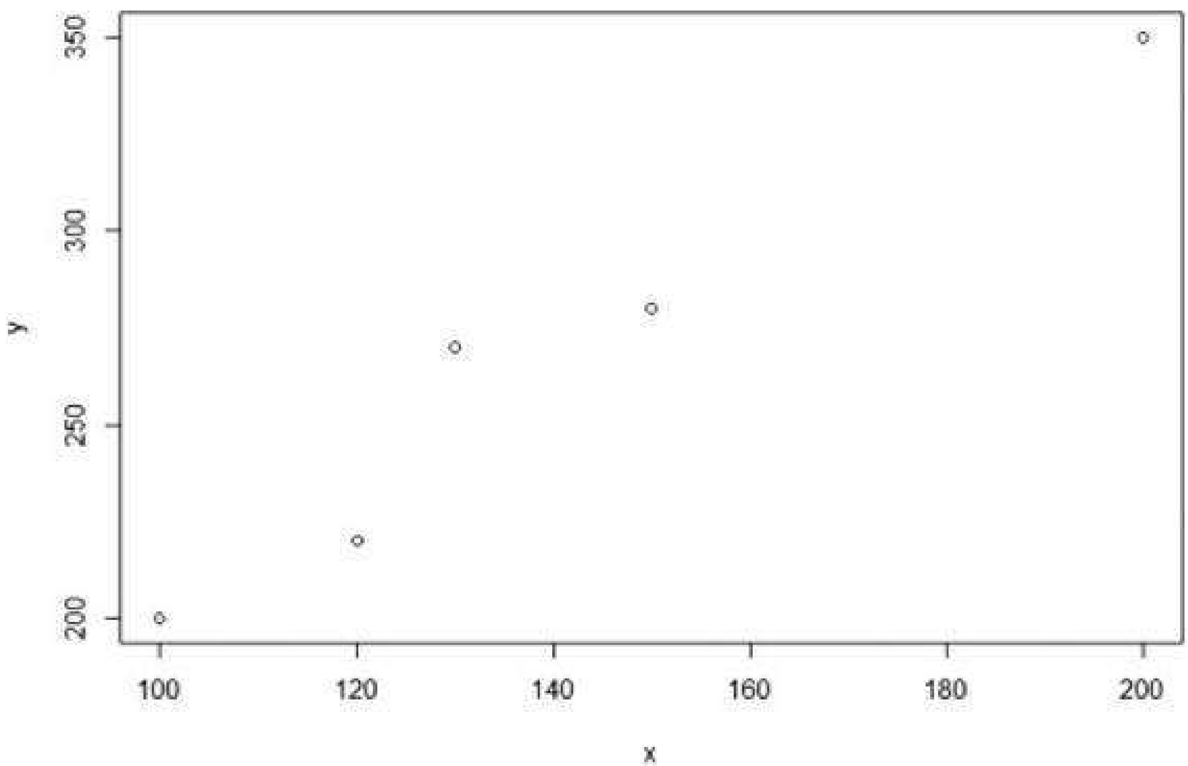
- **type**—type of plot, using “l” for line, “p” for scatterplot, “h” for histogram (columns) etc.
- **main**—main chart title
- **xlab**—X axes labels
- **ylab**—Y axes labels

- **`ylim`**—Y axes limits
- **`xlim`**—X axes limits

The simple example below tries to illustrate the use of **`plot()`** in R:

```
# Plotting data in R
x <- c(100, 120, 130, 150, 200)
y <- c(200, 220, 270, 280, 350)
plot(x,y)
```

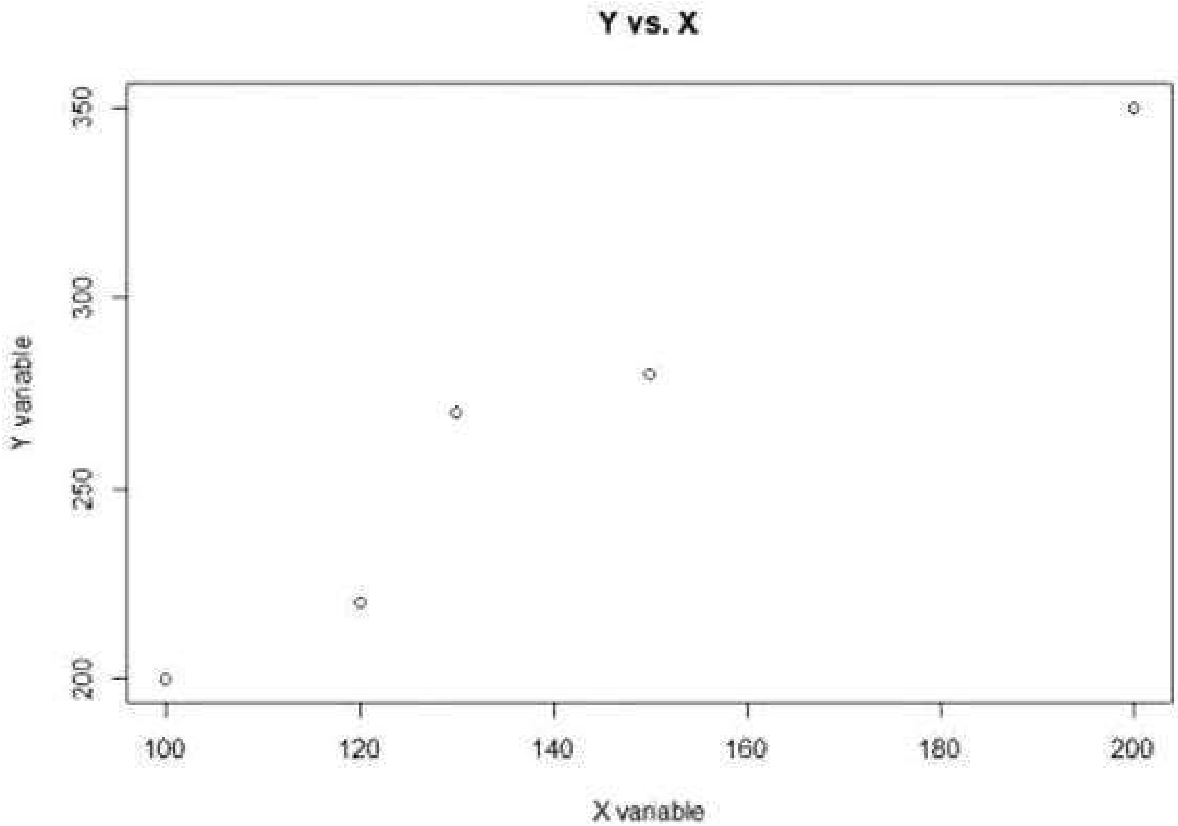
This example gives:



While the code

```
plot(x,y, main = "Y vs. X", xlab = "X variable", ylab = "Y variable")
```

Gives:



33.9 The apply Function

The apply function in R is very useful. When we have a table and want to perform a calculation on each row (or column) and not on the whole table, we use the apply function.

It is the R equivalent of creating a function in Excel and then "dragging" it for multiple rows or columns. The most basic structure of the apply function is as follows:

```
apply(X, MARGIN, FUN)
```

Here, **X** is an array or a table (matrix/ data frame), **MARGIN** is either rows (1) or columns (2), and **FUN** is the function to apply. For example:

```
apply(my_df, 1, sum)
```

In the example below we create a simple 3 × 3 matrix that look like this:

	V1	V2	V3
1	1	2	3
2	4	5	6
3	7	8	9

We then use the apply function to calculate the sums for rows and for columns:

```
#33.9  using the apply function
data_matrix <- matrix(c(1, 2, 3,
                        4, 5, 6,
                        7, 8, 9), nrow=3, byrow=TRUE)
row_sums <- apply(data_matrix, 1, sum)
colum_sums <- apply(data_matrix, 2, sum)
```

The results are:

```
> row_sums
[1]  6 15 24
> colum_sums
[1] 12 15 18
>
```

33.10 The lapply and sapply Functions

The "lapply" function is similar to the apply function but is performed on a list instead of an array or matrix. In practical terms, this means you can perform a function on multiple tables or other data structures. As a result, you will get a list. For example, we can take a list of two matrices and apply a rowSums function on each one of them:

```
#33.10  The lapply and sapply functions
data_matrix1 <- matrix(c(1, 2, 3,
                         4, 5, 6,
                         7, 8, 9), nrow=3, byrow=TRUE)

data_matrix2 <- matrix(c(2, 4, 6,
                         8, 10, 12,
                         14, 16, 18), nrow=3, byrow=TRUE)
# Create a list
data_list <- list(data_matrix1, data_matrix2)
lapply(data_list, rowSums)
```

The result is a list of two arrays:

```
> lapply(data_list, rowSums)
[[1]]
[1]  6 15 24

[[2]]
[1] 12 30 48
```

The “sapply” function is like “lapply,” but instead of getting a list as a result, you will get a matrix. For example, we can apply the “sapply” function on **`data_list`** as defined above:

```
87 sapply(data_list, rowSums)
```

This returns the following matrix:

```
> sapply(data_list, rowSums)
     [,1] [,2]
[1,]    6   12
[2,]   15   30
[3,]   24   48
```

---

1. 1. See, for example, Zuur, Ieno, and Meesters (2009).
2. 2. See https://rstudio.com/.
3. 3. For Excel users: Data type is somewhat similar to cell format.

Selected References

Note: This bibliography is not intended to be extensive. We give the references mentioned in *Financial Modeling* and occasionally add books and articles that may help readers expand their horizons to more advanced topics. Sometimes the same reference appears several times in different sections. Sometimes we give the references for a number of chapters together, and at other times we reference single chapters.

Chapters 1–5: Corporate Finance and Valuation

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